BITSAT: 50 Full Mocks

TEJAS200 | Private Review Edition

Each test: 130 main questions + 12 optional bonus questions.
Answer keys are grouped after Mock 50. Time: 180 minutes per test.

Independent correctness and difficulty validation remain incomplete.
This compilation does not certify 100% correctness or measured 10% harder.

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Bookmarks open individual tests. Printed page numbers restart within each mock.
Recorded small-grid print limits: M35 Q84, M37 Q139, M38 Q138. Zoom when needed.

BITSAT Mock 01

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A nonrelativistic particle of fixed charge and mass moves perpendicular to a uniform magnetic field. Its kinetic energy is multiplied by 16 and the magnetic field is multiplied by 7. Find the new trajectory radius divided by the original radius.

(A) $\frac{1}{7}$ (B) $\frac{4}{7}$

(C) 28 (D) $\frac{1}{28}$

2. A drop of mass 4.48×10^{-17} kg is suspended in a vertical field of magnitude 700 N/C. It has excess electrons. One excess electron is removed without changing its mass. Find its immediate downward acceleration, taking $g = 10\ \text{m/s}^2$ and $e = 1.6 \times 10^{-19}$ C.

(A) $\frac{10}{3}\ \text{m/s}^2$ (B) $\frac{5}{2}\ \text{m/s}^2$

(C) $\frac{5}{8}\ \text{m/s}^2$ (D) $2\ \text{m/s}^2$

3. Two bodies of masses 2 kg and 5 kg move toward each other at 10 m/s and 3 m/s and stick together. Half the lost kinetic energy heats the combined mass uniformly; each has specific heat 50 J/(kg K). Find temperature rise.

(A) $\frac{169}{280}$ K (B) $\frac{1}{392}$ K

(C) $\frac{169}{490}$ K (D) $\frac{169}{980}$ K

4. Water flows steadily from a pipe section of area A at 3 m/s to a section of area $A/4$ located 4 m higher. Neglect viscosity. Find $p_1 - p_2$. Take density 1000 kg/m^3 and $g = 10\ \text{m/s}^2$.

(A) 27500 Pa (B) $\frac{143125}{4}$ Pa

(C) 107500 Pa (D) 112000 Pa

5. A projectile of mass 12 kg is launched with speed 40 m/s, $\cos\theta = 3/5$. At its highest point it splits into masses $M/3$ and $2M/3$. The lighter fragment moves horizontally forward at 30 m/s. Find the signed horizontal velocity of the heavier fragment immediately after the explosion.

(A) 45 m/s (B) 51 m/s

(C) 42 m/s (D) 21 m/s

6. Two masses 4 kg and 3 kg are connected by a taut string over a fixed, smooth pulley and accelerate freely. The string has uniform linear density μ, but its total mass is negligible compared with either attached mass. A second string has linear density 3μ and is held at a tension equal to the weight of a 1 kg mass. Find the ratio of the transverse wave speeds relative to their respective strings.

(A) 3 (B) $2\sqrt{3}$

(C) $\frac{72}{7}$ (D) $\frac{6\sqrt{14}}{7}$

7. A block released from rest takes time t_s to slide a fixed distance down a smooth incline and time $t_r = (7/2)t_s$ down an otherwise identical rough incline. The incline has $\tan\theta = \frac{4}{3}$. Find the kinetic-friction coefficient.

(A) $\frac{60}{49}$ (B) $\frac{135}{196}$

(C) $\frac{20}{21}$ (D) 15

8. A plane wave passes through a thin prism away from edges. Separately, a pointlike aperture produces an outgoing spherical wavefront of radius 7 cm, which propagates another 11 cm in a homogeneous medium. Which pair correctly describes the prism wavefront and the later aperture wavefront?

(A) Approximately plane and unchanged in direction; spherical with radius 7 cm.

(B) Spherical; plane with infinite radius.

(C) Approximately plane and tilted; spherical with radius 18 cm.

(D) Approximately plane and tilted; spherical with radius 4 cm.

9. A source emitting 400 Hz and a listener approach each other, each at 50 m/s relative to still air. Sound speed is 300 m/s. Find heard frequency before they pass.

(A) 560 Hz (B) 480 Hz

(C) 600 Hz (D) $\frac{1400}{3}$ Hz

10. A platform of moment of inertia 3 kg m^2 carries a person of mass 4 kg at radius 3 m and rotates at 6 rad/s. The person walks to half the stated initial radius; external torque is zero. Find final angular speed. Treat the person as a point mass.

(A) 78 rad/s (B) $\frac{39}{2}$ rad/s

(C) 24 rad/s (D) $\frac{24}{13}$ rad/s

11. The 4th harmonic of an open pipe of length 4 m equals the 4th allowed resonance of a closed pipe. Both have the same sound speed and negligible end correction. Find the closed-pipe length.

(A) 2 m (B) $\frac{5}{2}$ m

(C) $\frac{9}{2}$ m (D) $\frac{7}{2}$ m

12. A balanced Wheatstone bridge has arms AB=AD=6 ohms and BC=DC=12 ohms. A resistor of arbitrary positive resistance connects B to D. Find equivalent resistance between A and C.

(A) 9 Ω (B) 18 Ω

(C) 15 Ω (D) 12 Ω

13. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 20 percent. The measured increase in its absolute temperature is 132 K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) 330 K (B) 3300 K

(C) 300 K (D) 132 K

14. A parallel-plate capacitor has vacuum capacitance C_0. A dielectric of constant 3 fills a fraction $\frac{1}{2}$ of the plate area and the entire gap in that region; the remainder stays vacuum. The voltage is fixed. Find the fraction of total stored energy residing in the dielectric-filled region.

(A) $\frac{1}{2}$ (B) 3

(C) $\frac{3}{4}$ (D) $\frac{1}{4}$

15. A standing wave is $y = 2\sin(3x)\cos(4t)$ cm with x in metres and t in seconds. At a point where $3x = \pi/6$, find the particle-speed magnitude when $4t = \pi/3$.

(A) $\frac{3\sqrt{3}}{2}$ cm/s (B) $2\sqrt{3}$ cm/s

(C) 8 cm/s (D) $\frac{\sqrt{3}}{2}$ cm/s

16. A galvanometer has resistance 20 ohms and full-scale current $\frac{1}{250}$ A. It is to measure a total current 4 times its own full-scale current. Find the required parallel shunt resistance.

(A) 5 Ω (B) 80 Ω

(C) $\frac{20}{3}$ Ω (D) 20 Ω

17. A uniformly charged nonconducting sphere has radius R and surface potential V_s relative to infinity. Find potential at $r = R/4$ divided by V_s.

(A) $\frac{47}{32}$ (B) $\frac{3}{2}$

(C) $\frac{1}{4}$ (D) $\frac{15}{16}$

18. A straight smooth wire descends at 30° to horizontal through a vertical height 7 m. A bead is released from rest at the top. Find its time to reach the bottom. Take $g = 10$ m/s^2.

(A) $\frac{\sqrt{70}}{10}$ s (B) $\frac{4\sqrt{35}}{5}$ s

(C) $\frac{\sqrt{35}}{5}$ s (D) $\frac{2\sqrt{35}}{5}$ s

19. A metal wire has linear temperature coefficient $\frac{1}{200}$ per degree Celsius referred to 0°C. Its dimensions are fixed and it stays across a constant voltage. Find power at 70°C divided by power at 0°C.

(A) $\frac{20}{13}$ (B) $\frac{20}{27}$

(C) $\frac{13}{20}$ (D) $\frac{400}{729}$

20. A plane wave in vacuum travels along $+x$. At one instant its electric field is $330\hat{k}$ V/m. If the electric amplitude is 3 times this instantaneous value, which gives the instantaneous magnetic field? Use $c = 3 \times 10^8$ m/s.

(A) $-110 \times 10^{-8}\hat{j}$ T (B) $+110 \times 10^{-8}\hat{i}$ T

(C) $-330 \times 10^{-8}\hat{j}$ T (D) $+110 \times 10^{-8}\hat{j}$ T

21. A one-electron ion with nuclear charge $Z = 1$ emits when its orbit changes. The ratio of initial to final orbit radii is $\frac{25}{4}$, and its final principal quantum number is 2. What is $R\lambda$ for the emitted photon? Here R is the hydrogen Rydberg constant.

(A) 25 (B) $\frac{100}{29}$

(C) $\frac{100}{21}$ (D) 4

22. A wheel initially rotates at 5 rad/s. A constant opposing angular acceleration brings it to rest after 2 s. What angle does it turn through during the last half of this stopping interval?

(A) $\frac{5}{8}$ rad (B) $\frac{5}{4}$ rad

(C) $\frac{15}{4}$ rad (D) 5 rad

23. An alpha particle begins far from a fixed nucleus of atomic number 29 with kinetic energy 8 MeV. Before its head-on approach, a retarding electric potential removes 4 MeV of kinetic energy. Find its distance of closest approach in fm, assuming only Coulomb repulsion near the nucleus. Use $ke^2 = 1.44$ MeV fm.

(A) $\frac{174}{25}$ fm (B) $\frac{2088}{25}$ fm

(C) $\frac{261}{25}$ fm (D) $\frac{522}{25}$ fm

24. A spherical blackbody's absolute temperature is multiplied by 4 while its radius is divided by 4. Neglect incoming background radiation. Find its emitted-power multiplier.

(A) 256 (B) 16

(C) $\frac{1}{4}$ (D) 64

25. A wire carries $i(t) = 3 + 6\sin\omega t$ A for many cycles. Find its rms current.

(A) $3\sqrt{3}$ A (B) 3 A

(C) 9 A (D) $\frac{3\sqrt{10}}{2}$ A

26. In YDSE a sheet of index 2 and thickness 8 micrometres covers one slit. A second sheet of the same thickness and index 3/2 covers the other. With wavelength $\frac{4}{5}$ micrometres, find the magnitude of central-fringe shift in units of fringe width.

(A) $\frac{5}{2}$ (B) 5

(C) 20 (D) 15

27. A satellite of mass m is moved from a circular orbit of radius R to one of radius 4R about mass M. Find the required increase in total orbital energy in units of $GMm/(2R)$.

(A) $\frac{5}{4}$ (B) $\frac{1}{4}$

(C) $\frac{3}{2}$ (D) $\frac{3}{4}$

28. A coil of inductance 4 H carries current $i(t) = 4t^2$ A. Find the magnitude of its self-induced emf at $t = 2$ s.

(A) 128 V (B) 4 V

(C) 64 V (D) 32 V

29. Two wires of the same material and length L, with radii r and 6r, are joined in series and stretched by a force F. Find their total stored elastic energy divided by the energy stored in the thinner wire alone under force F.

(A) $\frac{37}{36}$ (B) 2

(C) $\frac{36}{37}$ (D) $\frac{7}{6}$

30. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 4 kΩ to a terminal at +13 V, and its cathode connects to a terminal at 0 V. Find the current from the +13 V terminal in mA.



(A) $\frac{123}{40}$ mA
(B) $\frac{137}{40}$ mA
(C) $\frac{13}{4}$ mA
(D) $\frac{29}{10}$ mA

Chemistry

31. A shallow lake receives phosphate-rich runoff. Algal biomass rises, and dissolved oxygen later falls as the bloom decays. Which intervention most directly interrupts the initiating chemical cause?

(A) Remove atmospheric nitrogen above the lake.
(B) Add more phosphate to support fish.
(C) Reduce the phosphate input.
(D) Increase carbon monoxide in the water.

32. propene reacts with HBr,without peroxides. Which pair gives the major constitutional product and the intermediate whose greater stability fixes its orientation? Ignore stereoisomer counting.

(A) 2-bromopropane; primary carbocation
(B) 2-bromopropane; secondary carbocation
(C) 1-bromopropane; primary carbocation
(D) 1-bromopropane; secondary carbocation

33. Excess water reacts completely with $\frac{1}{100}$ mol CaC_2. Which gas identity/amount pair follows from the balanced hydrolysis?

(A) C_2H_2, $\frac{1}{100}$ mol.
(B) C_2H_2, $\frac{1}{200}$ mol.
(C) O_2, $\frac{1}{200}$ mol.
(D) CO_2, $\frac{1}{100}$ mol.

34. nitrobenzene is treated with Sn/HCl,then basification. Which free organic product follows the nitro-group transformation and retained C-N connectivity?

(A) aniline
(B) phenol
(C) benzamide
(D) benzylamine

35. Compare $MgSO_4$ and $BaSO_4$ in water at the same temperature. Which member is generally more soluble,and which explanation avoids applying one universal down-group rule to all salts?

(A) $MgSO_4$; sulfate solubility generally decreases down group2 as hydration weakening outweighs lattice change.

(B) $BaSO_4$; every group2salt becomes more soluble as the cation grows.

(C) Solubility is fixed solely by formula mass,with no lattice or hydration contribution.

(D) They must be equally soluble because both cations have charge+2.

36. A cathode receives $\frac{2}{25}$mol electrons. Its only products are Cu from Cu^{2+} and $\frac{1}{125}$mol H_2. Cu M64. Deposited copper mass in g is

(A) $\frac{128}{25}$
(B) $\frac{64}{25}$
(C) $\frac{288}{125}$
(D) $\frac{256}{125}$

37. Fresh aqueous solutions of $[Co(NH_3)_5Br]SO_4$ and its ionisation isomer $[Co(NH_3)_5SO_4]Br$ are tested immediately; coordinated ligands do not exchange on the test timescale. Which paired observation is expected?

(A) Both give immediate Ag halide because every halide inside or outside brackets is free.

(B) $[Co(NH_3)_5Br]SO_4$ immediately precipitates its coordinated halide while $[Co(NH_3)_5SO_4]Br$ retains its external halide.

(C) $[Co(NH_3)_5Br]SO_4$ gives $BaSO_4$ with excess Ba^{2+}; $[Co(NH_3)_5SO_4]Br$ gives AgBr with excess Ag^{+}.

(D) Neither gives a precipitate because coordination suppresses counter-ion dissociation.

38. In the gas phase,N_2 is converted into N_2^{+} by changing one electron. Under the elementary MO model, which final bond-order/unpaired-electron pair follows?

(A) Bond order 3; 0 unpaired electrons.

(B) Bond order $\frac{5}{2}$; 1 unpaired electron.

(C) Bond order $\frac{7}{2}$; 1 unpaired electrons.

(D) Bond order $\frac{5}{2}$; 0 unpaired electrons.

39. A model solvent has Tb=400K,molar mass100g/mol and molar vaporisation enthalpy32000J/mol. UseR=8.00J $mol^{-1}K^{-1}$ and Kb=R Tb^2 M/(1000ΔH),M in g/mol. A $\frac{3}{100}$mol/kg A_2B solution is50% dissociated into3ions. Boiling temperature in K is

(A) $\frac{20003}{50}$

(B) $\frac{10006}{25}$

(C) $\frac{10003}{25}$

(D) $\frac{10009}{25}$

40. For two van der Waals gases,(Tc/Pc)A is2times(Tc/Pc)B. UsingTc=8a/(27Rb),Pc=a/(27b^2),Vc=3b andTB=a/(Rb),which pair(Vc,A/Vc,B;TB,A/Tc,A)is correct?

(A) (4;27/8).

(B) (2;8/27).

(C) (2;27/8).

(D) (1/2;27/8).

41. An ideal mixture initially has$\frac{1}{10}$molA and$\frac{3}{10}$molB. A selective step removes half ofB and noA. At the new equilibriumT,V,totalpressure is8bar. Partialpressure ofA,in bar,is

(A) $\frac{16}{5}$

(B) 20

(C) 4.0

(D) 2

42. diethyl adipate,EtOOC-(CH2)4-COOEt undergoes NaOEt/EtOH,then acidification only. Under the stated intramolecular closure pathway,which isolated product matches both ring size and work-up endpoint?

(A) cyclohexanone

(B) cyclopentanone

(C) ethyl 2-oxocyclohexane-1-carboxylate

(D) ethyl 2-oxocyclopentane-1-carboxylate

43. A $\frac{3}{100}$ mol FeC_2O_4 inventory is analysed. A fraction $\frac{1}{4}$ of Fe^{2+} has already become Fe^{3+},but every oxalate remains unchanged. Subsequent complete oxidation gives Fe^{3+} and CO_2. Electron equivalents released,in mol,are

(A) $\frac{9}{400}$

(B) $\frac{39}{400}$

(C) $\frac{3}{50}$

(D) $\frac{33}{400}$

44. In ethane viewed along C-C,a selected front C-H bond is at0°. In conformation A,the rear C-H directions are60°,180°,300°; B is obtained by rotating only the rear carbon through120°. Which is correct?

(A) Both are staggered,but B is a configurational stereoisomer.

(B) A is eclipsed and B staggered.

(C) A and B are equivalent staggered conformations.

(D) B is eclipsed because120° is not60°.

45. Compare H_2 and D_2 at equal temperature. Which statement preserves the distinction between mass/nuclear effects and electronic structure?

(A) H_2 has the greater root-mean-square speed; their ordinary molecular electron counts agree.

(B) An isotope with more neutrons must have a different number of protons and belong to another element.

(C) Changing H to D adds an electron and therefore changes the oxidation state of every bonded atom.

(D) Every physical property and reaction rate is identical because isotopes have the same atomic number.

46. Reduction potentials M^{2+}/M for metals A,B,C,D are respectively $-\frac{7}{10}$, $-\frac{3}{10}$, $\frac{3}{10}$, $\frac{4}{5}$V. Which paired claim is correct under standard conditions?

(A) A is the strongest oxidising ion because the corresponding metal is most reducing.

(B) A cannot reduce D^{2+} because its own reduction potential is negative.

(C) D is the strongest reducing metal because its reduction potential is greatest.

(D) A is the strongest reducing metal;D^{2+} oxidises A spontaneously.

47. 15 mL of a gaseous hydrocarbon is exploded with 130 mL O_2. All volumes are at the same T,P. After water condenses,the dry volume is lower than the initial total by 45 mL. KOH then removes 60 mL CO_2. The hydrocarbon formula is

(A) C_4H_8

(B) C_5H_8

(C) C_4H_6

(D) C_4H_{10}

48. A binary liquid mixture is observed to have a minimum-boiling azeotrope at a fixed pressure. Which paired conclusion is valid?

(A) Its vapour composition differs from its liquid composition at the azeotrope,so repeated distillation necessarily gives both pure components.

(B) The azeotrope proves no intermolecular interactions exist in either pure liquid.

(C) It shows negative deviation,and every liquid mixture must have such an azeotrope.

(D) It shows positive deviation in the relevant composition region;ordinary fractional distillation cannot pass through the azeotropic composition.

49. benzal chloride,C6H5CHCl2 is subjected to aqueous hydrolysis. Which product or product pair follows the named reaction and side-chain oxidation state?

(A) benzyl alcohol

(B) benzaldehyde

(C) chlorobenzene

(D) benzoic acid

50. Complete hydrolysis of sucrose gives an equimolar pair of monosaccharides. Which pair and carbonyl classes are correct?

(A) Two glucose molecules, both aldohexoses.

(B) Glucose and galactose, both ketohexoses.

(C) Glucose(an aldohexose)and fructose(a ketohexose).

(D) Ribose and deoxyribose, both hexoses.

51. Compare ethoxide and acetate as oxygen nucleophiles towards the same unhindered methyl substrate in one fixed solvent. Ignore concentration differences. Which pairing of the stronger nucleophile and the main electronic reason is most appropriate?

(A) acetate; the more stabilised anion necessarily attacks faster

(B) ethoxide; it is weaker because a localised lone pair cannot attack

(C) ethoxide; ethoxide has a localised negative charge whereas acetate delocalises it over two oxygens

(D) acetate; resonance always increases local charge on oxygen

52. A 15 g $CaCO_3$/inert mixture is80.0% $CaCO_3$ by mass. It receives $\frac{3}{25}$ mol HCl. For $CaCO_3+2HCl\rightarrow CaCl_2+CO_2+H_2O$,90.0% of formed CO_2 is collected. Collected CO_2 amount in mol is

(A) $\frac{6}{125}$

(B) $\frac{3}{50}$

(C) $\frac{3}{25}$

(D) $\frac{27}{500}$

53. A freshly assayed 10-volume H_2O_2 solution releases 10 L O_2 at STP per litre on complete decomposition. Only80.0% of the evolved oxygen reaches a burner that requires0.560 L O_2 at STP. The solution volume needed,in mL,is

(A) 28

(B) 70

(C) 140

(D) 56

54. Acidic permanganate oxidises SO_2 to HSO_4^- and becomes Mn^{2+}. With excess SO_2, $\frac{1}{20}$ mol MnO_4^- and only $\frac{1}{80}$ mol available H^+ are supplied. Count the net H^+ in the balanced reaction. Maximum Mn^{2+} formed,in mol,is

(A) $\frac{1}{40}$

(B) $\frac{1}{160}$

(C) $\frac{1}{640}$

(D) $\frac{1}{20}$

55. A1.00 L vessel initially contains $\frac{3}{20}$ mol A,$\frac{3}{10}$ mol B and $\frac{3}{100}$ mol C,with no D. For $A+2B\rightleftharpoons 2C+D$, equilibrium C amount is $\frac{9}{100}$ mol. Numerical K_c using molar concentrations is

(A) $\frac{9}{256}$

(B) $\frac{9}{128}$

(C) $\frac{9}{64}$

(D) $\frac{3}{112}$

56. For ethanal and benzaldehyde,which structural-reactivity comparison is correct under ordinary enolate/aldol conditions?

(A) only benzaldehyde self-aldolises

(B) ethanal self-aldolises;benzaldehyde does not

(C) neither self-aldolises

(D) both self-aldolise

57. In a regime where attraction is negligible,Z=1+pb/(RT). The Z-versus-p slope for gasA at300K is2times that for gasB at600K. Ratio bA/bB is

(A) $\frac{1}{2}$ (B) 2

(C) 4 (D) 1

58. At one T,K_1=4 for A⇌B,and K_2=9 for B⇌C. Numerical equilibrium constant for2C⇌2A is

(A) $\frac{1}{97}$ (B) $\frac{1}{1296}$

(C) $\frac{1}{36}$ (D) 36

59. A Fischer projection of tartaric acid has its four-carbon chain vertical (COOH at top and COOH at bottom). At C2,OH is on the right; at C3,OH is on the right; each other horizontal group is H. Horizontal bonds point toward the observer. Which classification and configuration pair is correct?

(A) chiral; (2R,3R) (B) chiral; (2R,3S)

(C) meso; (2R,3S) (D) chiral; (2S,3S)

60. A carbon-containing organic compound contains nitrogen and sulfur among N,S and halogens. Sodium fusion is performed with sufficient excess sodium to cleave any thiocyanate formed. Which product/test pairing is consistent?

(A) Elemental chlorine; a Prussian-blue response.

(B) NaCN and Na2S; Prussian-blue nitrogen test and violet nitroprusside sulfur test.

(C) Na2SO4; a violet nitroprusside response for sulfate.

(D) NaNO3; a brown-ring test directly on the organic compound.

English Proficiency

61. Read the sentence: After automation, two stages of the procedure became redundant because the same check was already performed elsewhere. Choose the closest meaning of 'redundant' as used in the sentence.

(A) too expensive (B) poorly designed

(C) impossible to carry out (D) no longer needed

62. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The volunteers are going to distribute the translated guide at the entrance.

(A) The translated guide is going to be distributed by the volunteers at the entrance.

(B) The volunteers are going to be distributed by the translated guide at the entrance.

(C) The translated guide was going to be distributed by the volunteers at the entrance.

(D) The translated guide is going to be translated by the volunteers at the entrance.

63. He learns the craft by working under a skilled maker, with instruction built into the work. What is he?

(A) contractor (B) apprentice

(C) consultant (D) supervisor

64. The principle rejects the use of war to resolve disagreements and insists on peaceful methods. Which word names it?

(A) isolation (B) pacifism

(C) militarism (D) patriotism

Read the passage for Question 65

The class kept two columns in its notebook. In the first it recorded what had actually been seen; in the second it proposed explanations. A bent stem belonged in the first column, but the claim that wind had bent it belonged in the second unless the cause had also been observed. This separation did not forbid explanations. It made them easier to revise when new evidence arrived, because a failed explanation did not require the students to erase the original observation. During discussion, the teacher returned to this separation whenever a confident guess began to be treated as a recorded fact. The notebook thus became a record not only of conclusions, but also of the difference between the evidence and the account offered for it.

65. What does 'this separation' refer to in the passage?

(A) Removing every explanation once a new observation is recorded.

(B) Dividing the class into students who observe and students who explain.

(C) Keeping observed facts distinct from proposed explanations of them.

(D) Separating bent stems from plants that have not been touched by wind.

66. Read the sentence: The cautious investor checked the risks before making any commitment. Choose the opposite in meaning of 'cautious' as used in the sentence.

(A) reckless (B) thorough

(C) hesitant (D) careful

67. Choose the most coherent order of the four sentences. A. The additional reminders reduced the number of residents caught unprepared. B. Many stored water in advance, but some did not see the notice. C. Residents were told that the water supply would stop for repairs. D. To reach those households, volunteers delivered printed reminders.

(A) C-B-D-A (B) B-C-D-A

(C) C-B-A-D (D) C-D-B-A

68. Read the sentence: The council kept a reserve rather than spend the entire grant; its decision was prudent. Choose the closest meaning of 'prudent' as used in the sentence.

(A) unnecessarily timid (B) reckless

(C) financially generous (D) carefully cautious

69. Choose the most coherent order of the four sentences. A. For the next meeting, the chairs were placed in a circle. B. The arrangement better suited the activity because participants could see one another. C. Participants at the back found it difficult to join a discussion. D. The meeting room's chairs were arranged in straight rows.

(A) D-C-B-A (B) D-C-A-B

(C) D-A-C-B (D) C-D-A-B

70. Read the sentence: Viewing the event through the lens of the local accounts changed the historian's interpretation. What does the figurative expression mean in this context?

(A) The accounts removed the need for independent evidence.
(B) The accounts proved every other interpretation false.
(C) The accounts supplied a perspective through which the event was understood.
(D) The accounts made the event visible for the first time.

Logical Reasoning

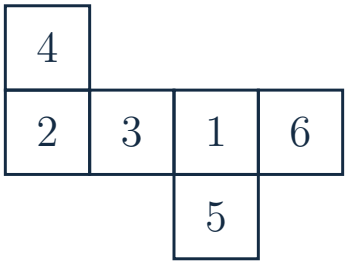
71. Preserve the exact relation and its order: Frog : Amphibian :: ? : Arachnid. Choose the missing term.

(A) Frog
(B) Spider
(C) Whale
(D) Sparrow

72. Two people differ in age by 4 years. The product of their ages 2 years ago was 725. What will the older person's age be 2 years from now?

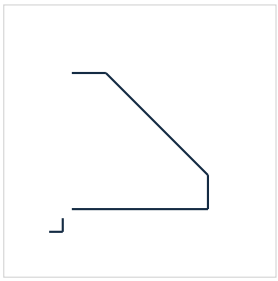
(A) 37
(B) 29
(C) 31
(D) 33

73. Fold the shown net into a cube. Give (label opposite 4, label opposite 2) in that order.

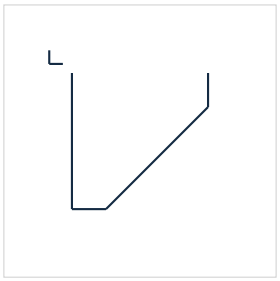


(A) $(5, 1)$
(B) $(5, 2)$
(C) $(5, 4)$
(D) $(5, 3)$

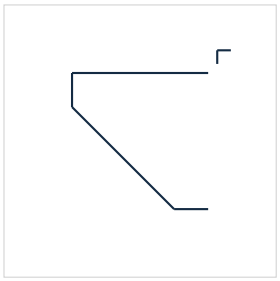
74. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1



2

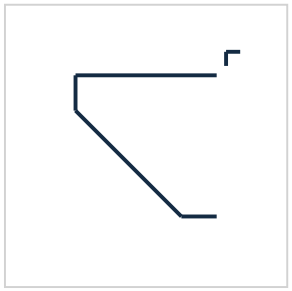


3

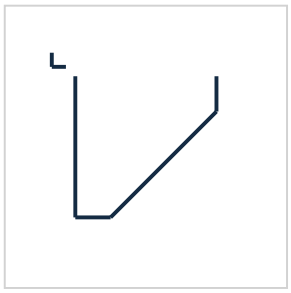
(A) Pattern U



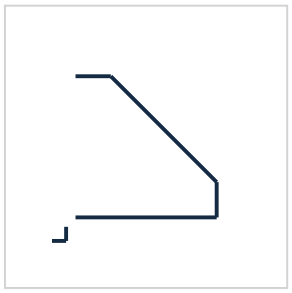
(B) Pattern T



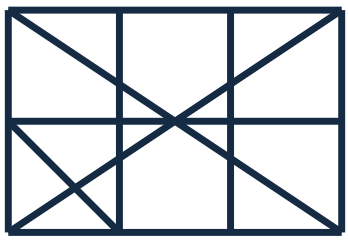
(C) Pattern Q



(D) Pattern P



75. How many triangles are formed by all the drawn segments?

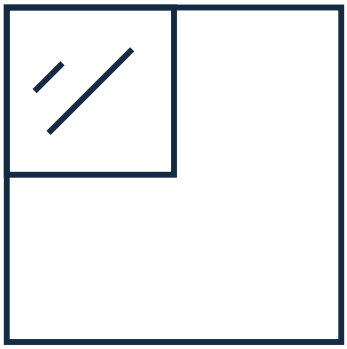


(A) 22
(B) 28
(C) 32
(D) 54

76. In a row, Asha is 9th from the left and 36th from the right. Bina is 9th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 22
(B) 29
(C) 23
(D) 2

77. Starting with a unit square sheet, fold right to left along the vertical midline, then fold bottom to top along the horizontal midline. The folded paper is cut along both short drawn slits. Select the pattern after complete unfolding.



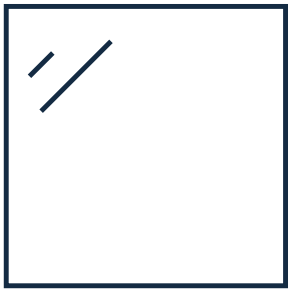
(A) Pattern M



(B) Pattern Q



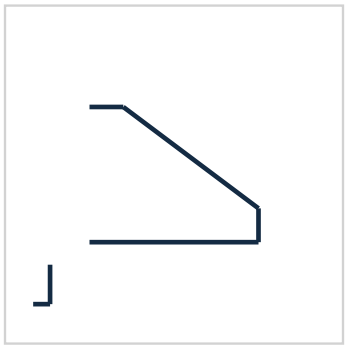
(C) Pattern K



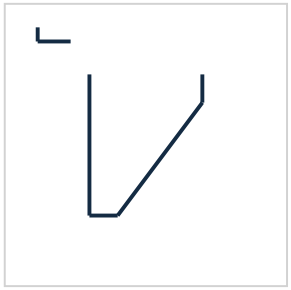
(D) Pattern W



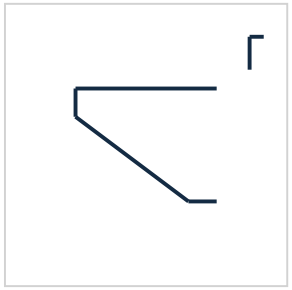
78. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



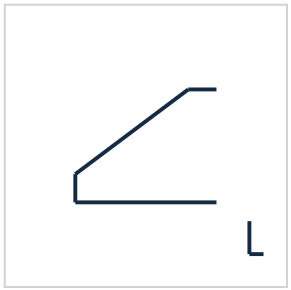
(A) Pattern X



(B) Pattern M



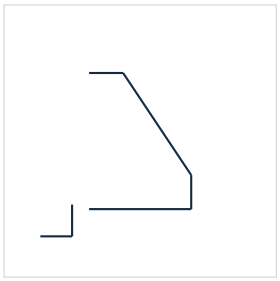
(C) Pattern R



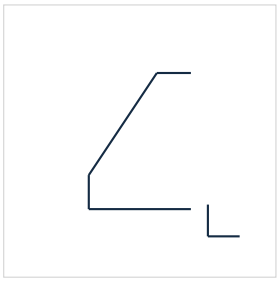
(D) Pattern Z



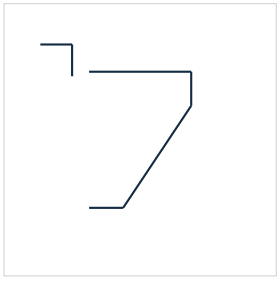
79. Select the missing bottom-right tile. Tiles reflect across the shared horizontal and vertical centre lines; there is no turning.



top left



top right



bottom left

?

bottom right

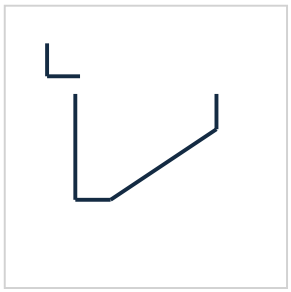
(A) Pattern L



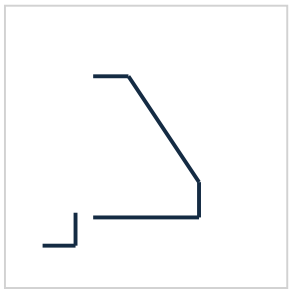
(B) Pattern T



(C) Pattern M



(D) Pattern Z



80. In this series the first differences themselves form an arithmetic progression. Find the missing internal term: 10, 12, ?, 19, 24, 30, 37.

(A) 17

(B) 13

(C) 16

(D) 15

81. Preserve the exact relation and its order: Bee : Swarm :: ? : Team. Choose the missing term.

(A) Wolf (B) Bee

(C) Lion (D) Player

82. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 6 and Z has code 31. Decode the separated values 17, 6, 18, 21.

(A) LAMP (B) MBNQ

(C) PMAL (D) KZLO

83. Consider only these statements; universal statements alone do not imply existence. Some designers are designers; All bakers are hikers. Conclusions: I. All designers are bakers. II. Some designers are not bakers. Which conclusion or complementary alternative must hold?

(A) Only II (B) Only I

(C) Either I or II follows, but neither is individually certain (D) Both I and II

84. A walker goes 8 m north, then 11 m west, then turns right and walks 2 m. In which direction is the starting point from the walker?

(A) North-east (B) South-east

(C) North-west (D) South-west

85. Preserve the exact relation and its order: Barometer : Atmospheric pressure :: ? : Electric current. Choose the missing term.

(A) Barometer (B) Ammeter

(C) Voltmeter (D) Thermometer

86. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?



1 2 3 4

(A) 1 (B) 2

(C) 4 (D) 3

87. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 7, 14, 19, ?, 43, 86, 91.

(A) 38 (B) 43

(C) 33 (D) 48

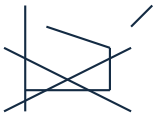
88. Three numbered triples satisfy both $y = x^2 + 4$ and $z - y = 3$. Which triple fails at least one condition? 1: (7,53,56), 2: (10,104,107), 3: (14,200,204), 4: (11,125,128).

(A) 4 (B) 2

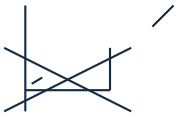
(C) 1 (D) 3

89. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





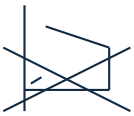
host 1



host 2



host 3



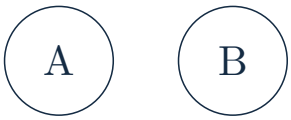
host 4

(A) 1 (B) 2

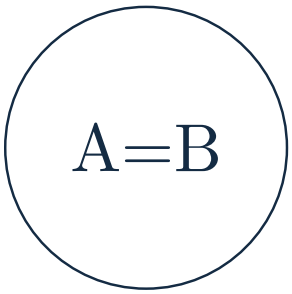
(C) 3 (D) 4

90. Choose the diagram that represents all these conditions, including the strictness information: All A are B, all B are A, and A is nonempty.

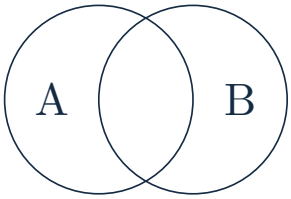
(A) Pattern W



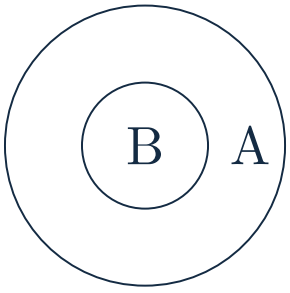
(B) Pattern R



(C) Pattern K



(D) Pattern X



Mathematics

91. A box contains 4 red and 5 blue distinct balls. Three are drawn uniformly without replacement. Given that at least one drawn ball is blue, find the probability that exactly two are red.

(A) $\frac{5}{8}$ (B) $\frac{5}{14}$

(C) $\frac{15}{41}$ (D) $\frac{3}{8}$

92. 15 distinct people sit at labelled round tables of sizes 9,6. Two specified people must be at different tables, and the first specified person must be at the 9-seat table. Rotations are identical. Count the seatings.

(A) 12454041600 (B) 6227020800

(C) 336259123200 (D) 24216192000

93. Let $y\prime + 5y = 3x$ and $y(0) = 3$. Find $y(1) + 1y\prime(1)$.

(A) $\frac{78}{25e^5} + \frac{12}{25}$ (B) $\frac{27}{25} - \frac{312}{25e^5}$

(C) $\frac{27}{25}$ (D) $\frac{312}{25e^5} + \frac{27}{25}$

94. Maximize Z=3x+4y subject to x≥0, y≥3, x+y≤9, x+2y≤11.

(A) 29 (B) 30

(C) 27 (D) 22

95. How many integers x satisfy |(x-6)+(x-10)|<|x-6|+|x-10|?

(A) 3 (B) 9

(C) 4 (D) 5

96. Evaluate $\int_0^{\pi/2} \sin(2x)\left[1 + 1\sin^2(2x)\right] \frac{\sin^6 x}{\sin^6 x + \cos^6 x}\, dx$.

(A) $\frac{5}{3}$ (B) $\frac{1}{2}$

(C) $\frac{5}{6}$ (D) 1

97. f(x)=(x^2-64)/(x-8) for x≠8, and f(8)=k. If f is differentiable at 8, find k+f′(8).

(A) 1 (B) 17

(C) 16 (D) 65

98. Let $P(x) = (1 + \sqrt{2}\,x)^8 + (1 - \sqrt{2}\,x)^8$. Find the sum of all nonconstant coefficients of P.

(A) 1154 (B) 1150

(C) 577 (D) 1152

99. Unit vectors a,b,c satisfy |(a×b)·c|=1. Find $|a+3b-5c|^2$.

(A) 65 (B) 9

(C) 34 (D) 35

100. Let R=[sin($\frac{11\pi}{102}$)+sin($\frac{23\pi}{102}$)]/[cos($\frac{11\pi}{102}$)+cos($\frac{23\pi}{102}$)]. Find (1-R^2)/(1+R^2).

(A) $\frac{3}{2}$ (B) $\frac{\sqrt{3}}{2}$

(C) $-\frac{1}{2}$ (D) $\frac{1}{2}$

101. Positive integers $a < b$ satisfy $a + b = 18$ and $3 \mid a$. Find the maximum value of ab.

(A) 18 (B) 72

(C) 81 (D) 45

102. The curve y=x^3+5x divides the disc x^2+y^2 ≤49. Find the area of the part satisfying y≥x^3+5x.

(A) $\frac{49\pi}{2}$ (B) $5 + \frac{49\pi}{2}$

(C) $\frac{49\pi}{4}$ (D) 49π

103. The four lines ±x/8±y/14=1 form a rhombus with vertices on the coordinate axes. Find its area.

(A) 112 (B) 224

(C) 448 (D) 56

104. F′(x)=$(x^2 + 5x + 5)\, e^x$ and F(0)=3. Find F(1)-2F(0).

(A) $-5 + 6e$ (B) $-2 + 6e$

(C) $1 + 6e$ (D) $-6e - 1$

105. Evaluate $\sum_{j=1}^{\infty} 3j(\frac{1}{6})^j$.

(A) $\frac{3}{25}$ (B) $\frac{18}{25}$

(C) $\frac{33}{25}$ (D) $\frac{18}{5}$

106. Evaluate $\displaystyle\sum_{\substack{0\le r\le 6\\ r \text{ even}}} \frac{\binom{6}{r}}{(r+1)(r+2)}$.

(A) $\frac{64}{7}$ (B) $\frac{247}{56}$

(C) $\frac{127}{56}$ (D) $\frac{99}{56}$

107. Two X and one Y cost 27; one X and three Y cost 31. Find the cost of one X plus one Y.

(A) 58 (B) 27

(C) 17 (D) 31

108. The line through P=(4,4,5) is perpendicular to directions (1,0,1),(0,1,1). It meets the plane z=0 at Q. Find Q_x+Q_y.

(A) −2 (B) 18

(C) 8 (D) 13

109. The hyperbola x^2/4-y^2/25=1 has eccentricity e. Find the eccentricity e′ of its conjugate hyperbola.

(A) $\frac{\sqrt{29}}{2}$ (B) $\sqrt{29}$

(C) $\frac{2\sqrt{29}}{29}$ (D) $\frac{\sqrt{29}}{5}$

110. $\theta \in(\pi/2,\pi)$, $\sin\theta=\frac{9\sqrt{85}}{85}$. Find $\cos(\theta/2)$.

(A) $-\frac{\sqrt{14450-340\sqrt{85}}}{170}$ (B) $-\frac{\sqrt{85}}{85}$

(C) $\frac{\sqrt{340\sqrt{85}+14450}}{170}$ (D) $\frac{\sqrt{14450-340\sqrt{85}}}{170}$

111. $x=t^2$, $y=t^3$ for $t>0$. Find d^2y/dx^2 at $t=5$.

(A) $\frac{3}{20}$ (B) $\frac{3}{2}$

(C) 15 (D) $\frac{3}{10}$

112. $a_j = 2 + 4(j - 1)$ and $f(x)=(3/2)x^2+bx+4$. If $f(1)=f(-1)$, find $f'(a_7)$.

(A) 78 (B) 26

(C) 90 (D) 82

113. Parabola $y=7(x-2)(x-3)$ meets the axes at three distinct points. A circle passes through these points. Find its other y-intercept.

(A) $-\frac{1}{7}$ (B) 1

(C) 7 (D) $\frac{1}{7}$

114. For $36x^2+144y^2=5184$, find the distance between its two directrices.

(A) 24 (B) $12\sqrt{3}$

(C) $8\sqrt{3}$ (D) $16\sqrt{3}$

115. Four observations are 9,9,x,y with population mean 9 and variance 4/2. Find xy.

(A) 85 (B) 81

(C) 154 (D) 77

116. $A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$ and $A^{-1}=pI+qA$. Find p+q.

(A) $-\frac{7}{15}$ (B) $\frac{7}{8}$

(C) $\frac{7}{15}$ (D) $\frac{8}{15}$

117. $f(x)=5x+4$. Find $f(f(4))$.

(A) 576 (B) 44

(C) 24 (D) 124

118. The line $r=(1,0,0)+s(t,1,2)$ is parallel to $x-7y+z=9$. Find the squared length of its direction vector.

(A) 30 (B) 54

(C) $\sqrt{30}$ (D) 28

119. $A = \begin{pmatrix} 3 & k \\ 0 & 4 \end{pmatrix}$ is symmetric. Find $\mathrm{tr}(A^TA)$.

(A) 25 (B) 49

(C) 7 (D) 12

120. For integer n, consider: "If 32 divides n, then 8 divides n." Which is its contrapositive?

(A) If 32 does not divide n, then 8 does not divide n.
(B) 32 divides n and 8 does not divide n.
(C) If 8 does not divide n, then 32 does not divide n.
(D) If 8 divides n, then 32 divides n.

121. Find the area enclosed by y=x(6-x) and y=4x.

(A) $\frac{76}{3}$
(B) $-\frac{4}{3}$
(C) $\frac{4}{3}$
(D) 36

122. A focal chord of y^2=8x has one endpoint parameter t=2. Find its length.

(A) $\frac{25}{2}$
(B) $\frac{9}{2}$
(C) 10
(D) $\frac{17}{2}$

123. α,β are nonzero roots of x^2-17x+70=0. Find the sum of $1+1/\alpha$ and $1+1/\beta$.

(A) $\frac{157}{70}$
(B) $\frac{87}{70}$
(C) $\frac{17}{70}$
(D) $\frac{104}{17}$

124. A rectangle has perimeter 16. Its length is constrained to be at least three times its width. Find its maximum area.

(A) 8
(B) 16
(C) 12
(D) 4

125. How many integers x satisfy $4^{-x} - (520)2^{-x} + 4096 \leq 0$?

(A) 10
(B) 6
(C) 7
(D) 5

126. If |z-11|=4, find the minimum of $|z|^2$.

(A) 105
(B) 49
(C) 7
(D) 225

127. Let k make lim as x$\to\infty$ of [$(6x^2+3x+1)/(x+2)$-kx] finite. Find that finite limit.

(A) -9
(B) 12
(C) 15
(D) 3

128. $F(x) = \int_0^{x^2} (t-16)(t-36)dt$. How many local minima of F lie in [-6,6]?

(A) 2
(B) 5
(C) 3
(D) 1

129. From two points on the same ray from a tower foot, separated by 2, elevations are 30° and 45°. Find the distance from the farther point to the foot.

(A) $1+\sqrt{3}$
(B) $\sqrt{3}+3$
(C) $\sqrt{3}$
(D) 2

130. How many integers x in [-9,13] satisfy (|x-2|-6)/(|x-2|-10)>0?

(A) 11 (B) 13

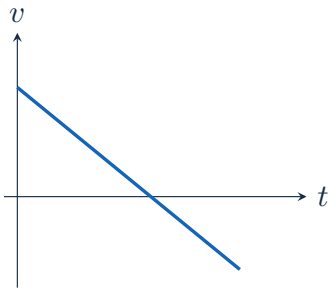
(C) 15 (D) 23

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A particle has velocity $v(0) = 3$ m/s, and constant acceleration makes $v(6) = -6$ m/s. What total distance does it travel in this interval?



(A) 36 m
(B) 12 m
(C) 15 m
(D) 3 m

132. A fixed amount of ideal gas has its absolute temperature multiplied by 3 and volume by 4. Find the final pressure divided by the initial pressure.

(A) $\frac{3}{4}$
(B) 3
(C) $\frac{1}{4}$
(D) $\frac{3}{16}$

133. A parallel-plate capacitor has vacuum capacitance C_0. A dielectric of constant 7 fills exactly half the gap thickness over the entire plate area; the other half-gap is vacuum. Find the equivalent capacitance divided by C_0.

(A) $\frac{4}{7}$
(B) $\frac{7}{4}$
(C) 14
(D) 4

Chemistry

134. Concentrated sulfuric acid is electrolysed under the specified anodic conditions that form a peroxo acid. Sulfur remains in oxidation state+6. Which product and oxidation change are consistent?

(A) $H_2S_2O_8$;the two O atoms in its O–O link have oxidation state-1.

(B) $H_2S_2O_7$;it contains an O–O peroxide link.

(C) $H_2S_2O_8$;sulfur is oxidised from+6to+7.

(D) H_2SO_3;sulfur is oxidised from+6to+4.

135. On controlled heating, $\frac{1}{100}$ mol $KMnO_4$ follows $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$. Which pair ($O_2$ amount,origin of the intense purple colour of the starting d^0 ion) is correct?

(A) ($\frac{1}{100}$ mol,a d-d transition of MnVII).
(B) ($\frac{1}{200}$ mol,a d-d transition of MnVII).
(C) ($\frac{1}{200}$ mol,charge-transfer absorption).
(D) ($\frac{1}{400}$ mol,charge-transfer absorption).

136. 2-bromo-2-methylbutane is heated with sodium ethoxide under elimination-favouring conditions. Which major constitutional product is expected,despite the presence of an alkoxide that could be used in a Williamson synthesis with a suitable different halide?

(A) the alkane formed by replacing halogen with H

(B) 2-methylbut-2-ene

(C) the corresponding tertiary alkoxide substitution product

(D) 2-methylbut-1-ene

Logical Reasoning

137. A word is coded as the sum of its forward alphabet positions plus 2 times its length. The code of S?ONE is 83. Which letter replaces the question mark?

(A) T

(B) S

(C) U

(D) V

138. All five people have different heights. Charu is shorter than Esha. Asha is shorter than Dev. Esha is shorter than Asha. Dev is shorter than Bina. If Bina is left out, who is 2nd shortest among the remaining people?

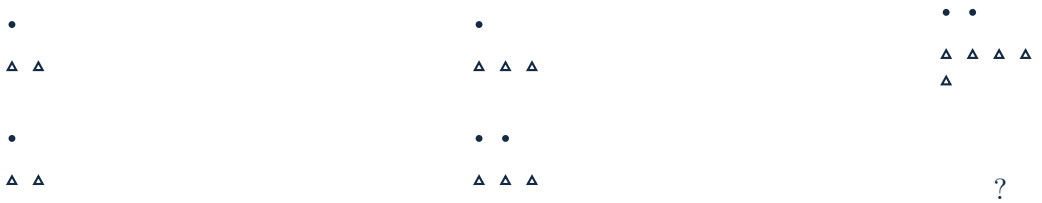
(A) Charu

(B) Esha

(C) Asha

(D) Bina

139. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern T

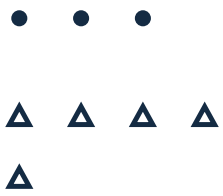
(B) Pattern Y



(C) Pattern V



(D) Pattern U





Mathematics

140. Evaluate $\sum_{j=1}^{7} j(j+2)$.

(A) 56 (B) 196

(C) 140 (D) 840

141. Circles $x^2+y^2=144$ and $(x-14)^2+y^2=144$ meet at A,B. Find $|AB|^2$.

(A) $2\sqrt{95}$ (B) 772

(C) 380 (D) 576

142. Three distinguishable fair dice are thrown. Given that the three faces are not all identical, find the probability that their sum is 12.

(A) $\frac{4}{35}$ (B) $\frac{25}{216}$

(C) $\frac{31}{35}$ (D) $\frac{5}{42}$

BITSAT Mock 02

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A 5 kg block is launched at 10 m/s on top of a 5 kg block initially at rest on a smooth floor. The coefficient of kinetic friction between them is $\frac{2}{5}$. Find the distance slid by the upper block relative to the lower before slipping ceases. Take $g = 10$ m/s^2.

(A) $\frac{75}{8}$ m
(B) $\frac{25}{4}$ m
(C) $\frac{25}{8}$ m
(D) $\frac{25}{2}$ m

2. A coil of inductance 7 H carries current $i(t) = 5t^2$ A. Find the magnitude of its self-induced emf at $t = 3$ s.

(A) 35 V
(B) $\frac{30}{7}$ V
(C) 630 V
(D) 210 V

3. Light of vacuum wavelength 200 nm passes from a medium of index 5 into one of index 6. Find the ratio of its frequencies in the second and first media.

(A) 1
(B) $\frac{36}{25}$
(C) $\frac{6}{5}$
(D) $\frac{5}{6}$

4. A small planar detector initially faces an isotropic point source normally at distance r. It is moved to a concentric sphere whose enclosed volume is 8 times as large, and tilted so that the cosine of the angle between its normal and the incident ray is $\frac{4}{5}$. Its area is unchanged. What is the ratio of intercepted powers, final to initial?

(A) $\frac{16}{5}$
(B) $\frac{1}{5}$
(C) $\frac{1}{10}$
(D) $\frac{1}{4}$

5. In an ideal intrinsic semiconductor at zero temperature, the band gap is $\frac{9}{5}$ eV. Neglect phonons, excitons and all transitions within a band. Two beams have photon energies $\frac{7}{5}$ eV and $\frac{11}{5}$ eV. Which can create a valence-to-conduction electron-hole pair by a single-photon absorption?

(A) Neither beam
(B) Both beams
(C) Only the higher-energy beam
(D) Only the lower-energy beam

6. A ferromagnetic material has Curie temperature 550 K. It is tested at 530 K and 560 K in a weak applied field. Which description is appropriate?

(A) It becomes more strongly spontaneously ferromagnetic at the higher temperature.

(B) Spontaneous domain order is possible at the lower temperature; above the Curie temperature the material is paramagnetic.

(C) Its Curie temperature changes to match either test temperature.

(D) It must be diamagnetic at both temperatures.

7. A particle has position-dependent velocity components $\vec{v} = 3x\,\hat{i} - 3y\,\hat{j}$ in SI units. Given $dx/dt = v_x$ and $dy/dt = v_y$, what is its acceleration magnitude at $(2, 4)$ m?

(A) $6\sqrt{5}\,\mathrm{m/s^2}$ (B) $18\sqrt{5}\,\mathrm{m/s^2}$

(C) $54\,\mathrm{m/s^2}$ (D) $36\,\mathrm{m/s^2}$

8. A network is defined by $P = \mathrm{NOR}(A, B)$, $Q = \mathrm{NOR}(A, P)$ and $Y = \mathrm{AND}(Q, \overline{B})$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 1$ and $A = 0$?

(A) 2 (B) 0

(C) 1 (D) 3

9. A particle of mass 3 kg starts from rest and experiences $F(t) = 4t$ N along its motion for 3 s. Find the work done by the force.

(A) 54 J (B) 24 J

(C) 108 J (D) 216 J

10. A rigid ladder of length 25 m slides with its ends on a floor and a vertical wall. At the instant its top is at height $4L/5$, the foot moves away at 6 m/s. What is the downward speed of the top?

(A) 6 m/s (B) 8 m/s

(C) $\frac{9}{2}$ m/s (D) $\frac{24}{5}$ m/s

11. In a series RLC circuit, ideal rms voltmeters read 60 V across R, 50 V across L and 30 V across C. Find source rms voltage.

(A) 60 V (B) $20\sqrt{10}$ V

(C) 100 V (D) $10\sqrt{70}$ V

12. A single slit of width 6 mm is illuminated by wavelength 300 nm. A screen is 4 m away. Find the separation of the second minima on opposite sides of the central maximum, in the small-angle approximation.

(A) 800 mm (B) $\frac{3}{5}$ mm

(C) $\frac{4}{5}$ mm (D) $\frac{1}{5}$ mm

13. A pure radioactive sample has activity A at time 8 hours after preparation and $A/8$ at time 20 hours after preparation. At what time, measured from preparation, will its activity be one sixteenth of the activity measured at 8 hours? Assume exponential decay with a constant decay rate.

(A) 36 h (B) 8 h

(C) 72 h (D) 24 h

14. One mole of a monatomic ideal gas follows $pV^7 = \text{constant}$. Find the process molar heat capacity in units of R.

(A) $\frac{4}{3}$ (B) $\frac{5}{3}$

(C) $\frac{7}{6}$ (D) $\frac{5}{2}$

15. A 2 kg mass on a string passes the top of a vertical circle of radius 4 m with string tension 80 N. Find string tension at the bottom. Take $g = 10\ \text{m/s}^2$.

(A) 120 N (B) 160 N

(C) 80 N (D) 200 N

16. A uniform wire is stretched to $\frac{5}{4}$ times its original length at constant volume. Its resistivity increases by 30 percent during the process. Find final resistance divided by initial resistance.

(A) $\frac{104}{125}$ (B) $\frac{325}{128}$

(C) $\frac{65}{32}$ (D) $\frac{26}{25}$

17. A uniform disc of mass 3 kg and radius 2 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) $\frac{45}{8}\ \text{kg m}^2$ (B) $\frac{9}{2}\ \text{kg m}^2$

(C) $6\ \text{kg m}^2$ (D) $\frac{39}{8}\ \text{kg m}^2$

18. A pendulum bob of mass 5 kg and length 1 m passes the horizontal position moving downward at 2 m/s. Find the instantaneous power of all forces when $\sin\theta = 3/5$, $\cos\theta = 4/5$, with θ measured from the downward vertical. Take $g = 10\ \text{m/s}^2$.

(A) 120 W (B) $100\sqrt{5}$ W

(C) $60\sqrt{5}$ W (D) 60 W

19. Two identical uniform spheres touch. Their radii are multiplied by 10 while their density is divided by 10. By what factor does their gravitational attraction change?

(A) 1 (B) 1000000

(C) $\frac{1}{10000}$ (D) 100

20. A charge of $5\,\mu$C moves from potential 50 V to -50 V in an electrostatic field. Find the work done by an external agent if the charge starts and ends at rest and no other force does work.

(A) $-500\,\mu$J (B) $500\,\mu$J

(C) $-100\,\mu$J (D) $-250\,\mu$J

21. An astronomical telescope in normal adjustment has magnifying-power magnitude 6 and tube length 28 cm. Find the eyepiece focal length.

(A) 4 cm (B) $\frac{28}{5}$ cm

(C) $\frac{28}{37}$ cm (D) $\frac{4}{7}$ cm

22. Two ideal gases A and B have molar masses M and $5M$. If their rms speeds are equal, what is the ratio of their average translational kinetic energies per molecule, K_B/K_A?

(A) 1 (B) $\frac{\sqrt{5}}{5}$

(C) 25 (D) 5

23. A wire consists of a semicircular arc of radius 5 m carrying 4 A, joined to two straight radial segments that extend outward from its ends. Find the magnetic field magnitude at the arc centre in units of μ_0.

(A) $\frac{1}{5\pi}$ (B) $\frac{1}{10}$

(C) $\frac{1}{5}$ (D) $\frac{2}{5}$

24. Two large parallel plates are held at potentials +700 V and −700 V, with gap 8 mm. Neglect edge effects. Find field magnitude in the gap.

(A) 17500 V/m (B) 175000 V/m

(C) 175 V/m (D) 1750000 V/m

25. A normal-adjustment astronomical telescope has tube length 24 cm and magnifying-power magnitude 3. The eyepiece is kept, but the objective is changed so magnifying power becomes 5. By how much must the tube length increase to restore normal adjustment?

(A) 12 cm (B) 8 cm

(C) 48 cm (D) 36 cm

26. Here L is inductance, C capacitance, R resistance, V potential, I current and t time. Which expression has the same dimensions as both $(CV)^4$ and $(It)^4$?

(A) $(V/R)^4$ (B) $(RC)^4$

(C) $(LI)^4$ (D) $((V/R)t)^4$

27. A nonrelativistic charged particle initially has momentum $5qEt$ along a uniform electric field, where $t > 0$ is a fixed reference interval. Its charge is $+q$. What is $\lambda(7t)/\lambda(4t)$ after acceleration in the same field?

(A) $\frac{1}{4}$ (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{3}{4}$ (D) $\frac{5}{12}$

28. Two rods of equal length and cross-sectional area have thermal conductivities k and 12k. They are joined end-to-end with no contact resistance. Find the effective conductivity of the composite divided by k.

(A) 12 (B) $\frac{24}{13}$

(C) $\frac{2}{13}$ (D) 1

29. One mole of nonoverlapping identical spheres of radius r has total geometric material volume V. A second sample contains $\frac{4}{5}$ mole of spheres of radius $3r$. What is its total geometric material volume divided by V? Ignore packing voids.

(A) $\frac{108}{5}$ (B) 27

(C) $\frac{135}{4}$ (D) $\frac{1728}{125}$

30. A uniform massless spring of constant 80 N/m is cut into 3 equal parts. All parts are connected in parallel to support a mass 5 kg. Find its small-oscillation period.

(A) $\frac{\pi}{6}$ s (B) $\frac{\sqrt{3}\pi}{18}$ s

(C) $\frac{\sqrt{3}\pi}{2}$ s (D) $\frac{\sqrt{3}\pi}{6}$ s

Chemistry

31. D-Glucose and D-mannose differ only at C2. Which pair of conclusions is correct?

(A) They are enantiomers and always give different osazones. (B) They are epimers and form the same osazone.

(C) They are identical open-chain structures. (D) They are anomers differing at C1 of a ring.

32. A zero-order reaction remains valid until reactant depletion. Which pair is correct?

(A) Its k has concentration/time units;halving initial concentration halves its half-life.

(B) Its k has concentration/time units;halving concentration doubles half-life.

(C) Its k has inverse-time units;half-life is independent of initial concentration.

(D) A zero-order process cannot have a finite completion time.

33. Free proline has its nitrogen bonded to two carbon atoms and one hydrogen. Which description and contrast with glycine are appropriate?

(A) Its ring makes it a disaccharide. (B) It contains a peptide bond before any condensation.

(C) Its amino nitrogen is secondary; glycine has a primary amino group. (D) Its nitrogen is quaternary and lacks every N-H bond.

34. A student assigns four bonding pairs and two lone pairs to the central atom of XeF_4. After independently checking valence-electron count,which combined molecular shape/domain count is consistent?

(A) See-saw; five electron domains. (B) Octahedral molecular shape; six electron domains.

(C) Square planar; six electron domains. (D) Tetrahedral; four electron domains.

35. A cubic cell has one species at all corners and a different species at the body centre,as in CsCl. Which statement distinguishes it from an elemental BCC Bravais lattice?

(A) It is FCC because its two species have different radii.

(B) It is a primitive cubic Bravais lattice with a two-species basis;corner-to-body translation does not preserve chemical identity.

(C) It must be BCC because every body-centred drawing is a BCC Bravais lattice.

(D) It has no translational periodicity because a primitive cell cannot contain two species.

36. propanone reacts with HCN. Which product correctly reflects nucleophilic addition or addition-elimination at its carbonyl rather than a different reaction class?

(A) propan-2-ol (B) 2-methylpropanoic acid

(C) propanenitrile (D) 2-hydroxy-2-methylpropanenitrile

37. In a pressure range where the Freundlich law $x/m = kp^{1/2}$ is valid at fixed temperature, adsorbent B has 2 times the mass of A and is exposed to 4 times its pressure. Find the ratio of total adsorbed amounts x_B/x_A.

(A) $\frac{1}{4}$ (B) 2

(C) 8 (D) 4

38. For one reaction,ln K is5 at300 K and1 at600 K. Assume ln K is linear in1/T. At400 K,ln K equals

(A) 4 (B) 5

(C) 3 (D) 1

39. 1-nitropropane is treated with H2/Ni under complete nitro reduction. Which free organic product follows the nitro-group transformation and retained C-N connectivity?

(A) propan-1-amine (B) propanamide

(C) ethylamine (D) propan-2-amine

40. Consider the statements. I. NO emitted from an engine can be oxidised in air to NO2. II. A pollutant made by an atmospheric reaction must be classified as primary solely because its precursor came from an engine. Choose the correct truth pattern.

(A) I false; II true. (B) I false; II false.

(C) I true; II true. (D) I true; II false.

41. A HA/A^- buffer initially has pH=pKa+$\log_{10}2$. Adding $\frac{1}{50}$ mol HCl without volume change gives pH=pKa-$\log_{10}2$;neither component is exhausted. Total initial $HA+A^-$ amount,in mol,is

(A) $\frac{1}{50}$ (B) $\frac{3}{50}$

(C) $\frac{2}{25}$ (D) $\frac{1}{25}$

42. Assign the formal metal oxidation state in $[Fe(CN)_5(NO)]^{2-}$,using five CN^- and one linear NO^+. For nitrosyl species use the explicitly stated formal NO^+ convention rather than inferring spin from it.

(A) 2 (B) 1

(C) 3 (D) 4

43. Standard reduction potentials for A^+/A,B^+/B,C^+/C are $\frac{2}{5}$,$\frac{1}{5}$,$\frac{1}{2}$ V. Which pair of claims is correct under standard conditions?

(A) B^+ oxidises A spontaneously because $\frac{1}{5}$ V is positive.

(B) A^+ oxidises C with standard cell potential $\frac{1}{10}$ V.

(C) C^+ oxidises A,but doubling coefficients doubles the standard cell potential to $\frac{1}{5}$ V.

(D) C^+ oxidises A;doubling all coefficients leaves standard cell potential $\frac{1}{10}$ V unchanged.

44. Compare pH of A:HCl at$\frac{1}{1000}$M;B:HA,Ka=10^{-6},C=$\frac{1}{10}$M;C:HB,Ka=10^{-5},C=$\frac{1}{1000}$M. Neglect water and use weak-ionisation ordering. Increasing pH is

(A) A<B<C. (B) B<A<C.

(C) C<B<A. (D) A<C<B.

45. Salicylic acid has a phenolicOH and aCOOH group. In aspirin, the phenolic oxygen bears an acetyl group whileCOOH remains. Which change occurred?

(A) Conversion of COOH into an amide.
(B) Replacement of the aromatic ring by glucose.
(C) O-acetylation to form an ester.
(D) Hydrogenation of every aromatic bond.

46. An ideal gas executes A→B→C→D→A through (V/L,P/bar):A(1,1),B(4,1),C(4,3),D(1,3). IUPAC work-on convention applies. Heats on AB,BC,DA are 800,300,-200 J. Heat on CD in J is (1 L bar=100 J)

(A) -300
(B) -1500
(C) -600
(D) -900

47. Compare dilute solutions of 2-nitrophenol and 4-nitrophenol in a non-hydrogen-bonding solvent. Which conclusion joins structure with the expected effect of further dilution on their hydrogen bonding?

(A) 2-nitrophenol must have stronger intermolecular association because its internal acceptor is farther away.
(B) Both lose all H bonding equally because every H bond requires two molecules.
(C) Only 4-nitrophenol can make an intramolecular H bond because para groups form a shorter ring.
(D) 2-nitrophenol can retain an intramolecular H bond on dilution; intermolecular association of 4-nitrophenol is reduced.

48. With one equivalent of cold concentrated HI and isolation before prolonged further reaction,which initial cleavage pair is expected from anisole,C6H5-O-CH3?

(A) phenyl iodide and methyl iodide
(B) benzyl iodide and water
(C) methyl iodide and phenol
(D) iodobenzene and methanol

49. cyclopentene (C5H8) adds one equivalent of :CCl2 across its only C=C without rearrangement. Which pair gives the adduct formula and the number of independent rings? Count the original ring and the new three-membered ring separately.

(A) C6H8Cl2; 1
(B) C5H8Cl2; 1
(C) C6H10Cl2; 2
(D) C6H8Cl2; 2

50. Consider I: The F-F bond is weaker than the Cl-Cl bond. II: NF3 has a smaller resultant dipole moment than NH3. Which assessment correctly recognises both anomalous first-element behaviours?

(A) Both statements are correct.
(B) Only II is correct.
(C) Only I is correct.
(D) Neither statement is correct.

51. In a hypothetical rule,$l = 0, 1, \ldots, n$ is allowed. Keep subshell capacity2(2l+1),Hund and Pauli,and fill by n+l then lower n. For 19 electrons,the last occupied subshell and its electron count are

(A) $2p^1$.
(B) $2d^1$.
(C) $3p^1$.
(D) $3d^1$.

52. Classify Na2O, Al2O3, SO3, in that order, by their usual acid-base behaviour. Which sequence also predicts that the middle oxide reacts with both suitable acids and strong alkalis?

(A) Basic; amphoteric; acidic.
(B) Acidic; amphoteric; basic.
(C) Amphoteric; basic; acidic.
(D) Basic; acidic; amphoteric.

53. Fe_3O_4 is represented as a mixed oxide of FeO and Fe_2O_3. Which formal metal oxidation-state ratio satisfies the formula,with ordinary oxide oxygen -2?

(A) 2+:3+ = 1:2.

(B) 2+:3+ = 2:1.

(C) All metal atoms must be formally +8/3,with no mixed-valence bookkeeping possible.

(D) 2+:3+ = 1:1.

54. A cation selected from Cu2+,Zn2+,Fe3+ gives a pale-blue hydroxide precipitate when aqueous ammonia is added dropwise, followed by a deep-blue solution containing an ammine complex on adding excess ammonia. Which identification and oxidation-state conclusion is consistent?

(A) Cu2+; the metal oxidation state remains +2.

(B) Cu2+; ammonia necessarily reduces the metal by one oxidation unit.

(C) Fe3+; every hydroxide dissolves to a deep-blue complex.

(D) Zn2+; a deep-blue colour is the universal zinc confirmation.

55. A measured[A]-versus-time plot is linear,falling from $\frac{7}{100}$M at0s to$\frac{1}{20}$M at20.0s. At20s,volume is instantly doubled by solvent;the same zero-order rate constant then applies. Time after dilution to halve the new[A],in s,is

(A) 25

(B) 20

(C) 35

(D) $\frac{25}{2}$

56. A polymer mixture contains three molecules of molar mass 10000 g/mol for every one molecule of molar mass 30000 g/mol. Which pair (number-average molar mass, weight-average molar mass) is correct?

(A) (20000,15000) g/mol

(B) (20000,20000) g/mol

(C) (15000,20000) g/mol

(D) (10000,30000) g/mol

57. propene is subjected to BH3/THF,then H2O2/OH-. Which product follows the stated regioselectivity or oxidation/dehydration endpoint?

(A) propanal

(B) propan-1-ol

(C) propan-2-ol

(D) propane

58. For $[Fe(CN)_6]^{4-}$,use the usual ligand-field/VBT assignments at room temperature. Which complete assignment is consistent with charge balance and magnetic behaviour?

(A) d^7,octahedral,d^2sp^3,spin-only moment 0 BM.

(B) d^6,octahedral,d^2sp^3,spin-only moment 0 BM.

(C) d^6,tetrahedral,sp^3,spin-only moment 0 BM.

(D) d^6,octahedral,d^2sp^3,spin-only moment $\sqrt{3}$ BM.

59. A one-electron ion has nuclear charge Z=2. A bound level has the same energy as H at n=2. In that ion shell,maximum electrons allowed by Pauli in orbitals with$m_l = 0$ would be (ignore electron interactions for the energy comparison)

(A) 16

(B) 4

(C) 8

(D) 32

60. For hexa-2,4-diene, CH3-CH=CH-CH=CH-CH3, count distinct configurational geometrical isomers. Ignore conformers and photochemical interconversion. Where reversing the entire carbon chain identifies two E/Z assignments, count them once.

(A) 8 (B) 3

(C) 1 (D) 4

English Proficiency

61. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. beyond the marked landing point. Q. Such was the force R. that the small boat was carried S. of the unexpected current

(A) R-Q-S-P (B) P-S-R-Q

(C) Q-S-R-P (D) S-Q-R-P

62. Read the sentence: With reliable support, the small reading club began to flourish. Choose the opposite in meaning of 'flourish' as used in the sentence.

(A) adapt (B) decline

(C) spread (D) continue

63. Read the sentence: The glossary served as a bridge between the specialist report and its new readers. What does the figurative expression mean in this context?

(A) It concealed disagreements between readers. (B) It physically joined two parts of the book.

(C) It helped readers connect unfamiliar language with understandable meanings. (D) It replaced the need to read the report.

64. Choose the active-voice equivalent that retains the tense, participants and all stated restrictions. The fragile object was carried by two assistants, not by the guide.

(A) Two assistants, not the guide, carried the fragile object. (B) Two assistants carry the fragile object while the guide watches.

(C) The guide, not the assistants, carried the fragile object. (D) Two assistants carried the guide instead of the fragile object.

65. Choose the most coherent order of the four sentences. A. Their captions used inconsistent spellings for the same place. B. The curator checked the names against a common reference list. C. Visitors could then recognise connections that the varying spellings had previously obscured. D. The exhibition included photographs from several different collections.

(A) A-D-B-C (B) D-A-C-B

(C) D-B-A-C (D) D-A-B-C

66. Complete: The unexpectedly neat result still required _____ scrutiny before it could be trusted.

(A) adjacent (B) close

(C) near (D) proximate

67. Read the sentence: The coating was uniform in thickness across the entire panel. Choose the opposite in meaning of 'uniform' as used in the sentence.

(A) thin (B) continuous

(C) smooth (D) uneven

68. Read the sentence: The witness offered a plausible explanation, although it still needed to be checked. Choose the closest meaning of 'plausible' as used in the sentence.

(A) seemingly reasonable (B) conclusively proved

(C) deliberately false (D) widely repeated

69. Microorganisms can break down the material through biological action; the statement does not merely mean that it dissolves in water. Which adjective fits?

(A) biodegradable (B) soluble

(C) reusable (D) recyclable

70. Use the standard contemporary negative adjective. The response was childish rather than mature; it was _____.

(A) unmature (B) immature

(C) dismature (D) inmature

Logical Reasoning

71. Dev points to a photograph and says, "This person is the son of my father's only daughter." Dev is male, and the photographed person is male. How is the photographed person related to Dev?

(A) Daughter (B) Brother

(C) Son (D) Nephew

72. Consider only these statements; universal statements alone do not imply existence. No bakers are inventors; Some inventors are campers; Some bakers are not campers. Conclusions: I. No campers are inventors. II. Some campers are not bakers. Which conclusions follow necessarily?

(A) Only II (B) Only I

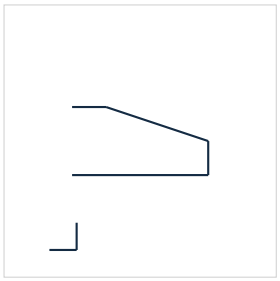
(C) Both I and II (D) Neither I nor II

73. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, TRAIN is coded as VTCKP. Which word is coded as RNCPV?

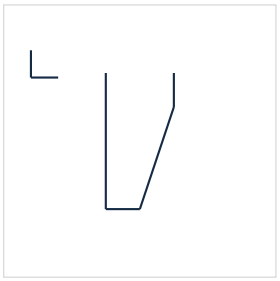
(A) QMBOU (B) PLANT

(C) OKZMS (D) TPERX

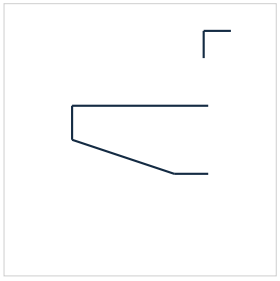
74. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

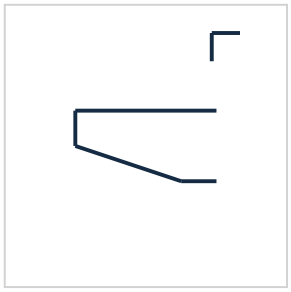


bottom left

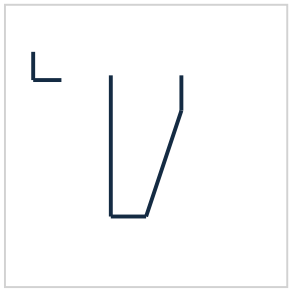
?

bottom right

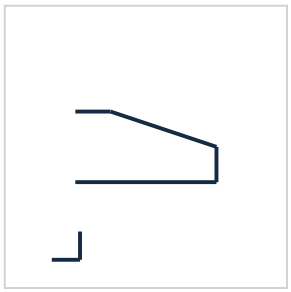
(A) Pattern T



(B) Pattern Z



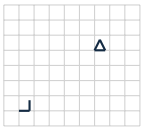
(C) Pattern L



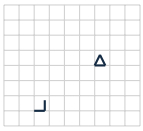
(D) Pattern U



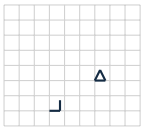
75. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



1

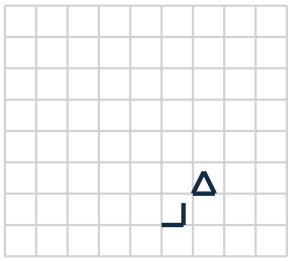


2

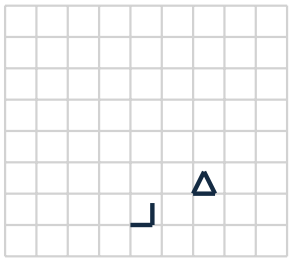


3

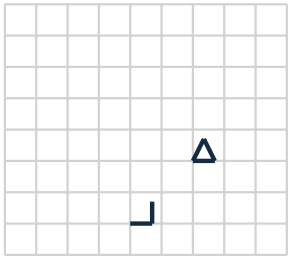
(A) Pattern N



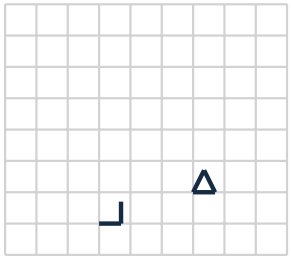
(B) Pattern T



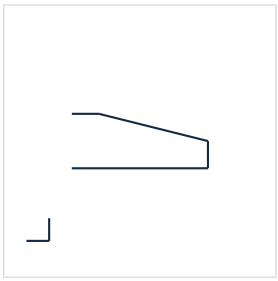
(C) Pattern Z



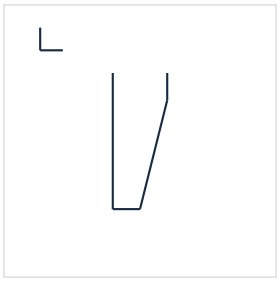
(D) Pattern V



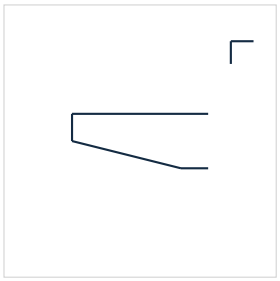
76. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

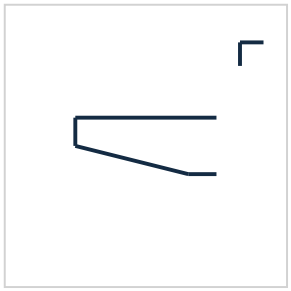


2

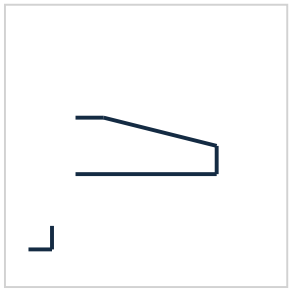


3

(A) Pattern Y



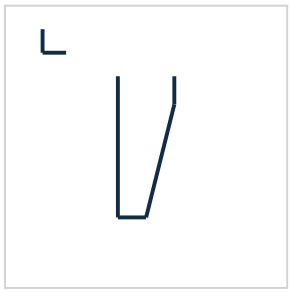
(B) Pattern R



(C) Pattern Z



(D) Pattern Q



77. The model triads are $(7, 16, 9)$ and $(9, 20, 11)$. In the intended rule the middle number is the sum of the outer numbers, and the outer-number difference is the same fixed positive value in every triad. Which option satisfies both relations?

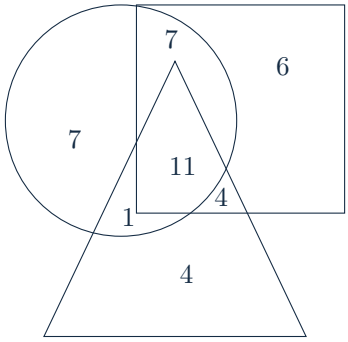
(A) $(6, 15, 9)$

(B) $(6, 14, 8)$

(C) $(6, 15, 8)$

(D) $(6, 16, 9)$

78. Circle=A, triangle=B, square=C. Each printed number is the count in that exact region. Find the total inside exactly one of A and C, but outside B.



(A) 6 (B) 7

(C) 13 (D) 14

79. All five people have different heights. Asha is shorter than Gita. Esha is shorter than Asha. Bina is shorter than Esha. Farah is shorter than Bina. If Gita is left out, who is 2nd shortest among the remaining people?

(A) Asha (B) Gita

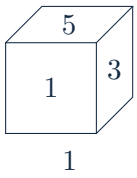
(C) Esha (D) Bina

80. $X\&Y$ means X is the mother of Y; $X@Y$ means X is the sister of Y. All names denote distinct people; a brother or sister is a full sibling, sharing both parents. Given $J\&E$ $E@F$ $F\&L$, how is J related to L?

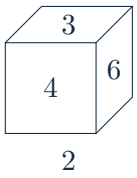
(A) Maternal grandmother (B) Father

(C) Mother (D) Maternal grandfather

81. The views show the same die. Give the ordered pair: (face opposite 1, face opposite 3). Do not assume opposite labels sum to seven.



1



2

(A) $(6, 3)$ (B) $(6, 4)$

(C) $(6, 1)$ (D) $(6, 2)$

82. 8 people sit around a circle facing its centre. Clockwise order fixes the convention; immediate left is clockwise. Dev is 2 places clockwise from Asha. Esha is 3 places clockwise from Asha. Farah is 5 places clockwise from Asha. Bina is 6 places clockwise from Asha. Hari is 7 places clockwise from Asha. Gita is not adjacent to Dev. Who sits immediately clockwise from Esha?

(A) Gita (B) Charu

(C) Bina (D) Asha

83. Three pairs contain exactly one mammal and one bird. Which pair does not?

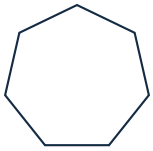
(A) Tiger / Whale (B) Bat / Eagle

(C) Cat / Sparrow (D) Horse / Crow

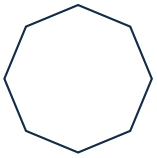
84. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

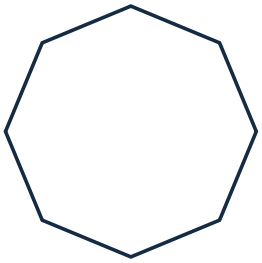


B

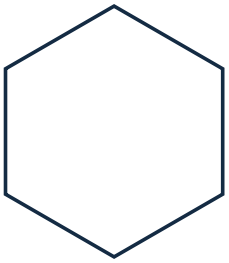


C

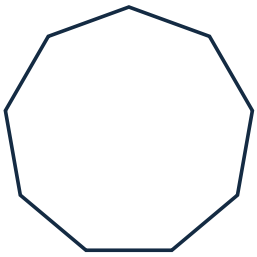
(A) Pattern L



(B) Pattern M



(C) Pattern V



(D) Pattern T



85. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: RTIM, TUFP, VVCS, ?, ZXWY, BYTB.

(A) WVYU (B) VZWX

(C) XWZV (D) YXAW

86. Every term from the third onward should equal the product of the two immediately preceding terms. Exactly one internal term (neither of the first two nor the last) was copied incorrectly: 1, 2, 2, 7, 8, 32. Which position contains the error? Positions begin at 1.

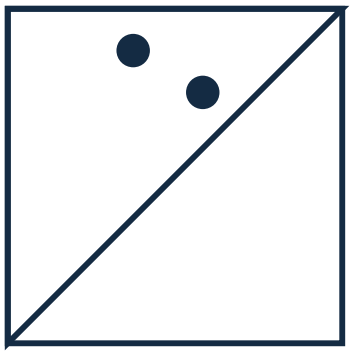
(A) 3 (B) 4

(C) 2 (D) 5

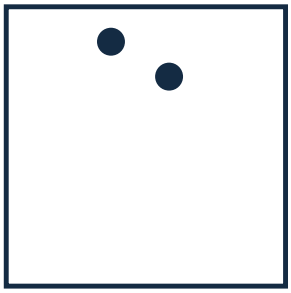
87. Preserve the exact relation and its order: Judge : Courtroom :: ? : Workshop. Choose the missing term.

(A) Mechanic
(B) Pilot
(C) Librarian
(D) Judge

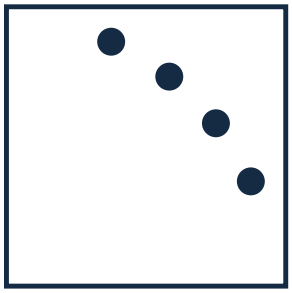
88. Starting with a unit square sheet, fold the part below the lower-left to upper-right diagonal onto the part above it. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



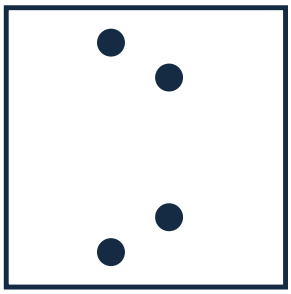
(A) Pattern V



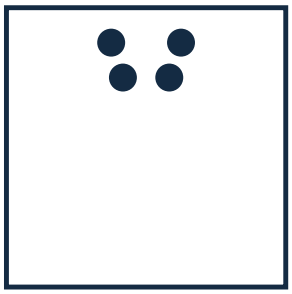
(B) Pattern Z



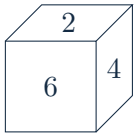
(C) Pattern N



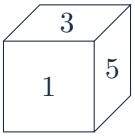
(D) Pattern U



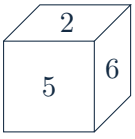
89. The views show the same die. Give the ordered pair: (face opposite 1, face opposite 3). Do not assume opposite labels sum to seven.



1



2



3

(A) $(2, 3)$
(B) $(2, 2)$
(C) $(2, 6)$
(D) $(2, 1)$

90. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 3, 12, ?, 68, 73, 292, 297.

(A) 21 (B) 7

(C) 17 (D) 43

Mathematics

91. Between positive a=36, b=121, let A be their arithmetic mean and G their geometric mean. Find A^2-G^2.

(A) $\frac{7225}{4}$ (B) 7225

(C) $\frac{85}{2}$ (D) $\frac{7225}{16}$

92. A complex number z satisfies z+conjugate(z)=14 and z-conjugate(z)=4i. Find z·conjugate(z).

(A) 25 (B) 106

(C) 53 (D) 212

93. How many discontinuities does (x-5)floor(x) have in (0,7)?

(A) 7 (B) 6

(C) 0 (D) 5

94. Under x=X+4, y=Y+3, the parabola $(y-3)^2=20(x-4)$ is normalized. Find the x coordinate of its focus in the original axes.

(A) −1 (B) 5

(C) 24 (D) 9

95. If z=2-4i, find |exp(iz)|.

(A) e^4 (B) e^2

(C) e^{-4} (D) 1

96. How many solutions in $0\leq x<4\pi$ satisfy $\sin^4x+\cos^4x=3/4$ and $\sin x>0$?

(A) 9 (B) 16

(C) 8 (D) 7

97. Positive a,b satisfy log(a+b)=log(3)+[log a+log b]/2 in one valid base. Find $(a^4+b^4)/(a^2b^2)$.

(A) 47 (B) 49

(C) 51 (D) 7

98. For $x\in[0,1]$, find the minimum of $(\arcsin x)^3+(\arccos x)^3$.

(A) $\frac{\pi^3}{16}$ (B) $\frac{\pi^3}{8}$

(C) $\frac{\pi^3}{64}$ (D) $\frac{\pi^3}{32}$

99. The line r=(3,2,3)+t(1,2,2) meets x+y+z=18 at Q. Find $|PQ|^2$ for P=(3,2,3).

(A) 18 (B) 36

(C) 110 (D) 6

100. Evaluate $\sum_{j=6}^{11} \binom{j}{2}$, with the convention $\binom{j}{r} = 0$ when j<r.

(A) 185 (B) 200

(C) 220 (D) 210

101. The system x+y+z=1, y+z=2, (k-7)z=k-9 has no solution for which k?

(A) 7 (B) 9

(C) −7 (D) 0

102. Write sin x $\cos^3$x+4$\sin^3$x cos x=P sin2x+Q sin4x. Find Q.

(A) $\frac{1}{8}$ (B) $\frac{5}{8}$

(C) $\frac{5}{4}$ (D) $-\frac{3}{8}$

103. f(x)=(x^2-100)/(x-10) for x≠10, and f(10)=k. If f is differentiable at 10, find k+f′(10).

(A) 11 (B) 21

(C) 20 (D) 22

104. $f(x)=x^3+2x$, $g=f^{-1}$. Find g″(33).

(A) $\frac{1}{18}$ (B) $\frac{18}{24389}$

(C) $-\frac{18}{841}$ (D) $-\frac{18}{24389}$

105. Find the area between y=(x-6)(x-13) and the x-axis for 0≤x≤16.

(A) $\frac{544}{3}$ (B) $\frac{887}{6}$

(C) $\frac{887}{3}$ (D) $-\frac{544}{3}$

106. Positive a,b,c form an AP with a=5. The numbers a,b,c+16/5 form a GP and b>a. Find c.

(A) 13 (B) 9

(C) 37 (D) −3

107. Find the coefficient of x^6 in (1-x)exp(2x).

(A) $\frac{4}{45}$ (B) $-\frac{8}{45}$

(C) $\frac{16}{45}$ (D) $-\frac{4}{15}$

108. Evaluate lim as x→0 of [sqrt(36+x)-6-x/12]/x^2.

(A) $-\frac{1}{288}$ (B) $\frac{1}{12}$

(C) $-\frac{1}{1728}$ (D) $\frac{1}{1728}$

109. Evaluate the limit of $[\cos(5x)-\cos(6x)]/x^2$ as $x\to 0$.

(A) $\frac{11}{2}$ (B) $-\frac{11}{2}$

(C) 11 (D) $\frac{1}{2}$

110. $f(x+1)=7x^2$. Find the area enclosed between $y=f(x)$ and $y=7(x-1)$.

(A) $\frac{7}{3}$ (B) $\frac{7}{2}$

(C) $-\frac{7}{6}$ (D) $\frac{7}{6}$

111. An invertible 4×4 matrix A has $\det A=34$. Find $\det(\text{adj } A)$.

(A) 136 (B) 39304

(C) 34 (D) $\frac{1}{34}$

112. Evaluate $\sin(2\arctan 3)$.

(A) $-\frac{3}{4}$ (B) $\frac{3\sqrt{10}}{10}$

(C) $\frac{3}{5}$ (D) $-\frac{3}{5}$

113. $x=4+9(1-t^2)/(1+t^2)$, $y=5+18t/(1+t^2)$, where t is real. Which point of the containing circle is not attained?

(A) $(4,-4)$ (B) $(4,14)$

(C) $(-5,5)$ (D) $(13,5)$

114. The letters A,A,B,C and 2 further distinct letters are arranged in a row. How many arrangements have nonadjacent As and have B,C adjacent in either order?

(A) 72 (B) 48

(C) 36 (D) 480

115. The region between $y=(x-5)(x-12)$ and the x-axis for $5\le x\le 12$ is cut by the line $x=c$, where $5<c<12$. Find c if the two resulting areas are equal.

(A) $\frac{7\sqrt{2}}{2}+5$ (B) $\frac{17}{2}$

(C) $5+\frac{7\cdot 2^{\frac{2}{3}}}{2}$ (D) $12-\frac{7\sqrt{2}}{2}$

116. How many integer t make $P=(t,0)$ strictly inside the larger segment cut by $x=2$ from $x^2+y^2=49$?

(A) 10 (B) 4

(C) 13 (D) 8

117. Chords of $x^2+y^2=36$ pass through (12,0). Find the radius of the circle containing their midpoint locus.

(A) 6 (B) 3

(C) 12 (D) 36

118. For f(x)=x^3-3kx^2+12x, how many integer k in [0,6] give a local maximum at a positive x?

(A) 6 (B) 5

(C) 4 (D) 3

119. The line r=(1,0,0)+s(t,1,2) is parallel to x-4y+z=6. Find the squared length of its direction vector.

(A) 7 (B) 3

(C) 21 (D) 9

120. X is binomial with mean 14 and variance $\frac{28}{3}$. Find its mode.

(A) 13 (B) 14

(C) 42 (D) 15

121. A positive integer-domain function satisfies f(m+n)=f(m)f(n), f(1)=5. Find $\sum_{j=1}^{6} f(j)$.

(A) 19530 (B) 15625

(C) 3906 (D) 93750

122. The chord of contact from P to $x^2/9$-$y^2/16$=1 is x=3/2. Find the product of the slopes of the two tangents from P.

(A) $-\frac{16}{27}$ (B) $-\frac{16}{9}$

(C) $\frac{16}{27}$ (D) $-\frac{16}{81}$

123. f(0)=0 and f(x)=$x^3\sin(1/x)$ for x≠0. Find f′(0).

(A) Does not exist (B) 0

(C) 1 (D) 3

124. A line through (8,5) has equal positive x- and y-intercepts. Find the sum of the coordinates of the midpoint of its intercept segment.

(A) 40 (B) $\frac{13}{2}$

(C) 13 (D) 26

125. Maximize Z=3x+4y subject to x≥0, y≥4, x+y≤12, x+2y≤15.

(A) 40 (B) 37

(C) 30 (D) 39

126. Find the sum of order and degree of $(d^2y/dx^2)^4+(dy/dx)^2+y=0$.

(A) 2 (B) 6

(C) 4 (D) 10

127. Find the sum of all solutions of $\sin^4 x + \cos^4 x = 3/4$ in $0 \le x < 2\pi$ that also satisfy $\sin x > 0$.

(A) $\frac{\pi}{2}$ (B) π

(C) 2π (D) 4π

128. In triangle ABC, A=2B, a/b=$\sqrt{3}$ and b=2. Find its area.

(A) 4 (B) $4\sqrt{3}$

(C) 2 (D) $2\sqrt{3}$

129. How many integer k in [0,110] make $x^2 - 16x + (55 - k) = 0$ have roots of opposite signs?

(A) 54 (B) 110

(C) 55 (D) 56

130. The hyperbola $x^2/4$-$y^2/9$=1 has eccentricity e. Find the eccentricity e′ of its conjugate hyperbola.

(A) $\frac{2\sqrt{13}}{13}$ (B) $\frac{\sqrt{13}}{2}$

(C) $\sqrt{13}$ (D) $\frac{\sqrt{13}}{3}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A uniform disc of initial mass 6 kg and radius 4 m has a concentric hole of radius $R/2$ removed. Find the remaining annulus's moment of inertia about a diameter lying in its plane.

(A) $\frac{45}{2}\,\mathrm{kg\,m^2}$ (B) $\frac{51}{2}\,\mathrm{kg\,m^2}$

(C) $24\,\mathrm{kg\,m^2}$ (D) $\frac{9}{2}\,\mathrm{kg\,m^2}$

132. A magnetic field is uniform inside a long solenoid of radius 6 m and negligible outside. It changes at 8 T/s. Find induced electric-field magnitude at radius $2R$ from the axis.

(A) 48 V/m (B) 12 V/m

(C) 24 V/m (D) 96 V/m

133. A series RLC circuit has $R = 70$ ohms, $X_L = 30$ ohms, and $X_C = 60$ ohms at the source frequency. An ideal source supplies 400 V rms. Find rms current.

(A) $\frac{20\sqrt{29}}{29}$ A (B) 4 A

(C) $\frac{20\sqrt{58}}{29}$ A (D) $\frac{20\sqrt{94}}{47}$ A

Chemistry

134. An octahedral Co(III) complex contains 1 neutral en ligands,with all remaining donor positions occupied by chloride. What is its coordination number and coordination-ion charge?

(A) CN 6; charge -1. (B) CN 7; charge -1.

(C) CN 6; charge -2. (D) CN 5; charge -1.

135. An internally consistent thermochemical comparison gives total resonance stabilisation of benzene as 36 kcal/mol (1 benzenoid rings) and naphthalene as 61 kcal/mol (2 rings). Which statement correctly separates total stabilisation from stabilisation per ring?

(A) Both quantities must rank compounds identically because each is aromatic. (B) Total: naphthalene; per ring: naphthalene.

(C) Total: naphthalene; per ring: benzene. (D) Total: benzene; per ring: benzene.

136. For electrophilic mononitration of benzene carrying NO2,which orientation/reason pair is correct? Consider intrinsic electronic directing effects,not an exact ortho:para ratio.

(A) meta; every electronegative substituent activates the ring

(B) meta; -M and -I withdrawal destabilise ortho/para sigma complexes particularly strongly

(C) ortho/para; -M and -I withdrawal destabilise ortho/para sigma complexes particularly strongly

(D) ortho/para; the substitution site is decided only by steric size

Logical Reasoning

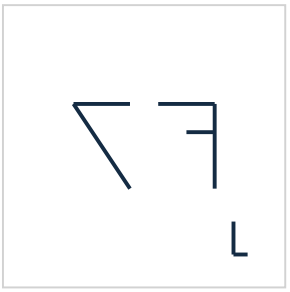
137. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



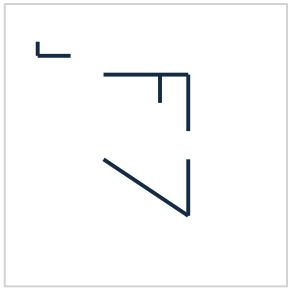
(A) Pattern R



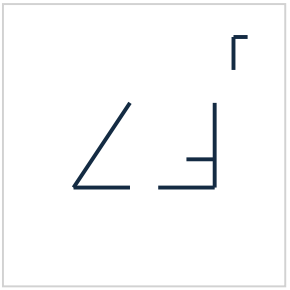
(B) Pattern Q



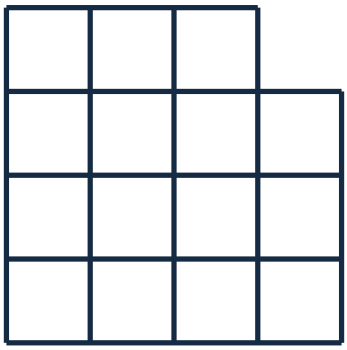
(C) Pattern K



(D) Pattern Z



138. How many rectangles, including squares are formed by all the drawn segments?



(A) 84

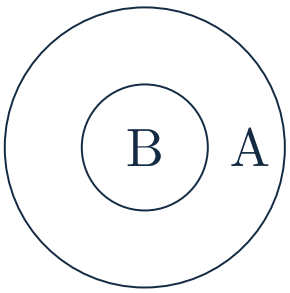
(B) 83

(C) 85

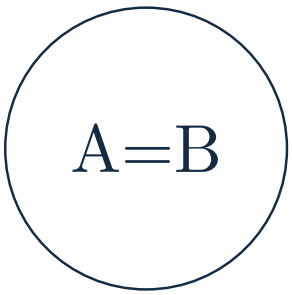
(D) 48

139. Choose the diagram that represents all these conditions, including the strictness information: Some A is B, some A is not B, and some B is not A.

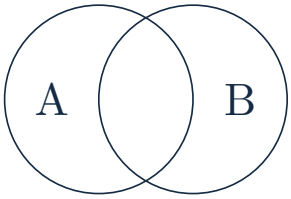
(A) Pattern U



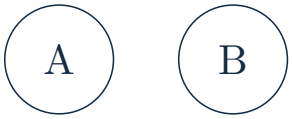
(B) Pattern L



(C) Pattern Q



(D) Pattern M



Mathematics

140. Let $a_j = 1/(6 + 4(j-1))$. Evaluate $\sum_{j=1}^{9} a_j a_{j+1}$.

(A) $\frac{1}{7}$ (B) $\frac{1}{28}$

(C) $\frac{2}{57}$ (D) $\frac{9}{38}$

141. How many real points of nondifferentiability does (x-3)|x-3|+|x-6| have?

(A) 2 (B) 0

(C) 1 (D) 3

142. y>0 satisfies $2yy'+8y^2=0$, y(0)=sqrt(5). Find $y^2(1)$.

(A) $\frac{5}{e^8}$ (B) $\frac{5}{e^4}$

(C) $5e^8$ (D) $\frac{\sqrt{5}}{e^4}$

BITSAT Mock 03

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A spring of constant 60 N/m is already stretched by $\frac{2}{5}$ m. How much additional work is required to increase its extension to $\frac{3}{5}$ m quasistatically?

(A) $\frac{6}{5}$ J
(B) 6 J
(C) $\frac{54}{5}$ J
(D) 12 J

2. A long solenoid has its turn number multiplied by 5 and length multiplied by 2, while its cross-sectional area and core permeability stay fixed. Find self-inductance multiplier.

(A) 50
(B) $\frac{25}{4}$
(C) $\frac{5}{2}$
(D) $\frac{25}{2}$

3. A block of mass 5 kg is fixed at rest while a particle of mass 7 kg moves with constant acceleration $\vec{a} = 4\hat{i} + 8\hat{j}$ m/s^2. What is the magnitude of acceleration of the centre of mass of the two-body system?

(A) 7 m/s^2
(B) $\frac{5\sqrt{5}}{3}$ m/s^2
(C) $\frac{7\sqrt{5}}{3}$ m/s^2
(D) $2\sqrt{5}$ m/s^2

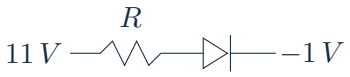
4. A series RLC circuit has $R = 30$ ohms, $X_L = 30$ ohms, and $X_C = 60$ ohms at the source frequency. An ideal source supplies 500 V rms. Find rms current.

(A) $\frac{25}{3}$ A
(B) $\frac{5\sqrt{10}}{3}$ A
(C) $\frac{50}{3}$ A
(D) $\frac{25\sqrt{2}}{3}$ A

5. A uniform block of density 700 kg/m^3 floats in a liquid. Then the liquid density is reduced to $\frac{4}{5}$ times 1000 kg/m^3, while the block's volume and density are unchanged. What fraction of the block is now submerged?

(A) $\frac{8}{7}$
(B) $\frac{4}{5}$
(C) $\frac{7}{8}$
(D) 1

6. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 4 kΩ to a terminal at +11 V, and its cathode connects to a terminal at -1 V. Find the current from the +11 V terminal in mA.



(A) $\frac{113}{40}$ mA
(B) 3 mA
(C) $\frac{53}{20}$ mA
(D) $\frac{103}{40}$ mA

7. A bar magnet of moment 6 A m^2 is rotated quasistatically from 60° to 120° relative to a uniform 6 T magnetic field. Find external work.

(A) 18 J
(B) 36 J
(C) $36\sqrt{3}$ J
(D) $18\sqrt{3}$ J

8. A platform of moment of inertia 5 kg m^2 carries a person of mass 2 kg at radius 3 m and rotates at 7 rad/s. The person walks to half the stated initial radius; external torque is zero. Find final angular speed. Treat the person as a point mass.

(A) $\frac{23}{2}$ rad/s
(B) 7 rad/s
(C) $\frac{322}{19}$ rad/s
(D) 28 rad/s

9. A gas receives 400 J of heat while its surroundings do 150 J of work on it. It then loses 200 J of heat at fixed volume. Find its total internal-energy change.

(A) 50 J
(B) 350 J
(C) 200 J
(D) 750 J

10. A liquid of volume 2 litres and bulk modulus 6000000000 Pa is subjected to an additional pressure of 5000000 Pa. Neglect changes in bulk modulus. Find the magnitude of volume decrease in cubic centimetres.

(A) $\frac{5}{3}$ cm^3
(B) $\frac{1}{600}$ cm^3
(C) $\frac{5000}{3}$ cm^3
(D) 2400000 cm^3

11. A conveyor receives 2 packets each second, each of mass $\frac{1}{10}$ kg and speed 4 m/s. It accelerates them to 10 m/s without changing elevation. Find the ideal mechanical power delivered to the packets.

(A) $\frac{42}{5}$ W
(B) $\frac{84}{5}$ W
(C) $\frac{6}{5}$ W
(D) $\frac{21}{5}$ W

12. A screw gauge advances 2.0 mm in four complete turns and has 100 circular divisions. With the jaws closed, its zero mark lies 4 divisions above the reference line; use the usual convention that a zero below the reference line gives positive zero error. With a wire, the main scale reads 8 mm and circular reading is 38 divisions. What is the corrected diameter?

(A) $\frac{821}{100}$ mm
(B) $\frac{1219}{100}$ mm
(C) $\frac{4583}{100}$ mm
(D) $\frac{817}{100}$ mm

13. A charge in a plane vacuum wave moves at speed $c/7$ along a direction making angle θ with the magnetic field, with $\cos\theta = 1/7$. Find the ratio of the electric-force magnitude to the magnetic-force magnitude.

(A) 49
(B) $\frac{49\sqrt{3}}{12}$
(C) $\frac{4\sqrt{3}}{49}$
(D) $4\sqrt{3}$

14. A 2 ohm resistor is connected in parallel with an unknown resistor across 4 V. The source supplies $\frac{32}{3}$ W in total. Find the unknown resistance.

(A) $\frac{24}{13}\,\Omega$
(B) $\frac{1}{2}\,\Omega$
(C) $\frac{6}{7}\,\Omega$
(D) $6\,\Omega$

15. An object moves slowly along the axis toward a concave mirror. At an instant its distance from the mirror is 60 cm; the mirror focal-length magnitude is 15 cm. Find the image-speed magnitude divided by object-speed magnitude at that instant.

(A) $\frac{1}{16}$
(B) $\frac{1}{25}$
(C) 9
(D) $\frac{1}{9}$

16. A Carnot engine initially runs between 600 K and 300 K. Its hot reservoir is raised by 20 K. By how much must the cold-reservoir temperature rise to keep efficiency unchanged?

(A) 10 K
(B) $\frac{300}{31}$ K
(C) 20 K
(D) 40 K

17. A 3 kg block moves along a horizontal rough surface from $x = 0$ to $x = 2$ m under a force $F(x) = 2x$ N parallel to motion. With $\mu_k = 1/5$ and $g = 10$ m/s^2, find the work done by the net force. The block has enough initial speed to keep moving forward.

(A) 16 J
(B) -8 J
(C) -4 J
(D) 4 J

18. A nucleus of mass number 24 splits into two equal nuclei. The parent has binding energy per nucleon $\frac{73}{10}$ MeV and each daughter has $\frac{79}{10}$ MeV. Ignoring other products, what energy is released in the split?

(A) $\frac{3}{5}$ MeV
(B) 204 MeV
(C) $\frac{72}{5}$ MeV
(D) $\frac{948}{5}$ MeV

19. Light is incident from index $\frac{11}{10}$ onto index $\frac{12}{5}$. At one incidence angle the reflected beam is completely plane polarised. Find the tangent of that angle.

(A) $\frac{24}{13}$
(B) $\frac{12}{5}$
(C) $\frac{24}{11}$
(D) $\frac{\sqrt{455}}{11}$

20. Two small identical conducting spheres initially carry positive charges $2q_0$ and $2q_0$ and are separated by distance d. They touch and are returned to the same separation. Find the final force divided by the initial force.

(A) 2
(B) $\frac{8}{9}$
(C) $\frac{3}{4}$
(D) 1

21. A particle has $v(0) = 6$ m/s and acceleration $a(t) = 10t$ m/s^2 for $0 \le t \le 3$ s. Find its average velocity over this interval.

(A) 21 m/s (B) 15 m/s

(C) 6 m/s (D) $\frac{57}{2}$ m/s

22. A wire carries $i(t) = 4 + 8\sin\omega t$ A for many cycles. Find its rms current.

(A) $4\sqrt{2}$ A (B) $4\sqrt{3}$ A

(C) 4 A (D) $2\sqrt{10}$ A

23. An ideal mixture contains 1 moles of monatomic gas and 4 moles of rigid diatomic gas. Vibrations are frozen. Find the heat-capacity ratio $\gamma = C_P/C_V$ of the mixture.

(A) $\frac{25}{23}$ (B) $\frac{27}{17}$

(C) $\frac{33}{23}$ (D) $\frac{7}{5}$

24. A body is dropped from rest at height 60 m above the ground. Find its speed when it is still 30 m above the ground. Neglect air resistance and take $g = 10$ m/s^2.

(A) $10\sqrt{6}$ m/s (B) $20\sqrt{3}$ m/s

(C) $20\sqrt{15}$ m/s (D) $10\sqrt{3}$ m/s

25. A thin straight tube of length 4 m and internal area $\frac{1}{2500}$ m^2 is completely filled with an incompressible liquid of density 700 kg/m^3. It rotates horizontally about one end at angular speed 6 rad/s. Ignoring gravity along the tube, find the pressure difference between its outer and inner ends.

(A) 201600 Pa (B) 50400 Pa

(C) 67200 Pa (D) 403200 Pa

26. A metal has work function 1 eV and is illuminated separately with photons of energies 9 eV and 33 eV. The fastest electrons enter the same perpendicular magnetic field. Find the second orbit radius divided by the first. Use nonrelativistic motion.

(A) $\frac{4\sqrt{2}}{3}$ (B) $\frac{\sqrt{85}}{5}$

(C) 2 (D) 4

27. A solid sphere of radius R has uniform density $2\rho_0$. A spherical cavity of radius $R/10$ is cut with its centre at distance $R/2$ from the original centre. Find field magnitude at the cavity centre in units of $\pi G\rho_0 R$.

(A) $\frac{16}{3}$ (B) $\frac{2}{3}$

(C) $\frac{8}{3}$ (D) $\frac{4}{3}$

28. A thin glass lens of refractive index 3/2 has focal length 250 cm in air. It is completely immersed in a liquid of index 4/3. Find its new focal length.

(A) 750 cm (B) 1000 cm

(C) $\frac{125}{2}$ cm (D) 250 cm

29. In a capillary experiment the tube radius is multiplied by 4, the liquid surface tension by 2, and density and contact angle remain unchanged. Find the new capillary rise divided by the original rise.

(A) 2 (B) $\frac{1}{4}$

(C) 8 (D) $\frac{1}{2}$

30. A uniform massless spring of constant 60 N/m is cut into 3 equal parts. All parts are connected in parallel to support a mass 4 kg. Find its small-oscillation period.

(A) $\frac{2\sqrt{5}\pi}{5}$ s (B) $\frac{2\sqrt{15}\pi}{45}$ s

(C) $\frac{2\sqrt{5}\pi}{45}$ s (D) $\frac{2\sqrt{5}\pi}{15}$ s

Chemistry

31. For $M(s)+2A^{+}\rightleftharpoons M^{2+}+2A(s), E^{\circ}=\frac{59}{2000}V$. Use$2.303RT/F=.0590V$. Equilibrium constant for the doubled reaction$2M+4A^{+}\rightleftharpoons 2M^{2+}+4A$ is

(A) 10000 (B) 100

(C) $\frac{1}{100}$ (D) 10

32. A $\frac{1}{50}$mol iron-ion inventory contains Fe^{2+}andFe^{3+}. Fraction$\frac{1}{2}$isFe^{3+}. All iron is quantitatively reduced to metal in the stated externally driven electron-balance model,with no side reactions. Electrons accepted, in mol,are

(A) $\frac{1}{20}$ (B) $\frac{7}{150}$

(C) $\frac{3}{100}$ (D) $\frac{4}{75}$

33. For NaH,which classification and consequence are mutually consistent?

(A) All hydrides contain free H+ ions in their solid lattices.

(B) Covalent; nonstoichiometry proves all hydrogen has been removed.

(C) ionic; H^{-} reacts with water to evolve H_2.

(D) Ionic; every example must be a gas composed of discrete molecules.

34. Nonrelativistic ions A,B start from rest. B has 4 times the mass and 2 times the positive charge of A,and is accelerated through 1 times A's potential difference. Ratioλ_B/λ_A is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) $\frac{\sqrt{2}}{4}$ (D) $2\sqrt{2}$

35. Compare methyl bromide and ethyl bromide toward the same cyanide nucleophile in a polar aprotic solvent under ordinary SN2 conditions. Which reacts faster or is the substituted site,and why?

(A) ethyl bromide; an sp2 C-Br bond behaves just like an unhindered methyl bromide

(B) ethyl bromide; steric hindrance accelerates backside attack

(C) methyl bromide; the mechanism requires formation of a free tertiary carbocation

(D) methyl bromide; Methyl bromide offers the least hindered backside approach.

36. A weak1:1electrolyte has$\alpha \ll 1$ and obeysKa$\approx C\alpha^2$. Dilute it by a factor4without changing total formula-unit amount or Ka,and remain in that approximation. Which pair(degreeα,totalfree-ion amount)is correct?

(A) α increases byfactor4,buttotalionsdecreasebyfactor4.

(B) Both remain unchanged because the number of solute formula units is fixed.

(C) Both increase by factor2.

(D) α decreases byfactor2,andtotalionsareunchanged.

37. A textbook deficiency table associates night blindness with one vitamin. Which vitamin and solubility class are correctly paired?

(A) Vitamin B1; water-soluble.

(B) Vitamin C; water-soluble.

(C) Vitamin B6; water-soluble.

(D) VitaminA; fat-soluble.

38. Construct one open linear peptide using each member of the following multiset exactly once: Pro, Gly, Ala, Val, Ser. The N-terminal residue isPro and the C-terminal residue is the only achiral amino acid. Count distinctN-to-C sequences; ignore stereoisomerism of the supplied residues.

(A) 24

(B) 7

(C) 6

(D) 120

39. In a hydrogen-like ion Z=1,an electron falls from n=6 to an s orbital with 3 radial nodes. With hydrogen ionisation energy13.6 eV and$hc = 1240$ eV nm,the emitted wavelength in nm is

(A) $\frac{55800}{17}$

(B) $\frac{89280}{17}$

(C) $\frac{44640}{17}$

(D) $\frac{11160}{17}$

40. An orbital has 1 angular nodes and 1 radial nodes. Maximum electrons in its entire subshell are

(A) 10

(B) 6

(C) 2

(D) 18

41. For the metal centre in CrO_4^{2-},Z=24. Determine the number of d electrons after formal oxidation-state assignment; use O=-2 in oxoanions.

(A) 1

(B) 0

(C) 6

(D) 2

42. Separate unknowns P and Q are each selected from Cu2+, Zn2+, Fe3+. Dropwise aqueous ammonia gives P a pale blue hydroxide precipitate and Q a white precipitate. Excess ammonia gives P a deep-blue ammine solution and Q a colourless ammine solution. Which identification is consistent, without invoking redox?

(A) P=Cu2+; Q=Zn2+; oxidation states unchanged.

(B) Both are the same ion because all hydroxides have identical ammonia behaviour.

(C) P=Cu2+; Q=Zn2+; both metals reduced to zero.

(D) P=Zn2+; Q=Cu2+; oxidation states unchanged.

43. Limiting molar conductivities of KCl,NaCl,KNO_3 are150,126,110 S cm^2/mol. A very dilute $NaNO_3$ solution measured in a cell of constant2.00cm^{-1} has conductance $\frac{43}{25000}$S. Neglect concentration corrections. $NaNO_3$ molarity is

(A) $\frac{1}{25}$ (B) $\frac{43}{1375}$

(C) $\frac{1}{50}$ (D) $\frac{2}{25}$

44. For isolated molecules of NH_3 and NF_3,which member has the larger permanent dipole moment,and which reasoning is sufficient?

(A) They must be equal because both contain polar covalent bonds.

(B) NH_3; N-H bond-resultant and lone-pair contribution reinforce in ammonia but largely oppose for NF3.

(C) NF_3; the number of polar bonds alone fixes the molecular dipole.

(D) The one with more atoms always has the larger dipole,independent of geometry.

45. At25C,a measurement gives$[OH^-]$=2×10^{-5}M. UseKw=10^{-14} and$\log_{10}2=\frac{301}{1000}$. Solution pH is

(A) $\frac{8699}{1000}$ (B) $\frac{9301}{1000}$

(C) $\frac{4699}{1000}$ (D) $\frac{5301}{1000}$

46. $M(OH)_2$ has molar solubility $\frac{3}{100000}$ M in pure water. Mix100mL0.00200M NaOH with100mL0.00100M HCl,then add excess solid $M(OH)_2$. Ignore its small contribution to OH^-. Final molar solubility is

(A) $\frac{27}{1000000000}$ (B) $\frac{27}{250000000}$

(C) $\frac{3}{100000}$ (D) $\frac{27}{62500000}$

47. At 300 K,combustion enthalpies of C(graphite) and CO(g) to CO_2(g) are respectively-393.5 and-283.0 kJ/mol. Treat gases ideally and use R=0.00800 kJ mol^{-1} K^{-1}. Constant-volume heat for C(graphite)+1/2 O_2(g)→CO(g),per mole CO,in kJ is

(A) $-\frac{1129}{10}$ (B) $-\frac{1117}{10}$

(C) $-\frac{1093}{10}$ (D) $-\frac{221}{2}$

48. For NO_3^-,use ordinary octet Lewis structures and resonance. What is the pair (average bond order of three equivalent N-O bonds,total lone pairs on all atoms of that same named species in one contributing structure)?

(A) ($\frac{4}{3}$, 7). (B) (1, 8).

(C) (2, 9). (D) ($\frac{4}{3}$, 8).

49. For SO_2,use explicitly octet-preserving charge-separated Lewis contributors and resonance. What is the pair (average bond order of the octet-preserving charge-separated resonance model for its two S-O bonds,total lone pairs on all atoms of that same named species in one contributing structure)?

(A) ($\frac{3}{2}$, 5). (B) (1, 6).

(C) (2, 7). (D) ($\frac{3}{2}$, 6).

50. Match two applications: I. a low-temperature coolant for superconducting equipment. II. an orange-red advertising discharge tube. Which ordered gas pair is appropriate?

(A) helium; neon.
(B) neon; helium.
(C) helium; argon.
(D) neon; neon.

51. For2A+B→3C+4D,the measured total disappearance rate of A is $\frac{7}{500}$M/s. A separate calibrated decay channel consumes A at $\frac{7}{2500}$M/s and produces no D. D appearance rate from the stated reaction,in M/s,is

(A) $\frac{28}{625}$
(B) $\frac{7}{250}$
(C) $\frac{7}{1250}$
(D) $\frac{14}{625}$

52. Benzoic acid is mononitrated,then ring brominated under electrophilic conditions. Assuming COOH and the first NO2 group retain their usual meta direction,which relative substitution pattern is favoured?

(A) COOH at1,NO2 at3,Br at5.
(B) COOH at1,NO2 at2,Br at4.
(C) COOH at1,NO2 at4,Br at2.
(D) COOH at1,NO2 at3,Br at2.

53. A carbon-containing organic compound contains nitrogen only among N,S and halogens. Sodium fusion is performed with limited sodium; if both N and S are present, thiocyanate formation is retained. Which product/test pairing is consistent?

(A) Na2SO4; a violet nitroprusside response for sulfate.
(B) NaCN; Prussian-blue nitrogen test.
(C) NaNO3; a brown-ring test directly on the organic compound.
(D) Elemental chlorine; a Prussian-blue response.

54. Ammonia from a $\frac{1}{5}$ g sample in a valid Kjeldahl determination is absorbed in $\frac{1}{50}$ mol HCl. Back-titration finds $\frac{1}{125}$ mol HCl left. Find nitrogen mass percentage, taking N=14.

(A) 84
(B) 140
(C) 56
(D) 168

55. cyclohexene (C6H10) adds one equivalent of :CCl2 across its only C=C without rearrangement. Which pair gives the adduct formula and the number of independent rings? Count the original ring and the new three-membered ring separately.

(A) C6H10Cl2; 1
(B) C7H10Cl2; 1
(C) C7H10Cl2; 2
(D) C7H12Cl2; 2

56. propene reacts with HBr with peroxide. Which pair gives the major constitutional product and the intermediate whose greater stability fixes its orientation? Ignore stereoisomer counting.

(A) 2-bromopropane; secondary radical
(B) 1-bromopropane; primary radical
(C) 1-bromopropane; secondary radical
(D) 2-bromopropane; primary radical

57. In a copper extraction stage, $\frac{1}{50}$ mol Cu_2S and $\frac{3}{100}$ mol Cu_2O react only by $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$. What amount of metallic copper, in mol, is formed?

(A) $\frac{7}{100}$
(B) $\frac{3}{25}$
(C) $\frac{3}{100}$
(D) $\frac{9}{100}$

58. 2-bromo-2-methylbutane is heated with potassium tert-butoxide under elimination-favouring conditions. Which major constitutional product is expected,despite the presence of an alkoxide that could be used in a Williamson synthesis with a suitable different halide?

(A) 2-methylbut-1-ene

(B) 2-methylbut-2-ene

(C) the corresponding tertiary alkoxide substitution product

(D) the alkane formed by replacing halogen with H

59. A mixture contains $\frac{3}{100}$ mol H_2SO_5 and $\frac{1}{50}$ mol $H_2S_2O_8$. Their structures contain one O–O link per molecule. All peroxide links react quantitatively with excess iodide in acid; each such link accepts two electrons. What amount of I_2, in mol, is liberated?

(A) $\frac{1}{20}$

(B) $\frac{7}{100}$

(C) $\frac{1}{40}$

(D) $\frac{2}{25}$

60. For propanal and 2,2-dimethylpropanal,which structural-reactivity comparison is correct under ordinary enolate/aldol conditions?

(A) both self-aldolise

(B) propanal self-aldolises;2,2-dimethylpropanal does not

(C) neither self-aldolises

(D) only 2,2-dimethylpropanal self-aldolises

English Proficiency

61. Read the sentence: The concise note explained the decision in a few well-chosen lines. Choose the opposite in meaning of 'concise' as used in the sentence.

(A) brief

(B) wordy

(C) clear

(D) direct

62. The attitude emphasises the possibility of success while acknowledging present difficulties. Which word names it?

(A) nostalgia

(B) optimism

(C) pessimism

(D) indifference

63. The publication lists available exhibits with descriptions so that a reader can identify them. What is it?

(A) narrative

(B) receipt

(C) catalogue

(D) itinerary

64. Choose the correctly spelt word.

(A) calandar

(B) calendarr

(C) callendar

(D) calendar

65. Choose the most coherent order of the four sentences. A. On comparing the notes, she noticed that the observation times differed greatly. B. She therefore repeated several visits at comparable times of day. C. A student wrote detailed field notes during every visit to the wetland. D. The new records allowed her to separate a timing effect from the change she had first suspected.

(A) C-A-D-B

(B) C-A-B-D

(C) C-B-A-D

(D) A-C-B-D

66. Read the sentence: Asked to choose between the proposals, she remained on the fence until more evidence arrived. What does 'on the fence' mean here?

(A) undecided between alternatives (B) unwilling to listen

(C) secretly opposed to both sides (D) excluded from the decision

67. Choose the meaning-preserving equivalent that retains the tense, participants and all stated restrictions. The print was too faint to be read without magnification.

(A) The print was faint enough to be read without magnification.

(B) The print was so faint that it could not be read even with magnification.

(C) The print was so faint that it could not be read without magnification.

(D) The print was unreadable because magnification was used.

68. Use the simple present to express the stated regular procedure. Choose the best replacement: Visitors who need assistance [reports] to the desk as soon as they arrive.

(A) report (B) has reported

(C) is reporting (D) was reporting

69. Read the sentence: A minute crack was visible only under magnification. Choose the closest meaning of 'minute' as used in the sentence.

(A) extremely small (B) recently formed

(C) precisely measured (D) likely to spread

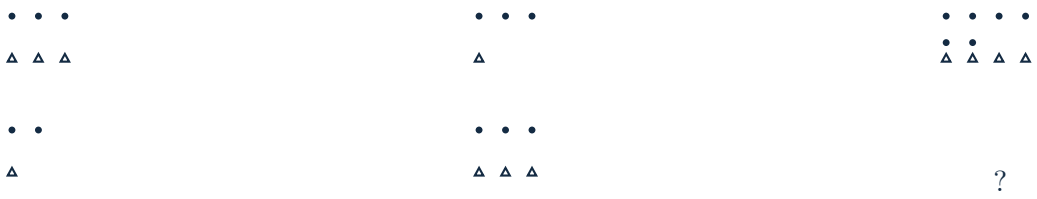
70. Read the sentence: Her explanation was consistent with the records; nothing in them contradicted it. Choose the closest meaning of 'consistent' as used in the sentence.

(A) independently proved (B) equally detailed

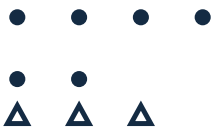
(C) identical in wording (D) in agreement

Logical Reasoning

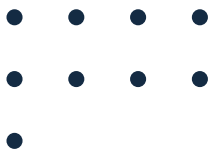
71. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



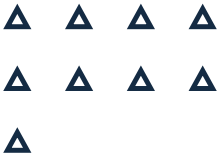
(A) Pattern V



(B) Pattern X



(C) Pattern R



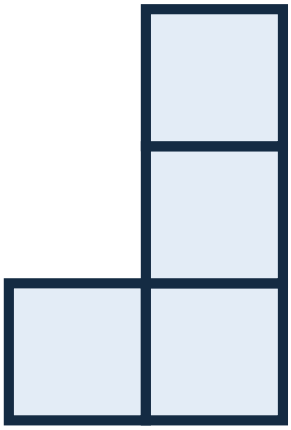
(D) Pattern W



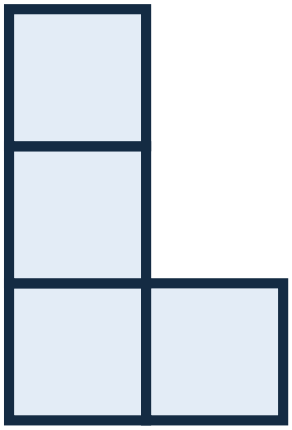
72. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



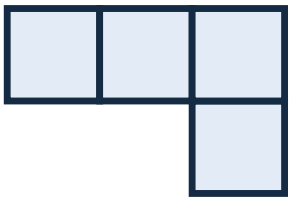
(A) Pattern M



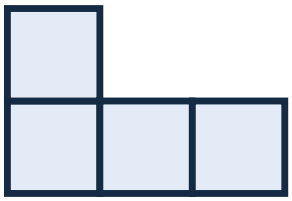
(B) Pattern K



(C) Pattern Z



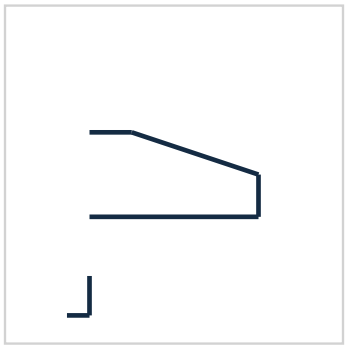
(D) Pattern R



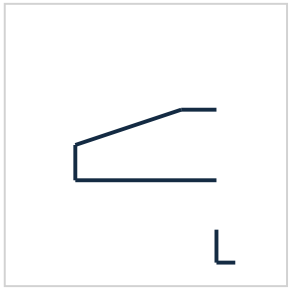
73. A community register uses the following three categories: campers, guides, artists. Its complete relevant assertions are in this short paragraph; do not import everyday assumptions. Some guides are not artists. No artists are campers. No guides are campers. Conclusions: I. No campers are guides. II. All campers are artists. Which conclusions follow necessarily?

(A) Only II
(B) Both I and II
(C) Only I
(D) Neither I nor II

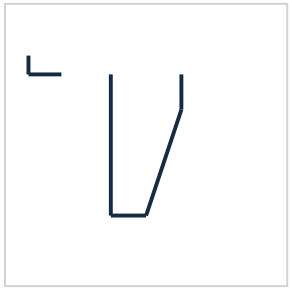
74. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



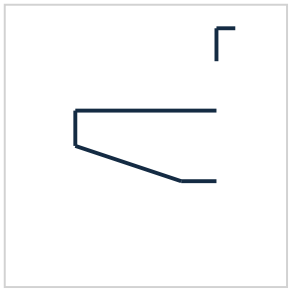
(A) Pattern T



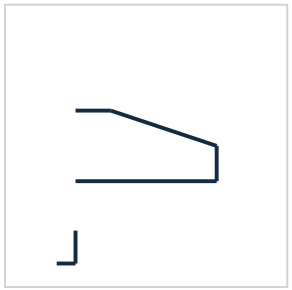
(B) Pattern R



(C) Pattern N



(D) Pattern X



75. Consider only these statements; universal statements alone do not imply existence. Some guides are guides; All artists are campers. Conclusions: I. All guides are artists. II. Some guides are not artists. Which conclusion or complementary alternative must hold?

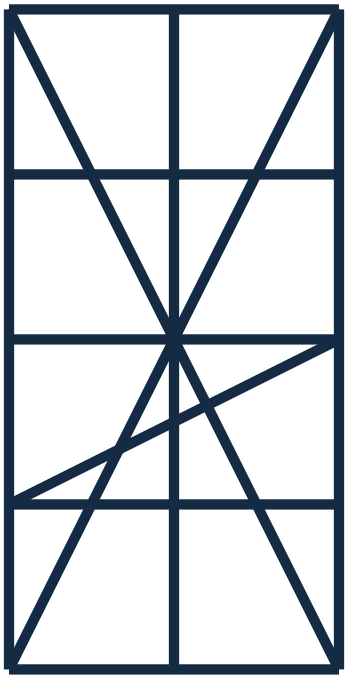
(A) Either I or II follows, but neither is individually certain

(B) Only I

(C) Both I and II

(D) Only II

76. How many triangles are formed by all the drawn segments?



(A) 44 (B) 41

(C) 79 (D) 45

77. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





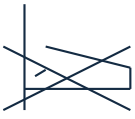
host 1



host 2



host 3

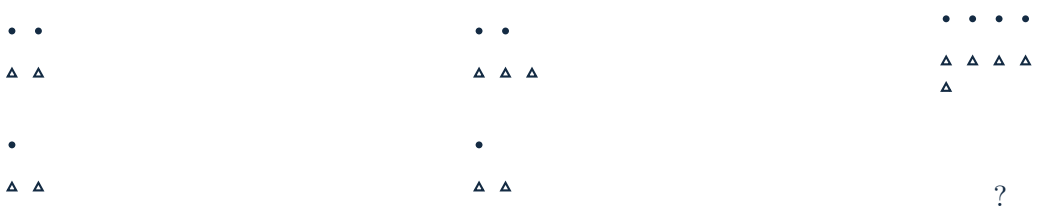


host 4

(A) 4 (B) 3

(C) 2 (D) 1

78. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.

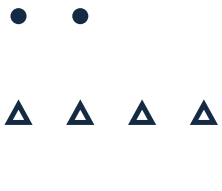


(A) Pattern M

(B) Pattern U



(C) Pattern Z

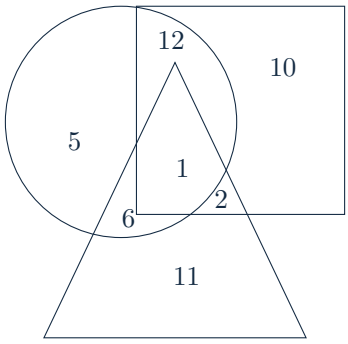


(D) Pattern X





79. Circle=A, triangle=B, square=C. Each printed number is the count in that exact region. Find the total inside A but outside B, whether inside or outside C.



(A) 5

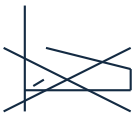
(B) 18

(C) 17

(D) 12

80. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

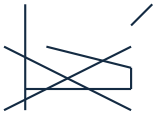




host 1



host 2



host 3



host 4

(A) 4 (B) 3

(C) 1 (D) 2

81. Preserve the exact relation and its order: Begin : Commence :: ? : Help. Choose the missing term.

(A) Assist (B) Begin

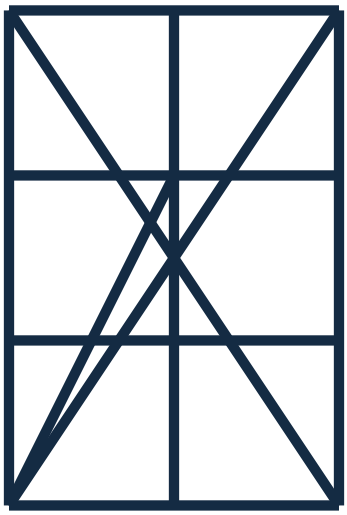
(C) Rapid (D) Silent

82. In this series the first differences form a geometric progression with positive integer ratio. Find the missing internal term: 2, 5, 11, ?, 47, 95, 191.

(A) 23 (B) 20

(C) 24 (D) 26

83. How many triangles are formed by all the drawn segments?



(A) 29 (B) 38

(C) 39 (D) 63

84. The age table uses the same two people throughout:

	younger	older
now	x	$x+5$
4 years ago	$x-4$	$x+1$

. The product of their ages 4 years ago was 374. What will the younger person's age be 3 years from now?

(A) 24 (B) 20

(C) 21 (D) 29

85. Bina receives 2 times as many tokens as Asha, and Charu receives 3 times as many as Bina. Their total is 144. After the allocation, Charu gives 2 tokens to Asha. How many tokens does Charu then retain?

(A) 14 (B) 98

(C) 94 (D) 96

86. R is the only brother of U; they are full siblings. L is the father of R, and U is married to V. All four are distinct people. How is L related to V?

(A) Maternal grandfather (B) Father

(C) Brother-in-law (D) Father-in-law

87. In a row, Asha is 5th from the left and 30th from the right. Bina is 7th from the right. Charu stands 2 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 22 (B) 19

(C) 21 (D) 20

88. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 13$ and $5 \mapsto 31$. Which input maps to 91?

(A) 14 (B) 15

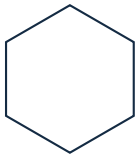
(C) 16 (D) 91

89. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, MANGO is coded as PDQJR. Which word is coded as VWRQH?

(A) TUPOF (B) STONE

(C) RSNMD (D) YZUTK

90. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

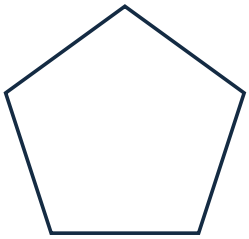


B

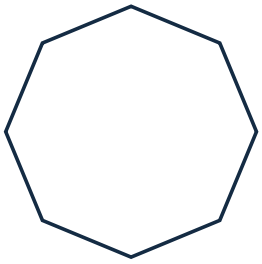


C

(A) Pattern Z



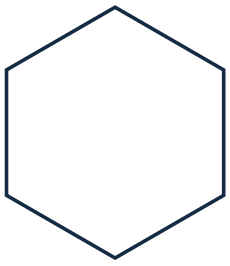
(B) Pattern R



(C) Pattern K



(D) Pattern M



Mathematics

91. A circular sector has radius 3 and arc length $\frac{3\pi}{5}$. Find its area.

(A) $\frac{27\pi}{10}$ (B) $\frac{9\pi}{10}$

(C) $\frac{\pi}{10}$ (D) $\frac{3\pi^2}{50}$

92. Find the population variance of 2+3, 4+3, ..., 22+3.

(A) 20 (B) 10

(C) 40 (D) 43

93. For R={(j,j+1):1≤j<6} on {1,...,6}, what is the minimum number of ordered pairs to add to make R transitive?

(A) 16 (B) 15

(C) 5 (D) 10

94. Evaluate $\sum_{j=1}^{\infty} 3j(\frac{1}{3})^j$.

(A) $\frac{15}{4}$ (B) $\frac{9}{2}$

(C) $\frac{3}{4}$ (D) $\frac{9}{4}$

95. The line 3x+4y=20 cuts $x^2+y^2=64$ in chord AB. Find $|AB|^2$.

(A) 144 (B) 192

(C) 320 (D) 224

96. Evaluate $\sum_{j=1}^{10} j(j+5)$.

(A) 275 (B) 385

(C) 3300 (D) 660

97. y′=x+y-1, y(0)=1. Find y′(1)-y(1).

(A) 0 (B) 1

(C) $-1+e$ (D) -1

98. The normal of slope 4 to $y^2=12x$ touches at P. Find the distance from P to the focus.

(A) 45 (B) 48

(C) 2601 (D) 51

99. a=(3,8,0), b=(8,-3,5). Find $|a+b|^2-|a-b|^2$.

(A) 192 (B) 171

(C) 220 (D) 0

100. Find the sum of order and degree of $(d^5y/dx^5)^4+(dy/dx)^2+y=0$.

(A) 13 (B) 14

(C) 9 (D) 5

101. Simplify [(p implies q) and (p implies not q)].

(A) q (B) not p

(C) not q (D) p

102. Evaluate the limit of $[\cos(4x)-\cos(6x)]/x^2$ as $x\to 0$.

(A) 1 (B) 10

(C) 20 (D) -10

103. Find the product of all real solutions of sqrt(7-log_3|x|)=log_3|x|-1.

(A) 729 (B) -729

(C) -27 (D) 0

104. α,β are roots of x^2-19x+84=0. The monic polynomial with roots α+1,β+1 is x^2-Ax+B. Find B.

(A) 7225 (B) 66

(C) 103 (D) 104

105. A=(5,0,4), B=(0,4,1), AP:PB=2:1 internally. Find $|AP|^2$.

(A) 50 (B) $\frac{200}{9}$

(C) $\frac{100}{3}$ (D) $\frac{50}{9}$

106. Vectors a,b satisfy $|a|^2$=5, $|b|^2$=17, a·b=6. Find $|2a-b|^2$.

(A) 13 (B) 15

(C) 37 (D) 61

107. If α,β are roots of x^2-13x+40=0, find $\alpha^2+\beta^2$ without solving the quadratic.

(A) 169 (B) 129

(C) 1600 (D) 89

108. $a_j = 6 + 5(j-1)$ and f(x)=(7/2)x^2+bx+5. If f(1)=f(-1), find f′(a_8).

(A) 322 (B) 41

(C) 287 (D) 292

109. Evaluate lim as n→ ∞ of $(3/n)\Sigma(3j/n)^2$, j=1,...,n.

(A) 9 (B) $\frac{27}{2}$

(C) 3 (D) $\frac{1}{3}$

110. The four lines ±x/8±y/12=1 form a rhombus with vertices on the coordinate axes. Find its area.

(A) 48 (B) 384

(C) 192 (D) 96

111. The homogeneous system 6x+y=0, kx+4y=0 has a nonzero solution. Find k+x^2/y^2.

(A) $\frac{865}{36}$ (B) $\frac{863}{36}$

(C) 24 (D) 60

112. Evaluate $\int_0^\infty dx/[(x^2+9)(x^2+36)]$.

(A) $\frac{\pi}{162}$ (B) $-\frac{\pi}{324}$

(C) $\frac{\pi}{324}$ (D) $\frac{\pi}{36}$

113. f(x)=4x+2. Find f(f(2)).

(A) 18 (B) 10

(C) 42 (D) 40

114. All tangents common to $x^2/36+y^2/9=1$ and $x^2+y^2=18$ touch the ellipse at points P_i. Find $\Sigma(P_i, x)^2$ over the four tangency points.

(A) 24 (B) 96

(C) 72 (D) 12

115. A sphere has $dV/dt=4$ cubic units/s. Find the rate of change of its surface area when $r=6$.

(A) $\frac{4}{3}$ (B) $\frac{2}{9}$

(C) $\frac{1}{36\pi}$ (D) $\frac{2}{3}$

116. Find the coefficient of x^4 in $(1+x+x^2)(1+x^2)^7$.

(A) 63 (B) 28

(C) 56 (D) 21

117. For $f(x)=x^3-12x$, find the difference between its local maximum value and local minimum value.

(A) 32 (B) 4

(C) 16 (D) -32

118. Let R(x) be the remainder when x^6 is divided by (x-1)(x-2). Find R(6).

(A) 316 (B) -309

(C) 378 (D) 46656

119. Two acute sun elevations $\theta_1<\theta_2$ satisfy $(16^5)^{sin^2\theta} + (16^5)^{cos^2\theta} = 2^5 + 2^15$. A vertical tower has height 3. Find the difference between its two shadow lengths.

(A) $3\sqrt{3}$ (B) $4\sqrt{3}$

(C) $2\sqrt{3}$ (D) 2

120. For integer n, consider: "If 20 divides n, then 4 divides n." Which is its contrapositive?

(A) If 4 divides n, then 20 divides n. (B) 20 divides n and 4 does not divide n.

(C) If 20 does not divide n, then 4 does not divide n. (D) If 4 does not divide n, then 20 does not divide n.

121. Box A contains 7 red and 3 blue balls; B contains 4 red and 5 blue balls. A fair coin chooses one box. Two independent draws with replacement from that box give exactly one red ball. Find the posterior probability that A was chosen.

(A) $\frac{2000}{3701}$ (B) $\frac{1701}{3701}$

(C) $\frac{63}{103}$ (D) $\frac{21}{100}$

122. For common tangents to $y^2=28x$ and $x^2+y^2=\frac{49}{6}$, find the sum of the squared slopes.

(A) 4 (B) -2

(C) 0 (D) 2

123. In how many ordered ways can three six-sided dice have sum 10, if the first die is known not to show 6?

(A) 26 (B) 27

(C) 24 (D) 48

124. How many solutions in $0 \leq x < 4\pi$ satisfy $\sin 2x = \sin x$ and $\cos x > 0$?

(A) 4 (B) 8

(C) 7 (D) 6

125. How many integers x are in the real domain of $\sqrt{x-5}\,\log(9-x)$?

(A) 3 (B) 5

(C) 4 (D) 9

126. How many four-digit numbers divisible by 6 can be formed from 0,1,2,3,4,5 without repeating a digit?

(A) 156 (B) 64

(C) 104 (D) 52

127. An equilateral triangle has vertex O and symmetric other vertices on $y^2=8x$. A circle has the side opposite O as diameter. Find the minimum distance from the parabola focus to this circle.

(A) $22-8\sqrt{3}$ (B) 22

(C) $8\sqrt{3}+22$ (D) $24-8\sqrt{3}$

128. On A={0,1,...,9}, define aRb if 2 divides a-b. How many ordered pairs are in R?

(A) 50 (B) 10

(C) 100 (D) 40

129. $A = \begin{bmatrix} 0 & 7 & -2 \\ -7 & 0 & 4 \\ 2 & -4 & 0 \end{bmatrix}$. Find det(I+A)-(49+4+16).

(A) 69 (B) 1

(C) -1 (D) 0

130. $F'(x)=\frac{3(x^2-1)(x^2+1)^2}{x^4}$ and F(1)=2. Find F(2)-2F(1).

(A) $\frac{61}{8}$ (B) $-\frac{77}{8}$

(C) $\frac{45}{8}$ (D) $\frac{77}{8}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A projectile is launched at 35 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{24}{25}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ m/s^2.

(A) $\frac{1764}{25}$ m

(B) $\frac{588}{25}$ m

(C) $\frac{1176}{25}$ m

(D) $\frac{588}{125}$ m

132. A metal wire has linear temperature coefficient $\frac{3}{500}$ per degree Celsius referred to 0°C. Its dimensions are fixed and it stays across a constant voltage. Find power at 70°C divided by power at 0°C.

(A) $\frac{50}{71}$

(B) $\frac{2500}{5041}$

(C) 1

(D) $\frac{21}{71}$

133. A lossless coating of refractive index $\frac{3}{2}$ lies between air and a substrate of still higher index. For normally incident vacuum wavelength 600 nm, find the smallest positive film thickness giving destructive interference between its two reflected rays. Assume reflections at both boundaries each gain a phase reversal.

(A) 300 nm

(B) $\frac{200}{3}$ nm

(C) 100 nm

(D) 200 nm

Chemistry

134. For $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$,consider only atoms of the element undergoing disproportionation. Fraction of those atoms oxidised is

(A) $\frac{2}{3}$

(B) $\frac{1}{3}$

(C) $\frac{1}{6}$

(D) $\frac{1}{2}$

135. For $MgCO_3$, $CaCO_3$, $SrCO_3$, $BaCO_3$,which increasing thermal-stability order and explanation is correct?

(A) $MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$; larger cations distort carbonate more strongly.

(B) $MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$; increasing cation size lowers carbonate-ion polarisation.

(C) All are equally stable because the cation charge is+2.

(D) $MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$; the cations become smaller down the group.

136. In water,ignoring special ortho steric effects,which decreasing basicity order is most appropriate for 4-methoxyaniline,aniline,4-nitroaniline?

(A) 4-nitroaniline > aniline >4-methoxyaniline

(B) 4-methoxyaniline >4-nitroaniline > aniline

(C) 4-methoxyaniline > aniline >4-nitroaniline

(D) aniline >4-methoxyaniline >4-nitroaniline

Logical Reasoning

137. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 5, 10, 13, 26, ?, 58, 61.

(A) 39 (B) 29

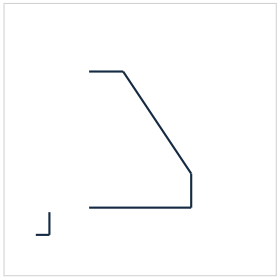
(C) 52 (D) 58

138. Preserve the exact relation and its order: Sodium : Alkali metal :: ? : Reptile. Choose the missing term.

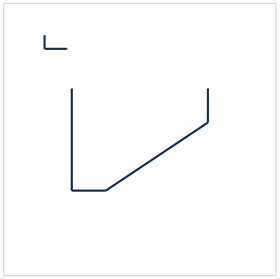
(A) Whale (B) Frog

(C) Lizard (D) Sparrow

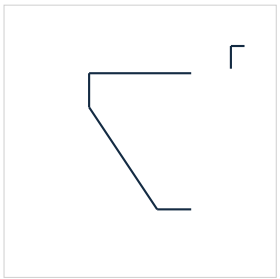
139. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right



bottom left

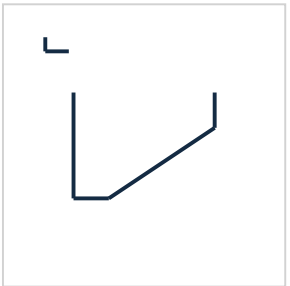
?

bottom right

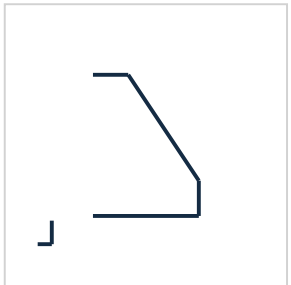
(A) Pattern Z



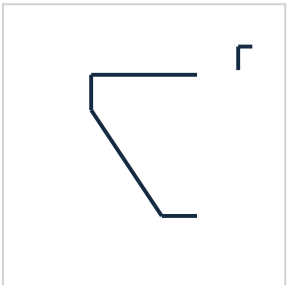
(B) Pattern N



(C) Pattern K



(D) Pattern M



Mathematics

140. f(x)=5x+1 and f(g(x))=5x+11. Find g(g(5)).

(A) 7 (B) 29

(C) 9 (D) 14

141. For a positive integer n, the remainder when x^n is divided by $(x-1)(x-2)$ is $31x-30$. Find n.

(A) 5 (B) 31

(C) 32 (D) 2

142. From two points on the same ray from a tower foot, separated by 8, elevations are 30° and 45°. Find the distance from the farther point to the foot.

(A) $4\sqrt{3}$ (B) $4\sqrt{3}+12$

(C) 8 (D) $4+4\sqrt{3}$

BITSAT Mock 04

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A particle is projected vertically outward from the surface of a nonrotating planet with speed $v_e/10$, where v_e is surface escape speed. Neglect atmosphere. Find maximum height divided by planet radius.

(A) $\frac{1}{100}$ (B) $\frac{1}{10}$

(C) $\frac{1}{9}$ (D) $\frac{1}{99}$

2. Two isolated conducting spheres of radii R and 2R are far apart and connected by a fine wire. Their total positive charge is Q. After equilibrium the wire is removed. Find the larger sphere's charge divided by Q.

(A) $\frac{1}{2}$ (B) $\frac{4}{5}$

(C) $\frac{2}{3}$ (D) 2

3. A geometrically similar solid structure is enlarged by a factor 2 in every linear dimension. Its material density is reduced by a factor 4. Its own weight is carried by a similarly scaled base. Find base compressive-stress multiplier.

(A) 1 (B) 2

(C) 8 (D) $\frac{1}{2}$

4. An active transistor has $\alpha = \frac{79}{80}$. Its collector resistor is 3 kΩ and the measured collector-resistor drop is 18 V. Neglect leakage. What base current in mA is required?

(A) $\frac{237}{40}$ mA (B) $\frac{480}{6241}$ mA

(C) $\frac{6}{79}$ mA (D) $\frac{3}{40}$ mA

5. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 4 kΩ to a terminal at +7 V, and its cathode connects to a terminal at -1 V. Find the current from the +7 V terminal in mA.



(A) $\frac{87}{40}$ mA (B) 2 mA

(C) $\frac{73}{40}$ mA (D) $\frac{63}{40}$ mA

6. A coil of inductance 8 H carries current $i(t) = 2t^2$ A. Find the magnitude of its self-induced emf at $t = 1$ s.

(A) 256 V
(B) $\frac{1}{2}$ V
(C) 16 V
(D) 32 V

7. A conductor in electrostatic equilibrium contains a sealed cavity with a point charge +7q inside. The conductor has total charge zero. Which statement is correct?

(A) Both induced surface charges are zero.
(B) Inner-surface induced charge is -7q and outer-surface charge is +7q.
(C) The inner surface carries +7q and the outer surface -7q.
(D) The entire conductor must acquire net charge +7q.

8. A solid sphere rolls without slipping from rest through vertical height 10 m. A thin ring does so through vertical height 3 m. Find the ratio of their squared centre-of-mass speeds, sphere to ring.

(A) $\frac{10}{3}$
(B) $\frac{100}{21}$
(C) $\frac{20}{3}$
(D) $\frac{21}{100}$

9. A uniform-density spherical planet has radius R and surface gravitational acceleration g. At what depth below the surface is gravity $g/5$? Give the depth as a fraction of R.

(A) $\frac{4}{5}$
(B) 4
(C) $\frac{24}{25}$
(D) 1

10. A ball is dropped from 16 m onto a horizontal fixed floor. At every impact the coefficient of restitution is $\frac{3}{5}$. Neglect air resistance. What maximum height does it reach after its second impact?

(A) $\frac{256}{625}$ m
(B) $\frac{48}{5}$ m
(C) $\frac{1296}{625}$ m
(D) $\frac{432}{125}$ m

11. A galvanometer has resistance 50 ohms and full-scale current $\frac{1}{200}$ A. It is to measure a total current 4 times its own full-scale current. Find the required parallel shunt resistance.

(A) $\frac{50}{3}\,\Omega$
(B) $10\,\Omega$
(C) $150\,\Omega$
(D) $50\,\Omega$

12. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 10 percent. The measured increase in its absolute temperature is $\frac{651}{10}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) 6510 K
(B) 310 K
(C) $\frac{31000}{121}$ K
(D) 651 K

13. A 5 kg block rests on a 3 kg block on a smooth horizontal floor. A force 48 N pulls the lower block. There is no slipping between the blocks. During a displacement 2 m, what work does friction do on the upper block?

(A) 60 J
(B) 160 J
(C) 36 J
(D) 96 J

14. A particle executes SHM with $v^2 = 144 - 16x^2$ in SI units. Find its displacement magnitude when kinetic energy is three times potential energy.

(A) $3\,\text{m}$ (B) $\sqrt{3}\,\text{m}$

(C) $\frac{3}{2}\,\text{m}$ (D) $\frac{3\sqrt{3}}{2}\,\text{m}$

15. A boat moves at 10 m/s relative to water. The current is 6 m/s and river width is 48 m. The boat lands directly opposite its starting point. Find crossing time.

(A) $6\,\text{s}$ (B) $\frac{15}{2}\,\text{s}$

(C) $\frac{12\sqrt{34}}{17}\,\text{s}$ (D) $8\,\text{s}$

16. A gas receives 300 J of heat while its surroundings do 150 J of work on it. It then loses 150 J of heat at fixed volume. Find its total internal-energy change.

(A) $600\,\text{J}$ (B) $300\,\text{J}$

(C) $150\,\text{J}$ (D) $450\,\text{J}$

17. A system has total mass 8 kg, centre-of-mass speed 3 m/s and kinetic energy 20 J measured in the CM frame. Which statement is correct?

(A) The laboratory kinetic energy is 20 J. (B) The laboratory kinetic energy is 56 J.

(C) The CM-frame kinetic energy must be zero. (D) The laboratory kinetic energy is 36 J.

18. A one-electron ion with nuclear charge $Z = 2$ emits when its orbit changes. The ratio of initial to final orbit radii is $\frac{49}{4}$, and its final principal quantum number is 2. What is $R\lambda$ for the emitted photon? Here R is the hydrogen Rydberg constant.

(A) $\frac{7}{10}$ (B) $\frac{49}{45}$

(C) $\frac{45}{49}$ (D) 1

19. A diamond network has four outer resistors AB, BC, CD and DA, each 2 ohms, and a bridge BD also 2 ohms. The DA arm additionally contains an ideal diode with anode at D and cathode at A. A source keeps A positive relative to C. Find equivalent resistance between A and C.

(A) $\frac{10}{3}\,\Omega$ (B) $\frac{8}{3}\,\Omega$

(C) $1\,\Omega$ (D) $6\,\Omega$

20. In YDSE wavelength is 400 nm, screen distance 2 m and slit separation 3 mm. Find the separation on the same side of the centre between the 4th bright fringe and the 6th dark fringe. Count the first dark fringe as order 1.

(A) $\frac{8}{15}\,\text{mm}$ (B) $\frac{2}{5}\,\text{mm}$

(C) $\frac{16}{15}\,\text{mm}$ (D) $\frac{1}{5}\,\text{mm}$

21. A series RLC circuit has $R = 80$ ohms, $X_L = 30$ ohms, and $X_C = 20$ ohms at the source frequency. An ideal source supplies 200 V rms. Find rms current.

(A) $\frac{20\sqrt{77}}{77}\,\text{A}$ (B) $\frac{20\sqrt{89}}{89}\,\text{A}$

(C) $\frac{20}{9}\,\text{A}$ (D) $\frac{4\sqrt{65}}{13}\,\text{A}$

22. The quantity $Z = x^3 y^1 / \sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 3, 1 and 2, respectively. To first order, what is the maximum percentage error in Z?

(A) 9 (B) 10

(C) 11 (D) 6

23. A cyclist rides straight outward at 6 m/s for 6 s, waits 4 s, then returns to the start. The whole journey must last 12 s. What constant return speed is required?

(A) 18 m/s (B) 3 m/s

(C) 6 m/s (D) $\frac{9}{2}$ m/s

24. A fixed amount of ideal gas has its absolute temperature multiplied by 4 and volume by 4. Find the final pressure divided by the initial pressure.

(A) $\frac{1}{4}$ (B) 4

(C) 16 (D) 1

25. Two ideal cells of emfs 15 V and 1 V oppose each other in a loop containing two resistors each 3 ohms. Find the current magnitude.

(A) $\frac{1}{6}$ A (B) $\frac{7}{3}$ A

(C) $\frac{14}{3}$ A (D) $\frac{5}{2}$ A

26. A ray in a medium of index $\frac{11}{5}$ is incident on a medium of index $\frac{11}{10}$. Find the sine of the critical-angle threshold for total internal reflection.

(A) $\frac{5}{11}$ (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{11}{10}$ (D) $\frac{1}{2}$

27. A Newtonian liquid of viscosity $\frac{1}{10}$ Pa s has velocity profile $v(y) = 7y + 2y^2$ in SI units. Find the shear-stress magnitude at $y = 2$ m.

(A) $\frac{3}{2}$ Pa (B) 15 Pa

(C) $\frac{7}{10}$ Pa (D) $\frac{2}{5}$ Pa

28. A uniform disc of mass 6 kg and radius 4 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) 39 kg m^2 (B) 45 kg m^2

(C) 57 kg m^2 (D) 36 kg m^2

29. A rigid ladder of length 20 m slides with its ends on a floor and a vertical wall. At the instant its top is at height $4L/5$, the foot moves away at 2 m/s. What is the downward speed of the top?

(A) $\frac{8}{5}$ m/s (B) 2 m/s

(C) $\frac{6}{5}$ m/s (D) $\frac{3}{2}$ m/s

30. A wire consists of a semicircular arc of radius 6 m carrying 3 A, joined to two straight radial segments that extend outward from its ends. Find the magnetic field magnitude at the arc centre in units of μ_0.

(A) $\frac{1}{4\pi}$ (B) $\frac{2+\pi}{8\pi}$

(C) $\frac{1}{8}$ (D) $\frac{1}{8\pi}$

Chemistry

31. A mixture contains NO2, ozone, PAN and unreacted hydrocarbon vapour. Which pair consists of secondary oxidants characteristic of photochemical smog?

(A) Hydrocarbon vapour and nitrogen.

(B) Ozone and PAN.

(C) NO2 and argon.

(D) Soot and water only.

32. Classify CaO, ZnO, CO2, in that order, by their usual acid-base behaviour. Which sequence also predicts that the middle oxide reacts with both suitable acids and strong alkalis?

(A) Amphoteric; basic; acidic.

(B) Basic; amphoteric; acidic.

(C) Acidic; amphoteric; basic.

(D) Basic; acidic; amphoteric.

33. The ions N^{3-}, O^{2-}, F^{-}, Na^{+} each have 10 electrons. Which decreasing-radius order and explanation is consistent?

(A) All radii are equal because electron numbers are equal.

(B) $O^{2-} > F^{-} > Na^{+} > N^{3-}$; positive charge always gives the largest ion.

(C) $N^{3-} > O^{2-} > F^{-} > Na^{+}$; increasing nuclear charge contracts the same electron cloud.

(D) $N^{3-} < O^{2-} < F^{-} < Na^{+}$; larger nuclear charge expands the cloud.

34. Which combined classification/reactivity description is correct for CH3CH(OH)CH2CH3?

(A) primary and iodoform-positive

(B) secondary and iodoform-negative

(C) secondary and iodoform-positive

(D) tertiary and iodoform-negative

35. Which pair identifies the usual alpha-helix stabilising interaction and the groups that form it?

(A) Ionic bonds between glucose units.

(B) Covalent bonds between every pair of side-chain methyl groups.

(C) Hydrogen bonds between backboneC=O and N-H groups.

(D) Hydrogen bonds only between free terminal COOH groups.

36. For o-nitrophenol and p-nitrophenol,which is expected to show greater steam volatility on the usual intramolecular-versus-intermolecular hydrogen-bonding argument?

(A) neither can hydrogen-bond because both have an aromatic ring

(B) o-nitrophenol

(C) p-nitrophenol

(D) both must have identical volatility because they are isomers

37. At fixed T,a reaction has K=16 and currently Q=16. Which joint conclusion is correct for this equilibrium mixture?

(A) $\Delta_r G = 0$ while$\Delta_r G^\circ < 0$;equilibrium does not require K=1.

(B) Both$\Delta_r G$ and$\Delta_r G^\circ$ must be zero because the reaction is reversible.

(C) $\Delta_r G < 0$ because its standard value is negative,even when Q=K.

(D) $\Delta_r G^\circ > 0$ because K exceeds1.

38. Fresh aqueous solutions of $[Co(NH_3)_5Cl]SO_4$ and its ionisation isomer $[Co(NH_3)_5SO_4]Cl$ are tested immediately; coordinated ligands do not exchange on the test timescale. Which paired observation is expected?

(A) Neither gives a precipitate because coordination suppresses counter-ion dissociation.

(B) Both give immediate Ag halide because every halide inside or outside brackets is free.

(C) $[Co(NH_3)_5Cl]SO_4$ gives $BaSO_4$ with excess Ba^{2+}; $[Co(NH_3)_5SO_4]Cl$ gives AgCl with excess Ag^{+}.

(D) $[Co(NH_3)_5Cl]SO_4$ immediately precipitates its coordinated halide while $[Co(NH_3)_5SO_4]Cl$ retains its external halide.

39. For ethanol and propan-1-ol,which test/observation assignment follows the actual carbinol class or oxidisable structural feature? Use ordinary room-temperature tests unless the named sequence specifies otherwise.

(A) Lucas test;only propan-1-ol gives immediate turbidity

(B) neutral FeCl3;only ethanol gives violet colour

(C) iodoform test;ethanol gives a yellow precipitate

(D) iodoform test;only propan-1-ol gives a yellow precipitate

40. benzoic acid is treated in separate stages with SOCl2,excess NH3,and Br2/NaOH. Which final product and carbon-accounting description are correct?

(A) benzylamine;all carbons retained

(B) aniline;the carbonyl carbon remains as CH2

(C) the original acid;amide nitrogen is never retained

(D) aniline;one carbon fewer than the starting acid

41. A closed system goes from state1to2 by path A with q=500 J and work on the system w=-100 J. It returns from2to1 along a different path with q=-700 J. IUPAC work on the return path,in J,is

(A) 400

(B) -300

(C) 100

(D) 300

42. On dilution of one electrolyte,the analytical concentration falls by a factor9while measured conductivity falls by a factor3. Cell geometry is unchanged. Ratio of final to initial molar conductivity is

(A) 9

(B) $\frac{1}{9}$

(C) $\frac{1}{3}$

(D) 3

43. Within an additive covalent-radius model,a single X-X bond is 120 pm and an X-H bond is 95 pm. A linear H-X-X-H skeleton is assumed with collinear single bonds. Its terminal H-to-H distance,in pm,is

(A) 380

(B) 310

(C) 215

(D) 430

44. $\frac{48}{25}$ g pure NH_4NO_2 decomposes by $NH_4NO_2 \rightarrow N_2 + 2H_2O$. Only80.0% of the theoretical dry N_2 is recovered. Find its STP volume in litres, using22.4L/mol and NH_4NO_2=64g/mol.

(A) $\frac{336}{625}$

(B) $\frac{672}{625}$

(C) $\frac{84}{125}$

(D) $\frac{21}{25}$

45. An octahedral complex has d^3 configuration and $\Delta_o = 100$ kJ mol^{-1}. Ignoring pairing energy,its d-electron energy relative to the unsplit barycentre,in kJ mol^{-1},is

(A) 120 (B) -220

(C) -20 (D) -120

46. Evaluate two proposed explanations: I, close nonbonding electron-pair repulsion contributes to the weak F-F bond relative to Cl-Cl; II, fluorine's small size guarantees the most exothermic electron-gain enthalpy in Group17.

(A) Both incorrect. (B) I incorrect; II correct.

(C) Both correct. (D) I correct; II incorrect.

47. For isolated Sc^{3+} (Z=21),use the high-spin spin-only approximation. Which pair (d-electron count, magnetic moment in BM) follows after removing electrons from 4s before 3d?

(A) (0,0). (B) $(1,\sqrt{3})$.

(C) (0,1). (D) $(0,\sqrt{3})$.

48. In an antifluorite-derived defect model,X^{2-}ions occupy every FCC anion site. Only7/8of the tetrahedral cation sites are occupied by M,which can be+1or+2. With no other charge compensation, fraction of occupied M ions in+2state is

(A) $\frac{2}{7}$ (B) $\frac{6}{7}$

(C) $\frac{1}{7}$ (D) $\frac{1}{8}$

49. In a textbook classification, an antiseptic is applied to living tissue while a disinfectant is used on nonliving surfaces. Which statement preserves that distinction?

(A) The application and suitable concentration matter; the same chemical may have different roles.

(B) All antimicrobial chemicals are food antioxidants.

(C) Antiseptics are defined as artificial sweeteners.

(D) Every disinfectant is suitable for living tissue at every concentration.

50. A photographic fixer contains $\frac{3}{100}$ mol free $S_2O_3^{2-}$ and receives $\frac{1}{50}$ mol unexposed AgBr. Use $AgBr + 2S_2O_3^{2-} \rightarrow [Ag(S_2O_3)_2]^{3-} + Br^-$. Treat complexation as complete until a reagent is exhausted. AgBr remaining,in mol,is

(A) $\frac{1}{200}$ (B) 0

(C) $\frac{1}{100}$ (D) $\frac{1}{50}$

51. In a pressure range where the Freundlich law $x/m = kp^{1/3}$ is valid at fixed temperature, adsorbent B has 3 times the mass of A and is exposed to 27 times its pressure. Find the ratio of total adsorbed amounts x_B/x_A.

(A) 81 (B) 27

(C) 9 (D) 1

52. A carbon-containing organic compound contains sulfur only among N,S and halogens. Sodium fusion is performed with sufficient excess sodium to cleave any thiocyanate formed. Which product/test pairing is consistent?

(A) Na2S; violet nitroprusside response.

(B) Elemental chlorine; a Prussian-blue response.

(C) NaNO3; a brown-ring test directly on the organic compound.

(D) Na2SO4; a violet nitroprusside response for sulfate.

53. $\frac{3}{100}$ mol borax is acidified completely. Each borax formula unit gives four H_3BO_3 molecules;75.0% of the boric acid is isolated. With glycerol present, the isolated boric acid is titrated completely as a monobasic acid. What amount of NaOH, in mol, is required?

(A) $\frac{9}{400}$

(B) $\frac{9}{100}$

(C) $\frac{9}{200}$

(D) $\frac{4}{25}$

54. Bond energies $D(X_2):D(Y_2):D(XY)=2:3:4$. Forming2.00 mol XY(g) from the gaseous elements releases 300 kJ. $D(Y_2)$,in kJ/mol,is

(A) 600

(B) 300

(C) 150.0

(D) 400

55. A binary liquid mixture is observed to have a maximum-boiling azeotrope at a fixed pressure. Which paired conclusion is valid?

(A) Its vapour composition differs from its liquid composition at the azeotrope,so repeated distillation necessarily gives both pure components.

(B) It shows negative deviation in the relevant composition region;ordinary fractional distillation cannot pass through the azeotropic composition.

(C) The azeotrope proves no intermolecular interactions exist in either pure liquid.

(D) It shows positive deviation,and every liquid mixture must have such an azeotrope.

56. Two model nonvolatile solutes A(M180) and B(M60) are dissolved separately in equal solvent masses. Equal solute masses are used. A remains molecular;fraction $\frac{1}{5}$ of B formula units is present in dimers. All solutions obey the stated ideal colligative model. Ratio $\Delta T_f(B)/\Delta T_f(A)$ is

(A) 3

(B) $\frac{10}{3}$

(C) $\frac{9}{10}$

(D) $\frac{27}{10}$

57. For aspirin represented as ortho-HOOC-C6H4-O-COCH3, which pair of functional groups remains present?

(A) Primary amine and ketone.

(B) Nitrile and polyamide.

(C) PhenolicOH and aldehyde.

(D) Carboxylic acid and ester.

58. In ideal FCC packing,the tetrahedral void at(a/4,a/4,a/4)is compared with the nearest octahedral void. Atoms touch along face diagonals. Ratio(nearest void-to-void distance)/(atomic radius)is

(A) $\frac{\sqrt{6}}{2}$

(B) $\frac{\sqrt{3}}{2}$

(C) $\frac{\sqrt{2}}{4}$

(D) $\frac{\sqrt{3}}{4}$

59. Compounds A and B have molecular formulae $(CH_2O)_2$ and $(CH_2O)_3$. Pure equal-mass samples are compared. Which joint statement is necessarily true,independently of functional-group identity?

(A) The samples contain equal total oxygen-atom amounts,but A contains $\frac{3}{2}$ times as many molecules as B.

(B) The oxygen-atom amounts are in ratio $\frac{2}{3}$,while molecule amounts are equal.

(C) Both samples contain equal molecule amounts because their empirical formulae agree.

(D) A contains $\frac{2}{3}$ times as many molecules as B,and has a different oxygen mass percentage.

60. An ideal gas executes A→B→C→D→A through (V/L,P/bar):A(1,1),B(5,1),C(5,3),D(1,3). IUPAC work-on convention applies. Heats on AB,BC,DA are 400,300,-200 J. Heat on CD in J is (1 L bar=100 J)

(A) 300 (B) -1300

(C) -800 (D) -500

English Proficiency

61. Choose the meaning-preserving equivalent that retains the tense, participants and all stated restrictions. He continued working in spite of the repeated interruptions.

(A) Although he was interrupted repeatedly, he continued working.

(B) He stopped working whenever an interruption occurred.

(C) He continued working only while he was being interrupted.

(D) He continued working because he was interrupted repeatedly.

62. In the fictional story, no opponent can defeat the champion. Which adjective fits?

(A) infallible (B) invincible

(C) invulnerable (D) irresistible

63. Read the sentence: The sign was meant in its literal sense, not as a figure of speech. Choose the opposite in meaning of 'literal' as used in the sentence.

(A) direct (B) figurative

(C) exact (D) ordinary

64. Choose the active-voice equivalent that retains the tense, participants and all stated restrictions. Only the rear door was locked by the caretaker before the inspection began.

(A) The caretaker locked only the rear door after the inspection began.

(B) The caretaker had locked only the rear door before the inspection began.

(C) Only the caretaker locked the rear door before the inspection began.

(D) The caretaker locked only the rear door before the inspection began.

65. Complete: The volunteer _____ found the missing register has already informed the office.

(A) who (B) which

(C) whose (D) whom

Read the passage for Question 66

A town wanted benches for a busy square. Two designs cost the same, and both looked attractive in the supplier's drawings. Rather than choose immediately, the council installed one example of each for a short trial. People used both, but the trial revealed that one design left too little space for a walking aid beside a seated person. The council chose the other design. The trial did not establish that its choice was best for every possible setting: a quiet garden might impose different requirements. It did, however, provide evidence about a requirement that the drawings had not made easy to judge. The useful question was not simply whether the trial produced a winner, but whether it exposed the practical distinction relevant to this particular decision.

66. Which conclusion is best supported by the passage?

(A) A limited trial can inform a local choice without establishing a universally superior design.

(B) Drawings provide no useful information about any design.

(C) Equal prices make practical trials unnecessary.

(D) The selected design must be the best choice for every public space.

67. Which ONE of the following words is misspelt?

(A) visible (B) separate

(C) committee (D) occurence

68. Read the sentence: The room was austere: a plain desk, a wooden chair and no decoration. Choose the closest meaning of 'austere' as used in the sentence.

(A) uncomfortably small (B) ornate

(C) severely plain (D) poorly maintained

69. Choose the most coherent order of the four sentences. A. The final plan kept the garden within that area instead of abandoning it altogether. B. The assessment showed that only a smaller area could safely be used. C. Before approving it, the committee asked an engineer to assess the added load. D. The building manager proposed a garden on the roof.

(A) D-B-C-A (B) D-C-B-A

(C) D-C-A-B (D) C-D-B-A

70. Read the sentence: The team was elated when the final result confirmed its success. Choose the opposite in meaning of 'elated' as used in the sentence.

(A) relieved (B) excited

(C) dejected (D) proud

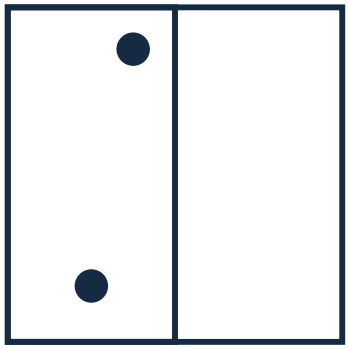
Logical Reasoning

71. Preserve the exact relation and its order: Surgeon : Scalpel :: ? : Saw. Choose the missing term.

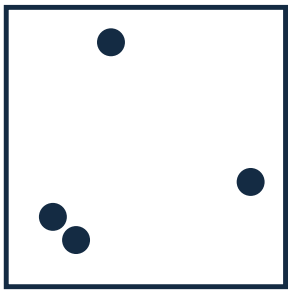
(A) Carpenter (B) Painter

(C) Tailor (D) Surgeon

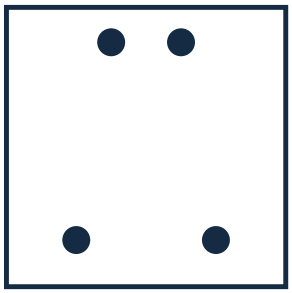
72. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



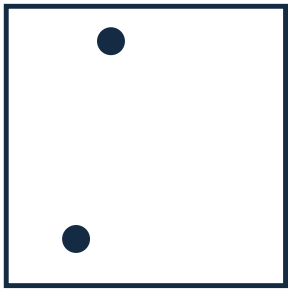
(A) Pattern Z



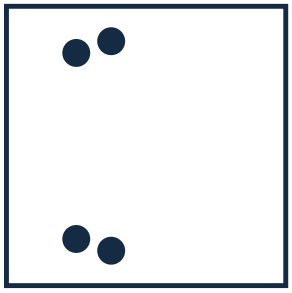
(B) Pattern K



(C) Pattern P



(D) Pattern V



73. Consider only these statements; universal statements alone do not imply existence. All joggers are designers; No designers are joggers; Some designers are artists. Conclusions: I. Some artists are not joggers. II. All artists are joggers. Which conclusions follow necessarily?

(A) Neither I nor II

(B) Both I and II

(C) Only II

(D) Only I

74. $X\%Y$ means X is the mother of Y; $X@Y$ means X is the sister of Y. All names denote distinct people; a brother or sister is a full sibling, sharing both parents. Given $V\%S$ $S@F$ $F\%U$, how is V related to U?

(A) Father

(B) Maternal grandfather

(C) Maternal grandmother

(D) Mother

75. In this code, @ means $\times$, # means $+$, & means $\div$, % means $-$. After substituting meanings, evaluate ((6#5)@(5&2)); printed parentheses retain their usual grouping.

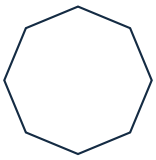
(A) $\frac{53}{2}$ (B) $\frac{57}{2}$

(C) $\frac{55}{2}$ (D) $\frac{-55}{2}$

76. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

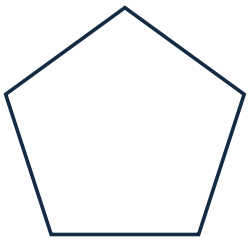


B

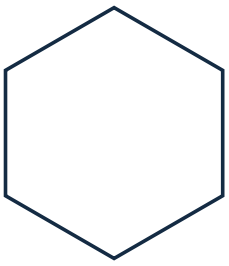


C

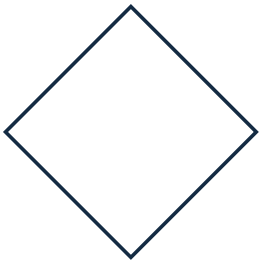
(A) Pattern Q



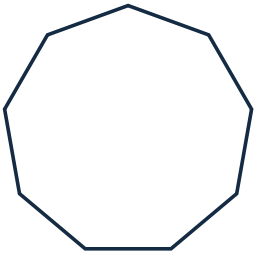
(B) Pattern Z



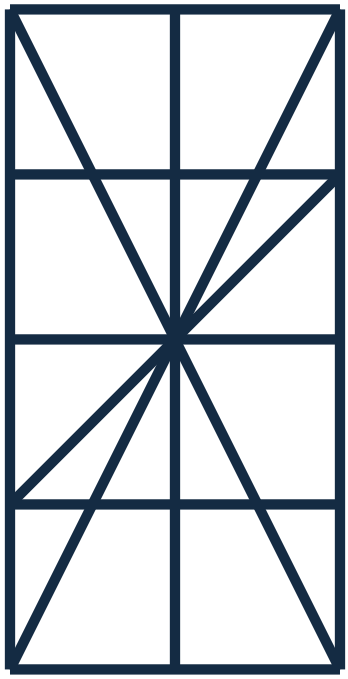
(C) Pattern R



(D) Pattern X



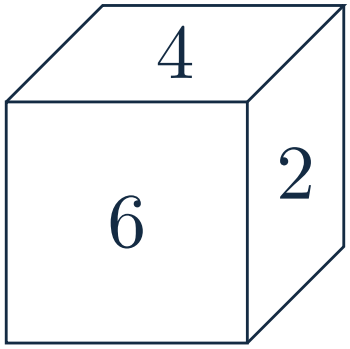
77. How many triangles are formed by all the drawn segments?



(A) 34 (B) 40

(C) 44 (D) 32

78. This standard die has opposite labels summing to seven. Roll it once to the right, so its present top face becomes the right face. Give the new (top, front) labels.



(A) $(2, 6)$ (B) $(5, 1)$

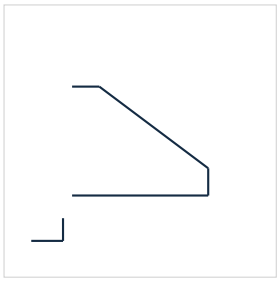
(C) $(5, 6)$ (D) $(4, 6)$

79. In this series the first differences themselves form an arithmetic progression. Find the missing internal term: 2, 6, ?, 23, 36, 52, 71.

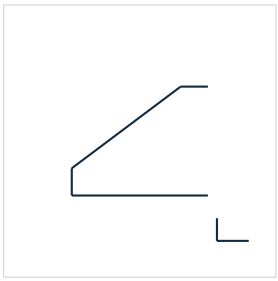
(A) 17 (B) 13

(C) 9 (D) 14

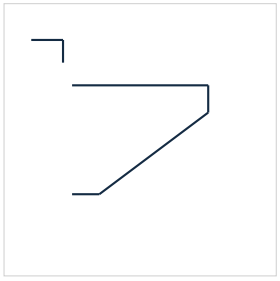
80. Select the missing bottom-right tile. Tiles reflect across the shared horizontal and vertical centre lines; there is no turning.



top left



top right



bottom left

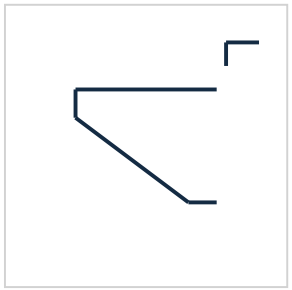
?

bottom right

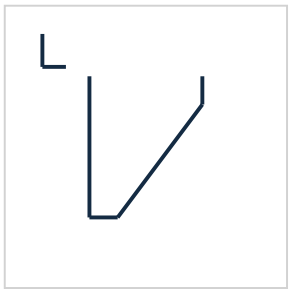
(A) Pattern U



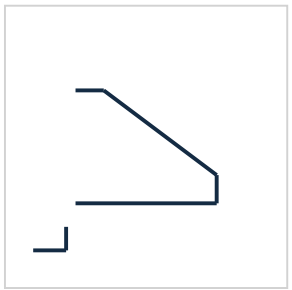
(B) Pattern L



(C) Pattern X



(D) Pattern Q

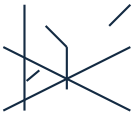


81. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 1
(B) 4
(C) 3
(D) 2

82. Consider only these statements; universal statements alone do not imply existence. Some designers are designers; All hikers are joggers. Conclusions: I. All designers are hikers. II. Some designers are not hikers. Which conclusion or complementary alternative must hold?

(A) Both I and II
(B) Either I or II follows, but neither is individually certain
(C) Only II
(D) Only I

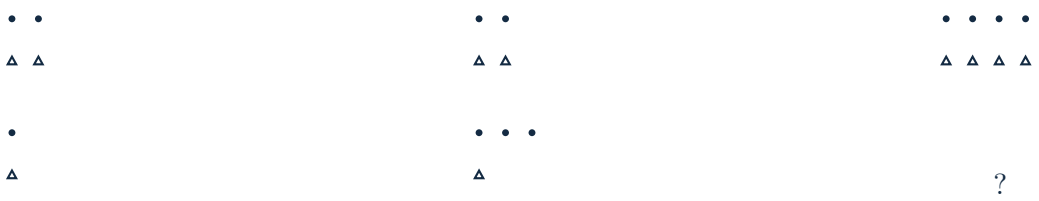
83. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: UPOS, RMMT, OJKU, ?, IDGW, FAEX.

(A) LGIV
(B) VIGL
(C) KFHU
(D) MHJW

84. All five people have different heights. Gita is shorter than Dev. Asha is shorter than Farah. Dev is shorter than Charu. Charu is shorter than Asha. If Charu is left out, who is 3rd shortest among the remaining people?

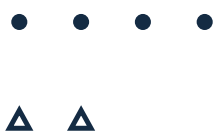
(A) Charu
(B) Asha
(C) Farah
(D) Gita

85. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern Z

(B) Pattern T



(C) Pattern K

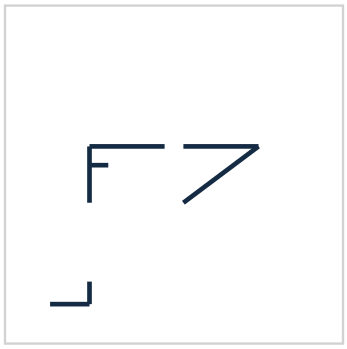


(D) Pattern N

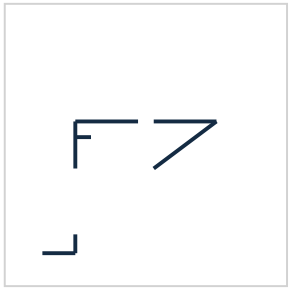




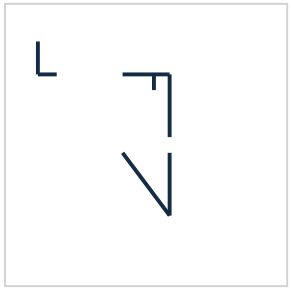
86. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



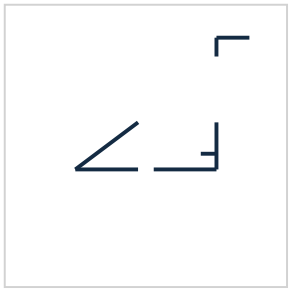
(A) Pattern R



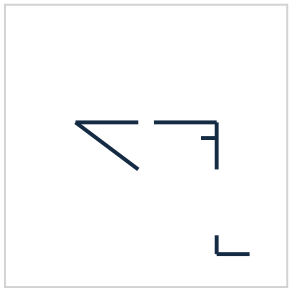
(B) Pattern T



(C) Pattern N



(D) Pattern L



87. For this question compare only cyclic forward alphabet gaps, with $Z \to A$ counting as one step. Three clusters share the same gap vector. Which numbered cluster differs? 1: STVW, 2: TUWX, 3: IJLM, 4: KLOP.

(A) 4

(B) 1

(C) 2

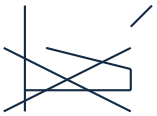
(D) 3

88. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





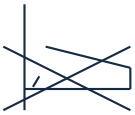
host 1



host 2



host 3



host 4

(A) 1 (B) 2

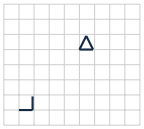
(C) 3 (D) 4

89. The recorded average of 8 readings is 42. One entry recorded as 48 should have been 44. After that correction, another reading of 19 is replaced by 31. What is the final average?

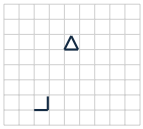
(A) $\frac{87}{2}$ (B) $\frac{83}{2}$

(C) 44 (D) 43

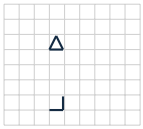
90. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



1

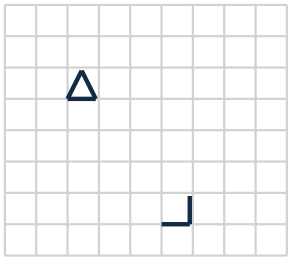


2

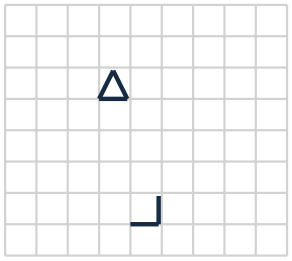


3

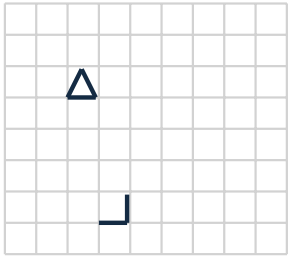
(A) Pattern Q



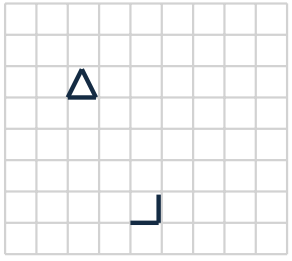
(B) Pattern T



(C) Pattern Z



(D) Pattern Y



Mathematics

91. A,B are acute, A+B=$\pi/4$ and tan A=$\frac{1}{2}$. Find tan(A-B).

(A) $\frac{1}{3}$ (B) $-\frac{1}{7}$

(C) $\frac{1}{6}$ (D) $\frac{1}{7}$

92. An invertible 4×4 matrix A has det A=14. Find det(adj A).

(A) 56 (B) 537824

(C) 2744 (D) 14

93. In $(1+x)^{13}$, find the ratio of the coefficient of x^6 to the coefficient of x^5.

(A) $\frac{9}{5}$ (B) 8

(C) $\frac{4}{3}$ (D) $\frac{3}{4}$

94. Positive integers $a < b$ satisfy $a + b = 20$ and $3 \mid a$. Find the maximum value of ab.

(A) 20 (B) 84

(C) 99 (D) 100

95. α,β are nonzero roots of x^2-9x+14=0. Find the sum of $1+1/\alpha$ and $1+1/\beta$.

(A) $\frac{23}{14}$ (B) $\frac{32}{9}$

(C) $\frac{9}{14}$ (D) $\frac{37}{14}$

96. a=(t,1,0), b=(1,6,5), a·b=12. Find the squared area of the parallelogram spanned by a,b.

(A) $5\sqrt{86}$ (B) 2294

(C) $\frac{1075}{2}$ (D) 2150

97. If |z-11|=5, find the minimum of $|z|^2$.

(A) 36 (B) 96

(C) 6 (D) 256

98. f(x)=sin(4x)/x+3 for x<0; f(x)=k cos x for x≥0. Find k for continuity at zero.

(A) 4 (B) 1

(C) 3 (D) 7

99. For $A = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$ and C=$7A^{-1}$, find C_{22}.

(A) 2 (B) $\frac{2}{7}$

(C) -1 (D) 4

100. A line has direction (1,2,2) and a plane normal (2,-1,1). Find the square of the sine of the angle between the line and plane.

(A) $\frac{2}{27}$ (B) $\frac{25}{27}$

(C) $\frac{\sqrt{6}}{9}$ (D) 1

101. |U|=90, |A|=23, |B|=22, |A∩B|=9. Find the number of elements belonging to both A,B or to neither.

(A) 27 (B) 54

(C) 81 (D) 63

102. The letters A,A,B,C and 7 further distinct letters are arranged in a row. How many arrangements have nonadjacent As and have B,C adjacent in either order?

(A) 5806080 (B) 725760

(C) 3628800 (D) 2903040

103. For $(d^2y/dx^2)^5+(y')^2+x=0$, find order+degree if the degree is defined; otherwise select "Degree is not defined".

(A) 7 (B) 8

(C) Degree is not defined (D) 10

104. For f(x)=|x-5|/(x-8), on which maximal open intervals is f strictly decreasing?

(A) $(5,8)\cup(8,\infty)$ (B) $(-\infty,5)$

(C) $(5,\infty)$ (D) $(-\infty,8)$

105. The same loaded die has odd-face probability $\frac{5}{6}$. Three independent throws are made. Given that the first two have an even sum, find the probability that the total of all three is odd.

(A) $\frac{35}{54}$ (B) $\frac{5}{6}$

(C) $\frac{1}{6}$ (D) $\frac{5}{18}$

106. A line through P=(4,6) has direction (3,4). It meets 3x+4y=86 at Q. Find $|PQ|^2$.

(A) 10 (B) 100

(C) 50 (D) 4

107. In a group of 60, 36 study A, 30 study B, and 6 study neither. How many study exactly one of A, B?

(A) 42 (B) 66

(C) 54 (D) 12

108. Find the area bounded by y=x, y=49/x, and x=49 in the first quadrant.

(A) 1176 (B) $1176-49\log(7)$

(C) $294-49\log(7)$ (D) $49\log(7)$

109. $F'(x)=\frac{(x+1)^2}{(x+2)^4}$ and F(0)=5. Find F(5)-2F(0).

(A) $-\frac{42545}{8232}$ (B) $-\frac{39775}{8232}$

(C) $\frac{42545}{8232}$ (D) $\frac{1385}{8232}$

110. ABC is a right triangle on level ground with AB=16, AC=16, angle A=90°. A vertical tower stands at midpoint M of BC. Its top elevation from A is 60°. Find its height.

(A) $16\sqrt{6}$ (B) $8\sqrt{6}$

(C) $8\sqrt{3}$ (D) $8\sqrt{2}$

111. Two distinguishable fair dice show X,Y. Given X+Y≥6, find the probability that at least one of X, Y is even.

(A) $\frac{5}{9}$ (B) $\frac{3}{13}$

(C) $\frac{14}{13}$ (D) $\frac{10}{13}$

112. An observer at height 3 sees a tower foot at depression 30° and top at elevation 60°. Find the tower height.

(A) 9 (B) 12

(C) $3\sqrt{3}$ (D) 6

113. Let $a_j = 1/(4 + 5(j - 1))$. Evaluate $\sum_{j=1}^{7} a_j a_{j+1}$.

(A) $\frac{3}{68}$ (B) $\frac{7}{156}$

(C) $\frac{7}{34}$ (D) $\frac{35}{156}$

114. For x∈[0,1], find the minimum of $2(\arcsin x)^3+(\arccos x)^3$.

(A) $\frac{\pi^3}{8}$ (B) $\frac{3\pi^3}{64}$

(C) $\frac{\pi^3}{4\left(1+\sqrt{2}\right)^2}$ (D) $-\frac{\sqrt{2}\pi^3}{4}+\frac{3\pi^3}{8}$

115. Find the population variance of 4+2, 8+2, ..., 44+2.

(A) 160 (B) 10

(C) 40 (D) 162

116. How many integer k in [0,36] make $x^2 - 9x + (18 - k) = 0$ have roots of opposite signs?

(A) 17 (B) 19

(C) 36 (D) 18

117. Evaluate the integral from 0 to $\pi/2$ of $(5-3\sin x)/(25-9\sin^2 x)\,dx$.

(A) $\frac{\operatorname{atan}(2)}{2}$ (B) $\operatorname{atan}\left(\frac{1}{2}\right)$

(C) $\frac{\pi}{10}$ (D) $\frac{\operatorname{atan}\left(\frac{1}{2}\right)}{2}$

118. Let $z=(1+i)^{17}$. Find Re(z)+Im(z).

(A) 256 (B) -512

(C) 512 (D) $256\sqrt{2}$

119. X takes values [-4, 0, 5] with probabilities 1/6,1/3,1/2. Let Y=4X+9. Find $E(Y^2)-[E(Y)]^2$.

(A) $\frac{1700}{9}$ (B) $\frac{2429}{9}$

(C) $\frac{728}{3}$ (D) $\frac{425}{9}$

120. The tangent to y=x^2+4x at P has slope 10. Find the slope of the normal at P.

(A) -10 (B) $\frac{1}{10}$

(C) $-\frac{1}{10}$ (D) 10

121. r=pa+qb, a=(1,0,0), b=(1,1,0), r·a=9, r·b=15. Find $|r|^2$.

(A) 45 (B) $3\sqrt{13}$

(C) 117 (D) 15

122. A positive integer-domain function satisfies f(m+n)=f(m)f(n), f(1)=3. Find $\sum_{j=1}^{5} f(j)$.

(A) 243 (B) 363

(C) 121 (D) 1215

123. A complex number z satisfies z+conjugate(z)=4 and z-conjugate(z)=6i. Find z·conjugate(z).

(A) $\sqrt{13}$ (B) 40

(C) 13 (D) 1

124. On $(0,\pi/2)$, define g(x)=sin x/x. Which description is correct?

(A) Strictly increasing (B) Decreasing then increasing

(C) Constant (D) Strictly decreasing

125. Positive a,b,c form an AP with a=2. The numbers a,b,c+9/2 form a GP and b>a. Find c.

(A) -4 (B) 8

(C) 5 (D) 20

126. A=(3,4), B=(3,-4); P lies on x^2+y^2=49 with PA=PB and P_x>0. Find PA^2.

(A) 49 (B) 116

(C) 32 (D) 16

127. A box contains 4 red and 3 blue distinct balls. Three are drawn uniformly without replacement. Given that at least one drawn ball is blue, find the probability that exactly two are red.

(A) $\frac{18}{35}$ (B) $\frac{18}{31}$

(C) $\frac{9}{2}$ (D) $\frac{13}{31}$

128. $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ sin\theta & \cos\theta \end{pmatrix}$, $0<\theta<\pi/2$, $\tan\theta=\frac{3}{8}$. Find tr(R(θ)^2).

(A) $\frac{110}{73}$ (B) 2

(C) $\frac{55}{73}$ (D) $\frac{96}{73}$

129. Evaluate $\sum_{j=0}^{3} \binom{3}{j}\binom{9}{2-j}$.

(A) 55 (B) 198

(C) 66 (D) 220

130. Evaluate $\int_0^{(\pi/2)}$ [((sin x+cos x)^2-1)/(2sin x cos x)]sin x dx, using the continuous extension at the endpoints.

(A) 1

(B) 0

(C) 2

(D) $\frac{\pi}{2}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. 27 identical uncharged liquid drops merge into one drop without evaporation. Find the fraction of the initial surface energy released; surface tension is constant.

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$

(C) $\frac{26}{27}$ (D) 2

132. A triangular p-V cycle has vertices (V_0, p_0), $(V_0 + 2\,\mathrm{L}, p_0)$ and $(V_0, p_0 + 6\,\mathrm{kPa})$. Traversed in that listed order and then back, find net work done by the gas.

(A) $-3\,\mathrm{J}$ (B) $12\,\mathrm{J}$

(C) $-6\,\mathrm{J}$ (D) $-12\,\mathrm{J}$

133. A pure radioactive sample has activity A at time 5 hours after preparation and $A/8$ at time 20 hours after preparation. At what time, measured from preparation, will its activity be one thirty-second of the activity measured at 5 hours? Assume exponential decay with a constant decay rate.

(A) 165 h (B) 30 h

(C) 17.5 h (D) 90 h

Chemistry

134. For isolated gaseous species,consider $O^- + e^- \rightarrow O^{2-}$. Which judgement also remains consistent with the Cl/F electron-gain anomaly?

(A) Every first and second electron gain has the same sign and magnitude for a group.

(B) The stated step is endothermic; first electron gain by Cl is more exothermic than by F.

(C) The stated step is endothermic; first electron gain by F is more exothermic than by Cl because F is smaller.

(D) The stated step is exothermic; every electron addition releases energy.

135. Compare finalpH: I=10mL $\frac{1}{10}$M HCl+20mL $\frac{1}{10}$M NaOH;II=50mL $\frac{1}{10}$M H_2SO_4+50mL $\frac{1}{10}$M NaOH; III=10mL $\frac{1}{10}$M HCl+10mL $\frac{1}{10}$M NaOH. Treat both sulfuric protons as fully neutralised. Increasing pH is

(A) II<I<III. (B) I<III<II.

(C) II<III<I. (D) III<II<I.

136. A disaccharide gives glucose and galactose on complete hydrolysis. Which identity and reducing classification fit the intact sugar?

(A) Lactose; reducing. (B) Maltose; nonreducing.

(C) Sucrose; nonreducing. (D) Cellulose; a disaccharide.

Logical Reasoning

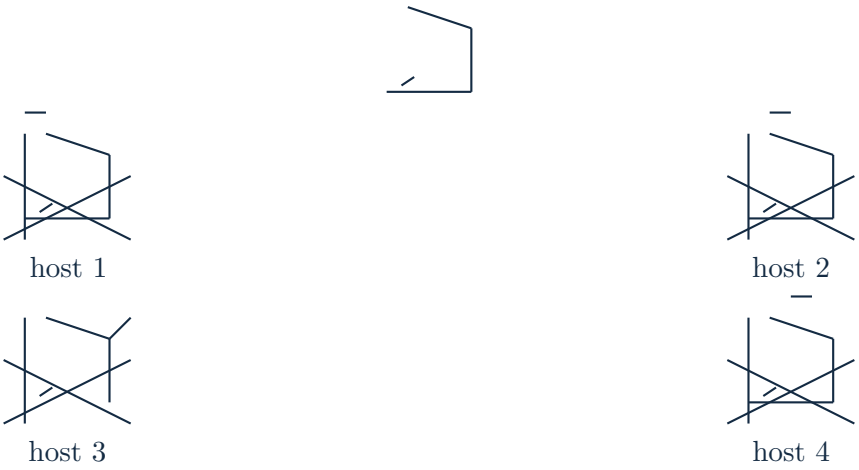
137. V and J are full siblings; J and K are also full siblings. G is the father of V. All four people are distinct. How is G related to K?

(A) Uncle
(B) Father
(C) Relationship cannot be determined
(D) Grandfather

138. 6 people sit around a circle facing its centre. Clockwise order fixes the convention; immediate left is clockwise. Farah is 2 places clockwise from Asha. Bina is 3 places clockwise from Asha. Charu is 5 places clockwise from Asha. Esha is not adjacent to Bina. Who sits immediately clockwise from Bina?

(A) Dev
(B) Charu
(C) Bina
(D) Asha

139. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.



(A) 4
(B) 2
(C) 3
(D) 1

Mathematics

140. Let $f(x)=\sin(x+\pi/6)$ restricted to $[-2\pi/3,\pi/3]$, and let $g=f^{-1}$. Find $g'(1/2)$.

(A) $\frac{1}{2}$
(B) $-\frac{2\sqrt{3}}{3}$
(C) $\frac{2\sqrt{3}}{3}$
(D) $\frac{\sqrt{3}}{2}$

141. For $f(x)=x^3-3kx^2+27x$, how many integer k in $[0,7]$ give a local maximum at a positive x?

(A) 4
(B) 7
(C) 5
(D) 3

142. $f''=g''$, $f(0)-g(0)=3$, $f'(0)-g'(0)=5$. Find $\int_0^2[f(x)-g(x)]dx$.

(A) 6
(B) 16
(C) 10
(D) 13

BITSAT Mock 05

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A deuteron of mass $2u$, charge $+e$, and an alpha particle of mass $4u$, charge $+2e$, start from rest. The deuteron is accelerated through potential $5V$ and the alpha particle through V. In the nonrelativistic regime, find λ_α/λ_d.

(A) $\frac{\sqrt{5}}{2}$ (B) $\frac{\sqrt{5}}{10}$

(C) $\frac{5}{4}$ (D) $\frac{2\sqrt{5}}{5}$

2. A platform of moment of inertia 6 kg m^2 carries a person of mass 2 kg at radius 2 m and rotates at 2 rad/s. The person walks to half the stated initial radius; external torque is zero. Find final angular speed. Treat the person as a point mass.

(A) $\sqrt{7}\,\text{rad/s}$ (B) $\frac{7}{2}\,\text{rad/s}$

(C) $2\,\text{rad/s}$ (D) $8\,\text{rad/s}$

3. An active transistor has $\alpha = \frac{49}{50}$. Its collector resistor is 4 kΩ and the measured collector-resistor drop is 16 V. Neglect leakage. What base current in mA is required?

(A) $\frac{4}{49}\,\text{mA}$ (B) $\frac{98}{25}\,\text{mA}$

(C) $\frac{200}{49}\,\text{mA}$ (D) $\frac{2}{25}\,\text{mA}$

4. A monatomic ideal gas expands reversibly and adiabatically until its pressure becomes 1/248832 of its initial value. Find final temperature divided by initial temperature.

(A) 1 (B) 144

(C) $\frac{1}{144}$ (D) $\frac{1}{12}$

5. A wire carries $i(t) = 7 + 4\sin\omega t$ A for many cycles. Find its rms current.

(A) $2\sqrt{2}\,\text{A}$ (B) $\sqrt{65}\,\text{A}$

(C) $7\,\text{A}$ (D) $\sqrt{57}\,\text{A}$

6. Current rises linearly from 0 to 7 A over 5 s and then falls linearly to 0 over another 5 s. It remains zero for the next 10 s. Find the average current over the entire 20 s interval.

(A) $\frac{7}{8}$ A (B) $\frac{7}{4}$ A

(C) $\frac{7}{2}$ A (D) 7 A

7. A car negotiates a level circular turn at its maximum non-slipping speed u when $\mu_s = \frac{3}{5}$. Rain lowers the coefficient to $\frac{3}{10}$ and the turn radius stays fixed. Find the new maximum speed divided by u.

(A) $\frac{\sqrt{2}}{2}$ (B) $\frac{\sqrt{30}}{10}$

(C) $\sqrt{2}$ (D) $\frac{3}{10}$

8. A block is placed on the vertical inner wall of a trolley. The trolley accelerates horizontally so the wall pushes the block. With coefficient of static friction $\frac{3}{10}$, what minimum trolley acceleration prevents vertical slipping? Take $g = 10$ m/s^2.

(A) $\frac{100}{3}$ m/s^2 (B) $\frac{100}{13}$ m/s^2

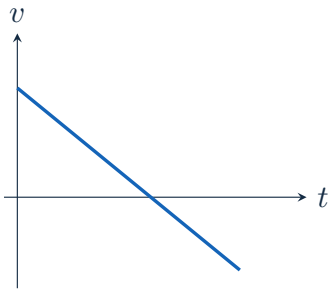
(C) 10 m/s^2 (D) $\frac{10\sqrt{30}}{3}$ m/s^2

9. A conveyor receives 3 packets each second, each of mass $\frac{1}{5}$ kg and speed 7 m/s. It accelerates them to 9 m/s without changing elevation. Find the ideal mechanical power delivered to the packets.

(A) $\frac{48}{5}$ W (B) $\frac{243}{10}$ W

(C) $\frac{96}{5}$ W (D) $\frac{16}{5}$ W

10. A particle has velocity $v(0) = 6$ m/s, and constant acceleration makes $v(5) = -2$ m/s. What total distance does it travel in this interval?



(A) $\frac{45}{4}$ m (B) $\frac{25}{2}$ m

(C) 30 m (D) $\frac{5}{4}$ m

11. 216 identical conducting liquid drops merge without losing charge. Before merging each has potential V and surface charge density σ. Find the final surface charge density divided by σ.

(A) 216 (B) $\frac{1}{6}$

(C) 6 (D) 36

12. A spherical blackbody's absolute temperature is multiplied by 2 while its radius is divided by 3. Neglect incoming background radiation. Find its emitted-power multiplier.

(A) $\frac{16}{27}$ (B) $\frac{9}{16}$

(C) $\frac{2}{9}$ (D) $\frac{16}{9}$

13. Water flows steadily from a pipe section of area A at 3 m/s to a section of area $A/2$ located 2 m higher. Neglect viscosity. Find $p_1 - p_2$. Take density 1000 kg/m^3 and $g = 10$ m/s^2.

(A) 6500 Pa (B) 20000 Pa

(C) 29000 Pa (D) 33500 Pa

14. Two identical converging thin lenses of focal length 20 cm are separated by 100 cm. A parallel beam enters the first. Find the distance of the final image from the second lens, positive on the outgoing side.

(A) 25 cm (B) $\frac{80}{3}$ cm

(C) 20 cm (D) 80 cm

15. A travelling wave is $y = 1\sin[4(x - \frac{35}{2}t) + \pi/4]$ mm with x in metres and t in seconds. What is its frequency?

(A) $\frac{140}{\pi}$ Hz (B) $\frac{2}{\pi}$ Hz

(C) $\frac{35}{\pi}$ Hz (D) $\frac{70}{\pi}$ Hz

16. A wheel initially rotates at 12 rad/s. A constant opposing angular acceleration brings it to rest after 3 s. What angle does it turn through during the last half of this stopping interval?

(A) 18 rad (B) $\frac{9}{2}$ rad

(C) 9 rad (D) 36 rad

17. A 3 kg block moving at 7 m/s compresses a spring of constant 120 N/m. During compression, a damper removes 1/5 of the block's initial kinetic energy. Find maximum compression; gravity does no work.

(A) $\frac{7\sqrt{6}}{30}$ m (B) $\frac{7\sqrt{10}}{20}$ m

(C) $\frac{49}{50}$ m (D) $\frac{7\sqrt{2}}{10}$ m

18. An insulated vessel containing one mole of an ideal gas of molar mass $\frac{3}{50}$ kg/mol moves at bulk speed 80 m/s and is brought to rest. All bulk kinetic energy becomes internal energy. The gas has 3 active degrees of freedom. Find temperature rise. Use $R = 8.314$ J/(mol K).

(A) $\frac{128000}{4157}$ K (B) $\frac{64000}{4157}$ K

(C) $\frac{32000}{4157}$ K (D) $\frac{128000}{12471}$ K

19. Two resistors R and 2R are in series across an ideal 9 V source. A branch consisting of a 5 microfarad capacitor in series with a finite resistor connects across the 2R resistor. At DC steady state, find the capacitor charge.

(A) 45 μC (B) 90 μC

(C) $\frac{45}{2}$ μC (D) 30 μC

20. In YDSE a sheet of index 1 and thickness 3 micrometres covers one slit. A second sheet of the same thickness and index 3/2 covers the other. With wavelength $\frac{4}{5}$ micrometres, find the magnitude of central-fringe shift in units of fringe width.

(A) $\frac{15}{4}$ (B) $\frac{8}{15}$

(C) $\frac{75}{8}$ (D) $\frac{15}{8}$

21. Two uniform thin rods meet at a right angle at the origin. Both have length 5 m. The rod along +x has linear density 2 times that of the rod along +y. A force parallel to the y-axis must produce pure translation. At what x-coordinate must its line of action lie?

(A) $\frac{10}{3}$ m (B) $\frac{5}{3}$ m

(C) $\frac{5}{6}$ m (D) $\frac{5}{2}$ m

22. A particle has $v(0) = 6$ m/s and acceleration $a(t) = 12t$ m/s^2 for $0 \le t \le 5$ s. Find its average velocity over this interval.

(A) 56 m/s (B) 106 m/s

(C) 156 m/s (D) 6 m/s

23. Two simple pendulums execute 7 and 6 small oscillations respectively in the same time at the same place. Find the first pendulum's length divided by the second's.

(A) $\frac{1}{49}$ (B) $\frac{216}{343}$

(C) $\frac{36}{49}$ (D) $\frac{49}{36}$

24. A nonrelativistic particle of fixed charge and mass moves perpendicular to a uniform magnetic field. Its kinetic energy is multiplied by 25 and the magnetic field is multiplied by 2. Find the new trajectory radius divided by the original radius.

(A) $\frac{2}{5}$ (B) $\frac{5}{2}$

(C) $\frac{1}{10}$ (D) $\frac{25}{2}$

25. A standing wave is $y = 4\sin(2x)\cos(5t)$ cm with x in metres and t in seconds. At a point where $2x = \pi/6$, find the particle-speed magnitude when $5t = \pi/3$.

(A) $\frac{5\sqrt{3}}{2}$ cm/s (B) $2\sqrt{3}$ cm/s

(C) $5\sqrt{3}$ cm/s (D) $\sqrt{3}$ cm/s

26. A uniform-density planet has radius R. Gravity at a depth d equals gravity at a height $(7-1)R$ above the surface. Find d/R.

(A) $\frac{48}{49}$ (B) $\frac{1}{49}$

(C) 1 (D) $\frac{6}{7}$

27. A point at the bottom of two plane liquid layers is viewed normally from air. The upper layer has thickness 9 cm and index 3/2; the lower has thickness 8 cm and index 4/3. Find its apparent depth below the top surface in the paraxial limit.

(A) $\frac{34}{3}$ cm (B) $\frac{51}{4}$ cm

(C) 6 cm (D) 12 cm

28. A nucleus of mass number 24 splits into two equal nuclei. The parent has binding energy per nucleon $\frac{15}{2}$ MeV and each daughter has $\frac{77}{10}$ MeV. Ignoring other products, what energy is released in the split?

(A) $\frac{24}{5}$ MeV (B) $\frac{12}{5}$ MeV

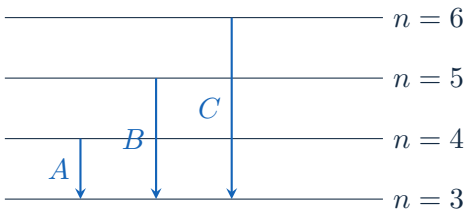
(C) 180 MeV (D) $\frac{924}{5}$ MeV

29. A parallel-plate capacitor has vacuum capacitance C_0. A dielectric of constant 6 fills a fraction $\frac{1}{2}$ of the plate area and the entire gap in that region; the remainder stays vacuum. The voltage is fixed. Find the fraction of total stored energy residing in the dielectric-filled region.

(A) $\frac{1}{2}$ (B) 3

(C) $\frac{6}{7}$ (D) 6

30. The diagram shows downward hydrogen transitions A, B and C from $n = 4$, 5 and 6, respectively, ending at $n = 3$. A filter transmits only photons with energy strictly above that of transition B. Which transmitted line has the longest wavelength?



(A) No transmitted line (B) B

(C) C (D) A

Chemistry

31. A mixture contains $\frac{1}{25}$ mol H_2SO_5 and $\frac{3}{100}$ mol $H_2S_2O_8$. Their structures contain one O–O link per molecule. All peroxide links react quantitatively with excess iodide in acid; each such link accepts two electrons. What amount of I_2, in mol, is liberated?

(A) $\frac{7}{50}$ (B) $\frac{7}{100}$

(C) $\frac{1}{25}$ (D) $\frac{1}{10}$

32. For $CaCO_3$, $SrCO_3$, $BaCO_3$,which increasing thermal-stability order and explanation is correct?

(A) $CaCO_3 < SrCO_3 < BaCO_3$; the cations become smaller down the group.

(B) $CaCO_3 < SrCO_3 < BaCO_3$; increasing cation size lowers carbonate-ion polarisation.

(C) $CaCO_3 > SrCO_3 > BaCO_3$; larger cations distort carbonate more strongly.

(D) All are equally stable because the cation charge is+2.

33. Construct one open linear peptide using each member of the following multiset exactly once: Pro, Gly, Ala, Leu, Thr, Val. The C-terminal residue isPro and the N-terminal residue is the only achiral amino acid. Count distinctN-to-C sequences; ignore stereoisomerism of the supplied residues.

(A) 25 (B) 24

(C) 720 (D) 120

34. For one reaction,ln K is8 at300 K and3 at600 K. Assume ln K is linear in1/T. At400 K,ln K equals

(A) $\frac{13}{2}$ (B) 3

(C) $\frac{11}{2}$ (D) 8

35. A hot saturated solution contains 100g water and60.0g solute per100g water. Cooling to a temperature where solubility is20.0g per100g water also removes 10g water by evaporation. No solute evaporates,and crystals are anhydrous. Mass crystallised in g is

(A) 18 (B) 38

(C) 42 (D) 40

36. For $[FeF_6]^{3-}$,use the usual ligand-field/VBT assignments at room temperature. Which complete assignment is consistent with charge balance and magnetic behaviour?

(A) d^6,octahedral,sp^3d^2,spin-only moment $\sqrt{35}$ BM.

(B) d^5,octahedral,sp^3d^2,spin-only moment $\sqrt{35}$ BM.

(C) d^5,octahedral,sp^3d^2,spin-only moment $4\sqrt{3}$ BM.

(D) d^5,tetrahedral,sp^3,spin-only moment $\sqrt{35}$ BM.

37. In a central-field description of a many-electron atom with no external field,which joint statement is correct?

(A) The3p ml=+1 state must emit a photon to become3p ml=-1 because ml changes.

(B) 4s precedes3d by the n+l filling rule;different ml states within one3p subshell remain degenerate.

(C) The n+l rule makes all orbitals with equal n+l exactly degenerate.

(D) 3d precedes4s because principal n alone determines every orbital energy.

38. For the metal centre in MnO_4^-,Z=25. Determine the number of d electrons after formal oxidation-state assignment; use O=-2 in oxoanions.

(A) 0 (B) 7

(C) 2 (D) 1

39. For ethanal with HCN,which product or mechanistic conclusion is correct under the stated conditions?

(A) CH3CH(OH)CN (B) CH3COOH

(C) CH3CH2CN (D) CH3CH2OH

40. Acidic permanganate oxidises SO_2 to HSO_4^- and becomes Mn^{2+}. With excess SO_2, $\frac{1}{50}$ mol MnO_4^- and only $\frac{1}{200}$ mol available H^+ are supplied. Count the net H^+ in the balanced reaction. Maximum Mn^{2+} formed,in mol,is

(A) $\frac{1}{400}$ (B) $\frac{1}{50}$

(C) $\frac{1}{1600}$ (D) $\frac{1}{100}$

41. In the trigonal-bipyramidal electron-domain model of SF_4,there are 1 central lone pairs. Which placement rule explains the observed arrangement,including the first lone pair?

(A) Place lone pairs equatorially first: an equatorial site has two 90-degree neighbours,whereas an axial site has three.

(B) Place the first lone pair axially because three 90-degree contacts are less repulsive than two.

(C) Place lone pairs only where they form the smallest possible angle to every bond.

(D) Use arbitrary sites because lone-pair/bond-pair and bond-pair/bond-pair repulsions are identical.

42. A gas above water has total pressure6bar,including1.00bar water vapour. Oxygen is25.0% of the dry gas by moles. Henry law is p_{O2}=KHx_{O2},KH=40000bar. Dissolved oxygen mole fraction is

(A) $\frac{3}{80000}$ (B) $\frac{1}{8000}$

(C) $\frac{1}{32000}$ (D) $\frac{1}{160000}$

43. A classical Rutherford electron is accelerated in its orbit. Which paired conclusion correctly distinguishes that model from Bohr stationary states?

(A) Rutherford quantises angular momentum and thereby predicts every line spectrum.

(B) Rutherford predicts stable nonradiating orbits because acceleration is zero.

(C) Bohr explains stationary orbits by continuously radiating at fixed energy.

(D) Classical electrodynamics predicts radiation and orbital instability;Bohr postulates no radiation in a stationary orbit.

44. For an gas-phase H_2O_2 molecule,which combined statement about local geometry and overall conformation is correct?

(A) All four atoms must lie in one plane because only sigma bonds are present.

(B) Each O is linear because it is bonded to only two atoms,and the skeleton is straight.

(C) The O-O bond is a double bond,so both O centres are planar with one lone pair.

(D) Each O has two bonds and two lone pairs; the H-O-O-H skeleton is nonplanar/open-book.

45. One mole of an acyclic monoalkene gives one mole methanal and one mole butan-2-one on complete reductive ozonolysis (O3,then Zn/H2O),with no other carbonyl products. Which parent and carbon-skeleton assignment fits?

(A) pent-2-ene (B) 2-methylbut-2-ene

(C) 2-methylbut-1-ene (D) 3-methylbut-1-ene

46. Equal volumes of $\frac{1}{5}$M $K_4[Fe(CN)_6]$ and $\frac{1}{5}$M glucose are mixed. In the final solution,the salt is50% dissociated into4K^+ and one intact complex anion;glucose does not dissociate. Use RT=24.0 L atm/mol. Osmotic pressure in atm is

(A) $\frac{24}{5}$ (B) $\frac{48}{5}$

(C) $\frac{72}{5}$ (D) $\frac{96}{5}$

47. Assign the formal metal oxidation state in $[Fe(H_2O)_5(NO)]^{2+}$,using five neutral H_2O and one formal NO^+. For nitrosyl species use the explicitly stated formal NO^+ convention rather than inferring spin from it.

(A) 2 (B) 3

(C) 1 (D) 0

48. An element has cubicformsAandB. Ahas an FCC lattice with fraction $\frac{3}{4}$of normal atomic sites occupied; Bhas fully occupiedBCC. Edge lengths obey aA=2aB,and atomic mass is unchanged. DensityratioρA/ρB is

(A) $\frac{1}{4}$ (B) $\frac{3}{8}$

(C) $\frac{3}{16}$ (D) 3

49. D-Glucose and D-galactose differ only at C4. Under normal osazone formation, which conclusion follows?

(A) They must give one identical osazone because every epimer does.

(B) They are enantiomers because one centre differs.

(C) They remain distinguishable as osazones becauseC4 is not erased.

(D) They differ only at the anomeric centre.

50. p-Aminophenol is converted into p-HO-C6H4-NH-COCH3 while retaining the phenolicOH. Which transformation gives that analgesic structure?

(A) N-acetylation of the amino group.

(B) O-acetylation of the phenolicOH only.

(C) Addition of chlorine to all ring carbons.

(D) Reduction of every carbonyl to methane.

51. Two unchanged capillaries initially carry a liquid and a dilute gas at fixed pressure drops. On heating,the calibrated liquid viscosity halves while gas viscosity doubles. For each capillary,$Q \propto 1/\eta$ in the stated regime. Factor by which Qliquid/Qgas changes is

(A) 1

(B) 2

(C) $\frac{1}{4}$

(D) 4

52. A freshly assayed 15-volume H_2O_2 solution releases 15 L O_2 at STP per litre on complete decomposition. Only80.0% of the evolved oxygen reaches a burner that requires0.560 L O_2 at STP. The solution volume needed,in mL,is

(A) $\frac{140}{3}$

(B) $\frac{280}{3}$

(C) $\frac{56}{3}$

(D) $\frac{112}{3}$

53. styrene,Ph-CH=CH2 reacts with HBr,without peroxides. Which pair gives the major constitutional product and the intermediate whose greater stability fixes its orientation? Ignore stereoisomer counting.

(A) Ph-CH(Br)-CH3; primary carbocation

(B) Ph-CH2-CH2Br; benzylic carbocation

(C) Ph-CH2-CH2Br; primary carbocation

(D) Ph-CH(Br)-CH3; benzylic carbocation

54. Compare ethyl bromide and isopropyl bromide toward the same cyanide nucleophile in a polar aprotic solvent under ordinary SN2 conditions. Which reacts faster or is the substituted site,and why?

(A) ethyl bromide; A primary centre is less hindered than a secondary centre with the same leaving group.

(B) isopropyl bromide; an sp2 C-Br bond behaves just like an unhindered methyl bromide

(C) isopropyl bromide; steric hindrance accelerates backside attack

(D) ethyl bromide; the mechanism requires formation of a free tertiary carbocation

55. At 350 K,combustion enthalpies of C(graphite) and CO(g) to CO_2(g) are respectively-393.5 and-283.0 kJ/mol. Treat gases ideally and use R=0.00800 kJ mol^{-1} K^{-1}. Constant-volume heat for C(graphite)+1/2 O_2(g)→CO(g),per mole CO,in kJ is

(A) $-\frac{1133}{10}$

(B) $-\frac{221}{2}$

(C) $-\frac{1091}{10}$

(D) $-\frac{1119}{10}$

56. A textbook agent is described as reducing pain and fever without being classified as a narcotic. Which function pair matches the two effects?

(A) Analgesic and antipyretic.
(B) Antacid and disinfectant.
(C) Antibiotic and polymer.
(D) Antiseptic and artificial sweetener.

57. A textbook deficiency table associates beriberi with one vitamin. Which vitamin and solubility class are correctly paired?

(A) Vitamin B12; water-soluble.
(B) VitaminD; fat-soluble.
(C) Vitamin B2; water-soluble.
(D) Vitamin B1; water-soluble.

58. At the same external pressure,choose the higher normal-boiling member in each pair:ethanol and dimethyl ether;n-pentane and neopentane.

(A) Equal in both pairs because molecular formula or group membership alone fixes boiling point.
(B) Ethanol; n-pentane.
(C) Ethanol; neopentane.
(D) Dimethyl ether; n-pentane.

59. A solution initially contains $\frac{1}{100}$ mol Fe^{2+}. Before titration, $\frac{3}{10}$ of it is oxidised quantitatively to Fe^{3+}. The volume in mL of0.0200 M $KMnO_4$ needed for the remaining Fe^{2+} in acid is

(A) 70
(B) 350
(C) 30
(D) 100

60. A student assigns three bonding pairs and two lone pairs to the central atom of ClF_3. After independently checking valence-electron count,which combined molecular shape/domain count is consistent?

(A) Trigonal planar; three electron domains.
(B) T-shaped; five electron domains.
(C) Trigonal pyramidal; four electron domains.
(D) Square pyramidal; six electron domains.

English Proficiency

61. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The host serves the main course after the guests have finished the starter.

(A) The main course is served before the guests have finished the starter.
(B) The main course is served by the host after the guests have finished the starter.
(C) The starter is served by the main course after the host has finished the guests.
(D) The main course was served by the host after the guests finished the starter.

62. Read the sentence: The durable cover survived years of frequent use. Choose the opposite in meaning of 'durable' as used in the sentence.

(A) reusable
(B) fragile
(C) hard
(D) sturdy

63. Read the sentence: The account was partial, presenting only the evidence favourable to one side. Choose the opposite in meaning of 'partial' as used in the sentence.

(A) impartial (B) supportive

(C) selective (D) incomplete

64. Choose the most coherent order of the four sentences. A. The investigation eliminated an ordinary explanation before treating the rise as evidence of a fault. B. She first checked whether more rooms had been occupied. C. When occupancy proved unchanged, she inspected the pipes and found a leak. D. The caretaker noticed an unusual rise in the building's water use.

(A) D-B-A-C (B) D-C-B-A

(C) B-D-C-A (D) D-B-C-A

65. Read the sentence: The report offered a dispassionate assessment, separating the evidence from the writer's anger. Choose the closest meaning of 'dispassionate' as used in the sentence.

(A) unemotional and objective (B) lacking conviction

(C) unwilling to reach a conclusion (D) indifferent to consequences

66. Every voter supports the motion, with no abstention or opposition. Which noun names the complete agreement?

(A) unanimity (B) uniformity

(C) popularity (D) majority

67. Choose the most coherent order of the four sentences. A. The expedition tested two types of footwear on a short local walk. B. One pair seemed comfortable until the path became wet. C. The testers added a wet-surface trial before choosing equipment for the longer journey. D. The choice then reflected the conditions likely to be faced rather than only the easiest part of the route.

(A) A-B-D-C (B) B-A-C-D

(C) A-B-C-D (D) A-C-B-D

68. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The team is going to replace the broken latch rather than the whole door.

(A) The whole door, rather than the broken latch, is going to be replaced by the team.

(B) The broken latch and the whole door are going to be replaced by the team.

(C) The broken latch replaced the whole door for the team.

(D) The broken latch, rather than the whole door, is going to be replaced by the team.

69. The picture is formed from many small coloured tiles fitted together, not painted on a single surface. What is it?

(A) sketch (B) mosaic

(C) engraving (D) mural

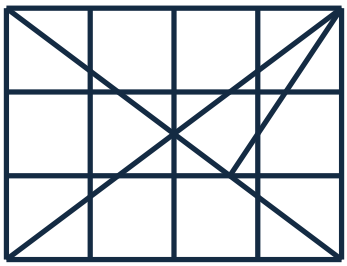
70. Read the sentence: The mediator proposed a fair division that gave neither side an unjust advantage. Choose the closest meaning of 'fair' as used in the sentence.

(A) equal in every detail (B) easy to administer

(C) equitable (D) generous

Logical Reasoning

71. How many triangles are formed by all the drawn segments?



(A) 53
(B) 41
(C) 42
(D) 57

72. Three pairs contain exactly one metal and one nonmetal. Which pair does not?

(A) Silver / Chlorine
(B) Gold / Carbon
(C) Lead / Zinc
(D) Aluminium / Sulfur

73. A word is coded as the sum of its forward alphabet positions plus 2 times its length. The code of MAN?O is 60. Which letter replaces the question mark?

(A) I
(B) H
(C) F
(D) G

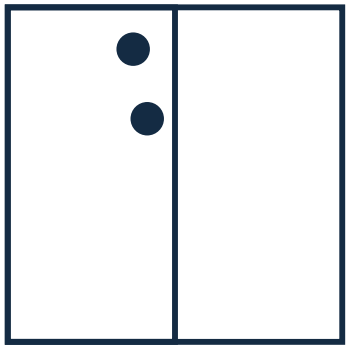
74. In a row, Asha is 6th from the left and 28th from the right. Bina is 6th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 18
(B) 19
(C) 17
(D) 24

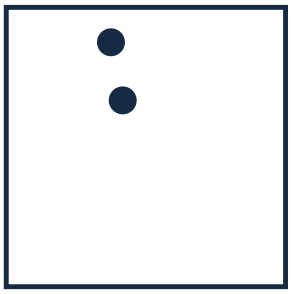
75. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 3 and Z has code 28. Decode the separated values 5, 10, 3, 11, 20.

(A) RIAHC
(B) DIBJS
(C) CHAIR
(D) BGZHQ

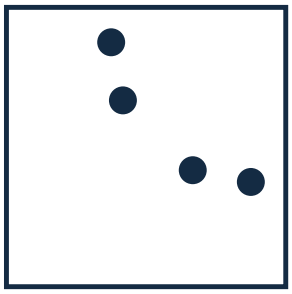
76. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



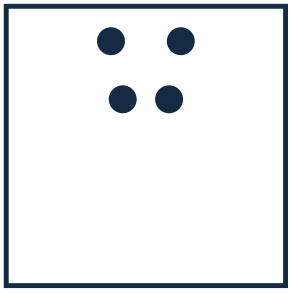
(A) Pattern W



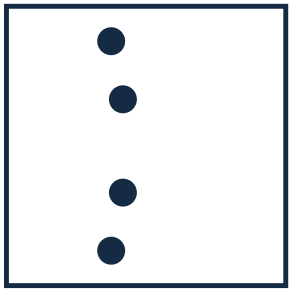
(B) Pattern X



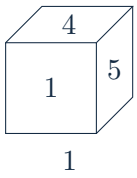
(C) Pattern P



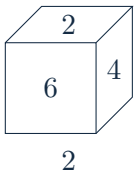
(D) Pattern T



77. The views show the same die. Give the ordered pair: (face opposite 6, face opposite 2). Do not assume opposite labels sum to seven.



1



2

(A) $(5, 3)$

(B) $(5, 1)$

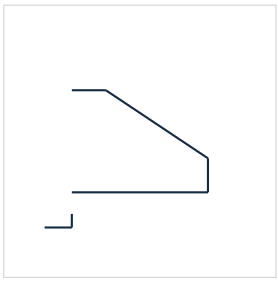
(C) $(5, 4)$

(D) $(5, 2)$

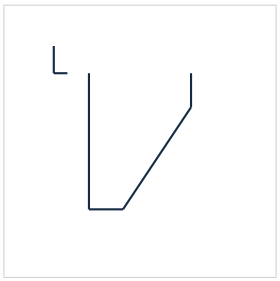
78. A community register uses the following three categories: guides, hikers, bakers. Its complete relevant assertions are in this short paragraph; do not import everyday assumptions. All hikers are guides. No hikers are bakers. All guides are hikers. Conclusions: I. No guides are bakers. II. Some bakers are not hikers. Which conclusions follow necessarily?

(A) Both I and II
(B) Only I
(C) Only II
(D) Neither I nor II

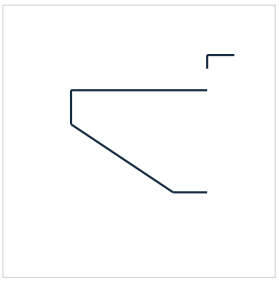
79. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1



2

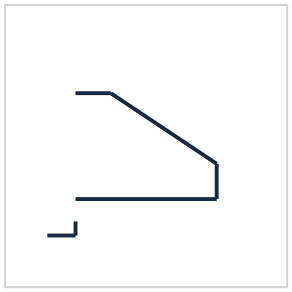


3

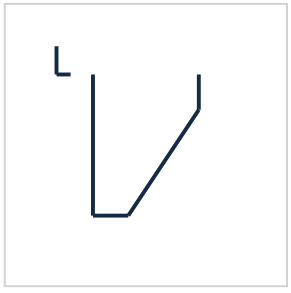
(A) Pattern W



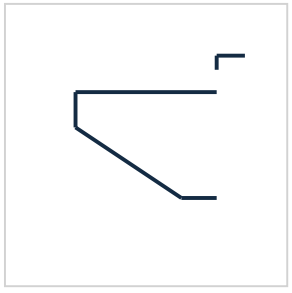
(B) Pattern L



(C) Pattern Y



(D) Pattern X



80. Vijay points to a photograph and says, "This person is the daughter of my father's only son." Vijay is male, and the photographed person is female. How is the photographed person related to Vijay?

(A) Son
(B) Brother
(C) Sister
(D) Daughter

81. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 2, 8, 17, ?, 77, 308, 317.

(A) 104 (B) 68

(C) 59 (D) 26

82. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

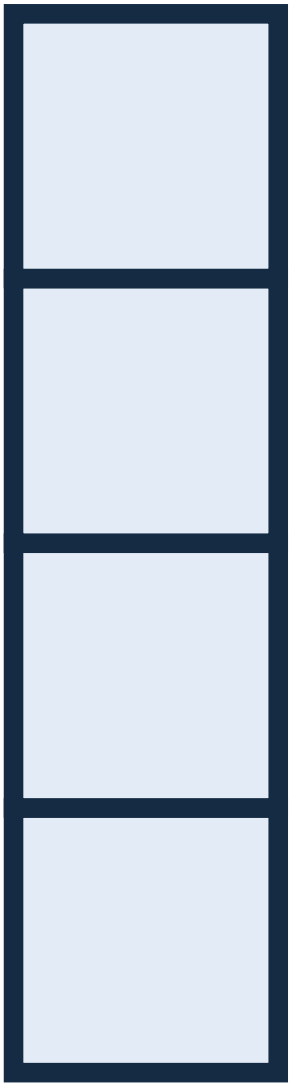


piece 1



piece 2

(A) Pattern P



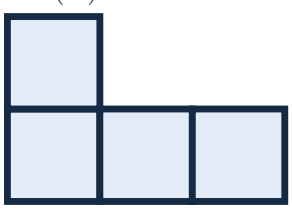
(B) Pattern W



(C) Pattern Z



(D) Pattern V



83. Consider only these statements; universal statements alone do not imply existence. Some guides are guides; All campers are designers. Conclusions: I. Some guides are campers. II. No guides are campers. Which conclusion or complementary alternative must hold?

(A) Only II

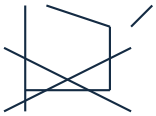
(B) Either I or II follows, but neither is individually certain

(C) Only I

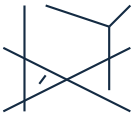
(D) Both I and II

84. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

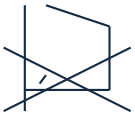




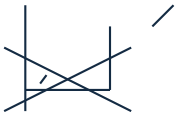
host 1



host 2



host 3



host 4

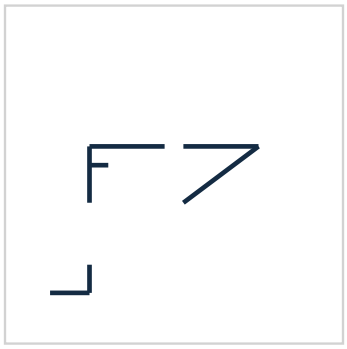
(A) 3

(B) 4

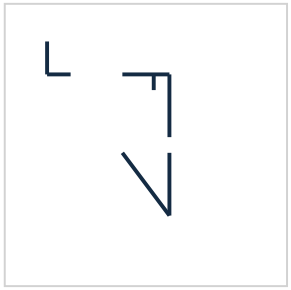
(C) 1

(D) 2

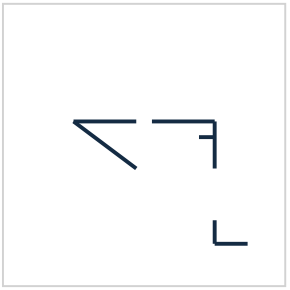
85. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



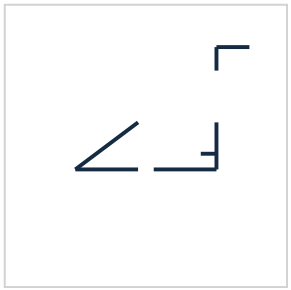
(A) Pattern U



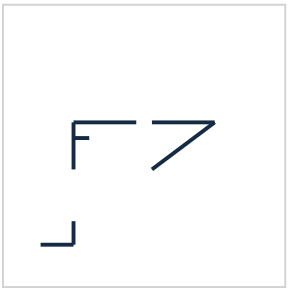
(B) Pattern Y



(C) Pattern Z



(D) Pattern V



86. All five people have different heights. Charu is shorter than Farah. Hari is shorter than Esha. Asha is shorter than Hari. Esha is shorter than Charu. If Asha is left out, who is 2nd shortest among the remaining people?

(A) Hari

(B) Farah

(C) Esha

(D) Asha

87. Each rectangle is drawn to the same local scale. In three numbered figures the interior straight line divides the rectangle into equal areas. Which does not?



1



2



3



4

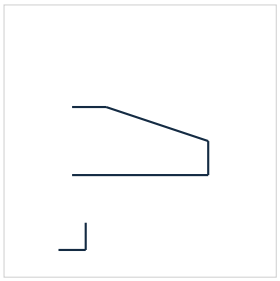
(A) 4

(B) 1

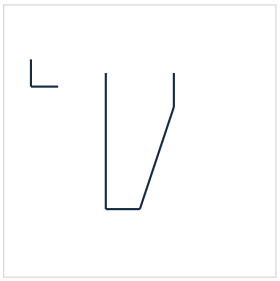
(C) 2

(D) 3

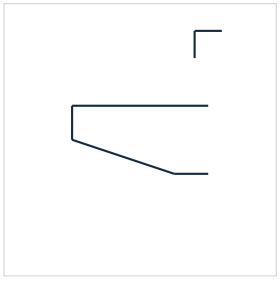
88. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

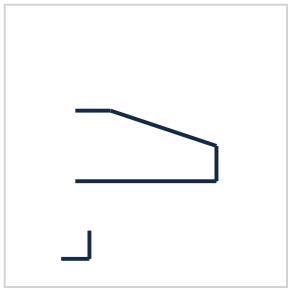


bottom left

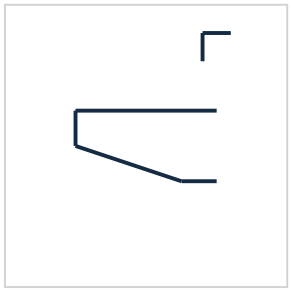
?

bottom right

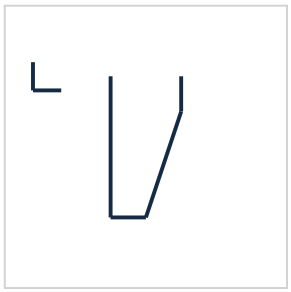
(A) Pattern W



(B) Pattern M



(C) Pattern V



(D) Pattern U



89. Preserve the exact relation and its order: Ammeter : Electric current :: ? : Elapsed time. Choose the missing term.

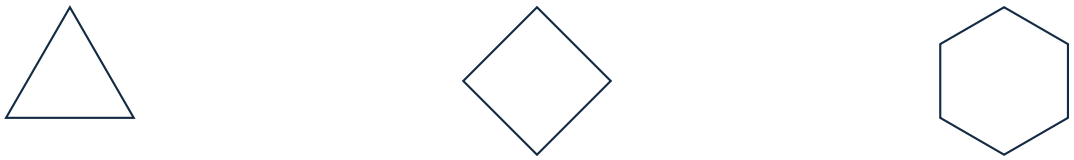
(A) Stopwatch

(B) Barometer

(C) Thermometer

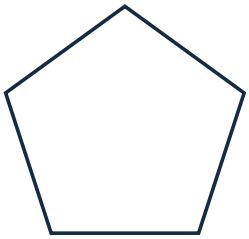
(D) Ammeter

90. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?

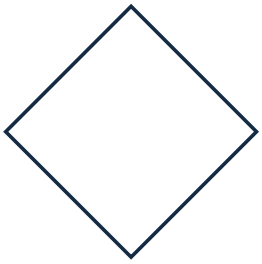


A B C

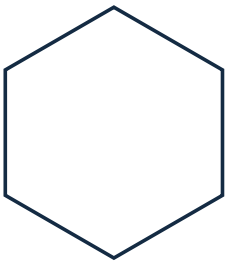
(A) Pattern K



(B) Pattern P



(C) Pattern R



(D) Pattern L



Mathematics

91. F′(x)=$(x^2+6x+8)\,e^x$ and F(0)=4. Find F(1)-2F(0).

(A) $-9e$

(B) $-8+9e$

(C) $9e$

(D) $-4+9e$

92. The line y=mx through O bisects the area of the triangle with vertices O,(7,5),(4,4). Assume the vertices are noncollinear. Find m.

(A) $\frac{1}{3}$

(B) $\frac{9}{11}$

(C) $\frac{4}{7}$

(D) $\frac{11}{9}$

93. 10 distinct people stand in a line. Three specified people cannot occupy positions 1,2 or 10. How many orders are possible?

(A) 5040

(B) 1693440

(C) 1058400

(D) 176400

94. On A={0,1,...,5}, define aRb if 5 divides a-b. How many ordered pairs are in R?

(A) 2 (B) 36

(C) 6 (D) 8

95. y>0 satisfies $2yy'+12y^2=0$, y(0)=sqrt(2). Find $y^2(1)$.

(A) $\frac{\sqrt{2}}{e^6}$ (B) $2e^{12}$

(C) $\frac{2}{e^6}$ (D) $\frac{2}{e^{12}}$

96. Let $R=[\sin(\frac{2\pi}{21})+\sin(\frac{5\pi}{21})]/[\cos(\frac{2\pi}{21})+\cos(\frac{5\pi}{21})]$. Find $(1-R^2)/(1+R^2)$.

(A) $\frac{3}{2}$ (B) $-\frac{1}{2}$

(C) $\frac{1}{2}$ (D) $\frac{\sqrt{3}}{2}$

97. Evaluate $\int_0^{\pi/2} \sin(2x)\left[1+2\sin^2(2x)\right]\frac{\sin^4 x}{\sin^4 x+\cos^4 x}\,dx$.

(A) $\frac{7}{6}$ (B) $\frac{1}{2}$

(C) $\frac{7}{3}$ (D) $\frac{3}{2}$

98. a=(10,8,0), b=(8,-10,6). Find $|a+b|^2-|a-b|^2$.

(A) 364 (B) 32800

(C) 0 (D) -144

99. For $f(x)=x^3-48x$, find the difference between its local maximum value and local minimum value.

(A) 8 (B) 256

(C) 128 (D) -256

100. For $P(x)=\prod_{j=4}^{9}(x+j)$, find the coefficient of x^3 in $P(x)/(x+4)$.

(A) 625 (B) 485

(C) 609 (D) 970

101. The line 3x+4y=15 cuts $x^2+y^2=81$ in chord AB. Find $|AB|^2$.

(A) 324 (B) 306

(C) 225 (D) 288

102. For 3x+4y+4=0 and 4x+3y+5=0, find the squared distance from the origin to the angle bisector of positive slope.

(A) 1 (B) $\frac{1}{2}$

(C) $\frac{81}{98}$ (D) $\frac{\sqrt{2}}{2}$

103. A rectangle has perimeter 48. Its length is constrained to be at least three times its width. Find its maximum area.

(A) 108 (B) 72

(C) 144 (D) 36

104. Between positive a=49, b=81, let A be their arithmetic mean and G their geometric mean. Find A^2-G^2.

(A) 4225 (B) 16

(C) 256 (D) 3969

105. A={j∈{1,...,72}:sin(jπ/2)=0}, B={j∈{1,...,72}:sin(jπ/3)=0}. Find |(A∪B)\(A∩B)|.

(A) 48 (B) 36

(C) 12 (D) 60

106. Let $y\prime + 2y = 4x$ and $y(0) = 2$. Find $y(1) + \frac{11}{10}y\prime(1)$.

(A) $\frac{16}{5}$ (B) $\frac{3}{e^2} + 1$

(C) $\frac{18}{5e^2} + \frac{16}{5}$ (D) $\frac{16}{5} - \frac{18}{5e^2}$

107. A sphere has dV/dt=5 cubic units/s. Find the rate of change of its surface area when r=5.

(A) 2 (B) $\frac{1}{20\pi}$

(C) $\frac{2}{5}$ (D) 1

108. f(x)=x^3+4x, g=f^{-1}. Find $g''(16)$.

(A) $\frac{1}{12}$ (B) $-\frac{3}{1024}$

(C) $\frac{3}{1024}$ (D) $-\frac{3}{64}$

109. Find the product of all real solutions of sqrt(13-log_2|x|)=log_2|x|-1.

(A) −256 (B) 0

(C) 256 (D) −16

110. Evaluate $\sum_{j=1}^{10} j(j+4)$.

(A) 605 (B) 385

(C) 3245 (D) 220

111. Find the squared distance from O to the plane through (2,0,0),(0,4,0),(0,0,2).

(A) 24 (B) $\frac{9}{16}$

(C) $\frac{4}{5}$ (D) $\frac{16}{9}$

112. In how many ordered ways can three six-sided dice have sum 14, if the first die is known not to show 6?

(A) 14 (B) 20

(C) 10 (D) 15

113. Evaluate lim as x→0 of [sqrt(9+x)-3-x/6]/x^2.

(A) $-\frac{1}{72}$ (B) $\frac{1}{216}$

(C) $\frac{1}{6}$ (D) $-\frac{1}{216}$

114. Two distinguishable fair dice show X,Y. Given X+Y≥9, find the probability that at least one of X, Y is even.

(A) $\frac{1}{4}$ (B) $\frac{1}{10}$

(C) $\frac{6}{5}$ (D) $\frac{9}{10}$

115. The system x+y+z=1, y+z=2, (k-6)z=k-9 has no solution for which k?

(A) 0 (B) 9

(C) 6 (D) -6

116. $A = \begin{pmatrix} 0 & -6 \\ 6 & 0 \end{pmatrix}$, B=(I-A)(I+A)$^{-1}$. Find det B.

(A) 1 (B) $\frac{1}{37}$

(C) 37 (D) -1

117. Group A has mean 70, SD 7. Every observation is increased by 70. Find its new coefficient of variation in percent.

(A) 20 (B) 5

(C) 35 (D) 10

118. Evaluate the integral from 0 to $\pi/2$ of [(sin x+cos x)/(sec x+csc x)](sin x+3 cos x) dx, using continuous extension at the endpoints.

(A) 2 (B) 1

(C) 4 (D) $\frac{4}{3}$

119. How many integer k in [0,80] make $x^2 - 13x + (40 - k) = 0$ have roots of opposite signs?

(A) 39 (B) 41

(C) 40 (D) 80

120. For the roots α,β of $z^2 - (-4 + 8i)z + (-24 - 32i) = 0$, find $|\alpha\text{-}\beta|^2$.

(A) 80 (B) 1600

(C) 40 (D) 160

121. A 6-element set contains distinct a,b. How many subsets contain a and exclude b?

(A) 16 (B) 32

(C) 62 (D) 64

122. z_1=7+5i, z_2=10+5i, z_3=10+8i. Find the smaller angle between z_2-z_1 and z_3-z_1.

(A) 0 (B) $\frac{\pi}{2}$

(C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{4}$

123. From two points on the same ray from a tower foot, separated by 5, elevations are 30° and 45°. Find the distance from the farther point to the foot.

(A) 5 (B) $\frac{5\sqrt{3}}{2}$

(C) $\frac{5\sqrt{3}}{2} + \frac{15}{2}$ (D) $\frac{5}{2} + \frac{5\sqrt{3}}{2}$

124. $A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}$ and A^{-1}=pI+qA. Find p+q.

(A) $\frac{6}{11}$ (B) $-\frac{6}{11}$

(C) $\frac{7}{11}$ (D) $\frac{1}{11}$

125. Find the area enclosed by y=x(6-x) and y=2x.

(A) $-\frac{32}{3}$ (B) $\frac{32}{3}$

(C) $\frac{104}{3}$ (D) 36

126. The locus |z-7|=|Re(z)+7| intersects the horizontal line Im(z)=16. Find the real part of the point.

(A) $\frac{64}{7}$ (B) $\frac{128}{7}$

(C) $\frac{4}{7}$ (D) $-\frac{64}{7}$

127. O=(0,0,0), A=(9,0,0), B=(1,5,0). Find the squared area of triangle OAB.

(A) $\frac{2025}{2}$ (B) $\frac{45}{2}$

(C) 2025 (D) $\frac{2025}{4}$

128. How many solutions in $0 \le x < 6\pi$ satisfy sin2x=sin x and cosx>0?

(A) 10 (B) 9

(C) 12 (D) 6

129. Three distinguishable fair dice are thrown. Given that the three faces are not all identical, find the probability that their sum is 9.

(A) $\frac{5}{42}$ (B) $\frac{31}{35}$

(C) $\frac{4}{35}$ (D) $\frac{25}{216}$

130. In $(1+x)^{11}$, find the ratio of the coefficient of x^8 to the coefficient of x^7.

(A) 2 (B) 4

(C) $\frac{1}{2}$ (D) $\frac{5}{7}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A mass 5 kg moves with $\vec{r}(t) = 1t\hat{i} + 3t^2\hat{j}$ m. Find the z-component of angular momentum about the origin at $t = 4$ s.

(A) $480\,\text{kg}\,\text{m}^2/\text{s}$ (B) $60\,\text{kg}\,\text{m}^2/\text{s}$

(C) $240\,\text{kg}\,\text{m}^2/\text{s}$ (D) $120\,\text{kg}\,\text{m}^2/\text{s}$

132. Two sound sources produce wavelengths $\frac{1}{2}$ m and $\frac{1}{3}$ m in the same gas at one temperature. The sound speed there is 400 m/s. At a second temperature the sound speed is 3/2 times larger, while both source frequencies stay unchanged. Find the new beat frequency.

(A) 600 Hz (B) 400 Hz

(C) 1000 Hz (D) $\frac{800}{3}$ Hz

133. Two equal-intensity coherent beams have maximum combined intensity 20 units. Find intensity when their path difference is $\lambda/3$.

(A) 5 (B) 20

(C) 15 (D) 10

Chemistry

134. A first-order reaction hasEa/R=6000K andfrequency factorA=(ln2)e^{20}/60s^{-1}. At what nearest whole-kelvin temperature is75.0% completion reached in60.0s? Useln2=.693.

(A) 311 (B) 330

(C) 300 (D) 290

135. CO binds to a low-oxidation-state metal. If metal-to-CO pi back-donation increases while the ligand remains bound through carbon,which paired bond changes and donation directions are consistent?

(A) Metal donates into a CO bonding orbital,so C-O strengthens while M-C weakens.

(B) CO donates a carbon lone pair to metal; metal donates into CO pi-star. M-C strengthens while C-O weakens.

(C) CO is only a sigma donor; back-donation leaves the C-O force constant unchanged.

(D) CO pi electrons are donated only from oxygen to metal,and both M-C and C-O necessarily strengthen.

136. Two air samples contain similar NOx and reactive hydrocarbons. One is illuminated and the other is kept dark. The illuminated sample develops more ozone and eye-irritating oxidants. Which explanation is appropriate?

(A) Ozone is simply emitted unchanged from every exhaust.

(B) Sulphur dioxide alone proves ozone is a primary pollutant.

(C) Sunlight drives secondary photochemical pollutant formation.

(D) Darkness is the essential energy source for photochemical smog.

Logical Reasoning

137. 6 people sit around a circle facing its centre. Clockwise order fixes the convention; immediate left is clockwise. Charu is 2 places clockwise from Asha. Farah is 3 places clockwise from Asha. Esha is 5 places clockwise from Asha. Dev is not adjacent to Charu. Who sits immediately clockwise from Farah?

(A) Charu (B) Dev

(C) Bina (D) Asha

138. Two people differ in age by 9 years. The product of their ages 3 years ago was 792. What will the older person's age be 4 years from now?

(A) 37 (B) 36

(C) 40 (D) 49

139. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

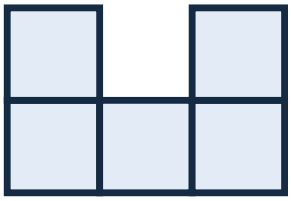


piece 1

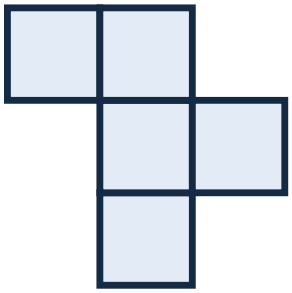


piece 2

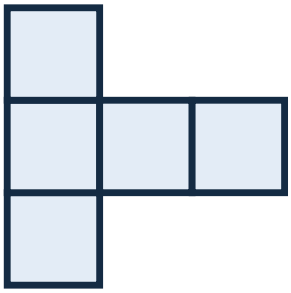
(A) Pattern U



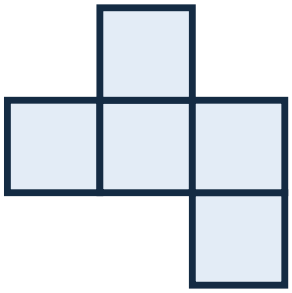
(B) Pattern R



(C) Pattern W



(D) Pattern V



Mathematics

140. Positive a,b,c form an AP with a=5. The numbers a,b,c+9/5 form a GP and b>a. Find c.

(A) 11 (B) -1

(C) 23 (D) 8

141. For the loop $y^2=x^2(4-x)$, find the area of the portion with $x \geq 4/2$.

(A) $\frac{128}{15}$

(B) $\frac{56\sqrt{2}}{15}$

(C) $\frac{112\sqrt{2}}{15}$

(D) $\frac{256}{15} - \frac{112\sqrt{2}}{15}$

142. The four lines $\pm x/6 \pm y/10=1$ form a rhombus with vertices on the coordinate axes. Find its area.

(A) 30

(B) 120

(C) 60

(D) 240

BITSAT Mock 06

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. The 2nd harmonic of an open pipe of length 4 m equals the 2nd allowed resonance of a closed pipe. Both have the same sound speed and negligible end correction. Find the closed-pipe length.

(A) 5 m (B) 6 m

(C) 4 m (D) 3 m

2. A projectile of mass 12 kg is launched with speed 20 m/s, $\cos\theta = 3/5$. At its highest point it splits into masses $M/3$ and $2M/3$. The lighter fragment moves horizontally forward at 17 m/s. Find the signed horizontal velocity of the heavier fragment immediately after the explosion.

(A) 12 m/s (B) $\frac{19}{2}$ m/s

(C) $\frac{43}{2}$ m/s (D) 19 m/s

3. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 1 kΩ to a terminal at +9 V, and its cathode connects to a terminal at 0 V. Find the current from the +9 V terminal in mA.



(A) $\frac{97}{10}$ mA (B) $\frac{83}{10}$ mA

(C) $\frac{38}{5}$ mA (D) 9 mA

4. A nucleus of mass number 20 splits into two equal nuclei. The parent has binding energy per nucleon $\frac{73}{10}$ MeV and each daughter has $\frac{77}{10}$ MeV. Ignoring other products, what energy is released in the split?

(A) 146 MeV (B) 8 MeV

(C) 162 MeV (D) 4 MeV

5. A magnetic field is uniform inside a long solenoid of radius 2 m and negligible outside. It changes at 8 T/s. Find induced electric-field magnitude at radius $5R$ from the axis.

(A) 8 V/m (B) 40 V/m

(C) $\frac{8}{5}$ V/m (D) $\frac{16}{5}$ V/m

6. A wheel initially rotates at 11 rad/s. A constant opposing angular acceleration brings it to rest after 7 s. What angle does it turn through during the last half of this stopping interval?

(A) $\frac{77}{8}$ rad (B) $\frac{77}{2}$ rad

(C) $\frac{77}{4}$ rad (D) $\frac{231}{8}$ rad

7. A nonrelativistic particle of fixed charge and mass moves perpendicular to a uniform magnetic field. Its kinetic energy is multiplied by 16 and the magnetic field is multiplied by 2. Find the new trajectory radius divided by the original radius.

(A) 4 (B) 2

(C) $\frac{1}{8}$ (D) 8

8. A potentiometer wire of resistance 70 ohms and length 500 cm is driven by an ideal 40 V source through a 40 ohm resistor and a silicon diode in series. The diode is forward biased with constant drop 0.7 V. A test cell of emf 1 V draws no current at balance. Find balance length, assuming it lies within the wire.

(A) $\frac{25}{2}$ cm (B) $\frac{1320500}{2751}$ cm

(C) $\frac{55000}{2751}$ cm (D) $\frac{275}{14}$ cm

9. A particle of mass 1 kg starts from rest and experiences $F(t) = 2t$ N along its motion for 2 s. Find the work done by the force.

(A) 4 J (B) $\frac{16}{3}$ J

(C) 16 J (D) 8 J

10. A mass 2 kg moves with $\vec{r}(t) = 3t\hat{i} + 5t^2\hat{j}$ m. Find the z-component of angular momentum about the origin at $t = 4$ s.

(A) 480 kg m^2/s (B) 120 kg m^2/s

(C) 1440 kg m^2/s (D) 960 kg m^2/s

11. A spring of constant 50 N/m is already stretched by $\frac{3}{10}$ m. How much additional work is required to increase its extension to $\frac{7}{10}$ m quasistatically?

(A) 10 J (B) 4 J

(C) $\frac{9}{4}$ J (D) 6 J

12. A network is defined by $P = \mathrm{NOR}(A, B)$, $Q = \mathrm{NOR}(A, P)$ and $Y = \mathrm{OR}(Q, B)$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 0$ and $P = 0$?

(A) 3 (B) 0

(C) 1 (D) 2

13. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 25 percent. The measured increase in its absolute temperature is 180 K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) 720 K (B) $\frac{2880}{41}$ K

(C) 320 K (D) 2880 K

14. A straight smooth wire descends at 45° to horizontal through a vertical height 4 m. A bead is released from rest at the top. Find its time to reach the bottom. Take $g = 10\ \text{m/s}^2$.

(A) $\frac{2\sqrt{5}}{5}$ s (B) $\frac{2\sqrt[4]{2}\sqrt{5}}{5}$ s

(C) $\frac{2\sqrt[3]{4}\sqrt{5}}{5}$ s (D) $\frac{2\sqrt{10}}{5}$ s

15. A charge of $4\,\mu$C moves from potential 70 V to -70 V in an electrostatic field. Find the work done by an external agent if the charge starts and ends at rest and no other force does work.

(A) $-8\,\mu$J (B) $-560\,\mu$J

(C) $0\,\mu$J (D) $280\,\mu$J

16. Unpolarised light passes three ideal polarisers with axes at 0°, 30°, and 105° relative to a common reference. Find final intensity divided by incident intensity.

(A) $\frac{3}{8}$ (B) $\frac{3}{16} - \frac{3\sqrt{3}}{32}$

(C) $-\frac{\sqrt{6}}{16} + \frac{3\sqrt{2}}{16}$ (D) $\frac{1}{4} - \frac{\sqrt{3}}{8}$

17. A series RLC circuit has $R = 60$ ohms, $X_L = 60$ ohms, and $X_C = 50$ ohms at the source frequency. An ideal source supplies 400 V rms. Find rms current.

(A) $\frac{40\sqrt{37}}{37}$ A (B) 40 A

(C) $\frac{20\sqrt{74}}{37}$ A (D) $\frac{40\sqrt{97}}{97}$ A

18. A thin glass lens of refractive index 3/2 has focal length 60 cm in air. It is completely immersed in a liquid of index 4/3. Find its new focal length.

(A) 240 cm (B) 80 cm

(C) 180 cm (D) 15 cm

19. A fixed amount of ideal gas has its absolute temperature multiplied by 4 and volume by 2. Find the final pressure divided by the initial pressure.

(A) 4 (B) 1

(C) $\frac{1}{4}$ (D) 2

20. A vertical uniform cylinder of mass 6 kg and cross-sectional area $\frac{1}{50}\ \text{m}^2$ floats in a liquid of density $600\ \text{kg/m}^3$. It is pushed down slightly and released. Its ends remain on opposite sides of the free surface throughout. Find the period of vertical oscillation. Take $g = 10\ \text{m/s}^2$ and neglect damping.

(A) $\frac{\sqrt{10}\pi}{10}$ s (B) $\frac{\sqrt{10}\pi}{5}$ s

(C) $\frac{\sqrt{5}\pi}{5}$ s (D) $4\sqrt{5}\pi$ s

21. A spherical planet rotates uniformly. Its rotation reduces effective gravity at the equator by 19 m/s^2. By how much does rotation reduce the radial component of effective gravity at latitude 60°?

(A) $\frac{19}{2}\ \text{m/s}^2$ (B) $\frac{57}{4}\ \text{m/s}^2$

(C) $19\ \text{m/s}^2$ (D) $\frac{19}{4}\ \text{m/s}^2$

22. Two rods of equal length and cross-sectional area have thermal conductivities k and 9k. They are joined end-to-end with no contact resistance. Find the effective conductivity of the composite divided by k.

(A) $\frac{9}{10}$ (B) 1

(C) 9 (D) $\frac{9}{5}$

23. A screw gauge advances 2.0 mm in four complete turns and has 50 circular divisions. With the jaws closed, its zero mark lies 4 divisions above the reference line; use the usual convention that a zero below the reference line gives positive zero error. With a wire, the main scale reads 6 mm and circular reading is 11 divisions. What is the corrected diameter?

(A) $\frac{1693}{100}$ mm (B) $\frac{1011}{100}$ mm

(C) $\frac{123}{20}$ mm (D) $\frac{313}{50}$ mm

24. A dipole of fixed moment 6 C m is rotated quasistatically from 60° to 120° relative to a uniform electric field 2 N/C. Find external work.

(A) $12\sqrt{3}\,\mathrm{J}$ (B) $12\,\mathrm{J}$

(C) $6\,\mathrm{J}$ (D) $4\pi\,\mathrm{J}$

25. A long solenoid has 400 turns per metre and carries 2 A. It is completely filled by a linear magnetic core of relative permeability 2. Find magnetisation M.

(A) $800\,\mathrm{A/m}$ (B) $1600\,\mathrm{A/m}$

(C) $400\,\mathrm{A/m}$ (D) $2400\,\mathrm{A/m}$

26. A triangular p-V cycle has vertices (V_0, p_0), $(V_0 + 5\,\mathrm{L}, p_0)$ and $(V_0, p_0 + 7\,\mathrm{kPa})$. Traversed in that listed order and then back, find net work done by the gas.

(A) $\frac{-35}{2}$ J (B) $-\frac{35}{4}$ J

(C) $-35\,\mathrm{J}$ (D) $0\,\mathrm{J}$

27. The same metal is illuminated successively at wavelengths λ and $\lambda/2$. Its maximum photoelectron kinetic energies are 6 eV and 18 eV, respectively. What is the ratio of threshold wavelength to λ?

(A) $\frac{1}{3}$ (B) 2

(C) $\frac{1}{2}$ (D) $\frac{2}{3}$

28. A rigid ladder of length 15 m slides with its ends on a floor and a vertical wall. At the instant its top is at height $4L/5$, the foot moves away at 6 m/s. What is the downward speed of the top?

(A) $\frac{9}{2}\,\mathrm{m/s}$ (B) $\frac{24}{5}\,\mathrm{m/s}$

(C) $\frac{18}{5}\,\mathrm{m/s}$ (D) $8\,\mathrm{m/s}$

29. A Newtonian liquid of viscosity $\frac{1}{2}$ Pa s has velocity profile $v(y) = 6y + 1y^2$ in SI units. Find the shear-stress magnitude at $y = 2$ m.

(A) $10\,\mathrm{Pa}$ (B) $8\,\mathrm{Pa}$

(C) $5\,\mathrm{Pa}$ (D) $4\,\mathrm{Pa}$

30. Here L is inductance, C capacitance, R resistance, V potential, I current and t time. Which expression has the same dimensions as both $(V/I)^3$ and $(L/(RC))^3$?

(A) $(CV)^3$ (B) $(LI)^3$

(C) $(RC)^3$ (D) $(\sqrt{L/C})^3$

Chemistry

31. For isolated gaseous species,consider $S^- + e^- \rightarrow S^{2-}$. Which judgement also remains consistent with the Cl/F electron-gain anomaly?

(A) The stated step is endothermic; first electron gain by F is more exothermic than by Cl because F is smaller.

(B) The stated step is endothermic; first electron gain by Cl is more exothermic than by F.

(C) Every first and second electron gain has the same sign and magnitude for a group.

(D) The stated step is exothermic; every electron addition releases energy.

32. For2A+B→3C+4D,the measured total disappearance rate of A is $\frac{2}{125}$M/s. A separate calibrated decay channel consumes A at $\frac{2}{625}$M/s and produces no D. D appearance rate from the stated reaction,in M/s,is

(A) $\frac{16}{625}$ (B) $\frac{32}{625}$

(C) $\frac{4}{625}$ (D) $\frac{4}{125}$

33. A cation selected from Cu2+,Zn2+,Fe3+ gives a white hydroxide precipitate when aqueous ammonia is added dropwise, followed by a colourless ammine-complex solution on adding excess ammonia. Which identification and oxidation-state conclusion is consistent?

(A) Cu2+; colourless excess-ammonia solutions are its normal confirmation.

(B) Zn2+; the metal oxidation state remains +2.

(C) Zn2+; ammonia necessarily reduces the metal to zero.

(D) Fe3+; its hydroxide always dissolves in excess ammonia.

34. Consider I: NH3 has a higher boiling point than PH3. II: BCl3 is an electron-pair acceptor. Which assessment correctly recognises both anomalous first-element behaviours?

(A) Both statements are correct. (B) Neither statement is correct.

(C) Only II is correct. (D) Only I is correct.

35. One mole of a disaccharide gives two moles of glucose and no other monosaccharide on complete hydrolysis. Which identity and water requirement fit?

(A) Fructose; one mole of water. (B) Lactose; no water.

(C) Maltose; one mole of water per mole hydrolysed. (D) Sucrose; two moles of water.

36. For a reaction,$\log_{10} K_p$ is 2 at300 K and $\frac{16}{3}$ at600 K. The plot is linear in1/T. Using2.303forln10,find the positive magnitude$|\Delta H^\circ|/R$ in K.

(A) 2303 (B) 2000

(C) 4606 (D) 9212

37. A first-order reaction has already consumed 25%of its initial A. Additional time needed to leave12.5% of the original A,expressed in half-lives,is ($\log_2 3$=1.585)

(A) $\frac{83}{200}$ (B) 3

(C) $\frac{517}{200}$ (D) 4

38. A diprotic acid H_2A is0.100M;its first dissociation is complete and$Ka2=10^{-5}$. Added HCl is $\frac{2}{5}$M in the final solution. Neglect the second-step contribution to H^+. Equilibrium$[A^{2-}]$in M is

(A) $\frac{1}{100000}$ (B) $\frac{1}{500000}$

(C) $\frac{1}{20000}$ (D) $\frac{1}{400000}$

39. An ideal gas executes A→B→C→D→A through (V/L,P/bar):A(1,1),B(4,1),C(4,2),D(1,2). IUPAC work-on convention applies. Heats on AB,BC,DA are 600,300,-200 J. Heat on CD in J is (1 L bar=100 J)

(A) −400 (B) −300

(C) −700 (D) −1000

40. Match two applications: I. a low-temperature coolant for superconducting equipment. II. an inert atmosphere for welding reactive hot metals. Which ordered gas pair is appropriate?

(A) neon; argon. (B) helium; argon.

(C) helium; xenon. (D) argon; helium.

41. For an ordinary diprotic acid with Ka1>Ka2,compare its conjugate bases HA^- and A^{2-} at one temperature. Which joint order is consistent withKw=KaKb for each conjugate pair?

(A) Both conjugate bases haveKb=Kw/(Ka1Ka2). (B) Ka1>Ka2,but$Kb(A^{2-})>Kb(HA^-)$.

(C) Ka1=Ka2 because both protons belong to one molecule. (D) Ka1>Ka2 thereforeKb$(HA^-)>Kb(A^{2-})$.

42. For planar cyclopentadienyl anion,choose the correct classification and the decisive reason. Use the explicitly stated geometry.

(A) antiaromatic; every cyclic compound obeys 4n+2

(B) nonaromatic; an ion can never be aromatic

(C) aromatic; five p orbitals are continuous and the anionic lone pair supplies two electrons

(D) aromatic; count only the neutral double bonds and ignore charge or geometry

43. For$2A(g)\rightleftharpoons A_2(g)$,the numerical K_c is1.00 at a T where RT=4 L bar/mol. Find numerical K_p for$1/2\ A_2(g)\rightleftharpoons A(g)$,using matching concentration and pressure conventions.

(A) 4 (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) 2

44. A $\frac{1}{25}$ mol FeC_2O_4 inventory is analysed. A fraction $\frac{1}{4}$ of Fe^{2+} has already become Fe^{3+},but every oxalate remains unchanged. Subsequent complete oxidation gives Fe^{3+} and CO_2. Electron equivalents released,in mol,are

(A) $\frac{7}{100}$ (B) $\frac{19}{100}$

(C) $\frac{3}{25}$ (D) $\frac{11}{100}$

45. 4-nitrobenzaldehyde is first treated with NaBH4 under ordinary aldehyde-selective conditions and worked up,then its nitro group is completely reduced with Fe/HCl and the product basified. Which final product follows both reductions?

(A) 4-aminobenzyl alcohol
(B) 4-aminobenzaldehyde
(C) 4-nitrobenzyl alcohol
(D) 3-aminobenzyl alcohol

46. Under the usual qualitative group trends,compare the ease of alkali-metal electron loss in Li/Na and the oxidising tendency of F_2/Cl_2. Which conclusion is consistent?

(A) Na loses its electron more readily; Cl_2 is the stronger halogen oxidant.

(B) Li loses its electron more readily; Cl_2 is the stronger halogen oxidant.

(C) Na loses its outer electron more readily; F_2 is the stronger halogen oxidant.

(D) Both trends must increase down a group because atoms become larger.

47. Excess water reacts completely with $\frac{1}{100}$ mol CaH_2. Which gas identity/amount pair follows from the balanced hydrolysis?

(A) H_2, $\frac{1}{100}$ mol.
(B) O_2, $\frac{1}{200}$ mol.
(C) CO_2, $\frac{1}{100}$ mol.
(D) H_2, $\frac{1}{50}$ mol.

48. In a pressure range where the Freundlich law $x/m = kp^{1/4}$ is valid at fixed temperature, adsorbent B has 4 times the mass of A and is exposed to 256 times its pressure. Find the ratio of total adsorbed amounts x_B/x_A.

(A) 16
(B) 4
(C) 64
(D) $\frac{1}{16}$

49. An orbital has 2 angular nodes and 1 radial nodes. Maximum electrons in its entire subshell are

(A) 6
(B) 10
(C) 32
(D) 14

50. Compare 1-bromopropane and neopentyl bromide toward the same cyanide nucleophile in a polar aprotic solvent under ordinary SN2 conditions. Which reacts faster or is the substituted site,and why?

(A) neopentyl bromide; steric hindrance accelerates backside attack

(B) 1-bromopropane; the mechanism requires formation of a free tertiary carbocation

(C) 1-bromopropane; Neopentyl bromide is formally primary but beta branching severely obstructs the SN2 transition state.

(D) neopentyl bromide; an sp2 C-Br bond behaves just like an unhindered methyl bromide

51. p-cresol undergoes Reimer-Tiemann formylation with CHCl3/aqueous NaOH and acidic work-up. Use ordinary ortho preference where an ortho site is available. Which major formylated product follows the unoccupied OH-directed sites?

(A) 2-hydroxy-5-methylbenzaldehyde
(B) 3-hydroxy-4-methylbenzaldehyde
(C) the corresponding benzoic acid formed by replacement of OH
(D) the product obtained by replacing a methyl carbon with CHO

52. Which choice combines a biodegradable polyester with a chemically sound reason for its degradability?

(A) PHBV; hydrolysable ester groups occur in its chain.

(B) PTFE; its C-F bonds are ester linkages.

(C) Polythene; every C-C chain is rapidly hydrolysed by water.

(D) PVC; the C-Cl group alone makes every chain a peptide.

53. Bond energies $D(X_2):D(Y_2):D(XY)=2:3:4$. Forming2.00 mol XY(g) from the gaseous elements releases 240 kJ. $D(Y_2)$,in kJ/mol,is

(A) 320

(B) 240

(C) 480

(D) 120.0

54. Compare dilute solutions of 2-hydroxyacetophenone and 4-hydroxyacetophenone in a non-hydrogen-bonding solvent. Which conclusion joins structure with the expected effect of further dilution on their hydrogen bonding?

(A) Only 4-hydroxyacetophenone can make an intramolecular H bond because para groups form a shorter ring.

(B) Both lose all H bonding equally because every H bond requires two molecules.

(C) 2-hydroxyacetophenone must have stronger intermolecular association because its internal acceptor is farther away.

(D) 2-hydroxyacetophenone can retain an intramolecular H bond on dilution; intermolecular association of 4-hydroxyacetophenone is reduced.

55. For $MgCO_3$, $CaCO_3$, $BaCO_3$,which increasing thermal-stability order and explanation is correct?

(A) $MgCO_3 > CaCO_3 > BaCO_3$; larger cations distort carbonate more strongly.

(B) $MgCO_3 < CaCO_3 < BaCO_3$; the cations become smaller down the group.

(C) All are equally stable because the cation charge is+2.

(D) $MgCO_3 < CaCO_3 < BaCO_3$; increasing cation size lowers carbonate-ion polarisation.

56. Compare pH of A:HCl at$\frac{1}{500}$M;B:HA,Ka=10^{-6},C=$\frac{1}{5}$M;C:HB,Ka=10^{-5},C=$\frac{1}{500}$M. Neglect water and use weak-ionisation ordering. Increasing pH is

(A) C<B<A.

(B) A<B<C.

(C) A<C<B.

(D) B<A<C.

57. Which systematic name correctly describes the condensed connectivity HOOC-CH(Br)-CH2-COOCH3,including principal-group priority and numbering?

(A) methyl 3-bromo-4-hydroxy-4-oxobutanoate

(B) 3-bromo-4-methoxy-4-oxobutanoic acid

(C) 2-bromo-4-methoxybutanedioic acid

(D) 2-bromo-4-methoxy-4-oxobutanoic acid

58. A saturated acyclic monocarbonyl compound X has formula C4H8O and gives iodoform but not Tollens. A separate sample is reduced with NaBH4 and protonated. Which alcohol is produced?

(A) butan-2-ol

(B) butanal

(C) butan-1-ol

(D) 2-methylpropan-1-ol

59. 2-bromo-2-methylpentane is heated with sodium ethoxide under elimination-favouring conditions. Which major constitutional product is expected,despite the presence of an alkoxide that could be used in a Williamson synthesis with a suitable different halide?

(A) the corresponding tertiary alkoxide substitution product
(B) 2-methylpent-1-ene
(C) 2-methylpent-2-ene
(D) the alkane formed by replacing halogen with H

60. Which reagent/name pair is suitable when a primary alkyl chloride must become an iodide without changing its skeleton? The target carbon framework must be retained.

(A) NaI in dry acetone;Finkelstein reaction
(B) Na/dry ether;Sandmeyer reaction
(C) AgCN;Swarts reaction
(D) CuCl/HCl;Wurtz reaction

English Proficiency

61. Choose the active-voice equivalent that retains the tense, participants and all stated restrictions. The revised list was sent to the visitors by the secretary, but the old list was not.

(A) The visitors sent the revised list to the secretary, but not the old list.

(B) The secretary sent the revised list to the visitors, but not the old list.

(C) The secretary sends the revised list to the visitors, but not the old list.

(D) The secretary sent both lists to the visitors.

62. Choose the best replacement for the bracketed part: The role involves checking records, answering enquiries and [to prepare summaries].

(A) preparation summaries
(B) summaries to prepare
(C) preparing summaries
(D) prepared summaries

63. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. until those meanings were made explicit. Q. Since the two accounts used R. different meanings of the same term, S. their apparent disagreement could not be assessed

(A) S-Q-R-P
(B) R-Q-S-P
(C) P-R-S-Q
(D) Q-R-S-P

64. The action is legal, meaning permitted by law. Choose the opposite of legal in this sense.

(A) legalistic
(B) illegal
(C) legislative
(D) legitimate

65. Read the sentence: The encouraging results should temper the team's pessimism, without making it overconfident. Choose the closest meaning of 'temper' as used in the sentence.

(A) remove completely
(B) conceal
(C) moderate
(D) justify

66. Choose the correctly spelt word.

(A) forty
(B) fortey
(C) fourthy
(D) fourty

67. The cover may safely be cleaned in water rather than being discarded after use. Which adjective fits?

(A) absorbent (B) disposable

(C) waterproof (D) washable

68. Read the sentence: Although three proposals were rejected, the tenacious volunteer continued searching for a workable plan. Choose the closest meaning of 'tenacious' as used in the sentence.

(A) persistent (B) argumentative

(C) easily discouraged (D) fortunate

69. Complete: When the latch was released, the gate swung _____ enough for the cart to pass.

(A) wide (B) widen

(C) widely (D) width

70. Read the sentence: The committee was receptive to new suggestions and invited further proposals. Choose the opposite in meaning of 'receptive' as used in the sentence.

(A) open-minded (B) unresponsive

(C) attentive (D) interested

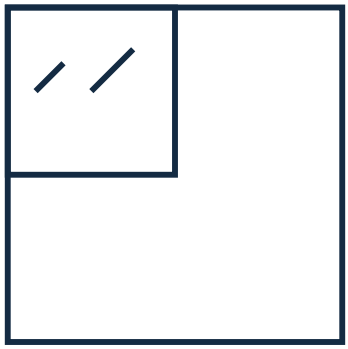
Logical Reasoning

71. A walker goes 7 m north, then 12 m west, then turns right and walks 6 m. In which direction is the starting point from the walker?

(A) North-west (B) South-west

(C) North-east (D) South-east

72. Starting with a unit square sheet, fold right to left along the vertical midline, then fold bottom to top along the horizontal midline. The folded paper is cut along both short drawn slits. Select the pattern after complete unfolding.



(A) Pattern P



(B) Pattern Z



(C) Pattern T



(D) Pattern Y



73. Consider only these statements; universal statements alone do not imply existence. All bakers are guides; Some bakers are not joggers; Some bakers are guides. Conclusions: I. Some guides are not joggers. II. All guides are joggers. Which conclusions follow necessarily?

(A) Both I and II

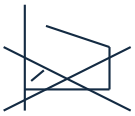
(B) Only I

(C) Only II

(D) Neither I nor II

74. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





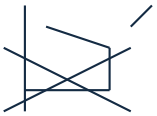
host 1



host 2



host 3



host 4

(A) 4 (B) 2

(C) 1 (D) 3

75. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 11$ and $5 \mapsto 26$. Which input maps to 56?

(A) 12 (B) 56

(C) 10 (D) 11

76. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

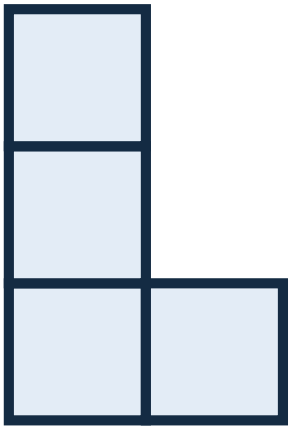


piece 1



piece 2

(A) Pattern T



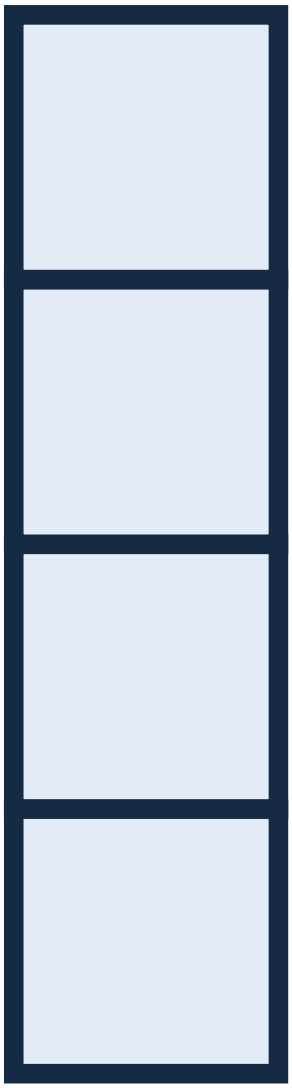
(B) Pattern Q



(C) Pattern K



(D) Pattern W



77. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: ADNK, CGPN, ?, GMTT, IPVW, KSXZ.

(A) FKSR (B) QRJE

(C) DIQP (D) EJRQ

78. $X@Y$ means X is the mother of Y; $X\#Y$ means X is the brother of Y; $X\%Y$ means X is the father of Y. All names denote distinct people; a brother or sister is a full sibling, sharing both parents. Given $L@A$ $A\#P$ $P\%E$, how is L related to E?

(A) Mother (B) Paternal grandmother

(C) Father (D) Maternal grandfather

79. For this question compare only cyclic forward alphabet gaps, with $Z \to A$ counting as one step. Three clusters share the same gap vector. Which numbered cluster differs? 1: WZBE, 2: NQRU, 3: VYZC, 4: CFGJ.

(A) 4 (B) 2

(C) 1 (D) 3

80. A code shifts letters cyclically through A–Z. Odd positions share one fixed shift and even positions share another. For example, CLOUD is coded as EIQRF. Which word is coded as OXPDQ?

(A) MANGO (B) QURAS

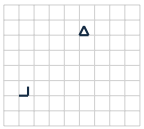
(C) NBOHP (D) LZMFN

81. C and A are full siblings; A and U are also full siblings. T is the mother of C. All four people are distinct. How is T related to U?

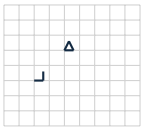
(A) Grandmother (B) Aunt

(C) Mother (D) Relationship cannot be determined

82. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



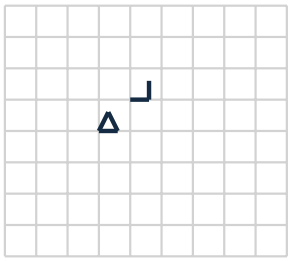
1



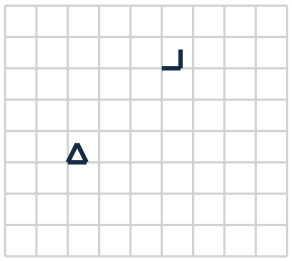
2

3

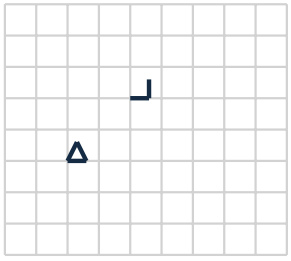
(A) Pattern L



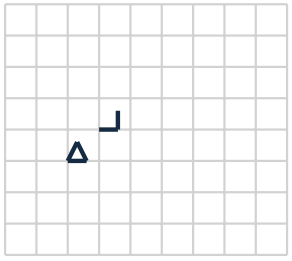
(B) Pattern M



(C) Pattern N



(D) Pattern K

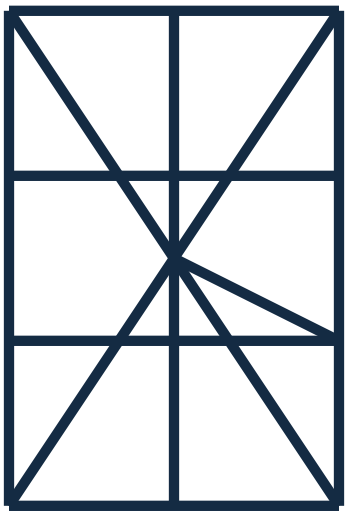


83. In this series the first differences themselves form an arithmetic progression. Find the missing internal term: 9, 13, 21, ?, 49, 69, 93.

(A) 37 (B) 33

(C) 34 (D) 29

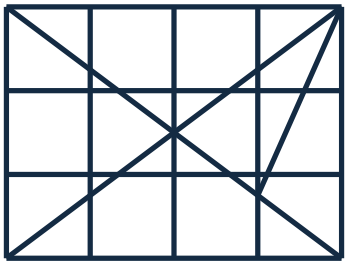
84. How many triangles are formed by all the drawn segments?



(A) 52 (B) 27

(C) 21 (D) 31

85. How many triangles are formed by all the drawn segments?



(A) 53
(B) 57
(C) 56
(D) 98

86. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

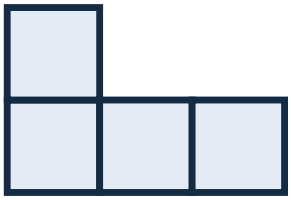


piece 1

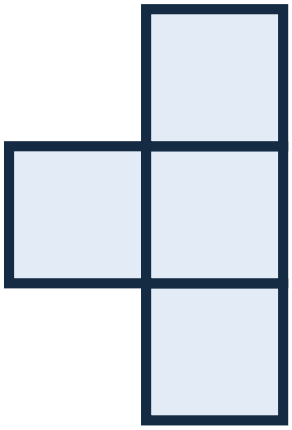


piece 2

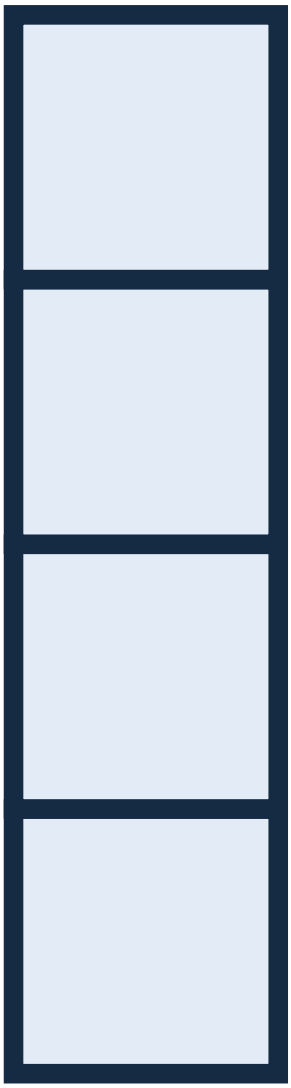
(A) Pattern R



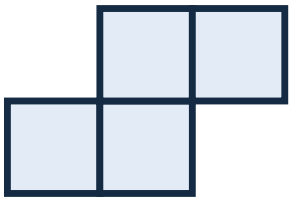
(B) Pattern X



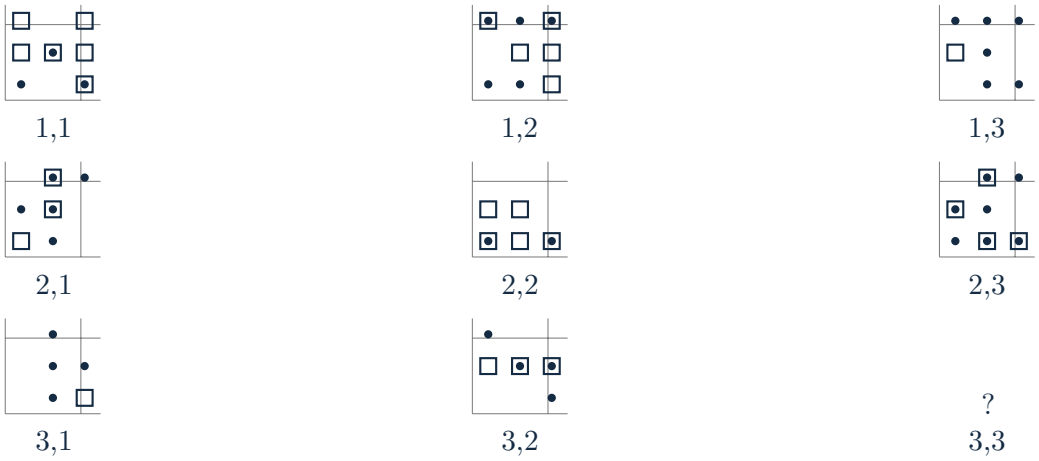
(C) Pattern K



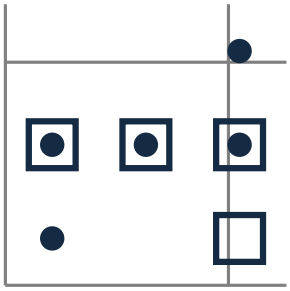
(D) Pattern V



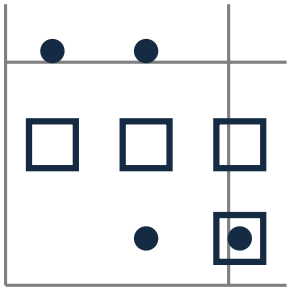
87. In each row, superimpose the first two cells to obtain the third. Two identical marks in the same position cancel; a dot and a hollow square do not cancel each other. Choose the missing cell.



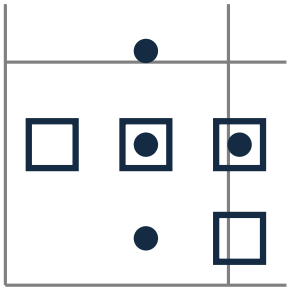
(A) Pattern Y



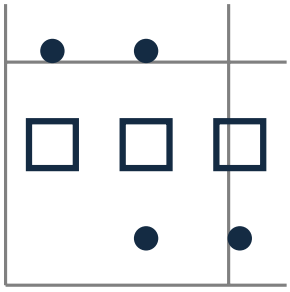
(B) Pattern Z



(C) Pattern W



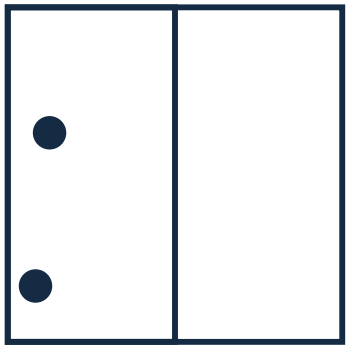
(D) Pattern R



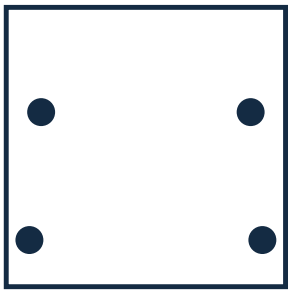
88. A community register uses the following three categories: inventors, campers, editors. Its complete relevant assertions are in this short paragraph; do not import everyday assumptions. Some campers are editors. No inventors are campers. Some editors are not campers. Conclusions: I. Some campers are not editors. II. No campers are inventors. Which conclusions follow necessarily?

(A) Only II
(B) Neither I nor II
(C) Only I
(D) Both I and II

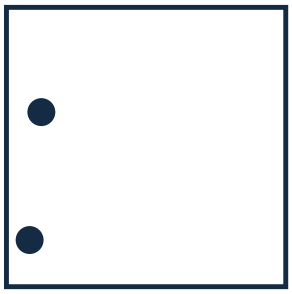
89. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



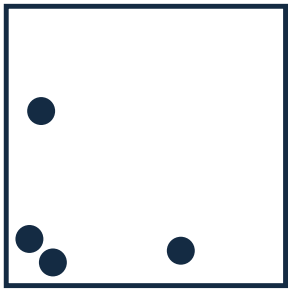
(A) Pattern W



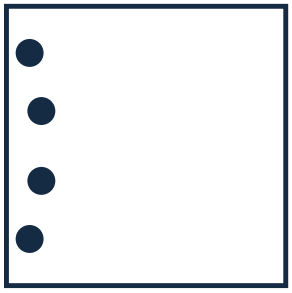
(B) Pattern Z



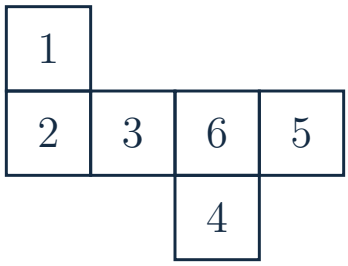
(C) Pattern T



(D) Pattern P



90. Fold the shown net into a cube. Give (label opposite 1, label opposite 2) in that order.



(A) $(4, 1)$
(B) $(4, 2)$
(C) $(4, 3)$
(D) $(4, 6)$

Mathematics

91. For f(x)=x^3-3kx^2+27x, how many integer k in [0,5] give a local maximum at a positive x?

(A) 5 (B) 3

(C) 1 (D) 2

92. $a_j = 7 + 4(j-1)$ and f(x)=(8/2)x^2+bx+4. If f(1)=f(-1), find f′(a_7).

(A) 31 (B) 248

(C) 252 (D) 280

93. For integer n, consider: "If 24 divides n, then 4 divides n." Which is its contrapositive?

(A) If 24 does not divide n, then 4 does not divide n. (B) If 4 divides n, then 24 divides n.

(C) 24 divides n and 4 does not divide n. (D) If 4 does not divide n, then 24 does not divide n.

94. A=(6,5,4), O=(0,0,0). Find the squared length of the orthogonal projection of OA on the line through O with direction (1,2,2).

(A) 13 (B) 576

(C) 8 (D) 64

95. For a=(1,5,0), b=(0,1,6), c=(4,0,1), evaluate (a×c)·(a+b).

(A) 0 (B) −120

(C) −121 (D) 121

96. In a group of 20, 12 study A, 10 study B, and 2 study neither. How many study exactly one of A,B?

(A) 4 (B) 18

(C) 22 (D) 14

97. f(x)=5x+7. Find f(f(7)).

(A) 189 (B) 42

(C) 217 (D) 224

98. How many integers k make $x^2 + kx + 16 = 0$ have two distinct non-real roots?

(A) 15 (B) 7

(C) 17 (D) 16

99. Let f(x)=sin x+3 cos x. Let [L,L+π] be the interval containing 0 in its interior on which f is increasing and bijective onto[-sqrt(10),sqrt(10)]. If g is its inverse, findL+g′(1).

(A) $-\frac{\pi}{2} - \operatorname{atan}(3) - \frac{1}{3}$

(B) $-\frac{\pi}{2} - \operatorname{atan}(3) + \frac{1}{3}$

(C) $-\frac{\pi}{2} - \operatorname{atan}(3) + \frac{\sqrt{10}}{10}$

(D) $-\operatorname{atan}(3) + \frac{1}{3} + \frac{\pi}{2}$

100. Positive integers $a < b$ satisfy $a + b = 22$ and $3 \mid a$. Find the maximum value of ab.

(A) 22 (B) 117

(C) 121 (D) 96

101. In triangle ABC, A=2B, a/b=$\sqrt{3}$ and b=9. Find its area.

(A) $\frac{81\sqrt{3}}{2}$ (B) $81\sqrt{3}$

(C) 18 (D) $\frac{81}{2}$

102. F′(x)=$\frac{1}{x^2} - \frac{1}{x^2\sqrt{x^2+1}}$ and F(1)=3. Find F(2)-2F(1).

(A) $-\frac{5}{2} - \sqrt{2} + \frac{\sqrt{5}}{2}$ (B) $-\sqrt{2} + \frac{1}{2} + \frac{\sqrt{5}}{2}$

(C) $-\frac{7}{2} - \frac{\sqrt{5}}{2} + \sqrt{2}$ (D) $-\sqrt{2} + \frac{\sqrt{5}}{2} + \frac{7}{2}$

103. If α,β are roots of $x^2 - 2kx + k^2 - 36 = 0$ and k∈[-3,3], find the maximum of $\alpha^2+\beta^2$.

(A) 72 (B) −36

(C) 90 (D) 81

104. $A = \begin{bmatrix} 7 & 1 \\ 1 & 3 \end{bmatrix}$ and A^{-1}=pI+qA. Find p+q.

(A) 180 (B) $-\frac{9}{20}$

(C) $\frac{9}{20}$ (D) $\frac{1}{20}$

105. Under x=X+2, y=Y+3, the parabola $(y-3)^2=36(x-2)$ is normalized. Find the x coordinate of its focus in the original axes.

(A) 11 (B) 9

(C) 38 (D) −7

106. Evaluate the limit of $[\cos(4x)-\cos(2x)]/x^2$ as x→0.

(A) −12 (B) −1

(C) −6 (D) 6

107. From a point at horizontal distance 4 from a tower, its elevation is 2θ, where $\tan\theta=\frac{9}{13}$ and $0<\theta<45°$. Find the height.

(A) $\frac{36}{13}$ (B) $\frac{176}{117}$

(C) $\frac{72}{13}$ (D) $\frac{117}{11}$

108. A polynomial f of degree at most 2 satisfies f′(x)+f′(-x)=6, f″(x)=4, and f(0)=2. Find f(2).

(A) 14 (B) 13

(C) 16 (D) 12

109. Circles $x^2+y^2=64$ and $(x-6)^2+y^2=64$ meet at A,B. Find $|AB|^2$.

(A) 256 (B) 220

(C) 55 (D) 184

110. $\log_2 a=3$, $\log_a b=6$, $\log_b c=5$. Find $\log_4(c^2/a)$.

(A) 90 (B) $\frac{177}{2}$

(C) 177 (D) $\frac{183}{2}$

111. A=(4,0,5), B=(0,2,1), AP:PB=2:1 internally. Find $|AP|^2$.

(A) 4 (B) 24

(C) 16 (D) 36

112. If $|z-13|=7$, find the minimum of $|z|^2$.

(A) 400 (B) 120

(C) 36 (D) 6

113. The chord of contact from P to $x^2/144+y^2/36=1$ is $x/12+y/6=1$. Find $|OP|^2$.

(A) $6\sqrt{5}$ (B) 180

(C) 18 (D) 108

114. Let $z=(1+i)^9$. Find $\mathrm{Re}(z)+\mathrm{Im}(z)$.

(A) 32 (B) -32

(C) $16\sqrt{2}$ (D) 16

115. Group A has mean 50, SD 5. Every observation is increased by 50. Find its new coefficient of variation in percent.

(A) 20 (B) 10

(C) 5 (D) 25

116. Evaluate $\sum_{j=7}^{9} \binom{j}{1}$, with the convention $\binom{j}{r} = 0$ when j<r.

(A) 45 (B) 30

(C) 17 (D) 24

117. Find the sum of order and degree of $(d^6y/dx^6)^2+(dy/dx)^2+y=0$.

(A) 12 (B) 10

(C) 2 (D) 8

118. X takes values [-4, 0, 3] with probabilities 1/6,1/3,1/2. Let Y=4X+7. Find $E(Y^2)-[E(Y)]^2$.

(A) $\frac{932}{9}$ (B) $\frac{233}{9}$

(C) $\frac{344}{3}$ (D) $\frac{1373}{9}$

119. Let A,B be the intersections of x=4 with $x^2/4-y^2/9=1$. Find the area of triangle OAB.

(A) $24\sqrt{3}$ (B) $12\sqrt{3}$

(C) 12 (D) $3\sqrt{3}$

120. Evaluate $\sum_{j=1}^{\infty} 4j(\frac{1}{5})^j$.

(A) 5 (B) 1

(C) $\frac{5}{9}$ (D) $\frac{5}{4}$

121. $A = \begin{bmatrix} 0 & 5 & -6 \\ -5 & 0 & 5 \\ 6 & -5 & 0 \end{bmatrix}$. Find det(I+A)-(25+36+25).

(A) 1 (B) -1

(C) 86 (D) 0

122. A circular sector has radius 8 and arc length 2π. Find its area.

(A) $\frac{\pi^2}{4}$ (B) 8

(C) 16π (D) 8π

123. Let $y'=(x+y)/(x+y+1)$, with $y(0)=1$. Find the positive value of x at whichx+y=4.

(A) $\frac{\log(3)}{4}+\frac{3}{2}$ (B) $\frac{5}{2}-\frac{\log(3)}{4}$

(C) $\frac{\log(3)}{2}+\frac{3}{2}$ (D) $\frac{\log(3)}{2}+3$

124. A box contains 7 red and 7 blue distinct balls. Three are drawn uniformly without replacement. Given that at least one drawn ball is blue, find the probability that exactly two are red.

(A) $\frac{21}{5}$ (B) $\frac{21}{47}$

(C) $\frac{26}{47}$ (D) $\frac{21}{52}$

125. Evaluate $\int$ from 0 to $\pi/2$ of $[\sin^2x/(\sin^2x+\cos^2x)+\sin^2(\pi/2-x)/(\sin^2(\pi/2-x)+\cos^2(\pi/2-x))]x\,dx$.

(A) $\frac{\pi^2}{4}$ (B) $\frac{\pi^2}{8}$

(C) $\frac{\pi}{4}$ (D) $\frac{\pi^2}{16}$

126. Find the coefficient of x^6 in $(1+x+x^2)(1+x^2)^7$.

(A) 56 (B) 70

(C) 35 (D) 105

127. Evaluate $\int_0^{\infty} dx/[(x^2+36)(x^2+64)]$.

(A) $\frac{\pi}{1344}$ (B) $-\frac{\pi}{1344}$

(C) $\frac{\pi}{96}$ (D) $\frac{\pi}{672}$

128. A parallelogram has opposite vertices (3,4) and (7,10), and another vertex (0,2). Find the sum of coordinates of the fourth vertex.

(A) 26 (B) 24

(C) 22 (D) 12

129. For x∈[π/3,π/2], find the maximum of sin x/x.

(A) $\frac{\sqrt{3}}{2}$ (B) 1

(C) $\frac{2\sqrt{3}\pi}{9}$ (D) $\frac{3\sqrt{3}}{2\pi}$

130. Find the area enclosed by y=x(10-x) and y=5x.

(A) $\frac{875}{6}$ (B) $\frac{125}{6}$

(C) $-\frac{125}{6}$ (D) $\frac{500}{3}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A particle has position-dependent velocity components $\vec{v} = 4x\,\hat{i} - 4y\,\hat{j}$ in SI units. Given $dx/dt = v_x$ and $dy/dt = v_y$, what is its acceleration magnitude at $(3, 5)$ m?

(A) $128\,\mathrm{m/s^2}$ (B) $16\sqrt{34}\,\mathrm{m/s^2}$

(C) $4\sqrt{34}\,\mathrm{m/s^2}$ (D) $48\,\mathrm{m/s^2}$

132. A soap bubble initially of radius 4 cm is expanded quasistatically until its volume is 8 times the initial volume. Surface tension remains $\frac{1}{50}$ N/m. Find work against surface tension, in units of $\pi \times 10^{-4}$ J.

(A) $\frac{448}{25}$ (B) $\frac{64}{25}$

(C) $\frac{192}{25}$ (D) $\frac{256}{25}$

133. A source emitting 600 Hz and a listener approach each other, each at 50 m/s relative to still air. Sound speed is 300 m/s. Find heard frequency before they pass.

(A) 840 Hz (B) 720 Hz

(C) $\frac{3000}{7}$ Hz (D) 700 Hz

Chemistry

134. At fixed T,a reaction has K=25 and currently Q=25. Which joint conclusion is correct for this equilibrium mixture?

(A) $\Delta_r G^\circ > 0$ because K exceeds1.

(B) $\Delta_r G = 0$ while$\Delta_r G^\circ < 0$;equilibrium does not require K=1.

(C) Both$\Delta_r G$ and$\Delta_r G^\circ$ must be zero because the reaction is reversible.

(D) $\Delta_r G < 0$ because its standard value is negative,even when Q=K.

135. Under specified conditions, ethanol vapour over heated copper gives ethanal, while over heated alumina it gives ethene. Which conclusion follows about catalytic selectivity and equilibrium?

(A) Catalyst identity can favour different reaction pathways; a catalyst does not alter the equilibrium constant of a specified reaction.

(B) The two outcomes show that activation barriers are irrelevant.

(C) Every catalyst must produce the same product because equilibrium constants are unchanged.

(D) The catalyst changes the equilibrium constant of each specified reaction.

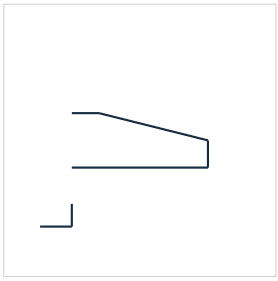
136. benzotrichloride,C6H5CCl3 is subjected to aqueous hydrolysis. Which product or product pair follows the named reaction and side-chain oxidation state?

(A) phenol (B) benzyl alcohol

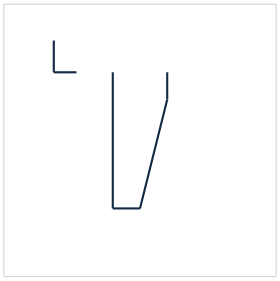
(C) benzoic acid (D) benzaldehyde

Logical Reasoning

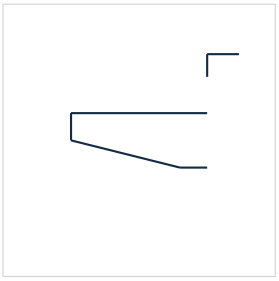
137. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

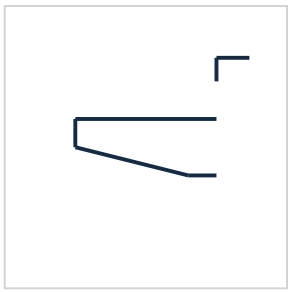


2

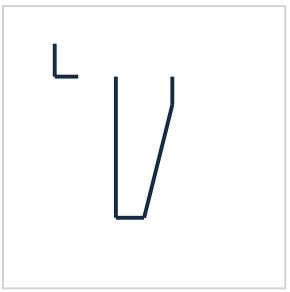


3

(A) Pattern P



(B) Pattern R



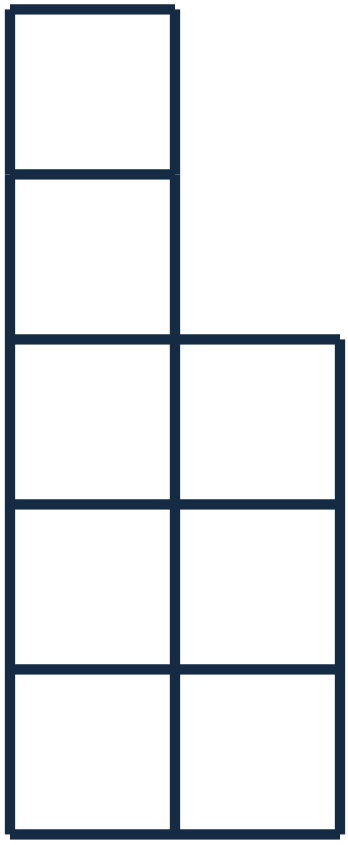
(C) Pattern Z



(D) Pattern V



138. How many rectangles, including squares are formed by all the drawn segments?



(A) 28 (B) 27

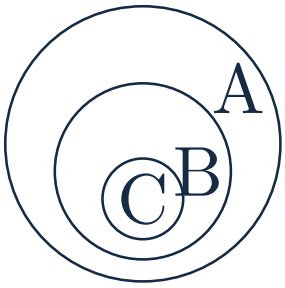
(C) 22 (D) 26

139. Choose the diagram that represents all these conditions, including the strictness information: Some A is B, some A is not B, and some B is not A. No C is A or B; C is nonempty.

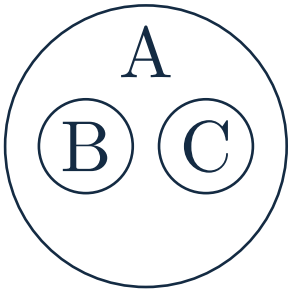
(A) Pattern W



(B) Pattern V



(C) Pattern Z



(D) Pattern R



Mathematics

140. In $(1+x)^{14}$, find the ratio of the coefficient of x^9 to the coefficient of x^8.

(A) $\frac{2}{3}$ (B) $\frac{7}{8}$

(C) $\frac{3}{2}$ (D) 6

141. Find the squared distance between 6x+8y=14 and 3x+4y=10.

(A) 9 (B) $\frac{9}{100}$

(C) $\frac{9}{25}$ (D) $\frac{3}{5}$

142. An observer at height 2 sees a tower foot at depression 30° and top at elevation 60°. Find the tower height.

(A) $2\sqrt{3}$ (B) 8

(C) 4 (D) 6

BITSAT Mock 07

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A liquid of volume 2 litres and bulk modulus 4000000000 Pa is subjected to an additional pressure of 3000000 Pa. Neglect changes in bulk modulus. Find the magnitude of volume decrease in cubic centimetres.

(A) $1500\,\text{cm}^3$ (B) $\frac{3}{2000}\,\text{cm}^3$

(C) $\frac{8000000}{3}\,\text{cm}^3$ (D) $\frac{3}{2}\,\text{cm}^3$

2. For a particle moving in one dimension, $v = 7 - 2x$ in SI units while $v > 0$. What is its acceleration at $x = \frac{7}{4}$ m?

(A) $-14\,\text{m/s}^2$ (B) $-7\,\text{m/s}^2$

(C) $-2\,\text{m/s}^2$ (D) $\frac{7}{2}\,\text{m/s}^2$

3. A uniform block of density 300 kg/m^3 floats in a liquid. Then the liquid density is reduced to $\frac{4}{5}$ times 1000 kg/m^3, while the block's volume and density are unchanged. What fraction of the block is now submerged?

(A) $\frac{3}{8}$ (B) $\frac{6}{25}$

(C) $\frac{5}{8}$ (D) $\frac{3}{10}$

4. A pure radioactive sample has activity A at time 4 hours after preparation and $A/8$ at time 10 hours after preparation. At what time, measured from preparation, will its activity be one sixteenth of the activity measured at 4 hours? Assume exponential decay with a constant decay rate.

(A) 18 h (B) 10 h

(C) 12 h (D) 8 h

5. A coil has 20 turns, area $\frac{1}{25}$ m^2, and rotates at 8 rad/s in a uniform 3 T field. At an instant the angle between its area vector and the field is 30°. Find instantaneous-emf magnitude.

(A) $\frac{48\sqrt{3}}{5}$ V (B) $\frac{48}{5}$ V

(C) $\frac{96}{5}$ V (D) $\frac{6}{5}$ V

6. A stretched string is divided by a bridge into lengths in the ratio 1 : 6. Tension and linear density are unchanged. The original fundamental is 70 Hz. Find the fundamental of the longer segment.

(A) 490 Hz (B) 70 Hz

(C) $\frac{245}{3}$ Hz (D) 98 Hz

7. In YDSE wavelength is 600 nm, screen distance 3 m and slit separation 2 mm. Find the separation on the same side of the centre between the 3rd bright fringe and the 4th dark fringe. Count the first dark fringe as order 1.

(A) $\frac{9}{20}$ mm (B) $\frac{27}{10}$ mm

(C) $\frac{27}{20}$ mm (D) $\frac{63}{20}$ mm

8. A balanced Wheatstone bridge has arms AB=AD=2 ohms and BC=DC=10 ohms. A resistor of arbitrary positive resistance connects B to D. Find equivalent resistance between A and C.

(A) 11 Ω (B) 12 Ω

(C) $\frac{10}{3}$ Ω (D) 6 Ω

9. A nonrelativistic particle of fixed charge and mass moves perpendicular to a uniform magnetic field. Its kinetic energy is multiplied by 9 and the magnetic field is multiplied by 5. Find the new trajectory radius divided by the original radius.

(A) $\frac{3}{5}$ (B) 3

(C) $\frac{9}{5}$ (D) $\frac{1}{15}$

10. A converging thin lens forms a real image with magnification magnitude 5. The object-image separation is 30 cm. Find focal length.

(A) $\frac{15}{2}$ cm (B) $\frac{75}{8}$ cm

(C) 5 cm (D) $\frac{25}{6}$ cm

11. A nonrelativistic charged particle initially has momentum $4qEt$ along a uniform electric field, where $t > 0$ is a fixed reference interval. Its charge is $+q$. What is $\lambda(6t)/\lambda(2t)$ after acceleration in the same field?

(A) $\frac{9}{25}$ (B) $\frac{2}{5}$

(C) $\frac{3}{5}$ (D) $\frac{\sqrt{15}}{5}$

12. A silicon diode and a 6 ohm resistor are in series across 16 V with the diode anode toward the positive terminal. In the stated piecewise model the diode is open in reverse bias and has constant 0.7 V drop in forward conduction. Find the current.

(A) $\frac{51}{40}$ A (B) $\frac{51}{20}$ A

(C) $\frac{167}{60}$ A (D) $\frac{73}{30}$ A

13. Light of vacuum wavelength 800 nm passes from a medium of index 2 into one of index 3. Find the ratio of its frequencies in the second and first media.

(A) $\frac{2}{3}$ (B) 1

(C) $\frac{1}{3}$ (D) $\frac{9}{4}$

14. A one-electron ion with nuclear charge $Z = 1$ absorbs a photon of energy $\frac{52}{5}$ eV. The ejected electron has maximum kinetic energy 7 eV. From which principal quantum number was the electron removed? Use $E_n = -13.6Z^2/n^2$ eV and neglect recoil.

(A) 1 (B) 4

(C) 3 (D) 2

15. A rigid diatomic ideal gas with vibrations frozen expands at constant pressure. It does 140 J work. Find the increase in internal energy and select its value.

(A) 490 J (B) 350 J

(C) 210 J (D) 56 J

16. A Carnot engine initially runs between 800 K and 400 K. Its hot reservoir is raised by 20 K. By how much must the cold-reservoir temperature rise to keep efficiency unchanged?

(A) 20 K (B) $\frac{400}{41}$ K

(C) 10 K (D) 40 K

17. A uniform-density planet has radius R. Gravity at a depth d equals gravity at a height $(5-1)R$ above the surface. Find d/R.

(A) $\frac{24}{25}$ (B) $\frac{124}{125}$

(C) $\frac{1}{25}$ (D) 1

18. A 7 ohm resistor is connected in parallel with an unknown resistor across 14 V. The source supplies 35 W in total. Find the unknown resistance.

(A) $\frac{63}{5}\,\Omega$ (B) $28\,\Omega$

(C) $\frac{28}{5}\,\Omega$ (D) $\frac{196}{33}\,\Omega$

19. In a series RLC circuit, ideal rms voltmeters read 30 V across R, 70 V across L and 10 V across C. Find source rms voltage.

(A) $10\sqrt{59}$ V (B) $10\sqrt{73}$ V

(C) 110 V (D) $30\sqrt{5}$ V

20. An ideal mixture contains 4 moles of monatomic gas and 2 moles of rigid diatomic gas. Vibrations are frozen. Find the heat-capacity ratio $\gamma = C_P/C_V$ of the mixture.

(A) $\frac{17}{11}$ (B) $\frac{7}{5}$

(C) $\frac{11}{17}$ (D) $\frac{5}{3}$

21. A uniform massless spring of constant 40 N/m is cut into 5 equal parts. All parts are connected in parallel to support a mass 5 kg. Find its small-oscillation period.

(A) $\frac{\sqrt{2}}{20}$ s (B) $\frac{\sqrt{10}\pi}{2}$ s

(C) $\frac{\sqrt{2}\pi}{10}$ s (D) $\frac{\sqrt{2}\pi}{2}$ s

22. Two uniform thin rods meet at a right angle at the origin. Both have length 8 m. The rod along +x has linear density 3 times that of the rod along +y. A force parallel to the y-axis must produce pure translation. At what x-coordinate must its line of action lie?

(A) 3 m (B) 1 m

(C) 12 m (D) 4 m

23. Two masses 1 kg and 3 kg are connected by a light string over a smooth pulley and accelerate freely. A transverse pulse travels in the taut straight part of this string, of linear density μ. Compare its speed to that in a second string of density 3μ held at tension g N numerically equal to 1 kg weight. Find v_1/v_2.

(A) $\frac{3\sqrt{2}}{2}$ (B) $\frac{\sqrt{6}}{2}$

(C) $\frac{\sqrt{2}}{3}$ (D) $\frac{\sqrt{2}}{2}$

24. A rod of length 2 m pivots at its lower end. A peg constrained to move horizontally stays in contact with its upper tip. At an instant the rod makes an angle with $\sin\theta = 3/5$ to the horizontal and the tip's horizontal speed has magnitude 2 m/s. Find the rod's angular-speed magnitude.

(A) 1 rad/s (B) $\frac{4}{5}$ rad/s

(C) $\frac{5}{4}$ rad/s (D) $\frac{5}{3}$ rad/s

25. A 5 kg block is pulled by a force of 25 N at an angle with $\cos\theta = 4/5$, $\sin\theta = 3/5$ above a horizontal floor. The block slides; $\mu_k = \frac{3}{10}$. Find the net horizontal force. Take $g = 10$ m/s^2.

(A) $\frac{61}{2}$ N (B) 20 N

(C) 6 N (D) $\frac{19}{2}$ N

26. A 3 kg block moves along a horizontal rough surface from $x = 0$ to $x = 3$ m under a force $F(x) = 2x$ N parallel to motion. With $\mu_k = 1/5$ and $g = 10$ m/s^2, find the work done by the net force. The block has enough initial speed to keep moving forward.

(A) 9 J (B) 27 J

(C) -9 J (D) -3 J

27. A vacuum electromagnetic wave of average intensity 40 W/m^2 passes through a device that reduces its electric-field amplitude by a factor 4. The transmitted wave falls normally on an area 7 cm^2. What average power reaches the area?

(A) $\frac{56}{125}$ W (B) $\frac{7}{4000}$ W

(C) $\frac{7}{1000}$ W (D) $\frac{5}{2}$ W

28. A particle has potential $U(x, y) = 2x^2y + 5y$ J in SI units. Find the force magnitude at $(2, 3)$ m.

(A) $\sqrt{313}$ N (B) 39 N

(C) $\sqrt{745}$ N (D) 37 N

29. A uniformly charged nonconducting sphere has radius R and surface potential V_s relative to infinity. Find potential at $r = R/15$ divided by V_s.

(A) $\frac{337}{225}$ (B) $\frac{1}{15}$

(C) $\frac{3}{2}$ (D) 15

30. The quantity $Z = x^3y^1/\sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 1, 4 and 6, respectively. To first order, what is the maximum percentage error in Z?

(A) 7 (B) 10

(C) 11 (D) $\sqrt{34}$

Chemistry

31. In a regime where attraction is negligible,Z=1+pb/(RT). The Z-versus-p slope for gasA at300K is3times that for gasB at600K. Ratio bA/bB is

(A) $\frac{1}{3}$ (B) 6

(C) $\frac{3}{2}$ (D) 3

32. On controlled heating,$\frac{1}{50}$ mol $KMnO_4$ follows $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$. Which pair ($O_2$ amount,origin of the intense purple colour of the starting d^0 ion) is correct?

(A) ($\frac{1}{100}$ mol,charge-transfer absorption). (B) ($\frac{1}{50}$ mol,a d-d transition of MnVII).

(C) ($\frac{1}{200}$ mol,charge-transfer absorption). (D) ($\frac{1}{100}$ mol,a d-d transition of MnVII).

33. For oxide formation, both normalised per mole of O_2, $\Delta G_A^\circ = -300 + 0.10T$ and $\Delta G_B^\circ = -360 + 0.20T$ in kJ. Ignoring kinetic limitations, at what temperatures is B metal a favourable standard-state reducing agent for oxide A?

(A) T < 600 K (B) T > 600 K

(C) At all positive temperatures (D) Only T = 600 K

34. A Fischer projection of tartaric acid has its four-carbon chain vertical (COOH at top and COOH at bottom). At C2,OH is on the left; at C3,OH is on the left; each other horizontal group is H. Horizontal bonds point toward the observer. Which classification and configuration pair is correct?

(A) chiral; (2S,3S) (B) chiral; (2R,3R)

(C) chiral; (2S,3R) (D) meso; (2S,3R)

35. A chlorofluorocarbon is stable enough to reach the stratosphere, where ultraviolet light can release chlorine radicals. Which pairing correctly connects the material to its principal textbook atmospheric concern?

(A) CFC; direct neutralisation of every acid rain droplet. (B) CFC; compulsory production of soil nitrate.

(C) CFC; catalytic ozone depletion. (D) CFC; prevention of all radical reactions.

36. Co_3O_4 is represented as a mixed oxide of CoO and Co_2O_3. Which formal metal oxidation-state ratio satisfies the formula,with ordinary oxide oxygen -2?

(A) All metal atoms must be formally +8/3,with no mixed-valence bookkeeping possible.

(B) 2+:3+ = 1:2.

(C) 2+:3+ = 2:1.

(D) 2+:3+ = 1:1.

37. For CaH_2,which classification and consequence are mutually consistent?

(A) Covalent; nonstoichiometry proves all hydrogen has been removed.

(B) ionic; two hydridic H atoms give two H_2 per formula unit on hydrolysis.

(C) Ionic; every example must be a gas composed of discrete molecules.

(D) All hydrides contain free H+ ions in their solid lattices.

38. Compare the elementary octahedral d-d splitting transitions of A=$[CrF_6]^{3-}$,B=$[Cr(H_2O)_6]^{3+}$, C=$[Cr(NH_3)_6]^{3+}$. For this comparison use only the usual spectrochemical ordering and $E = hc/\lambda$. Which ordering of absorbed wavelengths follows?

(A) $\lambda_A < \lambda_B < \lambda_C$.

(B) $\lambda_A > \lambda_B > \lambda_C$.

(C) $\lambda_A = \lambda_B = \lambda_C$ because metal oxidation states match.

(D) $\lambda_B > \lambda_C > \lambda_A$.

39. For propan-2-ol and propan-1-ol,which test/observation assignment follows the actual carbinol class or oxidisable structural feature? Use ordinary room-temperature tests unless the named sequence specifies otherwise.

(A) iodoform test;only propan-2-ol is positive

(B) iodoform test;only propan-1-ol is positive

(C) FeCl3;only propan-2-ol gives violet colour

(D) Lucas test;both give immediate turbidity

40. Separate unknowns P and Q are each selected from Cu2+, Zn2+, Fe3+. Dropwise aqueous ammonia gives P a pale blue hydroxide precipitate and Q a reddish brown precipitate. Excess ammonia gives P a deep-blue ammine solution and Q an undissolved reddish-brown hydroxide. Which identification is consistent, without invoking redox?

(A) P=Cu2+; Q=Fe3+; oxidation states unchanged.

(B) P=Fe3+; Q=Cu2+; oxidation states unchanged.

(C) P=Cu2+; Q=Fe3+; both metals reduced to zero.

(D) Both are the same ion because all hydroxides have identical ammonia behaviour.

41. For electrophilic mononitration of benzene carrying CHO,which orientation/reason pair is correct? Consider intrinsic electronic directing effects,not an exact ortho:para ratio.

(A) meta; the carbonyl group withdraws by resonance and induction

(B) meta; every electronegative substituent activates the ring

(C) ortho/para; the substitution site is decided only by steric size

(D) ortho/para; the carbonyl group withdraws by resonance and induction

42. For a completely filled n=3 shell,how many electrons have$m_l = 0$ and$m_s = +1/2$?

(A) 3

(B) 18

(C) 9

(D) 6

43. A $\frac{25}{2}$ g $CaCO_3$/inert mixture is80.0% $CaCO_3$ by mass. It receives $\frac{3}{25}$ mol HCl. For $CaCO_3$+2HCl→$CaCl_2$+CO_2+H_2O,90.0% of formed CO_2 is collected. Collected CO_2 amount in mol is

(A) $\frac{3}{50}$

(B) $\frac{3}{25}$

(C) $\frac{27}{500}$

(D) $\frac{1}{10}$

44. A rubber is a copolymer of buta-1,3-diene and styrene. Which identification avoids confusing the aromatic comonomer with a nitrile?

(A) Buna-S.

(B) Nylon-6.

(C) Neoprene.

(D) Buna-N.

45. Two hydrogen-like bound states A and B have nuclear charges ZA=1 and ZB=3, and radii rB=3rA. The states obey the allowed integer-principal-quantum-number Bohr model. What is the ratio of binding-energy magnitudes |EB|/|EA|?

(A) 2 (B) 1/2

(C) 1 (D) 3

46. A cobalt(III) chloride ammine contains six NH_3/Cl ligands in an octahedral sphere. Each mole of salt immediately gives 1 mol AgCl with excess $AgNO_3$. Its molar concentration of freely moving ions relative to formula units,under complete dissociation with no ligand exchange,is

(A) 3 (B) 1

(C) 4 (D) 2

47. At fixedT,liquidXhas saturationpressure0.100bar andRT=20.0Lbar/mol. A1.00L vessel contains$\frac{1}{100}$molXtotal,initially withliquidpresent. Accessible volume is raised to3.00L withoutXloss. Neglect liquidvolume. FinalXpressure in bar is

(A) $\frac{1}{10}$ (B) $\frac{1}{15}$

(C) $\frac{1}{5}$ (D) $\frac{1}{30}$

48. An isolated cathode compartment contains $\frac{3}{25}$mol HA and $\frac{3}{25}$mol NaA,pKa5.00,in a supported NaCl electrolyte. Current5.00A flows for 1158s. Each electron generates one OH^-;no anode product enters, buffer loss or volume change occurs. F=96500C/mol andlog3=.477. FinalpH is

(A) 5 (B) $\frac{4523}{1000}$

(C) 13 (D) $\frac{5477}{1000}$

49. cyclohexene is treated with chloroform,CHCl3 and strong base under dihalocarbene-generation conditions. Which joint statement correctly identifies the haloalkane step and the hydrocarbon-skeleton change?

(A) A free methyl cation adds; the original C=C remains intact.

(B) :CCl2 adds by replacing a ring carbon,so total carbon count is unchanged.

(C) Beta elimination generates an alkene from the haloform; the ring size is unchanged.

(D) Alpha elimination generates :CCl2; the alkene gains one carbon and one additional ring.

50. A primary-standard sample of pure anhydrous Na_2CO_3 weighs $\frac{53}{250}$ g. It is titrated with0.100M HCl. Find the additional acid volume in mL needed after the first carbonate-to-hydrogencarbonate equivalence point to reach complete conversion to CO_2/water. Take Na_2CO_3=106g/mol.

(A) 10 (B) 40

(C) 41 (D) 20

51. Atoms form ideal touching FCC spheres of radius r and cell edge a. Fractional coordinates are measured in a. Using the straight displacement between the two specified points (0,0,0) and (1/2,1/2,0), without replacing it by another periodic image, find $(\text{distance}/r)^2$. These points describe atom centre to nearest atom centre.

(A) 5 (B) 4

(C) 2 (D) 8

52. A student assigns two bonding pairs and three lone pairs to the central atom of I_3^-. After independently checking valence-electron count,which combined molecular shape/domain count is consistent?

(A) Trigonal planar; five electron domains.
(B) Linear; two electron domains.
(C) Linear; five electron domains.
(D) Bent; four electron domains.

53. 2-nitropropane is treated with complete reduction,then free-base work-up. Which free organic product follows the nitro-group transformation and retained C-N connectivity?

(A) propan-1-amine
(B) propanone
(C) propan-2-amine
(D) trimethylamine

54. diethyl pimelate,EtOOC-(CH2)5-COOEt undergoes NaOEt/EtOH,then acidification only. Under the stated intramolecular closure pathway,which isolated product matches both ring size and work-up endpoint?

(A) cyclohexanone
(B) cyclopentanone
(C) ethyl 2-oxocyclopentane-1-carboxylate
(D) ethyl 2-oxocyclohexane-1-carboxylate

55. A two-step profile has energies in kJ/mol:reactants0,intermediate20,products-10,first transition state40,second transition state70. With comparable prefactors,which pair(step with larger forward barrier, highest reverse barrier measured from products)is correct?

(A) Step1;80kJ/mol.
(B) Step2;80kJ/mol.
(C) Step2;70kJ/mol.
(D) Step1;40kJ/mol.

56. A weak monoprotic acid at C=$\frac{9}{10000}$M has measured degree of dissociation $\frac{1}{10}$. Use the exact concentration relationKa=Cα^2/(1-α). At the same T,analytical concentration required for degree $\frac{1}{20}$ is

(A) $\frac{9}{5000}$
(B) $\frac{19}{5000}$
(C) $\frac{9}{2500}$
(D) $\frac{9}{40000}$

57. At 250 K,combustion enthalpies of C(graphite) and CO(g) to CO_2(g) are respectively-393.5 and-283.0 kJ/mol. Treat gases ideally and use R=0.00800 kJ mol^{-1} K^{-1}. Constant-volume heat for C(graphite)+1/2 O_2(g)→CO(g),per mole CO,in kJ is

(A) $-\frac{223}{2}$
(B) $-\frac{219}{2}$
(C) $-\frac{225}{2}$
(D) $-\frac{221}{2}$

58. A mixture contains $\frac{1}{20}$ mol H_2SO_5 and $\frac{1}{25}$ mol $H_2S_2O_8$. Their structures contain one O–O link per molecule. All peroxide links react quantitatively with excess iodide in acid; each such link accepts two electrons. What amount of I_2, in mol, is liberated?

(A) $\frac{7}{50}$
(B) $\frac{13}{100}$
(C) $\frac{9}{100}$
(D) $\frac{9}{50}$

59. In the trigonal-bipyramidal electron-domain model of ClF_3,there are 2 central lone pairs. Which placement rule explains the observed arrangement,including the first lone pair?

(A) Place the first lone pair axially because three 90-degree contacts are less repulsive than two.
(B) Use arbitrary sites because lone-pair/bond-pair and bond-pair/bond-pair repulsions are identical.
(C) Place lone pairs only where they form the smallest possible angle to every bond.
(D) Place lone pairs equatorially first: an equatorial site has two 90-degree neighbours,whereas an axial site has three.

60. Match two applications: I. a low-temperature coolant for superconducting equipment. II. an intense flash lamp. Which ordered gas pair is appropriate?

(A) helium; helium.
(B) helium; xenon.
(C) neon; xenon.
(D) xenon; helium.

English Proficiency

61. Choose the most coherent order of the four sentences. A. Researchers looking for a particular song had to open many unrelated entries. B. The archive stored recordings under the names of the performers. C. The archivist added a second index arranged by song title. D. The same collection could then answer a different kind of search without being physically rearranged.

(A) A-B-C-D
(B) B-A-D-C
(C) B-C-A-D
(D) B-A-C-D

62. The organisers had to _____ the demonstration until the replacement part arrived.

(A) put out
(B) put down
(C) put off
(D) put through

63. Complete the question tag: The two machines have never started together, _____?

(A) did they
(B) have they
(C) haven't they
(D) has it

64. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. in their labelled folders, Q. As long as the originals remain R. the staff can trace each scan S. back to its source.

(A) P-Q-R-S
(B) R-Q-P-S
(C) Q-P-R-S
(D) S-P-R-Q

65. Read the sentence: Keeping a reserve was a sensible precaution given the uncertain delivery date. Choose the closest meaning of 'sensible' as used in the sentence.

(A) practical and reasonable
(B) based entirely on physical sensation
(C) easily affected emotionally
(D) readily noticeable

66. The editor receives the author's unpublished text for consideration. Which word names the submitted work?

(A) bibliography
(B) edition
(C) review
(D) manuscript

67. The speaker states that she believes no deity exists, rather than merely saying that the answer is unknown. What is she?

(A) atheist
(B) agnostic
(C) traditionalist
(D) theist

68. Which figure of speech is chiefly illustrated by the line? I have waited an eternity for this short meeting to begin.

(A) personification (B) hyperbole

(C) simile (D) apostrophe

69. Read the sentence: The article's diffuse argument wandered through several unrelated points. Choose the opposite in meaning of 'diffuse' as used in the sentence.

(A) tightly focused (B) extended

(C) uncertain (D) wide-ranging

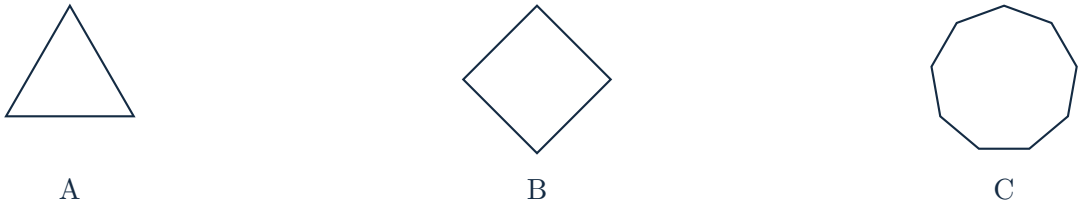
70. Use the standard contemporary negative adjective. The reply lacked ordinary courtesy and was therefore _____.

(A) inpolite (B) impolite

(C) dispolite (D) unpolite

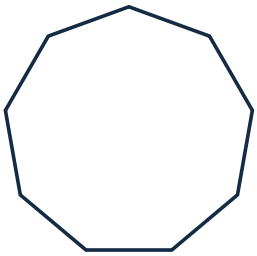
Logical Reasoning

71. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?

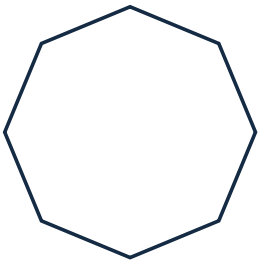


A B C

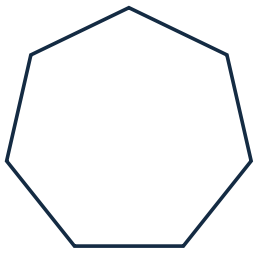
(A) Pattern V



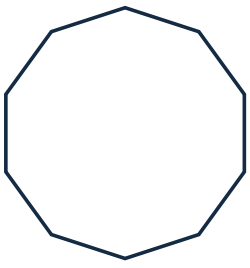
(B) Pattern Q



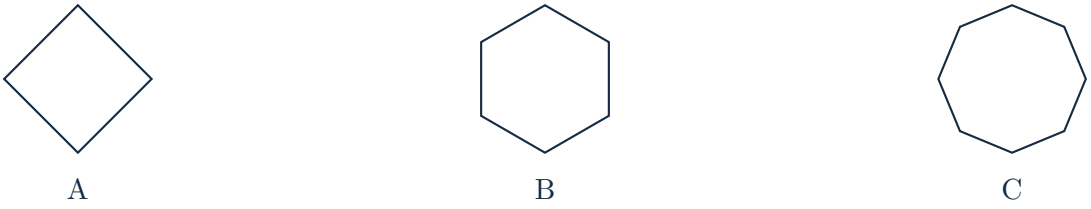
(C) Pattern R



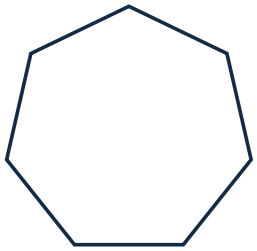
(D) Pattern L



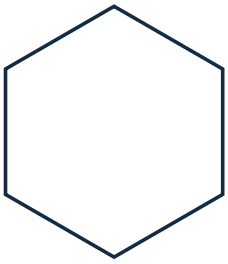
72. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



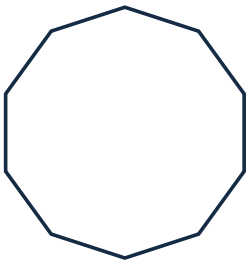
(A) Pattern Q



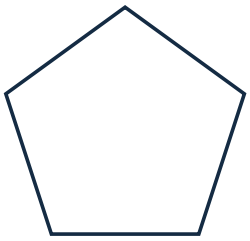
(B) Pattern M



(C) Pattern R



(D) Pattern V



73. The model triads are $(8, 21, 13)$ and $(10, 25, 15)$. In the intended rule the middle number is the sum of the outer numbers, and the outer-number difference is the same fixed positive value in every triad. Which option satisfies both relations?

(A) $(7, 20, 12)$ (B) $(7, 19, 12)$

(C) $(7, 20, 13)$ (D) $(7, 21, 13)$

74. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



(A) Pattern P



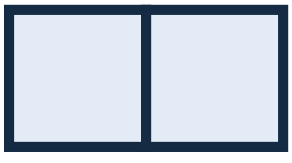
(B) Pattern X



(C) Pattern R



(D) Pattern Z



75. The recorded average of 12 readings is 40. One entry recorded as 56 should have been 51. After that correction, another reading of 24 is replaced by 34. What is the final average?

(A) $\frac{475}{12}$ (B) $\frac{497}{12}$

(C) $\frac{485}{12}$ (D) $\frac{245}{6}$

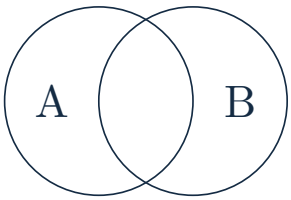
76. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: YNK, AON, CPQ, ?, GRW, ISZ.

(A) TQE (B) FRU

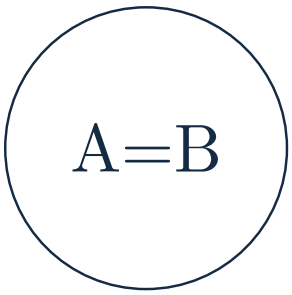
(C) DPS (D) EQT

77. Choose the diagram that represents all these conditions, including the strictness information: Both A and B are nonempty, and no A is B.

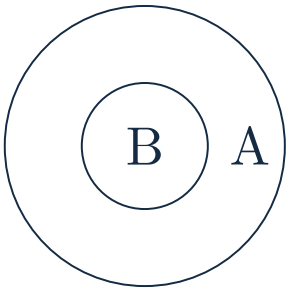
(A) Pattern T



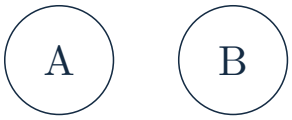
(B) Pattern N



(C) Pattern M



(D) Pattern L

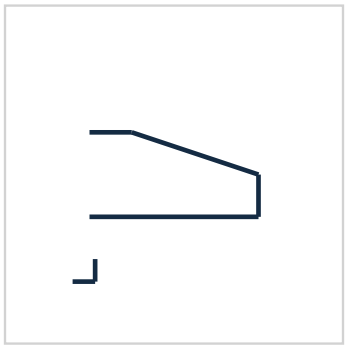


78. 6 people sit around a circle facing its centre. Clockwise order fixes the convention; immediate left is clockwise. Bina is 1 places clockwise from Asha. Dev is 2 places clockwise from Asha. Charu is 5 places clockwise from Asha. Esha is not adjacent to Charu. Who sits immediately clockwise from Esha?

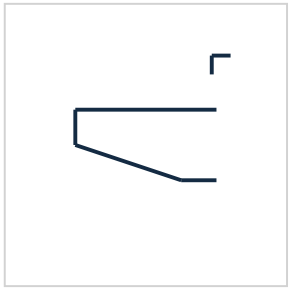
(A) Bina (B) Farah

(C) Asha (D) Charu

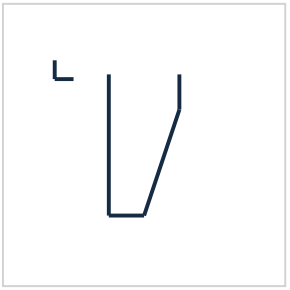
79. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



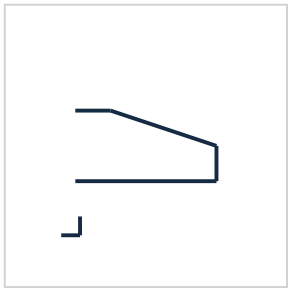
(A) Pattern N



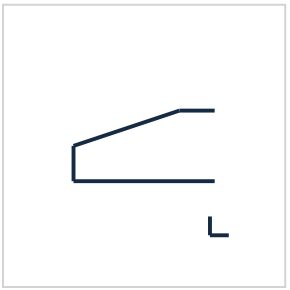
(B) Pattern V



(C) Pattern L



(D) Pattern Z



80. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 2

(B) 3

(C) 1

(D) 4

81. The increase from the left cell to the right cell is the same in each row, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.





?

(A) Pattern Z

(B) Pattern N

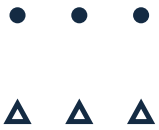




(C) Pattern Q

(D) Pattern W





82. Preserve the exact relation and its order: Genuine : Authentic :: ? : Commence. Choose the missing term.

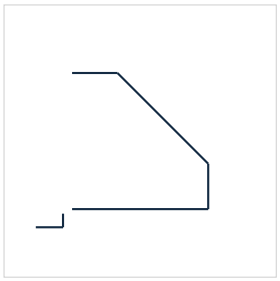
(A) Courageous

(B) Rapid

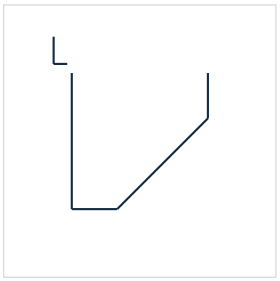
(C) Begin

(D) Silent

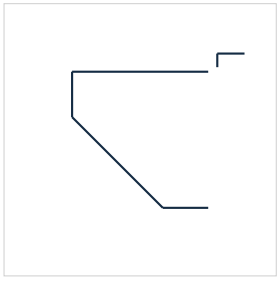
83. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right



bottom left

?

bottom right

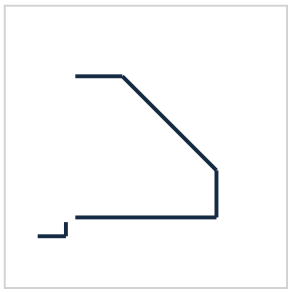
(A) Pattern N



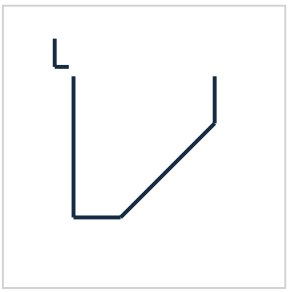
(B) Pattern Q



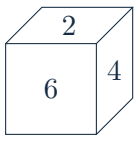
(C) Pattern K



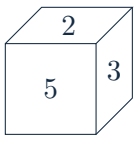
(D) Pattern M



84. The views show the same die. Give the ordered pair: (face opposite 2, face opposite 6). Do not assume opposite labels sum to seven.



1



2

(A) $(1, 5)$

(B) $(1, 3)$

(C) $(1, 1)$

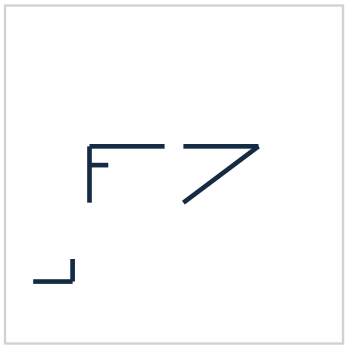
(D) $(1, 2)$

85. Two people differ in age by 4 years. The product of their ages 2 years ago was 285. What will the older person's age be 2 years from now?

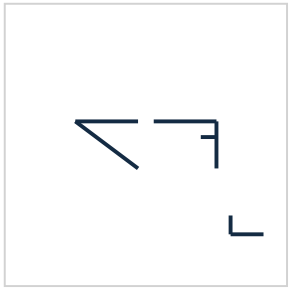
(A) 21 (B) 27

(C) 23 (D) 19

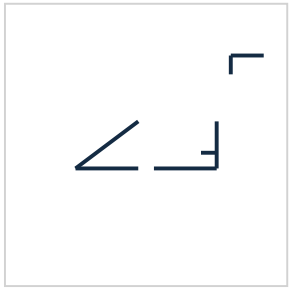
86. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



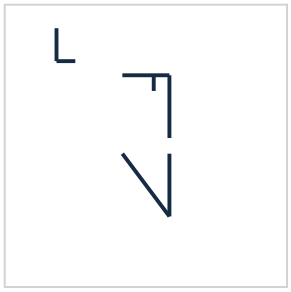
(A) Pattern M



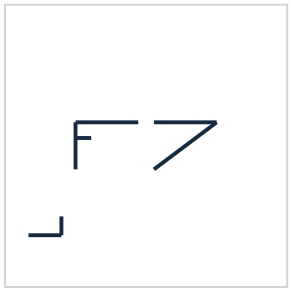
(B) Pattern K



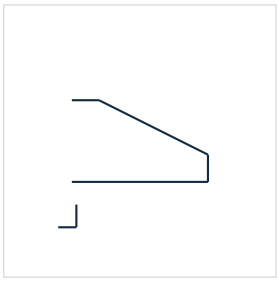
(C) Pattern P



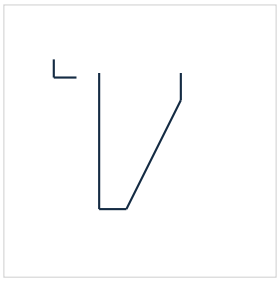
(D) Pattern V



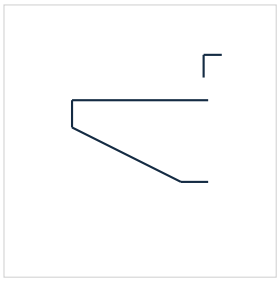
87. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

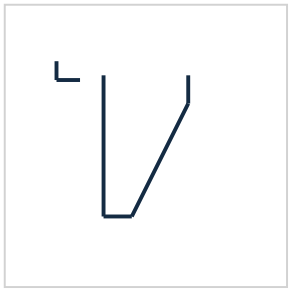


2

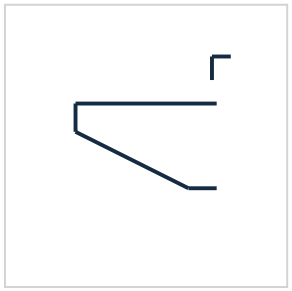


3

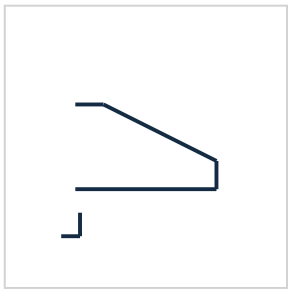
(A) Pattern Q



(B) Pattern N



(C) Pattern K



(D) Pattern T



88. Consider only these statements; universal statements alone do not imply existence. Some guides are guides; All inventors are bakers. Conclusions: I. All guides are inventors. II. Some guides are not inventors. Which conclusion or complementary alternative must hold?

(A) Only I

(B) Only II

(C) Either I or II follows, but neither is individually certain

(D) Both I and II

89. In a row, Asha is 9th from the left and 21st from the right. Bina is 10th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 7

(B) 6

(C) 1

(D) 9

90. A word is coded as the sum of its forward alphabet positions plus 5 times its length. The code of PLAN? is 88. Which letter replaces the question mark?

(A) T (B) S

(C) U (D) Y

Mathematics

91. A={j∈{1,...,24}:sin(jπ/2)=0}, B={j∈{1,...,24}:sin(jπ/3)=0}. Find |(A∪B)\(A∩B)|.

(A) 12 (B) 20

(C) 4 (D) 16

92. f(x)=3x+1 and f(g(x))=3x+19. Find g(g(3)).

(A) 18 (B) 21

(C) 9 (D) 15

93. The line y=mx through O bisects the area of the triangle with vertices O,(5,3),(3,3). Assume the vertices are noncollinear. Find m.

(A) $\frac{3}{5}$ (B) $\frac{3}{4}$

(C) 0 (D) $\frac{4}{3}$

94. Let k make lim as x→ ∞ of [$(6x^2+4x+1)/(x+5)$-kx] finite. Find that finite limit.

(A) 34 (B) 4

(C) 30 (D) -26

95. Evaluate $\sum_{j=1}^{10} j(j+3)$.

(A) 385 (B) 165

(C) 550 (D) 3190

96. For x>0, xy′-y=x and y(1)=2. Find y(4)/4-y(1).

(A) $\log(256)+6$ (B) $\log(4)+2$

(C) $-\log(4)$ (D) $\log(4)$

97. $R(\theta)=\begin{pmatrix}\cos\theta & -\sin\theta \\ sin\theta & \cos\theta\end{pmatrix}$, 0<θ<π/2, tanθ=$\frac{3}{5}$. Find tr(R(θ)^2).

(A) $\frac{30}{17}$ (B) $\frac{16}{17}$

(C) 2 (D) $\frac{8}{17}$

98. Between positive a=25, b=64, let A be their arithmetic mean and G their geometric mean. Find A^2-G^2.

(A) $\frac{39}{2}$ (B) $\frac{7921}{4}$

(C) 1521 (D) $\frac{1521}{4}$

99. 11 distinct people stand in a line. Three specified people cannot occupy positions 1,2 or 11. How many orders are possible?

(A) 2257920 (B) 13547520

(C) 20321280 (D) 40320

100. A complex number z satisfies z+conjugate(z)=12 and z-conjugate(z)=8i. Find z·conjugate(z).

(A) $2\sqrt{13}$ (B) 104

(C) 52 (D) 4

101. X earns 7 and uses 2 labour hours; Y earns 4 and uses 3 hours. There are 110 hours and at least 3 items of Y must be made. Which model is correct?

(A) Minimize $7x + 4y$; $2x + 3y \leq 110$, $y \geq 3$ (B) Maximize $7x + 4y$; $2x + 3y \geq 110$, $x, y \geq 0$

(C) Maximize $4x + 7y$; $2x + 3y \leq 110$, $y \leq 3$ (D) Maximize $7x + 4y$; $2x + 3y \leq 110$, $x \geq 0$, $y \geq 3$

102. Positive a,b,c form an AP with a=7. The numbers a,b,c+4/7 form a GP and b>a. Find c.

(A) 9 (B) 11

(C) 15 (D) 3

103. How many discontinuities does (x-3)floor(x) have in (0,7)?

(A) 5 (B) 6

(C) 0 (D) 7

104. z=11+3i. Find 1/z+1/conjugate(z).

(A) $\frac{3}{65}$ (B) $\frac{11}{65}$

(C) $\frac{1}{22}$ (D) 22

105. A,B are acute, A+B=$\pi/4$ and tan A=$\frac{7}{12}$. Find tan(A-B).

(A) $\frac{73}{228}$ (B) $\frac{5}{19}$

(C) $-\frac{73}{263}$ (D) $\frac{73}{263}$

106. Four observations are 7,7,x,y with population mean 7 and variance 4/2. Find xy.

(A) 53 (B) 90

(C) 45 (D) 49

107. a=(5,1,0), b=(t,2,1). If a$\perp$b, find $|5b|^2$.

(A) 129 (B) $\frac{129}{25}$

(C) 104 (D) 125

108. Three distinguishable fair dice are thrown. Given that exactly two faces are equal, find the probability that their sum is at least 10.

(A) $\frac{3}{5}$ (B) $\frac{1}{4}$

(C) $\frac{19}{30}$ (D) $\frac{2}{5}$

109. Evaluate sin(2arctan 4).

(A) $-\frac{8}{15}$ (B) $\frac{4\sqrt{17}}{17}$

(C) $\frac{8}{17}$ (D) $-\frac{8}{17}$

110. In $(5x^2 - x^{-1})^7$, find the ratio of the coefficient of x^5 to that of x^8.

(A) $\frac{1}{3}$ (B) -3

(C) $-\frac{1}{3}$ (D) $\frac{5}{3}$

111. The midpoints of the sides of a triangle are (3,0),(0,3),(3,3). Find the sum of coordinates of its circumcentre.

(A) 0 (B) 6

(C) 12 (D) 4

112. A rectangle has vertices (±x,±y) on $x^2/36+y^2/9=1$, with $x \geq 6\sqrt{3}/2$. Find its maximum area.

(A) $36\sqrt{3}$ (B) $18\sqrt{3}$

(C) 18 (D) 36

113. α,β are roots of x^2-17x+70=0. The monic polynomial with roots $\alpha+1,\beta+1$ is x^2-Ax+B. Find B.

(A) 54 (B) 5041

(C) 88 (D) 71

114. The tangent to $y=x^2+5x$ at P has slope 9. Find the slope of the normal at P.

(A) $\frac{1}{9}$ (B) -9

(C) 9 (D) $-\frac{1}{9}$

115. L_1:r=t(1,0,1), L_2:r=(6,0,0)+s(0,1,1). Find their shortest distance.

(A) 6 (B) 2

(C) 0 (D) $2\sqrt{3}$

116. $F'(x)=\frac{5(x^2-1)(x^2+1)^4}{x^6}$ and F(1)=2. Find F(2)-2F(1).

(A) $\frac{2165}{32}$ (B) $\frac{2037}{32}$

(C) $-\frac{2165}{32}$ (D) $\frac{2101}{32}$

117. $x=6+10(1-t^2)/(1+t^2)$, $y=4+20t/(1+t^2)$, where t is real. Which point of the containing circle is not attained?

(A) $(-4, 4)$ (B) $(6, 14)$

(C) $(6, -6)$ (D) $(16, 4)$

118. ABC is a right triangle on level ground with AB=18, AC=18, angle A=90°. A vertical tower stands at midpoint M of BC. Its top elevation from A is 60°. Find its height.

(A) $9\sqrt{3}$ (B) $18\sqrt{6}$

(C) $9\sqrt{2}$ (D) $9\sqrt{6}$

119. y>0 satisfies 2yy′+12y^2=0, y(0)=sqrt(3). Find y^2(1).

(A) $\frac{3}{e^6}$ (B) $\frac{\sqrt{3}}{e^6}$

(C) $3e^{12}$ (D) $\frac{3}{e^{12}}$

120. From two points on the same ray from a tower foot, separated by 6, elevations are 30° and 45°. Find the distance from the farther point to the foot.

(A) $3\sqrt{3}+9$ (B) $3\sqrt{3}$

(C) $3+3\sqrt{3}$ (D) 6

121. Evaluate $\sum_{r=0}^{6} r(r-1)\binom{6}{r}$.

(A) 960 (B) 576

(C) 480 (D) 192

122. Two distinguishable fair dice show X,Y. Given X+Y≥8, find the probability that at least one of X, Y is even.

(A) $\frac{4}{5}$ (B) $\frac{1}{3}$

(C) $\frac{6}{5}$ (D) $\frac{1}{5}$

123. The normal of slope 6 to y^2=12x touches at P. Find the distance from P to the focus.

(A) 108 (B) 111

(C) 12321 (D) 105

124. The letters A,A,B,C and 4 further distinct letters are arranged in a row. How many arrangements have nonadjacent As and have B,C adjacent in either order?

(A) 3600 (B) 1800

(C) 1440 (D) 2400

125. Points on r=(2,2,0)+t(1,0,0) lie at distance 3 from (2,2,0). Find the sum of the squares of their x coordinates.

(A) 8 (B) 18

(C) 26 (D) 16

126. For $A=\begin{bmatrix}7 & 1\\ 1 & 2\end{bmatrix}$ and C=13A^{-1}, find C$_{22}$.

(A) 7 (B) $\frac{7}{13}$

(C) 2 (D) -1

127. The curve y=x^3+5x divides the disc x^2+y^2 ≤81. Find the area of the part satisfying y≥x^3+5x.

(A) 81π (B) $\frac{81\pi}{2}$

(C) $\frac{81\pi}{4}$ (D) $5+\frac{81\pi}{2}$

128. How many integer t make P=(t,0) strictly inside the larger segment cut by x=2 from $x^2+y^2=64$?

(A) 5 (B) 15

(C) 9 (D) 11

129. A=(7,3,0), C=(10,9,3). B lies on AC with AB:BC=2:1. Find $|OB|^2$, where O is the origin.

(A) 134 (B) 190

(C) 54 (D) 58

130. How many integers k make $6\sin x+8\cos x+2=k$ have exactly two distinct solutions in $0\leq x<2\pi$?

(A) 20 (B) 21

(C) 19 (D) 9

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A 4 kg block rests on a 8 kg block on a smooth horizontal floor. A force 96 N pulls the lower block. There is no slipping between the blocks. During a displacement 6 m, what work does friction do on the upper block?

(A) 192 J (B) 288 J

(C) 576 J (D) 384 J

132. A T-frame consists of a vertical rod from $(0, 0)$ to $(0, 8)$ and a horizontal rod of the same length centred at its top. The horizontal rod has 4 times the vertical rod's linear density. A horizontal force must cause pure translation. Find the required force-line height above the origin.

(A) 36 m (B) $\frac{36}{5}$ m

(C) $\frac{24}{5}$ m (D) 4 m

133. An active transistor has $\alpha = \frac{99}{100}$. Its collector resistor is 4 kΩ and the measured collector-resistor drop is 20 V. Neglect leakage. What base current in mA is required?

(A) 500 mA (B) $\frac{500}{99}$ mA

(C) $\frac{5}{99}$ mA (D) $\frac{1}{20}$ mA

Chemistry

134. CO binds to a more electron-rich metal centre. If metal-to-CO pi back-donation increases while the ligand remains bound through carbon,which paired bond changes and donation directions are consistent?

(A) CO donates a carbon lone pair to metal; metal donates into CO pi-star. M-C strengthens while C-O weakens.

(B) CO is only a sigma donor; back-donation leaves the C-O force constant unchanged.

(C) Metal donates into a CO bonding orbital,so C-O strengthens while M-C weakens.

(D) CO pi electrons are donated only from oxygen to metal,and both M-C and C-O necessarily strengthen.

135. Assign the formal metal oxidation state in $[Co(CO)_4]^-$,using four neutral CO. For nitrosyl species use the explicitly stated formal NO^+ convention rather than inferring spin from it.

(A) 0 (B) −1

(C) −2 (D) 1

136. For hepta-2,4-diene, CH3-CH=CH-CH=CH-CH2-CH3, count distinct configurational geometrical isomers. Ignore conformers and photochemical interconversion. Where reversing the entire carbon chain identifies two E/Z assignments, count them once.

(A) 8 (B) 3

(C) 4 (D) 5

Logical Reasoning

137. Preserve the exact relation and its order: Tailor : Needle :: ? : Telescope. Choose the missing term.

(A) Tailor (B) Astronomer

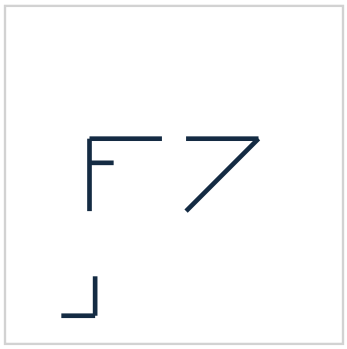
(C) Carpenter (D) Painter

138. A community register uses the following three categories: inventors, farmers, artists. Its complete relevant assertions are in this short paragraph; do not import everyday assumptions. No inventors are farmers. Some inventors are not artists. Some artists are farmers. Conclusions: I. Some farmers are artists. II. No farmers are inventors. Which conclusions follow necessarily?

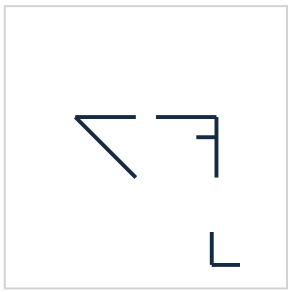
(A) Only II (B) Only I

(C) Both I and II (D) Neither I nor II

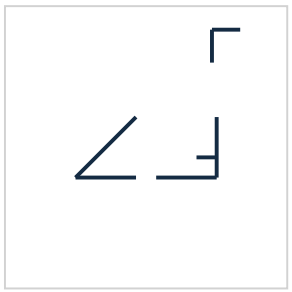
139. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



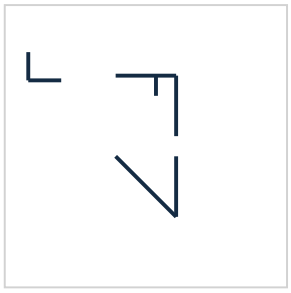
(A) Pattern X



(B) Pattern R



(C) Pattern Z



(D) Pattern Q



Mathematics

140. 14 distinct people sit at labelled round tables of sizes 9,5. Two specified people must be at different tables, and the first specified person must be at the 9-seat table. Rotations are identical. Count the seatings.

(A) 1937295360
(B) 479001600
(C) 21555072000
(D) 958003200

141. Let y′=(x+y)/(x+y+1), with y(0)=1. Find the positive value of x at whichx+y=5.

(A) $\frac{\log\left(\frac{11}{3}\right)}{4} + 2$
(B) $-\frac{\log(11)}{4} + \frac{\log(3)}{4} + 2$
(C) $\log\left(\frac{\sqrt{33}}{3}\right) + 2$
(D) $\log\left(\frac{\sqrt{33}}{3}\right) + 4$

142. For a=(1,2,0), b=(0,1,2), c=(4,0,1), evaluate (a×c)·(a+b).

(A) 17
(B) 0
(C) −17
(D) −16

BITSAT Mock 08

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A solid sphere rolls without slipping from rest through vertical height 11 m. A thin ring does so through vertical height 9 m. Find the ratio of their squared centre-of-mass speeds, sphere to ring.

(A) $\frac{110}{63}$ (B) $\frac{\sqrt{770}}{21}$
(C) $\frac{77}{45}$ (D) $\frac{55}{63}$

2. A projectile is launched at 15 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{15}{16}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ m/s^2.

(A) $\frac{81}{10}$ m (B) 9 m
(C) $\frac{27}{20}$ m (D) $\frac{27}{5}$ m

3. A spring of constant 40 N/m is already stretched by $\frac{1}{10}$ m. How much additional work is required to increase its extension to $\frac{1}{5}$ m quasistatically?

(A) $\frac{4}{5}$ J (B) $\frac{2}{5}$ J
(C) $\frac{1}{5}$ J (D) $\frac{3}{5}$ J

4. A car negotiates a level circular turn at its maximum non-slipping speed u when $\mu_s = \frac{11}{10}$. Rain lowers the coefficient to $\frac{2}{5}$ and the turn radius stays fixed. Find the new maximum speed divided by u.

(A) $\frac{\sqrt{10}}{5}$ (B) $\frac{4}{11}$
(C) $\frac{2\sqrt{11}}{11}$ (D) $\frac{\sqrt{11}}{2}$

5. Two parallel plane mirrors separated by 3 cm extend from axial position 0 to $\frac{27}{2}$ cm. A ray enters midway between them at axial position 0, making 45° with the normal to the mirrors. How many reflections occur before it passes the far edge? Exclude a contact exactly on the exit edge.

(A) 4 (B) 6
(C) 2 (D) 5

6. The accelerating potential of an X-ray tube is increased from 25 kV to 100 kV, while its tube current is reduced by a factor 4. Ignoring target heating and recoil, what is the new cutoff wavelength in nm? Use $hc = 1240$ eV nm.

(A) $\frac{124}{625}$ nm (B) $\frac{31}{2500}$ nm

(C) $\frac{31}{625}$ nm (D) $\frac{62}{625}$ nm

7. A 1 kg block initially moves at 3 m/s on a horizontal rough floor. A forward constant force of 6 N acts for 4 s. With $\mu_k = 1/5$ and $g = 10$ m/s^2, find net work during this interval.

(A) 176 J (B) 360 J

(C) 264 J (D) 16 J

8. A pure radioactive sample has activity A at time 12 hours after preparation and $A/8$ at time 24 hours after preparation. At what time, measured from preparation, will its activity be one thirty-second of the activity measured at 12 hours? Assume exponential decay with a constant decay rate.

(A) $\frac{32}{3}$ h (B) 44 h

(C) 32 h (D) 140 h

9. A charge of 7 μC moves from potential 70 V to -70 V in an electrostatic field. Find the work done by an external agent if the charge starts and ends at rest and no other force does work.

(A) 490 μJ (B) -490 μJ

(C) 980 μJ (D) -980 μJ

10. A galvanometer has resistance 80 ohms and full-scale current $\frac{1}{1000}$ A. It is to measure a total current 4 times its own full-scale current. Find the required parallel shunt resistance.

(A) $\frac{320}{3}\,\Omega$ (B) $80\,\Omega$

(C) $\frac{80}{3}\,\Omega$ (D) $240\,\Omega$

11. A projectile is launched at 15 m/s. At its highest point its kinetic energy is $\frac{1}{2}$ times its initial kinetic energy. Find its maximum height. Neglect drag and take $g = 10$ m/s^2.

(A) $\frac{45}{4} - \frac{45\sqrt{2}}{8}$ m (B) $\frac{45}{8}$ m

(C) $\frac{135}{16}$ m (D) $\frac{45}{4}$ m

12. One mole of nonoverlapping identical spheres of radius r has total geometric material volume V. A second sample contains $\frac{2}{5}$ mole of spheres of radius $2r$. What is its total geometric material volume divided by V? Ignore packing voids.

(A) 20 (B) $\frac{8}{5}$

(C) $\frac{16}{5}$ (D) $\frac{1}{20}$

13. Two rods of equal length and cross-sectional area have thermal conductivities k and 8k. They are joined end-to-end with no contact resistance. Find the effective conductivity of the composite divided by k.

(A) $\frac{2}{9}$ (B) $\frac{16}{9}$

(C) 1 (D) 9

14. A wire carries $i(t) = 4 + 5\sin\omega t$ A for many cycles. Find its rms current.

(A) $\frac{\sqrt{114}}{2}$ A (B) $\frac{\sqrt{82}}{2}$ A

(C) $\frac{5\sqrt{2}}{2}$ A (D) $\sqrt{41}$ A

15. A stretched string is divided by a bridge into lengths in the ratio 1 : 4. Tension and linear density are unchanged. The original fundamental is 20 Hz. Find the fundamental of the longer segment.

(A) 100 Hz (B) 5 Hz

(C) 20 Hz (D) 25 Hz

16. A diamond network has four outer resistors AB, BC, CD and DA, each 3 ohms, and a bridge BD also 3 ohms. The DA arm additionally contains an ideal diode with anode at D and cathode at A. A source keeps A positive relative to C. Find equivalent resistance between A and C.

(A) 3 Ω (B) 9 Ω

(C) $\frac{3}{2}$ Ω (D) 5 Ω

17. A hydrogen atom emits a photon in the $3 \to 2$ transition. A hydrogen-like ion of nuclear charge $Z = 4$ emits a photon of exactly the same wavelength and ends in level $n_f = 8$. What is its initial principal quantum number?

(A) 48 (B) 12

(C) 7 (D) 8

18. A satellite of mass m is moved from a circular orbit of radius R to one of radius 11R about mass M. Find the required increase in total orbital energy in units of $GMm/(2R)$.

(A) $\frac{20}{11}$ (B) $\frac{10}{11}$

(C) $\frac{120}{121}$ (D) $\frac{12}{11}$

19. A pendulum bob of mass 4 kg and length 3 m moves downward at a position with $\cos\theta = 4/5$, $\sin\theta = 3/5$ relative to the downward vertical. String tension is 240 N. Find instantaneous power of all forces. Take $g = 10$ m/s^2.

(A) $144\sqrt{5}$ W (B) $64\sqrt{39}$ W

(C) $48\sqrt{39}$ W (D) $80\sqrt{39}$ W

20. An object moves slowly along the axis toward a concave mirror. At an instant its distance from the mirror is 120 cm; the mirror focal-length magnitude is 20 cm. Find the image-speed magnitude divided by object-speed magnitude at that instant.

(A) 25 (B) $\frac{1}{25}$

(C) $\frac{1}{36}$ (D) $\frac{1}{5}$

21. A 2 kg mass on a string passes the top of a vertical circle of radius 5 m with string tension 60 N. Find string tension at the bottom. Take $g = 10$ m/s^2.

(A) 160 N (B) 180 N

(C) 20 N (D) 100 N

22. A wire consists of a semicircular arc of radius 4 m carrying 6 A, joined to two straight radial segments that extend outward from its ends. Find the magnetic field magnitude at the arc centre in units of μ_0.

(A) $\frac{3}{8}$ (B) $\frac{3}{4}$

(C) $\frac{3}{4\pi}$ (D) $\frac{3}{8\pi}$

23. A thin glass lens of refractive index 3/2 has focal length 160 cm in air. It is completely immersed in a liquid of index 4/3. Find its new focal length.

(A) 640 cm (B) $\frac{640}{3}$ cm

(C) 40 cm (D) 480 cm

24. A particle has $v(0) = 5$ m/s and acceleration $a(t) = 10t$ m/s^2 for $0 \le t \le 3$ s. Find its average velocity over this interval.

(A) 20 m/s (B) $\frac{55}{2}$ m/s

(C) 35 m/s (D) 5 m/s

25. A nucleus of mass number 28 splits into two equal nuclei. The parent has binding energy per nucleon $\frac{71}{10}$ MeV and each daughter has $\frac{73}{10}$ MeV. Ignoring other products, what energy is released in the split?

(A) $\frac{994}{5}$ MeV (B) $\frac{14}{5}$ MeV

(C) 210 MeV (D) $\frac{28}{5}$ MeV

26. A metal wire has linear temperature coefficient $\frac{1}{250}$ per degree Celsius referred to 0°C. Its dimensions are fixed and it stays across a constant voltage. Find power at 70°C divided by power at 0°C.

(A) $\frac{7}{32}$ (B) $\frac{25}{32}$

(C) $\frac{32}{25}$ (D) $\frac{25}{18}$

27. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 15 percent. The measured increase in its absolute temperature is $\frac{4257}{40}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) $\frac{132000}{529}$ K (B) $\frac{17457}{40}$ K

(C) $\frac{1419}{4}$ K (D) 330 K

28. Two large parallel plates are held at potentials +200 V and −200 V, with gap 5 mm. Neglect edge effects. Find field magnitude in the gap.

(A) 800000 V/m (B) 80 V/m

(C) 80000 V/m (D) 40000 V/m

29. A drop of mass 3.84×10^{-17} kg is suspended in a vertical field of magnitude 300 N/C. It has excess electrons. One excess electron is removed without changing its mass. Find its immediate downward acceleration, taking $g = 10$ m/s^2 and $e = 1.6 \times 10^{-19}$ C.

(A) $\frac{5}{32}$ m/s^2 (B) $\frac{10}{7}$ m/s^2

(C) $\frac{5}{4}$ m/s^2 (D) $\frac{35}{4}$ m/s^2

30. A wheel initially rotates at 12 rad/s. A constant opposing angular acceleration brings it to rest after 7 s. What angle does it turn through during the last half of this stopping interval?

(A) $\frac{21}{2}$ rad (B) 42 rad

(C) 84 rad (D) $\frac{63}{2}$ rad

Chemistry

31. $E^\circ(M^{4+}/M)=\frac{2}{5}V$ for4electrons,and$E^\circ(M^{4+}/M^{2+})=\frac{3}{5}V$ for2electrons. A Z^{2+}/Z couple is-0.800V. Standard cell potential for$Z+M^{2+}\rightarrow Z^{2+}+M$ is

(A) $\frac{6}{5}$ (B) $\frac{7}{5}$

(C) $\frac{13}{10}$ (D) 1

32. On controlled heating,$\frac{3}{100}$ mol $KMnO_4$ follows $2KMnO_4\rightarrow K_2MnO_4+MnO_2+O_2$. Which pair ($O_2$ amount,origin of the intense purple colour of the starting d^0 ion) is correct?

(A) ($\frac{3}{200}$ mol,charge-transfer absorption). (B) ($\frac{3}{200}$ mol,a d-d transition of MnVII).

(C) ($\frac{3}{100}$ mol,a d-d transition of MnVII). (D) ($\frac{3}{400}$ mol,charge-transfer absorption).

33. An octahedral Co(III) complex contains 2 neutral en ligands,with all remaining donor positions occupied by chloride. What is its coordination number and coordination-ion charge?

(A) CN 6; charge +1. (B) CN 8; charge +1.

(C) CN 4; charge +1. (D) CN 6; charge -1.

34. Excess water reacts completely with $\frac{1}{100}$ mol NaH. Which gas identity/amount pair follows from the balanced hydrolysis?

(A) CO_2, $\frac{1}{100}$ mol. (B) H_2, $\frac{1}{200}$ mol.

(C) H_2, $\frac{1}{100}$ mol. (D) O_2, $\frac{1}{200}$ mol.

35. An ideal mixture initially has$\frac{1}{10}$molA and$\frac{2}{5}$molB. A selective step removes half ofB and noA. At the new equilibriumT,V,totalpressure is6bar. Partialpressure ofA,in bar,is

(A) 3.0 (B) 2

(C) 18 (D) $\frac{6}{5}$

36. 2-methylpropene is subjected to water/dilute acid. Which product follows the stated regioselectivity or oxidation/dehydration endpoint?

(A) propanone (B) 2-methylpropanal

(C) 2-methylpropan-1-ol (D) 2-methylpropan-2-ol

37. A weak1:1electrolyte has$\alpha \ll 1$ and obeys$Ka\approx C\alpha^2$. Dilute it by a factor9without changing total formula-unit amount or Ka,and remain in that approximation. Which pair(degreeα,totalfree-ion amount)is correct?

(A) α increases byfactor9,buttotalionsdecreasebyfactor9.

(B) α decreases byfactor3,andtotalionsareunchanged.

(C) Both increase by factor3.

(D) Both remain unchanged because the number of solute formula units is fixed.

38. 2-bromo-2-methylpentane is heated with potassium tert-butoxide under elimination-favouring conditions. Which major constitutional product is expected,despite the presence of an alkoxide that could be used in a Williamson synthesis with a suitable different halide?

(A) the alkane formed by replacing halogen with H

(B) 2-methylpent-2-ene

(C) the corresponding tertiary alkoxide substitution product

(D) 2-methylpent-1-ene

39. Which paired classification is valid:BF_3accepts an electron pair fromNH_3,whileNH_3can also accept $H^+$from water?

(A) $BF_3$is a Lewis acid without donating a proton;$NH_3$is both a Lewis base and a Bronsted base in these reactions.

(B) $BF_3$is a Bronsted acid because every Lewis acid must contain a removable H.

(C) $NH_3$is an Arrhenius strong base because proton acceptance is complete in water.

(D) NH_3acts as a Lewis acid when it donates its lone pair toBF_3.

40. In a copper extraction stage, $\frac{3}{100}$ mol Cu_2S and $\frac{1}{25}$ mol Cu_2O react only by $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$. What amount of metallic copper, in mol, is formed?

(A) $\frac{1}{25}$

(B) $\frac{3}{25}$

(C) $\frac{9}{50}$

(D) $\frac{1}{10}$

41. An internally consistent thermochemical comparison gives total resonance stabilisation of benzene as 36 kcal/mol (1 benzenoid rings) and anthracene as 84 kcal/mol (3 rings). Which statement correctly separates total stabilisation from stabilisation per ring?

(A) Total: benzene; per ring: benzene.

(B) Total: anthracene; per ring: benzene.

(C) Total: anthracene; per ring: anthracene.

(D) Both quantities must rank compounds identically because each is aromatic.

42. A $\frac{1}{5}$ g organic sample contains chlorine as its only halogen. Carius analysis gives $\frac{574}{8875}$ g dry AgCl, but this is only 80.0% of the theoretical AgCl because of a known collection loss. Find chlorine mass percentage. Use Ag=108, Cl=35.5.

(A) 10

(B) $\frac{2870}{71}$

(C) 8

(D) $\frac{25}{2}$

43. Two model nonvolatile solutes A(M180) and B(M60) are dissolved separately in equal solvent masses. Equal solute masses are used. A remains molecular;fraction $\frac{2}{5}$ of B formula units is present in dimers. All solutions obey the stated ideal colligative model. Ratio $\Delta T_f(B)/\Delta T_f(A)$ is

(A) $\frac{4}{5}$

(B) 3

(C) $\frac{12}{5}$

(D) $\frac{15}{4}$

44. An element has cubicformsAandB. Ahas an FCC lattice with fraction$\frac{7}{8}$of normal atomic sites occupied; Bhas fully occupiedBCC. Edge lengths obey aA=2aB,and atomic mass is unchanged. Densityratio$\rho A/\rho B$ is

(A) $\frac{7}{16}$

(B) $\frac{7}{2}$

(C) $\frac{7}{32}$

(D) $\frac{1}{4}$

45. A Fischer projection of tartaric acid has its four-carbon chain vertical (COOH at top and COOH at bottom). At C2,OH is on the right; at C3,OH is on the left; each other horizontal group is H. Horizontal bonds point toward the observer. Which classification and configuration pair is correct?

(A) chiral; (2R,3R)
(B) chiral; (2S,3S)
(C) meso; (2R,3S)
(D) meso; (2R,3R)

46. $\frac{1}{25}$ mol borax is acidified completely. Each borax formula unit gives four H_3BO_3 molecules;75.0% of the boric acid is isolated. With glycerol present, the isolated boric acid is titrated completely as a monobasic acid. What amount of NaOH, in mol, is required?

(A) $\frac{9}{100}$
(B) $\frac{3}{25}$
(C) $\frac{9}{25}$
(D) $\frac{4}{25}$

47. Liquid A is immiscible with water, steam-volatile and decomposes near its normal boiling point. A recovered solid impurity sublimes while a nonvolatile salt does not. Which sequence separates A first and then the solid impurity without ordinary high-temperature boiling?

(A) Fractional distillation; steam distillation of the salt.
(B) Simple distillation; filtration of dissolved salt.
(C) Steam distillation; sublimation.
(D) Sublimation of A; simple distillation of the salt.

48. styrene,Ph-CH=CH2 reacts with HBr with peroxide. Which pair gives the major constitutional product and the intermediate whose greater stability fixes its orientation? Ignore stereoisomer counting.

(A) Ph-CH2-CH2Br; primary radical
(B) Ph-CH(Br)-CH3; primary radical
(C) Ph-CH2-CH2Br; benzylic radical
(D) Ph-CH(Br)-CH3; benzylic radical

49. For $M(s)+2A^{+}\rightleftharpoons M^{2+}+2A(s), E^{\circ}=\frac{59}{1000}V$. Use$2.303RT/F=.0590V$. Equilibrium constant for the doubled reaction$2M+4A^{+}\rightleftharpoons 2M^{2+}+4A$ is

(A) 10000
(B) 100
(C) 100000000
(D) $\frac{1}{10000}$

50. For hexa-1,4-diene, CH2=CH-CH2-CH=CH-CH3, count distinct configurational geometrical isomers. Ignore conformers and photochemical interconversion. Where reversing the entire carbon chain identifies two E/Z assignments, count them once.

(A) 2
(B) 1
(C) 4
(D) 3

51. A cation selected from Cu2+,Zn2+,Fe3+ gives a reddish-brown hydroxide precipitate when aqueous ammonia is added dropwise, followed by a precipitate that does not dissolve appreciably in excess ammonia under the usual test conditions on adding excess ammonia. Which identification and oxidation-state conclusion is consistent?

(A) Zn2+; a deep-blue colour is the universal zinc confirmation.
(B) Fe3+; the metal oxidation state remains +3.
(C) Fe3+; ammonia necessarily reduces the metal by one oxidation unit.
(D) Fe3+; every hydroxide dissolves to a deep-blue complex.

52. Classify MgO, Al2O3, P4O10, in that order, by their usual acid-base behaviour. Which sequence also predicts that the middle oxide reacts with both suitable acids and strong alkalis?

(A) Amphoteric; basic; acidic.
(B) Acidic; amphoteric; basic.
(C) Basic; amphoteric; acidic.
(D) Basic; acidic; amphoteric.

53. Two hydrogen-like bound states A and B have nuclear charges ZA=1 and ZB=4, and radii rB=1rA. The states obey the allowed integer-principal-quantum-number Bohr model. What is the ratio of binding-energy magnitudes |EB|/|EA|?

(A) 2
(B) 12
(C) 8
(D) 4

54. For the metal centre in Fe^{3+},Z=26. Determine the number of d electrons after formal oxidation-state assignment; use O=-2 in oxoanions.

(A) 6
(B) 5
(C) 7
(D) 4

55. For o-hydroxybenzaldehyde and p-hydroxybenzaldehyde,which is expected to show greater steam volatility on the usual intramolecular-versus-intermolecular hydrogen-bonding argument?

(A) p-hydroxybenzaldehyde
(B) o-hydroxybenzaldehyde
(C) neither can hydrogen-bond because both have an aromatic ring
(D) both must have identical volatility because they are isomers

56. For isolated Ti^{3+} (Z=22),use the high-spin spin-only approximation. Which pair (d-electron count, magnetic moment in BM) follows after removing electrons from 4s before 3d?

(A) (1,1).
(B) $(1,2\sqrt{2})$.
(C) $(2,2\sqrt{2})$.
(D) $(1,\sqrt{3})$.

57. A HA/A^- buffer initially has pH=pKa+$\log_{10}2$. Adding $\frac{1}{25}$ mol HCl without volume change gives pH=pKa-$\log_{10}2$;neither component is exhausted. Total initial $HA+A^-$ amount,in mol,is

(A) $\frac{1}{25}$
(B) $\frac{4}{25}$
(C) $\frac{2}{25}$
(D) $\frac{3}{25}$

58. Under the usual qualitative group trends,compare the ease of alkali-metal electron loss in Na/K and the oxidising tendency of Cl_2/Br_2. Which conclusion is consistent?

(A) Na loses its electron more readily; Br_2 is the stronger halogen oxidant.
(B) K loses its outer electron more readily; Cl_2 is the stronger halogen oxidant.
(C) Both trends must increase down a group because atoms become larger.
(D) K loses its electron more readily; Br_2 is the stronger halogen oxidant.

59. For2A+B→3C+4D,the measured total disappearance rate of A is $\frac{1}{100}$M/s. A separate calibrated decay channel consumes A at $\frac{1}{500}$M/s and produces no D. D appearance rate from the stated reaction,in M/s,is

(A) $\frac{1}{50}$
(B) $\frac{1}{250}$
(C) $\frac{4}{125}$
(D) $\frac{2}{125}$

60. A mixture contains $\frac{3}{50}$ mol H_2SO_5 and $\frac{1}{20}$ mol $H_2S_2O_8$. Their structures contain one O–O link per molecule. All peroxide links react quantitatively with excess iodide in acid; each such link accepts two electrons. What amount of I_2, in mol, is liberated?

(A) $\frac{11}{100}$ (B) $\frac{4}{25}$

(C) $\frac{3}{50}$ (D) $\frac{11}{50}$

English Proficiency

61. Use an indefinite article for a previously unmentioned, nonspecific referent; no particular one has been identified. Complete: The panel appointed _____ independent observer who had no connection with either party.

(A) no article (B) a

(C) an (D) the

62. Choose the meaning-preserving equivalent that retains the tense, participants and all stated restrictions. The explanation was clear enough for a beginner to follow.

(A) The explanation was so clear that a beginner could follow it. (B) The beginner followed every clear explanation.

(C) The explanation was too clear for a beginner to follow. (D) Only a beginner could follow the clear explanation.

63. The conclusion is logical, meaning consistent with the stated reasoning. Choose the opposite in that sense.

(A) logical (B) logistical

(C) logarithmic (D) illogical

64. Read the sentence: The barrier helped to diminish the noise reaching the reading room. Choose the opposite in meaning of 'diminish' as used in the sentence.

(A) reduce (B) redirect

(C) measure (D) increase

65. A witness supplies a written statement formally sworn to be true; the question asks for that specific document, not testimony in general. Which term fits?

(A) affidavit (B) certificate

(C) verdict (D) petition

66. Read the sentence: The question was pertinent because its answer would determine whether the plan could proceed. Choose the closest meaning of 'pertinent' as used in the sentence.

(A) particularly difficult (B) directly relevant

(C) politely phrased (D) asked at the right moment

67. Complete: Because _____ rain was expected, the organisers moved the demonstration indoors.

(A) weighty (B) heavy

(C) dense-bodied (D) burdened

68. At the meeting, she immediately renders each spoken statement into another language. What is her role?

(A) translator (B) editor

(C) interpreter (D) stenographer

69. Choose the most coherent order of the four sentences. A. The organisers then issued a short notice containing only the change. B. The organisers sent a long programme to every conference participant. C. Several participants missed a change buried in the middle of the document. D. Separating the new information from the full programme made it much harder to overlook.

(A) C-B-A-D (B) B-C-A-D

(C) B-A-C-D (D) B-C-D-A

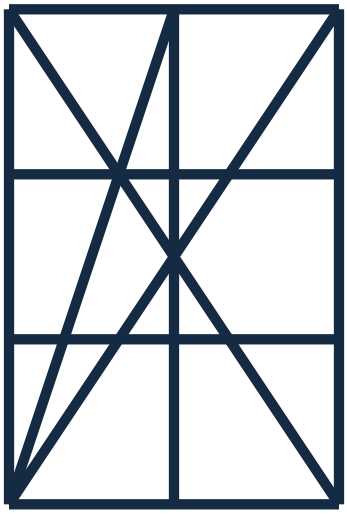
70. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. to mistake a precaution Q. the less likely a beginner is R. The more carefully the instructions are read, S. for an optional suggestion.

(A) R-Q-P-S (B) Q-R-P-S

(C) P-R-Q-S (D) S-Q-P-R

Logical Reasoning

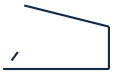
71. How many triangles are formed by all the drawn segments?



(A) 38 (B) 31

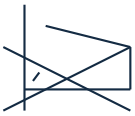
(C) 26 (D) 42

72. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

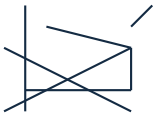




host 1



host 2



host 3



host 4

(A) 1 (B) 4

(C) 2 (D) 3

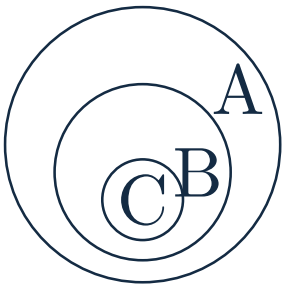
73. In a row, Asha is 9th from the left and 29th from the right. Bina is 6th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 19 (B) 25

(C) 21 (D) 20

74. Choose the diagram that represents all these conditions, including the strictness information: No A is B, no B is C, and no A is C. Each class is nonempty.

(A) Pattern R



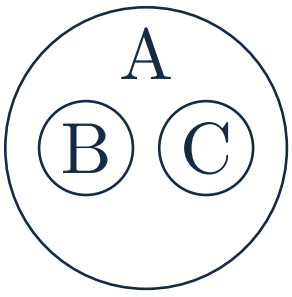
(B) Pattern L



(C) Pattern Z



(D) Pattern X



75. Preserve the exact relation and its order: Neon : Noble gas :: ? : Reptile. Choose the missing term.

(A) Whale (B) Lizard

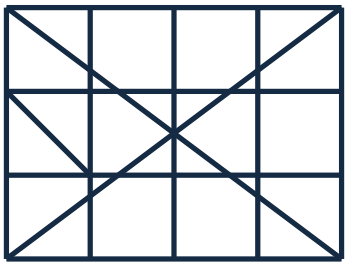
(C) Sparrow (D) Frog

76. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 28 and Z has code 3. Decode the separated values 13, 17, 28, 15, 9.

(A) TNALP (B) PLANT

(C) QMBOU (D) OKZMS

77. How many triangles are formed by all the drawn segments?



(A) 46 (B) 38

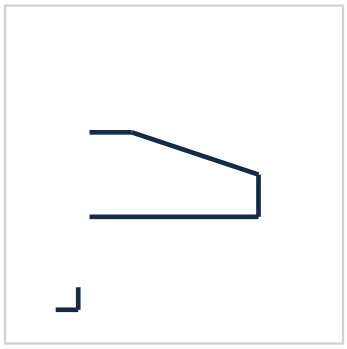
(C) 84 (D) 32

78. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 5, 10, 15, ?, 35, 70, 75.

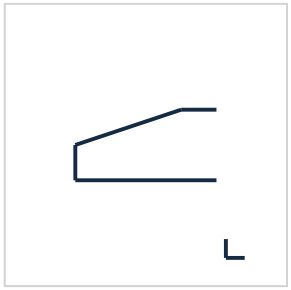
(A) 25 (B) 40

(C) 20 (D) 30

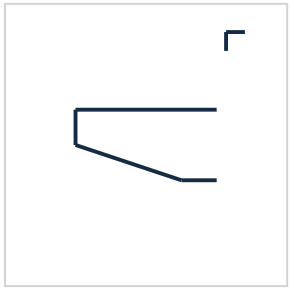
79. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



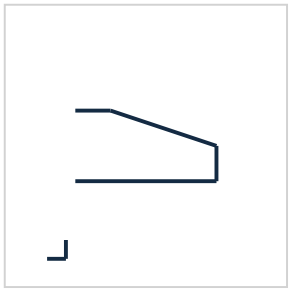
(A) Pattern R



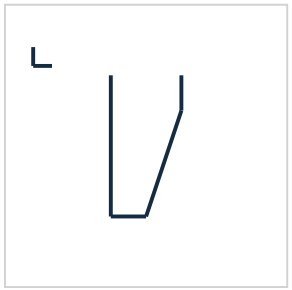
(B) Pattern M



(C) Pattern U



(D) Pattern L



80. Consider only these statements; universal statements alone do not imply existence. Some editors are hikers; Some editors are not hikers; All editors are designers. Conclusions: I. Some editors are designers. II. No hikers are editors. Which conclusions follow necessarily?

(A) Only I

(B) Only II

(C) Both I and II

(D) Neither I nor II

81. Vijay points to a photograph and says, "This person is the son of the only daughter of my father's only son." Vijay is male, and the photographed person is male. How is the photographed person related to Vijay?

(A) Brother

(B) Son

(C) Grandson

(D) Daughter

82. A person makes these turns in order: 90 degrees left, 90 degrees left, 90 degrees right, 180 degrees. The final heading is west. What was the heading immediately after turn 3?

(A) West (B) South

(C) East (D) North

83. Bina receives 3 times as many tokens as Asha, and Charu receives 2 times as many as Bina. Their total is 100. After the allocation, Bina gives 5 tokens to Asha. How many tokens does Bina then retain?

(A) 30 (B) 5

(C) 35 (D) 25

84. Preserve the exact relation and its order: Balance : Mass :: ? : Speed. Choose the missing term.

(A) Barometer (B) Speedometer

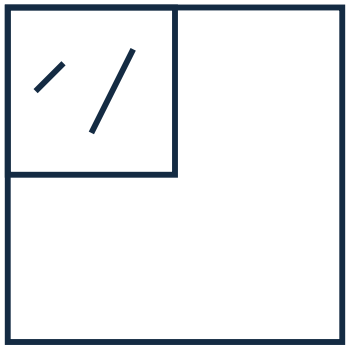
(C) Ammeter (D) Thermometer

85. The model triads are $(11, 29, 18)$ and $(13, 33, 20)$. In the intended rule the middle number is the sum of the outer numbers, and the outer-number difference is the same fixed positive value in every triad. Which option satisfies both relations?

(A) $(10, 27, 17)$ (B) $(10, 28, 18)$

(C) $(10, 28, 17)$ (D) $(10, 29, 18)$

86. Starting with a unit square sheet, fold right to left along the vertical midline, then fold bottom to top along the horizontal midline. The folded paper is cut along both short drawn slits. Select the pattern after complete unfolding.



(A) Pattern P



(B) Pattern V



(C) Pattern W



(D) Pattern Z



87. Three pairs contain exactly one mammal and one bird. Which pair does not?

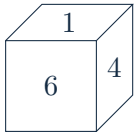
(A) Elephant / Ostrich

(B) Whale / Sparrow

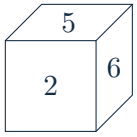
(C) Dog / Horse

(D) Cat / Owl

88. The views show the same die. Give the ordered pair: (face opposite 4, face opposite 6). Do not assume opposite labels sum to seven.



1



2

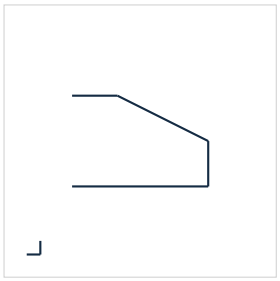
(A) $(5, 4)$

(B) $(5, 3)$

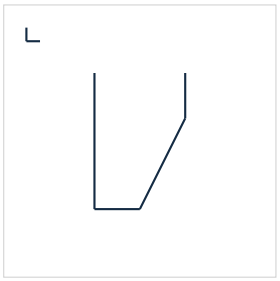
(C) $(5, 1)$

(D) $(5, 2)$

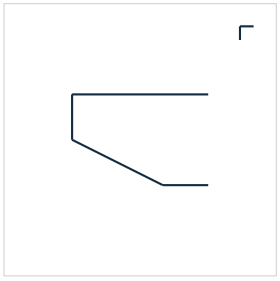
89. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

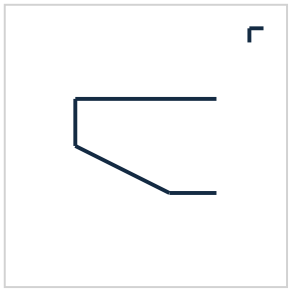


bottom left

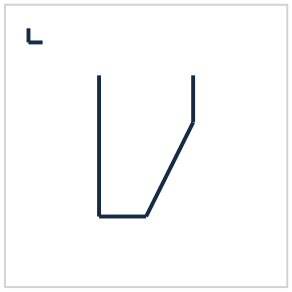
?

bottom right

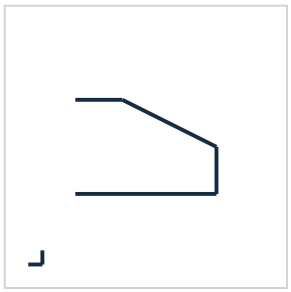
(A) Pattern Z



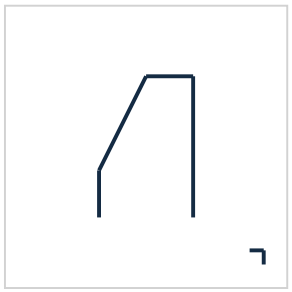
(B) Pattern R



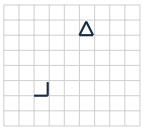
(C) Pattern V



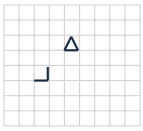
(D) Pattern Y



90. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



1

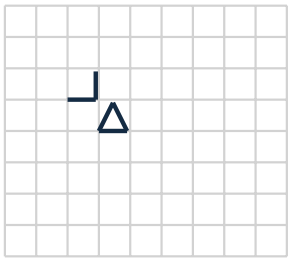


2

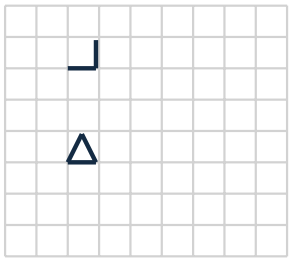


3

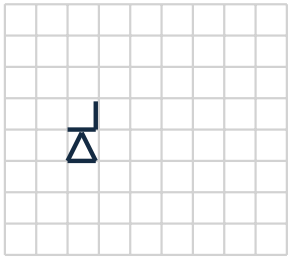
(A) Pattern R



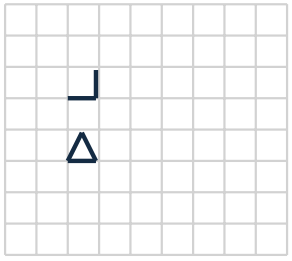
(B) Pattern Q



(C) Pattern M



(D) Pattern X



Mathematics

91. For f(x)=|x-4|/(x-9), on which maximal open intervals is f strictly decreasing?

(A) $(-\infty, 4)$

(B) $(4, 9) \cup (9, \infty)$

(C) $(-\infty, 9)$

(D) $(4, \infty)$

92. A box contains 9 red and 5 blue distinct balls. Three are drawn uniformly without replacement. Given that at least one drawn ball is blue, find the probability that exactly two are red.

(A) $\frac{5}{14}$

(B) $\frac{9}{14}$

(C) $\frac{15}{7}$

(D) $\frac{45}{91}$

93. The parabola $(y-4)^2=16(x-3)$ has focus F and vertex V. A circle centred at F passes through V. Find the tangent length from P=(15,12).

(A) 112

(B) $4\sqrt{7}$

(C) $8\sqrt{2}$

(D) $4\sqrt{3}$

94. Find the area enclosed by y=x(15-x) and y=6x.

(A) $-\frac{243}{2}$

(B) $\frac{243}{2}$

(C) $\frac{1053}{2}$

(D) $\frac{1125}{2}$

95. A chord of $x^2+y^2=2500$ touches $x^2/25-y^2/9=1$. Its nonzero midpoint (h,k) satisfies h=k. Find h^2+k^2.

(A) 4 (B) 8

(C) 16 (D) 17

96. From two points on the same ray from a tower foot, separated by 3, elevations are 30° and 45°. Find the distance from the farther point to the foot.

(A) $\frac{3\sqrt{3}}{2}+\frac{9}{2}$ (B) 3

(C) $\frac{3}{2}+\frac{3\sqrt{3}}{2}$ (D) $\frac{3\sqrt{3}}{2}$

97. On A={0,1,...,8}, define aRb if 4 divides a-b. How many ordered pairs are in R?

(A) 21 (B) 12

(C) 81 (D) 9

98. A circle passes through O=(0,0), A=(8,0), B=(0,4). Find the length of the tangent from P=(16,0).

(A) $8\sqrt{2}$ (B) 128

(C) 16 (D) 8

99. Evaluate the limit of $[\cos(4x)-\cos(3x)]/x^2$ as $x\to 0$.

(A) $-\frac{1}{2}$ (B) $-\frac{7}{2}$

(C) -7 (D) $\frac{7}{2}$

100. How many integers k make $24\sin x+32\cos x+5=k$ have exactly two distinct solutions in $0\le x<2\pi$?

(A) 79 (B) 81

(C) 39 (D) 80

101. For the loop $y^2=x^2(9-x)$, find the area of the portion with $x\ge 9/2$.

(A) $\frac{567\sqrt{2}}{20}$ (B) $\frac{324}{5}$

(C) $\frac{567\sqrt{2}}{10}$ (D) $\frac{648}{5}-\frac{567\sqrt{2}}{10}$

102. Find the coefficient of x^9 in $(1-x)\exp(2x)$.

(A) $\frac{4}{2835}$ (B) $\frac{22}{2835}$

(C) $-\frac{2}{405}$ (D) $-\frac{2}{315}$

103. Evaluate $\sum_{j=1}^{10} j(j+2)$.

(A) 495 (B) 385

(C) 3135 (D) 110

104. There are 3 Boolean variables p_1,...,p 3. How many truth assignments make [(p_1 implies p_2) and (p_1 implies not p_2)] true?

(A) 7 (B) 4

(C) 2 (D) 8

105. 4 distinct objects are distributed into three nonempty unlabelled boxes. How many distributions are possible?

(A) 18 (B) 36

(C) 13 (D) 6

106. α,β are nonzero roots of x^2-10x+21=0. Find the sum of $1+1/\alpha$ and $1+1/\beta$.

(A) $\frac{31}{21}$ (B) $\frac{41}{10}$

(C) $\frac{52}{21}$ (D) $\frac{10}{21}$

107. f(x)=(x^2-4)/(x-2) for x≠2, and f(2)=k. If f is differentiable at 2, find k+f′(2).

(A) 8 (B) 4

(C) 3 (D) 5

108. Find the sum of order and degree of $(d^4y/dx^4)^2+(dy/dx)^2+y=0$.

(A) 6 (B) 8

(C) 2 (D) 4

109. A line has direction (1,2,2). A plane has normal (2,1,2). Let θ be their acute angle. Find $\tan^2\theta$.

(A) $\frac{64}{17}$ (B) $\frac{64}{81}$

(C) $\frac{81}{64}$ (D) $\frac{17}{81}$

110. For a=(1,7,0), b=(0,1,5), c=(5,0,1), evaluate (a×c)·(a+b).

(A) 0 (B) −175

(C) −176 (D) 176

111. For x∈[0,1], find the minimum of $3(\arcsin x)^3+(\arccos x)^3$.

(A) $\frac{\pi^3}{8}$ (B) $\frac{3\pi^3}{8\left(1-\sqrt{3}\right)^2}$

(C) $\frac{3\pi^3}{8\left(1+\sqrt{3}\right)^2}$ (D) $-\frac{\sqrt{3}\pi^3}{16}+\frac{\pi^3}{8}$

112. X has mean 9, variance 16. Y=aX+4, a>0, has mean 40. Find Var(Y).

(A) 64 (B) 260

(C) 1552 (D) 256

113. If |z-6|=2, find the minimum of $|z|^2$.

(A) 64 (B) 4

(C) 16 (D) 32

114. Find the product of all real solutions of sqrt(7-log_4|x|)=log_4|x|-1.

(A) 4096 (B) 0

(C) −4096 (D) −64

115. How many integer k in [0,72] make $x^2 - 13x + (36 - k) = 0$ have roots of opposite signs?

(A) 35 (B) 72

(C) 37 (D) 36

116. X is binomial with mean 10 and variance $\frac{20}{3}$. Find its mode.

(A) 10 (B) 9

(C) 11 (D) 30

117. Let $P(x) = (1 + \sqrt{2}x)^9 + (1 - \sqrt{2}x)^9$. Find the sum of all nonconstant coefficients of P.

(A) 2786 (B) 2782

(C) 1393 (D) 2784

118. Between positive a=9, b=49, let A be their arithmetic mean and G their geometric mean. Find A^2-G^2.

(A) 100 (B) 20

(C) 400 (D) 441

119. For f(x)=x^3-108x, find the difference between its local maximum value and local minimum value.

(A) −864 (B) 12

(C) 432 (D) 864

120. Evaluate lim as n→ ∞ of $(2/n)\Sigma(2j/n)^2$, j=1,...,n.

(A) $\frac{1}{3}$ (B) $\frac{4}{3}$

(C) 4 (D) $\frac{8}{3}$

121. Disjoint A,B have sizes 5,6. Find |(A×A)∪(B×B)|.

(A) 30 (B) 121

(C) 61 (D) 60

122. How many integers x are in the real domain of $\sqrt{x-7}\,\log(10-x)$?

(A) 3 (B) 10

(C) 4 (D) 2

123. How many integers x in [-1,15] satisfy (|x-7|-4)/(|x-7|-7)>0?

(A) 11 (B) 9

(C) 17 (D) 7

124. 13 distinct people sit at labelled round tables of sizes 9,4. Two specified people must be at different tables, and the first specified person must be at the 9-seat table. Rotations are identical. Count the seatings.

(A) 79833600 (B) 172972800

(C) 1437004800 (D) 39916800

125. A=(4,0,5), B=(0,5,1), AP:PB=2:1 internally. Find $|AP|^2$.

(A) 57 (B) $\frac{19}{3}$

(C) 38 (D) $\frac{76}{3}$

126. An equilateral triangle has vertex (0,6) and opposite side on y=0. Find the squared inradius.

(A) 36 (B) 16

(C) 4 (D) 2

127. Write sin x $\cos^3$x+9$\sin^3$x cos x=P sin2x+Q sin4x. Find Q.

(A) $\frac{5}{4}$ (B) -1

(C) $\frac{1}{8}$ (D) $\frac{5}{2}$

128. $A = \begin{bmatrix} 2 & 1 \\ 1 & 5 \end{bmatrix}$ and A^{-1}=pI+qA. Find p+q.

(A) $\frac{2}{3}$ (B) $\frac{1}{9}$

(C) 54 (D) $\frac{8}{9}$

129. F′(x)=$(x^2 + 7x + 10)\, e^x$ and F(0)=5. Find F(1)-2F(0).

(A) $11e$ (B) $-5 + 11e$

(C) $-11e$ (D) $-10 + 11e$

130. The homogeneous system 5x+y=0, kx+6y=0 has a nonzero solution. Find k+x^2/y^2.

(A) $\frac{749}{25}$ (B) $\frac{751}{25}$

(C) 55 (D) 30

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A cyclist rides straight outward at 3 m/s for 5 s, waits 2 s, then returns to the start. The whole journey must last 10 s. What constant return speed is required?

(A) 3 m/s (B) $\frac{5}{3}$ m/s

(C) 5 m/s (D) $\frac{15}{8}$ m/s

132. A fixed ideal gas has 3 active degrees of freedom per molecule. Assertion: its molar internal-energy change is $\frac{3}{2}R\Delta T$. Reason: its heat-capacity ratio is $\frac{5}{3}$. Choose the best statement.

(A) Both are true, but no thermodynamic relation connects them.

(B) Both are true, and using $C_V = R/(\gamma - 1)$ the reason explains the assertion.

(C) The assertion is true and the reason is false.

(D) The assertion is false and the reason is true.

133. A photon and a nonrelativistic particle have equal de Broglie wavelengths. The particle speed is $c/11$. If the photon then loses half its energy while the particle momentum is unchanged, what is the ratio of particle kinetic energy to the photon's final energy?

(A) $\frac{1}{11}$ (B) 11

(C) $\frac{2}{11}$ (D) $\frac{1}{22}$

Chemistry

134. An ideal gas executes A→B→C→D→A through (V/L,P/bar):A(1,1),B(5,1),C(5,2),D(1,2). IUPAC work-on convention applies. Heats on AB,BC,DA are 800,300,-200 J. Heat on CD in J is (1 L bar=100 J)

(A) −1300 (B) −400

(C) −500 (D) −900

135. Compare pH of A:HCl at $\frac{3}{1000}$M;B:HA,Ka=10^{-6},C=$\frac{3}{10}$M;C:HB,Ka=10^{-5},C=$\frac{3}{1000}$M. Neglect water and use weak-ionisation ordering. Increasing pH is

(A) B<A<C. (B) A<B<C.

(C) A<C<B. (D) C<B<A.

136. A $\frac{3}{100}$mol iron-ion inventory contains Fe^{2+}andFe^{3+}. Fraction$\frac{3}{4}$isFe^{3+}. All iron is quantitatively reduced to metal in the stated externally driven electron-balance model,with no side reactions. Electrons accepted,in mol,are

(A) $\frac{51}{700}$ (B) $\frac{27}{350}$

(C) $\frac{33}{400}$ (D) $\frac{27}{400}$

Logical Reasoning

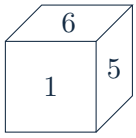
137. Three pairs contain exactly one string instrument and one wind instrument. Which pair does not?

(A) Guitar / Oboe
(B) Ukulele / Trombone
(C) Harp / Violin
(D) Cello / Trumpet

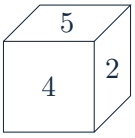
138. Consider only these statements; universal statements alone do not imply existence. Some editors are not artists; Some joggers are artists. Conclusions: I. Some artists are joggers. II. All artists are editors. Which conclusions follow necessarily?

(A) Only I
(B) Neither I nor II
(C) Only II
(D) Both I and II

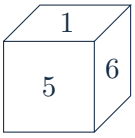
139. The views show the same die. Give the ordered pair: (face opposite 1, face opposite 5). Do not assume opposite labels sum to seven.



1



2



3

(A) $(2, 1)$
(B) $(2, 3)$
(C) $(2, 2)$
(D) $(2, 4)$

Mathematics

140. All tangents common to $x^2/144+y^2/36=1$ and $x^2+y^2=\frac{504}{5}$ touch the ellipse at points P_i. Find $\Sigma(P_i,x)^2$ over the four tangency points.

(A) $\frac{3456}{7}$
(B) $\frac{864}{7}$
(C) $\frac{144}{7}$
(D) $\frac{2016}{5}$

141. Box A contains 5 red and 3 blue balls; B contains 5 red and 5 blue balls. A fair coin chooses one box. Two independent draws with replacement from that box give exactly one red ball. Find the posterior probability that A was chosen.

(A) $\frac{16}{31}$
(B) $\frac{15}{31}$
(C) $\frac{5}{9}$
(D) $\frac{15}{64}$

142. Let $\delta=\pi/10$. Evaluate $\sin\delta\ \Sigma[1/(\sin(j\delta)\sin((j+1)\delta))]$, $j=2,\ldots,4$.

(A) $-\frac{\sqrt{10\sqrt{5}+25}}{5}$
(B) $\sqrt{5-2\sqrt{5}}$
(C) $\frac{\sqrt{10\sqrt{5}+25}}{5}$
(D) $\sqrt{2\sqrt{5}+5}$

BITSAT Mock 09

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A projectile is launched at 30 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{8}{9}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ m/s^2.

(A) $\frac{144}{5}$ m (B) $\frac{288}{5}$ m

(C) $\frac{432}{5}$ m (D) 48 m

2. A circular parallel-plate capacitor of plate radius 4 cm is charged at 54 mA. Neglect fringing. What displacement current passes through the annular part of the gap between radii 1 cm and 4 cm, centred on the plate axis?

(A) $\frac{27}{8}$ mA (B) $\frac{405}{8}$ mA

(C) 810 mA (D) $\frac{135}{2}$ mA

3. An active transistor has $\alpha = \frac{39}{40}$. Its collector resistor is 5 kΩ and the measured collector-resistor drop is 30 V. Neglect leakage. What base current in mA is required?

(A) $\frac{80}{13}$ mA (B) $\frac{3}{20}$ mA

(C) $\frac{2}{13}$ mA (D) 6 mA

4. A soap bubble initially of radius 6 cm is expanded quasistatically until its volume is 64 times the initial volume. Surface tension remains $\frac{1}{20}$ N/m. Find work against surface tension, in units of $\pi \times 10^{-4}$ J.

(A) 216 (B) $\frac{1152}{5}$

(C) $\frac{4536}{5}$ (D) $\frac{72}{5}$

5. A satellite of mass m is moved from a circular orbit of radius R to one of radius 10R about mass M. Find the required increase in total orbital energy in units of $GMm/(2R)$.

(A) $\frac{1}{10}$ (B) $\frac{9}{10}$

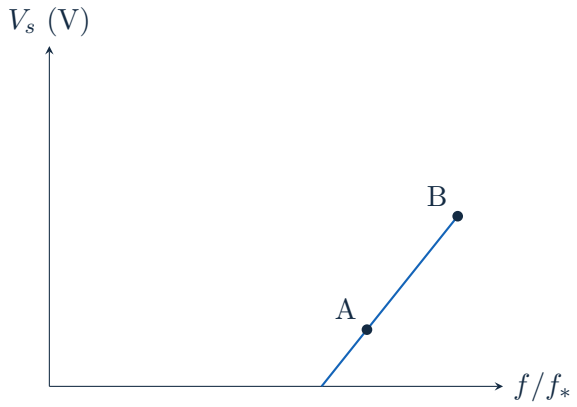
(C) $\frac{9}{5}$ (D) $\frac{11}{10}$

6. A nonrelativistic particle of fixed charge and mass moves perpendicular to a uniform magnetic field. Its kinetic energy is multiplied by 16 and the magnetic field is multiplied by 3. Find the new trajectory radius divided by the original radius.

(A) 4 (B) $\frac{2}{3}$

(C) $\frac{1}{3}$ (D) $\frac{4}{3}$

7. For a metal, the stopping-potential graph has points A $(7f_*, 3\,\text{V})$ and B $(9f_*, 9\,\text{V})$. The electronic gain of the vertical recording channel is later found to be 3: its recorded voltage is 3 times the true stopping voltage. What is hf_*/e in volts?



(A) $\frac{1}{9}$ V (B) 2 V

(C) 1 V (D) 9 V

8. An astronomical telescope in normal adjustment has magnifying-power magnitude 8 and tube length 54 cm. Find the eyepiece focal length.

(A) 48 cm (B) 6 cm

(C) $\frac{27}{4}$ cm (D) $\frac{2}{3}$ cm

9. A series RLC circuit has $R = 60$ ohms, $X_L = 30$ ohms, and $X_C = 70$ ohms at the source frequency. An ideal source supplies 400 V rms. Find rms current.

(A) 4 A (B) $\frac{20\sqrt{94}}{47}$ A

(C) $\frac{20\sqrt{13}}{13}$ A (D) $\frac{10\sqrt{34}}{17}$ A

10. A rod of length 7 m pivots at its lower end. A peg constrained to move horizontally stays in contact with its upper tip. At an instant the rod makes an angle with $\sin\theta = 3/5$ to the horizontal and the tip's horizontal speed has magnitude 3 m/s. Find the rod's angular-speed magnitude.

(A) 3 rad/s (B) $\frac{12}{35}$ rad/s

(C) $\frac{9}{35}$ rad/s (D) $\frac{5}{7}$ rad/s

11. A 8 kg block on a rough horizontal table is tied over an ideal pulley to a hanging 4 kg mass. The hanging mass moves downward and $\mu_k = 1/5$. Find string tension. Take $g = 10$ m/s^2.

(A) 80 N (B) 32 N

(C) 40 N (D) 48 N

12. A ball is dropped from 28 m onto a horizontal fixed floor. At every impact the coefficient of restitution is $\frac{2}{5}$. Neglect air resistance. What maximum height does it reach after its second impact?

(A) $\frac{7168}{390625}$ m (B) $\frac{112}{25}$ m

(C) $\frac{448}{625}$ m (D) 28 m

13. A plane wave in vacuum travels along $+x$. At one instant its electric field is $240\hat{k}$ V/m. If the electric amplitude is 4 times this instantaneous value, which gives the instantaneous magnetic field? Use $c = 3 \times 10^8$ m/s.

(A) $+80 \times 10^{-8}\hat{j}$ T (B) $+80 \times 10^{-8}\hat{i}$ T

(C) $-320 \times 10^{-8}\hat{j}$ T (D) $-80 \times 10^{-8}\hat{j}$ T

14. An ideal mixture contains 4 moles of monatomic gas and 3 moles of rigid diatomic gas; vibrations are frozen. For this mixture, use $v_s = \sqrt{\gamma RT/\bar{M}}$ and $v_{rms,eff} = \sqrt{3RT/\bar{M}}$. Find $v_s/v_{rms,eff}$.

(A) $\frac{\sqrt{41}}{9}$ (B) $\frac{\sqrt{123}}{9}$

(C) $\frac{9\sqrt{41}}{41}$ (D) $\frac{41}{81}$

15. A lossless coating of refractive index $\frac{5}{2}$ lies between air and a substrate of still higher index. For normally incident vacuum wavelength 400 nm, find the smallest positive film thickness giving destructive interference between its two reflected rays. Assume reflections at both boundaries each gain a phase reversal.

(A) 20 nm (B) 40 nm

(C) 16 nm (D) 120 nm

16. The 4th harmonic of an open pipe of length 7 m equals the 4th allowed resonance of a closed pipe. Both have the same sound speed and negligible end correction. Find the closed-pipe length.

(A) $\frac{49}{4}$ m (B) 4 m

(C) $\frac{49}{8}$ m (D) $\frac{63}{8}$ m

17. A triangular p-V cycle has vertices (V_0, p_0), $(V_0 + 3\,\text{L}, p_0)$ and $(V_0, p_0 + 3\,\text{kPa})$. Traversed in that listed order and then back, find net work done by the gas.

(A) -9 J (B) 0 J

(C) $\frac{9}{2}$ J (D) $\frac{-9}{2}$ J

18. A dipole of fixed moment 5 C m is rotated quasistatically from 60° to 120° relative to a uniform electric field 2 N/C. Find external work.

(A) $10\sqrt{3}$ J (B) 10 J

(C) 5 J (D) $\frac{10\pi}{3}$ J

19. A uniform-density spherical planet has radius R and surface gravitational acceleration g. At what depth below the surface is gravity $g/9$? Give the depth as a fraction of R.

(A) $\frac{8}{9}$ (B) 1

(C) $\frac{1}{9}$ (D) $\frac{2}{3}$

20. A straight conducting rod of length 1 m rotates in a plane perpendicular to a uniform 4 T field about one end at angular speed 5 rad/s. Find the emf magnitude between its ends.

(A) 10 V (B) 20 V

(C) −10 V (D) 0 V

21. Two ferromagnetic samples have similar remanent flux density. Their coercive fields are 600 A/m and 4200 A/m. A reversing field of magnitude 2400 A/m will act in service. Which sample is better at retaining its original magnetisation against this field, based only on the stated coercivities?

(A) Neither can have nonzero remanence. (B) Both are necessarily demagnetised equally.

(C) The sample with coercive field 600 A/m. (D) The sample with coercive field 4200 A/m.

22. A wheel initially rotates at 11 rad/s. A constant opposing angular acceleration brings it to rest after 6 s. What angle does it turn through during the last half of this stopping interval?

(A) $\frac{33}{4}$ rad (B) 66 rad

(C) $\frac{99}{4}$ rad (D) $\frac{33}{2}$ rad

23. A network is defined by $P = \mathrm{NAND}(A, B)$, $Q = \mathrm{NAND}(A, P)$ and $Y = \mathrm{AND}(Q, B)$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 1$ and $A = 1$?

(A) 0 (B) 2

(C) 3 (D) 1

24. A mass 2 kg is attached to two massless springs in series with spring constants 80 N/m and 160 N/m. Find the small-oscillation period on a smooth horizontal surface.

(A) $\frac{\sqrt{15}\pi}{15}$ s (B) $\frac{\sqrt{10}\pi}{10}$ s

(C) $\frac{\sqrt{15}}{20}$ s (D) $\frac{\sqrt{15}\pi}{10}$ s

25. A spherical blackbody's absolute temperature is multiplied by 2 while its radius is divided by 2. Neglect incoming background radiation. Find its emitted-power multiplier.

(A) $\frac{1}{4}$ (B) 8

(C) 4 (D) 16

26. A 6 ohm resistor is connected in parallel with an unknown resistor across 12 V. The source supplies $\frac{144}{5}$ W in total. Find the unknown resistance.

(A) $\frac{360}{67}\,\Omega$ (B) $1\,\Omega$

(C) $30\,\Omega$ (D) $11\,\Omega$

27. A capacitor C initially charged to 40 V is connected positive-to-positive in parallel with an uncharged capacitor 3C; the battery is disconnected. Find the fraction of initial electrostatic energy lost.

(A) $\frac{9}{16}$ (B) $\frac{3}{4}$

(C) 1 (D) $\frac{15}{16}$

28. A quantity with the dimensions of length is written $G^a h^b c^d$. Given $[G] = M^{-1}L^3T^{-2}$, $[h] = ML^2T^{-1}$, $[c] = LT^{-1}$, find $a + 2b - d$.

(A) $\frac{5}{2}$ (B) 3

(C) $\frac{3}{2}$ (D) 2

29. Two coherent beams yield maximum-to-minimum interference intensity ratio $\frac{49}{25}$. Find the larger individual beam intensity divided by the smaller.

(A) 36 (B) 49

(C) $\frac{1}{36}$ (D) $\frac{1369}{144}$

30. A silicon diode and a 6 ohm resistor are in series across 6 V with the diode anode toward the positive terminal. In the stated piecewise model the diode is open in reverse bias and has constant 0.7 V drop in forward conduction. Find the current.

(A) $\frac{53}{60}$ A (B) $\frac{159}{5}$ A

(C) $\frac{67}{60}$ A (D) 1 A

Chemistry

31. At the same external pressure,choose the higher normal-boiling member in each pair:water and hydrogen sulfide;n-butane and isobutane.

(A) Water; n-butane.

(B) Equal in both pairs because molecular formula or group membership alone fixes boiling point.

(C) Hydrogen sulfide; n-butane.

(D) Water; isobutane.

32. OxideI contains8.00gX and4.00gO;oxideII contains4.00gX and4.00gO. Xhasatomicmass32,O16. Which pair(formulaI,formulaII)and fixed-X oxygen-mass ratio is consistent?

(A) XO,XO_2;oxygenII:I=2:1. (B) XO,XO;oxygenII:I=1:1.

(C) X_2O,XO_2;oxygenII:I=4:1. (D) XO_2,XO;oxygenII:I=1:2.

33. Water hardness is due only to dissolved $Ca(HCO_3)_2$ and $CaSO_4$. Which treatment statement correctly distinguishes the two contributions?

(A) Filtration of the clear untreated solution removes dissolved hardness ions without any precipitation.

(B) Boiling removes the bicarbonate contribution; washing soda can remove the remaining Ca2+ as carbonate.

(C) Boiling alone necessarily removes every dissolved sulfate and chloride quantitatively.

(D) Adding NaCl precipitates both calcium and magnesium because all chlorides are insoluble.

34. At fixed temperature a single adsorbent obeys x/m=k $p^{(1/n)}$. In experiment B the adsorbent mass is 2 times that in A and pressure is 4 times that in A. Total adsorbed amount is 4 times that in A. What is 1/n?

(A) 3/2 (B) 1/2

(C) 1/4 (D) 2

35. An octahedral complex has high-spin d^4 configuration and $\Delta_o = 150$ kJ mol^{-1}. Ignoring pairing energy,its d-electron energy relative to the unsplit barycentre,in kJ mol^{-1},is

(A) 60 (B) -90

(C) 90 (D) -240

36. Compare H_2O and D_2O as nuclear-reactor moderators. Which statement preserves the distinction between mass/nuclear effects and electronic structure?

(A) An isotope with more neutrons must have a different number of protons and belong to another element.

(B) Both can slow neutrons,but D_2O has much smaller neutron absorption and is used as heavy-water moderator.

(C) Every physical property and reaction rate is identical because isotopes have the same atomic number.

(D) Changing H to D adds an electron and therefore changes the oxidation state of every bonded atom.

37. A student assigns four bonding pairs and one lone pair to the central atom of SF_4. After independently checking valence-electron count,which combined molecular shape/domain count is consistent?

(A) See-saw; five electron domains.

(B) Square planar; six electron domains.

(C) Trigonal pyramidal; five electron domains.

(D) Tetrahedral; four electron domains.

38. At fixed T,a reaction has K=4 and currently Q=4. Which joint conclusion is correct for this equilibrium mixture?

(A) $\Delta_r G = 0$ while$\Delta_r G^\circ < 0$;equilibrium does not require K=1.

(B) $\Delta_r G^\circ > 0$ because K exceeds1.

(C) Both$\Delta_r G$ and$\Delta_r G^\circ$ must be zero because the reaction is reversible.

(D) $\Delta_r G < 0$ because its standard value is negative,even when Q=K.

39. A freshly assayed 20-volume H_2O_2 solution releases 20 L O_2 at STP per litre on complete decomposition. Only80.0% of the evolved oxygen reaches a burner that requires0.560 L O_2 at STP. The solution volume needed,in mL,is

(A) 70

(B) 35

(C) 14

(D) 28

40. Reduction potentials M^{2+}/M for metals A,B,C,D are respectively $-\frac{3}{5}$, $-\frac{1}{5}$, $\frac{2}{5}$, $\frac{9}{10}$V. Which paired claim is correct under standard conditions?

(A) A cannot reduce D^{2+} because its own reduction potential is negative.

(B) D is the strongest reducing metal because its reduction potential is greatest.

(C) A is the strongest reducing metal;D^{2+} oxidises A spontaneously.

(D) A is the strongest oxidising ion because the corresponding metal is most reducing.

41. An ideal H_2/N_2 mixture atT has mean molar mass$\frac{43}{2}$g/mol ($H_2$2,$N_2$28). Define the ensemble mean of v^2 by molecule-number weighting. Ratio of this mean to pureN_2at the sameT is

(A) $\frac{56}{43}$

(B) $\frac{43}{56}$

(C) $\frac{17}{4}$

(D) $\frac{\sqrt{17}}{2}$

42. Compounds A and B have molecular formulae $(CH_2O)_3$ and $(CH_2O)_4$. Pure equal-mass samples are compared. Which joint statement is necessarily true,independently of functional-group identity?

(A) A contains $\frac{3}{4}$ times as many molecules as B,and has a different oxygen mass percentage.

(B) The samples contain equal total oxygen-atom amounts,but A contains $\frac{4}{3}$ times as many molecules as B.

(C) The oxygen-atom amounts are in ratio $\frac{3}{4}$,while molecule amounts are equal.

(D) Both samples contain equal molecule amounts because their empirical formulae agree.

43. A mixture contains $\frac{7}{100}$ mol H_2SO_5 and $\frac{3}{50}$ mol $H_2S_2O_8$. Their structures contain one O–O link per molecule. All peroxide links react quantitatively with excess iodide in acid; each such link accepts two electrons. What amount of I_2, in mol, is liberated?

(A) $\frac{13}{50}$ (B) $\frac{13}{100}$

(C) $\frac{1}{5}$ (D) $\frac{19}{100}$

44. Two glucopyranoses differ only in the configuration at the hemiacetal carbonC1. Which description and solution behaviour fit?

(A) Anomers; they may interconvert through the open-chain form. (B) Identical forms; no new centre was created on cyclisation.

(C) Constitutional isomers; the carbon skeleton changes. (D) Enantiomers; all stereocentres are inverted.

45. For a reaction,$\log_{10}K_p$ is 2 at300 K and 7 at600 K. The plot is linear in1/T. Using2.303forln10,find the positive magnitude$|\Delta H^\circ|/R$ in K.

(A) $\frac{6909}{2}$ (B) 13818

(C) 6909 (D) 3000

46. Compare neutral H,D and T atoms. Which statement preserves the distinction between mass/nuclear effects and electronic structure?

(A) All have one electron and one proton,but neutron counts are0,1 and2.

(B) Changing H to D adds an electron and therefore changes the oxidation state of every bonded atom.

(C) Every physical property and reaction rate is identical because isotopes have the same atomic number.

(D) An isotope with more neutrons must have a different number of protons and belong to another element.

47. Compare4p,5s,3d,4s using the stated n+l rule with lower n breaking ties. Which complete order is correct?

(A) 4s<4p<3d<5s. (B) 3d<4s<4p<5s.

(C) 4s<3d<4p<5s. (D) 4s<5s<4p<3d.

48. Nonrelativistic ions A,B start from rest. B has 9 times the mass and 2 times the positive charge of A,and is accelerated through 2 times A's potential difference. Ratioλ_B/λ_A is

(A) $\frac{1}{9}$ (B) 6

(C) $\frac{\sqrt{2}}{6}$ (D) $\frac{1}{6}$

49. For CO_3^{2-},use ordinary octet Lewis structures and resonance. What is the pair (average bond order of three equivalent C-O bonds,total lone pairs on all atoms of that same named species in one contributing structure)?

(A) (1, 8).
(B) (2, 9).
(C) ($\frac{4}{3}$, 8).
(D) ($\frac{4}{3}$, 7).

50. For propanone and benzophenone,which structural-reactivity comparison is correct under ordinary enolate/aldol conditions?

(A) propanone self-aldolises;benzophenone does not
(B) only benzophenone self-aldolises
(C) neither self-aldolises
(D) both self-aldolise

51. An ideal mixture initially has$\frac{1}{10}$molA and$\frac{2}{5}$molB. A selective step removes half ofB and noA. At the new equilibriumT,V,totalpressure is4bar. Partialpressure ofA,in bar,is

(A) $\frac{4}{3}$
(B) 2.0
(C) $\frac{4}{5}$
(D) 12

52. Acidic permanganate oxidises SO_2 to HSO_4^- and becomes Mn^{2+}. With excess SO_2, $\frac{1}{25}$ mol MnO_4^- and only $\frac{1}{100}$ mol available H^+ are supplied. Count the net H^+ in the balanced reaction. Maximum Mn^{2+} formed,in mol,is

(A) $\frac{1}{25}$
(B) $\frac{1}{200}$
(C) $\frac{1}{800}$
(D) $\frac{1}{50}$

53. For one reaction,ln K is7 at300 K and2 at600 K. Assume ln K is linear in1/T. At400 K,ln K equals

(A) 7
(B) 2
(C) $\frac{11}{2}$
(D) $\frac{9}{2}$

54. A photographic fixer contains $\frac{3}{50}$ mol free $S_2O_3^{2-}$ and receives $\frac{1}{25}$ mol unexposed AgBr. Use $AgBr+2S_2O_3^{2-}\rightarrow[Ag(S_2O_3)_2]^{3-}+Br^-$. Treat complexation as complete until a reagent is exhausted. AgBr remaining,in mol,is

(A) $\frac{1}{25}$
(B) 0
(C) $\frac{1}{50}$
(D) $\frac{1}{100}$

55. A $\frac{43}{125}$ g volatile sample displaces $\frac{1280}{13}$ mL wet air at 300 K and 790 mmHg. Water-vapour pressure is 30 mmHg. Using STP=273 K,760 mmHg and22.4 L/mol,the sample molar mass, rounded to the nearest whole g/mol, is

(A) 164
(B) 83
(C) 2179
(D) 86

56. A model reproduces a positive central nucleus and most alpha particles passing through foil,but supplies no allowed energies. Which additional observation remains unexplained by that information alone?

(A) The fact that most alpha particles pass through,which the model explicitly reproduces.
(B) Discrete line spectra and long-term electronic stability.
(C) The existence of a central positive charge,which the stated scattering already supports.
(D) The alpha-particle charge,which follows from electronic quantum numbers.

57. Bond energies $D(X_2):D(Y_2):D(XY)=1:3:4$. Forming2.00 mol XY(g) from the gaseous elements releases 320 kJ. $D(Y_2)$,in kJ/mol,is

(A) 120.0 (B) 480

(C) 240 (D) 320

58. Acetic acidC2H4O2 fitsCx(H2O)y, but is not a carbohydrate. Which conclusion follows?

(A) Carbonhydrates contain no carbonyl chemistry.

(B) Every carbohydrate must be an aldehyde with noOH.

(C) Every compound with C: H: O=1:2:1 is a carbohydrate.

(D) The empirical hydrate pattern alone is insufficient for carbohydrate classification.

59. A1.00 L vessel initially contains $\frac{1}{5}$ mol A,$\frac{2}{5}$ mol B and $\frac{1}{25}$ mol C,with no D. For $A+2B\rightleftharpoons 2C+D$, equilibrium C amount is $\frac{3}{25}$ mol. Numerical K_c using molar concentrations is

(A) $\frac{9}{64}$ (B) $\frac{9}{128}$

(C) $\frac{9}{256}$ (D) $\frac{3}{112}$

60. Classify the two operations: I. Removal of dissolved electrolyte through a semipermeable membrane. II. addition of a little suitable electrolyte to convert a fresh precipitate into a sol. Select the correct paired classification.

(A) Peptisation; dialysis (B) Coagulation; purification

(C) Preparation by chemical reaction; Brownian motion (D) Dialysis; peptisation

English Proficiency

61. Choose the most coherent order of the four sentences. A. The office replaced a technical notice with simpler wording. B. The writer changed the passive sentence to name the responsible person directly. C. The final notice reduced uncertainty about action, not just difficulty in vocabulary. D. A test reader still misunderstood who was required to act.

(A) A-D-B-C (B) A-B-D-C

(C) D-A-B-C (D) A-D-C-B

62. Read the sentence: The map was accurate, showing the paths exactly where they were found. Choose the opposite in meaning of 'accurate' as used in the sentence.

(A) detailed (B) inaccurate

(C) recent (D) precise

63. Read the sentence: The factory was compliant with the new rule, having met every stated requirement. Choose the closest meaning of 'compliant' as used in the sentence.

(A) unchanged by regulation (B) cooperative in spirit only

(C) approved for all activities (D) conforming

64. Complete: The investigator was asked to distinguish the observation _____ the proposed explanation.

(A) from (B) to

(C) with (D) at

65. Choose the active-voice equivalent that retains the tense, participants and all stated restrictions. The parcel was not opened by the assistant until the owner arrived.

(A) The owner did not open the parcel until the assistant arrived. (B) The assistant did not open the parcel until the owner arrived.

(C) The assistant opened the parcel before the owner arrived. (D) The assistant had not opened the parcel when the owner left.

66. Read the sentence: The team's shared purpose acted as a compass when several methods were available. What does the figurative expression mean in this context?

(A) It supplied a precise timetable for the work. (B) It guaranteed that every method would succeed.

(C) It guided choices without prescribing every individual step. (D) It prevented the team from changing direction.

67. The sentence can reasonably be understood in two different ways even though every word is readable. Which noun identifies the problem?

(A) falsehood (B) illegibility

(C) redundancy (D) ambiguity

68. Read the sentence: The reception was lavish, with elaborate decorations and far more food than the guests required. Choose the closest meaning of 'lavish' as used in the sentence.

(A) carefully planned (B) hospitable

(C) extravagant (D) widely attended

69. Choose the most coherent order of the four sentences. A. The desk lamp made the page bright but left the screen surrounded by glare. B. Moving the lamp changed the angle at which its light reached the screen. C. The reflection disappeared even though enough light remained on the page. D. The user first reduced the brightness without solving the reflection.

(A) A-D-B-C (B) D-A-B-C

(C) A-B-D-C (D) A-D-C-B

70. Use the standard contemporary negative adjective. The figures do not match the records, so the summary is _____.

(A) disaccurate (B) inaccurate

(C) imaccurate (D) unaccurate

Logical Reasoning

71. A community register uses the following three categories: guides, farmers, artists. Its complete relevant assertions are in this short paragraph; do not import everyday assumptions. No artists are guides. No guides are artists. All artists are guides. Conclusions: I. All artists are farmers. II. All farmers are guides. Which conclusions follow necessarily?

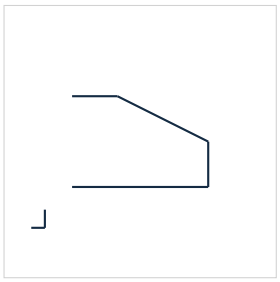
(A) Neither I nor II

(B) Only I

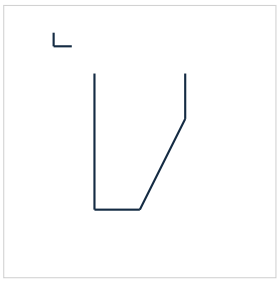
(C) Only II

(D) Both I and II

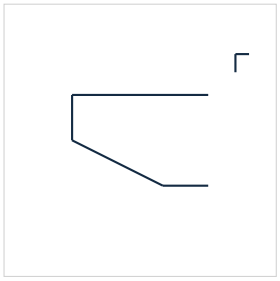
72. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

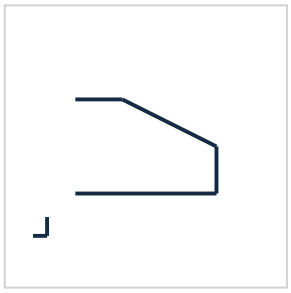


bottom left

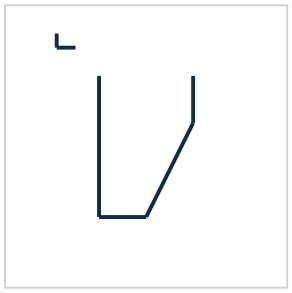
?

bottom right

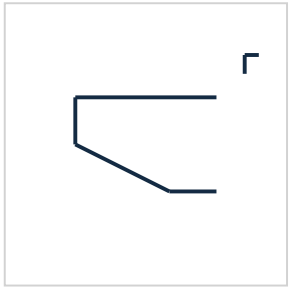
(A) Pattern Z



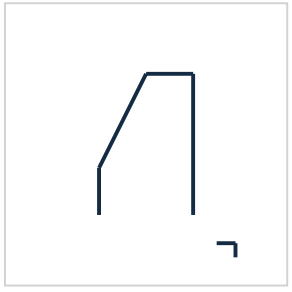
(B) Pattern P



(C) Pattern M



(D) Pattern N



73. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: PWJJ, MTNN, ?, GNVV, DKZZ, AHDD.

(A) RRQJ (B) IPQQ

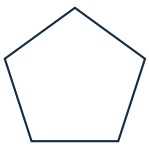
(C) JQRR (D) KRSS

74. The model triads are $(11, 27, 16)$ and $(13, 31, 18)$. In the intended rule the middle number is the sum of the outer numbers, and the outer-number difference is the same fixed positive value in every triad. Which option satisfies both relations?

(A) $(10, 26, 16)$ (B) $(10, 27, 16)$

(C) $(10, 26, 15)$ (D) $(10, 25, 15)$

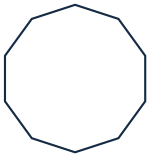
75. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

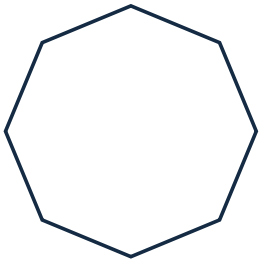


B

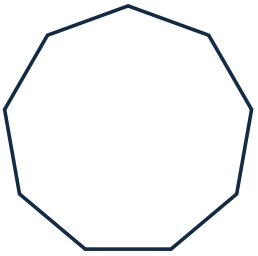


C

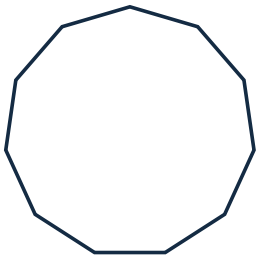
(A) Pattern P



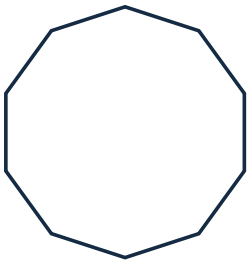
(B) Pattern U



(C) Pattern V



(D) Pattern L



76. All five people have different heights. Hari is shorter than Gita. Bina is shorter than Farah. Asha is shorter than Hari. Farah is shorter than Asha. If Bina is left out, who is 3rd shortest among the remaining people?

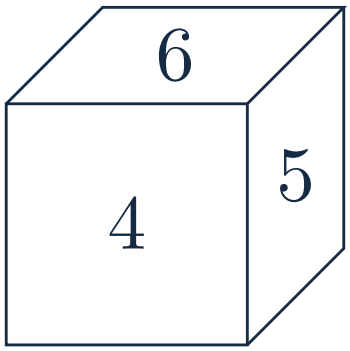
(A) Hari (B) Farah

(C) Asha (D) Gita

77. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 29 and Z has code 4. Decode the separated values 28, 12, 21, 27, 19.

(A) BRICK
(B) CSJDL
(C) KCIRB
(D) AQHBJ

78. This standard die has opposite labels summing to seven. Roll it once to the right, so its present top face becomes the right face. Give the new (top, front) labels.

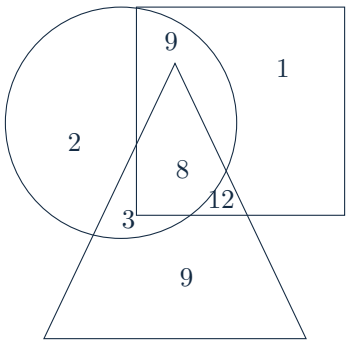


(A) $(2, 4)$
(B) $(6, 4)$
(C) $(2, 3)$
(D) $(5, 4)$

79. $X\%Y$ means X is the mother of Y; $X\#Y$ means X is the brother of Y; $X@Y$ means X is the father of Y. All names denote distinct people; a brother or sister is a full sibling, sharing both parents. Given $A\%H$ $H\#B$ $B@T$, how is A related to T?

(A) Mother
(B) Maternal grandfather
(C) Paternal grandmother
(D) Father

80. Circle=A, triangle=B, square=C. Each printed number is the count in that exact region. Find the total inside A but outside B, whether inside or outside C.



(A) 12
(B) 2
(C) 9
(D) 11

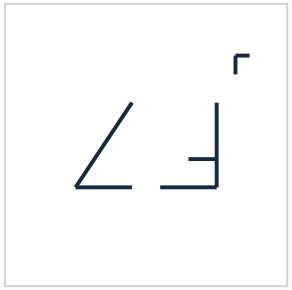
81. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



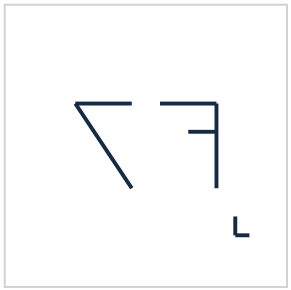
(A) Pattern L



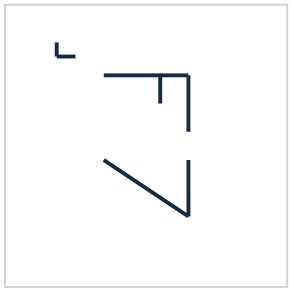
(B) Pattern U



(C) Pattern Z



(D) Pattern V



82. Each rectangle is drawn to the same local scale. In three numbered figures the interior straight line divides the rectangle into equal areas. Which does not?



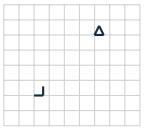
(A) 2
(B) 1
(C) 3
(D) 4

83. Preserve the exact relation and its order: Genuine : Authentic :: ? : Old. Choose the missing term.

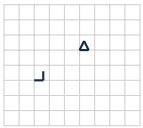
(A) Ancient (B) Silent

(C) Rapid (D) Begin

84. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



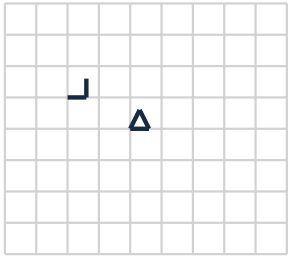
1



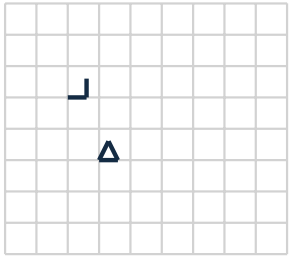
2

3

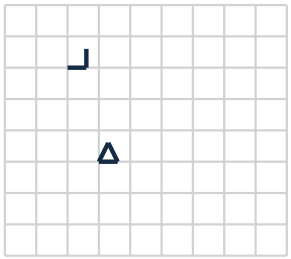
(A) Pattern K



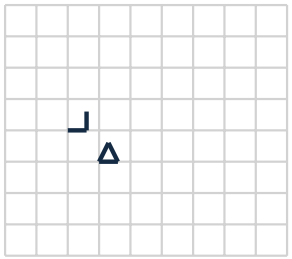
(B) Pattern L



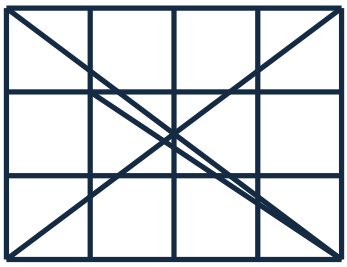
(C) Pattern M



(D) Pattern W



85. How many triangles are formed by all the drawn segments?



(A) 66 (B) 54

(C) 67 (D) 63

86. 8 people sit around a circle facing its centre. Clockwise order fixes the convention; immediate left is clockwise. Dev is 1 places clockwise from Asha. Bina is 2 places clockwise from Asha. Farah is 3 places clockwise from Asha. Hari is 4 places clockwise from Asha. Esha is 6 places clockwise from Asha. Gita is not adjacent to Hari. Who sits immediately clockwise from Esha?

(A) Asha (B) Gita

(C) Bina (D) Charu

87. The age table uses the same two people throughout:

	younger	older
now	x	$x+11$
4 years ago	$x-4$	$x+7$

. The product of their ages 4 years ago was 840. What will the older person's age be 4 years from now?

(A) 43 (B) 32

(C) 54 (D) 39

88. In this series the first differences are a positive integer multiple of consecutive squares $1, 4, 9, 16, \ldots$. Find the missing internal term: 11, 15, 31, ?, 131, 231, 375.

(A) 71 (B) 67

(C) 63 (D) 68

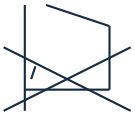
89. Consider only these statements; universal statements alone do not imply existence. Some joggers are joggers; All hikers are bakers. Conclusions: I. All joggers are hikers. II. Some joggers are not hikers. Which conclusion or complementary alternative must hold?

(A) Only I (B) Only II

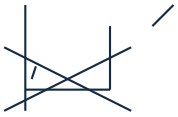
(C) Either I or II follows, but neither is individually certain (D) Both I and II

90. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

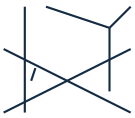




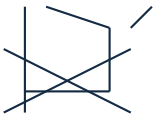
host 1



host 2



host 3



host 4

(A) 3 (B) 4

(C) 2 (D) 1

Mathematics

91. y>0 satisfies 2yy′+6y^2=0, y(0)=sqrt(3). Find y^2(1).

(A) $\frac{\sqrt{3}}{e^3}$ (B) $\frac{3}{e^6}$

(C) $3e^6$ (D) $\frac{3}{e^3}$

92. How many integers x satisfy |(x-6)+(x-9)|<|x-6|+|x-9|?

(A) 4 (B) 3

(C) 2 (D) 8

93. An invertible 4×4 matrix A has det A=7. Find det(adj A).

(A) 28 (B) 2401

(C) 343 (D) 7

94. A=(2,5,4), B=(4,5,9). Find the squared length of the component of AB perpendicular to direction (1,2,2).

(A) 13 (B) 29

(C) 16 (D) 25

95. Let R=[sin($\frac{\pi}{12}$)+sin($\frac{\pi}{4}$)]/[cos($\frac{\pi}{12}$)+cos($\frac{\pi}{4}$)]. Find (1-R^2)/(1+R^2).

(A) $\frac{3}{2}$ (B) $\frac{1}{2}$

(C) $-\frac{1}{2}$ (D) $\frac{\sqrt{3}}{2}$

96. For integer n, consider: "If 24 divides n, then 6 divides n." Which is its contrapositive?

(A) 24 divides n and 6 does not divide n. (B) If 6 does not divide n, then 24 does not divide n.

(C) If 6 divides n, then 24 divides n. (D) If 24 does not divide n, then 6 does not divide n.

97. An observer at height 6 sees a tower foot at depression 30° and top at elevation 60°. Find the tower height.

(A) 18 (B) 12

(C) 24 (D) $6\sqrt{3}$

98. The four lines ±x/14±y/4=1 form a rhombus with vertices on the coordinate axes. Find its area.

(A) 224 (B) 28

(C) 56 (D) 112

99. Event E has likelihood $\frac{2}{5}$ under source A and $\frac{3}{5}$ under source B. Its total probability is $\frac{8}{15}$. If the source is A with probability r, find r.

(A) $\frac{2}{3}$ (B) $\frac{2}{5}$

(C) $\frac{8}{15}$ (D) $\frac{1}{3}$

100. If α,β are roots of x^2-19x+84=0, find $\alpha^2+\beta^2$ without solving the quadratic.

(A) 168 (B) 7056

(C) 277 (D) 193

101. How many integers k make $x^2 + kx + 49 = 0$ have two distinct non-real roots?

(A) 27 (B) 28

(C) 29 (D) 13

102. In $(3x^2 - x^{-1})^8$, find the ratio of the coefficient of x^4 to that of x^7.

(A) $-\frac{5}{12}$ (B) $-\frac{12}{5}$

(C) $\frac{5}{4}$ (D) $\frac{5}{12}$

103. A circular sector has radius 8 and arc length $\frac{4\pi}{3}$. Find its area.

(A) $\frac{16\pi}{3}$ (B) 64π

(C) $\frac{\pi^2}{9}$ (D) $\frac{128\pi}{3}$

104. f(x)=3x+5. Find f(f(5)).

(A) 65 (B) 70

(C) 60 (D) 55

105. z_1=6+5i, z_2=11+5i, z_3=11+10i. Find the smaller angle between z_2-z_1 and z_3-z_1.

(A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{4}$

(C) 0 (D) $\frac{\pi}{4}$

106. X takes values [-2, 0, 5] with probabilities 1/6,1/3,1/2. Let Y=2X+7. Find $E(Y^2)$-$[E(Y)]^2$.

(A) $\frac{746}{9}$ (B) $\frac{305}{18}$

(C) $\frac{158}{3}$ (D) $\frac{305}{9}$

107. The tangent to $y=x^2+2x$ at P has slope 12. Find the slope of the normal at P.

(A) $-\frac{1}{12}$ (B) 12

(C) -144 (D) $\frac{1}{12}$

108. $f(x)=x^3+3x$, $g=f^{-1}$. Find $g''(234)$.

(A) $\frac{4}{151959}$ (B) $-\frac{4}{151959}$

(C) $\frac{1}{36}$ (D) $-\frac{4}{1369}$

109. For common tangents to $y^2=16x$ and $x^2+y^2=\frac{8}{15}$, find the sum of the squared slopes.

(A) 0 (B) 5

(C) -5 (D) 10

110. Maximize Z=3x+4y subject to $x\geq 0$, $y\geq 4$, $x+y\leq 8$, $x+2y\leq 11$.

(A) 27 (B) 22

(C) 28 (D) 25

111. a=(4,0,0), b=(0,3,0), c=(0,0,2), p=a+b. Evaluate [p×(b×c)]·(a+c).

(A) −36 (B) 36

(C) 0 (D) −180

112. For f(x)=|x-3|/(x-7), on which maximal open intervals is f strictly decreasing?

(A) $(-\infty, 3)$ (B) $(3, \infty)$

(C) $(3, 7) \cup (7, \infty)$ (D) $(-\infty, 7)$

113. For the loop $y^2=x^2(7-x)$, find the area of the portion with $x \geq 7/2$.

(A) $\frac{196\sqrt{7}}{15}$ (B) $\frac{343\sqrt{14}}{30}$

(C) $-\frac{343\sqrt{14}}{30} + \frac{392\sqrt{7}}{15}$ (D) $\frac{343\sqrt{14}}{60}$

114. The same loaded die has odd-face probability $\frac{3}{4}$. Three independent throws are made. Given that the first two have an even sum, find the probability that the total of all three is odd.

(A) $\frac{3}{4}$ (B) $\frac{3}{8}$

(C) $\frac{9}{16}$ (D) $\frac{1}{4}$

115. Let $a_j = 1/(3 + 2(j-1))$. Evaluate $\sum_{j=1}^{6} a_j a_{j+1}$.

(A) $\frac{6}{13}$ (B) $\frac{5}{39}$

(C) $\frac{2}{15}$ (D) $\frac{4}{15}$

116. Vectors a,b satisfy $|a|^2=26$, $|b|^2=17$, a·b=9. Find $|2a-b|^2$.

(A) 157 (B) 121

(C) 85 (D) 51

117. $f(x)=x^2+5x$ for $x \leq 6$, and $f(x)=px+q$ for $x>6$. If f is differentiable at 6, find p-q.

(A) −19 (B) 66

(C) 53 (D) 17

118. For $16x^2+64y^2=1024$, find the distance between its two directrices.

(A) 16 (B) $\frac{16\sqrt{3}}{3}$

(C) $\frac{32\sqrt{3}}{3}$ (D) $8\sqrt{3}$

119. The line Re(z)=4 divides the disc $|z| \leq 8$. Find larger-segment area minus smaller-segment area.

(A) $16\sqrt{3} + \frac{128\pi}{3}$ (B) $-16\sqrt{3} + \frac{64\pi}{3}$

(C) $32\sqrt{3} + \frac{64\pi}{3}$ (D) $-\frac{64\pi}{3} - 32\sqrt{3}$

120. Evaluate lim as $n \to \infty$ of $(6/n)\Sigma(6j/n)^2$, j=1,...,n.

(A) 12 (B) 72

(C) 108 (D) $\frac{1}{3}$

121. Chords of $x^2+y^2=64$ pass through (16,0). Find the radius of the circle containing their midpoint locus.

(A) 8 (B) 16

(C) 64 (D) 4

122. Evaluate lim as x→0 of [sqrt(4+x)-2-x/4]/x^2.

(A) $-\frac{1}{64}$ (B) $\frac{1}{4}$

(C) $-\frac{1}{32}$ (D) $\frac{1}{64}$

123. How many solutions in $0\leq x<14\pi$ satisfy sin2x=sin x and cosx>0?

(A) 14 (B) 22

(C) 21 (D) 28

124. What is the maximum number of distinct intersection points formed by 7 circles and 3 straight lines, with no coincident curves?

(A) 129 (B) 66

(C) 84 (D) 87

125. For $A=\begin{bmatrix}3 & 1\\ 1 & 4\end{bmatrix}$ and $C=11A^{-1}$, find C_{22}.

(A) 4 (B) 3

(C) $\frac{3}{11}$ (D) -1

126. How many pairs $1\leq m<n\leq 11$ satisfy both $5 \mid (6^m+9^n)$ and $3 \mid (m+n)$?

(A) 20 (B) 30

(C) 11 (D) 10

127. Positive a,b,c form an AP with a=2. The numbers a,b,c+25/2 form a GP and b>a. Find c.

(A) −8 (B) 12

(C) 7 (D) 52

128. The system x+y+z=1, y+z=2, (k-7)z=k-11 has no solution for which k?

(A) 0 (B) 7

(C) 11 (D) −7

129. $F'(x)=\frac{(x+1)^2}{(x+2)^4}$ and F(0)=4. Find F(6)-2F(0).

(A) $-\frac{2141}{512}$ (B) $-\frac{1955}{512}$

(C) $\frac{93}{512}$ (D) $\frac{2141}{512}$

130. How many integers x satisfy $4^{-x}-(544)2^{-x}+16384\leq 0$?

(A) 3 (B) 10

(C) 4 (D) 5

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. Here L is inductance, C capacitance, R resistance, V potential, I current and t time. Which expression has the same dimensions as both $(V/I)^4$ and $(L/(RC))^4$?

(A) $(CV)^4$ (B) $(\sqrt{L/C})^4$

(C) $(RC)^4$ (D) $(LI)^4$

132. A monatomic ideal gas expands reversibly and adiabatically until its pressure becomes 1/3125 of its initial value. Find final temperature divided by initial temperature.

(A) $\frac{1}{25}$ (B) 25

(C) 1 (D) $\frac{1}{125}$

133. A particle of mass 5 kg and charge 2 C initially has speed 8 m/s. It moves through a uniform electric field 6 N/C with a net displacement 3 m along the field. A simultaneous magnetic field bends its trajectory. Neglect other forces. Find final speed.

(A) $\frac{76}{5}$ m/s (B) $\frac{6\sqrt{10}}{5}$ m/s

(C) $\frac{14\sqrt{10}}{5}$ m/s (D) 8 m/s

Chemistry

134. In the gas phase,O_2 is converted into O_2^+ by changing one electron. Under the elementary MO model,which final bond-order/unpaired-electron pair follows?

(A) Bond order $\frac{5}{2}$; 2 unpaired electrons. (B) Bond order $\frac{5}{2}$; 1 unpaired electron.

(C) Bond order 2; 2 unpaired electrons. (D) Bond order $\frac{3}{2}$; 1 unpaired electrons.

135. For A(g)⇌B(g),an equilibrium contains $\frac{3}{20}$ mol A and $\frac{3}{10}$ mol B. Remove $\frac{3}{50}$ mol B and add $\frac{3}{20}$ mol A at the same T. All remain gases in a fixed vessel. Final B amount in mol is

(A) $\frac{9}{20}$ (B) $\frac{27}{100}$

(C) $\frac{9}{25}$ (D) $\frac{6}{25}$

136. Phenol undergoes controlled mononitration with dilute nitric acid. The resulting ortho/para products are separated by steam distillation. Which isomer is enriched in the steam distillate,and why does its formation remain consistent with OH directing effects?

(A) o-Nitrophenol; OH directs ortho/para,and the ortho product forms an intramolecular hydrogen bond.

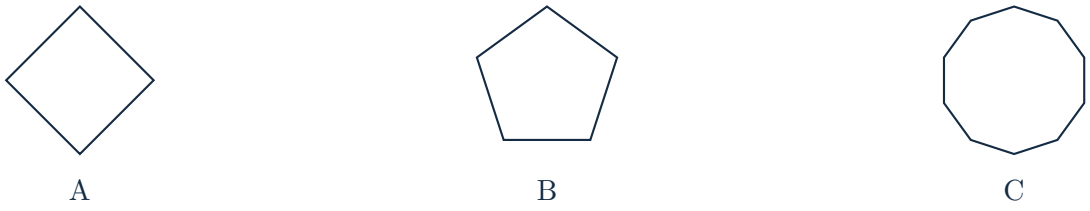
(B) p-Nitrophenol; only para substitution is allowed by OH.

(C) p-Nitrophenol; intermolecular hydrogen bonding increases its volatility.

(D) m-Nitrophenol; OH is meta directing because oxygen is electronegative.

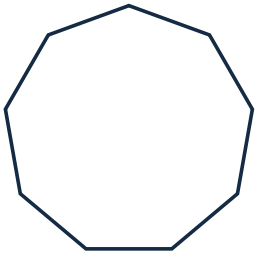
Logical Reasoning

137. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?

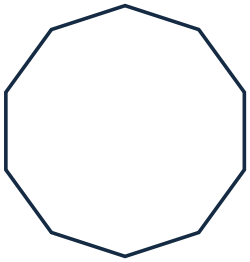


A B C

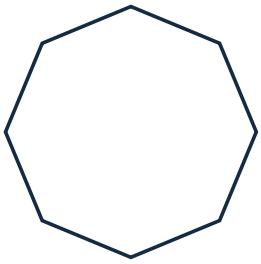
(A) Pattern M



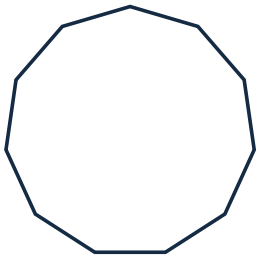
(B) Pattern U



(C) Pattern W



(D) Pattern X



138. In this code, @ means $\times$, # means $+$, & means $\div$, % means $-$. After substituting meanings, evaluate ((6#3)@(2&3)); printed parentheses retain their usual grouping.

(A) 7 (B) 6

(C) 5 (D) -6

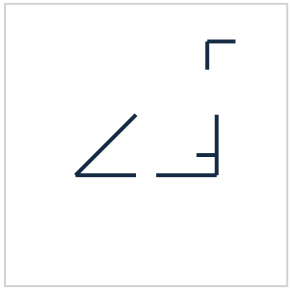
139. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



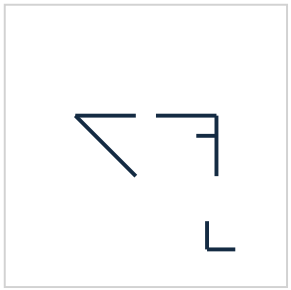
(A) Pattern Z



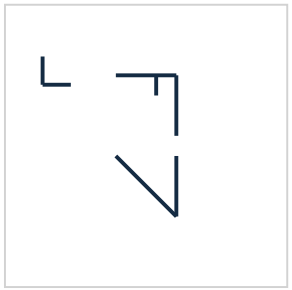
(B) Pattern P



(C) Pattern U



(D) Pattern T



Mathematics

140. $|U|=150$, $|A|=41$, $|B|=34$, $|A\cap B|=15$. Find the number of elements belonging to both A,B or to neither.

(A) 105 (B) 45

(C) 135 (D) 90

141. The curve $y=x^3+2x$ divides the disc $x^2+y^2 \leq 36$. Find the area of the part satisfying $y\geq x^3+2x$.

(A) $2+18\pi$ (B) 9π

(C) 18π (D) 36π

142. Parabola $y=6(x-3)(x-4)$ meets the axes at three distinct points. A circle passes through these points. Find its other y-intercept.

(A) $-\frac{1}{6}$ (B) $\frac{1}{6}$

(C) 1 (D) 6

BITSAT Mock 10

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A ray in a medium of index $\frac{21}{10}$ is incident on a medium of index 1. Find the sine of the critical-angle threshold for total internal reflection.

(A) $\frac{\sqrt{341}}{21}$ (B) $\frac{21}{10}$

(C) $\frac{10}{21}$ (D) 1

2. In an ideal intrinsic semiconductor at zero temperature, the band gap is 1 eV. Neglect phonons, excitons and all transitions within a band. Two beams have photon energies $\frac{4}{5}$ eV and $\frac{6}{5}$ eV. Which can create a valence-to-conduction electron-hole pair by a single-photon absorption?

(A) Only the lower-energy beam (B) Only the higher-energy beam

(C) Neither beam (D) Both beams

3. A solid sphere of radius R has uniform density $9\rho_0$. A spherical cavity of radius $R/10$ is cut with its centre at distance $R/6$ from the original centre. Find field magnitude at the cavity centre in units of $\pi G\rho_0 R$.

(A) $\frac{8}{81}$ (B) 2

(C) $\frac{8}{729}$ (D) 8

4. A projectile of mass 9 kg is launched with speed 25 m/s, $\cos\theta = 3/5$. At its highest point it splits into masses $M/3$ and $2M/3$. The lighter fragment moves horizontally forward at 20 m/s. Find the signed horizontal velocity of the heavier fragment immediately after the explosion.

(A) $\frac{25}{2}$ m/s (B) $\frac{65}{2}$ m/s

(C) $\frac{55}{2}$ m/s (D) 20 m/s

5. 6 charges each equal to q occupy vertices of a regular 6-gon of circumradius R. One charge is removed. Find the remaining field magnitude at the centre in units of kq/R^2.

(A) $\sqrt{5}$ (B) 5

(C) 1 (D) $\frac{1}{5}$

6. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 30 percent. The measured increase in its absolute temperature is $\frac{1173}{5}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) 340 K
(B) $\frac{2873}{5}$ K
(C) $\frac{34000}{169}$ K
(D) $\frac{1173}{5}$ K

7. A fixed amount of ideal gas has its absolute temperature multiplied by 5 and volume by 2. Find the final pressure divided by the initial pressure.

(A) $\frac{5}{2}$
(B) $\frac{5}{4}$
(C) 5
(D) $\frac{1}{2}$

8. A soap bubble initially of radius 3 cm is expanded quasistatically until its volume is 27 times the initial volume. Surface tension remains $\frac{7}{100}$ N/m. Find work against surface tension, in units of $\pi \times 10^{-4}$ J.

(A) $\frac{3276}{25}$
(B) $\frac{1008}{25}$
(C) $\frac{126}{25}$
(D) $\frac{252}{25}$

9. Two large parallel plates are held at potentials +400 V and −400 V, with gap 2 mm. Neglect edge effects. Find field magnitude in the gap.

(A) 400000 V/m
(B) 400 V/m
(C) 4000000 V/m
(D) 200000 V/m

10. A wheel initially rotates at 10 rad/s. A constant opposing angular acceleration brings it to rest after 3 s. What angle does it turn through during the last half of this stopping interval?

(A) 15 rad
(B) $\frac{15}{8}$ rad
(C) $\frac{15}{2}$ rad
(D) $\frac{15}{4}$ rad

11. A standing wave is $y = 2\sin(2x)\cos(4t)$ cm with x in metres and t in seconds. At a point where $2x = \pi/6$, find the particle-speed magnitude when $4t = \pi/3$.

(A) 8 cm/s
(B) $4\sqrt{3}$ cm/s
(C) $2\sqrt{3}$ cm/s
(D) $\sqrt{3}$ cm/s

12. A 3 kg block is pulled by a force of 15 N at an angle with $\cos\theta = 4/5$, $\sin\theta = 3/5$ above a horizontal floor. The block slides; $\mu_k = \frac{3}{10}$. Find the net horizontal force. Take $g = 10$ m/s^2.

(A) $\frac{87}{10}$ N
(B) 3 N
(C) $\frac{18}{5}$ N
(D) $\frac{57}{10}$ N

13. A plane wave passes through a thin prism away from edges. Separately, a pointlike aperture produces an outgoing spherical wavefront of radius 10 cm, which propagates another 8 cm in a homogeneous medium. Which pair correctly describes the prism wavefront and the later aperture wavefront?

(A) Approximately plane and tilted; spherical with radius 18 cm.

(B) Approximately plane and tilted; spherical with radius 2 cm.

(C) Approximately plane and unchanged in direction; spherical with radius 10 cm.

(D) Spherical; plane with infinite radius.

14. A uniform disc of mass 3 kg and radius 3 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) $\frac{27}{2}\,\mathrm{kg\,m^2}$ (B) $\frac{513}{32}\,\mathrm{kg\,m^2}$

(C) $\frac{405}{32}\,\mathrm{kg\,m^2}$ (D) $\frac{351}{32}\,\mathrm{kg\,m^2}$

15. Two wires of the same material and length L, with radii r and 7r, are joined in series and stretched by a force F. Find their total stored elastic energy divided by the energy stored in the thinner wire alone under force F.

(A) $\frac{1}{49}$ (B) 2

(C) $\frac{50}{49}$ (D) $\frac{49}{50}$

16. One mole of a monatomic ideal gas follows $pV^3 = \text{constant}$. Find the process molar heat capacity in units of R.

(A) $\frac{5}{2}$ (B) 1

(C) $\frac{3}{2}$ (D) $\frac{7}{6}$

17. A screw gauge advances 2.0 mm in four complete turns and has 100 circular divisions. With the jaws closed, its zero mark lies 2 divisions below the reference line; use the usual convention that a zero below the reference line gives positive zero error. With a wire, the main scale reads 7 mm and circular reading is 21 divisions. What is the corrected diameter?

(A) $\frac{1421}{200}$ mm (B) $\frac{142099}{20000}$ mm

(C) $\frac{1021}{200}$ mm (D) $\frac{1419}{200}$ mm

18. A series RLC circuit has $R = 70$ ohms, $X_L = 60$ ohms, and $X_C = 30$ ohms at the source frequency. An ideal source supplies 300 V rms. Find rms current.

(A) $\frac{15\sqrt{94}}{47}$ A (B) 3 A

(C) $\frac{3\sqrt{130}}{13}$ A (D) $\frac{15\sqrt{58}}{29}$ A

19. A particle moves with constant acceleration. It travels 43 m in the (7)th second and 49 m in the 8th second. What was its initial velocity?

(A) 49 m/s (B) 1 m/s

(C) 7 m/s (D) 4 m/s

20. A wire consists of a semicircular arc of radius 7 m carrying 3 A, joined to two straight radial segments that extend outward from its ends. Find the magnetic field magnitude at the arc centre in units of μ_0.

(A) $\frac{3(2+\pi)}{28\pi}$ (B) $\frac{3}{14}$

(C) $\frac{3}{28}$ (D) $\frac{3}{56}$

21. A Newtonian liquid of viscosity $\frac{1}{2}$ Pa s has velocity profile $v(y) = 7y + 5y^2$ in SI units. Find the shear-stress magnitude at $y = 4$ m.

(A) $\frac{47}{2}$ Pa (B) 5 Pa

(C) 54 Pa (D) $\frac{27}{2}$ Pa

22. A nucleus of mass number 28 splits into two equal nuclei. The parent has binding energy per nucleon $\frac{71}{10}$ MeV and each daughter has $\frac{15}{2}$ MeV. Ignoring other products, what energy is released in the split?

(A) $\frac{994}{5}$ MeV (B) $\frac{56}{5}$ MeV

(C) $\frac{2}{5}$ MeV (D) $\frac{28}{5}$ MeV

23. A square conducting loop in a constant perpendicular field 2 T has side length $\ell(t) = 6t$ m. Find induced-emf magnitude at $t = 4$ s, assuming the loop expands uniformly.

(A) 576 V (B) 48 V

(C) 1152 V (D) 288 V

24. A particle starts from its SHM mean position moving in the positive direction with period 5 s. Find the first time its displacement equals half its positive amplitude.

(A) $\frac{5}{24}$ s (B) $\frac{5}{12}$ s

(C) $\frac{5}{4}$ s (D) $\frac{25}{12}$ s

25. A spring of constant 20 N/m is already stretched by $\frac{2}{5}$ m. How much additional work is required to increase its extension to $\frac{7}{10}$ m quasistatically?

(A) $\frac{9}{10}$ J (B) $\frac{33}{10}$ J

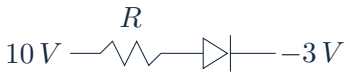
(C) $\frac{12}{5}$ J (D) $\frac{8}{5}$ J

26. A network is defined by $P = \text{NAND}(A, B)$, $Q = \text{NOR}(A, P)$ and $Y = \text{NOR}(Q, B)$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 0$ and $B = 1$?

(A) 2 (B) 3

(C) 1 (D) 0

27. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 4 kΩ to a terminal at +10 V, and its cathode connects to a terminal at -3 V. Find the current from the +10 V terminal in mA.



(A) $\frac{13}{4}$ mA (B) $\frac{123}{40}$ mA

(C) $\frac{29}{10}$ mA (D) $\frac{93}{40}$ mA

28. A uniform wire is stretched to $\frac{5}{4}$ times its original length at constant volume. Its resistivity increases by 5 percent during the process. Find final resistance divided by initial resistance.

(A) $\frac{25}{16}$ (B) $\frac{525}{256}$

(C) $\frac{21}{16}$ (D) $\frac{105}{64}$

29. A gas receives 600 J of heat while its surroundings do 100 J of work on it. It then loses 300 J of heat at fixed volume. Find its total internal-energy change.

(A) 700 J (B) 1000 J

(C) 400 J (D) 200 J

30. A projectile is launched at 30 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{24}{25}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ m/s^2.

(A) $\frac{144}{5}$ m (B) $\frac{432}{125}$ m

(C) $\frac{432}{25}$ m (D) $\frac{864}{25}$ m

Chemistry

31. Paracetamol is represented here as p-HO-C6H4-NHCOCH3. Which functional-group assignment follows directly?

(A) Phenol and amide. (B) Ether and nitrile only.

(C) Aldehyde and primary amine. (D) Carboxylic acid and ester.

32. A sample has recurring -CO-NH- groups in its backbone. Which identification and interaction are consistent with a conventional nylon fibre?

(A) Polyalkene; only C-C bonds are present. (B) Polyamide; interchain hydrogen bonding is possible.

(C) Polysaccharide; every nitrogen is a glycosidic oxygen. (D) Polyester; no N-H donor is present.

33. Excess water reacts completely with $\frac{1}{100}$ mol $CaCN_2$. Which gas identity/amount pair follows from the balanced hydrolysis?

(A) O_2, $\frac{1}{200}$ mol. (B) CO_2, $\frac{1}{100}$ mol.

(C) NH_3, $\frac{1}{100}$ mol. (D) NH_3, $\frac{1}{50}$ mol.

34. For the reaction $2M + O_2 \rightarrow 2MO$, take $\Delta H^\circ = -200$ kJ and $\Delta S^\circ = -0.200$ kJ K^{-1} over the stated range. Find ΔG° in kJ at 700 K and the sign of its slope against temperature.

(A) -200 kJ; zero slope (B) 60 kJ; negative slope

(C) -340 kJ; negative slope (D) -60 kJ; positive slope

35. Classify Na2O, ZnO, SO2, in that order, by their usual acid-base behaviour. Which sequence also predicts that the middle oxide reacts with both suitable acids and strong alkalis?

(A) Acidic; amphoteric; basic. (B) Basic; amphoteric; acidic.

(C) Amphoteric; basic; acidic. (D) Basic; acidic; amphoteric.

36. An aldopentoseFischer projection has CHO at top, CH2OH at bottom, and OH sides at C2, C3, C4 respectively R, R, R, whereR means right andL means left. Oxidise both terminal groups to COOH without changing internal configurations. Which D/L assignment for the sugar and classification of the diacid are correct?

(A) L; meso(optically inactive). (B) D; meso(optically inactive).

(C) D; chiral. (D) D/L is fixed by the sign of rotation alone.

37. Samples I,II,III contain respectively $\frac{1}{10}$ mol CO_2,$\frac{3}{20}$ mol CO and $\frac{3}{40}$ mol NH_3. Count all atoms,and count molecules separately. Which comparison is correct?

(A) Total atoms are equal; molecules obey III>I>II. (B) Total atom amounts are equal; molecule amounts obey II>I>III.

(C) Total atom amounts obey II>I>III; molecule amounts are equal. (D) Both total atoms and molecules are equal.

38. Bond energies $D(X_2):D(Y_2):D(XY)=1:2:3$. Forming2.00 mol XY(g) from the gaseous elements releases 360 kJ. $D(Y_2)$,in kJ/mol,is

(A) 360 (B) 240

(C) 480 (D) 120.0

39. A school chemistry table lists dilute0.2%phenol as antiseptic and 1%phenol as disinfectant. Which conclusion follows from that table alone?

(A) The two solutions contain different elements.

(B) Phenol is an artificial sweetener at both concentrations.

(C) The more concentrated solution is therefore suitable for every tissue.

(D) Changing concentration can change the listed application of the same compound.

40. For a neutral atom with ground-state configuration $[Ne]3s^23p^5$,which (period,group,block) assignment is correct?

(A) (3,17,p). (B) (2,17,p).

(C) (3,17,d). (D) (3,16,p).

41. A synthetic rubber incorporates -CH2-CH(CN)- units with diene-derived units. Which comonomer pair is consistent?

(A) Buta-1,3-diene and styrene. (B) Adipic acid and hexamethylenediamine.

(C) Chloroprene alone. (D) Buta-1,3-diene and acrylonitrile.

42. For isolated V^{3+} (Z=23),use the high-spin spin-only approximation. Which pair (d-electron count, magnetic moment in BM) follows after removing electrons from 4s before 3d?

(A) $(3,\sqrt{15})$. (B) (2,2).

(C) $(2,2\sqrt{2})$. (D) $(2,\sqrt{15})$.

43. A phenolic compound has molar mass 110 g/mol. A $\frac{11}{10}$ g sample is completely O-acetylated without ring substitution,and dry product mass is $\frac{97}{50}$ g. How many acetylatable OH groups occur per molecule? Each OH→OCOCH3 replacement adds42 g/mol.

(A) 1 (B) 3

(C) 2 (D) 4

44. Mn_3O_4 is represented as a mixed oxide of MnO and Mn_2O_3. Which formal metal oxidation-state ratio satisfies the formula,with ordinary oxide oxygen -2?

(A) 2+:3+ = 2:1.

(B) 2+:3+ = 1:1.

(C) All metal atoms must be formally +8/3,with no mixed-valence bookkeeping possible.

(D) 2+:3+ = 1:2.

45. On controlled heating, $\frac{1}{25}$ mol $KMnO_4$ follows $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$. Which pair ($O_2$ amount,origin of the intense purple colour of the starting d^0 ion) is correct?

(A) ($\frac{1}{50}$ mol,charge-transfer absorption).
(B) ($\frac{1}{100}$ mol,charge-transfer absorption).
(C) ($\frac{1}{25}$ mol,a d-d transition of MnVII).
(D) ($\frac{1}{50}$ mol,a d-d transition of MnVII).

46. Limiting molar conductivities of KCl,NaCl,KNO_3 are150,126,110 S cm^2/mol. A very dilute $NaNO_3$ solution measured in a cell of constant2.00cm^{-1} has conductance $\frac{43}{100000}$S. Neglect concentration corrections. $NaNO_3$ molarity is

(A) $\frac{1}{100}$
(B) $\frac{43}{5500}$
(C) $\frac{1}{50}$
(D) $\frac{1}{200}$

47. A first-order reaction has already consumed 50%of its initial A. Additional time needed to leave12.5% of the original A,expressed in half-lives,is ($\log_2 3 = 1.585$)

(A) 4
(B) 1
(C) 3
(D) 2

48. Two unknown fluids P and Q are warmed modestly without changing phase. P becomes less viscous, while Q becomes more viscous. One is an ordinary molecular liquid and the other a dilute gas, away from critical conditions. Which assignment and mechanism fit both observations?

(A) P is the gas; Q is the liquid; both trends are caused by stronger covalent bonds.

(B) P is the liquid; Q is the gas; liquid cohesion and gas momentum transfer give opposite trends.

(C) Both are liquids because all fluids become more viscous on heating.

(D) Both are gases because all fluids become less viscous on heating.

49. An internal proline residue in a peptide has noN-H on its acylated ring nitrogen. Why can this disrupt a regular alpha-helix hydrogen-bond pattern?

(A) It supplies two additionalN-H donors at that nitrogen.
(B) It replaces every carbonyl oxygen with chlorine.
(C) It hydrolyses the whole protein automatically.
(D) It removes one backboneN-H donor and constrains the local chain geometry.

50. An H_2SO_4 stock is29.0% by mass. A20.0mL aliquot is diluted to 40mL,and the diluted molarity is $\frac{9}{5}$M. H_2SO_4 M98. Stock density in g/mL is

(A) $\frac{441}{725}$
(B) $\frac{441}{1250}$
(C) $\frac{882}{725}$
(D) $\frac{1764}{725}$

51. A $\frac{1}{50}$mol iron-ion inventory contains Fe^{2+}andFe^{3+}. Fraction$\frac{3}{4}$isFe^{3+}. All iron is quantitatively reduced to metal in the stated externally driven electron-balance model,with no side reactions. Electrons accepted, in mol,are

(A) $\frac{11}{200}$
(B) $\frac{9}{175}$
(C) $\frac{9}{200}$
(D) $\frac{1}{20}$

52. A solution hasΛm100S cm^2/mol,conductance $\frac{1}{500}$S and cell constant2.00cm^{-1}. Density1.010g/mL and solute molar mass100g/mol. Its molality in mol/kg is

(A) $\frac{1}{25}$ (B) $\frac{4}{101}$

(C) $\frac{20}{503}$ (D) $\frac{20}{507}$

53. 2-methylpropene reacts with HBr,without peroxides. Which pair gives the major constitutional product and the intermediate whose greater stability fixes its orientation? Ignore stereoisomer counting.

(A) 2-bromo-2-methylpropane; primary carbocation (B) 2-bromo-2-methylpropane; tertiary carbocation

(C) 1-bromo-2-methylpropane; primary carbocation (D) 1-bromo-2-methylpropane; tertiary carbocation

54. Evaluate independently. I: N_2^+ has lower bond order than N_2. II: NO_2 is an odd-electron species.

(A) Neither I nor II is correct. (B) Only II is correct.

(C) Both I and II are correct. (D) Only I is correct.

55. A HA/A^- buffer initially has pH=pKa+$\log_{10}2$. Adding $\frac{1}{100}$ mol HCl without volume change gives pH=pKa-$\log_{10}2$;neither component is exhausted. Total initial HA+A^- amount,in mol,is

(A) $\frac{3}{100}$ (B) $\frac{1}{50}$

(C) $\frac{1}{100}$ (D) $\frac{1}{25}$

56. For NO_2^-,use ordinary octet Lewis structures and resonance. What is the pair (average bond order of two equivalent N-O bonds,total lone pairs on all atoms of that same named species in one contributing structure)?

(A) ($\frac{3}{2}$, 6). (B) ($\frac{3}{2}$, 5).

(C) (2, 7). (D) (1, 6).

57. $M(OH)_2$ has molar solubility $\frac{1}{50000}$ M in pure water. Mix100mL0.00200M NaOH with100mL0.00100M HCl,then add excess solid $M(OH)_2$. Ignore its small contribution to OH^-. Final molar solubility is

(A) $\frac{1}{125000000}$ (B) $\frac{1}{31250000}$

(C) $\frac{1}{7812500}$ (D) $\frac{1}{50000}$

58. Compare $Mg(OH)_2$ and $Ba(OH)_2$ in water at the same temperature. Which member is generally more soluble,and which explanation avoids applying one universal down-group rule to all salts?

(A) $Mg(OH)_2$; every group2salt becomes more soluble as the cation grows.

(B) $Ba(OH)_2$; hydroxide solubility generally increases down group2 because lattice weakening dominates.

(C) They must be equally soluble because both cations have charge+2.

(D) Solubility is fixed solely by formula mass,with no lattice or hydration contribution.

59. butan-2-ol is subjected to concentrated H2SO4,heat. Which product follows the stated regioselectivity or oxidation/dehydration endpoint?

(A) mainly but-2-ene (B) butanal

(C) but-1-ene only (D) butane

60. For two van der Waals gases,(Tc/Pc)A is3times(Tc/Pc)B. UsingTc=8a/(27Rb),Pc=a/(27b^2),Vc=3b andTB=a/(Rb),which pair(Vc,A/Vc,B;TB,A/Tc,A)is correct?

(A) (9;27/8). (B) (3;8/27).

(C) (3;27/8). (D) (1/3;27/8).

English Proficiency

61. Read the sentence: The vote was unanimous: every member supported the resolution. Choose the closest meaning of 'unanimous' as used in the sentence.

(A) in complete agreement (B) strongly expressed

(C) held without debate (D) legally valid

62. Choose the correctly spelt word.

(A) argument (B) arguemment

(C) arguement (D) argumant

63. Read the sentence: The inquisitive child kept asking how the mechanism worked. Choose the closest meaning of 'inquisitive' as used in the sentence.

(A) intrusive (B) talkative

(C) curious (D) suspicious

64. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The panel is to consider the appeal only after receiving the additional record.

(A) The appeal is to be considered by the panel only after the panel receives the additional record.

(B) Only the appeal is to consider the panel after the additional record.

(C) The appeal has been considered by the panel without the additional record.

(D) The appeal is to be considered before the panel receives the additional record.

65. Choose the most coherent order of the four sentences. A. A strip fitted around that door reduced the incoming noise. B. The room finally provided the quiet setting it had been designed to offer. C. The college opened a room intended for silent study. D. At first, its door allowed every corridor conversation to be heard inside.

(A) C-D-B-A (B) C-A-D-B

(C) C-D-A-B (D) D-C-A-B

66. Read the sentence: The letter used a formal style suitable for an official application. Choose the opposite in meaning of 'formal' as used in the sentence.

(A) detailed (B) informal

(C) respectful (D) conventional

67. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. Did the committee approve the shorter route rather than the original one?

(A) Is the shorter route rather than the original one approved by the committee?

(B) Was the shorter route, rather than the original one, approved by the committee?

(C) Were both routes approved by the committee?

(D) Was the original route, rather than the shorter one, approved by the committee?

68. After use, the material can be processed to make a new product rather than simply used again unchanged. Which adjective fits?

(A) reusable (B) recyclable

(C) renewable (D) repairable

69. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. the report is ready Q. which still needs a source check, R. Except for the final appendix, S. to be sent to the printers.

(A) S-Q-P-R (B) P-R-Q-S

(C) Q-R-P-S (D) R-Q-P-S

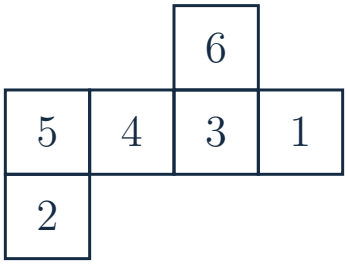
70. Read the sentence: Their enduring friendship survived the long separation. Choose the opposite in meaning of 'enduring' as used in the sentence.

(A) loyal (B) close

(C) unchanged (D) short-lived

Logical Reasoning

71. Fold the shown net into a cube. Give (label opposite 6, label opposite 1) in that order.



(A) $(2, 3)$ (B) $(2, 5)$

(C) $(2, 4)$ (D) $(2, 6)$

72. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 9$ and $5 \mapsto 21$. Which input maps to 33?

(A) 8 (B) 33

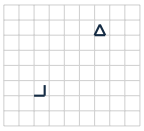
(C) 7 (D) 9

73. A walker goes 5 m east, then 6 m north, then turns left and walks 1 m. In which direction is the starting point from the walker?

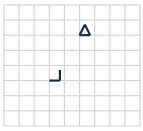
(A) South-east (B) South-west

(C) North-west (D) North-east

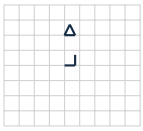
74. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



1

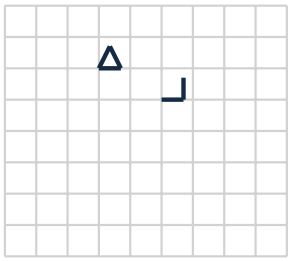


2

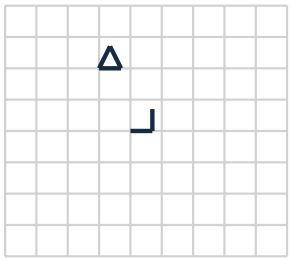


3

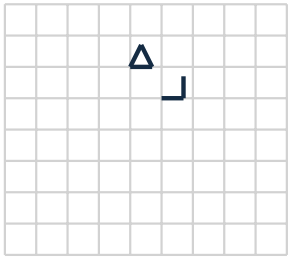
(A) Pattern Y



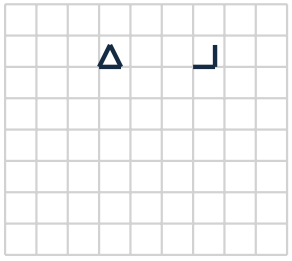
(B) Pattern W



(C) Pattern M



(D) Pattern X



75. The age table uses the same two people throughout:

	younger	older
now	x	$x+7$
2 years ago	$x-2$	$x+5$

. The product of their ages 2 years ago was 368. What will the older person's age be 6 years from now?

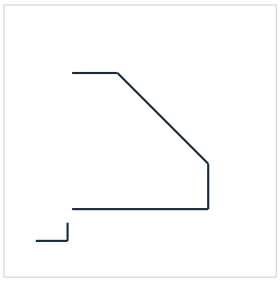
(A) 25

(B) 38

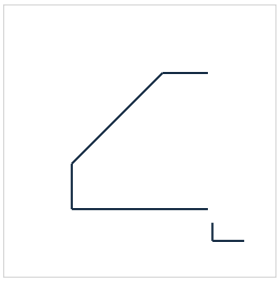
(C) 29

(D) 31

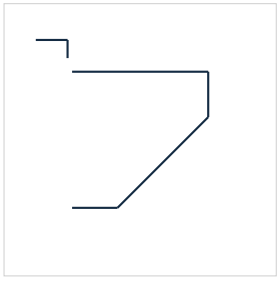
76. Select the missing bottom-right tile. Tiles reflect across the shared horizontal and vertical centre lines; there is no turning.



top left



top right

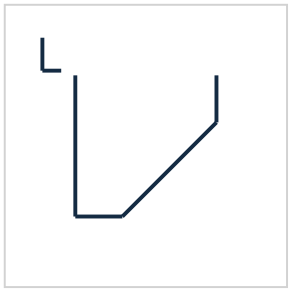


bottom left

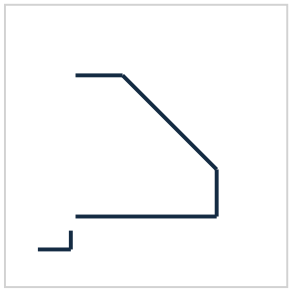
?

bottom right

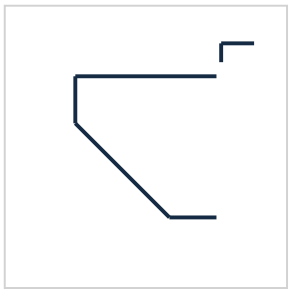
(A) Pattern X



(B) Pattern P



(C) Pattern T



(D) Pattern M



77. Each rectangle is drawn to the same local scale. In three numbered figures the interior straight line divides the rectangle into equal areas. Which does not?



(A) 2
(B) 1
(C) 3
(D) 4

78. The increase from the left cell to the right cell is the same in each row, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern Y
(B) Pattern P



(C) Pattern K



(D) Pattern Q





79. For this question compare only cyclic forward alphabet gaps, with $Z \to A$ counting as one step. Three clusters share the same gap vector. Which numbered cluster differs? 1: PTWA, 2: SWYC, 3: GKMQ, 4: OSUY.

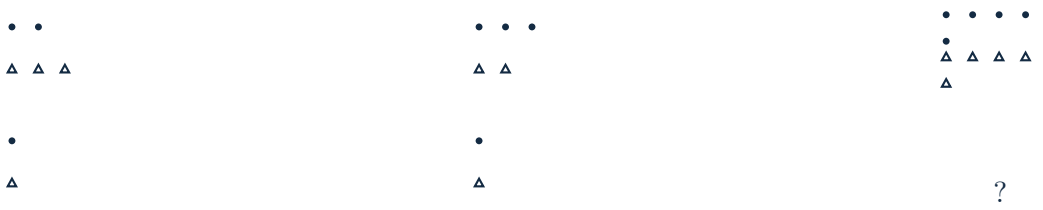
(A) 4
(B) 3
(C) 1
(D) 2

80. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 2, 4, ?, 20, 26, 52, 58.

(A) 6 (B) 10

(C) 14 (D) 8

81. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



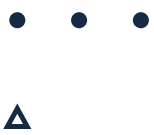
(A) Pattern Y



(B) Pattern Q



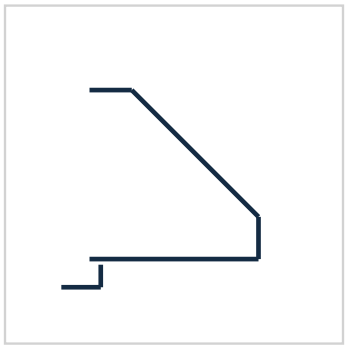
(C) Pattern P



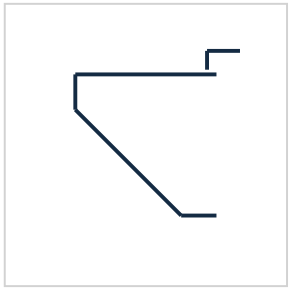
(D) Pattern K



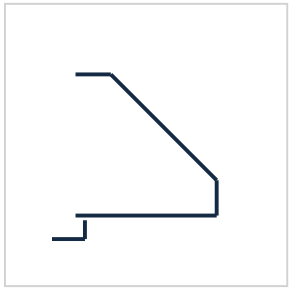
82. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



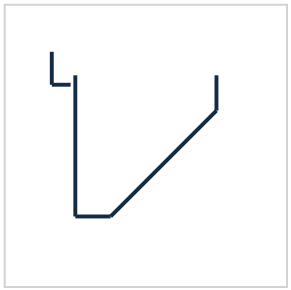
(A) Pattern Q



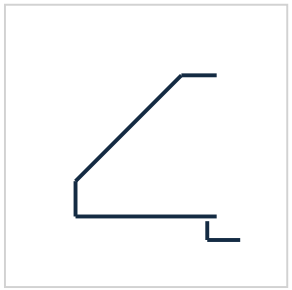
(B) Pattern V



(C) Pattern N



(D) Pattern K



83. Preserve the exact relation and its order: Neon : Noble gas :: ? : Mammal. Choose the missing term.

(A) Whale
(B) Frog
(C) Lizard
(D) Sparrow

84. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?

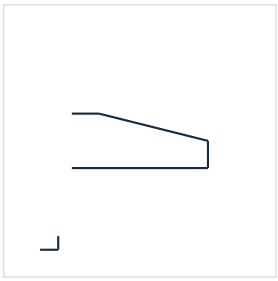


(A) 4
(B) 3
(C) 2
(D) 1

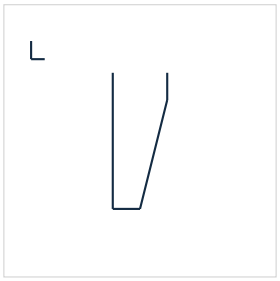
85. Preserve the exact relation and its order: Pilot : Cockpit :: ? : Kitchen. Choose the missing term.

(A) Chef
(B) Librarian
(C) Pilot
(D) Judge

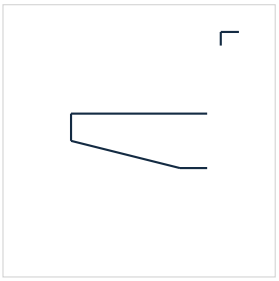
86. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

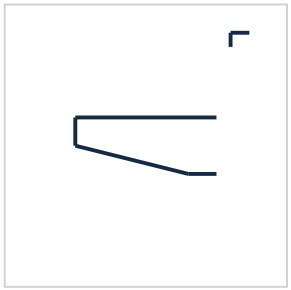


2

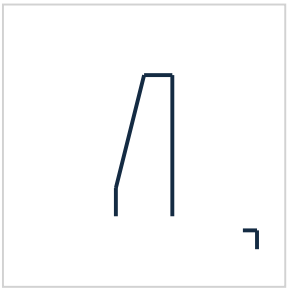


3

(A) Pattern V



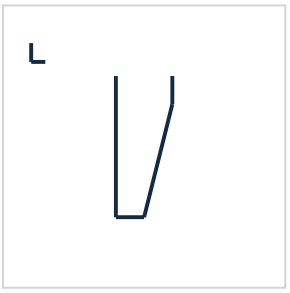
(B) Pattern K



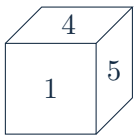
(C) Pattern W



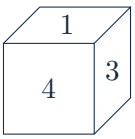
(D) Pattern L



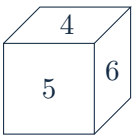
87. The views show the same die. Give the ordered pair: (face opposite 6, face opposite 3). Do not assume opposite labels sum to seven.



1



2



3

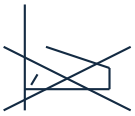
(A) $(1, 2)$
(B) $(1, 5)$
(C) $(1, 1)$
(D) $(1, 3)$

88. A walker goes 8 m north, then 7 m west, then turns right and walks 11 m. In which direction is the walker from the starting point?

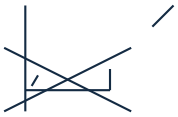
(A) South-west
(B) North-west
(C) North-east
(D) South-east

89. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





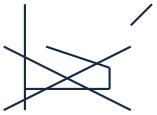
host 1



host 2



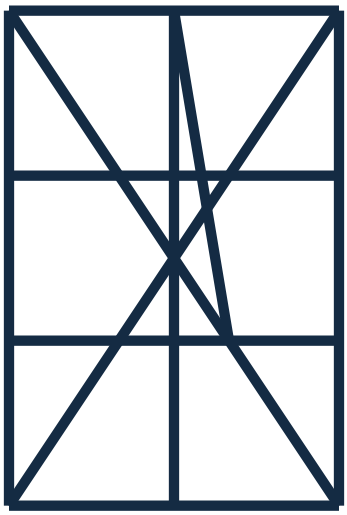
host 3



host 4

(A) 1
(B) 3
(C) 4
(D) 2

90. How many triangles are formed by all the drawn segments?



(A) 58
(B) 35
(C) 36
(D) 32

Mathematics

91. From two points on the same ray from a tower foot, separated by 7, elevations are 30° and 45°. Find the distance from the farther point to the foot.

(A) $\frac{7\sqrt{3}}{2}+\frac{21}{2}$ (B) $\frac{7\sqrt{3}}{2}$

(C) $\frac{7}{2}+\frac{7\sqrt{3}}{2}$ (D) 7

92. For $A=\begin{bmatrix}7 & 1\\ 1 & 3\end{bmatrix}$ and C=20A^{-1}, find C_{22}.

(A) $\frac{7}{20}$ (B) -1

(C) 3 (D) 7

93. $A=\begin{pmatrix}2 & k\\ 0 & 5\end{pmatrix}$ is symmetric. Find tr(A^{T}A).

(A) 29 (B) 49

(C) 7 (D) 10

94. A sphere has dV/dt=4 cubic units/s. Find the rate of change of its surface area when r=5.

(A) $\frac{1}{25\pi}$ (B) $\frac{8}{25}$

(C) $\frac{8}{5}$ (D) $\frac{4}{5}$

95. f(x)=(x^2-36)/(x-6) for x$\neq$6, and f(6)=k. If f is differentiable at 6, find k+f′(6).

(A) 13 (B) 12

(C) 14 (D) 7

96. Two distinguishable fair dice show X,Y. Given X+Y$\geq$10, find the probability that at least one of X, Y is even.

(A) $\frac{4}{3}$ (B) $\frac{5}{6}$

(C) $\frac{1}{6}$ (D) $\frac{5}{36}$

97. A focal chord of y^2=28x has one endpoint parameter t=4. Find its length.

(A) $\frac{1575}{16}$ (B) $\frac{1799}{16}$

(C) $\frac{2023}{16}$ (D) $\frac{119}{2}$

98. Unit vectors a,b,c satisfy |(a$\times$b)$\cdot$c|=1. Find |a+6b-3c|^2.

(A) 10 (B) 16

(C) 46 (D) 45

99. An observer at height 4 sees a tower foot at depression 30° and top at elevation 60°. Find the tower height.

(A) 8 (B) 16

(C) $4\sqrt{3}$ (D) 12

100. f(x)=sin(6x)/x+2 for x<0; f(x)=k cos x for x≥0. Find k for continuity at zero.

(A) 4 (B) 2

(C) 6 (D) 8

101. How many integer k in [0,132] make $x^2 - 17x + (66 - k) = 0$ have roots of opposite signs?

(A) 132 (B) 66

(C) 67 (D) 65

102. Find the area enclosed by y=x(8-x) and y=2x.

(A) 84 (B) -36

(C) 36 (D) $\frac{256}{3}$

103. For x∈[π/4,π/2], find the maximum of sin x/x.

(A) $\frac{2\sqrt{2}}{\pi}$ (B) $\frac{\sqrt{2}\pi}{4}$

(C) $\frac{4}{\pi}$ (D) 1

104. The hyperbola $x^2/9-y^2/25=1$ has eccentricity e. Find the eccentricity e′ of its conjugate hyperbola.

(A) $\frac{\sqrt{34}}{5}$ (B) $\frac{3\sqrt{34}}{34}$

(C) $\sqrt{34}$ (D) $\frac{\sqrt{34}}{3}$

105. How many integers x satisfy $4^{-x} - (80)2^{-x} + 1024 \leq 0$?

(A) 3 (B) 2

(C) 1 (D) 7

106. How many real points of nondifferentiability does (x-2)|x-2|+|x-4| have?

(A) 3 (B) 2

(C) 0 (D) 1

107. The normal of slope 6 to $y^2=24x$ touches at P. Find the distance from P to the focus.

(A) 216 (B) 222

(C) 49284 (D) 210

108. A polynomial f of degree at most 2 satisfies f′(x)+f′(-x)=6, f″(x)=8, and f(0)=2. Find f(2).

(A) 21 (B) 24

(C) 22 (D) 16

109. A,B are acute, A+B=π/4 and tan A=$\frac{5}{7}$. Find tan(A-B).

(A) $\frac{23}{47}$ (B) $\frac{23}{42}$

(C) $\frac{1}{6}$ (D) $-\frac{23}{47}$

110. In how many ordered ways can three six-sided dice have sum 13, if the first die is known not to show 6?

(A) 30 (B) 20

(C) 21 (D) 15

111. a=(1,0,0), a×c=(0,-3,3), a·c=6. Find $|c+a|^2$.

(A) 66 (B) 55

(C) 54 (D) 67

112. Let R(x) be the remainder when x^8 is divided by (x-1)(x-2). Find R(4).

(A) 65536 (B) 1020

(C) −761 (D) 766

113. For $x^2/100+y^2/25=1$, find the area enclosed by the conic containing the midpoints of all chords through its right focus.

(A) $\frac{75\pi}{2}$ (B) $\frac{25\sqrt{3}\pi}{4}$

(C) 50π (D) $\frac{75\pi}{8}$

114. For common tangents to $y^2=28x$ and $x^2+y^2=\frac{49}{12}$, find the sum of the squared slopes.

(A) 3 (B) −3

(C) 6 (D) 0

115. The line y=mx through O bisects the area of the triangle with vertices O,(8,5),(5,5). Assume the vertices are noncollinear. Find m.

(A) $\frac{10}{13}$ (B) $\frac{5}{8}$

(C) $\frac{13}{10}$ (D) 0

116. In a group of 30, 18 study A, 15 study B, and 3 study neither. How many study exactly one of A, B?

(A) 33 (B) 6

(C) 21 (D) 27

117. Evaluate $\displaystyle\int_0^{\pi/2} \sin(2x)\left[1+3\sin^2(2x)\right]\frac{\sin^7 x}{\sin^7 x+\cos^7 x}\,dx.$

(A) $\frac{3}{2}$ (B) $\frac{1}{2}$

(C) 3 (D) 2

118. Evaluate the limit of $[\cos(2x)-\cos(6x)]/x^2$ as $x\to 0$.

(A) −16 (B) 32

(C) 2 (D) 16

119. On A={0,1,...,7}, define aRb if 3 divides a-b. How many ordered pairs are in R?

(A) 8 (B) 64

(C) 14 (D) 22

120. Evaluate $\int_0^\infty dx/[(x^2+16)(x^2+49)]$.

(A) $-\frac{\pi}{616}$ (B) $\frac{\pi}{308}$

(C) $\frac{\pi}{616}$ (D) $\frac{\pi}{56}$

121. $F'(x)=(x^2+6x+7)\,e^x$ and F(0)=4. Find F(1)-2F(0).

(A) $-8e-1$ (B) $1+8e$

(C) $-3+8e$ (D) $-7+8e$

122. Find the area between y=(x-9)(x-16) and the x-axis for $0\le x\le 18$.

(A) 486 (B) −486

(C) $\frac{1801}{3}$ (D) $\frac{1801}{6}$

123. How many solutions in $0\le x<9\pi$ satisfy $\sin^4 x+\cos^4 x=3/4$ and sin x>0?

(A) 36 (B) 20

(C) 21 (D) 18

124. Positive integers $a<b$ satisfy $a+b=24$ and $3\mid a$. Find the maximum value of ab.

(A) 135 (B) 24

(C) 144 (D) 108

125. Evaluate $\sum_{j=1}^{\infty} 6j(\frac{1}{5})^j$.

(A) $\frac{3}{2}$ (B) $\frac{15}{8}$

(C) $\frac{27}{8}$ (D) $\frac{15}{2}$

126. Find the squared distance from O to the plane through (9,0,0),(0,2,0),(0,0,6).

(A) $\frac{47}{162}$ (B) $\frac{162}{47}$

(C) 121 (D) $\frac{9}{7}$

127. A box contains 4 red and 4 blue distinct balls. Three are drawn uniformly without replacement. Given that at least one drawn ball is blue, find the probability that exactly two are red.

(A) $\frac{3}{7}$ (B) $\frac{6}{13}$

(C) 6 (D) $\frac{7}{13}$

128. 9 distinct people sit at labelled round tables of sizes 4,5. Two specified people must be at different tables, and the first specified person must be at the 4-seat table. Rotations are identical. Count the seatings.

(A) 100800 (B) 5040

(C) 10080 (D) 18144

129. ABC is a right triangle on level ground with AB=6, AC=6, angle A=90°. A vertical tower stands at midpoint M of BC. Its top elevation from A is 60°. Find its height.

(A) $6\sqrt{6}$ (B) $3\sqrt{3}$

(C) $3\sqrt{6}$ (D) $3\sqrt{2}$

130. a=(3,2,0), b=(2,-3,3). Find $|a+b|^2$-$|a-b|^2$.

(A) 35 (B) -9

(C) 0 (D) 48

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. 64 identical uncharged liquid drops merge into one drop without evaporation. Find the fraction of the initial surface energy released; surface tension is constant.

(A) 3 (B) $\frac{3}{4}$

(C) 1 (D) $\frac{63}{64}$

132. A travelling wave is $y = 1\sin[2(x - 40t) + \pi/4]$ mm with x in metres and t in seconds. What is its frequency?

(A) $\frac{40}{\pi}$ Hz (B) $\frac{1}{\pi}$ Hz

(C) $\frac{20}{\pi}$ Hz (D) $\frac{80}{\pi}$ Hz

133. A pure radioactive sample has activity A at time 7 hours after preparation and $A/8$ at time 28 hours after preparation. At what time, measured from preparation, will its activity be one sixteenth of the activity measured at 7 hours? Assume exponential decay with a constant decay rate.

(A) 119 h (B) 56 h

(C) 35 h (D) $\frac{35}{3}$ h

Chemistry

134. At one T,ample solids establish $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$,$K_p$=4 bar,and $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$, K_p=1 bar. All named solids remain. He has partial pressure 3 bar. Equilibrium CO mole fraction is

(A) $\frac{1}{3}$ (B) $\frac{4}{9}$

(C) $\frac{2}{9}$ (D) $\frac{1}{8}$

135. An100mL sample contains $\frac{2}{25}$ mol HA,pKa=6,and $\frac{1}{25}$ mol HCl. Add100mL $\frac{6}{5}$ M NaOH. Use usual weak-base salt hydrolysis and $\log_{10}(\frac{2}{5}) = -\frac{199}{500}$. Final pH is

(A) $\frac{9801}{1000}$ (B) 7

(C) 6 (D) $\frac{6801}{500}$

136. Compare benzyl bromide and bromobenzene toward the same cyanide nucleophile in a polar aprotic solvent under ordinary SN2 conditions. Which reacts faster or is the substituted site,and why?

(A) benzyl bromide; the mechanism requires formation of a free tertiary carbocation

(B) benzyl bromide; Benzylic sp3 carbon supports substitution; aryl sp2 C-Br does not undergo ordinary backside SN2.

(C) bromobenzene; an sp2 C-Br bond behaves just like an unhindered methyl bromide

(D) bromobenzene; steric hindrance accelerates backside attack

Logical Reasoning

137. Preserve the exact relation and its order: Speedometer : Speed :: ? : Temperature. Choose the missing term.

(A) Ammeter
(B) Barometer
(C) Thermometer
(D) Voltmeter

138. V and E are full siblings; E and H are also full siblings. M is the mother of V. All four people are distinct. How is M related to H?

(A) Grandmother
(B) Mother
(C) Aunt
(D) Relationship cannot be determined

139. Each of Hari, Charu, Bina visits on one different day (Monday, Tuesday or Wednesday) and buys one different fruit (Orange, Peach, Pear). Hari visits on Wednesday, and Charu on Monday. The visitor on Wednesday buys peach. Charu does not buy pear. Which fruit does Charu buy?

(A) Orange
(B) No unique assignment
(C) Peach
(D) Pear

Mathematics

140. For the roots α,β of $z^2 - (3+9i)z + (-36+27i) = 0$, find $|\alpha\text{-}\beta|^2$.

(A) 90
(B) 2025
(C) 45
(D) 180

141. $A = \begin{bmatrix} 0 & 3 & -2 \\ -3 & 0 & 4 \\ 2 & -4 & 0 \end{bmatrix}$. Find det(I+A)-(9+4+16).

(A) 0
(B) 29
(C) 1
(D) -1

142. The line r=(4,4,4)+t(1,2,2) and plane 2x-y=8 are given. Find the squared minimum distance between them.

(A) $\frac{4\sqrt{5}}{5}$
(B) $\frac{16}{5}$
(C) 16
(D) 0

BITSAT Mock 11

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. Two masses 5 kg and 2 kg are connected by a light string over a smooth pulley and accelerate freely. A transverse pulse travels in the taut straight part of this string, of linear density μ. Compare its speed to that in a second string of density 4μ held at tension g N numerically equal to 1 kg weight. Find v_1/v_2.

(A) $\frac{80}{7}$ (B) $\frac{4\sqrt{35}}{7}$

(C) $2\sqrt{7}$ (D) $2\sqrt{5}$

2. An active transistor has $\alpha = \frac{49}{50}$. Its collector resistor is 2 kΩ and the measured collector-resistor drop is 6 V. Neglect leakage. What base current in mA is required?

(A) 150 mA (B) $\frac{3}{49}$ mA

(C) 3 mA (D) $\frac{147}{50}$ mA

3. A projectile of mass 18 kg is launched with speed 40 m/s, $\cos\theta = 3/5$. At its highest point it splits into masses $M/3$ and $2M/3$. The lighter fragment moves horizontally forward at 27 m/s. Find the signed horizontal velocity of the heavier fragment immediately after the explosion.

(A) 45 m/s (B) $\frac{99}{2}$ m/s

(C) 27 m/s (D) $\frac{45}{2}$ m/s

4. A one-electron ion with nuclear charge $Z = 3$ absorbs a photon of energy $\frac{213}{20}$ eV. The ejected electron has maximum kinetic energy 3 eV. From which principal quantum number was the electron removed? Use $E_n = -13.6Z^2/n^2$ eV and neglect recoil.

(A) 4 (B) 3

(C) 12 (D) 5

5. Two bodies of masses 4 kg and 3 kg move toward each other at 7 m/s and 3 m/s and stick together. Half the lost kinetic energy heats the combined mass uniformly; each has specific heat 40 J/(kg K). Find temperature rise.

(A) $\frac{15}{98}$ K (B) $\frac{15}{49}$ K

(C) $\frac{361}{7840}$ K (D) $\frac{15}{56}$ K

6. Photons of energy 19 eV eject electrons from a surface. The fastest electrons enter a uniform magnetic field perpendicular to their velocities. In this field an electron with kinetic energy 3 eV has orbit radius r. The observed maximum orbit radius is $2r$. What is the surface work function? Neglect recoil.

(A) 12 eV (B) 16 eV

(C) 7 eV (D) $\frac{7}{4}$ eV

7. An astronomical telescope in normal adjustment has magnifying-power magnitude 4 and tube length 15 cm. Find the eyepiece focal length.

(A) 3 cm (B) $\frac{15}{4}$ cm

(C) $\frac{3}{5}$ cm (D) 12 cm

8. A cell balances on a potentiometer at 70 cm on open circuit and at 40 cm when a 6 ohm resistor is connected across it. The potentiometer gradient is unchanged. Find the cell's internal resistance.

(A) 6 Ω (B) $\frac{24}{7}$ Ω

(C) $\frac{21}{2}$ Ω (D) $\frac{9}{2}$ Ω

9. Water flows steadily from a pipe section of area A at 2 m/s to a section of area $A/4$ located 4 m higher. Neglect viscosity. Find $p_1 - p_2$. Take density 1000 kg/m^3 and $g = 10$ m/s^2.

(A) 52000 Pa (B) 10000 Pa

(C) 70000 Pa (D) 38125 Pa

10. A 3 kg block is launched at 5 m/s on top of a 2 kg block initially at rest on a smooth floor. The coefficient of kinetic friction between them is $\frac{1}{10}$. Find the distance slid by the upper block relative to the lower before slipping ceases. Take $g = 10$ m/s^2.

(A) 8 m (B) $\frac{25}{2}$ m

(C) 3 m (D) 5 m

11. A series RLC circuit has $R = 50$ ohms, $X_L = 40$ ohms, and $X_C = 70$ ohms at the source frequency. An ideal source supplies 300 V rms. Find rms current.

(A) 10 A (B) $\frac{15\sqrt{146}}{73}$ A

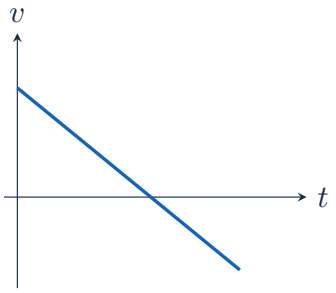
(C) $\frac{15\sqrt{17}}{17}$ A (D) $\frac{15\sqrt{34}}{17}$ A

12. A circular parallel-plate capacitor of plate radius 3 cm is charged at 24 mA. Neglect fringing. What displacement current passes through the annular part of the gap between radii 2 cm and 3 cm, centred on the plate axis?

(A) $\frac{32}{3}$ mA (B) $\frac{40}{3}$ mA

(C) 8 mA (D) 24 mA

13. A particle has velocity $v(0) = 5$ m/s, and constant acceleration makes $v(5) = -3$ m/s. What total distance does it travel in this interval?



(A) $\frac{125}{16}$ m
(B) $\frac{85}{8}$ m
(C) 20 m
(D) 25 m

14. A particle has potential $U(x, y) = 5x^2y + 5y$ J in SI units. Find the force magnitude at $(2, 2)$ m.

(A) $5\sqrt{41}$ N
(B) $5\sqrt{89}$ N
(C) 50 N
(D) $20\sqrt{5}$ N

15. A particle executes SHM at frequency 7 Hz. At t=0 it is at its positive extreme. In the interval $0 <= t < 1$ second, how many times does its kinetic energy equal its potential energy?

(A) 28
(B) 15
(C) 29
(D) 14

16. A pure radioactive sample has activity A at time 14 hours after preparation and $A/8$ at time 35 hours after preparation. At what time, measured from preparation, will its activity be one thirty-second of the activity measured at 14 hours? Assume exponential decay with a constant decay rate.

(A) 147 h
(B) 49 h
(C) 238 h
(D) 70 h

17. A mass 5 kg moves with $\vec{r}(t) = 3t\hat{i} + 2t^2\hat{j}$ m. Find the z-component of angular momentum about the origin at $t = 3$ s.

(A) 270 kg m^2/s
(B) 810 kg m^2/s
(C) 135 kg m^2/s
(D) 90 kg m^2/s

18. Two identical capacitors each 6 microfarads are connected in series across an ideal 20 V battery. One capacitor undergoes permanent conducting breakdown, becoming a short circuit. Find the additional charge drawn from the battery after the new steady state is reached.

(A) 120 μC
(B) 60 μC
(C) 40 μC
(D) 30 μC

19. A balanced Wheatstone bridge has arms AB=AD=5 ohms and BC=DC=20 ohms. A resistor of arbitrary positive resistance connects B to D. Find equivalent resistance between A and C.

(A) 25 Ω
(B) 8 Ω
(C) $\frac{25}{2}$ Ω
(D) 4 Ω

20. A 3 kg block moves along a horizontal rough surface from $x = 0$ to $x = 2$ m under a force $F(x) = 5x$ N parallel to motion. With $\mu_k = 1/5$ and $g = 10$ m/s^2, find the work done by the net force. The block has enough initial speed to keep moving forward.

(A) -2 J (B) 8 J

(C) 2 J (D) 22 J

21. In YDSE a sheet of index $\frac{5}{2}$ and thickness 5 micrometres covers one slit. A second sheet of the same thickness and index 3/2 covers the other. With wavelength $\frac{3}{10}$ micrometres, find the magnitude of central-fringe shift in units of fringe width.

(A) $\frac{125}{3}$ (B) $\frac{100}{3}$

(C) $\frac{50}{3}$ (D) 25

22. A liquid of volume 2 litres and bulk modulus 6000000000 Pa is subjected to an additional pressure of 3000000 Pa. Neglect changes in bulk modulus. Find the magnitude of volume decrease in cubic centimetres.

(A) 1 cm^3 (B) $\frac{1}{1000}$ cm^3

(C) $\frac{1}{2}$ cm^3 (D) 1000 cm^3

23. A square conducting loop in a constant perpendicular field 5 T has side length $\ell(t) = 3t$ m. Find induced-emf magnitude at $t = 4$ s, assuming the loop expands uniformly.

(A) 60 V (B) 720 V

(C) 360 V (D) 180 V

24. An object moves slowly along the axis toward a concave mirror. At an instant its distance from the mirror is 100 cm; the mirror focal-length magnitude is 25 cm. Find the image-speed magnitude divided by object-speed magnitude at that instant.

(A) $\frac{1}{3}$ (B) 9

(C) $\frac{1}{9}$ (D) $\frac{1}{25}$

25. A long solenoid has its turn number multiplied by 3 and length multiplied by 2, while its cross-sectional area and core permeability stay fixed. Find self-inductance multiplier.

(A) $\frac{2}{9}$ (B) $\frac{9}{2}$

(C) $\frac{3}{2}$ (D) $\frac{9}{4}$

26. A galvanometer has resistance 50 ohms and full-scale current $\frac{1}{500}$ A. It is to measure a total current 5 times its own full-scale current. Find the required parallel shunt resistance.

(A) $\frac{125}{2}\,\Omega$ (B) $50\,\Omega$

(C) $200\,\Omega$ (D) $\frac{25}{2}\,\Omega$

27. A uniform-density planet has radius R. Gravity at a depth d equals gravity at a height $(2-1)R$ above the surface. Find d/R.

(A) 1 (B) $\frac{7}{8}$

(C) $\frac{1}{2}$ (D) $\frac{3}{4}$

28. Two isolated conducting spheres of radii R and 3R are far apart and connected by a fine wire. Their total positive charge is Q. After equilibrium the wire is removed. Find the larger sphere's charge divided by Q.

(A) $\frac{3}{4}$ (B) $\frac{1}{3}$

(C) $\frac{9}{10}$ (D) $\frac{27}{28}$

29. One mole of a monatomic ideal gas follows $pV^6 =$ constant. Find the process molar heat capacity in units of R.

(A) $\frac{3}{2}$ (B) $\frac{4}{3}$

(C) $\frac{5}{2}$ (D) $\frac{13}{10}$

30. A long solenoid has 300 turns per metre and carries 6 A. It is completely filled by a linear magnetic core of relative permeability 4. Find magnetisation M.

(A) 1800 A/m (B) 18 A/m

(C) 5400 A/m (D) 1350 A/m

Chemistry

31. Consider I: The N-N single bond is unusually weak relative to the P-P single bond. II: Nitrogen does not form an ordinary neutral NCl5 analogue of PCl5. Which assessment correctly recognises both anomalous first-element behaviours?

(A) Only II is correct. (B) Neither statement is correct.

(C) Only I is correct. (D) Both statements are correct.

32. A student assigns five bonding pairs and one lone pair to the central atom of BrF_5. After independently checking valence-electron count,which combined molecular shape/domain count is consistent?

(A) Square pyramidal; six electron domains. (B) Pentagonal planar; six electron domains.

(C) Octahedral molecular shape; six electron domains. (D) Trigonal bipyramidal; five electron domains.

33. In a hydrogen-like ion Z=2,an electron falls from n=5 to an s orbital with 2 radial nodes. With hydrogen ionisation energy13.6 eV and$hc = 1240$ eV nm,the emitted wavelength in nm is

(A) $\frac{174375}{544}$ (B) $\frac{174375}{136}$

(C) $\frac{174375}{2176}$ (D) $\frac{19375}{34}$

34. On dilution of one electrolyte,the analytical concentration falls by a factor4while measured conductivity falls by a factor2. Cell geometry is unchanged. Ratio of final to initial molar conductivity is

(A) 2 (B) $\frac{1}{4}$

(C) 4 (D) $\frac{1}{2}$

35. For octa-2,4,6-triene, CH3-CH=CH-CH=CH-CH=CH-CH3, count distinct configurational geometrical isomers. Ignore conformers and photochemical interconversion. Where reversing the entire carbon chain identifies two E/Z assignments, count them once.

(A) 8 (B) 6

(C) 7 (D) 16

36. A 24g nonvolatile nonassociating solute in1.000kg water raises boiling point by $\frac{2}{15}$K;Kb=1.00Kkg/mol. Another 12g of the same solute is added without water loss. Exact relative vapour-pressure lowering is (waterM18)

(A) $\frac{3}{1253}$ (B) $\frac{9}{2500}$

(C) $\frac{1}{180}$ (D) $\frac{9}{2509}$

37. At25C,a measurement gives$[OH^-]=3\times10^{-4}$M. UseKw=10^{-14} and$\log_{10}3=\frac{477}{1000}$. Solution pH is

(A) $\frac{10477}{1000}$ (B) $\frac{4477}{1000}$

(C) $\frac{3523}{1000}$ (D) $\frac{9523}{1000}$

38. For electrophilic mononitration of benzene carrying COOH,which orientation/reason pair is correct? Consider intrinsic electronic directing effects,not an exact ortho:para ratio.

(A) meta; the carboxyl group withdraws electron density by resonance and induction

(B) ortho/para; the carboxyl group withdraws electron density by resonance and induction

(C) ortho/para; the substitution site is decided only by steric size

(D) meta; every electronegative substituent activates the ring

39. In an antifluorite-derived crystal, X2- occupies all FCC anion sites. A measured fraction 1/3 of occupied M cations is M2+ and the rest M+. Only tetrahedral cation vacancies compensate charge; no interstitials or anion defects occur. What fraction of all tetrahedral sites is vacant?

(A) 1/6 (B) 1/4

(C) 3/4 (D) 1/8

40. Equal volumes of $\frac{3}{20}$M $K_4[Fe(CN)_6]$ and $\frac{3}{20}$M glucose are mixed. In the final solution,the salt is50% dissociated into4K^+ and one intact complex anion;glucose does not dissociate. Use RT=24.0 L atm/mol. Osmotic pressure in atm is

(A) $\frac{18}{5}$ (B) $\frac{72}{5}$

(C) $\frac{36}{5}$ (D) $\frac{54}{5}$

41. A textbook deficiency table associates cheilosis with one vitamin. Which vitamin and solubility class are correctly paired?

(A) Vitamin B2; water-soluble. (B) Vitamin B6; water-soluble.

(C) Vitamin C; water-soluble. (D) VitaminK; fat-soluble.

42. For isolated gaseous species,consider $Cl(g)+e^-\rightarrow Cl^-(g)$. Which judgement also remains consistent with the Cl/F electron-gain anomaly?

(A) Every first and second electron gain has the same sign and magnitude for a group.

(B) The stated step is endothermic; first electron gain by F is more exothermic than by Cl because F is smaller.

(C) The stated step is exothermic; first electron gain by Cl is more exothermic than by F.

(D) The stated step is endothermic; every electron addition releases energy.

43. For one reaction,ln K is2 at300 K and−1 at600 K. Assume ln K is linear in1/T. At400 K,ln K equals

(A) 2 (B) $\frac{3}{2}$

(C) −1 (D) $\frac{1}{2}$

44. $\frac{93}{50}$ g aniline is acetylated to acetanilide in80.0% isolated molar yield. The isolated acetanilide is then mononitrated and gives p-nitroacetanilide in75.0% isolated molar yield. Both steps are1:1in substrate moles. Find final product mass in g. Use molar masses: aniline93, acetanilide135, p-nitroacetanilide180 g/mol.

(A) $\frac{18}{5}$ (B) $\frac{27}{10}$

(C) $\frac{54}{25}$ (D) $\frac{72}{25}$

45. For benzyl cation,C6H5CH2+;tert-butyl cation;(CH3)2CH+,which decreasing stability order and its underlying class distinction is correct? Neglect solvent-specific reversals.

(A) tert-butyl > benzyl > isopropyl (B) benzyl > isopropyl > tert-butyl

(C) benzyl > tert-butyl > isopropyl (D) isopropyl > tert-butyl > benzyl

46. 2-methylpropene reacts with HBr with peroxide. Which pair gives the major constitutional product and the intermediate whose greater stability fixes its orientation? Ignore stereoisomer counting.

(A) 2-bromo-2-methylpropane; primary radical (B) 1-bromo-2-methylpropane; primary radical

(C) 2-bromo-2-methylpropane; tertiary radical (D) 1-bromo-2-methylpropane; tertiary radical

47. In the trigonal-bipyramidal electron-domain model of I_3^-,there are 3 central lone pairs. Which placement rule explains the observed arrangement,including the first lone pair?

(A) Place the first lone pair axially because three 90-degree contacts are less repulsive than two.

(B) Place lone pairs equatorially first: an equatorial site has two 90-degree neighbours,whereas an axial site has three.

(C) Use arbitrary sites because lone-pair/bond-pair and bond-pair/bond-pair repulsions are identical.

(D) Place lone pairs only where they form the smallest possible angle to every bond.

48. An ideal gas executes A→B→C→D→A through (V/L,P/bar):A(1,1),B(4,1),C(4,2),D(1,2). IUPAC work-on convention applies. Heats on AB,BC,DA are 400,300,-200 J. Heat on CD in J is (1 L bar=100 J)

(A) −300 (B) −500

(C) −800 (D) −200

49. At fixed T,a reaction has K=9 and currently Q=9. Which joint conclusion is correct for this equilibrium mixture?

(A) $\Delta_r G < 0$ because its standard value is negative,even when Q=K.

(B) $\Delta_r G = 0$ while$\Delta_r G^\circ < 0$;equilibrium does not require K=1.

(C) Both$\Delta_r G$ and$\Delta_r G^\circ$ must be zero because the reaction is reversible.

(D) $\Delta_r G^\circ > 0$ because K exceeds1.

50. An octahedral complex has high-spin d^5 configuration and $\Delta_o = 200$ kJ mol^{-1}. Ignoring pairing energy,its d-electron energy relative to the unsplit barycentre,in kJ mol^{-1},is

(A) 0
(B) 400
(C) −200
(D) 200

51. A $\frac{3}{100}$mol iron-ion inventory contains Fe^{2+}andFe^{3+}. Fraction$\frac{1}{2}$isFe^{3+}. All iron is quantitatively reduced to metal in the stated externally driven electron-balance model,with no side reactions. Electrons accepted,in mol,are

(A) $\frac{9}{200}$
(B) $\frac{3}{40}$
(C) $\frac{2}{25}$
(D) $\frac{7}{100}$

52. cyclopentene is treated with bromoform,CHBr3 and strong base under dihalocarbene-generation conditions. Which joint statement correctly identifies the haloalkane step and the hydrocarbon-skeleton change?

(A) Beta elimination generates an alkene from the haloform; the ring size is unchanged.

(B) :CBr2 adds by replacing a ring carbon,so total carbon count is unchanged.

(C) A free methyl cation adds; the original C=C remains intact.

(D) Alpha elimination generates :CBr2; the alkene gains one carbon and one additional ring.

53. propan-1-ol is converted with PBr3 to its primary alkyl bromide,then heated with potassium tert-butoxide. Which product results,and does the PBr3 step itself change its carbon skeleton?

(A) the corresponding alkane;no C-O bond change
(B) propene; no,the carbon skeleton is retained
(C) the corresponding ether;SN2 at a tertiary carbon is required
(D) propene; yes,the carbon chain is cleaved

54. An ion has 18 electrons,22 neutrons and charge +2. Which combination gives its element,mass number and neutral-element group correctly?

(A) An element with Z=18;A=40;group18.
(B) Ca;A=40;group2.
(C) Ca;A=42;group18.
(D) Ca;A=42;group2.

55. For methyl ethanoate,label A=carbonyl C=O,B=acyl C-O,C=alkoxy O-C(sp3). Which bond order/length statement correctly distinguishes the two C-O single bonds shown in a Lewis formula?

(A) B has partial double-bond character,and length A < B < C.
(B) C has partial double-bond character,and length A < C < B.
(C) A is longest because oxygen is more electronegative.
(D) B and C must have identical lengths because both are drawn single.

56. Sucrose joins glucoseC1 to fructoseC2 through oxygen. Both named carbons were carbonyl carbons before cyclisation. Which consequence follows?

(A) A free glucoseC1 hemiacetal necessarily remains.

(B) Hydrolysis yields only galactose.

(C) Both anomeric centres are tied in the link, so intact sucrose is nonreducing.

(D) The sugar has no oxygen atoms.

57. At 300 K, ΔH for C(graphite)+O2(g)→CO2(g) is -380 kJ/mol and for CO(g)+(1/2)O2(g)→CO2(g) is -270 kJ/mol. Carbon is converted only into CO, producing 1/2 mol CO. Assuming ideal gases and R=0.008314 kJ/mol/K, what is total ΔU in kJ? Use the stated reaction extents.

(A) -1112471/40000
(B) -1112471/10000
(C) -1112471/20000
(D) 1112471/20000

58. Assign the formal metal oxidation state in $[Mn(CO)_6]^+$,using six neutral CO. For nitrosyl species use the explicitly stated formal NO^+ convention rather than inferring spin from it.

(A) 0
(B) 1
(C) 2
(D) 3

59. Rain downwind of fuel combustion contains added sulphate and nitrate and is appreciably more acidic than ordinary carbon-dioxide-equilibrated rain. Which precursor pair best accounts for this change?

(A) Ozone and helium only.
(B) Nitrogen and argon.
(C) Methane and neon only.
(D) Sulphur dioxide and nitrogen oxides.

60. H_2SO_5has one O–O peroxide link:two O atoms are-1,and its other three O atoms are-2. WithH+1, sulfur oxidation state is

(A) 5
(B) 6
(C) 4
(D) 8

English Proficiency

61. Read the sentence: The brief exchange planted a seed that later grew into a full research project. What does the figurative expression mean in this context?

(A) It delayed the beginning of the project.
(B) It supplied an initial idea that developed over time.
(C) It guaranteed the project's eventual success.
(D) It provided all the evidence needed for the project.

62. Read the sentence: After checking the final page, the tired editors decided to call it a day. What does 'call it a day' mean here?

(A) arrange work for the next morning
(B) finish the entire project permanently
(C) stop working for the time being
(D) declare the day a holiday

63. Read the sentence: The normally docile animal followed the handler without resistance. Choose the opposite in meaning of 'docile' as used in the sentence.

(A) tame
(B) obedient
(C) quiet
(D) unruly

64. Even after the setback, he expects the next attempt to succeed, though he accepts that failure is possible. Which word describes his outlook?

(A) perfectionist
(B) optimist
(C) cynic
(D) pessimist

65. The chart gives the complete conventional set of letters of the script, not merely a list of words. What does it show?

(A) glossary (B) vocabulary

(C) syllabus (D) alphabet

66. Read the sentence: At first the apparent agreement seemed complete, but private objections soon emerged. Choose the opposite in meaning of 'apparent' as used in the sentence.

(A) actual (B) initial

(C) outward (D) obvious

67. Complete: The technician to _____ the fault was reported arrived before noon.

(A) whom (B) whose

(C) who (D) which

68. Choose the active-voice equivalent that retains the tense, participants and all stated restrictions. The invitation was accepted by the speaker only after the travel arrangements were confirmed.

(A) The speaker accepted the invitation only after the travel arrangements were confirmed.

(B) The speaker has accepted the invitation only after the travel arrangements are confirmed.

(C) The speaker accepted the invitation before the travel arrangements were confirmed.

(D) Only the speaker accepted the invitation after confirming every arrangement personally.

69. Read the sentence: The editor made a judicious choice after weighing both relevance and available space. Choose the closest meaning of 'judicious' as used in the sentence.

(A) formally justified by a rule (B) showing sound judgement

(C) relating to a court of law (D) inclined to condemn people

70. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. on the evidence already available Q. the committee will have to decide R. to its members. S. Unless the missing record is found,

(A) P-S-Q-R (B) R-Q-P-S

(C) S-Q-P-R (D) Q-S-P-R

Logical Reasoning

71. Two people differ in age by 8 years. The product of their ages 3 years ago was 768. What will the younger person's age be 4 years from now?

(A) 27 (B) 31

(C) 39 (D) 28

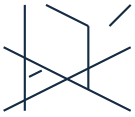
72. A community register uses the following three categories: guides, inventors, artists. Its complete relevant assertions are in this short paragraph; do not import everyday assumptions. No artists are inventors. Some artists are guides. Some guides are not inventors. Conclusions: I. Some guides are artists. II. Some artists are inventors. Which conclusions follow necessarily?

(A) Only II (B) Only I

(C) Neither I nor II (D) Both I and II

73. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

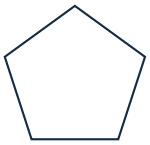
(A) 4

(B) 2

(C) 3

(D) 1

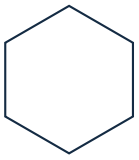
74. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

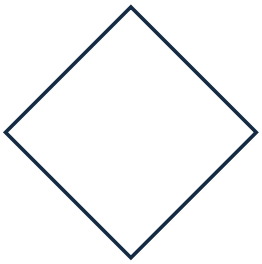


B

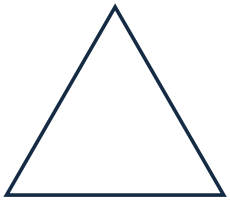


C

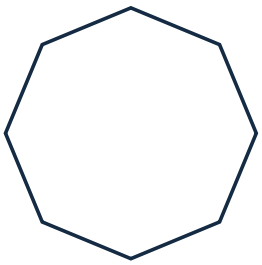
(A) Pattern V



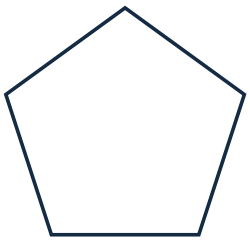
(B) Pattern K



(C) Pattern P



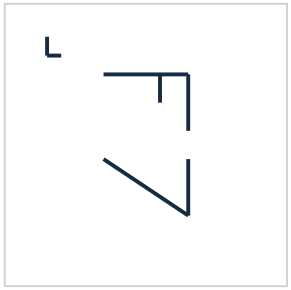
(D) Pattern N



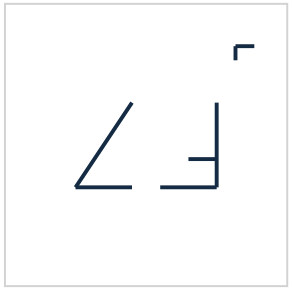
75. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



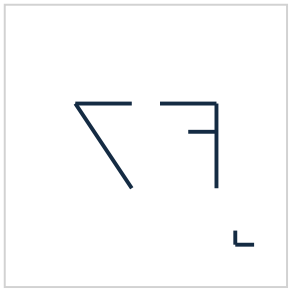
(A) Pattern U



(B) Pattern T



(C) Pattern Y



(D) Pattern N



76. Three pairs contain exactly one mammal and one bird. Which pair does not?

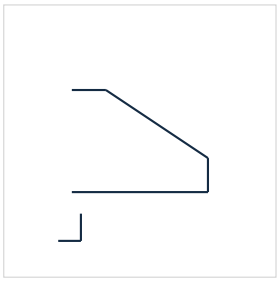
(A) Dog / Crow

(B) Bat / Ostrich

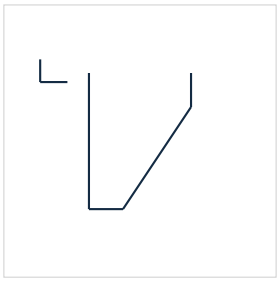
(C) Whale / Sparrow

(D) Horse / Rabbit

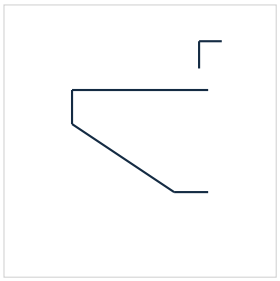
77. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

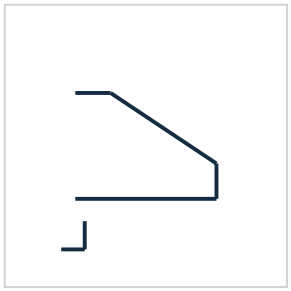


2



3

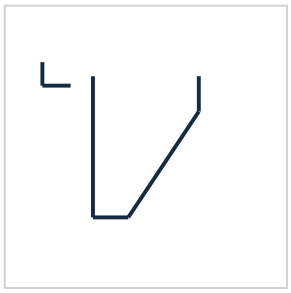
(A) Pattern P



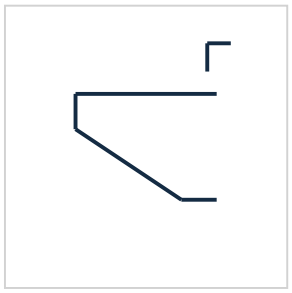
(B) Pattern W



(C) Pattern Q



(D) Pattern Y



78. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: QTX, OUY, ?, KWA, IXB, GYC.

(A) NWA (B) LUY

(C) ZVM (D) MVZ

79. In a row, Asha is 6th from the left and 36th from the right. Bina is 7th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 25 (B) 31

(C) 27 (D) 1

80. In this series the first differences themselves form an arithmetic progression. Find the missing internal term: 13, 16, ?, 34, 49, 68, 91.

(A) 24 (B) 26

(C) 23 (D) 20

81. Preserve the exact relation and its order: Rapid : Swift :: ? : Authentic. Choose the missing term.

(A) Genuine (B) Silent

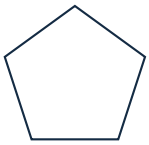
(C) Rapid (D) Begin

82. Anu points to a photograph and says, "This person is my mother's only son." Anu is female, and the photographed person is male. How is the photographed person related to Anu?

(A) Son (B) Sister

(C) Brother (D) Daughter

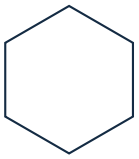
83. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

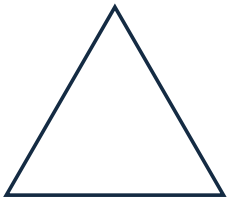


B

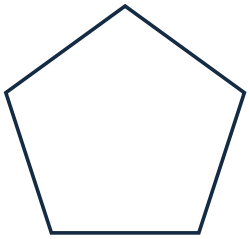


C

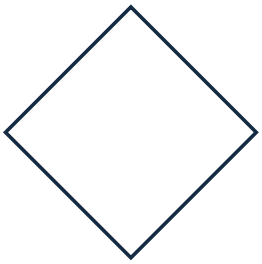
(A) Pattern W



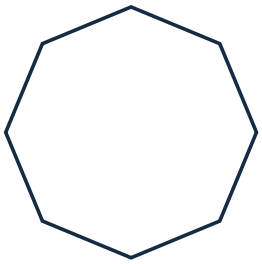
(B) Pattern T



(C) Pattern N



(D) Pattern U

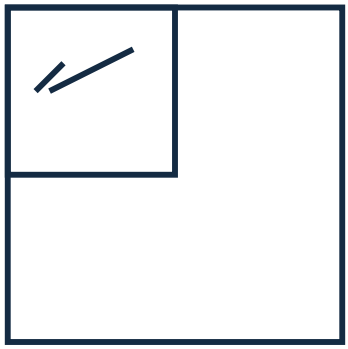


84. A word is coded as the sum of its forward alphabet positions plus 6 times its length. The code of RIV?R is 102. Which letter replaces the question mark?

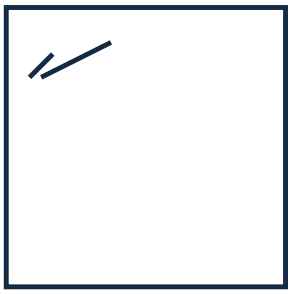
(A) D (B) K

(C) F (D) E

85. Starting with a unit square sheet, fold right to left along the vertical midline, then fold bottom to top along the horizontal midline. The folded paper is cut along both short drawn slits. Select the pattern after complete unfolding.



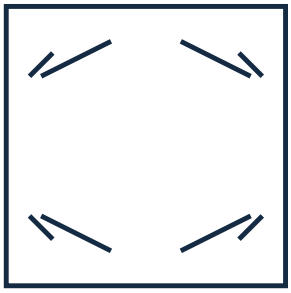
(A) Pattern X



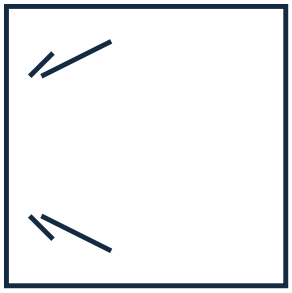
(B) Pattern K



(C) Pattern M



(D) Pattern R



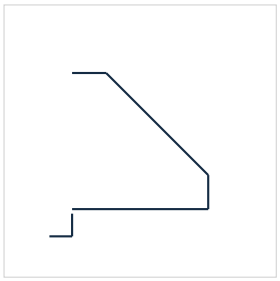
86. All five people have different heights. Charu is shorter than Hari. Gita is shorter than Farah. Asha is shorter than Charu. Farah is shorter than Asha. If Asha is left out, who is 2nd shortest among the remaining people?

(A) Gita
(B) Charu
(C) Asha
(D) Farah

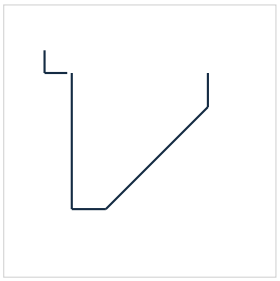
87. A code shifts letters cyclically through A–Z. Odd positions share one fixed shift and even positions share another. For example, LAMP is coded as OXPM. Which word is coded as UFYBU?

(A) RIVER
(B) SJWFS
(C) XCBYX
(D) QHUDQ

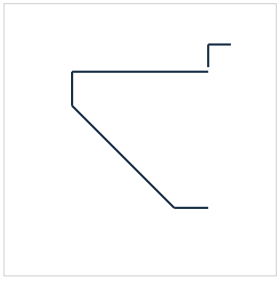
88. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

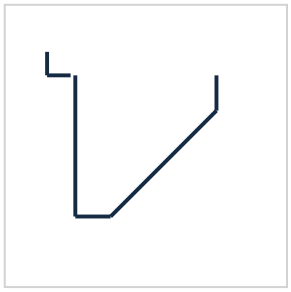


bottom left

?

bottom right

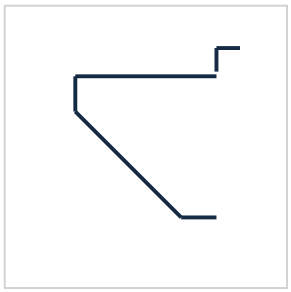
(A) Pattern P



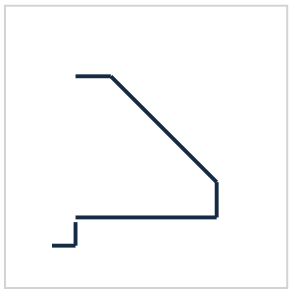
(B) Pattern K



(C) Pattern M



(D) Pattern T



89. Consider only these statements; universal statements alone do not imply existence. All editors are inventors; Some editors are not bakers; Some inventors are not editors. Conclusions: I. All bakers are editors. II. Some editors are inventors. Which conclusions follow necessarily?

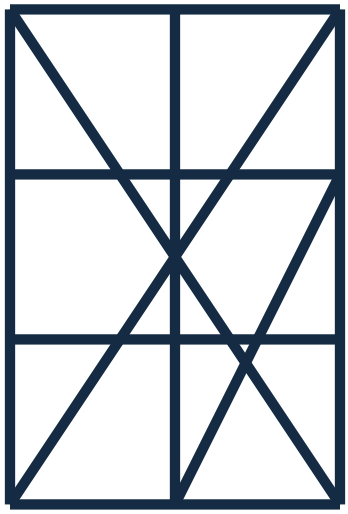
(A) Neither I nor II

(B) Only II

(C) Only I

(D) Both I and II

90. How many triangles are formed by all the drawn segments?



(A) 31
(B) 59
(C) 34
(D) 35

Mathematics

91. Group A has mean 80, SD 8. Every observation is increased by 80. Find its new coefficient of variation in percent.

(A) 10
(B) 20
(C) 40
(D) 5

92. In $(6x^2 - x^{-1})^7$, find the ratio of the coefficient of x^{-1} to that of x^2.

(A) $\frac{1}{10}$
(B) -10
(C) $\frac{3}{5}$
(D) $-\frac{1}{10}$

93. Find the squared distance between 6x+8y=16 and 3x+4y=15.

(A) $\frac{49}{25}$
(B) 49
(C) $\frac{7}{5}$
(D) $\frac{49}{100}$

94. How many integers k make $x^2 + kx + 25 = 0$ have two distinct non-real roots?

(A) 19
(B) 9
(C) 20
(D) 21

95. For the loop $y^2=x^2(8-x)$, find the area of the portion with $x \geq 8/2$.

(A) $\frac{896}{15}$
(B) $\frac{512\sqrt{2}}{15}$
(C) $-\frac{896}{15} + \frac{1024\sqrt{2}}{15}$
(D) $\frac{448}{15}$

96. There are 4 Boolean variables p_1,...,p 4. How many truth assignments make [(p_1 implies p_2) and (p_1 implies not p_2)] true?

(A) 15 (B) 8

(C) 16 (D) 4

97. Evaluate the integral from 0 to $\pi/2$ of [(sin x+cos x)/(sec x+csc x)](sin x+4 cos x) dx, using continuous extension at the endpoints.

(A) $\frac{10}{3}$ (B) $\frac{5}{3}$

(C) $\frac{4}{3}$ (D) 5

98. The system x+y+z=1, y+z=2, (k-4)z=k-10 has no solution for which k?

(A) 0 (B) 4

(C) 10 (D) -4

99. Let f(x)=sin x+4 cos x. Let [L,L+π] be the interval containing 0 in its interior on which f is increasing and bijective onto[-sqrt(17),sqrt(17)]. If g is its inverse, findL+$g'(1)$.

(A) $-\frac{\pi}{2} - \operatorname{atan}(4) + \frac{1}{4}$ (B) $-\frac{\pi}{2} - \operatorname{atan}(4)$

(C) $-\operatorname{atan}(4) + \frac{1}{4} + \frac{\pi}{2}$ (D) $-\frac{\pi}{2} - \operatorname{atan}(4) + 4$

100. Two distinguishable fair dice show X,Y. Given X+Y$\geq$7, find the probability that at least one of X, Y is even.

(A) $\frac{8}{7}$ (B) $\frac{1}{2}$

(C) $\frac{6}{7}$ (D) $\frac{1}{7}$

101. f(x)=3x+7. Find f(f(7)).

(A) 49 (B) 91

(C) 77 (D) 84

102. For $f(x)=x^3$-27x, find the difference between its local maximum value and local minimum value.

(A) 54 (B) 108

(C) 6 (D) -108

103. a=(8,6,0), b=(6,-8,4). Find $|a+b|^2$-$|a-b|^2$.

(A) -16 (B) -112

(C) 0 (D) 11600

104. Evaluate lim as x$\to$0 of [sqrt(25+x)-5-x/10]/x^2.

(A) $\frac{1}{10}$ (B) $-\frac{1}{200}$

(C) $\frac{1}{1000}$ (D) $-\frac{1}{1000}$

105. A rectangle has perimeter 32. Its length is constrained to be at least three times its width. Find its maximum area.

(A) 48 (B) 16

(C) 32 (D) 64

106. Find the coefficient of x^8 in (1-x)exp(4x).

(A) $-\frac{1024}{315}$ (B) $\frac{512}{105}$

(C) $\frac{512}{315}$ (D) $-\frac{512}{315}$

107. P=(6,4,4) is reflected in the x-axis to P′. Find $|PP'|^2$.

(A) 272 (B) 32

(C) 128 (D) 64

108. The line 3x+4y=20 cuts $x^2+y^2=100$ in chord AB. Find $|AB|^2$.

(A) 84 (B) 336

(C) 400 (D) 464

109. $x=t^2$, $y=t^3$ for t>0. Find d^2y/dx^2 at t=2.

(A) $\frac{3}{4}$ (B) $\frac{3}{2}$

(C) 6 (D) $\frac{3}{8}$

110. A={j∈{1,...,84}:sin(jπ/2)=0}, B={j∈{1,...,84}:sin(jπ/3)=0}. Find |(A∪B)\(A∩B)|.

(A) 14 (B) 56

(C) 42 (D) 70

111. $a_j = 2 + 4(j-1)$ and $f(x)=(3/2)x^2+bx+5$. If f(1)=f(-1), find f′(a_8).

(A) 102 (B) 95

(C) 30 (D) 90

112. A=(4,5,0), C=(7,11,3). B lies on AC with AB:BC=2:1. Find $|OB|^2$, where O is the origin.

(A) 41 (B) 121

(C) 179 (D) 54

113. The chord of contact from P to $x^2/9-y^2/36=1$ is x=3/2. Find the product of the slopes of the two tangents from P.

(A) $\frac{4}{3}$ (B) -1

(C) $-\frac{4}{9}$ (D) $-\frac{4}{3}$

114. Find the coefficient of x^4 in $(1+x+x^2)(1+x^2)^9$.

(A) 45 (B) 120

(C) 108 (D) 36

115. The locus |z-5|=|Re(z)+5| intersects the horizontal line Im(z)=12. Find the real part of the point.

(A) $\frac{72}{5}$ (B) $\frac{3}{5}$

(C) $\frac{36}{5}$ (D) $-\frac{36}{5}$

116. F′(x)=$\frac{5}{(x+1)(x+6)}$ and F(0)=5. Find F(1)-2F(0).

(A) $-5+\log\left(\frac{12}{7}\right)$ (B) $\log\left(\frac{12}{7}\right)+5$

(C) $-5-\log(12)+\log(7)$ (D) $\log\left(\frac{12}{7}\right)$

117. Let $a_j = 1/(5+3(j-1))$. Evaluate $\sum_{j=1}^{8} a_j a_{j+1}$.

(A) $\frac{7}{130}$ (B) $\frac{24}{145}$

(C) $\frac{8}{145}$ (D) $\frac{4}{13}$

118. An invertible 4×4 matrix A has det A=3. Find det(adj A).

(A) 12 (B) 3

(C) 27 (D) $\frac{1}{3}$

119. y′+11y=exp(-11x), y(0)=2. Find exp(11)y(1)-2.

(A) $\frac{3}{e^{11}}$ (B) -1

(C) 1 (D) 3

120. Let $y\prime + 6y = 5x$ and $y(0) = 5$. Find $y(1) + \frac{6}{5}y\prime(1)$.

(A) $\frac{1147}{36e^6} + \frac{61}{36}$ (B) $\frac{185}{36e^6} + \frac{25}{36}$

(C) $\frac{61}{36} - \frac{1147}{36e^6}$ (D) $\frac{61}{36}$

121. Evaluate the product of $\cos(2^j\pi/65)$ for j=0,...,5.

(A) $\frac{1}{64}$ (B) $-\frac{1}{64}$

(C) $\frac{1}{32}$ (D) $\frac{1}{65}$

122. x=2+5(1-t^2)/(1+t^2), y=3+10t/(1+t^2), where t is real. Which point of the containing circle is not attained?

(A) $(7,3)$ (B) $(2,8)$

(C) $(2,-2)$ (D) $(-3,3)$

123. How many three-element subsets of {1,2,...,15} contain both an odd and an even number?

(A) 364 (B) 196

(C) 420 (D) 399

124. $\log_2$a=2, log_a b=3, log_b c=3. Find $\log_4(c^2/a)$.

(A) 19 (B) 18

(C) 34 (D) 17

125. A line through P=(3,6) has direction (3,4). It meets 3x+4y=183 at Q. Find $|PQ|^2$.

(A) 36 (B) 30

(C) 150 (D) 900

126. If z=5-6i, find |exp(iz)|.

(A) e^5 (B) e^6

(C) e^{-6} (D) 1

127. A complex number z satisfies z+conjugate(z)=4 and z-conjugate(z)=12i. Find z·conjugate(z).

(A) 160 (B) 40

(C) 148 (D) 80

128. Maximize Z=3x+4y subject to x≥0, y≥6, x+y≤16, x+2y≤21.

(A) 53 (B) 51

(C) 54 (D) 42

129. Write sin x $\cos^3$x+2$\sin^3$x cos x=P sin2x+Q sin4x. Find Q.

(A) $-\frac{1}{8}$ (B) $\frac{1}{8}$

(C) $\frac{3}{8}$ (D) $\frac{3}{4}$

130. Find the sum of order and degree of $(d^3y/dx^3)^2+(dy/dx)^2+y=0$.

(A) 2 (B) 7

(C) 5 (D) 6

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. Two sound sources produce wavelengths $\frac{1}{3}$ m and $\frac{1}{4}$ m in the same gas at one temperature. The sound speed there is 400 m/s. At a second temperature the sound speed is 3/2 times larger, while both source frequencies stay unchanged. Find the new beat frequency.

(A) 400 Hz

(B) 2800 Hz

(C) 1200 Hz

(D) 600 Hz

132. A parallel-plate capacitor has vacuum capacitance C_0. A dielectric of constant 5 fills a fraction $\frac{1}{2}$ of the plate area and the entire gap in that region; the remainder stays vacuum. The voltage is fixed. Find the fraction of total stored energy residing in the dielectric-filled region.

(A) $\frac{1}{6}$

(B) $\frac{5}{6}$

(C) $\frac{5}{2}$

(D) $\frac{1}{2}$

133. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 1 kΩ to a terminal at +12 V, and its cathode connects to a terminal at -1 V. Find the current from the +12 V terminal in mA.



(A) $\frac{137}{10}$ mA

(B) $\frac{113}{10}$ mA

(C) $\frac{123}{10}$ mA

(D) 13 mA

Chemistry

134. Two oxides of X contain 32 g X combined respectively with 32 g O and 48 g O. Atomic masses are X=32, O=16. Which empirical-formula conclusion jointly follows from mole ratios and the law of multiple proportions?

(A) Both are XO because X masses are equal.

(B) First empirical formula XO2; second empirical formula XO3; fixed-X oxygen mass ratio second:first=3/2.

(C) No formula can be inferred even though both atomic masses and both compositions are supplied.

(D) First oxide XO3; second oxide XO2; fixed-X oxygen mass ratio second:first=2/3.

135. 2,6-dimethylphenol undergoes Reimer-Tiemann formylation with CHCl3/aqueous NaOH and acidic work-up. Use ordinary ortho preference where an ortho site is available. Which major formylated product follows the unoccupied OH-directed sites?

(A) 4-hydroxy-3,5-dimethylbenzaldehyde

(B) the product obtained by replacing a methyl carbon with CHO

(C) the corresponding benzoic acid formed by replacement of OH

(D) 2-hydroxy-3,5-dimethylbenzaldehyde

136. Construct one open linear peptide using each member of the following multiset exactly once: Pro, Gly, Ala, Val, Ser, Leu. The N-terminal residue isPro, the C-terminal residue is the only achiral amino acid, and Ala immediately precedesVal. Count distinctN-to-C sequences; ignore stereoisomerism of the supplied residues.

(A) 120 (B) 24

(C) 6 (D) 720

Logical Reasoning

137. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 3, 9, 11, 33, ?, 105, 107.

(A) 55 (B) 35

(C) 101 (D) 97

138. Q is the only child of C, and Q is a parent of F. C is the maternal grandmother of F. What can be concluded about Q's gender?

(A) Female (B) The stated relationships are inconsistent

(C) Cannot be determined (D) Male

139. 6 people sit around a circle facing its centre. Clockwise order fixes the convention; immediate left is clockwise. Charu is 1 places clockwise from Asha. Dev is 3 places clockwise from Asha. Esha is 4 places clockwise from Asha. Farah is not adjacent to Dev. Who sits immediately clockwise from Esha?

(A) Charu (B) Bina

(C) Farah (D) Asha

Mathematics

140. Evaluate $\sum_{r=0}^{7} r(r-1)\binom{7}{r}$.

(A) 1344 (B) 2688

(C) 448 (D) 1568

141. The line y=mx through O bisects the area of the triangle with vertices O,(6,3),(7,7). Assume the vertices are noncollinear. Find m.

(A) $\frac{7}{6}$ (B) $\frac{13}{10}$

(C) $\frac{10}{13}$ (D) 4

142. A=(3,4,4), O=(0,0,0). Find the squared length of the orthogonal projection of OA on the line through O with direction (1,2,2).

(A) $\frac{8}{9}$ (B) $\frac{361}{9}$

(C) $\frac{19}{3}$ (D) 361

BITSAT Mock 12

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. The quantity $Z = x^3y^2/\sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 1, 1 and 4, respectively. To first order, what is the maximum percentage error in Z?

(A) 4 (B) $\sqrt{17}$

(C) 7 (D) 6

2. A Carnot engine initially runs between 900 K and 200 K. Its hot reservoir is raised by 40 K. By how much must the cold-reservoir temperature rise to keep efficiency unchanged?

(A) $\frac{400}{47}$ K (B) 40 K

(C) $\frac{80}{9}$ K (D) $\frac{280}{9}$ K

3. A particle moves with constant acceleration. It travels 49/2 m in the (7)th second and 55/2 m in the 8th second. What was its initial velocity?

(A) $\frac{7}{2}$ m/s (B) 3 m/s

(C) 5 m/s (D) $\frac{13}{2}$ m/s

4. A network is defined by $P = \mathrm{NOR}(A, B)$, $Q = \mathrm{NAND}(A, P)$ and $Y = \mathrm{NAND}(Q, B)$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 0$ and $A = 0$?

(A) 3 (B) 1

(C) 2 (D) 0

5. A bar magnet of moment 3 A m^2 is rotated quasistatically from 60° to 120° relative to a uniform 6 T magnetic field. Find external work.

(A) 6π J (B) 18 J

(C) $18\sqrt{3}$ J (D) 9 J

6. A 9 V source feeds a circuit containing one permanent 4 ohm resistor and a second identical resistor connected in parallel through an ideal photo-switch. The switch closes only if photon energy is at least the semiconductor band gap. Band gap is 1 eV and photon energy is $\frac{3}{2}$ eV. Find source current; each active resistor sees the full source voltage.

(A) $\frac{9}{2}$ A (B) $\frac{9}{8}$ A

(C) 2 A (D) $\frac{9}{4}$ A

7. A wire consists of a semicircular arc of radius 2 m carrying 8 A, joined to two straight radial segments that extend outward from its ends. Find the magnetic field magnitude at the arc centre in units of μ_0.

(A) $\frac{1}{\pi}$ (B) 2

(C) $\frac{2+\pi}{\pi}$ (D) 1

8. Two ideal cells of emfs 18 V and 4 V oppose each other in a loop containing two resistors each 3 ohms. Find the current magnitude.

(A) 3 A (B) $\frac{11}{3}$ A

(C) $\frac{14}{3}$ A (D) $\frac{7}{3}$ A

9. A diamond network has four outer resistors AB, BC, CD and DA, each 4 ohms, and a bridge BD also 4 ohms. The DA arm additionally contains an ideal diode with anode at D and cathode at A. A source keeps A positive relative to C. Find equivalent resistance between A and C.

(A) $2\,\Omega$ (B) $\frac{16}{3}\,\Omega$

(C) $\frac{20}{3}\,\Omega$ (D) $12\,\Omega$

10. A particle of mass 4 kg starts from rest and experiences $F(t) = 1t$ N along its motion for 3 s. Find the work done by the force.

(A) $\frac{81}{16}$ J (B) $\frac{81}{32}$ J

(C) $\frac{27}{2}$ J (D) $\frac{81}{8}$ J

11. Two identical converging thin lenses of focal length 25 cm are separated by 100 cm. A parallel beam enters the first. Find the distance of the final image from the second lens, positive on the outgoing side.

(A) 75 cm (B) $\frac{75}{2}$ cm

(C) $\frac{100}{3}$ cm (D) 25 cm

12. A solid sphere rolls without slipping from rest through vertical height 11 m. A thin ring does so through vertical height 6 m. Find the ratio of their squared centre-of-mass speeds, sphere to ring.

(A) $\frac{55}{21}$ (B) $\frac{\sqrt{1155}}{21}$

(C) $\frac{11}{3}$ (D) $\frac{11}{6}$

13. A plane wave in vacuum travels along $+x$. At one instant its electric field is $450\hat{k}$ V/m. If the electric amplitude is 3 times this instantaneous value, which gives the instantaneous magnetic field? Use $c = 3 \times 10^8$ m/s.

(A) $+150 \times 10^{-8}\hat{i}$ T (B) $+150 \times 10^{-8}\hat{j}$ T

(C) $-150 \times 10^{-8}\hat{j}$ T (D) $-450 \times 10^{-8}\hat{j}$ T

14. In a series RLC circuit, ideal rms voltmeters read 40 V across R, 40 V across L and 10 V across C. Find source rms voltage.

(A) 30 V (B) $10\sqrt{41}$ V

(C) $10\sqrt{33}$ V (D) 50 V

15. For a positive constant A, which expression describes SHM about its own equilibrium position?

(A) $x = Ae^{8t}$.
(B) $x = A\sin(8t) + A\sin(16t)$.
(C) $x = A\sin^2(8t)$.
(D) $x = A\sin(8t^2)$.

16. A source emitting 300 Hz and a listener approach each other, each at 20 m/s relative to still air. Sound speed is 300 m/s. Find heard frequency before they pass.

(A) $\frac{2400}{7}$ Hz
(B) $\frac{2250}{7}$ Hz
(C) 340 Hz
(D) $\frac{4500}{13}$ Hz

17. A one-electron ion with nuclear charge $Z = 2$ emits when its orbit changes. The ratio of initial to final orbit radii is 16, and its final principal quantum number is 1. What is $R\lambda$ for the emitted photon? Here R is the hydrogen Rydberg constant.

(A) $\frac{4}{15}$
(B) $\frac{4}{17}$
(C) $\frac{1}{3}$
(D) 4

18. A capacitor of vacuum capacitance 4 microfarads remains connected to a 6 V battery while a dielectric of constant 5 is inserted fully. Find the increase in stored electrostatic energy.

(A) 576 μJ
(B) 288 μJ
(C) $\frac{288}{5}$ μJ
(D) 360 μJ

19. A particle has $v(0) = 1$ m/s and acceleration $a(t) = 8t$ m/s^2 for $0 \le t \le 5$ s. Find its average velocity over this interval.

(A) 1 m/s
(B) 41 m/s
(C) $\frac{100}{3}$ m/s
(D) $\frac{103}{3}$ m/s

20. A block is placed on the vertical inner wall of a trolley. The trolley accelerates horizontally so the wall pushes the block. With coefficient of static friction $\frac{2}{5}$, what minimum trolley acceleration prevents vertical slipping? Take $g = 10$ m/s^2.

(A) 25 m/s^2
(B) 10 m/s^2
(C) $\frac{50}{7}$ m/s^2
(D) 4 m/s^2

21. An insulated vessel containing one mole of an ideal gas of molar mass $\frac{7}{100}$ kg/mol moves at bulk speed 30 m/s and is brought to rest. All bulk kinetic energy becomes internal energy. The gas has 5 active degrees of freedom. Find temperature rise. Use $R = 8.314$ J/(mol K).

(A) $\frac{6300}{4157}$ K
(B) $\frac{31500}{4157}$ K
(C) $\frac{3150}{4157}$ K
(D) $\frac{4200}{4157}$ K

22. A coil of inductance 8 H carries current $i(t) = 6t^2$ A. Find the magnitude of its self-induced emf at $t = 4$ s.

(A) 1536 V
(B) 384 V
(C) 48 V
(D) 768 V

23. Light is incident from index $\frac{7}{5}$ onto index 2. At one incidence angle the reflected beam is completely plane polarised. Find the tangent of that angle.

(A) 2 (B) $\frac{10}{7}$

(C) $\frac{3}{7}$ (D) $\frac{\sqrt{51}}{7}$

24. A Newtonian liquid of viscosity $\frac{3}{10}$ Pa s has velocity profile $v(y) = 4y + 2y^2$ in SI units. Find the shear-stress magnitude at $y = 2$ m.

(A) $\frac{6}{5}$ Pa (B) $\frac{12}{5}$ Pa

(C) $\frac{24}{5}$ Pa (D) $\frac{18}{5}$ Pa

25. A charge of $4\,\mu$C moves from potential 50 V to -50 V in an electrostatic field. Find the work done by an external agent if the charge starts and ends at rest and no other force does work.

(A) $-100\,\mu$J (B) $-400\,\mu$J

(C) $400\,\mu$J (D) $-8\,\mu$J

26. A spherical blackbody's absolute temperature is multiplied by 3 while its radius is divided by 3. Neglect incoming background radiation. Find its emitted-power multiplier.

(A) 1 (B) 27

(C) $\frac{1}{3}$ (D) 9

27. Light of vacuum wavelength 600 nm passes from a medium of index 3 into one of index 4. Find the ratio of its frequencies in the second and first media.

(A) $\frac{1}{4}$ (B) 4

(C) 1 (D) $\frac{3}{4}$

28. A satellite of mass m is moved from a circular orbit of radius R to one of radius 13R about mass M. Find the required increase in total orbital energy in units of $GMm/(2R)$.

(A) $\frac{12}{13}$ (B) $\frac{24}{13}$

(C) 12 (D) $\frac{168}{169}$

29. A nonrelativistic charged particle initially has momentum $1qEt$ along a uniform electric field, where $t > 0$ is a fixed reference interval. Its charge is $+q$. What is $\lambda(6t)/\lambda(5t)$ after acceleration in the same field?

(A) $\frac{1}{7}$ (B) $\frac{36}{49}$

(C) $\frac{\sqrt{42}}{7}$ (D) $\frac{6}{7}$

30. A uniform thin rod of length 4 m is hinged at one end and released from rest in a horizontal position. Find the magnitude of its initial angular acceleration. Take $g = 10$ m/s^2.

(A) $\frac{15}{4}$ rad/s^2 (B) 60 rad/s^2

(C) $\frac{15}{2}$ rad/s^2 (D) $\frac{5}{2}$ rad/s^2

Chemistry

31. For O_3,use ordinary octet Lewis structures and resonance. What is the pair (average bond order of two equivalent O-O bonds,total lone pairs on all atoms of that same named species in one contributing structure)?

(A) $(\frac{3}{2}, 6)$.
(B) (2, 7).
(C) $(\frac{3}{2}, 5)$.
(D) (1, 6).

32. aniline is acetylated at nitrogen,then subjected to controlled bromination,then its N-acetyl group is hydrolysed. Use the usual para preference relative to NHCOCH3 where free. Which principal final product results?

(A) bromobenzene
(B) 3-bromoaniline
(C) 2,4,6-tribromoaniline
(D) 4-bromoaniline

33. For isolated molecules of H_2O and CO_2,which member has the larger permanent dipole moment,and which reasoning is sufficient?

(A) The one with more atoms always has the larger dipole,independent of geometry.

(B) H_2O; Water is bent so its bond dipoles do not cancel; linear carbon dioxide has opposite equal bond dipoles.

(C) They must be equal because both contain polar covalent bonds.

(D) CO_2; the number of polar bonds alone fixes the molecular dipole.

34. benzoyl chloride is subjected to H2/Pd-BaSO4(poisoned catalyst). Which product or product pair follows the named reaction and side-chain oxidation state?

(A) benzyl alcohol
(B) benzoic acid
(C) toluene
(D) benzaldehyde

35. An ideal mixture initially has $\frac{1}{10}$molA and $\frac{3}{10}$molB. A selective step removes half ofB and noA. At the new equilibriumT,V,totalpressure is6bar. Partialpressure ofA,in bar,is

(A) 15
(B) $\frac{12}{5}$
(C) 3.0
(D) $\frac{3}{2}$

36. Two hydrogen-like bound states A and B have nuclear charges ZA=1 and ZB=2, and radii rB=2rA. The states obey the allowed integer-principal-quantum-number Bohr model. What is the ratio of binding-energy magnitudes |EB|/|EA|?

(A) 1/2
(B) 1
(C) 3
(D) 2

37. A first-order reaction has already consumed 75%of its initial A. Additional time needed to leave12.5% of the original A,expressed in half-lives,is ($\log_2 3$=1.585)

(A) 3
(B) 2
(C) 4
(D) 1

38. In a regime where attraction is negligible,Z=1+pb/(RT). The Z-versus-p slope for gasA at300K is4times that for gasB at600K. Ratio bA/bB is

(A) 4 (B) 2

(C) $\frac{1}{4}$ (D) 8

39. At a fixed temperature K=4. A reaction mixture initially has Q=16. Which direction lowers Gibbs energy, and which composition change must occur before equilibrium can be reached? Use ΔrG=RT ln(Q/K).

(A) Neither direction whenever K is not one; equilibrium requires K=1. (B) reverse; Q must decrease to K.

(C) Reverse in every case because the standard reaction is exergonic. (D) Forward in every case because K exceeds one.

40. butan-2-one reacts with NH2OH. Which product correctly reflects nucleophilic addition or addition-elimination at its carbonyl rather than a different reaction class?

(A) butanoic acid (B) ethyl propyl ether

(C) butan-2-one oxime (D) butan-2-ol

41. Heating disrupts the native folding of a protein without hydrolysing peptide bonds. Which distinction follows?

(A) Every amino acid must become glucose.

(B) Secondary/tertiary organisation can change while the amino-acid sequence remains.

(C) All peptide bonds must break whenever folding changes.

(D) Primary sequence is defined only by hydrogen-bond shape.

42. Compare dilute solutions of 2-hydroxybenzoic acid and 4-hydroxybenzoic acid in a non-hydrogen-bonding solvent. Which conclusion joins structure with the expected effect of further dilution on their hydrogen bonding?

(A) Both lose all H bonding equally because every H bond requires two molecules.

(B) 2-hydroxybenzoic acid can retain an intramolecular H bond on dilution; intermolecular association of 4-hydroxybenzoic acid is reduced.

(C) 2-hydroxybenzoic acid must have stronger intermolecular association because its internal acceptor is farther away.

(D) Only 4-hydroxybenzoic acid can make an intramolecular H bond because para groups form a shorter ring.

43. Dry air is taken to contain21.0% O2by volume. From 100L of this air, an isolation process recovers80.0% of its oxygen into a pure stream. All gas volumes refer to the same T,P. Find recovered O2volume in litres.

(A) $\frac{316}{5}$ (B) $\frac{105}{4}$

(C) 21 (D) $\frac{84}{5}$

44. A photographic fixer contains $\frac{9}{100}$ mol free $S_2O_3^{2-}$ and receives $\frac{3}{50}$ mol unexposed AgBr. Use $AgBr+2S_2O_3^{2-}\rightarrow[Ag(S_2O_3)_2]^{3-}+Br^-$. Treat complexation as complete until a reagent is exhausted. AgBr remaining,in mol,is

(A) $\frac{3}{200}$
(B) $\frac{3}{50}$
(C) $\frac{3}{100}$
(D) 0

45. On polar silica with an unchanged weakly polar eluent, benzyl alcohol is oxidised to benzaldehyde. In a given separation the aldehyde travels farther than the alcohol. Which assignment of Rf values is consistent with that observation?

(A) Alcohol0.25; aldehyde0.25.
(B) Alcohol0.60; aldehyde0.25.
(C) Alcohol0.25; aldehyde0.60.
(D) Alcohol1.25; aldehyde0.60.

46. A one-electron ion has nuclear charge Z=3. A bound level has the same energy as H at n=2. In that ion shell,maximum electrons allowed by Pauli in orbitals with$m_l = 0$ would be (ignore electron interactions for the energy comparison)

(A) 12
(B) 36
(C) 72
(D) 6

47. In the gas phase,O_2 is converted into O_2^- by changing one electron. Under the elementary MO model,which final bond-order/unpaired-electron pair follows?

(A) Bond order 2; 2 unpaired electrons.
(B) Bond order $\frac{3}{2}$; 2 unpaired electrons.
(C) Bond order $\frac{3}{2}$; 1 unpaired electron.
(D) Bond order $\frac{5}{2}$; 1 unpaired electrons.

48. An ordinary liquid and a dilute gas are cooled over a range in which neither changes phase. Which prediction is consistent with the distinct origins of viscosity?

(A) Liquid viscosity decreases; gas viscosity increases.
(B) Both viscosities increase because viscosity has one universal temperature dependence.
(C) Liquid viscosity increases; gas viscosity decreases.
(D) Both viscosities remain constant until a phase transition occurs.

49. A mixture contains 1/50 mol total of the ionisation isomers [Co(NH3)5Br]SO4 and [Co(NH3)5SO4]Br. An immediately tested aliquot representing the entire inventory gives 2/125 mol AgBr with excess Ag+. Coordinated ligands do not exchange, and no other bromide exists. What mole fraction of the inventory was [Co(NH3)5Br]SO4?

(A) 2/5
(B) 1/10
(C) -1/5
(D) 1/5

50. diethyl adipate,EtOOC-(CH2)4-COOEt undergoes NaOEt,then exhaustive aqueous acid hydrolysis and heat. Under the stated intramolecular closure pathway,which isolated product matches both ring size and work-up endpoint?

(A) cyclopentanone
(B) cyclohexanone
(C) pentanoic acid
(D) cyclopentane

51. A hot saturated solution contains 150g water and60.0g solute per100g water. Cooling to a temperature where solubility is20.0g per100g water also removes 15g water by evaporation. No solute evaporates,and crystals are anhydrous. Mass crystallised in g is

(A) 60 (B) 57

(C) 27 (D) 63

52. Compare the elementary octahedral d-d splitting transitions of A=$[CoF_6]^{3-}$,B=$[Co(NH_3)_6]^{3+}$, C=$[Co(CN)_6]^{3-}$. For this comparison use only the usual spectrochemical ordering and $E = hc/\lambda$. Which ordering of absorbed wavelengths follows?

(A) $\lambda_A > \lambda_B > \lambda_C$. (B) $\lambda_A < \lambda_B < \lambda_C$.

(C) $\lambda_A = \lambda_B = \lambda_C$ because metal oxidation states match. (D) $\lambda_B > \lambda_C > \lambda_A$.

53. A textbook deficiency table associates convulsions associated with deficiency with one vitamin. Which vitamin and solubility class are correctly paired?

(A) Vitamin B6; water-soluble. (B) VitaminD; fat-soluble.

(C) Vitamin B12; water-soluble. (D) VitaminA; fat-soluble.

54. The ions O^{2-}, F^-, Na^+, Mg^{2+} each have 10 electrons. Which decreasing-radius order and explanation is consistent?

(A) $F^- > Na^+ > Mg^{2+} > O^{2-}$; positive charge always gives the largest ion.

(B) $O^{2-} < F^- < Na^+ < Mg^{2+}$; larger nuclear charge expands the cloud.

(C) $O^{2-} > F^- > Na^+ > Mg^{2+}$; increasing nuclear charge contracts the same electron cloud.

(D) All radii are equal because electron numbers are equal.

55. In water,ignoring special ortho steric effects,which decreasing basicity order is most appropriate for 4-methylaniline,aniline,4-nitroaniline?

(A) 4-methylaniline >4-nitroaniline > aniline (B) 4-methylaniline > aniline >4-nitroaniline

(C) aniline >4-methylaniline >4-nitroaniline (D) 4-nitroaniline > aniline >4-methylaniline

56. Consider the statements. I. For a perfectly monodisperse polymer, Mw/Mn=1. II. A distribution with Mw/Mn>1 contains only one exact molecular mass. Choose the correct truth pattern.

(A) I true; II false. (B) I false; II false.

(C) I false; II true. (D) I true; II true.

57. For isolated Cr^{3+} (Z=24),use the high-spin spin-only approximation. Which pair (d-electron count, magnetic moment in BM) follows after removing electrons from 4s before 3d?

(A) $(3,\sqrt{15})$. (B) (3,3).

(C) $(4,2\sqrt{6})$. (D) $(3,2\sqrt{6})$.

58. For one reaction,ln K is3 at300 K and0 at600 K. Assume ln K is linear in1/T. At400 K,ln K equals

(A) 3 (B) 0

(C) $\frac{3}{2}$ (D) $\frac{5}{2}$

59. For a reaction,$\log_{10}K_p$ is 2 at300 K and $\frac{11}{3}$ at600 K. The plot is linear in1/T. Using2.303forln10,find the positive magnitude$|\Delta H^\circ|/R$ in K.

(A) $\frac{2303}{2}$ (B) 4606

(C) 2303 (D) 1000

60. Consider the statements. I. Aerobic decomposition of a large dead algal bloom can consume dissolved oxygen. II. Adding more phosphate is a direct way to remove the initiating nutrient surplus. Choose the correct truth pattern.

(A) I true; II true. (B) I false; II true.

(C) I false; II false. (D) I true; II false.

English Proficiency

61. Complete: The conclusion is _____, not because it has been proved false, but because the available evidence does not yet justify confidence in it.

(A) irrefutable (B) tentative

(C) definitive (D) conclusive

62. Read the sentence: In that dense passage, many closely packed ideas appear in a few lines. Choose the opposite in meaning of 'dense' as used in the sentence.

(A) difficult (B) loosely developed

(C) compressed (D) technical

63. Choose the most coherent order of the four sentences. A. Several young plants grew poorly despite receiving enough water. B. A week of observations showed that the corner received very little direct light. C. The next planting used species better suited to the conditions actually recorded there. D. The school planted a garden in a corner shaded by the main building.

(A) D-A-C-B (B) A-D-B-C

(C) D-B-A-C (D) D-A-B-C

64. The mark cannot be erased from the surface once it has formed. Which adjective fits?

(A) indelible (B) permanent-looking

(C) invisible (D) illegible

65. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The clerk records each correction before returning the form.

(A) Each correction has been recorded by the clerk before returning the form.

(B) Each correction is recorded after the form is returned.

(C) Every form records the clerk before it corrects the return.

(D) Each correction is recorded by the clerk before the form is returned.

66. Read the sentence: The speaker was assertive, stating her needs clearly without insulting anyone. Choose the closest meaning of 'assertive' as used in the sentence.

(A) unwilling to compromise (B) forcefully aggressive

(C) confidently direct (D) eager to interrupt

67. Which figure of speech is chiefly illustrated by the line? The footnote was a window into a debate the main text barely mentioned.

(A) metaphor
(B) apostrophe
(C) alliteration
(D) simile

68. Choose the meaning-preserving equivalent that retains the tense, participants and all stated restrictions. Despite having little time, the team verified the key figures.

(A) Because the team had little time, it did not verify the key figures.
(B) Although the team had little time, it verified the key figures.
(C) The team verified all figures because it had plenty of time.
(D) The team verified the key figures only when it had little time.

69. Choose the complete and grammatical sentence.

(A) Had the alarm sounded earlier, the visitors would have left sooner.
(B) Had the alarm sound earlier, the visitors would left sooner.
(C) Had sounded the alarm earlier, would have left the visitors sooner.
(D) Had the alarm sounded earlier, would have left sooner.

70. Read the sentence: The empty shelves presented a stark contrast to the fully stocked shop next door. Choose the closest meaning of 'stark' as used in the sentence.

(A) simple and attractive
(B) unexpected and brief
(C) sharp and striking
(D) unfair and misleading

Logical Reasoning

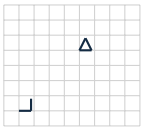
71. 8 people sit around a circle facing its centre. Clockwise order fixes the convention; immediate left is clockwise. Dev is 1 places clockwise from Asha. Hari is 3 places clockwise from Asha. Charu is 5 places clockwise from Asha. Farah is 6 places clockwise from Asha. Bina is 7 places clockwise from Asha. Gita is not adjacent to Dev. Who sits immediately clockwise from Dev?

(A) Charu
(B) Bina
(C) Asha
(D) Esha

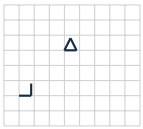
72. The recorded average of 7 readings is 26. One entry recorded as 39 should have been 37. After that correction, another reading of 20 is replaced by 31. What is the final average?

(A) $\frac{198}{7}$
(B) $\frac{193}{7}$
(C) $\frac{191}{7}$
(D) $\frac{180}{7}$

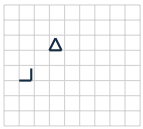
73. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



1

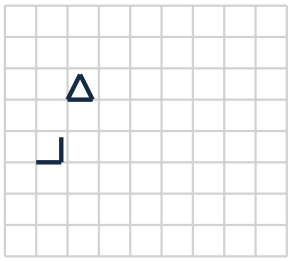


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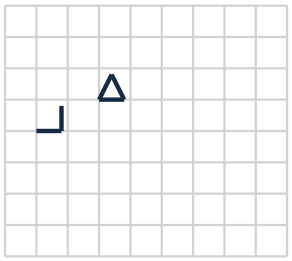


3

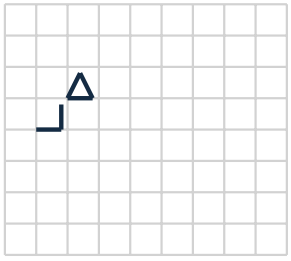
(A) Pattern V



(B) Pattern U



(C) Pattern P



(D) Pattern W



74. In this code, @ means $\times$, # means $\div$, & means $+$, % means $-$. After substituting meanings, evaluate ((7#6)@(5&4)); printed parentheses retain their usual grouping.

(A) $\frac{-21}{2}$

(B) $\frac{21}{2}$

(C) $\frac{23}{2}$

(D) $\frac{19}{2}$

75. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

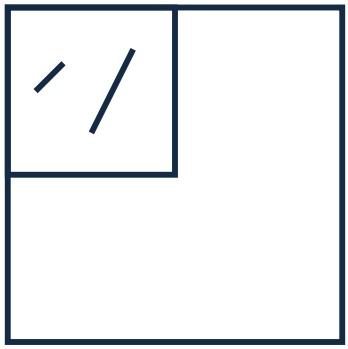
(A) 3

(B) 4

(C) 1

(D) 2

76. Starting with a unit square sheet, fold right to left along the vertical midline, then fold bottom to top along the horizontal midline. The folded paper is cut along both short drawn slits. Select the pattern after complete unfolding.



(A) Pattern K



(B) Pattern T



(C) Pattern Z



(D) Pattern Q



77. All five people have different heights. Farah is shorter than Bina. Asha is shorter than Esha. Esha is shorter than Hari. Bina is shorter than Asha. If Asha is left out, who is 3rd shortest among the remaining people?

(A) Farah
(B) Esha
(C) Hari
(D) Bina

78. Consider only these statements; universal statements alone do not imply existence. No hikers are bakers; No joggers are bakers; All joggers are bakers. Conclusions: I. No bakers are joggers. II. All hikers are joggers. Which conclusions follow necessarily?

(A) Only I
(B) Only II
(C) Both I and II
(D) Neither I nor II

79. A and S are full siblings; S and R are also full siblings. H is the mother of A. All four people are distinct. How is H related to R?

(A) Aunt

(B) Mother

(C) Grandmother

(D) Relationship cannot be determined

80. The model triads are $(10, 24, 14)$ and $(12, 28, 16)$. In the intended rule the middle number is the sum of the outer numbers, and the outer-number difference is the same fixed positive value in every triad. Which option satisfies both relations?

(A) $(9, 23, 13)$

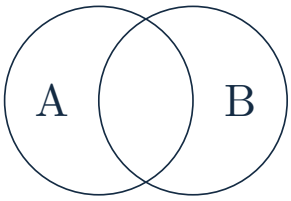
(B) $(9, 24, 14)$

(C) $(9, 23, 14)$

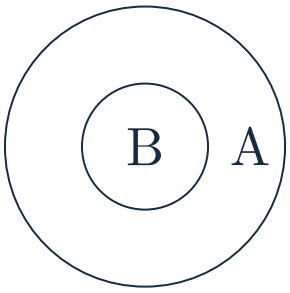
(D) $(9, 22, 13)$

81. Choose the diagram that represents all these conditions, including the strictness information: Both A and B are nonempty, and no A is B.

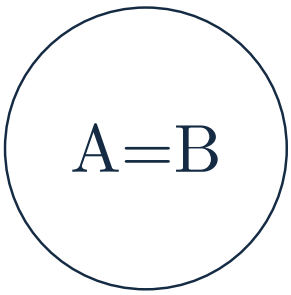
(A) Pattern W



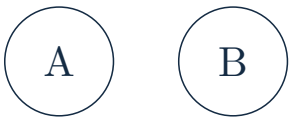
(B) Pattern Q



(C) Pattern N



(D) Pattern R



82. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

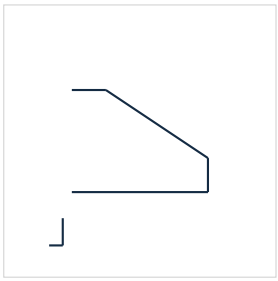
(A) 2

(B) 3

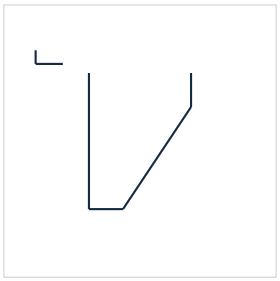
(C) 1

(D) 4

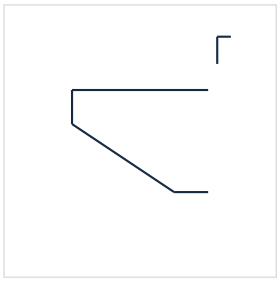
83. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

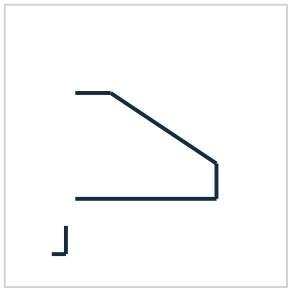


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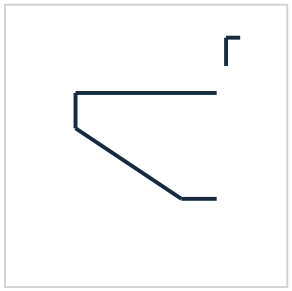


3

(A) Pattern N



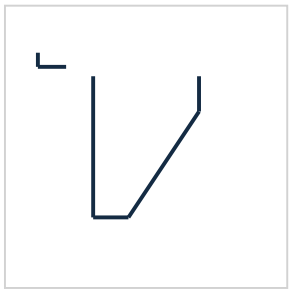
(B) Pattern Z



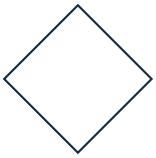
(C) Pattern M



(D) Pattern K



84. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

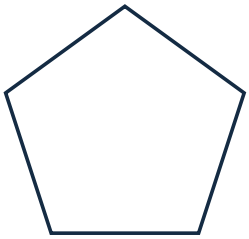


B

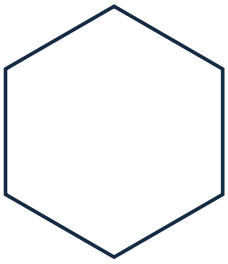


C

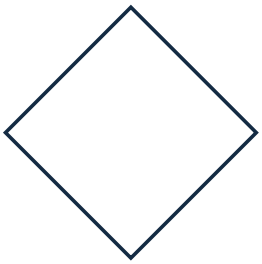
(A) Pattern M



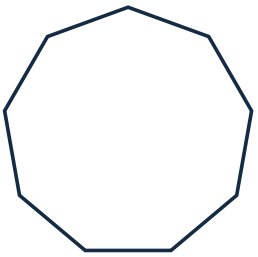
(B) Pattern Z



(C) Pattern T



(D) Pattern N



85. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: ENEK, GLHO, ?, KHNW, MFQA, ODTE.

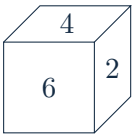
(A) IJKS

(B) SKJI

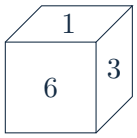
(C) HIJR

(D) JKLT

86. The views show the same die. Give the ordered pair: (face opposite 2, face opposite 6). Do not assume opposite labels sum to seven.



1



2

(A) $(3, 3)$

(B) $(3, 2)$

(C) $(3, 5)$

(D) $(3, 1)$

87. Preserve the exact relation and its order: Barometer : Atmospheric pressure :: ? : Temperature. Choose the missing term.

(A) Thermometer (B) Voltmeter

(C) Barometer (D) Ammeter

88. Consider only these statements; universal statements alone do not imply existence. Some campers are campers; All artists are designers. Conclusions: I. Some campers are artists. II. No campers are artists. Which conclusion or complementary alternative must hold?

(A) Only I (B) Either I or II follows, but neither is individually certain

(C) Both I and II (D) Only II

89. For this question compare only cyclic forward alphabet gaps, with $Z \to A$ counting as one step. Three clusters share the same gap vector. Which numbered cluster differs? 1: OPRS, 2: NOQR, 3: ABEF, 4: DEGH.

(A) 4 (B) 2

(C) 1 (D) 3

90. In this series the first differences themselves form an arithmetic progression. Find the missing internal term: 9, 10, 14, ?, 31, 44, 60.

(A) 20 (B) 22

(C) 21 (D) 23

Mathematics

91. Find the maximum of $\sin^2 x \cos x$ on $[0,\pi/2]$.

(A) $\frac{2\sqrt{3}}{9}$ (B) $\frac{1}{2}$

(C) $\frac{1}{3}$ (D) $\frac{\sqrt{3}}{3}$

92. X takes values [-3, 0, 2] with probabilities 1/6,1/3,1/2. Let Y=3X+5. Find $E(Y^2)$-$[E(Y)]^2$.

(A) $\frac{39}{4}$ (B) $\frac{217}{4}$

(C) $\frac{117}{4}$ (D) $\frac{63}{2}$

93. y'-y/x=1 for x>0, y(1)=8. Find y(6)/6-y(1).

(A) $\log(6)$ (B) $\log(6) + 8$

(C) $-\log(6)$ (D) $6\log(6) + 48$

94. Find the area enclosed by y=x(12-x) and y=3x.

(A) $\frac{567}{2}$ (B) $\frac{243}{2}$

(C) 288 (D) $-\frac{243}{2}$

95. A 7-element set contains distinct a,b. How many subsets contain a and exclude b?

(A) 32 (B) 128

(C) 64 (D) 126

96. How many integers x satisfy |(x-5)+(x-8)|<|x-5|+|x-8|?

(A) 4 (B) 2

(C) 7 (D) 3

97. Evaluate lim as n→ ∞ of (5/n)Σ(5j/n)^2, j=1,...,n.

(A) $\frac{125}{3}$ (B) $\frac{1}{3}$

(C) $\frac{25}{3}$ (D) $\frac{125}{2}$

98. The line r=(1,0,0)+s(t,1,2) is parallel to x-6y+z=8. Find the squared length of its direction vector.

(A) 19 (B) 41

(C) 21 (D) $\sqrt{21}$

99. Evaluate $\sum_{j=1}^{\infty} 3j(\frac{1}{5})^j$.

(A) $\frac{3}{16}$ (B) $\frac{27}{16}$

(C) $\frac{15}{16}$ (D) $\frac{15}{4}$

100. Evaluate $\int_0^{\infty}$ dx/[(x^2+36)(x^2+81)].

(A) $-\frac{\pi}{1620}$ (B) $\frac{\pi}{810}$

(C) $\frac{\pi}{1620}$ (D) $\frac{\pi}{108}$

101. Box A contains 4 red and 3 blue balls; B contains 6 red and 5 blue balls. A fair coin chooses one box. Two independent draws with replacement from that box give exactly one red ball. Find the posterior probability that A was chosen.

(A) $\frac{12}{49}$ (B) $\frac{245}{487}$

(C) $\frac{242}{487}$ (D) $\frac{22}{43}$

102. Let k make lim as x→ ∞ of [(2x^2+3x+1)/(x+4)-kx] finite. Find that finite limit.

(A) 8 (B) −5

(C) 11 (D) 3

103. Let $\alpha=\pi/12$, $\beta=\pi/11$. Evaluate arccos(-cosαcosβ-sinαsinβ).

(A) $\frac{133\pi}{132}$ (B) $\frac{\pi}{132}$

(C) $-\frac{\pi}{132}$ (D) $\frac{131\pi}{132}$

104. Positive integers $a < b$ satisfy $a + b = 26$ and $3 \mid a$. Find the maximum value of ab.

(A) 26 (B) 168

(C) 153 (D) 169

105. Circles x^2+y^2=81 and (x-14)^2+y^2=81 meet at A,B. Find |AB|^2.

(A) $8\sqrt{2}$ (B) 324

(C) 16 (D) 128

106. $A = \begin{pmatrix} 7 & k \\ 0 & 6 \end{pmatrix}$ is symmetric. Find tr($A^T A$).

(A) 169 (B) 85

(C) 13 (D) 42

107. The homogeneous system 5x+y=0, kx+5y=0 has a nonzero solution. Find $k+x^2/y^2$.

(A) $\frac{624}{25}$ (B) 50

(C) 25 (D) $\frac{626}{25}$

108. f(x)=sin(5x)/x+4 for x<0; f(x)=k cos x for x≥0. Find k for continuity at zero.

(A) 4 (B) 1

(C) 9 (D) 5

109. How many integers x satisfy $4^{-x} - (4224)2^{-x} + 524288 \leq 0$?

(A) 4 (B) 6

(C) 13 (D) 5

110. a=(2,4,0), b=(4,-2,3). Find $|a+b|^2$-$|a-b|^2$.

(A) 48 (B) −9

(C) 0 (D) 64

111. Let y′=(x+y)/(x+y+1), with y(0)=1. Find the positive value of x at whichx+y=6.

(A) $\frac{\log\left(\frac{13}{3}\right)}{4} + \frac{5}{2}$ (B) $\frac{7}{2} - \frac{\log\left(\frac{13}{3}\right)}{4}$

(C) $\frac{5}{2} - \frac{\log\left(\frac{13}{3}\right)}{4}$ (D) $\log(6)$

112. Find the area between y=(x-8)(x-11) and the x-axis for 0≤x≤16.

(A) $\frac{1024}{3}$ (B) $\frac{1051}{3}$

(C) $-\frac{1024}{3}$ (D) $\frac{1051}{6}$

113. Maximize Z=3x+4y subject to x≥0, y≥4, x+y≤13, x+2y≤16.

(A) 43 (B) 42

(C) 32 (D) 40

114. If |z-8|=4, find the minimum of $|z|^2$.

(A) 16 (B) 4

(C) 144 (D) 48

115. Find the coefficient of x^{13} in (1+5x)exp(8x).

(A) $\frac{1073741824}{2027025}$ (B) $\frac{536870912}{6081075}$

(C) $\frac{4898947072}{6081075}$ (D) $\frac{67108864}{93555}$

116. All tangents common to $x^2/64+y^2/16=1$ and $x^2+y^2=\frac{128}{3}$ touch the ellipse at points P_i. Find $\Sigma(P_i,x)^2$ over the four tangency points.

(A) $\frac{512}{3}$ (B) $\frac{640}{3}$

(C) $\frac{32}{3}$ (D) $\frac{160}{3}$

117. How many solutions in $0\le x<7\pi$ satisfy $\sin^4x+\cos^4x=3/4$ and $\sin x>0$?

(A) 28 (B) 14

(C) 17 (D) 16

118. Evaluate the integral from 0 to 2π of $(5-2\sin x)(1+\cos x)/(25-4\sin^2x)\,dx$.

(A) $\frac{\sqrt{21}\pi}{42}$ (B) $-\log\left(\frac{\sqrt{35}}{5}\right)+\frac{2\sqrt{21}\pi}{21}$

(C) $\frac{2\sqrt{21}\pi}{21}$ (D) $\log\left(\frac{\sqrt{35}}{5}\right)+\frac{2\sqrt{21}\pi}{21}$

119. $f(x)=4x+1$ and $f(g(x))=4x+9$. Find $g(g(4))$.

(A) 12 (B) 6

(C) 20 (D) 8

120. A=(2,6,3), B=(4,6,9). Find the squared length of the component of AB perpendicular to direction (1,2,2).

(A) 40 (B) $\frac{106}{3}$

(C) $\frac{164}{9}$ (D) $\frac{196}{9}$

121. For $f(x)=x^3-3kx^2+48x$, how many integer k in [0,6] give a local maximum at a positive x?

(A) 3 (B) 1

(C) 6 (D) 2

122. How many integers x in [-2,14] satisfy $(|x-6|-3)/(|x-6|-7)>0$?

(A) 5 (B) 9

(C) 17 (D) 7

123. In triangle ABC, A=2B, $a/b=\sqrt{3}$ and b=8. Find its area.

(A) $32\sqrt{3}$ (B) 32

(C) 16 (D) $64\sqrt{3}$

124. The four lines $\pm x/6\pm y/12=1$ form a rhombus with vertices on the coordinate axes. Find its area.

(A) 144 (B) 72

(C) 288 (D) 36

125. What is the maximum number of distinct intersection points formed by 5 circles and 6 straight lines, with no coincident curves?

(A) 95 (B) 80

(C) 65 (D) 115

126. Evaluate $\sum_{j=0}^{3} \binom{3}{j}\binom{11}{2-j}$.

(A) 91 (B) 273

(C) 364 (D) 78

127. An observer at height 7 sees a tower foot at depression 30° and top at elevation 60°. Find the tower height.

(A) 21 (B) 28

(C) $7\sqrt{3}$ (D) 14

128. Find the product of all real solutions of sqrt(7-log_6|x|)=log_6|x|-1.

(A) -216 (B) 0

(C) 46656 (D) -46656

129. Find the population variance of 6+5, 12+5, ..., 36+5.

(A) $\frac{35}{12}$ (B) 110

(C) $\frac{35}{2}$ (D) 105

130. Chords of $x^2+y^2=49$ pass through (14,0). Find the radius of the circle containing their midpoint locus.

(A) $\frac{7}{2}$ (B) 7

(C) 14 (D) 49

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A drop of mass 1.024×10^{-16} kg is suspended in a vertical field of magnitude 800 N/C. It has excess electrons. One excess electron is removed without changing its mass. Find its immediate downward acceleration, taking $g = 10\ \text{m/s}^2$ and $e = 1.6 \times 10^{-19}$ C.

(A) $10\,\text{m/s}^2$ (B) $\frac{5}{4}\,\text{m/s}^2$

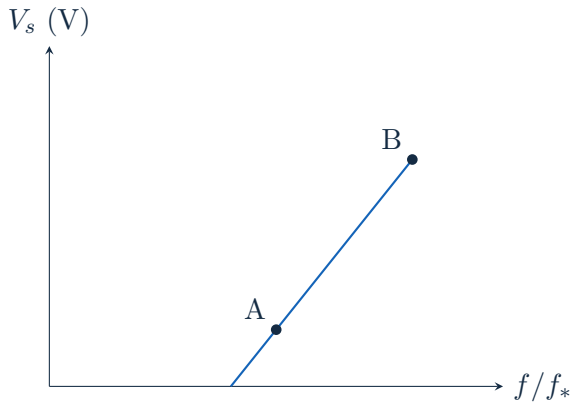
(C) $\frac{10}{9}\,\text{m/s}^2$ (D) $\frac{10}{7}\,\text{m/s}^2$

132. A compound microscope is used with final image at infinity. Its objective focal length is 1 cm, eyepiece focal length 6 cm, and effective optical tube length 80 cm. Under the standard long-tube approximation, find magnifying-power magnitude. Use least distance of distinct vision 25 cm.

(A) $\frac{25}{6}$ (B) $\frac{96}{5}$

(C) $\frac{1000}{3}$ (D) $\frac{2000}{7}$

133. For a metal, the stopping-potential graph has points A $(5f_*, 3\,\text{V})$ and B $(8f_*, 12\,\text{V})$. The electronic gain of the vertical recording channel is later found to be 2: its recorded voltage is 2 times the true stopping voltage. What is hf_*/e in volts?



(A) $\frac{3}{2}\,\text{V}$ (B) $6\,\text{V}$

(C) $\frac{9}{2}\,\text{V}$ (D) $\frac{3}{4}\,\text{V}$

Chemistry

134. A student assigns two bonding domains and one lone pair to the central atom of NO_2^-. After independently checking valence-electron count,which combined molecular shape/domain count is consistent?

(A) Linear; two electron domains.

(B) Trigonal planar molecular shape; three electron domains.

(C) Bent; three electron domains.

(D) T-shaped; five electron domains.

135. Assign the formal metal oxidation state in $[Ni(CN)_4]^{2-}$,using four CN^-. For nitrosyl species use the explicitly stated formal NO^+ convention rather than inferring spin from it.

(A) 3 (B) 2

(C) 1 (D) 4

136. 3-nitrobenzaldehyde is first treated with NaBH4 under ordinary aldehyde-selective conditions and worked up,then its nitro group is completely reduced with Fe/HCl and the product basified. Which final product follows both reductions?

(A) 3-aminobenzyl alcohol (B) 3-nitrobenzyl alcohol

(C) 4-aminobenzyl alcohol (D) 3-aminobenzaldehyde

Logical Reasoning

137. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 5, 15, 17, 51, ?, 159, 161.

(A) 151 (B) 49

(C) 53 (D) 153

138. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: ZHDI, CLHL, FPLO, ?, LXTU, OBXX.

(A) HSOQ (B) ITPR

(C) RPTI (D) JUQS

139. $X@Y$ means X is the father of Y; $X\%Y$ means X is the sister of Y; $X\#Y$ means X is the mother of Y. All names denote distinct people; a brother or sister is a full sibling, sharing both parents. Given $F@L\ L\%S\ S\#Q$, how is F related to Q?

(A) Maternal grandfather (B) Paternal grandfather

(C) Mother (D) Father

Mathematics

140. 5 distinct objects are distributed into three nonempty unlabelled boxes. How many distributions are possible?

(A) 150 (B) 25

(C) 40 (D) 75

141. The curve $y=x^3+2x$ divides the disc $x^2+y^2 \leq 9$. Find the area of the part satisfying $y \geq x^3+2x$.

(A) $\frac{9\pi}{2}$ (B) $\frac{9\pi}{4}$

(C) 9π (D) $2+\frac{9\pi}{2}$

142. For $9x^2+36y^2=324$, find the distance between its two directrices.

(A) 12 (B) $6\sqrt{3}$

(C) $8\sqrt{3}$ (D) $4\sqrt{3}$

BITSAT Mock 13

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A liquid of volume 2 litres and bulk modulus 3000000000 Pa is subjected to an additional pressure of 3000000 Pa. Neglect changes in bulk modulus. Find the magnitude of volume decrease in cubic centimetres.

(A) $2000000\,\text{cm}^3$ (B) $1\,\text{cm}^3$

(C) $\frac{1}{500}\,\text{cm}^3$ (D) $2\,\text{cm}^3$

2. A projectile is launched at 15 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{3}{4}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ m/s^2.

(A) $\frac{54}{5}$ m (B) 18 m

(C) $\frac{108}{5}$ m (D) $\frac{81}{5}$ m

3. A square conducting loop in a constant perpendicular field 1 T has side length $\ell(t) = 3t$ m. Find induced-emf magnitude at $t = 3$ s, assuming the loop expands uniformly.

(A) 27 V (B) 81 V

(C) 9 V (D) 54 V

4. A mass 5 kg starts at rest on a smooth line. A force increases linearly from zero to 5 N over 6 s, so $F(t) = (5/6)t$ N in this interval. Find its displacement at the end of the interval.

(A) 18 m (B) 6 m

(C) 3 m (D) 30 m

5. Two coherent beams yield maximum-to-minimum interference intensity ratio 9. Find the larger individual beam intensity divided by the smaller.

(A) $\frac{25}{16}$ (B) 9

(C) 5 (D) 4

6. A projectile is launched at 15 m/s. At its highest point its kinetic energy is $\frac{7}{10}$ times its initial kinetic energy. Find its maximum height. Neglect drag and take $g = 10$ m/s^2.

(A) $\frac{45}{4}$ m (B) $\frac{27}{8}$ m

(C) $\frac{45}{4} - \frac{9\sqrt{70}}{8}$ m (D) $\frac{459}{80}$ m

7. A 6 kg block is pulled by a force of 30 N at an angle with $\cos\theta = 4/5$, $\sin\theta = 3/5$ above a horizontal floor. The block slides; $\mu_k = \frac{1}{5}$. Find the net horizontal force. Take $g = 10$ m/s^2.

(A) $\frac{108}{5}$ N (B) $\frac{78}{5}$ N

(C) $\frac{42}{5}$ N (D) $\frac{54}{5}$ N

8. A boat moves at 15 m/s relative to water. The current is 9 m/s and river width is 96 m. The boat lands directly opposite its starting point. Find crossing time.

(A) 8 s (B) 4 s

(C) 16 s (D) $\frac{32}{5}$ s

9. Two identical uniform spheres touch. Their radii are multiplied by 9 while their density is divided by 9. By what factor does their gravitational attraction change?

(A) 81 (B) $\frac{1}{6561}$

(C) 1 (D) 531441

10. A 1 kg block moving at 7 m/s compresses a spring of constant 120 N/m. During compression, a damper removes 1/4 of the block's initial kinetic energy. Find maximum compression; gravity does no work.

(A) $\frac{7\sqrt{6}}{24}$ m (B) $\frac{49}{160}$ m

(C) $\frac{7\sqrt{10}}{40}$ m (D) $\frac{7\sqrt{5}}{40}$ m

11. In a series RLC circuit, ideal rms voltmeters read 60 V across R, 70 V across L and 20 V across C. Find source rms voltage.

(A) $10\sqrt{61}$ V (B) 60 V

(C) 150 V (D) 110 V

12. A coil of inductance 5 H carries current $i(t) = 3t^2$ A. Find the magnitude of its self-induced emf at $t = 2$ s.

(A) 15 V (B) 120 V

(C) 30 V (D) 60 V

13. An active transistor has $\alpha = \frac{39}{40}$. Its collector resistor is 5 kΩ and the measured collector-resistor drop is 15 V. Neglect leakage. What base current in mA is required?

(A) $\frac{3}{40}$ mA (B) 3 mA

(C) $\frac{1}{13}$ mA (D) $\frac{117}{40}$ mA

14. A particle executes SHM at frequency 8 Hz. At t=0 it is at its positive extreme. In the interval $0 <= t < 1$ second, how many times does its kinetic energy equal its potential energy?

(A) 32 (B) 16

(C) 17 (D) 8

15. A pipe of cross-sectional area 4A carries incompressible fluid at 6 m/s and feeds 6 identical holes, each area A, with equal flow. Find speed from each hole.

(A) 1 m/s (B) 24 m/s

(C) 4 m/s (D) 9 m/s

16. A drop of mass 8.96×10^{-17} kg is suspended in a vertical field of magnitude 800 N/C. It has excess electrons. One excess electron is removed without changing its mass. Find its immediate downward acceleration, taking $g = 10$ m/s^2 and $e = 1.6 \times 10^{-19}$ C.

(A) 10 m/s^2 (B) $\frac{10}{7}$ m/s^2

(C) $\frac{5}{4}$ m/s^2 (D) $\frac{60}{7}$ m/s^2

17. A triangular p-V cycle has vertices (V_0, p_0), $(V_0 + 3\,\text{L}, p_0)$ and $(V_0, p_0 + 2\,\text{kPa})$. Traversed in that listed order and then back, find net work done by the gas.

(A) -6 J (B) 3 J

(C) $-\frac{3}{2}$ J (D) -3 J

18. A series RLC circuit has $R = 80$ ohms, $X_L = 70$ ohms, and $X_C = 30$ ohms at the source frequency. An ideal source supplies 300 V rms. Find rms current.

(A) $\frac{3\sqrt{10}}{4}$ A (B) $\frac{15\sqrt{41}}{41}$ A

(C) $\frac{3\sqrt{5}}{2}$ A (D) $\frac{15\sqrt{122}}{61}$ A

19. A gas receives 500 J of heat while its surroundings do 50 J of work on it. It then loses 250 J of heat at fixed volume. Find its total internal-energy change.

(A) 800 J (B) 300 J

(C) 200 J (D) 550 J

20. A vacuum electromagnetic wave of average intensity 120 W/m^2 passes through a device that reduces its electric-field amplitude by a factor 4. The transmitted wave falls normally on an area 5 cm^2. What average power reaches the area?

(A) $\frac{3}{200}$ W (B) $\frac{3}{800}$ W

(C) $\frac{24}{25}$ W (D) $\frac{75}{2}$ W

21. A horizontal platform of moment of inertia 2 kg m^2 carries a small slider of mass 5 kg at radius 5 m and rotates at 7 rad/s. The slider moves radially along a slot to radius 2.5 m. External torque about the rotation axis is zero. Treat the slider as a point mass. Find the final angular speed.

(A) $\frac{508}{19}$ rad/s (B) $\frac{889}{2}$ rad/s

(C) $\frac{2\sqrt{16891}}{19}$ rad/s (D) $\frac{931}{508}$ rad/s

22. A projectile of mass 15 kg is launched with speed 40 m/s, $\cos\theta = 3/5$. At its highest point it splits into masses $M/3$ and $2M/3$. The lighter fragment moves horizontally forward at 28 m/s. Find the signed horizontal velocity of the heavier fragment immediately after the explosion.

(A) 20 m/s (B) 28 m/s

(C) 22 m/s (D) 44 m/s

23. A soap bubble initially of radius 6 cm is expanded quasistatically until its volume is 8 times the initial volume. Surface tension remains $\frac{7}{100}$ N/m. Find work against surface tension, in units of $\pi \times 10^{-4}$ J.

(A) $\frac{756}{25}$ (B) $\frac{1512}{25}$

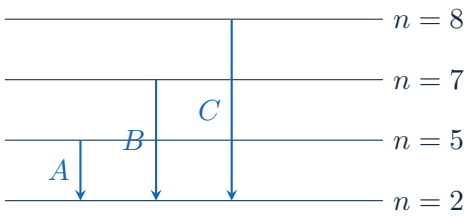
(C) $\frac{2016}{25}$ (D) $\frac{504}{25}$

24. One mole of a monatomic ideal gas follows $pV^5 = \text{constant}$. Find the process molar heat capacity in units of R.

(A) $\frac{5}{2}$ (B) $\frac{13}{10}$

(C) $\frac{5}{4}$ (D) $\frac{7}{4}$

25. The diagram shows downward hydrogen transitions A, B and C from $n = 5$, 7 and 8, respectively, ending at $n = 2$. A filter transmits only photons with energy strictly above that of transition B. Which transmitted line has the longest wavelength?



(A) B (B) C

(C) A (D) No transmitted line

26. A hydrogen atom emits a photon in the $3 \to 2$ transition. A hydrogen-like ion of nuclear charge $Z = 3$ emits a photon of exactly the same wavelength and ends in level $n_f = 6$. What is its initial principal quantum number?

(A) 9 (B) 3

(C) 27 (D) 1

27. Two thin prisms in air are placed with opposite orientations. Their indices are 3/2 and 5/3 and their small prism angles are A and B. For zero net mean deviation, find B/A.

(A) $\frac{3}{4}$ (B) $\frac{10}{9}$

(C) $\frac{3}{7}$ (D) $\frac{1}{2}$

28. A nonrelativistic particle of fixed charge and mass moves perpendicular to a uniform magnetic field. Its kinetic energy is multiplied by 4 and the magnetic field is multiplied by 7. Find the new trajectory radius divided by the original radius.

(A) 2 (B) $\frac{2}{49}$

(C) $\frac{\sqrt{2}}{7}$ (D) $\frac{2}{7}$

29. A small-angle pendulum has period T in a stationary lift. In a vertically accelerating lift its period is 2T. Find the downward lift acceleration divided by g.

(A) $\frac{1}{2}$ (B) $\frac{5}{4}$

(C) $\frac{3}{4}$ (D) $\frac{1}{4}$

30. A wire carries $i(t) = 5 + 4\sin\omega t$ A for many cycles. Find its rms current.

(A) 5 A (B) 9 A

(C) $\sqrt{41}$ A (D) $\sqrt{33}$ A

Chemistry

31. Classify CaO, Al2O3, N2O5, in that order, by their usual acid-base behaviour. Which sequence also predicts that the middle oxide reacts with both suitable acids and strong alkalis?

(A) Basic; amphoteric; acidic. (B) Acidic; amphoteric; basic.

(C) Basic; acidic; amphoteric. (D) Amphoteric; basic; acidic.

32. A freshly assayed 25-volume H_2O_2 solution releases 25 L O_2 at STP per litre on complete decomposition. Only80.0% of the evolved oxygen reaches a burner that requires0.560 L O_2 at STP. The solution volume needed,in mL,is

(A) 56 (B) $\frac{112}{5}$

(C) $\frac{56}{5}$ (D) 28

33. In a copper extraction stage, $\frac{1}{25}$ mol Cu_2S and $\frac{1}{20}$ mol Cu_2O react only by $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$. What amount of metallic copper, in mol, is formed?

(A) $\frac{3}{20}$ (B) $\frac{1}{20}$

(C) $\frac{13}{100}$ (D) $\frac{6}{25}$

34. A textbook antiseptic formulation combines chloroxylenol and terpineol. Which statement is chemically consistent?

(A) Its role proves that it contains only sucrose.

(B) It is necessarily a dipeptide sweetener.

(C) It is a polymerisation catalyst because it is a mixture.

(D) It is classified by antimicrobial use, rather than by sweetness or pain relief.

35. At 320 K, ΔH for C(graphite)+O2(g)→CO2(g) is -390 kJ/mol and for CO(g)+(1/2)O2(g)→CO2(g) is -275 kJ/mol. Carbon is converted only into CO, producing 1 mol CO. Assuming ideal gases and R=0.008314 kJ/mol/K, what is total ΔU in kJ? Use the stated reaction extents.

(A) 363532/3125 (B) -181766/3125

(C) -363532/3125 (D) -727064/3125

36. A Fischer projection of tartaric acid has its four-carbon chain vertical (COOH at top and COOH at bottom). At C2,OH is on the left; at C3,OH is on the right; each other horizontal group is H. Horizontal bonds point toward the observer. Which classification and configuration pair is correct?

(A) meso; (2R,3S) (B) chiral; (2R,3R)

(C) chiral; (2S,3S) (D) meso; (2S,3S)

37. A phenolic compound has molar mass 126 g/mol. A $\frac{63}{10}$ g sample is completely O-acetylated without ring substitution,and dry product mass is $\frac{63}{5}$ g. How many acetylatable OH groups occur per molecule? Each OH→OCOCH3 replacement adds42 g/mol.

(A) 3 (B) 2

(C) 1 (D) 4

38. After sufficient equilibration, phenolphthalein is colourless in an acidic solution, pink in moderately alkaline solution, and can fade in a large excess of strong alkali. Which explanation consistently distinguishes its two colourless states?

(A) Acid favours a closed lactone form; very strong alkali can form a different colourless species by hydroxide addition.

(B) Both colourless states prove that the solution is acidic.

(C) The moderately alkaline pink colour is a precipitation reaction.

(D) Strong alkali necessarily restores the acidic lactone by donating protons.

39. TheFischerOH pattern at C2-C5 isR-L-R-R for sugarA andL-R-L-L for sugarB, with CHO at top and CH2OH at bottom. What relation and D/L labels fit?

(A) C2 epimers; bothD. (B) Enantiomers; A isD andB isL.

(C) Anomers; bothL. (D) Identical structures; D/L depends only onCHO.

40. An internally consistent thermochemical comparison gives total resonance stabilisation of benzene as 36 kcal/mol (1 benzenoid rings) and phenanthrene as 92 kcal/mol (3 rings). Which statement correctly separates total stabilisation from stabilisation per ring?

(A) Total: benzene; per ring: benzene. (B) Total: phenanthrene; per ring: phenanthrene.

(C) Total: phenanthrene; per ring: benzene. (D) Both quantities must rank compounds identically because each is aromatic.

41. A carbon-containing organic compound contains chlorine only among N,S and halogens. Sodium fusion is performed with limited sodium; if both N and S are present, thiocyanate formation is retained. Which product/test pairing is consistent?

(A) Elemental chlorine; a Prussian-blue response. (B) NaCl; white AgCl after appropriate acidification.

(C) NaNO3; a brown-ring test directly on the organic compound. (D) Na2SO4; a violet nitroprusside response for sulfate.

42. A hypothetical atom obeys classical electrodynamics, with an electron continually moving on a circular orbit. Which extra assumption in Bohr's model prevents the classical stability problem?

(A) The electron has zero charge. (B) Certain stationary states do not radiate despite the orbital picture.

(C) Every circular motion has zero acceleration. (D) The nucleus continuously replaces radiated energy by an unspecified fuel.

43. For acetylacetone and propanone,which structural-reactivity comparison is correct under ordinary enolate/aldol conditions?

(A) neither has an alpha H

(B) acetylacetone has the more acidic carbon-bound H

(C) propanone has the more acidic carbon-bound H

(D) their acidity must be identical

44. An energy density obeys u=(3/2) n kT,whereu isJ/m^3 andn counts particles per m^3. WithNA in mol^{-1},which units for k andR=NAk are correct?

(A) k:J $mol^{-1}K^{-1}$;R:J K^{-1}.

(B) k:Pa K^{-1};R:Pa $mol^{-1}K^{-1}$.

(C) k:J K^{-1};R:J $mol^{-1}K^{-1}$.

(D) Both k andR have units J K^{-1} becauseNA is dimensionless.

45. A mixture contains $\frac{2}{25}$ mol H_2SO_5 and $\frac{7}{100}$ mol $H_2S_2O_8$. Their structures contain one O–O link per molecule. All peroxide links react quantitatively with excess iodide in acid; each such link accepts two electrons. What amount of I_2, in mol, is liberated?

(A) $\frac{7}{100}$

(B) $\frac{3}{20}$

(C) $\frac{2}{25}$

(D) $\frac{3}{40}$

46. Match two applications: I. an orange-red advertising discharge tube. II. a low-temperature coolant for superconducting equipment. Which ordered gas pair is appropriate?

(A) neon; helium.

(B) helium; neon.

(C) argon; helium.

(D) neon; neon.

47. A nonvolatile solute in 15mol water gives relative vapour-pressure lowering1/101. Remove $\frac{15}{2}$mol water without solute loss. Under exact Raoult mole fractions,new relative lowering is

(A) $\frac{2}{101}$

(B) $\frac{1}{51}$

(C) $\frac{1}{101}$

(D) $\frac{1}{50}$

48. 1,2-dibromopropane is treated with excess NaNH2 in liquid NH3,then exactly one equivalent of methyl iodide per initially formed acetylide,and finally aqueous HgSO4/H2SO4. Work-ups are appropriate between stages. What final carbonyl product is obtained?

(A) propanal

(B) butan-1-ol

(C) butanal

(D) butan-2-one

49. Anisole is cleaved with one equivalent of cold concentrated HI. The alkyl iodide fragment is separated and treated with KCN in a suitable aprotic solvent. Which final organic product comes from that fragment?

(A) iodobenzene,C6H5I

(B) benzonitrile,C6H5CN

(C) ethanenitrile,CH3CN

(D) methyl isocyanide,CH3NC

50. A HA/A^- buffer initially has pH=pKa+$\log_{10}2$. Adding $\frac{3}{100}$ mol HCl without volume change gives pH=pKa-$\log_{10}2$;neither component is exhausted. Total initial $HA+A^-$ amount,in mol,is

(A) $\frac{3}{50}$

(B) $\frac{3}{25}$

(C) $\frac{9}{100}$

(D) $\frac{3}{100}$

51. A one-electron ion has nuclear charge Z=4. A bound level has the same energy as H at n=2. In that ion shell,maximum electrons allowed by Pauli in orbitals with$m_l = 0$ would be (ignore electron interactions for the energy comparison)

(A) 16
(B) 128
(C) 64
(D) 8

52. In lactose, galactoseC1 is joined to glucoseC4. Which observation is consistent with this linkage before hydrolysis?

(A) GlucoseC4 becomes an aldehyde carbon by definition.
(B) The glucoseC1 centre remains free and can supply a reducing end.
(C) BothC1 centres must be bound to each other.
(D) Lactose becomes nonreducing solely because it is a disaccharide.

53. One mole of an acyclic saturated hydrocarbon consumes 8 mol O2 on complete combustion. How many hydrogen atoms are present in one molecule,and what initial relation must be used?

(A) 10; n+1 = oxygen moles
(B) 14; (3n+1)/2 = oxygen moles
(C) 16; 2n+2 = oxygen moles
(D) 12; (3n+1)/2 = oxygen moles

54. A cathode receives $\frac{1}{10}$mol electrons. Its only products are Cu from Cu^{2+} and $\frac{1}{100}$mol H_2. Cu M64. Deposited copper mass in g is

(A) $\frac{16}{5}$
(B) $\frac{72}{25}$
(C) $\frac{32}{5}$
(D) $\frac{64}{25}$

55. Benzoic acid is mononitrated. The isolated acid is esterified with an alcohol A chosen from methanol, ethanol,propan-1-ol and propan-2-ol. A is primary and gives a positive iodoform test. Which final ester fits both the ring-directing and alcohol-test constraints?

(A) ethyl 3-nitrobenzoate
(B) propan-2-yl 3-nitrobenzoate
(C) ethyl 4-nitrobenzoate
(D) propyl 3-nitrobenzoate

56. In a pressure range where the Freundlich law $x/m = kp^{1/2}$ is valid at fixed temperature, adsorbent B has 5 times the mass of A and is exposed to 25 times its pressure. Find the ratio of total adsorbed amounts x_B/x_A.

(A) 125
(B) 25
(C) $\frac{1}{25}$
(D) 5

57. Water hardness is due only to dissolved $Mg(HCO_3)_2$ and $MgCl_2$. Which treatment statement correctly distinguishes the two contributions?

(A) Filtration of the clear untreated solution removes dissolved hardness ions without any precipitation.

(B) Adding NaCl precipitates both calcium and magnesium because all chlorides are insoluble.

(C) Boiling alone necessarily removes every dissolved sulfate and chloride quantitatively.

(D) Boiling can remove temporary hardness; washing soda can remove remaining magnesium hardness through precipitation.

58. A mixture contains one mole each of phenol,ethanol and ethanoic acid. Treat with excess NaHCO3. What is the total amount,in mol,of CO2? Assume all stated acid-base reactions go to completion and no side reactions.

(A) 2 (B) 1/2

(C) 3/2 (D) 1

59. For2A+B→3C+4D,the measured total disappearance rate of A is $\frac{3}{250}$M/s. A separate calibrated decay channel consumes A at $\frac{3}{1250}$M/s and produces no D. D appearance rate from the stated reaction,in M/s,is

(A) $\frac{12}{625}$ (B) $\frac{3}{625}$

(C) $\frac{3}{125}$ (D) $\frac{24}{625}$

60. A mixture contains 3/100 mol total of the ionisation isomers [Co(NH3)5Br]SO4 and [Co(NH3)5SO4]Br. An immediately tested aliquot representing the entire inventory gives 9/500 mol AgBr with excess Ag+. Coordinated ligands do not exchange, and no other bromide exists. What mole fraction of the inventory was [Co(NH3)5Br]SO4?

(A) 1/5 (B) -2/5

(C) 2/5 (D) 4/5

English Proficiency

61. Read the sentence: Before accepting the proposal, the panel will scrutinise its assumptions and calculations. Choose the closest meaning of 'scrutinise' as used in the sentence.

(A) summarise briefly (B) examine closely

(C) revise completely (D) criticise publicly

62. Choose the most coherent order of the four sentences. A. The student corrected those marks but repeated the same errors in new work. B. The tutor first marked errors in every sentence of the draft. C. The later exercise tested understanding rather than the ability to copy a supplied repair. D. The tutor then asked the student to explain the rule behind each correction.

(A) B-A-D-C (B) A-B-D-C

(C) B-A-C-D (D) B-D-A-C

63. He supports the goal but adjusts the plan to the funds and staff actually available. Which word best fits this outlook?

(A) pessimist (B) idealist

(C) visionary (D) realist

64. Read the sentence: The message was innocuous, containing nothing likely to cause harm or offence. Choose the opposite in meaning of 'innocuous' as used in the sentence.

(A) mild (B) harmful

(C) trivial (D) ordinary

65. Read the sentence: By evening the wind began to abate, and the crew could finally hear one another. Choose the closest meaning of 'abate' as used in the sentence.

(A) change direction
(B) arrive suddenly
(C) become less intense
(D) intensify

66. Choose the most coherent order of the four sentences. A. The mapmaker added a legend clarifying each symbol. B. With that explanation available, the same route could be followed without the earlier confusion. C. The walking map used several symbols that were not explained. D. Visitors consequently mistook a private gate for a public entrance.

(A) C-D-A-B
(B) C-D-B-A
(C) D-C-A-B
(D) C-A-D-B

67. Read the sentence: The frequent interruptions repeatedly broke her concentration. Choose the opposite in meaning of 'frequent' as used in the sentence.

(A) disruptive
(B) short
(C) regular
(D) rare

68. The second check adds nothing because the identical check has already been completed elsewhere. Which noun identifies this unnecessary duplication?

(A) scarcity
(B) ambiguity
(C) redundancy
(D) inconsistency

69. Complete: The folder, _____ label had faded, was placed beside the scanner.

(A) whom
(B) whose
(C) which
(D) who

70. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The inspector had sealed the container before the courier arrived.

(A) The container had been sealed by the inspector before the courier arrived.
(B) The inspector had been sealed by the container before the courier arrived.
(C) The container has been sealed by the inspector before the courier arrives.
(D) The container had been sealed after the courier arrived.

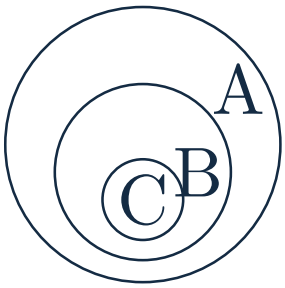
Logical Reasoning

71. Choose the diagram that represents all these conditions, including the strictness information: Some A is B, some A is not B, and some B is not A. No C is A or B; C is nonempty.

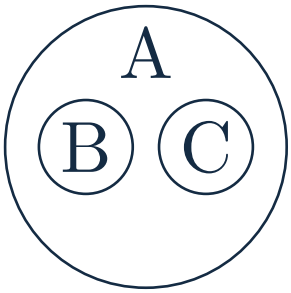
(A) Pattern Y



(B) Pattern X



(C) Pattern Z



(D) Pattern K



72. A person makes these turns in order: 90 degrees right, 90 degrees right, 90 degrees left, 180 degrees. The final heading is north. What was the heading immediately after turn 1?

(A) West

(B) North

(C) South

(D) East

73. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.



host 1



host 2



host 3



host 4



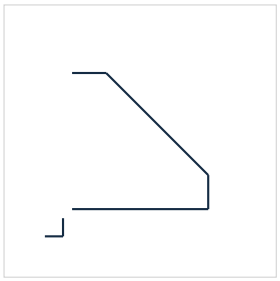
(A) 2

(B) 1

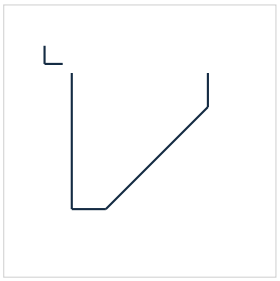
(C) 3

(D) 4

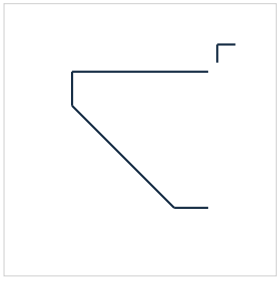
74. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

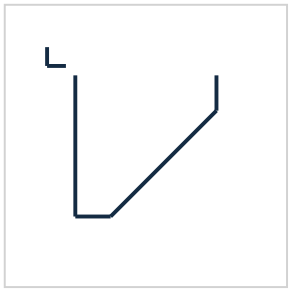


bottom left

?

bottom right

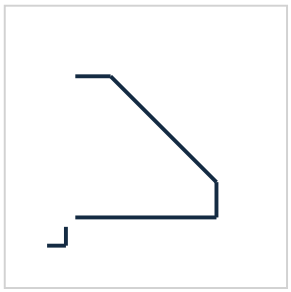
(A) Pattern W



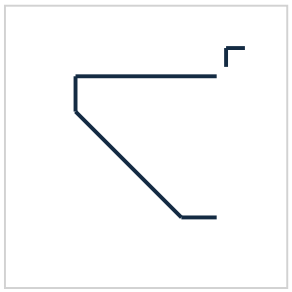
(B) Pattern V



(C) Pattern Q



(D) Pattern U



75. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 29 and Z has code 4. Decode the separated values 10, 12, 29, 21, 16.

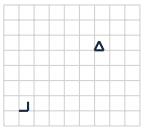
(A) TRAIN

(B) NIART

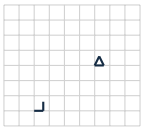
(C) SQZHM

(D) USBJO

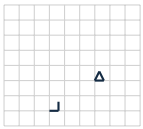
76. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



1

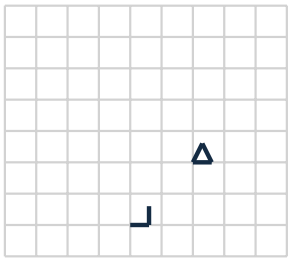


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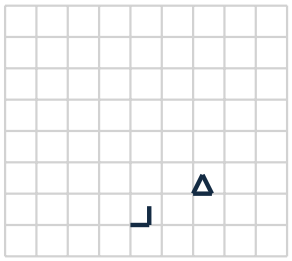


3

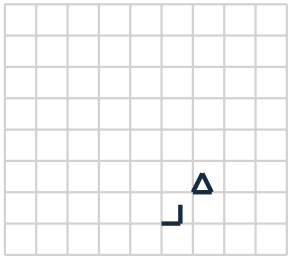
(A) Pattern V



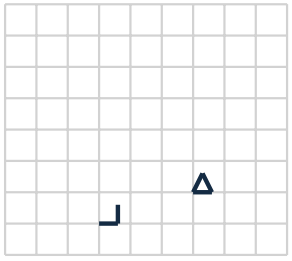
(B) Pattern N



(C) Pattern T



(D) Pattern X



77. E and Q are full siblings; Q and U are also full siblings. J is the mother of E. All four people are distinct. How is J related to U?

(A) Aunt

(B) Grandmother

(C) Mother

(D) Relationship cannot be determined

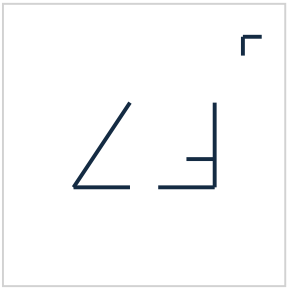
78. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



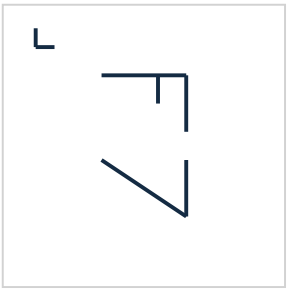
(A) Pattern L



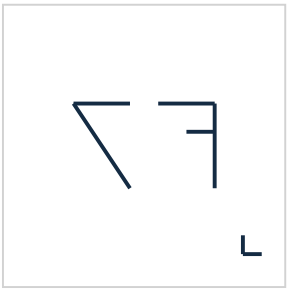
(B) Pattern Q



(C) Pattern Y



(D) Pattern K



79. Preserve the exact relation and its order: Wolf : Pack :: ? : Fleet. Choose the missing term.

(A) Lion
(B) Wolf
(C) Ship
(D) Bee

80. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern N



(B) Pattern T



(C) Pattern M



(D) Pattern Q



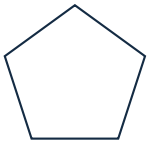
81. 8 people sit around a circle facing its centre. Clockwise order fixes the convention; immediate left is clockwise. Dev is 1 places clockwise from Asha. Hari is 4 places clockwise from Asha. Gita is 5 places clockwise from Asha. Esha is 6 places clockwise from Asha. Charu is 7 places clockwise from Asha. Farah is not adjacent to Hari. Who sits immediately clockwise from Dev?

(A) Farah
(B) Charu
(C) Bina
(D) Asha

82. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: PNS, TQQ, ?, BWM, FZK, JCI.

(A) WSN
(B) XTO
(C) OTX
(D) YUP

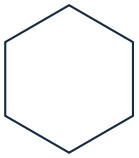
83. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

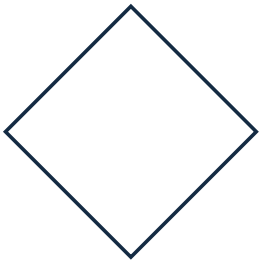


B

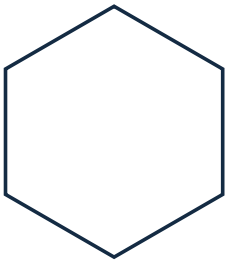


C

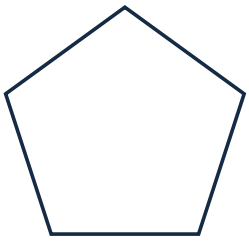
(A) Pattern Q



(B) Pattern Y



(C) Pattern Z



(D) Pattern K

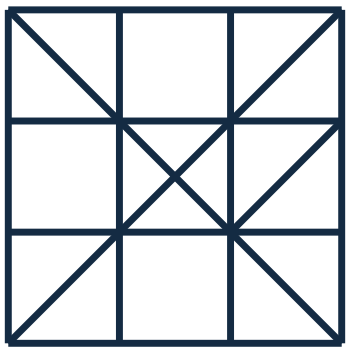


84. In this series the first differences are a positive integer multiple of consecutive squares $1, 4, 9, 16, \ldots$. Find the missing internal term: 5, 7, 15, 33, ?, 115, 187.

(A) 66 (B) 67

(C) 63 (D) 65

85. How many triangles are formed by all the drawn segments?



(A) 22 (B) 32

(C) 72 (D) 36

86. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

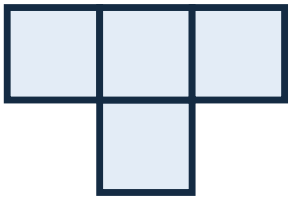


piece 1

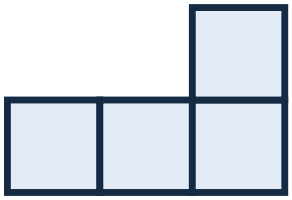


piece 2

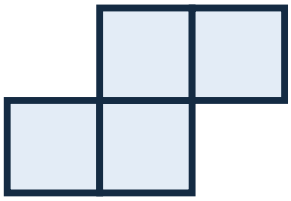
(A) Pattern X



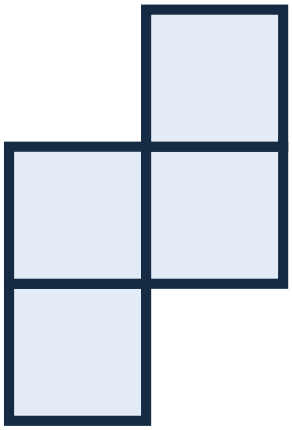
(B) Pattern Q



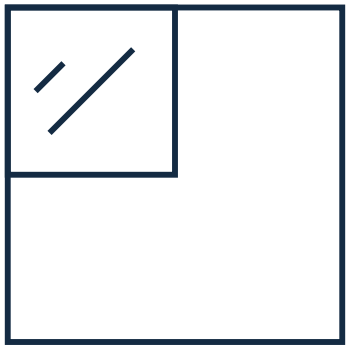
(C) Pattern N



(D) Pattern P



87. Starting with a unit square sheet, fold right to left along the vertical midline, then fold bottom to top along the horizontal midline. The folded paper is cut along both short drawn slits. Select the pattern after complete unfolding.



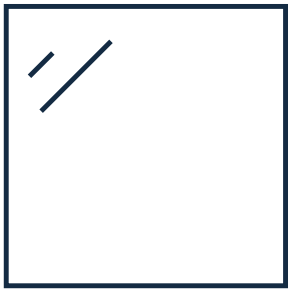
(A) Pattern U



(B) Pattern W



(C) Pattern Z



(D) Pattern Y



88. Each rectangle is drawn to the same local scale. In three numbered figures the interior straight line divides the rectangle into equal areas. Which does not?



1



2



3



4

(A) 3

(B) 4

(C) 1

(D) 2

89. In a row, Asha is 12th from the left and 35th from the right. Bina is 7th from the right. Charu stands 4 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 23 (B) 24

(C) 25 (D) 8

90. A community register uses the following three categories: designers, guides, artists. Its complete relevant assertions are in this short paragraph; do not import everyday assumptions. Some guides are not designers. No artists are designers. All artists are designers. Conclusions: I. No guides are artists. II. Some artists are not guides. Which conclusions follow necessarily?

(A) Only II (B) Only I

(C) Both I and II (D) Neither I nor II

Mathematics

91. The letters A,A,B,C and 3 further distinct letters are arranged in a row. How many arrangements have nonadjacent As and have B,C adjacent in either order?

(A) 720 (B) 288

(C) 240 (D) 480

92. The tangent to $y=x^2+5x$ at P has slope 11. Find the slope of the normal at P.

(A) $-\frac{1}{11}$ (B) -11

(C) 11 (D) -121

93. The system $x+y+z=1$, $y+z=2$, $(2\sin\theta-1)z=6$ has $0\le\theta<12\pi$. How many θ make it inconsistent?

(A) 6 (B) 24

(C) 12 (D) 0

94. For $x\in[0,1]$, find the minimum of $4(\arcsin x)^3+(\arccos x)^3$.

(A) $\frac{\pi^3}{72}$ (B) $\frac{5\pi^3}{64}$

(C) $\frac{\pi^3}{18}$ (D) $\frac{\pi^3}{2}$

95. A line through (6,7) has equal positive x- and y-intercepts. Find the sum of the coordinates of the midpoint of its intercept segment.

(A) 13 (B) 42

(C) 26 (D) $\frac{13}{2}$

96. Evaluate lim as $x\to0$ of $[\text{sqrt}(16+x)-4-x/8]/x^2$.

(A) $\frac{1}{8}$ (B) $\frac{1}{512}$

(C) $-\frac{1}{128}$ (D) $-\frac{1}{512}$

97. Positive a,b,c form an AP with a=6. The numbers a,b,c+3/2 form a GP and b>a. Find c.

(A) 24 (B) 12

(C) 9 (D) 0

98. In $(1+x)^{14}$, find the ratio of the coefficient of x^{13} to the coefficient of x^{12}.

(A) $\frac{1}{4}$ (B) $\frac{13}{2}$

(C) $\frac{2}{13}$ (D) 2

99. For $0 \leq \theta < 14\pi$, roots of $x^2-3x+(1-2\sin\theta)=0$ have opposite signs. How many connected intervals of θ satisfy this?

(A) 14 (B) 8

(C) 1 (D) 7

100. A sphere has $dV/dt=2$ cubic units/s. Find the rate of change of its surface area when $r=3$.

(A) $\frac{1}{18\pi}$ (B) $\frac{2}{3}$

(C) $\frac{4}{9}$ (D) $\frac{4}{3}$

101. A box contains 6 red and 4 blue distinct balls. Three are drawn uniformly without replacement. Given that at least one drawn ball is blue, find the probability that exactly two are red.

(A) $\frac{3}{5}$ (B) $\frac{1}{2}$

(C) 3 (D) $\frac{2}{5}$

102. Unit vectors a,b,c satisfy $|(a\times b)\cdot c|=1$. Find $|a+5b-2c|^2$.

(A) 50 (B) 16

(C) 64 (D) 30

103. Under $x=X+2$, $y=Y+5$, the parabola $(y-5)^2=32(x-2)$ is normalized. Find the x coordinate of its focus in the original axes.

(A) 10 (B) 34

(C) -6 (D) 8

104. Evaluate $\sum_{j=1}^{7} j(j+3)$.

(A) 224 (B) 84

(C) 140 (D) 868

105. $z=8+5i$. Find $1/z+1/\text{conjugate}(z)$.

(A) $\frac{16}{89}$ (B) $\frac{10}{89}$

(C) 16 (D) $\frac{1}{16}$

106. For integer n, consider: "If 15 divides n, then 5 divides n." Which is its contrapositive?

(A) 15 divides n and 5 does not divide n.

(B) If 5 does not divide n, then 15 does not divide n.

(C) If 5 divides n, then 15 divides n.

(D) If 15 does not divide n, then 5 does not divide n.

107. A={j∈{1,...,24}:i^j=1}, B={j∈{1,...,24}:$(-1)^j$=1}. Find |B\A|.

(A) 18 (B) 3

(C) 12 (D) 6

108. $A = \begin{bmatrix} 4 & 4 \\ -4 & 2 \end{bmatrix}$, K=(A-$A^T$)/2. Find tr($K^2$).

(A) −32 (B) 32

(C) 0 (D) −16

109. y′+y/(x+1)=1 for x≥0, y(0)=2. Find (8)y(7)-y(0).

(A) $\frac{67}{16}$ (B) $\frac{63}{2}$

(C) $\frac{67}{2}$ (D) $-\frac{63}{2}$

110. A chord of $x^2+y^2=8100$ touches $x^2/81-y^2/25=1$. Its nonzero midpoint (h,k) satisfies h=k. Find h^2+k^2.

(A) 56 (B) 14

(C) 28 (D) 53

111. Evaluate $\sum_{r=0}^{9} r(r-1)\binom{9}{r}$.

(A) 2304 (B) 9216

(C) 10368 (D) 18432

112. Circles $x^2+y^2=64$ and $(x-4)^2+y^2=64$ meet at A,B. Find $|AB|^2$.

(A) 240 (B) 224

(C) $4\sqrt{15}$ (D) 144

113. For the loop $y^2=x^2(5-x)$, find the area of the portion with x≥5/2.

(A) $-\frac{35\sqrt{10}}{6}+\frac{40\sqrt{5}}{3}$ (B) $\frac{20\sqrt{5}}{3}$

(C) $\frac{35\sqrt{10}}{12}$ (D) $\frac{35\sqrt{10}}{6}$

114. Positive a,b satisfy log(a+b)=log(4)+[log a+log b]/2 in one valid base. Find $(a^4+b^4)/(a^2b^2)$.

(A) 14 (B) 196

(C) 194 (D) 198

115. Evaluate the limit of $[\cos(3x)-\cos(4x)]/x^2$ as x→0.

(A) $\frac{1}{2}$ (B) $\frac{7}{2}$

(C) 7 (D) $-\frac{7}{2}$

116. For sqrt(d^2y/dx^2)=$(y')^5$+x, find order+degree if the degree is defined; otherwise select "Degree is not defined".

(A) 10 (B) Degree is not defined

(C) 3 (D) 8

117. X is binomial with mean 4 and variance $\frac{8}{3}$. Find its mode.

(A) 12 (B) 4

(C) 3 (D) 5

118. In triangle ABC, A=2B, a/b=$\sqrt{3}$ and b=7. Find its area.

(A) 14 (B) $\frac{49\sqrt{3}}{2}$

(C) $\frac{49}{2}$ (D) $49\sqrt{3}$

119. Evaluate $\sum_{j=7}^{13}\binom{j}{1}$, with the convention $\binom{j}{r}=0$ when j<r.

(A) 70 (B) 76

(C) 63 (D) 91

120. A,B are acute, A+B=$\pi/4$ and tan A=$\frac{6}{11}$. Find tan(A-B).

(A) $-\frac{47}{217}$ (B) $\frac{5}{17}$

(C) $\frac{47}{217}$ (D) $\frac{47}{187}$

121. F′(x)=$\frac{(x+1)^2}{(x+2)^4}$ and F(0)=3. Find F(6)-2F(0).

(A) $\frac{1629}{512}$ (B) $\frac{93}{512}$

(C) $-\frac{1629}{512}$ (D) $-\frac{1443}{512}$

122. Write sin x $\cos^3$x+3$\sin^3$x cos x=P sin2x+Q sin4x. Find Q.

(A) 1 (B) $-\frac{1}{4}$

(C) $\frac{1}{8}$ (D) $\frac{1}{2}$

123. Let $D(t)=\det\begin{pmatrix}8+t & 1+t\\ 1+t & 6+t\end{pmatrix}$. Find D′(0).

(A) 94 (B) 47

(C) 12 (D) −12

124. How many integers x are in the real domain of $\sqrt{x-3}\,\log(6-x)$?

(A) 4 (B) 6

(C) 2 (D) 3

125. If α,β are roots of $x^2-2kx+k^2-25=0$ and k∈[-3,3], find the maximum of α^2+β^2.

(A) 50 (B) 68

(C) −14 (D) 59

126. Evaluate $\int_0^\infty$ dx/[(x^2+25)(x^2+81)].

(A) $\frac{\pi}{90}$ (B) $-\frac{\pi}{1260}$

(C) $\frac{\pi}{1260}$ (D) $\frac{\pi}{630}$

127. z_1=4+3i, z_2=9+3i, z_3=9+8i. Find the smaller angle between z_2-z_1 and z_3-z_1.

(A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$

(C) $\frac{\pi}{2}$ (D) 0

128. The line r=(4,5,3)+t(1,2,2) meets x+y+z=37 at Q. Find $|PQ|^2$ for P=(4,5,3).

(A) 15 (B) 225

(C) 475 (D) 45

129. Let $a_j = 1/(2 + 2(j - 1))$. Evaluate $\sum_{j=1}^{5} a_j a_{j+1}$.

(A) $\frac{1}{2}$ (B) $\frac{5}{12}$

(C) $\frac{1}{5}$ (D) $\frac{5}{24}$

130. Four observations have Σ(x-7)^2=164, Σ(x-9)^2=164. Find their population variance.

(A) $2\sqrt{10}$ (B) 160

(C) 40 (D) 41

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. Two identical capacitors each 6 microfarads are connected in series across an ideal 40 V battery. One capacitor undergoes permanent conducting breakdown, becoming a short circuit. Find the additional charge drawn from the battery after the new steady state is reached.

(A) $120\,\mu$C
(B) $80\,\mu$C
(C) $240\,\mu$C
(D) $60\,\mu$C

132. A series RLC circuit is resonant at angular frequency ω_0. Its capacitance is then multiplied by 121 while L stays fixed. What angular frequency restores resonance, as a fraction of ω_0?

(A) $\frac{\sqrt{11}}{11}$
(B) $\frac{1}{11}$
(C) $\frac{1}{121}$
(D) 1

133. Two equal-intensity coherent beams have maximum combined intensity 8 units. Find intensity when their path difference is $\lambda/4$.

(A) 2
(B) 8
(C) 4
(D) $4\sqrt{2}$

Chemistry

134. One electron has(n,l,ml,ms)=(3,2,1,+1/2). A second electron must occupy the same spatial orbital. Which set for the second is allowed?

(A) (3,2,1,0).
(B) (3,2,1,-1/2).
(C) (3,2,3,-1/2).
(D) (3,2,1,+1/2).

135. At a fixed temperature K=9. A reaction mixture initially has Q=9/4. Which direction lowers Gibbs energy, and which composition change must occur before equilibrium can be reached? Use ΔrG=RT ln(Q/K).

(A) Reverse in every case because the standard reaction is exergonic.
(B) Forward in every case because K exceeds one.
(C) forward; Q must increase to K.
(D) Neither direction whenever K is not one; equilibrium requires K=1.

136. Compare H_2 and D_2 dissociation in an elementary isotope comparison. Which statement preserves the distinction between mass/nuclear effects and electronic structure?

(A) D_2 has a slightly larger dissociation energy from its vibrational ground state; both have the same electronic bond order.

(B) Every physical property and reaction rate is identical because isotopes have the same atomic number.

(C) Changing H to D adds an electron and therefore changes the oxidation state of every bonded atom.

(D) An isotope with more neutrons must have a different number of protons and belong to another element.

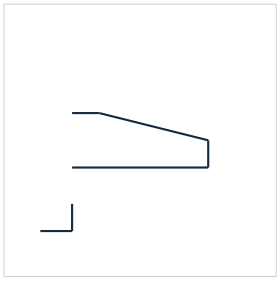
Logical Reasoning

137. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 19$ and $5 \mapsto 37$. Which input maps to 73?

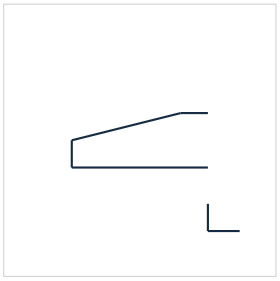
(A) 11 (B) 73

(C) 12 (D) 10

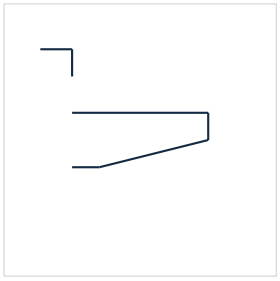
138. Select the missing bottom-right tile. Tiles reflect across the shared horizontal and vertical centre lines; there is no turning.



top left



top right

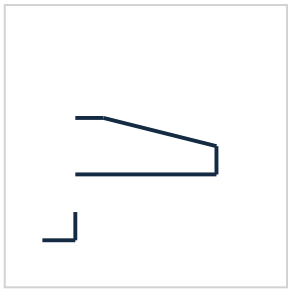


bottom left

?

bottom right

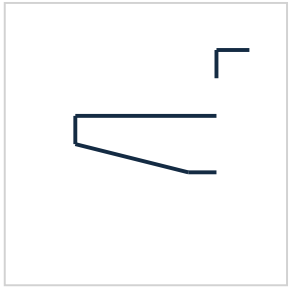
(A) Pattern L



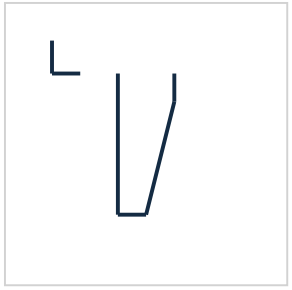
(B) Pattern V



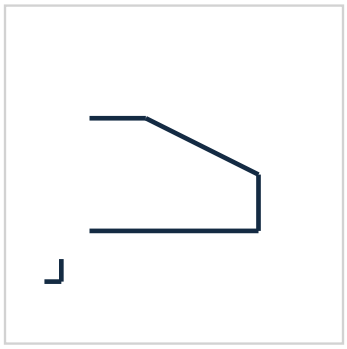
(C) Pattern P



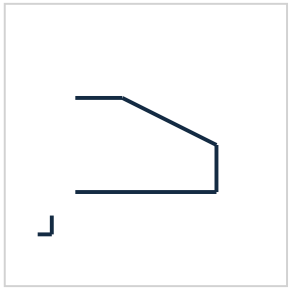
(D) Pattern Z



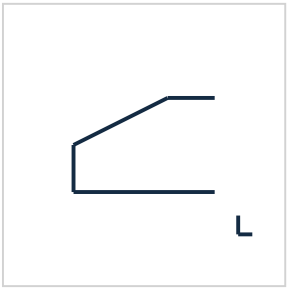
139. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



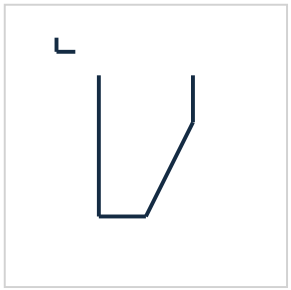
(A) Pattern Z



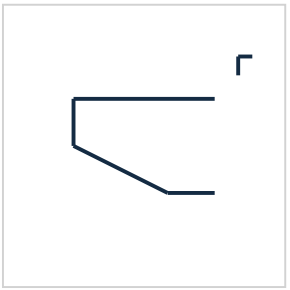
(B) Pattern V



(C) Pattern X



(D) Pattern Y



Mathematics

140. In $(1+x)^n$, the coefficients of x^3 and x^4 are equal, where $n \geq 4$. Find n.

(A) 6 (B) 4

(C) 7 (D) 8

141. Find the coefficient of x^5 in (1-x)exp(2x).

(A) $\frac{14}{15}$ (B) $-\frac{2}{5}$

(C) $-\frac{2}{3}$ (D) $\frac{4}{15}$

142. There are 5 Boolean variables p_1,...,p 5. How many truth assignments make [(p_1 implies p_2) and (p_1 implies not p_2)] true?

(A) 16 (B) 8

(C) 32 (D) 31

BITSAT Mock 14

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A point at the bottom of two plane liquid layers is viewed normally from air. The upper layer has thickness 6 cm and index 3/2; the lower has thickness 28 cm and index 4/3. Find its apparent depth below the top surface in the paraxial limit.

(A) 25 cm
(B) $\frac{68}{3}$ cm
(C) $\frac{139}{3}$ cm
(D) 21 cm

2. Charges +2 C at x=0 and +4 C at x=2 m have a nonzero net charge. The coordinate origin is shifted from x=0 to the old coordinate x=2 m, with axis direction unchanged. Find the new signed x-component of electric dipole moment.

(A) −4 C m
(B) −8 C m
(C) 0 C m
(D) 4 C m

3. A satellite of mass m is moved from a circular orbit of radius R to one of radius 15R about mass M. Find the required increase in total orbital energy in units of $GMm/(2R)$.

(A) 14
(B) $\frac{224}{225}$
(C) $\frac{14}{15}$
(D) $\frac{16}{15}$

4. A 2 kg block rests on a 4 kg block on a smooth horizontal floor. A force 30 N pulls the lower block. There is no slipping between the blocks. During a displacement 4 m, what work does friction do on the upper block?

(A) 60 J
(B) 120 J
(C) 80 J
(D) 40 J

5. In YDSE wavelength is 700 nm, screen distance 6 m and slit separation 2 mm. Find the separation on the same side of the centre between the 5th bright fringe and the 8th dark fringe. Count the first dark fringe as order 1.

(A) $\frac{21}{4}$ mm
(B) $\frac{63}{10}$ mm
(C) $\frac{147}{20}$ mm
(D) $\frac{105}{4}$ mm

6. A parallel-plate capacitor has vacuum capacitance C_0. A dielectric of constant 4 fills a fraction $\frac{1}{2}$ of the plate area and the entire gap in that region; the remainder stays vacuum. The voltage is fixed. Find the fraction of total stored energy residing in the dielectric-filled region.

(A) $\frac{1}{5}$ (B) $\frac{1}{2}$

(C) 4 (D) $\frac{4}{5}$

7. Light of vacuum wavelength 300 nm passes from a medium of index 2 into one of index 3. Find the ratio of its frequencies in the second and first media.

(A) $\frac{4}{9}$ (B) $\frac{1}{3}$

(C) $\frac{3}{2}$ (D) 1

8. An active transistor has $\alpha = \frac{79}{80}$. Its collector resistor is 5 kΩ and the measured collector-resistor drop is 10 V. Neglect leakage. What base current in mA is required?

(A) $\frac{2}{79}$ mA (B) $\frac{79}{40}$ mA

(C) $\frac{160}{6241}$ mA (D) 160 mA

9. A uniform-density spherical planet has radius R and surface gravitational acceleration g. At what depth below the surface is gravity $g/2$? Give the depth as a fraction of R.

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$

(C) $1 - \frac{\sqrt{2}}{2}$ (D) 1

10. A rigid ladder of length 20 m slides with its ends on a floor and a vertical wall. At the instant its top is at height $4L/5$, the foot moves away at 6 m/s. What is the downward speed of the top?

(A) 6 m/s (B) $\frac{9}{2}$ m/s

(C) 8 m/s (D) $\frac{24}{5}$ m/s

11. The accelerating potential of an X-ray tube is increased from 10 kV to 40 kV, while its tube current is reduced by a factor 4. Ignoring target heating and recoil, what is the new cutoff wavelength in nm? Use $hc = 1240$ eV nm.

(A) $\frac{31}{250}$ nm (B) $\frac{31}{1000}$ nm

(C) 31 nm (D) $\frac{31}{4000}$ nm

12. A direct vernier has 20 divisions equal to 19 main-scale divisions; 1 MSD is 1 mm. Its signed zero error is $\frac{-1}{10}$ mm. For an object the main scale is 7 mm and the 3rd vernier division coincides. Find the corrected reading.

(A) $\frac{199}{20}$ mm (B) $\frac{1431}{200}$ mm

(C) $\frac{141}{20}$ mm (D) $\frac{29}{4}$ mm

13. A geometrically similar solid structure is enlarged by a factor 3 in every linear dimension. Its material density is reduced by a factor 2. Its own weight is carried by a similarly scaled base. Find base compressive-stress multiplier.

(A) $\frac{3}{4}$ (B) $\frac{3}{2}$

(C) $\frac{27}{2}$ (D) 6

14. A source emitting 500 Hz and a listener approach each other, each at 20 m/s relative to still air. Sound speed is 300 m/s. Find heard frequency before they pass.

(A) $\frac{1700}{3}$ Hz (B) 500 Hz

(C) $\frac{7500}{13}$ Hz (D) $\frac{4000}{7}$ Hz

15. A sinusoidal source of peak $\frac{57}{10}$ V feeds a back-to-back series pair of identical Zeners through a 1 kΩ current-limiting resistor. The output is across the diode pair. Each Zener has reverse breakdown 3 V and forward drop 0.7 V. What is the peak-to-peak output voltage? Use the piecewise constant-drop model.

(A) $\frac{7}{5}$ V (B) $\frac{37}{5}$ V

(C) $\frac{57}{5}$ V (D) $\frac{44}{5}$ V

16. A pure radioactive sample has activity A at time 9 hours after preparation and $A/8$ at time 18 hours after preparation. At what time, measured from preparation, will its activity be one sixteenth of the activity measured at 9 hours? Assume exponential decay with a constant decay rate.

(A) 21 h (B) 30 h

(C) 7 h (D) 18 h

17. A photon and a nonrelativistic particle have equal de Broglie wavelengths. The particle speed is $c/21$. If the photon then loses half its energy while the particle momentum is unchanged, what is the ratio of particle kinetic energy to the photon's final energy?

(A) $\frac{1}{441}$ (B) 1

(C) $\frac{1}{21}$ (D) 21

18. An ideal mixture contains 4 moles of monatomic gas and 3 moles of rigid diatomic gas. Vibrations are frozen. Find the heat-capacity ratio $\gamma = C_P/C_V$ of the mixture.

(A) $\frac{7}{5}$ (B) $\frac{41}{27}$

(C) $\frac{5}{3}$ (D) $\frac{27}{41}$

19. A 1 kg mass on a string passes the top of a vertical circle of radius 3 m with string tension 30 N. Find string tension at the bottom. Take $g = 10$ m/s^2.

(A) 80 N (B) 30 N

(C) 90 N (D) 10 N

20. A standing wave is $y = 3\sin(5x)\cos(2t)$ cm with x in metres and t in seconds. At a point where $5x = \pi/6$, find the particle-speed magnitude when $2t = \pi/3$.

(A) $3\sqrt{3}$ cm/s (B) 3 cm/s

(C) 6 cm/s (D) $\frac{3\sqrt{3}}{2}$ cm/s

21. A nucleus of mass number 20 splits into two equal nuclei. The parent has binding energy per nucleon $\frac{73}{10}$ MeV and each daughter has $\frac{15}{2}$ MeV. Ignoring other products, what energy is released in the split?

(A) 150 MeV (B) $\frac{1}{5}$ MeV

(C) 4 MeV (D) 154 MeV

22. A screw gauge advances 2.0 mm in four complete turns and has 100 circular divisions. With the jaws closed, its zero mark lies 2 divisions below the reference line; use the usual convention that a zero below the reference line gives positive zero error. With a wire, the main scale reads 5 mm and circular reading is 37 divisions. What is the corrected diameter?

(A) $\frac{207}{40}$ mm (B) $\frac{8361}{200}$ mm

(C) $\frac{134}{25}$ mm (D) $\frac{637}{200}$ mm

23. A 2 kg block on a rough horizontal table is tied over an ideal pulley to a hanging 8 kg mass. The hanging mass moves downward and $\mu_k = 1/5$. Find string tension. Take $g = 10$ m/s^2.

(A) 64 N (B) $\frac{96}{5}$ N

(C) 4 N (D) 20 N

24. A stretched string is divided by a bridge into lengths in the ratio 1 : 2. Tension and linear density are unchanged. The original fundamental is 40 Hz. Find the fundamental of the longer segment.

(A) 40 Hz (B) $\frac{80}{3}$ Hz

(C) 60 Hz (D) 20 Hz

25. A charge 4 C moves with $\vec{v} = 2\hat{i} + 5\hat{j}$ m/s in $\vec{B} = 1\hat{i} + 4\hat{j}$ T. What is the signed z-component of its magnetic force?

(A) 12 N (B) 32 N

(C) 52 N (D) 88 N

26. A charge of 5 μC moves from potential 60 V to -60 V in an electrostatic field. Find the work done by an external agent if the charge starts and ends at rest and no other force does work.

(A) $-600\,\mu$J (B) $600\,\mu$J

(C) $0\,\mu$J (D) $-300\,\mu$J

27. Two ferromagnetic samples have similar remanent flux density. Their coercive fields are 1200 A/m and 9600 A/m. A reversing field of magnitude 5400 A/m will act in service. Which sample is better at retaining its original magnetisation against this field, based only on the stated coercivities?

(A) Neither can have nonzero remanence. (B) Both are necessarily demagnetised equally.

(C) The sample with coercive field 9600 A/m. (D) The sample with coercive field 1200 A/m.

28. A 4 ohm resistor is connected in parallel with an unknown resistor across 24 V. The source supplies 180 W in total. Find the unknown resistance.

(A) $\frac{36}{5}\,\Omega$ (B) $16\,\Omega$

(C) $\frac{16}{9}\,\Omega$ (D) $\frac{96}{29}\,\Omega$

29. A spring of constant 40 N/m is already stretched by $\frac{2}{5}$ m. How much additional work is required to increase its extension to $\frac{7}{10}$ m quasistatically?

(A) $\frac{66}{5}$ J (B) $\frac{16}{5}$ J

(C) $\frac{49}{5}$ J (D) $\frac{33}{5}$ J

30. A wheel initially rotates at 8 rad/s. A constant opposing angular acceleration brings it to rest after 3 s. What angle does it turn through during the last half of this stopping interval?

(A) $\frac{3}{2}$ rad (B) 24 rad

(C) 3 rad (D) 6 rad

Chemistry

31. Excess water reacts completely with $\frac{1}{50}$ mol CaC_2. Which gas identity/amount pair follows from the balanced hydrolysis?

(A) C_2H_2, $\frac{1}{50}$ mol. (B) C_2H_2, $\frac{1}{100}$ mol.

(C) CO_2, $\frac{1}{50}$ mol. (D) O_2, $\frac{1}{100}$ mol.

32. In a hypothetical rule,$l = 0, 1, \ldots, n$ is allowed. Keep subshell capacity2(2l+1),Hund and Pauli,and fill by n+l then lower n. For 20 electrons,the last occupied subshell and its electron count are

(A) $3p^2$. (B) $3d^2$.

(C) $2p^2$. (D) $2d^2$.

33. A peptideC-N bond is shorter and less freely rotating than an ordinary isolated amineC-N bond. Which explanation is chemically appropriate?

(A) The carbonyl oxygen has been removed from the peptide. (B) The peptide bond is a freely rotating ionic gap.

(C) Hydrogen bonding changes the covalentC-N link into a glycosidic bond. (D) Resonance gives the peptideC-N bond partial double-bond character.

34. ethyl bromide is reacted separately with AgCN and KCN. Which pair correctly assigns the principal organic products,in that order?

(A) ethyl cyanide,C2H5-CN; ethyl cyanide,C2H5-CN

(B) ethyl cyanide,C2H5-CN; ethyl isocyanide,C2H5-NC

(C) ethyl isocyanide,C2H5-NC; ethyl isocyanide,C2H5-NC

(D) ethyl isocyanide,C2H5-NC; ethyl cyanide,C2H5-CN

35. A0.0200 mol aniline feed gives p-nitroacetanilide through acetylation followed by mononitration of isolated acetanilide. The first isolated molar yield is80.0%; final isolated product mass is2.16 g (M=180 g/mol). Both transformations are1:1 in substrate moles. What is the second-step molar yield as a fraction?

(A) 3/4 (B) -3/4

(C) 3/2 (D) 3/8

36. Bond energies $D(X_2)$:$D(Y_2)$:D(XY)=2:3:4. Forming2.00 mol XY(g) from the gaseous elements releases 360 kJ. $D(Y_2)$,in kJ/mol,is

(A) 360 (B) 480

(C) 720 (D) 180.0

37. For $MgCO_3$, $SrCO_3$, $BaCO_3$,which increasing thermal-stability order and explanation is correct?

(A) All are equally stable because the cation charge is+2.

(B) $MgCO_3 < SrCO_3 < BaCO_3$; the cations become smaller down the group.

(C) $MgCO_3 > SrCO_3 > BaCO_3$; larger cations distort carbonate more strongly.

(D) $MgCO_3 < SrCO_3 < BaCO_3$; increasing cation size lowers carbonate-ion polarisation.

38. A carbon-containing organic sample contains N and S among nitrogen, sulfur and halogens. It undergoes sodium fusion with limited sodium. Which conversion and subsequent test protocol is chemically consistent? For a silver-halide test, use a separately prepared portion boiled with nitric acid to remove cyanide/sulfide interference where present.

(A) Unchanged covalent bonds; add silver nitrate directly to the untreated organic sample.

(B) Nitrate and sulfate in every case; all give the same Prussian-blue test.

(C) cyanide and sulfide; no further treatment is ever needed before adding silver nitrate.

(D) thiocyanate; blood-red Fe(III)-thiocyanate complex.

39. A model solvent has Tb=400K,molar mass100g/mol and molar vaporisation enthalpy32000J/mol. UseR=8.00J $mol^{-1}K^{-1}$ and Kb=R Tb^2 M/(1000ΔH),M in g/mol. A $\frac{1}{25}$mol/kg A_2B solution is50% dissociated into3ions. Boiling temperature in K is

(A) $\frac{10002}{25}$ (B) $\frac{10008}{25}$

(C) $\frac{10012}{25}$ (D) $\frac{10004}{25}$

40. Compare dilute solutions of 2-hydroxybenzaldehyde and 4-hydroxybenzaldehyde in a non-hydrogen-bonding solvent. Which conclusion joins structure with the expected effect of further dilution on their hydrogen bonding?

(A) Only 4-hydroxybenzaldehyde can make an intramolecular H bond because para groups form a shorter ring.

(B) 2-hydroxybenzaldehyde can retain an intramolecular H bond on dilution; intermolecular association of 4-hydroxybenzaldehyde is reduced.

(C) Both lose all H bonding equally because every H bond requires two molecules.

(D) 2-hydroxybenzaldehyde must have stronger intermolecular association because its internal acceptor is farther away.

41. For $[Co(NH_3)_6]^{3+}$,use the usual ligand-field/VBT assignments at room temperature. Which complete assignment is consistent with charge balance and magnetic behaviour?

(A) d^6,octahedral,d^2sp^3,spin-only moment $\sqrt{3}$ BM.

(B) d^6,octahedral,d^2sp^3,spin-only moment 0 BM.

(C) d^6,tetrahedral,sp^3,spin-only moment 0 BM.

(D) d^7,octahedral,d^2sp^3,spin-only moment 0 BM.

42. diethyl pimelate,EtOOC-(CH2)5-COOEt undergoes NaOEt,then exhaustive aqueous acid hydrolysis and heat. Under the stated intramolecular closure pathway,which isolated product matches both ring size and work-up endpoint?

(A) cyclohexane
(B) cyclohexanone
(C) cyclopentanone
(D) hexanoic acid

43. Reduction potentials M^{2+}/M for metals A,B,C,D are respectively $-\frac{4}{5}$, $-\frac{2}{5}$, $\frac{1}{5}$, $\frac{7}{10}$V. Which paired claim is correct under standard conditions?

(A) A is the strongest oxidising ion because the corresponding metal is most reducing.

(B) A is the strongest reducing metal;D^{2+} oxidises A spontaneously.

(C) A cannot reduce D^{2+} because its own reduction potential is negative.

(D) D is the strongest reducing metal because its reduction potential is greatest.

44. At one T,K_1=9 for A⇌B,and K_2=16 for B⇌C. Numerical equilibrium constant for2C⇌2A is

(A) $\frac{1}{20736}$
(B) $\frac{1}{337}$
(C) 144
(D) $\frac{1}{144}$

45. Compare allyl bromide and vinyl bromide toward the same cyanide nucleophile in a polar aprotic solvent under ordinary SN2 conditions. Which reacts faster or is the substituted site,and why?

(A) allyl bromide; Allylic sp3 substitution is feasible; the vinylic C-Br bond is not an ordinary SN2 site.

(B) vinyl bromide; steric hindrance accelerates backside attack

(C) allyl bromide; the mechanism requires formation of a free tertiary carbocation

(D) vinyl bromide; an sp2 C-Br bond behaves just like an unhindered methyl bromide

46. Two model nonvolatile solutes A(M180) and B(M60) are dissolved separately in equal solvent masses. Equal solute masses are used. A remains molecular;fraction $\frac{3}{5}$ of B formula units is present in dimers. All solutions obey the stated ideal colligative model. Ratio $\Delta T_f(B)/\Delta T_f(A)$ is

(A) $\frac{7}{10}$
(B) $\frac{21}{10}$
(C) 3
(D) $\frac{30}{7}$

47. For the metal centre in Cu^{2+},Z=29. Determine the number of d electrons after formal oxidation-state assignment; use O=-2 in oxoanions.

(A) 9
(B) 10
(C) 8
(D) 11

48. A phenolphthalein solution becomes pink on adding moderate base, then colourless after sufficient equilibration with a large excess of strong alkali. Does the final colour prove acidification?

(A) No; the pink state is a metal precipitate and simply settles out.

(B) No; strong alkali can form a different colourless hydroxide-addition species.

(C) Yes; every colourless phenolphthalein solution must be acidic.

(D) Yes; hydroxide is necessarily a proton donor under all conditions.

49. At a fixed temperature, a diprotic acid H2A has pKa1=3, pKa2=7, and pKw=14. Find (pKb of A2-, Kb(A2-)/Kb(HA-)). Use the conjugate pair appropriate to each basic species.

(A) (11, 10000)
(B) (11, 1/10000)
(C) (7, 1/10000)
(D) (7, 10000)

50. For isolated gaseous species,consider $F(g)+e^- \rightarrow F^-(g)$. Which judgement also remains consistent with the Cl/F electron-gain anomaly?

(A) The stated step is endothermic; every electron addition releases energy.

(B) The stated step is endothermic; first electron gain by F is more exothermic than by Cl because F is smaller.

(C) The stated step is exothermic; first electron gain by Cl is more exothermic than by F.

(D) Every first and second electron gain has the same sign and magnitude for a group.

51. In the stated high-pressure model Z=1+b p/(RT), gas A at 250 K and gas B at 400 K have Z-versus-p slopes in ratio 16/15. Their molar covolumes b do not change with temperature in this model. At what temperature must gas A be measured for its slope to equal that of B at 400 K?

(A) 600
(B) 250
(C) 800/3
(D) 400

52. $M(OH)_2$ has molar solubility $\frac{1}{100000}$ M in pure water. Mix100mL0.00200M NaOH with100mL0.00100M HCl,then add excess solid $M(OH)_2$. Ignore its small contribution to OH^-. Final molar solubility is

(A) $\frac{1}{250000000}$
(B) $\frac{1}{1000000000}$
(C) $\frac{1}{62500000}$
(D) $\frac{1}{100000}$

53. A cobalt(III) chloride ammine contains six NH_3/Cl ligands in an octahedral sphere. Each mole of salt immediately gives 2 mol AgCl with excess $AgNO_3$. Its molar concentration of freely moving ions relative to formula units,under complete dissociation with no ligand exchange,is

(A) 4
(B) 5
(C) 3
(D) 2

54. $\frac{1}{20}$ mol borax is acidified completely. Each borax formula unit gives four H_3BO_3 molecules;75.0% of the boric acid is isolated. With glycerol present, the isolated boric acid is titrated completely as a monobasic acid. What amount of NaOH, in mol, is required?

(A) $\frac{3}{20}$
(B) $\frac{1}{5}$
(C) $\frac{9}{80}$
(D) $\frac{9}{20}$

55. An internally consistent thermochemical comparison gives total resonance stabilisation of naphthalene as 61 kcal/mol (2 benzenoid rings) and anthracene as 84 kcal/mol (3 rings). Which statement correctly separates total stabilisation from stabilisation per ring?

(A) Total: naphthalene; per ring: naphthalene.
(B) Total: anthracene; per ring: anthracene.
(C) Total: anthracene; per ring: naphthalene.
(D) Both quantities must rank compounds identically because each is aromatic.

56. Which proposed routes give propan-2-ol as the stated major alcohol? I:propanone+NaBH4,then water;II:propene+H2O/H+;III:propanal+NaBH4,then water. Where no work-up is written after NaBH4, assume the usual protonating work-up.

(A) all three (B) I only

(C) I and II only (D) II andIII only

57. For HNO_3,use ordinary octet Lewis structures and resonance. What is the pair (average bond order of the two nonprotonated N-O bonds,total lone pairs on all atoms of that same named species in one contributing structure)?

(A) $(\frac{3}{2}, 6)$. (B) (2, 8).

(C) (1, 7). (D) $(\frac{3}{2}, 7)$.

58. An ideal gas executes A→B→C→D→A through (V/L,P/bar):A(1,1),B(3,1),C(3,2),D(1,2). IUPAC work-on convention applies. Heats on AB,BC,DA are 400,300,-200 J. Heat on CD in J is (1 L bar=100 J)

(A) −500 (B) −700

(C) −200 (D) −300

59. propanone reacts with 2,4-dinitrophenylhydrazine. Which product correctly reflects nucleophilic addition or addition-elimination at its carbonyl rather than a different reaction class?

(A) a 2,4-dinitrophenylhydrazone (B) an ether

(C) a carboxylate salt (D) a tertiary alcohol

60. A salt MF_2 dissolves only as $M^{2+}+2F^-$. Its pure-water molar solubility is $\frac{3}{10000}$M. A second saturated solution has imposed free $F^-=\frac{3}{1000}$M;ignore further F^- from dissolution. Dissolved M^{2+} concentration in that second solution is

(A) $\frac{3}{2500000}$ (B) $\frac{3}{1000000}$

(C) $\frac{3}{250000}$ (D) $\frac{3}{100000}$

English Proficiency

61. Choose the most coherent order of the four sentences. A. After those labels remained readable, they adopted the new material for the rest of the stock. B. The nursery labelled its young plants with handwritten cards. C. The staff tested a waterproof label on a small batch. D. Water blurred some of the writing before the plants were sold.

(A) B-C-D-A (B) B-D-A-C

(C) D-B-C-A (D) B-D-C-A

62. The term literate here means able to read and write. Choose its opposite.

(A) literary (B) literal

(C) illiterate (D) literate

63. Read the sentence: The survey was comprehensive, covering every department rather than a selected few. Choose the closest meaning of 'comprehensive' as used in the sentence.

(A) easy to repeat (B) broadly inclusive

(C) extremely detailed (D) entirely accurate

64. The volume brings together selected poems by several writers rather than narrating one writer's life. What is it?

(A) biography (B) dictionary

(C) chronicle (D) anthology

65. Choose the best completion: The revised guide is intended to be clearer, shorter and _____.

(A) more useful (B) using more

(C) of usefully (D) usefully

66. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The editor is to remove the duplicated paragraph without changing the conclusion.

(A) The editor has removed the duplicated paragraph and changed the conclusion.

(B) The duplicated paragraph is to be removed by changing the conclusion.

(C) The duplicated paragraph is to be removed by the editor without changing the conclusion.

(D) The conclusion is to be removed without changing the duplicated paragraph.

67. Read the sentence: The board decided to defer the decision until the safety report arrived. Choose the closest meaning of 'defer' as used in the sentence.

(A) cancel permanently (B) postpone

(C) discuss privately (D) delegate

68. Choose the correctly spelt word.

(A) rhythm (B) rhytm

(C) rythm (D) rhythym

69. Once completed, the change cannot be undone to restore the original state. Which adjective fits?

(A) irreversible (B) inevitable

(C) unpredictable (D) permanent-looking

70. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. but it also allowed Q. the staff to find each file R. without moving the neighbouring boxes. S. Not only did the new arrangement save space,

(A) P-S-Q-R (B) S-P-Q-R

(C) R-P-Q-S (D) Q-S-P-R

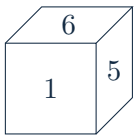
Logical Reasoning

71. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 7, 28, 31, 124, ?, 508, 511.

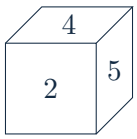
(A) 496 (B) 499

(C) 121 (D) 127

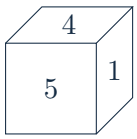
72. The views show the same die. Give the ordered pair: (face opposite 6, face opposite 1). Do not assume opposite labels sum to seven.



1



2



3

(A) $(4, 4)$ (B) $(4, 3)$

(C) $(4, 1)$ (D) $(4, 2)$

73. Three pairs contain exactly one metal and one nonmetal. Which pair does not?

(A) Aluminium / Nitrogen (B) Copper / Iodine

(C) Tin / Silver (D) Zinc / Oxygen

74. Preserve the exact relation and its order: Barometer : Atmospheric pressure :: ? : Humidity. Choose the missing term.

(A) Barometer (B) Hygrometer

(C) Thermometer (D) Ammeter

75. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 7$ and $5 \mapsto 16$. Which input maps to 37?

(A) 11 (B) 12

(C) 13 (D) 37

76. Bina receives 2 times as many tokens as Asha, and Charu receives 3 times as many as Bina. Their total is 171. After the allocation, Charu gives 3 tokens to Asha. How many tokens does Charu then retain?

(A) 111 (B) 117

(C) 16 (D) 114

77. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: VOG, SQI, PSK, ?, JWO, GYQ.

(A) MUM (B) NVN

(C) JWO (D) LTL

78. Consider only these statements; universal statements alone do not imply existence. No joggers are inventors; No joggers are farmers; All joggers are inventors. Conclusions: I. Some joggers are farmers. II. No farmers are joggers. Which conclusions follow necessarily?

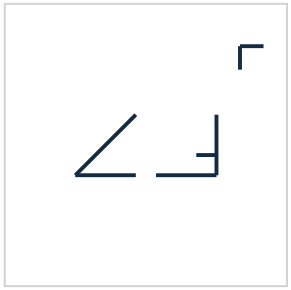
(A) Both I and II (B) Only I

(C) Only II (D) Neither I nor II

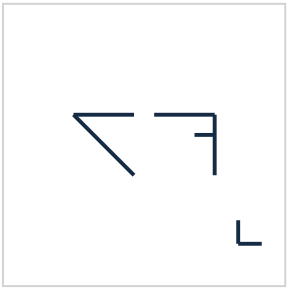
79. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



(A) Pattern V



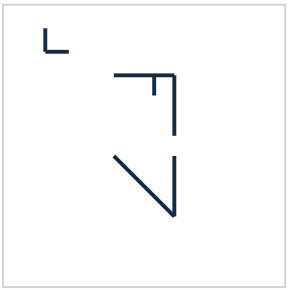
(B) Pattern N



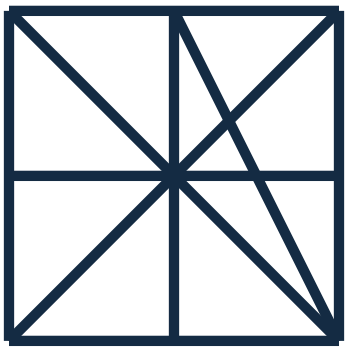
(C) Pattern Z



(D) Pattern R



80. How many triangles are formed by all the drawn segments?



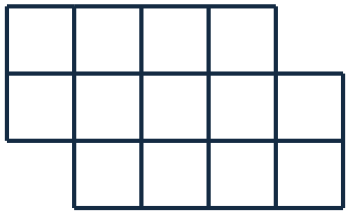
(A) 29

(B) 25

(C) 19

(D) 50

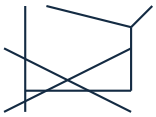
81. How many rectangles, including squares are formed by all the drawn segments?



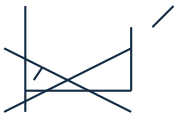
(A) 60
(B) 48
(C) 61
(D) 62

82. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





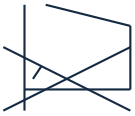
host 1



host 2



host 3



host 4

(A) 3
(B) 1
(C) 4
(D) 2

83. Preserve the exact relation and its order: Librarian : Library :: ? : Field. Choose the missing term.
(A) Judge
(B) Pilot
(C) Librarian
(D) Farmer

84. $X\$Y$ means X is the mother of Y; $X\#Y$ means X is the brother of Y; $X\&Y$ means X is the father of Y. All names denote distinct people; a brother or sister is a full sibling, sharing both parents. Given $B\$R\ R\#C\ C\&M\ M\#T$, how is B related to T?
(A) Paternal grandmother
(B) Maternal grandfather
(C) Father
(D) Mother

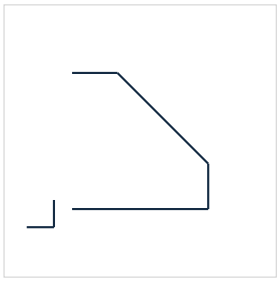
85. Consider only these statements; universal statements alone do not imply existence. Some hikers are hikers; All joggers are guides. Conclusions: I. All hikers are joggers. II. Some hikers are not joggers. Which conclusion or complementary alternative must hold?
(A) Only I
(B) Either I or II follows, but neither is individually certain
(C) Both I and II
(D) Only II

86. A word is coded as the sum of its forward alphabet positions plus 2 times its length. The code of MANG? is 60. Which letter replaces the question mark?

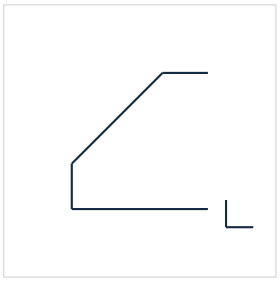
(A) P (B) O

(C) Q (D) N

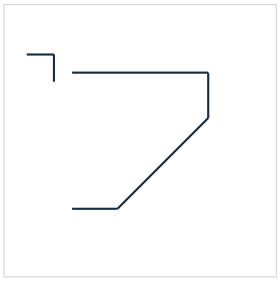
87. Select the missing bottom-right tile. Tiles reflect across the shared horizontal and vertical centre lines; there is no turning.



top left



top right

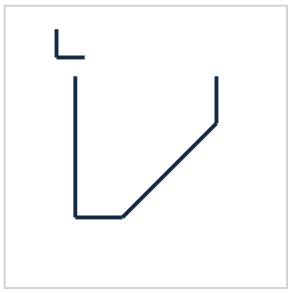


bottom left

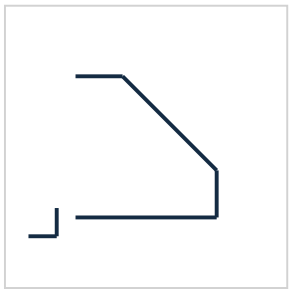
?

bottom right

(A) Pattern Y



(B) Pattern W



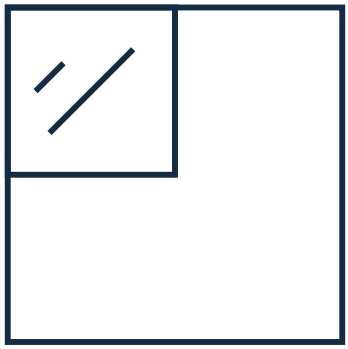
(C) Pattern L



(D) Pattern Z



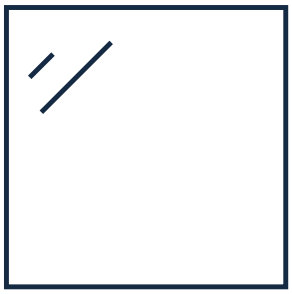
88. Starting with a unit square sheet, fold right to left along the vertical midline, then fold bottom to top along the horizontal midline. The folded paper is cut along both short drawn slits. Select the pattern after complete unfolding.



(A) Pattern N



(B) Pattern Z



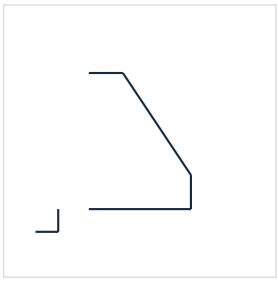
(C) Pattern X



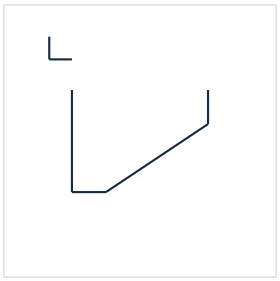
(D) Pattern L



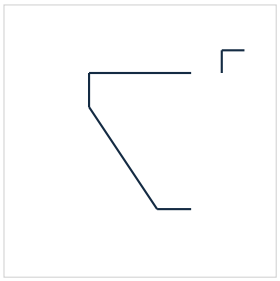
89. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

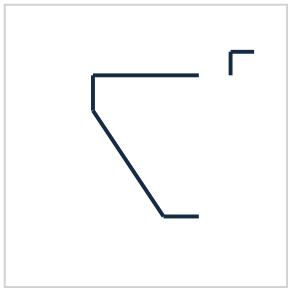


2

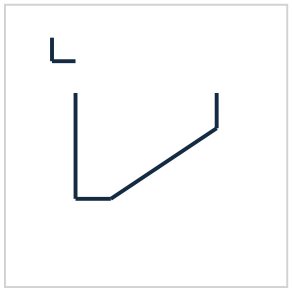


3

(A) Pattern T



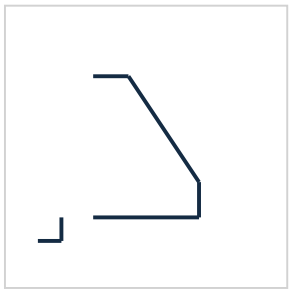
(B) Pattern Q



(C) Pattern W



(D) Pattern U



90. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

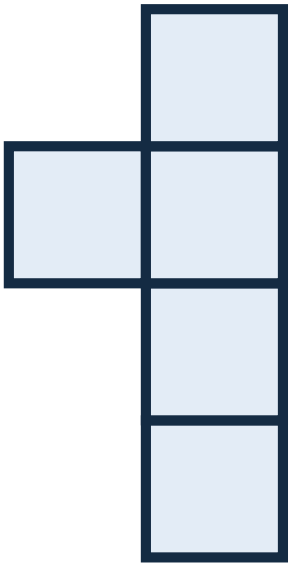


piece 1

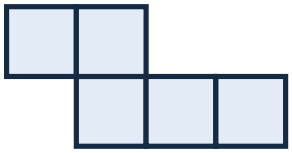


piece 2

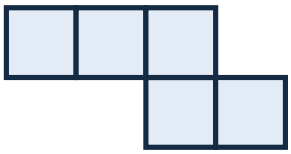
(A) Pattern Z



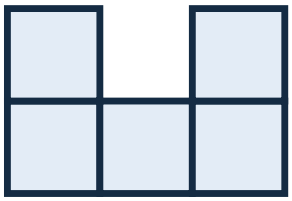
(B) Pattern T



(C) Pattern V



(D) Pattern Q



Mathematics

91. There are 4 Boolean variables p_1,...,p 4. Exactly 2 variables must be true. How many truth assignments satisfy [(p_1 implies p_2) and (p_1 implies not p_2)]?

(A) 3 (B) 16

(C) 1 (D) 2

92. Let R=[sin($\frac{7\pi}{78}$)+sin($\frac{19\pi}{78}$)]/[cos($\frac{7\pi}{78}$)+cos($\frac{19\pi}{78}$)]. Find (1-R^2)/(1+R^2).

(A) $\frac{3}{2}$ (B) $-\frac{1}{2}$

(C) $\frac{1}{2}$ (D) $\frac{\sqrt{3}}{2}$

93. B needs 3 days longer than A; together they need $\frac{40}{13}$ days. A works alone for $\frac{5}{2}$ days, then B joins. How many additional days finish the job?

(A) $\frac{5}{2}$ (B) $\frac{40}{13}$

(C) $\frac{15}{26}$ (D) $\frac{20}{13}$

94. Parabola y=7(x-4)(x-5) meets the axes at three distinct points. A circle passes through these points. Find its other y-intercept.

(A) 7
(B) 1
(C) $\frac{1}{7}$
(D) $-\frac{1}{7}$

95. If |z-9|=3, find the minimum of $|z|^2$.

(A) 36
(B) 144
(C) 72
(D) 6

96. For f(x)=|x-6|/(x-11), on which maximal open intervals is f strictly decreasing?

(A) $(6, \infty)$
(B) $(-\infty, 11)$
(C) $(-\infty, 6)$
(D) $(6, 11) \cup (11, \infty)$

97. For x,y>0 with sqrt(x)+sqrt(y)=8, find d^2y/dx^2 at x=36.

(A) $\frac{2}{3}$
(B) $\frac{1}{27}$
(C) $-\frac{1}{54}$
(D) $\frac{1}{54}$

98. X earns 7 and uses 2 labour hours; Y earns 2 and uses 3 hours. There are 90 hours and at least 3 items of Y must be made. Which model is correct?

(A) Maximize $2x + 7y$; $2x + 3y \leq 90$, $y \leq 3$
(B) Maximize $7x + 2y$; $2x + 3y \geq 90$, $x, y \geq 0$
(C) Minimize $7x + 2y$; $2x + 3y \leq 90$, $y \geq 3$
(D) Maximize $7x + 2y$; $2x + 3y \leq 90$, $x \geq 0$, $y \geq 3$

99. y>0 satisfies $2yy' + 4y^2 = 0$, y(0)=sqrt(3). Find $y^2(1)$.

(A) $\frac{\sqrt{3}}{e^2}$
(B) $\frac{3}{e^2}$
(C) $3e^4$
(D) $\frac{3}{e^4}$

100. Disjoint A,B have sizes 4,8. Find $|(A \times A) \cup (B \times B)|$.

(A) 64
(B) 12
(C) 144
(D) 80

101. α, β are nonzero roots of x^2-13x+36=0. Find the sum of $1+1/\alpha$ and $1+1/\beta$.

(A) $\frac{85}{36}$
(B) $\frac{13}{36}$
(C) $\frac{62}{13}$
(D) $\frac{49}{36}$

102. Let $z=(1+i)^{21}$. Find Re(z)+Im(z).

(A) -2048
(B) -1024
(C) $1024\sqrt{2}$
(D) 2048

103. A differentiable function f satisfies $f'(t) = (t - 49)(t - 121)$ for every real t. Let $g(x) = f(x^2)$. Which is the maximal open interval containing $x = -9$ on which $g'(x) > 0$?

(A) (-7, 0)
(B) (7, 11)
(C) (-11, -7)
(D) (-infinity, -11)

104. r=pa+qb, a=(1,0,0), b=(1,1,0), r·a=7, r·b=12. Find $|r|^2$.

(A) 29 (B) 74

(C) $\sqrt{74}$ (D) 12

105. Box A contains 7 red and 3 blue balls; B contains 3 red and 5 blue balls. A fair coin chooses one box. Two independent draws with replacement from that box give exactly one red ball. Find the posterior probability that A was chosen.

(A) $\frac{28}{43}$ (B) $\frac{112}{237}$

(C) $\frac{125}{237}$ (D) $\frac{21}{100}$

106. $F'(x)=\frac{1}{x^2}-\frac{1}{x^2\sqrt{x^2+1}}$ and F(1)=2. Find F(5)-2F(1).

(A) $-\sqrt{2}-\frac{6}{5}+\frac{\sqrt{26}}{5}$ (B) $-\sqrt{2}+\frac{4}{5}+\frac{\sqrt{26}}{5}$

(C) $-\sqrt{2}+\frac{\sqrt{26}}{5}+\frac{14}{5}$ (D) $-\frac{14}{5}-\frac{\sqrt{26}}{5}+\sqrt{2}$

107. ABC is a right triangle on level ground with AB=10, AC=10, angle A=90°. A vertical tower stands at midpoint M of BC. Its top elevation from A is 60°. Find its height.

(A) $10\sqrt{6}$ (B) $5\sqrt{6}$

(C) $5\sqrt{2}$ (D) $5\sqrt{3}$

108. Find the area bounded by y=x, y=64/x, and x=48 in the first quadrant.

(A) $64\log(6)$ (B) 1120

(C) $1120-64\log(6)$ (D) $320-64\log(6)$

109. 14 distinct people sit at labelled round tables of sizes 8,6. Two specified people must be at different tables, and the first specified person must be at the 8-seat table. Rotations are identical. Count the seatings.

(A) 1816214400 (B) 22992076800

(C) 479001600 (D) 958003200

110. $x=t^2$, $y=t^3$ for t>0. Find d^2y/dx^2 at t=6.

(A) 18 (B) $\frac{3}{2}$

(C) $\frac{1}{4}$ (D) $\frac{1}{8}$

111. a=(9,7,0), b=(7,-9,3). Find $|a+b|^2-|a-b|^2$.

(A) 269 (B) 0

(C) 18070 (D) −9

112. Evaluate $\int_0^\infty \log(2x)/(x^2+4)\,dx$.

(A) $\frac{\pi\log(2)}{2}$ (B) 0

(C) $\pi\log(2)$ (D) $\frac{\pi\log(2)}{4}$

113. The hyperbola $x^2/25-y^2/16=1$ has eccentricity e. Find the eccentricity e′ of its conjugate hyperbola.

(A) $\frac{5\sqrt{41}}{41}$ (B) $\frac{\sqrt{41}}{4}$

(C) $\sqrt{41}$ (D) $\frac{\sqrt{41}}{5}$

114. For $A=\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix}$ and $C=5A^{-1}$, find C_{22}.

(A) 2 (B) 3

(C) -1 (D) $\frac{2}{5}$

115. For $x>0$, $F\prime(x)=1/x^2-1/(x^2\sqrt{x^2+1})$. A function H is continuous at 0, satisfies $H(0)=0$, and has $H\prime(x)=x^2F\prime(x)$ for $x>0$. Find $H(\sqrt{3})$.

(A) $\log\left(\sqrt{3}+2\right)+\sqrt{3}$ (B) $-\log\left(\sqrt{3}+2\right)+\sqrt{3}$

(C) $-1+\sqrt{3}$ (D) $-\frac{\pi}{3}+\sqrt{3}$

116. How many integer t make P=(t,0) strictly inside the larger segment cut by x=7 from $x^2+y^2=169$?

(A) 25 (B) 21

(C) 19 (D) 5

117. Evaluate lim as $n\to\infty$ of $(4/n)\Sigma(4j/n)^2$, j=1,...,n.

(A) $\frac{1}{3}$ (B) $\frac{64}{3}$

(C) $\frac{16}{3}$ (D) 32

118. The four lines $\pm x/18\pm y/8=1$ form a rhombus with vertices on the coordinate axes. Find its area.

(A) 72 (B) 144

(C) 288 (D) 576

119. How many integers x satisfy $|(x-3)+(x-8)|<|x-3|+|x-8|$?

(A) 5 (B) 7

(C) 6 (D) 4

120. Positive integers $a<b$ satisfy $a+b=28$ and $3\mid a$. Find the maximum value of ab.

(A) 196 (B) 28

(C) 192 (D) 171

121. For $1<x<2$, $f(x)=\arctan(2x/(1-x^2))$. Find $f(x)-2\arctan x$.

(A) 0 (B) $-\pi$

(C) $-\frac{\pi}{2}$ (D) π

122. $F'(x)=(x^2+7x+9)\,e^x$ and $F(0)=5$. Find $F(1)-2F(0)$.

(A) $1+10e$ (B) $-4+10e$

(C) $-10e-1$ (D) $-9+10e$

123. Four observations are 7,7,x,y with population mean 7 and variance 16/2. Find xy.

(A) -15 (B) 33

(C) 65 (D) 17

124. Evaluate $\sum_{j=1}^{\infty} 5j(\frac{1}{4})^j$.

(A) $\frac{20}{3}$ (B) $\frac{35}{9}$

(C) $\frac{20}{9}$ (D) $\frac{4}{5}$

125. Find the area between y=(x-7)(x-14) and the x-axis for $0 \leq x \leq 17$.

(A) $\frac{767}{4}$ (B) $-\frac{1615}{6}$

(C) $\frac{1615}{6}$ (D) $\frac{767}{2}$

126. The line 3x+4y=10 cuts $x^2+y^2=64$ in chord AB. Find $|AB|^2$.

(A) 240 (B) $4\sqrt{15}$

(C) 196 (D) 60

127. For a=(1,7,0), b=(0,1,2), c=(4,0,1), evaluate $(a \times c) \cdot (a+b)$.

(A) -57 (B) 57

(C) -56 (D) 0

128. Between positive a=4, b=16, let A be their arithmetic mean and G their geometric mean. Find A^2-G^2.

(A) 36 (B) 68

(C) 6 (D) 144

129. Group A has mean 60, SD 6. Every observation is increased by 60. Find its new coefficient of variation in percent.

(A) 20 (B) 10

(C) 5 (D) 30

130. $f(x)=x^2+4x$ for $x \leq 4$, and $f(x)=px+q$ for $x>4$. If f is differentiable at 4, find p-q.

(A) -4 (B) 28

(C) 12 (D) 32

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A spherical planet rotates uniformly. Its rotation reduces effective gravity at the equator by 12 m/s^2. By how much does rotation reduce the radial component of effective gravity at latitude 60°?

(A) 9 m/s^2

(B) $\frac{3}{4}$ m/s^2

(C) 3 m/s^2

(D) 12 m/s^2

132. A uniformly charged nonconducting sphere has radius R and surface potential V_s relative to infinity. Find potential at $r = R/11$ divided by V_s.

(A) $\frac{181}{121}$

(B) $\frac{120}{121}$

(C) $\frac{3}{2}$

(D) 1

133. At normal incidence, an ideal absorber receives intensity I and an ideal reflector receives intensity $I/3$. Which statement gives the ratio of reflector pressure to absorber pressure, and the reason?

(A) 1/3; photons transfer momentum only on absorption.

(B) 2/3; reflection reverses the normal photon momentum.

(C) 3/2; pressure is inversely proportional to intensity.

(D) 23; reflected light transfers negative energy to the surface.

Chemistry

134. For one reaction,ln K is6 at300 K and1 at600 K. Assume ln K is linear in1/T. At400 K,ln K equals

(A) 6

(B) $\frac{9}{2}$

(C) $\frac{7}{2}$

(D) 1

135. Consider I: NF3 has a smaller resultant dipole moment than NH3. II: Oxygen normally has lower maximum covalency than sulfur in their common stable compounds. Which assessment correctly recognises both anomalous first-element behaviours?

(A) Both statements are correct.

(B) Only II is correct.

(C) Neither statement is correct.

(D) Only I is correct.

136. Arrange propane,dimethylamine,ethylamine and ethanol in increasing normal boiling point. The last three have similar molar masses;use their ordinary hydrogen-bonding behaviour.

(A) dimethylamine < propane < ethanol < ethylamine

(B) propane < dimethylamine < ethylamine < ethanol

(C) propane < ethylamine < dimethylamine < ethanol

(D) ethanol < ethylamine < dimethylamine < propane

Logical Reasoning

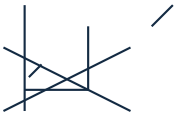
137. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, CHAIR is coded as EJCKT. Which word is coded as ENQWF?

(A) DMPVE (B) GPSYH

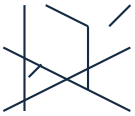
(C) CLOUD (D) BKNTC

138. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3

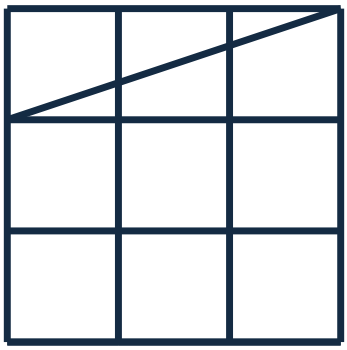


host 4

(A) 3 (B) 4

(C) 2 (D) 1

139. How many maximal straight connected strokes are formed by all the drawn segments?



(A) 11 (B) 9

(C) 10 (D) 8

Mathematics

140. The locus |z-2|=|Re(z)+2| intersects the horizontal line Im(z)=6. Find the real part of the point.

(A) $\frac{9}{2}$ (B) 9

(C) $-\frac{9}{2}$ (D) $\frac{3}{4}$

141. Let $a_j = 1/(2 + 5(j - 1))$. Evaluate $\sum_{j=1}^{5} a_j a_{j+1}$.

(A) $\frac{5}{22}$ (B) $\frac{5}{54}$

(C) $\frac{1}{11}$ (D) $\frac{25}{54}$

142. f(x+1)=$9x^2$. Find the area enclosed between y=f(x) and y=9(x-1).

(A) $\frac{9}{2}$ (B) 3

(C) $\frac{3}{2}$ (D) $-\frac{3}{2}$

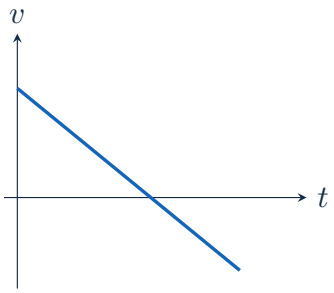
BITSAT Mock 15

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A particle has velocity $v(0) = 6$ m/s, and constant acceleration makes $v(5) = -5$ m/s. What total distance does it travel in this interval?



(A) $\frac{305}{11}$ m
(B) $\frac{125}{22}$ m
(C) $\frac{305}{22}$ m
(D) 25 m

2. A drop of mass 4.8×10^{-17} kg is suspended in a vertical field of magnitude 500 N/C. It has excess electrons. One excess electron is removed without changing its mass. Find its immediate downward acceleration, taking $g = 10$ m/s^2 and $e = 1.6 \times 10^{-19}$ C.

(A) $\frac{10}{7}$ m/s^2
(B) $\frac{5}{18}$ m/s^2
(C) 2 m/s^2
(D) $\frac{5}{3}$ m/s^2

3. A 5 kg block moving at 3 m/s compresses a spring of constant 160 N/m. During compression, a damper removes 1/3 of the block's initial kinetic energy. Find maximum compression; gravity does no work.

(A) $\frac{3}{16}$ m
(B) $\frac{\sqrt{3}}{4}$ m
(C) $\frac{\sqrt{15}}{20}$ m
(D) $\frac{3\sqrt{2}}{8}$ m

4. A particle has displacement $x = 3\cos(2\pi t/7)$ cm. Find its signed velocity at $t = \frac{7}{6}$ s.

(A) $-\frac{6\pi^2}{49}$ cm/s (B) $-\frac{3\sqrt{3}\pi}{7}$ cm/s

(C) $-\frac{6\pi}{7}$ cm/s (D) $\frac{3\sqrt{3}\pi}{7}$ cm/s

5. A uniform disc of mass 7 kg and radius 4 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) $\frac{133}{2}\,\mathrm{kg\,m^2}$ (B) $42\,\mathrm{kg\,m^2}$

(C) $\frac{105}{2}\,\mathrm{kg\,m^2}$ (D) $\frac{91}{2}\,\mathrm{kg\,m^2}$

6. One mole of nonoverlapping identical spheres of radius r has total geometric material volume V. A second sample contains $\frac{1}{10}$ mole of spheres of radius $4r$. What is its total geometric material volume divided by V? Ignore packing voids.

(A) $\frac{8}{125}$ (B) $\frac{32}{5}$

(C) 64 (D) $\frac{8}{5}$

7. In YDSE a sheet of index 2 and thickness 5 micrometres covers one slit. A second sheet of the same thickness and index 3/2 covers the other. With wavelength $\frac{2}{5}$ micrometres, find the magnitude of central-fringe shift in units of fringe width.

(A) $\frac{25}{2}$ (B) $\frac{25}{8}$

(C) $\frac{25}{4}$ (D) $\frac{4}{25}$

8. In an ideal intrinsic semiconductor at zero temperature, the band gap is $\frac{4}{5}$ eV. Neglect phonons, excitons and all transitions within a band. Two beams have photon energies $\frac{3}{10}$ eV and $\frac{13}{10}$ eV. Which can create a valence-to-conduction electron-hole pair by a single-photon absorption?

(A) Only the higher-energy beam (B) Only the lower-energy beam

(C) Neither beam (D) Both beams

9. A uniform-density spherical planet has radius R and surface gravitational acceleration g. At what depth below the surface is gravity $g/10$? Give the depth as a fraction of R.

(A) 1 (B) $\frac{9}{10}$

(C) 9 (D) $\frac{1}{10}$

10. A wire carries $i(t) = 6 + 6\sin\omega t$ A for many cycles. Find its rms current.

(A) $3\sqrt{6}\,\mathrm{A}$ (B) 6 A

(C) 12 A (D) $6\sqrt{2}\,\mathrm{A}$

11. A galvanometer has resistance 40 ohms and full-scale current $\frac{1}{500}$ A. It is to measure a total current 5 times its own full-scale current. Find the required parallel shunt resistance.

(A) $10\,\Omega$ (B) $50\,\Omega$

(C) $40\,\Omega$ (D) $8\,\Omega$

12. For steady laminar flow through a capillary, the radius is multiplied by 4 and length by 2. The volume flow rate must remain unchanged and viscosity is fixed. By what factor must the pressure difference be multiplied?

(A) $\frac{1}{128}$ (B) $\frac{1}{2}$

(C) $\frac{1}{8}$ (D) 1

13. Current rises linearly from 0 to 6 A over 6 s and then falls linearly to 0 over another 6 s. It remains zero for the next 12 s. Find the average current over the entire 24 s interval.

(A) 2 A (B) $\frac{3}{2}$ A

(C) 6 A (D) 3 A

14. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 40 percent. The measured increase in its absolute temperature is 336 K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) $\frac{1250}{7}$ K (B) 2100 K

(C) 350 K (D) 420 K

15. A fixed amount of ideal gas has its absolute temperature multiplied by 2 and volume by 3. Find the final pressure divided by the initial pressure.

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) 2 (D) $\frac{2}{9}$

16. A small pendulum hangs steadily inside a vehicle moving at constant speed 4 m/s around a horizontal circle of radius 20 m. Find the tangent of its angle to the vertical. Take $g = 10$ m/s^2.

(A) $\frac{1}{50}$ (B) $\frac{25}{2}$

(C) $\frac{2}{25}$ (D) $\frac{1}{250}$

17. 343 identical conducting liquid drops merge without losing charge. Before merging each has potential V and surface charge density σ. Find the final surface charge density divided by σ.

(A) 7 (B) $\frac{1}{49}$

(C) 49 (D) 343

18. A block of mass 5 kg is fixed at rest while a particle of mass 2 kg moves with constant acceleration $\vec{a} = 5\hat{i} + 10\hat{j}$ m/s^2. What is the magnitude of acceleration of the centre of mass of the two-body system?

(A) $5\sqrt{5}$ m/s^2 (B) $\frac{10}{7}$ m/s^2

(C) $\frac{10\sqrt{5}}{7}$ m/s^2 (D) $\frac{30}{7}$ m/s^2

19. A pendulum bob of mass 3 kg and length 5 m passes the horizontal position moving downward at 3 m/s. Find the instantaneous power of all forces when $\sin\theta = 3/5$, $\cos\theta = 4/5$, with θ measured from the downward vertical. Take $g = 10$ m/s^2.

(A) 54 W (B) $24\sqrt{89}$ W

(C) $18\sqrt{69}$ W (D) $18\sqrt{89}$ W

20. Two parallel plane mirrors separated by 4 cm extend from axial position 0 to 22 cm. A ray enters midway between them at axial position 0, making 45° with the normal to the mirrors. How many reflections occur before it passes the far edge? Exclude a contact exactly on the exit edge.

(A) 4 (B) 10

(C) 6 (D) 5

21. Two equal resistors of 8 ohms are in series across an ideal 40 V source. A voltmeter of resistance 8 ohms is connected across one resistor. Find its reading.

(A) $\frac{40}{3}$ V (B) 20 V

(C) 10 V (D) $\frac{80}{3}$ V

22. A network is defined by $P = \mathrm{NOR}(A, B)$, $Q = \mathrm{NAND}(A, P)$ and $Y = \mathrm{NOR}(Q, \overline{B})$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 1$ and $A = 0$?

(A) 1 (B) 2

(C) 0 (D) 3

23. A nonrelativistic charged particle initially has momentum $5qEt$ along a uniform electric field, where $t > 0$ is a fixed reference interval. Its charge is $+q$. What is $\lambda(5t)/\lambda(2t)$ after acceleration in the same field?

(A) $\frac{3}{10}$ (B) $\frac{1}{2}$

(C) $\frac{49}{100}$ (D) $\frac{7}{10}$

24. A travelling wave is $y = 1\sin[3(x - \frac{50}{3}t) + \pi/4]$ mm with x in metres and t in seconds. What is its frequency?

(A) $\frac{25}{3\pi}$ Hz (B) $\frac{3}{2\pi}$ Hz

(C) $\frac{50}{\pi}$ Hz (D) $\frac{25}{\pi}$ Hz

25. A uniform rod of mass 6 kg and length 4 m lies on a smooth horizontal table, free to rotate about a fixed vertical pivot at its centre. A point mass 3 kg moving perpendicular to the rod at 3 m/s strikes an end and sticks. Find the angular speed immediately afterward.

(A) $\frac{9}{10}$ rad/s (B) $\frac{9}{5}$ rad/s

(C) $\frac{9}{28}$ rad/s (D) $\frac{3}{2}$ rad/s

26. A triangular p-V cycle has vertices (V_0, p_0), $(V_0 + 3\,\mathrm{L}, p_0)$ and $(V_0, p_0 + 4\,\mathrm{kPa})$. Traversed in that listed order and then back, find net work done by the gas.

(A) 0 J (B) 12 J

(C) −6 J (D) 6 J

27. A long solenoid has its turn number multiplied by 5 and length multiplied by 3, while its cross-sectional area and core permeability stay fixed. Find self-inductance multiplier.

(A) $\frac{25}{3}$ (B) 25

(C) $\frac{3}{25}$ (D) $\frac{1}{3}$

28. A projectile is launched at 60 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{3}{4}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ m/s^2.

(A) $\frac{1296}{5}$ m (B) $\frac{1728}{5}$ m

(C) 288 m (D) $\frac{864}{5}$ m

29. A ball is dropped from 28 m onto a horizontal fixed floor. At every impact the coefficient of restitution is $\frac{4}{5}$. Neglect air resistance. What maximum height does it reach after its second impact?

(A) 28 m (B) $\frac{7168}{625}$ m

(C) $\frac{28}{625}$ m (D) $\frac{1835008}{390625}$ m

30. A pure radioactive sample has activity A at time 4 hours after preparation and $A/8$ at time 10 hours after preparation. At what time, measured from preparation, will its activity be one thirty-second of the activity measured at 4 hours? Assume exponential decay with a constant decay rate.

(A) 68 h (B) 14 h

(C) 42 h (D) 20 h

Chemistry

31. For electrophilic mononitration of benzene carrying OCH3,which orientation/reason pair is correct? Consider intrinsic electronic directing effects,not an exact ortho:para ratio.

(A) meta; oxygen lone-pair donation outweighs its inductive withdrawal for ring activation

(B) meta; the substitution site is decided only by steric size

(C) ortho/para; every electronegative substituent activates the ring

(D) ortho/para; oxygen lone-pair donation outweighs its inductive withdrawal for ring activation

32. An anode converts two sulfate-derived units into a peroxodisulfate unit with an O-O linkage, leaving sulfur at+6. Which atom-level oxidation explains electron release?

(A) No atom changes oxidation state when a peroxide bond forms.

(B) Each hydrogen is reduced from+1 to-1.

(C) Both sulfur atoms are oxidised from+6 to+7.

(D) Two oxide-like oxygen atoms are oxidised from-2 to-1 as the peroxide link forms.

33. Match two applications: I. an orange-red advertising discharge tube. II. an inert atmosphere for welding reactive hot metals. Which ordered gas pair is appropriate?

(A) neon; argon. (B) neon; xenon.

(C) argon; neon. (D) argon; argon.

34. Atoms form ideal touching FCC spheres of radius r and cell edge a. Fractional coordinates are measured in a. Using the straight displacement between the two specified points (0,0,0) and (1/2,1/2,1/2), without replacing it by another periodic image, find (distance/r)^2. These points describe corner atom centre to central octahedral site.

(A) 6 (B) 3

(C) 7 (D) 12

35. A0.400 g compound containing only C,H,O gives0.660 g CO2 and0.180 g H2O on complete combustion. Find its oxygen mass percentage. Use C12,H1,O16.

(A) 50 (B) 55

(C) 45 (D) 5

36. A solution has0.0100M M^{2+} and0.0100M N^{2+},with sulfideKsp10^{-20} and10^{-16}. SaturatedH_2S stays0.100M andKa1Ka2=10^{-21}. Ignore complexes and volume change. Choose pH giving at least99.9% M precipitation and no N precipitation.

(A) $\frac{15}{4}$ (B) $\frac{9}{2}$

(C) 2 (D) 3

37. Nonrelativistic ions A,B start from rest. B has 2 times the mass and 3 times the positive charge of A,and is accelerated through 4 times A's potential difference. Ratioλ_B/λ_A is

(A) $\frac{1}{2}$ (B) $\frac{\sqrt{2}}{4}$

(C) $2\sqrt{6}$ (D) $\frac{\sqrt{6}}{12}$

38. A pure KMnO4 portion gives 1/200 mol O2 on complete controlled heating by2KMnO4→K2MnO4+MnO2+O2. With KMnO4=158 g/mol, which pair gives the initial mass and a viable origin of its intense purple colour despite Mn(VII) being d0?

(A) 79/25 g; d-d absorption within an occupied d shell. (B) 79/50 g; charge-transfer absorption.

(C) 79/100 g; charge-transfer absorption. (D) 79/50 g; d-d absorption within an occupied d shell.

39. At 1 atm compare two separate pairs: (ethanol, dimethyl ether) and (water, hydrogen sulfide). A student selects dimethyl ether for the first higher-boiling substance because its formula mass is unchanged, and water for the second. Which assessment is consistent with intermolecular interactions?

(A) Both selections correct. (B) First selection wrong; second selection correct.

(C) Both selections wrong. (D) First selection correct; second selection wrong.

40. A student assigns three bonding domains and no lone pair to the central atom of SO_3. After independently checking valence-electron count,which combined molecular shape/domain count is consistent?

(A) Tetrahedral; four electron domains. (B) Trigonal planar; three electron domains.

(C) T-shaped; five electron domains. (D) Trigonal pyramidal; four electron domains.

41. A weak monoprotic acid at C=$\frac{9}{10000}$M has measured degree of dissociation $\frac{1}{20}$. Use the exact concentration relationKa=Cα^2/(1-α). At the same T,analytical concentration required for degree $\frac{1}{40}$ is

(A) $\frac{351}{95000}$ (B) $\frac{9}{5000}$

(C) $\frac{9}{40000}$ (D) $\frac{9}{2500}$

42. tert-butyl bromide is heated with sodium ethoxide under elimination-favouring conditions. Which major constitutional product is expected,despite the presence of an alkoxide that could be used in a Williamson synthesis with a suitable different halide?

(A) the alkane formed by replacing halogen with H (B) the corresponding tertiary alkoxide substitution product

(C) tert-butyl ethyl ether (D) 2-methylpropene

43. A gas evolved during an anion test turns lead acetate paper black. Which gas and chemical explanation match this complete observation?

(A) NH3; molecular oxygen causes the stated change.
(B) H2S; PbS accounts for the diagnostic change.
(C) N2; acid-base neutralisation always produces the stated colour.
(D) HCl; elemental chlorine is deposited.

44. A dilute non-hydrogen-bonding solvent separates molecules of2-nitrophenol and4-nitrophenol. Which comparison preserves the difference between intra- and intermolecular hydrogen bonding?

(A) An intramolecular hydrogen bond requires two different molecules.

(B) The ortho isomer can retain an intramolecular O-H···O interaction; dilution suppresses intermolecular association of the para isomer.

(C) Both isomers must form the same intramolecular ring despite their different substitution positions.

(D) Dilution necessarily breaks all covalent O-H bonds.

45. The ions F^-, Na^+, Mg^{2+}, Al^{3+} each have 10 electrons. Which decreasing-radius order and explanation is consistent?

(A) $Na^+ > Mg^{2+} > Al^{3+} > F^-$; positive charge always gives the largest ion.

(B) All radii are equal because electron numbers are equal.

(C) $F^- < Na^+ < Mg^{2+} < Al^{3+}$; larger nuclear charge expands the cloud.

(D) $F^- > Na^+ > Mg^{2+} > Al^{3+}$; increasing nuclear charge contracts the same electron cloud.

46. An amine of formula C2H7N gives a positive carbylamine test. It reacts with benzenesulphonyl chloride to give its monosulphonamide. Count symmetry-distinct proton environments in that product, including N-H. Treat rapid rotation as averaging,ignore accidental spectral overlap,and use an achiral medium. The count is

(A) 5
(B) 7
(C) 8
(D) 6

47. Excess water reacts completely with $\frac{1}{50}$ mol CaH_2. Which gas identity/amount pair follows from the balanced hydrolysis?

(A) H_2, $\frac{1}{50}$ mol.
(B) O_2, $\frac{1}{100}$ mol.
(C) H_2, $\frac{1}{25}$ mol.
(D) CO_2, $\frac{1}{50}$ mol.

48. ethanol is subjected to PCC under anhydrous conditions. Which product follows the stated regioselectivity or oxidation/dehydration endpoint?

(A) ethene
(B) ethanal
(C) ethane
(D) ethanoic acid

49. For $3Cl_2 + 6OH^- \rightarrow 5Cl^- + ClO_3^- + 3H_2O$,consider only atoms of the element undergoing disproportionation. Fraction of those atoms oxidised is

(A) $\frac{1}{6}$
(B) $\frac{1}{2}$
(C) $\frac{5}{6}$
(D) $\frac{1}{3}$

50. Assign the formal metal oxidation state in $[Fe(CO)_5]$,using five neutral CO. For nitrosyl species use the explicitly stated formal NO^+ convention rather than inferring spin from it.

(A) 2 (B) 1

(C) 0 (D) -1

51. A mixture contains $\frac{9}{100}$ mol H_2SO_5 and $\frac{2}{25}$ mol $H_2S_2O_8$. Their structures contain one O–O link per molecule. All peroxide links react quantitatively with excess iodide in acid; each such link accepts two electrons. What amount of I_2, in mol, is liberated?

(A) $\frac{17}{100}$ (B) $\frac{2}{25}$

(C) $\frac{17}{200}$ (D) $\frac{9}{100}$

52. At 340 K, ΔH for C(graphite)+O2(g)→CO2(g) is -400 kJ/mol and for CO(g)+(1/2)O2(g)→CO2(g) is -280 kJ/mol. Carbon is converted only into CO, producing 3/2 mol CO. Assuming ideal gases and R=0.008314 kJ/mol/K, what is total ΔU in kJ? Use the stated reaction extents.

(A) -18212007/50000 (B) -18212007/200000

(C) -18212007/100000 (D) 18212007/100000

53. cycloheptene (C7H12) adds one equivalent of :CBr2 across its only C=C without rearrangement. Which pair gives the adduct formula and the number of independent rings? Count the original ring and the new three-membered ring separately.

(A) C8H12Br2; 2 (B) C7H12Br2; 1

(C) C8H14Br2; 2 (D) C8H12Br2; 1

54. In the trigonal-bipyramidal electron-domain model of XeF_2,there are 3 central lone pairs. Which placement rule explains the observed arrangement,including the first lone pair?

(A) Place lone pairs only where they form the smallest possible angle to every bond.

(B) Place the first lone pair axially because three 90-degree contacts are less repulsive than two.

(C) Place lone pairs equatorially first: an equatorial site has two 90-degree neighbours,whereas an axial site has three.

(D) Use arbitrary sites because lone-pair/bond-pair and bond-pair/bond-pair repulsions are identical.

55. A 15 g $CaCO_3$/inert mixture is80.0% $CaCO_3$ by mass. It receives $\frac{9}{50}$ mol HCl. For $CaCO_3$+2HCl→$CaCl_2$+CO_2+H_2O,90.0% of formed CO_2 is collected. Collected CO_2 amount in mol is

(A) $\frac{9}{100}$ (B) $\frac{81}{1000}$

(C) $\frac{9}{50}$ (D) $\frac{3}{25}$

56. At fixedT,V,A(g)→P(g)+Q(g)+R(g)is first order,and an inert gas is present. Initial totalpressure is$\frac{2}{5}$bar;complete-decomposition pressure is$\frac{3}{5}$bar. At two half-lives,totalpressure in bar is

(A) $\frac{1}{2}$ (B) $\frac{3}{5}$

(C) $\frac{11}{20}$ (D) 1

57. In the gas phase,N_2 is converted into N_2^- by changing one electron. Under the elementary MO model,which final bond-order/unpaired-electron pair follows?

(A) Bond order $\frac{5}{2}$; 0 unpaired electrons.
(B) Bond order 3; 0 unpaired electrons.
(C) Bond order $\frac{7}{2}$; 1 unpaired electrons.
(D) Bond order $\frac{5}{2}$; 1 unpaired electron.

58. A $\frac{1}{100}$mol iron-ion inventory contains Fe^{2+}andFe^{3+}. Fraction$\frac{3}{4}$isFe^{3+}. All iron is quantitatively reduced to metal in the stated externally driven electron-balance model,with no side reactions. Electrons accepted,in mol,are

(A) $\frac{1}{40}$
(B) $\frac{11}{400}$
(C) $\frac{9}{350}$
(D) $\frac{9}{400}$

59. p-methylbenzoyl chloride is subjected to H2/Pd-BaSO4(poisoned catalyst). Which product or product pair follows the named reaction and side-chain oxidation state?

(A) benzaldehyde
(B) 4-methylbenzyl alcohol
(C) 4-methylbenzaldehyde
(D) terephthalic acid

60. One mole each of sucrose and lactose are completely hydrolysed. What is the mole ratio glucose: fructose: galactose in the products?

(A) 2:2:0.
(B) 1:1:2.
(C) 2:1:1.
(D) 1:2:1.

English Proficiency

61. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. whereas the second tested whether Q. The first trial tested whether R. it could move reliably under load. S. the mechanism could move,

(A) R-S-P-Q
(B) Q-S-P-R
(C) S-Q-P-R
(D) P-Q-S-R

62. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. Did the organisers reserve the front row for the visiting speakers?

(A) Was the front row being reserved for the visiting speakers by the organisers?
(B) Was the front row reserved for the visiting speakers by the organisers?
(C) Was the front row reserved by the visiting speakers for the organisers?
(D) Were the visiting speakers reserved for the front row by the organisers?

63. Read the sentence: The checklist was indispensable; the team could not safely proceed without it. Choose the opposite in meaning of 'indispensable' as used in the sentence.

(A) reliable
(B) useful
(C) essential
(D) dispensable

64. Read the sentence: During the repair, clean containers were scarce, so each one had to be shared. Choose the closest meaning of 'scarce' as used in the sentence.

(A) abundant
(B) in short supply
(C) difficult to carry
(D) carefully stored

65. Read the sentence: The difference was negligible and had no practical effect on the result. Choose the opposite in meaning of 'negligible' as used in the sentence.

(A) uncertain
(B) small
(C) substantial
(D) measurable

66. Considering ordinary voice volume, which word does not belong with the other three?

(A) mutter
(B) murmur
(C) whisper
(D) shout

67. Choose the correctly spelt word.

(A) maintenance
(B) maintanence
(C) maintainance
(D) maintenence

68. Read the sentence: The admission rules acted as a filter, excluding applications that lacked the required documents. What does the figurative expression mean in this context?

(A) They ranked all eligible applicants by merit.
(B) They improved every application that was submitted.
(C) They selected applications according to a stated condition.
(D) They hid applications from the assessors.

69. The doctrine says that whatever people do, the final outcome has already been fixed. Which word names it?

(A) fatalism
(B) optimism
(C) individualism
(D) empiricism

70. Use an indefinite article for a previously unmentioned, nonspecific referent; no particular one has been identified. Complete: The collection includes _____ hour-long recording, not merely a written summary.

(A) an
(B) a
(C) no article
(D) the

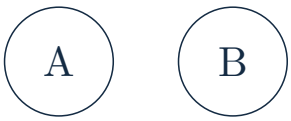
Logical Reasoning

71. All five people have different heights. Gita is shorter than Farah. Farah is shorter than Hari. Hari is shorter than Charu. Asha is shorter than Gita. If Farah is left out, who is 2nd shortest among the remaining people?

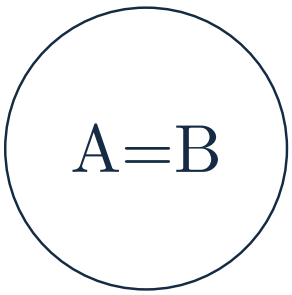
(A) Charu
(B) Gita
(C) Farah
(D) Asha

72. Choose the diagram that represents all these conditions, including the strictness information: All B are A; some A is not B, and B is nonempty.

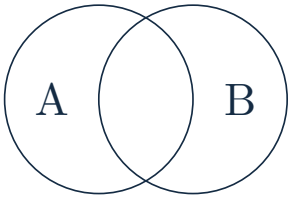
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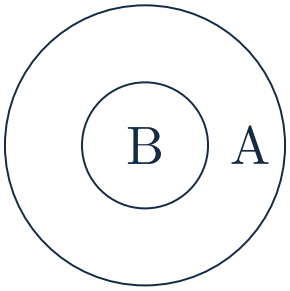
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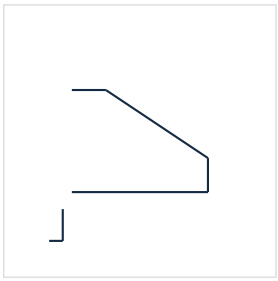
(C) Pattern T



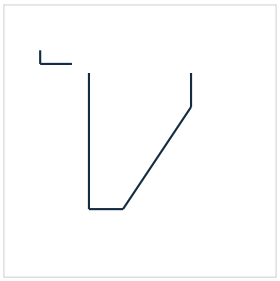
(D) Pattern U



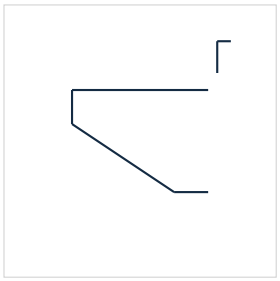
73. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

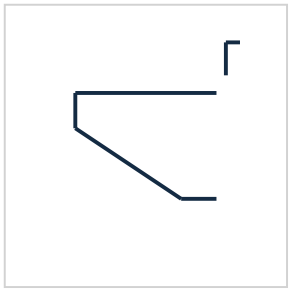


2

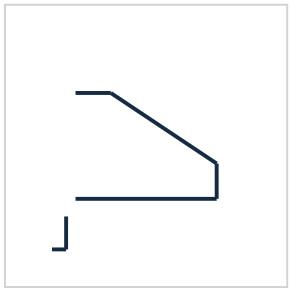


3

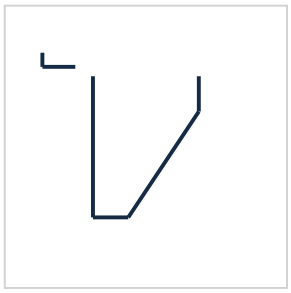
(A) Pattern T



(B) Pattern Y



(C) Pattern L



(D) Pattern N

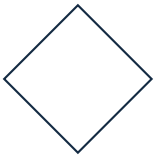


74. A word is coded as the sum of its forward alphabet positions plus 5 times its length. The code of MA?GO is 75. Which letter replaces the question mark?

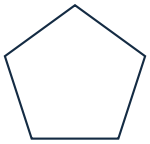
(A) S (B) M

(C) N (D) O

75. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

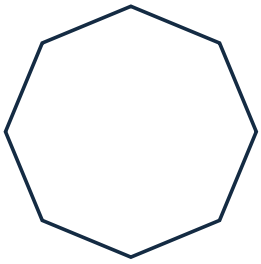


B

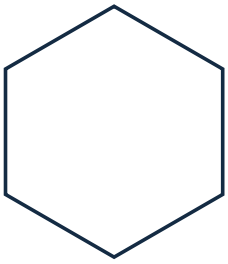


C

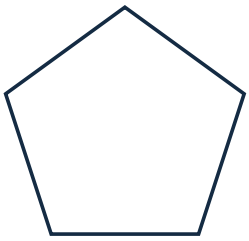
(A) Pattern X



(B) Pattern R



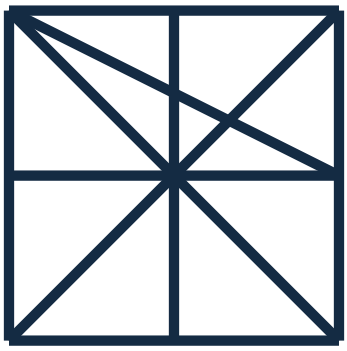
(C) Pattern V



(D) Pattern K



76. How many triangles are formed by all the drawn segments?

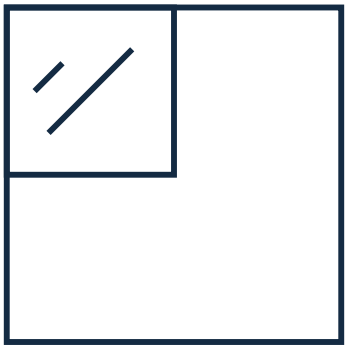


(A) 28
(B) 21
(C) 29
(D) 25

77. Two people differ in age by 11 years. The product of their ages 2 years ago was 432. What will the younger person's age be 4 years from now?

(A) 22
(B) 33
(C) 20
(D) 18

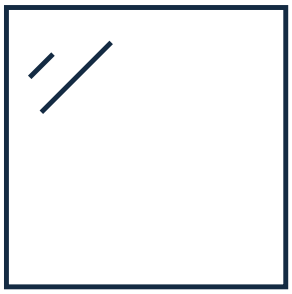
78. Starting with a unit square sheet, fold right to left along the vertical midline, then fold bottom to top along the horizontal midline. The folded paper is cut along both short drawn slits. Select the pattern after complete unfolding.



(A) Pattern W



(B) Pattern V



(C) Pattern L



(D) Pattern Y



79. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

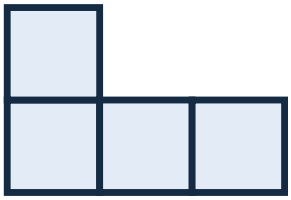


piece 1



piece 2

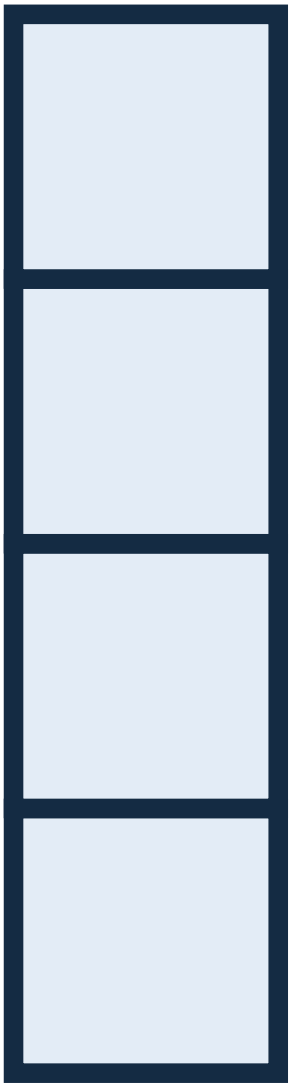
(A) Pattern Q



(B) Pattern L



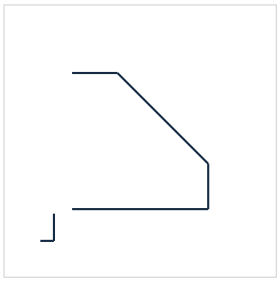
(C) Pattern Z



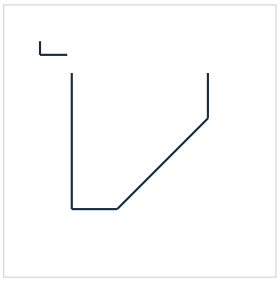
(D) Pattern T



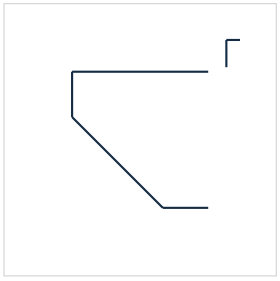
80. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

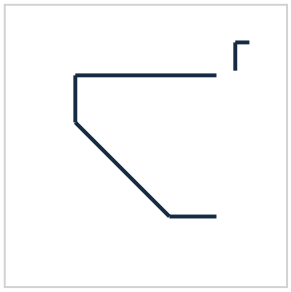


bottom left

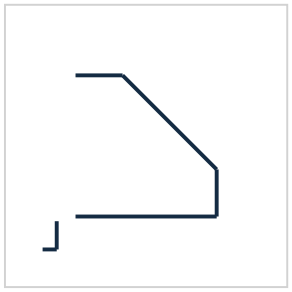
?

bottom right

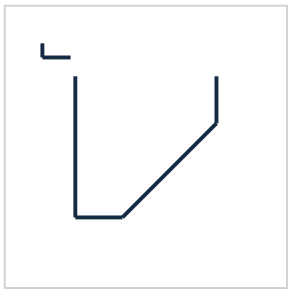
(A) Pattern W



(B) Pattern Q



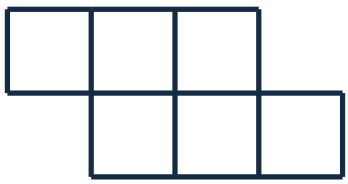
(C) Pattern V



(D) Pattern Z



81. How many rectangles, including squares are formed by all the drawn segments?



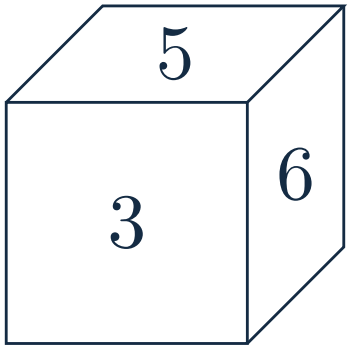
(A) 16

(B) 15

(C) 14

(D) 12

82. This standard die has opposite labels summing to seven. Roll it once to the right, so its present top face becomes the right face. Give the new (top, front) labels.



(A) $(6, 3)$ (B) $(1, 4)$

(C) $(5, 3)$ (D) $(1, 3)$

83. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 4, 8, 13, ?, 31, 62, 67.

(A) 18 (B) 31

(C) 26 (D) 36

84. Preserve the exact relation and its order: Chlorine : Halogen :: ? : Mammal. Choose the missing term.

(A) Sparrow (B) Whale

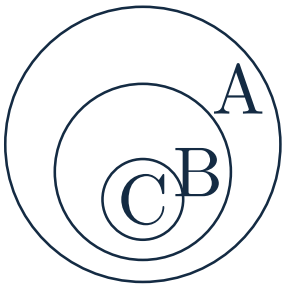
(C) Lizard (D) Frog

85. Choose the diagram that represents all these conditions, including the strictness information: Some A is B, some A is not B, and some B is not A. No C is A or B; C is nonempty.

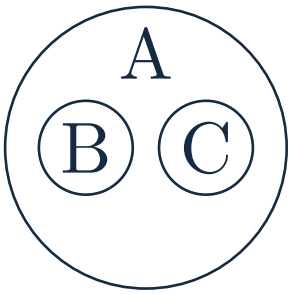
(A) Pattern L



(B) Pattern Q



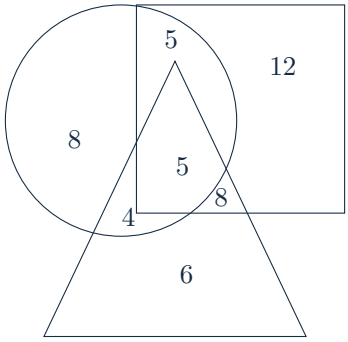
(C) Pattern V



(D) Pattern R



86. Circle=A, triangle=B, square=C. Each printed number is the count in that exact region. Find the total inside A but outside B, whether inside or outside C.



(A) 13 (B) 8

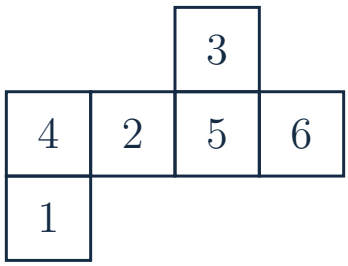
(C) 14 (D) 5

87. A community register uses the following three categories: editors, bakers, farmers. Its complete relevant assertions are in this short paragraph; do not import everyday assumptions. All farmers are editors. Some editors are farmers. Some editors are not farmers. Conclusions: I. Some farmers are editors. II. No editors are farmers. Which conclusions follow necessarily?

(A) Both I and II (B) Only I

(C) Neither I nor II (D) Only II

88. Fold the shown net into a cube. Give (label opposite 6, label opposite 5) in that order.



(A) $(2, 4)$ (B) $(2, 5)$

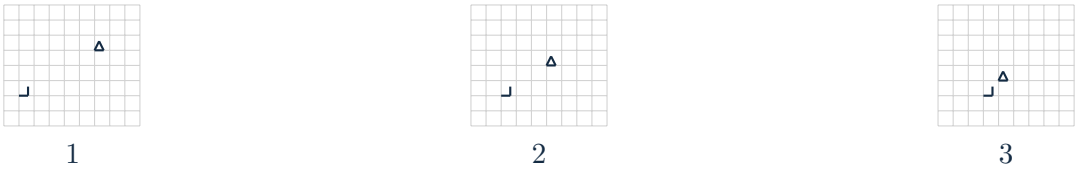
(C) $(2, 2)$ (D) $(2, 3)$

89. Consider only these statements; universal statements alone do not imply existence. No inventors are artists; No artists are campers; Some campers are inventors. Conclusions: I. Some inventors are not artists. II. All inventors are artists. Which conclusions follow necessarily?

(A) Only II (B) Both I and II

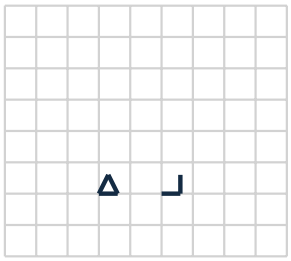
(C) Only I (D) Neither I nor II

90. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.

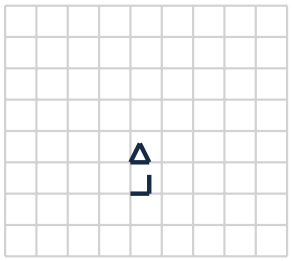


1 2 3

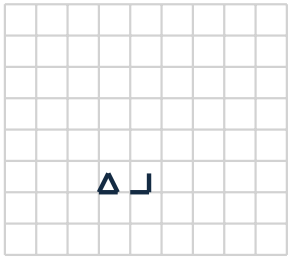
(A) Pattern K



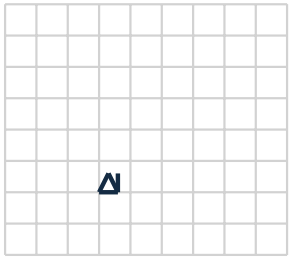
(B) Pattern V



(C) Pattern U



(D) Pattern P



Mathematics

91. What is the maximum number of distinct intersection points formed by 9 circles and 6 straight lines, with no coincident curves?

(A) 180 (B) 195

(C) 141 (D) 267

92. The homogeneous system $4x+y=0$, $kx+6y=0$ has a nonzero solution. Find $k+x^2/y^2$.

(A) 24 (B) $\frac{383}{16}$

(C) 40 (D) $\frac{385}{16}$

93. The system $x+y+z=1$, $y+z=2$, $(k-5)z=k-9$ has no solution for which k?

(A) 9 (B) 0

(C) -5 (D) 5

94. 9 distinct people stand in a line. Three specified people cannot occupy positions 1,2 or 9. How many orders are possible?

(A) 720 (B) 14400

(C) 151200 (D) 86400

95. In $(1+x)^8$, find the ratio of the coefficient of x^7 to the coefficient of x^6.

(A) $\frac{1}{2}$ (B) 2

(C) $\frac{2}{7}$ (D) $\frac{7}{2}$

96. X is binomial with mean 6 and variance 4. Find its mode.

(A) 6 (B) 18

(C) 5 (D) 7

97. How many solutions in $0 \le x < 16\pi$ satisfy $\sin 2x = \sin x$ and $\cos x > 0$?

(A) 32 (B) 25

(C) 24 (D) 16

98. α, β are roots of $x^2-13x+36=0$. The monic polynomial with roots $\alpha+1, \beta+1$ is x^2-Ax+B. Find B.

(A) 38 (B) 37

(C) 15 (D) 50

99. For $P(x) = \prod_{j=5}^{14}(x+j)$, find the coefficient of x^7 in $P(x)/(x+5)$.

(A) 4020 (B) 3570

(C) 3995 (D) 7140

100. A line has direction(1,5,2) and plane normal(2,1,5). If θ is the acute angle between the line and plane, find $\tan^2\theta$.

(A) $\frac{17}{30}$ (B) $\frac{289}{60}$

(C) $\frac{17}{611}$ (D) $\frac{289}{611}$

101. Find the sum of order and degree of $(d^3y/dx^3)^5+(dy/dx)^2+y=0$.

(A) 3 (B) 8

(C) 11 (D) 15

102. An invertible 4×4 matrix A has $\det A = 17$. Find $\det(\text{adj } A)$.

(A) 4913 (B) 68

(C) $\frac{1}{17}$ (D) 83521

103. How many integers k make $24\sin x+32\cos x+3=k$ have exactly two distinct solutions in $0 \le x < 2\pi$?

(A) 81 (B) 79

(C) 39 (D) 80

104. Let $y' + 6y = 2x$ and $y(0) = 5$. Find $y(1) + \frac{13}{10}y'(1)$.

(A) $\frac{1547}{45e^6} + \frac{32}{45}$ (B) $\frac{91}{18e^6} + \frac{5}{18}$

(C) $\frac{32}{45}$ (D) $\frac{32}{45} - \frac{1547}{45e^6}$

105. $\log_2 a=2$, log_a b=4, log_b c=4. Find $\log_4(c^2/a)$.

(A) 31 (B) 32

(C) 33 (D) 62

106. $a_j = 4 + 2(j-1)$ and $f(x)=(5/2)x^2+bx+2$. If f(1)=f(-1), find f′(a_5).

(A) 70 (B) 12

(C) 62 (D) 60

107. a=(4,6,0), b=(6,-4,2). Find $|a+b|^2-|a-b|^2$.

(A) 0 (B) 80

(C) 2912 (D) 108

108. Evaluate sin(2arctan 2).

(A) $-\frac{4}{5}$ (B) $\frac{2\sqrt{5}}{5}$

(C) $\frac{4}{5}$ (D) $-\frac{4}{3}$

109. Maximize Z=3x+4y subject to $x\geq 0$, $y\geq 7$, $x+y\leq 17$, $x+2y\leq 23$.

(A) 55 (B) 57

(C) 46 (D) 58

110. Evaluate the limit of $[\cos(5x)-\cos(3x)]/x^2$ as $x\to 0$.

(A) 8 (B) −16

(C) −8 (D) −1

111. For $f(x)=x^3-75x$, find the difference between its local maximum value and local minimum value.

(A) −500 (B) 10

(C) 500 (D) 250

112. The chord of contact from P to $x^2/36-y^2/4=1$ is x=6/2. Find the product of the slopes of the two tangents from P.

(A) $\frac{1}{27}$ (B) $-\frac{1}{27}$

(C) $-\frac{1}{81}$ (D) $-\frac{1}{36}$

113. From a point at horizontal distance 3 from a tower, its elevation is 2θ, where $\tan\theta=\frac{2}{5}$ and $0<\theta<45°$. Find the height.

(A) $\frac{6}{5}$ (B) $\frac{63}{20}$

(C) $\frac{12}{5}$ (D) $\frac{20}{7}$

114. Evaluate $\displaystyle\int_0^{\pi/2} \sin(2x)\left[1+4\sin^2(2x)\right]\frac{\sin^7 x}{\sin^7 x+\cos^7 x}\,dx$.

(A) $\frac{1}{2}$ (B) $\frac{11}{3}$

(C) $\frac{11}{6}$ (D) $\frac{5}{2}$

115. A positive integer-domain function satisfies f(m+n)=f(m)f(n), f(1)=5. Find $\sum_{j=1}^{8} f(j)$.

(A) 3125000 (B) 97656

(C) 390625 (D) 488280

116. Positive a,b,c form an AP with a=4. The numbers a,b,c+9/4 form a GP and b>a. Find c.

(A) 10 (B) -2

(C) 7 (D) 22

117. x=7+10(1-t^2)/(1+t^2), y=3+20t/(1+t^2), where t is real. Which point of the containing circle is not attained?

(A) $(7, -7)$ (B) $(-3, 3)$

(C) $(7, 13)$ (D) $(17, 3)$

118. For the roots α,β of $z^2 - (-3 + 7i)z + (-20 - 21i) = 0$, find $|\alpha\text{-}\beta|^2$.

(A) 58 (B) 841

(C) 116 (D) 29

119. Positive integers $a < b$ satisfy $a + b = 30$ and $3 \mid a$. Find the maximum value of ab.

(A) 225 (B) 216

(C) 30 (D) 189

120. A line through P=(6,7) has direction (3,4). It meets 3x+4y=96 at Q. Find |PQ|^2.

(A) 50 (B) 10

(C) 100 (D) 4

121. A complex number z satisfies z+conjugate(z)=6 and z-conjugate(z)=10i. Find z·conjugate(z).

(A) 34 (B) 109

(C) 136 (D) $\sqrt{34}$

122. A rectangle has perimeter 24. Its length is constrained to be at least three times its width. Find its maximum area.

(A) 27 (B) 36

(C) 9 (D) 18

123. In a group of 70, 42 study A, 35 study B, and 7 study neither. How many study exactly one of A, B?

(A) 63 (B) 49

(C) 77 (D) 14

124. Find the area enclosed by y=x(11-x) and y=7x.

(A) $-\frac{32}{3}$ (B) $\frac{32}{3}$

(C) $\frac{1331}{6}$ (D) $\frac{494}{3}$

125. Two acute sun elevations $\theta_1<\theta_2$ satisfy $(16^6)^{sin^2\theta} + (16^6)^{cos^2\theta} = 2^6 + 2^18$. A vertical tower has height 6. Find the difference between its two shadow lengths.

(A) $6\sqrt{3}$ (B) $8\sqrt{3}$

(C) $4\sqrt{3}$ (D) 4

126. There are 5 Boolean variables p_1,...,p 5. Exactly 3 variables must be true. How many truth assignments satisfy [(p_1 implies p_2) and (p_1 implies not p_2)]?

(A) 3 (B) 10

(C) 1 (D) 4

127. The chord of contact from P to $x^2/64+y^2/16=1$ is $x/8+y/4=1$. Find $|OP|^2$.

(A) 80 (B) $4\sqrt{5}$

(C) 48 (D) 12

128. A=(2,0,3), B=(0,2,1), AP:PB=2:1 internally. Find $|AP|^2$.

(A) $\frac{4}{3}$ (B) $\frac{16}{3}$

(C) 12 (D) 8

129. How many reflexive symmetric relations on a labelled 6-element set contain at least one off-diagonal ordered pair?

(A) 32766 (B) 16384

(C) 32767 (D) 32768

130. Two distinguishable fair dice show X,Y. Given $X+Y\geq 11$, find the probability that at least one of X,Y is even.

(A) $\frac{4}{3}$ (B) 0

(C) 1 (D) $\frac{1}{12}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A spherical blackbody's absolute temperature is multiplied by 4 while its radius is divided by 2. Neglect incoming background radiation. Find its emitted-power multiplier.

(A) 1 (B) 32

(C) 64 (D) 256

132. A point charge $9q$ is placed at a vertex of a cube. Find the electric flux through the three faces not meeting that vertex, combined, in units of q/ε_0.

(A) $\frac{9}{8}$ (B) $\frac{9}{4}$

(C) $\frac{3}{2}$ (D) $\frac{9}{16}$

133. Photons of energy 41 eV eject electrons from a surface. The fastest electrons enter a uniform magnetic field perpendicular to their velocities. In this field an electron with kinetic energy 3 eV has orbit radius r. The observed maximum orbit radius is $2r$. What is the surface work function? Neglect recoil.

(A) 53 eV (B) 29 eV

(C) 35 eV (D) 12 eV

Chemistry

134. Two oxides of X contain 14 g X combined respectively with 16 g O and 32 g O. Atomic masses are X=14, O=16. Which empirical-formula conclusion jointly follows from mole ratios and the law of multiple proportions?

(A) First oxide XO2; second oxide XO; fixed-X oxygen mass ratio second:first=1/2.

(B) First empirical formula XO; second empirical formula XO2; fixed-X oxygen mass ratio second:first=2.

(C) Both are XO because X masses are equal.

(D) No formula can be inferred even though both atomic masses and both compositions are supplied.

135. An octahedral complex has low-spin d^5 configuration and $\Delta_o = 250$ kJ mol^{-1}. Ignoring pairing energy,its d-electron energy relative to the unsplit barycentre,in kJ mol^{-1},is

(A) -500 (B) -750

(C) -250 (D) 500

136. Construct one open linear peptide using each member of the following multiset exactly once: Pro, Gly, Ala, Val, Ser, Thr. The N-terminal residue isPro and the C-terminal residue is the only achiral amino acid. Among the internal residues, Ala must occur beforeSer. Count distinctN-to-C sequences; ignore stereoisomerism of the supplied residues.

(A) 24 (B) 720

(C) 12 (D) 120

Logical Reasoning

137. The model triads are $(6, 19, 13)$ and $(8, 23, 15)$. In the intended rule the middle number is the sum of the outer numbers, and the outer-number difference is the same fixed positive value in every triad. Which option satisfies both relations?

(A) $(5, 18, 12)$ (B) $(5, 18, 13)$

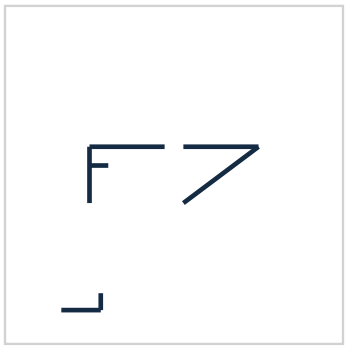
(C) $(5, 17, 12)$ (D) $(5, 19, 13)$

138. In a row, Asha is 8th from the left and 21st from the right. Bina is 9th from the right. Charu stands 2 places to the left of Bina. How many people stand strictly between Asha and Charu?

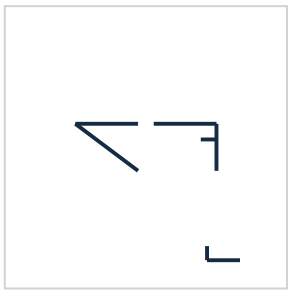
(A) 10 (B) 9

(C) 0 (D) 8

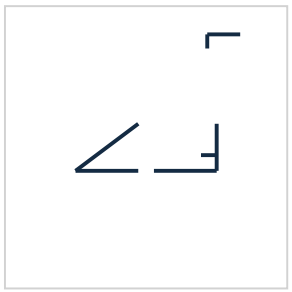
139. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



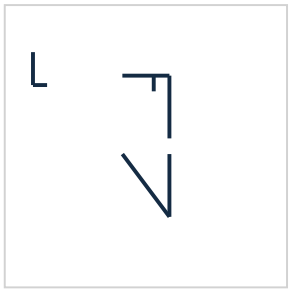
(A) Pattern Y



(B) Pattern U



(C) Pattern M



(D) Pattern R



Mathematics

140. |U|=70, |A|=17, |B|=18, |A∩B|=7. Find the number of elements belonging to both A,B or to neither.

(A) 63 (B) 42

(C) 49 (D) 21

141. If z=7-3i, find |exp(iz)|.

(A) e^3 (B) e^7

(C) e^{-3} (D) 1

142. Evaluate $\int_0^\infty dx/[(x^2+9)(x^2+25)]$.

(A) $-\frac{\pi}{240}$ (B) $\frac{\pi}{240}$

(C) $\frac{\pi}{30}$ (D) $\frac{\pi}{120}$

BITSAT Mock 16

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

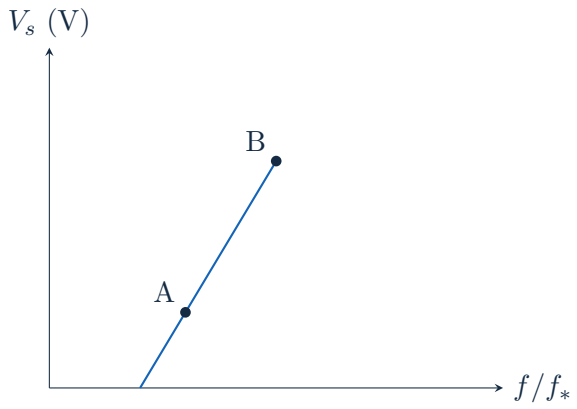
Physics

1. 125 equal liquid drops coalesce in a viscous liquid. The density difference is unchanged. If viscosity is then multiplied by 3, find the new terminal speed divided by that of one original drop. Assume Stokes flow remains valid.

(A) $\frac{1}{3}$ (B) $\frac{125}{3}$

(C) 75 (D) $\frac{25}{3}$

2. For a metal, the stopping-potential graph has points A $(3f_*, 4\,\text{V})$ and B $(5f_*, 12\,\text{V})$. The electronic gain of the vertical recording channel is later found to be 3: its recorded voltage is 3 times the true stopping voltage. What is hf_*/e in volts?



(A) 4 V (B) $\frac{4}{3}$ V

(C) $\frac{1}{12}$ V (D) $\frac{4}{5}$ V

3. 5 kg of water at 35°C is mixed in an insulated vessel with $\frac{735}{167}$ kg of ice at 0°C. Use $c_w = 4200$ J/(kg K), $L_f = 334000$ J/kg. Find the mass of ice remaining at equilibrium.

(A) $\frac{735}{334}$ kg (B) $\frac{230790}{80327}$ kg

(C) $\frac{2205}{668}$ kg (D) $\frac{735}{167}$ kg

4. 3 moles and 1 moles of the same ideal gas, initially at 300 K and 700 K, are connected in a rigid insulated system. There is no external work. Find equilibrium temperature, assuming constant molar heat capacity.

(A) 400 K (B) 600 K

(C) 500 K (D) 300 K

5. In a series RLC circuit, ideal rms voltmeters read 30 V across R, 40 V across L and 20 V across C. Find source rms voltage.

(A) 20 V (B) $10\sqrt{29}$ V

(C) $10\sqrt{13}$ V (D) $30\sqrt{5}$ V

6. A wire consists of a semicircular arc of radius 3 m carrying 7 A, joined to two straight radial segments that extend outward from its ends. Find the magnetic field magnitude at the arc centre in units of μ_0.

(A) $\frac{7}{6}$ (B) $\frac{7}{12}$

(C) $\frac{7}{6\pi}$ (D) $\frac{7}{24}$

7. Two sound sources produce wavelengths $\frac{1}{5}$ m and $\frac{1}{6}$ m in the same gas at one temperature. The sound speed there is 400 m/s. At a second temperature the sound speed is 3/2 times larger, while both source frequencies stay unchanged. Find the new beat frequency.

(A) 400 Hz (B) 600 Hz

(C) 4400 Hz (D) $\frac{800}{3}$ Hz

8. A uniform thin rod of length 5 m swings through small angles in a vertical plane about a fixed horizontal axis through one end. Find its period. Take $g = 10$ m/s^2.

(A) $\frac{2\sqrt{3}\pi}{3}$ s (B) π s

(C) $\frac{\sqrt{3}\pi}{3}$ s (D) $\sqrt{2}\pi$ s

9. A cyclist rides straight outward at 3 m/s for 4 s, waits 3 s, then returns to the start. The whole journey must last 8 s. What constant return speed is required?

(A) $\frac{12}{5}$ m/s (B) $\frac{3}{2}$ m/s

(C) 12 m/s (D) 3 m/s

10. A uniform wire is stretched to $\frac{5}{4}$ times its original length at constant volume. Its resistivity increases by 20 percent during the process. Find final resistance divided by initial resistance.

(A) $\frac{25}{16}$ (B) $\frac{15}{8}$

(C) $\frac{75}{32}$ (D) $\frac{5}{4}$

11. A 6 kg block on a rough horizontal table is tied over an ideal pulley to a hanging 2 kg mass. The hanging mass moves downward and $\mu_k = 1/5$. Find string tension. Take $g = 10$ m/s^2.

(A) 6 N (B) 18 N

(C) 22 N (D) 12 N

12. A ball is dropped from 32 m onto a horizontal fixed floor. At every impact the coefficient of restitution is $\frac{4}{5}$. Neglect air resistance. What maximum height does it reach after its second impact?

(A) $\frac{8192}{625}$ m (B) $\frac{128}{5}$ m

(C) $\frac{2097152}{390625}$ m (D) $\frac{32}{625}$ m

13. An astronomical telescope in normal adjustment has magnifying-power magnitude 6 and tube length 14 cm. Find the eyepiece focal length.

(A) $\frac{7}{3}$ cm (B) $\frac{2}{7}$ cm

(C) 2 cm (D) $\frac{14}{5}$ cm

14. A solid sphere rolls without slipping from rest through vertical height 6 m. A thin ring does so through vertical height 11 m. Find the ratio of their squared centre-of-mass speeds, sphere to ring.

(A) $\frac{6}{11}$ (B) $\frac{42}{55}$

(C) $\frac{77}{60}$ (D) $\frac{60}{77}$

15. A car negotiates a level circular turn at its maximum non-slipping speed u when $\mu_s = \frac{11}{10}$. Rain lowers the coefficient to $\frac{1}{10}$ and the turn radius stays fixed. Find the new maximum speed divided by u.

(A) $\frac{\sqrt{10}}{10}$ (B) $\sqrt{11}$

(C) $\frac{\sqrt{11}}{11}$ (D) $\frac{1}{11}$

16. The quantity $Z = x^3y^2/\sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 4, 4 and 6, respectively. To first order, what is the maximum percentage error in Z?

(A) 20 (B) 23

(C) $\sqrt{217}$ (D) 11

17. Two ferromagnetic samples have similar remanent flux density. Their coercive fields are 500 A/m and 2500 A/m. A reversing field of magnitude 1500 A/m will act in service. Which sample is better at retaining its original magnetisation against this field, based only on the stated coercivities?

(A) Neither can have nonzero remanence. (B) Both are necessarily demagnetised equally.

(C) The sample with coercive field 2500 A/m. (D) The sample with coercive field 500 A/m.

18. An ideal gas initially has pressure 400 Pa and volume 2 m^3. It is compressed reversibly and isothermally to $V/4$. Find work done on the gas.

(A) 2400 J (B) $400 \log(2)$ J

(C) $800 \ln 4$ J (D) 600 J

19. Two masses 2 kg and 3 kg are connected by a light string over a smooth pulley and accelerate freely. A transverse pulse travels in the taut straight part of this string, of linear density μ. Compare its speed to that in a second string of density 2μ held at tension g N numerically equal to 1 kg weight. Find v_1/v_2.

(A) $\frac{\sqrt{30}}{12}$ (B) $\frac{2\sqrt{15}}{5}$

(C) $\frac{2\sqrt{30}}{5}$ (D) 2

20. An active transistor has $\alpha = \frac{49}{50}$. Its collector resistor is 1 kΩ and the measured collector-resistor drop is 2 V. Neglect leakage. What base current in mA is required?

(A) 2 mA
(B) $\frac{100}{49}$ mA
(C) $\frac{100}{2401}$ mA
(D) $\frac{2}{49}$ mA

21. A uniformly charged nonconducting sphere has radius R and surface potential V_s relative to infinity. Find potential at $r = R/14$ divided by V_s.

(A) $\frac{587}{392}$
(B) $\frac{3}{2}$
(C) $\frac{195}{196}$
(D) $\frac{1}{14}$

22. A diamond network has four outer resistors AB, BC, CD and DA, each 6 ohms, and a bridge BD also 6 ohms. The DA arm additionally contains an ideal diode with anode at D and cathode at A. A source keeps A positive relative to C. Find equivalent resistance between A and C.

(A) 12 Ω
(B) 6 Ω
(C) 3 Ω
(D) 10 Ω

23. A point charge $24q$ is placed at a vertex of a cube. Find the electric flux through the three faces not meeting that vertex, combined, in units of q/ε_0.

(A) 4
(B) 24
(C) 1
(D) 3

24. A plane wave in vacuum travels along $+x$. At one instant its electric field is $90\hat{k}$ V/m. If the electric amplitude is 4 times this instantaneous value, which gives the instantaneous magnetic field? Use $c = 3 \times 10^8$ m/s.

(A) $+30 \times 10^{-8}\hat{i}$ T
(B) $+30 \times 10^{-8}\hat{j}$ T
(C) $-120 \times 10^{-8}\hat{j}$ T
(D) $-30 \times 10^{-8}\hat{j}$ T

25. A conveyor receives 5 packets each second, each of mass $\frac{1}{2}$ kg and speed 2 m/s. It accelerates them to 6 m/s without changing elevation. Find the ideal mechanical power delivered to the packets.

(A) 40 W
(B) 10 W
(C) 20 W
(D) 8 W

26. A coil of inductance 8 H carries current $i(t) = 3t^2$ A. Find the magnitude of its self-induced emf at $t = 2$ s.

(A) 192 V
(B) 96 V
(C) 12 V
(D) 768 V

27. Two wires of the same material and length L, with radii r and 13r, are joined in series and stretched by a force F. Find their total stored elastic energy divided by the energy stored in the thinner wire alone under force F.

(A) 170
(B) $\frac{170}{169}$
(C) $\frac{14}{13}$
(D) $\frac{169}{170}$

28. A mass 3 kg moves with $\vec{r}(t) = 1t\hat{i} + 4t^2\hat{j}$ m. Find the z-component of angular momentum about the origin at $t = 3$ s.

(A) $324\,\mathrm{kg\,m^2/s}$ (B) $36\,\mathrm{kg\,m^2/s}$

(C) $54\,\mathrm{kg\,m^2/s}$ (D) $108\,\mathrm{kg\,m^2/s}$

29. Light is incident from index $\frac{13}{10}$ onto index $\frac{11}{5}$. At one incidence angle the reflected beam is completely plane polarised. Find the tangent of that angle.

(A) $\frac{22}{13}$ (B) $\frac{13}{22}$

(C) $\frac{9}{13}$ (D) $\frac{3\sqrt{35}}{13}$

30. A solid sphere of radius R has uniform density $5\rho_0$. A spherical cavity of radius $R/10$ is cut with its centre at distance $R/11$ from the original centre. Find field magnitude at the cavity centre in units of $\pi G\rho_0 R$.

(A) $\frac{220}{3}$ (B) $\frac{20}{33}$

(C) $\frac{44}{3}$ (D) $\frac{4}{15}$

Chemistry

31. At25C,a measurement gives$[OH^-]$=5×10^{-6}M. UseKw=10^{-14} and$\log_{10}5$=$\frac{699}{1000}$. Solution pH is

(A) $\frac{5301}{1000}$ (B) $\frac{8699}{1000}$

(C) $\frac{6699}{1000}$ (D) $\frac{7301}{1000}$

32. In a pressure range where the Freundlich law $x/m = kp^{1/3}$ is valid at fixed temperature, adsorbent B has 6 times the mass of A and is exposed to 216 times its pressure. Find the ratio of total adsorbed amounts x_B/x_A.

(A) 6 (B) 36

(C) 1296 (D) 1

33. For 2A+B→3C+4D, A also disappears by an independent channel producing no C or D, and D is independently consumed at 1/2000 M/s. Measured total A disappearance is 1/200 M/s and net D appearance is 7/2000 M/s. What is the A disappearance rate through its independent channel, in M/s?

(A) 13/4000 (B) 3/1000

(C) 1/250 (D) 7/1000

34. For HCO_3^-,use ordinary octet Lewis structures and resonance. What is the pair (average bond order of only the two nonprotonated C-O bonds,total lone pairs on all atoms of that same named species in one contributing structure)?

(A) (2, 8). (B) ($\frac{3}{2}$, 7).

(C) (1, 7). (D) ($\frac{3}{2}$, 6).

35. A textbook deficiency table associates pernicious anaemia in the textbook deficiency table with one vitamin. Which vitamin and solubility class are correctly paired?

(A) Vitamin B12; water-soluble. (B) VitaminK; fat-soluble.

(C) Vitamin B1; water-soluble. (D) Vitamin C; water-soluble.

36. In a hydrogen-like ion Z=3,an electron falls from n=5 to an s orbital with 2 radial nodes. With hydrogen ionisation energy13.6 eV and$hc = 1240$ eV nm,the emitted wavelength in nm is

(A) $\frac{174375}{136}$ (B) $\frac{19375}{544}$

(C) $\frac{38750}{153}$ (D) $\frac{19375}{136}$

37. $\frac{3}{50}$ mol borax is acidified completely. Each borax formula unit gives four H_3BO_3 molecules;75.0% of the boric acid is isolated. With glycerol present, the isolated boric acid is titrated completely as a monobasic acid. What amount of NaOH, in mol, is required?

(A) $\frac{9}{50}$ (B) $\frac{6}{25}$

(C) $\frac{27}{50}$ (D) $\frac{8}{25}$

38. An internally consistent thermochemical comparison gives total resonance stabilisation of naphthalene as 61 kcal/mol (2 benzenoid rings) and phenanthrene as 92 kcal/mol (3 rings). Which statement correctly separates total stabilisation from stabilisation per ring?

(A) Total: phenanthrene; per ring: phenanthrene.

(B) Total: naphthalene; per ring: phenanthrene.

(C) Both quantities must rank compounds identically because each is aromatic.

(D) Total: phenanthrene; per ring: naphthalene.

39. A solution initially contains $\frac{1}{125}$ mol Fe^{2+}. Before titration, $\frac{1}{10}$ of it is oxidised quantitatively to Fe^{3+}. The volume in mL of0.0200 M $KMnO_4$ needed for the remaining Fe^{2+} in acid is

(A) 72 (B) 8

(C) 360 (D) 80

40. At 360 K, ΔH for C(graphite)+O2(g)→CO2(g) is -410 kJ/mol and for CO(g)+(1/2)O2(g)→CO2(g) is -285 kJ/mol. Carbon is converted only into CO, producing 2 mol CO. Assuming ideal gases and R=0.008314 kJ/mol/K, what is total ΔU in kJ? Use the stated reaction extents.

(A) -3162413/25000 (B) -3162413/12500

(C) 3162413/12500 (D) -3162413/6250

41. Na has a lower first ionisation enthalpy than Mg. Which statement about the second removal is consistent with the configurations after the first removal?

(A) Mg has the larger second ionisation enthalpy because Mg+ already has a noble-gas configuration.

(B) The second ionisation enthalpy of either element must be smaller than its own first ionisation enthalpy.

(C) Na must also have the lower second ionisation enthalpy because the first-IE order can never reverse.

(D) Na has the much larger second ionisation enthalpy because Na+ is [Ne],while Mg+ still has a3s electron.

42. A solution is5%by mass solute(M100g/mol)and hasdensity1.20g/mL. Which pair(molarity,molality)is correct?

(A) ($\frac{1}{2}$mol/L,$\frac{10}{19}$mol/kg).

(B) ($\frac{3}{5}$mol/L,$\frac{1}{2}$mol/kg).

(C) ($\frac{3}{5}$mol/L,$\frac{10}{19}$mol/kg).

(D) ($\frac{10}{19}$mol/L,$\frac{3}{5}$mol/kg).

43. A nearly monodisperse polythene sample has chain molar mass 28000 g/mol. Neglect end groups and take the ethene repeat mass as28 g/mol. Which conclusion follows, and which quantity is being counted?

(A) Degree of polymerisation28; number of polymer molecules.
(B) Degree of polymerisation1000; repeat units per chain.
(C) Degree of polymerisation500; carbon atoms per chain.
(D) Degree of polymerisation28000; monomer molecular mass.

44. Compounds A and B have molecular formulae $(CH_2O)_4$ and $(CH_2O)_5$. Pure equal-mass samples are compared. Which joint statement is necessarily true,independently of functional-group identity?

(A) The samples contain equal total oxygen-atom amounts,but A contains $\frac{5}{4}$ times as many molecules as B.

(B) Both samples contain equal molecule amounts because their empirical formulae agree.

(C) The oxygen-atom amounts are in ratio $\frac{4}{5}$,while molecule amounts are equal.

(D) A contains $\frac{4}{5}$ times as many molecules as B,and has a different oxygen mass percentage.

45. Consider the statements. I. Carbon monoxide and carbon dioxide have different effects on haemoglobin. II. A compound is harmless whenever it contains only elements normally present in the body. Choose the correct truth pattern.

(A) I true; II true.
(B) I false; II true.
(C) I true; II false.
(D) I false; II false.

46. Classify MgO, ZnO, SO3, in that order, by their usual acid-base behaviour. Which sequence also predicts that the middle oxide reacts with both suitable acids and strong alkalis?

(A) Amphoteric; basic; acidic.
(B) Basic; acidic; amphoteric.
(C) Acidic; amphoteric; basic.
(D) Basic; amphoteric; acidic.

47. A pure KMnO4 portion gives 1/100 mol O2 on complete controlled heating by2KMnO4→K2MnO4+MnO2+O2. With KMnO4=158 g/mol, which pair gives the initial mass and a viable origin of its intense purple colour despite Mn(VII) being d0?

(A) 79/50 g; charge-transfer absorption.
(B) 79/25 g; charge-transfer absorption.
(C) 158/25 g; d-d absorption within an occupied d shell.
(D) 79/25 g; d-d absorption within an occupied d shell.

48. Aspirin is acetylsalicylic acid, whereas morphine belongs to the opioid/narcotic analgesic class. Which comparison is correct in the textbook classification?

(A) Aspirin is a dipeptide antibiotic.
(B) Both are identical molecules.
(C) Both are analgesics, but aspirin is non-narcotic.
(D) Morphine is an artificial sweetener.

49. A mixture contains 1/25 mol total of the ionisation isomers [Co(NH3)5Br]SO4 and [Co(NH3)5SO4]Br. An immediately tested aliquot representing the entire inventory gives 2/125 mol AgBr with excess Ag+. Coordinated ligands do not exchange, and no other bromide exists. What mole fraction of the inventory was [Co(NH3)5Br]SO4?

(A) 6/5
(B) -3/5
(C) 3/5
(D) 3/10

50. For isolated molecules of cis-1,2-dichloroethene and trans-1,2-dichloroethene,which member has the larger permanent dipole moment,and which reasoning is sufficient?

(A) cis-1,2-dichloroethene; The trans planar isomer has inversion-related cancelling C-Cl bond moments; the cis isomer does not.

(B) The one with more atoms always has the larger dipole,independent of geometry.

(C) They must be equal because both contain polar covalent bonds.

(D) trans-1,2-dichloroethene; the number of polar bonds alone fixes the molecular dipole.

51. An element has cubicformsAandB. Ahas an FCC lattice with fraction $\frac{5}{8}$of normal atomic sites occupied; Bhas fully occupiedBCC. Edge lengths obey aA=2aB,and atomic mass is unchanged. DensityratioρA/ρB is

(A) $\frac{5}{2}$ (B) $\frac{1}{4}$

(C) $\frac{5}{16}$ (D) $\frac{5}{32}$

52. A first-order reaction has a constant half-life of 10 min. A sample is observed now and again 20 min later, when unreacted material equals 1/8 of the original amount present when the reaction began. How many minutes had elapsed before the first observation?

(A) 20 (B) 30

(C) 40 (D) 10

53. Compare protium and tritium atoms. Which statement preserves the distinction between mass/nuclear effects and electronic structure?

(A) Their neutral electron counts agree,but tritium is radioactive whereas protium is stable.

(B) An isotope with more neutrons must have a different number of protons and belong to another element.

(C) Changing H to D adds an electron and therefore changes the oxidation state of every bonded atom.

(D) Every physical property and reaction rate is identical because isotopes have the same atomic number.

54. For butan-2-ol and2-methylpropan-2-ol,which test/observation assignment follows the actual carbinol class or oxidisable structural feature? Use ordinary room-temperature tests unless the named sequence specifies otherwise.

(A) Lucas reagent;tertiary alcohol is immediate,secondary reacts more slowly

(B) iodoform;only the tertiary alcohol is positive

(C) Victor Meyer;both give red colour

(D) Lucas reagent;secondary is immediate and tertiary never reacts

55. A1.00 L vessel initially contains $\frac{1}{20}$ mol A,$\frac{1}{10}$ mol B and $\frac{1}{100}$ mol C,with no D. For A+2B$\rightleftharpoons$2C+D, equilibrium C amount is $\frac{3}{100}$ mol. Numerical K_c using molar concentrations is

(A) $\frac{9}{128}$ (B) $\frac{9}{64}$

(C) $\frac{3}{112}$ (D) $\frac{9}{256}$

56. An ideal gas executes A→B→C→D→A through (V/L,P/bar):A(1,1),B(3,1),C(3,4),D(1,4). IUPAC work-on convention applies. Heats on AB,BC,DA are 600,300,-200 J. Heat on CD in J is (1 L bar=100 J)

(A) -600

(B) -1300

(C) -100

(D) -700

57. For $M(s)+2A^{+}\rightleftharpoons M^{2+}+2A(s), E^{\circ}=\frac{177}{2000}V$. Use$2.303RT/F=.0590V$. Equilibrium constant for the doubled reaction$2M+4A^{+}\rightleftharpoons 2M^{2+}+4A$ is

(A) 1000000000000

(B) $\frac{1}{1000000}$

(C) 1000000

(D) 1000

58. A $\frac{1}{50}$mol iron-ion inventory contains Fe^{2+}andFe^{3+}. Fraction$\frac{1}{4}$isFe^{3+}. All iron is quantitatively reduced to metal in the stated externally driven electron-balance model,with no side reactions. Electrons accepted, in mol,are

(A) $\frac{1}{20}$

(B) $\frac{11}{200}$

(C) $\frac{9}{200}$

(D) $\frac{7}{125}$

59. Two Group2 cations have equal charge but radii r_A<r_B. In the usual carbonate-stability trend, which causal chain is consistent?

(A) Equal charge guarantees identical carbonate decomposition behaviour.

(B) A polarises carbonate less strongly, so its carbonate is less stable.

(C) B must polarise carbonate more strongly solely because it is larger.

(D) A polarises carbonate more strongly, so its carbonate is generally less thermally stable.

60. Pb_3O_4 is represented as a mixed oxide of PbO and PbO_2. Which formal metal oxidation-state ratio satisfies the formula,with ordinary oxide oxygen -2?

(A) 2+:4+ = 1:1.

(B) All metal atoms must be formally +8/3,with no mixed-valence bookkeeping possible.

(C) 2+:4+ = 2:1.

(D) 2+:4+ = 1:2.

English Proficiency

61. Read the sentence: The district had abundant rainfall that season, filling every storage tank. Choose the opposite in meaning of 'abundant' as used in the sentence.

(A) uneven

(B) scarce

(C) seasonal

(D) plentiful

62. Choose the active-voice equivalent that retains the tense, participants and all stated restrictions. The first draft was rejected by the editor, but the revised draft was retained.

(A) The editor rejected the first draft but retained the revised draft.

(B) The editor rejected both the first and the revised drafts.

(C) The editor retained the first draft but rejected the revised draft.

(D) The first draft rejected the editor but retained the revised draft.

63. Read the sentence: The agreement was called historic because it marked a major turning point, not merely because it occurred long ago. Choose the closest meaning of 'historic' as used in the sentence.

(A) concerning the study of past events
(B) important in history
(C) recorded in any archive
(D) very old in physical age

64. The committee approved a plan, but a separate team was appointed to _____ it.

(A) carry off
(B) carry over
(C) carry out
(D) carry away

65. The policy is defended as helping strangers for their own sake rather than as an investment in future favours. Which word names that motive?

(A) competition
(B) opportunism
(C) altruism
(D) egoism

66. Read the sentence: The discussion became stagnant, with no fresh ideas or decisions. Choose the opposite in meaning of 'stagnant' as used in the sentence.

(A) repetitive
(B) dynamic
(C) unproductive
(D) quiet

67. Read the sentence: The guide lists the current opening hours rather than those used last winter. Choose the closest meaning of 'current' as used in the sentence.

(A) presently applicable
(B) widely known
(C) recently announced
(D) regularly changing

68. Choose the correctly spelt word.

(A) comittee
(B) committe
(C) commitee
(D) committee

69. Choose the most coherent order of the four sentences. A. Because it stood beyond the actual turning, walkers often passed the path before seeing it. B. The sign was moved to a position visible before the junction. C. Its message became useful once people could read it in time to act. D. A sign at the junction pointed towards the lake.

(A) D-A-B-C
(B) D-A-C-B
(C) D-B-A-C
(D) A-D-B-C

70. Choose the most coherent order of the four sentences. A. The catalogue added descriptions of what each product was used for. B. The revised search could match a customer's need as well as a remembered label. C. The shop's catalogue allowed searches only by product name. D. Customers who knew an item's use but not its name often found nothing.

(A) D-C-A-B
(B) C-A-D-B
(C) C-D-B-A
(D) C-D-A-B

Logical Reasoning

71. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 4, 12, ?, 42, 44, 132, 134.

(A) 38 (B) 42

(C) 14 (D) 20

72. A person makes these turns in order: 180 degrees, 90 degrees left, 90 degrees right, 90 degrees right. The final heading is north. What was the heading immediately after turn 3?

(A) West (B) South

(C) East (D) North

73. Consider only these statements; universal statements alone do not imply existence. All editors are joggers; No joggers are campers. Conclusions: I. Some editors are campers. II. No editors are campers. Which conclusions follow necessarily?

(A) Only II (B) Only I

(C) Both I and II (D) Neither I nor II

74. In a row, Asha is 8th from the left and 30th from the right. Bina is 10th from the right. Charu stands 2 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 17 (B) 21

(C) 19 (D) 18

75. The increase from the left cell to the right cell is the same in each row, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern N



(B) Pattern Z



(C) Pattern X



(D) Pattern T

76. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 1 (B) 4

(C) 2 (D) 3

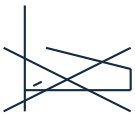
77. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 12$ and $5 \mapsto 21$. Which input maps to 36?

(A) 11 (B) 36

(C) 10 (D) 9

78. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

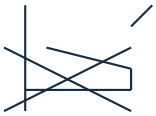




host 1



host 2



host 3



host 4

(A) 1 (B) 4

(C) 2 (D) 3

79. The recorded average of 11 readings is 25. One entry recorded as 43 should have been 39. After that correction, another reading of 15 is replaced by 27. What is the final average?

(A) $\frac{271}{11}$ (B) $\frac{283}{11}$

(C) $\frac{287}{11}$ (D) $\frac{294}{11}$

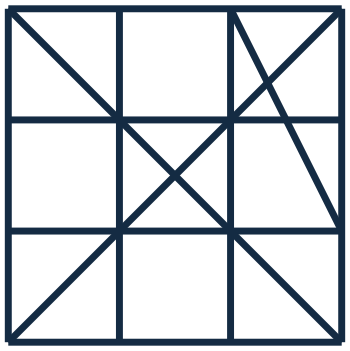
80. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: BPC, CRF, ?, EVL, FXO, GZR.

(A) ITD
(B) CSH
(C) EUJ
(D) DTI

81. In this code, @ means $\times$, # means $+$, & means $\div$, % means $-$. After substituting meanings, evaluate ((3#2)@(2&3)); printed parentheses retain their usual grouping.

(A) $\frac{13}{3}$
(B) $\frac{10}{3}$
(C) $\frac{-10}{3}$
(D) $\frac{7}{3}$

82. How many triangles are formed by all the drawn segments?



(A) 27
(B) 41
(C) 36
(D) 77

83. Preserve the exact relation and its order: Speedometer : Speed :: ? : Atmospheric pressure. Choose the missing term.

(A) Ammeter
(B) Voltmeter
(C) Barometer
(D) Thermometer

84. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target



piece 1



piece 2

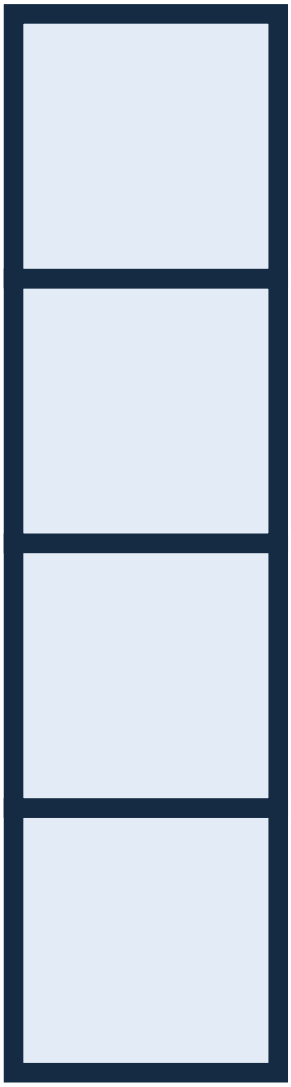
(A) Pattern V



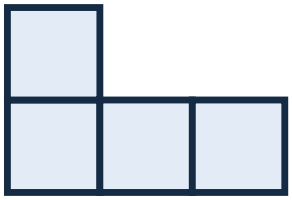
(B) Pattern U



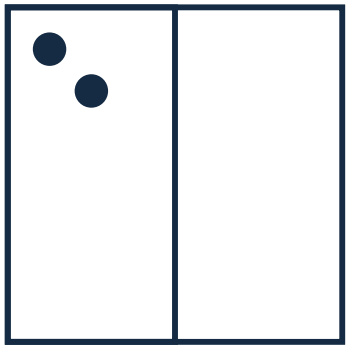
(C) Pattern M



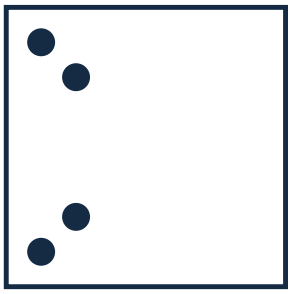
(D) Pattern P



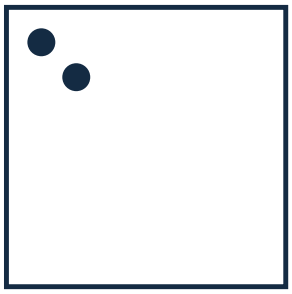
85. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



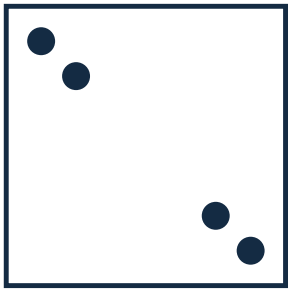
(A) Pattern W



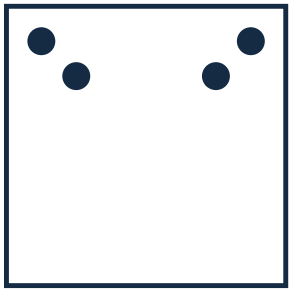
(B) Pattern M



(C) Pattern V



(D) Pattern L

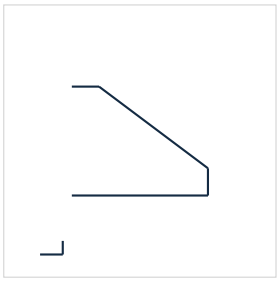


86. Each rectangle is drawn to the same local scale. In three numbered figures the interior straight line divides the rectangle into equal areas. Which does not?

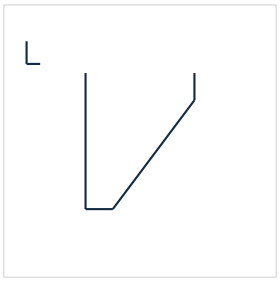


(A) 1
(B) 2
(C) 3
(D) 4

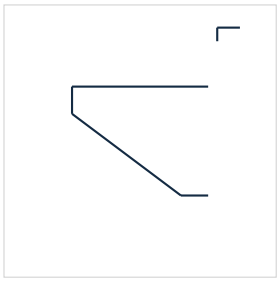
87. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

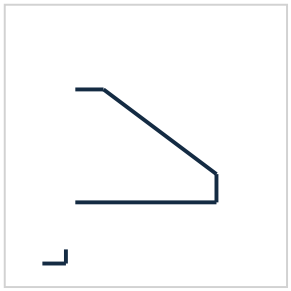


2



3

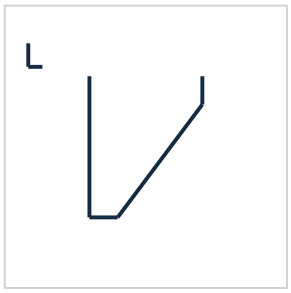
(A) Pattern V



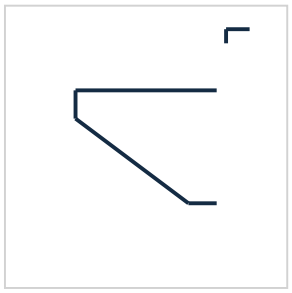
(B) Pattern L



(C) Pattern P



(D) Pattern U



88. $X\$Y$ means X is the father of Y; $X@Y$ means X is the sister of Y; $X\&Y$ means X is the mother of Y. All names denote distinct people; a brother or sister is a full sibling, sharing both parents. Given $C\$D$ $D@J$ $J\&T$, how is C related to T?

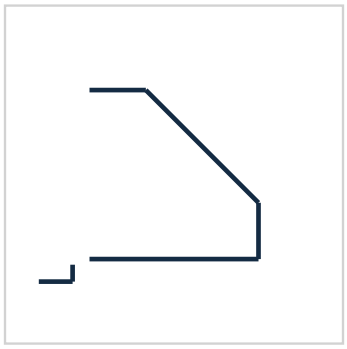
(A) Maternal grandfather

(B) Paternal grandfather

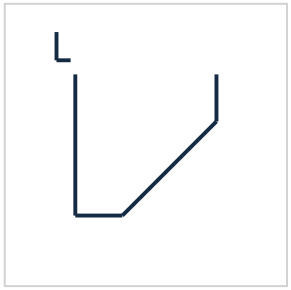
(C) Mother

(D) Father

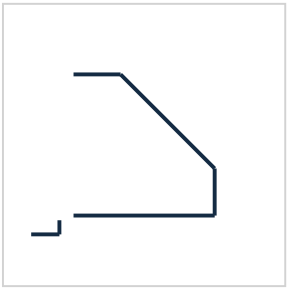
89. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



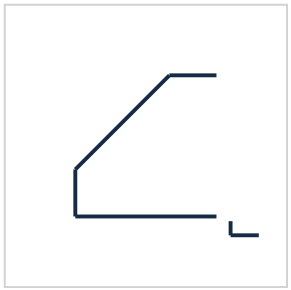
(A) Pattern L



(B) Pattern Q



(C) Pattern Z



(D) Pattern X



90. Preserve the exact relation and its order: Lizard : Reptile :: ? : Bird. Choose the missing term.

(A) Lizard (B) Frog

(C) Whale (D) Sparrow

Mathematics

91. The line r=(7,3,4)+t(1,2,2) meets x+y+z=29 at Q. Find $|PQ|^2$ for P=(7,3,4).

(A) 281 (B) 81

(C) 27 (D) 9

92. $y'+3y=\exp(-3x)$, $y(0)=2$. Find $\exp(3)y(1)-2$.

(A) -1 (B) 1

(C) $\frac{3}{e^3}$ (D) 3

93. f(x)=4x+6. Find f(f(6)).

(A) 54 (B) 30

(C) 900 (D) 126

94. A box contains 8 red and 4 blue distinct balls. Three are drawn uniformly without replacement. Given that at least one drawn ball is blue, find the probability that exactly two are red.

(A) $\frac{13}{41}$ (B) $\frac{28}{41}$

(C) 2 (D) $\frac{28}{55}$

95. How many solutions in $0 \leq x < 8\pi$ satisfy $\sin^4 x + \cos^4 x = 3/4$ and $\sin x > 0$?

(A) 16 (B) 15

(C) 32 (D) 17

96. Find the population variance of 2+5, 4+5, ..., 20+5.

(A) 33 (B) $\frac{33}{2}$

(C) $\frac{33}{4}$ (D) 38

97. How many integers x are in the real domain of $\sqrt{x-4}\,\log(6-x)$?

(A) 2 (B) 1

(C) 3 (D) 6

98. Let $R=[\sin(\frac{5\pi}{48})+\sin(\frac{11\pi}{48})]/[\cos(\frac{5\pi}{48})+\cos(\frac{11\pi}{48})]$. Find $(1-R^2)/(1+R^2)$.

(A) $\frac{\sqrt{3}}{2}$ (B) $-\frac{1}{2}$

(C) $\frac{3}{2}$ (D) $\frac{1}{2}$

99. B needs 2 days longer than A; together they need $\frac{15}{8}$ days. A works alone for $\frac{3}{2}$ days, then B joins. How many additional days finish the job?

(A) $\frac{3}{2}$ (B) $\frac{3}{8}$

(C) $\frac{15}{8}$ (D) $\frac{15}{16}$

100. $A = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$ and $A^{-1} = pI + qA$. Find p+q.

(A) $\frac{1}{7}$ (B) $\frac{5}{6}$

(C) $\frac{5}{7}$ (D) $-\frac{5}{7}$

101. The midpoints of the sides of a triangle are (3,0),(0,2),(3,2). Find the sum of coordinates of its circumcentre.

(A) 0 (B) 5

(C) 10 (D) $\frac{10}{3}$

102. An observer at height 9 sees a tower foot at depression 30° and top at elevation 60°. Find the tower height.

(A) 36 (B) 27

(C) 18 (D) $9\sqrt{3}$

103. X takes values [-6, 0, 4] with probabilities 1/6,1/3,1/2. Let Y=6X+10. Find $E(Y^2)$-$[E(Y)]^2$.

(A) 78 (B) 568

(C) 468 (D) 504

104. For x∈[π/5,π/2], find the maximum of sin x/x.

(A) $\frac{5\sqrt{10-2\sqrt{5}}}{4\pi}$ (B) $\frac{5}{\pi}$

(C) $\frac{2\sqrt{2}\pi}{5\sqrt{5-\sqrt{5}}}$ (D) $\frac{2}{\pi}$

105. α,β are roots of x^2-9x+18=0. The monic polynomial with roots α+1,β+1 is x^2-Ax+B. Find B.

(A) 27 (B) 28

(C) 121 (D) 361

106. $\log_2 a=5$, log_a b=6, log_b c=4. Find $\log_4(c^2/a)$.

(A) $\frac{235}{2}$ (B) 120

(C) $\frac{245}{2}$ (D) 235

107. For x>1 let f(x)=arctan(2 x/(1-x^2)), with arctan in (-π/2,π/2). Find the integral from 3 to 4 of f′(x) dx.

(A) $2\,\mathrm{atan}\left(\frac{1}{3}\right)$

(B) π

(C) $-2\,\mathrm{atan}\,(3)+2\,\mathrm{atan}\,(4)$

(D) $-\,\mathrm{atan}\,(3)+\mathrm{atan}\,(4)$

108. Evaluate $\sum_{j=1}^{\infty} 2j(\frac{1}{3})^j$.

(A) $\frac{5}{2}$ (B) $\frac{1}{2}$

(C) $\frac{3}{2}$ (D) 1

109. Let y′=(x+y)/(x+y+1), with y(0)=1. Find the positive value of x at whichx+y=7.

(A) $4-\frac{\log(5)}{4}$ (B) $\frac{\log(5)}{4}+3$

(C) $\frac{\log(5)}{2}+3$ (D) 3

110. Vectors a,b satisfy $|a|^2$=50, $|b|^2$=17, a·b=11. Find $|2a-b|^2$.

(A) 173 (B) 217

(C) 95 (D) 261

111. Positive a,b satisfy log(a+b)=log(8)+[log a+log b]/2 in one valid base. Find $(a^4+b^4)/(a^2b^2)$.

(A) 62 (B) 3844

(C) 3846 (D) 3842

112. The locus |z-3|=|Re(z)+3| intersects the horizontal line Im(z)=8. Find the real part of the point.

(A) $\frac{32}{3}$ (B) $-\frac{16}{3}$

(C) $\frac{2}{3}$ (D) $\frac{16}{3}$

113. Using digits 0,1,2,3,4,5,6,7 without repetition, how many 5-digit positive integers are divisible by 4, have first digit at most 2, and first digit smaller than last?

(A) 120 (B) 500

(C) 1320 (D) 300

114. In triangle ABC, A=2B, a/b=$\sqrt{3}$ and b=6. Find its area.

(A) 12 (B) $18\sqrt{3}$

(C) $36\sqrt{3}$ (D) 18

115. From two points on the same ray from a tower foot, separated by 9, elevations are 30° and 45°. Find the distance from the farther point to the foot.

(A) 9 (B) $\frac{9\sqrt{3}}{2}$

(C) $\frac{9\sqrt{3}}{2}+\frac{27}{2}$ (D) $\frac{9}{2}+\frac{9\sqrt{3}}{2}$

116. $\theta \in(\pi/2,\pi)$, $\sin\theta=\frac{4}{5}$. Find $\cos(\theta/2)$.

(A) $-\frac{3}{10}$ (B) $\frac{\sqrt{5}}{5}$

(C) $\frac{2\sqrt{5}}{5}$ (D) $-\frac{\sqrt{5}}{5}$

117. A=(7,2,2), O=(0,0,0). Find the squared length of the orthogonal projection of OA on the line through O with direction (1,2,2).

(A) 32 (B) 5

(C) 225 (D) 25

118. For $25x^2+100y^2=2500$, find the distance between its two directrices.

(A) $10\sqrt{3}$ (B) $\frac{20\sqrt{3}}{3}$

(C) $\frac{40\sqrt{3}}{3}$ (D) 20

119. Write $\sin x\cos^3x+7\sin^3x\cos x=P\sin2x+Q\sin4x$. Find Q.

(A) 1 (B) $\frac{1}{8}$

(C) $-\frac{3}{4}$ (D) 2

120. Box A contains 7 red and 3 blue balls; B contains 2 red and 5 blue balls. A fair coin chooses one box. Two independent draws with replacement from that box give exactly one red ball. Find the posterior probability that A was chosen.

(A) $\frac{1000}{2029}$ (B) $\frac{1029}{2029}$

(C) $\frac{49}{69}$ (D) $\frac{21}{100}$

121. A 5-element set contains distinct a,b. How many subsets contain a and exclude b?

(A) 16 (B) 32

(C) 8 (D) 30

122. Evaluate lim as x→0 of [sqrt(9+x)-3-x/6+x^2/216]/x^3.

(A) $\frac{1}{3888}$ (B) 0

(C) $\frac{1}{432}$ (D) $\frac{1}{1944}$

123. How many integers x satisfy $4^{-x} - (20)2^{-x} + 64 \leq 0$?

(A) 5 (B) 3

(C) 2 (D) 1

124. Let f(x)=sin x+5 cos x. Let [L,L+π] be the interval containing 0 in its interior on which f is increasing and bijective onto[-sqrt(26),sqrt(26)]. If g is its inverse, findL+$g'(1)$.

(A) $-\frac{\pi}{2} - \operatorname{atan}(5) + \frac{\sqrt{26}}{26}$

(B) $-\operatorname{atan}(5) + \frac{1}{5} + \frac{\pi}{2}$

(C) $-\frac{\pi}{2} + \frac{1}{5} + \operatorname{atan}(5)$

(D) $-\frac{\pi}{2} - \operatorname{atan}(5) + \frac{1}{5}$

125. a=(1,0,0), a×c=(0,-5,3), a·c=4. Find $|c+a|^2$.

(A) 58 (B) 51

(C) 59 (D) 50

126. Find the coefficient of x^4 in (1-x)exp(5x).

(A) $\frac{125}{24}$ (B) $\frac{375}{8}$

(C) $\frac{625}{24}$ (D) $-\frac{125}{6}$

127. Two distinguishable fair dice show X,Y. Given X+Y≥5, find the probability that X≠Y and XY is even.

(A) $\frac{4}{15}$ (B) $\frac{11}{18}$

(C) $\frac{11}{15}$ (D) $\frac{9}{13}$

128. The tangent to y=x^2+3x at P has slope 9. Find the slope of the normal at P.

(A) $-\frac{1}{9}$ (B) -9

(C) 9 (D) -81

129. y>0 satisfies yy′=x and y(0)=2. Find y(5).

(A) 7 (B) $\sqrt{14}$

(C) $-\sqrt{29}$ (D) $\sqrt{29}$

130. Evaluate the integral from 0 to $\pi/2$ of [(sin x+cos x)/(sec x+csc x)](sin x+5 cos x) dx, using continuous extension at the endpoints.

(A) 6

(B) $\frac{5}{3}$

(C) $-\frac{4}{3}$

(D) 2

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 5 percent. The measured increase in its absolute temperature is $\frac{369}{10}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) $\frac{3969}{10}$ K
(B) 360 K
(C) $\frac{369}{10}$ K
(D) 14760 K

132. The 4th harmonic of an open pipe of length 8 m equals the 4th allowed resonance of a closed pipe. Both have the same sound speed and negligible end correction. Find the closed-pipe length.

(A) 7 m
(B) 14 m
(C) 4 m
(D) $\frac{32}{7}$ m

133. Two equal-intensity coherent beams have maximum combined intensity 20 units. Find intensity when their path difference is $\lambda/6$.

(A) 60
(B) 20
(C) 15
(D) 10

Chemistry

134. An ion has 10 electrons,14 neutrons and charge +2. Which combination gives its element,mass number and neutral-element group correctly?

(A) Mg;A=24;group2.
(B) Mg;A=26;group2.
(C) An element with Z=10;A=24;group18.
(D) Mg;A=26;group18.

135. Two hydrogen-like bound states A and B have nuclear charges ZA=1 and ZB=3, and radii rB=3/4rA. The states obey the allowed integer-principal-quantum-number Bohr model. What is the ratio of binding-energy magnitudes |EB|/|EA|?

(A) 2
(B) 8
(C) 4
(D) 12

136. Compare pH of A:HCl at $\frac{1}{250}$M;B:HA,Ka=10^{-6},C=$\frac{2}{5}$M;C:HB,Ka=10^{-5},C=$\frac{1}{250}$M. Neglect water and use weak-ionisation ordering. Increasing pH is

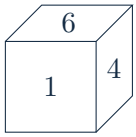
(A) A<B<C.
(B) C<B<A.
(C) A<C<B.
(D) B<A<C.

Logical Reasoning

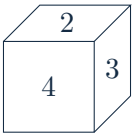
137. Preserve the exact relation and its order: Astronomer : Observatory :: ? : Courtroom. Choose the missing term.

(A) Judge
(B) Librarian
(C) Pilot
(D) Farmer

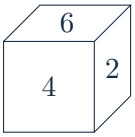
138. The views show the same die. Give the ordered pair: (face opposite 6, face opposite 4). Do not assume opposite labels sum to seven.



1



2



3

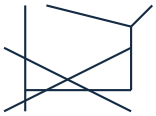
(A) $(3, 1)$
(B) $(3, 3)$
(C) $(3, 5)$
(D) $(3, 2)$

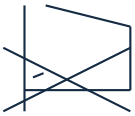
139. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.




host 1


host 2


host 3


host 4

(A) 1
(B) 4
(C) 3
(D) 2

Mathematics

140. 7 distinct people stand in a line. Three specified people cannot occupy positions 1,2 or 7. How many orders are possible?

(A) 576
(B) 24
(C) 1440
(D) 96

141. $F'(x)=\frac{3(x^2-1)(x^2+1)^2}{x^4}$ and $F(1)=4$. Find $F(2)-2F(1)$.

(A) $\frac{61}{8}$
(B) $\frac{93}{8}$
(C) $\frac{29}{8}$
(D) $-\frac{93}{8}$

142. For xy-7x-4y+28=0, find the squared length of the diagonal of the rectangle formed with the axes.

(A) 28 (B) 65

(C) $\sqrt{65}$ (D) 121

BITSAT Mock 17

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A particle has position-dependent velocity components $\vec{v} = 5x\,\hat{i} - 5y\,\hat{j}$ in SI units. Given $dx/dt = v_x$ and $dy/dt = v_y$, what is its acceleration magnitude at $(4, 4)$ m?

(A) $100\sqrt{2}\,\mathrm{m/s^2}$ (B) $200\,\mathrm{m/s^2}$

(C) $20\sqrt{2}\,\mathrm{m/s^2}$ (D) $100\,\mathrm{m/s^2}$

2. A soap bubble initially of radius 5 cm is expanded quasistatically until its volume is 27 times the initial volume. Surface tension remains $\frac{2}{25}$ N/m. Find work against surface tension, in units of $\pi \times 10^{-4}$ J.

(A) 16 (B) 128

(C) 144 (D) 416

3. A uniform-density planet has radius R. Gravity at a depth d equals gravity at a height $(3-1)R$ above the surface. Find d/R.

(A) $\frac{8}{9}$ (B) $\frac{1}{9}$

(C) $\frac{2}{3}$ (D) $\frac{26}{27}$

4. A spring of constant 20 N/m is already stretched by $\frac{1}{5}$ m. How much additional work is required to increase its extension to $\frac{1}{2}$ m quasistatically?

(A) $\frac{21}{10}$ J (B) $\frac{2}{5}$ J

(C) 3 J (D) $\frac{5}{2}$ J

5. A vacuum electromagnetic wave of average intensity 100 $\mathrm{W/m^2}$ passes through a device that reduces its electric-field amplitude by a factor 3. The transmitted wave falls normally on an area 3 $\mathrm{cm^2}$. What average power reaches the area?

(A) $\frac{100}{3}$ W (B) $\frac{100}{9}$ W

(C) $\frac{1}{300}$ W (D) $\frac{3}{100}$ W

6. A fixed amount of ideal gas has its absolute temperature multiplied by 2 and volume by 4. Find the final pressure divided by the initial pressure.

(A) $\frac{1}{4}$ (B) 8

(C) $\frac{1}{8}$ (D) $\frac{1}{2}$

7. A nucleus of mass number 20 splits into two equal nuclei. The parent has binding energy per nucleon $\frac{71}{10}$ MeV and each daughter has $\frac{77}{10}$ MeV. Ignoring other products, what energy is released in the split?

(A) 154 MeV (B) 166 MeV

(C) 12 MeV (D) 6 MeV

8. Photons of energy 47 eV eject electrons from a surface. The fastest electrons enter a uniform magnetic field perpendicular to their velocities. In this field an electron with kinetic energy 2 eV has orbit radius r. The observed maximum orbit radius is $3r$. What is the surface work function? Neglect recoil.

(A) $\frac{29}{9}$ eV (B) 29 eV

(C) 41 eV (D) $\frac{421}{9}$ eV

9. A 2 kg mass on a string passes the top of a vertical circle of radius 4 m with string tension 120 N. Find string tension at the bottom. Take $g = 10$ m/s^2.

(A) 200 N (B) 160 N

(C) 240 N (D) 220 N

10. A particle of mass 5 kg starts from rest and experiences $F(t) = 5t$ N along its motion for 5 s. Find the work done by the force.

(A) $\frac{3125}{2}$ J (B) $\frac{3125}{8}$ J

(C) $\frac{125}{2}$ J (D) $\frac{3125}{4}$ J

11. A cell balances on a potentiometer at 70 cm on open circuit and at 30 cm when a 5 ohm resistor is connected across it. The potentiometer gradient is unchanged. Find the cell's internal resistance.

(A) $\frac{15}{7}\,\Omega$ (B) $\frac{20}{3}\,\Omega$

(C) $\frac{20}{7}\,\Omega$ (D) $5\,\Omega$

12. A liquid of volume 2 litres and bulk modulus 7000000000 Pa is subjected to an additional pressure of 4000000 Pa. Neglect changes in bulk modulus. Find the magnitude of volume decrease in cubic centimetres.

(A) 3500000 cm^3 (B) $\frac{4}{7}$ cm^3

(C) $\frac{8}{7}$ cm^3 (D) $\frac{1}{875}$ cm^3

13. Two rods of equal length and cross-sectional area have thermal conductivities k and 11k. They are joined end-to-end with no contact resistance. Find the effective conductivity of the composite divided by k.

(A) 6 (B) $\frac{11}{6}$

(C) 1 (D) $\frac{11}{12}$

14. A ball is dropped from 8 m onto a horizontal fixed floor. At every impact the coefficient of restitution is $\frac{4}{5}$. Neglect air resistance. What maximum height does it reach after its second impact?

(A) $\frac{2048}{625}$ m (B) $\frac{512}{125}$ m

(C) 8 m (D) $\frac{32}{5}$ m

15. In an ideal intrinsic semiconductor at zero temperature, the band gap is $\frac{3}{2}$ eV. Neglect phonons, excitons and all transitions within a band. Two beams have photon energies $\frac{13}{10}$ eV and $\frac{17}{10}$ eV. Which can create a valence-to-conduction electron-hole pair by a single-photon absorption?

(A) Only the higher-energy beam (B) Both beams

(C) Neither beam (D) Only the lower-energy beam

16. A direct vernier has 20 divisions equal to 19 main-scale divisions; 1 MSD is 1 mm. Its signed zero error is $\frac{-1}{20}$ mm. For an object the main scale is 4 mm and the 11th vernier division coincides. Find the corrected reading.

(A) $\frac{111}{20}$ mm (B) $\frac{23}{5}$ mm

(C) $\frac{91}{20}$ mm (D) $\frac{9}{2}$ mm

17. A particle has displacement $x = 6\cos(2\pi t/4)$ cm. Find its signed velocity at $t = \frac{2}{3}$ s.

(A) $-\frac{3\pi}{2}$ cm/s (B) $\frac{3\sqrt{3}\pi}{2}$ cm/s

(C) -3π cm/s (D) $-\frac{3\sqrt{3}\pi}{2}$ cm/s

18. A lossless coating of refractive index $\frac{3}{2}$ lies between air and a substrate of still higher index. For normally incident vacuum wavelength 700 nm, find the smallest positive film thickness giving destructive interference between its two reflected rays. Assume reflections at both boundaries each gain a phase reversal.

(A) $\frac{1400}{9}$ nm (B) $\frac{700}{3}$ nm

(C) $\frac{700}{9}$ nm (D) $\frac{350}{3}$ nm

19. A ray in a medium of index $\frac{23}{10}$ is incident on a medium of index $\frac{13}{10}$. Find the sine of the critical-angle threshold for total internal reflection.

(A) $\frac{23}{13}$ (B) $\frac{6\sqrt{10}}{23}$

(C) $\frac{13}{23}$ (D) $\frac{10}{23}$

20. A parallel-plate capacitor has vacuum capacitance C_0. A dielectric of constant 4 fills a fraction $\frac{1}{3}$ of the plate area and the entire gap in that region; the remainder stays vacuum. The voltage is fixed. Find the fraction of total stored energy residing in the dielectric-filled region.

(A) $\frac{4}{3}$ (B) $\frac{2}{3}$

(C) $\frac{4}{5}$ (D) $\frac{1}{9}$

21. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 20 percent. The measured increase in its absolute temperature is $\frac{814}{5}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) 4070 K (B) $\frac{2664}{5}$ K

(C) $\frac{814}{5}$ K (D) 370 K

22. Two parallel plane mirrors separated by 2 cm extend from axial position 0 to 5 cm. A ray enters midway between them at axial position 0, making 45° with the normal to the mirrors. How many reflections occur before it passes the far edge? Exclude a contact exactly on the exit edge.

(A) 2 (B) 3

(C) 4 (D) 1

23. A particle has $v(0) = 2$ m/s and acceleration $a(t) = 12t$ m/s^2 for $0 \le t \le 2$ s. Find its average velocity over this interval.

(A) 14 m/s (B) 26 m/s

(C) 10 m/s (D) 8 m/s

24. A network is defined by $P = \mathrm{NOR}(A, B)$, $Q = \mathrm{NAND}(A, P)$ and $Y = \mathrm{NAND}(Q, B)$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 0$ and $P = 1$?

(A) 3 (B) 1

(C) 2 (D) 0

25. A wire carries $i(t) = 2 + 5\sin\omega t$ A for many cycles. Find its rms current.

(A) 2 A (B) 7 A

(C) $\sqrt{29}$ A (D) $\frac{\sqrt{66}}{2}$ A

26. A platform of moment of inertia 4 kg m^2 carries a person of mass 1 kg at radius 5 m and rotates at 2 rad/s. The person walks to half the stated initial radius; external torque is zero. Find final angular speed. Treat the person as a point mass.

(A) $\frac{41}{58}$ rad/s (B) 8 rad/s

(C) $\frac{4\sqrt{1189}}{41}$ rad/s (D) $\frac{232}{41}$ rad/s

27. A nonrelativistic particle of fixed charge and mass moves perpendicular to a uniform magnetic field. Its kinetic energy is multiplied by 4 and the magnetic field is multiplied by 3. Find the new trajectory radius divided by the original radius.

(A) $\frac{2}{3}$ (B) $\frac{1}{6}$

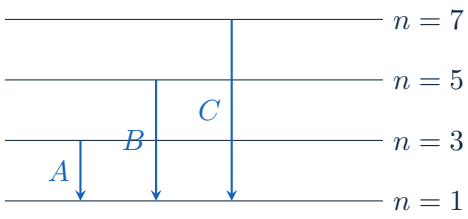
(C) $\frac{1}{3}$ (D) 6

28. A uniform disc of mass 8 kg and radius 4 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) 52 kg m^2 (B) 48 kg m^2

(C) 76 kg m^2 (D) 60 kg m^2

29. The diagram shows downward hydrogen transitions A, B and C from $n = 3$, 5 and 7, respectively, ending at $n = 1$. A filter transmits only photons with energy strictly above that of transition B. Which transmitted line has the longest wavelength?



(A) B
(B) A
(C) C
(D) No transmitted line

30. Two small identical conducting spheres initially carry positive charges $5q_0$ and $5q_0$ and are separated by distance d. They touch and are returned to the same separation. Find the final force divided by the initial force.

(A) $\frac{4}{5}$
(B) 1
(C) $\frac{5}{9}$
(D) $\frac{3}{5}$

Chemistry

31. In the usual planar structure of $Ni(dmg)_2$,each monoanionic dimethylglyoximate binds through two N atoms. Which pair is consistent with Ni(II) d^8 and this donor pattern?

(A) Tetrahedral and diamagnetic; four chelate rings.
(B) Octahedral and paramagnetic; two six-membered chelate rings.
(C) Square planar and diamagnetic; two five-membered chelate rings.
(D) Square planar with two unpaired electrons; no chelate ring.

32. Sucrose has formulaC12H22O11. A derivative is formed by replacing three OH groups by three Cl atoms, without changing the carbon skeleton. What is its formula?

(A) C12H25Cl3O14.
(B) C12H19Cl3O8.
(C) C9H19Cl3O8.
(D) C12H22Cl3O11.

33. A solution initially contains $\frac{3}{500}$ mol Fe^{2+}. Before titration, $\frac{1}{10}$ of it is oxidised quantitatively to Fe^{3+}. The volume in mL of0.0200 M $KMnO_4$ needed for the remaining Fe^{2+} in acid is

(A) 6
(B) 60
(C) 270
(D) 54

34. With one equivalent of cold concentrated HI and isolation before prolonged further reaction,which initial cleavage pair is expected from phenetole,C6H5-O-C2H5?

(A) ethene and iodobenzene
(B) benzyl iodide and methanol
(C) iodobenzene and ethanol
(D) ethyl iodide and phenol

35. In water,ignoring special ortho steric effects,which decreasing acid strength order is most appropriate for 4-methoxyanilinium,anilinium,4-nitroanilinium?

(A) 4-nitroanilinium >4-methoxyanilinium > anilinium
(B) 4-nitroanilinium > anilinium >4-methoxyanilinium
(C) anilinium >4-nitroanilinium >4-methoxyanilinium
(D) 4-methoxyanilinium > anilinium >4-nitroanilinium

36. 1-methylcyclohexene is treated with chloroform,CHCl3 and strong base under dihalocarbene-generation conditions. Which joint statement correctly identifies the haloalkane step and the hydrocarbon-skeleton change?

(A) Alpha elimination generates :CCl2; the alkene gains one carbon and one additional ring.

(B) Beta elimination generates an alkene from the haloform; the ring size is unchanged.

(C) A free methyl cation adds; the original C=C remains intact.

(D) :CCl2 adds by replacing a ring carbon,so total carbon count is unchanged.

37. For a reaction over300-600 K, assume temperature-independent ΔH°. The measured natural logarithm ln[K(600)/K(300)] is 5/3. Using ln(K2/K1)=-(ΔH°/R)(1/T2-1/T1), find ΔH°/R in K, including sign.

(A) -1000
(B) 2000
(C) 500
(D) 1000

38. but-1-ene reacts with HCl with peroxide. Which pair gives the major constitutional product and the intermediate whose greater stability fixes its orientation? Ignore stereoisomer counting.

(A) 1-chlorobutane; secondary carbocation
(B) 2-chlorobutane; secondary carbocation
(C) 2-chlorobutane; primary radical
(D) 1-chlorobutane; primary radical

39. D-Glucose, D-mannose andD-fructose share the same configurations from C3 onward. Which conclusion follows under excess phenylhydrazine osazone formation?

(A) Only glucose can form an osazone because ketoses never react.
(B) The reaction necessarily changes every stereocentre.
(C) All three must give three distinct osazones.
(D) All three can give the same osazone.

40. A photographic fixer contains $\frac{3}{25}$ mol free $S_2O_3^{2-}$ and receives $\frac{2}{25}$ mol unexposed AgBr. Use $AgBr+2S_2O_3^{2-}\rightarrow[Ag(S_2O_3)_2]^{3-}+Br^-$. Treat complexation as complete until a reagent is exhausted. AgBr remaining,in mol,is

(A) $\frac{2}{25}$
(B) $\frac{1}{25}$
(C) $\frac{1}{50}$
(D) 0

41. At fixedT,V,A(g)→P(g)+Q(g)+R(g)is first order,and an inert gas is present. Initial totalpressure is$\frac{3}{10}$bar;complete-decomposition pressure is$\frac{1}{2}$bar. At two half-lives,totalpressure in bar is

(A) $\frac{7}{20}$
(B) $\frac{3}{4}$
(C) $\frac{9}{20}$
(D) $\frac{2}{5}$

42. A carbon-containing organic sample contains N and S among nitrogen, sulfur and halogens. It undergoes sodium fusion with large excess sodium. Which conversion and subsequent test protocol is chemically consistent? For a silver-halide test, use a separately prepared portion boiled with nitric acid to remove cyanide/sulfide interference where present.

(A) thiocyanate; no further treatment is ever needed before adding silver nitrate.

(B) cyanide and sulfide; Prussian blue and violet nitroprusside response.

(C) Unchanged covalent bonds; add silver nitrate directly to the untreated organic sample.

(D) Nitrate and sulfate in every case; all give the same Prussian-blue test.

43. For $[Ni(CN)_4]^{2-}$,use the usual ligand-field/VBT assignments at room temperature. Which complete assignment is consistent with charge balance and magnetic behaviour?

(A) d^8,square planar,dsp^2,spin-only moment 0 BM.

(B) d^9,square planar,dsp^2,spin-only moment 0 BM.

(C) d^8,square planar,dsp^2,spin-only moment $\sqrt{3}$ BM.

(D) d^8,tetrahedral,sp^3,spin-only moment 0 BM.

44. A model M(OH)2 has molar solubility 1/50000 M in pure water; neglect water autoionisation there. In a buffered solution its measured molar solubility is 1/31250000 M. The buffer fixes free OH- and the solid's contribution to OH- is negligible. What is the imposed OH- concentration?

(A) 1/500

(B) 1/1000000

(C) 1/2000

(D) 1/1000

45. cyclohexanone reacts with NaBH4 followed by water. Which product correctly reflects nucleophilic addition or addition-elimination at its carbonyl rather than a different reaction class?

(A) cyclohexene

(B) cyclohexanol

(C) phenol

(D) cyclohexane

46. phenol undergoes CHCl3/NaOH,then acidification;then NaBH4. Which final product preserves all ring substituents and the selective reduction endpoint?

(A) 2-hydroxybenzoic acid

(B) 4-methoxybenzaldehyde

(C) 2-(hydroxymethyl)phenol

(D) benzyl alcohol

47. A first-order reaction has a constant half-life of 10 min. A sample is observed now and again 30 min later, when unreacted material equals 1/32 of the original amount present when the reaction began. How many minutes had elapsed before the first observation?

(A) 30

(B) 50

(C) 60

(D) 20

48. For butan-1-ol in the Victor Meyer sequence,which test/observation assignment follows the actual carbinol class or oxidisable structural feature? Use ordinary room-temperature tests unless the named sequence specifies otherwise.

(A) violet ferric-phenolate complex

(B) blue colour characteristic of the secondary sequence

(C) red colour from the primary-alcohol sequence

(D) no colour characteristic of the tertiary sequence

49. A reaction has the same Arrhenius prefactor over the range. Its rate constant doubles from300K to330K. Temperature where its rate constant is four times the300K value,in K,is

(A) 360 (B) 600

(C) $\frac{1100}{3}$ (D) 330

50. An H_2SO_4 stock is29.0% by mass. A20.0mL aliquot is diluted to 60mL,and the diluted molarity is $\frac{6}{5}$M. H_2SO_4 M98. Stock density in g/mL is

(A) $\frac{882}{725}$ (B) $\frac{294}{725}$

(C) $\frac{441}{1250}$ (D) $\frac{2646}{725}$

51. For the pair ethoxyethane and1-methoxypropane,which classification identifies the structural difference rather than merely stating that the formula agrees?

(A) metamers (B) geometrical isomers

(C) optical antipodes (D) functional isomers

52. Why can2-nitrophenol be more steam-volatile than4-nitrophenol in the usual introductory comparison?

(A) Intramolecular hydrogen bonding in the ortho isomer reduces intermolecular association relative to the para isomer.

(B) All ortho-substituted compounds necessarily have lower molar mass.

(C) Its molecular formula has fewer oxygen atoms.

(D) The para isomer contains no O-H group.

53. In the gas phase,B_2 is converted into B_2^+ by changing one electron. Under the elementary MO model, which final bond-order/unpaired-electron pair follows?

(A) Bond order $\frac{1}{2}$; 2 unpaired electrons. (B) Bond order $\frac{1}{2}$; 1 unpaired electron.

(C) Bond order $\frac{3}{2}$; 1 unpaired electrons. (D) Bond order 1; 2 unpaired electrons.

54. A weak monoprotic acid HA is the only electrolyte. Conductance is $\frac{1}{25000}$S in a cell of constant2.00cm^{-1};an independent measurement gives$[H^+]=\frac{1}{5000}$M. Assume $H^+=A^-$ and negligible water ionisation. Limiting molar conductivity Λ_0(HA),in S cm^2/mol,is

(A) 0.4 (B) 800

(C) 400 (D) 200.0

55. At 1 atm compare two separate pairs: (ethanol, dimethyl ether) and (n-pentane, neopentane). A student selects dimethyl ether for the first higher-boiling substance because its formula mass is unchanged, and n-pentane for the second. Which assessment is consistent with intermolecular interactions?

(A) Both selections correct. (B) Both selections wrong.

(C) First selection correct; second selection wrong. (D) First selection wrong; second selection correct.

56. Two amino acids condense to form a dipeptide, and the resulting molecules later form intermolecular hydrogen bonds. Which distinction is correct?

(A) The peptide connection is covalent; the later hydrogen bond is a separate interaction.

(B) Both are the sameCO-NH covalent bond.

(C) Only the hydrogen bond determines residue sequence.

(D) Condensation forms two glucose units.

57. For isolated Mn^{2+} (Z=25),use the high-spin spin-only approximation. Which pair (d-electron count, magnetic moment in BM) follows after removing electrons from 4s before 3d?

(A) (5,5).
(B) $(5,\sqrt{35})$.
(C) $(5,4\sqrt{3})$.
(D) $(6,4\sqrt{3})$.

58. Atoms form ideal touching FCC spheres of radius r and cell edge a. Fractional coordinates are measured in a. Using the straight displacement between the two specified points (1/4,1/4,1/4) and (0,0, 0), without replacing it by another periodic image, find $(\text{distance}/r)^2$. These points describe tetrahedral site to nearest atom centre.

(A) 3/2
(B) 3
(C) 5/2
(D) 3/4

59. methylcyclohexane undergoes low-conversion monobromination with Br2/light. Which is the major constitutional product? Use the usual strong tertiary-over-secondary-over-primary bromination selectivity.

(A) 1-bromo-1-methylcyclohexane
(B) bromomethylcyclohexane
(C) 2-bromo-1-methylcyclohexane
(D) 3-bromo-1-methylcyclohexane

60. In an antifluorite-derived crystal, X2- occupies all FCC anion sites. A measured fraction 1/4 of occupied M cations is M2+ and the rest M+. Only tetrahedral cation vacancies compensate charge; no interstitials or anion defects occur. What fraction of all tetrahedral sites is vacant?

(A) 4/5
(B) 1/5
(C) 1/10
(D) 1/8

English Proficiency

61. Choose the best replacement: She prefers [to check the original rather than relying on a copy].

(A) to checking the original rather than a copy
(B) checking the original instead relying on a copy
(C) checking the original than to rely on a copy
(D) checking the original to relying on a copy

Read the passage for Question 62

An office rewrote its instructions in shorter sentences and replaced several technical terms. Yet visitors continued to ask which desk they should approach. A member of staff watched several new visitors use the notice and found the missing link: the notice described the forms clearly but never said which form belonged at which desk. The next revision added that relationship. Questions at the entrance then became less frequent. It would be premature to credit short sentences alone, since they had been introduced before the improvement appeared. The episode suggests a more careful approach to clarity. A text may contain familiar words and still fail if it omits the connection a reader needs in order to act. Testing that connection requires attention to the reader's task, not only to the length of the sentences.

62. Which inference follows most reasonably from the office's experience?

(A) Improving sentence simplicity is insufficient when an essential relation needed for action is missing.
(B) Readers' questions show that familiar words should be replaced by technical ones.
(C) The final improvement proves that sentence length never affects understanding.
(D) Longer sentences always produce clearer practical instructions.

63. Read the sentence: After checking the cost and the time available, they judged the smaller scheme feasible. Choose the closest meaning of 'feasible' as used in the sentence.

(A) profitable (B) practicable

(C) already approved (D) desirable

64. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. Did the guide open the side entrance only after the queue formed?

(A) Was only the guide opening the side entrance after the queue formed?

(B) Was the side entrance opened by the guide only after the queue formed?

(C) Was the side entrance opened by the guide before the queue formed?

(D) Did the side entrance open the guide only after the queue formed?

65. Read the sentence: During the disagreement, the written agreement became an anchor for the discussion. What does the figurative expression mean in this context?

(A) It prevented any new idea from being considered. (B) It forced the participants to remain in one place.

(C) It made the discussion impossible to finish. (D) It provided a stable point of reference.

66. The final section lists the books and articles used in preparing the essay rather than explaining difficult words. What is it?

(A) bibliography (B) index

(C) appendix of illustrations (D) glossary

67. Choose the most coherent order of the four sentences. A. The coordinators asked donors to check labels before bringing items. B. The additional check reduced the time needed to sort unsuitable donations at the collection point. C. Several donated packets lacked readable dates. D. Volunteers organised a collection of packaged food.

(A) D-C-B-A (B) D-C-A-B

(C) C-D-A-B (D) D-A-C-B

68. Read the sentence: The courteous driver stopped to let the elderly pedestrian cross. Choose the opposite in meaning of 'courteous' as used in the sentence.

(A) patient (B) considerate

(C) discourteous (D) attentive

69. Read the sentence: The dispute had been dormant for years, but the new proposal brought it back into discussion. Choose the closest meaning of 'dormant' as used in the sentence.

(A) resolved (B) concealed

(C) inactive (D) forgotten by everyone

70. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. that several readers failed Q. The explanation was so compressed R. from the evidence presented. S. to see how the final claim followed

(A) Q-P-S-R (B) S-Q-P-R

(C) R-P-S-Q (D) P-Q-S-R

Logical Reasoning

71. R and P are full siblings; P and G are also full siblings. C is the father of R. All four people are distinct. How is C related to G?

(A) Grandfather (B) Father

(C) Uncle (D) Relationship cannot be determined

72. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

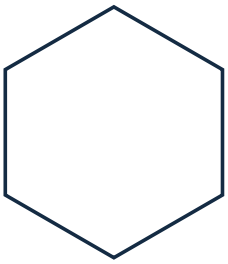


B

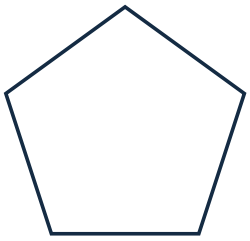


C

(A) Pattern R



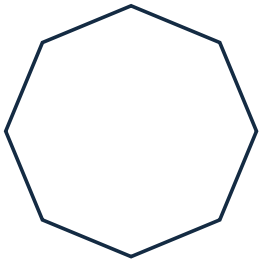
(B) Pattern T



(C) Pattern P



(D) Pattern Z



73. In a row, Asha is 5th from the left and 42nd from the right. Bina is 10th from the right. Charu stands 2 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 28 (B) 31

(C) 29 (D) 2

74. Preserve the exact relation and its order: Lion : Pride :: ? : Team. Choose the missing term.

(A) Player (B) Bee

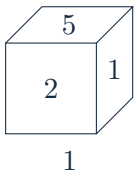
(C) Wolf (D) Lion

75. The model triads are $(6, 16, 10)$ and $(8, 20, 12)$. In the intended rule the middle number is the sum of the outer numbers, and the outer-number difference is the same fixed positive value in every triad. Which option satisfies both relations?

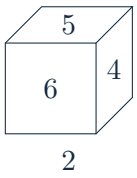
(A) $(5, 16, 10)$ (B) $(5, 15, 10)$

(C) $(5, 15, 9)$ (D) $(5, 14, 9)$

76. The views show the same die. Give the ordered pair: (face opposite 1, face opposite 5). Do not assume opposite labels sum to seven.



1



2

(A) $(4, 4)$ (B) $(4, 2)$

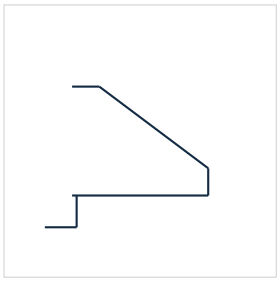
(C) $(4, 3)$ (D) $(4, 1)$

77. A word is coded as the sum of its forward alphabet positions plus 3 times its length. The code of PL?NT is 78. Which letter replaces the question mark?

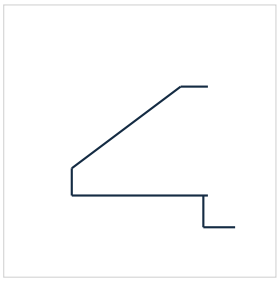
(A) Z (B) B

(C) A (D) D

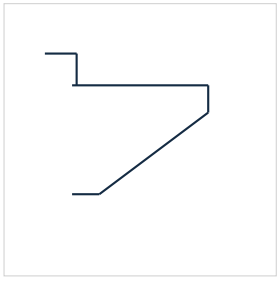
78. Select the missing bottom-right tile. Tiles reflect across the shared horizontal and vertical centre lines; there is no turning.



top left



top right

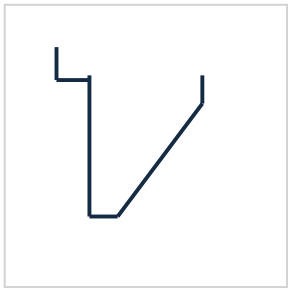


bottom left

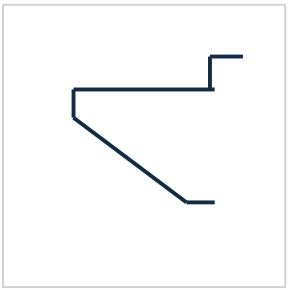
?

bottom right

(A) Pattern K



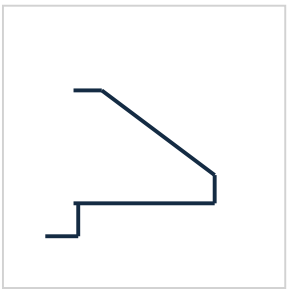
(B) Pattern P



(C) Pattern W



(D) Pattern Q



79. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 5, 15, ?, 63, 69, 207, 213.

(A) 9

(B) 63

(C) 25

(D) 21

80. The increase from the left cell to the right cell is the same in each row, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.





(A) Pattern M

(B) Pattern Q





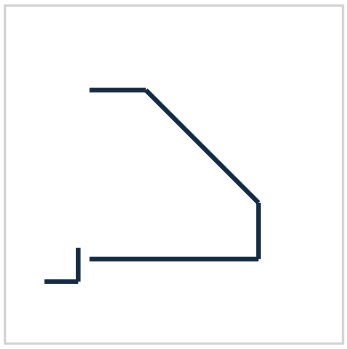
(C) Pattern R

(D) Pattern L

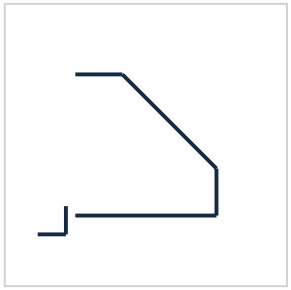




81. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



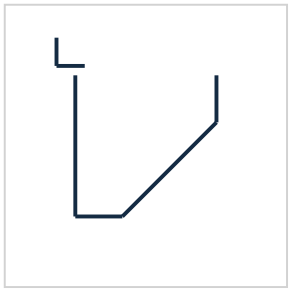
(A) Pattern T



(B) Pattern Q



(C) Pattern Y



(D) Pattern W



82. Three pairs contain exactly one metal and one nonmetal. Which pair does not?

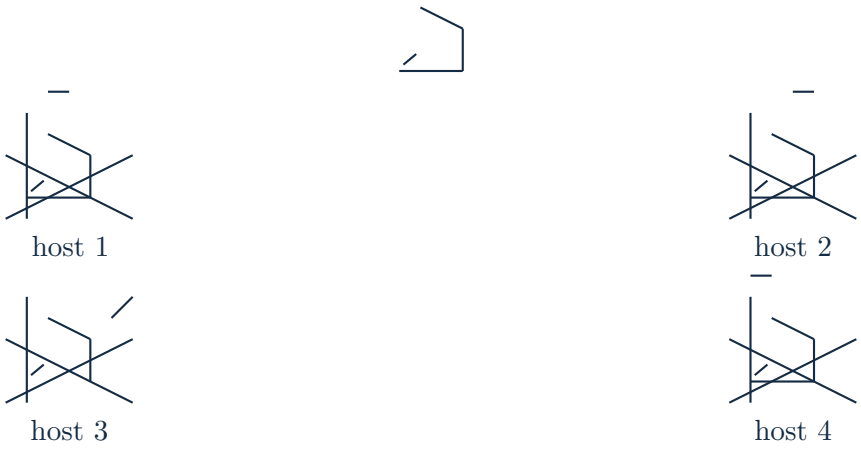
(A) Iron / Oxygen

(B) Zinc / Nitrogen

(C) Gold / Carbon

(D) Tin / Silver

83. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.



(A) 3
(B) 1
(C) 4
(D) 2

84. Consider only these statements; universal statements alone do not imply existence. No designers are campers; Some designers are editors. Conclusions: I. Some campers are editors. II. Some designers are not campers. Which conclusions follow necessarily?

(A) Both I and II
(B) Only I
(C) Only II
(D) Neither I nor II

85. For this question compare only cyclic forward alphabet gaps, with $Z \to A$ counting as one step. Three clusters share the same gap vector. Which numbered cluster differs? 1: XABE, 2: ILMP, 3: SVWZ, 4: GJLO.

(A) 1
(B) 3
(C) 2
(D) 4

86. Two people differ in age by 10 years. The product of their ages 4 years ago was 299. What will the older person's age be 6 years from now?

(A) 43
(B) 33
(C) 29
(D) 27

87. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 3 and Z has code 28. Decode the separated values 18, 14, 3, 16, 22.

(A) PLANT
(B) QMBOU
(C) TNALP
(D) OKZMS

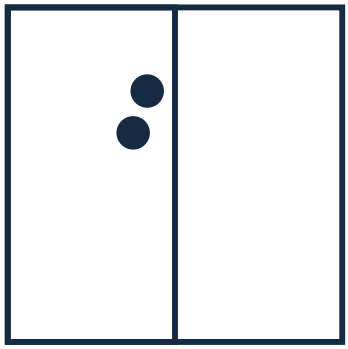
88. Five people occupy positions 1 to 5 from left to right, one per position, and wear five different shirt colours: red, blue, green, white and yellow. Dev is 1st from the left. Bina is 2nd from the left. Charu is 3rd from the left. Asha is 4th from the left. Esha is 5th from the left. The yellow shirt is in position 1. The green shirt is in position 2. The white shirt is in position 3. Esha does not wear blue. Which shirt does Esha wear?

(A) White
(B) Blue
(C) Green
(D) Red

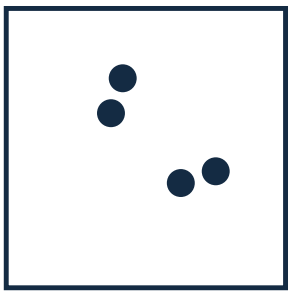
89. Consider only these statements; universal statements alone do not imply existence. Some editors are joggers; No guides are joggers; All guides are joggers. Conclusions: I. No joggers are guides. II. Some editors are not joggers. Which conclusions are individually possible without contradicting the statements?

(A) Both I and II
(B) Only I
(C) Neither I nor II
(D) Only II

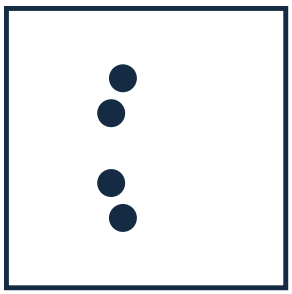
90. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



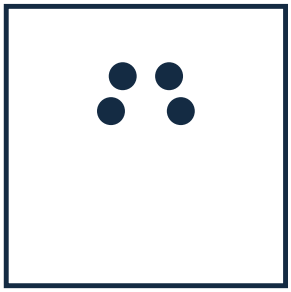
(A) Pattern M



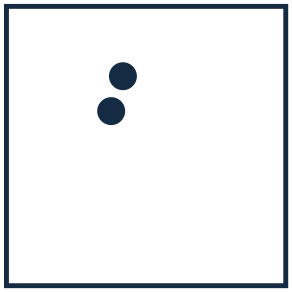
(B) Pattern P



(C) Pattern V



(D) Pattern L



Mathematics

91. The normal of slope 5 to $y^2=28x$ touches at P. Find the distance from P to the focus.

(A) 175
(B) 33124
(C) 168
(D) 182

92. $\log_2 a=5$, log_a b=6, log_b c=3. Find $\log_4(c^2/a)$.

(A) $\frac{175}{2}$ (B) $\frac{185}{2}$

(C) 175 (D) 90

93. How many pairs $1\le m<n\le 12$ satisfy both $5 \mid (6^m+9^n)$ and $3 \mid (m+n)$?

(A) 10 (B) 30

(C) 11 (D) 20

94. The tangent to $y = x^3 - 24x$ at $x = 5$ meets the two coordinate axes at A and B. Find the area of triangle OAB, where O is the origin.

(A) $\frac{31250}{51}$ (B) $\frac{15625}{51}$

(C) $\frac{62500}{51}$ (D) $\frac{1250}{3}$

95. L_1:r=t(1,0,1), L_2:r=(2,0,0)+s(0,1,1). Find their shortest distance.

(A) $\frac{2}{3}$ (B) $\frac{2\sqrt{3}}{3}$

(C) 0 (D) 2

96. Two acute sun elevations $\theta_1<\theta_2$ satisfy $(16^4)^{sin^2\theta} + (16^4)^{cos^2\theta} = 2^4 + 2^12$. A vertical tower has height 9. Find the difference between its two shadow lengths.

(A) $12\sqrt{3}$ (B) $9\sqrt{3}$

(C) 6 (D) $6\sqrt{3}$

97. Let $\alpha=\pi/5$, $\beta=\pi/4$. Evaluate $\arccos(-\cos\alpha\cos\beta-\sin\alpha\sin\beta)$.

(A) $-\frac{\pi}{20}$ (B) $\frac{19\pi}{20}$

(C) $\frac{\pi}{20}$ (D) $\frac{21\pi}{20}$

98. f(x)=5x+1 and f(g(x))=5x+36. Find g(g(5)).

(A) 24 (B) 19

(C) 12 (D) 39

99. Evaluate lim as $x\to 0$ of $[\text{sqrt}(16+x)-4-x/8+x^2/512]/x^3$.

(A) $-\frac{1}{512}$ (B) $\frac{1}{16384}$

(C) $\frac{1}{8192}$ (D) $\frac{1}{1024}$

100. A rectangle has vertices $(\pm x,\pm y)$ on $x^2/256+y^2/64=1$, with $x\ge 16\sqrt{3}/2$. Find its maximum area.

(A) 256 (B) 128

(C) $128\sqrt{3}$ (D) $256\sqrt{3}$

101. $x=2+4(1-t^2)/(1+t^2)$, $y=2+8t/(1+t^2)$, where t is real. Which point of the containing circle is not attained?

(A) $(2,-2)$ (B) $(-2,2)$

(C) $(2,6)$ (D) $(6,2)$

102. An invertible 4×4 matrix A has det A=11. Find det(adj A).

(A) 11 (B) 1331

(C) 161051 (D) 14641

103. From a point at horizontal distance 6 from a tower, its elevation is 2θ, where $\tan\theta=\frac{5}{11}$ and $0<\theta<45°$. Find the height.

(A) $\frac{288}{55}$ (B) $\frac{60}{11}$

(C) $\frac{55}{8}$ (D) $\frac{30}{11}$

104. If α,β are roots of $x^2-2kx+k^2-9=0$ and k∈[-2,2], find the maximum of $\alpha^2+\beta^2$.

(A) −2 (B) 26

(C) 22 (D) 18

105. The hyperbola $x^2/49-y^2/16=1$ has eccentricity e. Find the eccentricity e′ of its conjugate hyperbola.

(A) $\frac{\sqrt{65}}{7}$ (B) $\frac{\sqrt{65}}{4}$

(C) $\frac{7\sqrt{65}}{65}$ (D) $\sqrt{65}$

106. The line r=(2,2,5)+t(1,2,2) and plane 2x-y=4 are given. Find the squared minimum distance between them.

(A) 0 (B) 4

(C) $\frac{4}{5}$ (D) $\frac{2\sqrt{5}}{5}$

107. The line y=mx through O bisects the area of the triangle with vertices O,(3,6),(6,6). Assume the vertices are noncollinear. Find m.

(A) 2 (B) 0

(C) $\frac{3}{4}$ (D) $\frac{4}{3}$

108. $z_1=3+4i$, $z_2=7+4i$, $z_3=7+8i$. Find the smaller angle between z_2-z_1 and z_3-z_1.

(A) 0 (B) $\frac{3\pi}{4}$

(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

109. $F'(x)=\frac{(x+1)^2}{(x+2)^4}$ and F(0)=2. Find F(2)-2F(0).

(A) $-\frac{365}{192}$ (B) $\frac{403}{192}$

(C) $\frac{19}{192}$ (D) $-\frac{403}{192}$

110. Two groups of sizes 2,3 have means 4,10 and population variances 4,5. Find the combined population variance.

(A) $\frac{9}{2}$ (B) $\frac{23}{5}$

(C) $\frac{331}{25}$ (D) 71

111. In how many ordered ways can three six-sided dice have sum 12, if the first die is known not to show 6?

(A) 25 (B) 40

(C) 24 (D) 20

112. α,β are nonzero roots of x^2-8x+12=0. Find the sum of $1+1/\alpha$ and $1+1/\beta$.

(A) $\frac{5}{3}$ (B) $\frac{8}{3}$

(C) $\frac{2}{3}$ (D) $\frac{7}{2}$

113. How many reflexive symmetric relations on a labelled 7-element set contain at least one off-diagonal ordered pair?

(A) 2097152 (B) 2097150

(C) 2097151 (D) 1048576

114. Evaluate $\displaystyle\sum_{\substack{0\le r\le 8\\ r \text{ even}}} \frac{\binom{8}{r}}{(r+1)(r+2)}$.

(A) $\frac{511}{90}$ (B) $\frac{233}{45}$

(C) $\frac{1013}{90}$ (D) $\frac{256}{9}$

115. For $A=\begin{bmatrix}6 & 1\\ 1 & 5\end{bmatrix}$ and C=29A^{-1}, find C_{22}.

(A) 5 (B) -1

(C) $\frac{6}{29}$ (D) 6

116. Positive integers $a<b$ satisfy $a+b=32$ and $3 \mid a$. Find the maximum value of ab.

(A) 255 (B) 240

(C) 256 (D) 32

117. A complex number z satisfies z+conjugate(z)=6 and z-conjugate(z)=4i. Find z·conjugate(z).

(A) 25 (B) 1

(C) 13 (D) 26

118. For a=(1,5,0), b=(0,1,5), c=(2,0,1), evaluate (a×c)·(a+b).

(A) -51 (B) 0

(C) -50 (D) 51

119. Positive a,b,c form an AP with a=6. The numbers a,b,c+8/3 form a GP and b>a. Find c.

(A) 14 (B) -2

(C) 38 (D) 10

120. In $(1+x)^9$, find the ratio of the coefficient of x^3 to the coefficient of x^2.

(A) $\frac{3}{7}$ (B) 4

(C) 7 (D) $\frac{7}{3}$

121. Chords of $x^2+y^2=81$ pass through (18,0). Find the radius of the circle containing their midpoint locus.

(A) 18 (B) 81

(C) 9 (D) $\frac{9}{2}$

122. Three distinguishable fair dice are thrown. Given that exactly two faces are equal, find the probability that their sum is at least 11.

(A) $\frac{5}{24}$ (B) $\frac{8}{15}$

(C) $\frac{23}{45}$ (D) $\frac{1}{2}$

123. For the loop $y^2=x^2(6-x)$, find the area of the portion with $x \geq 6/2$.

(A) $\frac{48\sqrt{6}}{5}$ (B) $\frac{42\sqrt{3}}{5}$

(C) $\frac{84\sqrt{3}}{5}$ (D) $-\frac{84\sqrt{3}}{5}+\frac{96\sqrt{6}}{5}$

124. The curve $y=x^3+7x$ divides the disc $x^2+y^2 \leq 16$. Find the area of the part satisfying $y \geq x^3+7x$.

(A) $7+8\pi$ (B) 8π

(C) 4π (D) 16π

125. For lines $3x+4y+3=0$ and $4x+3y+7=0$, find the squared distance from(1,4) to their angle bisector of positive slope.

(A) 49 (B) $\frac{47}{2}$

(C) $\frac{51}{2}$ (D) $\frac{1}{2}$

126. The homogeneous system $7x+y=0$, $kx+3y=0$ has a nonzero solution. Find $k+x^2/y^2$.

(A) $\frac{1030}{49}$ (B) $\frac{1028}{49}$

(C) 21 (D) 70

127. Let $a_j = 1/(7+2(j-1))$. Evaluate $\sum_{j=1}^{10} a_j a_{j+1}$.

(A) $\frac{20}{189}$ (B) $\frac{2}{5}$

(C) $\frac{10}{189}$ (D) $\frac{9}{175}$

128. Let $y' + 3y = 5x$ and $y(0) = 3$. Find $y(1) + \frac{7}{5}y'(1)$.

(A) $\frac{512}{45e^3}+\frac{31}{9}$ (B) $\frac{32}{9e^3}+\frac{10}{9}$

(C) $\frac{31}{9}$ (D) $\frac{31}{9}-\frac{512}{45e^3}$

129. $f(x)=x^3+4x$, $g=f^{-1}$. Find $g''(39)$.

(A) $-\frac{18}{29791}$ (B) $\frac{1}{18}$

(C) $-\frac{18}{961}$ (D) $\frac{18}{29791}$

130. Evaluate $\displaystyle\int_0^{\pi/2} \sin(2x)\left[1+5\sin^2(2x)\right]\frac{\sin^3 x}{\sin^3 x+\cos^3 x}\,dx.$

(A) 3 (B) $\frac{13}{3}$

(C) $\frac{1}{2}$ (D) $\frac{13}{6}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A mass 5 kg initially moves in the positive direction at 3 m/s. A force opposite to that direction rises linearly from zero to 7 N and then falls linearly to zero over a total duration 4 s. Find its signed final momentum.

(A) 15 kg m/s (B) 29 kg m/s

(C) 1 kg m/s (D) 14 kg m/s

132. A Carnot engine initially runs between 600 K and 400 K. Its hot reservoir is raised by 60 K. By how much must the cold-reservoir temperature rise to keep efficiency unchanged?

(A) 90 K (B) 40 K

(C) 60 K (D) 20 K

133. An insulated vessel containing one mole of an ideal gas of molar mass $\frac{1}{50}$ kg/mol moves at bulk speed 20 m/s and is brought to rest. All bulk kinetic energy becomes internal energy. The gas has 5 active degrees of freedom. Find temperature rise. Use $R = 8.314$ J/(mol K).

(A) $\frac{800}{4157}$ K (B) $\frac{4000}{12471}$ K

(C) $\frac{4000}{4157}$ K (D) $\frac{1600}{4157}$ K

Chemistry

134. A primary-standard sample of pure anhydrous Na_2CO_3 weighs $\frac{159}{500}$ g. It is titrated with0.100M HCl. Find the additional acid volume in mL needed after the first carbonate-to-hydrogencarbonate equivalence point to reach complete conversion to CO_2/water. Take Na_2CO_3=106g/mol.

(A) 61 (B) 30

(C) 60 (D) 15

135. propan-2-ol is subjected to acidified dichromate. Which product follows the stated regioselectivity or oxidation/dehydration endpoint?

(A) propane (B) propanoic acid

(C) propanone (D) propanal

136. For ethyl acetoacetate and ethyl acetate,which structural-reactivity comparison is correct under ordinary enolate/aldol conditions?

(A) ethyl acetoacetate has the more acidic alpha H (B) neither has an alpha H

(C) ester groups prevent all enolisation (D) ethyl acetate has the more acidic alpha H

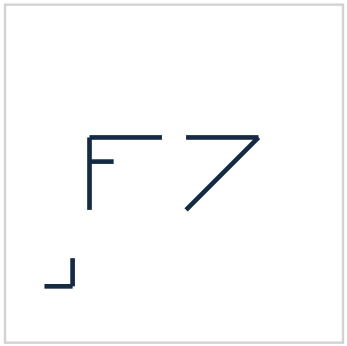
Logical Reasoning

137. Preserve the exact relation and its order: Carpenter : Saw :: ? : Whisk. Choose the missing term.

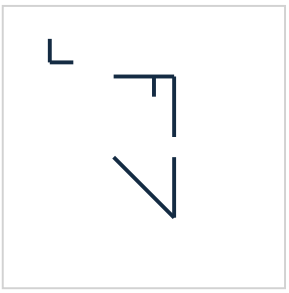
(A) Painter (B) Chef

(C) Tailor (D) Carpenter

138. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



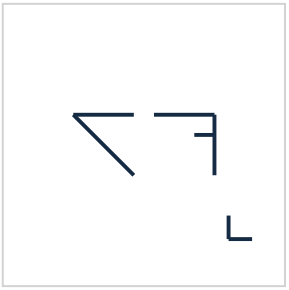
(A) Pattern R



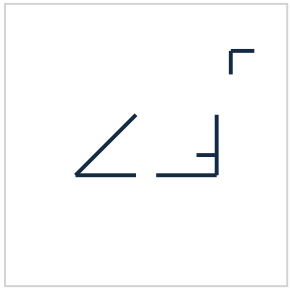
(B) Pattern P



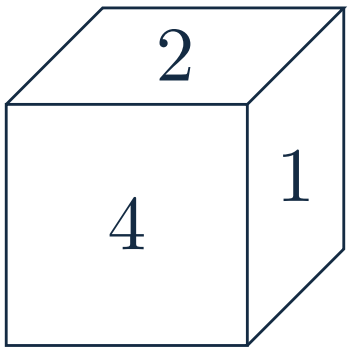
(C) Pattern Q



(D) Pattern N



139. This standard die has opposite labels summing to seven. Roll it once to the right, so its present top face becomes the right face. Give the new (top, front) labels.



(A) $(6, 4)$
(B) $(6, 3)$
(C) $(2, 4)$
(D) $(1, 4)$

Mathematics

140. Let k make lim as x→ ∞ of [$(4x^2+2x+1)/(x+3)$-kx] finite. Find that finite limit.

(A) -10
(B) 12
(C) 2
(D) 14

141. A vector has direction ratios (7,7,2). Find the sum of the squares of its y,z direction cosines.

(A) $\frac{\sqrt{5406}}{102}$
(B) $\frac{53}{102}$
(C) 53
(D) $\frac{49}{102}$

142. How many integers k make 24sin x+32cos x+4=k have exactly two distinct solutions in $0 \leq x < 2\pi$?

(A) 39
(B) 80
(C) 79
(D) 81

BITSAT Mock 18

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. Two large parallel plates are held at potentials +600 V and −600 V, with gap 7 mm. Neglect edge effects. Find field magnitude in the gap.

(A) $\frac{1200000}{7}$ V/m (B) $\frac{1200}{7}$ V/m

(C) $\frac{120000}{7}$ V/m (D) $\frac{12000000}{7}$ V/m

2. Two identical uniform spheres touch. Their radii are multiplied by 8 while their density is divided by 8. By what factor does their gravitational attraction change?

(A) $\frac{1}{64}$ (B) $\frac{1}{4096}$

(C) 262144 (D) 64

3. A particle executes SHM with $v^2 = 144 - 4x^2$ in SI units. Find its displacement magnitude when kinetic energy is three times potential energy.

(A) 3 m (B) $3\sqrt{3}$ m

(C) $2\sqrt{3}$ m (D) 6 m

4. At the highest point of its flight, at height 35 m, a projectile has horizontal velocity 25 m/s. It explodes into two equal fragments, both initially moving horizontally. One moves forward at 5 m/s. Find the horizontal distance travelled by the other fragment from the explosion point to the ground. Take $g = 10$ m/s^2.

(A) 315 m (B) $45\sqrt{7}$ m

(C) $25\sqrt{7}$ m (D) $55\sqrt{7}$ m

5. In YDSE wavelength is 500 nm, screen distance 4 m and slit separation 4 mm. Find the separation on the same side of the centre between the 4th bright fringe and the 8th dark fringe. Count the first dark fringe as order 1.

(A) $\frac{7}{4}$ mm (B) $\frac{15}{4}$ mm

(C) $\frac{7}{8}$ mm (D) $\frac{9}{4}$ mm

6. In an ideal intrinsic semiconductor at zero temperature, the band gap is $\frac{3}{2}$ eV. Neglect phonons, excitons and all transitions within a band. Two beams have photon energies $\frac{7}{5}$ eV and $\frac{8}{5}$ eV. Which can create a valence-to-conduction electron-hole pair by a single-photon absorption?

(A) Both beams
(B) Only the lower-energy beam
(C) Neither beam
(D) Only the higher-energy beam

7. A gas receives 400 J of heat while its surroundings do 50 J of work on it. It then loses 200 J of heat at fixed volume. Find its total internal-energy change.

(A) 450 J
(B) 250 J
(C) 150 J
(D) 200 J

8. A galvanometer has resistance 40 ohms and full-scale current $\frac{1}{1000}$ A. It is to measure a total current 4 times its own full-scale current. Find the required parallel shunt resistance.

(A) $40\,\Omega$
(B) $\frac{160}{3}\,\Omega$
(C) $160\,\Omega$
(D) $\frac{40}{3}\,\Omega$

9. In a series RLC circuit, ideal rms voltmeters read 70 V across R, 90 V across L and 30 V across C. Find source rms voltage.

(A) $10\sqrt{85}$ V
(B) $10\sqrt{193}$ V
(C) $10\sqrt{139}$ V
(D) 190 V

10. One mole of a monatomic ideal gas follows $pV^{11} = \text{constant}$. Find the process molar heat capacity in units of R.

(A) $\frac{11}{10}$
(B) $\frac{7}{5}$
(C) $\frac{31}{22}$
(D) $\frac{8}{5}$

11. A standing wave is $y = 3\sin(6x)\cos(4t)$ cm with x in metres and t in seconds. At a point where $6x = \pi/6$, find the particle-speed magnitude when $4t = \pi/3$.

(A) 6 cm/s
(B) $3\sqrt{3}$ cm/s
(C) $\frac{9\sqrt{3}}{2}$ cm/s
(D) 12 cm/s

12. A thin glass lens of refractive index 3/2 has focal length 80 cm in air. It is completely immersed in a liquid of index 4/3. Find its new focal length.

(A) 20 cm
(B) 240 cm
(C) 320 cm
(D) $\frac{320}{3}$ cm

13. A network is defined by $P = \text{NOR}(A, B)$, $Q = \text{NOR}(A, P)$ and $Y = \text{AND}(Q, \overline{B})$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 0$ and $A = 1$?

(A) 3
(B) 0
(C) 2
(D) 1

14. A 3 ohm resistor is connected in parallel with an unknown resistor across 18 V. The source supplies 162 W in total. Find the unknown resistance.

(A) $\frac{6}{5}\,\Omega$ (B) $1\,\Omega$

(C) $6\,\Omega$ (D) $5\,\Omega$

15. A boat moves at 20 m/s relative to water. The current is 12 m/s and river width is 60 m. The boat lands directly opposite its starting point. Find crossing time.

(A) $\frac{75}{16}\,\text{s}$ (B) $3\,\text{s}$

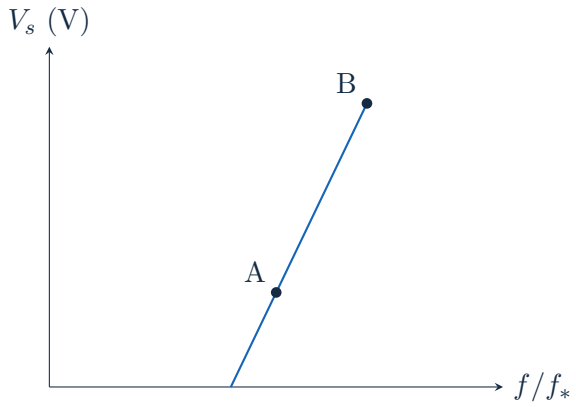
(C) $\frac{15}{4}\,\text{s}$ (D) $\frac{15}{2}\,\text{s}$

16. A rod of length 6 m pivots at its lower end. A peg constrained to move horizontally stays in contact with its upper tip. At an instant the rod makes an angle with $\sin\theta = 3/5$ to the horizontal and the tip's horizontal speed has magnitude 3 m/s. Find the rod's angular-speed magnitude.

(A) $\frac{5}{8}\,\text{rad/s}$ (B) $\frac{5}{6}\,\text{rad/s}$

(C) $\frac{3}{10}\,\text{rad/s}$ (D) $3\,\text{rad/s}$

17. For a metal, the stopping-potential graph has points A $(5f_*, 5\,\text{V})$ and B $(7f_*, 15\,\text{V})$. The electronic gain of the vertical recording channel is later found to be 3: its recorded voltage is 3 times the true stopping voltage. What is hf_*/e in volts?



(A) $\frac{5}{3}\,\text{V}$ (B) $15\,\text{V}$

(C) $\frac{5}{7}\,\text{V}$ (D) $\frac{1}{3}\,\text{V}$

18. A particle of mass 3 kg starts from rest. A force $\vec{F} = 2t\,\hat{i} + 4\,\hat{j}$ N acts for 2 s. What is the magnitude of its momentum at that instant?

(A) $4\,\text{kg m/s}$ (B) $12\,\text{kg m/s}$

(C) $4\sqrt{5}\,\text{kg m/s}$ (D) $8\sqrt{2}\,\text{kg m/s}$

19. A point charge $21q$ is placed at a vertex of a cube. Find the electric flux through the three faces not meeting that vertex, combined, in units of q/ε_0.

(A) $\frac{21}{8}$ (B) $\frac{7}{2}$

(C) $\frac{21}{16}$ (D) $\frac{7}{8}$

20. A point at the bottom of two plane liquid layers is viewed normally from air. The upper layer has thickness 6 cm and index 3/2; the lower has thickness 20 cm and index 4/3. Find its apparent depth below the top surface in the paraxial limit.

(A) $\frac{39}{2}$ cm (B) 15 cm

(C) 19 cm (D) $\frac{107}{3}$ cm

21. Current rises linearly from 0 to 6 A over 2 s and then falls linearly to 0 over another 2 s. It remains zero for the next 4 s. Find the average current over the entire 8 s interval.

(A) 2 A (B) $\frac{3}{2}$ A

(C) 6 A (D) $\frac{3}{4}$ A

22. A nucleus of mass number 28 splits into two equal nuclei. The parent has binding energy per nucleon $\frac{15}{2}$ MeV and each daughter has $\frac{81}{10}$ MeV. Ignoring other products, what energy is released in the split?

(A) $\frac{84}{5}$ MeV (B) 210 MeV

(C) $\frac{1134}{5}$ MeV (D) $\frac{1218}{5}$ MeV

23. A particle moves with constant acceleration. It travels 13 m in the (6)th second and 15 m in the 7th second. What was its initial velocity?

(A) 15 m/s (B) 1 m/s

(C) 3 m/s (D) 2 m/s

24. A plane wave in vacuum travels along $+x$. At one instant its electric field is $180\hat{k}$ V/m. If the electric amplitude is 4 times this instantaneous value, which gives the instantaneous magnetic field? Use $c = 3 \times 10^8$ m/s.

(A) $+60 \times 10^{-8}\hat{j}$ T (B) $-240 \times 10^{-8}\hat{j}$ T

(C) $-60 \times 10^{-8}\hat{j}$ T (D) $+60 \times 10^{-8}\hat{i}$ T

25. A particle has potential $U(x, y) = 4x^2y + 6y$ J in SI units. Find the force magnitude at $(4, 4)$ m.

(A) 198 N (B) 70 N

(C) $2\sqrt{2249}$ N (D) $2\sqrt{5321}$ N

26. Masses 4 kg and 7 kg hang from a light inextensible string over a pulley with moment of inertia I and radius R such that $I/R^2 = 3$ kg. There is no slipping or axle friction. Find acceleration magnitude. Take $g = 10$ m/s^2.

(A) $\frac{15}{7}$ m/s^2 (B) $\frac{30}{17}$ m/s^2

(C) $\frac{15}{4}$ m/s^2 (D) 10 m/s^2

27. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 10 percent. The measured increase in its absolute temperature is $\frac{399}{5}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) 798 K (B) $\frac{2299}{5}$ K

(C) $\frac{38000}{121}$ K (D) 380 K

28. Here L is inductance, C capacitance, R resistance, V potential, I current and t time. Which expression has the same dimensions as both $(CV)^1$ and $(It)^1$?

(A) $(V/R)^1$ (B) $((V/R)t)^1$

(C) $(LI)^1$ (D) $(RC)^1$

29. In a thin-foil scattering experiment, most alpha particles are barely deflected, but 22 of 500000 incident particles undergo large-angle scattering. Which paired inference is supported without assuming the Thomson model?

(A) Atoms contain no electrons; positive charge is uniformly distributed.

(B) Most atomic volume is nearly empty; positive charge is concentrated in a very small central region.

(C) Most atomic mass is spread uniformly; central charge is negative.

(D) Most atomic volume is positively charged; electrons cause the rare large deflections.

30. 125 identical uncharged liquid drops merge into one drop without evaporation. Find the fraction of the initial surface energy released; surface tension is constant.

(A) 1 (B) $\frac{1}{5}$

(C) 4 (D) $\frac{4}{5}$

Chemistry

31. Construct one open linear peptide using each member of the following multiset exactly once: Pro, Gly, Ala, Ser, Gly. The N-terminal residue isPro and the C-terminal residue isGly. The two Gly residues are indistinguishable; do not label them as different species. Count distinctN-to-C sequences; ignore stereoisomerism of the supplied residues.

(A) 7 (B) 6

(C) 24 (D) 120

32. A one-electron ion has nuclear charge Z=5. A bound level has the same energy as H at n=2. In that ion shell,maximum electrons allowed by Pauli in orbitals with$m_l = 0$ would be (ignore electron interactions for the energy comparison)

(A) 200 (B) 10

(C) 20 (D) 100

33. An100mL sample contains $\frac{1}{50}$ mol HA,pKa=5,and $\frac{1}{100}$ mol HCl. Add100mL $\frac{3}{10}$ M NaOH. Use usual weak-base salt hydrolysis and $\log_{10}(\frac{1}{10})=-1$. Final pH is

(A) 9.0 (B) 13

(C) 7 (D) 5

34. A textbook deficiency table associates scurvy with one vitamin. Which vitamin and solubility class are correctly paired?

(A) Vitamin B2; water-soluble. (B) VitaminA; fat-soluble.

(C) Vitamin C; water-soluble. (D) VitaminD; fat-soluble.

35. Ignoring special ortho effects,which decreasing acid-strength order is correct for p-nitrophenol,phenol, p-cresol?

(A) p-nitrophenol > p-cresol > phenol

(B) p-nitrophenol > phenol > p-cresol

(C) phenol > p-nitrophenol > p-cresol

(D) p-cresol > phenol > p-nitrophenol

36. An atomic emission spectrum contains separated lines. Which feature of Bohr's model addresses this observation, rather than merely describing a positive nucleus?

(A) Transitions between discrete stationary energies emit photons with hν equal to the energy difference.

(B) Each spectral line is emitted without any change in atomic energy.

(C) The nuclear charge alone forbids every possible electron transition.

(D) A continuous range of electron energies necessarily gives only discrete lines.

37. For $HCOO^-$,use ordinary octet Lewis structures and resonance. What is the pair (average bond order of the two C-O bonds,total lone pairs on all atoms of that same named species in one contributing structure)?

(A) ($\frac{3}{2}$, 5).

(B) (2, 6).

(C) ($\frac{3}{2}$, 4).

(D) (1, 5).

38. For$2A(g) \rightleftharpoons A_2(g)$,the numerical K_c is1.00 at a T where RT=25 L bar/mol. Find numerical K_p for$1/2\ A_2(g) \rightleftharpoons A(g)$,using matching concentration and pressure conventions.

(A) $\frac{1}{25}$

(B) $\frac{1}{5}$

(C) 5

(D) 25

39. Standard reduction potentials for A^+/A,B^+/B,C^+/C are $\frac{3}{5}$,$\frac{2}{5}$,$\frac{7}{10}$ V. Which pair of claims is correct under standard conditions?

(A) C^+ oxidises A,but doubling coefficients doubles the standard cell potential to $\frac{1}{5}$ V.

(B) A^+ oxidises C with standard cell potential $\frac{1}{10}$ V.

(C) B^+ oxidises A spontaneously because $\frac{2}{5}$ V is positive.

(D) C^+ oxidises A;doubling all coefficients leaves standard cell potential $\frac{1}{10}$ V unchanged.

40. Bond energies $D(X_2):D(Y_2):D(XY)=1:2:3$. Forming2.00 mol XY(g) from the gaseous elements releases 300 kJ. $D(Y_2)$,in kJ/mol,is

(A) 300

(B) 200

(C) 400

(D) 100.0

41. A weak1:1electrolyte has$\alpha \ll 1$ and obeys$Ka \approx C\alpha^2$. Dilute it by a factor16without changing total formula-unit amount or Ka,and remain in that approximation. Which pair(degreeα,totalfree-ion amount)is correct?

(A) α increases byfactor16,buttotalionsdecreasebyfactor16.

(B) Both remain unchanged because the number of solute formula units is fixed.

(C) Both increase by factor4.

(D) α decreases byfactor4,andtotalionsareunchanged.

42. For 2A+B→3C+4D, A also disappears by an independent channel producing no C or D, and D is independently consumed at 1/2000 M/s. Measured total A disappearance is 3/500 M/s and net D appearance is 11/2000 M/s. What is the A disappearance rate through its independent channel, in M/s?

(A) 9/1000 (B) 13/4000

(C) 1/250 (D) 3/1000

43. 4-nitrotoluene is treated with Fe/HCl,then basification. Which free organic product follows the nitro-group transformation and retained C-N connectivity?

(A) 4-aminobenzoic acid (B) 4-methylaniline

(C) benzylamine (D) 3-methylaniline

44. A student assigns four bonding pairs and no lone pair to the central atom of NH_4^+. After independently checking valence-electron count,which combined molecular shape/domain count is consistent?

(A) Trigonal pyramidal; four electron domains. (B) See-saw; five electron domains.

(C) Square planar; six electron domains. (D) Tetrahedral; four electron domains.

45. Consider the statements. I. Settling can remove suspended solids without removing every dissolved ion. II. Clear-looking water is therefore necessarily free from dissolved pollutants. Choose the correct truth pattern.

(A) I false; II true. (B) I true; II false.

(C) I false; II false. (D) I true; II true.

46. Consider I: BCl3 is an electron-pair acceptor. II: Carbon shows stronger ordinary catenation than silicon. Which assessment correctly recognises both anomalous first-element behaviours?

(A) Both statements are correct. (B) Only I is correct.

(C) Only II is correct. (D) Neither statement is correct.

47. Compounds A and B have molecular formulae $(CH_2O)_5$ and $(CH_2O)_6$. Pure equal-mass samples are compared. Which joint statement is necessarily true,independently of functional-group identity?

(A) The oxygen-atom amounts are in ratio $\frac{5}{6}$,while molecule amounts are equal.

(B) The samples contain equal total oxygen-atom amounts,but A contains $\frac{6}{5}$ times as many molecules as B.

(C) A contains $\frac{5}{6}$ times as many molecules as B,and has a different oxygen mass percentage.

(D) Both samples contain equal molecule amounts because their empirical formulae agree.

48. A freshly assayed 30-volume H_2O_2 solution releases 30 L O_2 at STP per litre on complete decomposition. Only80.0% of the evolved oxygen reaches a burner that requires0.560 L O_2 at STP. The solution volume needed,in mL,is

(A) $\frac{28}{3}$ (B) $\frac{70}{3}$

(C) $\frac{140}{3}$ (D) $\frac{56}{3}$

49. 1-bromopropane is taken through alcoholic KOH and heat;HBr without peroxide;Mg/dry ether; water,with suitable separate-stage work-ups. Identify the final organic product without adding or deleting carbon atoms except when the reagent itself introduces carbon.

(A) propane
(B) butane
(C) propan-2-ol
(D) propene

50. Both metals in $[Co(NH_3)_6][Cr(CN)_6]$ are formally +3. Which proposed partner preserves the total composition while exchanging ligand sets between two coordination spheres?

(A) $[Co(NH_3)_6][Cr(CN)_6]$ with only the counter-ion positions in the written formula reversed.

(B) The same salt with one ligand merely renamed; no atoms or donor attachments change.

(C) $[Co(CN)_6][Cr(NH_3)_6]$

(D) $[Co(NH_3)_5Cl]Cl_2$ formed by adding chloride to the sample.

51. A cathode receives $\frac{3}{25}$mol electrons. Its only products are Cu from Cu^{2+} and $\frac{3}{250}$mol H_2. Cu M64. Deposited copper mass in g is

(A) $\frac{384}{125}$
(B) $\frac{432}{125}$
(C) $\frac{192}{25}$
(D) $\frac{96}{25}$

52. The same element forms a fully occupied BCC phase A with cell edge a and an FCC phase B with edge 1a. The measured density ratio ρ_B/ρ_A is 3/5. All FCC sites have equal occupancy, atomic mass is unchanged, and vacancies are the only defects. What fraction of FCC sites is vacant?

(A) 13/20
(B) 7/30
(C) 7/20
(D) 7/10

53. but-1-ene reacts with HBr with peroxide. Which pair gives the major constitutional product and the intermediate whose greater stability fixes its orientation? Ignore stereoisomer counting.

(A) 2-bromobutane; secondary radical
(B) 2-bromobutane; primary radical
(C) 1-bromobutane; primary radical
(D) 1-bromobutane; secondary radical

54. A central atom has five electron domains: four bond pairs and one lone pair. In the elementary trigonal-bipyramidal VSEPR model, which placement and molecular shape minimise90° lone-pair interactions?

(A) No lone pair; trigonal bipyramidal molecular shape.
(B) Equatorial lone pair; seesaw molecular shape.
(C) Axial lone pair; square planar molecular shape.
(D) Equatorial lone pair; tetrahedral molecular shape.

55. heptane-2,6-dione,CH3CO(CH2)3COCH3 undergoes dilute base,then heat with dehydration. Under the stated intramolecular closure pathway,which isolated product matches both ring size and work-up endpoint?

(A) 3-methylcyclopent-2-en-1-one
(B) cycloheptanone
(C) 2-methylcyclohexanone
(D) 3-methylcyclohex-2-en-1-one

56. Two samples have similar linear repeat chemistry. SampleA consists of uncrosslinked chains; sampleB has an extensive permanent covalent network. Which qualitative difference is most defensible on reheating below decomposition?

(A) B must flow more freely because covalent links remove constraints.

(B) A andB must be different elements.

(C) A can exhibit thermoplastic flow; B cannot flow as independent chains.

(D) Both must hydrolyse because all polymers are esters.

57. At 380 K, ΔH for C(graphite)+O2(g)→CO2(g) is -420 kJ/mol and for CO(g)+(1/2)O2(g)→CO2(g) is -290 kJ/mol. Carbon is converted only into CO, producing 5/2 mol CO. Assuming ideal gases and R=0.008314 kJ/mol/K, what is total ΔU in kJ? Use the stated reaction extents.

(A) 6578983/20000

(B) -6578983/10000

(C) -6578983/40000

(D) -6578983/20000

58. In a hypothetical rule,$l = 0, 1, \ldots, n$ is allowed. Keep subshell capacity2(2l+1),Hund and Pauli,and fill by n+l then lower n. For 21 electrons,the last occupied subshell and its electron count are

(A) $3d^3$.

(B) $2p^3$.

(C) $2d^3$.

(D) $3p^3$.

59. A sample contains $\frac{3}{100}$ mol $KClO_3$ and a small amount of MnO_2 catalyst.75.0% of the chlorate decomposes by $2KClO_3 \rightarrow 2KCl + 3O_2$. The catalyst is recovered unchanged. Find O_2 formed in mol.

(A) $\frac{27}{800}$

(B) $\frac{27}{400}$

(C) $\frac{9}{200}$

(D) $\frac{9}{400}$

60. A pure KMnO4 portion gives 3/200 mol O2 on complete controlled heating by2KMnO4→K2MnO4+MnO2+O2. With KMnO4=158 g/mol, which pair gives the initial mass and a viable origin of its intense purple colour despite Mn(VII) being d0?

(A) 237/50 g; charge-transfer absorption.

(B) 237/50 g; d-d absorption within an occupied d shell.

(C) 237/25 g; d-d absorption within an occupied d shell.

(D) 237/100 g; charge-transfer absorption.

English Proficiency

61. The alarm can be heard from the next room, though it cannot be seen. Which adjective fits?

(A) legible

(B) visible

(C) vocal

(D) audible

62. Choose the correctly spelt word.

(A) separate

(B) sepparate

(C) seperrate

(D) seperate

63. Choose the most coherent order of the four sentences. A. The conductor located it in a note shared by two sections. B. Correcting that note improved the combined performance without changing either section's rhythm. C. The choir practised each part separately before its first joint rehearsal. D. When the parts were combined, the singers heard a clash that had gone unnoticed.

(A) C-D-B-A

(B) C-D-A-B

(C) D-C-A-B

(D) C-A-D-B

64. Read the sentence: The wall was intended as a permanent division, not a temporary screen. Choose the opposite in meaning of 'permanent' as used in the sentence.

(A) temporary (B) complete

(C) solid (D) fixed

65. Choose the pair with the same ordered relationship as AUTHOR : NOVEL.

(A) PAPER : PRINTER (B) ACTOR : AUDIENCE

(C) COMPOSER : SYMPHONY (D) READER : LIBRARY

66. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The librarian is going to display the restored map beside its damaged cover.

(A) The restored map is going to display the librarian beside its damaged cover.

(B) The damaged cover is going to be displayed instead of the restored map.

(C) The restored map is going to be displayed by the librarian beside its damaged cover.

(D) The restored map was displayed by the librarian beside its damaged cover.

67. Choose the most coherent order of the four sentences. A. The reading group initially discussed every chapter in publication order. B. The next discussion followed that list and compared the ideas directly. C. Members soon found that related arguments were scattered across several chapters. D. They drew up a thematic list linking those arguments.

(A) A-D-C-B (B) A-C-D-B

(C) C-A-D-B (D) A-C-B-D

68. Read the sentence: The permissive policy allowed considerable freedom in choosing methods. Choose the opposite in meaning of 'permissive' as used in the sentence.

(A) flexible (B) informal

(C) tolerant (D) restrictive

69. Choose the active-voice equivalent that retains the tense, participants and all stated restrictions. No fee was charged by the clinic for the initial consultation.

(A) The patient charged no fee to the clinic for the initial consultation. (B) The clinic charged no fee for the initial consultation.

(C) The clinic charged a small fee for the initial consultation. (D) The clinic did not charge a fee for any consultation.

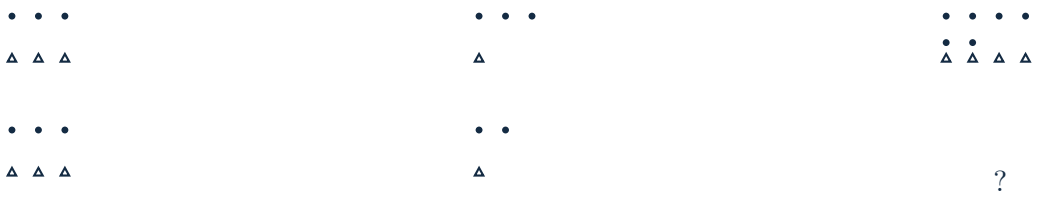
70. Read the sentence: Without a backup, the archive was vulnerable to a single equipment failure. Choose the closest meaning of 'vulnerable' as used in the sentence.

(A) susceptible to harm (B) dependent on outside help

(C) certain to be damaged (D) difficult to repair

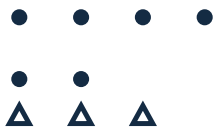
Logical Reasoning

71. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern Q

(B) Pattern L





(C) Pattern P

(D) Pattern U





72. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

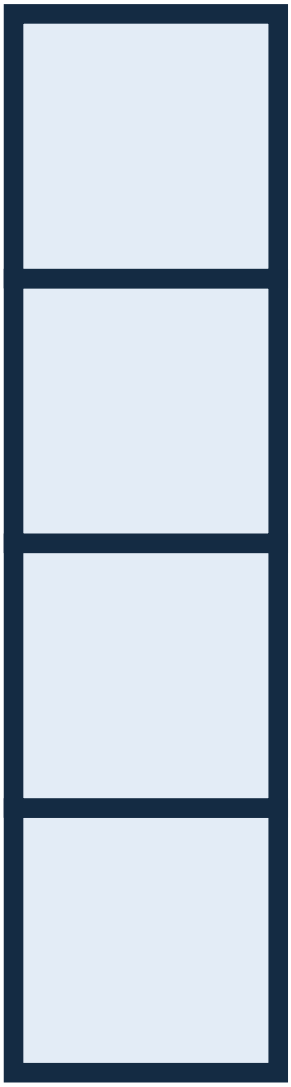


piece 1

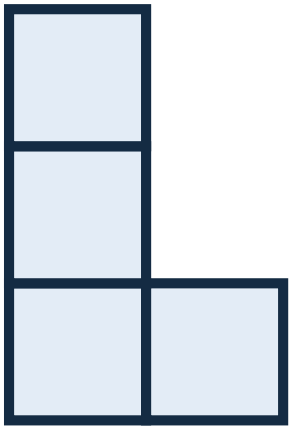


piece 2

(A) Pattern Z



(B) Pattern X



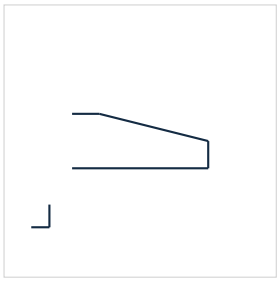
(C) Pattern Y



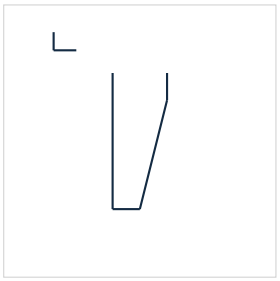
(D) Pattern K



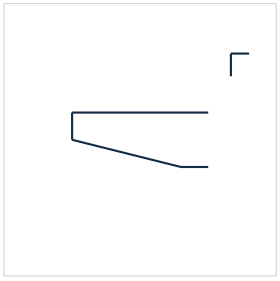
73. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

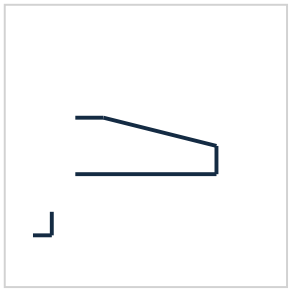


bottom left

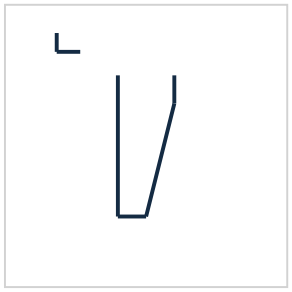
?

bottom right

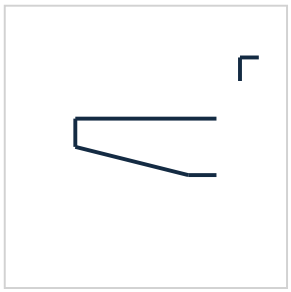
(A) Pattern M



(B) Pattern N



(C) Pattern P



(D) Pattern U



74. In a row, Asha is 12th from the left and 21st from the right. Bina is 10th from the right. Charu stands 4 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 6

(B) 8

(C) 5

(D) 7

75. Preserve the exact relation and its order: Rapid : Swift :: ? : Brave. Choose the missing term.

(A) Begin (B) Silent

(C) Courageous (D) Rapid

76. The recorded average of 10 readings is 31. One entry recorded as 53 should have been 51. After that correction, another reading of 23 is replaced by 32. What is the final average?

(A) $\frac{327}{10}$ (B) $\frac{317}{10}$

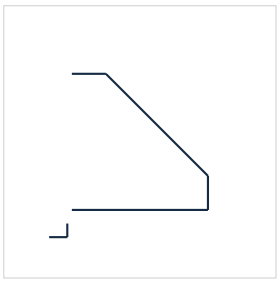
(C) $\frac{319}{10}$ (D) $\frac{154}{5}$

77. Consider only these statements; universal statements alone do not imply existence. Some artists are joggers; All artists are joggers. Conclusions: I. No editors are joggers. II. Some joggers are artists. Which conclusions follow necessarily?

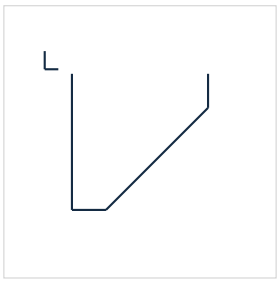
(A) Neither I nor II (B) Only I

(C) Only II (D) Both I and II

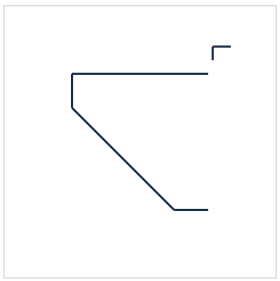
78. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1



2

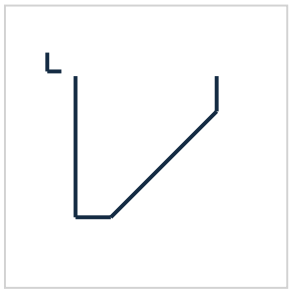


3

(A) Pattern T



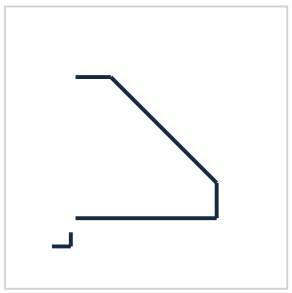
(B) Pattern R



(C) Pattern P



(D) Pattern Q

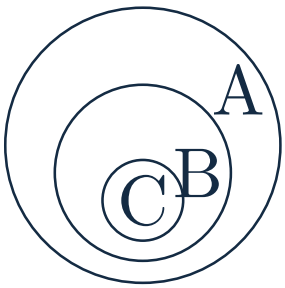


79. Choose the diagram that represents all these conditions, including the strictness information: No A is B, no B is C, and no A is C. Each class is nonempty.

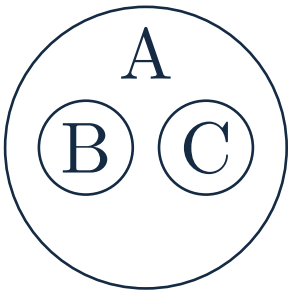
(A) Pattern Z



(B) Pattern V



(C) Pattern W



(D) Pattern R



80. Consider only these statements; universal statements alone do not imply existence. All campers are artists; All campers are bakers; Some artists are campers. Conclusions: I. All bakers are campers. II. All artists are bakers. Which conclusions are individually possible without contradicting the statements?

(A) Only II

(B) Both I and II

(C) Only I

(D) Neither I nor II

81. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

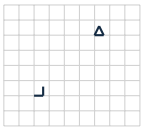
(A) 3

(B) 1

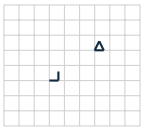
(C) 4

(D) 2

82. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



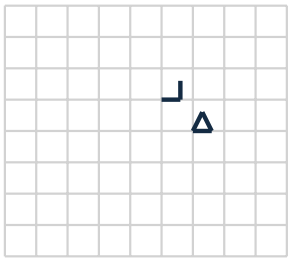
1



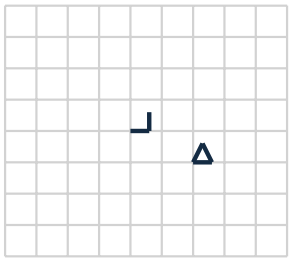
2

3

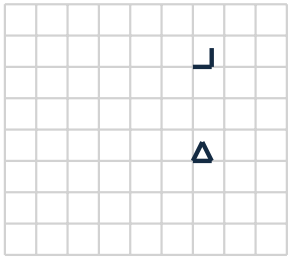
(A) Pattern M



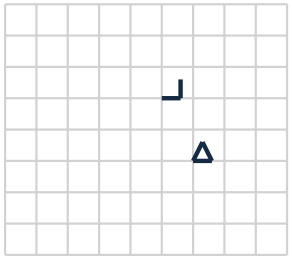
(B) Pattern Z



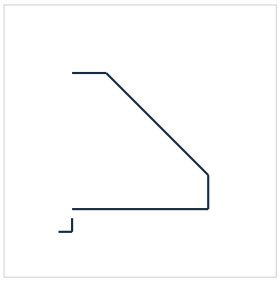
(C) Pattern V



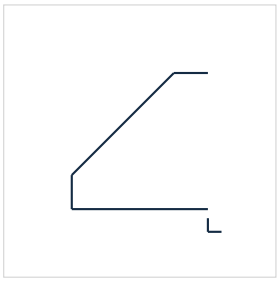
(D) Pattern P



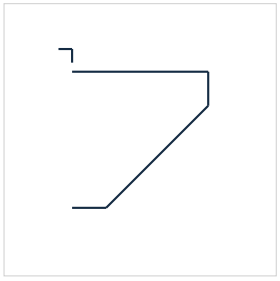
83. Select the missing bottom-right tile. Tiles reflect across the shared horizontal and vertical centre lines; there is no turning.



top left



top right

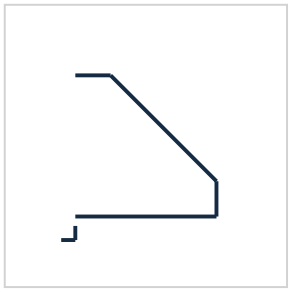


bottom left

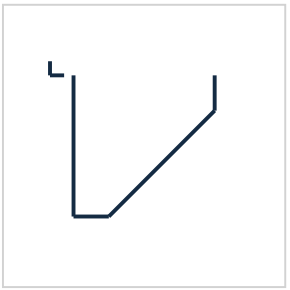
?

bottom right

(A) Pattern K



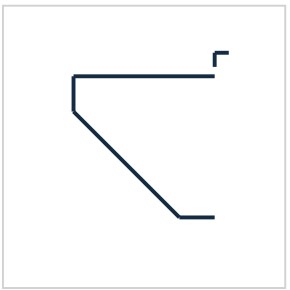
(B) Pattern P



(C) Pattern W

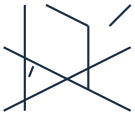


(D) Pattern V



84. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 2

(B) 4

(C) 3

(D) 1

85. Each rectangle is drawn to the same local scale. In three numbered figures the interior straight line divides the rectangle into equal areas. Which does not?



1



2



3



4

(A) 3

(B) 1

(C) 2

(D) 4

86. The increase from the left cell to the right cell is the same in each row, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



?

(A) Pattern N

(B) Pattern W





(C) Pattern Y

(D) Pattern M





87. A community register uses the following three categories: artists, editors, guides. Its complete relevant assertions are in this short paragraph; do not import everyday assumptions. No artists are editors. Some artists are not editors. Some editors are not artists. Conclusions: I. No editors are artists. II. Some artists are editors. Which conclusions follow necessarily?

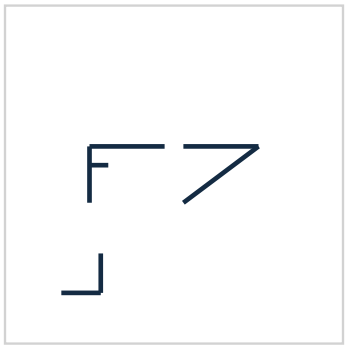
(A) Neither I nor II

(B) Only I

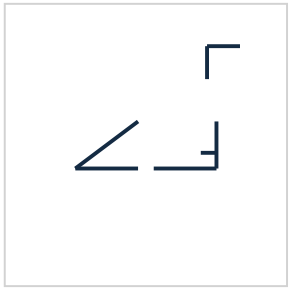
(C) Both I and II

(D) Only II

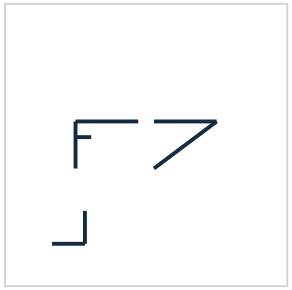
88. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



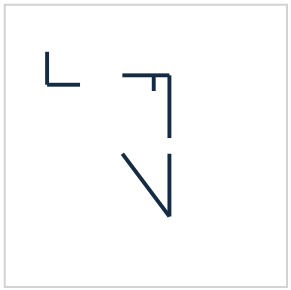
(A) Pattern Y



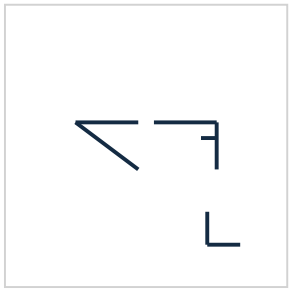
(B) Pattern M



(C) Pattern P



(D) Pattern V



89. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 2, 4, 11, 22, ?, 58, 65.

(A) 29 (B) 58

(C) 51 (D) 33

90. Five people occupy positions 1 to 5 from left to right, one per position, and wear five different shirt colours: red, blue, green, white and yellow. Esha is 1st from the left. Asha is 2nd from the left. Charu is 3rd from the left. Bina is 4th from the left. Dev is 5th from the left. The white shirt is in position 1. The green shirt is in position 3. The blue shirt is in position 5. Asha does not wear yellow. Which shirt does Asha wear?

(A) Red (B) Green

(C) White (D) Blue

Mathematics

91. $A = \begin{pmatrix} 4 & k \\ 0 & 3 \end{pmatrix}$ is symmetric. Find tr($A^T A$).

(A) 12 (B) 49

(C) 25 (D) 7

92. A point is equidistant from (8,0) and x=-8. Find the area between its locus, the y-axis, and y=8/2, y=8.

(A) $\frac{14}{3}$ (B) $\frac{2}{3}$

(C) $\frac{8}{3}$ (D) $\frac{16}{3}$

93. Under x=X+2, y=Y+5, the parabola $(y-5)^2=36(x-2)$ is normalized. Find the x coordinate of its focus in the original axes.

(A) -7 (B) 38

(C) 11 (D) 9

94. Find the sum of order and degree of $(d^2y/dx^2)^3+(dy/dx)^2+y=0$.

(A) 7 (B) 5

(C) 2 (D) 3

95. A circular sector has radius 11 and arc length $\frac{11\pi}{8}$. Find its area.

(A) $\frac{121\pi}{16}$ (B) $\frac{121\pi}{8}$

(C) $\frac{\pi}{16}$ (D) 121π

96. There are 6 Boolean variables p_1,...,p 6. How many truth assignments make [(p_1 implies p_2) and (p_1 implies not p_2)] true?

(A) 63 (B) 16

(C) 64 (D) 32

97. A line has direction (1,3,2). A plane has normal (2,1,3). Let θ be their acute angle. Find $\tan^2\theta$.

(A) $\frac{121}{75}$ (B) $\frac{75}{196}$

(C) $\frac{196}{121}$ (D) $\frac{121}{196}$

98. How many pairs $1\leq m<n\leq 7$ satisfy both $5 \mid (6^m+9^n)$ and $3 \mid (m+n)$?

(A) 8 (B) 5

(C) 12 (D) 4

99. Evaluate the limit of $[\cos(2x)-\cos(5x)]/x^2$ as $x\to 0$.

(A) 21 (B) $\frac{21}{2}$

(C) $-\frac{21}{2}$ (D) $\frac{3}{2}$

100. F′(x)=$2x-\frac{2}{x^3}$ and F(1)=3. Find F(2)-2F(1).

(A) $-\frac{3}{4}$ (B) $-\frac{21}{4}$

(C) $\frac{9}{4}$ (D) $\frac{21}{4}$

101. For the roots α,β of $z^2-(1+5i)z+(-12+5i)=0$, find $|\alpha\text{-}\beta|^2$.

(A) 13 (B) 52

(C) 26 (D) 169

102. How many integers x in [-2,12] satisfy (|x-5|-4)/(|x-5|-6)>0?

(A) 9 (B) 11

(C) 15 (D) 7

103. How many integer k in [0,108] make $x^2-15x+(54-k)=0$ have roots of opposite signs?

(A) 54 (B) 53

(C) 55 (D) 108

104. Maximize Z=3x+4y subject to x≥0, y≥3, x+y≤12, x+2y≤14.

(A) 39 (B) 36

(C) 38 (D) 28

105. Evaluate $\sum_{j=4}^{9}\binom{j}{3}$, with the convention $\binom{j}{r}=0$ when j<r.

(A) 205 (B) 209

(C) 120 (D) 210

106. Let $a_j=1/(5+2(j-1))$. Evaluate $\sum_{j=1}^{8}a_ja_{j+1}$.

(A) $\frac{8}{105}$ (B) $\frac{8}{19}$

(C) $\frac{7}{95}$ (D) $\frac{16}{105}$

107. A box contains 9 red and 7 blue distinct balls. Three are drawn uniformly without replacement. Given that at least one drawn ball is blue, find the probability that exactly two are red.

(A) $\frac{9}{20}$ (B) $\frac{8}{17}$

(C) 3 (D) $\frac{9}{17}$

108. In a group of 40, 24 study A, 20 study B, and 4 study neither. How many study exactly one of A, B?

(A) 8 (B) 36

(C) 28 (D) 44

109. For the ellipse $x^2/25+y^2/9=1$, find the distance from its right focus to its nearer directrix.

(A) 1 (B) $\frac{9}{4}$

(C) $\frac{25}{4}$ (D) $\frac{41}{4}$

110. The parabola $(y-3)^2=28(x-3)$ has focus F and vertex V. A circle centred at F passes through V. Find the tangent length from P=(24,17).

(A) $7\sqrt{3}$ (B) 343

(C) $14\sqrt{2}$ (D) $7\sqrt{7}$

111. Let $f(x)=x^3-27x+5$. Find the product of its local maximum and local minimum values.

(A) 3481 (B) -2916

(C) -8723 (D) -2891

112. The system $x+y+z=1$, $y+z=2$, $(k-3)z=k-7$ has no solution for which k?

(A) 0 (B) -3

(C) 7 (D) 3

113. Points on $r=(6,4,0)+t(1,0,0)$ lie at distance 2 from (6,4,0). Find the sum of the squares of their x coordinates.

(A) 8 (B) 144

(C) 80 (D) 72

114. X takes values 0,5,7 with probabilities 1/2,1/3,k. Find Var(X).

(A) $\frac{33}{2}$ (B) $\frac{289}{36}$

(C) $\frac{305}{36}$ (D) $\frac{\sqrt{305}}{6}$

115. $a=(5,0,0)$, $b=(0,6,0)$, $c=(0,0,3)$, $p=a+b$. Evaluate $[p\times(b\times c)]\cdot(a+c)$.

(A) 0 (B) -1224

(C) 324 (D) -324

116. Assuming $1+3+\cdots+(2n-1)=n^2$, the next-step sum is n^2+2n+1. At n=4, what square does this establish?

(A) 16 (B) 24

(C) 25 (D) 36

117. Find the squared distance between $6x+8y=16$ and $3x+4y=11$.

(A) $\frac{9}{100}$ (B) $\frac{9}{25}$

(C) 9 (D) $\frac{3}{5}$

118. A,B are acute, $A+B=\pi/4$ and $\tan A=\frac{4}{5}$. Find $\tan(A-B)$.

(A) $\frac{31}{45}$ (B) $-\frac{31}{49}$

(C) $\frac{31}{49}$ (D) $\frac{1}{9}$

119. On $A=\{0,1,\ldots,6\}$, define aRb if 5 divides a-b. How many ordered pairs are in R?

(A) 49 (B) 4

(C) 7 (D) 11

120. For f(x)=|x-2|/(x-6), on which maximal open intervals is f strictly decreasing?

(A) $(-\infty, 6)$ (B) $(2, 6) \cup (6, \infty)$

(C) $(2, \infty)$ (D) $(-\infty, 2)$

121. Evaluate $\sum_{j=1}^{9} j(j+4)$.

(A) 285 (B) 180

(C) 2205 (D) 465

122. In $(2x^2 - x^{-1})^8$, find the ratio of the coefficient of x^1 to that of x^4.

(A) $-\frac{5}{2}$ (B) $-\frac{2}{5}$

(C) $\frac{2}{5}$ (D) $\frac{4}{5}$

123. Evaluate $\sum_{j=0}^{3} \binom{3}{j}\binom{10}{2-j}$.

(A) 286 (B) 78

(C) 234 (D) 66

124. Find the area between y=(x-9)(x-13) and the x-axis for 0≤x≤16.

(A) $\frac{1264}{3}$ (B) $\frac{664}{3}$

(C) $\frac{1328}{3}$ (D) $-\frac{1264}{3}$

125. Circles $x^2+y^2=196$ and $(x-12)^2+y^2=196$ meet at A,B. Find $|AB|^2$.

(A) $8\sqrt{10}$ (B) 784

(C) 496 (D) 640

126. Four observations are 8,8,x,y with population mean 8 and variance 25/2. Find xy.

(A) 39 (B) 64

(C) 89 (D) 103

127. For a=(1,3,0), b=(0,1,6), c=(5,0,1), evaluate (a×c)·(a+b).

(A) 91 (B) −90

(C) −91 (D) 0

128. An AP has a_1=6, d=7. Find $a_1+a_3+\cdots+a_17$.

(A) 558 (B) 690

(C) 306 (D) 1116

129. $A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 3 \\ 3 & -3 & 0 \end{bmatrix}$. Find det(I+A)-(4+9+9).

(A) −1 (B) 1

(C) 0 (D) 22

130. Evaluate lim as x→0 of [sqrt(25+x)-5-x/10+x^2/1000]/x^3.

(A) $\frac{1}{50000}$

(B) $\frac{1}{25000}$

(C) 0

(D) $-\frac{1}{50000}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A uniform-density planet has radius R. Gravity at a depth d equals gravity at a height $(8-1)R$ above the surface. Find d/R.

(A) $\frac{511}{512}$
(B) $\frac{1}{64}$
(C) $\frac{63}{64}$
(D) 1

132. A monatomic ideal gas expands reversibly and adiabatically until its pressure becomes 1/32768 of its initial value. Find final temperature divided by initial temperature.

(A) $\frac{1}{64}$
(B) 64
(C) $\frac{1}{512}$
(D) $\frac{1}{1024}$

133. Two ideal cells of emfs 21 V and 4 V oppose each other in a loop containing two resistors each 4 ohms. Find the current magnitude.

(A) $\frac{21}{8}$ A
(B) $\frac{17}{8}$ A
(C) $\frac{25}{4}$ A
(D) $\frac{25}{8}$ A

Chemistry

134. Sucrose hydrolysis gives equal masses of glucose and fructose. Their specific rotations under the stated common conditions are+52.5°and-92.4°. Which net direction follows for the product mixture?

(A) No rotation because all monosaccharides are achiral.
(B) Optically inactive; equal masses always cancel.
(C) Levorotatory; the negative contribution has larger magnitude.
(D) Dextrorotatory; both products must inherit sucrose's sign.

135. Which pair contains only fat-soluble vitamins while avoiding the commonB12 exception to simplistic storage claims?

(A) VitaminsA andD.
(B) VitaminsB2 and B6.
(C) VitaminsB1 and C.
(D) VitaminsC and B12.

136. Two cysteine side-chain thiols form a disulphide bridge while the peptide backbone remains intact. Which change occurs?

(A) Every backbone amide is hydrolysed.
(B) Side-chainS-S connectivity changes without adding a new peptide bond.
(C) A glycosidic bond joins the two alpha carbons.
(D) Both sulphur atoms are removed as nitrogen gas.

Logical Reasoning

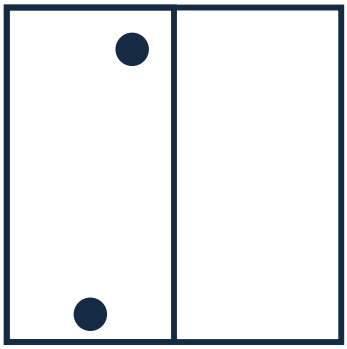
137. Preserve the exact relation and its order: Balance : Mass :: ? : Electric current. Choose the missing term.

(A) Voltmeter
(B) Thermometer
(C) Ammeter
(D) Barometer

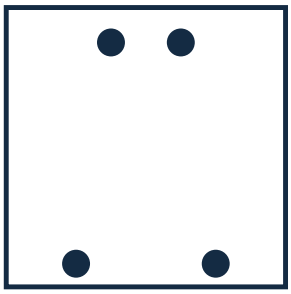
138. Consider only these statements; universal statements alone do not imply existence. Some hikers are hikers; All farmers are guides. Conclusions: I. Some hikers are farmers. II. No hikers are farmers. Which conclusion or complementary alternative must hold?

(A) Only I
(B) Either I or II follows, but neither is individually certain
(C) Both I and II
(D) Only II

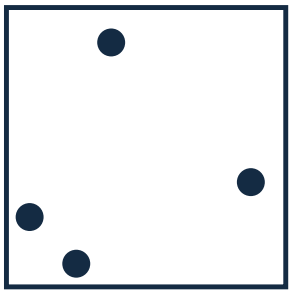
139. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



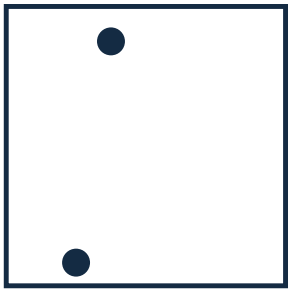
(A) Pattern U



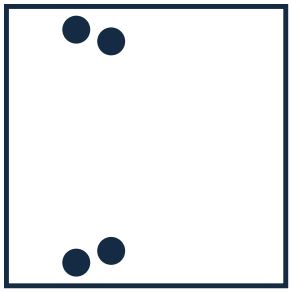
(B) Pattern Q



(C) Pattern Z



(D) Pattern N



Mathematics

140. $A = \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix}$. Find the (1,2) entry of $(I+A)^9$.

(A) 3 (B) 19683

(C) 27 (D) 9

141. Let k make lim as $x \to \infty$ of $[(6x^2+5x+1)/(x+3)-kx]$ finite. Find that finite limit.

(A) 18 (B) -13

(C) 23 (D) 5

142. ABC is a right triangle on level ground with AB=14, AC=14, angle A=90°. A vertical tower stands at midpoint M of BC. Its top elevation from A is 60°. Find its height.

(A) $7\sqrt{6}$ (B) $14\sqrt{6}$

(C) $7\sqrt{3}$ (D) $7\sqrt{2}$

BITSAT Mock 19

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A spring of constant 60 N/m is already stretched by $\frac{3}{10}$ m. How much additional work is required to increase its extension to $\frac{3}{5}$ m quasistatically?

(A) $\frac{81}{5}$ J
(B) $\frac{81}{10}$ J
(C) $\frac{27}{5}$ J
(D) $\frac{54}{5}$ J

2. One mole of a monatomic ideal gas follows $pV^{15} = \text{constant}$. Find the process molar heat capacity in units of R.

(A) $\frac{3}{2}$
(B) $\frac{5}{2}$
(C) $\frac{15}{14}$
(D) $\frac{10}{7}$

3. A travelling wave is $y = 4\sin[7(x - \frac{40}{7}t) + \pi/4]$ mm with x in metres and t in seconds. What is its frequency?

(A) $\frac{40}{\pi}$ Hz
(B) $\frac{20}{\pi}$ Hz
(C) $\frac{\pi}{20}$ Hz
(D) $\frac{7}{2\pi}$ Hz

4. A 5 kg block on a rough horizontal table is tied over an ideal pulley to a hanging 2 kg mass. The hanging mass moves downward and $\mu_k = 1/5$. Find string tension. Take $g = 10$ m/s^2.

(A) $\frac{50}{7}$ N
(B) $\frac{120}{7}$ N
(C) $\frac{160}{7}$ N
(D) 20 N

5. Two parallel plane mirrors separated by 5 cm extend from axial position 0 to $\frac{35}{2}$ cm. A ray enters midway between them at axial position 0, making 45° with the normal to the mirrors. How many reflections occur before it passes the far edge? Exclude a contact exactly on the exit edge.

(A) 1
(B) 3
(C) 4
(D) 5

6. A particle has position-dependent velocity components $\vec{v} = 4x\,\hat{i} - 4y\,\hat{j}$ in SI units. Given $dx/dt = v_x$ and $dy/dt = v_y$, what is its acceleration magnitude at $(6, 5)$ m?

(A) $176\,\mathrm{m/s^2}$ (B) $4\sqrt{61}\,\mathrm{m/s^2}$

(C) $80\,\mathrm{m/s^2}$ (D) $16\sqrt{61}\,\mathrm{m/s^2}$

7. In YDSE a sheet of index $\frac{5}{2}$ and thickness 8 micrometres covers one slit. A second sheet of the same thickness and index 3/2 covers the other. With wavelength $\frac{3}{5}$ micrometres, find the magnitude of central-fringe shift in units of fringe width.

(A) 8 (B) $\frac{100}{3}$

(C) $\frac{40}{3}$ (D) $\frac{20}{3}$

8. An ideal mixture contains 3 moles of monatomic gas and 1 moles of rigid diatomic gas. Vibrations are frozen. Find the heat-capacity ratio $\gamma = C_P/C_V$ of the mixture.

(A) $\frac{11}{7}$ (B) $\frac{5}{3}$

(C) $\frac{13}{9}$ (D) $\frac{7}{5}$

9. In an ideal intrinsic semiconductor at zero temperature, the band gap is $\frac{8}{5}$ eV. Neglect phonons, excitons and all transitions within a band. Two beams have photon energies $\frac{3}{2}$ eV and $\frac{17}{10}$ eV. Which can create a valence-to-conduction electron-hole pair by a single-photon absorption?

(A) Neither beam (B) Only the lower-energy beam

(C) Only the higher-energy beam (D) Both beams

10. A screw gauge advances 2.0 mm in four complete turns and has 100 circular divisions. With the jaws closed, its zero mark lies 5 divisions above the reference line; use the usual convention that a zero below the reference line gives positive zero error. With a wire, the main scale reads 4 mm and circular reading is 43 divisions. What is the corrected diameter?

(A) $\frac{1843}{200}$ mm (B) $\frac{106}{25}$ mm

(C) $\frac{891}{200}$ mm (D) $\frac{33721}{8000}$ mm

11. For a particle moving in one dimension, $v = 11 - 2x$ in SI units while $v > 0$. What is its acceleration at $x = \frac{11}{4}$ m?

(A) $-22\,\mathrm{m/s^2}$ (B) $-2\,\mathrm{m/s^2}$

(C) $0\,\mathrm{m/s^2}$ (D) $-11\,\mathrm{m/s^2}$

12. A ball is dropped from 8 m onto a horizontal fixed floor. At every impact the coefficient of restitution is $\frac{3}{5}$. Neglect air resistance. What maximum height does it reach after its second impact?

(A) $\frac{648}{625}$ m (B) $\frac{52488}{390625}$ m

(C) $\frac{24}{5}$ m (D) $\frac{72}{25}$ m

13. A coil of inductance 7 H carries current $i(t) = 6t^2$ A. Find the magnitude of its self-induced emf at $t = 3$ s.

(A) 252 V (B) 126 V

(C) 378 V (D) $\frac{36}{7}$ V

14. A long solenoid has 700 turns per metre and carries 6 A. It is completely filled by a linear magnetic core of relative permeability 4. Find magnetisation M.

(A) 12600 A/m (B) 21000 A/m

(C) 4200 A/m (D) 18 A/m

15. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 25 percent. The measured increase in its absolute temperature is $\frac{1755}{8}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) $\frac{1755}{2}$ K (B) 3510 K

(C) $\frac{3510}{41}$ K (D) 390 K

16. A wheel initially rotates at 7 rad/s. A constant opposing angular acceleration brings it to rest after 4 s. What angle does it turn through during the last half of this stopping interval?

(A) 28 rad (B) $\frac{7}{4}$ rad

(C) $\frac{7}{2}$ rad (D) 14 rad

17. A satellite of mass m is moved from a circular orbit of radius R to one of radius 5R about mass M. Find the required increase in total orbital energy in units of $GMm/(2R)$.

(A) $\frac{8}{5}$ (B) $\frac{6}{5}$

(C) $\frac{24}{25}$ (D) $\frac{4}{5}$

18. An object moves slowly along the axis toward a concave mirror. At an instant its distance from the mirror is 80 cm; the mirror focal-length magnitude is 20 cm. Find the image-speed magnitude divided by object-speed magnitude at that instant.

(A) $\frac{4}{3}$ (B) $\frac{1}{16}$

(C) $\frac{1}{9}$ (D) $\frac{1}{3}$

19. A nonrelativistic particle of fixed charge and mass moves perpendicular to a uniform magnetic field. Its kinetic energy is multiplied by 4 and the magnetic field is multiplied by 6. Find the new trajectory radius divided by the original radius.

(A) $\frac{1}{18}$ (B) $\frac{1}{3}$

(C) 12 (D) $\frac{2}{3}$

20. A metal wire has linear temperature coefficient $\frac{3}{1000}$ per degree Celsius referred to 0°C. Its dimensions are fixed and it stays across a constant voltage. Find power at 110°C divided by power at 0°C.

(A) $\frac{33}{133}$ (B) $\frac{100}{133}$

(C) $\frac{100}{67}$ (D) $\frac{133}{100}$

21. A charge of $2\,\mu$C moves from potential 60 V to -60 V in an electrostatic field. Find the work done by an external agent if the charge starts and ends at rest and no other force does work.

(A) $120\,\mu$J (B) $0\,\mu$J

(C) $-240\,\mu$J (D) $240\,\mu$J

22. A diamond network has four outer resistors AB, BC, CD and DA, each 9 ohms, and a bridge BD also 9 ohms. The DA arm additionally contains an ideal diode with anode at D and cathode at A. A source keeps A positive relative to C. Find equivalent resistance between A and C.

(A) $18\,\Omega$ (B) $\frac{27}{2}\,\Omega$

(C) $15\,\Omega$ (D) $9\,\Omega$

23. A Newtonian liquid of viscosity $\frac{1}{10}$ Pa s has velocity profile $v(y) = 6y + 1y^2$ in SI units. Find the shear-stress magnitude at $y = 4$ m.

(A) $\frac{7}{5}$ Pa (B) 1 Pa

(C) 4 Pa (D) 14 Pa

24. A 3 kg mass on a string passes the top of a vertical circle of radius 5 m with string tension 180 N. Find string tension at the bottom. Take $g = 10$ m/s^2.

(A) 360 N (B) 30 N

(C) 330 N (D) 180 N

25. A spherical blackbody's absolute temperature is multiplied by 2 while its radius is divided by 4. Neglect incoming background radiation. Find its emitted-power multiplier.

(A) $\frac{1}{8}$ (B) 16

(C) $\frac{1}{4}$ (D) 1

26. A small-angle pendulum has period T in a stationary lift. In a vertically accelerating lift its period is 8T. Find the downward lift acceleration divided by g.

(A) $\frac{63}{64}$ (B) $\frac{7}{8}$

(C) $\frac{1}{8}$ (D) 63

27. A wire carries $i(t) = 2 + 8\sin\omega t$ A for many cycles. Find its rms current.

(A) 6 A (B) $4\sqrt{2}$ A

(C) 2 A (D) 10 A

28. A circular parallel-plate capacitor of plate radius 4 cm is charged at 24 mA. Neglect fringing. What displacement current passes through the annular part of the gap between radii 3 cm and 4 cm, centred on the plate axis?

(A) $\frac{3}{2}$ mA (B) 42 mA

(C) $\frac{21}{2}$ mA (D) 24 mA

29. A drop of mass 5.12×10^{-17} kg is suspended in a vertical field of magnitude 800 N/C. It has excess electrons. One excess electron is removed without changing its mass. Find its immediate downward acceleration, taking $g = 10$ m/s^2 and $e = 1.6 \times 10^{-19}$ C.

(A) $\frac{5}{8}$ m/s^2 (B) 10 m/s^2

(C) $\frac{15}{2}$ m/s^2 (D) $\frac{5}{2}$ m/s^2

30. A wire consists of a semicircular arc of radius 4 m carrying 4 A, joined to two straight radial segments that extend outward from its ends. Find the magnetic field magnitude at the arc centre in units of μ_0.

(A) $\frac{2+\pi}{4\pi}$ (B) $\frac{1}{4}$

(C) $\frac{1}{8}$ (D) $\frac{1}{2\pi}$

Chemistry

31. Assign the formal metal oxidation state in $[Cr(CO)_6]$,using six neutral CO. For nitrosyl species use the explicitly stated formal NO^+ convention rather than inferring spin from it.

(A) 1 (B) 0

(C) 2 (D) -1

32. An ideal gas executes A→B→C→D→A through (V/L,P/bar):A(1,1),B(3,1),C(3,3),D(1,3). IUPAC work-on convention applies. Heats on AB,BC,DA are 400,300,-200 J. Heat on CD in J is (1 L bar=100 J)

(A) -900 (B) -400

(C) -100 (D) -500

33. Compare bromomethylcyclohexane and1-bromo-1-methylcyclohexane toward the same cyanide nucleophile in a polar aprotic solvent under ordinary SN2 conditions. Which reacts faster or is the substituted site,and why?

(A) 1-bromo-1-methylcyclohexane; an sp2 C-Br bond behaves just like an unhindered methyl bromide

(B) 1-bromo-1-methylcyclohexane; steric hindrance accelerates backside attack

(C) bromomethylcyclohexane; The exocyclic CH2Br is primary; the other C-Br centre is tertiary.

(D) bromomethylcyclohexane; the mechanism requires formation of a free tertiary carbocation

34. For the pair but-1-ene and but-2-ene,which classification identifies the structural difference rather than merely stating that the formula agrees?

(A) position isomers (B) functional isomers

(C) chain isomers (D) optical antipodes

35. A reaction has the same Arrhenius prefactor over the range. Its rate constant doubles from300K to360K. Temperature where its rate constant is four times the300K value,in K,is

(A) 600 (B) 420

(C) 450 (D) 360

36. In the gas phase,C_2 is converted into C_2^- by changing one electron. Under the elementary MO model, which final bond-order/unpaired-electron pair follows?

(A) Bond order $\frac{5}{2}$; 0 unpaired electrons. (B) Bond order $\frac{3}{2}$; 1 unpaired electrons.

(C) Bond order $\frac{5}{2}$; 1 unpaired electron. (D) Bond order 2; 0 unpaired electrons.

37. Compare D_2 and T_2 at equal temperature. Which statement preserves the distinction between mass/nuclear effects and electronic structure?

(A) An isotope with more neutrons must have a different number of protons and belong to another element.

(B) Changing H to D adds an electron and therefore changes the oxidation state of every bonded atom.

(C) D_2 has the greater root-mean-square speed; both neutral molecules have two electrons.

(D) Every physical property and reaction rate is identical because isotopes have the same atomic number.

38. An orbital has 3 angular nodes and 1 radial nodes. Maximum electrons in its entire subshell are

(A) 2 (B) 14

(C) 7 (D) 12

39. Water hardness is due only to dissolved $Ca(HCO_3)_2$ and $MgSO_4$. Which treatment statement correctly distinguishes the two contributions?

(A) Adding NaCl precipitates both calcium and magnesium because all chlorides are insoluble.

(B) Boiling removes the temporary carbonate-hardness contribution; ion exchange can remove the remaining ions.

(C) Boiling alone necessarily removes every dissolved sulfate and chloride quantitatively.

(D) Filtration of the clear untreated solution removes dissolved hardness ions without any precipitation.

40. A mixture contains $\frac{1}{10}$ mol H_2SO_5 and $\frac{9}{100}$ mol $H_2S_2O_8$. Their structures contain one O–O link per molecule. All peroxide links react quantitatively with excess iodide in acid; each such link accepts two electrons. What amount of I_2, in mol, is liberated?

(A) $\frac{19}{100}$ (B) $\frac{7}{25}$

(C) $\frac{1}{10}$ (D) $\frac{19}{50}$

41. A hot saturated solution contains 200g water and60.0g solute per100g water. Cooling to a temperature where solubility is20.0g per100g water also removes 20g water by evaporation. No solute evaporates,and crystals are anhydrous. Mass crystallised in g is

(A) 84 (B) 76

(C) 80 (D) 36

42. A 10 g $CaCO_3$/inert mixture is80.0% $CaCO_3$ by mass. It receives $\frac{9}{50}$ mol HCl. For $CaCO_3+2HCl\rightarrow CaCl_2+CO_2+H_2O$,90.0% of formed CO_2 is collected. Collected CO_2 amount in mol is

(A) $\frac{9}{50}$ (B) $\frac{9}{125}$

(C) $\frac{2}{25}$ (D) $\frac{9}{100}$

43. Two one-electron ions A and B satisfy Z_B/Z_A=2 and |E_B|/|E_A|=16 for their selected Bohr levels. In the ideal hydrogenic model $E\propto -Z^2/n^2$ and $r\propto n^2/Z$. What is r_B/r_A?

(A) 1/4 (B) -1/8

(C) 1/16 (D) 1/8

44. A sugar has its lowest chiralOH on the right in a conventionalFischer projection. No polarimetric measurement is supplied. Which assignment is justified?

(A) L configuration becauseOH is on the right.

(B) D configuration; the sign of optical rotation is not determined byD alone.

(C) Positive rotation; D is defined as clockwise rotation.

(D) Optical inactivity because the projection is planar.

45. Butan-2-one is reduced with NaBH4 and protonated. The isolated alcohol is then subjected to I2/ aqueous NaOH. Which pair gives the alcohol and the final diagnostic observation?

(A) 2-methylpropan-2-ol;yellow CHI3 precipitate

(B) butan-2-ol;yellow CHI3 precipitate

(C) butan-1-ol;yellow CHI3 precipitate

(D) butan-2-ol;no iodoform because every secondary alcohol is negative

46. For isolated molecules of SO_2 and SO_3,which member has the larger permanent dipole moment,and which reasoning is sufficient?

(A) SO_2; Bent SO2 has a nonzero resultant; the three equal vectors of trigonal-planar SO3 cancel.

(B) SO_3; the number of polar bonds alone fixes the molecular dipole.

(C) They must be equal because both contain polar covalent bonds.

(D) The one with more atoms always has the larger dipole,independent of geometry.

47. A student predicts MgCO3 is more thermally stable than BaCO3 because Mg2+ attracts the carbonate more strongly. Which correction identifies the relevant effect?

(A) The smaller Mg2+ polarises and destabilises carbonate more strongly; BaCO3 is generally more thermally stable.

(B) Ba2+ is smaller than Mg2+.

(C) Thermal stability is the same property as aqueous solubility and must have the same ordering.

(D) Stronger polarisation always stabilises an oxyanion against decomposition.

48. At a fixed temperature K=16. A reaction mixture initially has Q=16. Which direction lowers Gibbs energy, and which composition change must occur before equilibrium can be reached? Use ΔrG=RT ln(Q/K).

(A) neither direction has a driving force; Q must already equal K.

(B) Reverse in every case because the standard reaction is exergonic.

(C) Forward in every case because K exceeds one.

(D) Neither direction whenever K is not one; equilibrium requires K=1.

49. A diprotic acid H_2A is0.100M;its first dissociation is complete andKa2=10^{-5}. Added HCl is $\frac{1}{5}$M in the final solution. Neglect the second-step contribution to H^+. Equilibrium$[A^{2-}]$in M is

(A) $\frac{3}{100000}$

(B) $\frac{1}{100000}$

(C) $\frac{1}{200000}$

(D) $\frac{1}{300000}$

50. p-cresol undergoes CHCl3/NaOH,then acidification;then NaBH4. Which final product preserves all ring substituents and the selective reduction endpoint?

(A) 4-(hydroxymethyl)-2-methylphenol

(B) 2-hydroxy-5-methylbenzoic acid

(C) 4-methylbenzyl alcohol

(D) 2-(hydroxymethyl)-4-methylphenol

51. Acidic permanganate oxidises SO_2 to HSO_4^- and becomes Mn^{2+}. With excess SO_2, $\frac{3}{100}$ mol MnO_4^- and only $\frac{3}{400}$ mol available H^+ are supplied. Count the net H^+ in the balanced reaction. Maximum Mn^{2+} formed,in mol,is

(A) $\frac{3}{800}$

(B) $\frac{3}{100}$

(C) $\frac{3}{200}$

(D) $\frac{3}{3200}$

52. Tincture of iodine is an iodine-containing alcoholic preparation. Which pairing of constituent and textbook role is correct?

(A) Aspartame solution; disinfectant by definition.

(B) Buta-1,3-diene; antibiotic by definition.

(C) Iodine-containing preparation; antiseptic.

(D) Sodium benzoate solution; tranquilizer.

53. For CH_4,which classification and consequence are mutually consistent?

(A) Covalent; nonstoichiometry proves all hydrogen has been removed.

(B) Ionic; every example must be a gas composed of discrete molecules.

(C) All hydrides contain free H+ ions in their solid lattices.

(D) covalent; C-H bonds belong to a molecular hydride,not a lattice of H- ions.

54. For Ce^{4+} (Z=58),which configuration/chemical interpretation is consistent with its electron count and the commonly discussed +3 state?

(A) $4f^0$; reducing character toward+3.

(B) $4f^0$; oxidising character in the stated conversion toward the common +3 state.

(C) $4f^1$; neither accepts nor loses an electron in changing to+3.

(D) $4f^1$; oxidising character.

55. An octahedral complex has high-spin d^6 configuration and $\Delta_o = 100$ kJ mol^{-1}. Ignoring pairing energy,its d-electron energy relative to the unsplit barycentre,in kJ mol^{-1},is

(A) -140

(B) -40

(C) 40

(D) 60

56. Bond energies $D(X_2):D(Y_2):D(XY)=1:2:3$. Forming2.00 mol XY(g) from the gaseous elements releases 240 kJ. $D(Y_2)$,in kJ/mol,is

(A) 320

(B) 240

(C) 160

(D) 80.0

57. Separate unknowns P and Q are each selected from Cu2+, Zn2+, Fe3+. Dropwise aqueous ammonia gives P a white hydroxide precipitate and Q a pale blue precipitate. Excess ammonia gives P a colourless ammine solution and Q a deep-blue ammine solution. Which identification is consistent, without invoking redox?

(A) P=Zn2+; Q=Cu2+; both metals reduced to zero.

(B) P=Cu2+; Q=Zn2+; oxidation states unchanged.

(C) Both are the same ion because all hydroxides have identical ammonia behaviour.

(D) P=Zn2+; Q=Cu2+; oxidation states unchanged.

58. A weak monoprotic acid at C=$\frac{9}{10000}$M has measured degree of dissociation $\frac{1}{5}$. Use the exact concentration relationKa=Cα^2/(1-α). At the same T,analytical concentration required for degree $\frac{1}{10}$ is

(A) $\frac{81}{20000}$ (B) $\frac{9}{40000}$

(C) $\frac{9}{2500}$ (D) $\frac{9}{5000}$

59. A textbook deficiency table associates rickets with one vitamin. Which vitamin and solubility class are correctly paired?

(A) Vitamin B6; water-soluble. (B) VitaminK; fat-soluble.

(C) Vitamin B1; water-soluble. (D) VitaminD; fat-soluble.

60. A mixture contains a water-immiscible steam-volatile organic liquid, a sublimable solid and a nonvolatile salt. Which order matches the physical properties without assuming that the liquid survives heating to its normal boiling point?

(A) Filter the dissolved components from one clear homogeneous solution without any phase change.

(B) Distil the salt first because ionic salts are always the most volatile.

(C) Sublimation must turn every component directly into vapour at the same rate.

(D) Steam-distil the liquid first; sublime the volatile solid from the remaining dry solid mixture.

English Proficiency

61. Read the sentence: The omission was inadvertent; the clerk corrected it as soon as she noticed it. Choose the closest meaning of 'inadvertent' as used in the sentence.

(A) unintentional (B) unexplained

(C) unimportant (D) unavoidable

62. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. measured the narrow opening Q. before the inspection began. R. had been checked for accuracy S. The tool with which the technician

(A) R-S-P-Q (B) S-P-R-Q

(C) P-S-R-Q (D) Q-P-R-S

63. The device is designed to be moved easily from one workplace to another; no claim is made that it folds or floats. Which adjective fits?

(A) foldable (B) buoyant

(C) detachable (D) portable

64. Read the sentence: After weeks of uncertainty, the board gave a decisive answer that settled the matter. Choose the opposite in meaning of 'decisive' as used in the sentence.

(A) firm (B) inconclusive

(C) unpopular (D) prompt

65. Complete: This is the shelf on _____ the original manuscripts must remain.

(A) who (B) which

(C) whom (D) whose

66. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The instructor may cancel the outdoor trial if the weather worsens.

(A) The instructor may be cancelled by the outdoor trial if the weather worsens.

(B) The outdoor trial must be cancelled by the instructor if the weather worsens.

(C) The outdoor trial may be cancelled only if the instructor worsens the weather.

(D) The outdoor trial may be cancelled by the instructor if the weather worsens.

67. Read the sentence: The capacious bag held all the equipment with room to spare. Choose the opposite in meaning of 'capacious' as used in the sentence.

(A) heavy (B) durable

(C) cramped (D) spacious

68. Read the sentence: Reaching the isolated campsite was an arduous journey requiring sustained effort. Choose the closest meaning of 'arduous' as used in the sentence.

(A) strenuous (B) dangerous

(C) unfamiliar (D) unnecessarily long

69. Choose the most coherent order of the four sentences. A. People continued checking their watches instead of trusting it. B. The square's clock was repaired after months of showing the wrong time. C. In response to that continued watch-checking, the council displayed a notice explaining when the clock had been reset and checked. D. After that notice appeared, confidence returned gradually as the clock continued to agree with other reliable timepieces.

(A) B-A-C-D (B) B-A-D-C

(C) A-B-C-D (D) B-C-A-D

70. The author narrates her own life rather than another person's. What kind of book is it?

(A) biography (B) anthology

(C) autobiography (D) chronicle of a city

Logical Reasoning

71. The recorded average of 10 readings is 41. One entry recorded as 55 should have been 48. After that correction, another reading of 24 is replaced by 31. What is the final average?

(A) $\frac{417}{10}$ (B) 41

(C) 42 (D) $\frac{403}{10}$

72. In a row, Asha is 7th from the left and 40th from the right. Bina is 12th from the right. Charu stands 4 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 22 (B) 23

(C) 25 (D) 31

73. Preserve the exact relation and its order: Frog : Amphibian :: ? : Reptile. Choose the missing term.

(A) Whale (B) Frog

(C) Sparrow (D) Lizard

74. The model triads are $(7, 18, 11)$ and $(9, 22, 13)$. In the intended rule the middle number is the sum of the outer numbers, and the outer-number difference is the same fixed positive value in every triad. Which option satisfies both relations?

(A) $(6, 17, 11)$ (B) $(6, 17, 10)$

(C) $(6, 18, 11)$ (D) $(6, 16, 10)$

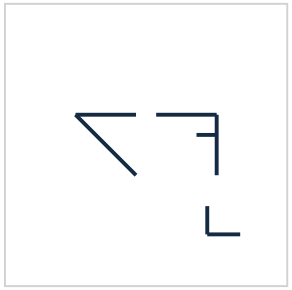
75. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



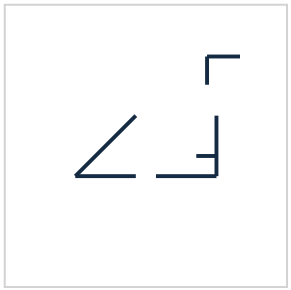
(A) Pattern W



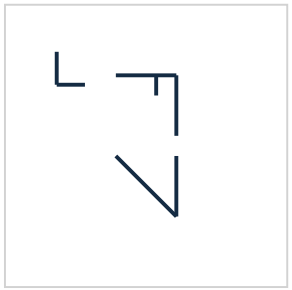
(B) Pattern Z



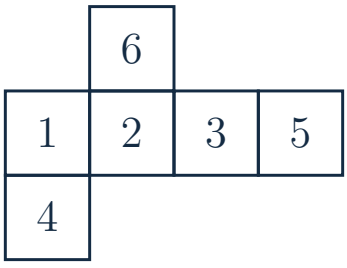
(C) Pattern M



(D) Pattern T



76. Fold the shown net into a cube. Give (label opposite 5, label opposite 1) in that order.



(A) $(2, 6)$

(B) $(2, 1)$

(C) $(2, 3)$

(D) $(2, 2)$

77. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 7, 21, 30, 90, ?, 297, 306.

(A) 270 (B) 297

(C) 150 (D) 99

78. Preserve the exact relation and its order: Astronomer : Observatory :: ? : Cockpit. Choose the missing term.

(A) Librarian (B) Pilot

(C) Farmer (D) Judge

79. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 12$ and $5 \mapsto 27$. Which input maps to 77?

(A) 14 (B) 77

(C) 16 (D) 15

80. $X@Y$ means X is the mother of Y; $X\&Y$ means X is the sister of Y. All names denote distinct people; a brother or sister is a full sibling, sharing both parents. Given $C@K$ $K\&T$ $T@M$, how is C related to M?

(A) Mother (B) Maternal grandmother

(C) Father (D) Maternal grandfather

81. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: CFU, DGV, ?, FIX, GJY, HKZ.

(A) DGV (B) WHE

(C) EHW (D) FIX

82. A code shifts letters cyclically through A–Z. Odd positions share one fixed shift and even positions share another. For example, CHAIR is coded as EGCHT. Which word is coded as VQCHP?

(A) SQZHM (B) XPEGR

(C) TRAIN (D) USBJO

83. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 3 (B) 2

(C) 1 (D) 4

84. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). The numerical component follows a constant arithmetic step. Find the missing internal term: SQSN2, QNWK7, OKAH12, ?, KEIB22, IBMY27.

(A) NIFF17 (B) LGDD17

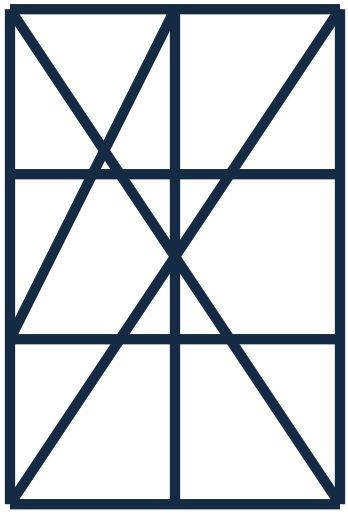
(C) 71EEHM (D) MHEE17

85. A walker goes 6 m north, then 11 m west, then turns right and walks 8 m. What is the square of the straight-line distance from the starting point, in square metres?

(A) 316 (B) 318

(C) 317 (D) 125

86. How many triangles are formed by all the drawn segments?



(A) 35 (B) 24

(C) 25 (D) 34

87. Five people occupy positions 1 to 5 from left to right, one per position, and wear five different shirt colours: red, blue, green, white and yellow. Dev is 1st from the left. Charu is 2nd from the left. Asha is 3rd from the left. Esha is 4th from the left. Bina is 5th from the left. The blue shirt is in position 2. The green shirt is in position 3. The yellow shirt is in position 4. Dev does not wear white. Which shirt does Dev wear?

(A) Green (B) Blue

(C) Red (D) White

88. Consider only these statements; universal statements alone do not imply existence. No editors are guides; Some guides are artists. Conclusions: I. Some artists are not editors. II. Some guides are not editors. Which conclusions follow necessarily?

(A) Both I and II (B) Only I

(C) Neither I nor II (D) Only II

89. Three pairs contain exactly one string instrument and one wind instrument. Which pair does not?

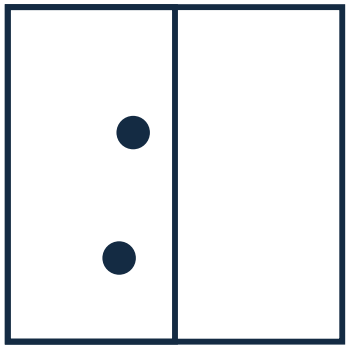
(A) Violin / Sitar

(B) Cello / Trumpet

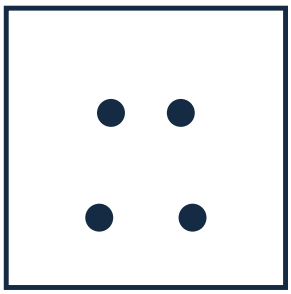
(C) Guitar / Saxophone

(D) Harp / Flute

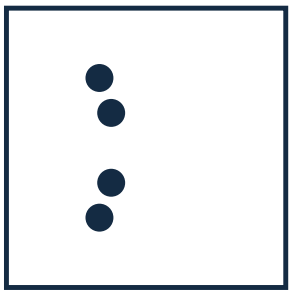
90. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



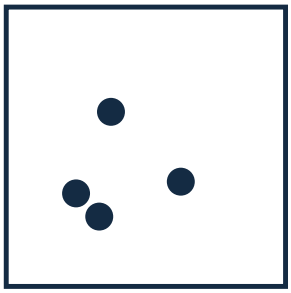
(A) Pattern U



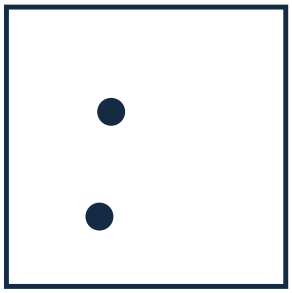
(B) Pattern Y



(C) Pattern N



(D) Pattern T



Mathematics

91. $F'(x)=-\frac{|x+1|}{(x+1)^3\sqrt{(x+1)^2+1}}$ and $F(0)=3$. Find $F(1)-2F(0)$.

(A) $-3-\sqrt{2}+\frac{\sqrt{5}}{2}$

(B) $-\sqrt{2}+\frac{\sqrt{5}}{2}$

(C) $-\sqrt{2}+\frac{\sqrt{5}}{2}+3$

(D) $-3-\frac{\sqrt{5}}{2}+\sqrt{2}$

92. $a_j = 5+5(j-1)$ and $f(x)=(6/2)x^2+bx+2$. If $f(1)=f(-1)$, find $f'(a_5)$.

(A) 150

(B) 152

(C) 25

(D) 180

93. An observer at height 5 sees a tower foot at depression 30° and top at elevation 60°. Find the tower height.

(A) $5\sqrt{3}$ (B) 15

(C) 20 (D) 10

94. a=(7,1,0), b=(t,3,1). If a⊥b, find $|7b|^2$.

(A) 490 (B) 450

(C) $\frac{499}{49}$ (D) 499

95. Evaluate $\sum_{j=1}^{\infty} 2j(\frac{1}{4})^j$.

(A) $\frac{2}{9}$ (B) $\frac{2}{3}$

(C) $\frac{32}{9}$ (D) $\frac{8}{9}$

96. z_1=2+4i, z_2=7+4i, z_3=7+9i. Find the smaller angle between z_2-z_1 and z_3-z_1.

(A) 0 (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{4}$

97. L_1:r=t(1,0,1), L_2:r=(5,0,0)+s(0,1,1). Find their shortest distance.

(A) 5 (B) $\frac{5}{3}$

(C) $\frac{5\sqrt{3}}{3}$ (D) 0

98. A line through P=(8,5) has direction (3,4). It meets 3x+4y=144 at Q. Find $|PQ|^2$.

(A) 100 (B) 16

(C) 20 (D) 400

99. Evaluate $\displaystyle\int_0^{\pi/2} \sin(2x)\left[1+6\sin^2(2x)\right]\frac{\sin^4 x}{\sin^4 x+\cos^4 x}\,dx$.

(A) 5 (B) $\frac{1}{2}$

(C) $\frac{7}{2}$ (D) $\frac{5}{2}$

100. For A={1,...,11}, how many (a,b)∈A×A satisfy a+b even and a<b?

(A) 30 (B) 60

(C) 50 (D) 25

101. The parabola $y=x^2+px+q$ passes through (0,9). The absolute value of its vertex ordinate is minimized over p≥0. Find p^2+q^2 at the minimum.

(A) 225 (B) 45

(C) 117 (D) 81

102. A sphere has dV/dt=5 cubic units/s. Find the rate of change of its surface area when r=6.

(A) $\frac{5}{144\pi}$ (B) $\frac{5}{6}$

(C) $\frac{5}{18}$ (D) $\frac{5}{3}$

103. Using digits 0,1,2,3,4,5,6 without repetition, how many 5-digit positive integers are divisible by 6 and have their first digit smaller than their last digit?

(A) 180 (B) 360

(C) 120 (D) 0

104. Let z=(1+i)^{25}. Find Re(z)+Im(z).

(A) 4096 (B) 8192

(C) -8192 (D) $4096\sqrt{2}$

105. Parabola y=2(x-4)(x-5) meets the axes at three distinct points. A circle passes through these points. Find its other y-intercept.

(A) 1 (B) $-\frac{1}{2}$

(C) 2 (D) $\frac{1}{2}$

106. A positive integer-domain function satisfies f(m+n)=f(m)f(n), f(1)=2. Find $\sum_{j=1}^{5} f(j)$.

(A) 160 (B) 32

(C) 62 (D) 31

107. In (1+x)^{10}, find the ratio of the coefficient of x^4 to the coefficient of x^3.

(A) $\frac{7}{4}$ (B) $\frac{8}{3}$

(C) 7 (D) $\frac{4}{7}$

108. $\log_2$a=7, log_a b=2, log_b c=2. Find $\log_4(c^2/a)$.

(A) 28 (B) $\frac{49}{2}$

(C) 49 (D) $\frac{63}{2}$

109. The homogeneous system 6x+y=0, kx+3y=0 has a nonzero solution. Find k+x^2/y^2.

(A) 18 (B) $\frac{647}{36}$

(C) 54 (D) $\frac{649}{36}$

110. For $A=\begin{bmatrix}6 & 1\\ 1 & 3\end{bmatrix}$ and C=17A^{-1}, find C$_{22}$.

(A) -1 (B) 6

(C) 3 (D) $\frac{6}{17}$

111. An AP has a=10,d=4; its first n terms sum to 192. Find a_(n+1).

(A) 42 (B) 24

(C) 46 (D) 38

112. Two distinguishable fair dice show X,Y. Given X+Y$\geq$6, find the probability that X$\neq$Y and XY is even.

(A) $\frac{9}{11}$ (B) $\frac{10}{13}$

(C) $\frac{9}{13}$ (D) $\frac{6}{13}$

113. Find the coefficient of x^6 in (1-x)exp(5x).

(A) $\frac{6875}{144}$ (B) $-\frac{625}{144}$

(C) $-\frac{625}{24}$ (D) $\frac{3125}{144}$

114. a=(6,4,0), b=(4,-6,3). Find $|a+b|^2$-$|a-b|^2$.

(A) 192 (B) 113

(C) −18 (D) 0

115. There are 7 Boolean variables p_1,...,p 7. How many truth assignments make [(p_1 implies p_2) and (p_1 implies not p_2)] true?

(A) 64 (B) 127

(C) 128 (D) 32

116. X is binomial with mean 12 and variance 8. Find its mode.

(A) 36 (B) 11

(C) 12 (D) 13

117. How many integer t make P=(t,0) strictly inside the larger segment cut by x=4 from $x^2+y^2=64$?

(A) 3 (B) 11

(C) 13 (D) 15

118. $A = \begin{bmatrix} 4 & 1 \\ 1 & 6 \end{bmatrix}$ and A^{-1}=pI+qA. Find p+q.

(A) $-\frac{9}{23}$ (B) $\frac{9}{23}$

(C) $\frac{11}{23}$ (D) $\frac{9}{10}$

119. y>0 satisfies $2yy'+6y^2=0$, y(0)=sqrt(4). Find $y^2(1)$.

(A) $\frac{4}{e^3}$ (B) $\frac{2}{e^3}$

(C) $\frac{4}{e^6}$ (D) $4e^6$

120. A circular sector has radius 3 and arc length $\frac{3\pi}{4}$. Find its area.

(A) $\frac{9\pi}{8}$ (B) $\frac{9}{8}$

(C) $\frac{\pi}{8}$ (D) $\frac{27\pi}{8}$

121. A rectangle has perimeter 40. Its length is constrained to be at least three times its width. Find its maximum area.

(A) 25 (B) 100

(C) 75 (D) 50

122. For $f(x)=x^3-3kx^2+75x$, how many integer k in [0,9] give a local maximum at a positive x?

(A) 4 (B) 5

(C) 9 (D) 3

123. Evaluate $\int_0^5$ x floor(x) dx.

(A) 30 (B) 10

(C) 35 (D) $\frac{125}{3}$

124. A chord of $x^2+y^2=8100$ touches $x^2/81-y^2/9=1$. Its nonzero midpoint (h,k) satisfies h=k. Find h^2+k^2.

(A) 18 (B) 36

(C) 45 (D) 72

125. Group A has mean 30, SD 3. Every observation is increased by 30. Find its new coefficient of variation in percent.

(A) 20 (B) 10

(C) 15 (D) 5

126. If α,β are roots of $x^2 - 2kx + k^2 - 9 = 0$ and k$\in$[-4,4], find the maximum of $\alpha^2+\beta^2$.

(A) 46 (B) 50

(C) 18 (D) 34

127. $x=t^2$, $y=t^3$ for t>0. Find d^2y/dx^2 at t=3.

(A) $\frac{1}{4}$ (B) $\frac{3}{2}$

(C) $\frac{1}{2}$ (D) 9

128. f(x)=5x+5. Find f(f(5)).

(A) 55 (B) 135

(C) 155 (D) 30

129. An equilateral triangle has vertex (0,12) and opposite side on y=0. Find the squared inradius.

(A) 16 (B) 144

(C) 4 (D) 64

130. For the loop $y^2=x^2(3-x)$, find the area of the portion with x$\geq$3/2.

(A) $\frac{21\sqrt{6}}{20}$ (B) $\frac{21\sqrt{6}}{10}$

(C) $\frac{12\sqrt{3}}{5}$ (D) $-\frac{21\sqrt{6}}{10}+\frac{24\sqrt{3}}{5}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A wheel with constant angular acceleration turns through 15 rad in its first 3 s and 33 rad in the next 3 s. Find its initial angular velocity.

(A) 2 rad/s (B) 5 rad/s

(C) 8 rad/s (D) 11 rad/s

132. Two coherent beams yield maximum-to-minimum interference intensity ratio $\frac{36}{25}$. Find the larger individual beam intensity divided by the smaller.

(A) $\frac{3721}{121}$ (B) 121

(C) $\frac{1}{121}$ (D) 144

133. In a thin-foil scattering experiment, most alpha particles are barely deflected, but 36 of 500000 incident particles undergo large-angle scattering. Which paired inference is supported without assuming the Thomson model?

(A) Most atomic mass is spread uniformly; central charge is negative.

(B) Most atomic volume is positively charged; electrons cause the rare large deflections.

(C) Most atomic volume is nearly empty; positive charge is concentrated in a very small central region.

(D) Atoms contain no electrons; positive charge is uniformly distributed.

Chemistry

134. Compare pH of A:HCl at$\frac{1}{200}$M;B:HA,Ka=10^{-6},C=$\frac{1}{2}$M;C:HB,Ka=10^{-5},C=$\frac{1}{200}$M. Neglect water and use weak-ionisation ordering. Increasing pH is

(A) A<B<C. (B) C<B<A.

(C) A<C<B. (D) B<A<C.

135. A pure KMnO4 portion gives 1/50 mol O2 on complete controlled heating by2KMnO4→K2MnO4+MnO2+O2. With KMnO4=158 g/mol, which pair gives the initial mass and a viable origin of its intense purple colour despite Mn(VII) being d0?

(A) 316/25 g; d-d absorption within an occupied d shell. (B) 158/25 g; charge-transfer absorption.

(C) 79/25 g; charge-transfer absorption. (D) 158/25 g; d-d absorption within an occupied d shell.

136. An amine of formula CH5N gives a positive carbylamine test. It reacts with benzenesulphonyl chloride to give its monosulphonamide. Count symmetry-distinct proton environments in that product, including N-H. Treat rapid rotation as averaging,ignore accidental spectral overlap,and use an achiral medium. The count is

(A) 4 (B) 6

(C) 5 (D) 7

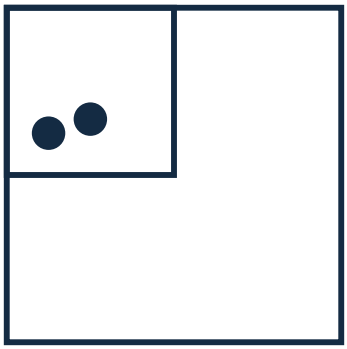
Logical Reasoning

137. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 7, 21, 25, 75, ?, 237, 241.

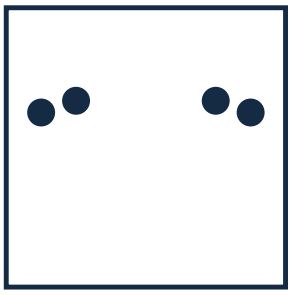
(A) 79 (B) 221

(C) 225 (D) 229

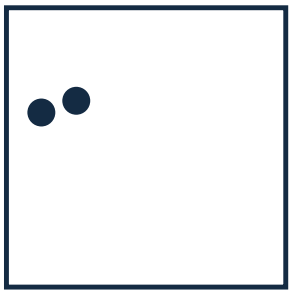
138. Starting with a unit square sheet, fold right to left along the vertical midline, then fold bottom to top along the horizontal midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



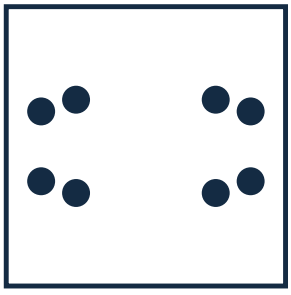
(A) Pattern R



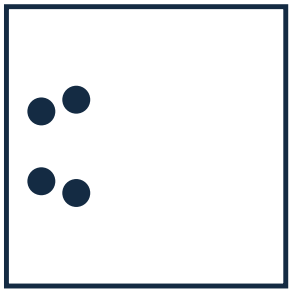
(B) Pattern N



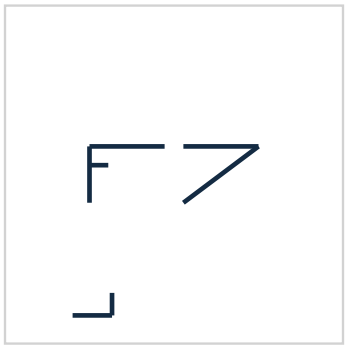
(C) Pattern U



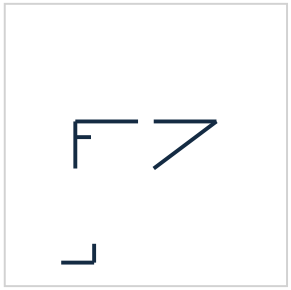
(D) Pattern Q



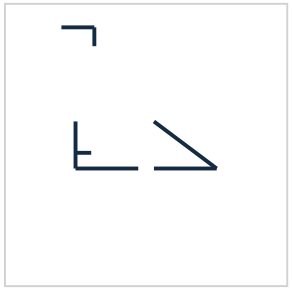
139. Choose the exact horizontal water reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



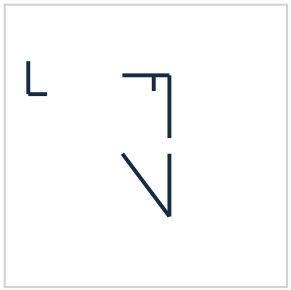
(A) Pattern U



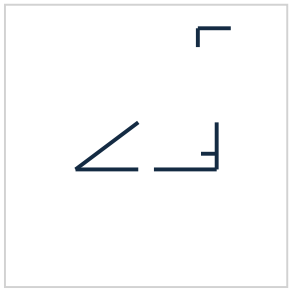
(B) Pattern T



(C) Pattern Q



(D) Pattern K



Mathematics

140. How many integers x satisfy $|(x-7)+(x-10)|<|x-7|+|x-10|$?

(A) 3 (B) 2

(C) 4 (D) 9

141. $y'-y/x=1$ for $x>0$, $y(1)=3$. Find $y(11)/11-y(1)$.

(A) $\log(11)+3$ (B) $-\log(11)$

(C) $\log(11)$ (D) $11\log(11)+33$

142. In triangle ABC, $A=2B$, $a/b=\sqrt{3}$ and $b=3$. Find its area.

(A) $\frac{9\sqrt{3}}{2}$ (B) $\frac{9}{2}$

(C) 6 (D) $9\sqrt{3}$

BITSAT Mock 20

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A rod of length 6 m pivots at its lower end. A peg constrained to move horizontally stays in contact with its upper tip. At an instant the rod makes an angle with $\sin\theta = 3/5$ to the horizontal and the tip's horizontal speed has magnitude 1 m/s. Find the rod's angular-speed magnitude.

(A) $\frac{1}{6}$ rad/s (B) $\frac{5}{18}$ rad/s

(C) $\frac{5}{24}$ rad/s (D) $\frac{1}{10}$ rad/s

2. Two equal-intensity coherent beams have maximum combined intensity 12 units. Find intensity when their path difference is $\lambda/4$.

(A) 6 (B) 3

(C) 12 (D) 24

3. A uniform disc of mass 3 kg and radius 5 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) $\frac{975}{32}$ kg m^2 (B) $\frac{1125}{32}$ kg m^2

(C) $\frac{225}{8}$ kg m^2 (D) $\frac{1425}{32}$ kg m^2

4. In YDSE wavelength is 400 nm, screen distance 3 m and slit separation 4 mm. Find the separation on the same side of the centre between the 3rd bright fringe and the 7th dark fringe. Count the first dark fringe as order 1.

(A) $\frac{21}{20}$ mm (B) $\frac{57}{20}$ mm

(C) $\frac{27}{20}$ mm (D) $\frac{6}{5}$ mm

5. A spring of constant 40 N/m is already stretched by $\frac{3}{10}$ m. How much additional work is required to increase its extension to $\frac{1}{2}$ m quasistatically?

(A) 5 J (B) $\frac{12}{5}$ J

(C) $\frac{16}{5}$ J (D) 4 J

6. The quantity $Z = x^3y^2/\sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 1, 2 and 4, respectively. To first order, what is the maximum percentage error in Z?

(A) 5 (B) 9

(C) 11 (D) 7

7. A nonrelativistic charged particle initially has momentum $3qEt$ along a uniform electric field, where $t > 0$ is a fixed reference interval. Its charge is $+q$. What is $\lambda(6t)/\lambda(2t)$ after acceleration in the same field?

(A) $\frac{9}{5}$ (B) $\frac{25}{81}$

(C) $\frac{5}{9}$ (D) $\frac{1}{3}$

8. A dipole of fixed moment 6 C m is rotated quasistatically from 60° to 120° relative to a uniform electric field 8 N/C. Find external work.

(A) 16π J (B) $48\sqrt{3}$ J

(C) $24\sqrt{3}$ J (D) 48 J

9. An insulated vessel containing one mole of an ideal gas of molar mass $\frac{3}{50}$ kg/mol moves at bulk speed 80 m/s and is brought to rest. All bulk kinetic energy becomes internal energy. The gas has 5 active degrees of freedom. Find temperature rise. Use $R = 8.314$ J/(mol K).

(A) $\frac{19200}{4157}$ K (B) $\frac{25600}{4157}$ K

(C) $\frac{192000}{4157}$ K (D) $\frac{38400}{4157}$ K

10. A point at the bottom of two plane liquid layers is viewed normally from air. The upper layer has thickness 21 cm and index 3/2; the lower has thickness 24 cm and index 4/3. Find its apparent depth below the top surface in the paraxial limit.

(A) 32 cm (B) $\frac{135}{4}$ cm

(C) 14 cm (D) $\frac{45}{2}$ cm

11. Light of vacuum wavelength 500 nm passes from a medium of index 2 into one of index 3. Find the ratio of its frequencies in the second and first media.

(A) $\frac{4}{9}$ (B) $\frac{1}{3}$

(C) 1 (D) $\frac{9}{4}$

12. A uniform-density spherical planet has radius R and surface gravitational acceleration g. At what depth below the surface is gravity $g/6$? Give the depth as a fraction of R.

(A) $1 - \frac{\sqrt{6}}{6}$ (B) 1

(C) $\frac{35}{36}$ (D) $\frac{5}{6}$

13. A source emitting 600 Hz and a listener approach each other, each at 30 m/s relative to still air. Sound speed is 300 m/s. Find heard frequency before they pass.

(A) $\frac{5400}{11}$ Hz (B) 600 Hz

(C) 750 Hz (D) $\frac{2200}{3}$ Hz

14. A stationary nucleus with mass number 32 and binding energy per nucleon $\frac{39}{5}$ MeV splits into two equal daughters, each having total binding energy $\frac{639}{5}$ MeV. Which statement is correct?

(A) The process absorbs 6 MeV; the daughters have larger binding energy per nucleon.

(B) The process releases 6 MeV; the daughters have larger binding energy per nucleon.

(C) No energy can be released because fission always requires a larger mass number than 32.

(D) The process releases 12 MeV; each daughter releases 6 MeV.

15. In an abrupt junction, the acceptor concentration is 625 times the donor concentration. Within the depletion approximation, which statement gives x_n/x_p and identifies the carriers mainly responsible for the small reverse current before breakdown?

(A) 625; minority carriers

(B) 1; majority carriers

(C) 25; majority carriers

(D) 1/625; minority carriers

16. A liquid of volume 2 litres and bulk modulus 2000000000 Pa is subjected to an additional pressure of 4000000 Pa. Neglect changes in bulk modulus. Find the magnitude of volume decrease in cubic centimetres.

(A) $1000000\,\mathrm{cm}^3$

(B) $\frac{1}{250}\,\mathrm{cm}^3$

(C) $4\,\mathrm{cm}^3$

(D) $2\,\mathrm{cm}^3$

17. 9 identical, noninteracting ideal-gas samples at the same temperature and pressure are combined without changing those conditions. Which pair both increases by a factor 9?

(A) Density and pressure.

(B) Pressure and temperature.

(C) Volume and internal energy.

(D) Temperature and molar heat capacity.

18. A mass 5 kg moves with $\vec{r}(t) = 3t\hat{i} + 5t^2\hat{j}$ m. Find the z-component of angular momentum about the origin at $t = 3$ s.

(A) $2025\,\mathrm{kg\,m^2/s}$

(B) $675\,\mathrm{kg\,m^2/s}$

(C) $\frac{675}{2}\,\mathrm{kg\,m^2/s}$

(D) $1350\,\mathrm{kg\,m^2/s}$

19. A plane wave in vacuum travels along $+x$. At one instant its electric field is $180\hat{k}$ V/m. If the electric amplitude is 2 times this instantaneous value, which gives the instantaneous magnetic field? Use $c = 3 \times 10^8$ m/s.

(A) $+60 \times 10^{-8}\hat{i}$ T

(B) $-120 \times 10^{-8}\hat{j}$ T

(C) $-60 \times 10^{-8}\hat{j}$ T

(D) $+60 \times 10^{-8}\hat{j}$ T

20. Here L is inductance, C capacitance, R resistance, V potential, I current and t time. Which expression has the same dimensions as both $(L/R)^2$ and $(RC)^2$?

(A) $(CV)^2$

(B) $(V/R)^2$

(C) $(LI)^2$

(D) $(\sqrt{LC})^2$

21. A nucleus of mass number 40 has binding energy per nucleon $\frac{37}{5}$ MeV. It splits into two equal fragments, each with binding energy per nucleon $\frac{39}{5}$ MeV. Neglect emitted-particle rest masses. Find energy release.

(A) 16 MeV

(B) 328 MeV

(C) 8 MeV

(D) 312 MeV

22. A projectile of mass 6 kg is launched with speed 35 m/s, $\cos\theta = 3/5$. At its highest point it splits into masses $M/3$ and $2M/3$. The lighter fragment moves horizontally forward at 24 m/s. Find the signed horizontal velocity of the heavier fragment immediately after the explosion.

(A) 21 m/s
(B) $\frac{39}{2}$ m/s
(C) $\frac{87}{2}$ m/s
(D) 24 m/s

23. Two coherent beams yield maximum-to-minimum interference intensity ratio $\frac{64}{49}$. Find the larger individual beam intensity divided by the smaller.

(A) 226
(B) 225
(C) $\frac{64}{49}$
(D) $\frac{12769}{225}$

24. A ball is dropped from 20 m onto a horizontal fixed floor. At every impact the coefficient of restitution is $\frac{2}{5}$. Neglect air resistance. What maximum height does it reach after its second impact?

(A) $\frac{16}{5}$ m
(B) $\frac{1024}{78125}$ m
(C) $\frac{64}{125}$ m
(D) $\frac{32}{25}$ m

25. A uniformly charged nonconducting sphere has radius R and surface potential V_s relative to infinity. Find potential at $r = R/7$ divided by V_s.

(A) $\frac{73}{49}$
(B) $\frac{1}{7}$
(C) $\frac{10}{7}$
(D) 7

26. A sinusoidal source of peak $\frac{87}{10}$ V feeds a back-to-back series pair of identical Zeners through a 3 kΩ current-limiting resistor. The output is across the diode pair. Each Zener has reverse breakdown 5 V and forward drop 0.7 V. What is the peak-to-peak output voltage? Use the piecewise constant-drop model.

(A) $\frac{64}{5}$ V
(B) $\frac{57}{5}$ V
(C) $\frac{87}{5}$ V
(D) 10 V

27. A 2 kg block is launched at 6 m/s on top of a 7 kg block initially at rest on a smooth floor. The coefficient of kinetic friction between them is $\frac{2}{5}$. Find the distance slid by the upper block relative to the lower before slipping ceases. Take $g = 10$ m/s^2.

(A) $\frac{63}{4}$ m
(B) 7 m
(C) 1 m
(D) $\frac{7}{2}$ m

28. A pendulum bob of mass 3 kg and length 3 m passes the horizontal position moving downward at 4 m/s. Find the instantaneous power of all forces when $\sin\theta = 3/5$, $\cos\theta = 4/5$, with θ measured from the downward vertical. Take $g = 10$ m/s^2.

(A) $36\sqrt{13}$ W
(B) 192 W
(C) 72 W
(D) 144 W

29. A particle of mass 1 kg starts from rest and experiences $F(t) = 4t$ N along its motion for 4 s. Find the work done by the force.

(A) 2048 J
(B) 512 J
(C) $\frac{512}{3}$ J
(D) 1024 J

30. Two identical capacitors each 2 microfarads are connected in series across an ideal 60 V battery. One capacitor undergoes permanent conducting breakdown, becoming a short circuit. Find the additional charge drawn from the battery after the new steady state is reached.

(A) $60\,\mu C$ (B) $30\,\mu C$

(C) $180\,\mu C$ (D) $40\,\mu C$

Chemistry

31. A $\frac{3}{100}$mol iron-ion inventory contains Fe^{2+}andFe^{3+}. Fraction$\frac{1}{4}$isFe^{3+}. All iron is quantitatively reduced to metal in the stated externally driven electron-balance model,with no side reactions. Electrons accepted,in mol,are

(A) $\frac{27}{400}$ (B) $\frac{3}{40}$

(C) $\frac{3}{400}$ (D) $\frac{33}{400}$

32. The internuclear axis is x. Two p_y orbitals overlap sideways. Their lobes facing each other above the axis have opposite phases; the lower facing lobes have the corresponding opposite signs. Which MO is produced,and what nodal property follows?

(A) A delta MO,because two perpendicular p orbitals always create two nodal planes.

(B) A pi antibonding MO; there is an additional node between nuclei besides the plane containing the axis.

(C) A sigma bonding MO,because every overlap along a line joining nuclei is sigma.

(D) A pi bonding MO; the facing-lobe phase test has been reversed.

33. For $M(s)+2A^{+}\rightleftharpoons M^{2+}+2A(s)$,$E^{\circ}=\frac{59}{500}$V. Use2.303RT/F=.0590V. Equilibrium constant for the doubled reaction$2M+4A^{+}\rightleftharpoons 2M^{2+}+4A$ is

(A) 10000 (B) 100000000

(C) $\frac{1}{100000000}$ (D) 10000000000000000

34. At fixed temperature a single adsorbent obeys $x/m=k\,p^{(1/n)}$. In experiment B the adsorbent mass is 3 times that in A and pressure is 8 times that in A. Total adsorbed amount is 6 times that in A. What is 1/n?

(A) 1/3 (B) 3

(C) 1/6 (D) 4/3

35. For two van der Waals gases,(Tc/Pc)A is4times(Tc/Pc)B. Using$Tc=8a/(27Rb)$,$Pc=a/(27b^2)$,$Vc=3b$ and$TB=a/(Rb)$,which pair(Vc,A/Vc,B;TB,A/Tc,A)is correct?

(A) (16;27/8). (B) (4;27/8).

(C) (1/4;27/8). (D) (4;8/27).

36. Excess water reacts completely with $\frac{1}{50}$ mol NaH. Which gas identity/amount pair follows from the balanced hydrolysis?

(A) H_2, $\frac{1}{100}$ mol. (B) H_2, $\frac{1}{50}$ mol.

(C) O_2, $\frac{1}{100}$ mol. (D) CO_2, $\frac{1}{50}$ mol.

37. Classify BaO, Al2O3, CO2, in that order, by their usual acid-base behaviour. Which sequence also predicts that the middle oxide reacts with both suitable acids and strong alkalis?

(A) Acidic; amphoteric; basic.
(B) Basic; acidic; amphoteric.
(C) Amphoteric; basic; acidic.
(D) Basic; amphoteric; acidic.

38. A Fischer projection of 2,3-dibromobutane has its four-carbon chain vertical (CH3 at top and CH3 at bottom). At C2,Br is on the right; at C3,Br is on the right; each other horizontal group is H. Horizontal bonds point toward the observer. Which classification and configuration pair is correct?

(A) chiral; (2S,3R)
(B) chiral; (2R,3R)
(C) meso; (2S,3R)
(D) chiral; (2S,3S)

39. For one reaction,ln K is6 at300 K and2 at600 K. Assume ln K is linear in1/T. At400 K,ln K equals

(A) 4
(B) 5
(C) 2
(D) 6

40. 2-methylaniline is diazotised at0-5°C,treated with CuCN,then its nitrile is fully hydrolysed and its benzylic side chain fully oxidised. Heating the resulting diacid gives which characteristic cyclic anhydride?

(A) terephthalic acid
(B) benzoic anhydride
(C) phthalic anhydride
(D) isophthalic acid

41. Which combined classification/reactivity description is correct for CH3CH2CH(OH)CH2CH3?

(A) primary and iodoform-negative
(B) secondary and iodoform-positive
(C) tertiary and iodoform-positive
(D) secondary and iodoform-negative

42. Five central-atom electron domains contain three bond pairs and two lone pairs. Where do the lone pairs lie in the introductory VSEPR minimum, and what is the resulting molecular shape?

(A) One axial and one equatorial; tetrahedral.
(B) Both equatorial; T-shaped.
(C) Both axial; trigonal planar.
(D) Both equatorial; linear.

43. For electrophilic mononitration of benzene carrying CH3,which orientation/reason pair is correct? Consider intrinsic electronic directing effects,not an exact ortho:para ratio.

(A) ortho/para; hyperconjugative and inductive donation stabilise ortho/para sigma complexes
(B) meta; hyperconjugative and inductive donation stabilise ortho/para sigma complexes
(C) ortho/para; every electronegative substituent activates the ring
(D) meta; the substitution site is decided only by steric size

44. An ideal linear chain is made by joining five molecules of one alpha-hydroxy acid HO-R-COOH, with no cyclisation. How many ester links and water molecules are formed?

(A) 4 ester links and 4 water molecules.
(B) 5 ester links and 5 water molecules.
(C) 4 ester links and 5 water molecules.
(D) 10 ester links andno water.

45. For H2SO5, take H=+1 and each sulfur=+6. Oxygen atoms are only ordinary oxide(-2) or peroxide(-1). Which pair gives (number of peroxide oxygen atoms, number of O-O peroxide links) per formula entity?

(A) (1, 1)
(B) (2, 1)
(C) (2, 2)
(D) (0, 0)

46. Beam A has wavelength$4\lambda_0$ and beam B has$5\lambda_0$. A carries 3 times B's total energy. Photon-number ratioN_A/N_B is

(A) 3
(B) $\frac{12}{5}$
(C) $\frac{4}{5}$
(D) $\frac{15}{4}$

47. In the stated high-pressure model Z=1+b p/(RT), gas A at 275 K and gas B at 420 K have Z-versus-p slopes in ratio 63/55. Their molar covolumes b do not change with temperature in this model. At what temperature must gas A be measured for its slope to equal that of B at 420 K?

(A) 560
(B) 315
(C) 275
(D) 420

48. In a octahedral crystal field,the upper set has 2 orbitals and the lower set has 3. Their separation is Δ. Which assignment preserves the five-orbital barycentre?

(A) e_g: $+\Delta$; t_{2g}: $-\Delta$.
(B) All five remain degenerate because the barycentre is unchanged.
(C) e_g: $\frac{3}{5}\Delta$; t_{2g}: $-\frac{2}{5}\Delta$.
(D) e_g: $\frac{2}{5}\Delta$; t_{2g}: $-\frac{3}{5}\Delta$.

49. Limiting molar conductivities of KCl,NaCl,KNO_3 are150,126,110 S cm^2/mol. A very dilute $NaNO_3$ solution measured in a cell of constant2.00cm^{-1} has conductance $\frac{129}{100000}$S. Neglect concentration corrections. $NaNO_3$ molarity is

(A) $\frac{129}{5500}$
(B) $\frac{3}{50}$
(C) $\frac{3}{200}$
(D) $\frac{3}{100}$

50. $\frac{7}{100}$ mol borax is acidified completely. Each borax formula unit gives four H_3BO_3 molecules;75.0% of the boric acid is isolated. With glycerol present, the isolated boric acid is titrated completely as a monobasic acid. What amount of NaOH, in mol, is required?

(A) $\frac{7}{25}$
(B) $\frac{21}{200}$
(C) $\frac{63}{400}$
(D) $\frac{21}{100}$

51. For oxide formation, both normalised per mole of O_2, $\Delta G^\circ_A = -305+0.10T$ and $\Delta G^\circ_B = -370+0.20T$ in kJ. Ignoring kinetic limitations, at what temperatures is B metal a favourable standard-state reducing agent for oxide A?

(A) At all positive temperatures
(B) T > 650 K
(C) T < 650 K
(D) Only T = 650 K

52. A protein segment forms an extended beta sheet rather than an alpha helix. Which interaction can still stabilise that secondary structure?

(A) Hydrogen bonding involving peptide-backboneC=O and N-H groups.
(B) Only conversion of amides into ethers.
(C) Only hydrolysis of every peptide bond.
(D) Formation of glucose-fructose glycosidic links.

53. Equal volumes of $\frac{1}{20}$M $K_4[Fe(CN)_6]$ and $\frac{1}{20}$M glucose are mixed. In the final solution,the salt is50% dissociated into4K^+ and one intact complex anion;glucose does not dissociate. Use RT=24.0 L atm/mol. Osmotic pressure in atm is

(A) $\frac{18}{5}$ (B) $\frac{24}{5}$

(C) $\frac{6}{5}$ (D) $\frac{12}{5}$

54. A combustion stack emits sulphur dioxide before any atmospheric oxidation occurs. Choose its classification at the point of emission and a later possible consequence.

(A) Secondary pollutant; formation only by ocean salts.

(B) Inert noble gas; no chemical change possible.

(C) Solid particulate; compulsory ozone formation.

(D) Primary pollutant; contribution to acid deposition.

55. At a fixed temperature K=25. A reaction mixture initially has Q=50. Which direction lowers Gibbs energy, and which composition change must occur before equilibrium can be reached? Use ΔrG=RT ln(Q/K).

(A) reverse; Q must decrease to K.

(B) Neither direction whenever K is not one; equilibrium requires K=1.

(C) Reverse in every case because the standard reaction is exergonic.

(D) Forward in every case because K exceeds one.

56. For Yb^{2+} (Z=70),which configuration/chemical interpretation is consistent with its electron count and the commonly discussed +3 state?

(A) $4f^14$; oxidising character toward+3.

(B) $4f^13$; neither accepts nor loses an electron in changing to+3.

(C) $4f^15$; reducing character.

(D) $4f^14$; reducing character in the stated conversion toward the common +3 state.

57. Separate unknowns P and Q are each selected from Cu2+, Zn2+, Fe3+. Dropwise aqueous ammonia gives P a white hydroxide precipitate and Q a reddish brown precipitate. Excess ammonia gives P a colourless ammine solution and Q an undissolved reddish-brown hydroxide. Which identification is consistent, without invoking redox?

(A) Both are the same ion because all hydroxides have identical ammonia behaviour.

(B) P=Fe3+; Q=Zn2+; oxidation states unchanged.

(C) P=Zn2+; Q=Fe3+; oxidation states unchanged.

(D) P=Zn2+; Q=Fe3+; both metals reduced to zero.

58. hexane-2,5-dione,CH3CO(CH2)2COCH3 undergoes dilute base,then heat with dehydration. Under the stated intramolecular closure pathway,which isolated product matches both ring size and work-up endpoint?

(A) cyclohexanone

(B) 2-methylcyclopentanone

(C) 3-methylcyclopent-2-en-1-one

(D) 3-methylcyclohex-2-en-1-one

59. Match two applications: I. an orange-red advertising discharge tube. II. an intense flash lamp. Which ordered gas pair is appropriate?

(A) neon; helium.
(B) argon; xenon.
(C) neon; xenon.
(D) xenon; neon.

60. A model M(OH)2 has molar solubility 3/100000 M in pure water; neglect water autoionisation there. In a buffered solution its measured molar solubility is 27/1000000000 M. The buffer fixes free OH- and the solid's contribution to OH- is negligible. What is the imposed OH- concentration?

(A) 1/500
(B) 1/1000
(C) 1/250000
(D) 1/250

English Proficiency

61. Choose the correctly spelt word.

(A) questionnare
(B) questionnaire
(C) questionaire
(D) questinnaire

62. Read the sentence: The review was shallow, noticing surface features but missing the central argument. Choose the opposite in meaning of 'shallow' as used in the sentence.

(A) critical
(B) simplified
(C) penetrating
(D) short

63. Choose the most coherent order of the four sentences. A. The organisers introduced a short form recording what each item had stopped doing. B. The community opened a repair cafe for damaged household objects. C. Visitors soon began arriving with items whose faults were difficult to describe. D. Repairers could then begin with useful symptoms rather than spend the first minutes guessing.

(A) C-B-A-D
(B) B-A-C-D
(C) B-C-A-D
(D) B-C-D-A

64. She treats fragile objects to stabilise them and preserve their original material rather than trading them. What is she?

(A) conservator
(B) auctioneer
(C) collector
(D) curator of sales

65. She continues to work towards a perfectly fair system even when others call that goal impractical. Which word best describes this outlook?

(A) idealist
(B) cynic
(C) opportunist
(D) materialist

66. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. Did the reviewer reject the claim because it lacked evidence?

(A) Was the reviewer rejected by the claim because it lacked evidence?
(B) Was the claim being rejected by the reviewer because it lacks evidence?
(C) Was the claim rejected although it had sufficient evidence?
(D) Was the claim rejected by the reviewer because it lacked evidence?

67. Read the sentence: The new lease allowed the workshop to expand into the adjoining rooms. Choose the opposite in meaning of 'expand' as used in the sentence.

(A) spread (B) move

(C) renovate (D) contract

68. Choose the meaning-preserving equivalent that retains the tense, participants and all stated restrictions. She checked the address again so that the parcel would not be sent to the wrong office.

(A) She checked the address again because the parcel had been sent to the wrong office.

(B) She checked the address again to prevent the parcel from being sent to the wrong office.

(C) She checked every address before any parcel was sent.

(D) She sent the parcel to the wrong office in order to check the address.

69. Which figure of speech is chiefly illustrated by the line? O silence, stay with me until this page is finished!

(A) apostrophe (B) hyperbole

(C) simile (D) pun

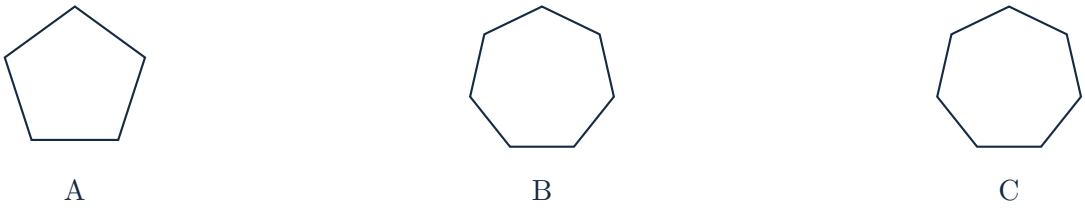
70. Read the sentence: The assistant was discreet about the confidential request and did not attract attention to it. Choose the closest meaning of 'discreet' as used in the sentence.

(A) separate and individually distinct (B) tactfully unobtrusive

(C) strictly divided (D) roughly estimated

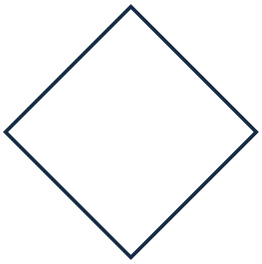
Logical Reasoning

71. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?

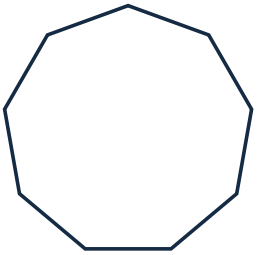


A B C

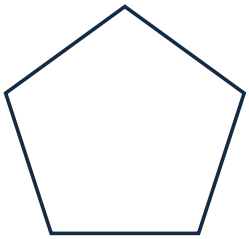
(A) Pattern N



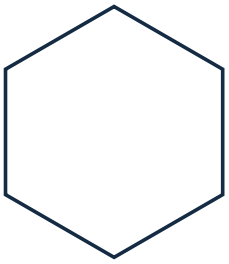
(B) Pattern L



(C) Pattern V



(D) Pattern Y



72. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

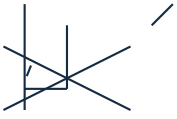




host 1



host 2



host 3



host 4

(A) 1 (B) 2

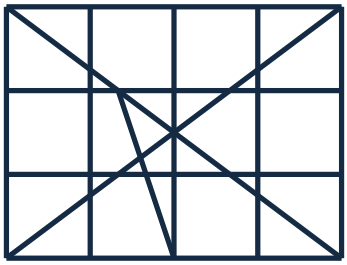
(C) 4 (D) 3

73. Bina receives 2 times as many tokens as Asha, and Charu receives 2 times as many as Bina. Their total is 126. After the allocation, Bina gives 6 tokens to Asha. How many tokens does Bina then retain?

(A) 42 (B) 36

(C) 30 (D) 12

74. How many triangles are formed by all the drawn segments?



(A) 92 (B) 50

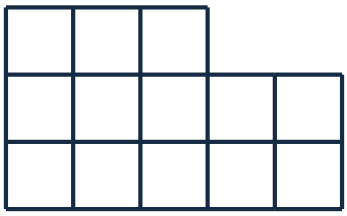
(C) 53 (D) 54

75. R and L are full siblings; L and G are also full siblings. K is the father of R. All four people are distinct. How is K related to G?

(A) Father (B) Uncle

(C) Grandfather (D) Relationship cannot be determined

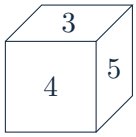
76. How many rectangles, including squares are formed by all the drawn segments?



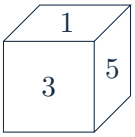
(A) 63 (B) 62

(C) 64 (D) 48

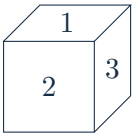
77. The views show the same die. Give the ordered pair: (face opposite 3, face opposite 5). Do not assume opposite labels sum to seven.



1



2

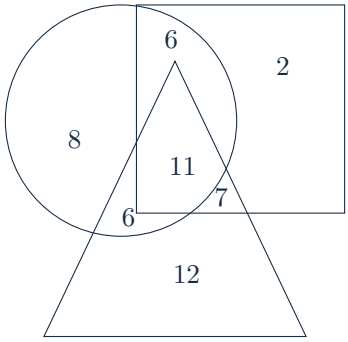


3

(A) $(6, 3)$ (B) $(6, 1)$

(C) $(6, 2)$ (D) $(6, 4)$

78. Circle=A, triangle=B, square=C. Each printed number is the count in that exact region. Find the total inside A but outside B, whether inside or outside C.



(A) 14 (B) 6

(C) 15 (D) 8

79. In a row, Asha is 9th from the left and 33rd from the right. Bina is 8th from the right. Charu stands 2 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 2 (B) 24

(C) 23 (D) 22

80. Consider only these statements; universal statements alone do not imply existence. Some hikers are not artists; All campers are artists. Conclusions: I. Some hikers are not campers. II. All hikers are artists. Which conclusions follow necessarily?

(A) Only II (B) Both I and II

(C) Neither I nor II (D) Only I

81. Preserve the exact relation and its order: Spider : Arachnid :: ? : Mammal. Choose the missing term.

(A) Frog (B) Whale

(C) Lizard (D) Sparrow

82. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, CLOUD is coded as GPSYH. Which word is coded as WXSRI?

(A) RSNMD (B) TUPOF

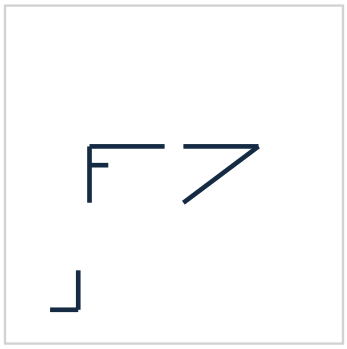
(C) ABWVM (D) STONE

83. Tara points to a photograph and says, "This person is the son of my father's only son." Tara is female, and the photographed person is male. How is the photographed person related to Tara?

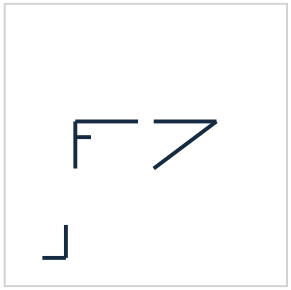
(A) Son (B) Daughter

(C) Brother (D) Nephew

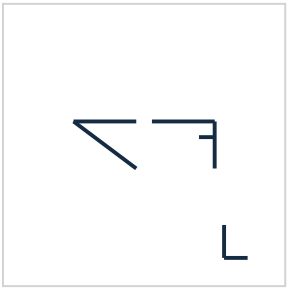
84. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



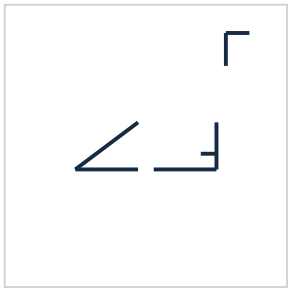
(A) Pattern M



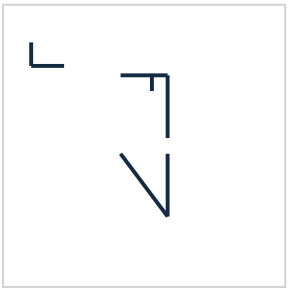
(B) Pattern Q



(C) Pattern W



(D) Pattern X



85. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 17$ and $5 \mapsto 35$. Which input maps to 71?

(A) 12 (B) 11

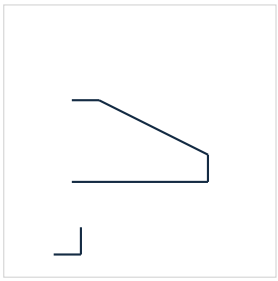
(C) 10 (D) 71

86. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 7, 14, 20, 40, ?, 92, 98.

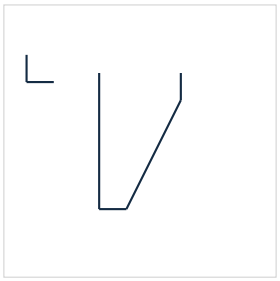
(A) 86 (B) 34

(C) 46 (D) 60

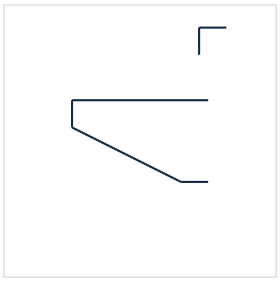
87. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

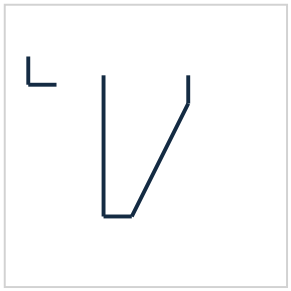


2



3

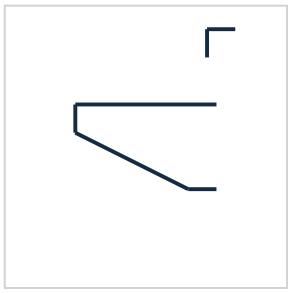
(A) Pattern U



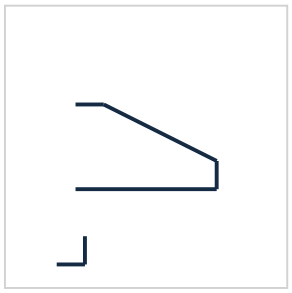
(B) Pattern L



(C) Pattern Y



(D) Pattern M



88. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: YUN, BYP, ?, HGT, KKV, NOX.

(A) ECR (B) DBQ

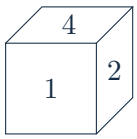
(C) RCE (D) FDS

89. For this question compare only cyclic forward alphabet gaps, with $Z \to A$ counting as one step. Three clusters share the same gap vector. Which numbered cluster differs? 1: KNOR, 2: LOPS, 3: VYZC, 4: FIKN.

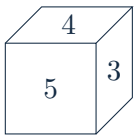
(A) 2 (B) 1

(C) 4 (D) 3

90. The views show the same die. Give the ordered pair: (face opposite 1, face opposite 2). Do not assume opposite labels sum to seven.



1



2

(A) $(5, 4)$
(B) $(5, 3)$
(C) $(5, 1)$
(D) $(5, 2)$

Mathematics

91. The line y=mx through O bisects the area of the triangle with vertices O,(3,2),(5,5). Assume the vertices are noncollinear. Find m.
(A) $\frac{8}{7}$
(B) $\frac{3}{2}$
(C) $\frac{5}{3}$
(D) $\frac{7}{8}$

92. Maximize Z=3x+4y subject to x≥0, y≥3, x+y≤11, x+2y≤13.
(A) 26
(B) 35
(C) 33
(D) 36

93. 8 distinct objects are distributed into three nonempty unlabelled boxes. How many distributions are possible?
(A) 2898
(B) 966
(C) 1093
(D) 5796

94. Between positive a=4, b=36, let A be their arithmetic mean and G their geometric mean. Find A^2-G^2.
(A) 328
(B) 256
(C) 64
(D) 400

95. If α,β are roots of x^2-20x+91=0, find $\alpha^2+\beta^2$ without solving the quadratic.
(A) 582
(B) 218
(C) 182
(D) 36

96. Let R=[sin($\frac{\pi}{10}$)+sin($\frac{7\pi}{30}$)]/[cos($\frac{\pi}{10}$)+cos($\frac{7\pi}{30}$)]. Find $(1\text{-}R^2)/(1+R^2)$.
(A) $\frac{1}{2}$
(B) $\frac{3}{2}$
(C) $-\frac{1}{2}$
(D) $\frac{\sqrt{3}}{2}$

97. Evaluate $\int_0^\infty dx/[(x^2+9)(x^2+64)]$.
(A) $\frac{\pi}{528}$
(B) $\frac{\pi}{264}$
(C) $\frac{\pi}{48}$
(D) $-\frac{\pi}{528}$

98. f(x)=x^3+5x, g=f^{-1}. Find g″(246).

(A) $\frac{1}{36}$ (B) $\frac{36}{1442897}$

(C) $-\frac{36}{1442897}$ (D) $-\frac{36}{12769}$

99. Let f(x)=x^3-48 x+6. Find the product of its local maximum and local minimum values.

(A) −49116 (B) −16348

(C) 16420 (D) 256

100. How many solutions in $0 \le x < 2\pi$ satisfy $\sin^4 x + \cos^4 x = 3/4$ and $\sin x > 0$?

(A) 8 (B) 3

(C) 4 (D) 5

101. Let R(x) be the remainder when x^4 is divided by (x-1)(x-2). Find R(7).

(A) 91 (B) 105

(C) 2401 (D) −83

102. For a=(1,4,0), b=(0,1,5), c=(4,0,1), evaluate (a×c)·(a+b).

(A) −80 (B) −81

(C) 0 (D) 81

103. Let $y\prime + 4y = 2x$ and $y(0) = 2$. Find $y(1) + \frac{3}{2}y\prime(1)$.

(A) $\frac{85}{8e^4} + \frac{9}{8}$ (B) $\frac{17}{8e^4} + \frac{3}{8}$

(C) $\frac{9}{8} - \frac{85}{8e^4}$ (D) $\frac{9}{8}$

104. 8 distinct people stand in a line. Three specified people cannot occupy positions 1,2 or 8. How many orders are possible?

(A) 1200 (B) 120

(C) 7200 (D) 14400

105. For the roots α,β of $z^2 - (3+7i)z + (-20+21i) = 0$, find $|\alpha\text{-}\beta|^2$.

(A) 58 (B) 116

(C) 29 (D) 841

106. Vectors a,b satisfy $|a|^2$=10, $|b|^2$=10, a·b=6. Find $|2a\text{-}b|^2$.

(A) 74 (B) 26

(C) 50 (D) 18

107. An AP has a_1=-27,d=3. How many of its first 14 terms are strictly positive?

(A) 10 (B) 11

(C) 3 (D) 4

108. Evaluate the integral from 0 to 2π of (6-3 sin x)(1+cos x)/(36-9 $\sin^2$x) dx.

(A) $-\frac{\log(3)}{3}+\frac{\log(2)}{3}+\frac{2\sqrt{3}\pi}{9}$

(B) $\frac{2\sqrt{5}\pi}{15}$

(C) $\frac{\sqrt{3}\pi}{18}$

(D) $\frac{2\sqrt{3}\pi}{9}$

109. Let k make lim as x$\to\infty$ of [($6x^2$+5x+1)/(x+2)-kx] finite. Find that finite limit.

(A) 12 (B) 5

(C) 17 (D) -7

110. Three distinguishable fair dice are thrown. Given that exactly two faces are equal, find the probability that their sum is at least 12.

(A) $\frac{1}{6}$ (B) $\frac{2}{5}$

(C) $\frac{3}{5}$ (D) $\frac{13}{30}$

111. How many integers x are in the real domain of $\sqrt{x-3}\,\log(7-x)$?

(A) 3 (B) 7

(C) 5 (D) 4

112. The normal of slope 4 to y^2=24x touches at P. Find the distance from P to the focus.

(A) 96 (B) 90

(C) 10404 (D) 102

113. How many integer t make P=(t,0) strictly inside the larger segment cut by x=2 from x^2+y^2=36?

(A) 9 (B) 3

(C) 7 (D) 11

114. Box A contains 5 red and 3 blue balls; B contains 4 red and 5 blue balls. A fair coin chooses one box. Two independent draws with replacement from that box give exactly one red ball. Find the posterior probability that A was chosen.

(A) $\frac{256}{499}$ (B) $\frac{45}{77}$

(C) $\frac{15}{64}$ (D) $\frac{243}{499}$

115. The letters A,A,B,C and 6 further distinct letters are arranged in a row. How many arrangements have nonadjacent As and have B,C adjacent in either order?

(A) 282240 (B) 2903040

(C) 362880 (D) 80640

116. $a_j = 4 + 2(j-1)$ and f(x)=(5/2)x^2+bx+4. If f(1)=f(-1), find f′(a_7).

(A) 90 (B) 16

(C) 84 (D) 80

117. The tangent to y=x^2+6x at P has slope 12. Find the slope of the normal at P.

(A) -12 (B) $-\frac{1}{12}$

(C) $-\frac{1}{144}$ (D) 12

118. Circles x^2+y^2=169 and (x-16)^2+y^2=169 meet at A,B. Find |AB|^2.

(A) 420 (B) 932

(C) 100 (D) 164

119. $A = \begin{bmatrix} 10 & 1 \\ 7 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix}$. Find tr(BA) without assuming AB=BA.

(A) -30 (B) 24

(C) 60 (D) 54

120. Evaluate $\sum_{r=0}^{10} r(r-1)\binom{10}{r}$.

(A) 23040 (B) 5120

(C) 25600 (D) 46080

121. The line r=(1,0,0)+s(t,1,2) is parallel to x-5y+z=7. Find the squared length of its direction vector.

(A) 14 (B) 12

(C) $\sqrt{14}$ (D) 30

122. Find the squared distance between 6x+8y=6 and 3x+4y=7.

(A) $\frac{4}{5}$ (B) $\frac{16}{25}$

(C) 16 (D) $\frac{4}{25}$

123. For integer n, consider: "If 6 divides n, then 3 divides n." Which is its contrapositive?

(A) If 3 divides n, then 6 divides n. (B) If 3 does not divide n, then 6 does not divide n.

(C) If 6 does not divide n, then 3 does not divide n. (D) 6 divides n and 3 does not divide n.

124. Four observations are 4,4,x,y with population mean 4 and variance 4/2. Find xy.

(A) 20 (B) 14

(C) 12 (D) 0

125. Let $P(x) = (1+\sqrt{2}x)^8 + (1-\sqrt{2}x)^8$. Find the sum of all nonconstant coefficients of P.

(A) 1154 (B) 1150

(C) 577 (D) 1152

126. If z=4-2i, find |exp(iz)|.

(A) e^{-2} (B) 1

(C) e^2 (D) e^4

127. Evaluate $5^{\log_5 6} + 2^{2(\log_2 5)(\log_5 6)}$.

(A) 12 (B) 41

(C) 42 (D) 36

128. Let y′=(x+y)/(x+y+1), with y(0)=1. Find the positive value of x at whichx+y=8.

(A) $\frac{\log\left(\frac{17}{3}\right)}{4} + \frac{7}{2}$ (B) $\log\left(\frac{\sqrt{51}}{3}\right) + 7$

(C) $\frac{9}{2} - \frac{\log\left(\frac{17}{3}\right)}{4}$ (D) $\frac{7}{2}$

129. How many solutions in $0 \leq x < 8\pi$ satisfy sin2x=sin x and cosx>0?

(A) 12 (B) 16

(C) 13 (D) 8

130. $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ sin\theta & \cos\theta \end{pmatrix}$, $0<\theta<\pi/2$, $\tan\theta=\frac{1}{4}$. Find tr(R$(\theta)^2$).

(A) $\frac{15}{17}$ (B) $\frac{16}{17}$

(C) $\frac{30}{17}$ (D) 2

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A satellite of mass m is moved from a circular orbit of radius R to one of radius 6R about mass M. Find the required increase in total orbital energy in units of $GMm/(2R)$.

(A) $\frac{1}{6}$ (B) $\frac{5}{6}$

(C) 5 (D) $\frac{35}{36}$

132. Two masses 2 kg and 6 kg are connected by a light string over a smooth pulley and accelerate freely. A transverse pulse travels in the taut straight part of this string, of linear density μ. Compare its speed to that in a second string of density 3μ held at tension g N numerically equal to 1 kg weight. Find v_1/v_2.

(A) 9 (B) $\sqrt{3}$

(C) 3 (D) $\frac{1}{3}$

133. A single slit of width 4 mm is illuminated by wavelength 400 nm. A screen is 5 m away. Find the separation of the second minima on opposite sides of the central maximum, in the small-angle approximation.

(A) 2 mm (B) $\frac{3}{2}$ mm

(C) $\frac{1}{2}$ mm (D) 4 mm

Chemistry

134. Bond energies D(X_2):D(Y_2):D(XY)=1:3:4. Forming2.00 mol XY(g) from the gaseous elements releases 480 kJ. D(Y_2),in kJ/mol,is

(A) 480 (B) 720

(C) 360 (D) 180.0

135. For 2A+B→3C+4D, A also disappears by an independent channel producing no C or D, and D is independently consumed at 1/2000 M/s. Measured total A disappearance is 7/1000 M/s and net D appearance is 3/400 M/s. What is the A disappearance rate through its independent channel, in M/s?

(A) 1/250 (B) 3/1000

(C) 13/4000 (D) 11/1000

136. For planar cyclobutadiene,choose the correct classification and the decisive reason. Use the explicitly stated geometry.

(A) antiaromatic; four continuous p orbitals contain four pi electrons

(B) aromatic; count only the neutral double bonds and ignore charge or geometry

(C) aromatic; every cyclic compound obeys 4n+2

(D) nonaromatic; an ion can never be aromatic

Logical Reasoning

137. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?



(A) 2 (B) 3

(C) 4 (D) 1

138. A word is coded as the sum of its forward alphabet positions plus 2 times its length. The code of BRIC? is 53. Which letter replaces the question mark?

(A) J (B) L

(C) K (D) M

139. Two people differ in age by 10 years. The product of their ages 4 years ago was 459. What will the younger person's age be 5 years from now?

(A) 26 (B) 36

(C) 21 (D) 22

Mathematics

140. The homogeneous system $6x+y=0$, $kx+2y=0$ has a nonzero solution. Find $k+x^2/y^2$.

(A) $\frac{431}{36}$ (B) 48

(C) $\frac{433}{36}$ (D) 12

141. X is binomial with mean 8 and variance $\frac{16}{3}$. Find its mode.

(A) 7 (B) 8

(C) 9 (D) 24

142. A,B are acute, $A+B=\pi/4$ and $\tan A=\frac{7}{13}$. Find $\tan(A-B)$.

(A) $\frac{31}{151}$ (B) $\frac{31}{130}$

(C) $\frac{3}{10}$ (D) $-\frac{31}{151}$

BITSAT Mock 21

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A 8 kg block on a rough horizontal table is tied over an ideal pulley to a hanging 6 kg mass. The hanging mass moves downward and $\mu_k = 1/5$. Find string tension. Take $g = 10\ \text{m/s}^2$.

(A) $48\,\text{N}$ (B) $16\,\text{N}$

(C) $\frac{288}{7}\,\text{N}$ (D) $\frac{176}{7}\,\text{N}$

2. A mass 3 kg is attached to two massless springs in series with spring constants 80 N/m and 160 N/m. Find the small-oscillation period on a smooth horizontal surface.

(A) $\frac{3\sqrt{10}\pi}{20}\,\text{s}$ (B) $\frac{\sqrt{30}\pi}{20}\,\text{s}$

(C) $\frac{3\sqrt{5}\pi}{10}\,\text{s}$ (D) $\frac{\sqrt{15}\pi}{10}\,\text{s}$

3. A galvanometer has resistance 50 ohms and full-scale current $\frac{3}{1000}$ A. It is to measure a total current 3 times its own full-scale current. Find the required parallel shunt resistance.

(A) $75\,\Omega$ (B) $25\,\Omega$

(C) $150\,\Omega$ (D) $\frac{25}{2}\,\Omega$

4. A stretched string is divided by a bridge into lengths in the ratio 1 : 6. Tension and linear density are unchanged. The original fundamental is 40 Hz. Find the fundamental of the longer segment.

(A) $56\,\text{Hz}$ (B) $\frac{70}{9}\,\text{Hz}$

(C) $\frac{20}{3}\,\text{Hz}$ (D) $\frac{140}{3}\,\text{Hz}$

5. A one-electron ion with nuclear charge $Z = 3$ absorbs a photon of energy $\frac{1487}{125}$ eV. The ejected electron has maximum kinetic energy 7 eV. From which principal quantum number was the electron removed? Use $E_n = -13.6Z^2/n^2$ eV and neglect recoil.

(A) 5 (B) 4

(C) 15 (D) $\frac{5}{3}$

6. A wire consists of a semicircular arc of radius 2 m carrying 5 A, joined to two straight radial segments that extend outward from its ends. Find the magnetic field magnitude at the arc centre in units of μ_0.

(A) $\frac{5}{4\pi}$ (B) $\frac{5}{16}$

(C) $\frac{5}{8}$ (D) $\frac{5(2+\pi)}{8\pi}$

7. A travelling wave is $y = 5\sin[6(x - \frac{35}{3}t) + \pi/4]$ mm with x in metres and t in seconds. What is its frequency?

(A) $\frac{35}{\pi}$ Hz (B) 70 Hz

(C) $\frac{210}{\pi}$ Hz (D) $\frac{70}{\pi}$ Hz

8. A uniform disc of mass 2 kg and radius 2 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) $\frac{19}{4}\,\mathrm{kg\,m^2}$ (B) $\frac{13}{4}\,\mathrm{kg\,m^2}$

(C) $3\,\mathrm{kg\,m^2}$ (D) $\frac{15}{4}\,\mathrm{kg\,m^2}$

9. Two ideal cells of emfs 6 V and 4 V oppose each other in a loop containing two resistors each 4 ohms. Find the current magnitude.

(A) $\frac{5}{2}$ A (B) $\frac{5}{4}$ A

(C) $\frac{1}{4}$ A (D) $\frac{3}{4}$ A

10. A ferromagnetic material has Curie temperature 600 K. It is tested at 590 K and 640 K in a weak applied field. Which description is appropriate?

(A) It must be diamagnetic at both temperatures.

(B) Spontaneous domain order is possible at the lower temperature; above the Curie temperature the material is paramagnetic.

(C) Its Curie temperature changes to match either test temperature.

(D) It becomes more strongly spontaneously ferromagnetic at the higher temperature.

11. A monatomic ideal gas expands reversibly and adiabatically until its pressure becomes 1/16807 of its initial value. Find final temperature divided by initial temperature.

(A) 49 (B) $\frac{1}{7}$

(C) $\frac{1}{16807}$ (D) $\frac{1}{49}$

12. A projectile is launched at 75 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{8}{9}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ $\mathrm{m/s^2}$.

(A) 300 m (B) 360 m

(C) 180 m (D) 540 m

13. A screw gauge advances 2.0 mm in four complete turns and has 50 circular divisions. With the jaws closed, its zero mark lies 3 divisions above the reference line; use the usual convention that a zero below the reference line gives positive zero error. With a wire, the main scale reads 6 mm and circular reading is 25 divisions. What is the corrected diameter?

(A) $\frac{311}{50}$ mm (B) $\frac{653}{100}$ mm

(C) $\frac{62503}{10000}$ mm (D) $\frac{157}{25}$ mm

14. A drop of mass 1.92×10^{-17} kg is suspended in a vertical field of magnitude 300 N/C. It has excess electrons. One excess electron is removed without changing its mass. Find its immediate downward acceleration, taking $g = 10\ \text{m/s}^2$ and $e = 1.6 \times 10^{-19}$ C.

(A) $10\,\text{m/s}^2$ (B) $2\,\text{m/s}^2$

(C) $\frac{15}{2}\,\text{m/s}^2$ (D) $\frac{5}{2}\,\text{m/s}^2$

15. A long solenoid has its turn number multiplied by 3 and length multiplied by 3, while its cross-sectional area and core permeability stay fixed. Find self-inductance multiplier.

(A) 27 (B) 3

(C) $\frac{1}{3}$ (D) 9

16. A ball is dropped from 30 m onto a fixed horizontal floor. Its coefficient of restitution is 1/2. Find its height above the floor at time $\frac{\sqrt{6}}{4}$ s after the first impact. Take $g = 10\ \text{m/s}^2$.

(A) $\frac{45}{8}\,\text{m}$ (B) $\frac{15}{2}\,\text{m}$

(C) $\frac{15}{8}\,\text{m}$ (D) $\frac{105}{8}\,\text{m}$

17. A network is defined by $P = \text{NOR}(A, B)$, $Q = \text{NOR}(A, P)$ and $Y = \text{OR}(Q, B)$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 1$ and $A = 0$?

(A) 1 (B) 3

(C) 0 (D) 2

18. In a series RLC circuit, ideal rms voltmeters read 40 V across R, 60 V across L and 10 V across C. Find source rms voltage.

(A) $10\sqrt{53}\,\text{V}$ (B) $40\,\text{V}$

(C) $10\sqrt{41}\,\text{V}$ (D) $90\,\text{V}$

19. A pipe of cross-sectional area 4A carries incompressible fluid at 5 m/s and feeds 6 identical holes, each area A, with equal flow. Find speed from each hole.

(A) $5\,\text{m/s}$ (B) $\frac{15}{2}\,\text{m/s}$

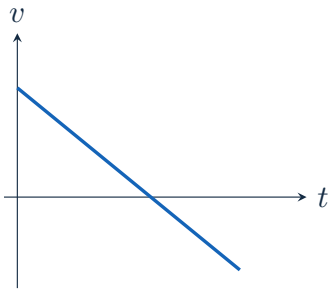
(C) $\frac{10}{3}\,\text{m/s}$ (D) $\frac{5}{6}\,\text{m/s}$

20. A solid sphere of radius R has uniform density $4\rho_0$. A spherical cavity of radius $R/10$ is cut with its centre at distance $R/3$ from the original centre. Find field magnitude at the cavity centre in units of $\pi G \rho_0 R$.

(A) $\frac{1}{3}$ (B) 16

(C) 4 (D) $\frac{16}{9}$

21. A particle has velocity $v(0) = 5$ m/s, and constant acceleration makes $v(5) = -5$ m/s. What total distance does it travel in this interval?



(A) 0 m
(B) $\frac{25}{4}$ m
(C) $\frac{25}{2}$ m
(D) 25 m

22. In YDSE a sheet of index 1 and thickness 7 micrometres covers one slit. A second sheet of the same thickness and index 3/2 covers the other. With wavelength $\frac{2}{5}$ micrometres, find the magnitude of central-fringe shift in units of fringe width.

(A) $\frac{175}{4}$
(B) $\frac{35}{2}$
(C) $\frac{35}{8}$
(D) $\frac{35}{4}$

23. A converging thin lens forms a real image with magnification magnitude 2. The object-image separation is 6 cm. Find focal length.

(A) $\frac{2}{3}$ cm
(B) $\frac{3}{2}$ cm
(C) 2 cm
(D) $\frac{4}{3}$ cm

24. A boat moves at 15 m/s relative to water. The current is 9 m/s and river width is 48 m. The boat lands directly opposite its starting point. Find crossing time.

(A) $\frac{8\sqrt{34}}{17}$ s
(B) 4 s
(C) $\frac{16}{3}$ s
(D) 8 s

25. A 6 V Zener regulator feeds a constant-current load of 4 mA. Supply varies from 12 to 16 V. Regulation requires Zener current at least 1 mA, and its safe maximum is 6 mA. What is the largest permissible series resistance? Assume ideal regulation and give resistance in Ω.

(A) 1200 Ω
(B) 2400 Ω
(C) 1000 Ω
(D) 1500 Ω

26. A coil of inductance 3 H carries current $i(t) = 6t^2$ A. Find the magnitude of its self-induced emf at $t = 4$ s.

(A) 72 V
(B) 144 V
(C) 576 V
(D) 18 V

27. Two identical uniform spheres touch. Their radii are multiplied by 15 while their density is divided by 15. By what factor does their gravitational attraction change?

(A) 1
(B) 225
(C) $\frac{1}{225}$
(D) 50625

28. A potentiometer wire of resistance 40 ohms and length 200 cm is driven by an ideal 25 V source through a 80 ohm resistor and a silicon diode in series. The diode is forward biased with constant drop 0.7 V. A test cell of emf 1 V draws no current at balance. Find balance length, assuming it lies within the wire.

(A) $\frac{2000}{243}$ cm (B) $\frac{2000}{81}$ cm

(C) $\frac{14200}{81}$ cm (D) 8 cm

29. A 2 kg block initially moves at 2 m/s on a horizontal rough floor. A forward constant force of 6 N acts for 1 s. With $\mu_k = 1/5$ and $g = 10$ m/s^2, find net work during this interval.

(A) 5 J (B) 15 J

(C) 18 J (D) 21 J

30. An ideal mixture contains 5 moles of monatomic gas and 3 moles of rigid diatomic gas. Vibrations are frozen. Find the heat-capacity ratio $\gamma = C_P/C_V$ of the mixture.

(A) $\frac{23}{15}$ (B) $\frac{47}{30}$

(C) $\frac{7}{5}$ (D) $\frac{15}{23}$

Chemistry

31. Use the supplied molar heat capacities:Cu(s)$3R$,Br_2(l)$9R$,Ar(g)$5R/2$. Molar masses are64,160,40 g/mol respectively. Order their specific heat capacities per gram increasingly.

(A) Br_2<Cu<Ar. (B) Cu<Br_2<Ar.

(C) Ar<Cu<Br_2. (D) Cu<Ar<Br_2.

32. At one T,K_1=4 for A⇌B,and K_2=25 for B⇌C. Numerical equilibrium constant for2C⇌2A is

(A) 100 (B) $\frac{1}{641}$

(C) $\frac{1}{100}$ (D) $\frac{1}{10000}$

33. A fixer initially receives 1/50 mol AgBr and an unknown amount of free thiosulfate. After complete stoichiometric complex formation, 1/100 mol AgBr remains and no free thiosulfate is left. Use AgBr+2S2O3^2-→[Ag(S2O3)2]^3-+Br-. What was the initial amount of thiosulfate?

(A) 1/50 (B) 3/100

(C) 1/25 (D) 1/100

34. Construct one open linear peptide using each member of the following multiset exactly once: Pro, Gly, Ala, Val, Ser. The N-terminal residue isPro. Exactly one terminus is occupied by the only achiral amino acid. Count distinctN-to-C sequences; ignore stereoisomerism of the supplied residues.

(A) 120 (B) 7

(C) 6 (D) 24

35. The sweetener sucralose contains carbon-bound chlorine atoms. Which conclusion avoids confusing a covalent substituent with an ionic salt?

(A) ReplacingOH by Cl must increase the oxygen count.

(B) Every chlorine atom necessarily occurs asNaCl.

(C) The carbon skeleton must disappear when chlorine enters.

(D) Chlorine content alone does not imply free chloride ions paired with sodium.

36. A saturated acyclic hydrocarbon has formula C5H12. All its hydrogen atoms are chemically equivalent by molecular symmetry. Which pair gives the hydrocarbon and the number of primary carbons?

(A) 2-methylbutane; 3

(B) 2,2-dimethylpropane; 4

(C) pentane; 2

(D) 2,2-dimethylpropane; 1

37. An ion has 18 electrons,20 neutrons and charge -1. Which combination gives its element,mass number and neutral-element group correctly?

(A) Cl;A=37;group17.

(B) An element with Z=18;A=38;group18.

(C) Cl;A=37;group18.

(D) Cl;A=38;group17.

38. Among CaCO3 and SrCO3, which member generally requires stronger heating to decompose, and why?

(A) SrCO3; the larger Sr2+ cation has lower carbonate-polarising power.

(B) SrCO3; Sr2+ has a higher charge than Ca2+.

(C) CaCO3; Ca2+ is larger than Sr2+.

(D) Both must decompose at the same temperature because their anions agree.

39. A first-order reaction hasEa/R=5400K andfrequency factorA=(ln2)e^{18}/60s^{-1}. At what nearest whole-kelvin temperature is75.0% completion reached in60.0s? Useln2=.693.

(A) 290

(B) 330

(C) 312

(D) 300

40. Consider I: Nitrogen does not form an ordinary neutral NCl5 analogue of PCl5. II: The F-F bond is weaker than the Cl-Cl bond. Which assessment correctly recognises both anomalous first-element behaviours?

(A) Both statements are correct.

(B) Only II is correct.

(C) Neither statement is correct.

(D) Only I is correct.

41. In an AX2E3 species with five electron domains, the molecule is linear. Which occupancy explains this within a trigonal-bipyramidal electron-domain model?

(A) There are no lone pairs because every linear molecule is AX2.

(B) Two equatorial lone pairs and three bond pairs are present.

(C) Three axial lone pairs leave two equatorial bond positions.

(D) Three equatorial lone pairs leave the two axial bond positions occupied.

42. In a hypothetical rule,$l = 0, 1, \ldots, n$ is allowed. Keep subshell capacity2(2l+1),Hund and Pauli,and fill by n+l then lower n. For 22 electrons,the last occupied subshell and its electron count are

(A) $3p^4$.

(B) $2p^4$.

(C) $2d^4$.

(D) $3d^4$.

43. For isolated Fe^{2+} (Z=26),use the high-spin spin-only approximation. Which pair (d-electron count, magnetic moment in BM) follows after removing electrons from 4s before 3d?

(A) $(6,2\sqrt{6})$.

(B) (6,4).

(C) $(7,\sqrt{35})$.

(D) $(6,\sqrt{35})$.

44. In a copper extraction stage, $\frac{1}{20}$ mol Cu_2S and $\frac{3}{50}$ mol Cu_2O react only by $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$. What amount of metallic copper, in mol, is formed?

(A) $\frac{9}{50}$ (B) $\frac{3}{50}$

(C) $\frac{4}{25}$ (D) $\frac{3}{10}$

45. 1-methylcyclohexene (C7H12) adds one equivalent of :CCl2 across its only C=C without rearrangement. Which pair gives the adduct formula and the number of independent rings? Count the original ring and the new three-membered ring separately.

(A) C7H12Cl2; 1 (B) C8H14Cl2; 2

(C) C8H12Cl2; 1 (D) C8H12Cl2; 2

46. A food extract contains exactly one of the listed two-component combinations. Using the usual qualitative tests under suitable conditions, its results are: Benedict on warming: positive; iodine: positive; Biuret: negative; Sudan III: negative. Which combination matches?

(A) glucose and albumin (B) starch and albumin

(C) glucose and starch (D) glucose and triolein

47. A rubber repeat contains a chlorine substituent inherited from 2-chlorobuta-1,3-diene. Which starting monomer and polymer pair is correct?

(A) Ethylene glycol; natural rubber. (B) Styrene; Buna-N.

(C) Vinyl chloride; nylon-6,6. (D) Chloroprene; neoprene.

48. In water,ignoring special ortho steric effects,which decreasing basicity order is most appropriate for aniline,benzylamine,4-nitroaniline?

(A) benzylamine >4-nitroaniline > aniline (B) aniline > benzylamine >4-nitroaniline

(C) benzylamine > aniline >4-nitroaniline (D) 4-nitroaniline > aniline > benzylamine

49. Match two applications: I. an inert atmosphere for welding reactive hot metals. II. a low-temperature coolant for superconducting equipment. Which ordered gas pair is appropriate?

(A) argon; neon. (B) argon; helium.

(C) helium; argon. (D) xenon; helium.

50. An energy density obeys u=(5/2) n kT,whereu isJ/m^3 andn counts particles per m^3. WithNA in mol^{-1},which units for k andR=NAk are correct?

(A) k:Pa K^{-1};R:Pa $mol^{-1}K^{-1}$. (B) k:J K^{-1};R:J $mol^{-1}K^{-1}$.

(C) k:J $mol^{-1}K^{-1}$;R:J K^{-1}. (D) Both k andR have units J K^{-1} becauseNA is dimensionless.

51. The internuclear axis is y. Two p_z orbitals overlap sideways. Their lobes facing each other above the axis have the same phases; the lower facing lobes have the corresponding opposite signs. Which MO is produced,and what nodal property follows?

(A) A pi antibonding MO; the facing-lobe phase test has been reversed.

(B) A pi bonding MO; the plane containing the axis is a node,but there is no additional internuclear node from phase cancellation.

(C) A sigma bonding MO,because every overlap along a line joining nuclei is sigma.

(D) A delta MO,because two perpendicular p orbitals always create two nodal planes.

52. An H_2SO_4 stock is29.0% by mass. A20.0mL aliquot is diluted to 80mL,and the diluted molarity is $\frac{9}{10}$M. H_2SO_4 M98. Stock density in g/mL is

(A) $\frac{3528}{725}$ (B) $\frac{882}{725}$

(C) $\frac{441}{1450}$ (D) $\frac{441}{1250}$

53. Acidic MnO4- oxidises SO2 to HSO4- and is reduced to Mn2+. The balanced reaction is 2MnO4-+5SO2+2H2O+H+→2Mn2++5HSO4-. Initially 11/1000 mol H+ and excess SO2 are available. After exactly 1/50 mol MnO4- is consumed, what amount of net H+ remains? Count net H+ from the displayed balanced equation.

(A) 1/2000 (B) 1/1000

(C) -1/1000 (D) 1/500

54. An unknown is one of sucrose, maltose or lactose. It is reducing before hydrolysis, and its complete hydrolysis gives only one kind of monosaccharide. Identify it and that product.

(A) Sucrose; fructose only. (B) Sucrose; galactose only.

(C) Maltose; glucose. (D) Lactose; glucose only.

55. An isolated NaCl cathode compartment of volume $\frac{2}{5}$L initially has $\frac{1}{25}$mol strong acid. Current5.00A flows for 1544s,F=96500C/mol. Only cathodic water reduction produces OH^-,one per electron,and no anode product enters. Volume is fixed. At25C,finalpH is

(A) 13 (B) 1

(C) 7 (D) 14

56. In a pressure range where the Freundlich law $x/m = kp^{1/4}$ is valid at fixed temperature, adsorbent B has 7 times the mass of A and is exposed to 2401 times its pressure. Find the ratio of total adsorbed amounts x_B/x_A.

(A) 7 (B) 16807

(C) 1 (D) 49

57. At25C,a measurement gives$[OH^-]=2\times10^{-8}$M. UseKw=10^{-14} and$\log_{10}2=\frac{301}{1000}$. Solution pH is

(A) $\frac{8301}{1000}$ (B) $\frac{6301}{1000}$

(C) $\frac{7699}{1000}$ (D) $\frac{5699}{1000}$

58. With one equivalent of cold concentrated HI and isolation before prolonged further reaction,which initial cleavage pair is expected from methyl isopropyl ether?

(A) propene and methyl iodide only (B) ethanol and ethyl iodide

(C) 2-iodopropane and methanol (D) methyl iodide and propan-2-ol

59. For one reaction,ln K is4 at300 K and0 at600 K. Assume ln K is linear in1/T. At400 K,ln K equals

(A) 3 (B) 0

(C) 2 (D) 4

60. For an isolated H_2O_2 molecule,which combined statement about local geometry and overall conformation is correct?

(A) Each O has two bonds and two lone pairs; the H-O-O-H skeleton is nonplanar/open-book.

(B) Each O is linear because it is bonded to only two atoms,and the skeleton is straight.

(C) The O-O bond is a double bond,so both O centres are planar with one lone pair.

(D) All four atoms must lie in one plane because only sigma bonds are present.

English Proficiency

61. Read the sentence: Without maintenance, the condition of the track will deteriorate. Choose the opposite in meaning of 'deteriorate' as used in the sentence.

(A) decline (B) improve

(C) be examined (D) remain uncertain

62. Which figure of speech is chiefly illustrated by the line? The stubborn lock refused every key we tried.

(A) hyperbole (B) apostrophe

(C) personification (D) simile

63. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. of the symbols on the existing one. Q. What the visitors needed R. but a clear explanation S. was not another map

(A) Q-S-R-P (B) S-Q-R-P

(C) P-S-R-Q (D) R-Q-S-P

64. Read the sentence: The ranking is provisional and may change after the appeals are considered. Choose the closest meaning of 'provisional' as used in the sentence.

(A) subject to revision (B) based on incomplete skill

(C) valid for one person (D) not officially circulated

65. Choose the active-voice equivalent that retains the tense, participants and all stated restrictions. The cracked panels were replaced by the technicians while the supervisor checked the wiring.

(A) The supervisor replaced the cracked panels while the technicians checked the wiring.

(B) The technicians were replacing the cracked panels when the supervisor checked the wiring.

(C) The technicians replaced the cracked panels after the supervisor checked the wiring.

(D) The technicians replaced the cracked panels while the supervisor checked the wiring.

66. The handwriting is legible, meaning clear enough to read. Choose its opposite.

(A) eligible (B) illegible

(C) legitimate (D) legible

67. She interprets pottery, tools and building remains to learn about earlier societies. What is she?

(A) geologist (B) archivist

(C) archaeologist (D) biographer

68. The group reorganises and resumes effective work after a serious setback. Which noun best names this capacity?

(A) rigidity (B) invulnerability

(C) permanence (D) resilience

69. Use the standard contemporary negative adjective. Under the stated conditions the task cannot be done; it is _____.

(A) dispossible (B) impossible

(C) unpossible (D) inpossible

70. Read the sentence: Asked why the trial failed, the researcher gave a candid account of her own mistakes. Choose the closest meaning of 'candid' as used in the sentence.

(A) frank (B) evasive

(C) accusatory (D) careless

Logical Reasoning

71. Consider only these statements; universal statements alone do not imply existence. No editors are joggers; All editors are joggers. Conclusions: I. Some hikers are not editors. II. All editors are hikers. Which conclusions follow necessarily?

(A) Neither I nor II (B) Both I and II

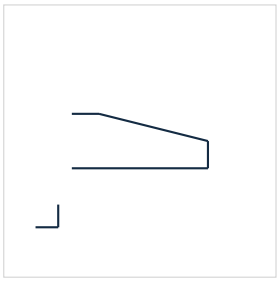
(C) Only II (D) Only I

72. In a row, Asha is 9th from the left and 21st from the right. Bina is 9th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

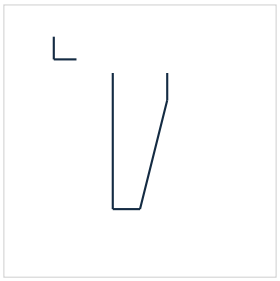
(A) 8 (B) 9

(C) 14 (D) 2

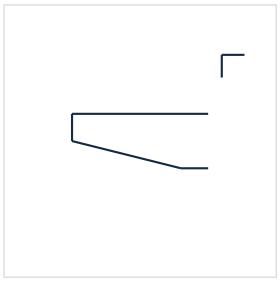
73. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

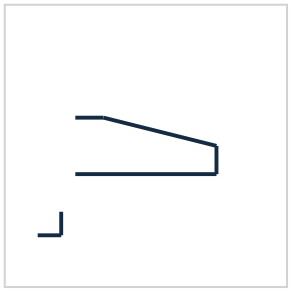


2

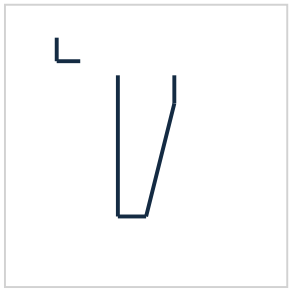


3

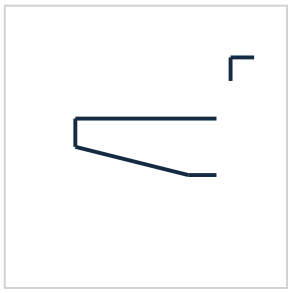
(A) Pattern L



(B) Pattern M



(C) Pattern T



(D) Pattern P



74. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 4

(B) 3

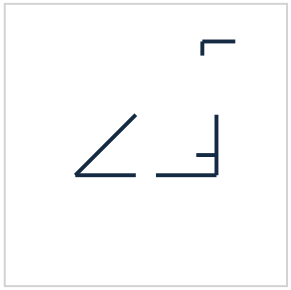
(C) 2

(D) 1

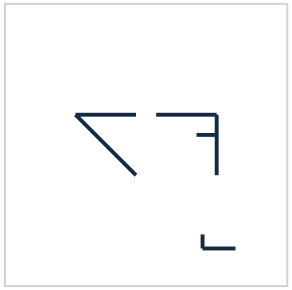
75. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



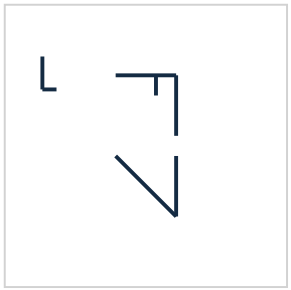
(A) Pattern X



(B) Pattern Z



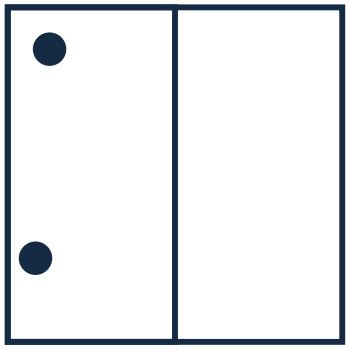
(C) Pattern M



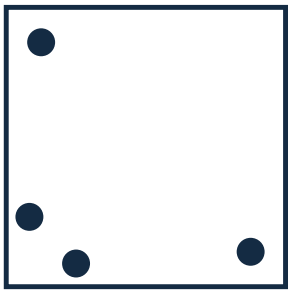
(D) Pattern U



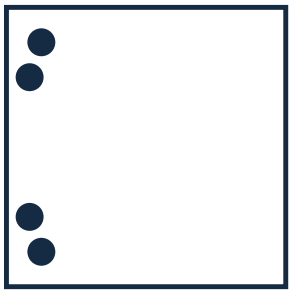
76. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



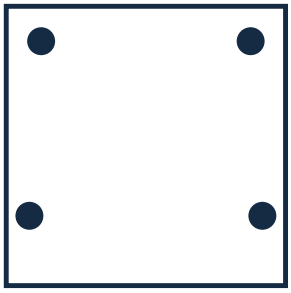
(A) Pattern W



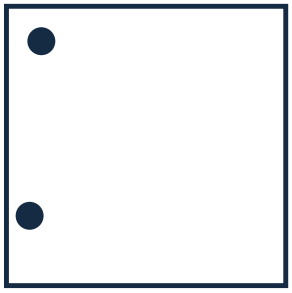
(B) Pattern M



(C) Pattern U



(D) Pattern Q



77. Five people occupy positions 1 to 5 from left to right, one per position, and wear five different shirt colours: red, blue, green, white and yellow. Bina is 1st from the left. Asha is 2nd from the left. Dev is 3rd from the left. Esha is 4th from the left. Charu is 5th from the left. The blue shirt is in position 1. The red shirt is in position 2. The yellow shirt is in position 3. Esha does not wear white. Which shirt does Esha wear?

(A) White (B) Blue

(C) Green (D) Red

78. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 2, 6, 9, ?, 30, 90, 93.

(A) 24 (B) 12

(C) 30 (D) 27

79. Each rectangle is drawn to the same local scale. In three numbered figures the interior straight line divides the rectangle into equal areas. Which does not?



(A) 1
(B) 4
(C) 2
(D) 3

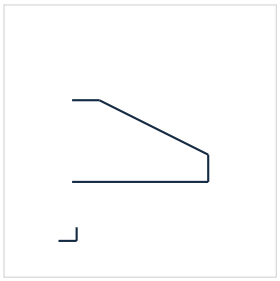
80. In this code, @ means $\times$, # means $+$, & means $-$, % means $\div$. After substituting meanings, evaluate ((3#4)@(3&3)); printed parentheses retain their usual grouping.

(A) -1
(B) 0
(C) 2
(D) 1

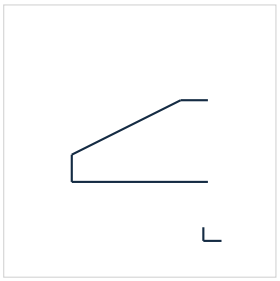
81. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 31 and Z has code 6. Decode the separated values 25, 14, 31, 16, 27.

(A) FQZOD
(B) GRAPE
(C) HSBQF
(D) EPARG

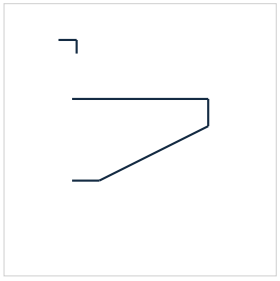
82. Select the missing bottom-right tile. Tiles reflect across the shared horizontal and vertical centre lines; there is no turning.



top left



top right

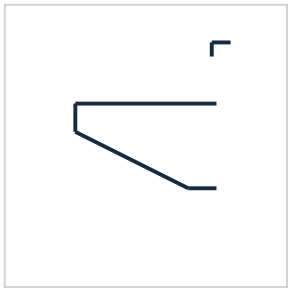


bottom left

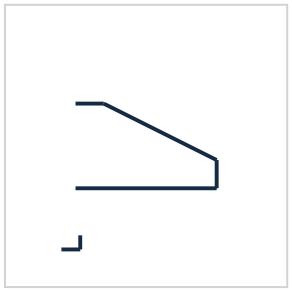
?

bottom right

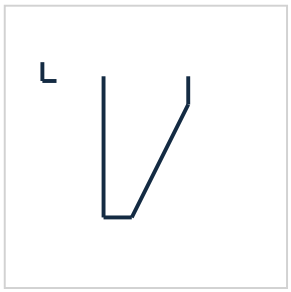
(A) Pattern K



(B) Pattern W



(C) Pattern Z



(D) Pattern L



83. Preserve the exact relation and its order: Wolf : Pack :: ? : Constellation. Choose the missing term.

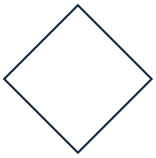
(A) Bee

(B) Star

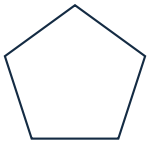
(C) Lion

(D) Wolf

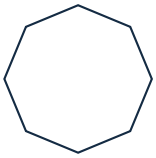
84. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

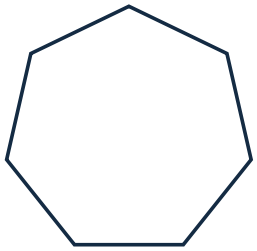


B

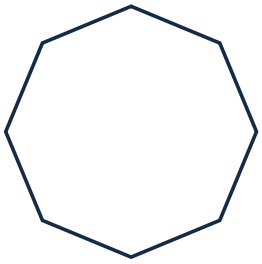


C

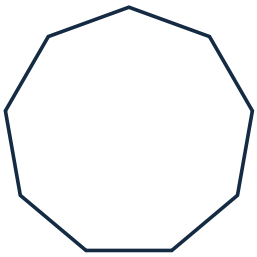
(A) Pattern L



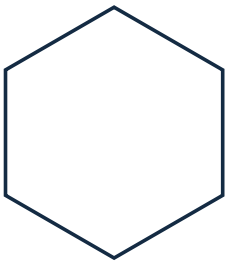
(B) Pattern M



(C) Pattern V



(D) Pattern K



85. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target



piece 1



piece 2

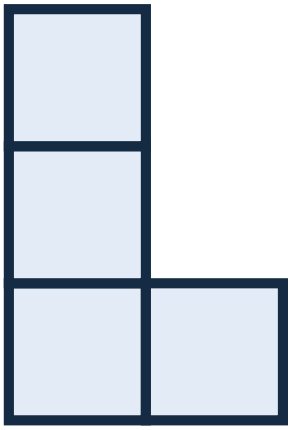
(A) Pattern N



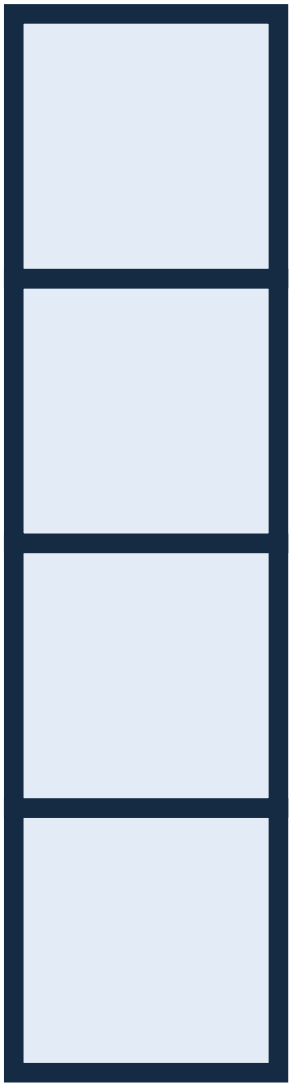
(B) Pattern Z



(C) Pattern K

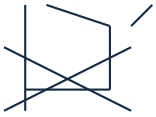


(D) Pattern Q

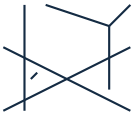


86. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

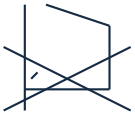




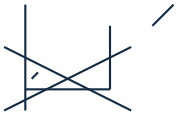
host 1



host 2



host 3



host 4

(A) 2

(B) 4

(C) 1

(D) 3

87. The increase from the left cell to the right cell is the same in each row, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.





?

(A) Pattern W



(B) Pattern V



(C) Pattern X



(D) Pattern M



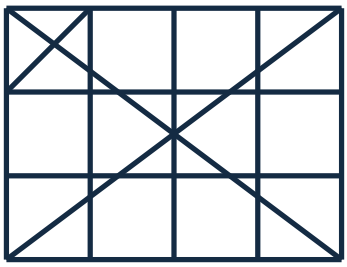
88. Consider only these statements; universal statements alone do not imply existence. Some guides are not hikers; Some guides are hikers; Some guides are not joggers. Conclusions: I. No joggers are guides. II. No guides are hikers. Which conclusions are individually possible without contradicting the statements?

(A) Both I and II
(B) Only II
(C) Neither I nor II
(D) Only I

89. V and A are full siblings; A and N are also full siblings. J is the mother of V. All four people are distinct. How is J related to N?

(A) Grandmother
(B) Mother
(C) Aunt
(D) Relationship cannot be determined

90. How many triangles are formed by all the drawn segments?



(A) 36
(B) 38
(C) 50
(D) 88

Mathematics

91. There are 8 Boolean variables p_1,...,p 8. How many truth assignments make [(p_1 implies p_2) and (p_1 implies not p_2)] true?

(A) 256
(B) 255
(C) 64
(D) 128

92. Find the area between y=(x-6)(x-9) and the x-axis for $0 \leq x \leq 13$.

(A) $-\frac{1001}{6}$
(B) $\frac{1055}{6}$
(C) $\frac{1001}{6}$
(D) $\frac{1055}{12}$

93. Evaluate the limit of $[\cos(2x)-\cos(3x)]/x^2$ as $x \to 0$.

(A) $\frac{5}{2}$
(B) $\frac{1}{2}$
(C) $-\frac{5}{2}$
(D) 5

94. The curve $y=x^3+7x$ divides the disc $x^2+y^2 \leq 64$. Find the area of the part satisfying $y \geq x^3+7x$.

(A) 64π
(B) 16π
(C) $7 + 32\pi$
(D) 32π

95. What is the maximum number of distinct intersection points formed by 5 circles and 4 straight lines, with no coincident curves?

(A) 60 (B) 46

(C) 66 (D) 86

96. How many integers k make 27sin x+36cos x+4=k have exactly two distinct solutions in $0 \le x < 2\pi$?

(A) 91 (B) 44

(C) 90 (D) 89

97. A=(5,2,3), O=(0,0,0). Find the squared length of the orthogonal projection of OA on the line through O with direction (1,2,2).

(A) 225 (B) 5

(C) 13 (D) 25

98. Evaluate $\int_0^\infty \log(6x)/(x^2+36)\,dx$.

(A) $\frac{\pi \log(6)}{12}$ (B) 0

(C) $\frac{\pi \log(6)}{6}$ (D) $\pi \log(6)$

99. $F'(x)=\frac{1}{x^2} - \frac{1}{x^2\sqrt{x^2+1}}$ and F(1)=2. Find F(2)-2F(1).

(A) $-\sqrt{2}+\frac{1}{2}+\frac{\sqrt{5}}{2}$ (B) $-\sqrt{2}+\frac{\sqrt{5}}{2}+\frac{5}{2}$

(C) $-\frac{5}{2}-\frac{\sqrt{5}}{2}+\sqrt{2}$ (D) $-\frac{3}{2}-\sqrt{2}+\frac{\sqrt{5}}{2}$

100. The same loaded die has odd-face probability $\frac{7}{8}$. Three independent throws are made. Given that the first two have an even sum, find the probability that the total of all three is odd.

(A) $\frac{1}{8}$ (B) $\frac{7}{32}$

(C) $\frac{7}{8}$ (D) $\frac{91}{128}$

101. The chord of contact from P to $x^2/4-y^2/9=1$ is x=2/2. Find the product of the slopes of the two tangents from P.

(A) $-\frac{3}{4}$ (B) $\frac{3}{4}$

(C) 1 (D) $-\frac{9}{4}$

102. On A={0,1,...,7}, define aRb if 5 divides a-b. How many ordered pairs are in R?

(A) 14 (B) 64

(C) 6 (D) 8

103. For $f(x)=x^3-3kx^2+108x$, how many integer k in [0,8] give a local maximum at a positive x?

(A) 3 (B) 8

(C) 2 (D) 1

104. Evaluate $\sum_{j=7}^{10} \binom{j}{1}$, with the convention $\binom{j}{r} = 0$ when j<r.

(A) 55 (B) 34

(C) 40 (D) 27

105. A complex number z satisfies z+conjugate(z)=4 and z-conjugate(z)=10i. Find z·conjugate(z).

(A) 29 (B) $\sqrt{29}$

(C) 104 (D) 58

106. For x∈[0,1], minimize $(\arcsin x)^3+13(\arccos x)^3$.

(A) $\frac{13\pi^3}{8\left(1-\sqrt{13}\right)^2}$ (B) $\frac{7\pi^3}{32}$

(C) $\frac{13\pi^3}{8}$ (D) $\frac{13\pi^3}{8\left(1+\sqrt{13}\right)^2}$

107. $f(x)=x^2+3x$ for $x\le 5$, and $f(x)=px+q$ for $x>5$. If f is differentiable at 5, find p-q.

(A) −12 (B) 40

(C) 38 (D) 13

108. Evaluate $\displaystyle\sum_{\substack{0\le r\le 10\\ r \text{ even}}} \frac{\binom{10}{r}}{(r+1)(r+2)}$.

(A) $\frac{1981}{132}$ (B) $\frac{2047}{132}$

(C) $\frac{1361}{44}$ (D) $\frac{1024}{11}$

109. What is the total length of the set satisfying $(\text{floor}(x)-6)(\text{floor}(x)-10)\le 0$?

(A) 5 (B) 16

(C) 6 (D) 4

110. For $P(x) = \prod_{j=6}^{10}(x+j)$, find the coefficient of x^2 in $P(x)/(x+6)$.

(A) 635 (B) 431

(C) 599 (D) 862

111. The locus $|z-4|=|\text{Re}(z)+4|$ intersects the horizontal line Im(z)=10. Find the real part of the point.

(A) $\frac{25}{4}$ (B) $\frac{25}{2}$

(C) $\frac{5}{8}$ (D) $-\frac{25}{4}$

112. An invertible 4×4 matrix A has det A=27. Find det(adj A).

(A) $\frac{1}{27}$ (B) 27

(C) 108 (D) 19683

113. An observer at height 8 sees a tower foot at depression 30° and top at elevation 60°. Find the tower height.

(A) 32 (B) 24

(C) 16 (D) $8\sqrt{3}$

114. Evaluate lim as x→0 of [sqrt(36+x)-6-x/12+x^2/1728]/x^3.

(A) $-\frac{1}{1728}$ (B) $\frac{1}{124416}$

(C) $\frac{1}{3456}$ (D) $-\frac{1}{124416}$

115. For R={(j,j+1):1≤j<7} on {1,...,7}, what is the minimum number of ordered pairs to add to make R transitive?

(A) 21 (B) 6

(C) 15 (D) 22

116. From two points on the same ray from a tower foot, separated by 4, elevations are 30° and 45°. Find the distance from the farther point to the foot.

(A) $2+2\sqrt{3}$ (B) $2\sqrt{3}+6$

(C) 4 (D) $2\sqrt{3}$

117. For f(x)=|x-4|/(x-6), on which maximal open intervals is f strictly decreasing?

(A) $(4,\infty)$ (B) $(-\infty,4)$

(C) $(-\infty,6)$ (D) $(4,6)\cup(6,\infty)$

118. x=6+9(1-t^2)/(1+t^2), y=3+18t/(1+t^2), where t is real. Which point of the containing circle is not attained?

(A) $(-3,3)$ (B) $(6,-6)$

(C) $(15,3)$ (D) $(6,12)$

119. Find the coefficient of x^5 in (1-x)exp(4x).

(A) $-\frac{32}{3}$ (B) $-\frac{32}{15}$

(C) $\frac{96}{5}$ (D) $\frac{128}{15}$

120. A,B are acute, A+B=$\pi/4$ and tan A=$\frac{4}{7}$. Find tan(A-B).

(A) $-\frac{23}{89}$ (B) $\frac{23}{77}$

(C) $\frac{23}{89}$ (D) $\frac{3}{11}$

121. Group A has 3 observations, mean 11 and population SD 2. Group B has 4 observations, mean 16 and population SD 3. Find the population variance of the combined data.

(A) $\frac{106}{7}$ (B) $\frac{324}{49}$

(C) $\frac{13}{2}$ (D) $\frac{636}{49}$

122. For $A=\begin{bmatrix}5 & 1\\ 1 & 3\end{bmatrix}$ and C=14A^{-1}, find C_{22}.

(A) -1 (B) 3

(C) 5 (D) $\frac{5}{14}$

123. $y'+y/(x+1)=1$ for $x\geq 0$, $y(0)=3$. Find $(11)y(10)-y(0)$.

(A) $\frac{63}{11}$ (B) 60

(C) -60 (D) 63

124. Evaluate $\sum_{j=1}^{6} j(j+5)$.

(A) 105 (B) 546

(C) 196 (D) 91

125. $f(x)=5x+6$. Find $f(f(6))$.

(A) 180 (B) 162

(C) 192 (D) 186

126. For $xy-4x-6y+24=0$, find the squared length of the diagonal of the rectangle formed with the axes.

(A) $2\sqrt{13}$ (B) 52

(C) 100 (D) 24

127. Find the squared distance from O to the plane through (3,0,0),(0,4,0),(0,0,5).

(A) $\frac{3600}{769}$ (B) $\frac{769}{3600}$

(C) $\frac{60}{47}$ (D) 50

128. X takes values [-7, 0, 2] with probabilities 1/6,1/3,1/2. Let $Y=7X+9$. Find $E(Y^2)-[E(Y)]^2$.

(A) $\frac{2989}{6}$ (B) $\frac{17885}{36}$

(C) $\frac{2555}{36}$ (D) $\frac{20801}{36}$

129. α,β are nonzero roots of $x^2-6x+8=0$. Find the sum of $1+1/\alpha$ and $1+1/\beta$.

(A) $\frac{11}{4}$ (B) $\frac{7}{4}$

(C) $\frac{10}{3}$ (D) $\frac{3}{4}$

130. The parabola $(y-5)^2=8(x-5)$ has focus F and vertex V. A circle centred at F passes through V. Find the tangent length from P=(11,9).

(A) $4\sqrt{2}$ (B) $2\sqrt{3}$

(C) $2\sqrt{7}$ (D) 28

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A car negotiates a level circular turn at its maximum non-slipping speed u when $\mu_s = \frac{4}{5}$. Rain lowers the coefficient to $\frac{2}{5}$ and the turn radius stays fixed. Find the new maximum speed divided by u.

(A) $\frac{\sqrt{2}}{2}$ (B) $\frac{2}{5}$

(C) $\frac{\sqrt{10}}{5}$ (D) 2

132. A Carnot engine initially runs between 600 K and 400 K. Its hot reservoir is raised by 20 K. By how much must the cold-reservoir temperature rise to keep efficiency unchanged?

(A) $\frac{400}{31}$ K (B) 20 K

(C) $\frac{40}{3}$ K (D) $\frac{20}{3}$ K

133. The 2nd harmonic of an open pipe of length 3 m equals the 2nd allowed resonance of a closed pipe. Both have the same sound speed and negligible end correction. Find the closed-pipe length.

(A) 3 m (B) $\frac{9}{4}$ m

(C) $\frac{3}{4}$ m (D) $\frac{3}{2}$ m

Chemistry

134. For $AgCl(s)+2NH_3 \rightleftharpoons [Ag(NH_3)_2]^+ + Cl^-$, $Ksp \times Kf = 1/4$. Before AgCl is added, a solution has $\frac{7}{500}$M silver-ammine complex, zero chloride and $\frac{7}{50}$M free NH_3. If all dissolved silver stays complexed and volume is fixed, the extra AgCl solubility s in M is

(A) $\frac{7}{200}$ (B) $\frac{7}{220}$

(C) $\frac{7}{100}$ (D) $\frac{7}{20}$

135. Compare ordinary water and heavy water. Which statement preserves the distinction between mass/nuclear effects and electronic structure?

(A) Heavy water has the larger normal boiling point; replacing H by D does not alter oxygen oxidation state.

(B) An isotope with more neutrons must have a different number of protons and belong to another element.

(C) Changing H to D adds an electron and therefore changes the oxidation state of every bonded atom.

(D) Every physical property and reaction rate is identical because isotopes have the same atomic number.

136. A solution initially contains $\frac{3}{250}$ mol Fe^{2+}. Before titration, $\frac{1}{5}$ of it is oxidised quantitatively to Fe^{3+}. The volume in mL of0.0200 M $KMnO_4$ needed for the remaining Fe^{2+} in acid is

(A) 480 (B) 120

(C) 96 (D) 24

Logical Reasoning

137. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: MRA, JPE, ?, DLM, AJQ, XHU.

(A) GNI (B) FMH

(C) HOJ (D) ING

138. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

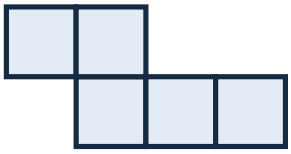


piece 1

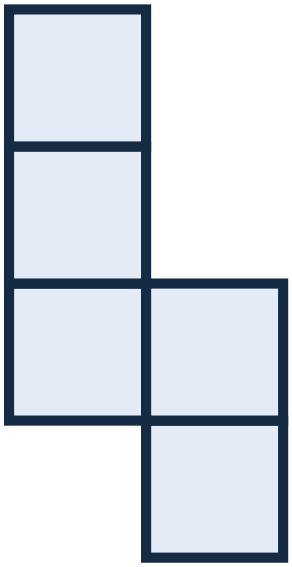


piece 2

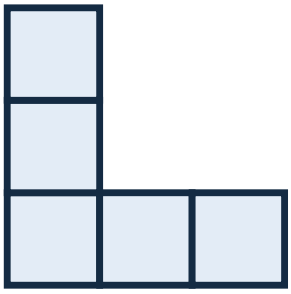
(A) Pattern U



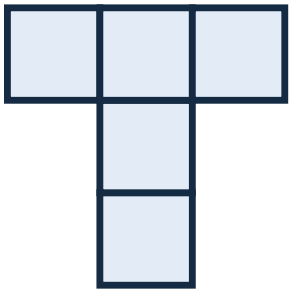
(B) Pattern M



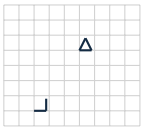
(C) Pattern X



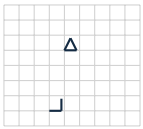
(D) Pattern R



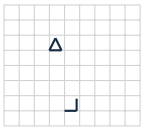
139. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



1

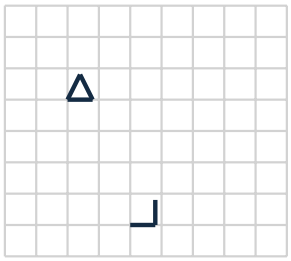


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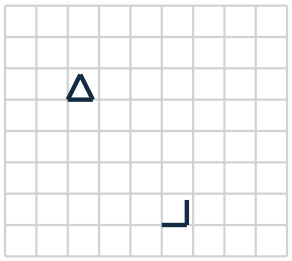


3

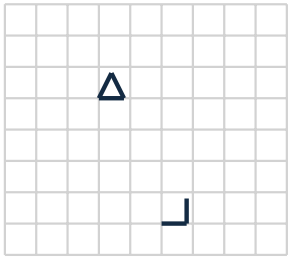
(A) Pattern W



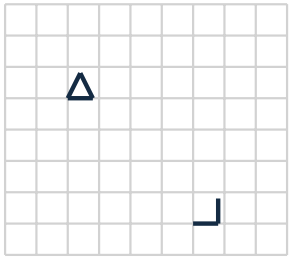
(B) Pattern R



(C) Pattern Y



(D) Pattern L



Mathematics

140. In $(1+x)^6$, find the ratio of the coefficient of x^3 to the coefficient of x^2.

(A) 4

(B) $\frac{4}{3}$

(C) $\frac{5}{2}$

(D) $\frac{3}{4}$

141. Between positive a=16, b=49, let A be their arithmetic mean and G their geometric mean. Find A^2-G^2.

(A) $\frac{1089}{4}$

(B) $\frac{33}{2}$

(C) $\frac{2657}{4}$

(D) 1089

142. Two acute sun elevations $\theta_1<\theta_2$ satisfy $(16^4)^{sin^2\theta} + (16^4)^{cos^2\theta} = 2^4 + 2^12$. A vertical tower has height 5. Find the difference between its two shadow lengths.

(A) $\frac{10}{3}$

(B) $5\sqrt{3}$

(C) $\frac{10\sqrt{3}}{3}$

(D) $\frac{20\sqrt{3}}{3}$

BITSAT Mock 22

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A gas receives 700 J of heat while its surroundings do 50 J of work on it. It then loses 350 J of heat at fixed volume. Find its total internal-energy change.

(A) 350 J (B) 750 J

(C) 400 J (D) 300 J

2. A particle has potential $U(x, y) = 4x^2y + 6y$ J in SI units. Find the force magnitude at $(3, 4)$ m.

(A) $6\sqrt{305}$ N (B) 42 N

(C) 138 N (D) $12\sqrt{73}$ N

3. A thin glass lens of refractive index 3/2 has focal length 70 cm in air. It is completely immersed in a liquid of index 4/3. Find its new focal length.

(A) $\frac{35}{2}$ cm (B) $\frac{280}{3}$ cm

(C) 210 cm (D) 280 cm

4. A magnetic field is uniform inside a long solenoid of radius 3 m and negligible outside. It changes at 8 T/s. Find induced electric-field magnitude at radius $4R$ from the axis.

(A) 12 V/m (B) $\frac{3}{4}$ V/m

(C) 3 V/m (D) 96 V/m

5. A 7 kg block is pulled by a force of 35 N at an angle with $\cos\theta = 4/5$, $\sin\theta = 3/5$ above a horizontal floor. The block slides; $\mu_k = \frac{1}{10}$. Find the net horizontal force. Take $g = 10$ m/s^2.

(A) $\frac{49}{10}$ N (B) $\frac{301}{10}$ N

(C) $\frac{231}{10}$ N (D) $\frac{329}{10}$ N

6. A car negotiates a level circular turn at its maximum non-slipping speed u when $\mu_s = \frac{6}{5}$. Rain lowers the coefficient to $\frac{1}{5}$ and the turn radius stays fixed. Find the new maximum speed divided by u.

(A) $\sqrt{6}$ (B) $\frac{\sqrt{6}}{6}$

(C) 1 (D) $\frac{\sqrt{5}}{5}$

7. Two ferromagnetic samples have similar remanent flux density. Their coercive fields are 200 A/m and 1600 A/m. A reversing field of magnitude 900 A/m will act in service. Which sample is better at retaining its original magnetisation against this field, based only on the stated coercivities?

(A) Both are necessarily demagnetised equally.
(B) The sample with coercive field 200 A/m.
(C) Neither can have nonzero remanence.
(D) The sample with coercive field 1600 A/m.

8. A nonrelativistic charged particle initially has momentum $2qEt$ along a uniform electric field, where $t > 0$ is a fixed reference interval. Its charge is $+q$. What is $\lambda(8t)/\lambda(6t)$ after acceleration in the same field?

(A) $\frac{3}{4}$
(B) $\frac{4}{5}$
(C) $\frac{2\sqrt{5}}{5}$
(D) $\frac{5}{4}$

9. 1728 identical conducting liquid drops merge without losing charge. Before merging each has potential V and surface charge density σ. Find the final surface charge density divided by σ.

(A) $\frac{1}{12}$
(B) 1728
(C) 1
(D) 12

10. A body is dropped from rest at height 240 m above the ground. Find its speed when it is still 40 m above the ground. Neglect air resistance and take $g = 10\ \mathrm{m/s^2}$.

(A) $20\sqrt{10}\,\mathrm{m/s}$
(B) $40\sqrt{3}\,\mathrm{m/s}$
(C) $4000\,\mathrm{m/s}$
(D) $20\sqrt{2}\,\mathrm{m/s}$

11. A mass 1 kg moves with $\vec{r}(t) = 1t\hat{i} + 3t^2\hat{j}$ m. Find the z-component of angular momentum about the origin at $t = 2$ s.

(A) $6\,\mathrm{kg\,m^2/s}$
(B) $24\,\mathrm{kg\,m^2/s}$
(C) $12\,\mathrm{kg\,m^2/s}$
(D) $36\,\mathrm{kg\,m^2/s}$

12. One mole of nonoverlapping identical spheres of radius r has total geometric material volume V. A second sample contains $\frac{3}{5}$ mole of spheres of radius $4r$. What is its total geometric material volume divided by V? Ignore packing voids.

(A) $\frac{192}{5}$
(B) $\frac{1728}{125}$
(C) $\frac{48}{5}$
(D) $\frac{12}{5}$

13. Two large parallel plates are held at potentials +500 V and −500 V, with gap 6 mm. Neglect edge effects. Find field magnitude in the gap.

(A) $\frac{5000000}{3}\,\mathrm{V/m}$
(B) $\frac{500000}{3}\,\mathrm{V/m}$
(C) $\frac{50000}{3}\,\mathrm{V/m}$
(D) $\frac{250000}{3}\,\mathrm{V/m}$

14. A standing wave is $y = 2\sin(2x)\cos(3t)$ cm with x in metres and t in seconds. At a point where $2x = \pi/6$, find the particle-speed magnitude when $3t = \pi/3$.

(A) $\frac{\sqrt{3}}{2}\,\mathrm{cm/s}$
(B) $\frac{3\sqrt{3}}{4}\,\mathrm{cm/s}$
(C) $\frac{3\sqrt{3}}{2}\,\mathrm{cm/s}$
(D) $3\,\mathrm{cm/s}$

15. Two small identical conducting spheres initially carry positive charges $4q_0$ and $2q_0$ and are separated by distance d. They touch and are returned to the same separation. Find the final force divided by the initial force.

(A) $\frac{9}{8}$ (B) 1

(C) 2 (D) $\frac{9}{4}$

16. An ideal mixture contains 3 moles of monatomic gas and 3 moles of rigid diatomic gas. Vibrations are frozen. Find the heat-capacity ratio $\gamma = C_P/C_V$ of the mixture.

(A) $\frac{13}{12}$ (B) $\frac{3}{2}$

(C) $\frac{23}{15}$ (D) $\frac{2}{3}$

17. A stationary nucleus with mass number 20 and binding energy per nucleon 8 MeV splits into two equal daughters, each having total binding energy 83.0 MeV. Which statement is correct?

(A) The process releases 6 MeV; the daughters have larger binding energy per nucleon.

(B) The process absorbs 6 MeV; the daughters have larger binding energy per nucleon.

(C) No energy can be released because fission always requires a larger mass number than 20.

(D) The process releases 12 MeV; each daughter releases 6 MeV.

18. A photon and a nonrelativistic particle have equal de Broglie wavelengths. The particle speed is $c/15$. If the photon then loses half its energy while the particle momentum is unchanged, what is the ratio of particle kinetic energy to the photon's final energy?

(A) $\frac{2}{15}$ (B) $\frac{1}{15}$

(C) 1 (D) 15

19. Current rises linearly from 0 to 2 A over 6 s and then falls linearly to 0 over another 6 s. It remains zero for the next 12 s. Find the average current over the entire 24 s interval.

(A) $\frac{2}{3}$ A (B) $\frac{1}{4}$ A

(C) $\frac{1}{2}$ A (D) 2 A

20. A 4 V Zener regulator feeds a constant-current load of 4 mA. Supply varies from 9 to 12 V. Regulation requires Zener current at least 2 mA, and its safe maximum is 7 mA. What is the largest permissible series resistance? Assume ideal regulation and give resistance in Ω.

(A) $\frac{2500}{3}\,\Omega$ (B) $\frac{4000}{3}\,\Omega$

(C) $\frac{8000}{11}\,\Omega$ (D) $1500\,\Omega$

21. A charge of $6\,\mu$C moves from potential 60 V to -60 V in an electrostatic field. Find the work done by an external agent if the charge starts and ends at rest and no other force does work.

(A) $360\,\mu$J (B) $720\,\mu$J

(C) $-12\,\mu$J (D) $-720\,\mu$J

22. Two waves of equal frequency, direction and amplitude 3 mm differ in phase by 60°. Find resultant amplitude.

(A) 6 mm (B) $3\sqrt{3}$ mm

(C) 3 mm (D) $3\sqrt{2}$ mm

23. Here L is inductance, C capacitance, R resistance, V potential, I current and t time. Which expression has the same dimensions as both $(CV)^3$ and $(It)^3$?

(A) $(V/R)^3$ (B) $(LI)^3$

(C) $(RC)^3$ (D) $((V/R)t)^3$

24. A 5 kg block moving at 7 m/s compresses a spring of constant 100 N/m. During compression, a damper removes 1/3 of the block's initial kinetic energy. Find maximum compression; gravity does no work.

(A) $\frac{7\sqrt{6}}{30}$ m (B) $\frac{7\sqrt{30}}{30}$ m

(C) $\frac{7\sqrt{15}}{15}$ m (D) $\frac{49}{30}$ m

25. In YDSE wavelength is 300 nm, screen distance 4 m and slit separation 3 mm. Find the separation on the same side of the centre between the 4th bright fringe and the 7th dark fringe. Count the first dark fringe as order 1.

(A) 1 mm (B) $\frac{13}{5}$ mm

(C) $\frac{6}{5}$ mm (D) $\frac{1}{2}$ mm

26. A soap bubble initially of radius 2 cm is expanded quasistatically until its volume is 27 times the initial volume. Surface tension remains $\frac{7}{100}$ N/m. Find work against surface tension, in units of $\pi \times 10^{-4}$ J.

(A) $\frac{448}{25}$ (B) $\frac{224}{25}$

(C) $\frac{56}{25}$ (D) $\frac{112}{25}$

27. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 1 kΩ to a terminal at +13 V, and its cathode connects to a terminal at 1 V. Find the current from the +13 V terminal in mA.



(A) $\frac{127}{10}$ mA (B) $\frac{123}{10}$ mA

(C) $\frac{113}{10}$ mA (D) $\frac{53}{5}$ mA

28. A source emitting 200 Hz and a listener approach each other, each at 30 m/s relative to still air. Sound speed is 300 m/s. Find heard frequency before they pass.

(A) 220 Hz (B) 240 Hz

(C) 200 Hz (D) $\frac{2200}{9}$ Hz

29. A thin equiconvex lens of glass index 3/2 has focal length 40 cm in air. It is cut by its midplane perpendicular to its axis to form two plano-convex halves. One half is immersed in liquid of index 4/3. Find its focal length in the liquid.

(A) 20 cm (B) 320 cm

(C) 160 cm (D) 40 cm

30. A one-electron ion with nuclear charge $Z = 1$ emits when its orbit changes. The ratio of initial to final orbit radii is 25, and its final principal quantum number is 1. What is $R\lambda$ for the emitted photon? Here R is the hydrogen Rydberg constant.

(A) 25 (B) 1

(C) $\frac{24}{25}$ (D) $\frac{25}{24}$

Chemistry

31. Compare finalpH: I=10mL $\frac{1}{5}$M HCl+20mL $\frac{1}{5}$M NaOH;II=50mL $\frac{1}{5}$M H_2SO_4+50mL $\frac{1}{5}$M NaOH; III=10mL $\frac{1}{5}$M HCl+10mL $\frac{1}{5}$M NaOH. Treat both sulfuric protons as fully neutralised. Increasing pH is

(A) I<III<II. (B) II<I<III.

(C) II<III<I. (D) III<II<I.

32. A mixture contains 1/20 mol total of the ionisation isomers [Co(NH3)5Br]SO4 and [Co(NH3)5SO4]Br. An immediately tested aliquot representing the entire inventory gives 1/100 mol AgBr with excess Ag+. Coordinated ligands do not exchange, and no other bromide exists. What mole fraction of the inventory was [Co(NH3)5Br]SO4?

(A) -4/5 (B) 4/5

(C) 8/5 (D) 2/5

33. Separate unknowns P and Q are each selected from Cu2+, Zn2+, Fe3+. Dropwise aqueous ammonia gives P a reddish brown hydroxide precipitate and Q a pale blue precipitate. Excess ammonia gives P an undissolved reddish-brown hydroxide and Q a deep-blue ammine solution. Which identification is consistent, without invoking redox?

(A) P=Cu2+; Q=Fe3+; oxidation states unchanged.

(B) P=Fe3+; Q=Cu2+; both metals reduced to zero.

(C) Both are the same ion because all hydroxides have identical ammonia behaviour.

(D) P=Fe3+; Q=Cu2+; oxidation states unchanged.

34. $\frac{2}{25}$ mol borax is acidified completely. Each borax formula unit gives four H_3BO_3 molecules;75.0% of the boric acid is isolated. With glycerol present, the isolated boric acid is titrated completely as a monobasic acid. What amount of NaOH, in mol, is required?

(A) $\frac{3}{25}$ (B) $\frac{6}{25}$

(C) $\frac{32}{75}$ (D) $\frac{3}{50}$

35. An octahedral complex has low-spin d^6 configuration and $\Delta_o = 150$ kJ mol^{-1}. Ignoring pairing energy,its d-electron energy relative to the unsplit barycentre,in kJ mol^{-1},is

(A) -360 (B) -210

(C) 360 (D) -510

36. Compare 1-bromobutane and2-bromobutane toward the same cyanide nucleophile in a polar aprotic solvent under ordinary SN2 conditions. Which reacts faster or is the substituted site,and why?

(A) 2-bromobutane; an sp2 C-Br bond behaves just like an unhindered methyl bromide

(B) 1-bromobutane; the mechanism requires formation of a free tertiary carbocation

(C) 1-bromobutane; The terminal primary C-Br site is less sterically crowded than the secondary site.

(D) 2-bromobutane; steric hindrance accelerates backside attack

37. An100mL sample contains $\frac{2}{25}$ mol HA,pKa=4,and $\frac{1}{25}$ mol HCl. Add100mL $\frac{6}{5}$ M NaOH. Use usual weak-base salt hydrolysis and $\log_{10}(\frac{2}{5})=-\frac{199}{500}$. Final pH is

(A) $\frac{8801}{1000}$ (B) $\frac{6801}{500}$

(C) 7 (D) 4

38. In a semiclassical one-electron orbit, de Broglie standing waves fit as$2\pi r=n\lambda$. With $\lambda=h/(mv)$, which Bohr condition follows?

(A) mvr=nh/(2π). (B) mvr=nh.

(C) The electron has no momentum because its wave is stationary. (D) mvr=h/(2πn^2).

39. Two paths connect the same equilibrium states. PathAhasq=500J,w=-100J using work-on signs. PathBhasq=700J. Which pair(workonB,internal-energychangeonB)is correct?

(A) (-300J,400J). (B) (-300J,0J).

(C) (300J,400J). (D) (-100J,600J).

40. A chlorine atom reacts with ozone to form ClO; ClO then reacts with an oxygen atom to regenerate chlorine. Why can one chlorine atom contribute to many ozone-loss events?

(A) Chlorine is regenerated and can repeat the catalytic cycle. (B) Only the oxygen atom is regenerated as the unchanged catalyst.

(C) Chlorine is consumed permanently after the first step. (D) The cycle converts ozone into more ozone overall.

41. For $[NiCl_4]^{2-}$,use the usual ligand-field/VBT assignments at room temperature. Which complete assignment is consistent with charge balance and magnetic behaviour?

(A) d^8,tetrahedral,sp^3,spin-only moment $\sqrt{15}$ BM.

(B) d^8,square planar,dsp^2,spin-only moment $2\sqrt{2}$ BM.

(C) d^9,tetrahedral,sp^3,spin-only moment $2\sqrt{2}$ BM.

(D) d^8,tetrahedral,sp^3,spin-only moment $2\sqrt{2}$ BM.

42. Use octet Lewis contributors for NO3- and $CO3^2$-; for SO2 exclude expanded-octet contributors. For HCOO-, HCO3- and HNO3, average only the two nonprotonated central-atom-O bonds. For other species average all central-atom bonds. Find (average bond order of the first minus that of the second, sum of total lone-pair counts in one contributor of each). A lone pair on any atom is counted once.

(A) (8/3, 16) (B) (0, 16)

(C) (0, 15) (D) (0, 17)

43. A closed system goes from state1to2 by path A with q=500 J and work on the system w=-300 J. It returns from2to1 along a different path with q=-700 J. IUPAC work on the return path,in J,is

(A) 300 (B) −500

(C) 500 (D) 200

44. The ions S^{2-}, Cl^{-}, K^{+}, Ca^{2+} each have 18 electrons. Which decreasing-radius order and explanation is consistent?

(A) $S^{2-} < Cl^{-} < K^{+} < Ca^{2+}$; larger nuclear charge expands the cloud.

(B) All radii are equal because electron numbers are equal.

(C) $Cl^{-} > K^{+} > Ca^{2+} > S^{2-}$; positive charge always gives the largest ion.

(D) $S^{2-} > Cl^{-} > K^{+} > Ca^{2+}$; increasing nuclear charge contracts the same electron cloud.

45. In a hydrogen-like ion Z=4,an electron falls from n=4 to an s orbital with 1 radial nodes. With hydrogen ionisation energy13.6 eV and$hc = 1240$ eV nm,the emitted wavelength in nm is

(A) $\frac{775}{102}$

(B) $\frac{24800}{51}$

(C) $\frac{1550}{17}$

(D) $\frac{1550}{51}$

46. A textbook deficiency table associates delayed blood clotting with one vitamin. Which vitamin and solubility class are correctly paired?

(A) Vitamin B2; water-soluble.

(B) Vitamin B12; water-soluble.

(C) VitaminA; fat-soluble.

(D) VitaminK; fat-soluble.

47. Two model nonvolatile solutes A(M180) and B(M60) are dissolved separately in equal solvent masses. Equal solute masses are used. A remains molecular;fraction $\frac{4}{5}$ of B formula units is present in dimers. All solutions obey the stated ideal colligative model. Ratio $\Delta T_f(B)/\Delta T_f(A)$ is

(A) $\frac{3}{5}$

(B) $\frac{9}{5}$

(C) 5

(D) 3

48. Use octet Lewis contributors for NO3- and NO2-; for SO2 exclude expanded-octet contributors. For HCOO-, HCO3- and HNO3, average only the two nonprotonated central-atom-O bonds. For other species average all central-atom bonds. Find (average bond order of the first minus that of the second, sum of total lone-pair counts in one contributor of each). A lone pair on any atom is counted once.

(A) (-1/6, 15)

(B) (-1/6, 13)

(C) (17/6, 14)

(D) (-1/6, 14)

49. Among constitutional monobromo derivatives with formula C4H9Br,how many give an alcohol with one stereogenic carbon on simple replacement Br→OH without changing the carbon skeleton? Count structures,not enantiomers.

(A) 3

(B) 2

(C) 4

(D) 1

50. A $\frac{1}{100}$mol iron-ion inventory contains Fe^{2+}andFe^{3+}. Fraction$\frac{1}{2}$isFe^{3+}. All iron is quantitatively reduced to metal in the stated externally driven electron-balance model,with no side reactions. Electrons accepted,in mol,are

(A) $\frac{3}{200}$

(B) $\frac{7}{300}$

(C) $\frac{1}{40}$

(D) $\frac{1}{200}$

51. A solution initially contains $\frac{3}{500}$ mol Fe^{2+}. Before titration, $\frac{3}{10}$ of it is oxidised quantitatively to Fe^{3+}. The volume in mL of0.0200 M $KMnO_4$ needed for the remaining Fe^{2+} in acid is

(A) 60 (B) 42

(C) 210 (D) 18

52. In the gas phase,O_2^- is converted into O_2^{2-} by changing one electron. Under the elementary MO model,which final bond-order/unpaired-electron pair follows?

(A) Bond order $\frac{3}{2}$; 1 unpaired electrons. (B) Bond order 2; 0 unpaired electrons.

(C) Bond order 1; 0 unpaired electrons. (D) Bond order 1; 1 unpaired electrons.

53. An open-chain aldohexose has CHO at C1, CH2OH at C6 and four independent stereogenic centres at C2-C5. With unlike end groups and no meso reduction, how many stereoisomers andD-series members are possible?

(A) 32 total;4 D-series. (B) 16 total;8 D-series.

(C) 8 total;8 D-series. (D) 16 total;16 D-series.

54. Classify SrO, ZnO, P4O10, in that order, by their usual acid-base behaviour. Which sequence also predicts that the middle oxide reacts with both suitable acids and strong alkalis?

(A) Basic; amphoteric; acidic. (B) Amphoteric; basic; acidic.

(C) Basic; acidic; amphoteric. (D) Acidic; amphoteric; basic.

55. An ideal mixture initially has$\frac{1}{5}$molA and$\frac{3}{10}$molB. A selective step removes half ofB and noA. At the new equilibriumT,V,totalpressure is4bar. Partialpressure ofA,in bar,is

(A) 2.0 (B) $\frac{16}{7}$

(C) $\frac{8}{5}$ (D) 7

56. Construct one open linear peptide using each member of the following multiset exactly once: Pro, Gly, Ala, Val, Ser, Thr. The N-terminal residue isPro, the C-terminal residue is the only achiral amino acid, and Val and Ser must not be adjacent. Count distinctN-to-C sequences; ignore stereoisomerism of the supplied residues.

(A) 12 (B) 120

(C) 720 (D) 24

57. As temperature approaches 0 K, which pair applies the introductory third law without extending it to every real sample?

(A) Frozen configurational disorder changes enthalpy but cannot contribute entropy.

(B) A perfect crystal has S=0 at every temperature below its melting point.

(C) A unique-ground-state perfect crystal has S→0; frozen disorder can leave residual S.

(D) Every pure crystal has S→0, even with frozen orientational disorder.

58. For electrophilic mononitration of benzene carrying Cl,which orientation/reason pair is correct? Consider intrinsic electronic directing effects,not an exact ortho:para ratio.

(A) ortho/para; every electronegative substituent activates the ring

(B) ortho/para; lone-pair resonance donation sets orientation despite overall inductive deactivation

(C) meta; lone-pair resonance donation sets orientation despite overall inductive deactivation

(D) meta; the substitution site is decided only by steric size

59. For butan-2-ol in the Victor Meyer sequence,which test/observation assignment follows the actual carbinol class or oxidisable structural feature? Use ordinary room-temperature tests unless the named sequence specifies otherwise.

(A) red colour characteristic of the primary sequence
(B) no colour characteristic of the tertiary sequence
(C) blue colour from the secondary-alcohol sequence
(D) yellow iodoform as the Victor Meyer endpoint

60. Within an additive covalent-radius model,a single X-X bond is 120 pm and an X-H bond is 100 pm. A linear H-X-X-H skeleton is assumed with collinear single bonds. Its terminal H-to-H distance,in pm,is

(A) 320
(B) 220
(C) 200
(D) 640

English Proficiency

61. Choose the most coherent order of the four sentences. A. A daytime visit revealed a stretch with no safe place to step aside. B. The nature club planned a walk after sunset. C. The leader changed the route before announcing the final meeting point. D. Participants therefore avoided a hazard that would have been harder to assess in darkness.

(A) B-A-D-C
(B) B-C-A-D
(C) A-B-C-D
(D) B-A-C-D

62. Read the sentence: The new machine cannot read the obsolete storage format used by the old archive. Choose the closest meaning of 'obsolete' as used in the sentence.

(A) temporarily unavailable
(B) legally restricted
(C) no longer in practical use
(D) physically damaged

63. Read the sentence: The committee had to curtail the programme because the building closed early. Choose the opposite in meaning of 'curtail' as used in the sentence.

(A) extend
(B) schedule
(C) revise
(D) shorten

64. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The staff are to inspect both rooms before admitting visitors.

(A) Both rooms are to inspect the staff before visitors are admitted.

(B) Both rooms are to be inspected by the staff before visitors are admitted.

(C) Both rooms were inspected by the staff before visitors were admitted.

(D) One room is to be inspected by the staff before visitors are admitted.

65. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. The first comparison ignored how recently rain had fallen at each site. B. The group collected river samples at several locations. C. Differences in the samples could then be interpreted with a relevant condition in view. D. The group added rainfall information to its records.

(A) Differences in the samples could then be interpreted with a relevant condition in view.

(B) The group collected river samples at several locations.

(C) The group added rainfall information to its records.

(D) The first comparison ignored how recently rain had fallen at each site.

66. Read the sentence: Although the plans were complete, he got cold feet just before making the public announcement. What does 'get cold feet' mean here?

(A) decided the plan was impossible
(B) became physically exhausted
(C) forgot the announcement
(D) became anxious and hesitant about proceeding

67. Read the sentence: They made a tentative booking while waiting for the final examination dates. Choose the closest meaning of 'tentative' as used in the sentence.

(A) reluctant
(B) careless
(C) provisional
(D) confidential

68. Read the sentence: The witness made a bold statement despite the pressure to remain silent. Choose the opposite in meaning of 'bold' as used in the sentence.

(A) direct
(B) timid
(C) public
(D) unusual

69. The witness's name is withheld, though the account itself is published. Which noun names that state?

(A) anonymity
(B) confidentiality
(C) ambiguity
(D) illiteracy

70. Choose the correctly spelt word.

(A) necessary
(B) necesary
(C) necessery
(D) neccessary

Logical Reasoning

71. Preserve the exact relation and its order: Player : Team :: ? : Constellation. Choose the missing term.

(A) Wolf
(B) Bee
(C) Lion
(D) Star

72. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 18$ and $5 \mapsto 36$. Which input maps to 60?

(A) 60
(B) 9
(C) 10
(D) 8

73. In a row, Asha is 12th from the left and 19th from the right. Bina is 12th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 3
(B) 9
(C) 4
(D) 5

74. Consider only these statements; universal statements alone do not imply existence. Some hikers are inventors; All hikers are inventors. Conclusions: I. Some inventors are hikers. II. Some hikers are not farmers. Which conclusions follow necessarily?

(A) Only I
(B) Both I and II
(C) Only II
(D) Neither I nor II

75. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target



piece 1

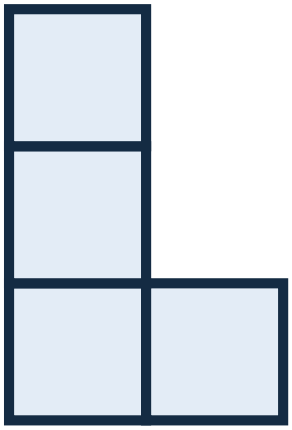


piece 2

(A) Pattern Q



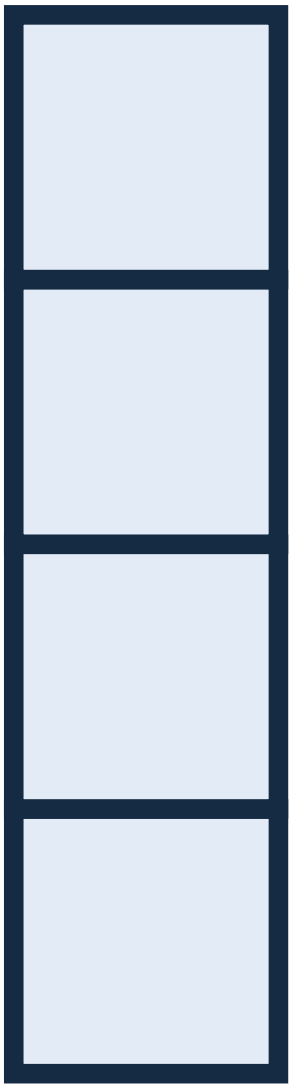
(B) Pattern X



(C) Pattern P



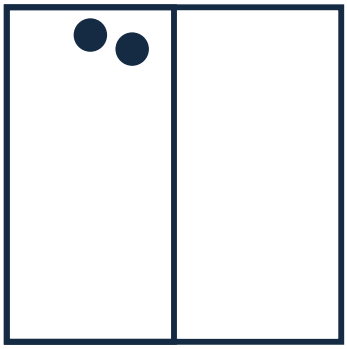
(D) Pattern L



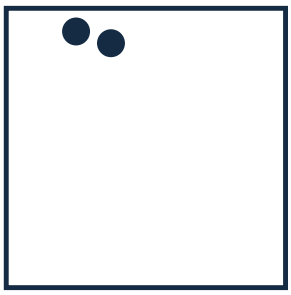
76. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 4 and Z has code 29. Decode the separated values 21, 12, 25, 8, 21.

(A) QHUDQ
(B) RIVER
(C) REVIR
(D) SJWFS

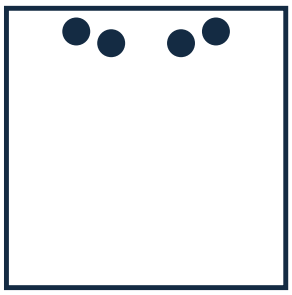
77. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



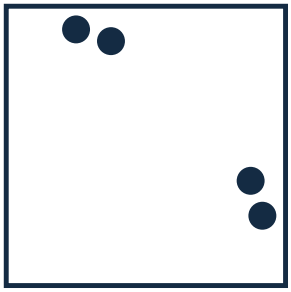
(A) Pattern V



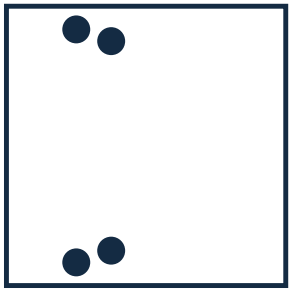
(B) Pattern X



(C) Pattern K



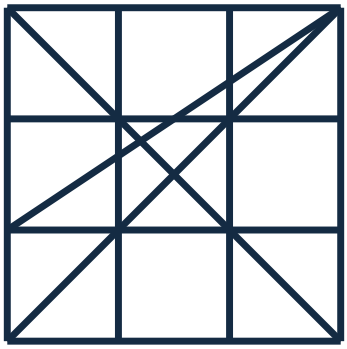
(D) Pattern M



78. Consider only these statements; universal statements alone do not imply existence. All inventors are editors; Some inventors are not farmers; No inventors are farmers. Conclusions: I. Some inventors are editors. II. All farmers are inventors. Which conclusions are individually possible without contradicting the statements?

(A) Neither I nor II
(B) Only II
(C) Both I and II
(D) Only I

79. How many triangles are formed by all the drawn segments?



(A) 40 (B) 51

(C) 98 (D) 55

80. Each letter position has its own fixed cyclic shift, applied identically in both pairs (Z wraps to A). Complete WDW : VBU :: GUE : ?.

(A) DSE (B) GTD

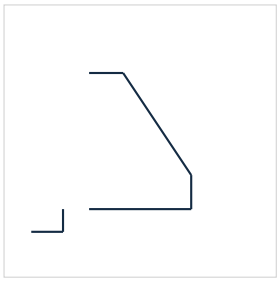
(C) FSC (D) ERB

81. A person makes these turns in order: 90 degrees right, 180 degrees, 90 degrees right, 180 degrees. The final heading is south. What was the heading immediately after turn 3?

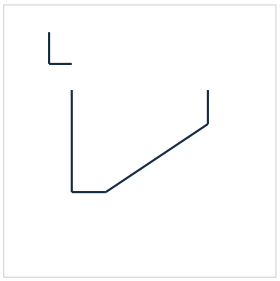
(A) North (B) West

(C) South (D) East

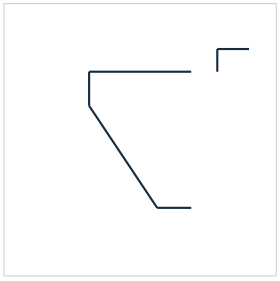
82. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

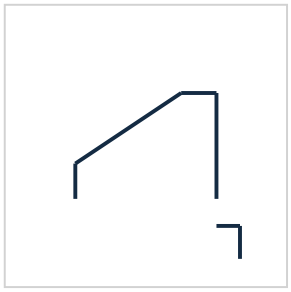


bottom left

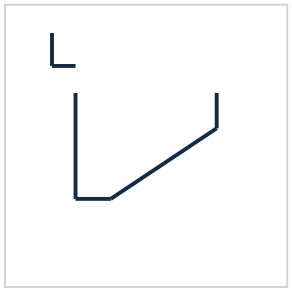
?

bottom right

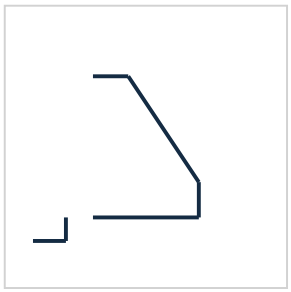
(A) Pattern K



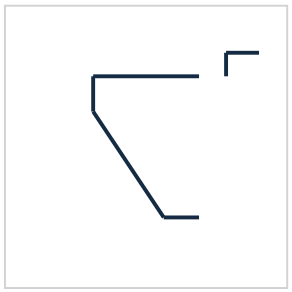
(B) Pattern U



(C) Pattern Y



(D) Pattern P



83. The recorded average of 7 readings is 28. One entry recorded as 39 should have been 37. After that correction, another reading of 24 is replaced by 33. What is the final average?

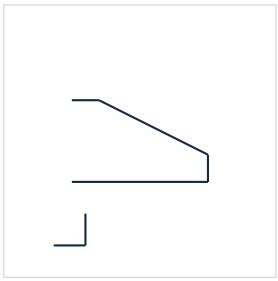
(A) 30

(B) $\frac{205}{7}$

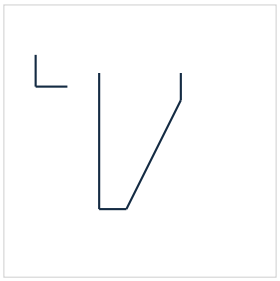
(C) 29

(D) $\frac{194}{7}$

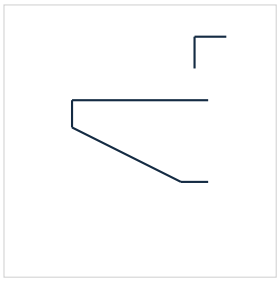
84. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

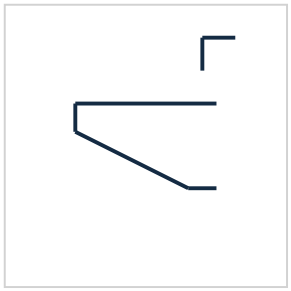


2



3

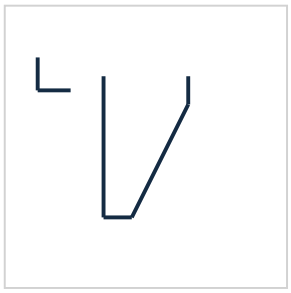
(A) Pattern M



(B) Pattern U



(C) Pattern X



(D) Pattern N



85. The increase from the left cell to the right cell is the same in each row, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.





?

(A) Pattern L

(B) Pattern X





(C) Pattern Q

(D) Pattern Y

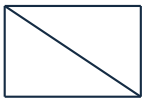




86. Each rectangle is drawn to the same local scale. In three numbered figures the interior straight line divides the rectangle into equal areas. Which does not?



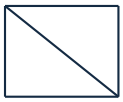
1



2



3



4

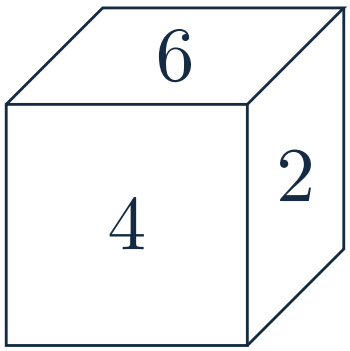
(A) 1

(B) 4

(C) 3

(D) 2

87. This standard die has opposite labels summing to seven. Roll it once to the right, so its present top face becomes the right face. Give the new (top, front) labels.

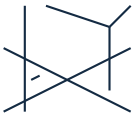


(A) $(5, 4)$ (B) $(2, 4)$

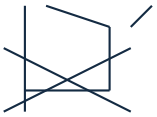
(C) $(6, 4)$ (D) $(5, 3)$

88. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

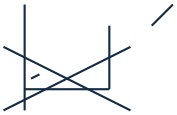




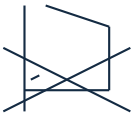
host 1



host 2



host 3



host 4

(A) 2 (B) 3

(C) 4 (D) 1

89. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 5, 20, 29, ?, 125, 500, 509.

(A) 152 (B) 125

(C) 38 (D) 116

90. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: ENZJ, HRDN, KVHR, ?, QDPZ, THTD.

(A) MYKU (B) VLZN

(C) OAMW (D) NZLV

Mathematics

91. For $0 \leq \theta < 10\pi$, roots of $x^2-6x+(1-2\sin\theta)=0$ have opposite signs. How many connected intervals of θ satisfy this?

(A) 5 (B) 10

(C) 6 (D) 1

92. How many real points of nondifferentiability does $(x-3)|x-3|+|x-5|$ have?

(A) 2 (B) 1

(C) 3 (D) 0

93. How many integers x are in the real domain of $\sqrt{x-4}\,\log(7-x)$?

(A) 3 (B) 4

(C) 2 (D) 7

94. The chord of contact from P to $x^2/16-y^2/16=1$ is $x=4/2$. Find the product of the slopes of the two tangents from P.

(A) $\frac{1}{3}$ (B) 0

(C) -1 (D) $-\frac{1}{3}$

95. $f(x)=x^3+4x$, $g=f^{-1}$. Find $g''(80)$.

(A) $\frac{3}{17576}$ (B) $-\frac{3}{338}$

(C) $-\frac{3}{17576}$ (D) $\frac{1}{24}$

96. The system $x+y+z=1$, $y+z=2$, $(k-2)z=k-6$ has no solution for which k?

(A) 6 (B) 0

(C) -2 (D) 2

97. Evaluate $\sin(2\arctan 6)$.

(A) $-\frac{12}{37}$ (B) $\frac{12}{37}$

(C) $\frac{6\sqrt{37}}{37}$ (D) $-\frac{12}{35}$

98. The chord of contact from P to $x^2/100+y^2/25=1$ is $x/10+y/5=1$. Find $|OP|^2$.

(A) 15 (B) 125

(C) $5\sqrt{5}$ (D) 75

99. Two different numbers are chosen uniformly from $\{1,\ldots,9\}$. Let X be their maximum. Find Var(X).

(A) $\frac{\sqrt{35}}{3}$ (B) $\frac{20}{3}$

(C) $\frac{145}{3}$ (D) $\frac{35}{9}$

100. Let R(x) be the remainder when x^6 is divided by $(x-1)(x-2)$. Find R(4).

(A) 252 (B) 190

(C) 4096 (D) -185

101. A line through P=(8,6) has direction (3,4). It meets 3x+4y=123 at Q. Find $|PQ|^2$.

(A) 15 (B) 75

(C) 9 (D) 225

102. Find the sum of order and degree of $(d^2y/dx^2)^5+(dy/dx)^2+y=0$.

(A) 12 (B) 5

(C) 7 (D) 10

103. A point is equidistant from (5,0) and x=-5. Find the area between its locus, the y-axis, and y=10/2, y=10.

(A) $\frac{25}{3}$ (B) $\frac{25}{12}$

(C) $\frac{50}{3}$ (D) $\frac{175}{12}$

104. A parallelogram has opposite vertices (3,7) and (9,13), and another vertex (0,3). Find the sum of coordinates of the fourth vertex.

(A) 16 (B) 29

(C) 35 (D) 32

105. 12 distinct people stand in a line. Three specified people cannot occupy positions 1,2 or 12. How many orders are possible?

(A) 362880 (B) 182891520

(C) 261273600 (D) 30481920

106. Evaluate $\int_0^\infty dx/[(x^2+36)(x^2+121)]$.

(A) $\frac{\pi}{2244}$ (B) $-\frac{\pi}{2244}$

(C) $\frac{\pi}{132}$ (D) $\frac{\pi}{1122}$

107. For $x\in[\pi/6,\pi/2]$, find the maximum of $\sin x/x$.

(A) $\frac{6}{\pi}$ (B) $\frac{3}{\pi}$

(C) $\frac{2}{\pi}$ (D) $\frac{\pi}{3}$

108. From a point at horizontal distance 5 from a tower, its elevation is 2θ, where $\tan\theta=\frac{4}{9}$ and $0<\theta<45°$. Find the height.

(A) $\frac{72}{13}$ (B) $\frac{40}{9}$

(C) $\frac{325}{72}$ (D) $\frac{20}{9}$

109. Find the population variance of 4+2, 8+2, ..., 28+2.

(A) 16 (B) 64

(C) 4 (D) 66

110. Let k make lim as $x\to\infty$ of $[(3x^2+5x+1)/(x+2)-kx]$ finite. Find that finite limit.

(A) 5 (B) 11

(C) -1 (D) 6

111. Write sin x $\cos^3$x+8$\sin^3$x cos x=P sin2x+Q sin4x. Find Q.

(A) $\frac{9}{8}$ (B) $\frac{9}{4}$

(C) $-\frac{7}{8}$ (D) $\frac{1}{8}$

112. The tangent to y=x^2+6x at P has slope 16. Find the slope of the normal at P.

(A) $-\frac{1}{256}$ (B) 16

(C) −16 (D) $-\frac{1}{16}$

113. A=(7,5,0), C=(10,11,3). B lies on AC with AB:BC=2:1. Find $|OB|^2$, where O is the origin.

(A) 74 (B) 54

(C) 166 (D) 230

114. For integer n, consider: "If 21 divides n, then 7 divides n." Which is its contrapositive?

(A) If 7 does not divide n, then 21 does not divide n. (B) If 21 does not divide n, then 7 does not divide n.

(C) 21 divides n and 7 does not divide n. (D) If 7 divides n, then 21 divides n.

115. For f(x)=x(x-11)(x-19), find the sum of all c∈(0,19) satisfying f′(c)=0.

(A) 20 (B) 209

(C) 30 (D) 10

116. If |z-12|=6, find the minimum of $|z|^2$.

(A) 6 (B) 36

(C) 324 (D) 108

117. y>0 satisfies 2yy′+6y^2=0, y(0)=sqrt(5). Find y^2(1).

(A) $\frac{\sqrt{5}}{e^3}$ (B) $\frac{5}{e^3}$

(C) $\frac{5}{e^6}$ (D) $5e^6$

118. Let f(x)=x^3-75 x+7. Find the product of its local maximum and local minimum values.

(A) −187451 (B) 66049

(C) 500 (D) −62451

119. Vectors a,b satisfy $|a|^2$=10, $|b|^2$=26, a·b=8. Find $|2a-b|^2$.

(A) 66 (B) 30

(C) 34 (D) 98

120. f(x)=3x+1 and f(g(x))=3x+10. Find g(g(3)).

(A) 9 (B) 6

(C) 15 (D) 12

121. Let $y\prime + 3y = 5x$ and $y(0) = 5$. Find $y(1) + \frac{8}{5}y\prime(1)$.

(A) $\frac{50}{9e^3} + \frac{10}{9}$ (B) $\frac{190}{9e^3} + \frac{34}{9}$

(C) $\frac{34}{9}$ (D) $\frac{34}{9} - \frac{190}{9e^3}$

122. $a_j = 6 + 3(j-1)$ and f(x)=(7/2)x^2+bx+3. If f(1)=f(-1), find f′(a_6).

(A) 147 (B) 21

(C) 168 (D) 150

123. Find the coefficient of x^6 in (1-x)exp(3x).

(A) $\frac{81}{80}$ (B) $-\frac{81}{40}$

(C) $\frac{243}{80}$ (D) $-\frac{81}{80}$

124. A line through (5,2) has equal positive x- and y-intercepts. Find the sum of the coordinates of the midpoint of its intercept segment.

(A) 7 (B) 10

(C) $\frac{7}{2}$ (D) 14

125. In a group of 50, 30 study A, 25 study B, and 5 study neither. How many study exactly one of A, B?

(A) 45 (B) 35

(C) 10 (D) 55

126. α,β are roots of x^2-18x+77=0. The monic polynomial with roots α+1,β+1 is x^2-Ax+B. Find B.

(A) 6084 (B) 95

(C) 96 (D) 78

127. Four observations are 13,13,x,y with population mean 13 and variance 25/2. Find xy.

(A) $\frac{313}{2}$ (B) 194

(C) 313 (D) 144

128. The line r=(5,6,3)+t(1,2,2) meets x+y+z=44 at Q. Find $|PQ|^2$ for P=(5,6,3).

(A) 670 (B) 54

(C) 324 (D) 18

129. How many singular 2×2 matrices with entries from {-1,0,1} have the sum of the squares of all entries equal to 2?

(A) 16 (B) 24

(C) 8 (D) 4

130. Between positive a=4, b=25, let A be their arithmetic mean and G their geometric mean. Find A^2-G^2.

(A) $\frac{841}{4}$ (B) $\frac{21}{2}$

(C) $\frac{441}{4}$ (D) $\frac{441}{16}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A rod of length 4 m pivots at its lower end. A peg constrained to move horizontally stays in contact with its upper tip. At an instant the rod makes an angle with $\sin\theta = 3/5$ to the horizontal and the tip's horizontal speed has magnitude 2 m/s. Find the rod's angular-speed magnitude.

(A) $\frac{3}{10}$ rad/s (B) $\frac{5}{6}$ rad/s

(C) $\frac{2}{5}$ rad/s (D) 2 rad/s

132. A uniform block of density 400 kg/m^3 floats in a liquid. Then the liquid density is reduced to $\frac{3}{4}$ times 1000 kg/m^3, while the block's volume and density are unchanged. What fraction of the block is now submerged?

(A) $\frac{8}{15}$ (B) $\frac{2}{5}$

(C) $\frac{15}{8}$ (D) 1

133. A stationary nucleus of mass number 224 emits an alpha particle whose kinetic energy is 9 MeV. Taking daughter and alpha masses proportional to $A-4$ and 4, find the total energy release Q, including daughter recoil.

(A) $\frac{9\sqrt{55}}{55} + 9$ MeV (B) 504 MeV

(C) $\frac{504}{55}$ MeV (D) $\frac{495}{56}$ MeV

Chemistry

134. For a completely filled n=4 shell,how many electrons have$m_l = 0$ and$m_s = +1/2$?

(A) 4 (B) 16

(C) 32 (D) 8

135. Consider I: Oxygen normally has lower maximum covalency than sulfur in their common stable compounds. II: NH3 has a higher boiling point than PH3. Which assessment correctly recognises both anomalous first-element behaviours?

(A) Neither statement is correct. (B) Both statements are correct.

(C) Only I is correct. (D) Only II is correct.

136. Of benzaldehyde and propanal,select the aldehyde that self-aldolises with dilute base. Reduce a separate pure sample of that selected aldehyde with NaBH4 and work up. Which alcohol forms?

(A) 2-methylpropan-1-ol (B) propan-2-ol

(C) propan-1-ol (D) benzyl alcohol

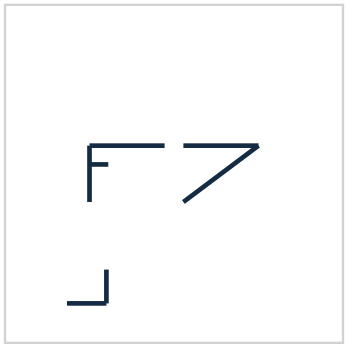
Logical Reasoning

137. A code shifts letters cyclically through A–Z. Odd positions share one fixed shift and even positions share another. For example, LAMP is coded as NXOM. Which word is coded as EECFT?

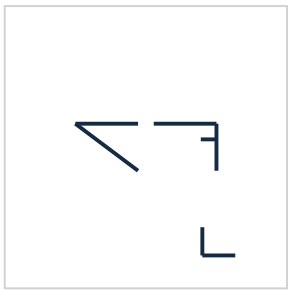
(A) BGZHQ (B) CHAIR

(C) GBECV (D) DIBJS

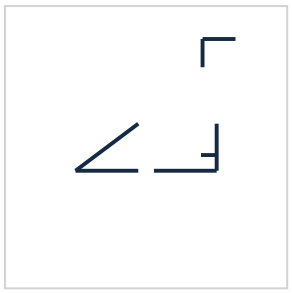
138. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



(A) Pattern P



(B) Pattern U



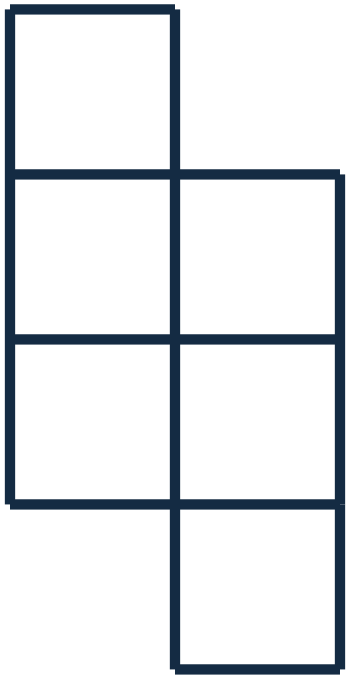
(C) Pattern Q



(D) Pattern Y



139. How many rectangles, including squares are formed by all the drawn segments?



(A) 16 (B) 14

(C) 15 (D) 12

Mathematics

140. How many integer k in [0,70] make $x^2 - 12x + (35 - k) = 0$ have roots of opposite signs?

(A) 35 (B) 34

(C) 36 (D) 70

141. In $(1+x)^{13}$, find the ratio of the coefficient of x^5 to the coefficient of x^4.

(A) 9 (B) $\frac{9}{5}$

(C) $\frac{5}{9}$ (D) $\frac{5}{2}$

142. There are 9 Boolean variables p_1,...,p 9. How many truth assignments make [(p_1 implies p_2) and (p_1 implies not p_2)] true?

(A) 511 (B) 512

(C) 256 (D) 128

BITSAT Mock 23

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A projectile is launched at 40 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{24}{25}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ m/s^2.

(A) $\frac{1536}{25}$ m (B) $\frac{768}{25}$ m

(C) $\frac{2304}{25}$ m (D) $\frac{768}{5}$ m

2. A stationary nucleus with mass number 32 and binding energy per nucleon $\frac{39}{5}$ MeV splits into two equal daughters, each having total binding energy $\frac{644}{5}$ MeV. Which statement is correct?

(A) The process releases 16 MeV; each daughter releases 8 MeV.

(B) The process absorbs 8 MeV; the daughters have larger binding energy per nucleon.

(C) The process releases 8 MeV; the daughters have larger binding energy per nucleon.

(D) No energy can be released because fission always requires a larger mass number than 32.

3. A spherical planet rotates uniformly. Its rotation reduces effective gravity at the equator by 6 m/s^2. By how much does rotation reduce the radial component of effective gravity at latitude 60°?

(A) $\frac{3}{2}$ m/s^2 (B) $\frac{3}{8}$ m/s^2

(C) $\frac{9}{2}$ m/s^2 (D) 6 m/s^2

4. A thin straight tube of length 4 m and internal area $\frac{1}{2500}$ m^2 is completely filled with an incompressible liquid of density 500 kg/m^3. It rotates horizontally about one end at angular speed 4 rad/s. Ignoring gravity along the tube, find the pressure difference between its outer and inner ends.

(A) 16000 Pa (B) 128000 Pa

(C) 64000 Pa (D) $\frac{64000}{3}$ Pa

5. A long solenoid has its turn number multiplied by 4 and length multiplied by 3, while its cross-sectional area and core permeability stay fixed. Find self-inductance multiplier.

(A) 48 (B) $\frac{16}{3}$

(C) $\frac{4}{3}$ (D) 16

6. A pendulum bob of mass 1 kg and length 1 m passes the horizontal position moving downward at 4 m/s. Find the instantaneous power of all forces when $\sin\theta = 3/5$, $\cos\theta = 4/5$, with θ measured from the downward vertical. Take $g = 10\ \text{m/s}^2$.

(A) $32\sqrt{2}\,\text{W}$ (B) $24\,\text{W}$

(C) $24\sqrt{2}\,\text{W}$ (D) $12\sqrt{7}\,\text{W}$

7. An astronomical telescope in normal adjustment has magnifying-power magnitude 5 and tube length 12 cm. Find the eyepiece focal length.

(A) $\frac{1}{3}\,\text{cm}$ (B) $\frac{6}{13}\,\text{cm}$

(C) $3\,\text{cm}$ (D) $2\,\text{cm}$

8. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 2 kΩ to a terminal at +12 V, and its cathode connects to a terminal at 0 V. Find the current from the +12 V terminal in mA.



(A) $\frac{53}{10}\,\text{mA}$ (B) $6\,\text{mA}$

(C) $\frac{113}{20}\,\text{mA}$ (D) $\frac{127}{20}\,\text{mA}$

9. A screw gauge advances 2.0 mm in four complete turns and has 50 circular divisions. With the jaws closed, its zero mark lies 4 divisions above the reference line; use the usual convention that a zero below the reference line gives positive zero error. With a wire, the main scale reads 8 mm and circular reading is 29 divisions. What is the corrected diameter?

(A) $\frac{147}{4}\,\text{mm}$ (B) $\frac{431}{50}\,\text{mm}$

(C) $\frac{829}{100}\,\text{mm}$ (D) $\frac{833}{100}\,\text{mm}$

10. A circular parallel-plate capacitor of plate radius 5 cm is charged at 12 mA. Neglect fringing. What displacement current passes through the annular part of the gap between radii 4 cm and 5 cm, centred on the plate axis?

(A) $\frac{27}{4}\,\text{mA}$ (B) $\frac{108}{25}\,\text{mA}$

(C) $12\,\text{mA}$ (D) $\frac{108}{5}\,\text{mA}$

11. Two wires of the same material and length L, with radii r and 5r, are joined in series and stretched by a force F. Find their total stored elastic energy divided by the energy stored in the thinner wire alone under force F.

(A) $\frac{6}{5}$ (B) 2

(C) $\frac{26}{25}$ (D) $\frac{25}{26}$

12. A spherical blackbody's absolute temperature is multiplied by 3 while its radius is divided by 2. Neglect incoming background radiation. Find its emitted-power multiplier.

(A) $\frac{81}{4}$ (B) $\frac{3}{4}$

(C) $\frac{81}{8}$ (D) $\frac{4}{81}$

13. Light of vacuum wavelength 700 nm passes from a medium of index 4 into one of index 5. Find the ratio of its frequencies in the second and first media.

(A) 1 (B) $\frac{25}{16}$

(C) 5 (D) $\frac{16}{25}$

14. A point charge $27q$ is placed at a vertex of a cube. Find the electric flux through the three faces not meeting that vertex, combined, in units of q/ε_0.

(A) $\frac{27}{4}$ (B) $\frac{9}{8}$

(C) $\frac{9}{2}$ (D) $\frac{27}{8}$

15. A parallel-plate capacitor has vacuum capacitance C_0. A dielectric of constant 7 fills a fraction $\frac{1}{2}$ of the plate area and the entire gap in that region; the remainder stays vacuum. The voltage is fixed. Find the fraction of total stored energy residing in the dielectric-filled region.

(A) $\frac{1}{2}$ (B) $\frac{7}{8}$

(C) $\frac{1}{8}$ (D) $\frac{7}{2}$

16. Two coaxial circular loops each radius 2 m carry equal currents 2 A in the same circulation sense. Their centres are separated by $2R$. Find field at the midpoint in units of μ_0.

(A) $\frac{\sqrt{2}}{2}$ (B) $\frac{\sqrt{2}}{8}$

(C) 1 (D) $\frac{\sqrt{2}}{4}$

17. A diamond network has four outer resistors AB, BC, CD and DA, each 7 ohms, and a bridge BD also 7 ohms. The DA arm additionally contains an ideal diode with anode at D and cathode at A. A source keeps A positive relative to C. Find equivalent resistance between A and C.

(A) $\frac{7}{2}\,\Omega$ (B) $\frac{35}{3}\,\Omega$

(C) $14\,\Omega$ (D) $21\,\Omega$

18. A metal wire has linear temperature coefficient $\frac{1}{1000}$ per degree Celsius referred to 0°C. Its dimensions are fixed and it stays across a constant voltage. Find power at 110°C divided by power at 0°C.

(A) $\frac{100}{89}$ (B) $\frac{111}{100}$

(C) $\frac{100}{111}$ (D) $\frac{89}{100}$

19. A spring of constant 70 N/m is already stretched by $\frac{1}{5}$ m. How much additional work is required to increase its extension to $\frac{3}{10}$ m quasistatically?

(A) $\frac{7}{4}$ J (B) $\frac{21}{10}$ J

(C) $\frac{7}{5}$ J (D) $\frac{7}{2}$ J

20. A galvanometer has resistance 40 ohms and full-scale current $\frac{1}{500}$ A. It is to measure a total current 2 times its own full-scale current. Find the required parallel shunt resistance.

(A) $40\,\Omega$ (B) $80\,\Omega$

(C) $20\,\Omega$ (D) $\frac{40}{3}\,\Omega$

21. Two uniform thin rods meet at a right angle at the origin. Both have length 2 m. The rod along +x has linear density 3 times that of the rod along +y. A force parallel to the y-axis must produce pure translation. At what x-coordinate must its line of action lie?

(A) 3 m (B) 1 m

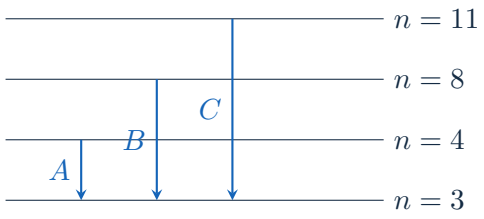
(C) $\frac{1}{2}$ m (D) $\frac{3}{4}$ m

22. A particle executes SHM at frequency 9 Hz. At t=0 it is at its positive extreme. In the interval $0 <= t < 1$ second, how many times does its kinetic energy equal its potential energy?

(A) 36 (B) 9

(C) 72 (D) 37

23. The diagram shows downward hydrogen transitions A, B and C from $n = 4$, 8 and 11, respectively, ending at $n = 3$. A filter transmits only photons with energy strictly above that of transition B. Which transmitted line has the longest wavelength?



(A) B (B) A

(C) No transmitted line (D) C

24. A rigid ladder of length 10 m slides with its ends on a floor and a vertical wall. At the instant its top is at height $4L/5$, the foot moves away at 6 m/s. What is the downward speed of the top?

(A) $\frac{24}{5}$ m/s (B) $\frac{9}{2}$ m/s

(C) 8 m/s (D) $\frac{18}{5}$ m/s

25. A triangular p-V cycle has vertices (V_0, p_0), $(V_0 + 4\,\text{L}, p_0)$ and $(V_0, p_0 + 7\,\text{kPa})$. Traversed in that listed order and then back, find net work done by the gas.

(A) 28 J (B) -14 J

(C) 0 J (D) 14 J

26. In a series RLC circuit, ideal rms voltmeters read 40 V across R, 40 V across L and 30 V across C. Find source rms voltage.

(A) $10\sqrt{17}$ V (B) 50 V

(C) $10\sqrt{65}$ V (D) 10 V

27. A 5 kg block on a rough horizontal table is tied over an ideal pulley to a hanging 3 kg mass. The hanging mass moves downward and $\mu_k = 1/5$. Find string tension. Take $g = 10$ m/s^2.

(A) 30 N (B) 10 N

(C) 50 N (D) $\frac{45}{2}$ N

28. A long solenoid has 500 turns per metre and carries 4 A. It is completely filled by a linear magnetic core of relative permeability 2. Find magnetisation M.

(A) 2000 A/m (B) 1000 A/m

(C) 4000 A/m (D) 6000 A/m

29. A uniform block of density 300 kg/m^3 floats in a liquid. Then the liquid density is reduced to $\frac{3}{4}$ times 1000 kg/m^3, while the block's volume and density are unchanged. What fraction of the block is now submerged?

(A) $\frac{9}{40}$ (B) 1

(C) $\frac{2}{5}$ (D) $\frac{3}{4}$

30. A particle is projected vertically outward from the surface of a nonrotating planet with speed $v_e/15$, where v_e is surface escape speed. Neglect atmosphere. Find maximum height divided by planet radius.

(A) $\frac{1}{226}$ (B) $\frac{1}{224}$

(C) $\frac{1}{225}$ (D) $\frac{1}{15}$

Chemistry

31. A weak1:1electrolyte has$\alpha \ll 1$ and obeysKa$\approx$Cα^2. Dilute it by a factor25without changing total formula-unit amount or Ka,and remain in that approximation. Which pair(degreeα,totalfree-ion amount)is correct?

(A) Both remain unchanged because the number of solute formula units is fixed.

(B) α increases byfactor25,buttotalionsdecreasebyfactor25.

(C) Both increase by factor5.

(D) α decreases byfactor5,andtotalionsareunchanged.

32. In the stated high-pressure model Z=1+b p/(RT), gas A at 300 K and gas B at 440 K have Z-versus-p slopes in ratio 88/75. Their molar covolumes b do not change with temperature in this model. At what temperature must gas A be measured for its slope to equal that of B at 440 K?

(A) 300 (B) 440

(C) 352 (D) 550

33. Assign the formal metal oxidation state in $[Fe(CN)_6]^{3-}$,using six CN^-. For nitrosyl species use the explicitly stated formal NO^+ convention rather than inferring spin from it.

(A) 2 (B) 5

(C) 3 (D) 4

34. For nona-2,4,6-triene, CH3-CH=CH-CH=CH-CH=CH-CH2-CH3, count distinct configurational geometrical isomers. Ignore conformers and photochemical interconversion. Where reversing the entire carbon chain identifies two E/Z assignments, count them once.

(A) 9 (B) 7

(C) 16 (D) 8

35. In the gas phase,N_2^+ is converted into N_2 by changing one electron. Under the elementary MO model, which final bond-order/unpaired-electron pair follows?

(A) Bond order $\frac{5}{2}$; 1 unpaired electrons.
(B) Bond order 3; 0 unpaired electrons.
(C) Bond order 3; 1 unpaired electrons.
(D) Bond order 2; 0 unpaired electrons.

36. A closed system goes from state1to2 by path A with q=700 J and work on the system w=-200 J. It returns from2to1 along a different path with q=-900 J. IUPAC work on the return path,in J,is

(A) 200
(B) -400
(C) 500
(D) 400

37. A one-electron ion of nuclear charge Z=1 has a selected shell which, if hypothetically filled subject to Pauli, contains 6 electrons in orbitals with ml=0. What is the magnitude of that shell's hydrogenic energy divided by the hydrogen ground-state energy magnitude? Ignore interactions in identifying the shell.

(A) -1/9
(B) 1/9
(C) 2/9
(D) 1/18

38. Limiting molar conductivities of KCl,NaCl,KNO_3 are150,126,110 S cm^2/mol. A very dilute $NaNO_3$ solution measured in a cell of constant2.00cm^{-1} has conductance $\frac{43}{50000}$S. Neglect concentration corrections. $NaNO_3$ molarity is

(A) $\frac{43}{2750}$
(B) $\frac{1}{100}$
(C) $\frac{1}{25}$
(D) $\frac{1}{50}$

39. A model oxide has composition MO9/8, where the subscript denotes O atoms per M atom. Oxygen is entirely O2- and metal sites are formally only M2+ or M3+. There are no other defects or ions. What fraction of M sites is M3+?

(A) 1/4
(B) 1/2
(C) -1/4
(D) 1/8

40. A fully dissociated HCl solution has concentration 1/1000 M. What analytical concentration of a monoprotic weak acid HA with Ka=1/1000000 gives approximately the same pH? Neglect water autoionisation and use [H+]$\approx \sqrt{KaC}$, which is to remain in its weak-ionisation regime.

(A) 2
(B) 1
(C) -1
(D) 1/2

41. P is a phenol described by 76.60%C,6.38%H,and the rest O;vapour density47 relative toH2. Which molecular formula and Kolbe-Schmitt product pair is consistent?

(A) C7H8O;4-methylbenzoic acid
(B) C6H6O;2-hydroxybenzoic acid
(C) C3H6O;3-hydroxybenzoic acid
(D) C6H12O;benzoic acid

42. Consider I: Carbon shows stronger ordinary catenation than silicon. II: The N-N single bond is unusually weak relative to the P-P single bond. Which assessment correctly recognises both anomalous first-element behaviours?

(A) Only II is correct.
(B) Both statements are correct.
(C) Only I is correct.
(D) Neither statement is correct.

43. A $\frac{1}{50}$ mol FeC_2O_4 inventory is analysed. A fraction $\frac{3}{4}$ of Fe^{2+} has already become Fe^{3+},but every oxalate remains unchanged. Subsequent complete oxidation gives Fe^{3+} and CO_2. Electron equivalents released,in mol,are

(A) $\frac{1}{200}$ (B) $\frac{1}{40}$

(C) $\frac{1}{25}$ (D) $\frac{9}{200}$

44. For O-(g)+e-→O2-(g), why can the process be endothermic even though oxygen is electronegative?

(A) A gas-phase anion has no electrons to repel the incoming electron.

(B) Electronegativity means every successive electron addition releases energy.

(C) The process changes oxygen into a different element.

(D) An electron is being added to an already negative isolated ion, so electron-ion repulsion must be overcome.

45. For $2H_2O_2 \rightarrow 2H_2O+O_2$,consider only atoms of the element undergoing disproportionation. Fraction of those atoms oxidised is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) $\frac{1}{3}$ (D) $\frac{1}{6}$

46. Excess water reacts completely with $\frac{1}{50}$ mol $CaCN_2$. Which gas identity/amount pair follows from the balanced hydrolysis?

(A) NH_3, $\frac{1}{25}$ mol. (B) CO_2, $\frac{1}{50}$ mol.

(C) O_2, $\frac{1}{100}$ mol. (D) NH_3, $\frac{1}{50}$ mol.

47. A nonvolatile solute in 5mol water gives relative vapour-pressure lowering1/101. Remove $\frac{5}{2}$mol water without solute loss. Under exact Raoult mole fractions,new relative lowering is

(A) $\frac{1}{101}$ (B) $\frac{2}{101}$

(C) $\frac{1}{51}$ (D) $\frac{1}{50}$

48. The internuclear axis is z. Two p_x orbitals overlap sideways. Their lobes facing each other above the axis have the same phases; the lower facing lobes have the corresponding opposite signs. Which MO is produced,and what nodal property follows?

(A) A sigma bonding MO,because every overlap along a line joining nuclei is sigma.

(B) A delta MO,because two perpendicular p orbitals always create two nodal planes.

(C) A pi bonding MO; the plane containing the axis is a node,but there is no additional internuclear node from phase cancellation.

(D) A pi antibonding MO; the facing-lobe phase test has been reversed.

49. A diprotic acid H_2A is0.100M;its first dissociation is complete and$Ka2=10^{-5}$. Added HCl is $\frac{3}{10}$M in the final solution. Neglect the second-step contribution to H^+. Equilibrium$[A^{2-}]$in M is

(A) $\frac{1}{400000}$ (B) $\frac{1}{100000}$

(C) $\frac{1}{300000}$ (D) $\frac{1}{25000}$

50. In a hypothetical rule,$l = 0, 1, \ldots, n$ is allowed. Keep subshell capacity2(2l+1),Hund and Pauli,and fill by n+l then lower n. For 23 electrons,the last occupied subshell and its electron count are

(A) $2p^5$.

(B) $3d^5$.

(C) $2d^5$.

(D) $3p^5$.

51. For the pair pentane and2-methylbutane,which classification identifies the structural difference rather than merely stating that the formula agrees?

(A) chain isomers

(B) tautomers

(C) geometrical isomers

(D) metamers

52. Consider the statements. I. Reducing sulphur in a fuel can lower its potential sulphur-dioxide emission. II. Filtering only visible dust guarantees elimination of gaseous sulphur dioxide. Choose the correct truth pattern.

(A) I false; II false.

(B) I true; II false.

(C) I true; II true.

(D) I false; II true.

53. A label identifies sodium benzoate as an additive used to limit spoilage. Which role and ion pair are consistent?

(A) Preservative; sodium cation and benzoate anion.

(B) Tranquilizer; sodium and nitrate ions.

(C) Analgesic; calcium and chloride ions.

(D) Synthetic rubber; carbon-only network.

54. Limiting Λ values in S cm^2/mol are KCl150,$K_2SO_4$300,$BaCl_2$250. A saturated $BaSO_4$ solution has conductivity $\frac{19}{2500000}$S/cm;pure-water conductivity is $\frac{1}{10000000}$S/cm. Neglect ion interactions. Ksp($BaSO_4$) is

(A) $\frac{361}{390625000000}$

(B) $\frac{3}{100000}$

(C) $\frac{9}{10000000000}$

(D) $\frac{9}{2500000000}$

55. Standard reduction potentials for A^+/A,B^+/B,C^+/C are $\frac{1}{2}, \frac{3}{10}, \frac{3}{5}$ V. Which pair of claims is correct under standard conditions?

(A) C^+ oxidises A;doubling all coefficients leaves standard cell potential $\frac{1}{10}$ V unchanged.

(B) A^+ oxidises C with standard cell potential $\frac{1}{10}$ V.

(C) C^+ oxidises A,but doubling coefficients doubles the standard cell potential to $\frac{1}{5}$ V.

(D) B^+ oxidises A spontaneously because $\frac{3}{10}$ V is positive.

56. For isolated Co^{2+} (Z=27),use the high-spin spin-only approximation. Which pair (d-electron count, magnetic moment in BM) follows after removing electrons from 4s before 3d?

(A) (7,3).

(B) $(8, 2\sqrt{6})$.

(C) $(7, 2\sqrt{6})$.

(D) $(7, \sqrt{15})$.

57. A sweetener is the methyl ester of the dipeptide formed from aspartic acid and phenylalanine. Which structural feature must it contain?

(A) Only carbon-carbon bonds and no nitrogen.

(B) A peptide amide bond and a methyl ester.

(C) A glycosidic bond between two glucose units.

(D) Three carbon-bound chlorine atoms by definition.

58. With one equivalent of cold concentrated HI and isolation before prolonged further reaction,which initial cleavage pair is expected from ethyl isopropyl ether?

(A) ethyl iodide and propan-2-ol

(B) 2-iodopropane and ethanol

(C) methyl iodide and butan-1-ol

(D) propane and ethanol

59. For isolated molecules of CH_3Cl and CCl_4,which member has the larger permanent dipole moment, and which reasoning is sufficient?

(A) The one with more atoms always has the larger dipole,independent of geometry.

(B) They must be equal because both contain polar covalent bonds.

(C) CCl_4; the number of polar bonds alone fixes the molecular dipole.

(D) CH_3Cl; The four equal tetrahedral C-Cl dipoles cancel in CCl4; unequal substituents in CH3Cl prevent cancellation.

60. Compare phenoxide and p-nitrophenoxide as oxygen nucleophiles towards the same unhindered methyl substrate in one fixed solvent. Ignore concentration differences. Which pairing of the stronger nucleophile and the main electronic reason is most appropriate?

(A) p-nitrophenoxide; the more stabilised anion necessarily attacks faster

(B) phenoxide; the nitro group further withdraws and delocalises negative charge

(C) p-nitrophenoxide; resonance always increases local charge on oxygen

(D) phenoxide; it is weaker because a localised lone pair cannot attack

English Proficiency

61. Complete: At the junction, continue _____; do not take either side path.

(A) straightness

(B) straight

(C) straighten

(D) straightly

62. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. The test prevented a procedure suitable for the coating from damaging the work beneath it. B. The restorer removed a small patch of discoloured coating from a painting. C. She chose a gentler method for the remaining surface. D. The test revealed that stronger treatment would also lift original colour.

(A) The restorer removed a small patch of discoloured coating from a painting.

(B) The test prevented a procedure suitable for the coating from damaging the work beneath it.

(C) She chose a gentler method for the remaining surface.

(D) The test revealed that stronger treatment would also lift original colour.

63. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. which of the similar folders Q. contained the records R. requested by the auditor. S. The assistant could not tell

(A) R-P-Q-S

(B) P-S-Q-R

(C) S-P-Q-R

(D) Q-S-P-R

64. Read the sentence: Despite their disagreement, both speakers maintained a cordial tone throughout the discussion. Choose the closest meaning of 'cordial' as used in the sentence.

(A) warm and courteous
(B) unusually enthusiastic
(C) completely approving
(D) strictly formal

65. Choose the meaning-preserving equivalent that retains the tense, participants and all stated restrictions. The road was so narrow that two buses could not pass each other.

(A) No bus could travel along the narrow road.
(B) The road was narrow enough for two buses to pass each other.
(C) Two buses could pass each other only because the road was narrow.
(D) The road was too narrow for two buses to pass each other.

66. The music gives him an affectionate longing for a period of his earlier life, rather than fear of the future. Which noun fits?

(A) nostalgia
(B) ambition
(C) anticipation
(D) anxiety

Read the passage for Question 67

The tutor stopped writing a full correction beside every error. Instead, she marked the point at which a student's reasoning first departed from the evidence and asked for a revised explanation. Some students initially preferred the old method because they could copy its corrections more quickly. The new task took longer, but it revealed whether a student could carry the repaired idea into another example. This did not mean that every difficulty should be welcomed. An unclear question or an unfamiliar word irrelevant to the lesson could obstruct learning without improving it. The tutor wanted effort directed at the intended reasoning, not effort spent guessing what the question meant. Her aim was therefore not to make work harder in every possible way, but to make the added demand serve an identifiable purpose.

67. What is the central distinction developed in the passage?

(A) The distinction between quick work, which is always superficial, and slow work, which is always valuable.
(B) The distinction between effort that develops the intended reasoning and difficulty that merely obstructs it.
(C) The distinction between written correction, which never helps, and unsupported struggle, which always helps.
(D) The distinction between students who enjoy copying and students who should receive no help.

68. The word moral here means ethically right, rather than merely concerned with ethics. Choose its opposite.

(A) moral
(B) immoral
(C) morale
(D) moralistic

69. Read the sentence: The night supervisor remained vigilant, checking even faint signs of a leak. Choose the closest meaning of 'vigilant' as used in the sentence.

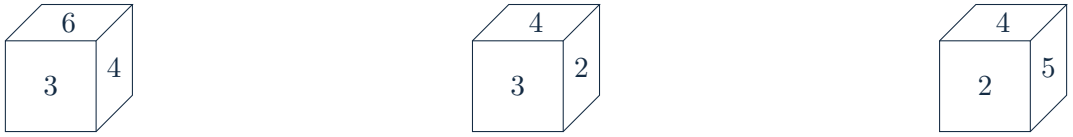
(A) unreasonably suspicious
(B) continuously anxious
(C) unusually strict
(D) watchfully alert

70. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. After the figures had been converted Q. the two reports proved R. to be in agreement. S. to a common unit,

(A) R-S-Q-P
(B) Q-P-S-R
(C) P-S-Q-R
(D) S-P-Q-R

Logical Reasoning

71. The views show the same die. Give the ordered pair: (face opposite 6, face opposite 4). Do not assume opposite labels sum to seven.

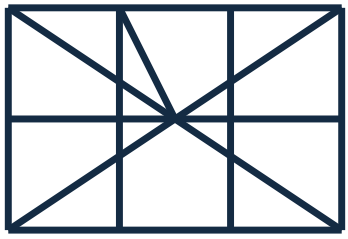


(A) $(2, 4)$
(B) $(2, 1)$
(C) $(2, 3)$
(D) $(2, 2)$

72. Bina receives 2 times as many tokens as Asha, and Charu receives 3 times as many as Bina. Their total is 81. After the allocation, Bina gives 6 tokens to Asha. How many tokens does Bina then retain?

(A) 3
(B) 12
(C) 18
(D) 24

73. How many triangles are formed by all the drawn segments?



(A) 22
(B) 31
(C) 52
(D) 21

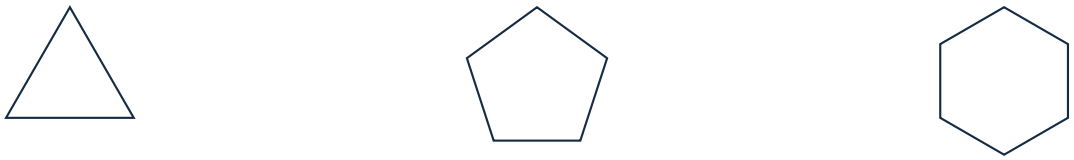
74. In a row, Asha is 12th from the left and 26th from the right. Bina is 9th from the right. Charu stands 2 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 14
(B) 13
(C) 15
(D) 4

75. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 3, 9, ?, 42, 47, 141, 146.

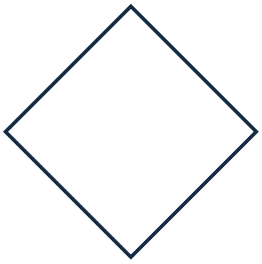
(A) 32
(B) 42
(C) 27
(D) 14

76. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?

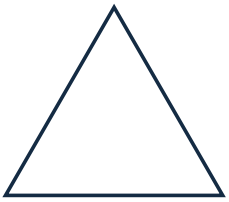


A B C

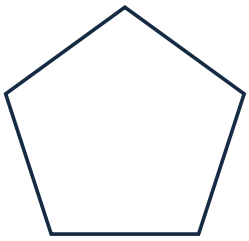
(A) Pattern Q



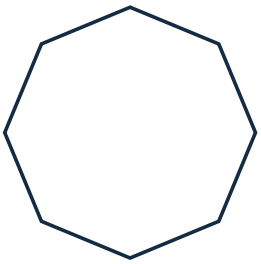
(B) Pattern W



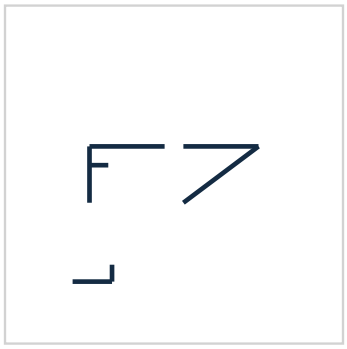
(C) Pattern V



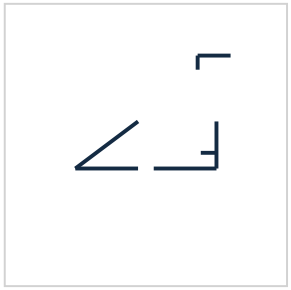
(D) Pattern M



77. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



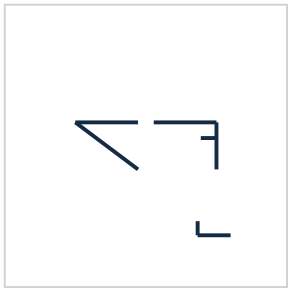
(A) Pattern K



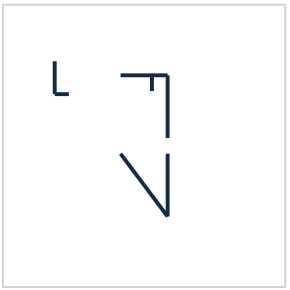
(B) Pattern T



(C) Pattern R



(D) Pattern L



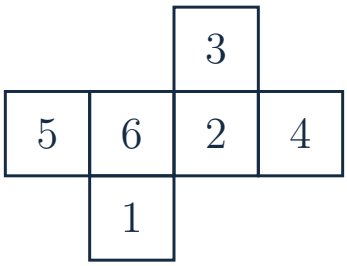
78. Preserve the exact relation and its order: Scarce : Rare :: ? : Authentic. Choose the missing term.

(A) Rapid
(B) Begin
(C) Silent
(D) Genuine

79. Ravi points to a photograph and says, "This person's only daughter is my mother." Ravi is male, and the photographed person is male. How is the photographed person related to Ravi?

(A) Son
(B) Daughter
(C) Grandfather
(D) Brother

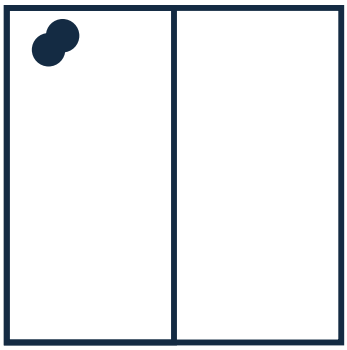
80. Fold the shown net into a cube. Give (label opposite 1, label opposite 2) in that order.



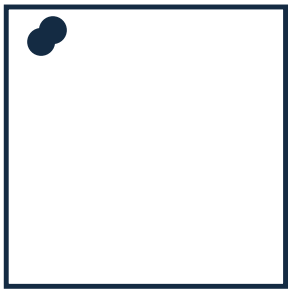
(A) $(3, 2)$ (B) $(3, 3)$

(C) $(3, 5)$ (D) $(3, 6)$

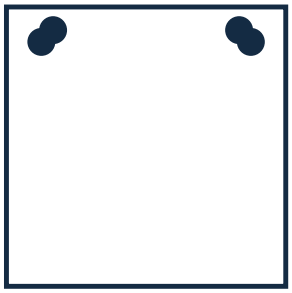
81. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



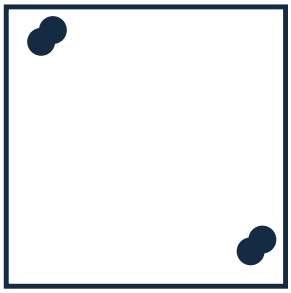
(A) Pattern X



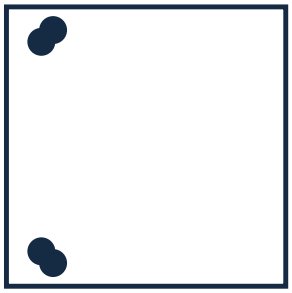
(B) Pattern P



(C) Pattern W



(D) Pattern Q



82. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: JHE, MEG, ?, SYK, VVM, YSO.

(A) PBI (B) QCJ

(C) IBP (D) OAH

83. Consider only these statements; universal statements alone do not imply existence. All inventors are farmers; No farmers are hikers. Conclusions: I. No inventors are hikers. II. Some inventors are not hikers. Which conclusions follow necessarily?

(A) Both I and II (B) Neither I nor II

(C) Only II (D) Only I

84. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 3 (B) 4

(C) 2 (D) 1

85. Three pairs contain exactly one mammal and one bird. Which pair does not?

(A) Rabbit / Penguin (B) Horse / Duck

(C) Bat / Whale (D) Cat / Eagle

86. A walker goes 12 m south, then 5 m east, then turns right and walks 8 m. What is the square of the straight-line distance from the starting point, in square metres?

(A) 425 (B) 41

(C) 424 (D) 426

87. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 11$ and $5 \mapsto 26$. Which input maps to 51?

(A) 11 (B) 51

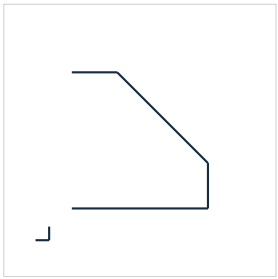
(C) 9 (D) 10

88. In this code, @ means $-$, # means $\div$, & means $\times$, % means $+$. After substituting meanings, evaluate ((3#6)@(5&3)); printed parentheses retain their usual grouping.

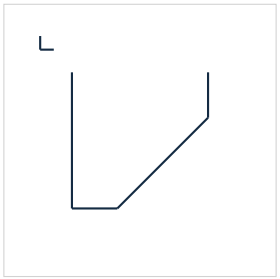
(A) $\frac{-27}{2}$ (B) $\frac{-29}{2}$

(C) $\frac{-31}{2}$ (D) $\frac{29}{2}$

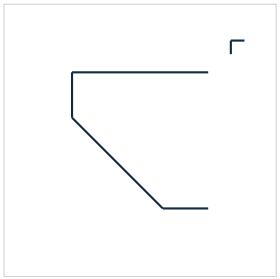
89. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1



2

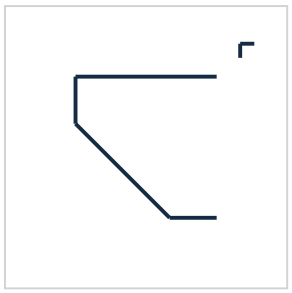


3

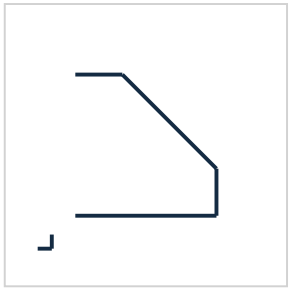
(A) Pattern L



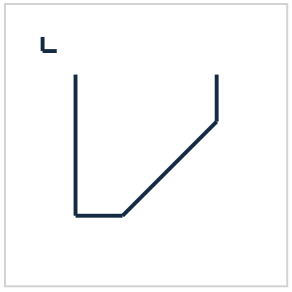
(B) Pattern K



(C) Pattern M



(D) Pattern N



90. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

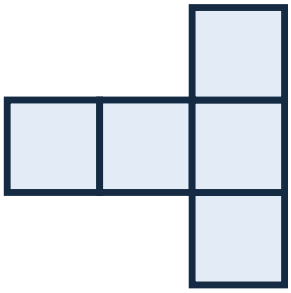


piece 1

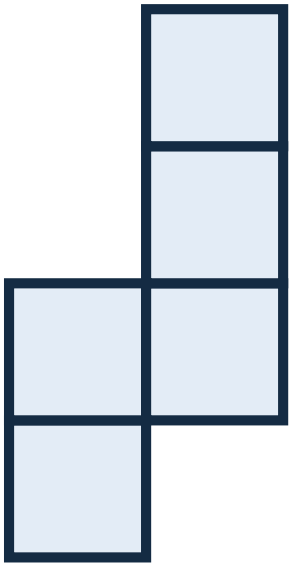


piece 2

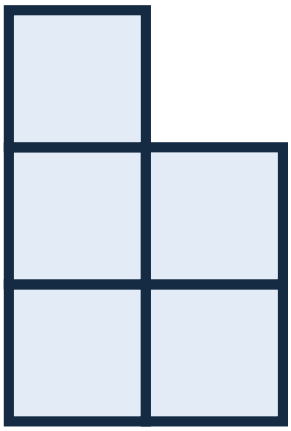
(A) Pattern L



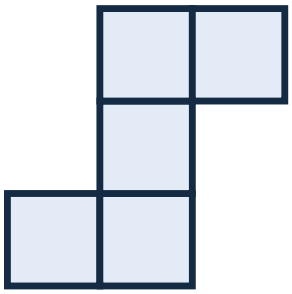
(B) Pattern R



(C) Pattern Y



(D) Pattern X



Mathematics

91. Evaluate $\sum_{j=0}^{3} \binom{3}{j}\binom{11}{4-j}$.

(A) 2002

(B) 3003

(C) 1001

(D) 715

92. For common tangents to $y^2=12x$ and $x^2+y^2=\frac{3}{4}$, find the sum of the squared slopes.

(A) -3

(B) 3

(C) 0

(D) 6

93. The locus |z-6|=|Re(z)+6| intersects the horizontal line Im(z)=14. Find the real part of the point.

(A) $-\frac{49}{6}$ (B) $\frac{7}{12}$

(C) $\frac{49}{3}$ (D) $\frac{49}{6}$

94. Let z=$(1+i)^{29}$. Find Re(z)+Im(z).

(A) −32768 (B) 32768

(C) −16384 (D) $16384\sqrt{2}$

95. Chords of $x^2+y^2=100$ pass through (20,0). Find the radius of the circle containing their midpoint locus.

(A) 5 (B) 20

(C) 100 (D) 10

96. What is the maximum number of distinct intersection points formed by 6 circles and 7 straight lines, with no coincident curves?

(A) 114 (B) 165

(C) 135 (D) 93

97. Circles $x^2+y^2=36$ and $(x-4)^2+y^2=36$ meet at A,B. Find $|AB|^2$.

(A) 128 (B) 112

(C) 160 (D) 32

98. A sphere has dV/dt=4 cubic units/s. Find the rate of change of its surface area when r=3.

(A) $\frac{8}{9}$ (B) $\frac{4}{3}$

(C) $\frac{1}{9\pi}$ (D) $\frac{8}{3}$

99. Let $y\prime + 6y = 5x$ and $y(0) = 4$. Find $y(1) + \frac{17}{10}y\prime(1)$.

(A) $\frac{149}{36e^6} + \frac{25}{36}$ (B) $\frac{19}{9}$

(C) $\frac{19}{9} - \frac{3427}{90e^6}$ (D) $\frac{3427}{90e^6} + \frac{19}{9}$

100. An AP has a_1=3, d=7. Find $a_1+a_3+\cdots+a_17$.

(A) 594 (B) 531

(C) 1003 (D) 1062

101. Find the area bounded by y=x, y=25/x, and x=15 in the first quadrant.

(A) $50 - 25\log(3)$ (B) 100

(C) $100 - 25\log(3)$ (D) $\log(847288609443)$

102. $A = \begin{bmatrix} 5 & 1 \\ 1 & 2 \end{bmatrix}$ and A^{-1}=pI+qA. Find p+q.

(A) $\frac{7}{9}$ (B) $\frac{6}{7}$

(C) $\frac{2}{3}$ (D) $\frac{8}{9}$

103. For the loop $y^2=x^2(4-x)$, find its area within the strip $4/3\le x\le 8/3$.

(A) $-\frac{256\sqrt{3}}{135}+\frac{128\sqrt{6}}{45}$ (B) $-\frac{512\sqrt{3}}{135}+\frac{256\sqrt{6}}{45}$

(C) $\frac{256\sqrt{6}}{45}$ (D) $-\frac{32\sqrt{3}}{27}+\frac{64\sqrt{6}}{27}$

104. ABC is a right triangle on level ground with AB=4, AC=4, angle A=90°. A vertical tower stands at midpoint M of BC. Its top elevation from A is 60°. Find its height.

(A) $2\sqrt{2}$ (B) $2\sqrt{3}$

(C) $2\sqrt{6}$ (D) $4\sqrt{6}$

105. Let k make lim as $x\to\infty$ of $[(2x^2+5x+1)/(x+4)-kx]$ finite. Find that finite limit.

(A) 13 (B) -3

(C) 5 (D) 8

106. Positive a,b satisfy $\log(a+b)=\log(9)+[\log a+\log b]/2$ in one valid base. Find $(a^4+b^4)/(a^2b^2)$.

(A) 79 (B) 6239

(C) 6241 (D) 6243

107. If $|z-10|=4$, find the minimum of $|z|^2$.

(A) 36 (B) 196

(C) 6 (D) 84

108. $A=\{j\in\{1,...,36\}:\sin(j\pi/2)=0\}$, $B=\{j\in\{1,...,36\}:\sin(j\pi/3)=0\}$. Find $|(A\cup B)\setminus(A\cap B)|$.

(A) 24 (B) 18

(C) 6 (D) 30

109. How many integers k make $21\sin x+28\cos x+3=k$ have exactly two distinct solutions in $0\le x<2\pi$?

(A) 71 (B) 34

(C) 70 (D) 69

110. For $x\in[0,1]$, find the minimum of $5(\arcsin x)^3+(\arccos x)^3$.

(A) $\frac{3\pi^3}{32}$ (B) $\frac{5\pi^3}{8\left(1-\sqrt{5}\right)^2}$

(C) $-\frac{5\pi^3}{32}+\frac{5\sqrt{5}\pi^3}{64}$ (D) $\frac{5\pi^3}{8\left(1+\sqrt{5}\right)^2}$

111. X is binomial with mean 4 and variance 8/3. Find P(X=0).

(A) $\frac{2048}{531441}$ (B) $\frac{527345}{531441}$

(C) $\frac{16}{81}$ (D) $\frac{4096}{531441}$

112. A positive integer-domain function satisfies $f(m+n)=f(m)f(n)$, $f(1)=2$. Find $\sum_{j=1}^{6} f(j)$.

(A) 63 (B) 384

(C) 64 (D) 126

113. The same loaded die has odd-face probability $\frac{6}{7}$. Three independent throws are made. Given that the first two have an even sum, find the probability that the total of all three is odd.

(A) $\frac{6}{7}$ (B) $\frac{234}{343}$

(C) $\frac{12}{49}$ (D) $\frac{1}{7}$

114. F′(x)=$2x-\frac{2}{x^3}$ and F(1)=2. Find F(2)-2F(1).

(A) $-\frac{17}{4}$ (B) $\frac{1}{4}$

(C) $\frac{17}{4}$ (D) $\frac{9}{4}$

115. The line r=(1,0,0)+s(t,1,2) is parallel to x-3y+z=5. Find the squared length of its direction vector.

(A) 4 (B) 14

(C) 6 (D) $\sqrt{6}$

116. An invertible 4×4 matrix A has det A=24. Find det(adj A).

(A) 7962624 (B) $\frac{1}{24}$

(C) 331776 (D) 13824

117. A rectangle has perimeter 36. Its length exceeds its width by at least 3. Find its maximum area.

(A) 45 (B) $\frac{315}{4}$

(C) $\frac{1287}{4}$ (D) $\frac{3\sqrt{35}}{2}$

118. f(x)=(x^2-121)/(x-11) for x≠11, and f(11)=k. If f is differentiable at 11, find k+f′(11).

(A) 23 (B) 24

(C) 1 (D) 22

119. A,B are acute, A+B=$\pi/4$ and tan A=$\frac{3}{7}$. Find tan(A-B).

(A) $\frac{1}{35}$ (B) $\frac{2}{5}$

(C) $\frac{1}{41}$ (D) $-\frac{1}{41}$

120. r=pa+qb, a=(1,0,0), b=(1,1,0), r·a=10, r·b=14. Find $|r|^2$.

(A) 14 (B) 116

(C) $2\sqrt{29}$ (D) 52

121. The four lines±x/6±y/9=1 bound a rhombus. Find the square of its inradius.

(A) $\frac{324}{25}$ (B) $\frac{18\sqrt{13}}{13}$

(C) 117 (D) $\frac{324}{13}$

122. $A=\begin{bmatrix} 0 & 2 & -6 \\ -2 & 0 & 5 \\ 6 & -5 & 0 \end{bmatrix}$. Find det(I+A)-(4+36+25).

(A) 1 (B) 65

(C) −1 (D) 0

123. Let $\delta=\pi/14$. Evaluate $\sin\delta\ \Sigma[1/(\sin(j\delta)\sin((j+1)\delta))]$, j=2,...,6.

(A) $\cot\left(\frac{\pi}{14}\right)$ (B) $\cot\left(\frac{\pi}{7}\right)$

(C) $\tan\left(\frac{\pi}{7}\right)$ (D) $-\cot\left(\frac{\pi}{7}\right)$

124. A convergent GP has first term 4, ratio $\frac{1}{5}$. Find the sum of all terms after its first 8 terms.

(A) $\frac{1}{78125}$ (B) $\frac{1}{390625}$

(C) $\frac{1}{312500}$ (D) $\frac{1}{15625}$

125. Maximize $Z=3x+4y$ subject to $x\geq0$, $y\geq5$, $x+y\leq14$, $x+2y\leq18$.

(A) 44 (B) 47

(C) 46 (D) 36

126. $f(x)=\sin(3x)/x+4$ for $x<0$; $f(x)=k\cos x$ for $x\geq0$. Find k for continuity at zero.

(A) 4 (B) 7

(C) 3 (D) -1

127. How many integers x satisfy $4^{-x}-(320)2^{-x}+16384\leq0$?

(A) 1 (B) 9

(C) 3 (D) 2

128. Find the coefficient of x^9 in $(1+6x)\exp(2x)$.

(A) $\frac{16}{405}$ (B) $\frac{4}{2835}$

(C) $\frac{4}{405}$ (D) $\frac{4}{105}$

129. Evaluate $\int_0^{(12\pi)}\sin^4x\,dx$.

(A) 0 (B) 6π

(C) $\frac{9\pi}{2}$ (D) $\frac{3\pi}{2}$

130. How many integers x satisfy $|(x-5)+(x-7)|<|x-5|+|x-7|$?

(A) 1 (B) 2

(C) 6 (D) 3

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. Two sound sources produce wavelengths $\frac{1}{2}$ m and $\frac{1}{3}$ m in the same gas at one temperature. The sound speed there is 500 m/s. At a second temperature the sound speed is 3/2 times larger, while both source frequencies stay unchanged. Find the new beat frequency.

(A) 2500 Hz

(B) 500 Hz

(C) 1250 Hz

(D) $\frac{1000}{3}$ Hz

132. A cell balances on a potentiometer at 60 cm on open circuit and at 20 cm when a 2 ohm resistor is connected across it. The potentiometer gradient is unchanged. Find the cell's internal resistance.

(A) $4\,\Omega$

(B) $\frac{4}{3}\,\Omega$

(C) $8\,\Omega$

(D) $6\,\Omega$

133. A network is defined by $P = \mathrm{NOR}(A, B)$, $Q = \mathrm{NAND}(A, P)$ and $Y = \mathrm{OR}(Q, B)$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 0$ and $P = 0$?

(A) 3

(B) 1

(C) 0

(D) 2

Chemistry

134. A freshly assayed 40-volume H_2O_2 solution releases 40 L O_2 at STP per litre on complete decomposition. Only80.0% of the evolved oxygen reaches a burner that requires0.560 L O_2 at STP. The solution volume needed,in mL,is

(A) 7

(B) 35

(C) $\frac{35}{2}$

(D) 14

135. Which observation is consistent with the ordinary MgCO3→CaCO3→SrCO3→BaCO3 trend and does not confuse thermal stability with solubility?

(A) Thermal stability generally increases as cation polarising power decreases; this does not force every salt's solubility to increase.

(B) Both thermal stability and every salt's solubility must follow molar mass alone.

(C) Cation radii decrease down Group2.

(D) All four carbonate anions have different formal charges.

136. Which systematic name correctly describes the condensed connectivity CH3COCH2CH2OH, including principal-group priority and numbering?

(A) 4-oxobutan-1-ol

(B) 4-hydroxybutan-2-one

(C) butane-2,4-diol

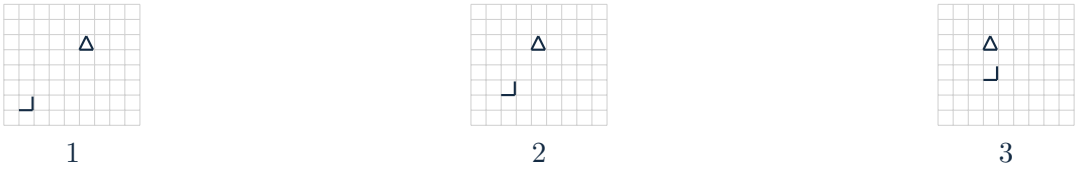
(D) 1-hydroxybutan-3-one

Logical Reasoning

137. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). The numerical component follows a constant arithmetic step. Find the missing internal term: XNN9, URO13, ?, OZQ21, LDR25, IHS29.

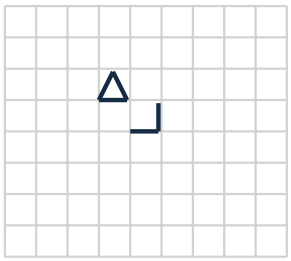
(A) SWQ17
(B) 71PVR
(C) RVP17
(D) QUO17

138. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.

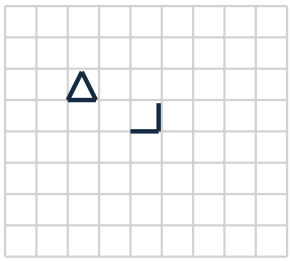


1 2 3

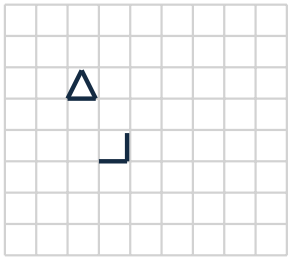
(A) Pattern V



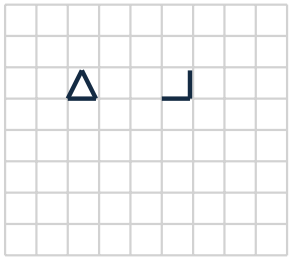
(B) Pattern N



(C) Pattern L

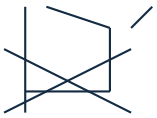


(D) Pattern W



139. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

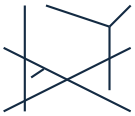




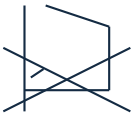
host 1



host 2



host 3



host 4

(A) 4
(B) 2
(C) 3
(D) 1

Mathematics

140. In how many ordered ways can three six-sided dice have sum 11, if the first die is known not to show 6?

(A) 46 (B) 23

(C) 26 (D) 27

141. For $x>0$, $(x^2y)'=0$ and $y(1)=3$. Find $y(3)\cdot 9$.

(A) 3 (B) 27

(C) $\frac{1}{3}$ (D) -3

142. Two points on opposite sides of a tower foot are 5 metres apart on a straight level line. The elevations of the tower top are 30° and 45°. Find the tower height.

(A) $\frac{5}{2}+\frac{5\sqrt{3}}{2}$ (B) $\frac{5\sqrt{3}}{6}$

(C) $-\frac{5}{2}+\frac{5\sqrt{3}}{2}$ (D) $\frac{5}{4}$

BITSAT Mock 24

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A satellite of mass m is moved from a circular orbit of radius R to one of radius 14R about mass M. Find the required increase in total orbital energy in units of $GMm/(2R)$.

(A) $\frac{13}{14}$ (B) $\frac{195}{196}$

(C) $\frac{15}{14}$ (D) 13

2. A stationary nucleus of mass number 216 emits an alpha particle whose kinetic energy is 6 MeV. Taking daughter and alpha masses proportional to $A-4$ and 4, find the total energy release Q, including daughter recoil.

(A) $\frac{324}{53}$ MeV (B) 6 MeV

(C) $\frac{6\sqrt{53}}{53} + 6$ MeV (D) 324 MeV

3. A coil of inductance 2 H carries current $i(t) = 1t^2$ A. Find the magnitude of its self-induced emf at $t = 1$ s.

(A) 2 V (B) 8 V

(C) 4 V (D) 1 V

4. Two equal-intensity coherent beams have maximum combined intensity 28 units. Find intensity when their path difference is $\lambda/6$.

(A) 21 (B) 28

(C) 14 (D) $14\sqrt{3}$

5. A 1 kg block is launched at 6 m/s on top of a 3 kg block initially at rest on a smooth floor. The coefficient of kinetic friction between them is $\frac{1}{10}$. Find the distance slid by the upper block relative to the lower before slipping ceases. Take $g = 10$ m/s^2.

(A) $\frac{27}{2}$ m (B) $\frac{9}{2}$ m

(C) 54 m (D) $\frac{27}{8}$ m

6. A particle has displacement $x = 3\cos(2\pi t/5)$ cm. Find its signed velocity at $t = \frac{5}{6}$ s.

(A) $-\frac{3\sqrt{3}\pi}{5}$ cm/s (B) $-\frac{3\sqrt{3}}{10}$ cm/s

(C) $-\frac{6\pi}{5}$ cm/s (D) $\frac{3\sqrt{3}\pi}{5}$ cm/s

7. A 4 ohm resistor is connected in parallel with an unknown resistor across 16 V. The source supplies 96 W in total. Find the unknown resistance.

(A) $\frac{64}{23}\,\Omega$ (B) $\frac{8}{3}\,\Omega$

(C) $8\,\Omega$ (D) $\frac{20}{3}\,\Omega$

8. A beam has measured power 16×10^{-7} W. A detector registers 5×10^{12} photons per second and detects a fraction $\frac{1}{2}$ of the incident photons, without altering the beam. Find its wavelength in nm. Use $e = 1.6 \times 10^{-19}$ C and $hc = 1240$ eV nm.

(A) 155 nm (B) 620 nm

(C) 2480 nm (D) 1240 nm

9. A monatomic ideal gas expands reversibly and adiabatically until its pressure becomes 1/59049 of its initial value. Find final temperature divided by initial temperature.

(A) 81 (B) 1

(C) $\frac{1}{729}$ (D) $\frac{1}{81}$

10. A boat moves at 10 m/s relative to water. The current is 6 m/s and river width is 36 m. The boat lands directly opposite its starting point. Find crossing time.

(A) $\frac{18}{5}$ s (B) 6 s

(C) $\frac{9}{2}$ s (D) $\frac{9}{4}$ s

11. A source emitting 700 Hz and a listener approach each other, each at 20 m/s relative to still air. Sound speed is 300 m/s. Find heard frequency before they pass.

(A) $\frac{1225}{2}$ Hz (B) $\frac{2240}{3}$ Hz

(C) 800 Hz (D) 700 Hz

12. An insulated vessel containing one mole of an ideal gas of molar mass $\frac{3}{100}$ kg/mol moves at bulk speed 70 m/s and is brought to rest. All bulk kinetic energy becomes internal energy. The gas has 5 active degrees of freedom. Find temperature rise. Use $R = 8.314$ J/(mol K).

(A) $\frac{14700}{4157}$ K (B) $\frac{7350}{4157}$ K

(C) $\frac{29400}{4157}$ K (D) $\frac{9800}{4157}$ K

13. A particle of mass 4 kg and charge 4 C initially has speed 4 m/s. It moves through a uniform electric field 6 N/C with a net displacement 2 m along the field. A simultaneous magnetic field bends its trajectory. Neglect other forces. Find final speed.

(A) $2\sqrt{10}$ m/s (B) $\sqrt{22}$ m/s

(C) $2\sqrt{7}$ m/s (D) 4 m/s

14. A network is defined by $P = \text{NAND}(A, B)$, $Q = \text{NOR}(A, P)$ and $Y = \text{NAND}(Q, B)$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 1$ and $A = 1$?

(A) 1 (B) 2

(C) 3 (D) 0

15. A spring of constant 40 N/m is already stretched by $\frac{1}{5}$ m. How much additional work is required to increase its extension to $\frac{3}{10}$ m quasistatically?

(A) $\frac{1}{5}$ J (B) 1 J

(C) $\frac{9}{5}$ J (D) $\frac{6}{5}$ J

16. A mass 3 kg initially moves in the positive direction at 5 m/s. A force opposite to that direction rises linearly from zero to 8 N and then falls linearly to zero over a total duration 4 s. Find its signed final momentum.

(A) 7 kg m/s (B) -1 kg m/s

(C) 1 kg m/s (D) 15 kg m/s

17. A ray in a medium of index $\frac{11}{5}$ is incident on a medium of index 1. Find the sine of the critical-angle threshold for total internal reflection.

(A) $\frac{6}{11}$ (B) $\frac{5}{11}$

(C) $\frac{11}{5}$ (D) 1

18. A particle has $v(0) = 5$ m/s and acceleration $a(t) = 14t$ m/s^2 for $0 \le t \le 5$ s. Find its average velocity over this interval.

(A) $\frac{175}{3}$ m/s (B) 75 m/s

(C) $\frac{185}{2}$ m/s (D) $\frac{190}{3}$ m/s

19. A vacuum electromagnetic wave of average intensity 50 W/m^2 passes through a device that reduces its electric-field amplitude by a factor 5. The transmitted wave falls normally on an area 7 cm^2. What average power reaches the area?

(A) 14 W (B) $\frac{7}{5000}$ W

(C) 2 W (D) $\frac{7}{50}$ W

20. A conveyor receives 3 packets each second, each of mass $\frac{1}{2}$ kg and speed 5 m/s. It accelerates them to 11 m/s without changing elevation. Find the ideal mechanical power delivered to the packets.

(A) 27 W (B) $\frac{363}{4}$ W

(C) 9 W (D) 72 W

21. A series circuit contains resistor R, two ideal inductors L_1, L_2, and a fixed-frequency AC source of angular frequency 40 rad/s. With both inductors active, current lags by 45°; when L_2 is shorted, it lags by 30°. If $R = 70$ ohms, find L_2.

(A) $\frac{7\sqrt{3}}{12} + \frac{7}{4}$ H (B) $\frac{7}{4} - \frac{7\sqrt{3}}{12}$ H

(C) $2800 - \frac{2800\sqrt{3}}{3}$ H (D) $\frac{7}{4}$ H

22. A solid sphere rolls without slipping from rest through vertical height 5 m. A thin ring does so through vertical height 2 m. Find the ratio of their squared centre-of-mass speeds, sphere to ring.

(A) $\frac{25}{7}$ (B) 5

(C) $\frac{25}{14}$ (D) $\frac{7}{2}$

23. A liquid of volume 2 litres and bulk modulus 4000000000 Pa is subjected to an additional pressure of 7000000 Pa. Neglect changes in bulk modulus. Find the magnitude of volume decrease in cubic centimetres.

(A) $3500\,\mathrm{cm}^3$ (B) $\frac{7}{2000}\,\mathrm{cm}^3$

(C) $\frac{7}{2}\,\mathrm{cm}^3$ (D) $\frac{8000000}{7}\,\mathrm{cm}^3$

24. A hydrogen atom emits a photon in the $4 \to 3$ transition. A hydrogen-like ion of nuclear charge $Z = 3$ emits a photon of exactly the same wavelength and ends in level $n_f = 9$. What is its initial principal quantum number?

(A) 36 (B) 4

(C) $\frac{4}{3}$ (D) 12

25. A soap bubble initially of radius 2 cm is expanded quasistatically until its volume is 8 times the initial volume. Surface tension remains $\frac{1}{20}$ N/m. Find work against surface tension, in units of $\pi \times 10^{-4}$ J.

(A) $\frac{56}{5}$ (B) $\frac{12}{5}$

(C) $\frac{24}{5}$ (D) $\frac{8}{5}$

26. 2 kg of water at 35°C is mixed in an insulated vessel with $\frac{294}{167}$ kg of ice at 0°C. Use $c_w = 4200$ J/(kg K), $L_f = 334000$ J/kg. Find the mass of ice remaining at equilibrium.

(A) $\frac{92316}{80327}\,\mathrm{kg}$ (B) $\frac{294}{167}\,\mathrm{kg}$

(C) $\frac{441}{334}\,\mathrm{kg}$ (D) $\frac{147}{167}\,\mathrm{kg}$

27. Two identical uniform spheres touch. Their radii are multiplied by 4 while their density is divided by 4. By what factor does their gravitational attraction change?

(A) 64 (B) 16

(C) $\frac{1}{256}$ (D) $\frac{1}{16}$

28. A rod of length 7 m pivots at its lower end. A peg constrained to move horizontally stays in contact with its upper tip. At an instant the rod makes an angle with $\sin\theta = 3/5$ to the horizontal and the tip's horizontal speed has magnitude 4 m/s. Find the rod's angular-speed magnitude.

(A) $\frac{4}{7}\,\mathrm{rad/s}$ (B) $\frac{12}{35}\,\mathrm{rad/s}$

(C) $\frac{5}{7}\,\mathrm{rad/s}$ (D) $\frac{20}{21}\,\mathrm{rad/s}$

29. A coil of 4 turns and area 5 m^2 carries 4 A in a uniform 6 T field. Its plane makes 30° with the field. Find torque magnitude.

(A) $240\,\mathrm{N\,m}$ (B) $480\,\mathrm{N\,m}$

(C) $240\sqrt{3}\,\mathrm{N\,m}$ (D) $60\sqrt{3}\,\mathrm{N\,m}$

30. In an abrupt junction, the acceptor concentration is 196 times the donor concentration. Within the depletion approximation, which statement gives x_n/x_p and identifies the carriers mainly responsible for the small reverse current before breakdown?

(A) 14; majority carriers

(B) 196; minority carriers

(C) 1/196; minority carriers

(D) 1; majority carriers

Chemistry

31. In a pressure range where the Freundlich law $x/m = kp^{1/2}$ is valid at fixed temperature, adsorbent B has 8 times the mass of A and is exposed to 64 times its pressure. Find the ratio of total adsorbed amounts x_B/x_A.

(A) 1

(B) 512

(C) 8

(D) 64

32. The ions P^{3-}, S^{2-}, Cl^-, K^+ each have 18 electrons. Which decreasing-radius order and explanation is consistent?

(A) $S^{2-} > Cl^- > K^+ > P^{3-}$; positive charge always gives the largest ion.

(B) All radii are equal because electron numbers are equal.

(C) $P^{3-} > S^{2-} > Cl^- > K^+$; increasing nuclear charge contracts the same electron cloud.

(D) $P^{3-} < S^{2-} < Cl^- < K^+$; larger nuclear charge expands the cloud.

33. A $\frac{27}{125}$ g volatile sample displaces $\frac{896}{13}$ mL wet air at 280 K and 770 mmHg. Water-vapour pressure is 10 mmHg. Using STP=273 K,760 mmHg and22.4 L/mol,the sample molar mass, rounded to the nearest whole g/mol, is

(A) 70

(B) 72

(C) 73

(D) 142

34. Match two applications: I. an inert atmosphere for welding reactive hot metals. II. an orange-red advertising discharge tube. Which ordered gas pair is appropriate?

(A) argon; neon.

(B) neon; argon.

(C) argon; argon.

(D) xenon; neon.

35. A0.500 g compound containing only C,H,O has40.0% oxygen by mass. Complete combustion produces11/12 g CO2. What mass of H2O in g is produced? Use C12,H1,O16.

(A) 9/20

(B) 9/10

(C) -9/20

(D) 9/40

36. At298K,a cell obeys E=0.600+$(-\frac{59}{1000})\log_{10}$Q in volts. Reverse the reaction and double every coefficient;define Q' for that new written reaction. Which pair(new E°,new slope of E versus logQ')is correct?

(A) $-\frac{3}{5}$V;$\frac{59}{2000}$V per log unit.

(B) $\frac{3}{5}$V;$-\frac{59}{1000}$V per log unit.

(C) $-\frac{6}{5}$V;$-\frac{59}{2000}$V per log unit.

(D) $-\frac{3}{5}$V;$-\frac{59}{2000}$V per log unit.

37. Compare methoxide and trifluoroethoxide as oxygen nucleophiles towards the same unhindered methyl substrate in one fixed solvent. Ignore concentration differences. Which pairing of the stronger nucleophile and the main electronic reason is most appropriate?

(A) methoxide; it is weaker because a localised lone pair cannot attack

(B) methoxide; CF3 withdraws electron density inductively from the nucleophilic oxygen

(C) trifluoroethoxide; resonance always increases local charge on oxygen

(D) trifluoroethoxide; the more stabilised anion necessarily attacks faster

38. Compare p-bromobenzyl bromide at its benzylic C-Br or aryl C-Br bond toward the same cyanide nucleophile in a polar aprotic solvent under ordinary SN2 conditions. Which reacts faster or is the substituted site,and why?

(A) the benzylic C-Br bond; the mechanism requires formation of a free tertiary carbocation

(B) the benzylic C-Br bond; Only the CH2Br carbon provides the accessible sp3 backside-attack site under ordinary cyanide SN2 conditions.

(C) the aryl C-Br bond; an sp2 C-Br bond behaves just like an unhindered methyl bromide

(D) the aryl C-Br bond; steric hindrance accelerates backside attack

39. A1.00 L vessel initially contains $\frac{1}{10}$ mol A,$\frac{1}{5}$ mol B and $\frac{1}{50}$ mol C,with no D. For A+2B⇌2C+D, equilibrium C amount is $\frac{3}{50}$ mol. Numerical K_c using molar concentrations is

(A) $\frac{9}{64}$ (B) $\frac{9}{128}$

(C) $\frac{9}{256}$ (D) $\frac{3}{112}$

40. For2A(g)⇌A_2(g),the numerical K_c is1.00 at a T where RT=16 L bar/mol. Find numerical K_p for1/2 A_2(g)⇌A(g),using matching concentration and pressure conventions.

(A) 4 (B) $\frac{1}{4}$

(C) 16 (D) $\frac{1}{16}$

41. cyclooctene (C8H14) adds one equivalent of :CBr2 across its only C=C without rearrangement. Which pair gives the adduct formula and the number of independent rings? Count the original ring and the new three-membered ring separately.

(A) C8H14Br2; 1 (B) C9H14Br2; 2

(C) C9H16Br2; 2 (D) C9H14Br2; 1

42. A model solvent has Tb=400K,molar mass100g/mol and molar vaporisation enthalpy32000J/mol. UseR=8.00J $mol^{-1}K^{-1}$ and Kb=R Tb^2 M/(1000ΔH),M in g/mol. A $\frac{1}{20}$mol/kg A_2B solution is50% dissociated into3ions. Boiling temperature in K is

(A) $\frac{2002}{5}$ (B) $\frac{2001}{5}$

(C) $\frac{4001}{10}$ (D) $\frac{2003}{5}$

43. A saturated hot solution initially contains60 g solute in100 g water. During cooling, water is lost by evaporation and46 g solute crystallises. Cold solubility is20 g per100 g water. No solute is lost except as the collected crystals. What mass of water evaporated in g?

(A) -30 (B) 60

(C) 30 (D) 15

44. Under the usual qualitative group trends,compare the ease of alkali-metal electron loss in K/Rb and the oxidising tendency of Br_2/I_2. Which conclusion is consistent?

(A) Rb loses its electron more readily; I_2 is the stronger halogen oxidant.

(B) Both trends must increase down a group because atoms become larger.

(C) K loses its electron more readily; I_2 is the stronger halogen oxidant.

(D) Rb loses its outer electron more readily; Br_2 is the stronger halogen oxidant.

45. 2-Methylpropene undergoes1:BH3/THF,2:H2O2/OH-,then the isolated alcohol is oxidised with PCC. Which final organic product is obtained?

(A) 2-methylpropan-2-ol

(B) 2-methylpropanal

(C) propanone

(D) 2-methylpropanoic acid

46. heptane-2,6-dione,CH3CO(CH2)3COCH3 undergoes intramolecular aldol addition at low temperature;isolate before dehydration. Under the stated intramolecular closure pathway,which isolated product matches both ring size and work-up endpoint?

(A) 3-methylcyclohex-2-en-1-one

(B) 3-hydroxy-3-methylcyclohexan-1-one

(C) 3-hydroxy-3-methylcyclopentan-1-one

(D) cycloheptane-1,3-diol

47. Equal masses of nonvolatile A (M=180) and B (M=60) are separately dissolved in equal solvent masses. A remains molecular, while B only dimerises. The measured freezing-depression ratio ΔTf(B)/ ΔTf(A) is 27/10. What fraction of the original B formula units is present in dimers? All solutions obey ideal particle counting.

(A) 1/5

(B) -1/5

(C) 1/10

(D) 2/5

48. Aspirin undergoes complete hydrolysis of its acetyl ester, without decarboxylation. Which products and recovered group are consistent?

(A) p-Aminophenol and ammonia; a nitrile is formed.

(B) Benzene and carbon dioxide only.

(C) Glucose and fructose; a glycosidic bond is broken.

(D) Salicylic acid and ethanoic acid; a free phenolicOH is restored.

49. A cathode receives $\frac{3}{50}$mol electrons. Its only products are Cu from Cu^{2+} and $\frac{3}{500}$mol H_2. Cu M64. Deposited copper mass in g is

(A) $\frac{192}{125}$

(B) $\frac{96}{25}$

(C) $\frac{48}{25}$

(D) $\frac{216}{125}$

50. A carbon-containing organic sample contains N only among nitrogen, sulfur and halogens. It undergoes sodium fusion with sufficient sodium. Which conversion and subsequent test protocol is chemically consistent? For a silver-halide test, use a separately prepared portion boiled with nitric acid to remove cyanide/sulfide interference where present.

(A) cyanide; Prussian blue after the complete nitrogen-test sequence.

(B) sulfide; no further treatment is ever needed before adding silver nitrate.

(C) Unchanged covalent bonds; add silver nitrate directly to the untreated organic sample.

(D) Nitrate and sulfate in every case; all give the same Prussian-blue test.

51. Evaluate independently. I: O_2^+ has higher bond order than O_2. II: ClF_3 has three equivalent F positions in a trigonal plane.

(A) Only II is correct.

(B) Both I and II are correct.

(C) Only I is correct.

(D) Neither I nor II is correct.

52. For standard M2+/M half-cells, E°_B-E°_A=2/5 V and E°_C-E°_B=1/5 V. B has E°=1/10 V. Which pair gives the strongest reducing metal among A,B,C and E°cell for A(s)+C2+→A2++C(s)?

(A) C; 3/5 V

(B) A; -3/5 V

(C) A; 3/5 V

(D) B; 1/5 V

53. For the isolated oxoanion ClO2-, choose its chlorine oxidation state and molecular geometry about chlorine.

(A) +3; linear.

(B) +5; trigonal planar.

(C) +1; bent.

(D) +3; bent.

54. For oxide formation, both normalised per mole of O_2, $\Delta G_A^\circ = -310+0.10T$ and $\Delta G_B^\circ = -380+0.20T$ in kJ. Ignoring kinetic limitations, at what temperatures is B metal a favourable standard-state reducing agent for oxide A?

(A) T > 700 K

(B) Only T = 700 K

(C) T < 700 K

(D) At all positive temperatures

55. For isolated molecules of SF_4 and SF_6,which member has the larger permanent dipole moment,and which reasoning is sufficient?

(A) They must be equal because both contain polar covalent bonds.

(B) The one with more atoms always has the larger dipole,independent of geometry.

(C) SF_6; the number of polar bonds alone fixes the molecular dipole.

(D) SF_4; See-saw SF4 is asymmetric; the six octahedral S-F vectors of SF6 cancel.

56. A student predicts aqueous behaviour of Li using only gas-phase first ionisation energy. Which correction is justified?

(A) The aqueous potential follows ionisation energy after hydration terms cancel.

(B) Li metal hydration, rather than Li+ hydration, controls the aqueous cycle.

(C) Strong Li+ hydration offsets costly atomisation and ionisation in the aqueous cycle.

(D) Strong Li+ hydration raises the free energy needed to form aqueous Li+.

57. Compounds A and B have molecular formulae $(CH_2O)_6$ and $(CH_2O)_7$. Pure equal-mass samples are compared. Which joint statement is necessarily true,independently of functional-group identity?

(A) The samples contain equal total oxygen-atom amounts,but A contains $\frac{7}{6}$ times as many molecules as B.

(B) A contains $\frac{6}{7}$ times as many molecules as B,and has a different oxygen mass percentage.

(C) The oxygen-atom amounts are in ratio $\frac{6}{7}$,while molecule amounts are equal.

(D) Both samples contain equal molecule amounts because their empirical formulae agree.

58. Two oxides of X contain 12 g X combined respectively with 16 g O and 32 g O. Atomic masses are X=12, O=16. Which empirical-formula conclusion jointly follows from mole ratios and the law of multiple proportions?

(A) Both are XO because X masses are equal.

(B) First empirical formula XO; second empirical formula XO2; fixed-X oxygen mass ratio second:first=2.

(C) No formula can be inferred even though both atomic masses and both compositions are supplied.

(D) First oxide XO2; second oxide XO; fixed-X oxygen mass ratio second:first=1/2.

59. $\frac{9}{100}$ mol borax is acidified completely. Each borax formula unit gives four H_3BO_3 molecules;75.0% of the boric acid is isolated. With glycerol present, the isolated boric acid is titrated completely as a monobasic acid. What amount of NaOH, in mol, is required?

(A) $\frac{81}{400}$ (B) $\frac{27}{100}$

(C) $\frac{27}{200}$ (D) $\frac{27}{400}$

60. tert-butyl chloride is heated with potassium tert-butoxide under elimination-favouring conditions. Which major constitutional product is expected,despite the presence of an alkoxide that could be used in a Williamson synthesis with a suitable different halide?

(A) the alkane formed by replacing halogen with H (B) di-tert-butyl ether

(C) the corresponding tertiary alkoxide substitution product (D) 2-methylpropene

English Proficiency

61. She uses specialised hand skills to make individual wooden instruments rather than operating a mass-production line. Which word fits?

(A) technician (B) inventor

(C) artisan (D) merchant

62. Read the sentence: They added another exit to hasten the departure of the crowd. Choose the opposite in meaning of 'hasten' as used in the sentence.

(A) delay (B) complete

(C) ease (D) organise

63. Choose the correctly spelt word.

(A) indepandent (B) independant

(C) indipendent (D) independent

64. Complete the sentence: Although the task was tiring, the volunteers continued to work _____ until it was finished.

(A) hard (B) hardness

(C) hardly (D) harden

65. Read the sentence: The engineer described the cracked support as a grave concern requiring immediate action. Choose the closest meaning of 'grave' as used in the sentence.

(A) serious
(B) lasting
(C) certain
(D) unexpected

66. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. or its key terms must be explained Q. Either the notice must be rewritten R. so that a new reader can understand S. what action is required.

(A) P-Q-R-S
(B) Q-P-R-S
(C) S-P-R-Q
(D) R-Q-P-S

67. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. Did the survey team record the water level before the rain began?

(A) Is the water level recorded by the survey team before the rain begins?
(B) Was the water level recorded after the rain began?
(C) Was the rain recorded by the water level before the team began?
(D) Was the water level recorded by the survey team before the rain began?

68. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The engineer is to test the revised model, not the earlier one.

(A) The revised model is testing the engineer.
(B) The revised model, not the earlier one, is to be tested by the engineer.
(C) The earlier model, not the revised one, is to be tested by the engineer.
(D) Both models are to be tested by the engineer.

69. She understands how the newcomer experiences the situation, rather than merely feeling sorry for him. Which noun fits?

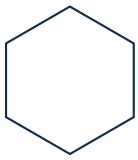
(A) pity
(B) imitation
(C) empathy
(D) agreement

70. Read the sentence: The agile player changed direction quickly without losing balance. Choose the opposite in meaning of 'agile' as used in the sentence.

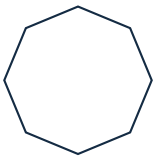
(A) swift
(B) clumsy
(C) energetic
(D) watchful

Logical Reasoning

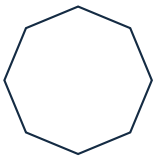
71. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

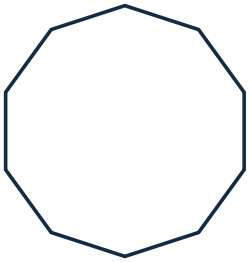


B

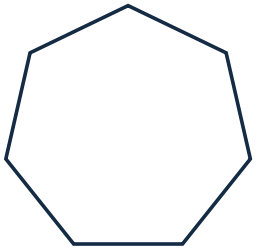


C

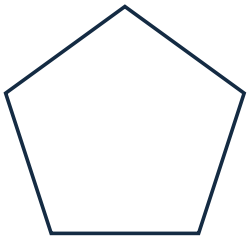
(A) Pattern M



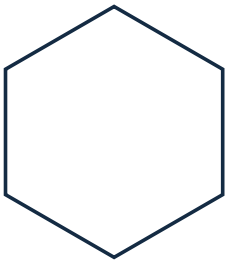
(B) Pattern Y



(C) Pattern Q



(D) Pattern T



72. Consider only these statements; universal statements alone do not imply existence. All farmers are inventors; No inventors are farmers. Conclusions: I. All farmers are designers. II. Some inventors are designers. Which conclusions follow necessarily?

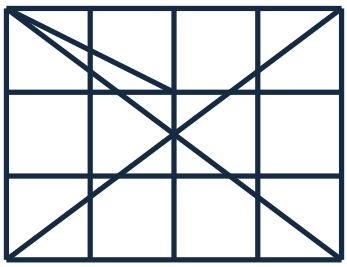
(A) Only I

(B) Both I and II

(C) Neither I nor II

(D) Only II

73. How many triangles are formed by all the drawn segments?



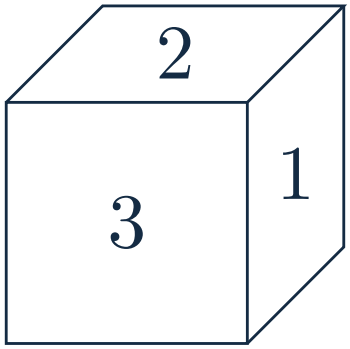
(A) 91

(B) 40

(C) 47

(D) 51

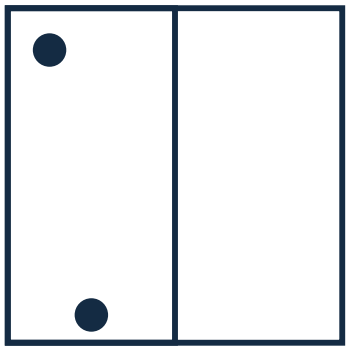
74. This standard die has opposite labels summing to seven. Roll it once to the right, so its present top face becomes the right face. Give the new (top, front) labels.



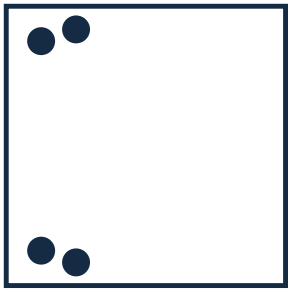
(A) $(1, 3)$ (B) $(6, 3)$

(C) $(6, 4)$ (D) $(2, 3)$

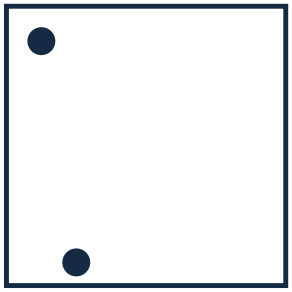
75. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



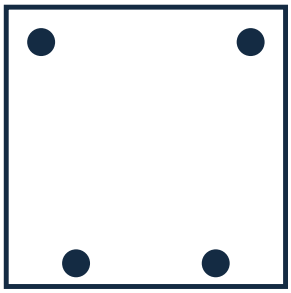
(A) Pattern Z



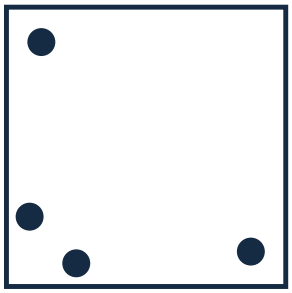
(B) Pattern K



(C) Pattern L



(D) Pattern N



76. Preserve the exact relation and its order: Player : Team :: ? : Swarm. Choose the missing term.

(A) Ship (B) Lion

(C) Bee (D) Wolf

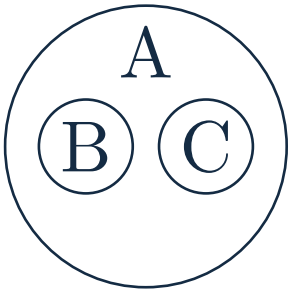
77. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 6, 24, ?, 128, 136, 544, 552.

(A) 32 (B) 16

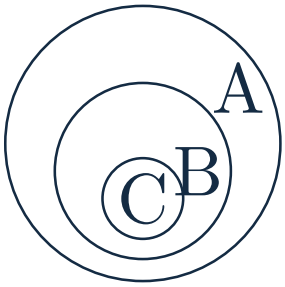
(C) 128 (D) 96

78. Choose the diagram that represents all these conditions, including the strictness information: All B and all C are A; no B is C. B and C are nonempty, and some A belongs to neither.

(A) Pattern K



(B) Pattern W



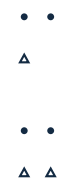
(C) Pattern U



(D) Pattern N



79. The increase from the left cell to the right cell is the same in each row, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.





(A) Pattern V

(B) Pattern T





(C) Pattern K

(D) Pattern X





80. Two people differ in age by 6 years. The product of their ages 5 years ago was 475. What will the younger person's age be 6 years from now?

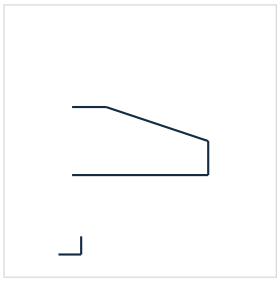
(A) 30

(B) 24

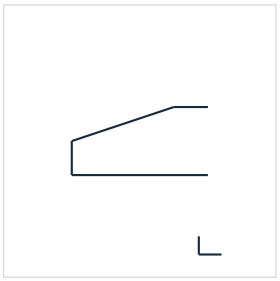
(C) 25

(D) 36

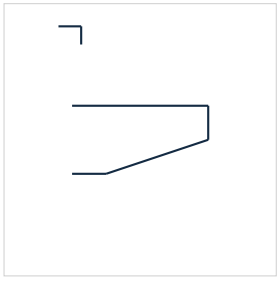
81. Select the missing bottom-right tile. Tiles reflect across the shared horizontal and vertical centre lines; there is no turning.



top left



top right



bottom left

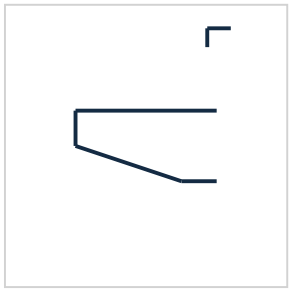
?

bottom right

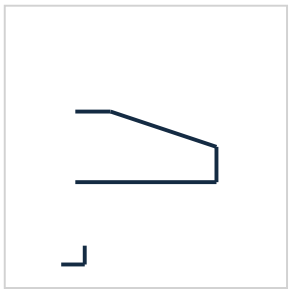
(A) Pattern L



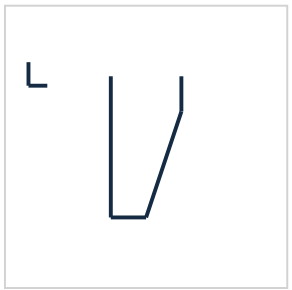
(B) Pattern X



(C) Pattern Y



(D) Pattern K



82. In a row, Asha is 10th from the left and 19th from the right. Bina is 12th from the right. Charu stands 2 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 5

(B) 8

(C) 3

(D) 4

83. Tara points to a photograph and says, "This person is the son of the only daughter of my father's only daughter." Tara is female, and the photographed person is male. How is the photographed person related to Tara?

(A) Son
(B) Brother
(C) Daughter
(D) Grandson

84. Three pairs contain exactly one mammal and one bird. Which pair does not?

(A) Cat / Penguin
(B) Bat / Dog
(C) Elephant / Eagle
(D) Whale / Sparrow

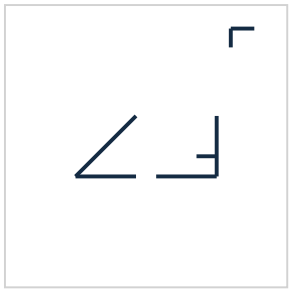
85. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



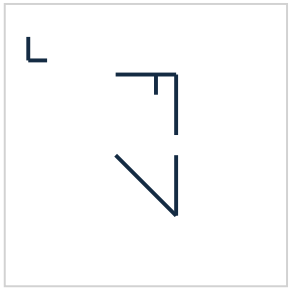
(A) Pattern V



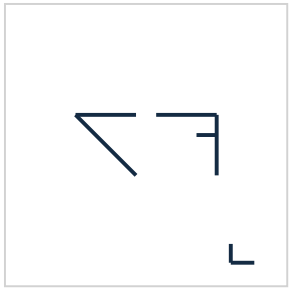
(B) Pattern Z



(C) Pattern X



(D) Pattern K

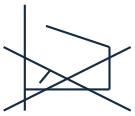


86. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 11$ and $5 \mapsto 20$. Which input maps to 44?

(A) 12
(B) 14
(C) 13
(D) 44

87. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





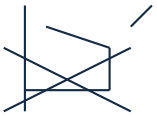
host 1



host 2



host 3

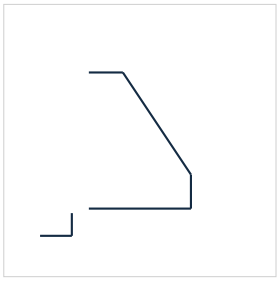


host 4

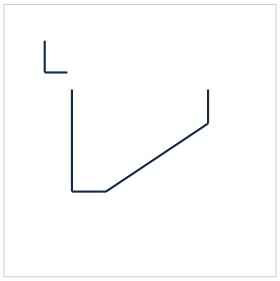
(A) 4 (B) 3

(C) 1 (D) 2

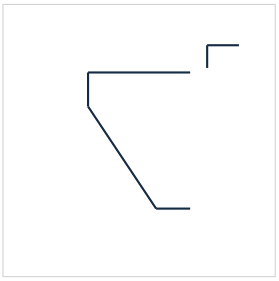
88. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

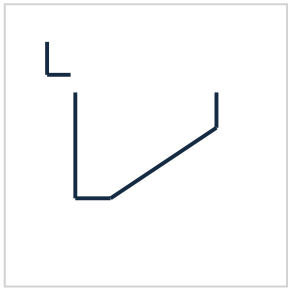


2



3

(A) Pattern N



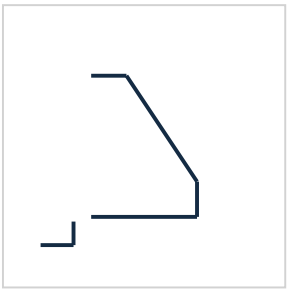
(B) Pattern X



(C) Pattern V



(D) Pattern P



89. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, TRAIN is coded as XVEMR. Which word is coded as KVETI?

(A) OZIXM (B) GRAPE

(C) HSBQF (D) FQZOD

90. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: CQTZ, GTWD, KWZH, ?, SCFP, WFIT.

(A) OZCL (B) LCZO

(C) NYBK (D) PADM

Mathematics

91. An observer at height 3 m sees the tower foot at depression 30° and its top at elevation 60°. The observer descends vertically to ground level. If the new elevation is α, find tanα.

(A) 12 (B) $\frac{4\sqrt{3}}{3}$

(C) $\frac{\sqrt{3}}{3}$ (D) $\frac{2\sqrt{3}}{3}$

92. The tangent to y=x^2+5x at P has slope 13. Find the slope of the normal at P.

(A) −169 (B) $\frac{1}{13}$

(C) $-\frac{1}{13}$ (D) −13

93. Let y′=(x+y)/(x+y+1), with y(0)=1. Find the positive value of x at whichx+y=9.

(A) $\log\left(\frac{\sqrt{57}}{3}\right)+8$ (B) $\frac{\log\left(\frac{19}{3}\right)}{4}+4$

(C) $\log\left(\frac{\sqrt{57}}{3}\right)+4$ (D) $\log(9)$

94. For integer n, consider: “If 30 divides n, then 6 divides n.” Which is its contrapositive?

(A) If 30 does not divide n, then 6 does not divide n. (B) If 6 divides n, then 30 divides n.

(C) If 6 does not divide n, then 30 does not divide n. (D) 30 divides n and 6 does not divide n.

95. An AP has a=8,d=5; its first n terms sum to 123. Find a_(n+1).

(A) 33 (B) 38

(C) 6 (D) $\frac{41}{2}$

96. $A=\begin{bmatrix}3 & 1\\ 1 & 4\end{bmatrix}$ and A^{-1}=pI+qA. Find p+q.

(A) $\frac{6}{11}$ (B) $-\frac{6}{11}$

(C) $\frac{8}{11}$ (D) $\frac{1}{11}$

97. A convergent GP has first term 7, ratio $\frac{1}{4}$. Find the sum of all terms after its first 7 terms.

(A) $\frac{38227}{4096}$ (B) $\frac{7}{16384}$

(C) $\frac{7}{12288}$ (D) $\frac{7}{3072}$

98. Let R=[sin($\frac{5\pi}{66}$)+sin($\frac{17\pi}{66}$)]/[cos($\frac{5\pi}{66}$)+cos($\frac{17\pi}{66}$)]. Find (1-R^2)/(1+R^2).

(A) $-\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{1}{2}$ (D) $\frac{3}{2}$

99. Two distinguishable fair dice show X,Y. Given X+Y≥7, find the probability that X≠Y and XY is even.

(A) $\frac{16}{21}$ (B) $\frac{5}{21}$

(C) $\frac{8}{9}$ (D) $\frac{6}{7}$

100. Evaluate Σ $j/2^j$ for j=1,...,9.

(A) $\frac{1013}{512}$ (B) 2

(C) $\frac{507}{256}$ (D) $\frac{9}{512}$

101. The curve $y=x^3+2x$ divides the disc $x^2+y^2 \leq 4$. Find the area of the part satisfying $y \geq x^3+2x$.

(A) 2π (B) $2+2\pi$

(C) 4π (D) π

102. Find the product of all real solutions of sqrt(13-log_6|x|)=log_6|x|-1.

(A) −1679616 (B) 1679616

(C) 0 (D) −1296

103. Let f(x)=sin x+6 cos x. Let [L,L+π] be the interval containing 0 in its interior on which f is increasing and bijective onto[-sqrt(37),sqrt(37)]. If g is its inverse, findL+$g'(1)$.

(A) $-\frac{\pi}{2} - \operatorname{atan}(6) - \frac{1}{6}$

(B) $-\frac{\pi}{2} - \operatorname{atan}(6) + \frac{1}{6}$

(C) $-\operatorname{atan}(6) + \frac{1}{6} + \frac{\pi}{2}$

(D) $-\frac{\pi}{2} + \frac{1}{6} + \operatorname{atan}(6)$

104. Evaluate the integral from 0 to π/2 of [(sin x+cos x)/(sec x+csc x)](sin x+6 cos x) dx, using continuous extension at the endpoints.

(A) 2 (B) $\frac{7}{3}$

(C) $-\frac{5}{3}$ (D) 7

105. z=7+6i. Find 1/z+1/conjugate(z).

(A) 14 (B) $\frac{12}{85}$

(C) $\frac{14}{85}$ (D) $\frac{1}{14}$

106. f(x)=(x^2-25)/(x-5) for x≠5, and f(5)=k. If f is differentiable at 5, find k+$f'(5)$.

(A) 10 (B) 12

(C) 11 (D) 20

107. The line r=(5,2,3)+t(1,2,2) and plane 2x-y=13 are given. Find the squared minimum distance between them.

(A) $\sqrt{5}$ (B) 5

(C) 0 (D) 25

108. F′(x)=$\frac{1}{x^2}$ − $\frac{1}{x^2\sqrt{x^2+1}}$ and F(1)=4. Find F(6)-2F(1).

(A) $-\sqrt{2}+\frac{\sqrt{37}}{6}+\frac{29}{6}$ (B) $-\sqrt{2}+\frac{5}{6}+\frac{\sqrt{37}}{6}$

(C) $-\frac{29}{6}-\frac{\sqrt{37}}{6}+\sqrt{2}$ (D) $-\frac{19}{6}-\sqrt{2}+\frac{\sqrt{37}}{6}$

109. In $(3x^2-x^{-1})^9$, find the ratio of the coefficient of x^3 to that of x^6.

(A) −3 (B) 1

(C) $-\frac{1}{3}$ (D) $\frac{1}{3}$

110. a=(7,0,0), b=(0,5,0), c=(0,0,2), p=a+b. Evaluate [p×(b×c)]·(a+c).

(A) −100 (B) 0

(C) 100 (D) −1325

111. Let $\alpha=\pi/10$, $\beta=\pi/9$. Evaluate arccos(-cosαcosβ-sinαsinβ).

(A) $-\frac{\pi}{90}$ (B) $\frac{\pi}{90}$

(C) $\frac{91\pi}{90}$ (D) $\frac{89\pi}{90}$

112. A=(7,6), B=(7,-6); P lies on $x^2+y^2=169$ with PA=PB and P_x>0. Find PA^2.

(A) 436 (B) 36

(C) 169 (D) 72

113. For the roots α,β of $z^2-(1+11i)z+(-60+11i)=0$, find $|\alpha-\beta|^2$.

(A) 244 (B) 3721

(C) 61 (D) 122

114. In how many ordered ways can three distinguishable fair dice have sum 11, show at least one 6, and have all three faces different?

(A) 15 (B) 18

(C) 2 (D) 12

115. F′(x)=$(x^2+5x+7)\,e^x$ and F(0)=3. Find F(1)-2F(0).

(A) $-7+8e$ (B) $1-8e$

(C) $-1+8e$ (D) $-4+8e$

116. α,β are roots of x^2-14x+40=0. The monic polynomial with roots α+1,β+1 is x^2-Ax+B. Find B.

(A) 55 (B) 41

(C) 27 (D) 256

117. Find the area between y=(x-2)(x-8) and the x-axis for $0 \le x \le 12$.

(A) -48 (B) 48

(C) 120 (D) 60

118. The normal of slope 3 to $y^2=20x$ touches at P. Find the distance from P to the focus.

(A) 40 (B) 2500

(C) 45 (D) 50

119. The line 3x+4y=25 cuts $x^2+y^2=100$ in chord AB. Find $|AB|^2$.

(A) 225 (B) 300

(C) 400 (D) $10\sqrt{3}$

120. How many solutions in $0 \le x < 12\pi$ satisfy sin2x=sin x and cosx>0?

(A) 12 (B) 19

(C) 24 (D) 18

121. Vectors a,b satisfy $|a|^2=37$, $|b|^2=5$, $a \cdot b=8$. Find $|2a-b|^2$.

(A) 121 (B) 185

(C) 63 (D) 153

122. A chord of $x^2+y^2=3600$ touches $x^2/36-y^2/4=1$. Its nonzero midpoint (h,k) satisfies h=k. Find h^2+k^2.

(A) 32 (B) 16

(C) 20 (D) 8

123. How many integers x are in the real domain of $\sqrt{x-3}\,\log(8-x)$?

(A) 4 (B) 8

(C) 6 (D) 5

124. The homogeneous system 2x+y=0, kx+5y=0 has a nonzero solution. Find $k+x^2/y^2$.

(A) 10 (B) $\frac{39}{4}$

(C) $\frac{41}{4}$ (D) 14

125. y>0 satisfies $2yy'+10y^2=0$, y(0)=sqrt(4). Find $y^2(1)$.

(A) $4e^{10}$ (B) $\frac{2}{e^5}$

(C) $\frac{4}{e^5}$ (D) $\frac{4}{e^{10}}$

126. A positive integer-domain function satisfies f(m+n)=f(m)f(n), f(1)=2. Find $\sum_{j=1}^{7} f(j)$.

(A) 128 (B) 254

(C) 127 (D) 896

127. A=(5,0,3), B=(0,5,1), AP:PB=2:1 internally. Find $|AP|^2$.

(A) 6 (B) 54

(C) 36 (D) 24

128. Box A contains 6 red and 3 blue balls; B contains 4 red and 5 blue balls. A fair coin chooses one box. Two independent draws with replacement from that box give exactly one red ball. Find the posterior probability that A was chosen.

(A) $\frac{10}{19}$ (B) $\frac{2}{9}$

(C) $\frac{9}{19}$ (D) $\frac{3}{5}$

129. How many solutions in $0 \le x < 6\pi$ satisfy $\sin^4 x + \cos^4 x = 3/4$ and $\sin x > 0$?

(A) 12 (B) 13

(C) 24 (D) 11

130. Using digits 0,1,2,3,4,5,6,7 without repetition, how many 4-digit positive integers are divisible by 3 and have their first digit smaller than their last digit?

(A) 209 (B) 210

(C) 498 (D) 214

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A 5 kg block rests on a 5 kg block on a smooth horizontal floor. A force 50 N pulls the lower block. There is no slipping between the blocks. During a displacement 2 m, what work does friction do on the upper block?

(A) 50 J
(B) 100 J
(C) −50 J
(D) 0 J

132. A stationary sonar source emits 400 Hz in a medium of sound speed 300 m/s. A flat target recedes directly away at 15 m/s and reflects the sound back. Find the echo frequency at the source. Use the two successive classical Doppler shifts.

(A) $\frac{8400}{19}$ Hz
(B) $\frac{8000}{21}$ Hz
(C) $\frac{7600}{21}$ Hz
(D) 380 Hz

133. Two coherent beams yield maximum-to-minimum interference intensity ratio $\frac{49}{36}$. Find the larger individual beam intensity divided by the smaller.

(A) $\frac{1}{169}$
(B) 169
(C) 196
(D) 170

Chemistry

134. A student argues that heating a dilute gas should lower its viscosity because the same is observed for a molecular liquid. What is the decisive flaw in this inference?

(A) Gas viscosity is dominated by momentum transfer, whereas liquid viscosity is strongly controlled by cohesive interactions.

(B) The viscosity of a dilute gas is always zero.

(C) Heating always strengthens intermolecular attractive forces in both phases.

(D) Every gas changes into a liquid as it is heated.

135. The internuclear axis is y. Two p_z orbitals overlap sideways. Their lobes facing each other above the axis have opposite phases; the lower facing lobes have the corresponding opposite signs. Which MO is produced,and what nodal property follows?

(A) A delta MO,because two perpendicular p orbitals always create two nodal planes.

(B) A pi bonding MO; the facing-lobe phase test has been reversed.

(C) A pi antibonding MO; there is an additional node between nuclei besides the plane containing the axis.

(D) A sigma bonding MO,because every overlap along a line joining nuclei is sigma.

136. A closed system goes from state1to2 by path A with q=900 J and work on the system w=-100 J. It returns from2to1 along a different path with q=-1100 J. IUPAC work on the return path,in J,is

(A) 100 (B) 300

(C) −300 (D) 800

Logical Reasoning

137. Each letter position has its own fixed cyclic shift, applied identically in both pairs (Z wraps to A). Complete CMV : AOY :: ZGB : ?.

(A) WHD (B) XIE

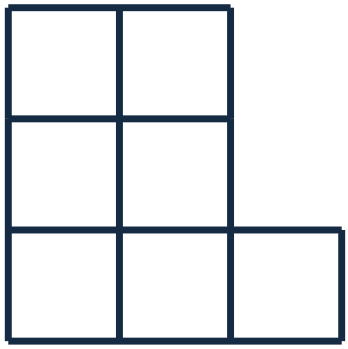
(C) YJF (D) ZIC

138. Three pairs contain exactly one string instrument and one wind instrument. Which pair does not?

(A) Harp / Cello (B) Ukulele / Flute

(C) Guitar / Oboe (D) Violin / Clarinet

139. How many rectangles, including squares are formed by all the drawn segments?



(A) 20 (B) 22

(C) 21 (D) 12

Mathematics

140. If α,β are roots of $x^2 - 2kx + k^2 - 9 = 0$ and k∈[-5,5], find the maximum of $\alpha^2+\beta^2$.

(A) 82 (B) 68

(C) 18 (D) 43

141. A line through P=(7,2) has direction (3,4). It meets 3x+4y=129 at Q. Find $|PQ|^2$.

(A) 100 (B) 16

(C) 400 (D) 20

142. In triangle ABC, A=2B, a/b=$\sqrt{3}$ and b=5. Find its area.

(A) $\frac{25\sqrt{3}}{2}$ (B) $\frac{25}{2}$

(C) $25\sqrt{3}$ (D) 10

BITSAT Mock 25

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A stretched string is divided by a bridge into lengths in the ratio $1:3$. Tension and linear density are unchanged. The original fundamental is 20 Hz. Find the fundamental of the longer segment.

(A) $\frac{80}{9}$ Hz (B) $\frac{20}{3}$ Hz

(C) $\frac{80}{3}$ Hz (D) 80 Hz

2. A wire consists of a semicircular arc of radius 8 m carrying 4 A, joined to two straight radial segments that extend outward from its ends. Find the magnetic field magnitude at the arc centre in units of μ_0.

(A) $\frac{1}{4\pi}$ (B) $\frac{1}{8}$

(C) $\frac{1}{16}$ (D) $\frac{1}{8\pi}$

3. An active transistor has $\alpha = \frac{49}{50}$. Its collector resistor is 1 kΩ and the measured collector-resistor drop is 6 V. Neglect leakage. What base current in mA is required?

(A) $\frac{300}{49}$ mA (B) $\frac{147}{25}$ mA

(C) $\frac{6}{49}$ mA (D) $\frac{3}{25}$ mA

4. A positive particle of charge 1 C starts with nonzero momentum along a uniform electric field 2 N/C. After 5 s it enters a perpendicular magnetic field of 4 T and follows a circle of radius $\frac{15}{4}$ m. Find its initial de Broglie wavelength in units of Planck constant h. Use $p_f = qBr$ and electric impulse qEt.

(A) $\frac{1}{15}$ (B) $\frac{1}{5}$

(C) $\frac{1}{10}$ (D) $\frac{1}{25}$

5. A cell balances on a potentiometer at 100 cm on open circuit and at 20 cm when a 6 ohm resistor is connected across it. The potentiometer gradient is unchanged. Find the cell's internal resistance.

(A) 36 Ω (B) 24 Ω

(C) 30 Ω (D) $\frac{24}{5}$ Ω

6. A series RC circuit is driven at fixed rms voltage. Multiplying angular frequency by 6 multiplies rms current by $\frac{6\sqrt{29}}{29}$. Find R^2/X_C^2, where X_C is the initial capacitive reactance.

(A) 36 (B) $\frac{23}{7}$

(C) $\frac{1}{4}$ (D) 4

7. A body is dropped from rest at height 120 m above the ground. Find its speed when it is still 30 m above the ground. Neglect air resistance and take $g = 10\ \text{m/s}^2$.

(A) $30\sqrt{2}\,\text{m/s}$ (B) $60\sqrt{5}\,\text{m/s}$

(C) $1800\,\text{m/s}$ (D) $20\sqrt{6}\,\text{m/s}$

8. A Carnot engine initially runs between 700 K and 400 K. Its hot reservoir is raised by 60 K. By how much must the cold-reservoir temperature rise to keep efficiency unchanged?

(A) $\frac{240}{7}\,\text{K}$ (B) $60\,\text{K}$

(C) $105\,\text{K}$ (D) $\frac{600}{19}\,\text{K}$

9. A series RLC circuit is resonant at angular frequency ω_0. Its capacitance is then multiplied by 169 while L stays fixed. What angular frequency restores resonance, as a fraction of ω_0?

(A) $\frac{1}{169}$ (B) $\frac{1}{13}$

(C) 1 (D) 13

10. Two identical capacitors each 5 microfarads are connected in series across an ideal 60 V battery. One capacitor undergoes permanent conducting breakdown, becoming a short circuit. Find the additional charge drawn from the battery after the new steady state is reached.

(A) $150\,\mu\text{C}$ (B) $450\,\mu\text{C}$

(C) $100\,\mu\text{C}$ (D) $75\,\mu\text{C}$

11. A mass 4 kg is attached to two massless springs in series with spring constants 30 N/m and 60 N/m. Find the small-oscillation period on a smooth horizontal surface.

(A) $\frac{2\sqrt{10}\pi}{5}\,\text{s}$ (B) $\frac{2\sqrt{5}\pi}{5}\,\text{s}$

(C) $\frac{4\sqrt{5}\pi}{15}\,\text{s}$ (D) $\frac{\sqrt{5}}{5}\,\text{s}$

12. A stationary nucleus with mass number 32 and binding energy per nucleon 8 MeV splits into two equal daughters, each having total binding energy 132.0 MeV. Which statement is correct?

(A) The process releases 8 MeV; the daughters have larger binding energy per nucleon.

(B) The process releases 16 MeV; each daughter releases 8 MeV.

(C) The process absorbs 8 MeV; the daughters have larger binding energy per nucleon.

(D) No energy can be released because fission always requires a larger mass number than 32.

13. A gas receives 400 J of heat while its surroundings do 100 J of work on it. It then loses 200 J of heat at fixed volume. Find its total internal-energy change.

(A) 200 J (B) 700 J

(C) 500 J (D) 300 J

14. A bar magnet of moment 4 A m^2 is rotated quasistatically from 60° to 120° relative to a uniform 7 T magnetic field. Find external work.

(A) 28 J (B) 14 J

(C) $28\sqrt{3}$ J (D) $\frac{28\pi}{3}$ J

15. A uniform disc of mass 2 kg and radius 3 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) $\frac{117}{16}$ kg m^2 (B) 9 kg m^2

(C) $\frac{171}{16}$ kg m^2 (D) $\frac{27}{4}$ kg m^2

16. A drop of mass 6.72×10^{-17} kg is suspended in a vertical field of magnitude 700 N/C. It has excess electrons. One excess electron is removed without changing its mass. Find its immediate downward acceleration, taking $g = 10$ m/s^2 and $e = 1.6 \times 10^{-19}$ C.

(A) $\frac{10}{7}$ m/s^2 (B) 2 m/s^2

(C) 10 m/s^2 (D) $\frac{5}{3}$ m/s^2

17. A uniform-density spherical planet has radius R and surface gravitational acceleration g. At what depth below the surface is gravity $g/14$? Give the depth as a fraction of R.

(A) $\frac{195}{196}$ (B) $\frac{1}{14}$

(C) $\frac{13}{14}$ (D) 1

18. For a circular-aperture telescope, observation wavelength is multiplied by 3. Its smallest resolvable angle must be reduced by a factor 4. By what factor must the objective diameter increase?

(A) $2\sqrt{3}$ (B) 12

(C) $\frac{4}{3}$ (D) 4

19. A fixed amount of ideal gas has its absolute temperature multiplied by 5 and volume by 3. Find the final pressure divided by the initial pressure.

(A) $\frac{5}{3}$ (B) $\frac{1}{3}$

(C) $\frac{3}{5}$ (D) $\frac{5}{9}$

20. Water flows steadily from a pipe section of area A at 4 m/s to a section of area $A/4$ located 2 m higher. Neglect viscosity. Find $p_1 - p_2$. Take density 1000 kg/m^3 and $g = 10$ m/s^2.

(A) 120000 Pa (B) 148000 Pa

(C) 140000 Pa (D) 100000 Pa

21. A travelling wave is $y = 2\sin[8(x - \frac{15}{4}t) + \pi/4]$ mm with x in metres and t in seconds. What is its frequency?

(A) $\frac{15}{8\pi}$ Hz (B) $\frac{15}{\pi}$ Hz

(C) 30 Hz (D) $\frac{120}{\pi}$ Hz

22. A cyclist rides straight outward at 4 m/s for 4 s, waits 2 s, then returns to the start. The whole journey must last 8 s. What constant return speed is required?

(A) 4 m/s (B) 2 m/s

(C) $\frac{8}{3}$ m/s (D) 8 m/s

23. A mass 1 kg moves with $\vec{r}(t) = 2t\hat{i} + 6t^2\hat{j}$ m. Find the z-component of angular momentum about the origin at $t = 1$ s.

(A) 12 kg m^2/s (B) 36 kg m^2/s

(C) 6 kg m^2/s (D) 24 kg m^2/s

24. A 6 V Zener regulator feeds a constant-current load of 4 mA. Supply varies from 14 to 16 V. Regulation requires Zener current at least 2 mA, and its safe maximum is 11 mA. What is the largest permissible series resistance? Assume ideal regulation and give resistance in Ω.

(A) $\frac{2000}{3}$ Ω (B) $\frac{1600}{3}$ Ω

(C) $\frac{7000}{3}$ Ω (D) $\frac{4000}{3}$ Ω

25. An alpha particle begins far from a fixed nucleus of atomic number 29 with kinetic energy 12 MeV. Before its head-on approach, a retarding electric potential removes 6 MeV of kinetic energy. Find its distance of closest approach in fm, assuming only Coulomb repulsion near the nucleus. Use $ke^2 = 1.44$ MeV fm.

(A) $\frac{174}{25}$ fm (B) $\frac{348}{25}$ fm

(C) $\frac{116}{25}$ fm (D) $\frac{1392}{25}$ fm

26. Two rods of equal length and cross-sectional area have thermal conductivities k and 14k. They are joined end-to-end with no contact resistance. Find the effective conductivity of the composite divided by k.

(A) $\frac{2}{15}$ (B) $\frac{15}{2}$

(C) $\frac{14}{15}$ (D) $\frac{28}{15}$

27. A particle of mass 5 kg starts from rest and experiences $F(t) = 4t$ N along its motion for 4 s. Find the work done by the force.

(A) $\frac{128}{5}$ J (B) 128 J

(C) $\frac{1024}{5}$ J (D) $\frac{512}{5}$ J

28. A ball is dropped from 10 m onto a fixed horizontal floor. Its coefficient of restitution is 1/2. Find its height above the floor at time $\frac{\sqrt{2}}{4}$ s after the first impact. Take $g = 10$ m/s^2.

(A) $\frac{5}{4}$ m (B) $\frac{25}{8}$ m

(C) $\frac{15}{8}$ m (D) $\frac{5}{2}$ m

29. A thin glass lens of refractive index 3/2 has focal length 200 cm in air. It is completely immersed in a liquid of index 4/3. Find its new focal length.

(A) 600 cm (B) 200 cm

(C) 800 cm (D) $\frac{800}{3}$ cm

30. A car negotiates a level circular turn at its maximum non-slipping speed u when $\mu_s = \frac{6}{5}$. Rain lowers the coefficient to $\frac{3}{10}$ and the turn radius stays fixed. Find the new maximum speed divided by u.

(A) 1
(B) 4
(C) $\frac{1}{2}$
(D) $\frac{3}{10}$

Chemistry

31. With one equivalent of cold concentrated HI and isolation before prolonged further reaction,which initial cleavage pair is expected from methyl tert-butyl ether?

(A) tert-butyl iodide and methanol
(B) iodobenzene and methanol
(C) methyl iodide and tert-butanol only
(D) ethyl iodide and propan-1-ol

32. For $[Cr(H_2O)_6]^{3+}$,use the usual ligand-field/VBT assignments at room temperature. Which complete assignment is consistent with charge balance and magnetic behaviour?

(A) d^3,tetrahedral,sp^3,spin-only moment $\sqrt{15}$ BM.
(B) d^4,octahedral,d^2sp^3,spin-only moment $\sqrt{15}$ BM.
(C) d^3,octahedral,d^2sp^3,spin-only moment $2\sqrt{6}$ BM.
(D) d^3,octahedral,d^2sp^3,spin-only moment $\sqrt{15}$ BM.

33. An open-chain ketohexose has CH2OH at C1, aC2 carbonyl and three stereogenic centres at C3-C5. Which count and configurational criterion are correct?

(A) 16 stereoisomers; C2 determinesD/L.
(B) 8 stereoisomers; C5 determinesD/L.
(C) 8 stereoisomers; optical sign alone definesD/L.
(D) 4 stereoisomers; the carbonyl oxygen determinesD/L.

34. Benzene in large excess is alkylated under strongly acidic conditions using butan-1-ol as the alkyl source. Assume a free-carbocation Friedel-Crafts pathway with hydride rearrangement before ring attack. Which major monoalkylbenzene and explanation are consistent with those explicit conditions?

(A) sec-Butylbenzene; the initially primary butyl cation rearranges to a secondary cation.
(B) n-Butylbenzene; no shift is possible in a four-carbon chain.
(C) Ethylbenzene; dehydration necessarily removes two carbons.
(D) tert-Butylbenzene; every four-carbon cation becomes tertiary without changing the skeleton.

35. A closed system goes from state1to2 by path A with q=700 J and work on the system w=-300 J. It returns from2to1 along a different path with q=-900 J. IUPAC work on the return path,in J,is

(A) 400
(B) 300
(C) 500
(D) -500

36. Water hardness is due only to dissolved $Mg(HCO_3)_2$ and $CaCl_2$. Which treatment statement correctly distinguishes the two contributions?

(A) Boiling alone necessarily removes every dissolved sulfate and chloride quantitatively.
(B) Boiling treats temporary hardness; washing soda can remove the remaining calcium ions.
(C) Adding NaCl precipitates both calcium and magnesium because all chlorides are insoluble.
(D) Filtration of the clear untreated solution removes dissolved hardness ions without any precipitation.

37. Nonrelativistic ions A and B start from rest and gain only electric kinetic energy. B has 2 times A's mass and 1 times its positive charge. To make λ_B/λ_A=1/2, what must V_B/V_A be?

(A) 3 (B) 2

(C) 4 (D) 1/2

38. For two van der Waals gases,(Tc/Pc)A is5times(Tc/Pc)B. UsingTc=8a/(27Rb),Pc=a/(27b^2),Vc=3b andTB=a/(Rb),which pair(Vc,A/Vc,B;TB,A/Tc,A)is correct?

(A) (25;27/8). (B) (5;8/27).

(C) (1/5;27/8). (D) (5;27/8).

39. For a reaction over300-600 K, assume temperature-independent ΔH°. The measured natural logarithm ln[K(600)/K(300)] is -10/3. Using ln(K2/K1)=-(ΔH°/R)(1/T2-1/T1), find ΔH°/R in K, including sign.

(A) 2000 (B) -4000

(C) -2000 (D) -1000

40. A model oxide has composition MO7/6, where the subscript denotes O atoms per M atom. Oxygen is entirely O2- and metal sites are formally only M2+ or M3+. There are no other defects or ions. What fraction of M sites is M3+?

(A) 1/6 (B) -1/3

(C) 1/3 (D) 2/3

41. For a simple closed system of fixed composition,$dG = VdP - SdT$. Which pair correctly identifies the two partial derivatives,including the held variables?

(A) $(\partial G/\partial P)_T = S$ and$(\partial G/\partial T)_P = V$. (B) $(\partial G/\partial P)_T = -V$ and$(\partial G/\partial T)_P = S$.

(C) $(\partial G/\partial P)_T = V$ and$(\partial G/\partial T)_P = -S$. (D) Both derivatives equal zero because G is a state function.

42. In an antifluorite-derived crystal, X2- occupies all FCC anion sites. A measured fraction 1/5 of occupied M cations is M2+ and the rest M+. Only tetrahedral cation vacancies compensate charge; no interstitials or anion defects occur. What fraction of all tetrahedral sites is vacant?

(A) 1/10 (B) 5/6

(C) 1/12 (D) 1/6

43. A solution initially contains $\frac{3}{500}$ mol Fe^{2+}. Before titration, $\frac{1}{5}$ of it is oxidised quantitatively to Fe^{3+}. The volume in mL of0.0200 M $KMnO_4$ needed for the remaining Fe^{2+} in acid is

(A) 48 (B) 60

(C) 12 (D) 240

44. In the stated high-pressure model Z=1+b p/(RT), gas A at 325 K and gas B at 460 K have Z-versus-p slopes in ratio 46/39. Their molar covolumes b do not change with temperature in this model. At what temperature must gas A be measured for its slope to equal that of B at 460 K?

(A) 325 (B) 1150/3

(C) 460 (D) 552

45. Compare ortho and para hydrogen. Which statement preserves the distinction between mass/nuclear effects and electronic structure?

(A) They differ in relative proton nuclear-spin orientation,not in electron spin bonding configuration.

(B) An isotope with more neutrons must have a different number of protons and belong to another element.

(C) Every physical property and reaction rate is identical because isotopes have the same atomic number.

(D) Changing H to D adds an electron and therefore changes the oxidation state of every bonded atom.

46. acetophenone,C6H5COCH3 reacts with HCN. Which product correctly reflects nucleophilic addition or addition-elimination at its carbonyl rather than a different reaction class?

(A) C6H5COOH

(B) C6H5-C(OH)(CN)-CH3

(C) C6H5CH2CH2OH

(D) C6H5CH2CN

47. A Hinsberg separation is applied to ethylamine. Which reagent identity and class-dependent outcome are correct?

(A) benzenesulphonic acid,C6H5SO3H;all classes give the same neutral product

(B) benzoyl chloride,C6H5COCl;a sulphonamide with N-H,soluble in alkali as its anion

(C) benzenesulphonyl chloride,C6H5SO2Cl;every amine gives an alkali-soluble N-H sulphonamide

(D) benzenesulphonyl chloride,C6H5SO2Cl;a sulphonamide with N-H,soluble in alkali as its anion

48. For isolated molecules of ClF_3 and BF_3,which member has the larger permanent dipole moment,and which reasoning is sufficient?

(A) ClF_3; T-shaped ClF3 retains a nonzero resultant,whereas equal planar BF3 vectors cancel.

(B) BF_3; the number of polar bonds alone fixes the molecular dipole.

(C) The one with more atoms always has the larger dipole,independent of geometry.

(D) They must be equal because both contain polar covalent bonds.

49. A laboratory note states: Interstitial carbides may be hard and retain metallic conductivity. Which judgement and reason are consistent?

(A) Incorrect because transition metals cannot exchange electrons or adsorb reactants.

(B) Correct because every transition-metal species has the same d-electron count.

(C) Incorrect: Small atoms occupy metal-lattice interstices; the compound need not behave as a molecular covalent solid.

(D) Correct: Small atoms occupy metal-lattice interstices; the compound need not behave as a molecular covalent solid.

50. benzaldehyde is subjected to concentrated aqueous NaOH. Which product or product pair follows the named reaction and side-chain oxidation state?

(A) benzyl alcohol only

(B) benzoin only

(C) benzyl alcohol and sodium benzoate

(D) benzene and sodium formate

51. Construct one open linear peptide using each member of the following multiset exactly once: Pro, Gly, Ala, Val, Ser. The C-terminal residue is the only achiral amino acid. Pro must occupy the N terminus, and Ala must not be immediately next to Pro. Count distinctN-to-C sequences; ignore stereoisomerism of the supplied residues.

(A) 6 (B) 4

(C) 120 (D) 24

52. At fixedT,V,A(g)→P(g)+Q(g)+R(g)is first order,and an inert gas is present. Initial totalpressure is$\frac{7}{10}$bar;complete-decomposition pressure is$\frac{3}{2}$bar. At two half-lives,totalpressure in bar is

(A) $\frac{3}{2}$ (B) $\frac{7}{5}$

(C) $\frac{13}{10}$ (D) 1

53. Acidic MnO4- oxidises SO2 to HSO4- and is reduced to Mn2+. The balanced reaction is 2MnO4- +5SO2+2H2O+H+→2Mn2++5HSO4-. Initially 17/1000 mol H+ and excess SO2 are available. After exactly 3/100 mol MnO4- is consumed, what amount of net H+ remains? Count net H+ from the displayed balanced equation.

(A) 1/1000 (B) 1/500

(C) 1/250 (D) -1/500

54. Atoms form ideal touching FCC spheres of radius r and cell edge a. Fractional coordinates are measured in a. Using the straight displacement between the two specified points (1/4,1/4,1/4) and (3/4,1/4,1/4), without replacing it by another periodic image, find (distance/r)^2. These points describe adjacent tetrahedral sites.

(A) 2 (B) 3

(C) 1 (D) 4

55. In a hydrogen-like ion Z=3,an electron falls from n=6 to an s orbital with 3 radial nodes. With hydrogen ionisation energy13.6 eV and$hc = 1240$ eV nm,the emitted wavelength in nm is

(A) $\frac{1240}{17}$ (B) $\frac{4960}{17}$

(C) $\frac{6200}{17}$ (D) $\frac{44640}{17}$

56. A weak monoprotic acid hasC=0.0100M andΛ0=400S cm^2/mol. Solute-only conductivity is $\frac{1}{12500}$S/cm. Useα=Λm/Λ0 andexactKa=Cα^2/(1-α). ItsKa is

(A) $\frac{1}{50}$ (B) $\frac{1}{250000}$

(C) $\frac{1}{245000}$ (D) $\frac{1}{490000}$

57. $\frac{1}{10}$ mol borax is acidified completely. Each borax formula unit gives four H_3BO_3 molecules;75.0% of the boric acid is isolated. With glycerol present, the isolated boric acid is titrated completely as a monobasic acid. What amount of NaOH, in mol, is required?

(A) $\frac{3}{40}$ (B) $\frac{3}{20}$

(C) $\frac{8}{15}$ (D) $\frac{3}{10}$

58. NH3 and PH3 are compared. Which pair of claims is consistent with the anomalous first Group15 member?

(A) NH3 has stronger conventional intermolecular hydrogen bonding and the higher normal boiling point.

(B) NH3 has no lone pair and cannot interact with a proton.

(C) PH3 forms the stronger conventional hydrogen-bond network and therefore boils higher.

(D) Their boiling points must agree because both have three H atoms.

59. A cured phenol-formaldehyde article does not soften into a freely flowing melt on reheating. Which structural explanation fits?

(A) Every chain is a free glucose monomer.

(B) It is a linear thermoplastic because curing removes all covalent bonds.

(C) It consists only of isolated ethene molecules.

(D) A cross-linked network restricts chain flow.

60. At fixed temperature a single adsorbent obeys x/m=k $p^{(1/n)}$. In experiment B the adsorbent mass is 4 times that in A and pressure is 16 times that in A. Total adsorbed amount is 8 times that in A. What is 1/n?

(A) 1/4

(B) 4

(C) 5/4

(D) 1/8

English Proficiency

61. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. but also on the resources needed Q. depends not only on the initial purchase R. Whether the proposal is affordable S. to maintain it over time.

(A) R-Q-P-S

(B) S-Q-P-R

(C) Q-R-P-S

(D) P-R-Q-S

62. The pattern is regular, with intervals following a consistent rule. Choose its opposite.

(A) irregular

(B) regulated

(C) regular

(D) regulatory

63. Read the sentence: Her ingenious arrangement solved the storage problem with a simple original device. Choose the closest meaning of 'ingenious' as used in the sentence.

(A) merely generous

(B) innocently candid

(C) inventively clever

(D) entirely genuine

64. Read the sentence: The dark clouds suggested that a storm was imminent. Choose the opposite in meaning of 'imminent' as used in the sentence.

(A) likely

(B) remote in time

(C) serious

(D) near

65. Read the sentence: The stain was a transient effect of the treatment and disappeared within minutes. Choose the closest meaning of 'transient' as used in the sentence.

(A) short-lived
(B) unexpected
(C) recurring
(D) lasting

66. She records the spoken proceedings rapidly rather than translating them into another language. What is she?

(A) commentator
(B) translator
(C) stenographer
(D) interpreter

67. Choose the meaning-preserving equivalent that retains the tense, participants and all stated restrictions. If you do not keep the receipt, you cannot claim the refund.

(A) You may claim the refund whether or not you keep the receipt.
(B) You cannot keep the receipt unless you claim the refund.
(C) If you keep the receipt, you will certainly receive the refund.
(D) Unless you keep the receipt, you cannot claim the refund.

68. Use the simple present to express the stated regular procedure. Choose the best replacement: The printer in the two offices [produce] a test page whenever it is restarted.

(A) have produced
(B) were producing
(C) are producing
(D) produces

69. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. The two figures then answered different questions instead of being treated as interchangeable. B. She separated moving time from total elapsed time. C. She later discovered that the recorded interval included a long stop at a crossing. D. The runner recorded the time taken on a new route.

(A) The two figures then answered different questions instead of being treated as interchangeable.

(B) The runner recorded the time taken on a new route.

(C) She separated moving time from total elapsed time.

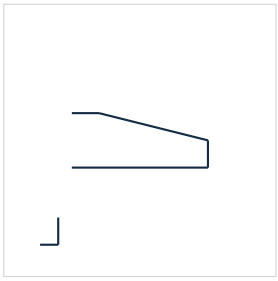
(D) She later discovered that the recorded interval included a long stop at a crossing.

70. The assessor applies the same standards without favouring a friend. Which noun names this quality?

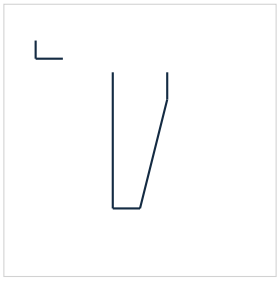
(A) ignorance
(B) impartiality
(C) indifference
(D) indecision

Logical Reasoning

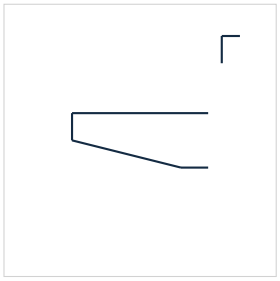
71. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

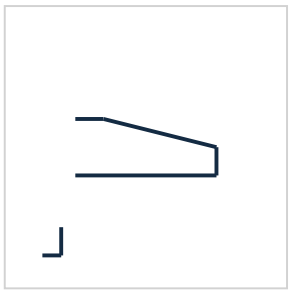


2

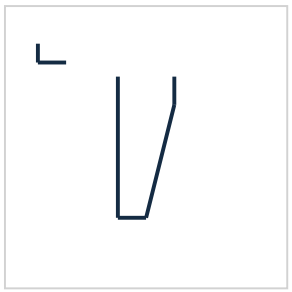


3

(A) Pattern T



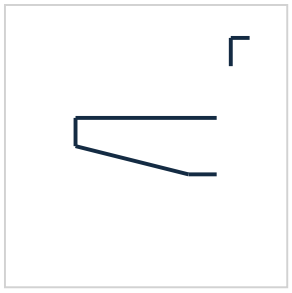
(B) Pattern R



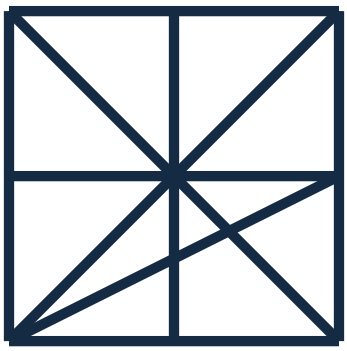
(C) Pattern N



(D) Pattern K

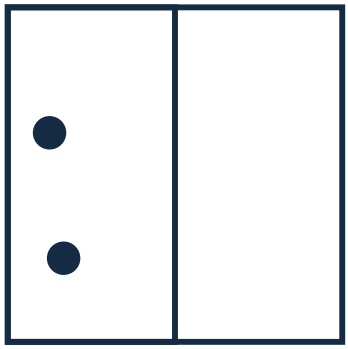


72. How many triangles are formed by all the drawn segments?

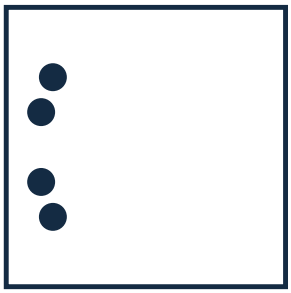


(A) 25
(B) 28
(C) 50
(D) 29

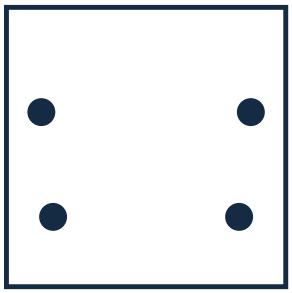
73. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



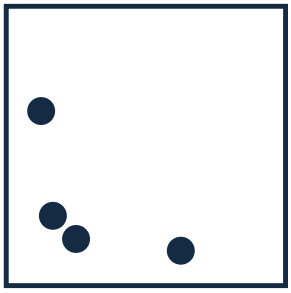
(A) Pattern M



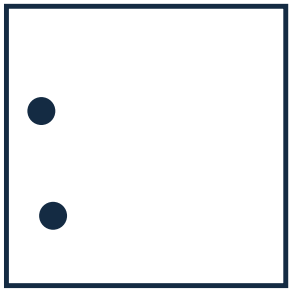
(B) Pattern W



(C) Pattern T



(D) Pattern Q



74. A person makes these turns in order: 180 degrees, 90 degrees left, 90 degrees right, 90 degrees left. The final heading is west. What was the heading immediately after turn 3?

(A) South

(B) East

(C) West

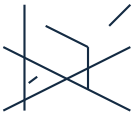
(D) North

75. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





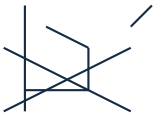
host 1



host 2



host 3



host 4

(A) 1
(B) 3
(C) 2
(D) 4

76. In this code, @ means +, # means ×, & means ÷, % means −. After substituting meanings, evaluate ((3#6)@(4&3)); printed parentheses retain their usual grouping.

(A) $\frac{-58}{3}$
(B) $\frac{55}{3}$
(C) $\frac{58}{3}$
(D) $\frac{61}{3}$

77. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 5, 10, 16, 32, ?, 76, 82.

(A) 38
(B) 48
(C) 58
(D) 64

78. Consider only these statements; universal statements alone do not imply existence. All designers are joggers; Some farmers are designers. Conclusions: I. No farmers are designers. II. Some designers are farmers. Which conclusions follow necessarily?

(A) Only II
(B) Both I and II
(C) Only I
(D) Neither I nor II

79. In a row, Asha is 7th from the left and 23rd from the right. Bina is 6th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 13
(B) 19
(C) 12
(D) 14

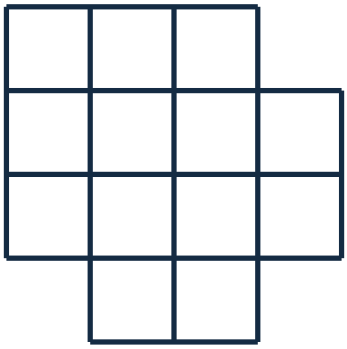
80. Consider only these statements; universal statements alone do not imply existence. All editors are hikers; Some editors are not inventors; Some inventors are not hikers. Conclusions: I. All hikers are inventors. II. Some editors are hikers. Which conclusions are individually possible without contradicting the statements?

(A) Only I
(B) Neither I nor II
(C) Only II
(D) Both I and II

81. Kiran points to a photograph and says, "This person is the son of my father's only daughter." Kiran is male, and the photographed person is male. How is the photographed person related to Kiran?

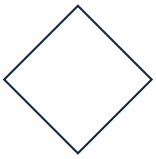
(A) Son
(B) Nephew
(C) Daughter
(D) Brother

82. How many rectangles, including squares are formed by all the drawn segments?



(A) 61
(B) 48
(C) 59
(D) 60

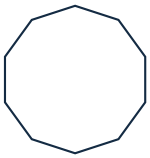
83. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

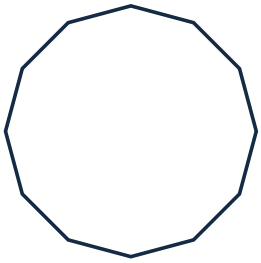


B

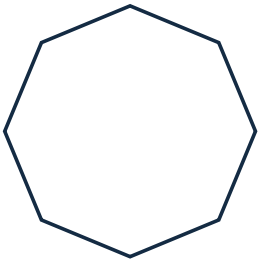


C

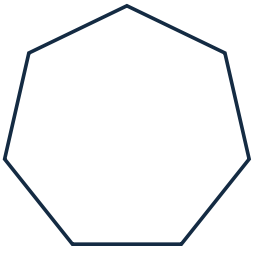
(A) Pattern M



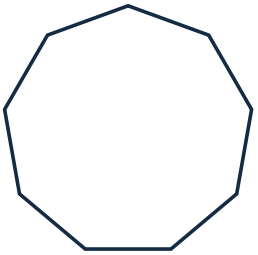
(B) Pattern P



(C) Pattern T



(D) Pattern X



84. Preserve the exact relation and its order: Silent : Quiet :: ? : Brave. Choose the missing term.

(A) Begin (B) Courageous

(C) Silent (D) Rapid

85. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

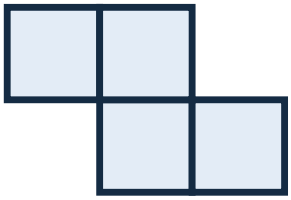


piece 1

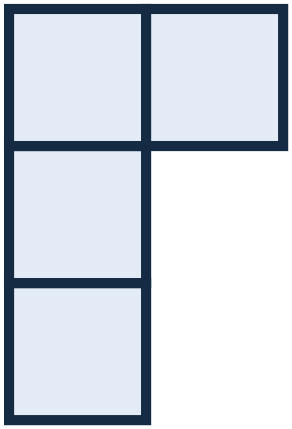


piece 2

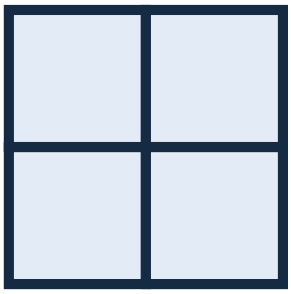
(A) Pattern L



(B) Pattern P



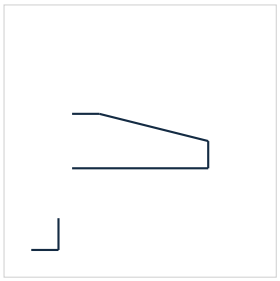
(C) Pattern R



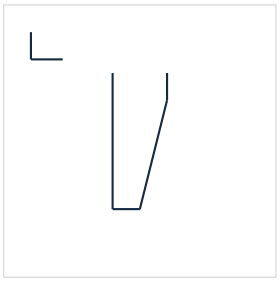
(D) Pattern Z



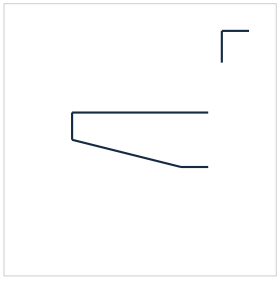
86. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right



bottom left

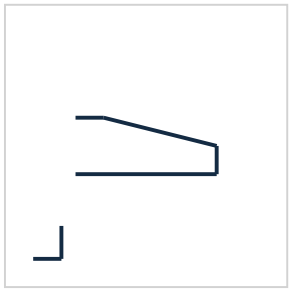
?

bottom right

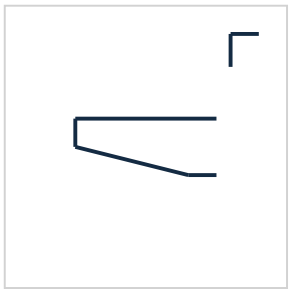
(A) Pattern X



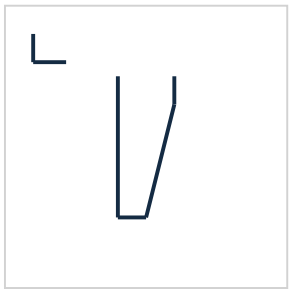
(B) Pattern L



(C) Pattern P



(D) Pattern W



87. Three pairs contain exactly one mammal and one bird. Which pair does not?

(A) Whale / Owl

(B) Rabbit / Sparrow

(C) Horse / Elephant

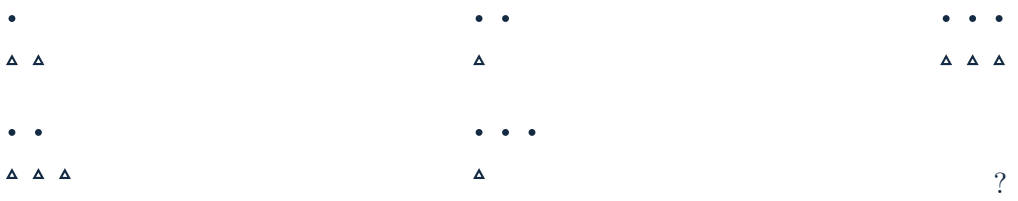
(D) Bat / Crow

88. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: GCJC, DFKE, ?, XLMI, UONK, RROM.

(A) BJMH (B) ZHKF

(C) GLIA (D) AILG

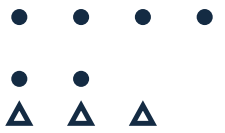
89. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern V (B) Pattern R



(C) Pattern T



(D) Pattern X



90. The recorded average of 9 readings is 36. One entry recorded as 43 should have been 41. After that correction, another reading of 20 is replaced by 28. What is the final average?

(A) $\frac{110}{3}$ (B) $\frac{113}{3}$

(C) $\frac{322}{9}$ (D) $\frac{332}{9}$

Mathematics

91. ABC is a right triangle on level ground with AB=8, AC=8, angle A=90°. A vertical tower stands at midpoint M of BC. Its top elevation from A is 60°. Find its height.

(A) $8\sqrt{6}$ (B) $4\sqrt{3}$

(C) $4\sqrt{6}$ (D) $4\sqrt{2}$

92. y′+3y=exp(-3x), y(0)=3. Find exp(3)y(1)-3.

(A) -1 (B) 1

(C) $\frac{4}{e^3}$ (D) 4

93. Evaluate lim as x→0 of $[(1+6x)/(1-2x)]^{(1/x)}$.

(A) e^8 (B) e^6

(C) 8 (D) e^4

94. A=(3,3,2), O=(0,0,0). Find the squared length of the orthogonal projection of OA on the line through O with direction (1,2,2).

(A) 169 (B) $\frac{169}{9}$

(C) $\frac{13}{3}$ (D) $\frac{29}{9}$

95. For the loop $y^2=x^2(5-x)$, find its area within the strip $5/3\leq x\leq 10/3$.

(A) $\frac{25\sqrt{30}}{27}+\frac{50\sqrt{15}}{27}$ (B) $-\frac{80\sqrt{15}}{27}+\frac{40\sqrt{30}}{9}$

(C) $\frac{40\sqrt{5}}{3}$ (D) $-\frac{20\sqrt{15}}{27}+\frac{40\sqrt{30}}{27}$

96. For $f(x)=x^3-3kx^2+75x$, how many integer k in [0,7] give a local maximum at a positive x?

(A) 2 (B) 7

(C) 3 (D) 1

97. Let y′=(x+y)/(x+y+1), with y(0)=1. Find the positive value of x at whichx+y=10.

(A) $\frac{9}{2}-\frac{\log(7)}{4}$ (B) $\frac{\log(7)}{4}+\frac{9}{2}$

(C) $\log(10)$ (D) $\frac{9}{2}$

98. The line y=mx through O bisects the area of the triangle with vertices O,(6,3),(4,4). Assume the vertices are noncollinear. Find m.

(A) $-\frac{1}{2}$ (B) $\frac{10}{7}$

(C) $\frac{7}{10}$ (D) $\frac{2}{3}$

99. A positive integer-domain function satisfies f(m+n)=f(m)f(n), f(1)=4. Find $\sum_{j=1}^{7} f(j)$.

(A) 114688 (B) 16384

(C) 21844 (D) 5461

100. If z=7-4i, find |exp(iz)|.

(A) e^4 (B) 1

(C) e^{-4} (D) e^7

101. Let $P(x)=(1+\sqrt{2}x)^7+(1-\sqrt{2}x)^7$. Find the sum of all nonconstant coefficients of P.

(A) 476 (B) 478

(C) 474 (D) 239

102. An AP has a_1=-30,d=5. How many of its first 11 terms are strictly positive?

(A) 8 (B) 4

(C) 7 (D) 3

103. $A = \begin{bmatrix} 0 & 5 & -5 \\ -5 & 0 & 5 \\ 5 & -5 & 0 \end{bmatrix}$. Find det(I+A)-(25+25+25).

(A) 1 (B) 75

(C) −1 (D) 0

104. For x,y>0 with sqrt(x)+sqrt(y)=9, find d^2y/dx^2 at x=25.

(A) $-\frac{9}{250}$ (B) $\frac{9}{125}$

(C) $\frac{9}{10}$ (D) $\frac{9}{250}$

105. α,β are roots of x^2-20x+91=0. The monic polynomial with roots α+1,β+1 is x^2-Ax+B. Find B.

(A) 111 (B) 92

(C) 112 (D) 8464

106. $F'(x)=\frac{6(x^2-1)(x^2+1)^5}{x^7}$ and F(1)=2. Find F(2)-2F(1).

(A) $\frac{11657}{64}$ (B) $\frac{11529}{64}$

(C) $-\frac{11657}{64}$ (D) $\frac{11401}{64}$

107. The parabola $(y-3)^2=28(x-4)$ has focus F and vertex V. A circle centred at F passes through V. Find the tangent length from P=(25,17).

(A) 343 (B) $14\sqrt{2}$

(C) $7\sqrt{3}$ (D) $7\sqrt{7}$

108. Let $R=[\sin(\frac{\pi}{15})+\sin(\frac{4\pi}{15})]/[\cos(\frac{\pi}{15})+\cos(\frac{4\pi}{15})]$. Find $(1-R^2)/(1+R^2)$.

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{3}{2}$

(C) $-\frac{1}{2}$ (D) $\frac{1}{2}$

109. On A={0,1,...,4}, define aRb if 2 divides a-b. How many ordered pairs are in R?

(A) 25 (B) 13

(C) 5 (D) 8

110. How many integer k in [0,30] make $x^2 - 8x + (15 - k) = 0$ have roots of opposite signs?

(A) 14 (B) 15

(C) 16 (D) 30

111. If α,β are roots of x^2-16x+55=0, find $\alpha^2+\beta^2$ without solving the quadratic.

(A) 36 (B) 110

(C) 146 (D) 256

112. $A = \begin{pmatrix} 4 & k \\ 0 & 2 \end{pmatrix}$ is symmetric. Find tr($A^T A$).

(A) 36 (B) 20

(C) 8 (D) 6

113. Two acute sun elevations $\theta_1<\theta_2$ satisfy $(16^6)^{sin^2\theta} + (16^6)^{cos^2\theta} = 2^6 + 2^18$. A vertical tower has height 4. Find the difference between its two shadow lengths.

(A) $\frac{8}{3}$ (B) $4\sqrt{3}$

(C) $\frac{16\sqrt{3}}{3}$ (D) $\frac{8\sqrt{3}}{3}$

114. Evaluate sin(2arctan 9).

(A) $\frac{9}{41}$ (B) $-\frac{9}{41}$

(C) $-\frac{9}{40}$ (D) $\frac{9\sqrt{82}}{82}$

115. Evaluate $\int_0^{(\pi/4)} \tan^5 x \sec^4 x \, dx$.

(A) $\frac{1}{8}$ (B) $\frac{1}{3}$

(C) $\frac{7}{24}$ (D) $\frac{1}{6}$

116. A convergent GP has first term 8, ratio $\frac{1}{4}$. Find the sum of all terms after its first 8 terms.

(A) $\frac{1}{4608}$ (B) $\frac{1}{1536}$

(C) $\frac{1}{49152}$ (D) $\frac{1}{6144}$

117. X takes values [-5, 0, 2] with probabilities 1/6,1/3,1/2. Let Y=5X+7. Find $E(Y^2)$-$[E(Y)]^2$.

(A) $\frac{5525}{36}$ (B) $\frac{1105}{36}$

(C) $\frac{7289}{36}$ (D) $\frac{925}{6}$

118. Circles $x^2+y^2=169$ and $(x-10)^2+y^2=169$ meet at A,B. Find $|AB|^2$.

(A) 276 (B) 476

(C) 576 (D) 256

119. $x=t^2$, $y=t^3$ for $t>0$. Find d^2y/dx^2 at t=4.

(A) $\frac{3}{16}$ (B) $\frac{3}{8}$

(C) $\frac{3}{2}$ (D) 12

120. Evaluate $\sum_{j=5}^{8} \binom{j}{3}$, with the convention $\binom{j}{r} = 0$ when j<r.

(A) 125 (B) 126

(C) 111 (D) 121

121. Find the squared distance between 6x+8y=18 and 3x+4y=15.

(A) $\frac{9}{25}$ (B) 36

(C) $\frac{36}{25}$ (D) $\frac{6}{5}$

122. A line has direction(t,1,k) and is parallel to both planes x-7 y+z=0 and 2 x+y-z=0. Find the squared length of its direction vector.

(A) 30 (B) 29

(C) $\frac{242}{9}$ (D) 1

123. Evaluate $\int_0^\infty dx/[(x^2+9)(x^2+49)]$.

(A) $\frac{\pi}{42}$ (B) $-\frac{\pi}{420}$

(C) $\frac{\pi}{210}$ (D) $\frac{\pi}{420}$

124. Evaluate $\Sigma\ j/2^j$ for j=1,...,3.

(A) $\frac{3}{2}$ (B) 2

(C) $\frac{11}{8}$ (D) $\frac{3}{8}$

125. For $49x^2+196y^2=9604$, find the distance between its two directrices.

(A) $14\sqrt{3}$ (B) 28

(C) $\frac{28\sqrt{3}}{3}$ (D) $\frac{56\sqrt{3}}{3}$

126. For $x\in[\pi/7,\pi/2]$, find the maximum of sin x/x.

(A) $\frac{2}{\pi}$ (B) $\frac{7\sin\left(\frac{\pi}{7}\right)}{\pi}$

(C) $\frac{\pi}{7\sin\left(\frac{\pi}{7}\right)}$ (D) $\sin\left(\frac{\pi}{7}\right)$

127. Three distinguishable fair dice are thrown. Given that exactly two faces are equal, find the probability that their sum is at least 13.

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{11}{30}$ (D) $\frac{5}{36}$

128. Evaluate the integral from 0 to 2π of $(7\text{-}4\sin x)(1+\cos x)/(49\text{-}16\sin^2 x)\,dx$.

(A) $-\frac{\log(11)}{4}+\frac{\log(7)}{4}+\frac{2\sqrt{33}\pi}{33}$

(B) $\frac{2\sqrt{33}\pi}{33}$

(C) $\frac{\sqrt{33}\pi}{66}$

(D) $-\frac{\log(7)}{4}+\frac{\log(11)}{4}+\frac{2\sqrt{33}\pi}{33}$

129. Evaluate $\sum_{j=0}^{3}\binom{3}{j}\binom{10}{3-j}$.

(A) 858 (B) 715

(C) 220 (D) 286

130. Maximize $Z=3x+4y$ subject to $x\geq 0$, $y\geq 7$, $x+y\leq 14$, $x+2y\leq 20$.

(A) 48 (B) 49

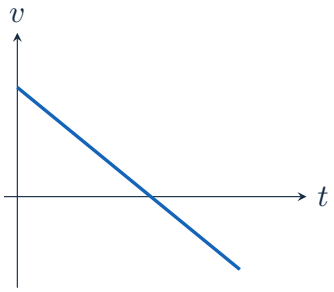
(C) 46 (D) 40

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A particle has velocity $v(0) = 4$ m/s, and constant acceleration makes $v(2) = -5$ m/s. What total distance does it travel in this interval?



(A) $\frac{82}{9}$ m
(B) $\frac{41}{9}$ m
(C) 10 m
(D) $\frac{25}{9}$ m

132. Current rises linearly from 0 to 5 A over 5 s and then falls linearly to 0 over another 5 s. It remains zero for the next 10 s. Find the average current over the entire 20 s interval.

(A) $\frac{5}{4}$ A
(B) $\frac{5}{8}$ A
(C) $\frac{5}{3}$ A
(D) 5 A

133. In an abrupt junction, the acceptor concentration is 784 times the donor concentration. Within the depletion approximation, which statement gives x_n/x_p and identifies the carriers mainly responsible for the small reverse current before breakdown?

(A) 28; majority carriers
(B) 1/784; minority carriers
(C) 784; minority carriers
(D) 1; majority carriers

Chemistry

134. In a pressure range where the Freundlich law $x/m = kp^{1/3}$ is valid at fixed temperature, adsorbent B has 9 times the mass of A and is exposed to 729 times its pressure. Find the ratio of total adsorbed amounts x_B/x_A.

(A) 81
(B) 729
(C) 6561
(D) 1

135. A freshly assayed 50-volume H_2O_2 solution releases 50 L O_2 at STP per litre on complete decomposition. Only80.0% of the evolved oxygen reaches a burner that requires0.560 L O_2 at STP. The solution volume needed,in mL,is

(A) $\frac{56}{5}$
(B) $\frac{28}{5}$
(C) 14
(D) 28

136. For benzaldehyde and formaldehyde,which structural-reactivity comparison is correct under ordinary enolate/aldol conditions?

(A) formaldehyde has an alpha carbon bearing two H
(B) both lack alpha H and can undergo Cannizzaro in suitable concentrated base
(C) both self-aldolise
(D) benzaldehyde alone self-aldolises

Logical Reasoning

137. The whole has 9 unit positions. Part A covers the first 4; part B covers 5 positions, overlapping A in exactly 1 positions as shown. How many positions remain uncovered by either part?


whole


part A positions


part B positions

(A) 1
(B) 2
(C) −1
(D) 0

138. Three numbered triples satisfy both $y = x^2 + 7$ and $z - y = 4$. Which triple fails at least one condition? 1: (4,23,27), 2: (14,203,207), 3: (6,43,48), 4: (12,151,155).

(A) 1
(B) 3
(C) 4
(D) 2

139. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, CLOUD is coded as FORXG. Which word is coded as WUDLQ?

(A) ZXGOT
(B) USBJO
(C) TRAIN
(D) SQZHM

Mathematics

140. An increasing positive GP has first three terms with product 216 and first-plus-third sum 15. Find its fourth term.

(A) 24
(B) 6
(C) $\frac{3}{2}$
(D) 12

141. Evaluate lim as n→ ∞ of (3/n)Σ|3 j/n-3/3|, j=1,...,n.

(A) $\frac{1}{2}$
(B) 2
(C) $\frac{5}{2}$
(D) $\frac{9}{2}$

142. How many solutions in $0 \le x < 3\pi$ satisfy $\sin^4 x + \cos^4 x = 3/4$ and $\sin x > 0$?

(A) 12
(B) 8
(C) 9
(D) 6

BITSAT Mock 26

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. Two large parallel plates are held at potentials +800 V and −800 V, with gap 5 mm. Neglect edge effects. Find field magnitude in the gap.

(A) 160000 V/m (B) 320 V/m

(C) 320000 V/m (D) 3200000 V/m

2. A nucleus of mass number 24 splits into two equal nuclei. The parent has binding energy per nucleon $\frac{73}{10}$ MeV and each daughter has $\frac{77}{10}$ MeV. Ignoring other products, what energy is released in the split?

(A) $\frac{972}{5}$ MeV (B) $\frac{48}{5}$ MeV

(C) $\frac{924}{5}$ MeV (D) $\frac{24}{5}$ MeV

3. A boat moves at 30 m/s relative to water. The current is 18 m/s and river width is 84 m. The boat lands directly opposite its starting point. Find crossing time.

(A) $\frac{7}{4}$ s (B) $\frac{14}{3}$ s

(C) $\frac{7}{2}$ s (D) $\frac{35}{8}$ s

4. A lossless coating of refractive index $\frac{5}{2}$ lies between air and a substrate of still higher index. For normally incident vacuum wavelength 700 nm, find the smallest positive film thickness giving destructive interference between its two reflected rays. Assume reflections at both boundaries each gain a phase reversal.

(A) 210 nm (B) 28 nm

(C) 70 nm (D) 140 nm

5. A solid sphere rolls without slipping from rest through vertical height 5 m. A thin ring does so through vertical height 9 m. Find the ratio of their squared centre-of-mass speeds, sphere to ring.

(A) $\frac{25}{63}$ (B) $\frac{5}{9}$

(C) $\frac{50}{63}$ (D) $\frac{63}{50}$

6. A monatomic ideal gas expands reversibly and adiabatically until its pressure becomes 1/7776 of its initial value. Find final temperature divided by initial temperature.

(A) $\frac{1}{6}$ (B) 1

(C) $\frac{1}{7776}$ (D) $\frac{1}{36}$

7. Light is incident from index $\frac{11}{10}$ onto index $\frac{21}{10}$. At one incidence angle the reflected beam is completely plane polarised. Find the tangent of that angle.

(A) $\frac{21}{10}$ (B) $\frac{21}{11}$

(C) $\frac{11}{10}$ (D) $\frac{10}{11}$

8. A mass 4 kg initially moves in the positive direction at 4 m/s. A force opposite to that direction rises linearly from zero to 6 N and then falls linearly to zero over a total duration 6 s. Find its signed final momentum.

(A) 18 kg m/s (B) 2 kg m/s

(C) −2 kg m/s (D) 16 kg m/s

9. A uniform block of density 700 kg/m^3 floats in a liquid. Then the liquid density is reduced to $\frac{3}{4}$ times 1000 kg/m^3, while the block's volume and density are unchanged. What fraction of the block is now submerged?

(A) $\frac{7}{10}$ (B) $\frac{14}{15}$

(C) 1 (D) $\frac{3}{4}$

10. A stretched string is divided by a bridge into lengths in the ratio 1 : 3. Tension and linear density are unchanged. The original fundamental is 40 Hz. Find the fundamental of the longer segment.

(A) $\frac{160}{3}$ Hz (B) $\frac{160}{9}$ Hz

(C) 40 Hz (D) 160 Hz

11. A small-angle pendulum has period T in a stationary lift. In a vertically accelerating lift its period is 13T. Find the downward lift acceleration divided by g.

(A) $\frac{1}{13}$ (B) $\frac{144}{169}$

(C) $\frac{1}{169}$ (D) $\frac{168}{169}$

12. A plane wave in vacuum travels along $+x$. At one instant its electric field is $210\hat{k}$ V/m. If the electric amplitude is 3 times this instantaneous value, which gives the instantaneous magnetic field? Use $c = 3 \times 10^8$ m/s.

(A) $+70 \times 10^{-8}\hat{j}$ T (B) $+70 \times 10^{-8}\hat{i}$ T

(C) $-70 \times 10^{-8}\hat{j}$ T (D) $-210 \times 10^{-8}\hat{j}$ T

13. A magnetic field is uniform inside a long solenoid of radius 5 m and negligible outside. It changes at 3 T/s. Find induced electric-field magnitude at radius $3R$ from the axis.

(A) $\frac{5}{6}$ V/m (B) 5 V/m

(C) $\frac{45}{2}$ V/m (D) $\frac{5}{2}$ V/m

14. Three long parallel wires are equally spaced by 4 m along a straight line. Outer wires each carry 8 A in the same direction; the middle carries 6 A in the opposite direction. Find the net force per unit length on the middle wire.

(A) $-\frac{6}{\pi}$ N/m (B) 0 N/m

(C) $\frac{12}{\pi}$ N/m (D) $\frac{4}{\pi}$ N/m

15. A car negotiates a level circular turn at its maximum non-slipping speed u when $\mu_s = \frac{1}{2}$. Rain lowers the coefficient to $\frac{1}{5}$ and the turn radius stays fixed. Find the new maximum speed divided by u.

(A) $\frac{1}{5}$ (B) $\frac{2}{5}$

(C) $\frac{\sqrt{5}}{5}$ (D) $\frac{\sqrt{10}}{5}$

16. A particle has $v(0) = 4$ m/s and acceleration $a(t) = 12t$ m/s^2 for $0 \le t \le 3$ s. Find its average velocity over this interval.

(A) 31 m/s (B) 22 m/s

(C) 4 m/s (D) 18 m/s

17. A one-electron ion with nuclear charge $Z = 2$ emits when its orbit changes. The ratio of initial to final orbit radii is 36, and its final principal quantum number is 1. What is $R\lambda$ for the emitted photon? Here R is the hydrogen Rydberg constant.

(A) $\frac{9}{35}$ (B) $\frac{9}{37}$

(C) $\frac{35}{9}$ (D) $\frac{1}{4}$

18. A point at the bottom of two plane liquid layers is viewed normally from air. The upper layer has thickness 15 cm and index 3/2; the lower has thickness 12 cm and index 4/3. Find its apparent depth below the top surface in the paraxial limit.

(A) 19 cm (B) $\frac{27}{2}$ cm

(C) $\frac{81}{4}$ cm (D) 10 cm

19. At the highest point of its flight, at height 10 m, a projectile has horizontal velocity 15 m/s. It explodes into two equal fragments, both initially moving horizontally. One moves forward at 1 m/s. Find the horizontal distance travelled by the other fragment from the explosion point to the ground. Take $g = 10$ m/s^2.

(A) 58 m (B) $15\sqrt{2}$ m

(C) 29 m (D) $29\sqrt{2}$ m

20. Two ideal cells of emfs 21 V and 1 V oppose each other in a loop containing two resistors each 3 ohms. Find the current magnitude.

(A) $\frac{10}{3}$ A (B) $\frac{22}{3}$ A

(C) $\frac{7}{2}$ A (D) $\frac{1}{6}$ A

21. A source emitting 700 Hz and a listener approach each other, each at 50 m/s relative to still air. Sound speed is 300 m/s. Find heard frequency before they pass.

(A) 500 Hz (B) 980 Hz

(C) 840 Hz (D) $\frac{2450}{3}$ Hz

22. A long solenoid has its turn number multiplied by 5 and length multiplied by 4, while its cross-sectional area and core permeability stay fixed. Find self-inductance multiplier.

(A) 25 (B) $\frac{25}{4}$

(C) $\frac{5}{4}$ (D) $\frac{4}{25}$

23. A power source sends 6000 W into a transmission line at 700 V. The line resistance is 6 ohms and this input power is fixed. Find resistive line loss.

(A) $\frac{2160000}{7}$ W (B) $\frac{360}{7}$ W

(C) $\frac{21600}{49}$ W (D) $\frac{245000}{3}$ W

24. The accelerating potential of an X-ray tube is increased from 10 kV to 20 kV, while its tube current is reduced by a factor 2. Ignoring target heating and recoil, what is the new cutoff wavelength in nm? Use $hc = 1240$ eV nm.

(A) $\frac{31}{500}$ nm (B) 62 nm

(C) $\frac{31}{250}$ nm (D) $\frac{31\sqrt{2}}{250}$ nm

25. A projectile is launched at 20 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{15}{16}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ m/s^2.

(A) $\frac{48}{5}$ m (B) $\frac{192}{5}$ m

(C) 16 m (D) $\frac{12}{5}$ m

26. A photon and a nonrelativistic particle have equal de Broglie wavelengths. The particle speed is $c/18$. If the photon then loses half its energy while the particle momentum is unchanged, what is the ratio of particle kinetic energy to the photon's final energy?

(A) $\frac{1}{9}$ (B) 18

(C) $\frac{1}{72}$ (D) $\frac{1}{18}$

27. A platform of moment of inertia 3 kg m^2 carries a person of mass 2 kg at radius 3 m and rotates at 4 rad/s. The person walks to half the stated initial radius; external torque is zero. Find final angular speed. Treat the person as a point mass.

(A) 4 rad/s (B) $\frac{56}{5}$ rad/s

(C) 16 rad/s (D) 28 rad/s

28. A 5 kg block on a rough horizontal table is tied over an ideal pulley to a hanging 8 kg mass. The hanging mass moves downward and $\mu_k = 1/5$. Find string tension. Take $g = 10$ m/s^2.

(A) $\frac{480}{13}$ N (B) 80 N

(C) $\frac{350}{13}$ N (D) 10 N

29. A homogeneous wire has Young modulus 8000000000 Pa and uniform elastic strain $\frac{3}{1000}$. Find stored elastic energy per unit volume.

(A) 108 J/m^3 (B) 24000000 J/m^3

(C) 72000 J/m^3 (D) 36000 J/m^3

30. An ideal mixture contains 1 moles of monatomic gas and 5 moles of rigid diatomic gas; vibrations are frozen. For this mixture, use $v_s = \sqrt{\gamma RT/\bar{M}}$ and $v_{rms,eff} = \sqrt{3RT/\bar{M}}$. Find $v_s/v_{rms,eff}$.

(A) $\frac{\sqrt{210}}{21}$ (B) $\frac{\sqrt{39}}{9}$

(C) $\frac{\sqrt{105}}{15}$ (D) $\frac{10}{21}$

Chemistry

31. The same element forms a fully occupied BCC phase A with cell edge a and an FCC phase B with edge 3/2a. The measured density ratio ρ_B/ρ_A is 32/135. All FCC sites have equal occupancy, atomic mass is unchanged, and vacancies are the only defects. What fraction of FCC sites is vacant?

(A) 1/5 (B) 7/10

(C) 3/10 (D) 3/5

32. A first-order reaction has a constant half-life of 10 min. A sample is observed now and again 40 min later, when unreacted material equals 1/128 of the original amount present when the reaction began. How many minutes had elapsed before the first observation?

(A) 80 (B) 30

(C) 70 (D) 40

33. A sample contains $\frac{1}{25}$ mol $KClO_3$ and a small amount of MnO_2 catalyst.75.0% of the chlorate decomposes by $2KClO_3 \rightarrow 2KCl + 3O_2$. The catalyst is recovered unchanged. Find O_2 formed in mol.

(A) $\frac{9}{100}$ (B) $\frac{9}{200}$

(C) $\frac{3}{100}$ (D) $\frac{3}{50}$

34. In an idealised Ostwald-process material balance, $\frac{3}{100}$ mol NH_3 is fed.90.0% of its nitrogen reaches the NO/NO_2 intermediate stream, and80.0% of that stream nitrogen is recovered as HNO_3. Find HNO_3 amount in mol. Each nitrogen atom is tracked once; recycled intermediates are not double-counted.

(A) $\frac{27}{1250}$ (B) $\frac{3}{125}$

(C) $\frac{51}{2000}$ (D) $\frac{27}{1000}$

35. A glucopyranose residue has a free OH at C1. Which property is consistent with that hemiacetal centre?

(A) It must be sucrose. (B) It contains no oxygen.

(C) It is necessarily locked as an acetal with no ring opening. (D) It can equilibrate with an open-chain aldehydic form.

36. For $AgCl(s)+2NH_3 \rightleftharpoons [Ag(NH_3)_2]^+ + Cl^-$,Ksp×Kf=1/4. Before AgCl is added,a solution has $\frac{3}{250}$M silver-ammine complex,zero chloride and $\frac{3}{25}$M free NH_3. If all dissolved silver stays complexed and volume is fixed,the extra AgCl solubility s in M is

(A) $\frac{3}{110}$ (B) $\frac{3}{100}$

(C) $\frac{3}{50}$ (D) $\frac{3}{10}$

37. In an antifluorite-derived crystal, X2- occupies all FCC anion sites. A measured fraction 2/3 of occupied M cations is M2+ and the rest M+. Only tetrahedral cation vacancies compensate charge; no interstitials or anion defects occur. What fraction of all tetrahedral sites is vacant?

(A) 1/5 (B) 3/5

(C) 2/5 (D) 1/3

38. For a fixed-composition closed system dG=V dP-S dT. At a state use V=2 m3, S=3 J/K and a local path with dT/dP=1 K/Pa. Find dG/dP along this path in m3. The supplied values are a formal local thermodynamic model; 1 Pa m3=1 J.

(A) -1 (B) -2

(C) 1 (D) -1/2

39. An octahedral complex has d^8 configuration and $\Delta_o = 200$ kJ mol^{-1}. Ignoring pairing energy,its d-electron energy relative to the unsplit barycentre,in kJ mol^{-1},is

(A) -240 (B) -40

(C) 240 (D) -440

40. Consider the statement: Pb(II)is relatively stabilised compared with Pb(IV)down Group14. Which explanation or consequence correctly accounts for this trend rather than applying an unqualified atomic-mass rule?

(A) Hydrogen bonding is necessarily stronger for every heavier hydride.

(B) The inert-pair effect can make Pb(IV)compounds oxidising toward species that reduce them to Pb(II).

(C) Every increase in atomic mass must increase all molecular bond strengths.

(D) The ns2pair is increasingly forced into all bonds down every group.

41. For the metal centre in Zn^{2+},Z=30. Determine the number of d electrons after formal oxidation-state assignment; use O=-2 in oxoanions.

(A) 10 (B) 11

(C) 12 (D) 9

42. A student expects persistent sodium-metal deposition during ordinary electrolysis of aqueous NaCl, using only gas-phase first ionisation energy. Which correction to that isolation prediction is justified?

(A) Ordinary aqueous NaCl electrolysis deposits Na before water can be reduced.

(B) Aqueous NaCl gives cathodic Na because Na+ migrates towards the cathode.

(C) Ordinary aqueous NaCl electrolysis reduces water to H2 rather than isolating Na.

(D) Molten and aqueous NaCl have identical competing cathodic reduction processes.

43. Why does the compact size of fluorine not guarantee a more exothermic first electron-gain enthalpy than chlorine?

(A) Fluorine is not a halogen.

(B) The added electron experiences greater crowding and repulsion in F's compact2p shell.

(C) The comparison concerns only second electron gain by F- and Cl-.

(D) Chlorine has a smaller principal valence shell than fluorine.

44. For 2A+B→3C+4D, A also disappears by an independent channel producing no C or D, and D is independently consumed at 1/2000 M/s. Measured total A disappearance is 1/125 M/s and net D appearance is 19/2000 M/s. What is the A disappearance rate through its independent channel, in M/s?

(A) 13/1000
(B) 3/1000
(C) 13/4000
(D) 1/250

45. At one T,P,a binary reaction R⇌P has a Gibbs-energy-versus-product-fraction curve. Pure R has G=20 kJ,pure P50 kJ,and the unique minimum G=10 kJ occurs at product fraction $\frac{1}{4}$. Which inference follows from both endpoints and the interior minimum?

(A) Positive pure-state change forces the equilibrium to be pure R.

(B) The pure-product minus pure-reactant standard change is positive,yet the equilibrium contains product.

(C) The minimum at an interior composition means both pure endpoints have equal G.

(D) G must decrease for a displacement from the minimum toward either pure endpoint.

46. For isolated Ni^{2+} (Z=28),use the high-spin spin-only approximation. Which pair (d-electron count, magnetic moment in BM) follows after removing electrons from 4s before 3d?

(A) (8,2).
(B) $(8,\sqrt{15})$.
(C) $(8,2\sqrt{2})$.
(D) $(9,\sqrt{15})$.

47. Why can PbO2 oxidise a reducing reagent while being converted to a Pb(II) compound?

(A) The oxidation number of oxygen must be+2 in every oxide.

(B) Lead must increase from+4 to+6 to act as an oxidant.

(C) Higher atomic mass alone forces all compounds to be reducing.

(D) Pb(II) is relatively stabilised by the inert-pair effect compared with Pb(IV).

48. For 2-methylpropan-2-ol in the Victor Meyer sequence,which test/observation assignment follows the actual carbinol class or oxidisable structural feature? Use ordinary room-temperature tests unless the named sequence specifies otherwise.

(A) blue because it is branched
(B) no characteristic red or blue colour
(C) violet because all OH compounds behave as phenols
(D) red because it has three methyl groups

49. 4-methylbenzoic acid is treated in separate stages with SOCl2,excess NH3,and Br2/NaOH. Which final product and carbon-accounting description are correct?

(A) the original acid;amide nitrogen is never retained
(B) 4-methylbenzylamine;all carbons retained
(C) 4-methylaniline;the carbonyl carbon remains as CH2
(D) 4-methylaniline;one carbon fewer than the starting acid

50. For $[CoF_6]^{3-}$,use the usual ligand-field/VBT assignments at room temperature. Which complete assignment is consistent with charge balance and magnetic behaviour?

(A) d^6,tetrahedral,sp^3,spin-only moment $2\sqrt{6}$ BM.

(B) d^7,octahedral,sp^3d^2,spin-only moment $2\sqrt{6}$ BM.

(C) d^6,octahedral,sp^3d^2,spin-only moment $2\sqrt{6}$ BM.

(D) d^6,octahedral,sp^3d^2,spin-only moment $\sqrt{35}$ BM.

51. A $\frac{2}{5}$ g organic sample contains chlorine as its only halogen. Carius analysis gives $\frac{6314}{44375}$ g dry AgCl, but this is only 80.0% of the theoretical AgCl because of a known collection loss. Find chlorine mass percentage. Use Ag=108, Cl=35.5.

(A) $\frac{44}{5}$

(B) $\frac{55}{4}$

(C) $\frac{3157}{71}$

(D) 11

52. A polluted-air exposure impairs oxygen transport even though the air still contains substantial oxygen. Which mechanism most directly explains the specific danger of carbon monoxide?

(A) It converts every oxygen molecule into ozone before inhalation.

(B) It binds haemoglobin strongly and competes with oxygen transport.

(C) It removes all nitrogen from the lungs.

(D) It is harmless because carbon dioxide is a normal metabolic product.

53. For neutral H2 and HD, take isotope mass numbers H=1, D=2, T=3 and neglect mass defects. Their gases have equal temperature. Which pair gives ($\langle v^2\rangle$_H2/$\langle v^2\rangle$_HD, number of neutrons in one HD molecule minus that in one H2 molecule)?

(A) (2/3, 1)

(B) (3/2, 3)

(C) (3/2, 1)

(D) (3/2, 0)

54. Glucose is first oxidised with bromine water to gluconic acid, then strongly oxidised with nitric acid. Which functional-group count changes from the first product to the final saccharic acid?

(A) Five amide bonds are formed.

(B) Every carbon atom becomes a separate methane molecule.

(C) Carboxyl groups increase from 1 to 2.

(D) Carboxyl groups decrease from 2 to 0.

55. A $\frac{18}{125}$ g volatile sample displaces $\frac{640}{13}$ mL wet air at 300 K and 790 mmHg. Water-vapour pressure is 30 mmHg. Using STP=273 K,760 mmHg and22.4 L/mol,the sample molar mass, rounded to the nearest whole g/mol, is

(A) 72

(B) 69

(C) 1824

(D) 138

56. Which pair distinguishes glutamic acid from glutamine by its side-chain functional group?

(A) Glutamine has an extra thiol; glutamic acid has Cl.

(B) Both contain a phenolic ring.

(C) Both have exactly the sameCONH2 side chain.

(D) Glutamic acid has an extraCOOH; glutamine hasCONH2.

57. A one-electron ion of nuclear charge Z=2 has a selected shell which, if hypothetically filled subject to Pauli, contains 8 electrons in orbitals with ml=0. What is the magnitude of that shell's hydrogenic energy divided by the hydrogen ground-state energy magnitude? Ignore interactions in identifying the shell.

(A) 1/4 (B) 1/2

(C) 1/8 (D) -1/4

58. A fixer initially receives 3/100 mol AgBr and an unknown amount of free thiosulfate. After complete stoichiometric complex formation, 1/100 mol AgBr remains and no free thiosulfate is left. Use AgBr+2S2O3^{2}-→[Ag(S2O3)2]^{3}-+Br-. What was the initial amount of thiosulfate?

(A) 1/50 (B) 3/50

(C) 1/20 (D) 1/25

59. A model oxide has composition MO6/5, where the subscript denotes O atoms per M atom. Oxygen is entirely O2- and metal sites are formally only M2+ or M3+. There are no other defects or ions. What fraction of M sites is M3+?

(A) 4/5 (B) -2/5

(C) 2/5 (D) 1/5

60. 2-methylaniline is acetylated at nitrogen,then subjected to controlled bromination,then its N-acetyl group is hydrolysed. Use the usual para preference relative to NHCOCH3 where free. Which principal final product results?

(A) 4-bromo-2-methylaniline (B) 2,4,6-tribromoaniline

(C) 3-bromo-2-methylaniline (D) 4-bromotoluene

Read the passage for Question 61

When the students recorded memories of the old station, they expected to produce a single smooth history. Instead, the interviews disagreed about some dates and gave different emphasis to the same events. Their first impulse was to remove the differences. A tutor suggested another approach: check dates where records survived, and make unresolved differences visible rather than silently choose one speaker. The final exhibition included verified dates, short extracts from several accounts and notes explaining what remained uncertain. It was less tidy than the students' original plan, but visitors could see both what the project had established and how people remembered it. The students had not abandoned accuracy. They had stopped treating a seamless account as its only possible appearance.

English Proficiency

61. Which statement best expresses the passage's main idea?

(A) Personal memories should always replace surviving written records.

(B) An exhibition is successful chiefly when it presents a simple uninterrupted story.

(C) A responsible history can distinguish verified facts from differing memories without disguising unresolved uncertainty.

(D) A historical account is accurate only when every disagreement has been removed.

62. Read the sentence: The signals were simultaneous, reaching both displays at exactly the same time. Choose the opposite in meaning of 'simultaneous' as used in the sentence.

(A) instantaneous
(B) successive
(C) identical
(D) synchronised

63. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. Empty outlines on the board then made missing tools immediately noticeable. B. A workshop kept losing time while workers searched for shared tools. C. Workers could check the collection before starting instead of discovering gaps halfway through a job. D. The supervisor marked a fixed place for each tool on a board.

(A) The supervisor marked a fixed place for each tool on a board.

(B) A workshop kept losing time while workers searched for shared tools.

(C) Workers could check the collection before starting instead of discovering gaps halfway through a job.

(D) Empty outlines on the board then made missing tools immediately noticeable.

64. Choose the meaning-preserving equivalent that retains the tense, participants and all stated restrictions. The instructions were too vague for the group to act on them confidently.

(A) The instructions were so vague that the group could not act on them confidently.

(B) The group could not act on any instructions confidently.

(C) The instructions were vague enough for the group to act confidently.

(D) The group refused to act because it disliked all instructions.

65. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. Only after comparing both versions Q. that a sentence had disappeared R. did the editor notice S. from the revised copy.

(A) R-P-Q-S
(B) Q-P-R-S
(C) S-R-Q-P
(D) P-R-Q-S

66. The business owner devotes a substantial part of his wealth to public libraries and hospitals. Which word identifies this role?

(A) philanthropist
(B) entrepreneur
(C) custodian
(D) beneficiary

67. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The organiser is going to postpone only the outdoor session.

(A) Only the organiser is going to postpone the outdoor session.
(B) All the sessions are going to be postponed by the organiser.
(C) Only the outdoor session is going to be postponed by the organiser.
(D) The outdoor session has been postponed by the organiser.

68. Choose the pair with the same ordered relationship as KEY : LOCK.

(A) PAGE : BOOK
(B) ROAD : JOURNEY
(C) DOOR : HINGE
(D) PASSWORD : ACCOUNT

69. Read the sentence: Rain fell in sporadic bursts rather than continuously through the afternoon. Choose the closest meaning of 'sporadic' as used in the sentence.

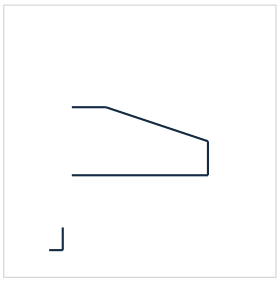
(A) irregular and occasional
(B) brief but predictable
(C) confined to one location
(D) frequent and heavy

70. The investigation concerns the arrangement of organs and bodily structures rather than the historical origins of words. Which field is it?

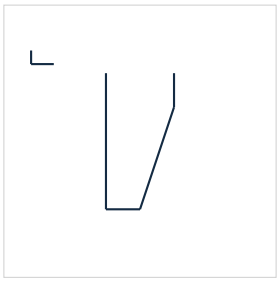
(A) ecology
(B) anatomy
(C) semantics
(D) meteorology

Logical Reasoning

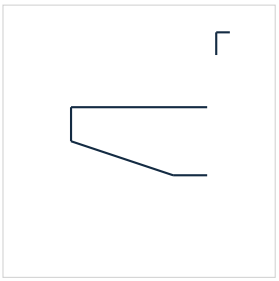
71. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1



2

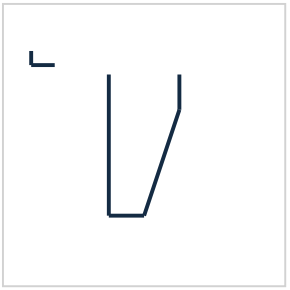


3

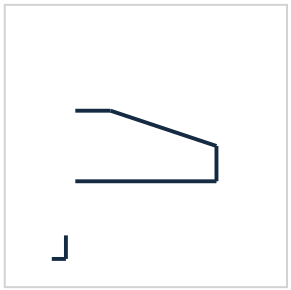
(A) Pattern V



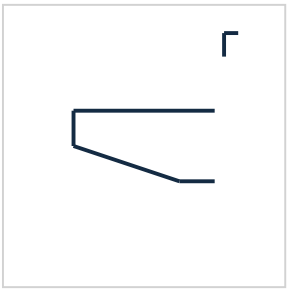
(B) Pattern W



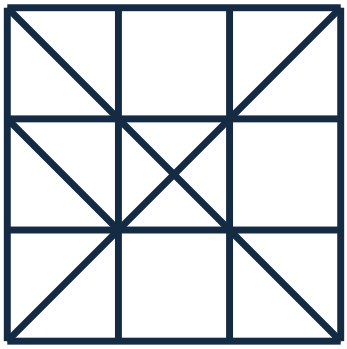
(C) Pattern R



(D) Pattern L



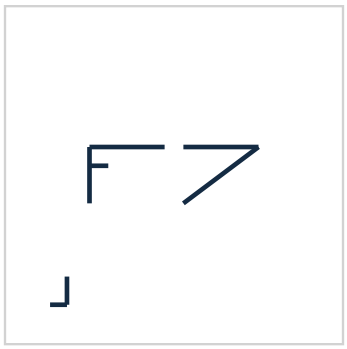
72. How many triangles are formed by all the drawn segments?



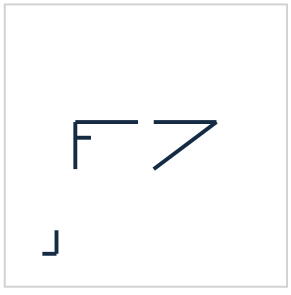
(A) 32 (B) 36

(C) 72 (D) 22

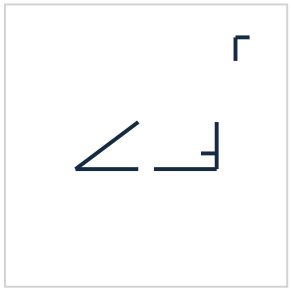
73. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



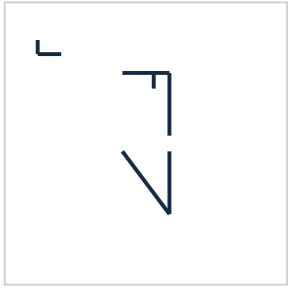
(A) Pattern P



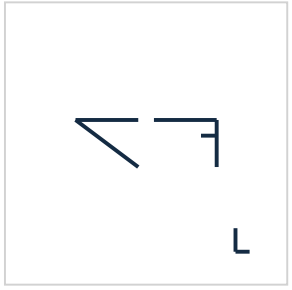
(B) Pattern M



(C) Pattern X



(D) Pattern R



74. Preserve the exact relation and its order: Lion : Pride :: ? : Fleet. Choose the missing term.

(A) Wolf (B) Bee

(C) Lion (D) Ship

75. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 4, 8, ?, 28, 34, 68, 74.

(A) 22 (B) 10

(C) 14 (D) 16

76. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: VQFX, YRGB, BSHF, ?, HUJN, KVKR.

(A) DSHI (B) ETIJ

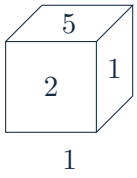
(C) FUJK (D) JITE

77. In a row, Asha is 11th from the left and 29th from the right. Bina is 6th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

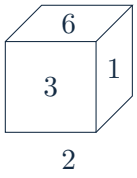
(A) 21 (B) 7

(C) 19 (D) 25

78. The views show the same die. Give the ordered pair: (face opposite 2, face opposite 3). Do not assume opposite labels sum to seven.



1



2

(A) $(3, 3)$ (B) $(3, 4)$

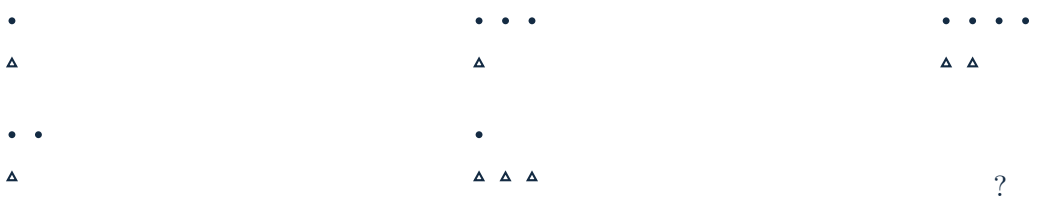
(C) $(3, 2)$ (D) $(3, 1)$

79. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 16$ and $5 \mapsto 31$. Which input maps to 61?

(A) 11 (B) 10

(C) 61 (D) 12

80. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern L



(B) Pattern K



(C) Pattern Z



(D) Pattern T



81. Each letter position has its own fixed cyclic shift, applied identically in both pairs (Z wraps to A). Complete AQB : DRC :: KTE : ?.

(A) NUF
(B) HUL
(C) MTE
(D) OVG

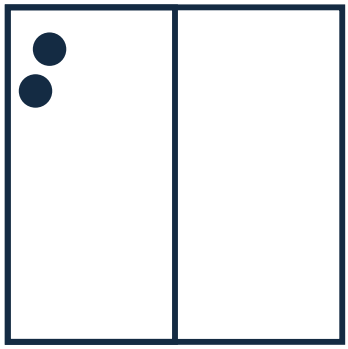
82. Dev points to a photograph and says, "This person is my mother's only daughter." Dev is male, and the photographed person is female. How is the photographed person related to Dev?

(A) Son
(B) Sister
(C) Daughter
(D) Brother

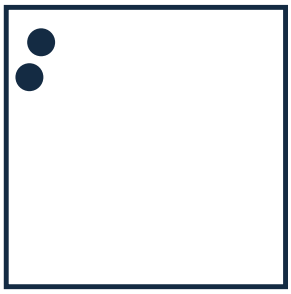
83. A walker goes 10 m north, then 12 m west, then turns left and walks 5 m. In which direction is the starting point from the walker?

(A) South-west
(B) North-west
(C) North-east
(D) South-east

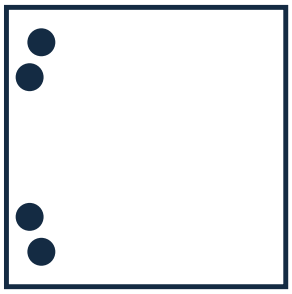
84. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



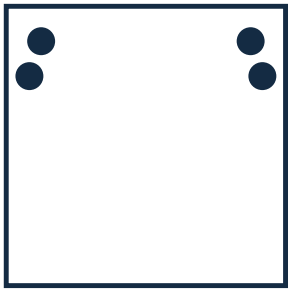
(A) Pattern M



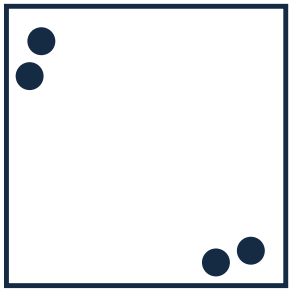
(B) Pattern V



(C) Pattern Z

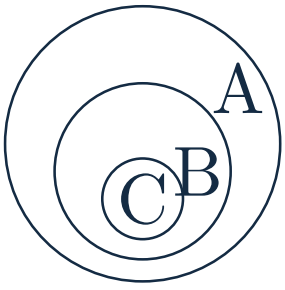


(D) Pattern W

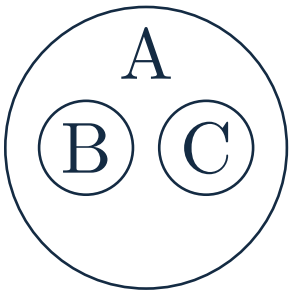


85. Choose the diagram that represents all these conditions, including the strictness information: No A is B, no B is C, and no A is C. Each class is nonempty.

(A) Pattern R



(B) Pattern V



(C) Pattern W



(D) Pattern L



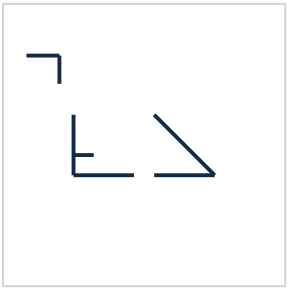
86. Choose the exact horizontal water reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



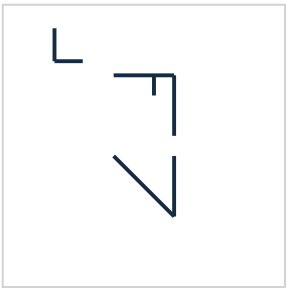
(A) Pattern R



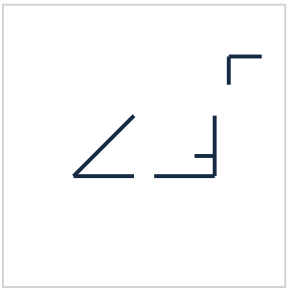
(B) Pattern L



(C) Pattern X

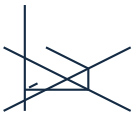


(D) Pattern Z

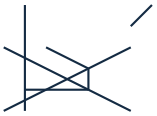


87. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 2

(B) 3

(C) 4

(D) 1

88. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?



1 2 3 4

(A) 2 (B) 1

(C) 4 (D) 3

89. In this code, @ means +, # means −, & means ×, % means ÷. After substituting meanings, evaluate ((7#4)@(2&5)); printed parentheses retain their usual grouping.

(A) 13 (B) 12

(C) −13 (D) 14

90. Consider only these statements; universal statements alone do not imply existence. All guides are campers; No guides are campers. Conclusions: I. All guides are joggers. II. Some campers are not guides. Which conclusions follow necessarily?

(A) Both I and II (B) Only I

(C) Only II (D) Neither I nor II

Mathematics

91. A,B are acute, A+B=$\pi/4$ and tan A=$\frac{2}{3}$. Find tan(A-B).

(A) $\frac{7}{15}$ (B) $\frac{7}{17}$

(C) $\frac{1}{5}$ (D) $-\frac{7}{17}$

92. X is binomial with mean 5 and variance 10/3. Find P(X=0).

(A) $\frac{16384}{14348907}$ (B) e^{-5}

(C) $\frac{32}{243}$ (D) $\frac{32768}{14348907}$

93. 8 distinct people stand in a line. Three specified people may occupy neither end position and no two of these three may be adjacent. How many orders are possible?

(A) 2400 (B) 40320

(C) 14400 (D) 2880

94. Two circles have centres (0,0),(13,0) and radii 5,4. Find the squared minimum distance between points on the two circles.

(A) 16 (B) 4

(C) 484 (D) 88

95. $A = \begin{bmatrix} 1 & 11 \\ 0 & 0 \end{bmatrix}$. For (I+A)^7=I+cA, find c.

(A) 128 (B) 127

(C) 7 (D) 6

96. Write sin x cos^{3}x+6sin^{3}x cos x=P sin2x+Q sin4x. Find Q.

(A) $\frac{7}{4}$ (B) $\frac{7}{8}$

(C) $-\frac{5}{8}$ (D) $\frac{1}{8}$

97. If |z-13|=5, find the minimum of $|z|^2$.

(A) 8 (B) 144

(C) 64 (D) 324

98. y′-y/x=1 for x>0, y(1)=7. Find y(7)/7-y(1).

(A) $7\log(7) + 49$ (B) $\log(7) + 7$

(C) $\log(7)$ (D) $-\log(7)$

99. In triangle ABC, A=2 B and a/b=sqrt 3. If b=4, find the area.

(A) $8\sqrt{3}$ (B) $-1 + 4\sqrt{3}$

(C) $1 + 4\sqrt{3}$ (D) $5 + 8\sqrt{3}$

100. Let $D(t) = \det \begin{pmatrix} 6+t & 1+t \\ 1+t & 8+t \end{pmatrix}$. Find D′(0).

(A) -16 (B) -12

(C) 12 (D) 47

101. How many solutions in 0≤x<18π satisfy sin2x=sin x and cosx>0?

(A) 27 (B) 28

(C) 36 (D) 18

102. Evaluate $\int_0^{\pi/2} \sin(2x)\left[1 + 7\sin^2(2x)\right] \frac{\sin^7 x}{\sin^7 x + \cos^7 x}\, dx$.

(A) $\frac{17}{6}$ (B) $\frac{17}{3}$

(C) 4 (D) $\frac{1}{2}$

103. How many integers x satisfy $4^{-x} - (4160)2^{-x} + 262144 \leq 0$?

(A) 6 (B) 7

(C) 13 (D) 5

104. There are 6 Boolean variables p_1,...,p 6. Exactly 1 variables must be true. How many truth assignments satisfy [(p_1 implies p_2) and (p_1 implies not p_2)]?

(A) 64 (B) 5

(C) 6 (D) 1

105. The letters A,A,B,C and 5 further distinct letters are arranged in a row. How many arrangements have nonadjacent As and have B,C adjacent in either order?

(A) 60480 (B) 30240

(C) 40320 (D) 21600

106. The circle $(x-(-1))^2+(y-(-2))^2=3025$ has a normal with direction ratio $3:4$. Its point of contact P satisfies $x_P>-1$. Find x_P+y_P.

(A) -80 (B) -14

(C) 52 (D) 74

107. For α=5π/17, β=5π/19, γ=5π/23, define B_ij=sin(t_i+t_j)+sin(t_i-t_j). Find det B.

(A) 1 (B) 0

(C) -1 (D) 2

108. Maximize Z=3x+4y subject to x≥0, y≥7, x+y≤16, x+2y≤22.

(A) 55 (B) 44

(C) 54 (D) 52

109. Three distinguishable fair dice are thrown. Given that exactly two faces are equal, find the probability that their sum is at least 14.

(A) $\frac{7}{30}$ (B) $\frac{4}{15}$

(C) $\frac{23}{30}$ (D) $\frac{7}{72}$

110. The chord of contact from P to $x^2/16+y^2/4=1$ is x/4+y/2=1. Find $|OP|^2$.

(A) 12 (B) 6

(C) 20 (D) $2\sqrt{5}$

111. Box A contains 3 red and 3 blue balls; B contains 6 red and 5 blue balls. A fair coin chooses one box. Two independent draws with replacement from that box give exactly one red ball. Find the posterior probability that A was chosen.

(A) $\frac{11}{23}$ (B) $\frac{121}{241}$

(C) $\frac{120}{241}$ (D) $\frac{1}{4}$

112. For integer n, consider: "If 40 divides n, then 8 divides n." Which is its contrapositive?

(A) If 40 does not divide n, then 8 does not divide n. (B) 40 divides n and 8 does not divide n.

(C) If 8 does not divide n, then 40 does not divide n. (D) If 8 divides n, then 40 divides n.

113. For f(x)=|x-6|/(x-9), on which maximal open intervals is f strictly decreasing?

(A) $(6,9)\cup(9,\infty)$ (B) $(6,\infty)$

(C) $(-\infty,9)$ (D) $(-\infty,6)$

114. Find the coefficient of x^7 in $(1+4x)\exp(8x)$.

(A) $\frac{65536}{45}$ (B) $\frac{131072}{63}$

(C) $\frac{131072}{315}$ (D) $\frac{65536}{35}$

115. A=(4,3,0), C=(7,9,3). B lies on AC with AB:BC=2:1. Find $|OB|^2$, where O is the origin.

(A) 139 (B) 89

(C) 54 (D) 25

116. A positive integer n satisfies $|\Pi(1+ji),\ j=1,\ldots,n|^2=44200$. Find n.

(A) 5 (B) 10

(C) 6 (D) 4

117. For x,y>0 with sqrt(x)+sqrt(y)=6, find d^2y/dx^2 at x=16.

(A) $\frac{3}{64}$ (B) $-\frac{3}{64}$

(C) $\frac{3}{32}$ (D) $\frac{3}{4}$

118. $f(x)=x^2+2x$ for $x\le 4$, and $f(x)=px+q$ for $x>4$. If f is differentiable at 4, find p-q.

(A) -6 (B) 24

(C) 10 (D) 26

119. Let $z=(1+i)^{13}$. Find Re(z)+Im(z).

(A) -64 (B) 128

(C) $64\sqrt{2}$ (D) -128

120. All tangents common to $x^2/100+y^2/25=1$ and $x^2+y^2=\frac{200}{3}$ touch the ellipse at points P_i. Find $\Sigma(P_i,x)^2$ over the four tangency points.

(A) $\frac{800}{3}$ (B) $\frac{250}{3}$

(C) $\frac{1000}{3}$ (D) $\frac{50}{3}$

121. For a=(1,7,0), b=(0,1,4), c=(3,0,1), evaluate $(a\times c)\cdot(a+b)$.

(A) 0 (B) 85

(C) -84 (D) -85

122. $f(t)=\sin t$ restricted to $[-\pi/2,\pi/2]$, $f(g(x))=\cos x$ on $[0,\pi]$. Find $g(g(\frac{\pi}{12}))$.

(A) $-\frac{\pi}{12}$ (B) $\frac{11\pi}{12}$

(C) $\frac{\pi}{12}$ (D) $\frac{5\pi}{12}$

123. The parabola $(y-3)^2=16(x-2)$ has focus F and vertex V. A circle centred at F passes through V. Find the tangent length from P=(14,11).

(A) $8\sqrt{2}$ (B) 112

(C) $4\sqrt{7}$ (D) $4\sqrt{3}$

124. $F'(x)=\frac{4(x^2-1)(x^2+1)^3}{x^5}$ and $F(1)=5$. Find $F(2)-2F(1)$.

(A) $\frac{289}{16}$ (B) $\frac{369}{16}$

(C) $-\frac{449}{16}$ (D) $\frac{449}{16}$

125. α,β are nonzero roots of $x^2-14x+45=0$. Find the sum of $1+1/\alpha$ and $1+1/\beta$.

(A) $\frac{14}{45}$ (B) $\frac{59}{45}$

(C) $\frac{73}{14}$ (D) $\frac{104}{45}$

126. In a group of 61, 31 studyA, 26 studyB and 10 study neither. A person is chosen uniformly from those who do not studyB. Find the probability the person studiesA.

(A) $\frac{31}{61}$ (B) $\frac{5}{6}$

(C) $\frac{31}{35}$ (D) $\frac{5}{7}$

127. How many x with $0\le x<3\pi$ satisfy $\sin^4x+\cos^4x=3/4$ and $\cos x>0$?

(A) 2 (B) 6

(C) 3 (D) 4

128. The hyperbola $x^2/36-y^2/4=1$ has eccentricity e. Find the eccentricity e' of its conjugate hyperbola.

(A) $\frac{3\sqrt{10}}{10}$ (B) $\frac{\sqrt{10}}{3}$

(C) $\sqrt{10}$ (D) $2\sqrt{10}$

129. 11 distinct people sit at labelled round tables of sizes 7,4. Two specified people must be at different tables, and the first specified person must be at the 7-seat table. Rotations are identical. Count the seatings.

(A) 362880 (B) 725760

(C) 10160640 (D) 1425600

130. Two points on $y^2=12x$ have parameters with product -1. Find the latus-rectum length of their midpoint locus.

(A) 12 (B) 6

(C) 3 (D) 24

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A wheel initially rotates at 7 rad/s. A constant opposing angular acceleration brings it to rest after 7 s. What angle does it turn through during the last half of this stopping interval?

(A) 49 rad
(B) $\frac{49}{8}$ rad
(C) $\frac{49}{16}$ rad
(D) $\frac{49}{2}$ rad

132. A soap bubble initially of radius 3 cm is expanded quasistatically until its volume is 27 times the initial volume. Surface tension remains $\frac{1}{50}$ N/m. Find work against surface tension, in units of $\pi \times 10^{-4}$ J.

(A) $\frac{288}{25}$
(B) $\frac{36}{25}$
(C) $\frac{144}{25}$
(D) $\frac{72}{25}$

133. A converging thin lens forms a real image with magnification magnitude 3. The object-image separation is 20 cm. Find focal length.

(A) 10 cm
(B) $\frac{5}{4}$ cm
(C) $\frac{15}{4}$ cm
(D) 5 cm

Chemistry

134. For weak-field high-spin Ti^{3+} and Cu^{2+},which statement follows in the spin-only approximation?

(A) They have the same moment $\sqrt{3}$ BM despite different d-electron counts.

(B) Their moments must differ because different elements cannot have equal unpaired-electron counts.

(C) They are both diamagnetic because ionic charge removes all d electrons.

(D) Equal moments prove equal atomic numbers and equal ionic radii.

135. Compare p-methoxyphenoxide and p-nitrophenoxide as oxygen nucleophiles towards the same unhindered methyl substrate in one fixed solvent. Ignore concentration differences. Which pairing of the stronger nucleophile and the main electronic reason is most appropriate?

(A) p-methoxyphenoxide; it is weaker because a localised lone pair cannot attack

(B) p-methoxyphenoxide; the donating methoxy group versus withdrawing nitro group changes oxygen electron availability

(C) p-nitrophenoxide; the more stabilised anion necessarily attacks faster

(D) p-nitrophenoxide; resonance always increases local charge on oxygen

136. A Hinsberg separation is applied to dimethylamine. Which reagent identity and class-dependent outcome are correct?

(A) benzoyl chloride,C6H5COCl;a sulphonamide without N-H,insoluble in alkali

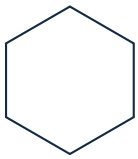
(B) benzenesulphonyl chloride,C6H5SO2Cl;every amine gives an alkali-soluble N-H sulphonamide

(C) benzenesulphonyl chloride,C6H5SO2Cl;a sulphonamide without N-H,insoluble in alkali

(D) benzenesulphonic acid,C6H5SO3H;all classes give the same neutral product

Logical Reasoning

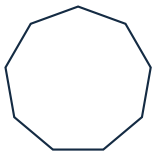
137. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

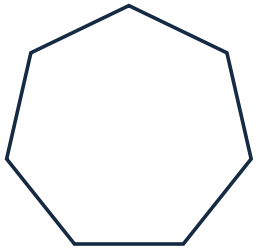


B

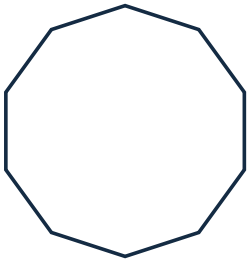


C

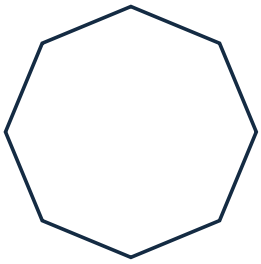
(A) Pattern R



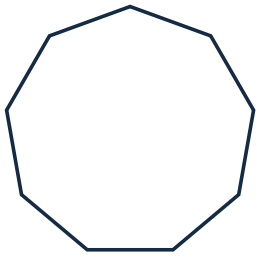
(B) Pattern N



(C) Pattern L



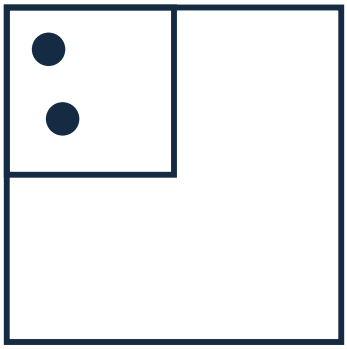
(D) Pattern K



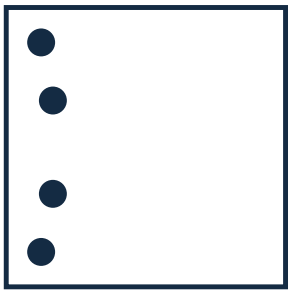
138. Bina receives 3 times as many tokens as Asha, and Charu receives 2 times as many as Bina. Their total is 200. After the allocation, Charu gives 5 tokens to Asha. How many tokens does Charu then retain?

(A) 115
(B) 15
(C) 120
(D) 125

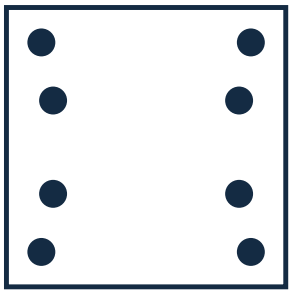
139. Starting with a unit square sheet, fold right to left along the vertical midline, then fold bottom to top along the horizontal midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



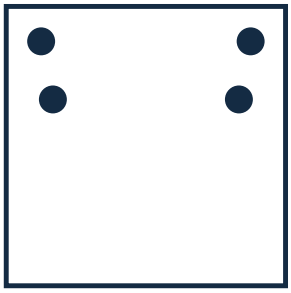
(A) Pattern N



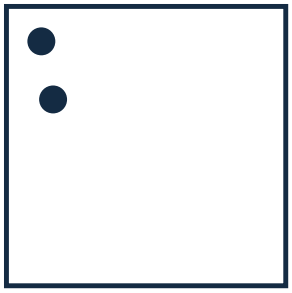
(B) Pattern W



(C) Pattern P



(D) Pattern Z



Mathematics

140. A sphere has dV/dt=2 cubic units/s. Find the rate of change of its surface area when r=6.

(A) $\frac{2}{3}$

(B) $\frac{1}{9}$

(C) $\frac{1}{3}$

(D) $\frac{1}{72\pi}$

141. Find the area between y=(x-3)(x-5) and the x-axis for $0 \leq x \leq 7$.

(A) 13

(B) 26

(C) $\frac{70}{3}$

(D) $-\frac{70}{3}$

142. The image of $y^2=24x$ under reflection in x-y=5 has equation $(x-5)^2=C(y+5)$. Find C.

(A) 6 (B) -24

(C) 24 (D) 44

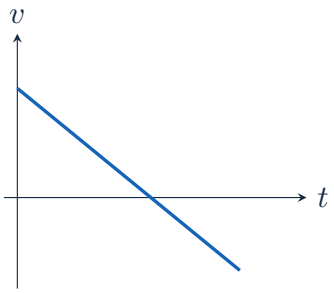
BITSAT Mock 27

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A particle has velocity $v(0) = 6$ m/s, and constant acceleration makes $v(5) = -4$ m/s. What total distance does it travel in this interval?



(A) 9 m
(B) 4 m
(C) 13 m
(D) 26 m

2. A metal wire has linear temperature coefficient $\frac{1}{1000}$ per degree Celsius referred to 0°C. Its dimensions are fixed and it stays across a constant voltage. Find power at 50°C divided by power at 0°C.

(A) $\frac{20}{19}$
(B) 1
(C) $\frac{20}{21}$
(D) $\frac{1}{21}$

3. A uniformly charged nonconducting sphere has radius R and surface potential V_s relative to infinity. Find potential at $r = R/13$ divided by V_s.

(A) 1
(B) $\frac{3}{2}$
(C) $\frac{19}{13}$
(D) $\frac{253}{169}$

4. A block is placed on the vertical inner wall of a trolley. The trolley accelerates horizontally so the wall pushes the block. With coefficient of static friction $\frac{1}{2}$, what minimum trolley acceleration prevents vertical slipping? Take $g = 10$ m/s^2.

(A) $10\sqrt{2}$ m/s^2
(B) 20 m/s^2
(C) 5 m/s^2
(D) 10 m/s^2

5. A spring of constant 20 N/m is already stretched by $\frac{3}{10}$ m. How much additional work is required to increase its extension to $\frac{3}{5}$ m quasistatically?

(A) $\frac{27}{10}$ J (B) $\frac{18}{5}$ J

(C) $\frac{27}{5}$ J (D) $\frac{9}{5}$ J

6. A long solid cylindrical conductor of radius 6 m carries uniformly distributed current 6 A. Find magnetic field magnitude at radial distance $R/5$ from the axis in units of $\mu_0/(2\pi)$.

(A) $\frac{1}{30}$ (B) $\frac{1}{25}$

(C) 5 (D) $\frac{1}{5}$

7. A solid sphere of radius R has uniform density $2\rho_0$. A spherical cavity of radius $R/10$ is cut with its centre at distance $R/5$ from the original centre. Find field magnitude at the cavity centre in units of $\pi G\rho_0 R$.

(A) $\frac{20}{3}$ (B) $\frac{5}{3}$

(C) $\frac{40}{3}$ (D) $\frac{8}{15}$

8. A uniform-density spherical planet has radius R and surface gravitational acceleration g. At what depth below the surface is gravity $g/3$? Give the depth as a fraction of R.

(A) $\frac{2}{3}$ (B) $1 - \frac{\sqrt{3}}{3}$

(C) 1 (D) 2

9. A dipole of fixed moment 2 C m is rotated quasistatically from 60° to 120° relative to a uniform electric field 7 N/C. Find external work.

(A) 14 J (B) 7 J

(C) $\frac{14\pi}{3}$ J (D) $7\sqrt{3}$ J

10. A 3 ohm resistor is connected in parallel with an unknown resistor across 15 V. The source supplies 90 W in total. Find the unknown resistance.

(A) 15 Ω (B) $\frac{1}{2}$ Ω

(C) $\frac{11}{2}$ Ω (D) $\frac{15}{11}$ Ω

11. Two rods of equal length and cross-sectional area have thermal conductivities k and 3k. They are joined end-to-end with no contact resistance. Find the effective conductivity of the composite divided by k.

(A) 1 (B) 3

(C) $\frac{3}{2}$ (D) 4

12. A car negotiates a level circular turn at its maximum non-slipping speed u when $\mu_s = \frac{9}{10}$. Rain lowers the coefficient to $\frac{2}{5}$ and the turn radius stays fixed. Find the new maximum speed divided by u.

(A) $\frac{2}{3}$ (B) $\frac{3}{2}$

(C) $\frac{9}{4}$ (D) $\frac{\sqrt{10}}{5}$

13. A network is defined by $P = \mathrm{NOR}(A, B)$, $Q = \mathrm{NOR}(A, P)$ and $Y = \mathrm{OR}(Q, \overline{B})$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 0$ and $A = 0$?

(A) 3 (B) 2

(C) 1 (D) 0

14. An insulated vessel containing one mole of an ideal gas of molar mass $\frac{3}{100}$ kg/mol moves at bulk speed 60 m/s and is brought to rest. All bulk kinetic energy becomes internal energy. The gas has 3 active degrees of freedom. Find temperature rise. Use $R = 8.314$ J/(mol K).

(A) $\frac{12000}{4157}$ K (B) $\frac{18000}{4157}$ K

(C) $\frac{54000}{4157}$ K (D) $\frac{9000}{4157}$ K

15. Two soap bubbles of radii 3 cm and 12 cm share a common thin soap film. Find the radius of curvature of the common film.

(A) 9 cm (B) 4 cm

(C) $\frac{15}{2}$ cm (D) 2 cm

16. A circular parallel-plate capacitor of plate radius 4 cm is charged at 16 mA. Neglect fringing. What displacement current passes through the annular part of the gap between radii 2 cm and 4 cm, centred on the plate axis?

(A) 48 mA (B) 24 mA

(C) 4 mA (D) 12 mA

17. A sinusoidal source of peak $\frac{97}{10}$ V feeds a back-to-back series pair of identical Zeners through a 3 kΩ current-limiting resistor. The output is across the diode pair. Each Zener has reverse breakdown 3 V and forward drop 0.7 V. What is the peak-to-peak output voltage? Use the piecewise constant-drop model.

(A) $\frac{37}{5}$ V (B) $\frac{97}{5}$ V

(C) 6 V (D) $\frac{23}{5}$ V

18. Photons of energy 20 eV eject electrons from a surface. The fastest electrons enter a uniform magnetic field perpendicular to their velocities. In this field an electron with kinetic energy 2 eV has orbit radius r. The observed maximum orbit radius is $2r$. What is the surface work function? Neglect recoil.

(A) 18 eV (B) 12 eV

(C) 28 eV (D) 3 eV

19. Two masses 3 kg and 3 kg are connected by a light string over a smooth pulley and accelerate freely. A transverse pulse travels in the taut straight part of this string, of linear density μ. Compare its speed to that in a second string of density 2μ held at tension g N numerically equal to 1 kg weight. Find v_1/v_2.

(A) 6 (B) $\frac{\sqrt{6}}{2}$

(C) $\sqrt{6}$ (D) $\sqrt{3}$

20. A converging thin lens has focal length 40 cm in air. Object and real-image distances are positive magnitudes. Which proposed reading pair is inconsistent?

(A) $u = 80$ cm, $v = 80$ cm. (B) $u = 80$ cm, $v = 120$ cm.

(C) $u = 120$ cm, $v = 60$ cm. (D) $u = 160$ cm, $v = \frac{160}{3}$ cm.

21. A plane wave passes through a thin prism away from edges. Separately, a pointlike aperture produces an outgoing spherical wavefront of radius 4 cm, which propagates another 15 cm in a homogeneous medium. Which pair correctly describes the prism wavefront and the later aperture wavefront?

(A) Spherical; plane with infinite radius.

(B) Approximately plane and tilted; spherical with radius 19 cm.

(C) Approximately plane and unchanged in direction; spherical with radius 4 cm.

(D) Approximately plane and tilted; spherical with radius 11 cm.

22. Two uniform thin rods meet at a right angle at the origin. Both have length 5 m. The rod along +x has linear density 4 times that of the rod along +y. A force parallel to the y-axis must produce pure translation. At what x-coordinate must its line of action lie?

(A) $\frac{5}{2}$ m

(B) $\frac{1}{2}$ m

(C) 2 m

(D) 4 m

23. A particle executes SHM at frequency 6 Hz. At t=0 it is at its positive extreme. In the interval $0 <= t < 1$ second, how many times does its kinetic energy equal its potential energy?

(A) 6

(B) 24

(C) 23

(D) 12

24. A 2 kg block moves along a horizontal rough surface from $x = 0$ to $x = 6$ m under a force $F(x) = 4x$ N parallel to motion. With $\mu_k = 1/5$ and $g = 10$ m/s^2, find the work done by the net force. The block has enough initial speed to keep moving forward.

(A) 48 J

(B) 72 J

(C) 8 J

(D) 96 J

25. Two parallel plane mirrors separated by 2 cm extend from axial position 0 to 13 cm. A ray enters midway between them at axial position 0, making 45° with the normal to the mirrors. How many reflections occur before it passes the far edge? Exclude a contact exactly on the exit edge.

(A) 7

(B) 6

(C) 8

(D) 12

26. A stretched string is divided by a bridge into lengths in the ratio 1 : 3. Tension and linear density are unchanged. The original fundamental is 50 Hz. Find the fundamental of the longer segment.

(A) 200 Hz

(B) $\frac{50}{3}$ Hz

(C) $\frac{200}{3}$ Hz

(D) 100 Hz

27. A series RC circuit is driven at fixed rms voltage. Multiplying angular frequency by 7 multiplies rms current by $\frac{7\sqrt{1105}}{221}$. Find R^2/X_C^2, where X_C is the initial capacitive reactance.

(A) 9

(B) 49

(C) $\frac{1}{9}$

(D) $\frac{31}{4}$

28. Water flows steadily from a pipe section of area A at 4 m/s to a section of area $A/3$ located 2 m higher. Neglect viscosity. Find $p_1 - p_2$. Take density 1000 kg/m^3 and $g = 10$ m/s^2.

(A) 92000 Pa

(B) 44000 Pa

(C) 84000 Pa

(D) 64000 Pa

29. A screw gauge advances 2.0 mm in four complete turns and has 50 circular divisions. With the jaws closed, its zero mark lies 2 divisions below the reference line; use the usual convention that a zero below the reference line gives positive zero error. With a wire, the main scale reads 5 mm and circular reading is 33 divisions. What is the corrected diameter?

(A) $\frac{333}{100}$ mm (B) $\frac{533}{100}$ mm

(C) $\frac{141}{25}$ mm (D) $\frac{531}{100}$ mm

30. A mass 5 kg moves with $\vec{r}(t) = 3t\hat{i} + 4t^2\hat{j}$ m. Find the z-component of angular momentum about the origin at $t = 1$ s.

(A) $180\,\mathrm{kg\,m^2/s}$ (B) $30\,\mathrm{kg\,m^2/s}$

(C) $120\,\mathrm{kg\,m^2/s}$ (D) $60\,\mathrm{kg\,m^2/s}$

Chemistry

31. A buffer contains 1/50 mol HA and 1/20 mol A-. After adding strong base without volume change it has pH=pKa+log10(6), with HA still present. How many moles of OH- were added? Quantitative neutralisation and the Henderson ratio model apply.

(A) 1/20 (B) 1/100

(C) 1/200 (D) 1/50

32. The internuclear axis is z. Two p_x orbitals overlap sideways. Their lobes facing each other above the axis have opposite phases; the lower facing lobes have the corresponding opposite signs. Which MO is produced,and what nodal property follows?

(A) A sigma bonding MO,because every overlap along a line joining nuclei is sigma.

(B) A pi bonding MO; the facing-lobe phase test has been reversed.

(C) A pi antibonding MO; there is an additional node between nuclei besides the plane containing the axis.

(D) A delta MO,because two perpendicular p orbitals always create two nodal planes.

33. CompareNH2-CH2-COOH with NH2-CH(CH3)-COOH. Which difference determines chirality at the alpha carbon?

(A) Glycine has twoH substituents; alanine has four different substituents.

(B) Both have four different substituents because each contains nitrogen.

(C) Only glycine is chiral at its alpha carbon.

(D) The carboxyl group alone makes every carbon chiral.

34. A carbon-containing organic sample contains S only among nitrogen, sulfur and halogens. It undergoes sodium fusion with sufficient sodium. Which conversion and subsequent test protocol is chemically consistent? For a silver-halide test, use a separately prepared portion boiled with nitric acid to remove cyanide/sulfide interference where present.

(A) cyanide; no further treatment is ever needed before adding silver nitrate.

(B) Unchanged covalent bonds; add silver nitrate directly to the untreated organic sample.

(C) Nitrate and sulfate in every case; all give the same Prussian-blue test.

(D) sulfide; violet colour with nitroprusside.

35. hexane-2,5-dione,CH3CO(CH2)2COCH3 undergoes intramolecular aldol addition at low temperature; isolate before dehydration. Under the stated intramolecular closure pathway,which isolated product matches both ring size and work-up endpoint?

(A) 3-hydroxy-3-methylcyclohexan-1-one

(B) 3-methylcyclopent-2-en-1-one

(C) 3-hydroxy-3-methylcyclopentan-1-one

(D) cyclohexane-1,3-diol

36. For I: decolourisation of sugar solution by activated charcoal; II: uniform dissolution of sugar in water, which classification correctly identifies surface accumulation and distinguishes it from a bulk process?

(A) Only II involves adsorption.

(B) I involves adsorption; II is not adsorption.

(C) Both are adsorption because both involve molecular interactions.

(D) Neither involves adsorption.

37. An ideal gas freely expands into vacuum in an insulated rigid enclosure,accessible gas volume increasing 2-fold. Which joint statement follows?

(A) The rigid enclosure forces the gas-accessible volume to remain unchanged.

(B) q=0,w=0 and temperature is unchanged;entropy increases,so the process is irreversible.

(C) The gas does work on the vacuum and its temperature necessarily falls.

(D) q=0 and w=0 imply entropy change is zero and the expansion is reversible.

38. For isolated molecules of XeF_4 and BrF_5,which member has the larger permanent dipole moment, and which reasoning is sufficient?

(A) XeF_4; the number of polar bonds alone fixes the molecular dipole.

(B) They must be equal because both contain polar covalent bonds.

(C) The one with more atoms always has the larger dipole,independent of geometry.

(D) BrF_5; Square-planar XeF4 has cancelling opposite bonds; square-pyramidal BrF5 retains an axial resultant.

39. For but-1-ene,cis-but-2-ene,trans-but-2-ene,which decreasing stability order also implies increasing heat released on complete hydrogenation to the same carbon skeleton where directly comparable?

(A) cis-but-2-ene > trans-but-2-ene > but-1-ene

(B) trans-but-2-ene > but-1-ene > cis-but-2-ene

(C) trans-but-2-ene > cis-but-2-ene > but-1-ene

(D) but-1-ene > cis-but-2-ene > trans-but-2-ene

40. An additive retards oxidative deterioration of a fatty food. Which classification separates its function from suppressing microbial growth?

(A) Antioxidant; it opposes oxidation rather than merely identifying a preservative as antimicrobial.

(B) Analgesic; it removes the sensation of pain.

(C) Nylon monomer; it necessarily makes a polyamide.

(D) Tranquilizer; it acts solely on the nervous system.

41. Use octet Lewis contributors for NO3- and O3; for SO2 exclude expanded-octet contributors. For HCOO-, HCO3- and HNO3, average only the two nonprotonated central-atom-O bonds. For other species average all central-atom bonds. Find (average bond order of the first minus that of the second, sum of total lone-pair counts in one contributor of each). A lone pair on any atom is counted once.

(A) (-1/6, 14)

(B) (17/6, 14)

(C) (-1/6, 13)

(D) (-1/6, 15)

42. An energy density obeys u=(7/2) n kT,whereu isJ/m^3 andn counts particles per m^3. WithNA in mol^{-1},which units for k andR=NAk are correct?

(A) k:J $mol^{-1}K^{-1}$;R:J K^{-1}.

(B) k:Pa K^{-1};R:Pa $mol^{-1}K^{-1}$.

(C) k:J K^{-1};R:J $mol^{-1}K^{-1}$.

(D) Both k andR have units J K^{-1} becauseNA is dimensionless.

43. 4-chlorobenzaldehyde is subjected to concentrated aqueous NaOH. Which product or product pair follows the named reaction and side-chain oxidation state?

(A) 4-chlorobenzyl alcohol and sodium 4-chlorobenzoate

(B) benzyl chloride and sodium benzoate

(C) 4-chlorobenzyl alcohol only

(D) 4-chlorophenol and sodium formate

44. A solution initially contains $\frac{1}{125}$ mol Fe^{2+}. Before titration, $\frac{1}{5}$ of it is oxidised quantitatively to Fe^{3+}. The volume in mL of0.0200 M $KMnO_4$ needed for the remaining Fe^{2+} in acid is

(A) 320

(B) 80

(C) 16

(D) 64

45. 2-bromo-2-methylhexane is heated with sodium ethoxide under elimination-favouring conditions. Which major constitutional product is expected,despite the presence of an alkoxide that could be used in a Williamson synthesis with a suitable different halide?

(A) the alkane formed by replacing halogen with H

(B) 2-methylhex-1-ene

(C) the corresponding tertiary alkoxide substitution product

(D) 2-methylhex-2-ene

46. An isolated cathode compartment contains $\frac{2}{25}$mol HA and $\frac{2}{25}$mol NaA,pKa5.00,in a supported NaCl electrolyte. Current5.00A flows for 772s. Each electron generates one OH^-;no anode product enters,buffer loss or volume change occurs. F=96500C/mol andlog3=.477. FinalpH is

(A) $\frac{5477}{1000}$

(B) 5

(C) $\frac{4523}{1000}$

(D) 13

47. A fully dissociated HCl solution has concentration 1/10000 M. What analytical concentration of a monoprotic weak acid HA with Ka=1/10000000 gives approximately the same pH? Neglect water autoionisation and use [H+]$\approx \sqrt{KaC}$, which is to remain in its weak-ionisation regime.

(A) -1/10

(B) 1/10

(C) 1/20

(D) 1/5

48. A closed system goes from state1to2 by path A with q=700 J and work on the system w=-100 J. It returns from2to1 along a different path with q=-900 J. IUPAC work on the return path,in J,is

(A) 100 (B) -300

(C) 300 (D) 600

49. During peroxide-initiated HBr addition to styrene,which propagation sequence explains the anti-Markovnikov major product?

(A) Br radical adds to terminal CH2 to give a benzylic radical; that radical abstracts H from HBr.

(B) Br radical adds to the benzylic carbon to give a primary radical,which is preferred.

(C) H radical first adds to terminal CH2,then the cation captures Br-.

(D) Peroxide forms a benzylic carbocation,then Br- attaches to terminal CH2.

50. Compare acetate and trifluoroacetate as oxygen nucleophiles towards the same unhindered methyl substrate in one fixed solvent. Ignore concentration differences. Which pairing of the stronger nucleophile and the main electronic reason is most appropriate?

(A) acetate; CF3 stabilises the carboxylate and lowers its basicity and attacking tendency

(B) acetate; it is weaker because a localised lone pair cannot attack

(C) trifluoroacetate; resonance always increases local charge on oxygen

(D) trifluoroacetate; the more stabilised anion necessarily attacks faster

51. Compare $CaSO_4$ and $BaSO_4$ in water at the same temperature. Which member is generally more soluble,and which explanation avoids applying one universal down-group rule to all salts?

(A) They must be equally soluble because both cations have charge+2.

(B) $BaSO_4$; every group2salt becomes more soluble as the cation grows.

(C) $CaSO_4$; the larger group2cation gives the much less soluble barium sulfate.

(D) Solubility is fixed solely by formula mass,with no lattice or hydration contribution.

52. A $\frac{3}{100}$ mol FeC_2O_4 inventory is analysed. A fraction $\frac{1}{2}$ of Fe^{2+} has already become Fe^{3+},but every oxalate remains unchanged. Subsequent complete oxidation gives Fe^{3+} and CO_2. Electron equivalents released,in mol,are

(A) $\frac{3}{50}$ (B) $\frac{3}{40}$

(C) $\frac{27}{200}$ (D) $\frac{21}{200}$

53. For a hypothetical MX(s),formation enthalpy is -400 kJ mol^{-1}. M atomisation is100,first ionisation500, X_2 bond dissociation200,and electron-gain enthalpy $X(g)+e^- \rightarrow X^-$ is-300,all in kJ mol^{-1}. The positive lattice dissociation enthalpy of MX is

(A) 700 (B) 900

(C) -800 (D) 800

54. A chemistry table groups compounds used to reduce anxiety and mental tension under one drug class. Which class and distinction are intended?

(A) Preservatives; all are added to food.

(B) Tranquilizers; the stated role differs from neutralising stomach acid.

(C) Sweeteners; all are sugars.

(D) Antacids; all act by binding hydrochloric acid.

55. A sample contains $\frac{1}{20}$ mol $KClO_3$ and a small amount of MnO_2 catalyst.75.0% of the chlorate decomposes by $2KClO_3 \rightarrow 2KCl + 3O_2$. The catalyst is recovered unchanged. Find O_2 formed in mol.

(A) $\frac{3}{40}$

(B) $\frac{9}{80}$

(C) $\frac{3}{80}$

(D) $\frac{9}{160}$

56. A student predicts aqueous behaviour of K using only gas-phase first ionisation energy. Which correction is justified?

(A) The fastest water reaction identifies the most negative standard potential.

(B) Hydration cancels from comparisons because all alkali ions have charge +1.

(C) Aqueous potentials combine atomisation, ionisation and hydration contributions.

(D) Aqueous potentials follow atomic radius once ionisation energy is included.

57. For a fixed-composition closed system dG=V dP-S dT. At a state use V=3 m3, S=4 J/K and a local path with dT/dP=-1 K/Pa. Find dG/dP along this path in m3. The supplied values are a formal local thermodynamic model; 1 Pa m3=1 J.

(A) 7

(B) 7/2

(C) 14

(D) -7

58. A food extract contains exactly one of the listed two-component combinations. Using the usual qualitative tests under suitable conditions, its results are: Benedict on warming: positive; iodine: negative; Biuret: positive; Sudan III: negative. Which combination matches?

(A) glucose and starch

(B) glucose and albumin

(C) starch and albumin

(D) glucose and triolein

59. Starting with toluene,apply hot alkaline KMnO4,then acidification,then concentrated HNO3/H2SO4. Which organic product description correctly tracks the functional group actually present at each step?

(A) o-nitrobenzoic acid

(B) p-nitrobenzoic acid

(C) p-nitrotoluene

(D) m-nitrobenzoic acid

60. A nonvolatile solute in 10mol water gives relative vapour-pressure lowering1/101. Remove 5mol water without solute loss. Under exact Raoult mole fractions,new relative lowering is

(A) $\frac{1}{101}$

(B) $\frac{1}{51}$

(C) $\frac{1}{50}$

(D) $\frac{2}{101}$

English Proficiency

61. Which ONE of the following words is misspelt?

(A) maintenance (B) possession

(C) pronunciation (D) concience

62. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. the researcher considered Q. under the same conditions. R. what evidence each explanation would predict S. Before deciding which explanation to accept,

(A) Q-P-R-S (B) S-P-R-Q

(C) P-S-R-Q (D) R-S-P-Q

63. Read the sentence: With the display stand missing, the group had to improvise using materials already in the room. Choose the closest meaning of 'improvise' as used in the sentence.

(A) repair an existing object (B) devise without prior preparation

(C) make a temporary copy (D) follow a simplified plan

64. Read the sentence: The coating was added to preserve the surface from decay. Choose the opposite in meaning of 'preserve' as used in the sentence.

(A) clean (B) protect

(C) destroy (D) retain

65. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. Did the assistant copy the names exactly as they appeared on the form?

(A) Were the names changed by the assistant from those on the form?

(B) Were the names copied by the assistant exactly as they appeared on the form?

(C) Was the names copied by the assistant exactly as they appeared on the form?

(D) Were the forms copied by the names exactly as the assistant appeared?

66. Read the sentence: The two guides are complementary: each supplies information missing from the other. Choose the closest meaning of 'complementary' as used in the sentence.

(A) identical in content (B) provided without charge

(C) expressing praise (D) mutually completing or enhancing

67. Her research concerns the structure and growth of plants, not the design of gardens. What is she?

(A) botanist (B) geologist

(C) gardener (D) zoologist

68. Read the sentence: The official tried to suppress the report and prevent its circulation. Choose the opposite in meaning of 'suppress' as used in the sentence.

(A) criticise (B) conceal

(C) publicise (D) revise

69. Read the sentence: He felt under the weather and stayed home rather than attend the crowded meeting. What does 'under the weather' mean here?

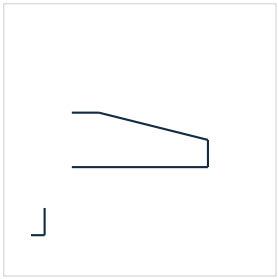
(A) slightly unwell
(B) concerned about the forecast
(C) afraid of the audience
(D) unable to find shelter

70. Read the sentence: The worked example was scaffolding that could be removed once learners understood the method. What does the figurative expression mean in this context?

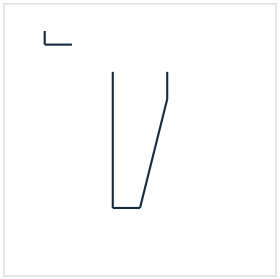
(A) It was temporary support for developing independent understanding.
(B) It was a permanent rule that learners must always copy.
(C) It was a decoration with no part in learning.
(D) It was proof that no further practice was necessary.

Logical Reasoning

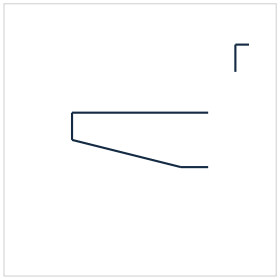
71. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

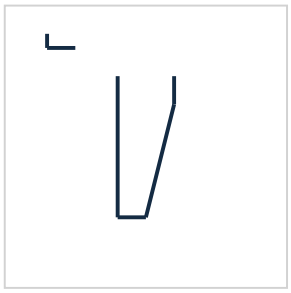


2

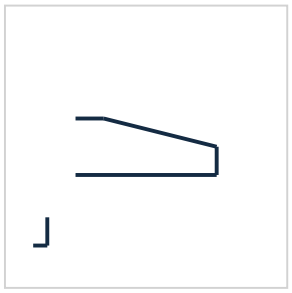


3

(A) Pattern V



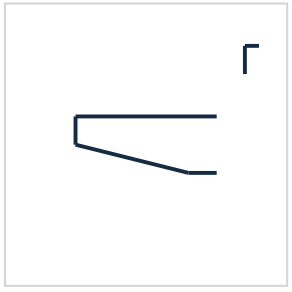
(B) Pattern Y



(C) Pattern R



(D) Pattern M



72. Three pairs contain exactly one string instrument and one wind instrument. Which pair does not?

(A) Sitar / Trombone
(B) Harp / Saxophone
(C) Ukulele / Violin
(D) Cello / Clarinet

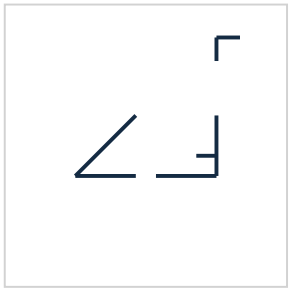
73. Bina receives 2 times as many tokens as Asha, and Charu receives 2 times as many as Bina. Their total is 77. After the allocation, Charu gives 5 tokens to Asha. How many tokens does Charu then retain?

(A) 49
(B) 6
(C) 39
(D) 44

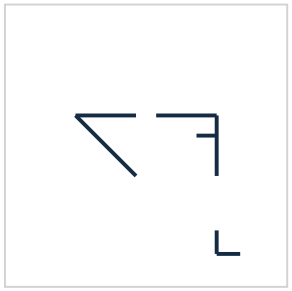
74. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



(A) Pattern L



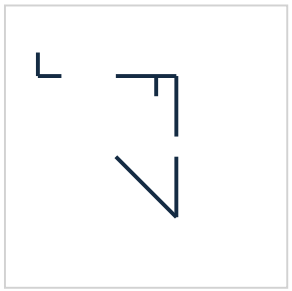
(B) Pattern U



(C) Pattern M



(D) Pattern R



75. Preserve the exact relation and its order: Genuine : Authentic :: ? : Rare. Choose the missing term.

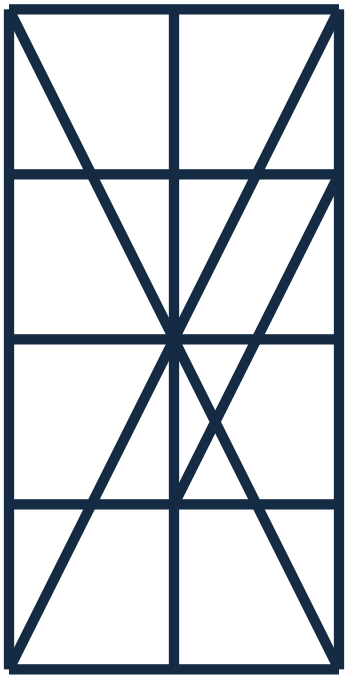
(A) Silent
(B) Scarce
(C) Begin
(D) Rapid

76. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, CLOUD is coded as HQTZI. Which word is coded as XYTSJ?

(A) RSNMD (B) CDYXO

(C) TUPOF (D) STONE

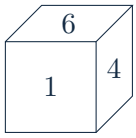
77. How many triangles are formed by all the drawn segments?



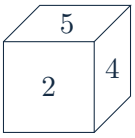
(A) 67 (B) 28

(C) 36 (D) 39

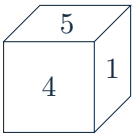
78. The views show the same die. Give the ordered pair: (face opposite 2, face opposite 4). Do not assume opposite labels sum to seven.



1



2



3

(A) $(1, 2)$ (B) $(1, 3)$

(C) $(1, 4)$ (D) $(1, 1)$

79. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 18$ and $5 \mapsto 36$. Which input maps to 66?

(A) 10 (B) 11

(C) 9 (D) 66

80. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 28 and Z has code 3. Decode the separated values 26, 17, 14, 8, 25.

(A) CLOUD

(B) DUOLC

(C) DMPVE

(D) BKNTC

81. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

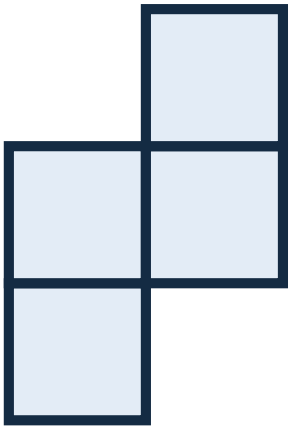


piece 1

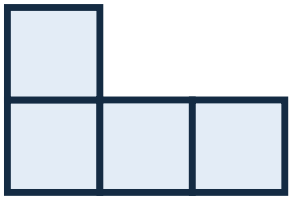


piece 2

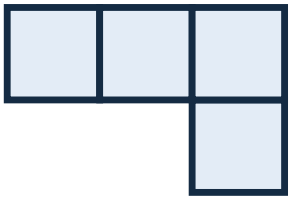
(A) Pattern V



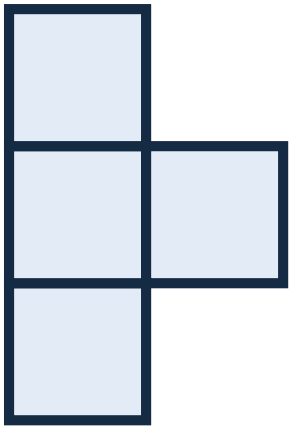
(B) Pattern L



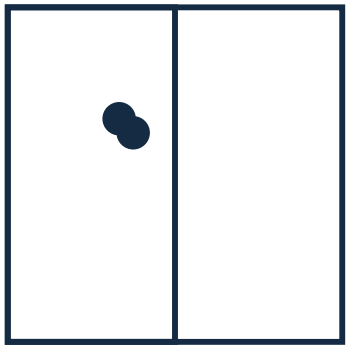
(C) Pattern Z



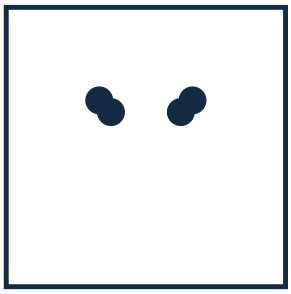
(D) Pattern Y



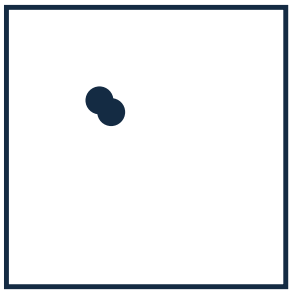
82. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



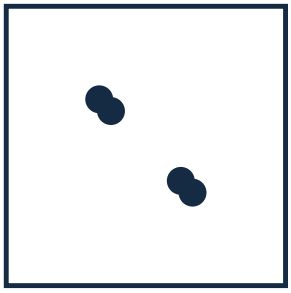
(A) Pattern L



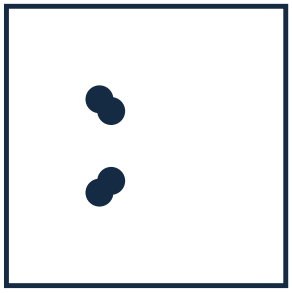
(B) Pattern T



(C) Pattern U



(D) Pattern N



83. Consider only these statements; universal statements alone do not imply existence. All campers are designers; Some guides are campers. Conclusions: I. Some guides are designers. II. All guides are designers. Which conclusions follow necessarily?

(A) Only II

(B) Only I

(C) Both I and II

(D) Neither I nor II

84. Tara points to a photograph and says, "This person is my mother's only son." Tara is female, and the photographed person is male. How is the photographed person related to Tara?

(A) Sister

(B) Brother

(C) Daughter

(D) Son

85. Each letter position has its own fixed cyclic shift, applied identically in both pairs (Z wraps to A). Complete IWQ : LUR :: PJH : ?.

(A) RGH (B) TIJ

(C) SHI (D) KHQ

86. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 7, 14, 21, ?, 49, 98, 105.

(A) 35 (B) 56

(C) 42 (D) 49

87. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

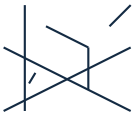




host 1



host 2



host 3

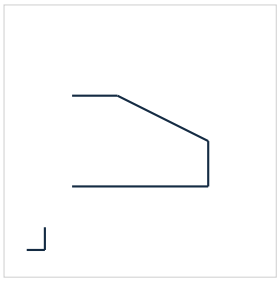


host 4

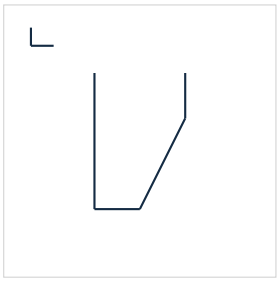
(A) 1 (B) 2

(C) 3 (D) 4

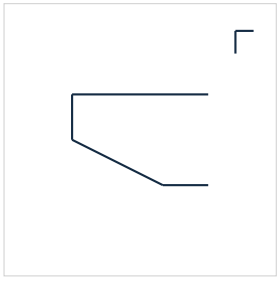
88. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

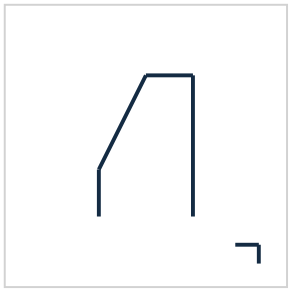


bottom left

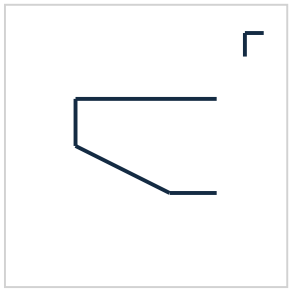
?

bottom right

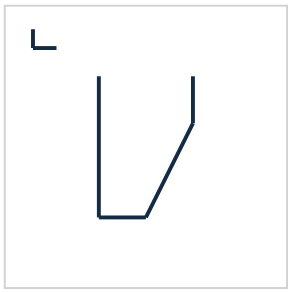
(A) Pattern P



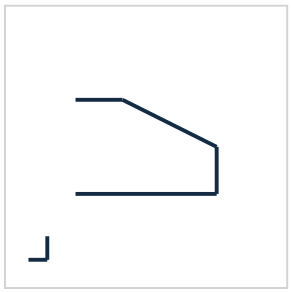
(B) Pattern V



(C) Pattern Z



(D) Pattern U



89. In a row, Asha is 12th from the left and 23rd from the right. Bina is 9th from the right. Charu stands 2 places to the left of Bina. How many people stand strictly between Asha and Charu?

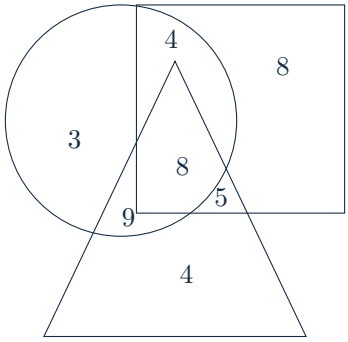
(A) 4

(B) 10

(C) 13

(D) 11

90. Circle=A, triangle=B, square=C. Each printed number is the count in that exact region. Find the total inside A but outside B, whether inside or outside C.



(A) 4 (B) 8

(C) 3 (D) 7

Mathematics

91. A complex number z satisfies z+conjugate(z)=14 and z-conjugate(z)=6i. Find z·conjugate(z).

(A) $\sqrt{58}$ (B) 85

(C) 58 (D) 116

92. F′(x)=$\frac{(x+1)^2}{(x+2)^4}$ and F(0)=4. Find F(3)-2F(0).

(A) $-\frac{3871}{1000}$ (B) $\frac{4129}{1000}$

(C) $-\frac{4129}{1000}$ (D) $\frac{129}{1000}$

93. Parabola y=7(x-6)(x-7) meets the axes at three distinct points. A circle passes through these points. Find its other y-intercept.

(A) 1 (B) $-\frac{1}{7}$

(C) $\frac{1}{7}$ (D) 7

94. For f(x)=x^3-3kx^2+75x, how many integer k in [0,10] give a local maximum at a positive x?

(A) 6 (B) 4

(C) 10 (D) 5

95. X earns 2 and uses 2 labour hours; Y earns 6 and uses 3 hours. There are 80 hours and at least 3 items of Y must be made. Which model is correct?

(A) Maximize $2x + 6y$; $2x + 3y \leq 80$, $x \geq 0$, $y \geq 3$ (B) Maximize $6x + 2y$; $2x + 3y \leq 80$, $y \leq 3$

(C) Maximize $2x + 6y$; $2x + 3y \geq 80$, $x, y \geq 0$ (D) Minimize $2x + 6y$; $2x + 3y \leq 80$, $y \geq 3$

96. An AP has a_1=7, d=7. Find a_1+a_3+$\cdots$+a_15.

(A) 448 (B) 896

(C) 252 (D) 343

97. For 3x+4y+7=0 and 4x+3y+4=0, find the squared distance from the origin to the angle bisector of positive slope.

(A) $\frac{121}{98}$ (B) $\frac{3\sqrt{2}}{2}$

(C) 9 (D) $\frac{9}{2}$

98. What is the maximum number of distinct intersection points formed by 9 circles and 3 straight lines, with no coincident curves?

(A) 201 (B) 129

(C) 126 (D) 102

99. Evaluate lim as n$\to\infty$ of (4/n)Σ|4 j/n-4/3|, j=1,...,n.

(A) 8 (B) $\frac{64}{9}$

(C) $\frac{40}{9}$ (D) $\frac{32}{9}$

100. $A = \begin{bmatrix} 7 & 1 \\ 0 & 2 \end{bmatrix}$. Find det($A^T$A).

(A) 28 (B) 196

(C) -196 (D) 14

101. Find the product of all real solutions of sqrt(13-log_8|x|)=log_8|x|-1.

(A) -16777216 (B) 0

(C) -4096 (D) 16777216

102. The same loaded die has odd-face probability $\frac{4}{5}$. Three independent throws are made. Given that the first two have an even sum, find the probability that the total of all three is odd.

(A) $\frac{8}{25}$ (B) $\frac{1}{5}$

(C) $\frac{76}{125}$ (D) $\frac{4}{5}$

103. How many x with $0\le x<4\pi$ satisfy $\sin^4x+\cos^4x=3/4$ and cos x>0?

(A) 16 (B) 0

(C) 8 (D) 4

104. X takes values [-6, 0, 3] with probabilities 1/6,1/3,1/2. Let Y=6X+9. Find $E(Y^2)-[E(Y)]^2$.

(A) 450 (B) 369

(C) $\frac{123}{2}$ (D) 378

105. Let k make lim as x$\to\infty$ of $[(2x^2+3x+1)/(x+5)-kx]$ finite. Find that finite limit.

(A) 13 (B) 10

(C) -7 (D) 3

106. Let f(x)=sin x+7 cos x. Let [L,L+π] be the interval containing 0 in its interior on which f is increasing and bijective onto[-sqrt(50),sqrt(50)]. If g is its inverse, findL+$g'(1)$.

(A) $-\frac{\pi}{2}+\frac{1}{7}+\operatorname{atan}(7)$

(B) $-\frac{\pi}{2}-\operatorname{atan}(7)$

(C) $-\frac{\pi}{2}-\operatorname{atan}(7)+\frac{\sqrt{2}}{10}$

(D) $-\frac{\pi}{2}-\operatorname{atan}(7)+\frac{1}{7}$

107. A convergent GP has first term 9, ratio $\frac{1}{9}$. Find the sum of all terms after its first 5 terms.

(A) $\frac{1}{5184}$ (B) $\frac{1}{6561}$

(C) $\frac{1}{52488}$ (D) $\frac{1}{5832}$

108. For x$\in[\pi/8,\pi/2]$, find the maximum of sin x/x.

(A) 1 (B) $\frac{\pi}{4\sqrt{2-\sqrt{2}}}$

(C) $\frac{2}{\pi}$ (D) $\frac{4\sqrt{2-\sqrt{2}}}{\pi}$

109. Find the coefficient of x^6 in $(1+x+x^2)(1+x^2)^{10}$.

(A) 330 (B) 165

(C) 120 (D) 360

110. The line y=mx through O bisects the area of the triangle with vertices O,(5,4),(4,4). Assume the vertices are noncollinear. Find m.

(A) $\frac{8}{9}$ (B) 0

(C) $\frac{9}{8}$ (D) $\frac{4}{5}$

111. Find the population variance of 2+2, 4+2, ..., 22+2.

(A) 10 (B) 40

(C) 20 (D) 42

112. α,β are roots of x^2-17x+66=0. The monic polynomial with roots α+1,β+1 is x^2-Ax+B. Find B.

(A) 84 (B) 67

(C) 361 (D) 50

113. ABC is a right triangle on level ground with AB=12, AC=12, angle A=90°. A vertical tower stands at midpoint M of BC. Its top elevation from A is 60°. Find its height.

(A) $6\sqrt{3}$ (B) $6\sqrt{2}$

(C) $12\sqrt{6}$ (D) $6\sqrt{6}$

114. Unit vectors a,b,c satisfy $|(a\times b)\cdot c|=1$. Find $|a+4b-2c|^2$.

(A) 37 (B) 20

(C) 21 (D) 7

115. Evaluate the integral from 0 to $\pi/2$ of [(sin x+cos x)/(sec x+csc x)](sin x+7 cos x) dx, using continuous extension at the endpoints.

(A) 4 (B) -2

(C) $\frac{8}{3}$ (D) $\frac{7}{3}$

116. In how many ordered ways can three distinguishable fair dice have sum 12, show at least one 6, and have all three faces different?

(A) 2 (B) 12

(C) 18 (D) 15

117. $A = \begin{bmatrix} 0 & 3 & -3 \\ -3 & 0 & 2 \\ 3 & -2 & 0 \end{bmatrix}$. Find det(I+A)-(9+9+4).

(A) 0 (B) 22

(C) -1 (D) 1

118. A line through P=(3,4) has direction (3,4). It meets 3x+4y=75 at Q. Find $|PQ|^2$.

(A) 10 (B) 50

(C) 100 (D) 4

119. If z=6-5i, find |exp(iz)|.

(A) e^5 (B) 1

(C) e^{-5} (D) e^6

120. How many integers x in [-4,14] satisfy (|x-5|-3)/(|x-5|-8)>0?

(A) 19 (B) 9

(C) 5 (D) 7

121. The four lines $\pm x/18 \pm y/12 = 1$ form a rhombus with vertices on the coordinate axes. Find its area.

(A) 216 (B) 108

(C) 432 (D) 864

122. On A={0,1,...,9}, define aRb if 5 divides a-b. How many ordered pairs are in R?

(A) 40 (B) 20

(C) 10 (D) 100

123. The curve $y=x^3+3x$ divides the disc $x^2+y^2 \leq 4$. Find the area of the part satisfying $y \geq x^3+3x$.

(A) 2π (B) $3 + 2\pi$

(C) 4π (D) π

124. Evaluate $\sum_{r=0}^{8} r(r-1)\binom{8}{r}$.

(A) 3584 (B) 7168

(C) 1024 (D) 4096

125. Evaluate $\Sigma\ j/2^j$ for j=1,...,11.

(A) 2 (B) $\frac{4083}{2048}$

(C) $\frac{11}{2048}$ (D) $\frac{1021}{512}$

126. r=pa+qb, a=(1,0,0), b=(1,1,0), r·a=9, r·b=13. Find $|r|^2$.

(A) 97 (B) 41

(C) $\sqrt{97}$ (D) 13

127. Let $D(t) = \det \begin{pmatrix} 2+t & 1+t \\ 1+t & 8+t \end{pmatrix}$. Find D′(0).

(A) 12 (B) 8

(C) −12 (D) 30

128. A chord of $x^2+y^2=12100$ touches $x^2/121-y^2/25=1$. Its nonzero midpoint (h,k) satisfies h=k. Find h^2+k^2.

(A) 96 (B) 48

(C) 24 (D) 73

129. Let A,B be the intersections of x=14 with $x^2/49-y^2/25=1$. Find the area of triangle OAB.

(A) 70 (B) $5\sqrt{3}$

(C) $70\sqrt{3}$ (D) $140\sqrt{3}$

130. How many solutions in $0\le x<3\pi$ satisfy sin 2 x=sin x and cos x>sin x?

(A) 2 (B) 4

(C) 6 (D) 3

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A rigid ladder of length 30 m slides with its ends on a floor and a vertical wall. At the instant its top is at height $4L/5$, the foot moves away at 3 m/s. What is the downward speed of the top?

(A) $\frac{12}{5}$ m/s (B) $\frac{9}{4}$ m/s

(C) $\frac{9}{5}$ m/s (D) 4 m/s

132. A triangular p-V cycle has vertices (V_0, p_0), $(V_0 + 3\,\text{L}, p_0)$ and $(V_0, p_0 + 5\,\text{kPa})$. Traversed in that listed order and then back, find net work done by the gas.

(A) $\frac{-15}{2}$ J (B) -15 J

(C) 15 J (D) $-\frac{15}{4}$ J

133. Two short magnetic dipoles of equal moment 4 A m^2 are arranged so that point P is at axial distance 3 m from the first and equatorial distance 3 m from the second. Their field vectors at P are perpendicular. Find resultant magnitude in units of $\mu_0/(4\pi)$.

(A) $\frac{4\sqrt{2}}{27}$ (B) $\frac{4}{27}$

(C) $\frac{4\sqrt{5}}{27}$ (D) $\frac{4\sqrt{5}}{9}$

Chemistry

134. In a hydrogen-like ion Z=3,an electron falls from n=4 to an s orbital with 1 radial nodes. With hydrogen ionisation energy13.6 eV and$hc = 1240$ eV nm,the emitted wavelength in nm is

(A) $\frac{6200}{459}$ (B) $\frac{24800}{459}$

(C) $\frac{24800}{153}$ (D) $\frac{24800}{51}$

135. H_2O and NH_4^+ are described by localised hybrid-orbital models. Which pair can be true without assigning the same molecular shape to both?

(A) H2O is sp because it has only two bonded atoms; NH4+ is sp3.

(B) Both are sp3 and therefore both must be tetrahedral molecules.

(C) Both central atoms are described as sp^3; H_2O is bent and NH_4^+ is tetrahedral.

(D) Both central atoms have two lone pairs because each needs an octet.

136. Within an additive covalent-radius model,a single X-X bond is 140 pm and an X-H bond is 110 pm. A linear H-X-X-H skeleton is assumed with collinear single bonds. Its terminal H-to-H distance,in pm,is

(A) 360 (B) 500

(C) 250 (D) 440

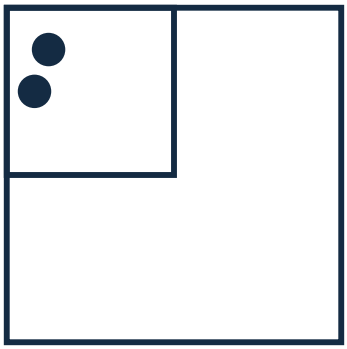
Logical Reasoning

137. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: YTHO, CVFR, ?, KZBX, OBZA, SDXD.

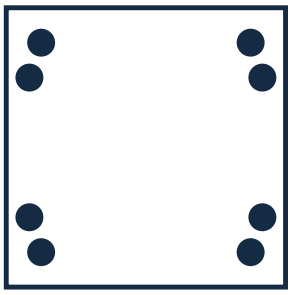
(A) UDXG (B) GXDU

(C) FWCT (D) HYEV

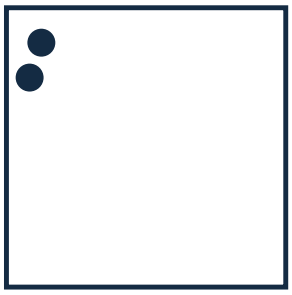
138. Starting with a unit square sheet, fold right to left along the vertical midline, then fold bottom to top along the horizontal midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



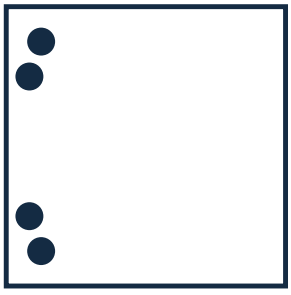
(A) Pattern L



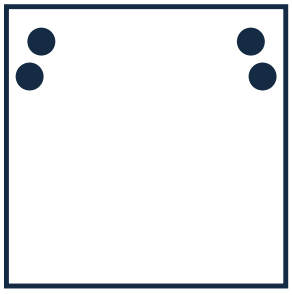
(B) Pattern W



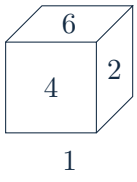
(C) Pattern M



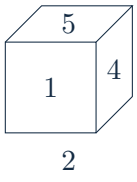
(D) Pattern T



139. The views show the same die. Give the ordered pair: (face opposite 4, face opposite 2). Do not assume opposite labels sum to seven.



1



2

(A) $(3, 2)$ (B) $(3, 1)$

(C) $(3, 5)$ (D) $(3, 3)$

Mathematics

140. F′(x)=$2x-\frac{2}{x^3}$ and F(1)=4. Find F(2)-2F(1).

(A) $-\frac{25}{4}$ (B) $-\frac{7}{4}$

(C) $\frac{25}{4}$ (D) $\frac{9}{4}$

141. a=(9,4,0), b=(4,-9,2). Find $|a+b|^2$-$|a-b|^2$.

(A) −8 (B) 288

(C) 0 (D) 9797

142. a=(6,0,0), b=(0,2,0), c=(0,0,4), p=a+b. Evaluate [p×(b×c)]·(a+c).

(A) −64 (B) −208

(C) 64 (D) 0

BITSAT Mock 28

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A conducting rod of length 4 m moves at constant speed 2 m/s on parallel rails in a uniform perpendicular field 8 T. Total circuit resistance is 3 ohms. Neglect self-inductance. Find the external force needed to maintain the motion.

(A) $\frac{4096}{3}$ N
(B) 64 N
(C) $\frac{256}{3}$ N
(D) $\frac{2048}{3}$ N

2. An ideal transformer has secondary-to-primary turns ratio 7. Its primary draws 3 A at 120 V while delivering power to a resistive secondary load. Find secondary current.

(A) 147 A
(B) 3 A
(C) $\frac{3}{7}$ A
(D) $\frac{3}{49}$ A

3. Two small identical conducting spheres initially carry positive charges $3q_0$ and $3q_0$ and are separated by distance d. They touch and are returned to the same separation. Find the final force divided by the initial force.

(A) $\frac{1}{3}$
(B) 4
(C) 1
(D) $\frac{3}{4}$

4. A rigid ladder of length 10 m slides with its ends on a floor and a vertical wall. At the instant its top is at height $4L/5$, the foot moves away at 4 m/s. What is the downward speed of the top?

(A) 3 m/s
(B) $\frac{16}{3}$ m/s
(C) $\frac{16}{5}$ m/s
(D) 4 m/s

5. A standing wave is $y = 6\sin(3x)\cos(2t)$ cm with x in metres and t in seconds. At a point where $3x = \pi/6$, find the particle-speed magnitude when $2t = \pi/3$.

(A) $3\sqrt{3}$ cm/s
(B) 6 cm/s
(C) $6\sqrt{3}$ cm/s
(D) 12 cm/s

6. A particle of mass 2 kg starts from rest and experiences $F(t) = 4t$ N along its motion for 2 s. Find the work done by the force.

(A) $\frac{32}{3}$ J (B) 8 J

(C) 16 J (D) 64 J

7. A charge of $4\,\mu$C moves from potential 60 V to -60 V in an electrostatic field. Find the work done by an external agent if the charge starts and ends at rest and no other force does work.

(A) $-240\,\mu$J (B) $240\,\mu$J

(C) $-120\,\mu$J (D) $-480\,\mu$J

8. A solid sphere rolls without slipping from rest through vertical height 8 m. A thin ring does so through vertical height 6 m. Find the ratio of their squared centre-of-mass speeds, sphere to ring.

(A) $\frac{4}{3}$ (B) $\frac{21}{40}$

(C) $\frac{2\sqrt{210}}{21}$ (D) $\frac{40}{21}$

9. One mole of nonoverlapping identical spheres of radius r has total geometric material volume V. A second sample contains $\frac{17}{20}$ mole of spheres of radius $2r$. What is its total geometric material volume divided by V? Ignore packing voids.

(A) $\frac{160}{17}$ (B) $\frac{17}{5}$

(C) $\frac{34}{5}$ (D) $\frac{4913}{1000}$

10. A balanced Wheatstone bridge has arms AB=AD=4 ohms and BC=DC=16 ohms. A resistor of arbitrary positive resistance connects B to D. Find equivalent resistance between A and C.

(A) $20\,\Omega$ (B) $\frac{16}{5}\,\Omega$

(C) $10\,\Omega$ (D) $18\,\Omega$

11. A fixed amount of ideal gas has its absolute temperature multiplied by 5 and volume by 4. Find the final pressure divided by the initial pressure.

(A) $\frac{1}{5}$ (B) 20

(C) 5 (D) $\frac{5}{4}$

12. Two ferromagnetic samples have similar remanent flux density. Their coercive fields are 200 A/m and 600 A/m. A reversing field of magnitude 400 A/m will act in service. Which sample is better at retaining its original magnetisation against this field, based only on the stated coercivities?

(A) The sample with coercive field 600 A/m. (B) Both are necessarily demagnetised equally.

(C) Neither can have nonzero remanence. (D) The sample with coercive field 200 A/m.

13. A uniform-density planet has radius R. Gravity at a depth d equals gravity at a height $(9-1)R$ above the surface. Find d/R.

(A) $\frac{80}{81}$ (B) $\frac{8}{9}$

(C) 1 (D) $\frac{1}{81}$

14. A particle is projected vertically outward from the surface of a nonrotating planet with speed $v_e/6$, where v_e is surface escape speed. Neglect atmosphere. Find maximum height divided by planet radius.

(A) $\frac{36}{35}$ (B) $\frac{1}{35}$

(C) $\frac{1}{36}$ (D) $\frac{1}{37}$

15. A liquid of volume 2 litres and bulk modulus 5000000000 Pa is subjected to an additional pressure of 5000000 Pa. Neglect changes in bulk modulus. Find the magnitude of volume decrease in cubic centimetres.

(A) $1\,\mathrm{cm}^3$ (B) $2\,\mathrm{cm}^3$

(C) $2000000\,\mathrm{cm}^3$ (D) $\frac{1}{500}\,\mathrm{cm}^3$

16. A circular parallel-plate capacitor of plate radius 2 cm is charged at 48 mA. Neglect fringing. What displacement current passes through the annular part of the gap between radii 1 cm and 2 cm, centred on the plate axis?

(A) 36 mA (B) 12 mA

(C) 24 mA (D) 48 mA

17. Two sound sources produce wavelengths $\frac{1}{3}$ m and $\frac{1}{4}$ m in the same gas at one temperature. The sound speed there is 300 m/s. At a second temperature the sound speed is 3/2 times larger, while both source frequencies stay unchanged. Find the new beat frequency.

(A) 900 Hz (B) 200 Hz

(C) 2100 Hz (D) 300 Hz

18. A 5 kg block is pulled by a force of 25 N at an angle with $\cos\theta = 4/5$, $\sin\theta = 3/5$ above a horizontal floor. The block slides; $\mu_k = \frac{1}{5}$. Find the net horizontal force. Take $g = 10\ \mathrm{m/s}^2$.

(A) 9 N (B) 20 N

(C) 18 N (D) 13 N

19. A uniform wire is stretched to $\frac{3}{2}$ times its original length at constant volume. Its resistivity increases by 15 percent during the process. Find final resistance divided by initial resistance.

(A) $\frac{23}{30}$ (B) $\frac{207}{80}$

(C) $\frac{23}{45}$ (D) $\frac{9}{4}$

20. A cell balances on a potentiometer at 100 cm on open circuit and at 30 cm when a 6 ohm resistor is connected across it. The potentiometer gradient is unchanged. Find the cell's internal resistance.

(A) $\frac{21}{5}\,\Omega$ (B) $14\,\Omega$

(C) $\frac{9}{5}\,\Omega$ (D) $20\,\Omega$

21. A uniformly charged nonconducting sphere has radius R and surface potential V_s relative to infinity. Find potential at $r = R/12$ divided by V_s.

(A) $\frac{431}{288}$ (B) 1

(C) $\frac{143}{144}$ (D) $\frac{1}{12}$

22. A series RLC circuit is resonant at angular frequency ω_0. Its capacitance is then multiplied by 144 while L stays fixed. What angular frequency restores resonance, as a fraction of ω_0?

(A) $\frac{1}{12}$ (B) $\frac{\sqrt{3}}{6}$

(C) 1 (D) 144

23. A boat moves at 25 m/s relative to water. The current is 15 m/s and river width is 48 m. The boat lands directly opposite its starting point. Find crossing time.

(A) $\frac{16}{5}$ s (B) $\frac{24}{5}$ s

(C) $\frac{24\sqrt{34}}{85}$ s (D) $\frac{12}{5}$ s

24. A wire carries $i(t) = 3 + 2\sin\omega t$ A for many cycles. Find its rms current.

(A) $\frac{\sqrt{26}}{2}$ A (B) $\sqrt{11}$ A

(C) $\sqrt{2}$ A (D) 5 A

25. A plane wave passes through a thin prism away from edges. Separately, a pointlike aperture produces an outgoing spherical wavefront of radius 14 cm, which propagates another 14 cm in a homogeneous medium. Which pair correctly describes the prism wavefront and the later aperture wavefront?

(A) Approximately plane and tilted; spherical with radius 0 cm.

(B) Approximately plane and tilted; spherical with radius 28 cm.

(C) Approximately plane and unchanged in direction; spherical with radius 14 cm.

(D) Spherical; plane with infinite radius.

26. A gas receives 300 J of heat while its surroundings do 100 J of work on it. It then loses 150 J of heat at fixed volume. Find its total internal-energy change.

(A) 50 J (B) 550 J

(C) 250 J (D) 400 J

27. The quantity $Z = x^3y^1/\sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 3, 2 and 6, respectively. To first order, what is the maximum percentage error in Z?

(A) 17 (B) 14

(C) $\sqrt{94}$ (D) 8

28. A plane wave in vacuum travels along $+x$. At one instant its electric field is $450\hat{k}$ V/m. If the electric amplitude is 2 times this instantaneous value, which gives the instantaneous magnetic field? Use $c = 3 \times 10^8$ m/s.

(A) $-300 \times 10^{-8}\hat{j}$ T (B) $-150 \times 10^{-8}\hat{j}$ T

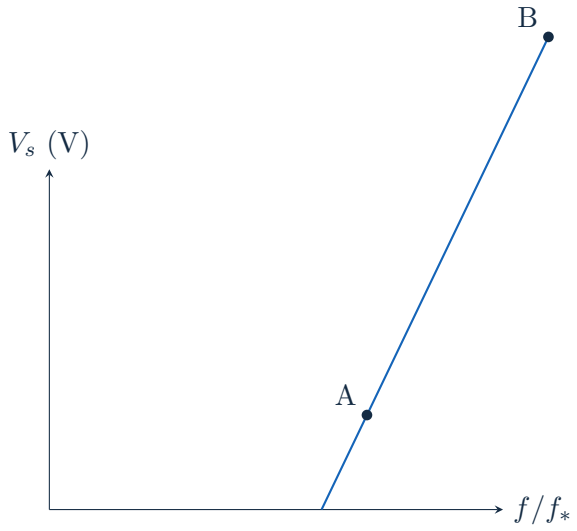
(C) $+150 \times 10^{-8}\hat{i}$ T (D) $+150 \times 10^{-8}\hat{j}$ T

29. 1000 identical conducting liquid drops merge without losing charge. Before merging each has potential V and surface charge density σ. Find the final surface charge density divided by σ.

(A) $\frac{1}{10}$ (B) 100

(C) 10 (D) 10000

30. For a metal, the stopping-potential graph has points A $(7f_*, 5\,\text{V})$ and B $(11f_*, 25\,\text{V})$. The electronic gain of the vertical recording channel is later found to be 3: its recorded voltage is 3 times the true stopping voltage. What is hf_*/e in volts?



(A) $\frac{5}{3}$ V
(B) 5 V
(C) 15 V
(D) $\frac{20}{3}$ V

Chemistry

31. E°(A+/A)=1/10 V and E°(B+/B)=3/10 V. For the stoichiometrically scaled standard reaction 2A(s)+2B+(aq)→2A+(aq)+2B(s), find ΔG° in kJ. Use F=96.5 kJ V-1 mol-1.

(A) -386/5
(B) -193/10
(C) 193/5
(D) -193/5

32. A model M(OH)2 has molar solubility 1/25000 M in pure water; neglect water autoionisation there. In a buffered solution its measured molar solubility is 1/35156250 M. The buffer fixes free OH- and the solid's contribution to OH- is negligible. What is the imposed OH- concentration?

(A) 3/500
(B) 9/1000000
(C) 3/2000
(D) 3/1000

33. A first-order reaction hasEa/R=7200K andfrequency factorA=(ln2)e^{24}/60s^{-1}. At what nearest whole-kelvin temperature is75.0% completion reached in60.0s? Useln2=.693.

(A) 309
(B) 290
(C) 330
(D) 300

34. A Fischer projection of 2,3-dibromobutane has its four-carbon chain vertical (CH3 at top and CH3 at bottom). At C2,Br is on the left; at C3,Br is on the left; each other horizontal group is H. Horizontal bonds point toward the observer. Which classification and configuration pair is correct?

(A) chiral; (2R,3S)
(B) chiral; (2S,3S)
(C) chiral; (2R,3R)
(D) meso; (2R,3S)

35. List all acyclic saturated carbon skeletons with formula C5H12. Count constitutional isomers,not conformers,and do not count alternative numbering of one skeleton twice. The count is

(A) 2
(B) 4
(C) 3
(D) 5

36. For NH_3,which classification and consequence are mutually consistent?

(A) covalent; N-H bonds belong to a molecular electron-rich hydride.
(B) Covalent; nonstoichiometry proves all hydrogen has been removed.
(C) Ionic; every example must be a gas composed of discrete molecules.
(D) All hydrides contain free H+ ions in their solid lattices.

37. With one equivalent of cold concentrated HI and isolation before prolonged further reaction,which initial cleavage pair is expected from ethyl tert-butyl ether?

(A) ethyl iodide and tert-butanol only
(B) but-1-ene and methanol
(C) tert-butyl iodide and ethanol
(D) two equivalents ethyl iodide

38. The same element forms a fully occupied BCC phase A with cell edge a and an FCC phase B with edge 2a. The measured density ratio ρ_B/ρ_A is 1/8. All FCC sites have equal occupancy, atomic mass is unchanged, and vacancies are the only defects. What fraction of FCC sites is vacant?

(A) 1/2
(B) 3/4
(C) 1/6
(D) 1/4

39. A freshly assayed 60-volume H_2O_2 solution releases 60 L O_2 at STP per litre on complete decomposition. Only80.0% of the evolved oxygen reaches a burner that requires0.560 L O_2 at STP. The solution volume needed,in mL,is

(A) $\frac{14}{3}$
(B) $\frac{28}{3}$
(C) $\frac{70}{3}$
(D) $\frac{35}{3}$

40. An ideal mixture initially has$\frac{3}{10}$molA and$\frac{2}{5}$molB. A selective step removes half ofB and noA. At the new equilibriumT,V,totalpressure is8bar. Partialpressure ofA,in bar,is

(A) $\frac{24}{7}$
(B) $\frac{24}{5}$
(C) 4.0
(D) $\frac{40}{3}$

41. Natural rubber is heated with a controlled amount of sulphur. Which structural change best explains reduced permanent flow without calling the product a new polyester?

(A) All C=C bonds are converted into peptide bonds.
(B) Every chain is cleaved into free glucose.
(C) Sulphur replaces every carbon atom in the backbone.
(D) Sulphur bridges link neighbouring rubber chains.

42. ethylcyclopentane undergoes low-conversion monobromination with Br2/light. Which is the major constitutional product? Use the usual strong tertiary-over-secondary-over-primary bromination selectivity.

(A) 2-bromo-1-ethylcyclopentane
(B) 1-(1-bromoethyl)cyclopentane
(C) 1-bromo-1-ethylcyclopentane
(D) bromomethylcyclopentane

43. A sample of but-2-yne is treated with H2/Lindlar catalyst. Which isolated product is consistent with the reagent limitation and carbon-skeleton change?

(A) cis-but-2-ene
(B) but-1-ene
(C) butane
(D) trans-but-2-ene

44. Methyl propanoate is treated with excess NH3 to form its primary amide,which is then converted with Br2/NaOH to a free amine A. Which pair correctly identifies A and the increasing normal-boiling-point order of dimethylamine,A and ethanol? Use their ordinary pure-compound behaviour.

(A) A=trimethylamine;A < dimethylamine < ethanol.
(B) A=ethylamine;dimethylamine < A < ethanol.
(C) A=propylamine;dimethylamine < ethanol < A.
(D) A=ethylamine;ethanol < A < dimethylamine.

45. An octahedral complex has d^9 configuration and $\Delta_o = 250$ kJ mol^{-1}. Ignoring pairing energy,its d-electron energy relative to the unsplit barycentre,in kJ mol^{-1},is

(A) 150
(B) −150
(C) 100
(D) −400

46. A model solvent has Tb=400K,molar mass100g/mol and molar vaporisation enthalpy32000J/mol. UseR=8.00J $mol^{-1}K^{-1}$ and Kb=R Tb^2 M/(1000ΔH),M in g/mol. A $\frac{1}{50}$mol/kg A_2B solution is50% dissociated into3ions. Boiling temperature in K is

(A) $\frac{10004}{25}$
(B) $\frac{10002}{25}$
(C) $\frac{10006}{25}$
(D) $\frac{10001}{25}$

47. For butan-2-one and 3,3-dimethylbutan-2-one,which structural-reactivity comparison is correct under ordinary enolate/aldol conditions?

(A) neither has alpha H
(B) only 3,3-dimethylbutan-2-one has alpha H
(C) only butan-2-one has alpha H
(D) both have at least one enolisable alpha-carbon site

48. An octahedral Co(III) complex contains 3 neutral en ligands,with all remaining donor positions occupied by chloride. What is its coordination number and coordination-ion charge?

(A) CN 9; charge +3.
(B) CN 3; charge +3.
(C) CN 6; charge +3.
(D) CN 6; charge +0.

49. Acidic MnO4- oxidises SO2 to HSO4- and is reduced to Mn2+. The balanced reaction is 2MnO4-+5SO2+2H2O+H+→2Mn2++5HSO4-. Initially 23/1000 mol H+ and excess SO2 are available. After exactly 1/25 mol MnO4- is consumed, what amount of net H+ remains? Count net H+ from the displayed balanced equation.

(A) 3/1000
(B) 3/500
(C) -3/1000
(D) 3/2000

50. A first-order reaction has a constant half-life of 10 min. A sample is observed now and again 20 min later, when unreacted material equals 1/64 of the original amount present when the reaction began. How many minutes had elapsed before the first observation?

(A) 70
(B) 60
(C) 20
(D) 40

51. For $[Fe(CN)_6]^{3-}$,use the usual ligand-field/VBT assignments at room temperature. Which complete assignment is consistent with charge balance and magnetic behaviour?

(A) d^5,tetrahedral,sp^3,spin-only moment $\sqrt{3}$ BM.

(B) d^5,octahedral,d^2sp^3,spin-only moment $2\sqrt{2}$ BM.

(C) d^5,octahedral,d^2sp^3,spin-only moment $\sqrt{3}$ BM.

(D) d^6,octahedral,d^2sp^3,spin-only moment $\sqrt{3}$ BM.

52. A weak 1:1 electrolyte obeys exact Ka=Cα^2/(1-α). In a closed dilution its dissociation degree changes from 1/20 to 1/10, while Ka and total formula-unit amount remain constant. By what factor does its volume increase? Do not replace 1-α by 1.

(A) 38/9 (B) 4

(C) 2 (D) 9/38

53. In a pressure range where the Freundlich law $x/m = kp^{1/4}$ is valid at fixed temperature, adsorbent B has 10 times the mass of A and is exposed to 10000 times its pressure. Find the ratio of total adsorbed amounts x_B/x_A.

(A) 1000 (B) 100

(C) $\frac{1}{100}$ (D) 100000

54. Ribose has C5H10O5 and 2-deoxyribose has C5H10O4. Which comparison is chemically correct?

(A) Both are hexoses.

(B) Removing oxygen necessarily removes a carbonyl carbon.

(C) Both are pentose-derived sugars; deoxy substitution removes one oxygen without changing five-carbon count.

(D) Deoxyribose must contain four carbons because it has four oxygens.

55. An ordinary liquid becomes easier to shear after moderate heating, with composition and phase unchanged. Which molecular interpretation is justified?

(A) All covalent bonds in the liquid must have broken.

(B) Thermal motion more readily permits molecular rearrangement against intermolecular cohesion.

(C) Its viscosity must have risen because every molecular collision transfers more momentum.

(D) Its lower viscosity proves that all intermolecular attractions have vanished.

56. For $2Cu^+ \rightarrow Cu^{2+} + Cu$,consider only atoms of the element undergoing disproportionation. Fraction of those atoms oxidised is

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{1}{4}$ (D) $\frac{1}{6}$

57. In the stated high-pressure model Z=1+b p/(RT), gas A at 350 K and gas B at 480 K have Z-versus-p slopes in ratio 288/245. Their molar covolumes b do not change with temperature in this model. At what temperature must gas A be measured for its slope to equal that of B at 480 K?

(A) 480 (B) 2880/7

(C) 560 (D) 350

58. propanone reacts with ethylene glycol and catalytic acid,remove water. Which product correctly reflects nucleophilic addition or addition-elimination at its carbonyl rather than a different reaction class?

(A) a nitrile

(B) a cyclic ketal

(C) an aldol polymer formed by replacing the oxygen with carbon

(D) a primary alcohol

59. At a fixed temperature, a diprotic acid H2A has pKa1=4, pKa2=8, and pKw=15. Find (pKb of A2-, Kb(A2-)/Kb(HA-)). Use the conjugate pair appropriate to each basic species.

(A) (7, 1/10000)

(B) (11, 1/10000)

(C) (7, 10000)

(D) (11, 10000)

60. Atoms form ideal touching FCC spheres of radius r and cell edge a. Fractional coordinates are measured in a. Using the straight displacement between the two specified points (1/2,1/2,1/2) and (1/2,0,0), without replacing it by another periodic image, find $(\text{distance}/r)^2$. These points describe two octahedral sites.

(A) 8

(B) 4

(C) 2

(D) 5

English Proficiency

61. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. it must reach the bridge Q. Whichever route the group chooses, R. is closed for the evening. S. before the temporary crossing

(A) P-Q-S-R

(B) R-P-S-Q

(C) S-Q-P-R

(D) Q-P-S-R

62. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. Did the technicians check both alarms during the inspection?

(A) Were some alarms checked by the technicians before the inspection?

(B) Were both technicians checked by the alarms during the inspection?

(C) Were both alarms checked by the technicians during the inspection?

(D) Was both alarms checked by the technicians during the inspection?

63. He officially represents his government in relations with another state rather than reporting on it as a journalist. Which role fits?

(A) ambassador

(B) traveller

(C) interpreter

(D) correspondent

64. Complete: The delivery arrived _____, after the office had already closed.

(A) latter

(B) late

(C) lateness

(D) lately

65. Read the sentence: Before sealing the parcel, the conservator made a meticulous record of every mark on the vase. Choose the closest meaning of 'meticulous' as used in the sentence.

(A) highly original
(B) needlessly slow
(C) careless
(D) extremely careful

66. The upper room is accessible by the open stairway. Choose the opposite of accessible in the sense of being reachable.

(A) inaccessible
(B) accessory
(C) accessible
(D) accidental

67. Read the sentence: The rules restrain the committee's power to spend without consulting members. Choose the closest meaning of 'restrain' as used in the sentence.

(A) limit
(B) question
(C) remove
(D) define

68. The investigation measures how speech sounds are produced and heard, rather than how words acquire meanings. Which field is it?

(A) etymology
(B) phonetics
(C) semantics
(D) syntax

69. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. She corrected the heading and repeated the comparison with the source records. B. The analyst checked the totals in the final table. C. The episode showed why checking a calculation alone does not establish that a table communicates it correctly. D. They were arithmetically correct but did not match the units stated in its heading.

(A) The episode showed why checking a calculation alone does not establish that a table communicates it correctly.

(B) The analyst checked the totals in the final table.

(C) She corrected the heading and repeated the comparison with the source records.

(D) They were arithmetically correct but did not match the units stated in its heading.

70. Use the simple present to express the stated regular procedure. Choose the best replacement: Neither of the two entrances [remain] open after the final visitor leaves each evening.

(A) have remained
(B) were remaining
(C) remains
(D) are remaining

Logical Reasoning

71. Preserve the exact relation and its order: Courageous : Brave :: ? : Commence. Choose the missing term.

(A) Begin
(B) Courageous
(C) Rapid
(D) Silent

72. Kiran points to a photograph and says, "This person is the daughter of my father's only daughter." Kiran is male, and the photographed person is female. How is the photographed person related to Kiran?

(A) Brother (B) Son

(C) Niece (D) Daughter

73. Consider only these statements; universal statements alone do not imply existence. Some artists are guides; Some guides are editors. Conclusions: I. All editors are guides. II. Some guides are artists. Which conclusions follow necessarily?

(A) Neither I nor II (B) Both I and II

(C) Only I (D) Only II

74. The age table uses the same two people throughout:

	younger	older
now	x	$x+9$
6 years ago	$x-6$	$x+3$

. The product of their ages 6 years ago was 442. What will the older person's age be 5 years from now?

(A) 37 (B) 31

(C) 46 (D) 32

75. Consider only these statements; universal statements alone do not imply existence. No hikers are artists; Some joggers are hikers; Some hikers are joggers. Conclusions: I. No artists are hikers. II. All hikers are artists. Which conclusions are individually possible without contradicting the statements?

(A) Both I and II (B) Only II

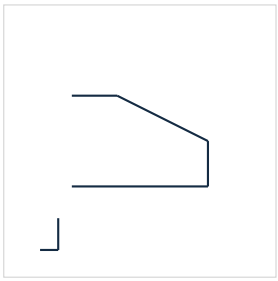
(C) Only I (D) Neither I nor II

76. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 7, 21, 28, 84, ?, 273, 280.

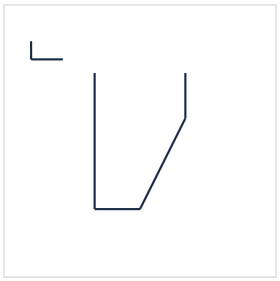
(A) 77 (B) 140

(C) 273 (D) 91

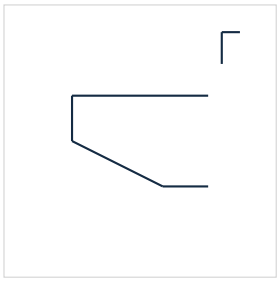
77. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

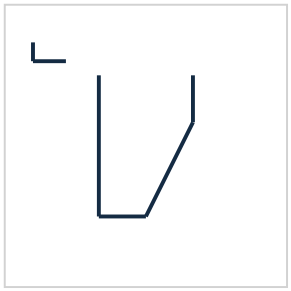


2

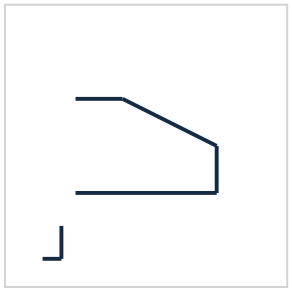


3

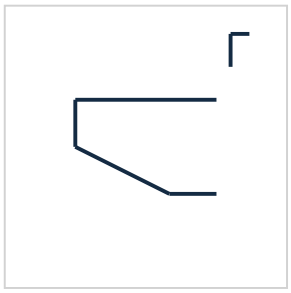
(A) Pattern Q



(B) Pattern P



(C) Pattern V



(D) Pattern Z



78. In a row, Asha is 6th from the left and 24th from the right. Bina is 10th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 12

(B) 10

(C) 16

(D) 9

79. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?



1



2



3



4

(A) 4

(B) 1

(C) 3

(D) 2

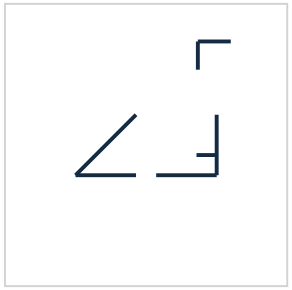
80. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



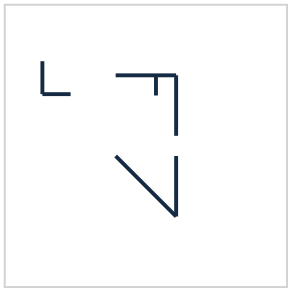
(A) Pattern U



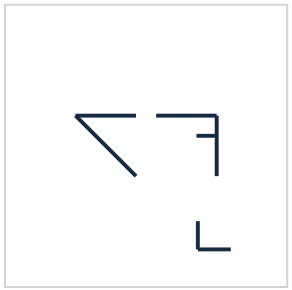
(B) Pattern X



(C) Pattern Q



(D) Pattern L



81. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, RIVER is coded as WNAJW. Which word is coded as UQFSY?

(A) PLANT

(B) ZVKXD

(C) OKZMS

(D) QMBOU

82. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



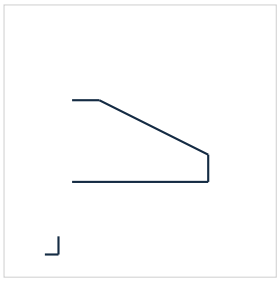
host 4

(A) 2
(B) 4
(C) 1
(D) 3

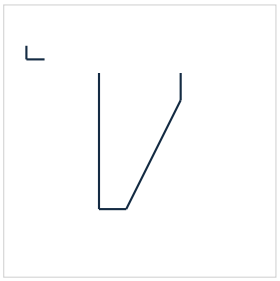
83. Bina receives 3 times as many tokens as Asha, and Charu receives 3 times as many as Bina. Their total is 208. After the allocation, Bina gives 2 tokens to Asha. How many tokens does Bina then retain?

(A) 48
(B) 50
(C) 46
(D) 14

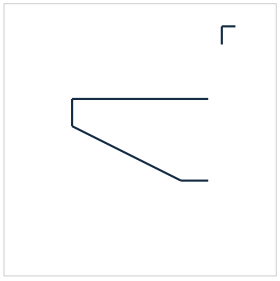
84. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right



bottom left

?

bottom right

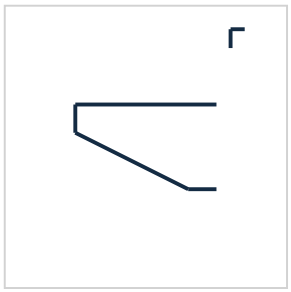
(A) Pattern Z



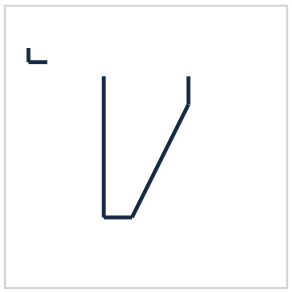
(B) Pattern L



(C) Pattern Q



(D) Pattern P



85. Three numbered triples satisfy both $y = x^2 + 6$ and $z - y = 7$. Which triple fails at least one condition? 1: (13,175,182), 2: (14,202,209), 3: (3,15,23), 4: (4,22,29).

(A) 2

(B) 3

(C) 1

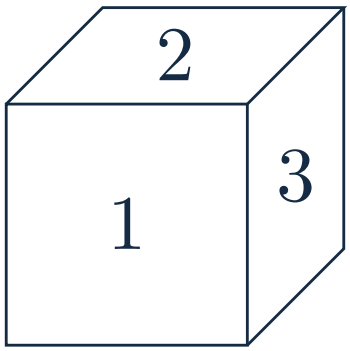
(D) 4

86. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: GYSK, DZQN, ?, XBMT, UCKW, RDIZ.

(A) ZZNP (B) BBPR

(C) AAOQ (D) QOAA

87. This standard die has opposite labels summing to seven. Roll it once to the right, so its present top face becomes the right face. Give the new (top, front) labels.



(A) $(3, 1)$ (B) $(4, 1)$

(C) $(4, 6)$ (D) $(2, 1)$

88. A walker goes 5 m west, then 10 m south, then turns left and walks 4 m. In which direction is the starting point from the walker?

(A) South-east (B) South-west

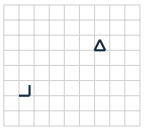
(C) North-west (D) North-east

89. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 6$ and $5 \mapsto 12$. Which input maps to 16?

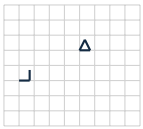
(A) 16 (B) 7

(C) 8 (D) 6

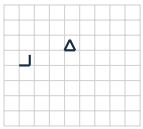
90. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



1

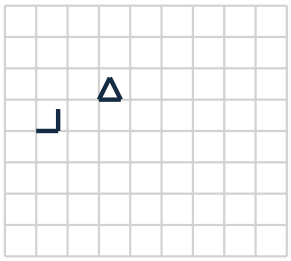


2

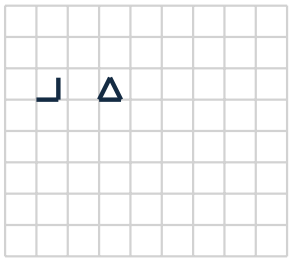


3

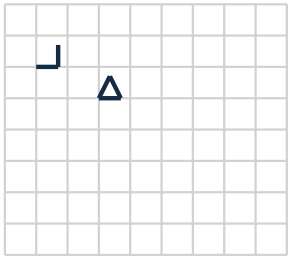
(A) Pattern V



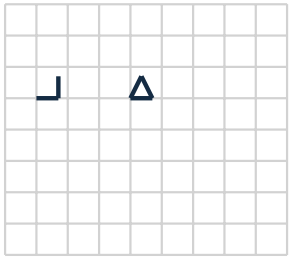
(B) Pattern R



(C) Pattern T



(D) Pattern Q



Mathematics

91. For $P(x) = \prod_{j=3}^{12}(x+j)$, find the coefficient of x^7 in $P(x)/(x+3)$.

(A) 2481 (B) 2490

(C) 4548 (D) 2274

92. ABC is right angled at A with AB=8,AC=10. A vertical tower stands at the midpointM ofBC. Its top elevation from A is 60°. Find the square of the tower height.

(A) 243 (B) 123

(C) 60 (D) 492

93. Three distinguishable fair dice are thrown. Given that exactly two faces are equal, find the probability that their sum is at least 15.

(A) $\frac{13}{15}$ (B) $\frac{2}{15}$

(C) $\frac{1}{6}$ (D) $\frac{1}{18}$

94. How many integer k in [0,96] make $x^2 - 14x + (48-k) = 0$ have roots of opposite signs?

(A) 47 (B) 96

(C) 49 (D) 48

95. L_1:r=t(1,0,1), L_2:r=(7,0,0)+s(0,1,1). Find their shortest distance.

(A) $\frac{7\sqrt{3}}{3}$ (B) $\frac{7}{3}$

(C) 7 (D) 0

96. $A = \begin{bmatrix} 2 & 1 \\ 0 & 4 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$. Find the (1,2) entry of $P^{-1}AP$.

(A) 9 (B) 1

(C) −8 (D) −7

97. f(x)=3x+1 and f(g(x))=3x+22. Find g(g(3)).

(A) 10 (B) 17

(C) 20 (D) 23

98. The normal of slope 6 to y^2=8x touches at P. Find the distance from P to the focus.

(A) 5476 (B) 72

(C) 70 (D) 74

99. How many integer t make P=(t,0) strictly inside the larger segment cut by x=3 from x^2+y^2=49?

(A) 11 (B) 3

(C) 13 (D) 9

100. For t≥0 define F(t)=lim as n→ ∞ of $n^{(-3)}\Sigma$ floor(tj^2), j=1,...,n. Evaluate Σ F($(1/4)^r$), r=0,...,∞.

(A) $\frac{1}{9}$ (B) $\frac{4}{3}$

(C) $\frac{4}{9}$ (D) $\frac{1}{3}$

101. The chord of contact from P to $x^2/36$-$y^2/16$=1 is x=6/2. Find the product of the slopes of the two tangents from P.

(A) $-\frac{4}{27}$ (B) $-\frac{4}{81}$

(C) $\frac{4}{27}$ (D) $-\frac{1}{9}$

102. How many integers x are in the real domain of $\sqrt{x-2}\,\log(7-x)$?

(A) 5 (B) 7

(C) 4 (D) 6

103. Find the squared distance between 6x+8y=4 and 3x+4y=4.

(A) $\frac{1}{25}$ (B) $\frac{2}{5}$

(C) $\frac{4}{25}$ (D) 4

104. Group A has 4 observations, mean 12 and population SD 2. Group B has 5 observations, mean 17 and population SD 3. Find the population variance of the combined data.

(A) $\frac{61}{9}$ (B) $\frac{286}{9}$

(C) $\frac{1049}{72}$ (D) $\frac{1049}{81}$

105. An AP has a=3,d=8; its first n terms sum to 248. Find a_(n+1).

(A) 31 (B) 8

(C) 67 (D) 75

106. For a=(1,7,0), b=(0,1,5), c=(3,0,1), evaluate (a×c)·(a+b).

(A) 106 (B) −106

(C) −105 (D) 0

107. f(x+1)=$3x^2$. Find the area enclosed between y=f(x) and y=3(x-1).

(A) $\frac{3}{2}$ (B) $-\frac{1}{2}$

(C) $\frac{1}{2}$ (D) 1

108. For roots α,β of x^2-2cosθ x+$\cos^2\theta$-9=0, find the minimum of $\alpha^2+\beta^2$ when $0\leq \theta \leq \pi/3$.

(A) 18 (B) 20

(C) 19 (D) $\frac{37}{2}$

109. How many integers k make 18sin x+24cos x+4=k have exactly two distinct solutions in $0\leq x<2\pi$?

(A) 61 (B) 60

(C) 59 (D) 29

110. $F'(x)=\frac{1}{x^2}-\frac{1}{x^2\sqrt{x^2+1}}$ and F(1)=2. Find F(3)-2F(1).

(A) $-\sqrt{2}+\frac{2}{3}+\frac{\sqrt{10}}{3}$ (B) $-\sqrt{2}+\frac{\sqrt{10}}{3}+\frac{8}{3}$

(C) $-\sqrt{2}-\frac{4}{3}+\frac{\sqrt{10}}{3}$ (D) $-\frac{8}{3}-\frac{\sqrt{10}}{3}+\sqrt{2}$

111. How many integers x satisfy |(x-3)+(x-7)|<|x-3|+|x-7|?

(A) 6 (B) 5

(C) 4 (D) 3

112. $\log_2$a=5, log_a b=6, log_b c=2. Find $\log_4(c^2/a)$.

(A) 115 (B) 60

(C) $\frac{115}{2}$ (D) $\frac{125}{2}$

113. Evaluate $\int_0^{\pi/2} \sin(2x)\left[1+8\sin^2(2x)\right]\frac{\sin^7 x}{\sin^7 x+\cos^7 x}\,dx$.

(A) $\frac{19}{6}$ (B) $\frac{1}{2}$

(C) $\frac{19}{3}$ (D) $\frac{9}{2}$

114. The line y=mx through O bisects the area of the triangle with vertices O,(8,3),(7,7). Assume the vertices are noncollinear. Find m.

(A) $\frac{2}{3}$ (B) −4

(C) $\frac{7}{8}$ (D) $\frac{3}{2}$

115. A=(2,4,3), B=(4,4,7). Find the squared length of the component of AB perpendicular to direction (1,2,2).

(A) $\frac{50}{3}$ (B) 20

(C) $\frac{100}{9}$ (D) $\frac{80}{9}$

116. The locus|z-3|=|Re(z)+2| meets Im(z)=5. Find $|z|^2$ at the intersection.

(A) 34 (B) 386

(C) 9 (D) 29

117. Two points on opposite sides of a tower foot are 6 metres apart on a straight level line. The elevations of the tower top are 30° and 45°. Find the tower height.

(A) $3+3\sqrt{3}$ (B) $-3+3\sqrt{3}$

(C) 3 (D) $\sqrt{3}$

118. For f(x)=|x-3|/(x-8), on which maximal open intervals is f strictly decreasing?

(A) $(3,8)\cup(8,\infty)$ (B) $(3,\infty)$

(C) $(-\infty,3)$ (D) $(-\infty,8)$

119. Evaluate lim as x→0 of $[(1+5x)/(1-3x)]^{(1/x)}$.

(A) e^2 (B) 8

(C) e^8 (D) e^5

120. In $(3x^2-x^{-1})^8$, find the ratio of the coefficient of x^7 to that of x^{10}.

(A) 2 (B) $\frac{2}{3}$

(C) $-\frac{3}{2}$ (D) $-\frac{2}{3}$

121. $f(x)=x^3+5x$, $g=f^{-1}$. Find $g''(150)$.

(A) $-\frac{3}{640}$ (B) $-\frac{3}{51200}$

(C) $\frac{1}{30}$ (D) $\frac{3}{51200}$

122. Vectors a,b satisfy $|a|^2=10$, $|b|^2=37$, a·b=9. Find $|2a-b|^2$.

(A) 77 (B) 41

(C) 113 (D) 39

123. Evaluate sin(2arctan 7).

(A) $\frac{7}{25}$ (B) $\frac{7\sqrt{2}}{10}$

(C) $-\frac{7}{24}$ (D) $-\frac{7}{25}$

124. Let $D(t)=\det\begin{pmatrix}8+t & 1+t\\ 1+t & 4+t\end{pmatrix}$. Find D′(0).

(A) 10 (B) −14

(C) 62 (D) 12

125. Chords of $x^2+y^2=16$ pass through (8,0). Find the radius of the circle containing their midpoint locus.

(A) 2 (B) 4

(C) 16 (D) 8

126. A convergent GP has first term 5, ratio $\frac{1}{9}$. Find the sum of all terms after its first 7 terms.

(A) $\frac{1}{1062882}$ (B) $\frac{5}{4251528}$

(C) $\frac{2989355}{531441}$ (D) $\frac{1}{4251528}$

127. A sphere has $dV/dt=2$ cubic units/s. Find the rate of change of its surface area when $r=5$.

(A) $\frac{4}{25}$ (B) $\frac{4}{5}$

(C) $\frac{2}{5}$ (D) $\frac{1}{50\pi}$

128. $y'+y/(x+1)=1$ for $x\geq 0$, $y(0)=3$. Find $(7)y(6)-y(0)$.

(A) $\frac{27}{7}$ (B) -24

(C) 24 (D) 27

129. For $(d^2y/dx^2)^5/(1+x^2)+y=0$, find order+degree if the degree is defined; otherwise select "Degree is not defined".

(A) 7 (B) 10

(C) 8 (D) Degree is not defined

130. Evaluate $\Sigma\, j/2^j$ for $j=1,\ldots,10$.

(A) $\frac{2037}{1024}$ (B) 2

(C) $\frac{509}{256}$ (D) $\frac{5}{512}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. Two identical uniform spheres touch. Their radii are multiplied by 12 while their density is divided by 12. By what factor does their gravitational attraction change?

(A) $\frac{1}{144}$ (B) 144

(C) 1728 (D) $\frac{1}{20736}$

132. A monatomic ideal gas expands reversibly and adiabatically until its pressure becomes 1/161051 of its initial value. Find final temperature divided by initial temperature.

(A) $\frac{1}{121}$ (B) 1

(C) $\frac{1}{11}$ (D) $\frac{1}{1331}$

133. In an abrupt junction, the acceptor concentration is 121 times the donor concentration. Within the depletion approximation, which statement gives x_n/x_p and identifies the carriers mainly responsible for the small reverse current before breakdown?

(A) 1/121; minority carriers (B) 11; majority carriers

(C) 121; minority carriers (D) 1; majority carriers

Chemistry

134. CO2 and SO2 both contain polar central-atom-oxygen bonds. Which statement follows from the isolated molecular geometries?

(A) CO2 is bent while SO2 is linear.

(B) Linear CO2 cancels its bond dipoles, whereas bent SO2 generally has a nonzero resultant.

(C) Both must be nonpolar because both have two oxygen atoms.

(D) Both must have equal dipole moments because O has one electronegativity.

135. For standard M2+/M half-cells, E°_B-E°_A=3/5 V and E°_C-E°_B=1/5 V. B has E°=1/5 V. Which pair gives the strongest reducing metal among A,B,C and E°cell for A(s)+C2+→A2++C(s)?

(A) B; 1/5 V (B) A; -4/5 V

(C) A; 4/5 V (D) C; 4/5 V

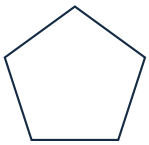
136. A model oxide has composition MO13/10, where the subscript denotes O atoms per M atom. Oxygen is entirely O2- and metal sites are formally only M2+ or M3+. There are no other defects or ions. What fraction of M sites is M3+?

(A) 3/5 (B) -3/5

(C) 3/10 (D) 6/5

Logical Reasoning

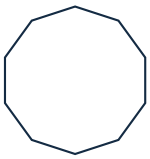
137. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

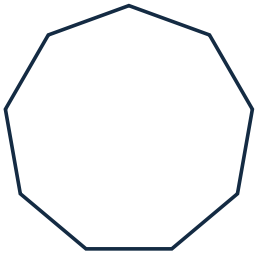


B

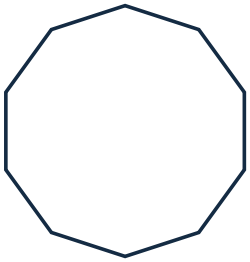


C

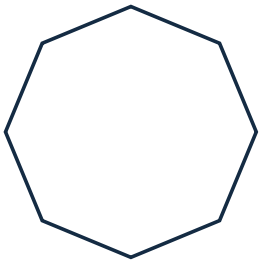
(A) Pattern T



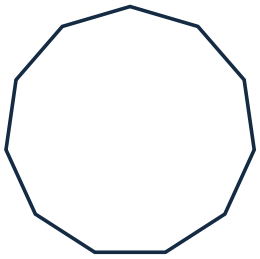
(B) Pattern Y



(C) Pattern N



(D) Pattern Z



138. S is the only child of N, and S is a parent of H. N is the paternal grandmother of H. What can be concluded about S's gender?

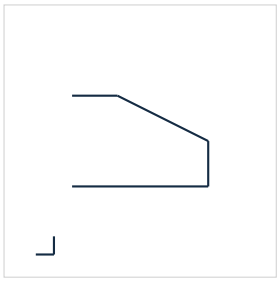
(A) The stated relationships are inconsistent

(B) Male

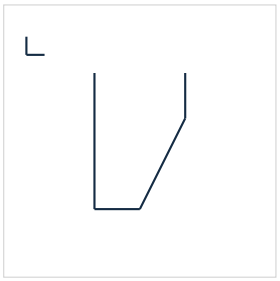
(C) Cannot be determined

(D) Female

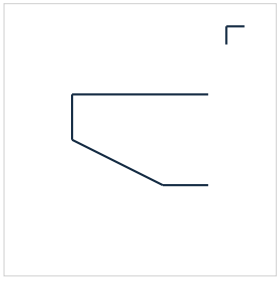
139. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

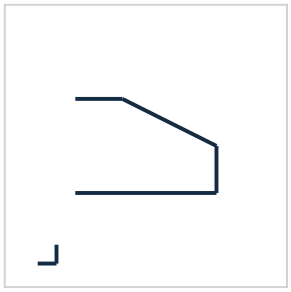


bottom left

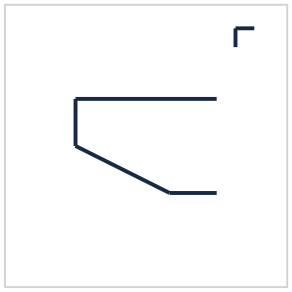
?

bottom right

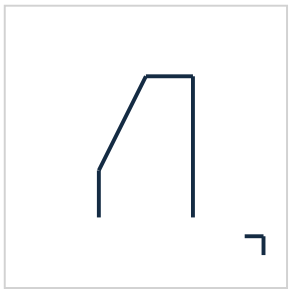
(A) Pattern Z



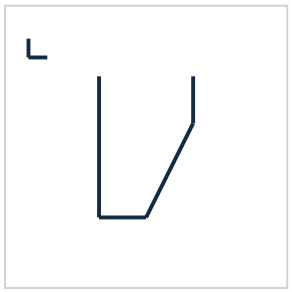
(B) Pattern T



(C) Pattern V



(D) Pattern W



Mathematics

140. For x>1 let f(x)=arctan(2 x/(1-x^2)), with arctan in (-π/2,π/2). Find the integral from 4 to 5 of f′(x) dx.

(A) π

(B) $-2\,\mathrm{atan}\,(4) + 2\,\mathrm{atan}\,(5)$

(C) $-\,\mathrm{atan}\,(4) + \mathrm{atan}\,(5)$

(D) $2\,\mathrm{atan}\left(\frac{1}{4}\right)$

141. a=(t,1,0), b=(1,4,2), a·b=7. Find the squared area of the parallelogram spanned by a,b.

(A) 161

(B) 210

(C) $\frac{161}{4}$

(D) $\sqrt{161}$

142. A line has direction (1,4,2). A plane has normal (2,1,4). Let θ be their acute angle. Find $\tan^2\theta$.

(A) $\frac{9}{4}$

(B) $\frac{5}{9}$

(C) $\frac{4}{5}$

(D) $\frac{4}{9}$

BITSAT Mock 29

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A cell balances on a potentiometer at 60 cm on open circuit and at 40 cm when a 3 ohm resistor is connected across it. The potentiometer gradient is unchanged. Find the cell's internal resistance.

(A) $\frac{15}{2}\,\Omega$ (B) $\frac{9}{2}\,\Omega$

(C) $\frac{3}{2}\,\Omega$ (D) $1\,\Omega$

2. A body is dropped from rest at height 105 m above the ground. Find its speed when it is still 35 m above the ground. Neglect air resistance and take $g = 10\ \text{m/s}^2$.

(A) $10\sqrt{7}\,\text{m/s}$ (B) $1400\,\text{m/s}$

(C) $20\sqrt{35}\,\text{m/s}$ (D) $10\sqrt{14}\,\text{m/s}$

3. Two coaxial circular loops each radius 4 m carry equal currents 4 A in the same circulation sense. Their centres are separated by $2R$. Find field at the midpoint in units of μ_0.

(A) $\frac{\sqrt{2}}{4}$ (B) 1

(C) $\frac{\sqrt{2}}{2}$ (D) $\frac{\sqrt{2}}{8}$

4. A mass 2 kg is attached to two massless springs in series with spring constants 50 N/m and 100 N/m. Find the small-oscillation period on a smooth horizontal surface.

(A) $\frac{\sqrt{6}}{10}\,\text{s}$ (B) $\frac{2\pi}{5}\,\text{s}$

(C) $\frac{\sqrt{2}\pi}{5}\,\text{s}$ (D) $\frac{\sqrt{6}\pi}{5}\,\text{s}$

5. A point at the bottom of two plane liquid layers is viewed normally from air. The upper layer has thickness 12 cm and index 3/2; the lower has thickness 8 cm and index 4/3. Find its apparent depth below the top surface in the paraxial limit.

(A) $14\,\text{cm}$ (B) $\frac{86}{3}\,\text{cm}$

(C) $8\,\text{cm}$ (D) $10\,\text{cm}$

6. A 1 kg block moving at 8 m/s compresses a spring of constant 140 N/m. During compression, a damper removes 1/3 of the block's initial kinetic energy. Find maximum compression; gravity does no work.

(A) $\frac{8\sqrt{105}}{105}$ m (B) $\frac{4\sqrt{105}}{105}$ m

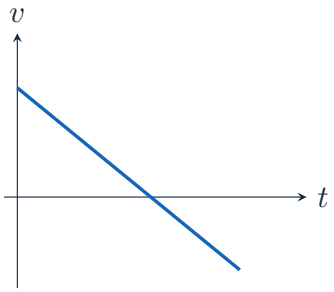
(C) $\frac{4\sqrt{210}}{105}$ m (D) $\frac{32}{105}$ m

7. A parallel-plate capacitor has vacuum capacitance C_0. A dielectric of constant 3 fills a fraction $\frac{1}{3}$ of the plate area and the entire gap in that region; the remainder stays vacuum. The voltage is fixed. Find the fraction of total stored energy residing in the dielectric-filled region.

(A) $\frac{2}{5}$ (B) 1

(C) $\frac{3}{2}$ (D) $\frac{3}{5}$

8. A particle has velocity $v(0) = 3$ m/s, and constant acceleration makes $v(3) = -6$ m/s. What total distance does it travel in this interval?



(A) $\frac{15}{2}$ m (B) $\frac{9}{2}$ m

(C) 6 m (D) $\frac{27}{2}$ m

9. A stationary nucleus with mass number 20 and binding energy per nucleon 8 MeV splits into two equal daughters, each having total binding energy 82.5 MeV. Which statement is correct?

(A) The process releases 5 MeV; the daughters have larger binding energy per nucleon.

(B) No energy can be released because fission always requires a larger mass number than 20.

(C) The process absorbs 5 MeV; the daughters have larger binding energy per nucleon.

(D) The process releases 10 MeV; each daughter releases 5 MeV.

10. A rod of length 3 m pivots at its lower end. A peg constrained to move horizontally stays in contact with its upper tip. At an instant the rod makes an angle with $\sin\theta = 3/5$ to the horizontal and the tip's horizontal speed has magnitude 2 m/s. Find the rod's angular-speed magnitude.

(A) $\frac{5}{6}$ rad/s (B) 2 rad/s

(C) $\frac{2}{5}$ rad/s (D) $\frac{10}{9}$ rad/s

11. A spherical planet rotates uniformly. Its rotation reduces effective gravity at the equator by 2 m/s^2. By how much does rotation reduce the radial component of effective gravity at latitude 60°?

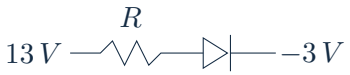
(A) 1 m/s^2 (B) 2 m/s^2

(C) $\frac{1}{2}$ m/s^2 (D) $\frac{1}{8}$ m/s^2

12. An insulated vessel containing one mole of an ideal gas of molar mass $\frac{7}{100}$ kg/mol moves at bulk speed 60 m/s and is brought to rest. All bulk kinetic energy becomes internal energy. The gas has 3 active degrees of freedom. Find temperature rise. Use $R = 8.314$ J/(mol K).

(A) $\frac{84000}{4157}$ K
(B) $\frac{126000}{4157}$ K
(C) $\frac{21000}{4157}$ K
(D) $\frac{42000}{4157}$ K

13. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 1 kΩ to a terminal at +13 V, and its cathode connects to a terminal at -3 V. Find the current from the +13 V terminal in mA.



(A) $\frac{153}{10}$ mA
(B) 16 mA
(C) $\frac{167}{10}$ mA
(D) $\frac{123}{10}$ mA

14. Two ferromagnetic samples have similar remanent flux density. Their coercive fields are 200 A/m and 800 A/m. A reversing field of magnitude 500 A/m will act in service. Which sample is better at retaining its original magnetisation against this field, based only on the stated coercivities?

(A) The sample with coercive field 200 A/m.
(B) The sample with coercive field 800 A/m.
(C) Both are necessarily demagnetised equally.
(D) Neither can have nonzero remanence.

15. A 5 kg mass on a string passes the top of a vertical circle of radius 3 m with string tension 350 N. Find string tension at the bottom. Take $g = 10$ m/s^2.

(A) 450 N
(B) 350 N
(C) 650 N
(D) 550 N

16. A converging thin lens forms a real image with magnification magnitude 5. The object-image separation is 18 cm. Find focal length.

(A) 15 cm
(B) $\frac{5}{2}$ cm
(C) $\frac{45}{8}$ cm
(D) $\frac{1}{2}$ cm

17. A particle has displacement $x = 5\cos(2\pi t/6)$ cm. Find its signed velocity at $t = 1$ s.

(A) $\frac{5\sqrt{3}\pi}{6}$ cm/s
(B) $-\frac{5\sqrt{3}}{12}$ cm/s
(C) $-\frac{5\sqrt{3}\pi}{6}$ cm/s
(D) $-\frac{5\pi^2}{18}$ cm/s

18. A ball is dropped from 8 m onto a horizontal fixed floor. At every impact the coefficient of restitution is $\frac{2}{5}$. Neglect air resistance. What maximum height does it reach after its second impact?

(A) $\frac{64}{125}$ m
(B) $\frac{2048}{390625}$ m
(C) $\frac{32}{25}$ m
(D) $\frac{128}{625}$ m

19. A one-electron ion with nuclear charge $Z = 3$ absorbs a photon of energy $\frac{737}{125}$ eV. The ejected electron has maximum kinetic energy 1 eV. From which principal quantum number was the electron removed? Use $E_n = -13.6Z^2/n^2$ eV and neglect recoil.

(A) 4
(B) 5
(C) 15
(D) 6

20. A 3 kg block on a rough horizontal table is tied over an ideal pulley to a hanging 4 kg mass. The hanging mass moves downward and $\mu_k = 1/5$. Find string tension. Take $g = 10\ \text{m/s}^2$.

(A) $\frac{144}{7}$ N (B) 40 N

(C) $\frac{102}{7}$ N (D) 32 N

21. A lossless coating of refractive index 2 lies between air and a substrate of still higher index. For normally incident vacuum wavelength 300 nm, find the smallest positive film thickness giving destructive interference between its two reflected rays. Assume reflections at both boundaries each gain a phase reversal.

(A) 75 nm (B) $\frac{75}{2}$ nm

(C) $\frac{75}{4}$ nm (D) $\frac{225}{2}$ nm

22. The 3rd harmonic of an open pipe of length 7 m equals the 3rd allowed resonance of a closed pipe. Both have the same sound speed and negligible end correction. Find the closed-pipe length.

(A) $\frac{35}{3}$ m (B) $\frac{35}{6}$ m

(C) $\frac{21}{5}$ m (D) $\frac{49}{6}$ m

23. A platform of moment of inertia 8 kg m^2 carries a person of mass 5 kg at radius 2 m and rotates at 7 rad/s. The person walks to half the stated initial radius; external torque is zero. Find final angular speed. Treat the person as a point mass.

(A) $\frac{14\sqrt{91}}{13}$ rad/s (B) $\frac{13}{4}$ rad/s

(C) $\frac{98}{9}$ rad/s (D) $\frac{196}{13}$ rad/s

24. A Newtonian liquid of viscosity $\frac{3}{10}$ Pa s has velocity profile $v(y) = 2y + 2y^2$ in SI units. Find the shear-stress magnitude at $y = 4$ m.

(A) $\frac{27}{5}$ Pa (B) 18 Pa

(C) 3 Pa (D) 12 Pa

25. An astronomical telescope in normal adjustment has magnifying-power magnitude 3 and tube length 20 cm. Find the eyepiece focal length.

(A) $\frac{5}{4}$ cm (B) 5 cm

(C) $\frac{20}{3}$ cm (D) 10 cm

26. A 1 kg block rests on a 6 kg block on a smooth horizontal floor. A force 42 N pulls the lower block. There is no slipping between the blocks. During a displacement 4 m, what work does friction do on the upper block?

(A) 84 J (B) 168 J

(C) 24 J (D) 28 J

27. A 5 V Zener regulator feeds a constant-current load of 6 mA. Supply varies from 12 to 15 V. Regulation requires Zener current at least 3 mA, and its safe maximum is 8 mA. What is the largest permissible series resistance? Assume ideal regulation and give resistance in Ω.

(A) $\frac{7000}{9}$ Ω (B) 500 Ω

(C) $\frac{3500}{3}$ Ω (D) $\frac{4000}{3}$ Ω

28. A charge of $5\,\mu$C moves from potential 40 V to -40 V in an electrostatic field. Find the work done by an external agent if the charge starts and ends at rest and no other force does work.

(A) $0\,\mu$J (B) $-400\,\mu$J

(C) $400\,\mu$J (D) $-10\,\mu$J

29. A triangular p-V cycle has vertices (V_0, p_0), $(V_0 + 6\,\text{L}, p_0)$ and $(V_0, p_0 + 5\,\text{kPa})$. Traversed in that listed order and then back, find net work done by the gas.

(A) -30 J (B) 0 J

(C) -15 J (D) 15 J

30. Two identical rods each mass 1 kg and length 5 m form an L along +x and +y from the origin on a smooth horizontal plane. A constant force of 2 N acts parallel to +x through the body's CM, which is initially at rest. Find the CM x-coordinate after 2 s.

(A) $\frac{5}{4}$ m (B) $\frac{13}{4}$ m

(C) $\frac{9}{4}$ m (D) 2 m

Chemistry

31. The ions Cl^-, K^+, Ca^{2+}, Sc^{3+} each have 18 electrons. Which decreasing-radius order and explanation is consistent?

(A) $Cl^- > K^+ > Ca^{2+} > Sc^{3+}$; increasing nuclear charge contracts the same electron cloud.

(B) $Cl^- < K^+ < Ca^{2+} < Sc^{3+}$; larger nuclear charge expands the cloud.

(C) All radii are equal because electron numbers are equal.

(D) $K^+ > Ca^{2+} > Sc^{3+} > Cl^-$; positive charge always gives the largest ion.

32. List all acyclic saturated carbon skeletons with formula C4H10. Count constitutional isomers,not conformers,and do not count alternative numbering of one skeleton twice. The count is

(A) 4 (B) 1

(C) 3 (D) 2

33. An ideal-gas equation is written p=nkT with n counting particles per m3. If n is replaced by molar density c in mol/m3, what coefficient must replace k?

(A) R=NAk, with units J mol-1 K-1. (B) R, with units J K-1 only.

(C) k/NA, with units J mol-1 K-1. (D) The same k because the mole has no unit.

34. In a hydrogen-like ion Z=4,an electron falls from n=5 to an s orbital with 2 radial nodes. With hydrogen ionisation energy13.6 eV and$hc = 1240$ eV nm,the emitted wavelength in nm is

(A) $\frac{174375}{8704}$ (B) $\frac{174375}{2176}$

(C) $\frac{174375}{136}$ (D) $\frac{19375}{136}$

35. An acyclic saturated monohydric alcohol C4H10O gives immediate turbidity with Lucas reagent at room temperature. It is converted to its chloride and then heated with potassium tert-butoxide. Exclude benzylic/allylic exceptions. Which final alkene is expected?

(A) 2-methylpropene (B) but-1-ene

(C) but-2-ene (D) cyclobutane

36. A hydrocarbon has empirical formula CH2 and independently measured molar mass 56 g/mol. Use C=12,H=1. Which molecular formula and interpretation are consistent?

(A) C4H8; molecular formula is 4 times the empirical formula unit.

(B) CH2; molar mass cannot distinguish an empirical formula from a molecular formula.

(C) C8H4; empirical and molecular ratios need not agree.

(D) C4H4; hydrogen subscripts do not scale.

37. For the metal centre in Ti^{4+},Z=22. Determine the number of d electrons after formal oxidation-state assignment; use O=-2 in oxoanions.

(A) 2

(B) 0

(C) 4

(D) 1

38. For an noncrystalline H_2O_2 molecule,which combined statement about local geometry and overall conformation is correct?

(A) All four atoms must lie in one plane because only sigma bonds are present.

(B) Each O has two bonds and two lone pairs; the H-O-O-H skeleton is nonplanar/open-book.

(C) Each O is linear because it is bonded to only two atoms,and the skeleton is straight.

(D) The O-O bond is a double bond,so both O centres are planar with one lone pair.

39. In a hypothetical atom, allowed l values are0,1,...,n. Subshell capacity remains2(2l+1), with filling by increasing n+l and then lower n. All subshells preceding 2d are filled, 2d contains 3 electrons, and no later subshell is occupied. How many electrons does the atom contain?

(A) -21

(B) 42

(C) 21/2

(D) 21

40. The internuclear axis is x. Two p_y orbitals overlap sideways. Their lobes facing each other above the axis have the same phases; the lower facing lobes have the corresponding opposite signs. Which MO is produced,and what nodal property follows?

(A) A pi bonding MO; the plane containing the axis is a node,but there is no additional internuclear node from phase cancellation.

(B) A sigma bonding MO,because every overlap along a line joining nuclei is sigma.

(C) A delta MO,because two perpendicular p orbitals always create two nodal planes.

(D) A pi antibonding MO; the facing-lobe phase test has been reversed.

41. butan-1-ol is converted with PBr3 to its primary alkyl bromide,then heated with potassium tert-butoxide. Which product results,and does the PBr3 step itself change its carbon skeleton?

(A) but-1-ene; yes,the carbon chain is cleaved

(B) the corresponding alkane;no C-O bond change

(C) but-1-ene; no,the carbon skeleton is retained

(D) the corresponding ether;SN2 at a tertiary carbon is required

42. A thermal assembly contains Cu(s) and He(g) in separate inert compartments at the same temperature. Helium is heated quasistatically at constant pressure under a freely moving piston. Containers have negligible heat capacity. Use the classical Dulong-Petit solid model for Cu, neglecting its Cp-Cv difference, and the ideal monatomic model for He. No phase change occurs. Molar masses are Cu64 and He4 g/mol. Over a small common temperature rise, the measured total heat capacity C divided by total Cu-plus-He mass m and gas constant R is C/(mR)=175/576 mol/g. What is the Cu mass fraction?

(A) 370/693 (B) 41/189

(C) 5/9 (D) 4/9

43. The same element forms a fully occupied BCC phase A with cell edge a and an FCC phase B with edge 5/2a. The measured density ratio ρ_B/ρ_A is 48/625. All FCC sites have equal occupancy, atomic mass is unchanged, and vacancies are the only defects. What fraction of FCC sites is vacant?

(A) 2/15 (B) 1/5

(C) 4/5 (D) 2/5

44. A closed system goes from state1to2 by path A with q=900 J and work on the system w=-200 J. It returns from2to1 along a different path with q=-1100 J. IUPAC work on the return path,in J,is

(A) 700 (B) 200

(C) 400 (D) −400

45. For a fixed-composition closed system dG=V dP-S dT. At a state use V=4 m3, S=2 J/K and a local path with dT/dP=2 K/Pa. Find dG/dP along this path in m3. The supplied values are a formal local thermodynamic model; 1 Pa m3=1 J.

(A) -1 (B) 2

(C) 1 (D) 0

46. Evaluate independently. I: XeF_4 is nonpolar despite polar Xe-F bonds. II: Each resonance contributor is a different molecule rapidly interconverting.

(A) Neither I nor II is correct. (B) Only I is correct.

(C) Only II is correct. (D) Both I and II are correct.

47. Two van der Waals gases obey Tc=8a/(27Rb) and Pc=a/(27b^2). Their measured ratios are Tc,A/Tc,B=3/2 and Pc,A/Pc,B=3/4. Find (a_A/a_B, b_A/b_B).

(A) (3, 2) (B) (3, 1)

(C) (6, 2) (D) (2, 3)

48. A cathode receives 3/25 mol electrons and deposits 32/25 g copper from Cu2+ (Cu=64 g/mol). Its only other product is H2. What amount of H2, in mol, is produced?

(A) 1/50 (B) 1/25

(C) 2/25 (D) -1/25

49. Limiting molar conductivities are KCl150, NaCl126 and KNO3 110 S cm2/mol. A very dilute 1/100 M NaNO3 solution has measured conductance 43/100000 S. Neglect finite-concentration corrections. What is the cell constant in cm-1?

(A) 1 (B) -2

(C) 2 (D) 4

50. Separate unknowns P and Q are each selected from Cu2+, Zn2+, Fe3+. Dropwise aqueous ammonia gives P a reddish brown hydroxide precipitate and Q a white precipitate. Excess ammonia gives P an undissolved reddish-brown hydroxide and Q a colourless ammine solution. Which identification is consistent, without invoking redox?

(A) Both are the same ion because all hydroxides have identical ammonia behaviour.

(B) P=Zn2+; Q=Fe3+; oxidation states unchanged.

(C) P=Fe3+; Q=Zn2+; both metals reduced to zero.

(D) P=Fe3+; Q=Zn2+; oxidation states unchanged.

51. For M+2A+⇌M2++2A, the equilibrium constant for the doubled reaction is10^4. At the same T use2.303RT/F=0.0590 V. What is E° of the original reaction in V?

(A) -59/1000

(B) 59/1000

(C) 59/2000

(D) 59/500

52. Nonrelativistic ions A and B start from rest and gain only electric kinetic energy. B has 3 times A's mass and 2 times its positive charge. To make λ_B/λ_A=1/3, what must V_B/V_A be?

(A) 4

(B) 2/3

(C) 3/2

(D) 9/2

53. A laboratory note states: A catalyst changes the equilibrium constant at fixed temperature. Which judgement and reason are consistent?

(A) Correct because every transition-metal species has the same d-electron count.

(B) Incorrect: A catalyst offers a lower-barrier path for both directions; it does not change thermodynamic equilibrium.

(C) Correct: A catalyst offers a lower-barrier path for both directions; it does not change thermodynamic equilibrium.

(D) Incorrect because transition metals cannot exchange electrons or adsorb reactants.

54. For CrO_2Cl_2,use two oxide oxygens and two chloride ligands. Which pair (Cr oxidation state,number of peroxide O-O links per formula entity) is correct?

(A) (+6,0).

(B) (+4,0).

(C) (+8,0).

(D) (+6,1).

55. A calorimeter holding50.0g water absorbs700J and rises2.00K. Water specific heat is4.00J $g^{-1}K^{-1}$; calorimeter heat capacity is unchanged. When it holds100g water and absorbs the same heat,temperature rise in K is

(A) 2

(B) $\frac{7}{4}$

(C) $\frac{14}{11}$

(D) 1

56. Which occurrence/extraction inference for NaCl is chemically consistent?

(A) It occurs in seawater/brine; sodium is too reactive to persist widely as free metal.

(B) The formula proves that oxygen is the only possible counter-ion in its natural deposits.

(C) The metal occurs chiefly as native metal because group1/2metals are exceptionally unreactive.

(D) The metal is obtained by ordinary water electrolysis without preferential hydrogen evolution.

57. A solution initially contains $\frac{3}{250}$ mol Fe^{2+}. Before titration, $\frac{3}{10}$ of it is oxidised quantitatively to Fe^{3+}. The volume in mL of0.0200 M $KMnO_4$ needed for the remaining Fe^{2+} in acid is

(A) 120 (B) 36

(C) 420 (D) 84

58. Use octet Lewis contributors for NO3- and HCOO-; for SO2 exclude expanded-octet contributors. For HCOO-, HCO3- and HNO3, average only the two nonprotonated central-atom-O bonds. For other species average all central-atom bonds. Find (average bond order of the first minus that of the second, sum of total lone-pair counts in one contributor of each). A lone pair on any atom is counted once.

(A) (-1/6, 14) (B) (-1/6, 13)

(C) (-1/6, 12) (D) (17/6, 13)

59. A fixer initially receives 1/25 mol AgBr and an unknown amount of free thiosulfate. After complete stoichiometric complex formation, 1/100 mol AgBr remains and no free thiosulfate is left. Use AgBr+2S2O3^{2}-→[Ag(S2O3)2]^{3}-+Br-. What was the initial amount of thiosulfate?

(A) 7/100 (B) 2/25

(C) 3/50 (D) 3/100

60. A 15 g $CaCO_3$/inert mixture is80.0% $CaCO_3$ by mass. It receives $\frac{11}{50}$ mol HCl. For $CaCO_3+2HCl \rightarrow CaCl_2+CO_2+H_2O$,90.0% of formed CO_2 is collected. Collected CO_2 amount in mol is

(A) $\frac{11}{50}$ (B) $\frac{3}{25}$

(C) $\frac{11}{100}$ (D) $\frac{99}{1000}$

English Proficiency

61. Read the sentence: The objection was valid because it exposed a genuine contradiction in the proposal. Choose the opposite in meaning of 'valid' as used in the sentence.

(A) forceful (B) unsound

(C) relevant (D) detailed

62. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. Did the curator photograph the label before removing it?

(A) Was the label photographed by the curator before it was removed? (B) Was the label photographed after it was removed?

(C) Had the label been photographed by the curator before it was removed? (D) Did the label photograph the curator before removing it?

63. Use the standard contemporary negative adjective. He would not wait for the explanation and repeatedly interrupted; he was _____.

(A) inpatient (B) dispatient

(C) impatient (D) unpatient

64. Read the sentence: The revised report was coherent: its examples supported a single clear argument. Choose the closest meaning of 'coherent' as used in the sentence.

(A) factually complete (B) written without hesitation

(C) universally accepted (D) logically connected

65. Which figure of speech is chiefly illustrated by the line? The empty chairs waited patiently for the audience.

(A) hyperbole (B) personification

(C) apostrophe (D) simile

66. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. No sooner had the team begun Q. brought the trial to a halt. R. than an unrelated power failure S. to test the revised mechanism

(A) S-P-R-Q (B) R-P-S-Q

(C) P-S-R-Q (D) Q-S-R-P

67. He draws pictures that explain or decorate a book's written content. Which role fits?

(A) illustrator (B) translator

(C) proofreader (D) publisher

68. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. could read them Q. without leaning across the display. R. The labels were enlarged S. so that visitors standing behind the barrier

(A) R-S-P-Q (B) P-R-S-Q

(C) Q-S-P-R (D) S-R-P-Q

69. Read the sentence: The account is credible because independent records support its central details. Choose the closest meaning of 'credible' as used in the sentence.

(A) credulous or easily persuaded (B) incredulous or disbelieving

(C) creditable merely for effort (D) believable

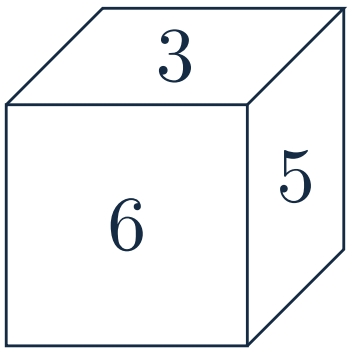
70. Use the simple present to express the stated regular procedure. Choose the best replacement for the bracketed words: Every evening, the caretaker [is checking] the empty rooms before locking the building.

(A) will have checked (B) checks

(C) had checked (D) has checked

Logical Reasoning

71. This standard die has opposite labels summing to seven. Roll it once to the right, so its present top face becomes the right face. Give the new (top, front) labels.



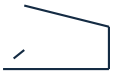
(A) $(2, 1)$ (B) $(5, 6)$

(C) $(3, 6)$ (D) $(2, 6)$

72. Consider only these statements; universal statements alone do not imply existence. No hikers are bakers; All bakers are hikers. Conclusions: I. All bakers are inventors. II. Some inventors are bakers. Which conclusions follow necessarily?

(A) Both I and II
(B) Neither I nor II
(C) Only I
(D) Only II

73. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

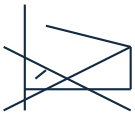




host 1



host 2



host 3



host 4

(A) 3
(B) 2
(C) 1
(D) 4

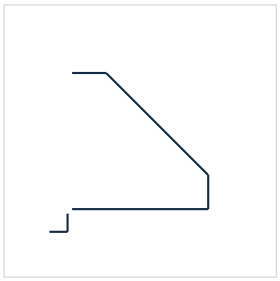
74. In a row, Asha is 9th from the left and 32nd from the right. Bina is 7th from the right. Charu stands 4 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 20
(B) 22
(C) 28
(D) 19

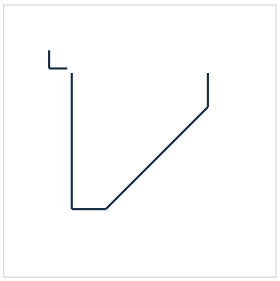
75. Tara points to a photograph and says, "This person's only daughter is my mother." Tara is female, and the photographed person is female. How is the photographed person related to Tara?

(A) Brother
(B) Son
(C) Grandmother
(D) Daughter

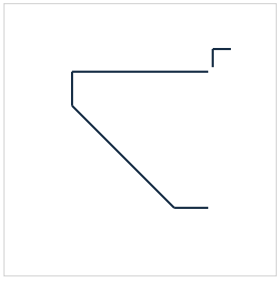
76. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

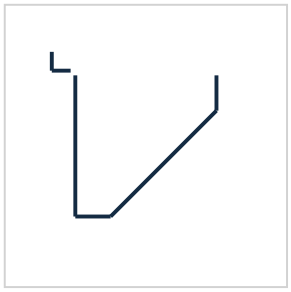


bottom left

?

bottom right

(A) Pattern M



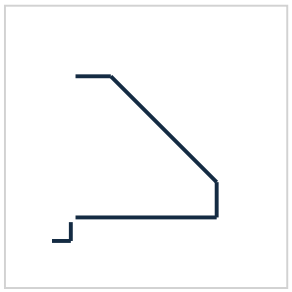
(B) Pattern X



(C) Pattern P



(D) Pattern K



77. Five people occupy positions 1 to 5 from left to right, one per position, and wear five different shirt colours: red, blue, green, white and yellow. Charu is 1st from the left. Bina is 2nd from the left. Esha is 3rd from the left. Asha is 4th from the left. Dev is 5th from the left. The blue shirt is in position 2. The green shirt is in position 3. The red shirt is in position 4. Dev does not wear white. Which shirt does Dev wear?

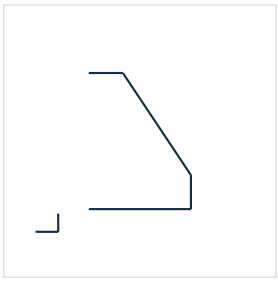
(A) Green

(B) Red

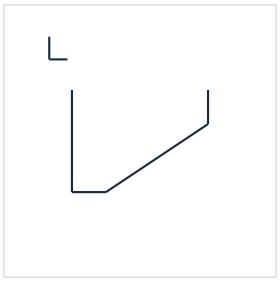
(C) Yellow

(D) Blue

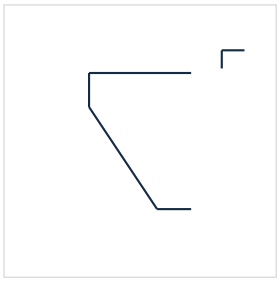
78. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

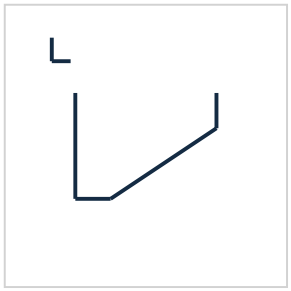


2



3

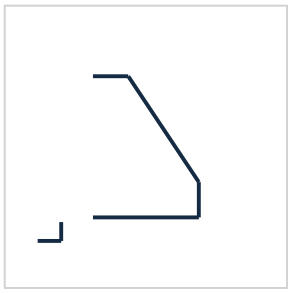
(A) Pattern P



(B) Pattern M



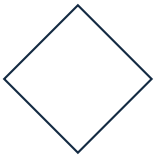
(C) Pattern T



(D) Pattern N



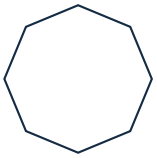
79. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

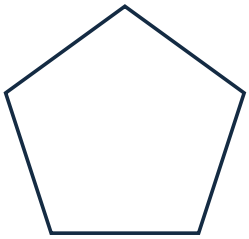


B



C

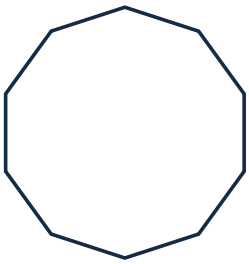
(A) Pattern N



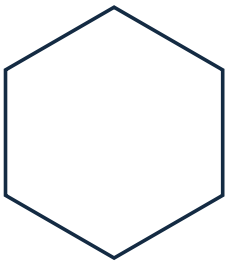
(B) Pattern R



(C) Pattern K

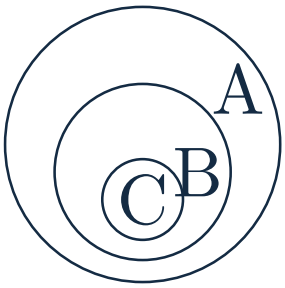


(D) Pattern M

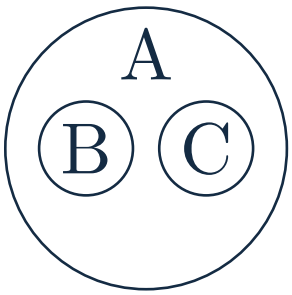


80. Choose the diagram that represents all these conditions, including the strictness information: Every C is B, every B is A; some A is not B, some B is not C, and C is nonempty.

(A) Pattern R



(B) Pattern X



(C) Pattern L



(D) Pattern Y



81. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target



piece 1



piece 2

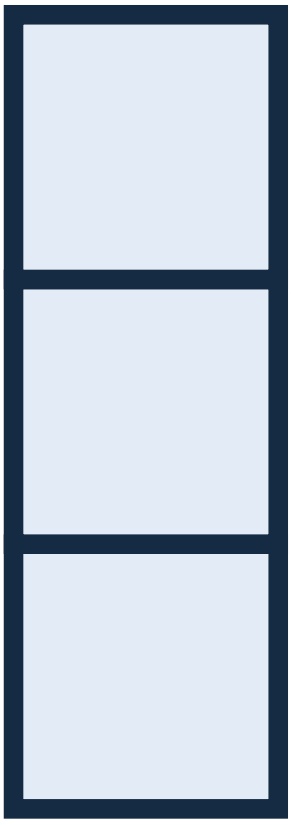
(A) Pattern K



(B) Pattern W



(C) Pattern T



(D) Pattern R



82. Preserve the exact relation and its order: Rapid : Swift :: ? : Help. Choose the missing term.

(A) Silent (B) Assist

(C) Begin (D) Rapid

83. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 5$ and $5 \mapsto 11$. Which input maps to 15?

(A) 8 (B) 7

(C) 6 (D) 15

84. Each letter position has its own fixed cyclic shift, applied identically in both pairs (Z wraps to A). Complete CUB : FWD :: GNX : ?.

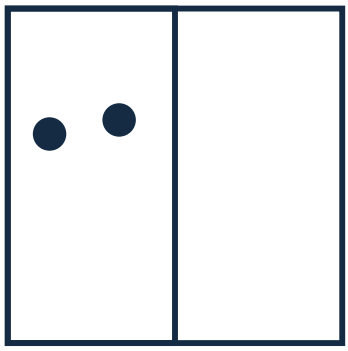
(A) KQA
(B) IOY
(C) API
(D) JPZ

85. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?

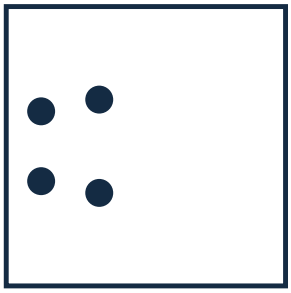


(A) 3
(B) 1
(C) 2
(D) 4

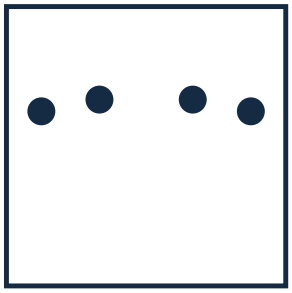
86. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



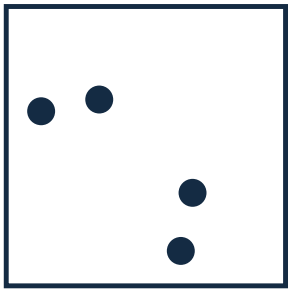
(A) Pattern N



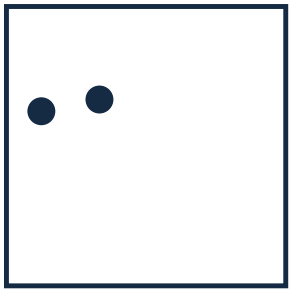
(B) Pattern Y



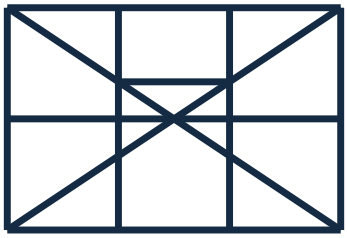
(C) Pattern X



(D) Pattern P



87. How many triangles are formed by all the drawn segments?



(A) 51 (B) 29

(C) 20 (D) 25

88. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 7, 14, 16, ?, 34, 68, 70.

(A) 34 (B) 18

(C) 30 (D) 32

89. A code shifts letters cyclically through A–Z. Odd positions share one fixed shift and even positions share another. For example, MANGO is coded as PYQER. Which word is coded as SJDLW?

(A) PLANT (B) VHGJZ

(C) OKZMS (D) QMBOU

90. Consider only these statements; universal statements alone do not imply existence. Some editors are campers; Some editors are not guides; All guides are editors. Conclusions: I. No campers are guides. II. All campers are guides. Which conclusions are individually possible without contradicting the statements?

(A) Neither I nor II (B) Both I and II

(C) Only II (D) Only I

Mathematics

91. For $\sin(d^3y/dx^3)+y=0$, find order+degree if the degree is defined; otherwise select "Degree is not defined".

(A) Degree is not defined (B) 9

(C) 8 (D) 15

92. For $f(x)=|x-2|/(x-7)$, on which maximal open intervals is f strictly decreasing?

(A) $(-\infty, 7)$ (B) $(2, 7) \cup (7, \infty)$

(C) $(2, \infty)$ (D) $(-\infty, 2)$

93. Two distinguishable fair dice show X,Y. Given $X+Y \geq 8$, find the probability that $X \neq Y$ and XY is even.

(A) $\frac{1}{3}$ (B) $\frac{5}{6}$

(C) $\frac{5}{18}$ (D) $\frac{2}{3}$

94. How many solutions in $0 \le x < 10\pi$ satisfy $\sin 2x = \sin x$ and $\cos x > 0$?

(A) 15 (B) 16

(C) 10 (D) 20

95. Let $z=(1+i)^7/(1-i)^2$. Find $\mathrm{Re}(z)+\mathrm{Im}(z)$.

(A) -32 (B) -8

(C) 8 (D) 0

96. An AP has first term 3, sum of its first 7 terms equal to 126. Find the sum of the squares of those 7 terms.

(A) 1879 (B) 63

(C) 15876 (D) 2968

97. S_n$=5n^2+3n$ is a sequence partial sum. Find $a_1 + a_3 + \cdots + a_{15}$.

(A) 638 (B) 664

(C) 344 (D) 624

98. $y'+4y=\exp(-4x)$, $y(0)=6$. Find $\exp(4)y(1)-6$.

(A) 1 (B) -1

(C) $\frac{7}{e^4}$ (D) 7

99. $A = \begin{bmatrix} 2 & 1 \\ 1 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 12 & 4 \\ 17 & 13 \end{bmatrix}$, and XA=B. Find X_{11}.

(A) $\frac{68}{11}$ (B) 5

(C) 12 (D) $-\frac{68}{11}$

100. For $4x^2+16y^2=64$, find the distance between its two directrices.

(A) $4\sqrt{3}$ (B) $\frac{8\sqrt{3}}{3}$

(C) 8 (D) $\frac{16\sqrt{3}}{3}$

101. A=(5,3,2), O=(0,0,0). Find the squared length of the orthogonal projection of OA on the line through O with direction (1,2,2).

(A) 25 (B) 225

(C) 13 (D) 5

102. A loaded die has odd-face probability 3/5. Three independent throws are made. Given the first two have an even sum, find the probability that all three faces are odd.

(A) $\frac{9}{13}$ (B) $\frac{27}{65}$

(C) $\frac{3}{5}$ (D) $\frac{27}{125}$

103. 12 distinct people sit at labelled round tables of sizes 5,7. Two specified people must be at different tables, and the first specified person must be at the 5-seat table. Rotations are identical. Count the seatings.

(A) 7257600
(B) 127008000
(C) 3628800
(D) 13685760

104. $A = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$, B=(I-A)(I+A)$^{-1}$. Find det B.

(A) 5
(B) 1
(C) $\frac{1}{5}$
(D) -1

105. The line 3x+4y=10 cuts $x^2+y^2=49$ in chord AB. Find $|AB|^2$.

(A) 196
(B) $6\sqrt{5}$
(C) 180
(D) 45

106. Evaluate $\sum_{j=5}^{7} \binom{j}{1}$, with the convention $\binom{j}{r} = 0$ when j<r.

(A) 28
(B) 22
(C) 13
(D) 18

107. $A = \begin{bmatrix} 8 & 4 \\ -4 & 4 \end{bmatrix}$, K=(A-A^T)/2. Find tr(K^2).

(A) 32
(B) -16
(C) -32
(D) 0

108. A line has direction(t,1,k) and is parallel to both planes x-10 y+z=0 and 2 x+y-z=0. Find the squared length of its direction vector.

(A) 121
(B) $\sqrt{59}$
(C) 59
(D) $\frac{491}{9}$

109. Evaluate lim as n$\to\infty$ of (5/n)Σ|5 j/n-5/3|, j=1,...,n.

(A) $\frac{125}{18}$
(B) $\frac{25}{18}$
(C) $\frac{25}{6}$
(D) $\frac{125}{9}$

110. Find the coefficient of x^8 in $(1+x+x^2)(1+x^2)^9$.

(A) 252
(B) 210
(C) 378
(D) 126

111. The line r=(7,5,5)+t(1,2,2) meets x+y+z=42 at Q. Find |PQ|^2 for P=(7,5,5).

(A) 15
(B) 594
(C) 225
(D) 45

112. Let $D(t) = \det \begin{pmatrix} 5+t & 1+t \\ 1+t & 4+t \end{pmatrix}$. Find D′(0).

(A) 11
(B) -7
(C) 38
(D) 7

113. Let R(x) be the remainder when x^7 is divided by (x-1)(x-2). Find R(7).

(A) 823543
(B) 763
(C) 889
(D) -755

114. An AP has a_1=-50,d=5. How many of its first 14 terms are strictly positive?

(A) 3
(B) 11
(C) 2
(D) 12

115. A convergent GP has first term 11, ratio $\frac{1}{7}$. Find the sum of all terms after its first 8 terms.

(A) $\frac{11}{4941258}$
(B) $\frac{10568800}{823543}$
(C) $\frac{11}{705894}$
(D) $\frac{11}{5764801}$

116. Find the coefficient of x^{12} in (1+5x)exp(8x).

(A) $\frac{134217728}{155925}$
(B) $\frac{570425344}{467775}$
(C) $\frac{33554432}{31185}$
(D) $\frac{67108864}{467775}$

117. Evaluate $\lim_{x\to 0} \frac{(1+x)^{11} - 1 - 11x - 4x^2}{x^2}$.

(A) 55
(B) 59
(C) 51
(D) 7

118. Evaluate $\int$ from 0 to $\pi/2$ of $[\sin^3 x/(\sin^3 x+\cos^3 x)+\sin^3(\pi/2\text{-}x)/(\sin^3(\pi/2\text{-}x)+\cos^3(\pi/2\text{-}x))]x\,dx$.

(A) $\frac{\pi^2}{4}$
(B) $\frac{\pi^2}{8}$
(C) $\frac{\pi^2}{16}$
(D) $\frac{\pi}{4}$

119. Let f(x)=sin x+8 cos x. Let [L,L+π] be the interval containing 0 in its interior on which f is increasing and bijective onto[-sqrt(65),sqrt(65)]. If g is its inverse, findL+g′(1).

(A) $-\frac{\pi}{2} - \operatorname{atan}(8) + 8$
(B) $-\frac{\pi}{2} - \operatorname{atan}(8) - \frac{1}{8}$
(C) $-\frac{\pi}{2} - \operatorname{atan}(8) + \frac{\sqrt{65}}{65}$
(D) $-\frac{\pi}{2} - \operatorname{atan}(8) + \frac{1}{8}$

120. In a group of 62, 32 studyA, 27 studyB and 10 study neither. A person is chosen uniformly from those who do not studyB. Find the probability the person studiesA.

(A) $\frac{7}{27}$
(B) $\frac{32}{35}$
(C) $\frac{5}{7}$
(D) $\frac{16}{31}$

121. Four observations are 6,6,x,y with population mean 6 and variance 4/2. Find xy.

(A) 32 (B) 28

(C) 36 (D) 68

122. f(x)=sin(4x)/x+5 for x<0; f(x)=k cos x for x≥0. Find k for continuity at zero.

(A) 9 (B) -1

(C) 4 (D) 5

123. α,β are roots of x^2-16x+63=0. The monic polynomial with roots $\alpha+1,\beta+1$ is x^2-Ax+B. Find B.

(A) 79 (B) 48

(C) 80 (D) 324

124. The letters A,A,B,C and 3 further distinct letters, one namedD, are arranged in a row. How many arrangements have nonadjacent As, adjacent B,C in either order, and D at an end?

(A) 96 (B) 72

(C) 576 (D) 144

125. α,β are nonzero roots of x^2-15x+50=0. Find the sum of $1+1/\alpha$ and $1+1/\beta$.

(A) $\frac{3}{10}$ (B) $\frac{16}{3}$

(C) $\frac{13}{10}$ (D) $\frac{23}{10}$

126. For x∈[π/9,π/2], find the maximum of sin x/x.

(A) $\cos\left(\frac{\pi}{9}\right)$ (B) $\frac{9\sin\left(\frac{\pi}{9}\right)}{\pi}$

(C) $\frac{\pi}{9\sin\left(\frac{\pi}{9}\right)}$ (D) 1

127. A positive integer-domain function satisfies f(m+n)=f(m)f(n), f(1)=5. Find $\sum_{j=1}^{4} f(j)$.

(A) 2500 (B) 156

(C) 625 (D) 780

128. Evaluate the integral from 0 to π/2 of [(sin x+cos x)/(sec x+csc x)](sin x+8 cos x) dx, using continuous extension at the endpoints.

(A) $\frac{8}{3}$ (B) 9

(C) 3 (D) $-\frac{7}{3}$

129. For $A=\begin{bmatrix}6 & 1\\ 1 & 4\end{bmatrix}$ and C=23A^{-1}, find C_{22}.

(A) 4 (B) 6

(C) -1 (D) $\frac{6}{23}$

130. r=pa+qb, a=(1,0,0), b=(1,1,0), r·a=5, r·b=7. Find $|r|^2$.

(A) $\sqrt{29}$ (B) 29

(C) 7 (D) 13

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A vacuum electromagnetic wave of average intensity 60 W/m^2 passes through a device that reduces its electric-field amplitude by a factor 4. The transmitted wave falls normally on an area 5 cm^2. What average power reaches the area?

(A) $\frac{15}{4}$ W
(B) $\frac{3}{1600}$ W
(C) $\frac{3}{100}$ W
(D) $\frac{3}{400}$ W

132. Photons of energy 38 eV eject electrons from a surface. The fastest electrons enter a uniform magnetic field perpendicular to their velocities. In this field an electron with kinetic energy 1 eV has orbit radius r. The observed maximum orbit radius is $3r$. What is the surface work function? Neglect recoil.

(A) 29 eV
(B) $\frac{29}{9}$ eV
(C) $\frac{341}{9}$ eV
(D) 47 eV

133. A sinusoidal source of peak $\frac{87}{10}$ V feeds a back-to-back series pair of identical Zeners through a 1 kΩ current-limiting resistor. The output is across the diode pair. Each Zener has reverse breakdown 4 V and forward drop 0.7 V. What is the peak-to-peak output voltage? Use the piecewise constant-drop model.

(A) $\frac{87}{5}$ V
(B) 8 V
(C) $\frac{47}{5}$ V
(D) $\frac{33}{5}$ V

Chemistry

134. A solution of one nonvolatile nonelectrolyte has exact relative vapour-pressure lowering 1/51. After removal of solvent only, the lowering becomes 3/53. The initial solvent amount was 10 mol, and Raoult's law holds without association. How many moles of solvent were removed?

(A) 20/9
(B) 20/3
(C) 103/15
(D) 10/3

135. For weak-field high-spin Cr^{3+} and Co^{2+},which statement follows in the spin-only approximation?

(A) They have the same moment $\sqrt{15}$ BM despite different d-electron counts.

(B) Equal moments prove equal atomic numbers and equal ionic radii.

(C) Their moments must differ because different elements cannot have equal unpaired-electron counts.

(D) They are both diamagnetic because ionic charge removes all d electrons.

136. A food extract contains exactly one of the listed two-component combinations. Using the usual qualitative tests under suitable conditions, its results are: Benedict on warming: negative; iodine: positive; Biuret: positive; Sudan III: negative. Which combination matches?

(A) glucose and albumin
(B) glucose and triolein
(C) starch and albumin
(D) glucose and starch

Logical Reasoning

137. A community register uses the following three categories: editors, campers, guides. Its complete relevant assertions are in this short paragraph; do not import everyday assumptions. Some campers are not guides. No editors are campers. No editors are guides. Conclusions: I. No guides are editors. II. Some campers are guides. Which conclusions follow necessarily?

(A) Only II

(B) Only I

(C) Both I and II

(D) Neither I nor II

138. Consider only these statements; universal statements alone do not imply existence. Some designers are designers; All artists are hikers. Conclusions: I. Some designers are artists. II. No designers are artists. Which conclusion or complementary alternative must hold?

(A) Only I

(B) Both I and II

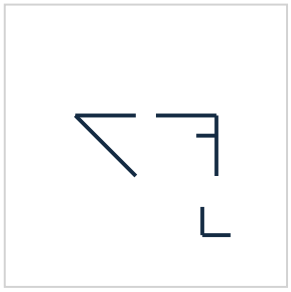
(C) Either I or II follows, but neither is individually certain

(D) Only II

139. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



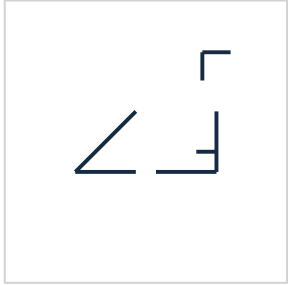
(A) Pattern X



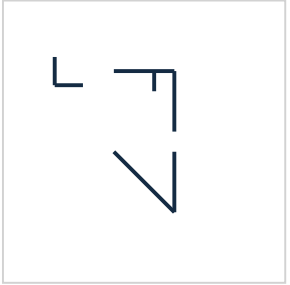
(B) Pattern Z



(C) Pattern L



(D) Pattern V



Mathematics

140. The parabola $(y-3)^2=24(x-2)$ has focus F and vertex V. A circle centred at F passes through V. Find the tangent length from P=(20,15).

(A) $6\sqrt{7}$ (B) 252

(C) $6\sqrt{3}$ (D) $12\sqrt{2}$

141. a=(11,5,0), b=(5,-11,2). Find $|a+b|^2-|a-b|^2$.

(A) 21900 (B) 0

(C) -4 (D) -384

142. Two acute sun elevations $\theta_1<\theta_2$ satisfy $(16^5)^{sin^2\theta} + (16^5)^{cos^2\theta} = 2^5 + 2^15$. A vertical tower has height 7. Find the difference between its two shadow lengths.

(A) $\frac{14}{3}$ (B) $7\sqrt{3}$

(C) $\frac{14\sqrt{3}}{3}$ (D) $\frac{28\sqrt{3}}{3}$

BITSAT Mock 30

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A magnetic field is uniform inside a long solenoid of radius 5 m and negligible outside. It changes at 5 T/s. Find induced electric-field magnitude at radius $5R$ from the axis.

(A) $\frac{5}{2}$ V/m (B) 5 V/m

(C) $\frac{1}{2}$ V/m (D) $\frac{125}{2}$ V/m

2. A series RC circuit is driven at fixed rms voltage. Multiplying angular frequency by 6 multiplies rms current by $\frac{3\sqrt{11}}{8}$. Find R^2/X_C^2, where X_C is the initial capacitive reactance.

(A) $\frac{19}{14}$ (B) $\frac{7}{4}$

(C) 36 (D) $\frac{4}{7}$

3. An ideal transformer has secondary-to-primary turns ratio 6. Its primary draws 3 A at 40 V while delivering power to a resistive secondary load. Find secondary current.

(A) $\frac{1}{2}$ A (B) $\frac{1}{12}$ A

(C) 3 A (D) 108 A

4. A photon and a nonrelativistic particle have equal de Broglie wavelengths. The particle speed is $c/14$. If the photon then loses half its energy while the particle momentum is unchanged, what is the ratio of particle kinetic energy to the photon's final energy?

(A) 1 (B) 14

(C) $\frac{1}{14}$ (D) $\frac{1}{56}$

5. A satellite of mass m is moved from a circular orbit of radius R to one of radius 3R about mass M. Find the required increase in total orbital energy in units of $GMm/(2R)$.

(A) 2 (B) $\frac{4}{3}$

(C) $\frac{1}{3}$ (D) $\frac{2}{3}$

6. A thin equiconvex lens of focal length 60 cm is cut by a plane through its principal axis into two equal halves. One half is used alone with the same optical axis. Find its focal length, ignoring diffraction and edge effects.

(A) $30\sqrt{2}\,\text{cm}$ (B) $60\,\text{cm}$

(C) $120\,\text{cm}$ (D) $240\,\text{cm}$

7. A 1 kg block is launched at 8 m/s on top of a 3 kg block initially at rest on a smooth floor. The coefficient of kinetic friction between them is $\frac{1}{5}$. Find the distance slid by the upper block relative to the lower before slipping ceases. Take $g = 10\ \text{m/s}^2$.

(A) $4\,\text{m}$ (B) $15\,\text{m}$

(C) $12\,\text{m}$ (D) $16\,\text{m}$

8. Two wires of the same material and length L, with radii r and 9r, are joined in series and stretched by a force F. Find their total stored elastic energy divided by the energy stored in the thinner wire alone under force F.

(A) $\frac{10}{9}$ (B) 81

(C) $\frac{82}{81}$ (D) 82

9. A stationary nucleus of mass number 224 emits an alpha particle whose kinetic energy is 4 MeV. Taking daughter and alpha masses proportional to $A-4$ and 4, find the total energy release Q, including daughter recoil.

(A) $\frac{4}{55}\,\text{MeV}$ (B) $4\,\text{MeV}$

(C) $\frac{224}{55}\,\text{MeV}$ (D) $\frac{55}{14}\,\text{MeV}$

10. An ideal mixture contains 5 moles of monatomic gas and 4 moles of rigid diatomic gas. Vibrations are frozen. Find the heat-capacity ratio $\gamma = C_P/C_V$ of the mixture.

(A) $\frac{35}{53}$ (B) $\frac{53}{35}$

(C) $\frac{37}{35}$ (D) $\frac{55}{37}$

11. A particle has displacement $x = 4\cos(2\pi t/3)$ cm. Find its signed velocity at $t = \frac{1}{2}$ s.

(A) $-\frac{4\sqrt{3}\pi}{3}\,\text{cm/s}$ (B) $-\frac{8\pi}{3}\,\text{cm/s}$

(C) $-\frac{2\sqrt{3}}{3}\,\text{cm/s}$ (D) $-\frac{4\pi}{3}\,\text{cm/s}$

12. A solid sphere of radius R has uniform density $6\rho_0$. A spherical cavity of radius $R/10$ is cut with its centre at distance $R/2$ from the original centre. Find field magnitude at the cavity centre in units of $\pi G\rho_0 R$.

(A) $\frac{2}{27}$ (B) $\frac{2}{9}$

(C) 4 (D) $\frac{1}{81}$

13. A gas receives 800 J of heat while its surroundings do 50 J of work on it. It then loses 400 J of heat at fixed volume. Find its total internal-energy change.

(A) $1250\,\text{J}$ (B) $350\,\text{J}$

(C) $400\,\text{J}$ (D) $450\,\text{J}$

14. 2197 identical conducting liquid drops merge without losing charge. Before merging each has potential V and surface charge density σ. Find the final surface charge density divided by σ.

(A) 13 (B) $\frac{1}{13}$

(C) 2197 (D) 28561

15. A network is defined by $P = \mathrm{NOR}(A, B)$, $Q = \mathrm{NOR}(A, P)$ and $Y = \mathrm{OR}(Q, \overline{B})$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 1$ and $P = 0$?

(A) 3 (B) 0

(C) 2 (D) 1

16. A pendulum bob of mass 1 kg and length 5 m passes the horizontal position moving downward at 6 m/s. Find the instantaneous power of all forces when $\sin\theta = 3/5$, $\cos\theta = 4/5$, with θ measured from the downward vertical. Take $g = 10$ m/s^2.

(A) $24\sqrt{6}$ W (B) 36 W

(C) $20\sqrt{29}$ W (D) $12\sqrt{29}$ W

17. In YDSE a sheet of index $\frac{5}{2}$ and thickness 8 micrometres covers one slit. A second sheet of the same thickness and index 3/2 covers the other. With wavelength $\frac{2}{5}$ micrometres, find the magnitude of central-fringe shift in units of fringe width.

(A) 50 (B) 20

(C) 30 (D) 40

18. Two ideal cells of emfs 12 V and 3 V oppose each other in a loop containing two resistors each 3 ohms. Find the current magnitude.

(A) $\frac{3}{2}$ A (B) 2 A

(C) $\frac{5}{2}$ A (D) 3 A

19. A mass 4 kg initially moves in the positive direction at 2 m/s. A force opposite to that direction rises linearly from zero to 2 N and then falls linearly to zero over a total duration 3 s. Find its signed final momentum.

(A) 2 kg m/s (B) 5 kg m/s

(C) 3 kg m/s (D) 11 kg m/s

20. A particle has $v(0) = 5$ m/s and acceleration $a(t) = 4t$ m/s^2 for $0 \le t \le 5$ s. Find its average velocity over this interval.

(A) 5 m/s (B) $\frac{65}{3}$ m/s

(C) 25 m/s (D) $\frac{115}{3}$ m/s

21. One mole of nonoverlapping identical spheres of radius r has total geometric material volume V. A second sample contains $\frac{7}{10}$ mole of spheres of radius $6r$. What is its total geometric material volume divided by V? Ignore packing voids.

(A) $\frac{756}{5}$ (B) $\frac{9261}{125}$

(C) $\frac{7}{2160}$ (D) $\frac{2160}{7}$

22. A travelling wave is $y = 3\sin[6(x - \frac{20}{3}t) + \pi/4]$ mm with x in metres and t in seconds. What is its frequency?

(A) $\frac{20}{\pi}$ Hz (B) $\frac{3}{\pi}$ Hz

(C) $\frac{40}{\pi}$ Hz (D) $\frac{\pi}{20}$ Hz

23. A mass 3 kg moves with $\vec{r}(t) = 3t\hat{i} + 2t^2\hat{j}$ m. Find the z-component of angular momentum about the origin at $t = 2$ s.

(A) $144\,\mathrm{kg\,m^2/s}$ (B) $216\,\mathrm{kg\,m^2/s}$

(C) $36\,\mathrm{kg\,m^2/s}$ (D) $72\,\mathrm{kg\,m^2/s}$

24. A coil of inductance 5 H carries current $i(t) = 1t^2$ A. Find the magnitude of its self-induced emf at $t = 1$ s.

(A) 5 V (B) 50 V

(C) 10 V (D) $\frac{2}{5}$ V

25. A uniform block of density 600 $\mathrm{kg/m^3}$ floats in a liquid. Then the liquid density is reduced to $\frac{4}{5}$ times 1000 $\mathrm{kg/m^3}$, while the block's volume and density are unchanged. What fraction of the block is now submerged?

(A) $\frac{4}{3}$ (B) 1

(C) $\frac{1}{4}$ (D) $\frac{3}{4}$

26. A particle of mass 2 kg and charge 3 C initially has speed 5 m/s. It moves through a uniform electric field 5 N/C with a net displacement 1 m along the field. A simultaneous magnetic field bends its trajectory. Neglect other forces. Find final speed.

(A) $2\sqrt{10}\,\mathrm{m/s}$ (B) $\frac{\sqrt{130}}{2}\,\mathrm{m/s}$

(C) $\sqrt{15}\,\mathrm{m/s}$ (D) $5\,\mathrm{m/s}$

27. Two rods of equal length and cross-sectional area have thermal conductivities k and 4k. They are joined end-to-end with no contact resistance. Find the effective conductivity of the composite divided by k.

(A) $\frac{5}{2}$ (B) $\frac{8}{5}$

(C) 1 (D) 4

28. A particle of mass 4 kg starts from rest and experiences $F(t) = 1t$ N along its motion for 5 s. Find the work done by the force.

(A) $\frac{625}{16}$ J (B) $\frac{125}{24}$ J

(C) $\frac{625}{8}$ J (D) $\frac{625}{32}$ J

29. A 6 V Zener regulator feeds a constant-current load of 6 mA. Supply varies from 14 to 18 V. Regulation requires Zener current at least 2 mA, and its safe maximum is 11 mA. What is the largest permissible series resistance? Assume ideal regulation and give resistance in Ω.

(A) $\frac{12000}{17}\,\Omega$ (B) $1000\,\Omega$

(C) $\frac{4000}{3}\,\Omega$ (D) $4000\,\Omega$

30. A particle of mass 1 kg follows $\vec{r}(t) = 2t^2\hat{i} + 6t\hat{j}$ m. Find the z-component of net torque about the origin at t=2 s.

(A) $-96\,\mathrm{N\,m}$ (B) $48\,\mathrm{N\,m}$

(C) $0\,\mathrm{N\,m}$ (D) $-48\,\mathrm{N\,m}$

Chemistry

31. A carbon-containing organic sample contains Cl only among nitrogen, sulfur and halogens. It undergoes sodium fusion with sufficient sodium. Which conversion and subsequent test protocol is chemically consistent? For a silver-halide test, use a separately prepared portion boiled with nitric acid to remove cyanide/sulfide interference where present.

(A) Nitrate and sulfate in every case; all give the same Prussian-blue test.

(B) chloride; white AgCl after nitric-acid treatment.

(C) Unchanged covalent bonds; add silver nitrate directly to the untreated organic sample.

(D) cyanide; no further treatment is ever needed before adding silver nitrate.

32. At fixedT,V,A(g)→P(g)+Q(g)+R(g)is first order,and an inert gas is present. Initial totalpressure is$\frac{3}{5}$bar;complete-decomposition pressure is$\frac{7}{5}$bar. At two half-lives,totalpressure in bar is

(A) $\frac{6}{5}$ (B) 1

(C) $\frac{7}{5}$ (D) $\frac{13}{10}$

33. An internally consistent thermochemical comparison gives total resonance stabilisation of anthracene as 84 kcal/mol (3 benzenoid rings) and phenanthrene as 92 kcal/mol (3 rings). Which statement correctly separates total stabilisation from stabilisation per ring?

(A) Both quantities must rank compounds identically because each is aromatic.

(B) Total: anthracene; per ring: phenanthrene.

(C) Total: phenanthrene; per ring: anthracene.

(D) Total: phenanthrene; per ring: phenanthrene.

34. How many listed compounds are secondary alcohols? CH3CH2OH, (CH3)2CHOH, (CH3)3COH, C6H5OH. Count compounds,not total OH groups.

(A) 0 (B) 1

(C) 2 (D) 3

35. Classify,in order,the carbon-halogen environments in CH2=CHCl,CH2=CHCH2Cl,C6H5CH2Cl, C6H5Cl. Classify by the carbon directly bonded to Cl,not by whether a double bond appears somewhere in the molecule.

(A) vinyl,allyl,benzyl,aryl (B) vinyl,benzyl,allyl,alkyl

(C) allyl,vinyl,aryl,benzyl (D) alkyl,aryl,vinyl,allyl

36. Equal volumes of $\frac{1}{10}$M $K_4[Fe(CN)_6]$ and $\frac{1}{10}$M glucose are mixed. In the final solution,the salt is50% dissociated into4K^+ and one intact complex anion;glucose does not dissociate. Use RT=24.0 L atm/mol. Osmotic pressure in atm is

(A) $\frac{12}{5}$ (B) $\frac{48}{5}$

(C) $\frac{36}{5}$ (D) $\frac{24}{5}$

37. A buffer contains 3/100 mol HA and 7/100 mol A-. After adding strong base without volume change it has pH=pKa+log10(9), with HA still present. How many moles of OH- were added? Quantitative neutralisation and the Henderson ratio model apply.

(A) 1/100 (B) 3/100

(C) 7/100 (D) 1/50

38. Ascorbic acid is readily oxidised. Which inference about vitamin C is consistent with both its name and standard class?

(A) It is vitaminK and necessarily insoluble in water.

(B) Its vitamin label means it cannot undergo a chemical reaction.

(C) It is vitaminD, a fat-soluble sterol solely because it is an acid.

(D) It is vitamin C, a water-soluble vitamin whose chemical state can change on oxidation.

39. Dry air is taken to contain21.0% O2by volume. From 120L of this air, an isolation process recovers80.0% of its oxygen into a pure stream. All gas volumes refer to the same T,P. Find recovered O2volume in litres.

(A) $\frac{1896}{25}$ (B) $\frac{126}{5}$

(C) $\frac{504}{25}$ (D) $\frac{63}{2}$

40. A first-order reaction has a constant half-life of 10 min. A sample is observed now and again 30 min later, when unreacted material equals 1/256 of the original amount present when the reaction began. How many minutes had elapsed before the first observation?

(A) 80 (B) 30

(C) 50 (D) 90

41. A one-electron ion of nuclear charge Z=3 has a selected shell which, if hypothetically filled subject to Pauli, contains 10 electrons in orbitals with ml=0. What is the magnitude of that shell's hydrogenic energy divided by the hydrogen ground-state energy magnitude? Ignore interactions in identifying the shell.

(A) 18/25 (B) 9/25

(C) 9/50 (D) -9/25

42. Compare phenoxide and benzenesulphonate as oxygen nucleophiles towards the same unhindered methyl substrate in one fixed solvent. Ignore concentration differences. Which pairing of the stronger nucleophile and the main electronic reason is most appropriate?

(A) benzenesulphonate; resonance always increases local charge on oxygen

(B) phenoxide; it is weaker because a localised lone pair cannot attack

(C) phenoxide; sulphonate spreads charge strongly over three oxygens

(D) benzenesulphonate; the more stabilised anion necessarily attacks faster

43. Methyl tert-butyl ether is cleaved with cold HI. The tert-butyl iodide fragment is isolated under conditions that retain it,then heated with a strong bulky base. Which final hydrocarbon forms?

(A) but-1-ene (B) 2-methylpropene

(C) but-2-ene (D) methane

44. For pentan-2-ol and pentan-3-ol,which test/observation assignment follows the actual carbinol class or oxidisable structural feature? Use ordinary room-temperature tests unless the named sequence specifies otherwise.

(A) Lucas reagent;only pentan-3-ol is tertiary

(B) iodoform test;only pentan-3-ol is positive

(C) FeCl3;both are phenols

(D) iodoform test;only pentan-2-ol is positive

45. A compound gives intense sweetness at a small concentration but is not added for antimicrobial action. Which role is supported by that description?

(A) Antiseptic solely because it dissolves.

(B) Tranquilizer solely because it is organic.

(C) Artificial sweetening agent.

(D) Disinfectant solely because it is synthetic.

46. A thermal assembly contains Cu(s) and He(g) in separate inert compartments at the same temperature. Helium is heated quasistatically at constant pressure under a freely moving piston. Containers have negligible heat capacity. Use the classical Dulong-Petit solid model for Cu, neglecting its Cp-Cv difference, and the ideal monatomic model for He. No phase change occurs. Molar masses are Cu64 and He4 g/mol. Over a small common temperature rise, the measured total heat capacity C divided by total Cu-plus-He mass m and gas constant R is C/(mR)=89/320 mol/g. What is the Cu mass fraction?

(A) 3/5

(B) 31/105

(C) 222/385

(D) 2/5

47. An ideal gas freely expands into vacuum in an insulated rigid enclosure,accessible gas volume increasing 3-fold. Which joint statement follows?

(A) The gas does work on the vacuum and its temperature necessarily falls.

(B) q=0 and w=0 imply entropy change is zero and the expansion is reversible.

(C) q=0,w=0 and temperature is unchanged;entropy increases,so the process is irreversible.

(D) The rigid enclosure forces the gas-accessible volume to remain unchanged.

48. $\frac{1}{25}$ mol Cl_2 is contacted with $\frac{3}{50}$ mol KOH in cold dilute solution. Only $Cl_2 + 2KOH \rightarrow KCl + KClO + H_2O$ occurs. Final solution volume is $\frac{1}{5}$ L. Find the KClOconcentration in mol/L, assuming no side reaction or product loss.

(A) $\frac{1}{20}$

(B) $\frac{3}{10}$

(C) $\frac{1}{5}$

(D) $\frac{3}{20}$

49. A standard one-electron cell constructed from AgCl(s)/Ag and Ag^+/Ag has $E^\circ = -\frac{649}{1000}$V for the net reaction $AgCl(s) \rightleftharpoons Ag^+ + Cl^-$. Use $\log_{10}K = E^\circ/.0590$. Place AgCl in a solution with imposed free$[Cl^-]$=0.0100M. Neglect added chloride from dissolution. Free$[Ag^+]$in M is

(A) $\frac{1}{10000000000000}$

(B) $\frac{1}{1000000000}$

(C) $\frac{1}{1000000000000}$

(D) $\frac{1}{100000000000}$

50. A first-order reaction hasEa/R=6600K andfrequency factorA=$(\ln 2)e^{22}/60s^{-1}$. At what nearest whole-kelvin temperature is75.0% completion reached in60.0s? Useln2=.693.

(A) 330

(B) 310

(C) 300

(D) 290

51. 2-bromo-2-methylhexane is heated with potassium tert-butoxide under elimination-favouring conditions. Which major constitutional product is expected,despite the presence of an alkoxide that could be used in a Williamson synthesis with a suitable different halide?

(A) 2-methylhex-2-ene

(B) 2-methylhex-1-ene

(C) the corresponding tertiary alkoxide substitution product

(D) the alkane formed by replacing halogen with H

52. A compound is classified in a chemistry text as a tranquilizer. Which conclusion does that label alone support?

(A) A nervous-system-related therapeutic role, not proof of antibacterial activity.

(B) It must be sodium benzoate.

(C) It must contain a glycosidic bond.

(D) It must kill every bacterium at every concentration.

53. A chemically pure solid retains two frozen orientations at very low temperature. Does purity alone justify setting its entropy to zero by the introductory third law?

(A) No; the perfect-crystal unique-ground-state premise has not been established.

(B) Yes; entropy of every solid is zero.

(C) Yes; purity forces one arrangement at every temperature.

(D) No; the third law requires every sample to be a gas.

54. For standard M2+/M half-cells, E°_B-E°_A=4/5 V and E°_C-E°_B=1/5 V. B has E°=3/10 V. Which pair gives the strongest reducing metal among A,B,C and E°cell for A(s)+C2+→A2++C(s)?

(A) B; 1/5 V

(B) A; 1 V

(C) A; -1 V

(D) C; 1 V

55. Evaluate independently. I: NO_2 is an odd-electron species. II: The central N formal charge in the usual nitrate contributor is zero.

(A) Only I is correct.

(B) Only II is correct.

(C) Neither I nor II is correct.

(D) Both I and II are correct.

56. For an isolated ClO2- ion, use the introductory VSEPR molecular description. Which pair gives chlorine's formal oxidation state and central-atom lone-pair count? Oxygen has oxidation state-2; use the usual lowest-formal-charge Lewis picture for this main-group ion.

(A) (+3, 3)

(B) (+5, 2)

(C) (+3, 2)

(D) (+1, 2)

57. A $\frac{3}{5}$ g organic sample contains chlorine as its only halogen. Carius analysis gives $\frac{10332}{44375}$ g dry AgCl, but this is only 80.0% of the theoretical AgCl because of a known collection loss. Find chlorine mass percentage. Use Ag=108, Cl=35.5.

(A) 12

(B) $\frac{3444}{71}$

(C) $\frac{48}{5}$

(D) 15

58. A solvent has Tb=400 K, M=100 g/mol and ΔHvap=32000 J/mol. Use R=8.00 and Kb=R Tb^2M/(1000ΔHvap). An A2B solution of analytical molality 1/50 mol/kg boils 14/125 K above pure solvent. Only dissociation A2B→2A+ +B2- occurs, with ideal dilute-particle counting. What fraction of A2B is dissociated?

(A) 1/15 (B) 1/10

(C) 1/5 (D) 3/5

59. For penta-1,3-diene, CH2=CH-CH=CH-CH3, count distinct configurational geometrical isomers. Ignore conformers and photochemical interconversion. Where reversing the entire carbon chain identifies two E/Z assignments, count them once.

(A) 1 (B) 4

(C) 3 (D) 2

60. Compare Sn(II) and Pb(II) under the qualitative Group14 inert-pair trend. Which conclusion is consistent?

(A) Sn(II) is more readily oxidised toward+4 than Pb(II), so Sn(II) compounds commonly show reducing character.

(B) The trend is caused by hydrogen bonding between bare metal ions.

(C) Pb(II) is necessarily the stronger reducing state because every heavier element loses all valence electrons more easily.

(D) Neither metal can show oxidation states other than+2.

English Proficiency

61. Which ONE of the following words is misspelt?

(A) independent (B) calendar

(C) pronounciation (D) argument

62. Read the sentence: His lucid explanation made a complicated procedure easy for the new staff to follow. Choose the closest meaning of 'lucid' as used in the sentence.

(A) informal and entertaining (B) clear and understandable

(C) brief and incomplete (D) confident but unsupported

63. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The organisers have reserved the side entrance for performers, not spectators.

(A) The side entrance has been reserved for spectators, not performers.

(B) The side entrance has been reserved by the organisers for performers, not spectators.

(C) The organisers have been reserved for the side entrance by the performers.

(D) The side entrance was being reserved by performers for the organisers.

64. She writes the life of a scientist using letters and interviews; the scientist is not the author. What is she?

(A) editor (B) biographer

(C) novelist (D) autobiographer

65. Which figure of speech is chiefly illustrated by the line? The narrow beam moved across the wall as steadily as the hand of a clock.

(A) apostrophe (B) metaphor

(C) simile (D) pun

66. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. The supervisor asked the group Q. before the summary was circulated. R. against the original records S. whether the figures had been checked

(A) R-P-S-Q (B) S-P-R-Q

(C) Q-S-R-P (D) P-S-R-Q

67. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. the visitors might have chosen Q. Had the warning been displayed earlier, R. a different entrance S. before the queue formed.

(A) Q-P-R-S (B) R-Q-P-S

(C) P-Q-R-S (D) S-P-R-Q

68. Complete: Her financial stake gave her a _____ interest in the outcome, so she did not assess the bids.

(A) dressed (B) clothed

(C) worn (D) vested

69. Read the sentence: The zealous campaigner pursued the cause with sustained enthusiasm. Choose the opposite in meaning of 'zealous' as used in the sentence.

(A) apathetic (B) committed

(C) energetic (D) persistent

70. Read the sentence: The new system will integrate the separate catalogues into one searchable record. Choose the opposite in meaning of 'integrate' as used in the sentence.

(A) separate (B) combine

(C) reorganise (D) update

Logical Reasoning

71. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 7, 28, ?, 148, 157, 628, 637.

(A) 37 (B) 49

(C) 148 (D) 112

72. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: DCJ, ADG, XED, ?, RGX, OHU.

(A) VGB (B) UFA

(C) AFU (D) TEZ

73. The age table uses the same two people throughout:

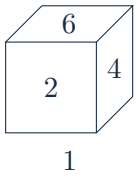
	younger	older
now	x	$x+11$
7 years ago	$x-7$	$x+4$

. The product of their ages 7 years ago was 242. What will the younger person's age be 4 years from now?

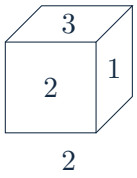
(A) 22 (B) 33

(C) 15 (D) 18

74. The views show the same die. Give the ordered pair: (face opposite 2, face opposite 3). Do not assume opposite labels sum to seven.



1



2

(A) $(5, 2)$ (B) $(5, 6)$

(C) $(5, 3)$ (D) $(5, 1)$

75. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

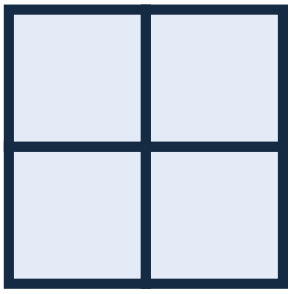


piece 1

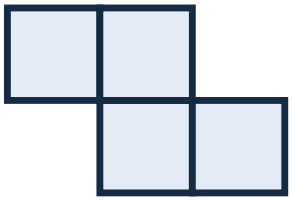


piece 2

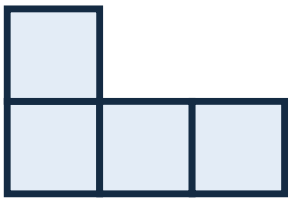
(A) Pattern U



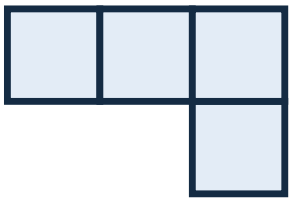
(B) Pattern L



(C) Pattern V



(D) Pattern Q



76. Nita points to a photograph and says, "This person is the son of my father's only daughter." Nita is female, and the photographed person is male. How is the photographed person related to Nita?

(A) Son (B) Brother

(C) Daughter (D) Sister

77. Preserve the exact relation and its order: Star : Constellation :: ? : Team. Choose the missing term.

(A) Lion (B) Wolf

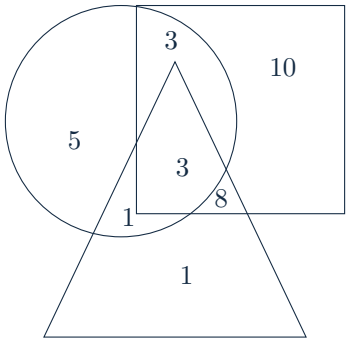
(C) Bee (D) Player

78. Three pairs contain exactly one mammal and one bird. Which pair does not?

(A) Rabbit / Crow (B) Cat / Eagle

(C) Elephant / Duck (D) Bat / Dog

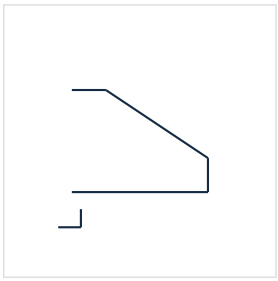
79. Circle=A, triangle=B, square=C. Each printed number is the count in that exact region. Find the total inside A but outside B, whether inside or outside C.



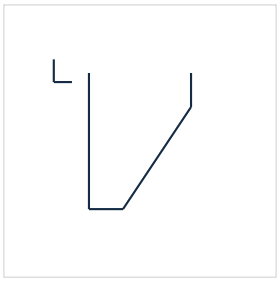
(A) 9 (B) 3

(C) 8 (D) 5

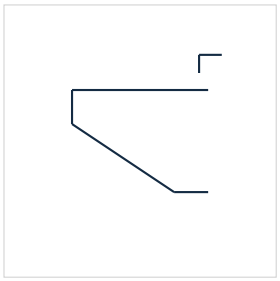
80. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

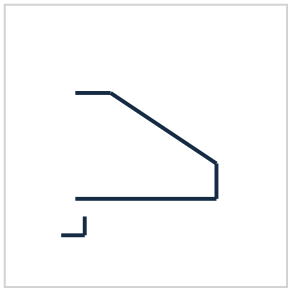


2

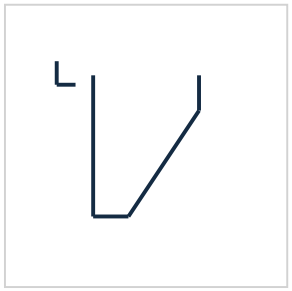


3

(A) Pattern Y



(B) Pattern T



(C) Pattern L



(D) Pattern V



81. In a row, Asha is 11th from the left and 20th from the right. Bina is 12th from the right. Charu stands 4 places to the left of Bina. How many people stand strictly between Asha and Charu?

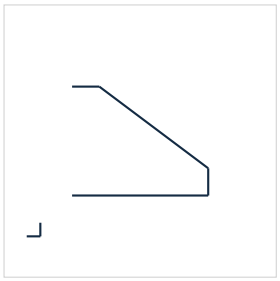
(A) 3

(B) 11

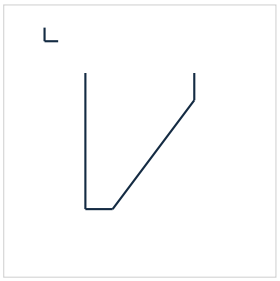
(C) 4

(D) 2

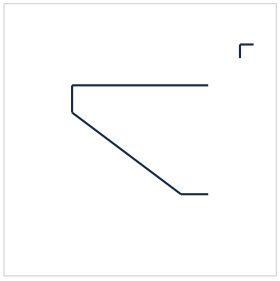
82. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

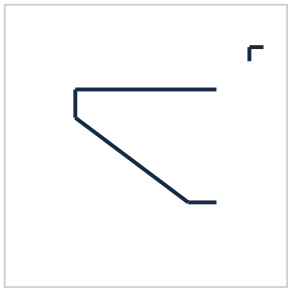


bottom left

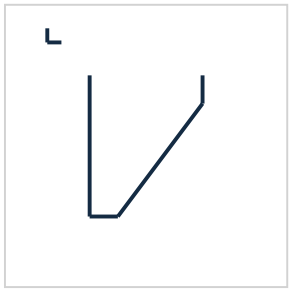
?

bottom right

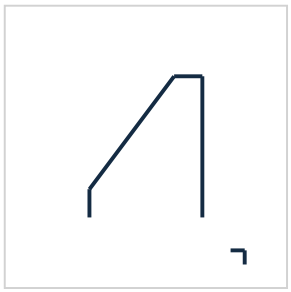
(A) Pattern Q



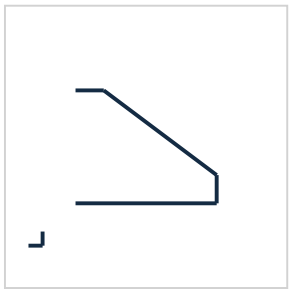
(B) Pattern Z



(C) Pattern X



(D) Pattern W



83. Five people occupy positions 1 to 5 from left to right, one per position, and wear five different shirt colours: red, blue, green, white and yellow. Bina is 1st from the left. Charu is 2nd from the left. Esha is 3rd from the left. Asha is 4th from the left. Dev is 5th from the left. The blue shirt is in position 1. The green shirt is in position 2. The white shirt is in position 5. Asha does not wear red. Which shirt does Asha wear?

(A) Red

(B) Green

(C) Blue

(D) Yellow

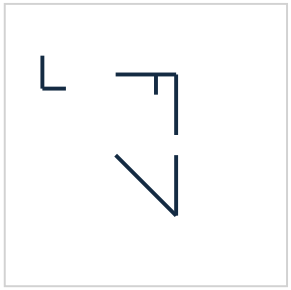
84. Consider only these statements; universal statements alone do not imply existence. No inventors are joggers; All hikers are inventors. Conclusions: I. No joggers are hikers. II. Some hikers are inventors. Which conclusions follow necessarily?

(A) Neither I nor II
(B) Both I and II
(C) Only II
(D) Only I

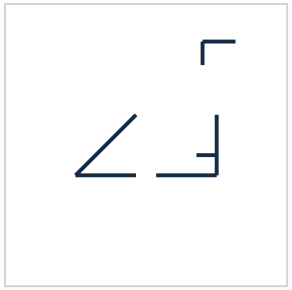
85. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



(A) Pattern P



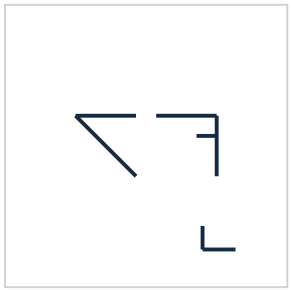
(B) Pattern V



(C) Pattern X



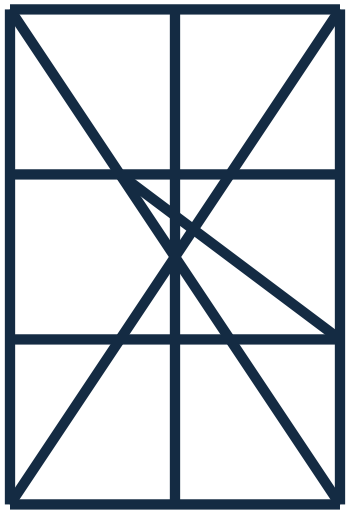
(D) Pattern U



86. Consider only these statements; universal statements alone do not imply existence. No hikers are designers; Some hikers are bakers; All designers are bakers. Conclusions: I. All bakers are designers. II. Some bakers are designers. Which conclusions are individually possible without contradicting the statements?

(A) Neither I nor II
(B) Both I and II
(C) Only II
(D) Only I

87. How many triangles are formed by all the drawn segments?

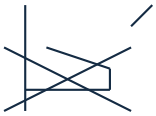


(A) 33 (B) 37

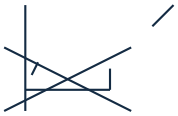
(C) 60 (D) 36

88. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

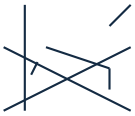




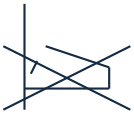
host 1



host 2



host 3



host 4

(A) 1 (B) 2

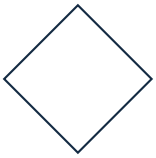
(C) 4 (D) 3

89. A code shifts letters cyclically through A–Z. Odd positions share one fixed shift and even positions share another. For example, STONE is coded as XPTJJ. Which word is coded as YNFES?

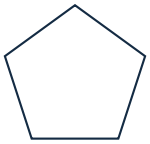
(A) DJKAX (B) TRAIN

(C) SQZHM (D) USBJO

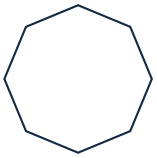
90. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

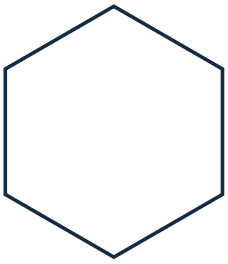


B



C

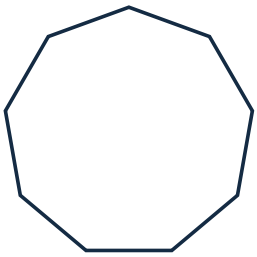
(A) Pattern W



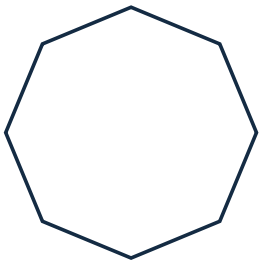
(B) Pattern Z



(C) Pattern R



(D) Pattern T



Mathematics

91. Evaluate $\sum_{j=0}^{3} \binom{3}{j}\binom{7}{3-j}$.

(A) 84

(B) 210

(C) 360

(D) 120

92. The parabola $(y-2)^2=16(x-6)$ has focus F and vertex V. A circle centred at F passes through V. Find the tangent length from P=(18,10).

(A) $4\sqrt{7}$

(B) 112

(C) $4\sqrt{3}$

(D) $8\sqrt{2}$

93. Evaluate sin(2arctan 8).

(A) $\frac{8\sqrt{65}}{65}$

(B) $\frac{16}{65}$

(C) $-\frac{16}{63}$

(D) $-\frac{16}{65}$

94. An AP has first term 5, sum of its first 9 terms equal to 189. Find the sum of the squares of those 9 terms.

(A) 4929 (B) 225

(C) 35721 (D) 3560

95. f(x)=5x+1 and f(g(x))=5x+31. Find g(g(5)).

(A) 37 (B) 17

(C) 22 (D) 11

96. f is cubic with leading coefficient 1/3, stationary points 2,4, and lim as x→0 [f(x)-4]/x=8. Find f(1).

(A) 12 (B) $\frac{16}{3}$

(C) 15.3333333333333 (D) $\frac{28}{3}$

97. For f(x)=|x-5|/(x-9), on which maximal open intervals is f strictly decreasing?

(A) $(-\infty, 9)$ (B) $(-\infty, 5)$

(C) $(5, \infty)$ (D) $(5, 9) \cup (9, \infty)$

98. f(x+1)=$8x^2$. Find the area enclosed between y=f(x) and y=8(x-1).

(A) $\frac{4}{3}$ (B) $-\frac{4}{3}$

(C) $\frac{8}{3}$ (D) 4

99. The chord of contact from P to $x^2/36+y^2/9=1$ is x/6+y/3=1. Find $|OP|^2$.

(A) 45 (B) 27

(C) 9 (D) $3\sqrt{5}$

100. Evaluate lim as x→0 of $[(1+3x)/(1-2x)]^{(1/x)}$.

(A) e^3 (B) e

(C) e^5 (D) 5

101. An increasing positive GP has first three terms with product 64 and first-plus-third sum 10. Find its fourth term.

(A) 1 (B) 8

(C) 16 (D) 4

102. Maximize Z=3x+4y subject to x≥0, y≥3, x+y≤8, x+2y≤10.

(A) 27 (B) 24

(C) 20 (D) 26

103. f(x)=x^3+3x, g=f^{-1}. Find g″(76).

(A) $\frac{1}{24}$ (B) $\frac{8}{44217}$

(C) $-\frac{8}{867}$ (D) $-\frac{8}{44217}$

104. A={j∈{1,...,60}:sin(jπ/2)=0}, B={j∈{1,...,60}:sin(jπ/3)=0}. Find |(A∪B)\(A∩B)|.

(A) 40 (B) 50

(C) 30 (D) 10

105. Two points on opposite sides of a tower foot are 7 metres apart on a straight level line. The elevations of the tower top are 30° and 45°. Find the tower height.

(A) $\frac{7\sqrt{3}}{6}$ (B) $-\frac{7}{2}+\frac{7\sqrt{3}}{2}$

(C) $\frac{7}{2}+\frac{7\sqrt{3}}{2}$ (D) $\frac{21}{2}-\frac{7\sqrt{3}}{2}$

106. A=(2,3,3), O=(0,0,0). Find the squared length of the orthogonal projection of OA on the line through O with direction (1,2,2).

(A) $\frac{14}{3}$ (B) $\frac{196}{9}$

(C) $\frac{2}{9}$ (D) 196

107. f(x)=5+$\int_0^x$ f(x-t)dt for x≥0. Find $\int_0^1$f(t)dt.

(A) 5 (B) $5e$

(C) $-1+e$ (D) $-5+5e$

108. Two observation points lie on the same side of a tower foot, on a straight level line, and are 8 metres apart. The nearer point sees the top at 45° and the farther point at 30°. Find the tower height.

(A) $4+4\sqrt{3}$ (B) $8+8\sqrt{3}$

(C) $-4+4\sqrt{3}$ (D) $8\sqrt{3}$

109. How many integers x in [2,14] satisfy (|x-8|-2)/(|x-8|-5)>0?

(A) 3 (B) 7

(C) 5 (D) 13

110. If lim as x→0 of [sin(2x)-ax]/x^3=b is finite, find a+b.

(A) $\frac{2}{3}$ (B) $\frac{10}{3}$

(C) $-\frac{4}{3}$ (D) 2

111. y′-y/x=1 for x>0, y(1)=4. Find y(5)/5-y(1).

(A) $\log(5)$ (B) $-\log(5)$

(C) $\log(5)+4$ (D) $5\log(5)+20$

112. For $x=t^2$,$y=t^3$,t>0, find d^3y/dx^3 at t=3.

(A) $\frac{1}{4}$ (B) $-\frac{1}{72}$

(C) $-\frac{1}{2}$ (D) $\frac{1}{72}$

113. S_n=$6n^2$+5n is a sequence partial sum. Find $a_1+a_3+\cdots+a_{17}$.

(A) 971 (B) 963

(C) 531 (D) 1017

114. L_1:r=t(1,0,1), L_2:r=(4,0,0)+s(0,1,1). Find their shortest distance.

(A) 0 (B) $\frac{4}{3}$

(C) $\frac{4\sqrt{3}}{3}$ (D) 4

115. $f(x)=x^2+2x$ for $x\leq 5$, and $f(x)=px+q$ for $x>5$. If f is differentiable at 5, find p-q.

(A) -13 (B) 35

(C) 37 (D) 12

116. The normal of slope 5 to $y^2=24x$ touches at P. Find the distance from P to the focus.

(A) 156 (B) 150

(C) 24336 (D) 144

117. 9 distinct people stand in a line. Three specified people may occupy neither end position and no two of these three may be adjacent. How many orders are possible?

(A) 43200 (B) 362880

(C) 151200 (D) 7200

118. How many integers x satisfy $|(x-7)+(x-9)|<|x-7|+|x-9|$?

(A) 2 (B) 8

(C) 3 (D) 1

119. A loaded die has odd-face probability 2/3. Three independent throws are made. Given the first two have an even sum, find the probability that all three faces are odd.

(A) $\frac{8}{27}$ (B) $\frac{2}{3}$

(C) $\frac{1}{15}$ (D) $\frac{8}{15}$

120. For $(d^2y/dx^2)^4+(y')^2+x=0$, find order+degree if the degree is defined; otherwise select "Degree is not defined".

(A) Degree is not defined (B) 7

(C) 6 (D) 8

121. Find the area bounded by $y=x$, $y=4/x$, and $x=14$ in the first quadrant.

(A) $24-4\log(7)$ (B) 96

(C) $\log(2401)$ (D) $96-4\log(7)$

122. $F'(x)=\frac{(x+1)^2}{(x+2)^4}$ and $F(0)=2$. Find F(6)-2F(0).

(A) $-\frac{1117}{512}$ (B) $-\frac{931}{512}$

(C) $\frac{93}{512}$ (D) $\frac{1117}{512}$

123. X earns 3 and uses 2 labour hours; Y earns 5 and uses 3 hours. There are 80 hours and at least 4 items of Y must be made. Which model is correct?

(A) Maximize $3x+5y$; $2x+3y\leq 80$, $x\geq 0$, $y\geq 4$ (B) Maximize $5x+3y$; $2x+3y\leq 80$, $y\leq 4$

(C) Minimize $3x+5y$; $2x+3y\leq 80$, $y\geq 4$ (D) Maximize $3x+5y$; $2x+3y\geq 80$, $x,y\geq 0$

124. For x,y>0 with sqrt(x)+sqrt(y)=10, find d^2y/dx^2 at x=36.

(A) $\frac{5}{108}$ (B) $\frac{5}{6}$

(C) $\frac{5}{216}$ (D) $-\frac{5}{216}$

125. An observer at height 4 m sees the tower foot at depression 45° and its top at elevation 60°. The observer descends vertically to ground level. If the new elevation is α, find $\tan\alpha$.

(A) 1 (B) $\sqrt{3}$

(C) $1+\sqrt{3}$ (D) $4+4\sqrt{3}$

126. Evaluate $\sum_{r=0}^{11} r(r-1)\binom{11}{r}$.

(A) 11264 (B) 56320

(C) 112640 (D) 61952

127. $A=\begin{bmatrix}2 & 8\\ -8 & 4\end{bmatrix}$, K=(A-$A^T$)/2. Find tr($K^2$).

(A) -64 (B) 128

(C) 0 (D) -128

128. Let $P(x)=(1+\sqrt{2}x)^{10}+(1-\sqrt{2}x)^{10}$. Find the sum of all nonconstant coefficients of P.

(A) 6722 (B) 6724

(C) 6726 (D) 3363

129. For $f(x)=|x-3|/(x-6)$, find all real x where f is differentiable and $f\prime(x)<-1/3$.

(A) (-∞,3) (B) (3,9)

(C) (6,9) (D) (3,6) ∪ (6,9)

130. For t≥0 define F(t)=lim as n→ ∞ of $n^{(-3)}\Sigma$ floor(tj^2), j=1,...,n. Evaluate Σ F($(1/5)^r$), r=0,...,∞.

(A) $\frac{1}{12}$ (B) $\frac{5}{4}$

(C) $\frac{5}{12}$ (D) $\frac{1}{3}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. One mole of a monatomic ideal gas follows $pV^{13} = \text{constant}$. Find the process molar heat capacity in units of R.

(A) $\frac{17}{12}$ (B) $\frac{19}{12}$

(C) $\frac{37}{26}$ (D) $\frac{13}{12}$

132. A charge 4 C moves with $\vec{v} = 4\hat{i} + 2\hat{j}$ m/s in $\vec{B} = 1\hat{i} + 6\hat{j}$ T. What is the signed z-component of its magnetic force?

(A) 8 N (B) 96 N

(C) 88 N (D) 104 N

133. A ray in a medium of index $\frac{8}{5}$ is incident on a medium of index $\frac{3}{2}$. Find the sine of the critical-angle threshold for total internal reflection.

(A) $\frac{1}{16}$ (B) $\frac{15}{16}$

(C) $\frac{5}{8}$ (D) $\frac{225}{256}$

Chemistry

134. Which claim cannot simultaneously be retained with the usual Bohr orbit picture as an exact description and the quantum uncertainty principle?

(A) A one-electron ion has discrete bound energies.

(B) An electron has both a perfectly sharp position on the orbit and a perfectly sharp momentum.

(C) A positive nucleus attracts an electron.

(D) An emitted photon carries the energy lost in a transition.

135. benzophenone reacts with NaBH4 then water. Which product correctly reflects nucleophilic addition or addition-elimination at its carbonyl rather than a different reaction class?

(A) benzoic acid (B) triphenylmethanol

(C) diphenylmethanol (D) benzyl alcohol

136. At its isoelectric point, a simple amino acid can be mainlyH3N+-CH(R)-COO-. Which distinction is correct?

(A) It has separated formal charges but zero net charge. (B) It has no charged atoms.

(C) Its net charge is+2. (D) Its net charge is-2.

Logical Reasoning

137. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). The numerical component follows a constant arithmetic step. Find the missing internal term: TXV2, XZW4, BBX6, ?, JFZ10, NHA12.

(A) FDY8 (B) ECX8

(C) 8YDF (D) GEZ8

138. Each of Charu, Gita, Farah visits on one different day (Monday, Tuesday or Wednesday) and buys one different fruit (Peach, Pear, Apple). Charu visits on Wednesday, and Gita on Monday. The visitor on Wednesday buys apple. Gita does not buy peach. Which fruit does Gita buy?

(A) Apple (B) No unique assignment

(C) Pear (D) Peach

139. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target



piece 1

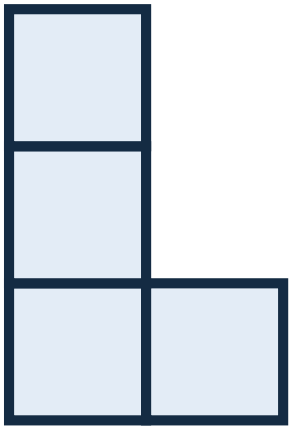


piece 2

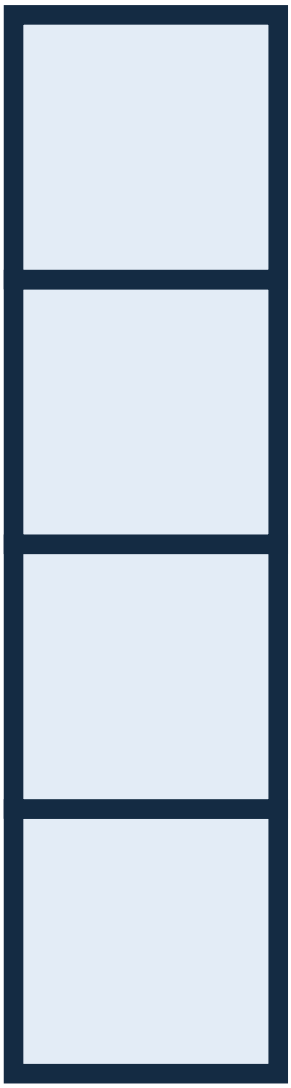
(A) Pattern L



(B) Pattern P



(C) Pattern N



(D) Pattern Q



Mathematics

140. Evaluate the integral from 0 to $\pi/2$ of [(sin x+cos x)/(sec x+csc x)](sin x+9 cos x) dx, using continuous extension at the endpoints.

(A) 10

(B) $\frac{10}{3}$

(C) $\frac{20}{3}$

(D) 3

141. Circles $x^2+y^2=100$ and $(x-16)^2+y^2=100$ meet at A,B. Find $|AB|^2$.

(A) 144

(B) 12

(C) 400

(D) 36

142. For $x\in[0,1]$, find the minimum of $6(\arcsin x)^3+(\arccos x)^3$.

(A) $-\frac{\sqrt{6}\pi^3}{100}+\frac{7\pi^3}{200}$

(B) $\frac{\pi^3}{8}$

(C) $\frac{3\pi^3}{4\left(1+\sqrt{6}\right)^2}$

(D) $-\frac{57\pi^3}{500}+\frac{27\sqrt{6}\pi^3}{500}$

BITSAT Mock 31

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A Carnot engine initially runs between 900 K and 300 K. Its hot reservoir is raised by 20 K. By how much must the cold-reservoir temperature rise to keep efficiency unchanged?

(A) 20 K (B) 60 K

(C) $\frac{150}{23}$ K (D) $\frac{20}{3}$ K

2. A spring of constant 80 N/m is already stretched by $\frac{2}{5}$ m. How much additional work is required to increase its extension to $\frac{7}{10}$ m quasistatically?

(A) $\frac{32}{5}$ J (B) $\frac{66}{5}$ J

(C) $\frac{84}{5}$ J (D) $\frac{132}{5}$ J

3. A rod of length 7 m pivots at its lower end. A peg constrained to move horizontally stays in contact with its upper tip. At an instant the rod makes an angle with $\sin\theta = 3/5$ to the horizontal and the tip's horizontal speed has magnitude 2 m/s. Find the rod's angular-speed magnitude.

(A) $\frac{6}{35}$ rad/s (B) $\frac{2}{7}$ rad/s

(C) $\frac{5}{14}$ rad/s (D) $\frac{10}{21}$ rad/s

4. A series circuit contains resistor R, two ideal inductors L_1, L_2, and a fixed-frequency AC source of angular frequency 50 rad/s. With both inductors active, current lags by 45°; when L_2 is shorted, it lags by 30°. If $R = 80$ ohms, find L_2.

(A) $\frac{8}{5} - \frac{8\sqrt{3}}{15}$ H (B) $\frac{8\sqrt{3}}{15} + \frac{8}{5}$ H

(C) $\frac{8}{5}$ H (D) $4000 - \frac{4000\sqrt{3}}{3}$ H

5. An ideal inductor of 1 H and a resistor of 7 ohms are in series with an ideal 40 V DC source after switching on. At the instant current reaches $\frac{2}{3}$ times its final value, find inductor voltage magnitude.

(A) 120 V (B) $\frac{40}{3}$ V

(C) 40 V (D) $\frac{40}{9}$ V

6. A sinusoidal source of peak $\frac{57}{10}$ V feeds a back-to-back series pair of identical Zeners through a 3 kΩ current-limiting resistor. The output is across the diode pair. Each Zener has reverse breakdown 3 V and forward drop 0.7 V. What is the peak-to-peak output voltage? Use the piecewise constant-drop model.

(A) $\frac{37}{5}$ V (B) $\frac{57}{5}$ V

(C) $\frac{23}{5}$ V (D) $\frac{37}{10}$ V

7. Two ideal cells of emfs 6 V and 3 V oppose each other in a loop containing two resistors each 6 ohms. Find the current magnitude.

(A) $\frac{3}{2}$ A (B) $\frac{3}{4}$ A

(C) $\frac{1}{4}$ A (D) $\frac{1}{2}$ A

8. A flat spiral coil has 40 turns whose radii are uniformly distributed from 3 cm to 6 cm. The same current 6 A flows in every turn. Approximating turns as circular and the distribution as continuous, find magnetic moment in units of $\pi \times 10^{-4}$ A m^2.

(A) 4860 (B) 5040

(C) 8640 (D) 126

9. A cyclist rides straight outward at 6 m/s for 3 s, waits 2 s, then returns to the start. The whole journey must last 6 s. What constant return speed is required?

(A) 6 m/s (B) $\frac{9}{2}$ m/s

(C) 18 m/s (D) 3 m/s

10. Light of vacuum wavelength 500 nm passes from a medium of index 3 into one of index 4. Find the ratio of its frequencies in the second and first media.

(A) $\frac{9}{16}$ (B) $\frac{3}{4}$

(C) $\frac{1}{4}$ (D) 1

11. A uniform block of density 500 kg/m^3 floats in a liquid. Then the liquid density is reduced to $\frac{4}{5}$ times 1000 kg/m^3, while the block's volume and density are unchanged. What fraction of the block is now submerged?

(A) $\frac{5}{8}$ (B) $\frac{8}{5}$

(C) $\frac{4}{5}$ (D) $\frac{2}{5}$

12. A 1 kg block moving at 7 m/s compresses a spring of constant 140 N/m. During compression, a damper removes 1/5 of the block's initial kinetic energy. Find maximum compression; gravity does no work.

(A) $\frac{7}{25}$ m (B) $\frac{\sqrt{7}}{10}$ m

(C) $\frac{\sqrt{7}}{5}$ m (D) $\frac{\sqrt{14}}{10}$ m

13. A uniform-density spherical planet has radius R and surface gravitational acceleration g. At what depth below the surface is gravity $g/12$? Give the depth as a fraction of R.

(A) $\frac{11}{12}$ (B) 11

(C) $\frac{143}{144}$ (D) 1

14. Here L is inductance, C capacitance, R resistance, V potential, I current and t time. Which expression has the same dimensions as both $(CV)^2$ and $(It)^2$?

(A) $(RC)^2$ (B) $((V/R)t)^2$

(C) $(V/R)^2$ (D) $(LI)^2$

15. A monatomic ideal gas expands reversibly and adiabatically until its pressure becomes 1/1024 of its initial value. Find final temperature divided by initial temperature.

(A) 1 (B) $\frac{1}{64}$

(C) $\frac{1}{16}$ (D) $\frac{1}{1024}$

16. Two identical converging thin lenses of focal length 15 cm are separated by 75 cm. A parallel beam enters the first. Find the distance of the final image from the second lens, positive on the outgoing side.

(A) 20 cm (B) $\frac{75}{4}$ cm

(C) 60 cm (D) $\frac{15}{2}$ cm

17. A stretched string is divided by a bridge into lengths in the ratio 1 : 6. Tension and linear density are unchanged. The original fundamental is 60 Hz. Find the fundamental of the longer segment.

(A) $\frac{360}{7}$ Hz (B) 60 Hz

(C) 420 Hz (D) 70 Hz

18. A ball is dropped from 20 m onto a horizontal fixed floor. At every impact the coefficient of restitution is $\frac{4}{5}$. Neglect air resistance. What maximum height does it reach after its second impact?

(A) 16 m (B) $\frac{4}{125}$ m

(C) $\frac{1024}{125}$ m (D) $\frac{64}{5}$ m

19. A 4 V Zener regulator feeds a constant-current load of 2 mA. Supply varies from 11 to 13 V. Regulation requires Zener current at least 3 mA, and its safe maximum is 10 mA. What is the largest permissible series resistance? Assume ideal regulation and give resistance in Ω.

(A) 1800 Ω (B) $\frac{7000}{3}$ Ω

(C) $\frac{1750}{3}$ Ω (D) 1400 Ω

20. A hydrogen atom emits a photon in the $5 \to 4$ transition. A hydrogen-like ion of nuclear charge $Z = 4$ emits a photon of exactly the same wavelength and ends in level $n_f = 16$. What is its initial principal quantum number?

(A) 16 (B) $\frac{5}{4}$

(C) 20 (D) 5

21. Two masses 2 kg and 2 kg are connected by a light string over a smooth pulley and accelerate freely. A transverse pulse travels in the taut straight part of this string, of linear density μ. Compare its speed to that in a second string of density 2μ held at tension g N numerically equal to 1 kg weight. Find v_1/v_2.

(A) 2 (B) $\sqrt{2}$

(C) $\frac{1}{2}$ (D) 4

22. A body is dropped from rest at height 45 m above the ground. Find its speed when it is still 15 m above the ground. Neglect air resistance and take $g = 10\ \text{m/s}^2$.

(A) $10\sqrt{3}\,\text{m/s}$ (B) $10\sqrt{6}\,\text{m/s}$

(C) $600\,\text{m/s}$ (D) $30\,\text{m/s}$

23. A liquid of volume 2 litres and bulk modulus 5000000000 Pa is subjected to an additional pressure of 8000000 Pa. Neglect changes in bulk modulus. Find the magnitude of volume decrease in cubic centimetres.

(A) $1250000\,\text{cm}^3$ (B) $\frac{16}{5}\,\text{cm}^3$

(C) $\frac{2}{625}\,\text{cm}^3$ (D) $3200\,\text{cm}^3$

24. A soap bubble initially of radius 5 cm is expanded quasistatically until its volume is 64 times the initial volume. Surface tension remains $\frac{1}{25}$ N/m. Find work against surface tension, in units of $\pi \times 10^{-4}$ J.

(A) 24 (B) 120

(C) 128 (D) 60

25. A stationary nucleus with mass number 24 and binding energy per nucleon $\frac{38}{5}$ MeV splits into two equal daughters, each having total binding energy $\frac{461}{5}$ MeV. Which statement is correct?

(A) The process releases 2 MeV; the daughters have larger binding energy per nucleon.

(B) The process releases 4 MeV; each daughter releases 2 MeV.

(C) No energy can be released because fission always requires a larger mass number than 24.

(D) The process absorbs 2 MeV; the daughters have larger binding energy per nucleon.

26. A uniform disc of mass 3 kg and radius 6 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) $\frac{513}{8}\,\text{kg m}^2$ (B) $\frac{351}{8}\,\text{kg m}^2$

(C) $\frac{405}{8}\,\text{kg m}^2$ (D) $54\,\text{kg m}^2$

27. Two simple pendulums execute 3 and 2 small oscillations respectively in the same time at the same place. Find the first pendulum's length divided by the second's.

(A) $\frac{3}{2}$ (B) $\frac{2}{3}$

(C) $\frac{1}{9}$ (D) $\frac{4}{9}$

28. A metal wire has linear temperature coefficient $\frac{1}{1000}$ per degree Celsius referred to 0°C. Its dimensions are fixed and it stays across a constant voltage. Find power at 30°C divided by power at 0°C.

(A) $\frac{103}{100}$ (B) $\frac{3}{103}$

(C) $\frac{100}{97}$ (D) $\frac{100}{103}$

29. Photons of energy 38 eV eject electrons from a surface. The fastest electrons enter a uniform magnetic field perpendicular to their velocities. In this field an electron with kinetic energy 3 eV has orbit radius r. The observed maximum orbit radius is $2r$. What is the surface work function? Neglect recoil.

(A) $\frac{13}{2}\,\text{eV}$ (B) $50\,\text{eV}$

(C) $26\,\text{eV}$ (D) $35\,\text{eV}$

30. Two small identical conducting spheres initially carry positive charges $5q_0$ and $2q_0$ and are separated by distance d. They touch and are returned to the same separation. Find the final force divided by the initial force.

(A) $\frac{49}{40}$ (B) $\frac{1}{8}$

(C) 1 (D) $\frac{9}{4}$

Chemistry

31. For neutral H2 and D2, take isotope mass numbers H=1, D=2, T=3 and neglect mass defects. Their gases have equal temperature. Which pair gives ($\langle v^2\rangle$_H2/$\langle v^2\rangle$_D2, number of neutrons in one D2 molecule minus that in one H2 molecule)?

(A) (2, 2) (B) (2, 4)

(C) (1/2, 2) (D) (2, 1)

32. A fixer initially receives 1/20 mol AgBr and an unknown amount of free thiosulfate. After complete stoichiometric complex formation, 1/100 mol AgBr remains and no free thiosulfate is left. Use AgBr+2S2O3^{2}-→[Ag(S2O3)2]^{3}-+Br-. What was the initial amount of thiosulfate?

(A) 9/100 (B) 2/25

(C) 1/10 (D) 1/25

33. A cold, humid industrial episode is dominated by soot, fog and sulphur dioxide. A student attributes it to the same warm-sunlight mechanism that generates ozone-rich summer smog. Which correction is best?

(A) Soot proves that the gas is pure ozone. (B) Classical smog contains neither particles nor sulphur compounds.

(C) The stated mixture requires intense summer sunlight to exist. (D) The stated mixture is characteristic of classical sulphurous smog.

34. 2-methylbutane undergoes low-conversion monobromination with Br2/light. Which is the major constitutional product? Use the usual strong tertiary-over-secondary-over-primary bromination selectivity.

(A) 1-bromo-3-methylbutane (B) 1-bromo-2-methylbutane

(C) 2-bromo-2-methylbutane (D) 2-bromo-3-methylbutane

35. For a hypothetical MX(s),formation enthalpy is -350 kJ mol^{-1}. M atomisation is100,first ionisation500, X_2 bond dissociation200,and electron-gain enthalpy X(g)+e$^-$→X$^-$ is-300,all in kJ mol^{-1}. The positive lattice dissociation enthalpy of MX is

(A) 850 (B) 750

(C) −750 (D) 650

36. The same element forms a fully occupied BCC phase A with cell edge a and an FCC phase B with edge 3a. The measured density ratio ρ_B/ρ_A is 7/135. All FCC sites have equal occupancy, atomic mass is unchanged, and vacancies are the only defects. What fraction of FCC sites is vacant?

(A) 3/20 (B) 17/20

(C) 3/10 (D) 1/10

37. Two van der Waals gases obey Tc=8a/(27Rb) and Pc=a/(27b^2). Their measured ratios are Tc,A/Tc,B=4/3 and Pc,A/Pc,B=4/9. Find (a_A/a_B, b_A/b_B).

(A) (8, 3)
(B) (4, 3)
(C) (3, 4)
(D) (4, 3/2)

38. In a simplified atmospheric sequence, nitrogen oxides undergo reactions leading to nitric acid in droplets. Which distinction is correct?

(A) Nitric acid must have been emitted whenever any nitrate is observed.

(B) The emitted NOx is precursor; nitric acid formed later is a secondary product.

(C) Nitrogen gas and nitric acid are the same pollutant class chemically.

(D) All atmospheric acids are particulate carbon.

39. Compare ethoxide and phenoxide as oxygen nucleophiles towards the same unhindered methyl substrate in one fixed solvent. Ignore concentration differences. Which pairing of the stronger nucleophile and the main electronic reason is most appropriate?

(A) phenoxide; resonance always increases local charge on oxygen
(B) ethoxide; it is weaker because a localised lone pair cannot attack
(C) phenoxide; the more stabilised anion necessarily attacks faster
(D) ethoxide; phenoxide delocalises charge into the ring unlike ethoxide

40. Which statement distinguishes cellulose from both a synthetic polyamide and a protein?

(A) It is a natural polysaccharide with glycosidic linkages.
(B) It is a protein made from alpha-amino acids.
(C) It is a polyalkene made only by ethene addition.
(D) It is a synthetic polyamide with peptide bonds.

41. Consider the statement: Bi(III)is relatively more stable than Bi(V)compared with the corresponding nitrogen oxidation states. Which explanation or consequence correctly accounts for this trend rather than applying an unqualified atomic-mass rule?

(A) The ns2pair is increasingly forced into all bonds down every group.

(B) The heavier group member more strongly favours retention of the ns2pair.

(C) Hydrogen bonding is necessarily stronger for every heavier hydride.

(D) Every increase in atomic mass must increase all molecular bond strengths.

42. In an idealised Ostwald-process material balance, $\frac{1}{25}$ mol NH_3 is fed.90.0% of its nitrogen reaches the NO/NO_2 intermediate stream, and80.0% of that stream nitrogen is recovered as HNO_3. Find HNO_3 amount in mol. Each nitrogen atom is tracked once; recycled intermediates are not double-counted.

(A) $\frac{18}{625}$
(B) $\frac{9}{250}$
(C) $\frac{4}{125}$
(D) $\frac{17}{500}$

43. Compare $Ca(OH)_2$ and $Sr(OH)_2$ in water at the same temperature. Which member is generally more soluble,and which explanation avoids applying one universal down-group rule to all salts?

(A) $Sr(OH)_2$; hydroxide solubility increases down group2.

(B) They must be equally soluble because both cations have charge+2.

(C) Solubility is fixed solely by formula mass,with no lattice or hydration contribution.

(D) $Ca(OH)_2$; every group2salt becomes more soluble as the cation grows.

44. Two one-electron ions A and B satisfy Z_B/Z_A=3 and |E_B|/|E_A|=81/4 for their selected Bohr levels. In the ideal hydrogenic model $E \propto -Z^2/n^2$ and $r \propto n^2/Z$. What is r_B/r_A?

(A) 2/27 (B) 8/27

(C) 4/27 (D) -4/27

45. For propanal with NaBH4,then water,which product or mechanistic conclusion is correct under the stated conditions?

(A) propan-1-ol (B) propan-2-ol

(C) propanoic acid (D) propane

46. 4-chlorobenzoic acid is treated in separate stages with SOCl2,excess NH3,and Br2/NaOH. Which final product and carbon-accounting description are correct?

(A) 4-chloroaniline;the carbonyl carbon remains as CH2 (B) 4-chlorobenzylamine;all carbons retained

(C) 4-chloroaniline;one carbon fewer than the starting acid (D) the original acid;amide nitrogen is never retained

47. For $AgCl(s)+2NH_3 \rightleftharpoons [Ag(NH_3)_2]^+ + Cl^-$,Ksp×Kf=1/4. Before AgCl is added,a solution has $\frac{1}{125}$M silver-ammine complex,zero chloride and $\frac{2}{25}$M free NH_3. If all dissolved silver stays complexed and volume is fixed,the extra AgCl solubility s in M is

(A) $\frac{1}{50}$ (B) $\frac{1}{25}$

(C) $\frac{1}{55}$ (D) $\frac{1}{5}$

48. Three separate portions of but-2-yne are treated with I:NaNH2/NH3,II:H2/Lindlar catalyst,III:Na/liquid NH3. Ignore over-reduction and compare the isolated main organic products. Which statement is correct?

(A) I releases H2;II gives trans-but-2-ene;III gives cis-but-2-ene.

(B) I does not form a terminal acetylide; II gives cis-but-2-ene; III gives trans-but-2-ene.

(C) I forms sodium but-2-ynide;II andIII both give cis-but-2-ene.

(D) I gives butane;II gives but-1-ene;III leaves the alkyne unchanged.

49. Both metals in $[Cr(NH_3)_6][Co(C_2O_4)_3]$ are formally +3. Which proposed partner preserves the total composition while exchanging ligand sets between two coordination spheres?

(A) The same salt with one ligand merely renamed; no atoms or donor attachments change.

(B) $[Co(NH_3)_6][Cr(C_2O_4)_3]$

(C) $[Cr(NH_3)_6][Co(C_2O_4)_3]$ with only the counter-ion positions in the written formula reversed.

(D) $[Co(NH_3)_5Cl]Cl_2$ formed by adding chloride to the sample.

50. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_y; same facing signs; 2: parallel p_y; opposite facing signs. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (2, 2) (B) (3, 1)

(C) (1, 3) (D) (2, 1)

51. Use octet Lewis contributors for NO3- and HCO3-; for SO2 exclude expanded-octet contributors. For HCOO-, HCO3- and HNO3, average only the two nonprotonated central-atom-O bonds. For other species average all central-atom bonds. Find (average bond order of the first minus that of the second, sum of total lone-pair counts in one contributor of each). A lone pair on any atom is counted once.

(A) (-1/6, 15)
(B) (-1/6, 14)
(C) (-1/6, 16)
(D) (17/6, 15)

52. Lysine has side chain-(CH2)4NH2. Which comparison with alanine is correct?

(A) The lysine side chain is a carboxylic acid.
(B) Alanine has the same extra amino site.
(C) Lysine must be achiral because it has nitrogen.
(D) Lysine has an additional basic amino site.

53. An open-chain sugar has six carbons, aC2 ketone and OH-bearing terminalCH2 groups. Which classification and reducing-test caution fit?

(A) Aldopentose; ketoses can never reduce any alkaline reagent.
(B) Disaccharide; every six-carbon formula contains two residues.
(C) Amino acid; the carbonyl proves nitrogen is present.
(D) Ketohexose; a positive alkaline reducing test is still possible.

54. For oxide formation, both normalised per mole of O_2, $\Delta G^\circ_A = -315{+}0.10T$ and $\Delta G^\circ_B = -390{+}0.20T$ in kJ. Ignoring kinetic limitations, at what temperatures is B metal a favourable standard-state reducing agent for oxide A?

(A) T $<$ 750 K
(B) At all positive temperatures
(C) T $>$ 750 K
(D) Only T = 750 K

55. Judge I: NO_2^- can bind through N or O. II: For comparable complexes of one metal,the chelate effect can favour a multidentate ligand without changing the metal oxidation state. Which is correct?

(A) Only II.
(B) Only I.
(C) Neither I nor II.
(D) Both I and II.

56. In a hydrogen-like ion Z=2,an electron falls from n=4 to an s orbital with 1 radial nodes. With hydrogen ionisation energy13.6 eV and$hc = 1240$ eV nm,the emitted wavelength in nm is

(A) $\frac{1550}{51}$
(B) $\frac{24800}{51}$
(C) $\frac{6200}{51}$
(D) $\frac{6200}{17}$

57. In an antifluorite-derived crystal, X2- occupies all FCC anion sites. A measured fraction 2/5 of occupied M cations is M2+ and the rest M+. Only tetrahedral cation vacancies compensate charge; no interstitials or anion defects occur. What fraction of all tetrahedral sites is vacant?

(A) 1/5
(B) 2/7
(C) 1/7
(D) 5/7

58. Starting with ethylbenzene,apply excess hot alkaline KMnO4,then acidification. Which organic product description correctly tracks the functional group actually present at each step?

(A) acetophenone
(B) benzaldehyde
(C) benzoic acid
(D) phenylacetic acid

59. Which pair correctly separates a nucleoside from a nucleotide?

(A) Nucleoside=base+sugar; nucleotide adds phosphate.
(B) Both necessarily contain a peptide bond between two amino acids.
(C) Nucleoside=protein+fat; nucleotide adds glucose.
(D) Nucleoside contains phosphate but nucleotide never does.

60. A $\frac{1}{50}$ mol FeC_2O_4 inventory is analysed. A fraction $\frac{1}{2}$ of Fe^{2+} has already become Fe^{3+},but every oxalate remains unchanged. Subsequent complete oxidation gives Fe^{3+} and CO_2. Electron equivalents released,in mol,are

(A) $\frac{1}{25}$
(B) $\frac{1}{100}$
(C) $\frac{7}{100}$
(D) $\frac{1}{20}$

English Proficiency

61. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The council must not publish the private addresses.

(A) The private addresses need not be published by the council.
(B) The private addresses must not be published by the council.
(C) The private addresses must be published by the council.
(D) The council must not be published by the private addresses.

62. He collects and reads books for the pleasure they give him, rather than trading them professionally. What is he?

(A) publisher
(B) bibliophile
(C) librarian
(D) bookseller

63. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. The secretary checked the notes and corrected that distinction. B. The final minutes accurately separated what had been proposed from what had been approved. C. One participant noticed that a suggestion had been recorded as a decision. D. The secretary drafted minutes immediately after the meeting.

(A) The final minutes accurately separated what had been proposed from what had been approved.

(B) The secretary checked the notes and corrected that distinction.

(C) The secretary drafted minutes immediately after the meeting.

(D) One participant noticed that a suggestion had been recorded as a decision.

64. Use the simple present to express the stated regular procedure. Choose the best replacement: Each of the lamps [are switched on] automatically at dusk every day.

(A) have been switched on
(B) is switched on
(C) were switched on
(D) is being switched on

65. Read the sentence: Once both supports had failed, the collapse was inevitable. Choose the closest meaning of 'inevitable' as used in the sentence.

(A) predictable to everyone
(B) irreversible
(C) immediate
(D) unavoidable

66. Read the sentence: The sample was deficient in the nutrient needed for normal growth. Choose the opposite in meaning of 'deficient' as used in the sentence.

(A) unbalanced (B) contaminated

(C) poorly stored (D) sufficiently supplied

67. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The team should have checked the backup before deleting the damaged file.

(A) The backup should have been checked by the team before the damaged file was deleted.

(B) The backup should be checked after the damaged file is deleted.

(C) The damaged file should have checked the team before deleting the backup.

(D) The backup has been checked by the team before deleting the damaged file.

68. The researcher traces where words came from and how their forms changed over centuries. Which field is this?

(A) semantics (B) orthography

(C) etymology (D) phonetics

69. Complete: Please turn _____ at the next doorway; the corridor to the left is closed.

(A) right (B) rightness

(C) rightly (D) righteous

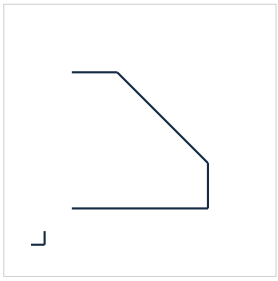
70. Read the sentence: The mediator tried to end the discord between the two clubs. Choose the opposite in meaning of 'discord' as used in the sentence.

(A) harmony (B) argument

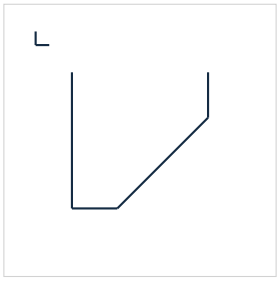
(C) competition (D) division

Logical Reasoning

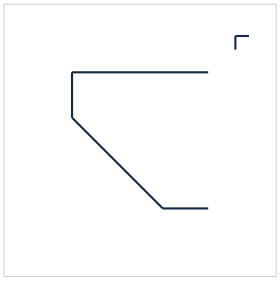
71. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

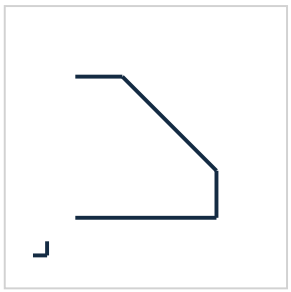


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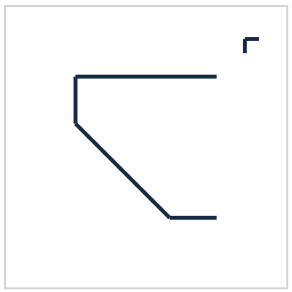


3

(A) Pattern N



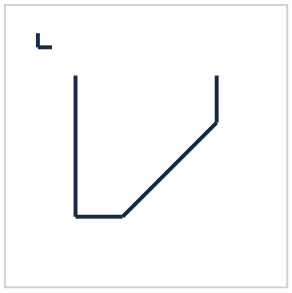
(B) Pattern R



(C) Pattern W



(D) Pattern P



72. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

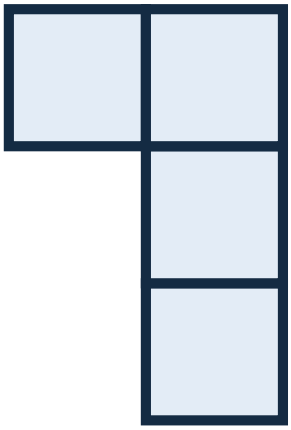


piece 1

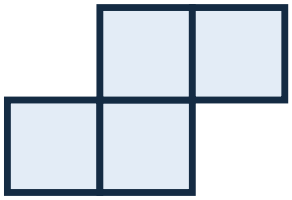


piece 2

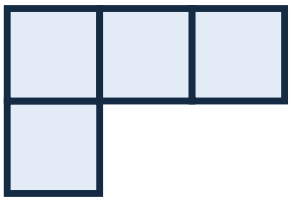
(A) Pattern M



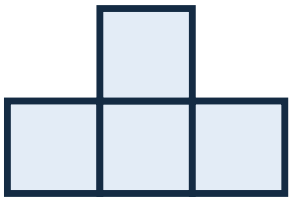
(B) Pattern V



(C) Pattern Q



(D) Pattern U



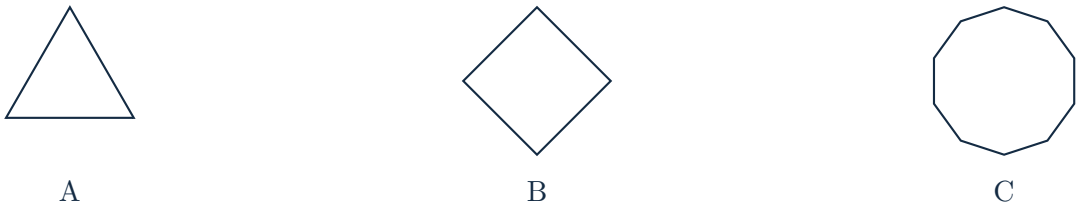
73. A code shifts letters cyclically through A–Z. Odd positions share one fixed shift and even positions share another. For example, RIVER is coded as VEZAV. Which word is coded as THEJX?

(A) OKZMS
(B) PLANT
(C) XDIFB
(D) QMBOU

74. Consider only these statements; universal statements alone do not imply existence. Some farmers are not editors; Some farmers are bakers. Conclusions: I. Some bakers are farmers. II. All farmers are editors. Which conclusions follow necessarily?

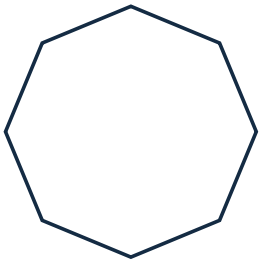
(A) Only I
(B) Neither I nor II
(C) Both I and II
(D) Only II

75. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?

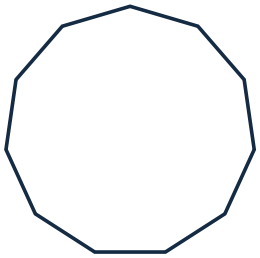


A B C

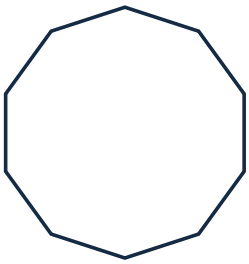
(A) Pattern W



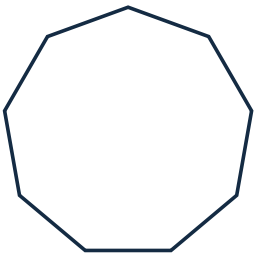
(B) Pattern R



(C) Pattern N



(D) Pattern P

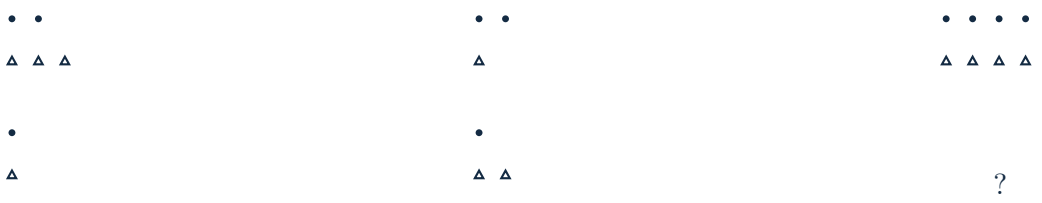


76. Tara points to a photograph and says, "This person is the son of my father's only daughter." Tara is female, and the photographed person is male. How is the photographed person related to Tara?

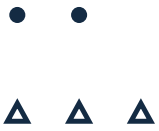
(A) Sister (B) Son

(C) Daughter (D) Brother

77. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



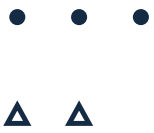
(A) Pattern M



(B) Pattern N



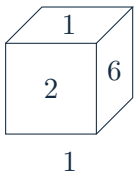
(C) Pattern W

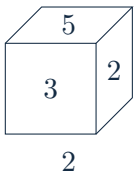


(D) Pattern Y



78. The views show the same die. Give the ordered pair: (face opposite 4, face opposite 5). Do not assume opposite labels sum to seven.





(A) $(2, 6)$
(B) $(2, 1)$
(C) $(2, 2)$
(D) $(2, 3)$

79. The age table uses the same two people throughout:

	younger	older
now	x	$x + 10$
2 years ago	$x - 2$	$x + 8$

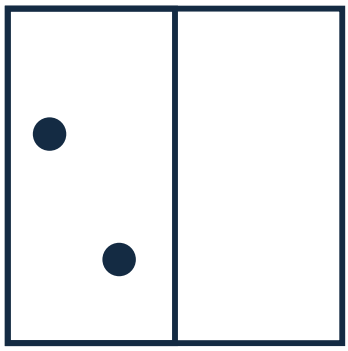
. The product of their ages 2 years ago was 459. What will the younger person's age be 6 years from now?

(A) 19
(B) 25
(C) 23
(D) 35

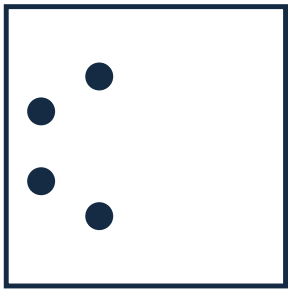
80. Preserve the exact relation and its order: Courageous : Brave :: ? : Authentic. Choose the missing term.

(A) Rapid
(B) Genuine
(C) Silent
(D) Begin

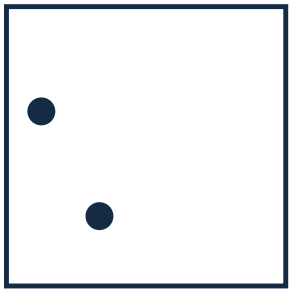
81. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



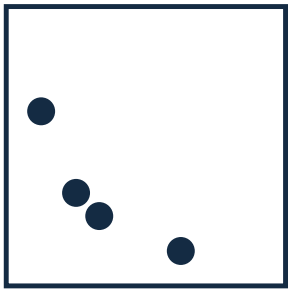
(A) Pattern U



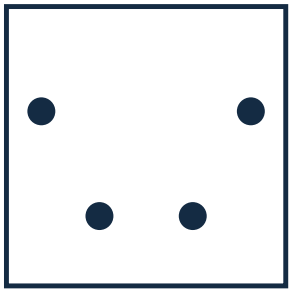
(B) Pattern T



(C) Pattern Q



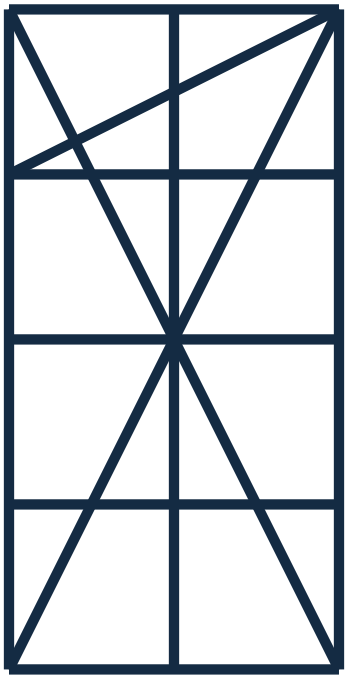
(D) Pattern X



82. Each of Hari, Bina, Esha visits on one different day (Monday, Tuesday or Wednesday) and buys one different fruit (Mango, Pear, Peach). Hari visits on Wednesday, and Bina on Tuesday. The visitor on Wednesday buys peach. Bina does not buy mango. Which fruit does Bina buy?

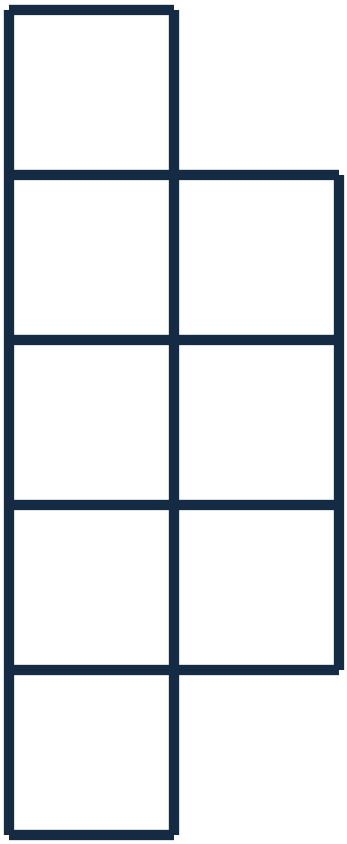
(A) No unique assignment
(B) Peach
(C) Pear
(D) Mango

83. How many triangles are formed by all the drawn segments?



(A) 43
(B) 77
(C) 39
(D) 34

84. How many rectangles, including squares are formed by all the drawn segments?

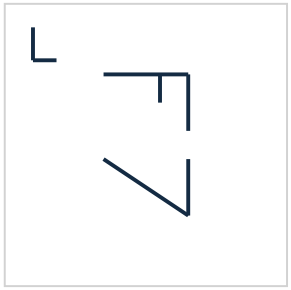


(A) 22
(B) 27
(C) 28
(D) 26

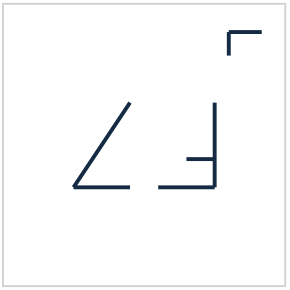
85. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



(A) Pattern Y



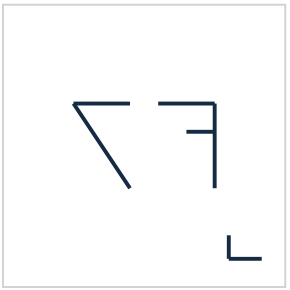
(B) Pattern X



(C) Pattern U



(D) Pattern L



86. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 2

(B) 1

(C) 3

(D) 4

87. Three pairs contain exactly one metal and one nonmetal. Which pair does not?

(A) Silver / Phosphorus (B) Tin / Sulfur

(C) Lead / Iron (D) Gold / Chlorine

88. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 3, 9, 14, ?, 47, 141, 146.

(A) 37 (B) 47

(C) 19 (D) 42

89. In a row, Asha is 8th from the left and 33rd from the right. Bina is 11th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 17 (B) 19

(C) 18 (D) 20

90. The whole has 9 unit positions. Part A covers the first 5; part B covers 5 positions, overlapping A in exactly 1 positions as shown. How many positions remain uncovered by either part?



whole



part A positions



part B positions

(A) -2 (B) 1

(C) 0 (D) -1

Mathematics

91. Group A has 5 observations, mean 13 and population SD 2. Group B has 6 observations, mean 18 and population SD 3. Find the population variance of the combined data.

(A) $\frac{1564}{121}$ (B) $\frac{74}{11}$

(C) 74 (D) $\frac{784}{121}$

92. There are 7 Boolean variables p_1,...,p 7. Exactly 2 variables must be true. How many truth assignments satisfy [(p_1 implies p_2) and (p_1 implies not p_2)]?

(A) 21 (B) 15

(C) 6 (D) 64

93. $P(A)=\frac{1}{4}$, $P(B)=\frac{7}{16}$, $P(A\cap B)=\frac{1}{8}$. Find P(neither A nor B).

(A) $\frac{3}{4}$ (B) $\frac{7}{16}$

(C) $\frac{1}{8}$ (D) $\frac{27}{64}$

94. Let R(x) be the remainder when x^5 is divided by (x-1)(x-2). Find R(4).

(A) 124 (B) 94

(C) 1024 (D) -89

95. A line through P=(7,3) has direction (3,4). It meets 3x+4y=83 at Q. Find $|PQ|^2$.

(A) 100 (B) 10

(C) 50 (D) 4

96. Find the squared distance between 6x+8y=18 and 3x+4y=11.

(A) 4 (B) $\frac{2}{5}$

(C) $\frac{4}{25}$ (D) $\frac{1}{25}$

97. An AP has a_1=8, d=8. Find $a_1+a_3+\cdots+a_15$.

(A) 576 (B) 648

(C) 512 (D) 960

98. How many integers k make $x^2 + kx + 36 = 0$ have two distinct non-real roots?

(A) 23 (B) 11

(C) 24 (D) 25

99. $x=3+7(1-t^2)/(1+t^2)$, $y=4+14t/(1+t^2)$, where t is real. Which point of the containing circle is not attained?

(A) $(3, -3)$ (B) $(10, 4)$

(C) $(3, 11)$ (D) $(-4, 4)$

100. $A=\{j\in\{1,...,48\}:\sin(j\pi/2)=0\}$, $B=\{j\in\{1,...,48\}:\sin(j\pi/3)=0\}$. Find $|(A\cup B)\setminus(A\cap B)|$.

(A) 40 (B) 8

(C) 32 (D) 24

101. For 3x+4y+7=0 and 4x+3y+5=0, find the squared distance from the origin to the angle bisector of positive slope.

(A) $\sqrt{2}$ (B) 2

(C) 4 (D) $\frac{72}{49}$

102. Positive a,b satisfy $\log(a+b)=\log(6)+[\log a+\log b]/2$ in one valid base. Find $(a^4+b^4)/(a^2b^2)$.

(A) 34 (B) 1156

(C) 1158 (D) 1154

103. The line r=(2,2,4)+t(1,2,2) and plane 2x-y=4 are given. Find the squared minimum distance between them.

(A) 0 (B) 4

(C) $\frac{2\sqrt{5}}{5}$ (D) $\frac{4}{5}$

104. For x>1 let $f(x)=\arctan(2x/(1-x^2))$, with arctan in $(-\pi/2,\pi/2)$. Find the integral from 5 to 6 of $f'(x)\,dx$.

(A) π

(B) $-2\,\mathrm{atan}\,(5) + 2\,\mathrm{atan}\,(6)$

(C) $2\,\mathrm{atan}\left(\frac{1}{5}\right)$

(D) $-\,\mathrm{atan}\,(5) + \mathrm{atan}\,(6)$

105. y′+y/(x+1)=1 for x≥0, y(0)=5. Find (4)y(3)-y(0).

(A) $\frac{25}{2}$ (B) $\frac{15}{2}$

(C) $\frac{25}{8}$ (D) $-\frac{15}{2}$

106. The four lines ±x/16±y/14=1 form a rhombus with vertices on the coordinate axes. Find its area.

(A) 448 (B) 224

(C) 896 (D) 112

107. In how many ordered ways can three distinguishable fair dice have sum 13, show at least one 6, and have all three faces different?

(A) 12 (B) 15

(C) 2 (D) 24

108. For $P(x) = \prod_{j=3}^{7}(x+j)$, find the coefficient of x^2 in $P(x)/(x+3)$.

(A) 236 (B) 179

(C) 358 (D) 245

109. Find the area between y=(x-4)(x-10) and the x-axis for 0≤x≤13.

(A) $-\frac{208}{3}$ (B) $\frac{208}{3}$

(C) $\frac{424}{3}$ (D) $\frac{212}{3}$

110. F′(x)=$(x^2+6x+6)\,e^x$ and F(0)=4. Find F(1)-2F(0).

(A) $2+7e$ (B) $-2+7e$

(C) $-6+7e$ (D) $-7e-2$

111. The circle $(x-(-1))^2+(y-(1))^2=400$ has a normal with direction ratio 3 : 4. Its point of contact P satisfies $x_P > -1$. Find $x_P + y_P$.

(A) 28 (B) −28

(C) 20 (D) −4

112. Two acute sun elevations $\theta_1<\theta_2$ satisfy $(16^6)^{sin^2\theta} + (16^6)^{cos^2\theta} = 2^6 + 2^18$. A vertical tower has height 5. Find the difference between its two shadow lengths.

(A) $\frac{10}{3}$ (B) $5\sqrt{3}$

(C) $\frac{10\sqrt{3}}{3}$ (D) $\frac{20\sqrt{3}}{3}$

113. An observer at height 5 m sees the tower foot at depression 30° and its top at elevation 60°. The observer descends vertically to ground level. If the new elevation is α, find tanα.

(A) $\sqrt{3}$ (B) $\frac{2\sqrt{3}}{3}$

(C) $\frac{\sqrt{3}}{3}$ (D) $\frac{4\sqrt{3}}{3}$

114. A loaded die has odd-face probability 5/7. Three independent throws are made. Given the first two have an even sum, find the probability that all three faces are odd.

(A) $\frac{125}{203}$ (B) $\frac{125}{273}$

(C) $\frac{5}{7}$ (D) $\frac{8}{203}$

115. A convergent GP has first term 3, ratio $\frac{1}{4}$. Find the sum of all terms after its first 5 terms.

(A) $\frac{1}{256}$ (B) $\frac{1}{1024}$

(C) $\frac{1}{192}$ (D) $\frac{3}{1280}$

116. z_1=4+5i, z_2=7+5i, z_3=7+8i. Find the smaller angle between z_2-z_1 and z_3-z_1.

(A) $\frac{\pi}{4}$ (B) 0

(C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{4}$

117. For f(x)=|x-4|/(x-7), on which maximal open intervals is f strictly decreasing?

(A) $(-\infty, 7)$ (B) $(4, \infty)$

(C) $(4, 7) \cup (7, \infty)$ (D) $(-\infty, 4)$

118. $P(A) = 9/22$, $P(B) = 4/11$, and $P(A \cap B) = 2/11$. Given that exactly one of A, B occurs, find the probability that A occurs.

(A) $\frac{9}{17}$ (B) $\frac{9}{13}$

(C) $\frac{5}{9}$ (D) $\frac{5}{13}$

119. Evaluate $\Sigma\ j/2^j$ for j=1,...,7.

(A) $\frac{7}{128}$ (B) $\frac{247}{128}$

(C) $\frac{31}{16}$ (D) 2

120. Two distinguishable fair dice show X,Y. Given X+Y≥9, find the probability that X≠Y and XY is even.

(A) $\frac{9}{10}$ (B) 1

(C) $\frac{2}{9}$ (D) $\frac{4}{5}$

121. $F'(x)=\frac{1}{x^2} - \frac{1}{x^2\sqrt{x^2+1}}$ and F(1)=5. Find F(6)-2F(1).

(A) $-\frac{35}{6} - \frac{\sqrt{37}}{6} + \sqrt{2}$ (B) $-\sqrt{2} + \frac{5}{6} + \frac{\sqrt{37}}{6}$

(C) $-\frac{25}{6} - \sqrt{2} + \frac{\sqrt{37}}{6}$ (D) $-\sqrt{2} + \frac{\sqrt{37}}{6} + \frac{35}{6}$

122. For f(x)=x(x-3)(x-5), find the sum of all c∈(0,5) satisfying f′(c)=0.

(A) 8 (B) $\frac{8}{3}$

(C) 15 (D) $\frac{16}{3}$

123. Find the coefficient of x^8 in $(1 + x + x^2)(1 + x^2)^6$.

(A) 15 (B) 21

(C) 35 (D) 45

124. Evaluate lim as n→ ∞ of (6/n)Σ|6 j/n-6/3|, j=1,...,n.

(A) 24 (B) 10

(C) 8 (D) 6

125. f(x)=6+$\int_0^x$ f(x-t)dt for x≥0. Find $\int_0^1$f(t)dt.

(A) $6e$ (B) $-6+6e$

(C) 6 (D) $-1+e$

126. What is the maximum number of distinct intersection points formed by 7 circles and 6 straight lines, with no coincident curves?

(A) 99 (B) 126

(C) 183 (D) 141

127. A convergent geometric progression has positive ratio r. Its sum to infinity is 40/7, and its first two terms have sum 45/8. Find r.

(A) $\frac{7}{8}$ (B) $\frac{3\sqrt{7}}{8}$

(C) $\frac{1}{8}$ (D) $\frac{1}{64}$

128. Let f(x)=sin x+9 cos x. Let [L,L+π] be the interval containing 0 in its interior on which f is increasing and bijective onto[-sqrt(82),sqrt(82)]. If g is its inverse, findL+g′(1).

(A) $-\frac{\pi}{2}-\operatorname{atan}(9)+\frac{1}{9}$

(B) $-\frac{\pi}{2}-\operatorname{atan}(9)+\frac{\sqrt{82}}{82}$

(C) $-\frac{\pi}{2}+\frac{1}{9}+\operatorname{atan}(9)$

(D) $-\frac{\pi}{2}-\operatorname{atan}(9)$

129. Let α=π/7, β=π/6. Evaluate arccos(-cosαcosβ-sinαsinβ).

(A) $\frac{43\pi}{42}$ (B) $-\frac{\pi}{42}$

(C) $\frac{\pi}{42}$ (D) $\frac{41\pi}{42}$

130. Chords of $x^2+y^2=169$ pass through (26,0). Find the radius of the circle containing their midpoint locus.

(A) $\frac{13}{2}$ (B) 169

(C) 26 (D) 13

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. The quantity $Z = x^3y^1/\sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 2, 2 and 6, respectively. To first order, what is the maximum percentage error in Z?

(A) 8 (B) 7

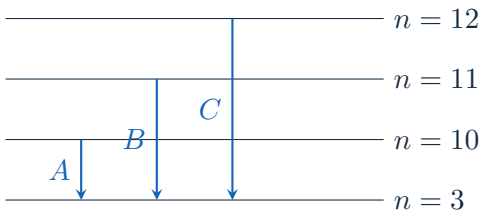
(C) 11 (D) 10

132. A 2 kg block moves along a horizontal rough surface from $x = 0$ to $x = 6$ m under a force $F(x) = 2x$ N parallel to motion. With $\mu_k = 1/5$ and $g = 10$ m/s^2, find the work done by the net force. The block has enough initial speed to keep moving forward.

(A) 12 J (B) 60 J

(C) 2 J (D) 36 J

133. The diagram shows downward hydrogen transitions A, B and C from $n = 10$, 11 and 12, respectively, ending at $n = 3$. A filter transmits only photons with energy strictly above that of transition B. Which transmitted line has the longest wavelength?



(A) B (B) C

(C) No transmitted line (D) A

Chemistry

134. For LiF,LiI and NaI,which comparison correctly separates anion polarisability from cation polarising power?

(A) LiI is more covalent than LiF and more covalent than NaI.

(B) LiF is most covalent because F has the largest electronegativity.

(C) All three have equal covalent character because all ions have unit charges.

(D) NaI is more covalent than LiI because a larger cation always distorts an anion more.

135. A perfect crystal with a unique ground state is compared with a configurationally disordered frozen solid. Which entropy statement is justified as T tends to0 K?

(A) Both must have equal nonzero entropy because their temperatures agree.

(B) The comparison proves that entropy is not a state function.

(C) The perfect crystal has the standard zero-entropy limit; the disordered solid need not meet that premise.

(D) Both must have zero entropy regardless of disorder.

136. Which reagent/name pair is suitable when an alkyl chloride must become the corresponding alkyl fluoride? The target carbon framework must be retained.

(A) CuBr/HBr;Sandmeyer reaction (B) AgF;Swarts reaction

(C) NaI/acetone;Finkelstein reaction (D) Na/dry ether;Fittig reaction

Logical Reasoning

137. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 15$ and $5 \mapsto 30$. Which input maps to 55?

(A) 55 (B) 11

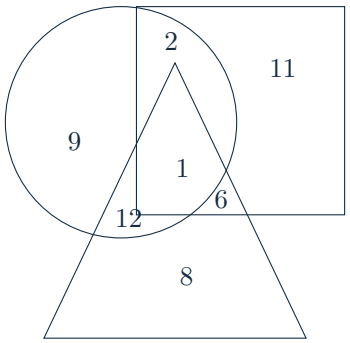
(C) 10 (D) 9

138. Five people occupy positions 1 to 5 from left to right, one per position, and wear five different shirt colours: red, blue, green, white and yellow. Dev is 1st from the left. Esha is 2nd from the left. Charu is 3rd from the left. Asha is 4th from the left. Bina is 5th from the left. The red shirt is in position 2. The green shirt is in position 3. The blue shirt is in position 4. Dev does not wear white. Which shirt does Dev wear?

(A) Yellow (B) Green

(C) Blue (D) Red

139. Circle=A, triangle=B, square=C. Each printed number is the count in that exact region. Find the total inside A but outside B, whether inside or outside C.



(A) 12 (B) 11

(C) 2 (D) 9

Mathematics

140. Find the product of all real solutions of sqrt(7-log_5|x|)=log_5|x|-1.

(A) -125 (B) -15625

(C) 15625 (D) 0

141. For sqrt(d^4y/dx^4)=$(y')^2$+x, find order+degree if the degree is defined; otherwise select "Degree is not defined".

(A) Degree is not defined (B) 8

(C) 5 (D) 7

142. The hyperbola $x^2/9\text{-}y^2/16=1$ has eccentricity e. Find the eccentricity e' of its conjugate hyperbola.

(A) $\frac{5}{4}$ (B) $\frac{5}{3}$

(C) 5 (D) $\frac{3}{5}$

BITSAT Mock 32

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. 2 kg of water at 40°C is mixed in an insulated vessel with $\frac{336}{167}$ kg of ice at 0°C. Use $c_w = 4200$ J/(kg K), $L_f = 334000$ J/kg. Find the mass of ice remaining at equilibrium.

(A) $\frac{56280}{41917}$ kg (B) $\frac{336}{167}$ kg

(C) $\frac{252}{167}$ kg (D) $\frac{168}{167}$ kg

2. A long solid cylindrical conductor of radius 4 m carries uniformly distributed current 6 A. Find magnetic field magnitude at radial distance $R/4$ from the axis in units of $\mu_0/(2\pi)$.

(A) $\frac{3}{8}$ (B) 6

(C) $\frac{3}{32}$ (D) 24

3. A rod of length 5 m pivots at its lower end. A peg constrained to move horizontally stays in contact with its upper tip. At an instant the rod makes an angle with $\sin\theta = 3/5$ to the horizontal and the tip's horizontal speed has magnitude 4 m/s. Find the rod's angular-speed magnitude.

(A) $\frac{4}{5}$ rad/s (B) $\frac{4}{3}$ rad/s

(C) $\frac{16}{25}$ rad/s (D) 4 rad/s

4. A stationary nucleus of mass number 224 emits an alpha particle whose kinetic energy is 8 MeV. Taking daughter and alpha masses proportional to $A-4$ and 4, find the total energy release Q, including daughter recoil.

(A) 8 MeV (B) $\frac{448}{55}$ MeV

(C) $\frac{8\sqrt{55}}{55} + 8$ MeV (D) 448 MeV

5. A uniform disc of mass 8 kg and radius 2 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) 13 kg m^2 (B) 19 kg m^2

(C) 16 kg m^2 (D) 15 kg m^2

6. A conveyor receives 8 packets each second, each of mass $\frac{1}{5}$ kg and speed 3 m/s. It accelerates them to 6 m/s without changing elevation. Find the ideal mechanical power delivered to the packets.

(A) $\frac{36}{5}$ W (B) $\frac{27}{10}$ W

(C) $\frac{108}{5}$ W (D) 36 W

7. A series circuit contains resistor R, two ideal inductors L_1, L_2, and a fixed-frequency AC source of angular frequency 50 rad/s. With both inductors active, current lags by 45°; when L_2 is shorted, it lags by 30°. If $R = 40$ ohms, find L_2.

(A) $-\frac{4}{5} + \frac{4\sqrt{3}}{5}$ H (B) $2000 - \frac{2000\sqrt{3}}{3}$ H

(C) $\frac{4}{5} - \frac{4\sqrt{3}}{15}$ H (D) $\frac{4}{5}$ H

8. One mole of nonoverlapping identical spheres of radius r has total geometric material volume V. A second sample contains $\frac{9}{10}$ mole of spheres of radius $5r$. What is its total geometric material volume divided by V? Ignore packing voids.

(A) $\frac{45}{2}$ (B) $\frac{9}{1250}$

(C) $\frac{225}{2}$ (D) $\frac{729}{8}$

9. In an abrupt junction, the acceptor concentration is 49 times the donor concentration. Within the depletion approximation, which statement gives x_n/x_p and identifies the carriers mainly responsible for the small reverse current before breakdown?

(A) 49; minority carriers (B) 7; majority carriers

(C) 1; majority carriers (D) 1/49; minority carriers

10. The accelerating potential of an X-ray tube is increased from 15 kV to 30 kV, while its tube current is reduced by a factor 2. Ignoring target heating and recoil, what is the new cutoff wavelength in nm? Use $hc = 1240$ eV nm.

(A) $\frac{31}{750}$ nm (B) $\frac{124}{375}$ nm

(C) $\frac{31}{1500}$ nm (D) $\frac{62}{375}$ nm

11. A network is defined by $P = \mathrm{NAND}(A, B)$, $Q = \mathrm{NAND}(A, P)$ and $Y = \mathrm{NOR}(Q, \overline{B})$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 0$ and $A = 0$?

(A) 0 (B) 3

(C) 1 (D) 2

12. Light is incident from index $\frac{7}{5}$ onto index $\frac{19}{10}$. At one incidence angle the reflected beam is completely plane polarised. Find the tangent of that angle.

(A) $\frac{19}{14}$ (B) $\frac{5}{14}$

(C) $\frac{7}{5}$ (D) $\frac{14}{19}$

13. The quantity $Z = x^3 y^1 / \sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 4, 1 and 6, respectively. To first order, what is the maximum percentage error in Z?

(A) 10 (B) 11

(C) 16 (D) 13

14. A circular parallel-plate capacitor of plate radius 2 cm is charged at 38 mA. Neglect fringing. What displacement current passes through the annular part of the gap between radii 1 cm and 2 cm, centred on the plate axis?

(A) 19 mA (B) 38 mA

(C) $\frac{19}{2}$ mA (D) $\frac{57}{2}$ mA

15. A point at the bottom of two plane liquid layers is viewed normally from air. The upper layer has thickness 9 cm and index 3/2; the lower has thickness 20 cm and index 4/3. Find its apparent depth below the top surface in the paraxial limit.

(A) 21 cm (B) 6 cm

(C) $\frac{241}{6}$ cm (D) $\frac{58}{3}$ cm

16. A 1 kg block moves along a horizontal rough surface from $x = 0$ to $x = 3$ m under a force $F(x) = 8x$ N parallel to motion. With $\mu_k = 1/5$ and $g = 10$ m/s^2, find the work done by the net force. The block has enough initial speed to keep moving forward.

(A) 66 J (B) 30 J

(C) 10 J (D) 36 J

17. A mass 2 kg is attached to two massless springs in series with spring constants 30 N/m and 60 N/m. Find the small-oscillation period on a smooth horizontal surface.

(A) $\frac{\sqrt{30}\pi}{15}$ s (B) $\frac{2\sqrt{10}\pi}{15}$ s

(C) $\frac{\sqrt{10}\pi}{5}$ s (D) $\frac{\sqrt{10}}{10}$ s

18. The 3rd harmonic of an open pipe of length 4 m equals the 3rd allowed resonance of a closed pipe. Both have the same sound speed and negligible end correction. Find the closed-pipe length.

(A) $\frac{14}{3}$ m (B) 2 m

(C) $\frac{12}{5}$ m (D) $\frac{10}{3}$ m

19. A wire consists of a semicircular arc of radius 4 m carrying 8 A, joined to two straight radial segments that extend outward from its ends. Find the magnetic field magnitude at the arc centre in units of μ_0.

(A) $\frac{1}{2\pi}$ (B) $\frac{1}{4}$

(C) $\frac{2+\pi}{2\pi}$ (D) $\frac{1}{2}$

20. A thin glass lens of refractive index 3/2 has focal length 210 cm in air. It is completely immersed in a liquid of index 4/3. Find its new focal length.

(A) $\frac{105}{2}$ cm (B) 630 cm

(C) 840 cm (D) 210 cm

21. 8 identical conducting liquid drops merge without losing charge. Before merging each has potential V and surface charge density σ. Find the final surface charge density divided by σ.

(A) 4 (B) 1

(C) 2 (D) 16

22. A bar magnet of moment 3 A m^2 is rotated quasistatically from 60° to 120° relative to a uniform 2 T magnetic field. Find external work.

(A) 3 J (B) 6 J

(C) $6\sqrt{3}$ J (D) 2π J

23. Two identical capacitors each 2 microfarads are connected in series across an ideal 20 V battery. One capacitor undergoes permanent conducting breakdown, becoming a short circuit. Find the additional charge drawn from the battery after the new steady state is reached.

(A) 20 μC (B) 60 μC

(C) 40 μC (D) 10 μC

24. A 4 kg block is launched at 4 m/s on top of a 4 kg block initially at rest on a smooth floor. The coefficient of kinetic friction between them is $\frac{1}{10}$. Find the distance slid by the upper block relative to the lower before slipping ceases. Take $g = 10$ m/s^2.

(A) 8 m (B) 6 m

(C) 2 m (D) 4 m

25. A standing wave is $y = 2\sin(6x)\cos(2t)$ cm with x in metres and t in seconds. At a point where $6x = \pi/6$, find the particle-speed magnitude when $2t = \pi/3$.

(A) $3\sqrt{3}$ cm/s (B) $\sqrt{3}$ cm/s

(C) $\frac{\sqrt{3}}{2}$ cm/s (D) $2\sqrt{3}$ cm/s

26. At the highest point of its flight, at height 40 m, a projectile has horizontal velocity 20 m/s. It explodes into two equal fragments, both initially moving horizontally. One moves forward at 4 m/s. Find the horizontal distance travelled by the other fragment from the explosion point to the ground. Take $g = 10$ m/s^2.

(A) $88\sqrt{2}$ m (B) $72\sqrt{2}$ m

(C) $32\sqrt{2}$ m (D) $8\sqrt{2}$ m

27. A Carnot engine initially runs between 500 K and 400 K. Its hot reservoir is raised by 20 K. By how much must the cold-reservoir temperature rise to keep efficiency unchanged?

(A) 16 K (B) $\frac{200}{13}$ K

(C) 25 K (D) 4 K

28. A charge of 6 μC moves from potential 70 V to -70 V in an electrostatic field. Find the work done by an external agent if the charge starts and ends at rest and no other force does work.

(A) $-140\,\mu$J (B) $-840\,\mu$J

(C) 840 μJ (D) $-420\,\mu$J

29. A body is dropped from rest at height 160 m above the ground. Find its speed when it is still 40 m above the ground. Neglect air resistance and take $g = 10$ m/s^2.

(A) $40\sqrt{2}$ m/s (B) 2400 m/s

(C) $20\sqrt{2}$ m/s (D) $20\sqrt{6}$ m/s

30. A uniform wire is stretched to $\frac{6}{5}$ times its original length at constant volume. Its resistivity increases by 40 percent during the process. Find final resistance divided by initial resistance.

(A) $\frac{7}{6}$
(B) $\frac{252}{125}$
(C) $\frac{108}{125}$
(D) $\frac{42}{25}$

Chemistry

31. CH3CH2CN is treated with complete aqueous acid hydrolysis. Which product satisfies carbon accounting and the reagent selectivity?

(A) ethanoic acid
(B) propan-1-ol
(C) propanoic acid
(D) propanone

32. For standard M2+/M half-cells, E°_B-E°_A=1 V and E°_C-E°_B=1/5 V. B has E°=2/5 V. Which pair gives the strongest reducing metal among A,B,C and E°cell for A(s)+C2+→A2++C(s)?

(A) A; -6/5 V
(B) C; 6/5 V
(C) A; 6/5 V
(D) B; 1/5 V

33. E°(A+/A)=1/5 V and E°(B+/B)=1/2 V. For the stoichiometrically scaled standard reaction 3A(s)+3B+(aq)→3A+(aq)+3B(s), find ΔG° in kJ. Use F=96.5 kJ V-1 mol-1.

(A) -1737/40
(B) -1737/10
(C) 1737/20
(D) -1737/20

34. Within an additive covalent-radius model,a single X-X bond is 140 pm and an X-H bond is 100 pm. A linear H-X-X-H skeleton is assumed with collinear single bonds. Its terminal H-to-H distance,in pm,is

(A) 340
(B) 200
(C) 480
(D) 680

35. A polymer formed by curing melamine with formaldehyde is used as a rigid thermosetting resin. Which classification is consistent with the curing step?

(A) Cross-linked condensation resin.
(B) Linear ethene-addition polymer containing no nitrogen.
(C) Natural protein made only from amino acids.
(D) Unreacted monomer mixture with no covalent network.

36. A student uses the third law to conclude ΔS=0 for every spontaneous reaction at300 K. What is the error?

(A) The third law says every reaction must be endothermic.
(B) Spontaneity forbids any entropy change anywhere.
(C) The law's zero-entropy reference concerns an ideal crystal at0 K, not every finite-temperature reaction entropy.
(D) The law states that all gases have zero entropy at300 K.

37. For Eu^{2+} (Z=63),which configuration/chemical interpretation is consistent with its electron count and the commonly discussed +3 state?

(A) $4f^7$; reducing character in the stated conversion toward the common +3 state.

(B) $4f^7$; oxidising character toward+3.

(C) $4f^6$; neither accepts nor loses an electron in changing to+3.

(D) $4f^8$; reducing character.

38. Which combined classification/reactivity description is correct for (CH3)3COH?

(A) primary; controlled oxidation gives an aldehyde without C-C cleavage

(B) tertiary; controlled oxidation gives a ketone without C-C cleavage

(C) secondary; controlled oxidation gives a ketone without C-C cleavage

(D) tertiary; oxidation, if forced, requires C-C cleavage

39. Condense the alphaCOOH of Ala with the alphaNH2 of Gly, leaving the other terminal groups unreacted. Which oriented dipeptide is formed?

(A) Ala-Gly.

(B) Gly-Gly.

(C) Gly-Ala.

(D) Ala-Ala.

40. Partial cleavage of a linear tetrapeptide gives overlapping fragmentsAla-Gly-Ser and Gly-Ser-Val. Independent terminal analysis givesAla at the N terminus. Which complete sequence fits both overlaps?

(A) Ala-Ser-Gly-Val.

(B) Ala-Gly-Ser-Val.

(C) Val-Gly-Ser-Ala.

(D) Ala-Gly-Val-Ser.

41. 2-ethylaniline is diazotised at0-5°C,treated with CuCN,then its nitrile is fully hydrolysed and its benzylic side chain fully oxidised. Heating the resulting diacid gives which characteristic cyclic anhydride?

(A) terephthalic acid

(B) isophthalic acid

(C) succinic anhydride

(D) phthalic anhydride

42. An100mL sample contains $\frac{1}{25}$ mol HA,pKa=6,and $\frac{1}{50}$ mol HCl. Add100mL $\frac{3}{5}$ M NaOH. Use usual weak-base salt hydrolysis and $\log_{10}(\frac{1}{5})=-\frac{699}{1000}$. Final pH is

(A) 6

(B) $\frac{19301}{2000}$

(C) $\frac{13301}{1000}$

(D) 7

43. For hexa-1,3,5-triene, CH2=CH-CH=CH-CH=CH2, count distinct configurational geometrical isomers. Ignore conformers and photochemical interconversion. Where reversing the entire carbon chain identifies two E/Z assignments, count them once.

(A) 4

(B) 1

(C) 3

(D) 2

44. For the reaction $2M + O_2 \rightarrow 2MO$, take $\Delta H^\circ = -210$ kJ and $\Delta S^\circ = -0.200$ kJ K^{-1} over the stated range. Find ΔG° in kJ at 725 K and the sign of its slope against temperature.

(A) 65 kJ; negative slope

(B) -65 kJ; positive slope

(C) -355 kJ; negative slope

(D) -210 kJ; zero slope

45. A laboratory note states: Finely divided iron is used in the Haber process. Which judgement and reason are consistent?

(A) Incorrect because transition metals cannot exchange electrons or adsorb reactants.

(B) Incorrect: Iron catalyses ammonia synthesis; it is not consumed as a stoichiometric nitrogen reagent.

(C) Correct because every transition-metal species has the same d-electron count.

(D) Correct: Iron catalyses ammonia synthesis; it is not consumed as a stoichiometric nitrogen reagent.

46. A compound has formula C4H8O and gives positive Tollens but negative iodoform. Restrict X to the candidate structures below. Which fits every clue?

(A) butan-2-one

(B) butanal

(C) ethanal

(D) propanone

47. A persistent fat-soluble pollutant is present at low concentration in water, but its concentration is much greater in a fish-eating predator. Which property pair makes this result plausible?

(A) High volatility with no contact with organisms.

(B) Slow degradation and accumulation through successive feeding levels.

(C) The pollutant is created solely by the predator from nitrogen gas.

(D) Instant decomposition and complete excretion at every level.

48. A model M(OH)2 has molar solubility 1/20000 M in pure water; neglect water autoionisation there. In a buffered solution its measured molar solubility is 1/32000000 M. The buffer fixes free OH- and the solid's contribution to OH- is negligible. What is the imposed OH- concentration?

(A) 1/62500

(B) 1/250

(C) 1/500

(D) 1/125

49. An ideal mixture initially has $\frac{1}{5}$molA and $\frac{3}{10}$molB. A selective step removes half ofB and noA. At the new equilibriumT,V,totalpressure is8bar. Partialpressure ofA,in bar,is

(A) 4.0

(B) $\frac{32}{7}$

(C) 14

(D) $\frac{16}{5}$

50. A repeat contains -NH-(CH2)6-NH-CO-(CH2)4-CO-. Which monomer pair and small by-product are consistent with direct acid-amine condensation?

(A) Isoprene and sulphur; carbon dioxide.

(B) Ethylene glycol and adipic acid; ammonia.

(C) Caprolactam and styrene; hydrogen chloride.

(D) Hexamethylenediamine and adipic acid; water.

51. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_y; same facing signs; 2: parallel p_x; same facing signs. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (1, 1)

(B) (0, 2)

(C) (1, 0)

(D) (2, 0)

52. Compare the energy for removing one more electron from Mn^{2+} and from Fe^{2+}. Which member is expected to resist that removal more strongly on the stated shell-stability basis?

(A) Mn^{2+}

(B) They must be identical because both are positive ions.

(C) The ion with the smaller written charge always has the larger removal energy.

(D) Fe^{2+}

53. Which comparison is correct for the standard DNA/RNA building blocks?

(A) DNA: deoxyribose/uracil; RNA: ribose/thymine

(B) DNA: ribose/thymine; RNA: deoxyribose/uracil

(C) DNA: deoxyribose/thymine; RNA: ribose/thymine

(D) DNA: deoxyribose/thymine; RNA: ribose/uracil

54. A sample of but-2-yne is treated with Na/liquid NH3. Which isolated product is consistent with the reagent limitation and carbon-skeleton change?

(A) trans-but-2-ene (B) cis-but-2-ene

(C) butane (D) but-1-ene

55. A 0.0100 mol portion of Cu2+ is first fully precipitated as M(OH)2 using aqueous ammonia and is then dissolved in sufficient excess ammonia to give the usual tetraammine species. Counting only metal-containing entities, what are (moles of metal in solution at the end, moles of NH3 bound in that ammine complex)? Assume quantitative steps and no metal redox.

(A) (0, 0.0400) (B) (0.0200, 0.0400)

(C) (0.0100, 0.0400) (D) (0.0100, 0.0200)

56. I:Semicarbazone formation needs an available nucleophilic amino nitrogen. II:Increasing acid strength without limit always accelerates it. Select the correct assessment.

(A) I false;II true. (B) Both true.

(C) I true;II false. (D) Both false.

57. A fully dissociated HCl solution has concentration 1/100000 M. What analytical concentration of a monoprotic weak acid HA with Ka=1/100000000 gives approximately the same pH? Neglect water autoionisation and use [H+]$\approx \sqrt{KaC}$, which is to remain in its weak-ionisation regime.

(A) 1/100 (B) 1/50

(C) -1/100 (D) 1/200

58. Acidifying 1/50 mol borax theoretically gives four H3BO3 per borax formula unit. The isolated boric acid requires 2/125 mol NaOH for complete titration in the presence of glycerol, where each H3BO3 behaves as monobasic. What fraction of the theoretical boric acid was recovered?

(A) 1/5 (B) -1/5

(C) 1/10 (D) 2/5

59. The ions Na^+, Mg^{2+}, Al^{3+} each have 10 electrons. Which decreasing-radius order and explanation is consistent?

(A) All radii are equal because electron numbers are equal.

(B) $Mg^{2+} > Al^{3+} > Na^+$; positive charge always gives the largest ion.

(C) $Na^+ > Mg^{2+} > Al^{3+}$; increasing nuclear charge contracts the same electron cloud.

(D) $Na^+ < Mg^{2+} < Al^{3+}$; larger nuclear charge expands the cloud.

60. A Hinsberg separation is applied to trimethylamine. Which reagent identity and class-dependent outcome are correct?

(A) benzenesulphonyl chloride,C6H5SO2Cl;every amine gives an alkali-soluble N-H sulphonamide

(B) benzenesulphonyl chloride,C6H5SO2Cl;no ordinary sulphonamide;the unchanged amine dissolves on acidification

(C) benzenesulphonic acid,C6H5SO3H;all classes give the same neutral product

(D) benzoyl chloride,C6H5COCl;no ordinary sulphonamide;the unchanged amine dissolves on acidification

English Proficiency

61. Read the sentence: The plants thrive in the well-drained beds. Choose the opposite in meaning of 'thrive' as used in the sentence.

(A) adapt

(B) wither

(C) grow

(D) spread

62. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The volunteers are arranging the chairs around the table.

(A) The chairs are being arranged around the table by the volunteers.

(B) The chairs have been arranged around the table by the volunteers.

(C) The chairs were being arranged around the table by the volunteers.

(D) The table is arranging the chairs around the volunteers.

63. Which figure of speech is chiefly illustrated by the line? Her careful questions were stepping-stones across the difficult argument.

(A) simile

(B) metaphor

(C) hyperbole

(D) apostrophe

64. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. Listening to it revealed a pause she had not noticed while speaking. B. Splitting that slide made the explanation smoother without making her speak faster. C. The speaker recorded a practice version of her presentation. D. She traced the pause to a slide that introduced too many ideas at once.

(A) The speaker recorded a practice version of her presentation.

(B) Splitting that slide made the explanation smoother without making her speak faster.

(C) Listening to it revealed a pause she had not noticed while speaking.

(D) She traced the pause to a slide that introduced too many ideas at once.

65. Read the sentence: The notice gave explicit instructions, leaving no doubt about which entrance to use. Choose the closest meaning of 'explicit' as used in the sentence.

(A) carefully justified (B) clearly stated

(C) widely distributed (D) strictly enforced

66. Use the standard contemporary negative adjective. The account deliberately concealed the truth and was therefore _____.

(A) imhonest (B) inhonest

(C) dishonest (D) unhonest

67. Read the sentence: The team proposed a novel way of arranging the archive, unlike any method used there before. Choose the closest meaning of 'novel' as used in the sentence.

(A) complex and ambitious (B) interesting but untested

(C) new and original (D) strange and impractical

68. Choose the pair that correctly completes both uses: The kite rose _____ above the roof, and its craftsmanship was _____ valued by collectors.

(A) high; highly (B) highly; high

(C) high; high (D) highly; highly

69. Complete the question tag: Let's check the original record before deciding, _____?

(A) will we (B) aren't we

(C) do we (D) shall we

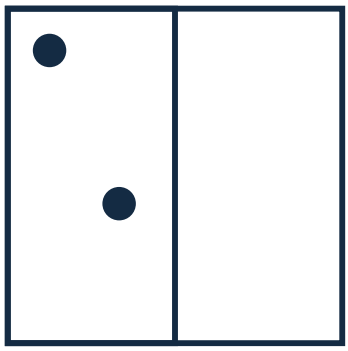
70. The organiser gives up her own advantage to help strangers and expects no reward. Which word best describes her?

(A) egoist (B) opportunist

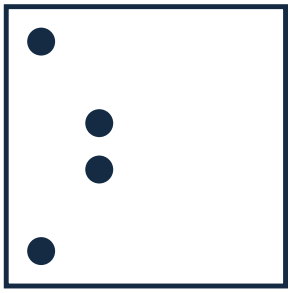
(C) spectator (D) altruist

Logical Reasoning

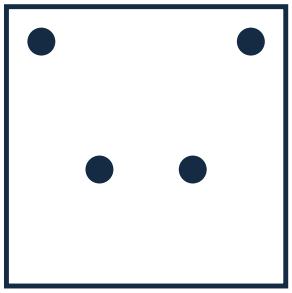
71. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



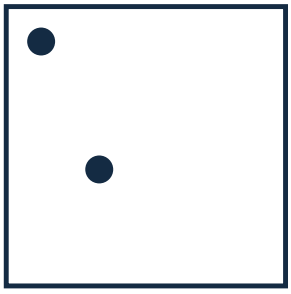
(A) Pattern U



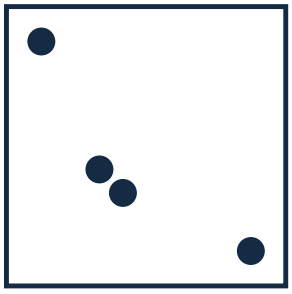
(B) Pattern W



(C) Pattern Y



(D) Pattern M

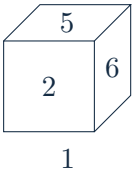


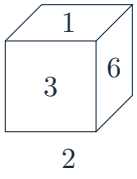
72. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 5, 10, 14, ?, 32, 64, 68.

(A) 24 (B) 18

(C) 32 (D) 28

73. The views show the same die. Give the ordered pair: (face opposite 4, face opposite 2). Do not assume opposite labels sum to seven.





(A) $(6, 4)$ (B) $(6, 2)$

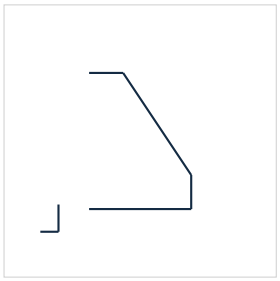
(C) $(6, 1)$ (D) $(6, 3)$

74. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, BRICK is coded as DTKEM. Which word is coded as OCPIQ?

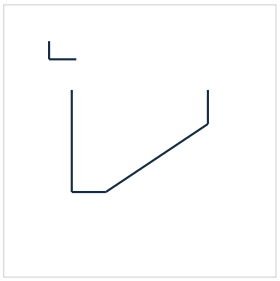
(A) QERKS (B) NBOHP

(C) MANGO (D) LZMFN

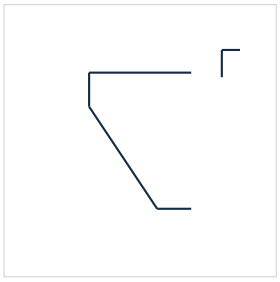
75. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right



bottom left

?

bottom right

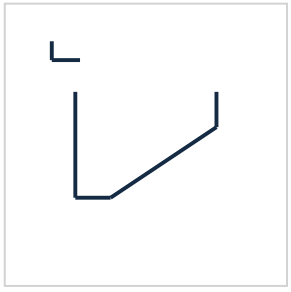
(A) Pattern Y



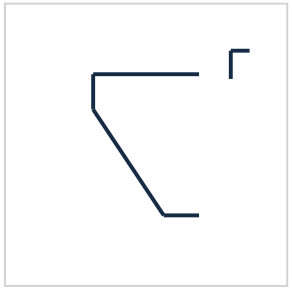
(B) Pattern K



(C) Pattern N



(D) Pattern Z

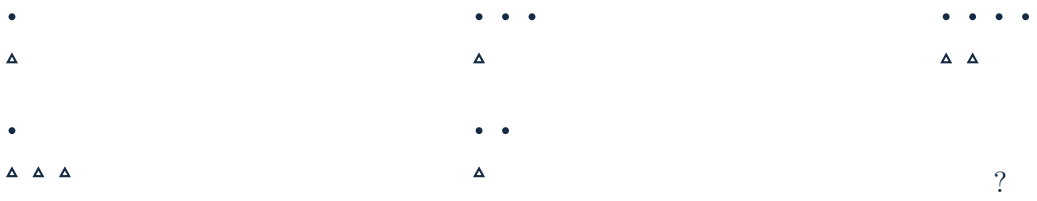


76. In a row, Asha is 12th from the left and 21st from the right. Bina is 10th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 7 (B) 4

(C) 13 (D) 8

77. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern N



(B) Pattern W



(C) Pattern Y



(D) Pattern L

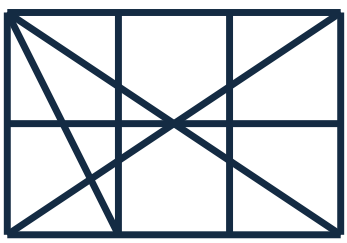


78. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: NLYX, RMVU, ?, ZOPO, DPML, HQJI.

(A) WOTS (B) RSNV

(C) UMRQ (D) VNSR

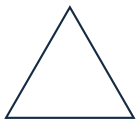
79. How many triangles are formed by all the drawn segments?



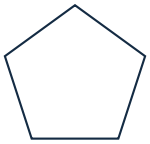
(A) 34 (B) 39

(C) 63 (D) 24

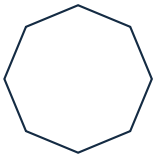
80. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

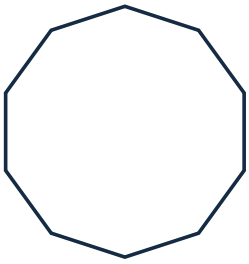


B



C

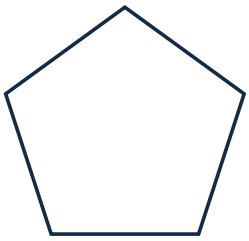
(A) Pattern U



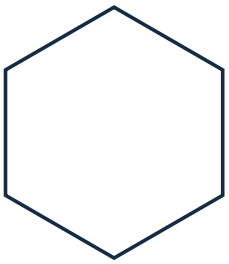
(B) Pattern M



(C) Pattern Q



(D) Pattern Y



81. The age table uses the same two people throughout:

	younger	older
now	x	$x+10$
5 years ago	$x-5$	$x+5$

. The product of their ages 5 years ago was 504. What will the older person's age be 4 years from now?

(A) 47 (B) 32

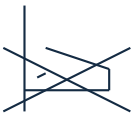
(C) 37 (D) 33

82. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





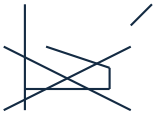
host 1



host 2



host 3



host 4

(A) 2
(B) 3
(C) 4
(D) 1

83. The whole has 9 unit positions. Part A covers the first 4; part B covers 4 positions, overlapping A in exactly 2 positions as shown. How many positions remain uncovered by either part?



whole



part A positions



part B positions

(A) −1
(B) 4
(C) 3
(D) 1

84. Preserve the exact relation and its order: Silent : Quiet :: ? : Old. Choose the missing term.

(A) Rapid
(B) Ancient
(C) Begin
(D) Silent

85. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?



1



2



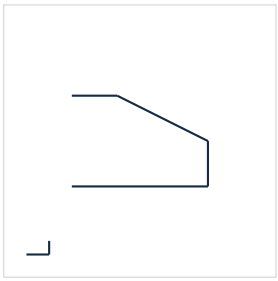
3



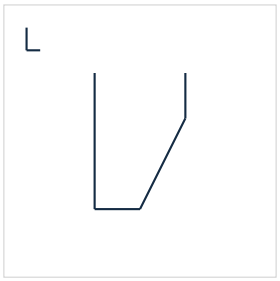
4

(A) 3
(B) 4
(C) 1
(D) 2

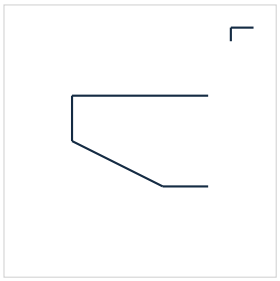
86. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

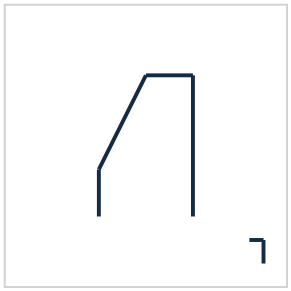


2

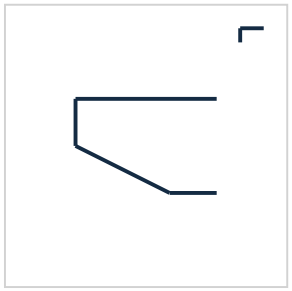


3

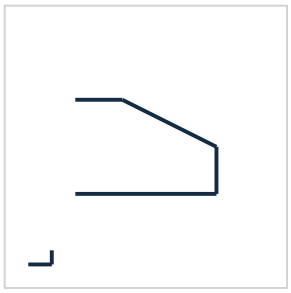
(A) Pattern L



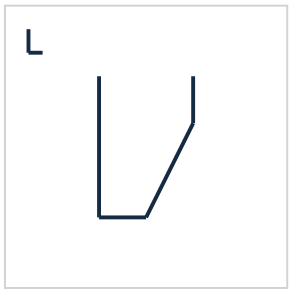
(B) Pattern X



(C) Pattern Z



(D) Pattern K



87. Consider only these statements; universal statements alone do not imply existence. Some campers are not bakers; All campers are inventors. Conclusions: I. Some inventors are not bakers. II. No inventors are bakers. Which conclusions follow necessarily?

(A) Only II

(B) Both I and II

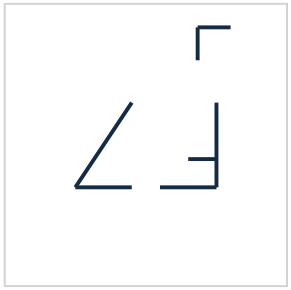
(C) Only I

(D) Neither I nor II

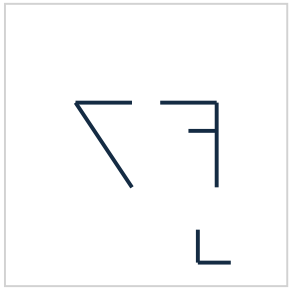
88. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



(A) Pattern P



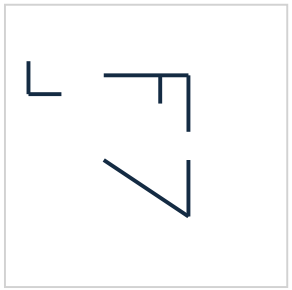
(B) Pattern Z



(C) Pattern W



(D) Pattern T



89. Consider only these statements; universal statements alone do not imply existence. Some inventors are not guides; All joggers are inventors; Some guides are not joggers. Conclusions: I. No inventors are joggers. II. Some joggers are guides. Which conclusions are individually possible without contradicting the statements?

(A) Only I

(B) Only II

(C) Neither I nor II

(D) Both I and II

90. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

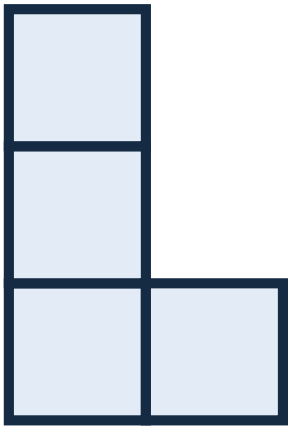


piece 1

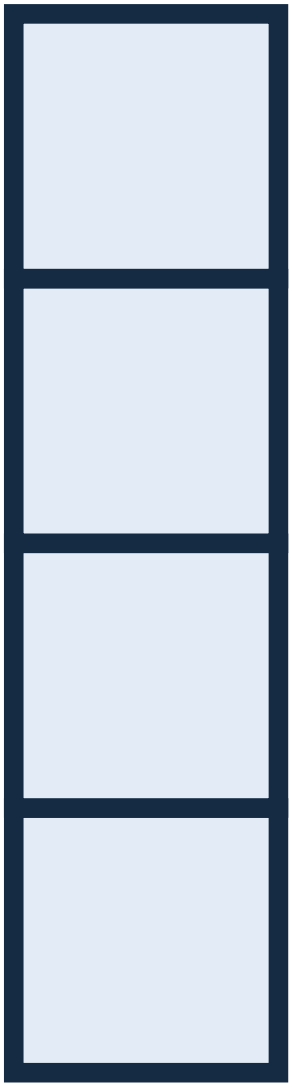


piece 2

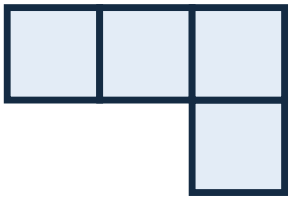
(A) Pattern K



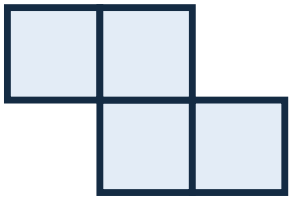
(B) Pattern T



(C) Pattern P



(D) Pattern U



Mathematics

91. F′(x)=$\frac{(x+1)^2}{(x+2)^4}$ and F(0)=5. Find F(2)-2F(0).

(A) $\frac{979}{192}$ (B) $\frac{19}{192}$

(C) $-\frac{941}{192}$ (D) $-\frac{979}{192}$

92. Two points on opposite sides of a tower foot are 9 metres apart on a straight level line. The elevations of the tower top are 30° and 45°. Find the tower height.

(A) $-\frac{9}{2}+\frac{9\sqrt{3}}{2}$ (B) $\frac{9}{2}+\frac{9\sqrt{3}}{2}$

(C) $\frac{27}{2}-\frac{9\sqrt{3}}{2}$ (D) $\frac{3\sqrt{3}}{2}$

93. Disjoint A,B have sizes 9,2. Find |(A×A)∪(B×B)|.

(A) 18 (B) 121

(C) 85 (D) 103

94. f(x+1)=5x^2. Find the area enclosed between y=f(x) and y=5(x-1).

(A) $-\frac{5}{6}$ (B) $\frac{5}{2}$

(C) $\frac{5}{3}$ (D) $\frac{5}{6}$

95. $A=\begin{bmatrix}0 & 4\\ 0 & 0\end{bmatrix}$. Find the (1,2) entry of (I+A)^9.

(A) 262144 (B) 9

(C) 4 (D) 36

96. y′+6y=exp(-6x), y(0)=7. Find exp(6)y(1)-7.

(A) $\frac{8}{e^6}$ (B) 1

(C) 8 (D) −1

97. Find the area enclosed by y=x(11-x) and y=2x.

(A) $-\frac{243}{2}$ (B) $\frac{1331}{6}$

(C) $\frac{441}{2}$ (D) $\frac{243}{2}$

98. Find the maximum of sin^3 x cos x on[0,π/2].

(A) $\frac{3}{8}$ (B) $\frac{3\sqrt{3}}{16}$

(C) $\frac{1}{2}$ (D) $\frac{27}{128}$

99. A=(2,5,5), B=(4,5,10). Find the squared length of the component of AB perpendicular to direction (1,2,2).

(A) 13 (B) 25

(C) 29 (D) 16

100. $F'(x)=\frac{6(x^2-1)(x^2+1)^5}{x^7}$ and F(1)=5. Find F(2)-2F(1).

(A) $\frac{11209}{64}$ (B) $-\frac{11849}{64}$

(C) $\frac{11849}{64}$ (D) $\frac{11529}{64}$

101. An AP has a=8,d=5; its first n terms sum to 90. Find a_(n+1).

(A) 33 (B) 18

(C) 28 (D) 38

102. If z=4-3i, find |exp(iz)|.

(A) e^3 (B) e^{-3}

(C) 1 (D) e^4

103. For the roots α,β of $z^2-(1+13i)z+(-84+13i)=0$, find $|\alpha-\beta|^2$.

(A) 85 (B) 7225

(C) 340 (D) 170

104. Write sin x $\cos^3$x+5$\sin^3$x cos x=P sin2x+Q sin4x. Find Q.

(A) $\frac{1}{8}$ (B) $\frac{3}{2}$

(C) $\frac{3}{4}$ (D) $-\frac{1}{2}$

105. How many integer t make P=(t,0) strictly inside the larger segment cut by x=7 from $x^2+y^2=144$?

(A) 4 (B) 20

(C) 18 (D) 23

106. Let D_n be the determinant of the 3×3 matrix with diagonal entries n and off-diagonal entries 1. Find ΣD_n for n=1,...,5.

(A) 190 (B) 225

(C) 180 (D) 560

107. For a=(1,5,0), b=(0,1,2), c=(5,0,1), evaluate (a×c)·(a+b).

(A) −50 (B) 0

(C) −51 (D) 51

108. The locus|z-4|=|Re(z)+3| meets Im(z)=6. Find $|z|^2$ at the intersection.

(A) $\frac{7897}{196}$ (B) $\frac{43}{14}$

(C) 52 (D) $\frac{8905}{196}$

109. f(x)=x^2+5x for x≤5, and f(x)=px+q for x>5. If f is differentiable at 5, find p-q.

(A) 40 (B) 15

(C) 50 (D) −10

110. Evaluate the limit of [cos(4x)-cos(5x)]/x^2 as x→0.

(A) $\frac{1}{2}$ (B) $-\frac{9}{2}$

(C) $\frac{9}{2}$ (D) 9

111. The line r=(3,3,2)+t(1,2,2) and plane 2x-y=6 are given. Find the squared minimum distance between them.

(A) 0 (B) $\frac{3\sqrt{5}}{5}$

(C) 9 (D) $\frac{9}{5}$

112. Find the area bounded by y=x, y=49/x, and x=21 in the first quadrant.

(A) 196 (B) $196 - 49\log(3)$

(C) $98 - 49\log(3)$ (D) $49\log(3)$

113. Let $D(t) = \det\begin{pmatrix} 11+t & 1+t \\ 1+t & 5+t \end{pmatrix}$. Find D′(0).

(A) −18 (B) 14

(C) 54 (D) −14

114. All tangents common to $x^2/16+y^2/4=1$ and $x^2+y^2=10$ touch the ellipse at points P_i. Find Σ(P_i, $x)^2$ over the four tangency points.

(A) $\frac{256}{5}$ (B) 40

(C) $\frac{64}{5}$ (D) $\frac{16}{5}$

115. X is binomial with n=9, p=1/3. Given X≥1, find P(X≥2).

(A) $\frac{2304}{19171}$ (B) $\frac{16867}{19171}$

(C) $\frac{19171}{19683}$ (D) $\frac{16867}{19683}$

116. How many integers x satisfy $4^{-x} - (2080)2^{-x} + 65536 \leq 0$?

(A) 6 (B) 12

(C) 7 (D) 5

117. How many integers k make 24sin x+32cos x+6=k have exactly two distinct solutions in 0≤x<2π?

(A) 79 (B) 80

(C) 39 (D) 81

118. 16 distinct people sit at labelled round tables of sizes 9,7. Two specified people must be at different tables, and the first specified person must be at the 9-seat table. Rotations are identical. Count the seatings.

(A) 332107776000 (B) 87178291200

(C) 174356582400 (D) 5492232345600

119. For xy-6x-4y+24=0, find the squared length of the diagonal of the rectangle formed with the axes.

(A) 24 (B) 52

(C) $2\sqrt{13}$ (D) 100

120. Unit vectors a,b,c satisfy $|(a\times b)\cdot c|=1$. Find $|a+7b-2c|^2$.

(A) 10 (B) 82

(C) $3\sqrt{6}$ (D) 54

121. How many solutions in $0\le x<4\pi$ satisfy sin 2 x=sin x and cos x>sin x?

(A) 2 (B) 6

(C) 8 (D) 4

122. A convergent GP has first term 9, ratio $\frac{1}{3}$. Find the sum of all terms after its first 4 terms.

(A) $\frac{1}{12}$ (B) $\frac{1}{6}$

(C) $\frac{1}{2}$ (D) $\frac{1}{54}$

123. f is cubic with leading coefficient 1/3, stationary points 6,11, and lim as $x\to 0$ [f(x)-3]/x=66. Find f(1).

(A) $\frac{347}{6}$ (B) 69

(C) $\frac{365}{6}$ (D) 77.8333333333333

124. Two acute sun elevations $\theta_1<\theta_2$ satisfy $(16^4)^{sin^2\theta} + (16^4)^{cos^2\theta} = 2^4 + 2^12$. A vertical tower has height 8. Find the difference between its two shadow lengths.

(A) $\frac{16}{3}$ (B) $\frac{32\sqrt{3}}{3}$

(C) $8\sqrt{3}$ (D) $\frac{16\sqrt{3}}{3}$

125. If α,β are roots of x^2-15x+50=0, find $\alpha^2+\beta^2$ without solving the quadratic.

(A) 225 (B) 2500

(C) 125 (D) 100

126. Parabola y=4(x-4)(x-5) meets the axes at three distinct points. A circle passes through these points. Find its other y-intercept.

(A) 1 (B) $\frac{1}{4}$

(C) $-\frac{1}{4}$ (D) 4

127. Vectors a,b satisfy $|a|^2$=37, $|b|^2$=37, a·b=12. Find $|2a-b|^2$.

(A) 137 (B) 87

(C) 185 (D) 233

128. Find the population variance of 2+5, 4+5, ..., 18+5.

(A) $\frac{95}{3}$ (B) $\frac{80}{3}$

(C) $\frac{40}{3}$ (D) $\frac{20}{3}$

129. There are 8 Boolean variables p_1,...,p 8. Exactly 3 variables must be true. How many truth assignments satisfy [(p_1 implies p_2) and (p_1 implies not p_2)]?

(A) 256 (B) 20

(C) 35 (D) 15

130. A={j∈{1,...,60}:i^j=1}, B={j∈{1,...,60}:$(-1)^j$=1}. Find |B\A|.

(A) 30 (B) 7

(C) 15 (D) 45

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A 5 kg block is pulled by a force of 25 N at an angle with $\cos\theta = 4/5$, $\sin\theta = 3/5$ above a horizontal floor. The block slides; $\mu_k = \frac{1}{10}$. Find the net horizontal force. Take $g = 10$ m/s^2.

(A) $\frac{33}{2}$ N
(B) 12 N
(C) $\frac{47}{2}$ N
(D) $\frac{43}{2}$ N

132. Three long parallel wires are equally spaced by 2 m along a straight line. Outer wires each carry 8 A in the same direction; the middle carries 5 A in the opposite direction. Find the net force per unit length on the middle wire.

(A) $\frac{8}{\pi}$ N/m
(B) 0 N/m
(C) $\frac{20}{\pi}$ N/m
(D) $-\frac{10}{\pi}$ N/m

133. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 4 kΩ to a terminal at +8 V, and its cathode connects to a terminal at -1 V. Find the current from the +8 V terminal in mA.



(A) $\frac{73}{40}$ mA
(B) $\frac{97}{40}$ mA
(C) $\frac{83}{40}$ mA
(D) $\frac{19}{10}$ mA

Chemistry

134. Acidic MnO4- oxidises SO2 to HSO4- and is reduced to Mn2+. The balanced reaction is 2MnO4- +5SO2+2H2O+H+→2Mn2++5HSO4-. Initially 29/1000 mol H+ and excess SO2 are available. After exactly 1/20 mol MnO4- is consumed, what amount of net H+ remains? Count net H+ from the displayed balanced equation.

(A) 1/125
(B) 1/250
(C) 1/500
(D) -1/250

135. A dry mixture contains only anhydrous MgCO3 and CaCO3. Use the ordinary Group2 carbonate stability trend. In an idealised selective-heating window, the less thermally stable carbonate decomposes completely by MCO3→MO+CO2, while the more stable carbonate remains unchanged. No other mass loss occurs. This is the stated model, not a specified real temperature. Atomic masses: Mg=24, Ca=40, C=12, O=16. The initial mixture mass is 117/25 g. The first-stage CO2 collected is 1/50 mol. What was the initial mass fraction of MgCO3?

(A) 2/5
(B) 22/117
(C) 14/39
(D) 50/117

136. For benzaldehyde with Tollens reagent,which product or mechanistic conclusion is correct under the stated conditions?

(A) benzoate under the alkaline reaction conditions, with metallic Ag

(B) benzene with silver chloride

(C) benzyl alcohol with metallic Ag

(D) phenol with CO2

Logical Reasoning

137. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). The numerical component follows a constant arithmetic step. Find the missing internal term: DAA6, EYE9, FWI12, ?, HSQ18, IQU21.

(A) FTL15

(B) GUM15

(C) HVN15

(D) 51MUG

138. Aman points to a photograph and says, "This person's only daughter is my mother." Aman is male, and the photographed person is male. How is the photographed person related to Aman?

(A) Grandfather

(B) Brother

(C) Son

(D) Daughter

139. Each of Asha, Farah, Gita visits on one different day (Monday, Tuesday or Wednesday) and buys one different fruit (Pear, Mango, Peach). Asha visits on Monday, and Farah on Wednesday. The visitor on Monday buys mango. Farah does not buy pear. Which fruit does Farah buy?

(A) No unique assignment

(B) Pear

(C) Peach

(D) Mango

Mathematics

140. $\log_2 a=4$, log_a b=6, log_b c=3. Find $\log_4(c^2/a)$.

(A) 74

(B) 140

(C) 70

(D) 72

141. For $A=\begin{bmatrix}5 & 1\\ 1 & 4\end{bmatrix}$ and $C=19A^{-1}$, find C_{22}.

(A) $\frac{5}{19}$

(B) 5

(C) 4

(D) -1

142. A=(5,4,0), C=(8,10,3). B lies on AC with AB:BC=2:1. Find $|OB|^2$, where O is the origin.

(A) 117

(B) 173

(C) 41

(D) 54

BITSAT Mock 33

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. Two parallel plane mirrors separated by 3 cm extend from axial position 0 to $\frac{33}{2}$ cm. A ray enters midway between them at axial position 0, making 45° with the normal to the mirrors. How many reflections occur before it passes the far edge? Exclude a contact exactly on the exit edge.

(A) 5 (B) 10

(C) 2 (D) 4

2. A quantity with the dimensions of mass is written $G^a h^b c^d$. Given $[G] = M^{-1}L^3T^{-2}$, $[h] = ML^2T^{-1}$, $[c] = LT^{-1}$, find $a + 2b - d$.

(A) 1 (B) $-\frac{1}{2}$

(C) 0 (D) -1

3. A 1 kg block rests on a 8 kg block on a smooth horizontal floor. A force 72 N pulls the lower block. There is no slipping between the blocks. During a displacement 6 m, what work does friction do on the upper block?

(A) 432 J (B) 54 J

(C) 48 J (D) 216 J

4. A dipole of fixed moment 3 C m is rotated quasistatically from 60° to 120° relative to a uniform electric field 2 N/C. Find external work.

(A) 2π J (B) 3 J

(C) 6 J (D) $6\sqrt{3}$ J

5. Water flows steadily from a pipe section of area A at 5 m/s to a section of area $A/2$ located 4 m higher. Neglect viscosity. Find $p_1 - p_2$. Take density 1000 kg/m^3 and $g = 10$ m/s^2.

(A) 77500 Pa (B) 37500 Pa

(C) 30625 Pa (D) 40000 Pa

6. A uniform disc of mass 4 kg and radius 4 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) $30\,\mathrm{kg\,m^2}$ (B) $26\,\mathrm{kg\,m^2}$

(C) $38\,\mathrm{kg\,m^2}$ (D) $24\,\mathrm{kg\,m^2}$

7. A parallel-plate capacitor has vacuum capacitance C_0. A dielectric of constant 2 fills a fraction $\frac{1}{3}$ of the plate area and the entire gap in that region; the remainder stays vacuum. The voltage is fixed. Find the fraction of total stored energy residing in the dielectric-filled region.

(A) $\frac{2}{3}$ (B) 1

(C) $\frac{1}{5}$ (D) $\frac{1}{2}$

8. For steady laminar flow through a capillary, the radius is multiplied by 3 and length by 2. The volume flow rate must remain unchanged and viscosity is fixed. By what factor must the pressure difference be multiplied?

(A) $\frac{2}{81}$ (B) $\frac{2}{9}$

(C) $\frac{2}{3}$ (D) $\frac{1}{81}$

9. A long solenoid has its turn number multiplied by 4 and length multiplied by 2, while its cross-sectional area and core permeability stay fixed. Find self-inductance multiplier.

(A) 4 (B) 8

(C) $\frac{1}{8}$ (D) 32

10. A 4 kg block on a rough horizontal table is tied over an ideal pulley to a hanging 3 kg mass. The hanging mass moves downward and $\mu_k = 1/5$. Find string tension. Take $g = 10\ \mathrm{m/s^2}$.

(A) 30 N (B) 40 N

(C) 8 N (D) $\frac{144}{7}$ N

11. A mass 1 kg initially moves in the positive direction at 6 m/s. A force opposite to that direction rises linearly from zero to 4 N and then falls linearly to zero over a total duration 2 s. Find its signed final momentum.

(A) $2\,\mathrm{kg\,m/s}$ (B) $10\,\mathrm{kg\,m/s}$

(C) $4\,\mathrm{kg\,m/s}$ (D) $6\,\mathrm{kg\,m/s}$

12. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 1 kΩ to a terminal at +11 V, and its cathode connects to a terminal at -4 V. Find the current from the +11 V terminal in mA.



(A) $\frac{157}{10}$ mA (B) 15 mA

(C) $\frac{103}{10}$ mA (D) $\frac{143}{10}$ mA

13. A small-angle pendulum has period T in a stationary lift. In a vertically accelerating lift its period is 14T. Find the downward lift acceleration divided by g.

(A) $\frac{13}{14}$ (B) $\frac{197}{196}$

(C) $\frac{195}{196}$ (D) $\frac{1}{14}$

14. A wire consists of a semicircular arc of radius 8 m carrying 2 A, joined to two straight radial segments that extend outward from its ends. Find the magnetic field magnitude at the arc centre in units of μ_0.

(A) $\frac{1}{16}$ (B) $\frac{1}{32}$

(C) $\frac{1}{8}$ (D) $\frac{2+\pi}{16\pi}$

15. Two equal-intensity coherent beams have maximum combined intensity 16 units. Find intensity when their path difference is $\lambda/3$.

(A) 16 (B) 8

(C) 4 (D) 12

16. A metal wire has linear temperature coefficient $\frac{1}{200}$ per degree Celsius referred to 0°C. Its dimensions are fixed and it stays across a constant voltage. Find power at 30°C divided by power at 0°C.

(A) $\frac{3}{23}$ (B) $\frac{23}{20}$

(C) $\frac{20}{23}$ (D) $\frac{17}{20}$

17. A coil of 4 turns and area 1 m^2 carries 3 A in a uniform 6 T field. Its plane makes 30° with the field. Find torque magnitude.

(A) $36\,\text{N m}$ (B) $36\sqrt{3}\,\text{N m}$

(C) $72\,\text{N m}$ (D) $9\sqrt{3}\,\text{N m}$

18. A particle is projected vertically outward from the surface of a nonrotating planet with speed $v_e/9$, where v_e is surface escape speed. Neglect atmosphere. Find maximum height divided by planet radius.

(A) $\frac{1}{80}$ (B) $\frac{1}{82}$

(C) 80 (D) $\frac{81}{80}$

19. A series circuit contains resistor R, two ideal inductors L_1, L_2, and a fixed-frequency AC source of angular frequency 70 rad/s. With both inductors active, current lags by 45°; when L_2 is shorted, it lags by 30°. If $R = 80$ ohms, find L_2.

(A) $-\frac{8}{7} + \frac{8\sqrt{3}}{7}\,\text{H}$ (B) $\frac{8}{7} - \frac{8\sqrt{3}}{21}\,\text{H}$

(C) $5600 - \frac{5600\sqrt{3}}{3}\,\text{H}$ (D) $\frac{8}{7}\,\text{H}$

20. An ideal mixture contains 3 moles of monatomic gas and 2 moles of rigid diatomic gas. Vibrations are frozen. Find the heat-capacity ratio $\gamma = C_P/C_V$ of the mixture.

(A) $\frac{5}{3}$ (B) $\frac{19}{29}$

(C) $\frac{31}{21}$ (D) $\frac{29}{19}$

21. A converging thin lens forms a real image with magnification magnitude 3. The object-image separation is 16 cm. Find focal length.

(A) 1 cm (B) 3 cm

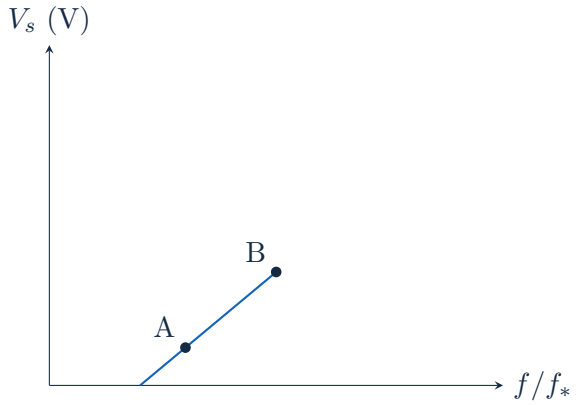
(C) 4 cm (D) 8 cm

22. A sinusoidal source of peak $\frac{87}{10}$ V feeds a back-to-back series pair of identical Zeners through a 1 kΩ current-limiting resistor. The output is across the diode pair. Each Zener has reverse breakdown 5 V and forward drop 0.7 V. What is the peak-to-peak output voltage? Use the piecewise constant-drop model.

(A) $\frac{64}{5}$ V (B) $\frac{57}{5}$ V

(C) $\frac{43}{5}$ V (D) $\frac{87}{5}$ V

23. For a metal, the stopping-potential graph has points A $(3f_*, 2\,\text{V})$ and B $(5f_*, 6\,\text{V})$. The electronic gain of the vertical recording channel is later found to be 2: its recorded voltage is 2 times the true stopping voltage. What is hf_*/e in volts?



(A) 1 V (B) $\frac{1}{4}$ V

(C) $\frac{1}{3}$ V (D) $\frac{3}{5}$ V

24. Two wires of the same material and length L, with radii r and 12r, are joined in series and stretched by a force F. Find their total stored elastic energy divided by the energy stored in the thinner wire alone under force F.

(A) $\frac{1}{144}$ (B) $\frac{13}{12}$

(C) $\frac{145}{144}$ (D) 2

25. Two small identical conducting spheres initially carry positive charges $7q_0$ and $5q_0$ and are separated by distance d. They touch and are returned to the same separation. Find the final force divided by the initial force.

(A) 1 (B) 5

(C) $\frac{5}{9}$ (D) $\frac{36}{35}$

26. A bar magnet of moment 5 A m^2 is rotated quasistatically from 60° to 120° relative to a uniform 4 T magnetic field. Find external work.

(A) 10 J (B) $\frac{20\pi}{3}$ J

(C) $10\sqrt{3}$ J (D) 20 J

27. A cyclist rides straight outward at 8 m/s for 5 s, waits 4 s, then returns to the start. The whole journey must last 10 s. What constant return speed is required?

(A) 4 m/s (B) 40 m/s

(C) 8 m/s (D) $\frac{20}{3}$ m/s

28. A one-electron ion with nuclear charge $Z = 1$ absorbs a photon of energy $\frac{693}{125}$ eV. The ejected electron has maximum kinetic energy 5 eV. From which principal quantum number was the electron removed? Use $E_n = -13.6Z^2/n^2$ eV and neglect recoil.

(A) 4 (B) 5

(C) 25 (D) 6

29. Two sound sources produce wavelengths $\frac{1}{4}$ m and $\frac{1}{5}$ m in the same gas at one temperature. The sound speed there is 300 m/s. At a second temperature the sound speed is 3/2 times larger, while both source frequencies stay unchanged. Find the new beat frequency.

(A) 1050 Hz (B) 2700 Hz

(C) 200 Hz (D) 300 Hz

30. A gas receives 500 J of heat while its surroundings do 150 J of work on it. It then loses 250 J of heat at fixed volume. Find its total internal-energy change.

(A) 400 J (B) 250 J

(C) 900 J (D) 100 J

Chemistry

31. A $\frac{25}{2}$ g $CaCO_3$/inert mixture is80.0% $CaCO_3$ by mass. It receives $\frac{9}{50}$ mol HCl. For $CaCO_3$+2HCl→$CaCl_2$+CO_2+H_2O,90.0% of formed CO_2 is collected. Collected CO_2 amount in mol is

(A) $\frac{81}{1000}$ (B) $\frac{9}{100}$

(C) $\frac{9}{50}$ (D) $\frac{1}{10}$

32. A dilute gas is warmed at fixed chemical composition and stays in the dilute regime. Its viscosity increases. Which conclusion explains the observation without treating the gas as a cohesive liquid?

(A) The gas must have frozen because high viscosity is possible only in solids.

(B) The increase proves formation of new covalent bonds between gas molecules.

(C) Faster molecular motion enhances transport of momentum between neighbouring moving layers.

(D) The gas viscosity is determined solely by liquid-like cohesive barriers, which heating always raises.

33. In a hypothetical atom, allowed l values are0,1,...,n. Subshell capacity remains2(2l+1), with filling by increasing n+l and then lower n. All subshells preceding 3p are filled, 3p contains 2 electrons, and no later subshell is occupied. How many electrons does the atom contain?

(A) 60 (B) 30

(C) -30 (D) 15

34. Why is an ordinary neutral NCl5 counterpart not analogous to familiar PCl5 in introductory valence-shell chemistry?

(A) Nitrogen has no valence electrons.

(B) Second-period nitrogen has an octet valence-shell constraint in ordinary stable compounds.

(C) Every atom in Group15 must have the same maximum covalency.

(D) Nitrogen is unable to form any covalent bond to chlorine.

35. A model predicts hydrogenic energies proportional to-Z^2/n^2 for any one-electron ion. Which proposed test stays within that model's intended scope?

(A) Assign every excited state the same energy because the nucleus is unchanged.

(B) Infer the exact spectrum of a many-electron atom from Z alone.

(C) Compare H, He+ and Li2+ transition energies for the same pair of n levels.

(D) Apply the formula unchanged to neutral helium including all electron interactions.

36. Under the usual qualitative group trends,compare the ease of alkali-metal electron loss in Li/K and the oxidising tendency of Cl_2/I_2. Which conclusion is consistent?

(A) Both trends must increase down a group because atoms become larger.

(B) K loses its electron more readily; I_2 is the stronger halogen oxidant.

(C) Li loses its electron more readily; I_2 is the stronger halogen oxidant.

(D) K loses its outer electron more readily; Cl_2 is the stronger halogen oxidant.

37. For ethyl propanoate,label A=carbonyl C=O,B=acyl C-O,C=alkoxy O-C(sp3). Which bond order/length statement correctly distinguishes the two C-O single bonds shown in a Lewis formula?

(A) B has partial double-bond character,and length A < B < C.

(B) A is longest because oxygen is more electronegative.

(C) B and C must have identical lengths because both are drawn single.

(D) C has partial double-bond character,and length A < C < B.

38. A closed system goes from state1to2 by path A with q=500 J and work on the system w=-200 J. It returns from2to1 along a different path with q=-700 J. IUPAC work on the return path,in J,is

(A) 300

(B) 400

(C) 200

(D) −400

39. At one T,P,a binary reaction R⇌P has a Gibbs-energy-versus-product-fraction curve. Pure R has G=20 kJ,pure P50 kJ,and the unique minimum G=10 kJ occurs at product fraction $\frac{1}{3}$. Which inference follows from both endpoints and the interior minimum?

(A) Positive pure-state change forces the equilibrium to be pure R.

(B) The minimum at an interior composition means both pure endpoints have equal G.

(C) G must decrease for a displacement from the minimum toward either pure endpoint.

(D) The pure-product minus pure-reactant standard change is positive,yet the equilibrium contains product.

40. Which occurrence/extraction inference for $CaCO_3$ is chemically consistent?

(A) The formula proves that oxygen is the only possible counter-ion in its natural deposits.

(B) The metal is obtained by ordinary water electrolysis without preferential hydrogen evolution.

(C) The metal occurs chiefly as native metal because group1/2metals are exceptionally unreactive.

(D) It occurs in limestone; calcium occurs as stable compounds rather than native metal.

41. For one molecule the mean translational energy is3kT/2. What is the energy for one mole of such molecules?

(A) 3R/(2T).

(B) 3kT/2 unchanged because counting molecules never changes an extensive energy.

(C) 3kT/(2NA).

(D) 3RT/2, because one mole contains NA molecules.

42. Which proposed routes give butan-2-ol as the stated major alcohol? I:butan-2-one+NaBH4; II:ethanal+ethylmagnesium bromide,then water;III:propanal+methylmagnesium bromide,then water. Where no work-up is written after NaBH4,assume the usual protonating work-up.

(A) I andII only

(B) II andIII only

(C) I only

(D) all three

43. BF3 is trigonal planar and NH3 is pyramidal. Which claim correctly uses geometry in comparing permanent dipoles?

(A) NH3 is planar and therefore its three vectors cancel.

(B) The symmetric BF3 bond-dipole vectors cancel; the NH3 resultant is nonzero.

(C) Both must be nonpolar because they each have three bonds.

(D) BF3 is necessarily more polar because B-F bonds are polar.

44. For $TiH_{1.7}$,which classification and consequence are mutually consistent?

(A) All hydrides contain free H+ ions in their solid lattices.

(B) interstitial/metallic; hydrogen occupies metal-lattice sites and nonstoichiometry is possible.

(C) Covalent; nonstoichiometry proves all hydrogen has been removed.

(D) Ionic; every example must be a gas composed of discrete molecules.

45. A food extract contains exactly one of the listed two-component combinations. Using the usual qualitative tests under suitable conditions, its results are: Benedict on warming: positive; iodine: negative; Biuret: negative; Sudan III: positive. Which combination matches?

(A) glucose and starch

(B) starch and albumin

(C) glucose and triolein

(D) glucose and albumin

46. For CrO_5,use one oxide O and two peroxide links. Which pair (Cr oxidation state,number of peroxide O-O links per formula entity) is correct?

(A) (+6,3).

(B) (+8,2).

(C) (+6,2).

(D) (+4,2).

47. For hepta-2,5-diene, CH3-CH=CH-CH2-CH=CH-CH3, count distinct configurational geometrical isomers. Ignore conformers and photochemical interconversion. Where reversing the entire carbon chain identifies two E/Z assignments, count them once.

(A) 3 (B) 4

(C) 8 (D) 1

48. For 1-bromobutane and2-bromobutane,which structural classification is correct when the actual halogen-bearing carbon and molecular formula are examined?

(A) enantiomers (B) conformers

(C) constitutional position isomers (D) chain isomers

49. A first-order reaction at 300 K leaves 1/2 of its initial reactant after100 s. Its Ea/(RT) at this temperature is 10. Use ln2=0.693 and k=A exp(-Ea/RT). Which is its frequency factor A in s-1?

(A) 693/1000 exp(10) (B) 693/100000 exp(-10)

(C) 100000/693 exp(10) (D) 693/100000 exp(10)

50. At25°C pH=9301/1000. Use pKw=14 and log10(2)=301/1000. What is [OH-] in mol/L?

(A) -1/50000 (B) 1/50000

(C) 1/25000 (D) 1/100000

51. The tetrahedral form of $[Ni(PPh_3)_2Cl_2]$ is isolated. Which comparison with square-planar $[Ni(CN)_4]^{2-}$ is correct under the elementary d^8 model?

(A) The phosphine/chloride form is paramagnetic with two unpaired electrons; the cyanide complex is diamagnetic.

(B) Both are diamagnetic because both contain NiII d8.

(C) The tetrahedral form has cis/trans isomers exactly like a square-planar MA2B2 complex.

(D) The cyanide complex is octahedral because cyanide is a strong-field ligand.

52. For oxide formation, both normalised per mole of O_2, $\Delta G^\circ_A = -320+0.10T$ and $\Delta G^\circ_B = -400+0.20T$ in kJ. Ignoring kinetic limitations, at what temperatures is B metal a favourable standard-state reducing agent for oxide A?

(A) At all positive temperatures (B) Only T = 800 K

(C) T < 800 K (D) T > 800 K

53. I:ethoxide+ethyl bromide;II:benzene+Br2/FeBr3;III:toluene+Cl2/light. Which complete mechanism assignment is correct?

(A) I nucleophilic substitution;II electrophilic substitution;III radical substitution.

(B) I radical substitution;II electrophilic addition;III electrophilic ring substitution.

(C) All three are electrophilic substitutions.

(D) I electrophilic addition;II radical addition;III nucleophilic substitution.

54. A pure anhydrous Na2CO3 standard contains 1/500 mol carbonate. In its HCl titration, the additional acid volume from the carbonate-to-hydrogencarbonate endpoint to the final CO2/water endpoint is 40 mL. What is the HCl molarity? Assume complete stoichiometric endpoints.

(A) 1/20
(B) 1/10
(C) -1/20
(D) 1/40

55. Which paired observations support the Li/Mg diagonal relationship without claiming that all group1properties become identical to group2?

(A) Both dissolve every fluoride and carbonate readily in water like all sodium salts.

(B) Both form nitrides directly; their carbonates decompose on heating to oxides and CO_2.

(C) Both form only superoxides in oxygen and their carbonates are perfectly heat-stable.

(D) Both have+2as their common ionic oxidation state.

56. Equal volumes of 1/5 M K4[Fe(CN)6] and 1/5 M glucose are mixed. The final osmotic pressure is 168/25 atm at RT=24 L atm/mol. Glucose stays molecular, and the salt dissociates only into4K+ and one intact complex anion. What fraction of salt formula units is dissociated in the final solution? Use ideal particle counting.

(A) -1/5
(B) 1/10
(C) 1/5
(D) 2/5

57. In a1.00 L vessel A+2B⇌2C+D. Initially A=3/10 mol, B=2/5 mol, D=0 and an unknown amount of C. At equilibrium D=1/100 mol and Kc=4/10469 using molar concentrations. What was the initial C amount in mol? All equilibrium amounts are positive.

(A) -1/50
(B) 1/100
(C) 1/25
(D) 1/50

58. Consider the statement: The usual basicity of Group15hydrides decreases from NH3to PH3to AsH3. Which explanation or consequence correctly accounts for this trend rather than applying an unqualified atomic-mass rule?

(A) Hydrogen bonding is necessarily stronger for every heavier hydride.

(B) The size and availability of the donor lone pair change down the group.

(C) Every increase in atomic mass must increase all molecular bond strengths.

(D) The ns2pair is increasingly forced into all bonds down every group.

59. ethylmagnesium bromide is treated with CO2,then acidic work-up. Which product satisfies carbon accounting and the reagent selectivity?

(A) propanoic acid
(B) ethanoic acid
(C) propan-1-ol
(D) butanoic acid

60. An isolated NaCl cathode compartment of volume $\frac{3}{10}$L initially has $\frac{3}{100}$mol strong acid. Current5.00A flows for 1158s,F=96500C/mol. Only cathodic water reduction produces OH^-,one per electron,and no anode product enters. Volume is fixed. At25C,finalpH is

(A) 13
(B) 7
(C) 14
(D) 1

English Proficiency

61. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. because it preserved details Q. Despite being offered a quicker method, R. the conservator chose the slower one S. that the quicker method would remove.

(A) Q-R-P-S
(B) P-Q-R-S
(C) S-R-P-Q
(D) R-Q-P-S

62. Read the sentence: The host asked a light question to break the ice before the difficult discussion. What does 'break the ice' mean here?

(A) end the discussion
(B) make a serious criticism
(C) make people feel less awkward initially
(D) expose a hidden disagreement

63. Read the sentence: Attendance at the safety briefing was mandatory. Choose the opposite in meaning of 'mandatory' as used in the sentence.

(A) scheduled
(B) optional
(C) required
(D) important

64. Read the sentence: The apparatus underwent rigorous testing before approval. Choose the opposite in meaning of 'rigorous' as used in the sentence.

(A) lax
(B) thorough
(C) repeated
(D) formal

Read the passage for Question 65

The first walking guide described every junction in detail. Its author was surprised when visitors still lost the route. Accompanying a new walker, she noticed that several descriptions became visible only after the walker had already needed to choose a direction. The information was accurate but badly timed. In the revised guide, a short warning appeared before each difficult turn, followed by the fuller description. The author retained the detail for readers who wanted it, while moving the action cue to the place where it could be used. The problem had not been a simple shortage of information. It concerned the relationship between the moment a decision had to be made and the moment the relevant information reached the reader.

65. What is the main lesson of the passage?

(A) Detailed descriptions should be removed from every practical guide.

(B) A warning is useful only after a mistaken turn has been taken.

(C) Walkers become lost chiefly because they dislike reading accurate information.

(D) Accurate information becomes more useful when it is available at the point at which a reader must act.

66. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The curator has not yet removed the temporary labels.

(A) The curator has not yet been removed by the temporary labels.
(B) The temporary labels have not yet been removed by the curator.
(C) The temporary labels had not yet been removed by the curator.
(D) The temporary labels have already been removed by the curator.

67. She creates images by recording them with a camera rather than drawing them by hand. What is she?

(A) illustrator (B) sculptor

(C) engraver (D) photographer

68. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. but to the time needed Q. The delay was due not to R. a shortage of volunteers S. to check the unfamiliar equipment.

(A) S-R-P-Q (B) Q-R-P-S

(C) R-Q-P-S (D) P-Q-R-S

69. Read the sentence: The report examines the economic effects of closing the local factory. Choose the closest meaning of 'economic' as used in the sentence.

(A) relating to the economy (B) sparing in the use of materials

(C) inexpensive to purchase (D) efficient at a small task

70. Read the sentence: Before commenting on the disputed record, the editor was circumspect and checked each source. Choose the closest meaning of 'circumspect' as used in the sentence.

(A) suspicious of everyone's motives (B) slow because of limited knowledge

(C) cautiously considerate of possible risks (D) unwilling to make any decision

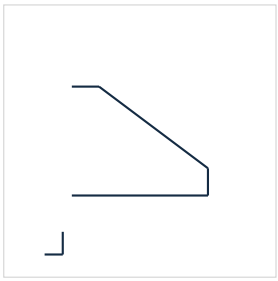
Logical Reasoning

71. Consider only these statements; universal statements alone do not imply existence. Some artists are not joggers; Some joggers are bakers. Conclusions: I. Some bakers are not joggers. II. Some bakers are joggers. Which conclusions follow necessarily?

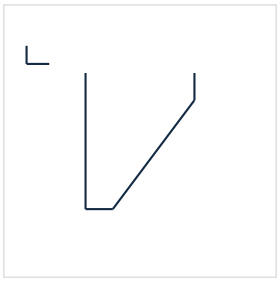
(A) Both I and II (B) Only II

(C) Only I (D) Neither I nor II

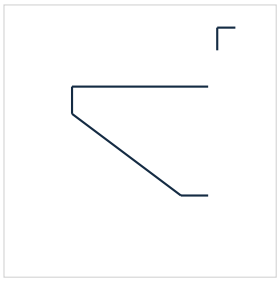
72. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

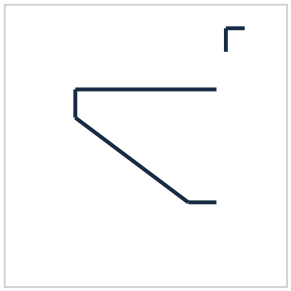


2



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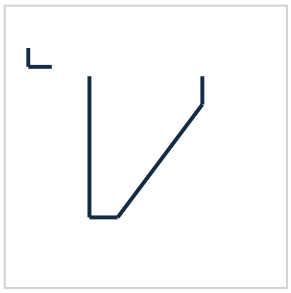
(A) Pattern Y



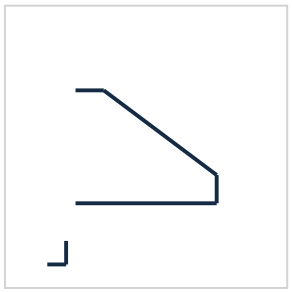
(B) Pattern W



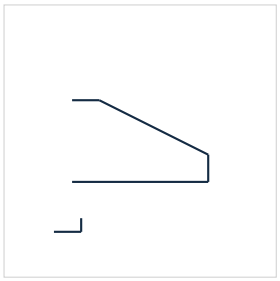
(C) Pattern M



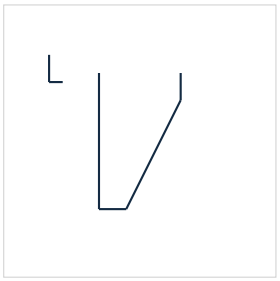
(D) Pattern Z



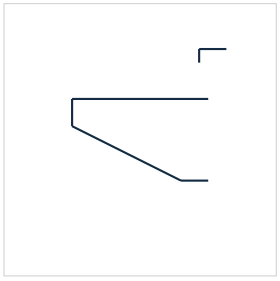
73. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

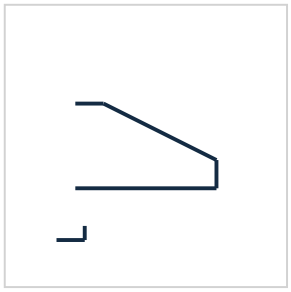


bottom left

?

bottom right

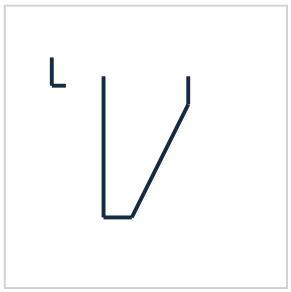
(A) Pattern Z



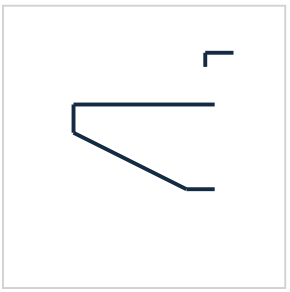
(B) Pattern K



(C) Pattern Y



(D) Pattern W



74. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 5, 20, 25, ?, 105, 420, 425.

(A) 105

(B) 120

(C) 30

(D) 100

75. Vijay points to a photograph and says, "This person is the son of my father's only son." Vijay is male, and the photographed person is male. How is the photographed person related to Vijay?

(A) Sister
(B) Daughter
(C) Brother
(D) Son

76. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





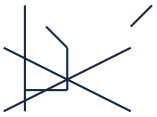
host 1



host 2



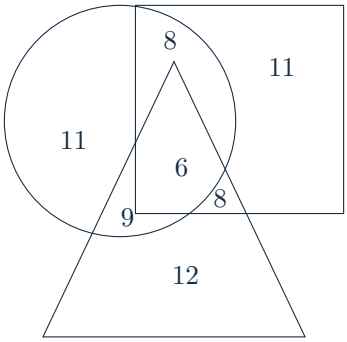
host 3



host 4

(A) 2
(B) 3
(C) 1
(D) 4

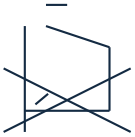
77. Circle=A, triangle=B, square=C. Each printed number is the count in that exact region. Find the total inside A but outside B, whether inside or outside C.



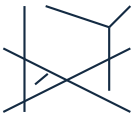
(A) 11
(B) 20
(C) 8
(D) 19

78. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

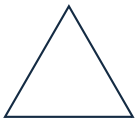
(A) 2

(B) 1

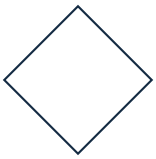
(C) 3

(D) 4

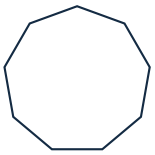
79. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

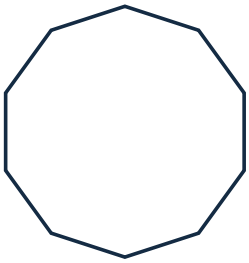


B

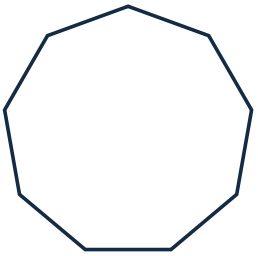


C

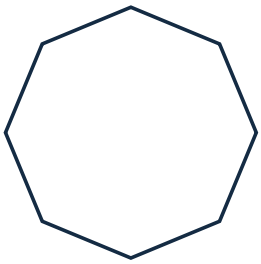
(A) Pattern V



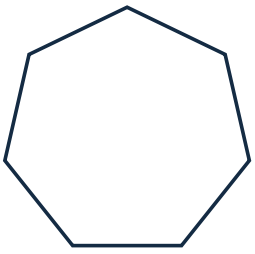
(B) Pattern R



(C) Pattern L



(D) Pattern T



80. Preserve the exact relation and its order: Ship : Fleet :: ? : Pride. Choose the missing term.

(A) Ship (B) Lion

(C) Wolf (D) Bee

81. In a row, Asha is 7th from the left and 39th from the right. Bina is 12th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 23 (B) 1

(C) 29 (D) 25

82. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?



1 2 3 4

(A) 1 (B) 3

(C) 2 (D) 4

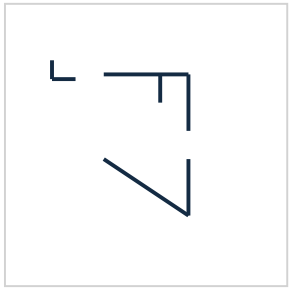
83. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



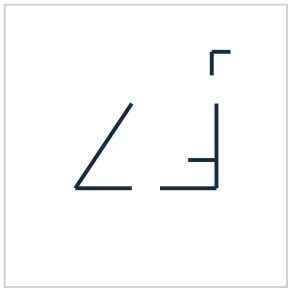
(A) Pattern U



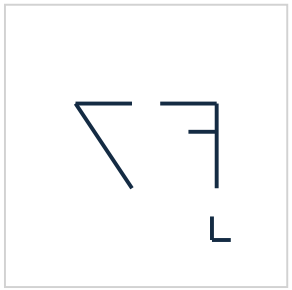
(B) Pattern Y



(C) Pattern Z



(D) Pattern W



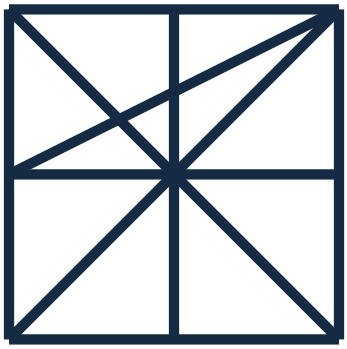
84. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, CLOUD is coded as FORXG. Which word is coded as EULFN?

(A) BRICK
(B) AQHBJ
(C) HXOIQ
(D) CSJDL

85. Consider only these statements; universal statements alone do not imply existence. No inventors are bakers; All editors are bakers; No inventors are editors. Conclusions: I. No bakers are editors. II. All inventors are bakers. Which conclusions are individually possible without contradicting the statements?

(A) Only I
(B) Only II
(C) Both I and II
(D) Neither I nor II

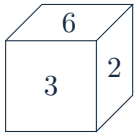
86. How many triangles are formed by all the drawn segments?



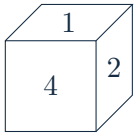
(A) 50 (B) 28

(C) 19 (D) 29

87. The views show the same die. Give the ordered pair: (face opposite 4, face opposite 1). Do not assume opposite labels sum to seven.



1



2

(A) $(3, 6)$ (B) $(3, 2)$

(C) $(3, 1)$ (D) $(3, 3)$

88. The whole has 12 unit positions. Part A covers the first 4; part B covers 3 positions, overlapping A in exactly 2 positions as shown. How many positions remain uncovered by either part?



whole



part A positions



part B positions

(A) 8 (B) 5

(C) 7 (D) 3

89. A person makes these turns in order: 180 degrees, 180 degrees, 90 degrees right, 180 degrees. The final heading is east. What was the heading immediately after turn 2?

(A) East (B) North

(C) South (D) West

90. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: YDKY, CEMB, ?, KGQH, OHSK, SIUN.

(A) GFOE (B) FEND

(C) HGPF (D) EOFG

Mathematics

91. Find the coefficient of x^4 in $(1+x+x^2)(1+x^2)^{11}$.

(A) 165 (B) 220

(C) 55 (D) 66

92. f(x)=sin(6x)/x+4 for x<0; f(x)=k cos x for x≥0. Find k for continuity at zero.

(A) 2 (B) 6

(C) 4 (D) 10

93. An AP has a_1=-36,d=6. How many of its first 10 terms are strictly positive?

(A) 2 (B) 3

(C) 4 (D) 8

94. For a=(1,3,0), b=(0,1,6), c=(3,0,1), evaluate (a×c)·(a+b).

(A) −55 (B) 55

(C) −54 (D) 0

95. A fair coin is tossed 5 times. Given that at least one of the first-3 and last-3 windows has exactly 2 heads, find the probability that both do.

(A) $\frac{5}{32}$ (B) $\frac{5}{24}$

(C) $\frac{19}{32}$ (D) $\frac{5}{19}$

96. A,B are acute, A+B=$\pi/4$ and tan A=$\frac{9}{11}$. Find tan(A-B).

(A) $\frac{79}{119}$ (B) $\frac{1}{10}$

(C) $-\frac{79}{119}$ (D) $\frac{79}{110}$

97. A chord of $x^2+y^2=4900$ touches $x^2/49-y^2/16=1$. Its nonzero midpoint (h,k) satisfies h=k. Find h^2+k^2.

(A) $\frac{33}{2}$ (B) 33

(C) $\frac{33}{4}$ (D) $\frac{65}{2}$

98. $f(x+1)=4x^2$. Find the area enclosed between y=f(x) and y=4(x-1).

(A) $\frac{4}{3}$ (B) 2

(C) $-\frac{2}{3}$ (D) $\frac{2}{3}$

99. For 3x+4y+4=0 and 4x+3y+6=0, find the squared distance from the origin to the angle bisector of positive slope.

(A) 2 (B) $\frac{50}{49}$

(C) $\sqrt{2}$ (D) 4

100. For f(x)=|x-4|/(x-8), on which maximal open intervals is f strictly decreasing?

(A) $(-\infty, 8)$ (B) $(4, \infty)$

(C) $(-\infty, 4)$ (D) $(4, 8) \cup (8, \infty)$

101. How many integers k make 6sin x+8cos x+3=k have exactly two distinct solutions in $0 \le x < 2\pi$?

(A) 20 (B) 21

(C) 19 (D) 9

102. A circular sector has radius 5 and arc length π. Find its area.

(A) $\frac{25\pi}{2}$ (B) $\frac{\pi}{10}$

(C) $\frac{5\pi}{2}$ (D) $\frac{5}{2}$

103. α,β are nonzero roots of x^2-7x+10=0. Find the sum of $1+1/\alpha$ and $1+1/\beta$.

(A) $\frac{17}{10}$ (B) $\frac{24}{7}$

(C) $\frac{27}{10}$ (D) $\frac{7}{10}$

104. Independent events A,B have probabilities $\frac{2}{9},\frac{5}{12}$. Given A∪B, find P(A∩B).

(A) $\frac{10}{59}$ (B) $\frac{10}{69}$

(C) $\frac{49}{59}$ (D) $\frac{5}{54}$

105. How many integers k make $6\sin 2x + 8(2\cos^2 x - 1) + 3 = k$ have exactly two distinct solutions in $0 \le x < \pi/4$?

(A) 1 (B) 19

(C) 2 (D) 3

106. $A = \begin{bmatrix} 11 & 1 \\ 8 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$. Find tr(BA) without assuming AB=BA.

(A) 49 (B) -19

(C) 45 (D) 26

107. Find the population variance of 2+5, 4+5, ..., 16+5.

(A) $\frac{21}{4}$ (B) 26

(C) 21 (D) $\frac{21}{2}$

108. The hyperbola $x^2/16$-$y^2/4$=1 has eccentricity e. Find the eccentricity e′ of its conjugate hyperbola.

(A) $\frac{2\sqrt{5}}{5}$ (B) $\sqrt{5}$

(C) $2\sqrt{5}$ (D) $\frac{\sqrt{5}}{2}$

109. For x>1 let f(x)=arctan(2 x/(1-x^2)), with arctan in $(-\pi/2,\pi/2)$. Find the integral from 6 to 7 of f′(x) dx.

(A) $-2\,\mathrm{atan}\,(6) + 2\,\mathrm{atan}\,(7)$

(B) $-\,\mathrm{atan}\,(6) + \mathrm{atan}\,(7)$

(C) $2\,\mathrm{atan}\left(\frac{1}{6}\right)$

(D) π

110. $z_1=2+4i$, $z_2=6+4i$, $z_3=6+8i$. Find the smaller angle between z_2-z_1 and z_3-z_1.

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$

(C) $\frac{3\pi}{4}$ (D) 0

111. Maximize $Z=3x+4y$ subject to $x\geq 0$, $y\geq 6$, $x+y\leq 9$, $x+2y\leq 14$.

(A) 32 (B) 28

(C) 33 (D) 30

112. For $x,y>0$ with $\sqrt{x}+\sqrt{y}=11$, find d^2y/dx^2 at $x=36$.

(A) $\frac{11}{216}$ (B) $\frac{11}{432}$

(C) $-\frac{11}{432}$ (D) $\frac{11}{12}$

113. Under $x=X+7$, $y=Y+4$, the parabola $(y-4)^2=36(x-7)$ is normalized. Find the x coordinate of its focus in the original axes.

(A) -2 (B) 43

(C) 16 (D) 9

114. $A=(7,2,2)$, $B=(14,2,4)$. Find the squared length of the component of AB perpendicular to direction $(1,2,2)$.

(A) $\frac{148}{3}$ (B) $\frac{356}{9}$

(C) $\frac{121}{9}$ (D) 53

115. $r=pa+qb$, $a=(1,0,0)$, $b=(1,1,0)$, $r\cdot a=5$, $r\cdot b=8$. Find $|r|^2$.

(A) 13 (B) $\sqrt{34}$

(C) 34 (D) 8

116. A line has direction$(t,1,k)$ and is parallel to both planes $x-13\,y+z=0$ and $2\,x+y-z=0$. Find the squared length of its direction vector.

(A) 126 (B) 97

(C) 1 (D) 98

117. $a=(4,0,0)$, $b=(0,6,0)$, $c=(0,0,4)$, $p=a+b$. Evaluate $[p\times(b\times c)]\cdot(a+c)$.

(A) 576 (B) -576

(C) -1152 (D) 0

118. Evaluate $\lim$ as $n\to\infty$ of $(7/n)\Sigma|7\,j/n-7/3|$, $j=1,\ldots,n$.

(A) $\frac{49}{18}$ (B) $\frac{98}{9}$

(C) $\frac{343}{9}$ (D) $\frac{245}{18}$

119. $|U|=70$, $|A|=19$, $|B|=16$, $|A\cap B|=7$. Find the number of elements belonging to both A,B or to neither.

(A) 49 (B) 63

(C) 21 (D) 42

120. Four observations are 8,8,x,y with population mean 8 and variance 4/2. Find xy.

(A) 60 (B) 62

(C) 64 (D) 68

121. A differentiable function f satisfies $f\prime(t) = (t-64)(t-144)$ for every real t. Let $g(x) = f(x^2)$. Which is the maximal open interval containing $x = -10$ on which $g\prime(x) > 0$?

(A) (8, 12) (B) (-12, -8)

(C) (-infinity, -12) (D) (-8, 0)

122. What is the sum of all integers attained by $4\sin^2 x - 4\sin x + 3$ for $-\pi/6 \le x \le \pi/2$?

(A) 20 (B) 14

(C) 18 (D) 65

123. If z=2-3i, find |exp(iz)|.

(A) e^2 (B) e^{-3}

(C) 1 (D) e^3

124. Evaluate sin(2arctan 5).

(A) $\frac{5\sqrt{26}}{26}$ (B) $\frac{5}{13}$

(C) $-\frac{5}{13}$ (D) $-\frac{5}{12}$

125. The normal of slope 4 to y^2=28x touches at P. Find the distance from P to the focus.

(A) 119 (B) 112

(C) 14161 (D) 105

126. Let z=$(1+i)^8/(1-i)^3$. Find Re(z)+Im(z).

(A) −64 (B) −8

(C) 0 (D) 4

127. Find the squared distance between 6x+8y=10 and 3x+4y=9.

(A) $\frac{4}{5}$ (B) $\frac{16}{25}$

(C) $\frac{4}{25}$ (D) 16

128. If α,β are roots of $x^2 - 2kx + k^2 - 9 = 0$ and k∈[-6,6], find the maximum of $\alpha^2+\beta^2$.

(A) 126 (B) 90

(C) 18 (D) 54

129. y′-y/x=1 for x>0, y(1)=7. Find y(2)/2-y(1).

(A) $-\log(2)$ (B) $2\log(2) + 14$

(C) $\log(2)$ (D) $\log(2) + 7$

130. Let $D(t) = \det \begin{pmatrix} 8+t & 1+t \\ 1+t & 7+t \end{pmatrix}$. Find $\mathrm{D}'(0)$.

(A) 15
(B) -17
(C) 110
(D) 13

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. The quantity $Z = x^3y^2/\sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 3, 3 and 2, respectively. To first order, what is the maximum percentage error in Z?

(A) 16 (B) $\sqrt{118}$

(C) 17 (D) 15

132. A stationary nucleus of mass number 216 emits an alpha particle whose kinetic energy is 9 MeV. Taking daughter and alpha masses proportional to $A-4$ and 4, find the total energy release Q, including daughter recoil.

(A) $\frac{9}{53}$ MeV (B) $\frac{53}{6}$ MeV

(C) $\frac{486}{53}$ MeV (D) $\frac{9\sqrt{53}}{53} + 9$ MeV

133. In an abrupt junction, the acceptor concentration is 841 times the donor concentration. Within the depletion approximation, which statement gives x_n/x_p and identifies the carriers mainly responsible for the small reverse current before breakdown?

(A) 1/841; minority carriers (B) 841; minority carriers

(C) 1; majority carriers (D) 29; majority carriers

Chemistry

134. A cathode receives 4/25 mol electrons and deposits 64/25 g copper from Cu2+ (Cu=64 g/mol). Its only other product is H2. What amount of H2, in mol, is produced?

(A) 1/25 (B) 2/25

(C) 1/50 (D) -1/25

135. A sample of 1,2-dibromoethane is treated with excess NaNH2,then aqueous protonation. Which isolated product is consistent with the reagent limitation and carbon-skeleton change?

(A) ethane (B) ethyne

(C) bromoethene (D) ethene

136. For methanal with concentrated aqueous NaOH,which product or mechanistic conclusion is correct under the stated conditions?

(A) ethanol and sodium acetate (B) ethanal and hydrogen

(C) methanol and sodium formate (D) methanol only

Logical Reasoning

137. A walker goes 8 m west, then 9 m south, then turns left and walks 5 m. What is the square of the straight-line distance from the starting point, in square metres?

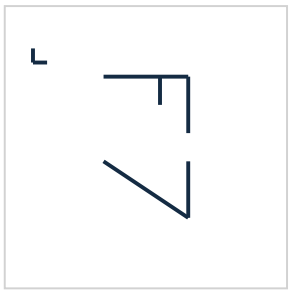
(A) 91 (B) 90

(C) 484 (D) 89

138. Choose the exact horizontal water reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



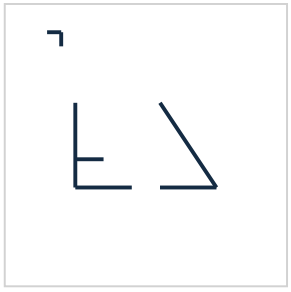
(A) Pattern M



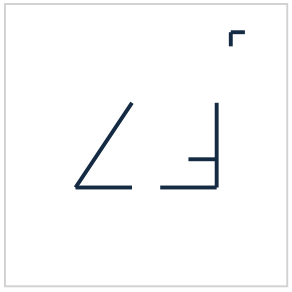
(B) Pattern P



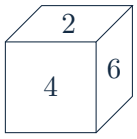
(C) Pattern T



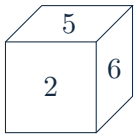
(D) Pattern W



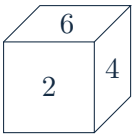
139. The views show the same die. Give the ordered pair: (face opposite 4, face opposite 5). Do not assume opposite labels sum to seven.



1



2



3

(A) (5, 4) (B) (5, 1)

(C) (5, 3) (D) (5, 2)

Mathematics

140. An equivalence relation has classes of sizes 8,3,4. How many unordered pairs of distinct related elements are there?

(A) 25 (B) 37

(C) 74 (D) 89

141. For $x=t^2, y=t^3, t>0$, find d^3y/dx^3 at $t=4$.

(A) $\frac{3}{512}$ (B) $-\frac{3}{8}$

(C) $-\frac{3}{512}$ (D) $-\frac{3}{64}$

142. For $(d^3y/dx^3)^5/(1+x^2)+y=0$, find order+degree if the degree is defined; otherwise select "Degree is not defined".

(A) 8 (B) 15

(C) 9 (D) Degree is not defined

BITSAT Mock 34

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A Newtonian liquid of viscosity $\frac{1}{10}$ Pa s has velocity profile $v(y) = 4y + 4y^2$ in SI units. Find the shear-stress magnitude at $y = 1$ m.

(A) $\frac{2}{5}$ Pa (B) $\frac{4}{5}$ Pa

(C) 12 Pa (D) $\frac{6}{5}$ Pa

2. Two ideal cells of emfs 6 V and 3 V oppose each other in a loop containing two resistors each 2 ohms. Find the current magnitude.

(A) $\frac{3}{4}$ A (B) $\frac{9}{2}$ A

(C) $\frac{9}{4}$ A (D) $\frac{3}{2}$ A

3. A uniform wire is stretched to $\frac{7}{5}$ times its original length at constant volume. Its resistivity increases by 10 percent during the process. Find final resistance divided by initial resistance.

(A) $\frac{441}{250}$ (B) $\frac{539}{250}$

(C) $\frac{55}{98}$ (D) $\frac{3773}{1250}$

4. A particle has displacement $x = 2\cos(2\pi t/5)$ cm. Find its signed velocity at $t = \frac{5}{6}$ s.

(A) $-\frac{4\pi}{5}$ cm/s (B) $-\frac{2\sqrt{3}\pi}{5}$ cm/s

(C) $-\frac{\sqrt{3}}{5}$ cm/s (D) $-\frac{2\pi}{5}$ cm/s

5. Two sound sources produce wavelengths $\frac{1}{5}$ m and $\frac{1}{6}$ m in the same gas at one temperature. The sound speed there is 500 m/s. At a second temperature the sound speed is 3/2 times larger, while both source frequencies stay unchanged. Find the new beat frequency.

(A) 750 Hz (B) 500 Hz

(C) 2000 Hz (D) $\frac{1000}{3}$ Hz

6. A body is dropped from rest at height 20 m above the ground. Find its speed when it is still 10 m above the ground. Neglect air resistance and take $g = 10$ m/s^2.

(A) 10 m/s (B) 200 m/s

(C) $20\sqrt{5}$ m/s (D) $10\sqrt{2}$ m/s

7. A stationary nucleus with mass number 20 and binding energy per nucleon $\frac{38}{5}$ MeV splits into two equal daughters, each having total binding energy 79 MeV. Which statement is correct?

(A) No energy can be released because fission always requires a larger mass number than 20.

(B) The process absorbs 6 MeV; the daughters have larger binding energy per nucleon.

(C) The process releases 12 MeV; each daughter releases 6 MeV.

(D) The process releases 6 MeV; the daughters have larger binding energy per nucleon.

8. A stationary nucleus of mass number 228 emits an alpha particle whose kinetic energy is 5 MeV. Taking daughter and alpha masses proportional to $A-4$ and 4, find the total energy release Q, including daughter recoil.

(A) $\frac{285}{56}$ MeV (B) $\frac{280}{57}$ MeV

(C) $\frac{5\sqrt{14}}{28}+5$ MeV (D) 285 MeV

9. A magnetic field is uniform inside a long solenoid of radius 6 m and negligible outside. It changes at 8 T/s. Find induced electric-field magnitude at radius $4R$ from the axis.

(A) 192 V/m (B) 96 V/m

(C) 6 V/m (D) $\frac{3}{2}$ V/m

10. A cyclist rides straight outward at 8 m/s for 8 s, waits 5 s, then returns to the start. The whole journey must last 16 s. What constant return speed is required?

(A) 8 m/s (B) $\frac{64}{11}$ m/s

(C) $\frac{64}{9}$ m/s (D) $\frac{64}{3}$ m/s

11. A uniform disc of mass 2 kg and radius 4 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) 12 kg m^2 (B) 13 kg m^2

(C) 19 kg m^2 (D) 16 kg m^2

12. A particle has position-dependent velocity components $\vec{v}=3x\,\hat{i}-3y\,\hat{j}$ in SI units. Given $dx/dt=v_x$ and $dy/dt=v_y$, what is its acceleration magnitude at $(2,2)$ m?

(A) 18 m/s^2 (B) $54\sqrt{2}$ m/s^2

(C) $18\sqrt{2}$ m/s^2 (D) 36 m/s^2

13. A network is defined by $P=\mathrm{NOR}(A,B)$, $Q=\mathrm{NOR}(A,P)$ and $Y=\mathrm{NOR}(Q,\overline{B})$. Each of the four input pairs is applied once. How many pairs satisfy both $Y=1$ and $P=0$?

(A) 1 (B) 0

(C) 3 (D) 2

14. A particle has potential $U(x,y)=5x^2y+6y$ J in SI units. Find the force magnitude at $(4,1)$ m.

(A) $2\sqrt{2249}$ N (B) 40 N

(C) $2\sqrt{1949}$ N (D) 86 N

15. Water flows steadily from a pipe section of area A at 5 m/s to a section of area $A/4$ located 3 m higher. Neglect viscosity. Find $p_1 - p_2$. Take density 1000 kg/m^3 and $g = 10$ m/s^2.

(A) 30000 Pa (B) 187500 Pa

(C) 230000 Pa (D) 217500 Pa

16. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 15 percent. The measured increase in its absolute temperature is 129 K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) 529 K (B) 400 K

(C) $\frac{51600}{929}$ K (D) 860 K

17. A 5 V Zener regulator feeds a constant-current load of 2 mA. Supply varies from 13 to 15 V. Regulation requires Zener current at least 3 mA, and its safe maximum is 9 mA. What is the largest permissible series resistance? Assume ideal regulation and give resistance in Ω.

(A) 4000 Ω (B) 1600 Ω

(C) $\frac{8000}{11}$ Ω (D) $\frac{10000}{11}$ Ω

18. A uniform disc of initial mass 3 kg and radius 3 m has a concentric hole of radius $R/2$ removed. Find the remaining annulus's moment of inertia about a diameter lying in its plane.

(A) $\frac{405}{64}$ kg m^2 (B) $\frac{81}{64}$ kg m^2

(C) $\frac{459}{64}$ kg m^2 (D) $\frac{27}{4}$ kg m^2

19. A series RC circuit is driven at fixed rms voltage. Multiplying angular frequency by 8 multiplies rms current by $\frac{8\sqrt{655}}{131}$. Find R^2/X_C^2, where X_C is the initial capacitive reactance.

(A) $\frac{2}{3} + \frac{16\sqrt{655}}{393}$ (B) $\frac{2}{3}$

(C) $\frac{3}{2}$ (D) $\frac{13}{27}$

20. Current rises linearly from 0 to 7 A over 2 s and then falls linearly to 0 over another 2 s. It remains zero for the next 4 s. Find the average current over the entire 8 s interval.

(A) $\frac{7}{8}$ A (B) 7 A

(C) $\frac{7}{4}$ A (D) $\frac{7}{3}$ A

21. A flat spiral coil has 30 turns whose radii are uniformly distributed from 3 cm to 4 cm. The same current 5 A flows in every turn. Approximating turns as circular and the distribution as continuous, find magnetic moment in units of $\pi \times 10^{-4}$ A m^2.

(A) 1850 (B) 1875

(C) 1350 (D) 2400

22. A gas receives 300 J of heat while its surroundings do 50 J of work on it. It then loses 150 J of heat at fixed volume. Find its total internal-energy change.

(A) 200 J (B) 100 J

(C) 150 J (D) 500 J

23. A 2 kg block is pulled by a force of 10 N at an angle with $\cos\theta = 4/5$, $\sin\theta = 3/5$ above a horizontal floor. The block slides; $\mu_k = \frac{1}{10}$. Find the net horizontal force. Take $g = 10\ \text{m/s}^2$.

(A) $\frac{7}{5}$ N
(B) $\frac{24}{5}$ N
(C) $\frac{33}{5}$ N
(D) 8 N

24. A platform of moment of inertia 4 kg m^2 carries a person of mass 5 kg at radius 2 m and rotates at 6 rad/s. The person walks to half the stated initial radius; external torque is zero. Find final angular speed. Treat the person as a point mass.

(A) $\frac{72}{7}$ rad/s
(B) $4\sqrt{6}$ rad/s
(C) 16 rad/s
(D) 36 rad/s

25. A stretched string is divided by a bridge into lengths in the ratio 1 : 5. Tension and linear density are unchanged. The original fundamental is 20 Hz. Find the fundamental of the longer segment.

(A) 120 Hz
(B) 20 Hz
(C) 24 Hz
(D) 30 Hz

26. A 2 kg block rests on a 7 kg block on a smooth horizontal floor. A force 27 N pulls the lower block. There is no slipping between the blocks. During a displacement 5 m, what work does friction do on the upper block?

(A) $\frac{135}{2}$ J
(B) 135 J
(C) 105 J
(D) 30 J

27. A long solenoid has its turn number multiplied by 4 and length multiplied by 4, while its cross-sectional area and core permeability stay fixed. Find self-inductance multiplier.

(A) 64
(B) $\frac{1}{4}$
(C) 16
(D) 4

28. Two small identical conducting spheres initially carry positive charges $6q_0$ and $3q_0$ and are separated by distance d. They touch and are returned to the same separation. Find the final force divided by the initial force.

(A) $\frac{3}{4}$
(B) $\frac{9}{8}$
(C) $\frac{1}{3}$
(D) $\frac{2}{3}$

29. A nonrelativistic particle with charge 5 C follows a circular orbit of radius 4 m in a perpendicular magnetic field 5 T. A photon has the same de Broglie wavelength as this particle. Find photon energy in units of speed of light c. Use $p = qBr$ for the particle and $E_\gamma = pc$.

(A) 100
(B) 20
(C) $\frac{1}{100}$
(D) 50

30. The 2nd harmonic of an open pipe of length 7 m equals the 2nd allowed resonance of a closed pipe. Both have the same sound speed and negligible end correction. Find the closed-pipe length.

(A) 7 m
(B) $\frac{35}{4}$ m
(C) $\frac{21}{4}$ m
(D) $\frac{21}{2}$ m

Chemistry

31. A one-electron ion of nuclear charge Z=4 has a selected shell which, if hypothetically filled subject to Pauli, contains 12 electrons in orbitals with ml=0. What is the magnitude of that shell's hydrogenic energy divided by the hydrogen ground-state energy magnitude? Ignore interactions in identifying the shell.

(A) 8/9

(B) -4/9

(C) 4/9

(D) 2/9

32. A closed system goes from state1to2 by path A with q=900 J and work on the system w=-300 J. It returns from2to1 along a different path with q=-1100 J. IUPAC work on the return path,in J,is

(A) 300

(B) -500

(C) 600

(D) 500

33. For$2A(g) \rightleftharpoons A_2(g)$,the numerical K_c is1.00 at a T where RT=9 L bar/mol. Find numerical K_p for$1/2\ A_2(g) \rightleftharpoons A(g)$,using matching concentration and pressure conventions.

(A) $\frac{1}{3}$

(B) 9

(C) $\frac{1}{9}$

(D) 3

34. One episode contains abundant ozone and PAN in sunlight; another contains SO2, smoke and fog in cold humid air. Which assignment is appropriate?

(A) Neither can contain a chemical pollutant.

(B) First: photochemical oxidising smog; second: classical sulphurous smog.

(C) Both must be identical because each reduces visibility.

(D) First: classical smog; second: ozone-only photochemistry.

35. For an ordinary covalent H_2O_2 molecule,which combined statement about local geometry and overall conformation is correct?

(A) The O-O bond is a double bond,so both O centres are planar with one lone pair.

(B) Each O has two bonds and two lone pairs; the H-O-O-H skeleton is nonplanar/open-book.

(C) Each O is linear because it is bonded to only two atoms,and the skeleton is straight.

(D) All four atoms must lie in one plane because only sigma bonds are present.

36. A mixture contains one mole each of phenol,ethanol and ethanoic acid. Treat with excess Na metal. What is the total amount,in mol,of H2? Assume all stated acid-base reactions go to completion and no side reactions.

(A) 3/2

(B) 2

(C) 1

(D) 1/2

37. A solution of one nonvolatile nonelectrolyte has exact relative vapour-pressure lowering 3/113. After removal of solvent only, the lowering becomes 9/119. The initial solvent amount was 11 mol, and Raoult's law holds without association. How many moles of solvent were removed?

(A) 229/30

(B) 22/9

(C) 11/3

(D) 22/3

38. In a1.00 L vessel A+2B⇌2C+D. Initially A=2/5 mol, B=1/2 mol, D=0 and an unknown amount of C. At equilibrium D=1/50 mol and Kc=49/40204 using molar concentrations. What was the initial C amount in mol? All equilibrium amounts are positive.

(A) 3/100
(B) 3/50
(C) -3/100
(D) 3/200

39. Nonrelativistic ions A and B start from rest and gain only electric kinetic energy. B has 4 times A's mass and 3 times its positive charge. To make $\lambda_B/\lambda_A=1/4$, what must V_B/V_A be?

(A) 3/4
(B) 16/3
(C) 4/3
(D) 5

40. A pure anhydrous Na2CO3 standard contains 1/250 mol carbonate. In its HCl titration, the additional acid volume from the carbonate-to-hydrogencarbonate endpoint to the final CO2/water endpoint is 40 mL. What is the HCl molarity? Assume complete stoichiometric endpoints.

(A) -1/10
(B) 1/20
(C) 1/10
(D) 1/5

41. How many listed compounds are secondary alcohols? C6H5CH(OH)CH3, CH3CH(OH)CH2CH3, C6H5CH2OH, HOCH2CH2OH. Count compounds,not total OH groups.

(A) 0
(B) 2
(C) 1
(D) 3

42. Low-density polythene has more branching than high-density polythene. Which explanation links this structural difference to packing?

(A) Branches hinder close chain packing.
(B) Branches necessarily convert every C-C bond into an ester.
(C) A branched polymer cannot contain carbon.
(D) Branching means each molecule must be a protein.

43. A solvent has Tb=400 K, M=100 g/mol and ΔHvap=32000 J/mol. Use R=8.00 and Kb=R Tb^2M/(1000ΔHvap). An A2B solution of analytical molality 3/100 mol/kg boils 27/125 K above pure solvent. Only dissociation A2B→2A+ +B2- occurs, with ideal dilute-particle counting. What fraction of A2B is dissociated?

(A) 2/15
(B) 1/5
(C) 7/10
(D) 2/5

44. For 2-methylpropanal with dilute base at low temperature,which product or mechanistic conclusion is correct under the stated conditions?

(A) only Cannizzaro is possible because it is an aldehyde
(B) self-aldol is impossible because branching removes every alpha H
(C) a ketone forms by loss of one carbon
(D) a self-aldol addition is possible because one alpha H remains

45. A Co(3) coordination entity has coordination number 6, exactly 2 fully bound en ligand(s), and chloride in all remaining donor sites. Take en as 2-dentate with charge 0; chloride is monodentate. Find (number of ligand entities, net charge of the coordination entity).

(A) (6, +2)
(B) (5, +1)
(C) (4, +1)
(D) (4, +0)

46. A $\frac{4}{5}$ g organic sample contains chlorine as its only halogen. Carius analysis gives $\frac{14924}{44375}$ g dry AgCl, but this is only 80.0% of the theoretical AgCl because of a known collection loss. Find chlorine mass percentage. Use Ag=108, Cl=35.5.

(A) $\frac{52}{5}$
(B) $\frac{65}{4}$
(C) $\frac{3731}{71}$
(D) 13

47. An ideal mixture initially has $\frac{3}{10}$molA and $\frac{2}{5}$molB. A selective step removes half ofB and noA. At the new equilibriumT,V,totalpressure is6bar. Partialpressure ofA,in bar,is

(A) $\frac{18}{5}$
(B) 3.0
(C) $\frac{18}{7}$
(D) 10

48. Which coordination name keeps the bracketed sphere,counter-ion and formal metal oxidation state consistent for $[Co(NH_3)_5Cl]Cl_2$?

(A) pentaamminechloridocobaltate(III) chloride
(B) pentaamminechloridocobalt(III) chloride
(C) pentaamminechloridocobalt(II) chloride
(D) pentaamminecobalt(III) trichloride

49. ethanol is heated under acid-catalysed dehydration conditions. Which major product follows removal of water while preserving the stated carbon framework?

(A) ethanal
(B) ethane
(C) propan-1-ol
(D) ethene

50. At fixedT,V,A(g)→P(g)+Q(g)+R(g)is first order,and an inert gas is present. Initial totalpressure is$\frac{2}{5}$bar;complete-decomposition pressure is1bar. At two half-lives,totalpressure in bar is

(A) $\frac{3}{4}$
(B) $\frac{17}{20}$
(C) $\frac{11}{20}$
(D) $\frac{7}{10}$

51. An orbital has 1 angular nodes and 2 radial nodes. Maximum electrons in its entire subshell are

(A) 10
(B) 2
(C) 6
(D) 32

52. An ideal gas freely expands into vacuum in an insulated rigid enclosure,accessible gas volume increasing 4-fold. Which joint statement follows?

(A) q=0 and w=0 imply entropy change is zero and the expansion is reversible.

(B) The rigid enclosure forces the gas-accessible volume to remain unchanged.

(C) q=0,w=0 and temperature is unchanged;entropy increases,so the process is irreversible.

(D) The gas does work on the vacuum and its temperature necessarily falls.

53. A calorimeter holding50.0g water absorbs600J and rises2.00K. Water specific heat is4.00J $g^{-1}K^{-1}$; calorimeter heat capacity is unchanged. When it holds100g water and absorbs the same heat,temperature rise in K is

(A) $\frac{6}{5}$ (B) 2

(C) 1 (D) $\frac{3}{2}$

54. A gas evolved during an anion test turns acidified dichromate from orange to green. Which gas and chemical explanation match this complete observation?

(A) N2; acid-base neutralisation always produces the stated colour. (B) SO2; Cr(III) accounts for the diagnostic change.

(C) HCl; elemental chlorine is deposited. (D) NH3; molecular oxygen causes the stated change.

55. Which pair distinguishes a first electron-gain process from electronegativity in a bond?

(A) First electron gain always begins with an anion.

(B) Electronegativity is the mass lost when a proton is removed.

(C) Cl(g)+e-→Cl-(g) is a gas-phase thermochemical process; electronegativity describes attraction for shared electrons.

(D) Both are exactly the same measurable enthalpy for all bonds.

56. A carbon-containing organic sample contains Br only among nitrogen, sulfur and halogens. It undergoes sodium fusion with sufficient sodium. Which conversion and subsequent test protocol is chemically consistent? For a silver-halide test, use a separately prepared portion boiled with nitric acid to remove cyanide/sulfide interference where present.

(A) bromide; pale-yellow AgBr after nitric-acid treatment.

(B) sulfide; no further treatment is ever needed before adding silver nitrate.

(C) Unchanged covalent bonds; add silver nitrate directly to the untreated organic sample.

(D) Nitrate and sulfate in every case; all give the same Prussian-blue test.

57. Classify the two operations: I. Removal of small ions accelerated by an electric field across a dialysing system. II. formation of a dispersed precipitate by adding an appropriate stabilising ion. Select the correct paired classification.

(A) Preparation by chemical reaction; Brownian motion (B) Electrodialysis; peptisation

(C) Peptisation; dialysis (D) Coagulation; purification

58. In a tetrahedral crystal field,the upper set has 3 orbitals and the lower set has 2. Their separation is Δ. Which assignment preserves the five-orbital barycentre?

(A) t_2: $\frac{2}{5}\Delta$; e: $-\frac{3}{5}\Delta$. (B) t_2: $\frac{3}{5}\Delta$; e: $-\frac{2}{5}\Delta$.

(C) All five remain degenerate because the barycentre is unchanged. (D) t_2: $+\Delta$; e: $-\Delta$.

59. A dry mixture contains only anhydrous CaCO3 and SrCO3. Use the ordinary Group2 carbonate stability trend. In an idealised selective-heating window, the less thermally stable carbonate decomposes completely by MCO3→MO+CO2, while the more stable carbonate remains unchanged. No other mass loss occurs. This is the stated model, not a specified real temperature. Atomic masses: Ca=40, Sr=88, C=12, O=16. The initial mixture mass is 149/25 g. The first-stage CO2 collected is 3/125 mol. This is 4/5 of the CO2 actually evolved; collection efficiency alone causes the loss. What was the initial mass fraction of SrCO3?

(A) 38/149
(B) 74/149
(C) 2/5
(D) 75/149

60. A river receives biodegradable organic effluent. Its biochemical oxygen demand increases while its dissolved oxygen falls. Which explanation connects the two observations?

(A) Microorganisms consume dissolved oxygen while decomposing the organic load.
(B) Dissolved oxygen must increase whenever microorganisms grow.
(C) BOD directly counts only dissolved metal ions.
(D) BOD measures the oxygen that the effluent necessarily produces.

English Proficiency

61. Her work examines rock layers and the processes that formed them, rather than predicting daily weather. What is she?

(A) geologist
(B) astronomer
(C) archaeologist
(D) meteorologist

62. Use the standard contemporary negative adjective. The mark cannot be seen under the stated conditions; it is _____.

(A) disvisible
(B) invisible
(C) unvisible
(D) imvisible

63. Read the sentence: The technicians hope to restore normal service before the library opens. Choose the closest meaning of 'restore' as used in the sentence.

(A) bring back
(B) improve beyond its previous state
(C) replace with something new
(D) maintain without interruption

64. Read the sentence: The reward was extrinsic, coming from outside the activity itself. Choose the opposite in meaning of 'extrinsic' as used in the sentence.

(A) external
(B) additional
(C) intrinsic
(D) financial

65. Read the sentence: At the crowded reception, the normally reserved poet spoke little about her private life. Choose the closest meaning of 'reserved' as used in the sentence.

(A) unwilling to meet anyone
(B) careful not to offend
(C) uncertain of her knowledge
(D) restrained in self-expression

66. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. The guide requested that every visitor Q. and the equipment was safe. R. until the demonstration had ended S. remain behind the marked line

(A) R-P-S-Q
(B) S-P-R-Q
(C) Q-S-R-P
(D) P-S-R-Q

Read the passage for Question 67

The residents agreed that the shared room needed a booking system, but they disagreed about what fairness required. Some wanted identical time slots for every group. Others pointed out that a brief reading circle and a rehearsal with equipment did not use the room in the same way. The final arrangement guaranteed each group a basic opportunity and allowed additional time to be requested with a reason. Requests and decisions were recorded openly. The system did not eliminate judgement, and an unpopular decision could still be challenged. Its advantage was that differences had to be explained rather than hidden behind apparently identical treatment. The residents came to see fairness less as a single pattern imposed on every situation and more as a process whose distinctions could be examined.

67. Which title best captures the passage's main argument?

(A) Why a shared room should be reserved for the largest group

(B) Replacing all judgement with an automatic booking rule

(C) Fairness through explainable distinctions rather than merely identical treatment

(D) Why equal time slots always produce unfair results

68. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. for staff moving equipment. Q. Visitors may use the side entrance R. provided that they keep S. the marked passage clear

(A) P-R-S-Q
(B) Q-R-S-P
(C) R-Q-S-P
(D) S-Q-R-P

69. The course examines the structure and use of language as a system, rather than teaching only one foreign language. Which field is it?

(A) linguistics
(B) translation
(C) calligraphy
(D) biography

70. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The guard will admit only visitors carrying passes.

(A) All visitors will be admitted by the guard.
(B) Only visitors carrying passes will be admitted by the guard.
(C) Only visitors carrying passes would have been admitted by the guard.
(D) Only the guard will admit visitors carrying passes.

Logical Reasoning

71. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?



(A) 2 (B) 1

(C) 4 (D) 3

72. A code shifts letters cyclically through A–Z. Odd positions share one fixed shift and even positions share another. For example, BRICK is coded as CQJBL. Which word is coded as UQBHO?

(A) USBJO (B) TRAIN

(C) VPCGP (D) SQZHM

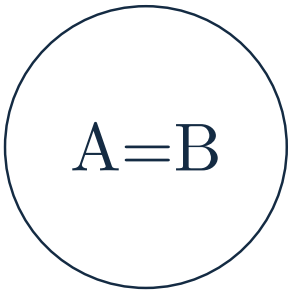
73. Preserve the exact relation and its order: Lion : Pride :: ? : Swarm. Choose the missing term.

(A) Ship (B) Wolf

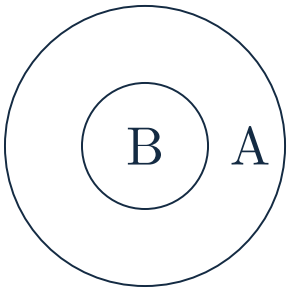
(C) Bee (D) Lion

74. Choose the diagram that represents all these conditions, including the strictness information: All B are A; some A is not B, and B is nonempty.

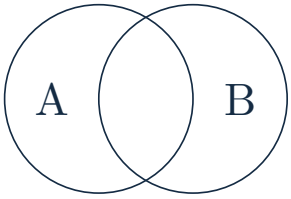
(A) Pattern U



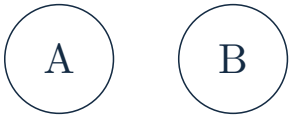
(B) Pattern Z



(C) Pattern L



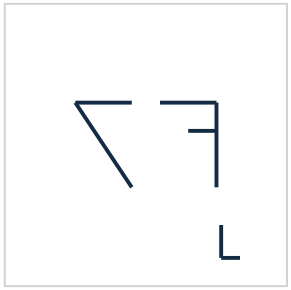
(D) Pattern R



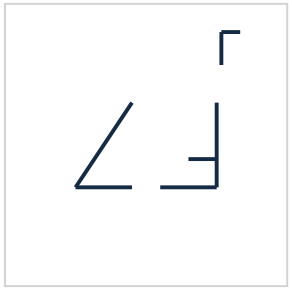
75. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



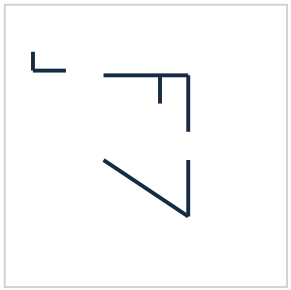
(A) Pattern Q



(B) Pattern X



(C) Pattern N



(D) Pattern P



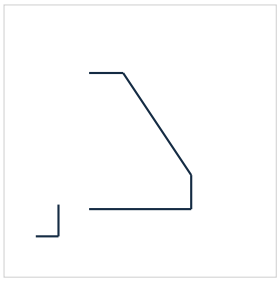
76. Consider only these statements; universal statements alone do not imply existence. No farmers are campers; All editors are farmers. Conclusions: I. No editors are farmers. II. No editors are campers. Which conclusions follow necessarily?

(A) Only I
(B) Neither I nor II
(C) Only II
(D) Both I and II

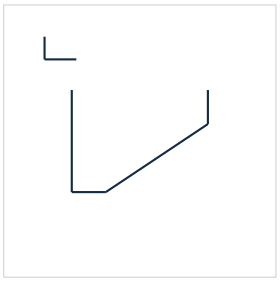
77. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 12$ and $5 \mapsto 24$. Which input maps to 64?

(A) 16
(B) 64
(C) 14
(D) 15

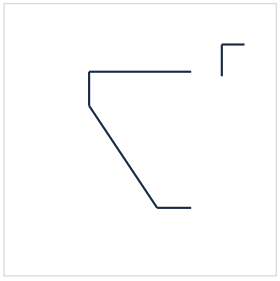
78. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

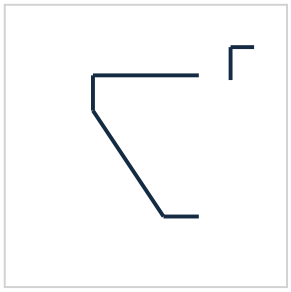


bottom left

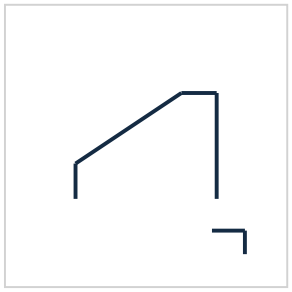
?

bottom right

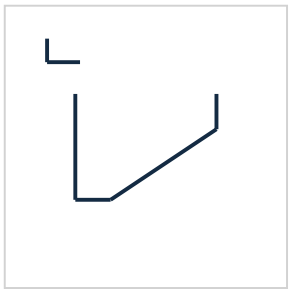
(A) Pattern R



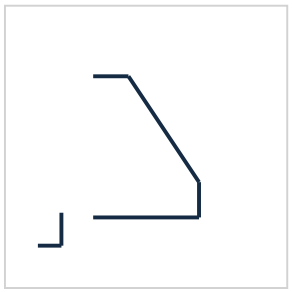
(B) Pattern U



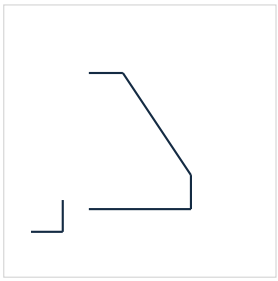
(C) Pattern M



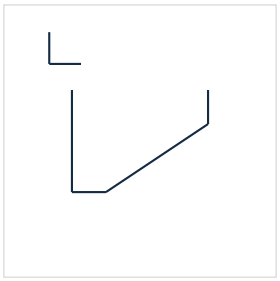
(D) Pattern Z



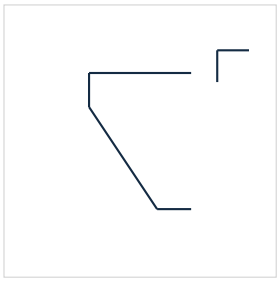
79. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1



2

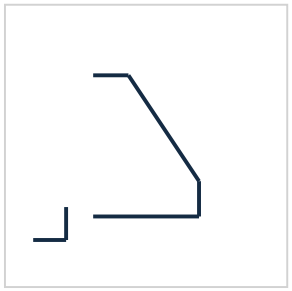


3

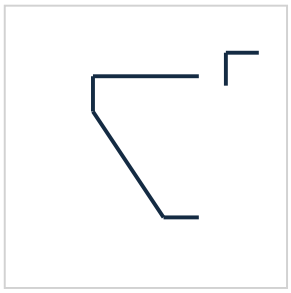
(A) Pattern U



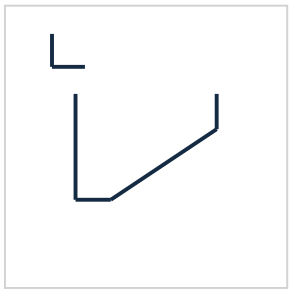
(B) Pattern X



(C) Pattern L



(D) Pattern V



80. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 6 and Z has code 31. Decode the separated values 24, 25, 20, 19, 10.

(A) STONE

(B) ENOTS

(C) TUPOF

(D) RSNMD

81. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 1 (B) 2

(C) 3 (D) 4

82. Nita points to a photograph and says, "This person's only daughter is my mother." Nita is female, and the photographed person is male. How is the photographed person related to Nita?

(A) Brother (B) Son

(C) Grandfather (D) Daughter

83. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 4, 8, 14, 28, ?, 68, 74.

(A) 42 (B) 68

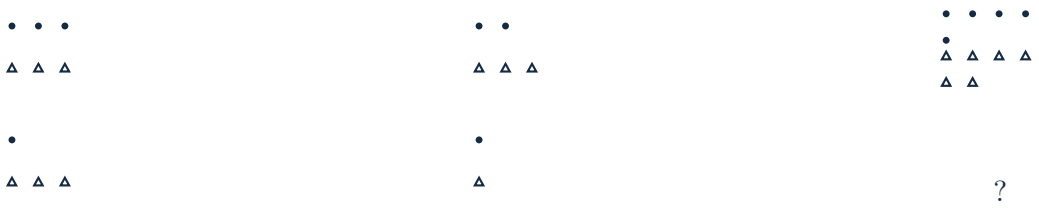
(C) 34 (D) 62

84. In a row, Asha is 6th from the left and 26th from the right. Bina is 6th from the right. Charu stands 4 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 3 (B) 15

(C) 14 (D) 17

85. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.

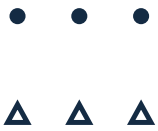


(A) Pattern N

(B) Pattern U



(C) Pattern Q

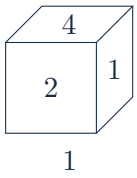


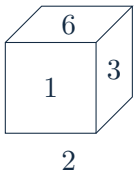
(D) Pattern K





86. The views show the same die. Give the ordered pair: (face opposite 2, face opposite 4). Do not assume opposite labels sum to seven.





(A) $(6, 2)$

(B) $(6, 3)$

(C) $(6, 1)$

(D) $(6, 4)$

87. Each letter position has its own fixed cyclic shift, applied identically in both pairs (Z wraps to A). Complete NGD : QEB :: IUI : ?.

(A) KRF

(B) MTH

(C) GSL

(D) LSG

88. The age table uses the same two people throughout:

	younger	older
now	x	$x+9$
7 years ago	$x-7$	$x+2$

. The product of their ages 7 years ago was 580. What will the older person's age be 5 years from now?

(A) 50 (B) 36

(C) 41 (D) 34

89. Five people occupy positions 1 to 5 from left to right, one per position, and wear five different shirt colours: red, blue, green, white and yellow. Charu is 1st from the left. Dev is 2nd from the left. Bina is 3rd from the left. Esha is 4th from the left. Asha is 5th from the left. The red shirt is in position 1. The white shirt is in position 3. The yellow shirt is in position 5. Esha does not wear green. Which shirt does Esha wear?

(A) Green (B) Red

(C) White (D) Blue

90. Three numbered triples satisfy both $y = x^2 + 4$ and $z - y = 2$. Which triple fails at least one condition? 1: (2,8,10), 2: (4,20,22), 3: (7,53,55), 4: (5,29,32).

(A) 2 (B) 1

(C) 3 (D) 4

Mathematics

91. There are 9 Boolean variables p_1,...,p 9. Exactly 1 variables must be true. How many truth assignments satisfy [(p_1 implies p_2) and (p_1 implies not p_2)]?

(A) 1 (B) 8

(C) 9 (D) 512

92. The line y=mx through O bisects the area of the triangle with vertices O,(2,5),(5,5). Assume the vertices are noncollinear. Find m.

(A) $\frac{10}{7}$ (B) 0

(C) $\frac{5}{2}$ (D) $\frac{7}{10}$

93. Independent events A,B have probabilities $\frac{5}{9}, \frac{5}{13}$. Given A∪B, find P(A∩B).

(A) $\frac{25}{117}$ (B) $\frac{12}{17}$

(C) $\frac{5}{22}$ (D) $\frac{5}{17}$

94. For x>0, xy′-y=x and y(1)=3. Find y(4)/4-y(1).

(A) $-\log(4)$ (B) $\log(4) + 3$

(C) $\log(4)$ (D) $\log(256) + 9$

95. A 7-element set contains distinct a,b. How many 3-element subsets contain exactly one of a,b and also contain a?

(A) 10 (B) 20

(C) 15 (D) 32

96. Let R(x) be the remainder when x^4 is divided by (x-1)(x-2). Find R(2).

(A) 16 (B) 17

(C) -13 (D) 30

97. Positive a,b satisfy log(a+b)=log(5)+[log a+log b]/2 in one valid base. Find $(a^4+b^4)/(a^2b^2)$.

(A) 529 (B) 527

(C) 531 (D) 23

98. The tangent to $y = x^3 - 6x$ at $x = 3$ meets the two coordinate axes at A and B. Find the area of triangle OAB, where O is the origin.

(A) 54 (B) $\frac{486}{7}$

(C) $\frac{243}{7}$ (D) $\frac{972}{7}$

99. $A = \begin{bmatrix} 1 & 6 \\ 0 & 0 \end{bmatrix}$. For $(I+A)^2=I+cA$, find c.

(A) 3 (B) 2

(C) 4 (D) 1

100. A line has direction (1,5,2). A plane has normal (2,1,5). Let θ be their acute angle. Find $\tan^2\theta$.

(A) $\frac{289}{611}$ (B) $\frac{289}{900}$

(C) $\frac{611}{900}$ (D) $\frac{900}{289}$

101. The chord of contact from P to $x^2/196+y^2/49=1$ is $x/14+y/7=1$. Find $|OP|^2$.

(A) 147 (B) 245

(C) $7\sqrt{5}$ (D) 21

102. Find the area bounded by y=x, y=49/x, and x=42 in the first quadrant.

(A) $245 - 49\log(6)$ (B) $\frac{1715}{2}$

(C) $49\log(6)$ (D) $\frac{1715}{2} - 49\log(6)$

103. How many x with $0 \le x < 5\pi$ satisfy $\sin^4x+\cos^4x=3/4$ and $\cos x>0$?

(A) 10 (B) 0

(C) 13 (D) 4

104. A=(2,3,0), C=(5,9,3). B lies on AC with AB:BC=2:1. Find $|OB|^2$, where O is the origin.

(A) 13 (B) 69

(C) 54 (D) 115

105. Evaluate the integral from 0 to 2π of $(8-5\sin x)(1+\cos x)/(64-25\sin^2 x)\,dx$.

(A) $\frac{\pi}{4}$ (B) $\frac{\sqrt{39}\pi}{39}$

(C) $\frac{2\sqrt{39}\pi}{39}$ (D) $\frac{2\sqrt{89}\pi}{89}$

106. Evaluate $\int_0^{(10\pi)} \sin^4 x\, dx$.

(A) $\frac{5\pi}{4}$ (B) 5π

(C) $\frac{15\pi}{4}$ (D) 0

107. An observer at height 6 m sees the tower foot at depression 45° and its top at elevation 60°. The observer descends vertically to ground level. If the new elevation is α, find $\tan\alpha$.

(A) $\sqrt{3}+2$ (B) $-1+\sqrt{3}$

(C) 1 (D) $1+\sqrt{3}$

108. An AP has a_1=5, d=8. Find $a_1+a_3+\cdots+a_9$.

(A) 270 (B) 225

(C) 185 (D) 116

109. An equivalence relation has classes of sizes 2,7,6. How many unordered pairs of distinct related elements are there?

(A) 105 (B) 1

(C) 74 (D) 37

110. Parabola y=6(x-4)(x-5) meets the axes at three distinct points. A circle passes through these points. Find its other y-intercept.

(A) 1 (B) $\frac{1}{6}$

(C) $-\frac{1}{6}$ (D) 6

111. Find the coefficient of x^7 in (1+6x)exp(2x).

(A) $\frac{176}{315}$ (B) $\frac{8}{45}$

(C) $\frac{8}{315}$ (D) $\frac{8}{15}$

112. Evaluate lim as x→0 of [sqrt(49+x)-7-x/14+x^2/2744]/x^3.

(A) $\frac{1}{268912}$ (B) 0

(C) $-\frac{1}{2744}$ (D) $\frac{1}{5488}$

113. A line through P=(6,4) has direction (3,4). It meets 3x+4y=134 at Q. Find $|PQ|^2$.

(A) 20 (B) 400

(C) 100 (D) 16

114. How many integers x in [-8,14] satisfy (|x-3|-6)/(|x-3|-10)>0?

(A) 13 (B) 11

(C) 23 (D) 15

115. Using digits 0,1,2,3,4,5 without repetition, how many 5-digit positive integers are divisible by 6 and have their first digit smaller than their last digit?

(A) 96 (B) 24

(C) 42 (D) 0

116. How many injective functions from {a,b,c} to a labelled 6-element set satisfy f(a)<f(b), when the codomain is naturally ordered?

(A) 30 (B) 60

(C) 120 (D) 20

117. Evaluate the product of $\cos(2^j\pi/33)$ for j=0,...,4.

(A) $\frac{1}{16}$ (B) $-\frac{1}{32}$

(C) $\frac{1}{33}$ (D) $\frac{1}{32}$

118. A tangent to $y = x^3 - 45x$ at a point with positive abscissa cuts the y-axis at -16. Find its slope.

(A) -33 (B) -45

(C) -39 (D) 33

119. The parabola $(y-4)^2=24(x-4)$ has focus F and vertex V. A circle centred at F passes through V. Find the tangent length from P=(22,16).

(A) 252 (B) $6\sqrt{7}$

(C) $12\sqrt{2}$ (D) $6\sqrt{3}$

120. $F(x) = \int_0^{x^2}(t-36)(t-64)dt$. How many local minima of F lie in [-8,8]?

(A) 1 (B) 2

(C) 5 (D) 3

121. f(x)=10x+8, g=f^{-1}. Find g(g(8)).

(A) 8 (B) 0

(C) $-\frac{2}{25}$ (D) $-\frac{4}{5}$

122. Let $D(t) = \det\begin{pmatrix} 10+t & 1+t \\ 1+t & 6+t \end{pmatrix}$. Find D′(0).

(A) 16 (B) 14

(C) 18 (D) 59

123. What is the maximum number of distinct intersection points formed by 4 circles and 3 straight lines, with no coincident curves?

(A) 36 (B) 27

(C) 39 (D) 51

124. The line r=(6,5,5)+t(1,2,2) meets x+y+z=41 at Q. Find $|PQ|^2$ for P=(6,5,5).

(A) 15 (B) 571

(C) 225 (D) 45

125. Evaluate $\int_0^\infty dx/[(x^2+16)(x^2+64)]$.

(A) $\frac{\pi}{64}$ (B) $-\frac{\pi}{768}$

(C) $\frac{\pi}{384}$ (D) $\frac{\pi}{768}$

126. $z_1=2+5i$, $z_2=6+5i$, $z_3=6+9i$. Find the smaller angle between z_2-z_1 and z_3-z_1.

(A) $\frac{3\pi}{4}$ (B) $\frac{\pi}{2}$

(C) 0 (D) $\frac{\pi}{4}$

127. An increasing positive GP has first three terms with product 1728 and first-plus-third sum 40. Find its fourth term.

(A) 36 (B) $\frac{4}{3}$

(C) 108 (D) 12

128. For $x\in[0,1]$, find the minimum of $7(\arcsin x)^3+(\arccos x)^3$.

(A) $-\frac{77\pi^3}{864}+\frac{35\sqrt{7}\pi^3}{864}$ (B) $\frac{7\pi^3}{8}$

(C) $\frac{\pi^3}{8}$ (D) $\frac{7\pi^3}{8\left(1+\sqrt{7}\right)^2}$

129. For $0\leq\theta<4\pi$, roots of $x^2-2x+(1-2\sin\theta)=0$ have opposite signs. How many connected intervals of θ satisfy this?

(A) 4 (B) 3

(C) 2 (D) 1

130. Evaluate $\sum_{j=0}^{3}\binom{3}{j}\binom{7}{2-j}$.

(A) 135 (B) 36

(C) 45 (D) 120

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A bridge divides a stretched string into two sections. Their fundamental frequencies are 300 Hz and 400 Hz under the same tension. Find the fundamental frequency of the full string if the bridge is removed without changing tension or total length.

(A) 300 Hz

(B) $\frac{1200}{7}$ Hz

(C) 1200 Hz

(D) 100 Hz

132. Two identical cells, each emf 4 V and internal resistance 2 ohms, are connected in parallel with like terminals joined and drive a load of 2 ohms. Find the current delivered by each cell.

(A) 2 A

(B) 1 A

(C) $\frac{2}{3}$ A

(D) $\frac{1}{2}$ A

133. A nucleus initially at rest with mass number 100 alpha-decays with total energy release 5 MeV. Taking alpha and daughter masses proportional to 4 and 96, find daughter recoil kinetic energy.

(A) $\frac{1}{5}$ MeV

(B) $\frac{5}{2}$ MeV

(C) 5 MeV

(D) $\frac{1}{125}$ MeV

Chemistry

134. A hydrocarbon has empirical formula CH2 and independently measured molar mass 70 g/mol. Use C=12,H=1. Which molecular formula and interpretation are consistent?

(A) C5H5; hydrogen subscripts do not scale.

(B) C5H10; molecular formula is 5 times the empirical formula unit.

(C) CH2; molar mass cannot distinguish an empirical formula from a molecular formula.

(D) C10H5; empirical and molecular ratios need not agree.

135. Use octet Lewis contributors for NO3- and HNO3; for SO2 exclude expanded-octet contributors. For HCOO-, HCO3- and HNO3, average only the two nonprotonated central-atom-O bonds. For other species average all central-atom bonds. Find (average bond order of the first minus that of the second, sum of total lone-pair counts in one contributor of each). A lone pair on any atom is counted once.

(A) (-1/6, 15)

(B) (-1/6, 14)

(C) (17/6, 15)

(D) (-1/6, 16)

136. A closed fullerene cage has 60carbon atoms. Every carbon has three neighbours,and every face is a pentagon or hexagon. The cage contains12pentagons. How many hexagons are present?

(A) 30

(B) 12

(C) 20

(D) 48

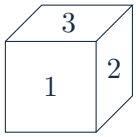
Logical Reasoning

137. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: DRK, AUH, ?, UAB, RDY, OGV.

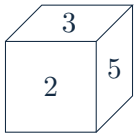
(A) EXX (B) XXE

(C) YYF (D) WWD

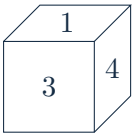
138. The views show the same die. Give the ordered pair: (face opposite 1, face opposite 6). Do not assume opposite labels sum to seven.



1



2



3

(A) $(5, 3)$ (B) $(5, 4)$

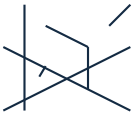
(C) $(5, 1)$ (D) $(5, 2)$

139. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 4 (B) 1

(C) 2 (D) 3

Mathematics

140. How many integers x satisfy $4^{-x} - (72)2^{-x} + 512 \leq 0$?

(A) 4 (B) 3

(C) 2 (D) 7

141. For f(x)=|x-6|/(x-8), on which maximal open intervals is f strictly decreasing?

(A) $(-\infty, 8)$ (B) $(6, 8) \cup (8, \infty)$

(C) $(6, \infty)$ (D) $(-\infty, 6)$

142. A circular sector has radius 3 and arc length $\frac{\pi}{2}$. Find its area.

(A) $\frac{\pi^2}{24}$
(B) $\frac{3\pi}{2}$
(C) $\frac{3\pi}{4}$
(D) 9π

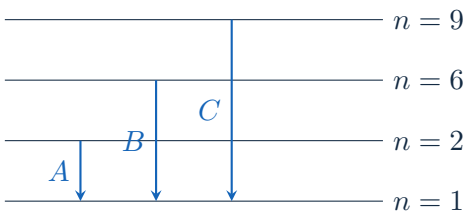
BITSAT Mock 35

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. The diagram shows downward hydrogen transitions A, B and C from $n = 2$, 6 and 9, respectively, ending at $n = 1$. A filter transmits only photons with energy strictly above that of transition B. Which transmitted line has the longest wavelength?



(A) No transmitted line
(B) B
(C) A
(D) C

2. A Carnot engine initially runs between 700 K and 300 K. Its hot reservoir is raised by 40 K. By how much must the cold-reservoir temperature rise to keep efficiency unchanged?

(A) $\frac{160}{7}$ K
(B) 40 K
(C) $\frac{280}{3}$ K
(D) $\frac{120}{7}$ K

3. A vacuum electromagnetic wave of average intensity 120 W/m^2 passes through a device that reduces its electric-field amplitude by a factor 4. The transmitted wave falls normally on an area 2 cm^2. What average power reaches the area?

(A) $\frac{3}{500}$ W
(B) $\frac{3}{2000}$ W
(C) $\frac{3}{125}$ W
(D) $\frac{15}{2}$ W

4. Light of vacuum wavelength 300 nm passes from a medium of index 5 into one of index 6. Find the ratio of its frequencies in the second and first media.

(A) 6
(B) 1
(C) $\frac{36}{25}$
(D) $\frac{25}{36}$

5. A solid sphere of radius R has uniform density $8\rho_0$. A spherical cavity of radius $R/10$ is cut with its centre at distance $R/7$ from the original centre. Find field magnitude at the cavity centre in units of $\pi G\rho_0 R$.

(A) $\frac{28}{3}$ (B) $\frac{1}{6}$

(C) $\frac{7}{48}$ (D) $\frac{32}{21}$

6. A 4 kg block is pulled by a force of 20 N at an angle with $\cos\theta = 4/5$, $\sin\theta = 3/5$ above a horizontal floor. The block slides; $\mu_k = \frac{3}{10}$. Find the net horizontal force. Take $g = 10$ m/s^2.

(A) $\frac{122}{5}$ N (B) $\frac{38}{5}$ N

(C) 4 N (D) $\frac{24}{5}$ N

7. 125 identical conducting liquid drops merge without losing charge. Before merging each has potential V and surface charge density σ. Find the final surface charge density divided by σ.

(A) 5 (B) 625

(C) 1 (D) $\frac{1}{25}$

8. Current rises linearly from 0 to 8 A over 2 s and then falls linearly to 0 over another 2 s. It remains zero for the next 4 s. Find the average current over the entire 8 s interval.

(A) 4 A (B) $\frac{8}{3}$ A

(C) 8 A (D) 2 A

9. A network is defined by $P = \mathrm{NOR}(A, B)$, $Q = \mathrm{NAND}(A, P)$ and $Y = \mathrm{NOR}(Q, B)$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 1$ and $B = 0$?

(A) 2 (B) 3

(C) 1 (D) 0

10. A long solid cylindrical conductor of radius 3 m carries uniformly distributed current 2 A. Find magnetic field magnitude at radial distance $R/6$ from the axis in units of $\mu_0/(2\pi)$.

(A) $\frac{1}{27}$ (B) $\frac{1}{54}$

(C) $\frac{1}{9}$ (D) $\frac{2}{3}$

11. A pendulum bob of mass 1 kg and length 5 m passes the horizontal position moving downward at 2 m/s. Find the instantaneous power of all forces when $\sin\theta = 3/5$, $\cos\theta = 4/5$, with θ measured from the downward vertical. Take $g = 10$ m/s^2.

(A) 48 W (B) $16\sqrt{21}$ W

(C) $12\sqrt{21}$ W (D) $20\sqrt{21}$ W

12. A boat moves at 25 m/s relative to water. The current is 15 m/s and river width is 84 m. The boat lands directly opposite its starting point. Find crossing time.

(A) $\frac{21}{5}$ s (B) $\frac{28}{5}$ s

(C) $\frac{42\sqrt{34}}{85}$ s (D) $\frac{84}{25}$ s

13. A projectile of mass 9 kg is launched with speed 35 m/s, $\cos\theta = 3/5$. At its highest point it splits into masses $M/3$ and $2M/3$. The lighter fragment moves horizontally forward at 26 m/s. Find the signed horizontal velocity of the heavier fragment immediately after the explosion.

(A) $\frac{79}{2}$ m/s (B) $\frac{37}{2}$ m/s

(C) $\frac{89}{2}$ m/s (D) 37 m/s

14. A pure radioactive sample has activity A at time 12 hours after preparation and $A/8$ at time 30 hours after preparation. At what time, measured from preparation, will its activity be one sixteenth of the activity measured at 12 hours? Assume exponential decay with a constant decay rate.

(A) 24 h (B) 36 h

(C) 54 h (D) 30 h

15. Masses 2 kg and 4 kg hang from a light inextensible string over a pulley with moment of inertia I and radius R such that $I/R^2 = 5$ kg. There is no slipping or axle friction. Find acceleration magnitude. Take $g = 10$ m/s^2.

(A) $\frac{20}{11}$ m/s^2 (B) 4 m/s^2

(C) 10 m/s^2 (D) $\frac{10}{3}$ m/s^2

16. A magnetic field is uniform inside a long solenoid of radius 6 m and negligible outside. It changes at 3 T/s. Find induced electric-field magnitude at radius $2R$ from the axis.

(A) $\frac{9}{2}$ V/m (B) 36 V/m

(C) $\frac{9}{4}$ V/m (D) 9 V/m

17. Two large parallel plates are held at potentials +300 V and −300 V, with gap 6 mm. Neglect edge effects. Find field magnitude in the gap.

(A) 100000 V/m (B) 10000 V/m

(C) 50000 V/m (D) 100 V/m

18. A particle of mass 5 kg starts from rest and experiences $F(t) = 3t$ N along its motion for 2 s. Find the work done by the force.

(A) $\frac{36}{5}$ J (B) $\frac{18}{5}$ J

(C) 6 J (D) $\frac{72}{5}$ J

19. A coil has 70 turns, area $\frac{1}{100}$ m^2, and rotates at 5 rad/s in a uniform 3 T field. At an instant the angle between its area vector and the field is 30°. Find instantaneous-emf magnitude.

(A) $\frac{21\sqrt{3}}{4}$ V (B) $\frac{21}{2}$ V

(C) $\frac{21}{4}$ V (D) $\frac{21}{20}$ V

20. A soap bubble initially of radius 5 cm is expanded quasistatically until its volume is 27 times the initial volume. Surface tension remains $\frac{1}{25}$ N/m. Find work against surface tension, in units of $\pi \times 10^{-4}$ J.

(A) 32 (B) 64

(C) 16 (D) 208

21. Two identical uniform spheres touch. Their radii are multiplied by 5 while their density is divided by 5. By what factor does their gravitational attraction change?

(A) 25 (B) $\frac{1}{25}$

(C) $\frac{1}{625}$ (D) 125

22. A spherical planet rotates uniformly. Its rotation reduces effective gravity at the equator by 8 $\mathrm{m/s^2}$. By how much does rotation reduce the radial component of effective gravity at latitude 60°?

(A) $\frac{1}{2}\,\mathrm{m/s^2}$ (B) $2\,\mathrm{m/s^2}$

(C) $8\,\mathrm{m/s^2}$ (D) $4\,\mathrm{m/s^2}$

23. A dipole of fixed moment 8 C m is rotated quasistatically from 60° to 120° relative to a uniform electric field 5 N/C. Find external work.

(A) 40 J (B) 20 J

(C) $20\sqrt{3}\,\mathrm{J}$ (D) $\frac{40\pi}{3}\,\mathrm{J}$

24. A solid cylinder of mass 5 kg rolls without slipping on a horizontal rough surface. A horizontal force 20 N is applied tangentially at its top. Find its centre-of-mass acceleration; $I = MR^2/2$.

(A) $16\,\mathrm{m/s^2}$ (B) $2\,\mathrm{m/s^2}$

(C) $4\,\mathrm{m/s^2}$ (D) $\frac{16}{3}\,\mathrm{m/s^2}$

25. A sinusoidal source of peak $\frac{107}{10}$ V feeds a back-to-back series pair of identical Zeners through a 1 kΩ current-limiting resistor. The output is across the diode pair. Each Zener has reverse breakdown 5 V and forward drop 0.7 V. What is the peak-to-peak output voltage? Use the piecewise constant-drop model.

(A) $\frac{7}{5}$ V (B) $\frac{107}{5}$ V

(C) $\frac{57}{5}$ V (D) 10 V

26. A liquid of volume 2 litres and bulk modulus 5000000000 Pa is subjected to an additional pressure of 3000000 Pa. Neglect changes in bulk modulus. Find the magnitude of volume decrease in cubic centimetres.

(A) $\frac{3}{2500}\,\mathrm{cm^3}$ (B) $1200\,\mathrm{cm^3}$

(C) $\frac{6}{5}\,\mathrm{cm^3}$ (D) $\frac{10000000}{3}\,\mathrm{cm^3}$

27. A particle has position-dependent velocity components $\vec{v} = 4x\,\hat{i} - 4y\,\hat{j}$ in SI units. Given $dx/dt = v_x$ and $dy/dt = v_y$, what is its acceleration magnitude at $(2, 5)$ m?

(A) $16\sqrt{29}\,\mathrm{m/s^2}$ (B) $112\,\mathrm{m/s^2}$

(C) $64\sqrt{29}\,\mathrm{m/s^2}$ (D) $4\sqrt{29}\,\mathrm{m/s^2}$

28. A circular parallel-plate capacitor of plate radius 5 cm is charged at 42 mA. Neglect fringing. What displacement current passes through the annular part of the gap between radii 1 cm and 5 cm, centred on the plate axis?

(A) $\frac{42}{25}\,\mathrm{mA}$ (B) $1008\,\mathrm{mA}$

(C) $42\,\mathrm{mA}$ (D) $\frac{1008}{25}\,\mathrm{mA}$

29. A 3 kg block moving at 5 m/s compresses a spring of constant 100 N/m. During compression, a damper removes 1/3 of the block's initial kinetic energy. Find maximum compression; gravity does no work.

(A) $\frac{\sqrt{3}}{2}$ m (B) $\frac{1}{2}$ m

(C) $\frac{\sqrt{2}}{2}$ m (D) 1 m

30. 5 moles of ideal gas with $\gamma = 7/5$ warm by 70 K in a rigid vessel. Find heat supplied in units of R.

(A) 1225 (B) 350

(C) 875 (D) 140

Chemistry

31. A dipeptideAla-Gly has one free alpha-amino terminus and one free alpha-carboxyl terminus, with uncharged side chains. At a statedpH, the terminus charges are+1and-1. Which count is correct?

(A) One terminal group; net charge-2. (B) Two terminal ionic groups; net terminal chargezero.

(C) No ionic groups; net chargezero. (D) Two peptide bonds; net charge+2.

32. A $\frac{29}{250}$ g volatile sample displaces $\frac{1792}{39}$ mL wet air at 280 K and 780 mmHg. Water-vapour pressure is 20 mmHg. Using STP=273 K,760 mmHg and22.4 L/mol,the sample molar mass, rounded to the nearest whole g/mol, is

(A) 60 (B) 115

(C) 57 (D) 58

33. Atoms form ideal touching FCC spheres of radius r and cell edge a. Fractional coordinates are measured in a. Using the straight displacement between the two specified points (0,0,0) and (1,0,0), without replacing it by another periodic image, find (distance/r)^2. These points describe two corner atom centres on an edge.

(A) 16 (B) 9

(C) 4 (D) 8

34. The ions K^+, Ca^{2+}, Sc^{3+} each have 18 electrons. Which decreasing-radius order and explanation is consistent?

(A) $K^+ > Ca^{2+} > Sc^{3+}$; increasing nuclear charge contracts the same electron cloud.

(B) $K^+ < Ca^{2+} < Sc^{3+}$; larger nuclear charge expands the cloud.

(C) All radii are equal because electron numbers are equal.

(D) $Ca^{2+} > Sc^{3+} > K^+$; positive charge always gives the largest ion.

35. methyl4-methylbenzoate is completely hydrolysed,acidified,then treated successively with SOCl2, excess NH3,and Br2/NaOH. Which amine is obtained,assuming ring substituents survive?

(A) aniline (B) 4-aminobenzoic acid

(C) 4-methylbenzylamine (D) 4-methylaniline

36. Use octet Lewis contributors for NO3- and SO2; for SO2 exclude expanded-octet contributors. For HCOO-, HCO3- and HNO3, average only the two nonprotonated central-atom-O bonds. For other species average all central-atom bonds. Find (average bond order of the first minus that of the second, sum of total lone-pair counts in one contributor of each). A lone pair on any atom is counted once.

(A) (-1/6, 13)
(B) (-1/6, 15)
(C) (-1/6, 14)
(D) (17/6, 14)

37. For oxide formation, both normalised per mole of O_2, $\Delta G^\circ_A = -325+0.10T$ and $\Delta G^\circ_B = -410+0.20T$ in kJ. Ignoring kinetic limitations, at what temperatures is B metal a favourable standard-state reducing agent for oxide A?

(A) At all positive temperatures
(B) T < 850 K
(C) Only T = 850 K
(D) T > 850 K

38. A thermal assembly contains Cu(s) and He(g) in separate inert compartments at the same temperature. Helium is heated quasistatically at constant pressure under a freely moving piston. Containers have negligible heat capacity. Use the classical Dulong-Petit solid model for Cu, neglecting its Cp-Cv difference, and the ideal monatomic model for He. No phase change occurs. Molar masses are Cu64 and He4 g/mol. Over a small common temperature rise, the measured total heat capacity C divided by total Cu-plus-He mass m and gas constant R is C/(mR)=135/512 mol/g. What is the Cu mass fraction?

(A) 3/8
(B) 19/56
(C) 5/8
(D) 185/308

39. A primary-standard sample of pure anhydrous Na_2CO_3 weighs $\frac{53}{125}$ g. It is titrated with0.100M HCl. Find the additional acid volume in mL needed after the first carbonate-to-hydrogencarbonate equivalence point to reach complete conversion to CO_2/water. Take Na_2CO_3=106g/mol.

(A) 20
(B) 40
(C) 80
(D) 81

40. Which occurrence/extraction inference for $MgCl_2$ is chemically consistent?

(A) The formula proves that oxygen is the only possible counter-ion in its natural deposits.

(B) It occurs in seawater/brine; magnesium can be recovered from dissolved salts.

(C) The metal occurs chiefly as native metal because group1/2metals are exceptionally unreactive.

(D) The metal is obtained by ordinary water electrolysis without preferential hydrogen evolution.

41. For benzyl chloride and chlorotoluene,which structural classification is correct when the actual halogen-bearing carbon and molecular formula are examined?

(A) the same compound under two names
(B) enantiomers
(C) constitutional isomers with benzylic versus aryl C-Cl
(D) conformers

42. Two one-electron ions A and B satisfy Z_B/Z_A=4 and |E_B|/|E_A|=256/9 for their selected Bohr levels. In the ideal hydrogenic model $E \propto -Z^2/n^2$ and $r \propto n^2/Z$. What is r_B/r_A?

(A) 9/64
(B) -9/64
(C) 9/128
(D) 9/32

43. Caprolactam undergoes ring-opening polymerisation. Which backbone and named product result?

(A) -CH2-CCl2-; cellulose.

(B) -NH-(CH2)5-CO-; nylon-6.

(C) -O-CH2-CH2-O-CO-; Buna-S.

(D) -CH2-CH(phenyl)-; PHBV.

44. In an antifluorite-derived crystal, X2- occupies all FCC anion sites. A measured fraction 3/5 of occupied M cations is M2+ and the rest M+. Only tetrahedral cation vacancies compensate charge; no interstitials or anion defects occur. What fraction of all tetrahedral sites is vacant?

(A) 3/10

(B) 3/8

(C) 5/8

(D) 3/16

45. A Cr(3) coordination entity has coordination number 6, exactly 2 fully bound oxalate ligand(s), and chloride in all remaining donor sites. Take oxalate as 2-dentate with charge -2; chloride is monodentate. Find (number of ligand entities, net charge of the coordination entity).

(A) (5, -3)

(B) (4, -3)

(C) (6, -2)

(D) (4, -4)

46. Compare the elementary octahedral d-d splitting transitions of A=$[FeF_6]^{3-}$,B=$[Fe(H_2O)_6]^{3+}$, C=$[Fe(CN)_6]^{3-}$. For this comparison use only the usual spectrochemical ordering and $E = hc/\lambda$. Which ordering of absorbed wavelengths follows?

(A) $\lambda_A > \lambda_B > \lambda_C$.

(B) $\lambda_A = \lambda_B = \lambda_C$ because metal oxidation states match.

(C) $\lambda_B > \lambda_C > \lambda_A$.

(D) $\lambda_A < \lambda_B < \lambda_C$.

47. An internally consistent thermochemical comparison gives total resonance stabilisation of naphthalene as 61 kcal/mol (2 benzenoid rings) and benzene as 36 kcal/mol (1 rings). Which statement correctly separates total stabilisation from stabilisation per ring?

(A) Total: naphthalene; per ring: benzene.

(B) Both quantities must rank compounds identically because each is aromatic.

(C) Total: naphthalene; per ring: naphthalene.

(D) Total: benzene; per ring: benzene.

48. I:Acidic hydrolysis of CH2=CH-OCH3 gives ethanal and methanol. II:The alkene carbon count means propanone must form. Select the correct assessment.

(A) I true;II false.

(B) Both false.

(C) Both true.

(D) I false;II true.

49. A solvent has Tb=400 K, M=100 g/mol and ΔHvap=32000 J/mol. Use R=8.00 and Kb=R Tb^2M/(1000ΔHvap). An A2B solution of analytical molality 1/25 mol/kg boils 44/125 K above pure solvent. Only dissociation A2B→2A+ +B2- occurs, with ideal dilute-particle counting. What fraction of A2B is dissociated?

(A) 1/5

(B) 3/10

(C) 3/5

(D) 4/5

50. Evaluate independently. I: ClF_3 has three equivalent F positions in a trigonal plane. II: In SF_4,axial and equatorial S-F bonds must be identical by symmetry.

(A) Neither I nor II is correct.

(B) Only I is correct.

(C) Only II is correct.

(D) Both I and II are correct.

51. At fixed temperature a single adsorbent obeys x/m=k $p^{(1/n)}$. In experiment B the adsorbent mass is 5 times that in A and pressure is 8 times that in A. Total adsorbed amount is 20 times that in A. What is 1/n?

(A) 5/3 (B) 3/2

(C) 1/3 (D) 2/3

52. Glucose is fully acetylated at its five OH groups. Its cyclic pentaacetate does not form an oxime with hydroxylamine under the standard test. Which conclusion is appropriate?

(A) Hydroxylamine detects only carbon-carbon single bonds.

(B) The product is necessarily an open-chain dialdehyde.

(C) Acetylation proves the original glucose contained noOH.

(D) The anomeric centre is locked as an acetal derivative, so no free CHO is supplied by ordinary ring opening.

53. Equal masses of nonvolatile A (M=180) and B (M=60) are separately dissolved in equal solvent masses. A remains molecular, while B only dimerises. The measured freezing-depression ratio ΔTf(B)/ ΔTf(A) is 12/5. What fraction of the original B formula units is present in dimers? All solutions obey ideal particle counting.

(A) -2/5 (B) 1/5

(C) 4/5 (D) 2/5

54. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_y; same facing signs; 2: parallel p_x; opposite facing signs. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (1, 1) (B) (1, 2)

(C) (0, 3) (D) (2, 1)

55. Count all distinct stereoisomers,including separately resolved enantiomers,for square-planar [Mabcd] and octahedral $[Ma_3b_3]$; a,b,c,d are distinct monodentate ligands and en is symmetric bidentate ethylene-diamine. What is the sum?

(A) 4 (B) 7

(C) 5 (D) 6

56. At one T,P,a binary reaction R⇌P has a Gibbs-energy-versus-product-fraction curve. Pure R has G=20 kJ,pure P50 kJ,and the unique minimum G=10 kJ occurs at product fraction $\frac{2}{5}$. Which inference follows from both endpoints and the interior minimum?

(A) The pure-product minus pure-reactant standard change is positive,yet the equilibrium contains product.

(B) G must decrease for a displacement from the minimum toward either pure endpoint.

(C) Positive pure-state change forces the equilibrium to be pure R.

(D) The minimum at an interior composition means both pure endpoints have equal G.

57. A Hinsberg separation is applied to propylamine. Which reagent identity and class-dependent outcome are correct?

(A) benzenesulphonic acid,C6H5SO3H;all classes give the same neutral product

(B) benzoyl chloride,C6H5COCl;a sulphonamide with N-H,soluble in alkali as its anion

(C) benzenesulphonyl chloride,C6H5SO2Cl;every amine gives an alkali-soluble N-H sulphonamide

(D) benzenesulphonyl chloride,C6H5SO2Cl;a sulphonamide with N-H,soluble in alkali as its anion

58. A 10 g $CaCO_3$/inert mixture is80.0% $CaCO_3$ by mass. It receives $\frac{11}{50}$ mol HCl. For $CaCO_3+2HCl \rightarrow CaCl_2+CO_2+H_2O$,90.0% of formed CO_2 is collected. Collected CO_2 amount in mol is

(A) $\frac{11}{100}$ (B) $\frac{2}{25}$

(C) $\frac{9}{125}$ (D) $\frac{11}{50}$

59. A sample contains $\frac{3}{50}$ mol $KClO_3$ and a small amount of MnO_2 catalyst.75.0% of the chlorate decomposes by $2KClO_3 \rightarrow 2KCl + 3O_2$. The catalyst is recovered unchanged. Find O_2 formed in mol.

(A) $\frac{27}{200}$ (B) $\frac{9}{100}$

(C) $\frac{27}{400}$ (D) $\frac{9}{200}$

60. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_y; same facing signs; 2: p_y on one centre and p_z on the other. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (2, 0) (B) (1, 0)

(C) (1, 1) (D) (0, 2)

English Proficiency

61. Read the sentence: A single verified counterexample was enough to refute the claim that every sample behaved alike. Choose the closest meaning of 'refute' as used in the sentence.

(A) question (B) dismiss

(C) qualify (D) disprove

62. Choose the pair with the same ordered relationship as SEED : PLANT.

(A) EGG : BIRD (B) NEST : TREE

(C) ROOT : SOIL (D) LEAF : BRANCH

63. In the described cricket match, he is the neutral official who rules on play, rather than a player or commentator. What is he?

(A) commentator (B) captain

(C) spectator (D) umpire

64. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The researchers will have completed the survey before the visitors return.

(A) The survey will have been completed by the researchers before the visitors return.

(B) The survey has been completed by the researchers before the visitors return.

(C) The survey will be completing the researchers before the visitors return.

(D) The survey will have been completed after the visitors return.

65. Complete the question tag: Nothing in these two reports contradicts the result, _____?

(A) do they (B) does it

(C) doesn't it (D) is it

66. Which figure of speech is chiefly illustrated by the line? The rumour spread through the town like ink through wet paper.

(A) apostrophe (B) personification

(C) simile (D) metaphor

67. Read the sentence: The stringent conditions allowed no exception to the safety requirement. Choose the opposite in meaning of 'stringent' as used in the sentence.

(A) demanding (B) lenient

(C) binding (D) precise

68. Use the simple present to express the stated regular procedure. Choose the best replacement: At the start of every session, the instructor [had repeated] the safety instruction.

(A) were repeating (B) has been repeated

(C) repeats (D) had been repeated

69. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. The charity placed a collection box near the entrance. B. Donors could then make a suitable choice before leaving an unusable object behind. C. It received many objects that the charity had no facilities to repair. D. A list of accepted items was displayed beside the box.

(A) The charity placed a collection box near the entrance.

(B) Donors could then make a suitable choice before leaving an unusable object behind.

(C) It received many objects that the charity had no facilities to repair.

(D) A list of accepted items was displayed beside the box.

70. Choose the most coherent order of the four sentences. A. Their final record preserved the memories while marking which dates remained uncertain. B. Those conflicting year assignments prompted the students to compare the accounts with dated photographs and timetables. C. The students recorded interviews about the history of the old station. D. Different speakers sometimes assigned the same event to different years.

(A) C-B-D-A (B) C-D-B-A

(C) D-C-B-A (D) C-D-A-B

Logical Reasoning

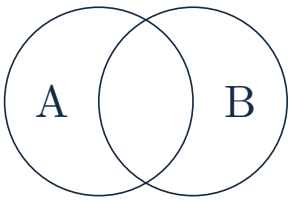
71. Nita points to a photograph and says, "This person is my mother's only son." Nita is female, and the photographed person is male. How is the photographed person related to Nita?

(A) Sister (B) Brother

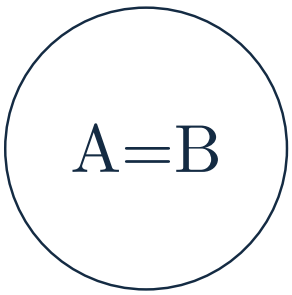
(C) Daughter (D) Son

72. Choose the diagram that represents all these conditions, including the strictness information: Both A and B are nonempty, and no A is B.

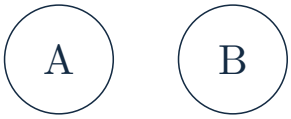
(A) Pattern X



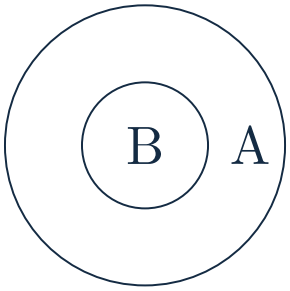
(B) Pattern R



(C) Pattern N



(D) Pattern V

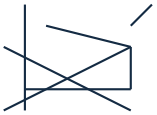


73. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

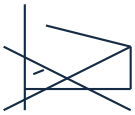




host 1



host 2



host 3



host 4

(A) 1
(B) 3
(C) 4
(D) 2

74. Preserve the exact relation and its order: Star : Constellation :: ? : Fleet. Choose the missing term.

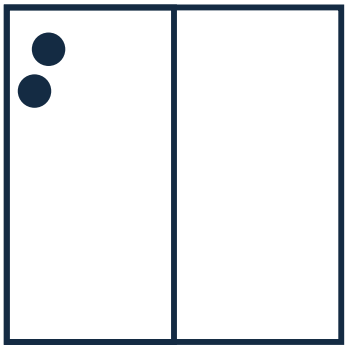
(A) Bee
(B) Lion
(C) Wolf
(D) Ship

75. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 13$ and $5 \mapsto 22$. Which input maps to 49?

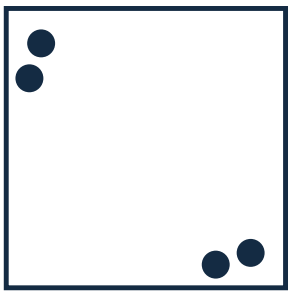
(A) 14 (B) 49

(C) 13 (D) 15

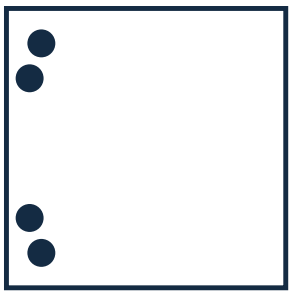
76. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



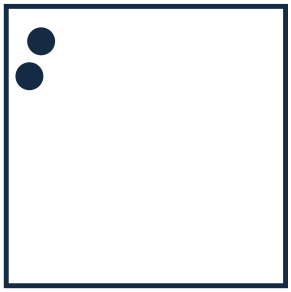
(A) Pattern K



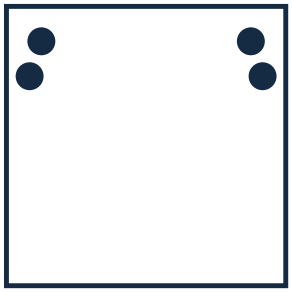
(B) Pattern Q



(C) Pattern Z



(D) Pattern V



77. Consider only these statements; universal statements alone do not imply existence. No artists are designers; All editors are designers. Conclusions: I. No artists are editors. II. All artists are editors. Which conclusions follow necessarily?

(A) Only II (B) Neither I nor II

(C) Both I and II (D) Only I

78. In a row, Asha is 6th from the left and 33rd from the right. Bina is 10th from the right. Charu stands 2 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 24 (B) 22

(C) 19 (D) 20

79. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 3, 6, ?, 22, 27, 54, 59.

(A) 11 (B) 7

(C) 22 (D) 12

80. The age table uses the same two people throughout:

	younger	older
now	x	$x+11$
3 years ago	$x-3$	$x+8$

. The product of their ages 3 years ago was 570. What will the older person's age be 4 years from now?

(A) 37 (B) 48

(C) 34 (D) 33

81. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: LGI, OJG, RME, ?, XSA, AVY.

(A) CPU (B) UPC

(C) TOB (D) VQD

82. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?

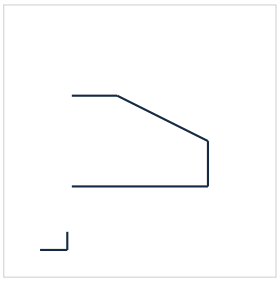


1 2 3 4

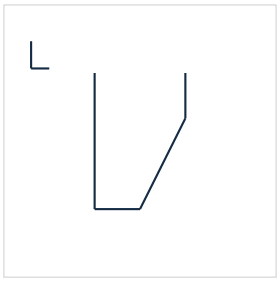
(A) 3 (B) 4

(C) 2 (D) 1

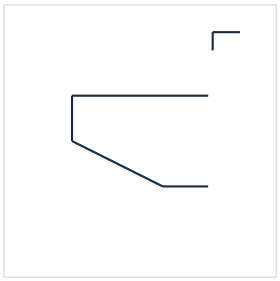
83. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

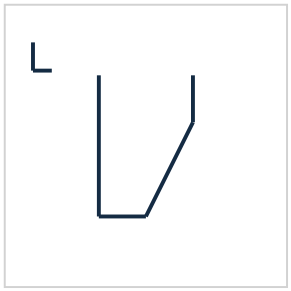


2



3

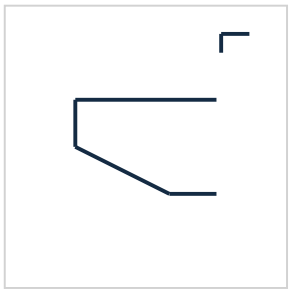
(A) Pattern K



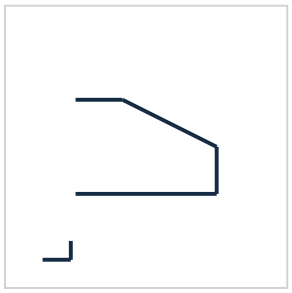
(B) Pattern P



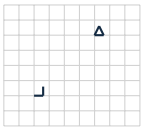
(C) Pattern L



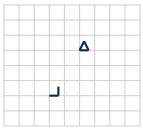
(D) Pattern N



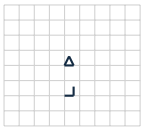
84. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



1

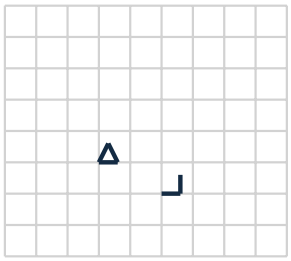


2

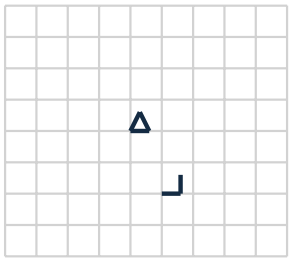


3

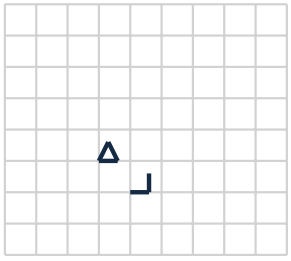
(A) Pattern W



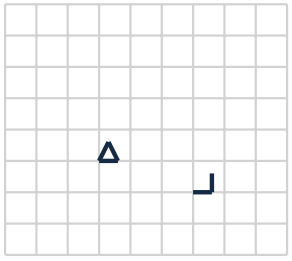
(B) Pattern Y



(C) Pattern V



(D) Pattern Z



85. Consider only these statements; universal statements alone do not imply existence. No inventors are farmers; All farmers are joggers; Some joggers are not farmers. Conclusions: I. All inventors are farmers. II. Some farmers are not joggers. Which conclusions are individually possible without contradicting the statements?

(A) Both I and II

(B) Only II

(C) Only I

(D) Neither I nor II

86. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). The numerical component follows a constant arithmetic step. Find the missing internal term: BFPL9, YHQI11, VJRF13, ?, PNTZ17, MPUW19.

(A) 51CSLS

(B) TMTD15

(C) SLSC15

(D) RKRB15

87. A walker goes 9 m west, then 8 m south, then turns left and walks 7 m. In which direction is the starting point from the walker?

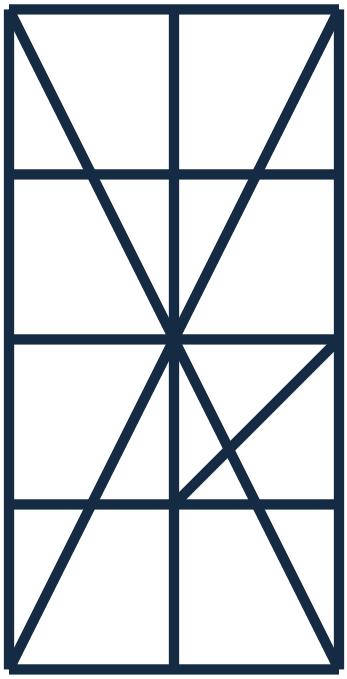
(A) South-west

(B) North-east

(C) North-west

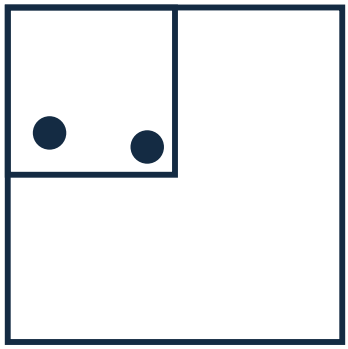
(D) South-east

88. How many triangles are formed by all the drawn segments?

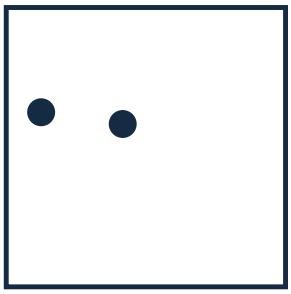


(A) 36
(B) 26
(C) 35
(D) 32

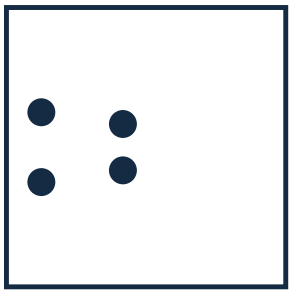
89. Starting with a unit square sheet, fold right to left along the vertical midline, then fold bottom to top along the horizontal midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



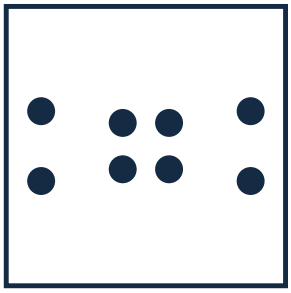
(A) Pattern V



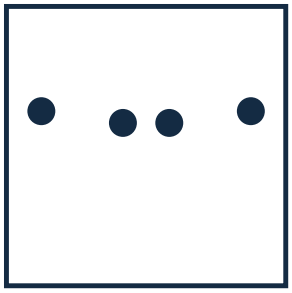
(B) Pattern N



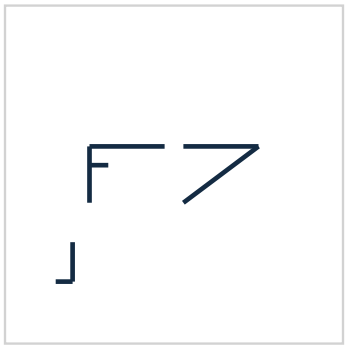
(C) Pattern M



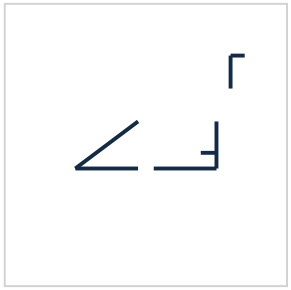
(D) Pattern W



90. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



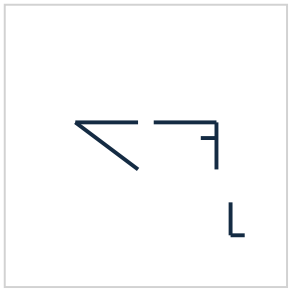
(A) Pattern N



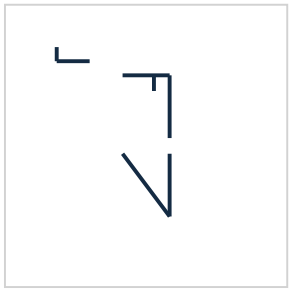
(B) Pattern Q



(C) Pattern L



(D) Pattern Z



Mathematics

91. The line $3x+4y=35$ cuts $x^2+y^2=121$ in chord AB. Find $|AB|^2$.

(A) 225 (B) 72

(C) 288 (D) 680

92. $P(A)=\frac{4}{7}$, $P(B)=\frac{1}{7}$, $P(A\cap B)=\frac{1}{14}$. Find P(neither A nor B).

(A) $\frac{2}{7}$ (B) $\frac{1}{14}$

(C) $\frac{18}{49}$ (D) $\frac{5}{14}$

93. Four observations are 7,7,x,y with population mean 7 and variance 9/2. Find xy.

(A) 13 (B) 31

(C) 89 (D) 40

94. Disjoint A,B have sizes 10,6. Find |(A×A)∪(B×B)|.

(A) 136 (B) 256

(C) 120 (D) 60

95. Evaluate lim as x→0 of $[(1+6x)/(1-5x)]^{(1/x)}$.

(A) 11 (B) e^{11}

(C) e^{6} (D) e

96. Let f(x)=|x-2|/(x-6) for x≠6. Within 0<x<10, what is the total length of the open intervals on which f′(x)<0?

(A) 6 (B) 9

(C) 8 (D) 2

97. An AP has a=9,d=2; its first n terms sum to 128. Find a_(n+1).

(A) 23 (B) 8

(C) 16 (D) 25

98. Let $\delta=\pi/18$. Evaluate $\sin\delta\ \Sigma[1/(\sin(j\delta)\sin((j+1)\delta))]$, j=2,...,8.

(A) $\cot\left(\frac{\pi}{9}\right)$ (B) $-\cot\left(\frac{\pi}{9}\right)$

(C) $\cot\left(\frac{\pi}{18}\right)$ (D) $\tan\left(\frac{\pi}{9}\right)$

99. 10 distinct people stand in a line. Three specified people may occupy neither end position and no two of these three may be adjacent. How many orders are possible?

(A) 604800 (B) 241920

(C) 100800 (D) 1693440

100. $f(x+1)=2x^2$. Find the area enclosed between y=f(x) and y=2(x-1).

(A) $-\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{1}{3}$ (D) 1

101. If α,β are roots of $x^2-2kx+k^2-4=0$ and k∈[-3,3], find the maximum of $\alpha^2+\beta^2$.

(A) 17 (B) 26

(C) 8 (D) 28

102. The hyperbola $x^2/25-y^2/9=1$ has eccentricity e. Find the eccentricity e′ of its conjugate hyperbola.

(A) $\frac{5\sqrt{34}}{34}$ (B) $\sqrt{34}$

(C) $\frac{\sqrt{34}}{5}$ (D) $\frac{\sqrt{34}}{3}$

103. If α,β are roots of $x^2-16x+63=0$, find $\alpha^2+\beta^2$ without solving the quadratic.

(A) 130 (B) 126

(C) 4 (D) 193

104. A convergent GP has first term 9, ratio $\frac{1}{3}$. Find the sum of all terms after its first 5 terms.

(A) $\frac{121}{9}$ (B) $\frac{1}{18}$

(C) $\frac{1}{27}$ (D) $\frac{1}{54}$

105. A=(5,5,5), O=(0,0,0). Find the squared length of the orthogonal projection of OA on the line through O with direction (1,2,2).

(A) 625 (B) $\frac{50}{9}$

(C) $\frac{25}{3}$ (D) $\frac{625}{9}$

106. Let k make lim as x$\to\infty$ of [$(5x^2+2x+1)/(x+4)$-kx] finite. Find that finite limit.

(A) -18 (B) 20

(C) 2 (D) 22

107. Evaluate $\int_0^7$ x floor(x) dx.

(A) 91 (B) $\frac{343}{3}$

(C) $\frac{203}{2}$ (D) 21

108. An equivalence relation has classes of sizes 7,5,5. How many unordered pairs of distinct related elements are there?

(A) 41 (B) 99

(C) 21 (D) 7

109. For f(x)=x(x-8)(x-13), find the sum of all c$\in$(0,13) satisfying f′(c)=0.

(A) 7 (B) 21

(C) 14 (D) 104

110. Find the product of all real solutions of sqrt(7-log_7|x|)=log_7|x|-1.

(A) -343 (B) -117649

(C) 117649 (D) 0

111. $A = \begin{bmatrix} 6 & 1 \\ 0 & 2 \end{bmatrix}$. Find det($A^T$A).

(A) -144 (B) 12

(C) 24 (D) 144

112. y′+y/(x+1)=1 for x$\geq$0, y(0)=2. Find (3)y(2)-y(0).

(A) 2 (B) 4

(C) 6 (D) -4

113. For x>0, xy′-y=x and y(1)=5. Find y(5)/5-y(1).

(A) $\log(5)$ (B) $-\log(5)$

(C) $5\log(5)+20$ (D) $\log(5)+5$

114. F′(x)=$(x^2+8x+11)\,e^x$ and F(0)=6. Find F(1)-2F(0).

(A) $1+12e$ (B) $-12e-1$

(C) $-5+12e$ (D) $-11+12e$

115. $\theta \in(\pi/2,\pi)$, $\sin\theta=\frac{5\sqrt{61}}{61}$. Find $\cos(\theta/2)$.

(A) $-\frac{3\sqrt{61}}{61}$ (B) $\frac{\sqrt{7442-732\sqrt{61}}}{122}$

(C) $\frac{\sqrt{732\sqrt{61}+7442}}{122}$ (D) $-\frac{\sqrt{7442-732\sqrt{61}}}{122}$

116. Let $D(t)=\det\begin{pmatrix}8+t & 1+t\\ 1+t & 3+t\end{pmatrix}$. Find D′(0).

(A) 9 (B) 23

(C) 13 (D) 46

117. Chords of $x^2+y^2=121$ pass through (22,0). Find the radius of the circle containing their midpoint locus.

(A) 121 (B) 22

(C) $\frac{11}{2}$ (D) 11

118. Let $P(x)=(1+\sqrt{2}x)^{11}+(1-\sqrt{2}x)^{11}$. Find the sum of all nonconstant coefficients of P.

(A) 16234 (B) 16236

(C) 8119 (D) 16238

119. Find the area between y=(x-3)(x-6) and the x-axis for $0\le x\le 12$.

(A) −144 (B) $\frac{153}{2}$

(C) 153 (D) 144

120. Evaluate $\int_0^{(\pi/4)} \tan^4x\,\sec^4x\,dx$.

(A) $\frac{12}{35}$ (B) $\frac{1}{5}$

(C) $\frac{1}{7}$ (D) $\frac{2}{5}$

121. Evaluate the product of $\cos(2^j\pi/17)$ for j=0,...,3.

(A) $\frac{1}{8}$ (B) $-\frac{1}{16}$

(C) $\frac{1}{17}$ (D) $\frac{1}{16}$

122. a=(7,0,0), b=(0,2,0), c=(0,0,2), p=a+b. Evaluate [p×(b×c)]·(a+c).

(A) 16 (B) −16

(C) 0 (D) −212

123. X earns 2 and uses 2 labour hours; Y earns 3 and uses 3 hours. There are 50 hours and at least 5 items of Y must be made. Which model is correct?

(A) Maximize $2x+3y$; $2x+3y\ge 50$, $x,y\ge 0$ (B) Minimize $2x+3y$; $2x+3y\le 50$, $y\ge 5$

(C) Maximize $2x+3y$; $2x+3y\le 50$, $x\ge 0$, $y\ge 5$ (D) Maximize $3x+2y$; $2x+3y\le 50$, $y\le 5$

124. $P(A) = 1/2$, $P(B) = 1/4$, and $P(A \cap B) = 1/8$. If X counts how many of A, B occur, find $\mathrm{Var}(X)$.

(A) $\frac{1}{16}$ (B) $\frac{7}{16}$

(C) $\frac{3}{4}$ (D) $\frac{3}{16}$

125. How many injective functions from {a,b,c} to a labelled 5-element set satisfy f(a)<f(b), when the codomain is naturally ordered?

(A) 60 (B) 20

(C) 30 (D) 10

126. Evaluate the sum of r(r-1)C(6,r)2^r over all even integers r with 0≤r≤6.

(A) 1230 (B) 1464

(C) 4920 (D) 4860

127. z=8+4i. Find 1/z+1/conjugate(z).

(A) 16 (B) $\frac{1}{10}$

(C) $\frac{1}{5}$ (D) $\frac{1}{16}$

128. L_1:r=t(1,0,1), L_2:r=(3,0,0)+s(0,1,1). Find their shortest distance.

(A) 1 (B) $\sqrt{3}$

(C) 3 (D) 0

129. For the roots α,β of $z^2 - (2 + 12i)z + (-70 + 24i) = 0$, find $|\alpha\text{-}\beta|^2$.

(A) 148 (B) 296

(C) 74 (D) 5476

130. The letters A,A,B,C and 4 further distinct letters, one namedD, are arranged in a row. How many arrangements have nonadjacent As, adjacent B,C in either order, and D at an end?

(A) 480 (B) 1440

(C) 4800 (D) 960

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A mass 3 kg is attached to two massless springs in series with spring constants 30 N/m and 60 N/m. Find the small-oscillation period on a smooth horizontal surface.

(A) $\frac{\sqrt{30}\pi}{5}$ s

(B) $\frac{\sqrt{15}\pi}{5}$ s

(C) $\frac{\sqrt{10}\pi}{5}$ s

(D) $\frac{\sqrt{30}\pi}{15}$ s

132. The accelerating potential of an X-ray tube is increased from 20 kV to 40 kV, while its tube current is reduced by a factor 2. Ignoring target heating and recoil, what is the new cutoff wavelength in nm? Use $hc = 1240$ eV nm.

(A) $\frac{31}{250}$ nm

(B) 31 nm

(C) $\frac{31}{1000}$ nm

(D) $\frac{31}{125}$ nm

133. A nucleus of mass number 24 splits into two equal nuclei. The parent has binding energy per nucleon $\frac{71}{10}$ MeV and each daughter has $\frac{15}{2}$ MeV. Ignoring other products, what energy is released in the split?

(A) $\frac{48}{5}$ MeV

(B) $\frac{948}{5}$ MeV

(C) $\frac{24}{5}$ MeV

(D) 180 MeV

Chemistry

134. In a hypothetical atom, allowed l values are0,1,...,n. Subshell capacity remains2(2l+1), with filling by increasing n+l and then lower n. All subshells preceding 4s are filled, 4s contains 1 electrons, and no later subshell is occupied. How many electrons does the atom contain?

(A) -35

(B) 35

(C) 70

(D) 35/2

135. A closed system follows states A→B→C→A. For A→B, heat q=300 J and work on the system w=-180 J. For C→A, q=-250 J and w=210 J. For B→C, q=-100 J. Find work on the system on B→C in J.

(A) 140

(B) -100

(C) 20

(D) -20

136. Limiting molar conductivities are KCl150, NaCl126 and KNO3 110 S cm2/mol. A very dilute 1/50 M NaNO3 solution has measured conductance 43/75000 S. Neglect finite-concentration corrections. What is the cell constant in cm-1?

(A) 3

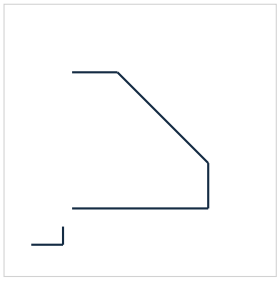
(B) -3

(C) 3/2

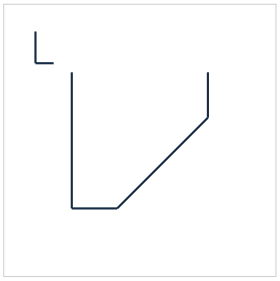
(D) 6

Logical Reasoning

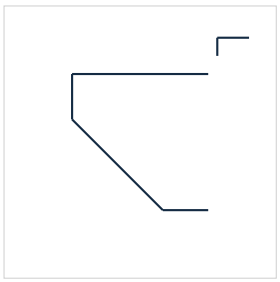
137. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right



bottom left

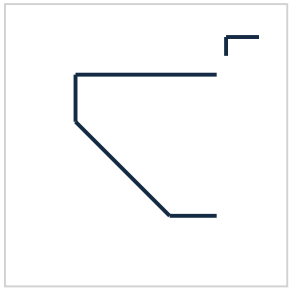
?

bottom right

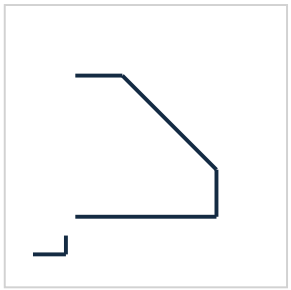
(A) Pattern N



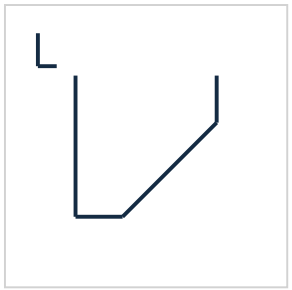
(B) Pattern R



(C) Pattern T



(D) Pattern W



138. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



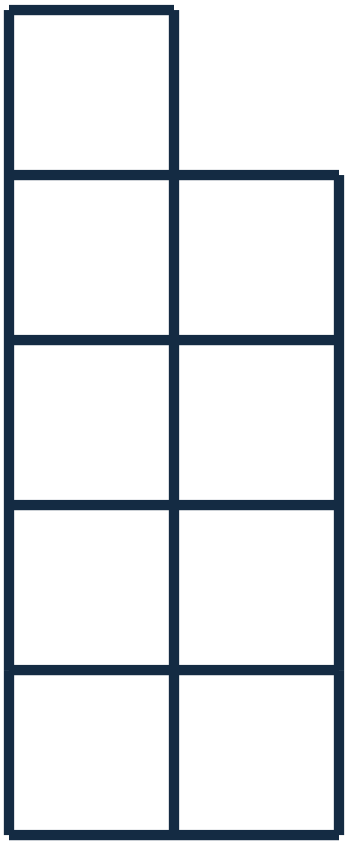
host 3



host 4

(A) 2
(B) 4
(C) 1
(D) 3

139. How many rectangles, including squares are formed by all the drawn segments?



(A) 34
(B) 22
(C) 35
(D) 36

Mathematics

140. Let R(x) be the remainder when x^7 is divided by (x-1)(x-2). Find R(4).

(A) 508 (B) -377

(C) 382 (D) 16384

141. Evaluate $\Sigma\, j/2^j$ for j=1,...,5.

(A) $\frac{57}{32}$ (B) 2

(C) $\frac{29}{16}$ (D) $\frac{5}{32}$

142. $F(x)=\int_x^{(x^2)}(t+2)^2dt$. Find $F'(3)$.

(A) 726 (B) 701

(C) 576 (D) 751

BITSAT Mock 36

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A series RC circuit is driven at fixed rms voltage. Multiplying angular frequency by 6 multiplies rms current by $\frac{12\sqrt{506}}{253}$. Find R^2/X_C^2, where X_C is the initial capacitive reactance.

(A) $7+\frac{84\sqrt{506}}{253}$ (B) 36

(C) $\frac{41}{7}$ (D) 7

2. The quantity $Z = x^2y^1/\sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 1, 2 and 6, respectively. To first order, what is the maximum percentage error in Z?

(A) 9 (B) 6

(C) 1 (D) 7

3. Two parallel plane mirrors separated by 1 cm extend from axial position 0 to $\frac{9}{2}$ cm. A ray enters midway between them at axial position 0, making 45° with the normal to the mirrors. How many reflections occur before it passes the far edge? Exclude a contact exactly on the exit edge.

(A) 4 (B) 2

(C) 8 (D) 5

4. A direct vernier has 20 divisions equal to 19 main-scale divisions; 1 MSD is 1 mm. Its signed zero error is $\frac{-1}{10}$ mm. For an object the main scale is 3 mm and the 7th vernier division coincides. Find the corrected reading.

(A) $\frac{39}{4}$ mm (B) $\frac{671}{200}$ mm

(C) $\frac{13}{4}$ mm (D) $\frac{69}{20}$ mm

5. A spherical blackbody's absolute temperature is multiplied by 4 while its radius is divided by 3. Neglect incoming background radiation. Find its emitted-power multiplier.

(A) $\frac{256}{9}$ (B) $\frac{16}{9}$

(C) $\frac{256}{27}$ (D) 256

6. A stretched string is divided by a bridge into lengths in the ratio $1:5$. Tension and linear density are unchanged. The original fundamental is 60 Hz. Find the fundamental of the longer segment.

(A) 72 Hz (B) 60 Hz

(C) 90 Hz (D) 360 Hz

7. 3 kg of water at 25°C is mixed in an insulated vessel with $\frac{315}{167}$ kg of ice at 0°C. Use $c_w = 4200$ J/(kg K), $L_f = 334000$ J/kg. Find the mass of ice remaining at equilibrium.

(A) $\frac{315}{334}$ kg (B) $\frac{945}{668}$ kg

(C) $\frac{315}{167}$ kg (D) $\frac{85680}{73313}$ kg

8. A hydrogen atom emits a photon in the $4 \to 3$ transition. A hydrogen-like ion of nuclear charge $Z = 2$ emits a photon of exactly the same wavelength and ends in level $n_f = 6$. What is its initial principal quantum number?

(A) 2 (B) 6

(C) 4 (D) 8

9. A satellite of mass m is moved from a circular orbit of radius R to one of radius 7R about mass M. Find the required increase in total orbital energy in units of $GMm/(2R)$.

(A) 6 (B) $\frac{12}{7}$

(C) $\frac{6}{7}$ (D) $\frac{1}{7}$

10. For a particle moving in one dimension, $v = 11 - 6x$ in SI units while $v > 0$. What is its acceleration at $x = \frac{11}{12}$ m?

(A) $33\,\mathrm{m/s^2}$ (B) $0\,\mathrm{m/s^2}$

(C) $-33\,\mathrm{m/s^2}$ (D) $-198\,\mathrm{m/s^2}$

11. An ideal diode and a 80 ohm resistor are in series with $v(t) = 60\sin\omega t$ V. The diode conducts only during the positive source half-cycle. Find average current over one complete cycle.

(A) $\frac{3}{8}$ A (B) $\frac{3}{4\pi}$ A

(C) $\frac{3}{8\pi}$ A (D) $\frac{3}{4}$ A

12. A stationary nucleus of mass number 216 emits an alpha particle whose kinetic energy is 8 MeV. Taking daughter and alpha masses proportional to $A-4$ and 4, find the total energy release Q, including daughter recoil.

(A) $\frac{220}{27}$ MeV (B) $\frac{8}{53}$ MeV

(C) $\frac{8\sqrt{53}}{53} + 8$ MeV (D) $\frac{432}{53}$ MeV

13. Two identical converging thin lenses of focal length 25 cm are separated by 75 cm. A parallel beam enters the first. Find the distance of the final image from the second lens, positive on the outgoing side.

(A) 50 cm (B) $\frac{25}{2}$ cm

(C) $\frac{50}{3}$ cm (D) 25 cm

14. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 30 percent. The measured increase in its absolute temperature is $\frac{2829}{10}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) $\frac{943}{2}$ K
(B) 410 K
(C) 943 K
(D) $\frac{6929}{10}$ K

15. Light is incident from index $\frac{13}{10}$ onto index $\frac{8}{5}$. At one incidence angle the reflected beam is completely plane polarised. Find the tangent of that angle.

(A) $\frac{8}{5}$
(B) $\frac{3}{13}$
(C) $\frac{16}{13}$
(D) $\frac{\sqrt{87}}{13}$

16. Three long parallel wires are equally spaced by 2 m along a straight line. Outer wires each carry 7 A in the same direction; the middle carries 5 A in the opposite direction. Find the net force per unit length on the middle wire.

(A) $-\frac{35}{4\pi}$ N/m
(B) 0 N/m
(C) $\frac{35}{8\pi}$ N/m
(D) $\frac{35}{2\pi}$ N/m

17. A series RLC circuit is resonant at angular frequency ω_0. Its capacitance is then multiplied by 25 while L stays fixed. What angular frequency restores resonance, as a fraction of ω_0?

(A) $\frac{1}{5}$
(B) $\frac{1}{25}$
(C) 25
(D) 1

18. A travelling wave is $y = 3\sin[8(x - \frac{5}{2}t) + \pi/4]$ mm with x in metres and t in seconds. What is its frequency?

(A) $\frac{\pi}{10}$ Hz
(B) $\frac{5}{4\pi}$ Hz
(C) $\frac{80}{\pi}$ Hz
(D) $\frac{10}{\pi}$ Hz

19. A bridge divides a stretched string into two sections. Their fundamental frequencies are 600 Hz and 200 Hz under the same tension. Find the fundamental frequency of the full string if the bridge is removed without changing tension or total length.

(A) 300 Hz
(B) 150 Hz
(C) 400 Hz
(D) 800 Hz

20. Photons of energy 40 eV eject electrons from a surface. The fastest electrons enter a uniform magnetic field perpendicular to their velocities. In this field an electron with kinetic energy 2 eV has orbit radius r. The observed maximum orbit radius is $3r$. What is the surface work function? Neglect recoil.

(A) 34 eV
(B) 22 eV
(C) 18 eV
(D) $\frac{22}{9}$ eV

21. A mass 4 kg initially moves in the positive direction at 6 m/s. A force opposite to that direction rises linearly from zero to 5 N and then falls linearly to zero over a total duration 4 s. Find its signed final momentum.

(A) 14 kg m/s
(B) 34 kg m/s
(C) 19 kg m/s
(D) 24 kg m/s

22. A vacuum electromagnetic wave of average intensity 120 W/m^2 passes through a device that reduces its electric-field amplitude by a factor 5. The transmitted wave falls normally on an area 4 cm^2. What average power reaches the area?

(A) $\frac{6}{125}$ W (B) $\frac{6}{3125}$ W

(C) $\frac{6}{625}$ W (D) $\frac{96}{5}$ W

23. A uniform wire is stretched to $\frac{3}{2}$ times its original length at constant volume. Its resistivity increases by 35 percent during the process. Find final resistance divided by initial resistance.

(A) $\frac{9}{4}$ (B) $\frac{729}{160}$

(C) $\frac{243}{80}$ (D) $\frac{3}{5}$

24. An ideal heater of power 6000 W supplies heat to 3 kg of ice initially at 0°C in an insulated vessel for 251 s. Find final temperature after complete melting. Use $L_f = 334000$ J/kg and $c_w = 4200$ J/(kg K); neglect vessel heat capacity.

(A) $\frac{84}{167}$ degree C (B) $\frac{2510}{21}$ degree C

(C) 40 degree C (D) $\frac{1675}{21}$ degree C

25. A Carnot engine initially runs between 800 K and 300 K. Its hot reservoir is raised by 60 K. By how much must the cold-reservoir temperature rise to keep efficiency unchanged?

(A) $\frac{75}{2}$ K (B) 60 K

(C) $\frac{900}{43}$ K (D) $\frac{45}{2}$ K

26. A projectile is launched at 30 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{15}{16}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ m/s^2.

(A) 54 m (B) $\frac{27}{5}$ m

(C) $\frac{108}{5}$ m (D) $\frac{162}{5}$ m

27. Two equal-intensity coherent beams have maximum combined intensity 12 units. Find intensity when their path difference is $\lambda/3$.

(A) 6 (B) 12

(C) 3 (D) 9

28. A particle is projected vertically outward from the surface of a nonrotating planet with speed $v_e/13$, where v_e is surface escape speed. Neglect atmosphere. Find maximum height divided by planet radius.

(A) $\frac{1}{168}$ (B) $\frac{1}{13}$

(C) $\frac{1}{12}$ (D) $\frac{1}{170}$

29. A particle has displacement $x = 2\cos(2\pi t/2)$ cm. Find its signed velocity at $t = \frac{1}{3}$ s.

(A) $\sqrt{3}\pi$ cm/s (B) $-\sqrt{3}\pi$ cm/s

(C) $-\pi^2$ cm/s (D) -2π cm/s

30. In YDSE wavelength is 400 nm, screen distance 4 m and slit separation 4 mm. Find the separation on the same side of the centre between the 3rd bright fringe and the 6th dark fringe. Count the first dark fringe as order 1.

(A) $\frac{11}{5}$ mm
(B) 1 mm
(C) $\frac{6}{5}$ mm
(D) $\frac{17}{5}$ mm

Chemistry

31. In a1.00 L vessel A+2B⇌2C+D. Initially A=1/2 mol, B=3/5 mol, D=0 and an unknown amount of C. At equilibrium D=3/100 mol and Kc=25/11421 using molar concentrations. What was the initial C amount in mol? All equilibrium amounts are positive.

(A) 2/25
(B) 1/50
(C) -1/25
(D) 1/25

32. Two monoanionic dimethylglyoximate ligands bind Ni2+ through two N donors each. In the usual planar Ni(dmg)2 complex, which inventory and electron conclusion is consistent?

(A) Four donor atoms and an unavoidable octahedral geometry.

(B) Each ligand donates only through oxygen, so no N donor is present.

(C) Two donor atoms, four chelate rings and a tetrahedral d0 centre.

(D) Four N donor atoms, two chelate rings and a diamagnetic square-planar d8 centre.

33. Two local viscosity-temperature graphs are recorded with no chemical or phase change: graph A slopes down and graph B slopes up. The samples are an ordinary molecular liquid and a dilute gas. Which explanation correctly accompanies the graph assignment?

(A) A is the liquid because heating helps overcome cohesive rearrangement barriers; B is the gas because momentum transport increases.

(B) B is the liquid because heating creates stronger permanent covalent crosslinks in any liquid.

(C) Either graph can be assigned from molecular formula alone, without knowing the phase.

(D) A is the gas because viscosity and temperature must always be inversely proportional.

34. I:ethene+Br2 in inert solvent;II:styrene+HBr/peroxide;III:benzene+CH3Cl/AlCl3. Which complete mechanism assignment is correct?

(A) I electrophilic substitution;II radical substitution;III electrophilic addition.

(B) I radical substitution;II electrophilic addition;III nucleophilic substitution.

(C) I electrophilic addition;II radical addition;III electrophilic substitution.

(D) All three are radical additions.

35. For $LaH_{2.8}$,which classification and consequence are mutually consistent?

(A) Ionic; every example must be a gas composed of discrete molecules.

(B) Covalent; nonstoichiometry proves all hydrogen has been removed.

(C) All hydrides contain free H+ ions in their solid lattices.

(D) interstitial/metallic; variable hydrogen occupancy can give nonstoichiometry.

36. List all acyclic saturated carbon skeletons with formula C6H14. Count constitutional isomers,not conformers,and do not count alternative numbering of one skeleton twice. The count is

(A) 5

(B) 7

(C) 4

(D) 6

37. H2O2 has an H-O-O-H skeleton with three sigma bonds and fourteen valence electrons. How many lone pairs are present in the whole molecule in its ordinary Lewis structure?

(A) Six; three on each oxygen.

(B) Four; two on each oxygen.

(C) Two; one on each oxygen.

(D) Zero, because all fourteen electrons are in the three sigma bonds.

38. An100mL sample contains $\frac{2}{25}$ mol HA,pKa=5,and $\frac{1}{25}$ mol HCl. Add100mL $\frac{6}{5}$ M NaOH. Use usual weak-base salt hydrolysis and $\log_{10}(\frac{2}{5})=-\frac{199}{500}$. Final pH is

(A) $\frac{6801}{500}$

(B) 7

(C) 5

(D) $\frac{9301}{1000}$

39. For benzaldehyde with ethylene glycol and catalytic acid,remove water,which product or mechanistic conclusion is correct under the stated conditions?

(A) benzyl alcohol

(B) an ester

(C) a cyclic acetal

(D) a cyanohydrin

40. A $\frac{129}{500}$ g volatile sample displaces $\frac{1024}{13}$ mL wet air at 320 K and 790 mmHg. Water-vapour pressure is 30 mmHg. Using STP=273 K,760 mmHg and22.4 L/mol,the sample molar mass, rounded to the nearest whole g/mol, is

(A) 83

(B) 86

(C) 80

(D) 2179

41. For neutral H2 and HT, take isotope mass numbers H=1, D=2, T=3 and neglect mass defects. Their gases have equal temperature. Which pair gives ($\langle v^2\rangle$_H2/$\langle v^2\rangle$_HT, number of neutrons in one HT molecule minus that in one H2 molecule)?

(A) (2, 4)

(B) (1/2, 2)

(C) (2, 2)

(D) (2, 1)

42. A water sample contains 1/500 mol Ca(HCO3)2 and 1/1000 mol CaSO4 and no other hardness source. On quantitative boiling, Ca(HCO3)2→CaCO3(s)+CO2+H2O; CaSO4 stays dissolved. Which pair gives the CaCO3 precipitate mass and remaining hardness type? CaCO3=100 g/mol.

(A) 1/10 g; only temporary hardness remains.

(B) 3/10 g; no hardness remains.

(C) Zero; boiling can never remove temporary hardness.

(D) 1/5 g; permanent hardness remains.

43. P is a phenol described by an empirical formula C6H6O and molar mass94 g/mol. Which molecular formula and Kolbe-Schmitt product pair is consistent?

(A) C6H12O;4-methylbenzoic acid

(B) C3H3O;benzoic acid

(C) C12H12O2;3-hydroxybenzoic acid

(D) C6H6O;2-hydroxybenzoic acid

44. Acidifying 3/100 mol borax theoretically gives four H3BO3 per borax formula unit. The isolated boric acid requires 6/125 mol NaOH for complete titration in the presence of glycerol, where each H3BO3 behaves as monobasic. What fraction of the theoretical boric acid was recovered?

(A) 4/5 (B) 2/5

(C) -2/5 (D) 1/5

45. A dry mixture contains only anhydrous SrCO3 and BaCO3. Use the ordinary Group2 carbonate stability trend. In an idealised selective-heating window, the less thermally stable carbonate decomposes completely by MCO3→MO+CO2, while the more stable carbonate remains unchanged. No other mass loss occurs. This is the stated model, not a specified real temperature. Atomic masses: Sr=88, Ba=137, C=12, O=16. The initial mixture mass is 234/25 g. The first-stage CO2 collected is 1/100 mol. What was the initial mass fraction of SrCO3?

(A) 197/936 (B) 11/234

(C) 1/5 (D) 37/234

46. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_y; opposite facing signs; 2: parallel p_x; same facing signs. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (0, 3) (B) (1, 1)

(C) (1, 2) (D) (2, 1)

47. Nonrelativistic ions A and B start from rest and gain only electric kinetic energy. B has 5 times A's mass and 4 times its positive charge. To make λ_B/λ_A=1/5, what must V_B/V_A be?

(A) 25/4 (B) 5/4

(C) 4/5 (D) 6

48. Dry air is taken to contain21.0% O2by volume. From 140L of this air, an isolation process recovers80.0% of its oxygen into a pure stream. All gas volumes refer to the same T,P. Find recovered O2volume in litres.

(A) $\frac{147}{4}$ (B) $\frac{588}{25}$

(C) $\frac{147}{5}$ (D) $\frac{2212}{25}$

49. Compare the energy for removing one more electron from Zn^{2+} and from Cu^{2+}. Which member is expected to resist that removal more strongly on the stated shell-stability basis?

(A) They must be identical because both are positive ions.

(B) The ion with the smaller written charge always has the larger removal energy.

(C) Zn^{2+}

(D) Cu^{2+}

50. For $Cr_2O_7^{2-}$,use seven ordinary oxide oxygens shared by two Cr. Which pair (Cr oxidation state, number of peroxide O-O links per formula entity) is correct?

(A) (+6,0). (B) (+6,1).

(C) (+4,0). (D) (+8,0).

51. Two van der Waals gases obey Tc=8a/(27Rb) and Pc=a/(27b^2). Their measured ratios are Tc,A/Tc,B=5/4 and Pc,A/Pc,B=5/16. Find (a_A/a_B, b_A/b_B).

(A) (5, 2)
(B) (10, 4)
(C) (5, 4)
(D) (4, 5)

52. A carbohydrate cannot be hydrolysed into simpler carbohydrate units. It has five carbons and an aldehyde in its open-chain form. Which complete classification fits?

(A) Polysaccharide and amino acid.
(B) Monosaccharide and aldopentose.
(C) Disaccharide and ketohexose.
(D) Dipeptide and aldotriose.

53. A 1g mixture of Al and an inert solid gives $\frac{9}{200}$mol $H_2$with excess aqueous NaOH by $2Al + 2NaOH + 6H_2O \rightarrow 2Na[Al(OH)_4] + 3H_2$. Find Almass percentage,using Al=27.

(A) $\frac{243}{2}$
(B) 81
(C) 100
(D) $\frac{81}{2}$

54. A polymer contains repeat units derived from glycine and 6-aminohexanoic acid. Which pair of classifications is appropriate?

(A) Polysiloxane; silicone rubber.
(B) Polyester; PET made from ethene only.
(C) Copolyamide; nylon-2-nylon-6.
(D) Polyalkene; polythene.

55. For Gd^{3+} (Z=64),which configuration/chemical interpretation is consistent with its electron count and the commonly discussed +3 state?

(A) 4f^7; the half-filled f shell is retained in Gd(III).
(B) 4f^6; the atomic5d electron remains.
(C) 4f^8; all three removed electrons were 6s.
(D) 4f^{10}; the ionic charge does not change electron count.

56. For standard M2+/M half-cells, E°_B-E°_A=6/5 V and E°_C-E°_B=1/5 V. B has E°=1/2 V. Which pair gives the strongest reducing metal among A,B,C and E°cell for A(s)+C2+→A2++C(s)?

(A) A; 7/5 V
(B) A; -7/5 V
(C) B; 1/5 V
(D) C; 7/5 V

57. For nonplanar tub-shaped cyclooctatetraene,choose the correct classification and the decisive reason. Use the explicitly stated geometry.

(A) nonaromatic; nonplanarity breaks the cyclic parallel-p-orbital requirement
(B) aromatic; count only the neutral double bonds and ignore charge or geometry
(C) nonaromatic; an ion can never be aromatic
(D) antiaromatic; every cyclic compound obeys 4n+2

58. An ideal mixture initially has$\frac{1}{10}$molA and$\frac{3}{10}$molB. A selective step removes half ofB and noA. At the new equilibriumT,V,totalpressure is4bar. Partialpressure ofA,in bar,is

(A) $\frac{8}{5}$
(B) 2.0
(C) 1
(D) 10

59. A Bi(V) reagent is reduced to Bi(III) in an oxidation reaction. Which periodic explanation is appropriate?

(A) Bi must gain five protons to reach+3.

(B) The reaction proves that all Group15 hydrides have equal basicity.

(C) The lower+3 state is relatively stabilised down Group15 by the inert-pair effect.

(D) The+5 state is necessarily the only allowed state because Bi is in Group15.

60. For M+2A+⇌M2++2A, the equilibrium constant for the doubled reaction is10^6. At the same T use2.303RT/F=0.0590 V. What is E° of the original reaction in V?

(A) 177/2000

(B) -177/2000

(C) 177/4000

(D) 177/1000

English Proficiency

61. Read the sentence: She was willing to lend the tool but drew the line at allowing anyone to alter it. What does 'draw the line' mean here?

(A) ended all cooperation

(B) set a limit to what she would permit

(C) issued a complete refusal

(D) recorded an exact measurement

62. Use the standard contemporary negative adjective. Choose the correctly formed adjective: Ignoring a known safety requirement without justification is _____.

(A) disresponsible

(B) irresponsible

(C) inresponsible

(D) unresponsible

63. She readily seeks company and conversation and feels refreshed after lively social gatherings. Which word fits?

(A) extrovert

(B) eccentric

(C) spectator

(D) introvert

64. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The editor does not alter quotations without consulting their sources.

(A) Quotations are not altered by the editor without consulting their sources.

(B) Quotations were not altered by the editor without consulting their sources.

(C) Quotations are altered by the editor without consulting their sources.

(D) The editor is not altered by quotations without consulting sources.

65. Read the sentence: Despite the damaged recording, the final sentence was intelligible. Choose the opposite in meaning of 'intelligible' as used in the sentence.

(A) audible

(B) recognisable

(C) brief

(D) incomprehensible

66. The research reconstructs earlier human life from tools, pottery and ruins rather than only from written chronicles. Which field is it?

(A) archaeology (B) geology

(C) meteorology (D) astronomy

67. Read the sentence: The artist used a subtle change in shade, noticeable only when the panels were placed together. Choose the closest meaning of 'subtle' as used in the sentence.

(A) faint because it was unfinished (B) accidental and irregular

(C) beautiful and unusual (D) delicate and not obvious

68. Read the sentence: The unsupported example would weaken rather than strengthen the argument. Choose the opposite in meaning of 'weaken' as used in the sentence.

(A) complicate (B) change

(C) strengthen (D) shorten

69. Read the sentence: The committee refused to sanction the purchase because it exceeded the approved budget. Choose the closest meaning of 'sanction' as used in the sentence.

(A) inspect (B) authorise

(C) punish (D) recommend

70. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. which shows the village Q. contains a map R. The book that the archivist recommended S. before the railway was built.

(A) Q-R-P-S (B) P-R-Q-S

(C) R-Q-P-S (D) S-Q-P-R

Logical Reasoning

71. The age table uses the same two people throughout:

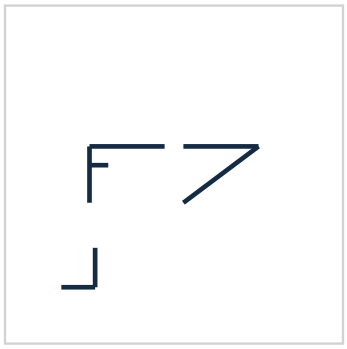
	younger	older
now	x	$x+12$
4 years ago	$x-4$	$x+8$

. The product of their ages 4 years ago was 640. What will the younger person's age be 5 years from now?

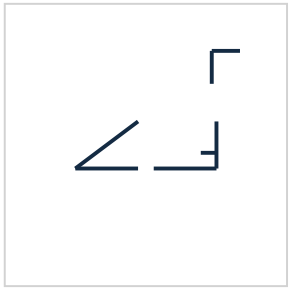
(A) 24 (B) 29

(C) 41 (D) 25

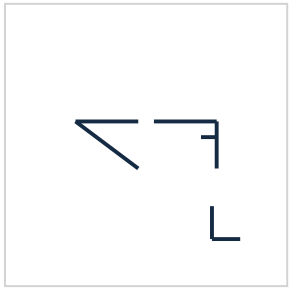
72. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



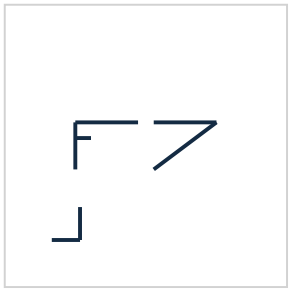
(A) Pattern K



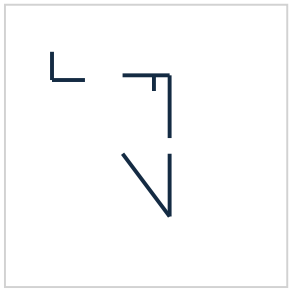
(B) Pattern X



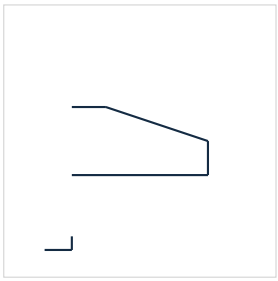
(C) Pattern M



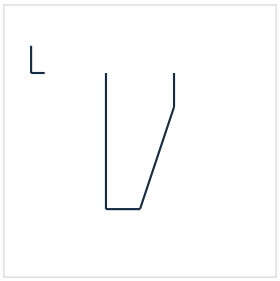
(D) Pattern T



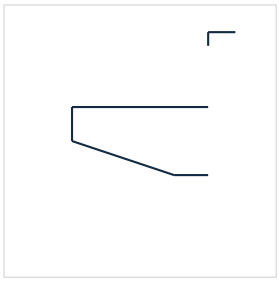
73. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

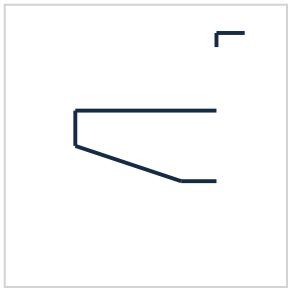


2

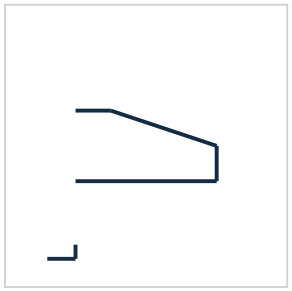


3

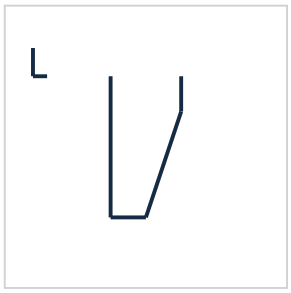
(A) Pattern X



(B) Pattern L



(C) Pattern M



(D) Pattern R



74. Three pairs contain exactly one mammal and one bird. Which pair does not?

(A) Bat / Owl (B) Cat / Eagle

(C) Elephant / Ostrich (D) Rabbit / Tiger

75. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, BRICK is coded as CSJDL. Which word is coded as MBNQ?

(A) LAMP (B) MBNQ

(C) KZLO (D) NCOR

76. In a row, Asha is 6th from the left and 32nd from the right. Bina is 6th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 22 (B) 21

(C) 24 (D) 23

77. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 4 and Z has code 29. Decode the separated values 19, 15, 4, 17, 23.

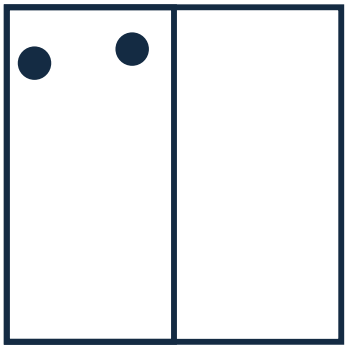
(A) QMBOU

(B) TNALP

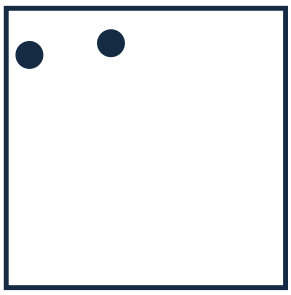
(C) PLANT

(D) OKZMS

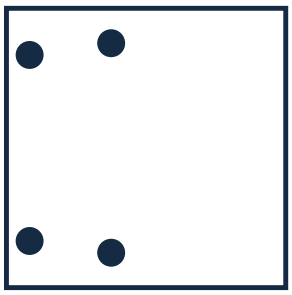
78. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



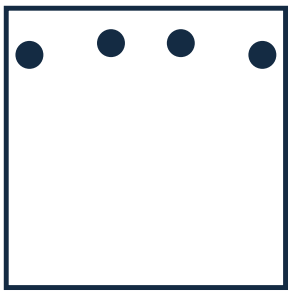
(A) Pattern K



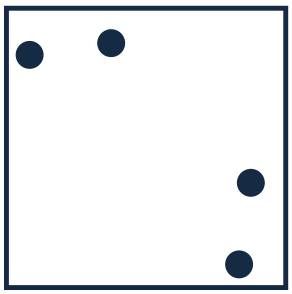
(B) Pattern Y



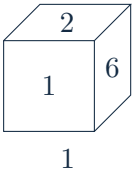
(C) Pattern U



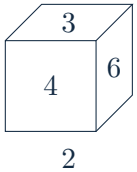
(D) Pattern W



79. The views show the same die. Give the ordered pair: (face opposite 3, face opposite 6). Do not assume opposite labels sum to seven.



1



2

(A) $(2, 2)$

(B) $(2, 3)$

(C) $(2, 5)$

(D) $(2, 1)$

80. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 2, 4, 12, 24, ?, 64, 72.

(A) 16 (B) 32

(C) 64 (D) 48

81. Mohan points to a photograph and says, "This person's only daughter is my mother." Mohan is male, and the photographed person is male. How is the photographed person related to Mohan?

(A) Grandfather (B) Son

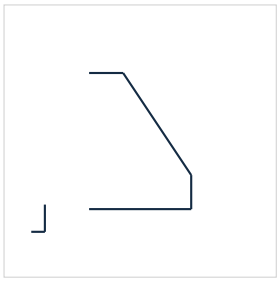
(C) Brother (D) Daughter

82. Consider only these statements; universal statements alone do not imply existence. Some farmers are guides; All campers are guides. Conclusions: I. Some guides are farmers. II. All farmers are campers. Which conclusions follow necessarily?

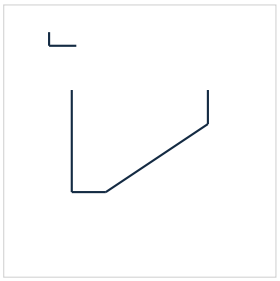
(A) Both I and II (B) Neither I nor II

(C) Only I (D) Only II

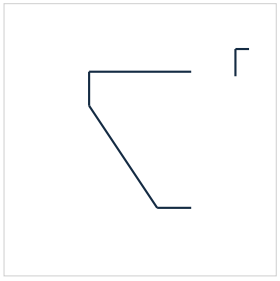
83. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

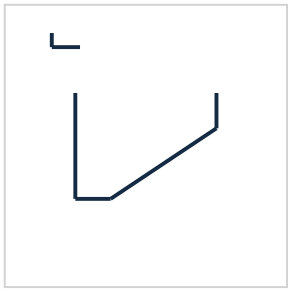


bottom left

?

bottom right

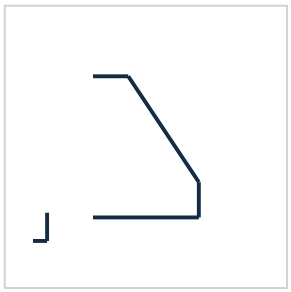
(A) Pattern W



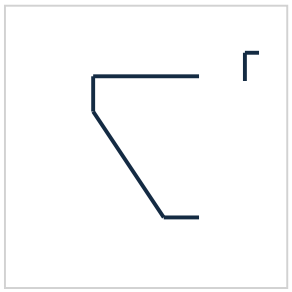
(B) Pattern M



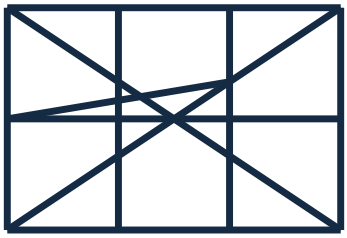
(C) Pattern Z



(D) Pattern L



84. How many triangles are formed by all the drawn segments?



(A) 22 (B) 35

(C) 32 (D) 36

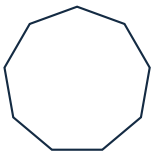
85. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

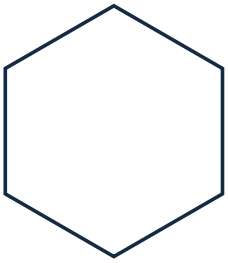


B



C

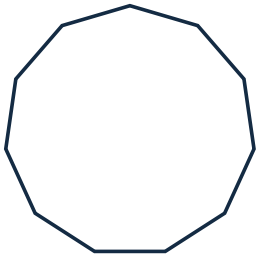
(A) Pattern L



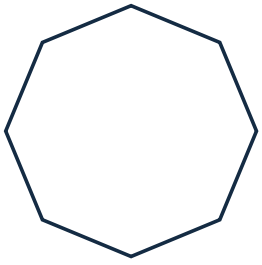
(B) Pattern Z



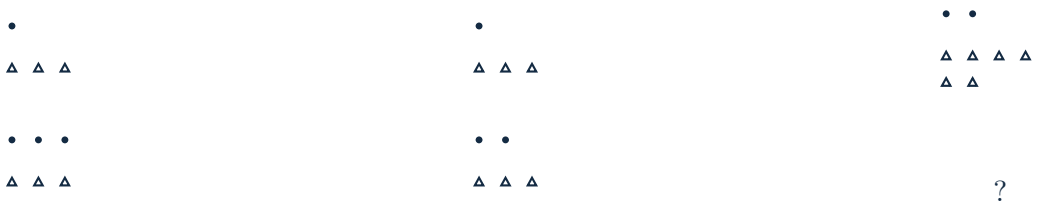
(C) Pattern Y



(D) Pattern P

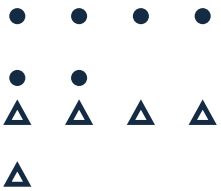


86. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern Q

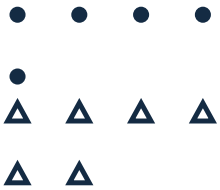
(B) Pattern M



(C) Pattern R



(D) Pattern Z



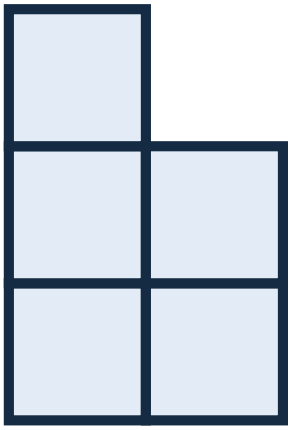
87. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.


target

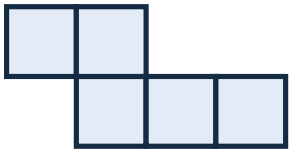

piece 1


piece 2

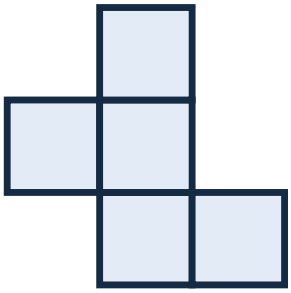
(A) Pattern Q



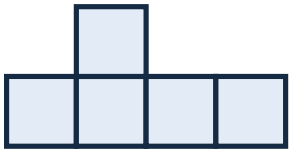
(B) Pattern V



(C) Pattern U



(D) Pattern K



88. The whole has 12 unit positions. Part A covers the first 5; part B covers 3 positions, overlapping A in exactly 2 positions as shown. How many positions remain uncovered by either part?


whole


part A positions


part B positions

(A) 4
(B) 2
(C) 7
(D) 6

89. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 6$ and $5 \mapsto 12$. Which input maps to 28?

(A) 13
(B) 12
(C) 28
(D) 14

90. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

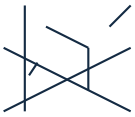




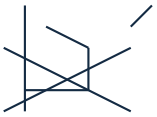
host 1



host 2



host 3



host 4

(A) 4
(B) 1
(C) 2
(D) 3

Mathematics

91. From a point at horizontal distance 5 from a tower, its elevation is 2θ, where $\tan\theta=\frac{2}{7}$ and $0<\theta<45°$. Find the height.

(A) $\frac{225}{28}$
(B) $\frac{20}{7}$
(C) $\frac{10}{7}$
(D) $\frac{28}{9}$

92. $P(A)=\frac{1}{6}$, $P(B)=\frac{1}{2}$, $P(A\cap B)=\frac{1}{12}$. Find P(neither A nor B).

(A) $\frac{1}{3}$
(B) $\frac{5}{12}$
(C) $\frac{1}{12}$
(D) $\frac{7}{12}$

93. The chord of contact from P to $x^2/25-y^2/9=1$ is $x=5/2$. Find the product of the slopes of the two tangents from P.

(A) $-\frac{9}{25}$
(B) $-\frac{9}{100}$
(C) $-\frac{3}{25}$
(D) $\frac{3}{25}$

94. $F'(x)=\frac{5(x^2-1)(x^2+1)^4}{x^6}$ and $F(1)=4$. Find $F(2)-2F(1)$.

(A) $\frac{2101}{32}$
(B) $\frac{2229}{32}$
(C) $-\frac{2229}{32}$
(D) $\frac{1973}{32}$

95. f is cubic with leading coefficient 1/3, stationary points 5,9, and $\lim_{x\to 0}\frac{f(x)-3}{x}=45$. Find f(1).

(A) $\frac{115}{3}$
(B) $\frac{124}{3}$
(C) 48
(D) 55.3333333333333

96. f(x+1)=6x^2. Find the area enclosed between y=f(x) and y=6(x-1).

(A) 1 (B) -1

(C) 3 (D) 2

97. A circular sector has radius 9 and arc length 3π. Find its area.

(A) 27π (B) $\frac{27}{2}$

(C) $\frac{243\pi}{2}$ (D) $\frac{27\pi}{2}$

98. S_n=6n^2+4n is a sequence partial sum. Find $a_1 + a_3 + \cdots + a_{17}$.

(A) 954 (B) 522

(C) 1008 (D) 970

99. For d^{3}y/dx^3+sin(d^{2}y/dx^2)+y=0, find order+degree if the degree is defined; otherwise select "Degree is not defined".

(A) Degree is not defined (B) 4

(C) 5 (D) 3

100. X is binomial with n=6, p=1/3. Given X≥1, find P(X≥2).

(A) $\frac{192}{665}$ (B) $\frac{665}{729}$

(C) $\frac{473}{665}$ (D) $\frac{473}{729}$

101. For $0\leq \theta<14\pi$, roots of x^2-4x+(1-2sinθ)=0 have opposite signs. How many connected intervals of θ satisfy this?

(A) 1 (B) 14

(C) 7 (D) 8

102. f(x)=sin(5x)/x+2 for x<0; f(x)=k cos x for x≥0. Find k for continuity at zero.

(A) 7 (B) 3

(C) 2 (D) 5

103. Independent events A,B have probabilities $\frac{3}{4}, \frac{1}{2}$. Given A∪B, find P(A∩B).

(A) $\frac{3}{8}$ (B) $\frac{3}{7}$

(C) $\frac{4}{7}$ (D) $\frac{3}{10}$

104. For t≥0 define F(t)=lim as n→ ∞ of n$^{(-3)}\Sigma$ floor(tj^2), j=1,...,n. Evaluate Σ F((1/6)^r), r=0,...,∞.

(A) $\frac{1}{15}$ (B) $\frac{1}{3}$

(C) $\frac{2}{5}$ (D) $\frac{6}{5}$

105. How many solutions in $0\leq x<5\pi$ satisfy sin 2 x=sin x and cos x>sin x?

(A) 7 (B) 5

(C) 8 (D) 2

106. The line r=(4,2,5)+t(1,2,2) and plane 2x-y=10 are given. Find the squared minimum distance between them.

(A) $\frac{16}{5}$ (B) 0

(C) $\frac{4\sqrt{5}}{5}$ (D) 16

107. Find the maximum of $\sin^4 x \cos x$ on$[0,\pi/2]$.

(A) $\frac{\sqrt{2}}{8}$ (B) 1

(C) $\frac{16\sqrt{5}}{125}$ (D) $\frac{\sqrt{5}}{5}$

108. 15 distinct people sit at labelled round tables of sizes 8,7. Two specified people must be at different tables, and the first specified person must be at the 8-seat table. Rotations are identical. Count the seatings.

(A) 12454041600 (B) 6227020800

(C) 23351328000 (D) 348713164800

109. For f(x)=|x-5|/(x-7), on which maximal open intervals is f strictly decreasing?

(A) $(-\infty, 5)$ (B) $(5, \infty)$

(C) $(-\infty, 7)$ (D) $(5, 7) \cup (7, \infty)$

110. An AP has a_1=-77,d=7. How many of its first 15 terms are strictly positive?

(A) 12 (B) 3

(C) 15 (D) 2

111. Evaluate lim as x→0 of $[(1+2x)/(1-5x)]^{(1/x)}$.

(A) 7 (B) e^2

(C) e^7 (D) e^{-3}

112. z=11+4i. Find 1/z+1/conjugate(z).

(A) $\frac{8}{137}$ (B) $\frac{1}{22}$

(C) 22 (D) $\frac{22}{137}$

113. $x=3+5(1-t^2)/(1+t^2)$, $y=2+10t/(1+t^2)$, where t is real. Which point of the containing circle is not attained?

(A) $(-2, 2)$ (B) $(8, 2)$

(C) $(3, -3)$ (D) $(3, 7)$

114. $f(x)=x^3+2x$, $g=f^{-1}$. Find $g''(135)$.

(A) $\frac{1}{30}$ (B) $-\frac{30}{5929}$

(C) $-\frac{30}{456533}$ (D) $\frac{30}{456533}$

115. |U|=150, |A|=40, |B|=35, |A∩B|=15. Find the number of elements belonging to both A,B or to neither.

(A) 90 (B) 105

(C) 135 (D) 45

116. If α,β are roots of x^2-10x+16=0, find $\alpha^2+\beta^2$ without solving the quadratic.

(A) 36 (B) 68

(C) 84 (D) 32

117. Group A has 6 observations, mean 14 and population SD 2. Group B has 7 observations, mean 19 and population SD 3. Find the population variance of the combined data.

(A) $\frac{412}{13}$ (B) $\frac{1089}{169}$

(C) $\frac{727}{52}$ (D) $\frac{2181}{169}$

118. Event E has likelihood $\frac{5}{7}$ under source A and $\frac{2}{7}$ under source B. Its total probability is $\frac{3}{7}$. If the source is A with probability r, find r.

(A) $\frac{5}{7}$ (B) $\frac{3}{7}$

(C) $\frac{1}{3}$ (D) $\frac{2}{3}$

119. Evaluate $\sum_{j=2}^{6}\binom{j}{1}$, with the convention $\binom{j}{r}=0$ when j<r.

(A) 20 (B) 7

(C) 21 (D) 18

120. f(x)=x^2+2x for x≤3, and f(x)=px+q for x>3. If f is differentiable at 3, find p-q.

(A) 8 (B) 17

(C) −1 (D) 15

121. Assuming 1+3+$\cdots$+(2n-1)=n^2, the next-step sum is n^2+2n+1. At n=11, what square does this establish?

(A) 143 (B) 144

(C) 121 (D) 169

122. y′+11y=exp(-11x), y(0)=4. Find exp(11)y(1)-4.

(A) 1 (B) $\frac{5}{e^{11}}$

(C) 5 (D) −1

123. Let $z=a+8i$, where $a>0$. If $1/z+1/\overline{z}=1/8$, find the sum of all possible values of a.

(A) 16 (B) 64

(C) 8 (D) 32

124. For $f(x)=|x-2|/(x-5)$, how many distinct real solutions does $f(x)=-2$ have?

(A) 2 (B) 3

(C) 0 (D) 1

125. Find the coefficient of x^6 in (1+4x)exp(7x).

(A) $\frac{117649}{144}$ (B) $\frac{16807}{30}$

(C) $\frac{521017}{720}$ (D) $\frac{117649}{720}$

126. F′(x)=$\frac{1}{x^2}-\frac{1}{x^2\sqrt{x^2+1}}$ and F(1)=5. Find F(2)-2F(1).

(A) $-\frac{11}{2}-\frac{\sqrt{5}}{2}+\sqrt{2}$ (B) $-\sqrt{2}+\frac{\sqrt{5}}{2}+\frac{11}{2}$

(C) $-\sqrt{2}+\frac{1}{2}+\frac{\sqrt{5}}{2}$ (D) $-\frac{9}{2}-\sqrt{2}+\frac{\sqrt{5}}{2}$

127. A positive integer n satisfies $|\Pi(1+ji),\ j=1,...,n|^2=10$. Find n.

(A) 2 (B) 4

(C) 1 (D) 3

128. Eliminate A,B from y=A exp(11x)+B exp(-11x).

(A) y″=0 (B) y″+121y=0

(C) y′-11y=0 (D) y″-121y=0

129. How many integers k make 21sin x+28cos x+5=k have exactly two distinct solutions in $0\leq x<2\pi$?

(A) 69 (B) 34

(C) 71 (D) 70

130. A tower is viewed from two points on the same level ray from its foot. The elevation at the farther point is θ and at the nearer point is 2θ, where $\tan\theta = 1/2$. The points are 5 metres apart. Find the tower height.

(A) $\frac{20}{3}$ (B) 10

(C) $\frac{5}{2}$ (D) 4

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. Two small identical conducting spheres initially carry positive charges $5q_0$ and $4q_0$ and are separated by distance d. They touch and are returned to the same separation. Find the final force divided by the initial force.

(A) $\frac{81}{80}$ (B) 1

(C) $\frac{5}{8}$ (D) $\frac{9}{16}$

132. A metal wire has linear temperature coefficient $\frac{3}{1000}$ per degree Celsius referred to 0°C. Its dimensions are fixed and it stays across a constant voltage. Find power at 40°C divided by power at 0°C.

(A) $\frac{3}{28}$ (B) $\frac{25}{28}$

(C) $\frac{625}{784}$ (D) $\frac{25}{22}$

133. 3 parallel branches each contain 5 identical uncoupled inductors L in series. A total steady current I divides equally among the branches. Find total stored magnetic energy divided by $LI^2/2$.

(A) $\frac{1}{15}$ (B) $\frac{3}{5}$

(C) $\frac{5}{3}$ (D) 5

Chemistry

134. E°(A+/A)=3/10 V and E°(B+/B)=7/10 V. For the stoichiometrically scaled standard reaction 4A(s)+4B+(aq)→4A+(aq)+4B(s), find ΔG° in kJ. Use F=96.5 kJ V-1 mol-1.

(A) 772/5 (B) -1544/5

(C) -772/5 (D) -386/5

135. A cathode receives 1/5 mol electrons and deposits 96/25 g copper from Cu2+ (Cu=64 g/mol). Its only other product is H2. What amount of H2, in mol, is produced?

(A) 1/25 (B) 1/50

(C) -1/25 (D) 2/25

136. butan-2-one reacts with semicarbazide under suitable mildly acidic conditions. Which product correctly reflects nucleophilic addition or addition-elimination at its carbonyl rather than a different reaction class?

(A) an alkyl halide (B) a semicarbazone

(C) an ether (D) a carboxylic ester

Logical Reasoning

137. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: LKI, PNM, ?, XTU, BWY, FZC.

(A) TQQ (B) URR

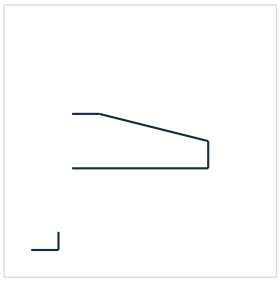
(C) QQT (D) SPP

138. Preserve the exact relation and its order: Ancient : Old :: ? : Swift. Choose the missing term.

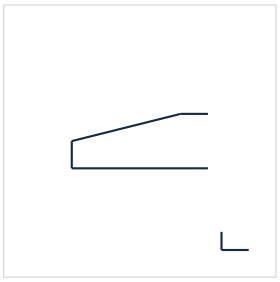
(A) Begin (B) Rapid

(C) Courageous (D) Silent

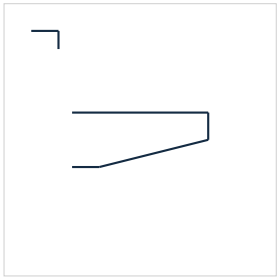
139. Select the missing bottom-right tile. Tiles reflect across the shared horizontal and vertical centre lines; there is no turning.



top left



top right

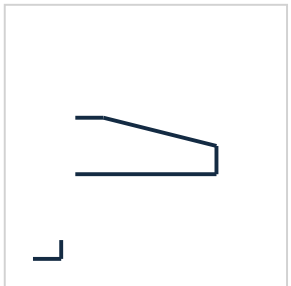


bottom left

?

bottom right

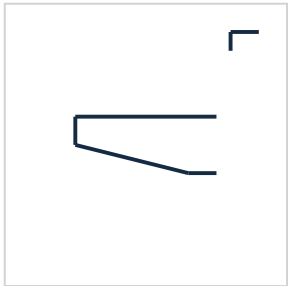
(A) Pattern K



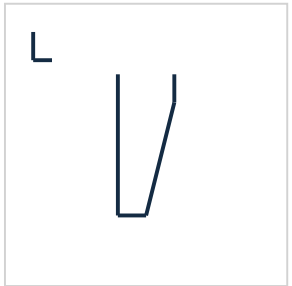
(B) Pattern Y



(C) Pattern T



(D) Pattern R



Mathematics

140. 9 distinct objects are distributed into three nonempty unlabelled boxes. How many distributions are possible?

(A) 3025 (B) 3280

(C) 18150 (D) 9075

141. For $x>0$, $(x^2y)'=0$ and $y(1)=3$. Find $y(2)\cdot 4$.

(A) 12 (B) $\frac{3}{4}$

(C) 3 (D) -3

142. Two distinguishable fair dice show X,Y. Given $X+Y\geq 10$, find the probability that $X\neq Y$ and XY is even.

(A) $\frac{5}{6}$ (B) $\frac{2}{3}$

(C) $\frac{1}{9}$ (D) 1

BITSAT Mock 37

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A vacuum electromagnetic wave of average intensity 100 $\mathrm{W/m^2}$ passes through a device that reduces its electric-field amplitude by a factor 2. The transmitted wave falls normally on an area 5 $\mathrm{cm^2}$. What average power reaches the area?

(A) $\frac{5}{4}$ W (B) $\frac{1}{80}$ W

(C) 25 W (D) $\frac{1}{40}$ W

2. Two coherent beams yield maximum-to-minimum interference intensity ratio 4. Find the larger individual beam intensity divided by the smaller.

(A) 3 (B) 16

(C) 9 (D) $\frac{1}{9}$

3. A particle has potential $U(x, y) = 6x^2y + 5y$ J in SI units. Find the force magnitude at $(4, 4)$ m.

(A) 404 N (B) 101 N

(C) $\sqrt{47065}$ N (D) 293 N

4. 4 identical, noninteracting ideal-gas samples at the same temperature and pressure are combined without changing those conditions. Which pair both increases by a factor 4?

(A) Temperature and molar heat capacity. (B) Density and pressure.

(C) Volume and internal energy. (D) Pressure and temperature.

5. A spherical planet rotates uniformly. Its rotation reduces effective gravity at the equator by 1 $\mathrm{m/s^2}$. By how much does rotation reduce the radial component of effective gravity at latitude 60°?

(A) $\frac{1}{2}$ $\mathrm{m/s^2}$ (B) 1 $\mathrm{m/s^2}$

(C) $\frac{1}{16}$ $\mathrm{m/s^2}$ (D) $\frac{1}{4}$ $\mathrm{m/s^2}$

6. A dipole of fixed moment 4 C m is rotated quasistatically from 60° to 120° relative to a uniform electric field 6 N/C. Find external work.

(A) 24 J (B) $12\sqrt{3}$ J

(C) 8π J (D) 12 J

7. A car negotiates a level circular turn at its maximum non-slipping speed u when $\mu_s = 1$. Rain lowers the coefficient to $\frac{2}{5}$ and the turn radius stays fixed. Find the new maximum speed divided by u.

(A) 1
(B) $\frac{\sqrt{10}}{5}$
(C) $\frac{2}{5}$
(D) $\frac{5}{2}$

8. In YDSE a sheet of index $\frac{3}{2}$ and thickness 6 micrometres covers one slit. A second sheet of the same thickness and index 3/2 covers the other. With wavelength $\frac{7}{10}$ micrometres, find the magnitude of central-fringe shift in units of fringe width.

(A) $\frac{60}{7}$
(B) 0
(C) $\frac{90}{7}$
(D) $\frac{30}{7}$

9. Two uniform thin rods meet at a right angle at the origin. Both have length 6 m. The rod along +x has linear density 5 times that of the rod along +y. A force parallel to the y-axis must produce pure translation. At what x-coordinate must its line of action lie?

(A) 5 m
(B) $\frac{5}{2}$ m
(C) $\frac{1}{2}$ m
(D) 15 m

10. A geometrically similar solid structure is enlarged by a factor 5 in every linear dimension. Its material density is reduced by a factor 3. Its own weight is carried by a similarly scaled base. Find base compressive-stress multiplier.

(A) $\frac{125}{3}$
(B) $\frac{5}{3}$
(C) 15
(D) $\frac{1}{3}$

11. An ideal transformer has secondary-to-primary turns ratio 5. Its primary draws 5 A at 140 V while delivering power to a resistive secondary load. Find secondary current.

(A) 5 A
(B) $\frac{1}{5}$ A
(C) 1 A
(D) 25 A

12. A ray in a medium of index 2 is incident on a medium of index $\frac{11}{10}$. Find the sine of the critical-angle threshold for total internal reflection.

(A) $\frac{11}{20}$
(B) $\frac{3\sqrt{31}}{20}$
(C) $\frac{11}{10}$
(D) $\frac{121}{400}$

13. A network is defined by $P = \mathrm{NAND}(A, B)$, $Q = \mathrm{NOR}(A, P)$ and $Y = \mathrm{NOR}(Q, B)$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 0$ and $P = 1$?

(A) 0
(B) 2
(C) 1
(D) 3

14. In two states of a dilute gas, sound measurements give frequency 600 Hz and wavelength 3 m initially, and 400 Hz and 5 m finally. The heat-capacity ratio and molecular number density stay fixed, while effective collision diameter is multiplied by 2. Use $v_s = f\lambda$, $v_{rms} = \sqrt{3/\gamma}v_s$, and collision frequency $\nu \propto nd^2v_{rms}$. Find the collision-frequency multiplier.

(A) $\frac{400}{81}$
(B) $\frac{10}{9}$
(C) $\frac{40}{9}$
(D) $\frac{20}{9}$

15. A parallel-plate capacitor has vacuum capacitance C_0. A dielectric of constant 7 fills a fraction $\frac{1}{3}$ of the plate area and the entire gap in that region; the remainder stays vacuum. The voltage is fixed. Find the fraction of total stored energy residing in the dielectric-filled region.

(A) $\frac{7}{3}$ (B) $\frac{7}{9}$

(C) $\frac{2}{9}$ (D) $\frac{1}{15}$

16. A projectile is launched at 45 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{24}{25}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ m/s^2.

(A) $\frac{972}{25}$ m (B) $\frac{972}{5}$ m

(C) $\frac{324}{5}$ m (D) $\frac{972}{125}$ m

17. A screw gauge advances 2.0 mm in four complete turns and has 50 circular divisions. With the jaws closed, its zero mark lies 2 divisions above the reference line; use the usual convention that a zero below the reference line gives positive zero error. With a wire, the main scale reads 6 mm and circular reading is 27 divisions. What is the corrected diameter?

(A) $\frac{25}{4}$ mm (B) $\frac{627}{100}$ mm

(C) $\frac{629}{100}$ mm (D) $\frac{827}{100}$ mm

18. In an abrupt junction, the acceptor concentration is 484 times the donor concentration. Within the depletion approximation, which statement gives x_n/x_p and identifies the carriers mainly responsible for the small reverse current before breakdown?

(A) 1/484; minority carriers (B) 1; majority carriers

(C) 22; majority carriers (D) 484; minority carriers

19. A one-electron ion with nuclear charge $Z = 2$ absorbs a photon of energy $\frac{37}{5}$ eV. The ejected electron has maximum kinetic energy 4 eV. From which principal quantum number was the electron removed? Use $E_n = -13.6Z^2/n^2$ eV and neglect recoil.

(A) 1 (B) 16

(C) 8 (D) 4

20. A charge 2 C moves with $\vec{v} = 5\hat{i} + 2\hat{j}$ m/s in $\vec{B} = 1\hat{i} + 5\hat{j}$ T. What is the signed z-component of its magnetic force?

(A) 46 N (B) 50 N

(C) 4 N (D) 30 N

21. A mass 3 kg is attached to two massless springs in series with spring constants 40 N/m and 80 N/m. Find the small-oscillation period on a smooth horizontal surface.

(A) $\frac{3\sqrt{5}\pi}{10}$ s (B) $\frac{3\sqrt{5}}{20}$ s

(C) $\frac{3\sqrt{10}\pi}{10}$ s (D) $\frac{\sqrt{10}\pi}{10}$ s

22. A point at the bottom of two plane liquid layers is viewed normally from air. The upper layer has thickness 18 cm and index 3/2; the lower has thickness 28 cm and index 4/3. Find its apparent depth below the top surface in the paraxial limit.

(A) 33 cm (B) 21 cm

(C) 23 cm (D) 12 cm

23. A 2 kg block moves along a horizontal rough surface from $x = 0$ to $x = 3$ m under a force $F(x) = 4x$ N parallel to motion. With $\mu_k = 1/5$ and $g = 10$ m/s^2, find the work done by the net force. The block has enough initial speed to keep moving forward.

(A) 18 J (B) 6 J

(C) 30 J (D) 24 J

24. An ideal mixture contains 1 moles of monatomic gas and 5 moles of rigid diatomic gas. Vibrations are frozen. Find the heat-capacity ratio $\gamma = C_P/C_V$ of the mixture.

(A) $\frac{7}{10}$ (B) $\frac{15}{14}$

(C) $\frac{13}{9}$ (D) $\frac{10}{7}$

25. A Newtonian liquid of viscosity $\frac{3}{10}$ Pa s has velocity profile $v(y) = 2y + 5y^2$ in SI units. Find the shear-stress magnitude at $y = 3$ m.

(A) 32 Pa (B) $\frac{153}{10}$ Pa

(C) $\frac{3}{5}$ Pa (D) $\frac{48}{5}$ Pa

26. Two ferromagnetic samples have similar remanent flux density. Their coercive fields are 400 A/m and 2400 A/m. A reversing field of magnitude 1400 A/m will act in service. Which sample is better at retaining its original magnetisation against this field, based only on the stated coercivities?

(A) The sample with coercive field 2400 A/m. (B) The sample with coercive field 400 A/m.

(C) Neither can have nonzero remanence. (D) Both are necessarily demagnetised equally.

27. A 3 kg block is launched at 6 m/s on top of a 7 kg block initially at rest on a smooth floor. The coefficient of kinetic friction between them is $\frac{3}{10}$. Find the distance slid by the upper block relative to the lower before slipping ceases. Take $g = 10$ m/s^2.

(A) $\frac{273}{50}$ m (B) $\frac{21}{5}$ m

(C) $\frac{63}{50}$ m (D) $\frac{9}{5}$ m

28. A uniform thin rod of length 7 m is hinged at one end and released from rest in a horizontal position. Find the magnitude of its initial angular acceleration. Take $g = 10$ m/s^2.

(A) $\frac{15}{7}$ rad/s^2 (B) $\frac{5}{7}$ rad/s^2

(C) $\frac{10}{7}$ rad/s^2 (D) 105 rad/s^2

29. A uniform disc of mass 5 kg and radius 5 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) $\frac{1875}{32}$ kg m^2 (B) $\frac{125}{2}$ kg m^2

(C) $\frac{375}{8}$ kg m^2 (D) $\frac{1625}{32}$ kg m^2

30. A charge of $3\,\mu$C moves from potential 50 V to -50 V in an electrostatic field. Find the work done by an external agent if the charge starts and ends at rest and no other force does work.

(A) $-100\,\mu$J (B) $300\,\mu$J

(C) $-150\,\mu$J (D) $-300\,\mu$J

Chemistry

31. An examination table separately lists a surface disinfectant and an antibiotic that acts against susceptible bacteria. Which inference is sound?

(A) Every surface disinfectant is automatically an orally suitable antibiotic.

(B) A disinfectant must contain a peptide bond.

(C) An antibiotic must be a nonreducing sugar.

(D) Both are antimicrobial categories, but their uses are not interchangeable merely because both inhibit microbes.

32. A $\frac{27}{125}$ g volatile sample displaces $\frac{1024}{13}$ mL wet air at 320 K and 790 mmHg. Water-vapour pressure is 30 mmHg. Using STP=273 K,760 mmHg and22.4 L/mol,the sample molar mass, rounded to the nearest whole g/mol, is

(A) 69 (B) 61

(C) 72 (D) 67

33. For the isolated oxoanion ClO3-, choose its chlorine oxidation state and molecular geometry about chlorine.

(A) +5; trigonal pyramidal. (B) +7; trigonal planar.

(C) +3; trigonal pyramidal. (D) +5; linear.

34. $E^\circ(A+/A)=2/5$ V and $E^\circ(B+/B)=9/10$ V. For the stoichiometrically scaled standard reaction $5A(s)+5B+(aq)\rightarrow 5A+(aq)+5B(s)$, find ΔG° in kJ. Use F=96.5 kJ V-1 mol-1.

(A) -965/2 (B) 965/4

(C) -965/8 (D) -965/4

35. A diol HO-CH2-CH2-OH is condensed with para-HOOC-C6H4-COOH. Which polymer linkage and name are correct?

(A) Only C-C; Buna-N. (B) Ester; PET/terylene.

(C) Amide; nylon-6,6. (D) Glycosidic; glycogen.

36. Acidifying 1/25 mol borax theoretically gives four H3BO3 per borax formula unit. The isolated boric acid requires 12/125 mol NaOH for complete titration in the presence of glycerol, where each H3BO3 behaves as monobasic. What fraction of the theoretical boric acid was recovered?

(A) 3/5 (B) 3/10

(C) -3/5 (D) 6/5

37. An internally consistent thermochemical comparison gives total resonance stabilisation of phenanthrene as 92 kcal/mol (3 benzenoid rings) and anthracene as 84 kcal/mol (3 rings). Which statement correctly separates total stabilisation from stabilisation per ring?

(A) Both quantities must rank compounds identically because each is aromatic. (B) Total: anthracene; per ring: phenanthrene.

(C) Total: phenanthrene; per ring: anthracene. (D) Total: phenanthrene; per ring: phenanthrene.

38. For octa-2,5-diene, CH3-CH=CH-CH2-CH=CH-CH2-CH3, count distinct configurational geometrical isomers. Ignore conformers and photochemical interconversion. Where reversing the entire carbon chain identifies two E/Z assignments, count them once.

(A) 4 (B) 3

(C) 5 (D) 8

39. A water sample contains 3/1000 mol Ca(HCO3)2 and 1/500 mol CaSO4 and no other hardness source. On quantitative boiling, Ca(HCO3)2→CaCO3(s)+CO2+H2O; CaSO4 stays dissolved. Which pair gives the CaCO3 precipitate mass and remaining hardness type? CaCO3=100 g/mol.

(A) Zero; boiling can never remove temporary hardness. (B) 3/20 g; only temporary hardness remains.

(C) 1/2 g; no hardness remains. (D) 3/10 g; permanent hardness remains.

40. Prolonged reduction of glucose withHI gives n-hexane. Which structural conclusion is supported, and which is not established by that observation alone?

(A) The chain is branched and every OH configuration is known.

(B) The six carbons form an unbranched chain; the completeOH stereochemistry is not fixed.

(C) The product proves that glucose originally had six carbonyl groups.

(D) Glucose contains an aromatic benzene ring.

41. For $AgCl(s)+2NH_3 \rightleftharpoons [Ag(NH_3)_2]^+ + Cl^-$,Ksp×Kf=1/4. Before AgCl is added,a solution has $\frac{1}{100}$M silver-ammine complex,zero chloride and $\frac{1}{10}$M free NH_3. If all dissolved silver stays complexed and volume is fixed,the extra AgCl solubility s in M is

(A) $\frac{1}{20}$ (B) $\frac{1}{4}$

(C) $\frac{1}{44}$ (D) $\frac{1}{40}$

42. A primary-standard sample of pure anhydrous Na_2CO_3 weighs $\frac{53}{100}$ g. It is titrated with0.100M HCl. Find the additional acid volume in mL needed after the first carbonate-to-hydrogencarbonate equivalence point to reach complete conversion to CO_2/water. Take Na_2CO_3=106g/mol.

(A) 101 (B) 50

(C) 25 (D) 100

43. For ideal gases at one T, $A(g) \rightleftharpoons 2B(g)$ has Kc=4 and numerical RT=2 in compatible units. What is Kp for$4B(g) \rightleftharpoons 2A(g)$? Use $Kp=Kc(RT)^{\Delta n}$ with the same standard-state convention throughout.

(A) 1/128 (B) -1/64

(C) 1/32 (D) 1/64

44. For a fixed-composition closed system dG=V dP-S dT. At a state use V=5 m3, S=3 J/K and a local path with dT/dP=-2 K/Pa. Find dG/dP along this path in m3. The supplied values are a formal local thermodynamic model; 1 Pa m3=1 J.

(A) 11 (B) 22

(C) -11 (D) 11/2

45. A nucleotide chain links the 3-prime oxygen of one sugar through phosphate to the 5-prime oxygen of the next. What is the connection, and what would its cleavage directly affect?

(A) A phosphodiester link; the covalent chain continuity.
(B) A glycosidic glucose-glucose bond; starch branching.
(C) A peptide link; only amino-acid side chains.
(D) A sulphur bridge; only cysteine crosslinks.

46. A buffer contains 1/25 mol HA and 9/100 mol A-. After adding strong base without volume change it has pH=pKa+log10(12), with HA still present. How many moles of OH- were added? Quantitative neutralisation and the Henderson ratio model apply.

(A) 3/200
(B) 9/100
(C) 1/25
(D) 3/100

47. For the displayed structurep-HO-C6H4-NHCOCH3, which change would reverse N-acetylation while preserving the ring substituent positions?

(A) Hydrolyse the amide to recover p-aminophenol and ethanoic acid.
(B) Hydrolyse a nonexistent ester to obtain two sugars.
(C) Oxidise a nonexistent aldehyde to form starch.
(D) Replace the aromatic ring by chloride ions.

48. NH3 is usually more basic than PH3 and AsH3. Which explanation addresses the actual property?

(A) Lone-pair availability for accepting a proton changes down the group; hydride basicity cannot be read from molar mass alone.

(B) Proton acceptance requires replacement of nitrogen by phosphorus.

(C) A molecule with a lone pair must have identical basicity to every other lone-pair molecule.

(D) Basicity is determined only by molecular mass.

49. Within an additive covalent-radius model,a single X-X bond is 130 pm and an X-H bond is 100 pm. A linear H-X-X-H skeleton is assumed with collinear single bonds. Its terminal H-to-H distance,in pm,is

(A) 230
(B) 330
(C) 200
(D) 660

50. A student identifies the uncertainty principle as an original Bohr postulate because both are discussed in atomic structure. Which correction is appropriate?

(A) Bohr postulated that every atomic energy is continuous.

(B) Bohr postulated quantised stationary states; uncertainty is a later quantum restriction on simultaneous observables.

(C) Uncertainty says that electrons have no measurable properties.

(D) Stationary states require a zero nuclear charge.

51. A Fischer projection of 2,3-dibromobutane has its four-carbon chain vertical (CH3 at top and CH3 at bottom). At C2,Br is on the right; at C3,Br is on the left; each other horizontal group is H. Horizontal bonds point toward the observer. Which classification and configuration pair is correct?

(A) chiral; (2R,3R)
(B) meso; (2S,3S)
(C) chiral; (2S,3S)
(D) meso; (2R,3S)

52. A $\frac{1}{20}$ mol FeC_2O_4 inventory is analysed. A fraction $\frac{1}{4}$ of Fe^{2+} has already become Fe^{3+},but every oxalate remains unchanged. Subsequent complete oxidation gives Fe^{3+} and CO_2. Electron equivalents released,in mol,are

(A) $\frac{7}{80}$ (B) $\frac{11}{80}$

(C) $\frac{19}{80}$ (D) $\frac{3}{20}$

53. An orbital has 2 angular nodes and 2 radial nodes. Maximum electrons in its entire subshell are

(A) 14 (B) 2

(C) 10 (D) 50

54. In a closed disproportionation, the only oxidation states of the redox-active element are initially Mn(VI) and finally Mn(VII) plus Mn(IV). No other element changes oxidation state. What fraction of the redox-active atoms ends in the higher oxidation state? Apply electron balance rather than counting other atoms.

(A) 4/3 (B) 1/3

(C) 2/3 (D) -2/3

55. A glycosidic linkage joins glucoseC1 and fructoseC2. Which structural reason explains nonreducing behaviour of the intact molecule?

(A) Both anomeric centres are consumed in the connection. (B) The formula contains no oxygen.

(C) A free glucoseC1OH remains by definition. (D) Neither named carbon was ever a carbonyl carbon.

56. CCl4 and CHCl3 are both tetrahedral about carbon. Why does identical central geometry not force identical molecular dipoles?

(A) Every tetrahedral molecule is nonpolar regardless of substituents.

(B) CCl4 has no polar C-Cl bond.

(C) CCl4 has four equivalent bonds with a cancelling vector sum; CHCl3 has inequivalent substituents and a nonzero resultant.

(D) CHCl3 must be linear because it contains hydrogen.

57. A gas energy density is u=(3/2)cRT with c in mol/m3. Which dimensions make this expression consistent?

(A) c has units of m3/mol. (B) Temperature must have energy units for this equation to be valid.

(C) R has J mol-1 K-1, giving u in J/m3. (D) R has J K-1, giving an unavoidable extra mole.

58. A solution contains $\frac{7}{1000}$ mol I_2. Addition of $\frac{1}{500}$ mol SO_3^{2-} consumes iodine by $I_2 + SO_3^{2-} + H_2O \rightarrow 2I^- + SO_4^{2-} + 2H^+$. The residual iodine is titrated by $I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$. Find the thiosulfate requirement in mol.

(A) $\frac{1}{200}$ (B) $\frac{1}{100}$

(C) $\frac{1}{250}$ (D) $\frac{7}{500}$

59. Nonrelativistic ions A and B start from rest and gain only electric kinetic energy. B has 6 times A's mass and 5 times its positive charge. To make λ_B/λ_A=1/6, what must V_B/V_A be?

(A) 36/5 (B) 5/6

(C) 7 (D) 6/5

60. At25°C pH=4301/500. Use pKw=14 and log10(4)=301/500. What is [OH-] in mol/L?

(A) -1/250000 (B) 1/250000

(C) 1/500000 (D) 1/125000

English Proficiency

61. The scholarship fund provides him with support; he is neither its donor nor its manager. What is he?

(A) sponsor (B) trustee

(C) beneficiary (D) benefactor

62. Which ONE of the following words is misspelt?

(A) restaurant (B) certanly

(C) successful (D) parallel

63. Read the sentence: The report linked the town's prosperity to its thriving local businesses. Choose the opposite in meaning of 'prosperity' as used in the sentence.

(A) hardship (B) growth

(C) wealth (D) employment

64. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. rather than the wording of its message, Q. the turn beside the bridge. R. It was the position of the sign, S. that caused visitors to miss

(A) P-R-S-Q (B) Q-P-S-R

(C) S-R-P-Q (D) R-P-S-Q

65. Read the sentence: The introduction was a map of the argument, showing where the main questions would be addressed. What does the figurative expression mean in this context?

(A) It reproduced the argument word for word. (B) It answered every question before the argument began.

(C) It outlined the organisation and direction of the argument. (D) It replaced the evidence in the later sections.

66. Complete: The express lift moved so _____ that the attendant reached the top before the stairs were cleared.

(A) fast (B) fastness

(C) fastest (D) fastly

67. Read the sentence: Among the grey folders, the bright orange one was conspicuous. Choose the closest meaning of 'conspicuous' as used in the sentence.

(A) deliberately displayed (B) easily noticed

(C) out of place in every respect (D) surprisingly valuable

68. The research examines animal classification and behaviour, rather than the earth's rocks or the weather. Which field is it?

(A) geology (B) meteorology

(C) botany (D) zoology

69. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. the carpenter examined its joints Q. could be repaired safely. R. Rather than discard the damaged table, S. to determine whether the frame

(A) R-P-S-Q (B) P-R-S-Q

(C) S-R-P-Q (D) Q-P-S-R

70. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The school is introducing the new procedure gradually rather than all at once.

(A) The new procedure is introducing the school gradually.

(B) The new procedure is being introduced gradually by the school rather than all at once.

(C) The new procedure was being introduced gradually by the school.

(D) The new procedure has been introduced all at once by the school.

Logical Reasoning

71. Consider only these statements; universal statements alone do not imply existence. All campers are guides; No guides are campers. Conclusions: I. No inventors are campers. II. Some inventors are not guides. Which conclusions follow necessarily?

(A) Only II (B) Both I and II

(C) Neither I nor II (D) Only I

72. Kiran points to a photograph and says, "This person is my mother's only daughter." Kiran is male, and the photographed person is female. How is the photographed person related to Kiran?

(A) Brother (B) Sister

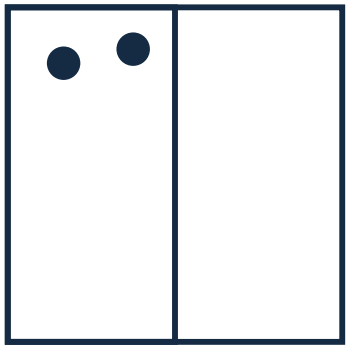
(C) Son (D) Daughter

73. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 6, 12, ?, 42, 51, 102, 111.

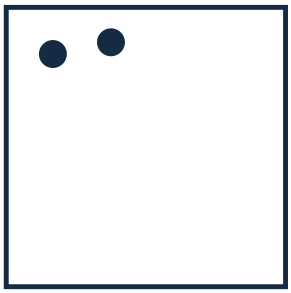
(A) 18 (B) 21

(C) 3 (D) 15

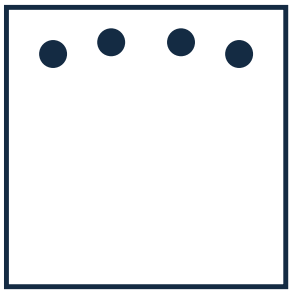
74. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



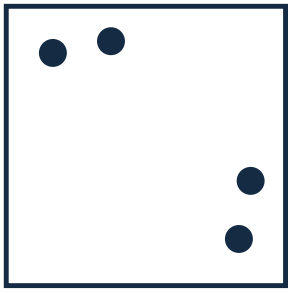
(A) Pattern Z



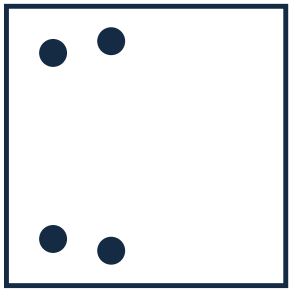
(B) Pattern R



(C) Pattern Q



(D) Pattern P



75. Three pairs contain exactly one string instrument and one wind instrument. Which pair does not?

(A) Violin / Trombone

(B) Ukulele / Trumpet

(C) Sitar / Oboe

(D) Guitar / Harp

76. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern R



(B) Pattern L



(C) Pattern T

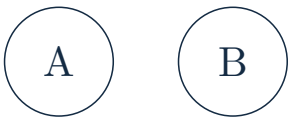


(D) Pattern M

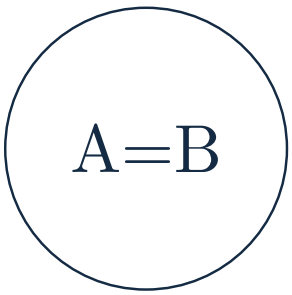


77. Choose the diagram that represents all these conditions, including the strictness information: Some A is B, some A is not B, and some B is not A.

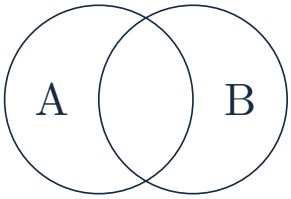
(A) Pattern N



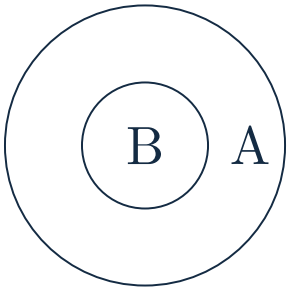
(B) Pattern L



(C) Pattern Q



(D) Pattern M

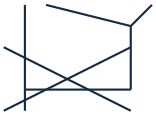


78. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

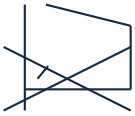




host 1



host 2



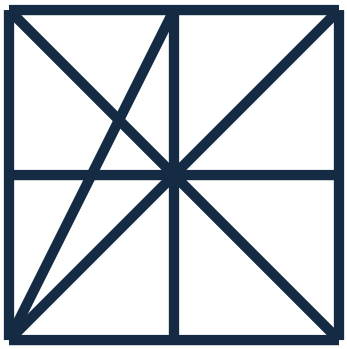
host 3



host 4

(A) 2
(B) 1
(C) 3
(D) 4

79. How many triangles are formed by all the drawn segments?

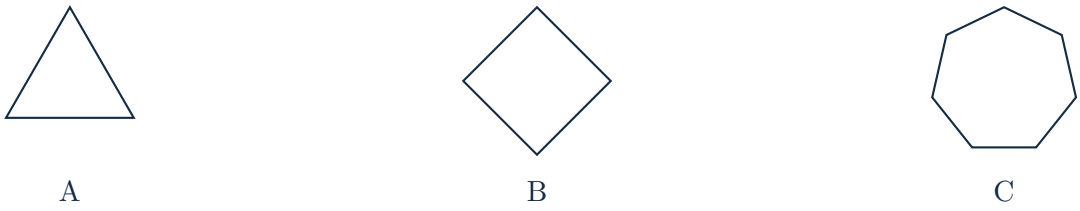


(A) 29
(B) 28
(C) 21
(D) 19

80. Preserve the exact relation and its order: Begin : Commence :: ? : Rare. Choose the missing term.

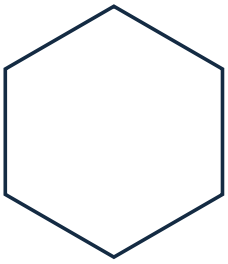
(A) Silent
(B) Rapid
(C) Scarce
(D) Begin

81. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?

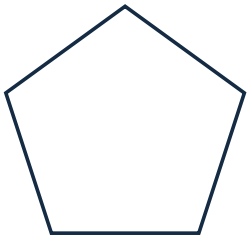


A B C

(A) Pattern X



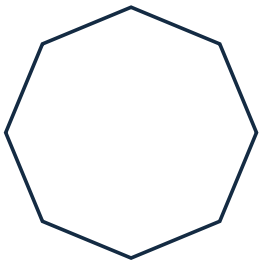
(B) Pattern T



(C) Pattern U



(D) Pattern Y



82. A walker goes 5 m south, then 9 m east, then turns left and walks 12 m. In which direction is the starting point from the walker?

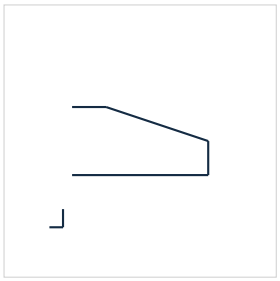
(A) North-west

(B) South-east

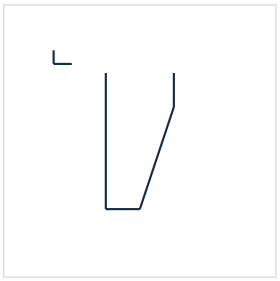
(C) South-west

(D) North-east

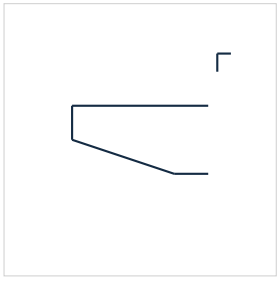
83. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

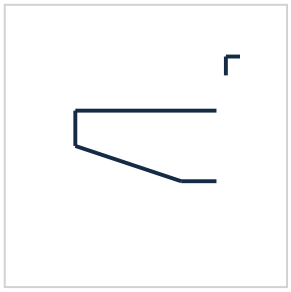


bottom left

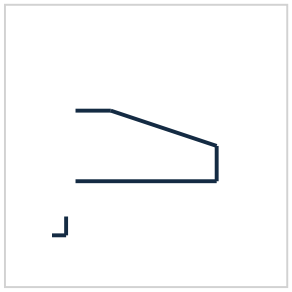
?

bottom right

(A) Pattern W



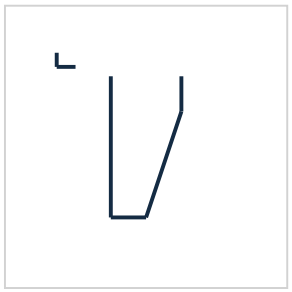
(B) Pattern Z



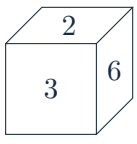
(C) Pattern Q



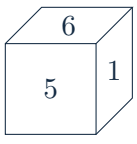
(D) Pattern X



84. The views show the same die. Give the ordered pair: (face opposite 5, face opposite 3). Do not assume opposite labels sum to seven.



1



2

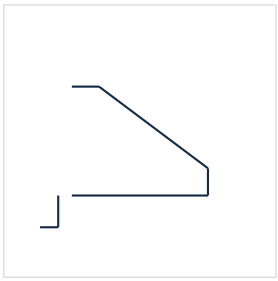
(A) $(2, 1)$

(B) $(2, 4)$

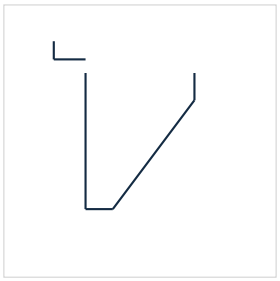
(C) $(2, 3)$

(D) $(2, 2)$

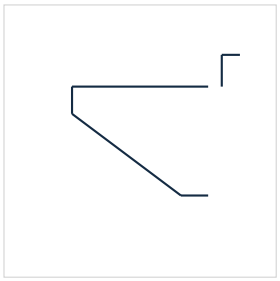
85. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

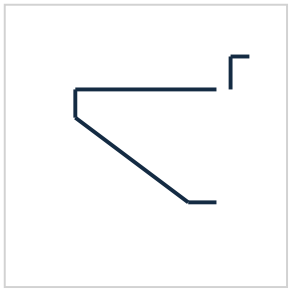


2

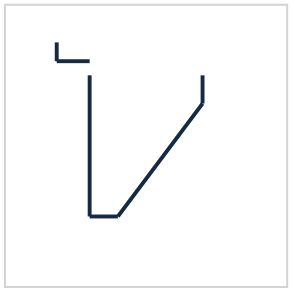


3

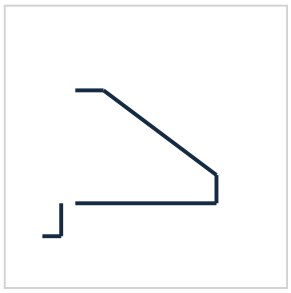
(A) Pattern R



(B) Pattern W



(C) Pattern K



(D) Pattern L



86. In a row, Asha is 6th from the left and 43rd from the right. Bina is 10th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 31 (B) 29

(C) 28 (D) 35

87. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: ZMZT, AJDV, ?, CDLZ, DAPB, EXTD.

(A) BGHX (B) XHGB

(C) AFGW (D) CHIY

88. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, GRAPE is coded as HSBQF. Which word is coded as SJWFS?

(A) TKXGT (B) SJWFS

(C) QHUDQ (D) RIVER

89. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



(A) Pattern Q



(B) Pattern U



(C) Pattern X



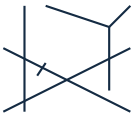
(D) Pattern Y



90. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.




host 1


host 2


host 3


host 4

(A) 2
(B) 1
(C) 3
(D) 4

Mathematics

91. An AP has first term 5, sum of its first 9 terms equal to 153. Find the sum of the squares of those 9 terms.

(A) 2300
(B) 3141
(C) 225
(D) 23409

92. Unit vectors a,b,c satisfy |(a×b)·c|=1. Find |a+7b-3c|^2.

(A) 11 (B) 58

(C) 101 (D) 59

93. The circle $(x-(-5))^2+(y-(-7))^2=900$ has a normal with direction ratio 3 : 4. Its point of contact P satisfies $x_P>-5$. Find x_P+y_P.

(A) 18 (B) −54

(C) 30 (D) −18

94. X takes values 0,4,6 with probabilities 1/2,1/3,k. Find Var(X).

(A) $\frac{34}{3}$ (B) $\frac{53}{9}$

(C) $\frac{49}{9}$ (D) $\frac{\sqrt{53}}{3}$

95. $A=\begin{bmatrix}2 & 1\\ 0 & 7\end{bmatrix}$, $P=\begin{bmatrix}1 & 4\\ 0 & 1\end{bmatrix}$. Find the (1,2) entry of $P^{-1}AP$.

(A) −19 (B) 21

(C) −20 (D) 1

96. For xy-7x-3y+21=0, find the squared length of the diagonal of the rectangle formed with the axes.

(A) 58 (B) 21

(C) $\sqrt{58}$ (D) 100

97. ABC is right angled at A with AB=10,AC=12. A vertical tower stands at the midpointM ofBC. Its top elevation from A is 60°. Find the square of the tower height.

(A) 61 (B) 183

(C) $\sqrt{183}$ (D) $61\sqrt{3}$

98. If α,β are roots of x^2-18x+77=0, find $\alpha^2+\beta^2$ without solving the quadratic.

(A) 478 (B) 247

(C) 170 (D) 16

99. A=(5,5,0), C=(8,11,3). B lies on AC with AB:BC=2:1. Find |OB|^2, where O is the origin.

(A) 50 (B) 54

(C) 134 (D) 194

100. Evaluate $\int_0^3 \Sigma\ \mathrm{arccot}((x+j)^2+(x+j)+1)dx$, j=0,...,1. Use arccot in $(0,\pi)$.

(A) $-3\,\mathrm{atan}\,(3)-2\,\mathrm{atan}\,(2)-\frac{\log(13)}{2}+\log(5)+5\,\mathrm{atan}\,(5)$

(B) $-2\,\mathrm{atan}\,(2)+\log\left(\frac{\sqrt{130}}{26}\right)+5\,\mathrm{atan}\,(5)$

(C) $-6\,\mathrm{atan}\,(3)-\frac{\pi}{2}+\log\left(\frac{20}{17}\right)+8\,\mathrm{atan}\,(4)$

(D) $-5\,\mathrm{atan}\,(5)-\log(5)+\frac{\log(13)}{2}+2\,\mathrm{atan}\,(2)+3\,\mathrm{atan}\,(3)$

101. $\theta \in(\pi/2,\pi)$, $\sin\theta=\frac{4\sqrt{41}}{41}$. Find $\cos(\theta/2)$.

(A) $\frac{\sqrt{410\sqrt{41}+3362}}{82}$ (B) $-\frac{5\sqrt{41}}{82}$

(C) $\frac{\sqrt{3362-410\sqrt{41}}}{82}$ (D) $-\frac{\sqrt{3362-410\sqrt{41}}}{82}$

102. The parabola $(y-2)^2=8(x-2)$ has focus F and vertex V. A circle centred at F passes through V. Find the tangent length from P=(8,6).

(A) $4\sqrt{2}$ (B) 28

(C) $2\sqrt{3}$ (D) $2\sqrt{7}$

103. A circular sector has radius 6 and arc length π. Find its area.

(A) $\frac{\pi^2}{12}$ (B) $\frac{\pi}{12}$

(C) 18π (D) 3π

104. $F'(x)=\frac{1}{x^2}-\frac{1}{x^2\sqrt{x^2+1}}$ and F(1)=3. Find F(3)-2F(1).

(A) $-\frac{11}{3}-\frac{\sqrt{10}}{3}+\sqrt{2}$ (B) $-\frac{7}{3}-\sqrt{2}+\frac{\sqrt{10}}{3}$

(C) $-\sqrt{2}+\frac{\sqrt{10}}{3}+\frac{11}{3}$ (D) $-\sqrt{2}+\frac{2}{3}+\frac{\sqrt{10}}{3}$

105. Let $\alpha=\pi/6$, $\beta=\pi/5$. Evaluate $\arccos(-\cos\alpha\cos\beta-\sin\alpha\sin\beta)$.

(A) $\frac{31\pi}{30}$ (B) $\frac{29\pi}{30}$

(C) $\frac{\pi}{30}$ (D) $-\frac{\pi}{30}$

106. For $x>1$ let $f(x)=\arctan(2x/(1-x^2))$, with arctan in $(-\pi/2,\pi/2)$. Find the integral from 7 to 8 of $f'(x)\,dx$.

(A) $2\,\mathrm{atan}\left(\frac{1}{7}\right)$

(B) $-2\,\mathrm{atan}\,(7)+2\,\mathrm{atan}\,(8)$

(C) $-\,\mathrm{atan}\,(7)+\mathrm{atan}\,(8)$

(D) π

107. 7 distinct objects are distributed into three nonempty unlabelled boxes. How many distributions are possible?

(A) 1806 (B) 364

(C) 301 (D) 903

108. $y'-y/x=1$ for $x>0$, $y(1)=5$. Find $y(3)/3-y(1)$.

(A) $\log(3)$ (B) $3\log(3)+15$

(C) $\log(3)+5$ (D) $-\log(3)$

109. A 8-element set contains distinct a,b. How many 4-element subsets contain exactly one of a,b and also contain a?

(A) 20 (B) 64

(C) 15 (D) 40

110. X earns 6 and uses 2 labour hours; Y earns 5 and uses 3 hours. There are 110 hours and at least 5 items of Y must be made. Which model is correct?

(A) Maximize $6x+5y$; $2x+3y \le 110$, $x \ge 0$, $y \ge 5$

(B) Minimize $6x+5y$; $2x+3y \le 110$, $y \ge 5$

(C) Maximize $5x+6y$; $2x+3y \le 110$, $y \le 5$

(D) Maximize $6x+5y$; $2x+3y \ge 110$, $x, y \ge 0$

111. A=(6,4,2), B=(12,4,6). Find the squared length of the component of AB perpendicular to direction (1,2,2).

(A) $\frac{196}{9}$

(B) 52

(C) $\frac{142}{3}$

(D) $\frac{272}{9}$

112. Let $D(t) = \det \begin{pmatrix} 10+t & 1+t \\ 1+t & 3+t \end{pmatrix}$. Find D′(0).

(A) 15

(B) -11

(C) 13

(D) 11

113. For $P(x) = \prod_{j=3}^{8}(x+j)$, find the coefficient of x^3 in $P(x)/(x+3)$.

(A) 445

(B) 710

(C) 355

(D) 436

114. Two different numbers are chosen uniformly from {1,...,8}. Let X be their maximum. Find Var(X).

(A) 3

(B) 39

(C) $\sqrt{3}$

(D) 6

115. Find the area bounded by y=x, y=81/x, and x=36 in the first quadrant.

(A) $\frac{1215}{2}$

(B) $162\log(2)$

(C) $243 - 162\log(2)$

(D) $\frac{1215}{2} - 162\log(2)$

116. Using digits 0,1,2,3,4,5,6 without repetition, how many 4-digit positive integers are divisible by 3 and have their first digit smaller than their last digit?

(A) 112

(B) 264

(C) 108

(D) 107

117. For A={1,...,7}, how many (a,b)∈A×A satisfy a+b even and a<b?

(A) 18

(B) 12

(C) 24

(D) 9

118. There are 10 Boolean variables p_1,...,p 10. Exactly 2 variables must be true. How many truth assignments satisfy [(p_1 implies p_2) and (p_1 implies not p_2)]?

(A) 36

(B) 1024

(C) 512

(D) 45

119. Find the coefficient of x^7 in (1+5x)exp(2x).

(A) $\frac{8}{315}$ (B) $\frac{148}{315}$

(C) $\frac{4}{9}$ (D) $\frac{16}{105}$

120. Evaluate $\int_0^4$ x floor(x) dx.

(A) $\frac{64}{3}$ (B) 14

(C) 17 (D) 6

121. For $A = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$ and $C=7A^{-1}$, find C_{22}.

(A) 4 (B) -1

(C) 2 (D) $\frac{4}{7}$

122. z=6+4i. Find 1/z+1/conjugate(z).

(A) 12 (B) $\frac{3}{13}$

(C) $\frac{1}{12}$ (D) $\frac{2}{13}$

123. An AP has a_1=4, d=8. Find $a_1+a_3+\cdots+a_17$.

(A) 1156 (B) 324

(C) 480 (D) 612

124. Find the coefficient of x^8 in $(1 + x + x^2)(1 + x^2)^{10}$.

(A) 462 (B) 630

(C) 330 (D) 210

125. Four observations are 11,11,x,y with population mean 11 and variance 16/2. Find xy.

(A) 105 (B) 89

(C) 137 (D) 121

126. For the loop $y^2=x^2(6-x)$, find its area within the strip $6/3 \le x \le 12/3$.

(A) $8 + 8\sqrt{2}$ (B) $\frac{32}{3} - \frac{8\sqrt{2}}{3}$

(C) $\frac{96}{5} - \frac{32\sqrt{2}}{5}$ (D) $\frac{192}{5} - \frac{64\sqrt{2}}{5}$

127. $\log_2 a=2$, log_a b=3, log_b c=5. Find $\log_4(c^2/a)$.

(A) 58 (B) 29

(C) 30 (D) 31

128. The tangent to $y = x^3 - 60x$ at $x = 8$ meets the two coordinate axes at A and B. Find the area of triangle OAB, where O is the origin.

(A) $\frac{65536}{33}$ (B) $\frac{8192}{3}$

(C) $\frac{131072}{33}$ (D) $\frac{262144}{33}$

129. A convergent GP has first term 11, ratio $\frac{1}{9}$. Find the sum of all terms after its first 8 terms.

(A) $\frac{11}{344373768}$ (B) $\frac{11}{38263752}$

(C) $\frac{11}{34012224}$ (D) $\frac{59189240}{4782969}$

130. How many integer t make P=(t,0) strictly inside the larger segment cut by x=2 from $x^2+y^2=25$?

(A) 2 (B) 8

(C) 9 (D) 6

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A 3 kg block rests on a 8 kg block on a smooth horizontal floor. A force 55 N pulls the lower block. There is no slipping between the blocks. During a displacement 3 m, what work does friction do on the upper block?

(A) 45 J
(B) $\frac{495}{8}$ J
(C) 120 J
(D) $\frac{165}{2}$ J

132. Water flows steadily from a pipe section of area A at 5 m/s to a section of area $A/4$ located 4 m higher. Neglect viscosity. Find $p_1 - p_2$. Take density 1000 kg/m^3 and $g = 10$ m/s^2.

(A) 240000 Pa
(B) 187500 Pa
(C) 227500 Pa
(D) 115000 Pa

133. Unpolarised light passes three ideal polarisers with axes at 0°, 30°, and 90° relative to a common reference. Find final intensity divided by incident intensity.

(A) $\frac{\sqrt{3}}{8}$
(B) $\frac{3}{32}$
(C) $\frac{3}{8}$
(D) $\frac{3}{16}$

Chemistry

134. Two van der Waals gases obey Tc=8a/(27Rb) and Pc=a/(27b^2). Their measured ratios are Tc,A/Tc,B=6/5 and Pc,A/Pc,B=6/25. Find (a_A/a_B, b_A/b_B).

(A) (6, 5/2)
(B) (12, 5)
(C) (6, 5)
(D) (5, 6)

135. A dry mixture contains only anhydrous MgCO3 and BaCO3. Use the ordinary Group2 carbonate stability trend. In an idealised selective-heating window, the less thermally stable carbonate decomposes completely by MCO3→MO+CO2, while the more stable carbonate remains unchanged. No other mass loss occurs. This is the stated model, not a specified real temperature. Atomic masses: Mg=24, Ba=137, C=12, O=16. The initial mixture mass is 1011/200 g and the solid residue after selective heating is 791/200 g. What was the initial mass fraction of BaCO3?

(A) 140/337
(B) 197/337
(C) 26/1011
(D) 3/8

136. For CH3COCH2COCH3 and CH3COCH3,which structural-reactivity comparison is correct under ordinary enolate/aldol conditions?

(A) the central CH2 in the diketone is more acidic
(B) all H in both are equally acidic
(C) acetone methyl H is always more acidic
(D) the central CH2 is not alpha to carbonyl

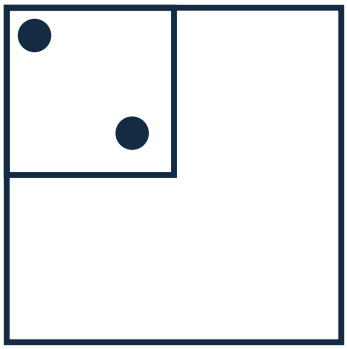
Logical Reasoning

137. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 17$ and $5 \mapsto 35$. Which input maps to 83?

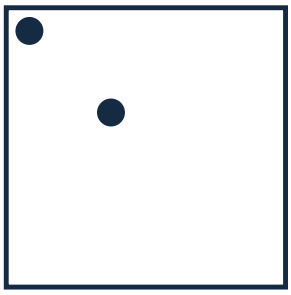
(A) 13 (B) 12

(C) 14 (D) 83

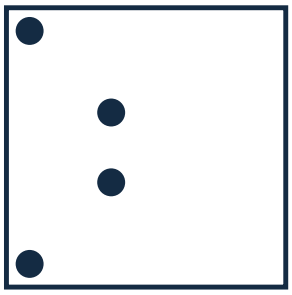
138. Starting with a unit square sheet, fold right to left along the vertical midline, then fold bottom to top along the horizontal midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



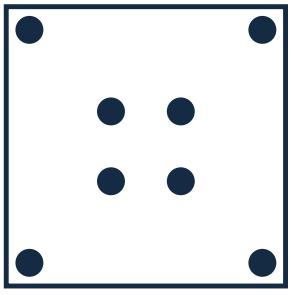
(A) Pattern R



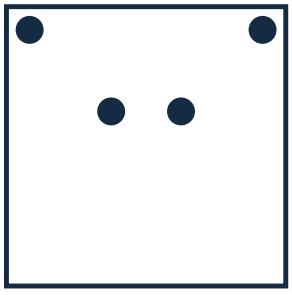
(B) Pattern M



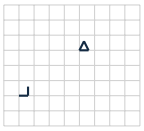
(C) Pattern Y



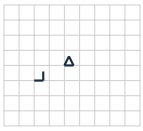
(D) Pattern U



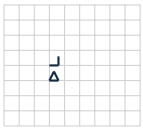
139. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



1

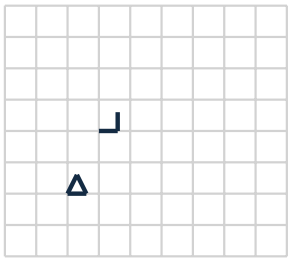


2

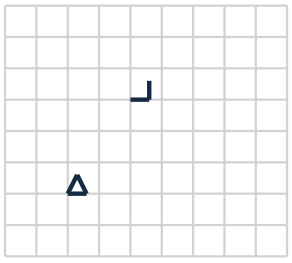


3

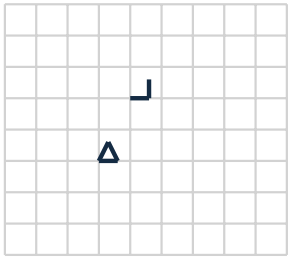
(A) Pattern U



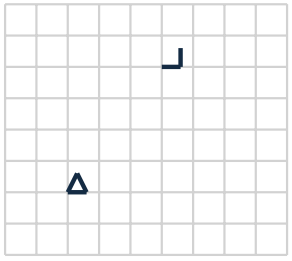
(B) Pattern M



(C) Pattern P



(D) Pattern L



Mathematics

140. $A = \begin{pmatrix} 0 & -4 \\ 4 & 0 \end{pmatrix}$, B=(I-A)(I+A)$^{-1}$. Find det B.

(A) 1

(B) 17

(C) -1

(D) $\frac{1}{17}$

141. Parabola y=4(x-2)(x-3) meets the axes at three distinct points. A circle passes through these points. Find its other y-intercept.

(A) $-\frac{1}{4}$

(B) 1

(C) $\frac{1}{4}$

(D) 4

142. A chord of $x^2+y^2=12100$ touches $x^2/121-y^2/36=1$. Its nonzero midpoint (h,k) satisfies h=k. Find h^2+k^2.

(A) $\frac{85}{4}$

(B) $\frac{85}{2}$

(C) 85

(D) $\frac{157}{2}$

BITSAT Mock 38

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A dipole of fixed moment 2 C m is rotated quasistatically from 60° to 120° relative to a uniform electric field 3 N/C. Find external work.

(A) 3 J
(B) 6 J
(C) 2π J
(D) $6\sqrt{3}$ J

2. In YDSE wavelength is 600 nm, screen distance 5 m and slit separation 5 mm. Find the separation on the same side of the centre between the 4th bright fringe and the 7th dark fringe. Count the first dark fringe as order 1.

(A) $\frac{21}{10}$ mm
(B) $\frac{12}{5}$ mm
(C) $\frac{3}{4}$ mm
(D) $\frac{3}{2}$ mm

3. Light of vacuum wavelength 200 nm passes from a medium of index 4 into one of index 5. Find the ratio of its frequencies in the second and first media.

(A) $\frac{1}{5}$
(B) 5
(C) $\frac{4}{5}$
(D) 1

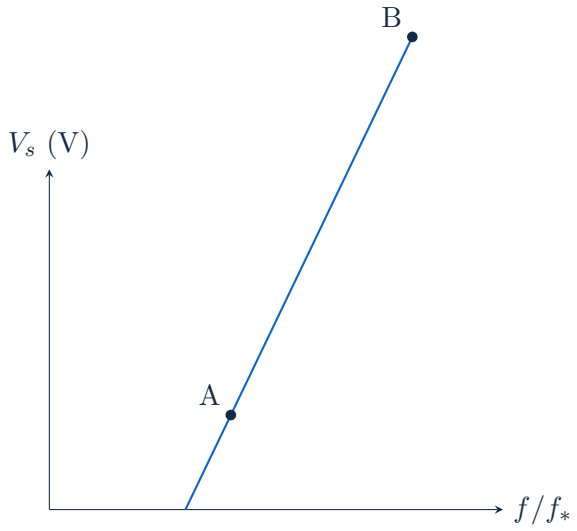
4. A car negotiates a level circular turn at its maximum non-slipping speed u when $\mu_s = \frac{6}{5}$. Rain lowers the coefficient to $\frac{1}{10}$ and the turn radius stays fixed. Find the new maximum speed divided by u.

(A) $\frac{\sqrt{3}}{6}$
(B) 1
(C) $\frac{\sqrt{10}}{10}$
(D) $\frac{1}{12}$

5. A particle is projected vertically outward from the surface of a nonrotating planet with speed $v_e/3$, where v_e is surface escape speed. Neglect atmosphere. Find maximum height divided by planet radius.

(A) $\frac{1}{8}$
(B) $\frac{1}{10}$
(C) 8
(D) $\frac{1}{9}$

6. For a metal, the stopping-potential graph has points A $(4f_*, 5\,\text{V})$ and B $(8f_*, 25\,\text{V})$. The electronic gain of the vertical recording channel is later found to be 2: its recorded voltage is 2 times the true stopping voltage. What is hf_*/e in volts?



(A) $\frac{5}{8}$ V
(B) $\frac{5}{2}$ V
(C) 10 V
(D) $\frac{25}{16}$ V

7. A bar magnet of moment 2 A m^2 is rotated quasistatically from 60° to 120° relative to a uniform 3 T magnetic field. Find external work.

(A) $6\sqrt{3}$ J
(B) 3 J
(C) 6 J
(D) $3\sqrt{3}$ J

8. A particle of mass 1 kg starts from rest and experiences $F(t) = 5t$ N along its motion for 4 s. Find the work done by the force.

(A) 800 J
(B) 1600 J
(C) 160 J
(D) 3200 J

9. At the highest point of its flight, at height 20 m, a projectile has horizontal velocity 30 m/s. It explodes into two equal fragments, both initially moving horizontally. One moves forward at 3 m/s. Find the horizontal distance travelled by the other fragment from the explosion point to the ground. Take $g = 10$ m/s^2.

(A) 6 m
(B) 114 m
(C) 126 m
(D) 228 m

10. A balanced Wheatstone bridge has arms AB=AD=3 ohms and BC=DC=9 ohms. A resistor of arbitrary positive resistance connects B to D. Find equivalent resistance between A and C.

(A) 6 Ω
(B) 24 Ω
(C) $\frac{21}{2}$ Ω
(D) $\frac{9}{4}$ Ω

11. The quantity $Z = x^2y^2/\sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 1, 2 and 4, respectively. To first order, what is the maximum percentage error in Z?

(A) 8 (B) 4

(C) 10 (D) 6

12. A converging thin lens forms a real image with magnification magnitude 4. The object-image separation is 30 cm. Find focal length.

(A) 10 cm (B) $\frac{24}{5}$ cm

(C) $\frac{15}{2}$ cm (D) 6 cm

13. A uniform rod of mass 2 kg and length 6 m lies on a smooth horizontal table, free to rotate about a fixed vertical pivot at its centre. A point mass 3 kg moving perpendicular to the rod at 9 m/s strikes an end and sticks. Find the angular speed immediately afterward.

(A) $\frac{54}{11}$ rad/s (B) $\frac{27}{38}$ rad/s

(C) $\frac{27}{11}$ rad/s (D) $\frac{27}{17}$ rad/s

14. A particle executes SHM at frequency 3 Hz. At t=0 it is at its positive extreme. In the interval $0 <= t < 1$ second, how many times does its kinetic energy equal its potential energy?

(A) 7 (B) 12

(C) 6 (D) 13

15. A block released from rest takes time t_s to slide a fixed distance down a smooth incline and time $t_r = (9/6)t_s$ down an otherwise identical rough incline. The incline has $\tan\theta = \frac{3}{4}$. Find the kinetic-friction coefficient.

(A) $\frac{3}{4}$ (B) $\frac{1}{3}$

(C) $\frac{5}{12}$ (D) $\frac{20}{27}$

16. A uniform disc of mass 3 kg and radius 4 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) 18 kg m^2 (B) $\frac{57}{2}$ kg m^2

(C) 24 kg m^2 (D) $\frac{39}{2}$ kg m^2

17. A series RLC circuit is resonant at angular frequency ω_0. Its capacitance is then multiplied by 49 while L stays fixed. What angular frequency restores resonance, as a fraction of ω_0?

(A) 7 (B) $\frac{\sqrt{7}}{7}$

(C) $\frac{1}{49}$ (D) $\frac{1}{7}$

18. An ideal mixture contains 2 moles of monatomic gas and 5 moles of rigid diatomic gas. Vibrations are frozen. Find the heat-capacity ratio $\gamma = C_P/C_V$ of the mixture.

(A) $\frac{39}{25}$ (B) $\frac{7}{5}$

(C) $\frac{45}{31}$ (D) $\frac{5}{3}$

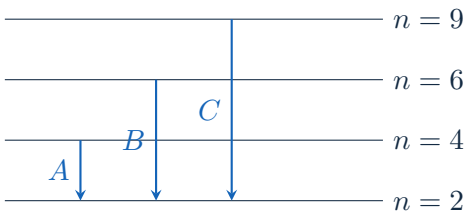
19. Two ideal cells of emfs 15 V and 2 V oppose each other in a loop containing two resistors each 5 ohms. Find the current magnitude.

(A) $\frac{13}{5}$ A
(B) $\frac{17}{5}$ A
(C) $\frac{1}{5}$ A
(D) $\frac{13}{10}$ A

20. Light is incident from index $\frac{13}{10}$ onto index 2. At one incidence angle the reflected beam is completely plane polarised. Find the tangent of that angle.

(A) $\frac{20}{7}$
(B) $\frac{20}{13}$
(C) $\frac{13}{10}$
(D) $\frac{\sqrt{231}}{13}$

21. The diagram shows downward hydrogen transitions A, B and C from $n = 4$, 6 and 9, respectively, ending at $n = 2$. A filter transmits only photons with energy strictly above that of transition B. Which transmitted line has the longest wavelength?



(A) B
(B) No transmitted line
(C) A
(D) C

22. In an abrupt junction, the acceptor concentration is 676 times the donor concentration. Within the depletion approximation, which statement gives x_n/x_p and identifies the carriers mainly responsible for the small reverse current before breakdown?

(A) 1; majority carriers
(B) 26; majority carriers
(C) 676; minority carriers
(D) 1/676; minority carriers

23. A source emitting 400 Hz and a listener approach each other, each at 30 m/s relative to still air. Sound speed is 300 m/s. Find heard frequency before they pass.

(A) 480 Hz
(B) $\frac{4400}{9}$ Hz
(C) 400 Hz
(D) $\frac{4000}{9}$ Hz

24. A uniform block of density 600 kg/m^3 floats in a liquid. Then the liquid density is reduced to $\frac{3}{4}$ times 1000 kg/m^3, while the block's volume and density are unchanged. What fraction of the block is now submerged?

(A) $\frac{4}{5}$
(B) $\frac{9}{20}$
(C) $\frac{1}{5}$
(D) $\frac{3}{5}$

25. Two coaxial circular loops each radius 5 m carry equal currents 6 A in the same circulation sense. Their centres are separated by $2R$. Find field at the midpoint in units of μ_0.

(A) $\frac{3\sqrt{2}}{10}$
(B) $\frac{3\sqrt{2}}{20}$
(C) $\frac{6\sqrt{5}}{125}$
(D) $\frac{6}{5}$

26. A particle has position-dependent velocity components $\vec{v} = 3x\,\hat{i} - 3y\,\hat{j}$ in SI units. Given $dx/dt = v_x$ and $dy/dt = v_y$, what is its acceleration magnitude at $(3, 3)$ m?

(A) $27\,\mathrm{m/s^2}$ (B) $54\,\mathrm{m/s^2}$

(C) $27\sqrt{2}\,\mathrm{m/s^2}$ (D) $81\sqrt{2}\,\mathrm{m/s^2}$

27. A 6 V Zener regulator feeds a constant-current load of 3 mA. Supply varies from 12 to 16 V. Regulation requires Zener current at least 2 mA, and its safe maximum is 9 mA. What is the largest permissible series resistance? Assume ideal regulation and give resistance in Ω.

(A) $2400\,\Omega$ (B) $3000\,\Omega$

(C) $1200\,\Omega$ (D) $2000\,\Omega$

28. A gas receives 500 J of heat while its surroundings do 100 J of work on it. It then loses 250 J of heat at fixed volume. Find its total internal-energy change.

(A) 600 J (B) 350 J

(C) 250 J (D) 850 J

29. A 3 kg block moves along a horizontal rough surface from $x = 0$ to $x = 6$ m under a force $F(x) = 7x$ N parallel to motion. With $\mu_k = 1/5$ and $g = 10\ \mathrm{m/s^2}$, find the work done by the net force. The block has enough initial speed to keep moving forward.

(A) 90 J (B) 15 J

(C) 216 J (D) 162 J

30. A coil of inductance 4 H carries current $i(t) = 2t^2$ A. Find the magnitude of its self-induced emf at $t = 3$ s.

(A) 3 V (B) 24 V

(C) 144 V (D) 48 V

Chemistry

31. At a fixed temperature, a diprotic acid H2A has pKa1=5, pKa2=9, and pKw=14. Find (pKb of A2-, Kb(A2-)/Kb(HA-)). Use the conjugate pair appropriate to each basic species.

(A) (5, 10000) (B) (5, 1/10000)

(C) (9, 1/10000) (D) (9, 10000)

32. At one T,P,a binary reaction R⇌P has a Gibbs-energy-versus-product-fraction curve. Pure R has G=20 kJ,pure P50 kJ,and the unique minimum G=10 kJ occurs at product fraction $\frac{3}{5}$. Which inference follows from both endpoints and the interior minimum?

(A) Positive pure-state change forces the equilibrium to be pure R.

(B) G must decrease for a displacement from the minimum toward either pure endpoint.

(C) The minimum at an interior composition means both pure endpoints have equal G.

(D) The pure-product minus pure-reactant standard change is positive,yet the equilibrium contains product.

33. At fixedT,V,A(g)→P(g)+Q(g)+R(g)is first order,and an inert gas is present. Initial totalpressure is$\frac{1}{5}$bar;complete-decomposition pressure is$\frac{2}{5}$bar. At two half-lives,totalpressure in bar is

(A) $\frac{7}{20}$ (B) $\frac{3}{10}$

(C) $\frac{1}{2}$ (D) $\frac{2}{5}$

34. E°(A+/A)=1/2 V and E°(B+/B)=11/10 V. For the stoichiometrically scaled standard reaction 6A(s)+6B+(aq)→6A+(aq)+6B(s), find ΔG° in kJ. Use F=96.5 kJ V-1 mol-1.

(A) -1737/10 (B) -1737/5

(C) 1737/5 (D) -3474/5

35. For closed-system paths A→B→C, q_AB=330 J, w_AB=-190 J, q_BC=-120 J and w_BC=50 J, with work defined on the system. A direct C→A path has w=195 J. What is its heat q in J?

(A) -125 (B) -405

(C) -265 (D) 265

36. A Fischer projection of 2,3-dibromobutane has its four-carbon chain vertical (CH3 at top and CH3 at bottom). At C2,Br is on the left; at C3,Br is on the right; each other horizontal group is H. Horizontal bonds point toward the observer. Which classification and configuration pair is correct?

(A) chiral; (2S,3S) (B) meso; (2R,3S)

(C) meso; (2R,3R) (D) chiral; (2R,3R)

37. I:Alcoholic KOH/heat favours beta elimination from bromoethane. II:Chlorobenzene undergoes ordinary methyl-like SN2 with aqueous OH- at room temperature. Choose the correct truth pattern.

(A) I true;II false. (B) I false;II true.

(C) Both false. (D) Both true.

38. A $\frac{1}{10}$L saturated AgCl solution is filtered to remove all solid. Ksp=10^{-10}. In a calibrated sequential cathode model,Ag^{+} deposits quantitatively before water reduction begins. Current $\frac{193}{200000}$A flows for 200s, F=96500C/mol. Ignore volume change. Final silver mass in g is

(A) $\frac{27}{25000}$ (B) $\frac{27}{250000}$

(C) $\frac{27}{500000}$ (D) $\frac{27}{125000}$

39. An isolated cathode compartment contains $\frac{1}{10}$mol HA and $\frac{1}{10}$mol NaA,pKa5.00,in a supported NaCl electrolyte. Current5.00A flows for 965s. Each electron generates one OH^{-};no anode product enters,buffer loss or volume change occurs. F=96500C/mol andlog3=.477. FinalpH is

(A) $\frac{4523}{1000}$ (B) $\frac{5477}{1000}$

(C) 5 (D) 13

40. For I: formation of a dye lake on freshly precipitated aluminium hydroxide; II: pink colour of phenolphthalein caused by its acid-base equilibrium, which classification correctly identifies surface accumulation and distinguishes it from a bulk process?

(A) Both are adsorption because both involve molecular interactions. (B) Neither involves adsorption.

(C) I involves adsorption; II is not adsorption. (D) Only II involves adsorption.

41. A $\frac{43}{100}$ g volatile sample displaces $\frac{1600}{13}$ mL wet air at 300 K and 780 mmHg. Water-vapour pressure is 20 mmHg. Using STP=273 K,760 mmHg and22.4 L/mol,the sample molar mass, rounded to the nearest whole g/mol, is

(A) 84

(B) 82

(C) 86

(D) 3268

42. For acetylsalicylic acid, represented asHOOC-C6H4-OCOCH3, which assertion follows from the shown groups?

(A) It has no oxygen atoms.

(B) It retainsCOOH but has no free phenolicO-H before ester hydrolysis.

(C) It retains a primaryNH2 group.

(D) Its ester proves that it is sucrose.

43. A Hinsberg separation is applied to diethylamine. Which reagent identity and class-dependent outcome are correct?

(A) benzenesulphonyl chloride,C6H5SO2Cl;every amine gives an alkali-soluble N-H sulphonamide

(B) benzenesulphonic acid,C6H5SO3H;all classes give the same neutral product

(C) benzoyl chloride,C6H5COCl;a sulphonamide without N-H,insoluble in alkali

(D) benzenesulphonyl chloride,C6H5SO2Cl;a sulphonamide without N-H,insoluble in alkali

44. Which additional statement is essential before using S→0 as T→0 for a solid in the usual introductory convention?

(A) Its molar mass exceeds100 g/mol.

(B) It is a perfect crystal with a unique ground state.

(C) Its sample mass is exactly one gram.

(D) It contains only one isotope, so all possible disorder is automatically excluded.

45. Consider the statements. I. The same chemical can have different listed antimicrobial applications at different concentrations. II. Therefore molecular identity alone makes all concentrations interchangeable. Choose the correct truth pattern.

(A) I false; II true.

(B) I false; II false.

(C) I true; II true.

(D) I true; II false.

46. A model M(OH)2 has molar solubility 3/50000 M in pure water; neglect water autoionisation there. In a buffered solution its measured molar solubility is 27/781250000 M. The buffer fixes free OH- and the solid's contribution to OH- is negligible. What is the imposed OH- concentration?

(A) 1/100

(B) 1/400

(C) 1/40000

(D) 1/200

47. Compare the energy for removing one more electron from Sc^{2+} and from Sc^{3+}. Which member is expected to resist that removal more strongly on the stated shell-stability basis?

(A) Sc^{2+}

(B) Sc^{3+}

(C) They must be identical because both are positive ions.

(D) The ion with the smaller written charge always has the larger removal energy.

48. For ethanal and methanal,which structural-reactivity comparison is correct under ordinary enolate/ aldol conditions?

(A) only methanal is a self-aldol donor

(B) only ethanal is a self-aldol enolate donor

(C) methanal has more alpha H than ethanal

(D) both are self-aldol donors

49. A sample of 1,2-dibromopropane is treated with excess NaNH2,then aqueous protonation. Which isolated product is consistent with the reagent limitation and carbon-skeleton change?

(A) propyne

(B) propane

(C) propadiene

(D) propene

50. A C5H11Br isomer gives2-methylbutane after Mg/dry ether followed by water. How many distinct constitutional bromides can satisfy this? Ignore stereoisomers.

(A) 3

(B) 2

(C) 4

(D) 5

51. For 1-chloropropane and1-bromopropane,which structural classification is correct when the actual halogen-bearing carbon and molecular formula are examined?

(A) position isomers

(B) different molecular formulae,not isomers

(C) enantiomers

(D) functional isomers

52. Which reagent/name pair is suitable when a benzenediazonium salt must become chlorobenzene? The target carbon framework must be retained.

(A) AgF;Swarts reaction

(B) Na/dry ether;Wurtz reaction

(C) CuCl/HCl;Sandmeyer reaction

(D) NaCl in water;Finkelstein reaction

53. Use octet Lewis contributors for $CO3^{2}$- and NO2-; for SO2 exclude expanded-octet contributors. For HCOO-, HCO3- and HNO3, average only the two nonprotonated central-atom-O bonds. For other species average all central-atom bonds. Find (average bond order of the first minus that of the second, sum of total lone-pair counts in one contributor of each). A lone pair on any atom is counted once.

(A) (-1/6, 13)

(B) (17/6, 14)

(C) (-1/6, 14)

(D) (-1/6, 15)

54. A first-order reaction at 320 K leaves 1/4 of its initial reactant after100 s. Its Ea/(RT) at this temperature is 11. Use ln2=0.693 and k=A exp(-Ea/RT). Which is its frequency factor A in s-1?

(A) 693/50000 exp(11)

(B) 693/500 exp(11)

(C) 693/50000 exp(-11)

(D) 50000/693 exp(11)

55. A two-step profile has energies in kJ/mol:reactants0,intermediate10,products-10,first transition state30,second transition state60. With comparable prefactors,which pair(step with larger forward barrier, highest reverse barrier measured from products)is correct?

(A) Step2;70kJ/mol.

(B) Step1;30kJ/mol.

(C) Step2;60kJ/mol.

(D) Step1;70kJ/mol.

56. A solvent has Tb=400 K, M=100 g/mol and ΔHvap=32000 J/mol. Use R=8.00 and Kb=R Tb^2M/(1000ΔHvap). An A2B solution of analytical molality 1/20 mol/kg boils 13/25 K above pure solvent. Only dissociation A2B→2A+ +B2- occurs, with ideal dilute-particle counting. What fraction of A2B is dissociated?

(A) 4/5 (B) 4/15

(C) 9/10 (D) 2/5

57. For oxide formation, both normalised per mole of O_2, $\Delta G_A^\circ = -330+0.10T$ and $\Delta G_B^\circ = -420+0.20T$ in kJ. Ignoring kinetic limitations, at what temperatures is B metal a favourable standard-state reducing agent for oxide A?

(A) T < 900 K (B) T > 900 K

(C) At all positive temperatures (D) Only T = 900 K

58. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_y; opposite facing signs; 2: parallel p_x; opposite facing signs. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (2, 2) (B) (0, 4)

(C) (1, 3) (D) (1, 2)

59. Limiting molar conductivities are KCl150, NaCl126 and KNO3 110 S cm2/mol. A very dilute 3/100 M NaNO3 solution has measured conductance 129/200000 S. Neglect finite-concentration corrections. What is the cell constant in cm-1?

(A) 8 (B) 2

(C) -4 (D) 4

60. An aldopentoseFischer projection has CHO at top, CH2OH at bottom, and OH sides at C2, C3, C4 respectively L, L, R, whereR means right andL means left. Oxidise both terminal groups to COOH without changing internal configurations. Which D/L assignment for the sugar and classification of the diacid are correct?

(A) D/L is fixed by the sign of rotation alone. (B) D; meso(optically inactive).

(C) D; chiral. (D) L; chiral.

English Proficiency

61. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The audit team has located the duplicate entry but has not deleted it.

(A) The duplicate entry had been located but was deleted by the audit team.

(B) The duplicate entry has been located and deleted by the audit team.

(C) The audit team has been located by the duplicate entry but has not deleted it.

(D) The duplicate entry has been located by the audit team but has not been deleted.

62. He preserves original letters and catalogues them for future researchers rather than writing a person's life story. What is he?

(A) biographer
(B) archivist
(C) publisher
(D) antiquarian

63. Read the sentence: The buyer was wary of the unusually cheap offer and checked its terms. Choose the opposite in meaning of 'wary' as used in the sentence.

(A) attentive
(B) unwary
(C) suspicious
(D) cautious

64. Read the sentence: The coach remained optimistic about the team's chances despite the setback. Choose the opposite in meaning of 'optimistic' as used in the sentence.

(A) patient
(B) confident
(C) hopeful
(D) pessimistic

65. Which ONE of the following words is misspelt?

(A) posession
(B) certainly
(C) surprise
(D) apparatus

66. The course studies plant structure and growth rather than the cultivation of gardens as a practical trade. Which field does it concern?

(A) botany
(B) zoology
(C) geology
(D) astronomy

67. Read the sentence: The new account was at odds with the dated record, contradicting its central claim. Choose the opposite in meaning of 'at odds with' as used here.

(A) based on
(B) in dispute with
(C) in agreement with
(D) unconnected with

68. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The organisers need not replace the sign if its wording is corrected.

(A) The sign need not be replaced by the organisers if its wording is corrected.

(B) The sign must not be replaced by the organisers if its wording is corrected.

(C) The sign must be replaced whether or not its wording is corrected.

(D) The organisers need not be replaced by the sign if its wording is corrected.

69. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. between the different sites. Q. before any comparison was made R. was sorted by location S. The material collected during the survey

(A) R-S-Q-P
(B) S-R-Q-P
(C) Q-S-R-P
(D) P-R-Q-S

70. Read the sentence: The committee will allocate part of the fund to equipment and the rest to training. Choose the closest meaning of 'allocate' as used in the sentence.

(A) collect from several sources

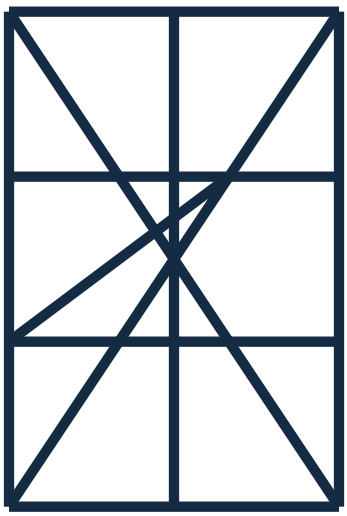
(B) divide into equal amounts

(C) assign for particular purposes

(D) calculate in advance

Logical Reasoning

71. How many triangles are formed by all the drawn segments?



(A) 37

(B) 33

(C) 23

(D) 36

72. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 7, 14, 17, ?, 37, 74, 77.

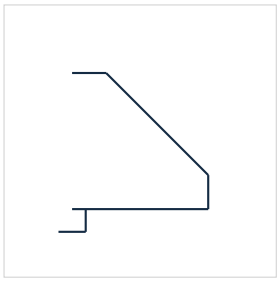
(A) 40

(B) 20

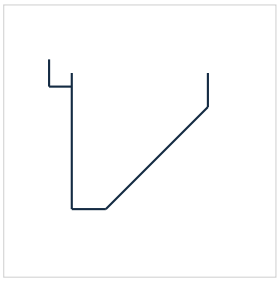
(C) 31

(D) 34

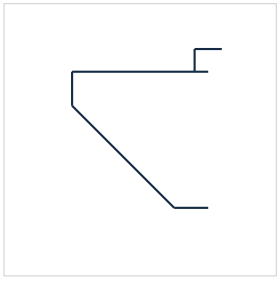
73. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

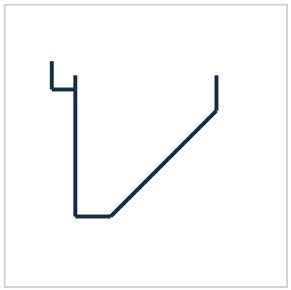


bottom left

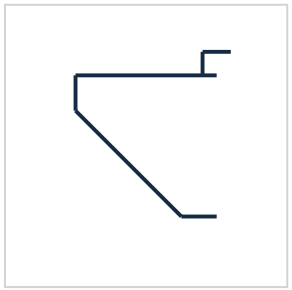
?

bottom right

(A) Pattern U



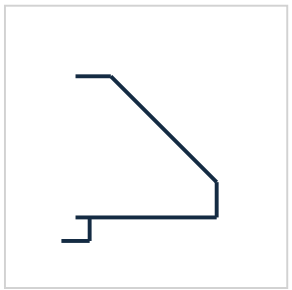
(B) Pattern V



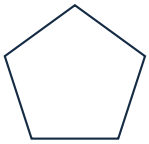
(C) Pattern T



(D) Pattern N



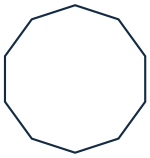
74. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

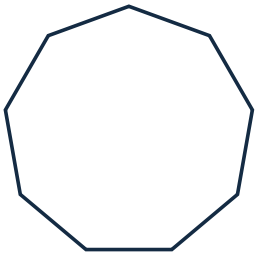


B



C

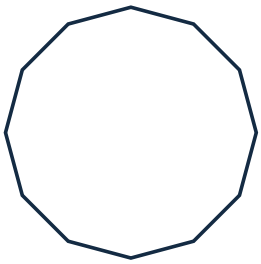
(A) Pattern K



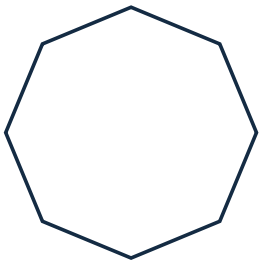
(B) Pattern Z



(C) Pattern V



(D) Pattern N



75. Consider only these statements; universal statements alone do not imply existence. No guides are hikers; Some hikers are not farmers. Conclusions: I. Some hikers are farmers. II. No hikers are guides. Which conclusions follow necessarily?

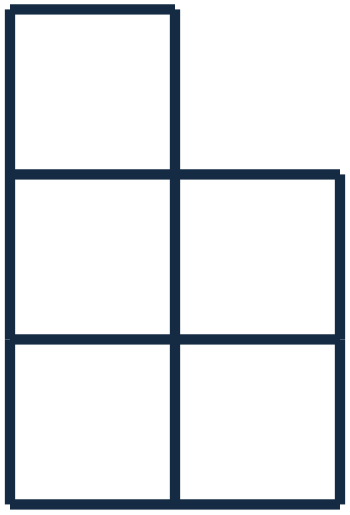
(A) Only I

(B) Only II

(C) Both I and II

(D) Neither I nor II

76. How many rectangles, including squares are formed by all the drawn segments?



(A) 13
(B) 5
(C) 11
(D) 12

77. Three pairs contain exactly one string instrument and one wind instrument. Which pair does not?
(A) Harp / Flute
(B) Guitar / Saxophone
(C) Ukulele / Oboe
(D) Violin / Cello

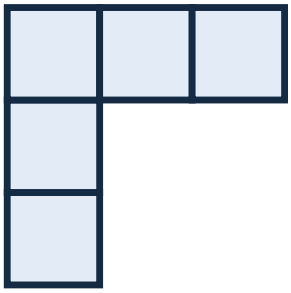
78. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.


target

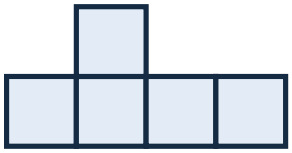

piece 1


piece 2

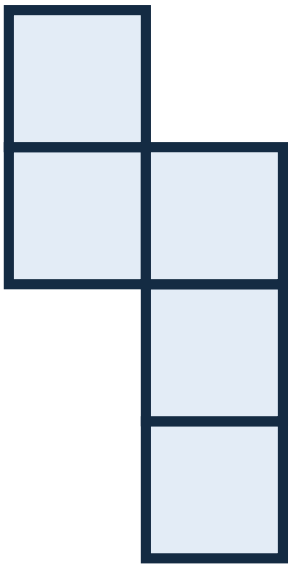
(A) Pattern R



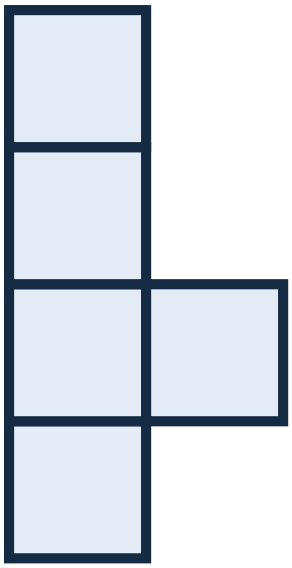
(B) Pattern U



(C) Pattern Q



(D) Pattern N



79. The whole has 11 unit positions. Part A covers the first 4; part B covers 5 positions, overlapping A in exactly 1 positions as shown. How many positions remain uncovered by either part?


whole


part A positions


part B positions

(A) 4
(B) 3
(C) 2
(D) 1

80. In a row, Asha is 9th from the left and 21st from the right. Bina is 7th from the right. Charu stands 2 places to the left of Bina. How many people stand strictly between Asha and Charu?

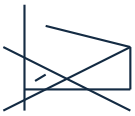
(A) 13
(B) 10
(C) 11
(D) 3

81. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

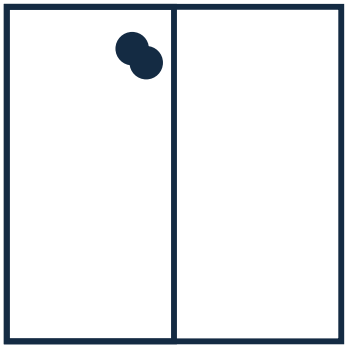
(A) 3

(B) 1

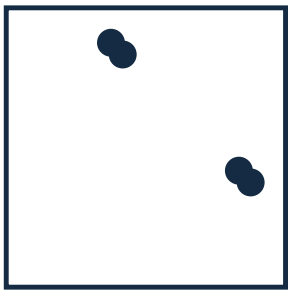
(C) 2

(D) 4

82. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



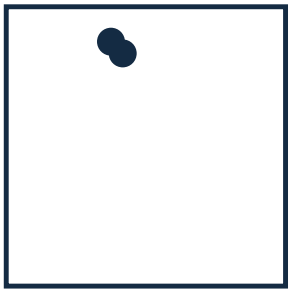
(A) Pattern P



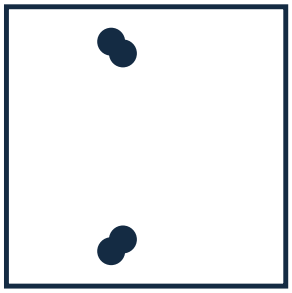
(B) Pattern Y



(C) Pattern U



(D) Pattern K



83. Five people occupy positions 1 to 5 from left to right, one per position, and wear five different shirt colours: red, blue, green, white and yellow. Bina is 1st from the left. Dev is 2nd from the left. Charu is 3rd from the left. Esha is 4th from the left. Asha is 5th from the left. The red shirt is in position 1. The white shirt is in position 4. The blue shirt is in position 5. Dev does not wear green. Which shirt does Dev wear?

(A) Green (B) Blue

(C) Red (D) Yellow

84. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, MANGO is coded as QERKS. Which word is coded as GPSYH?

(A) BKNTC (B) DMPVE

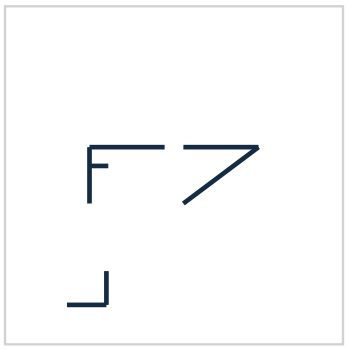
(C) CLOUD (D) KTWCL

85. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 16$ and $5 \mapsto 31$. Which input maps to 76?

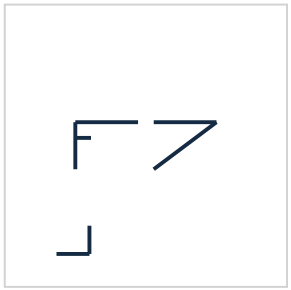
(A) 15 (B) 14

(C) 76 (D) 13

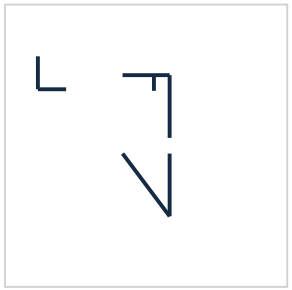
86. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



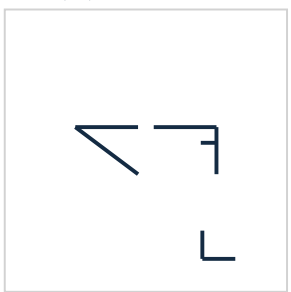
(A) Pattern X



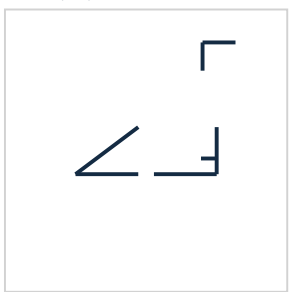
(B) Pattern T



(C) Pattern L



(D) Pattern Y



87. The age table uses the same two people throughout:

	younger	older
now	x	$x+6$
5 years ago	$x-5$	$x+1$

. The product of their ages 5 years ago was 616. What will the younger person's age be 4 years from now?

(A) 31 (B) 26

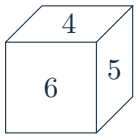
(C) 37 (D) 27

88. Consider only these statements; universal statements alone do not imply existence. Some designers are bakers; No bakers are hikers; Some designers are hikers. Conclusions: I. All hikers are designers. II. No designers are hikers. Which conclusions are individually possible without contradicting the statements?

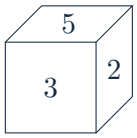
(A) Only I (B) Both I and II

(C) Neither I nor II (D) Only II

89. The views show the same die. Give the ordered pair: (face opposite 1, face opposite 2). Do not assume opposite labels sum to seven.



1

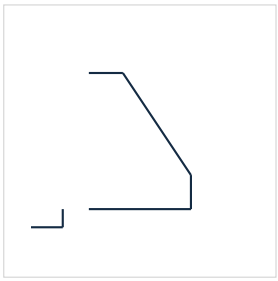


2

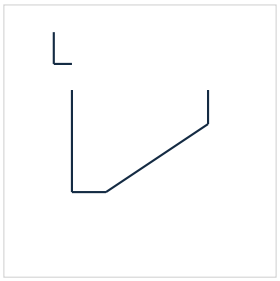
(A) $(5, 6)$ (B) $(5, 3)$

(C) $(5, 1)$ (D) $(5, 2)$

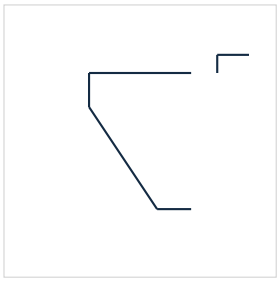
90. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1



2

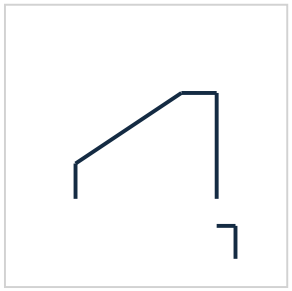


3

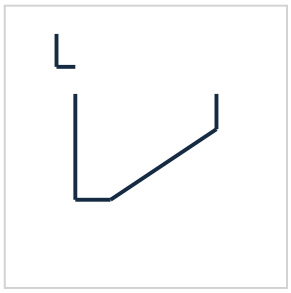
(A) Pattern T



(B) Pattern U



(C) Pattern P



(D) Pattern Y



Mathematics

91. A,B are acute, A+B=$\pi/4$ and tan A=$\frac{3}{5}$. Find tan(A-B).

(A) $\frac{7}{20}$ (B) $-\frac{7}{23}$

(C) $\frac{7}{23}$ (D) $\frac{1}{4}$

92. For x,y>0 with sqrt(x)+sqrt(y)=7, find d^2y/dx^2 at x=16.

(A) $\frac{7}{64}$ (B) $-\frac{7}{128}$

(C) $\frac{7}{128}$ (D) $\frac{7}{8}$

93. How many integers k make 9sin x+12cos x+6=k have exactly two distinct solutions in $0\le x<2\pi$?

(A) 14 (B) 29

(C) 31 (D) 30

94. Find the area bounded by y=x, y=16/x, and x=8 in the first quadrant.

(A) 24 (B) $16 - 16\log(2)$

(C) $\log(65536)$ (D) $24 - \log(65536)$

95. Find the squared distance between 6x+8y=4 and 3x+4y=6.

(A) $\frac{4}{5}$ (B) 16

(C) $\frac{16}{25}$ (D) $\frac{4}{25}$

96. r=pa+qb, a=(1,0,0), b=(1,1,0), r·a=7, r·b=10. Find $|r|^2$.

(A) 58 (B) 25

(C) 10 (D) $\sqrt{58}$

97. Evaluate the sum of r(r-1)C(7,r)2^r over all even integers r with $0 \le r \le 7$.

(A) 5082 (B) 40824

(C) 20496 (D) 20328

98. $A = \begin{bmatrix} 8 & 4 \\ -4 & 3 \end{bmatrix}$, K=(A-$A^T$)/2. Find tr($K^2$).

(A) 0 (B) -32

(C) 32 (D) -16

99. There are 11 Boolean variables p_1,...,p 11. Exactly 3 variables must be true. How many truth assignments satisfy [(p_1 implies p_2) and (p_1 implies not p_2)]?

(A) 2048 (B) 1024

(C) 36 (D) 120

100. Disjoint A,B have sizes 9,3. Find $|(A\times A)\cup(B\times B)|$.

(A) 90 (B) 27

(C) 54 (D) 144

101. An equivalence relation has classes of sizes 5,4,3. How many unordered pairs of distinct related elements are there?

(A) 50 (B) 38

(C) 10 (D) 19

102. Let $P(x) = (1+\sqrt{2}x)^6 + (1-\sqrt{2}x)^6$. Find the sum of all nonconstant coefficients of P.

(A) 196 (B) 194

(C) 198 (D) 99

103. Let R(x) be the remainder when x^7 is divided by (x-1)(x-2). Find R(6).

(A) -629 (B) 762

(C) 636 (D) 279936

104. A=(6,4,0), C=(9,10,3). B lies on AC with AB:BC=2:1. Find $|OB|^2$, where O is the origin.

(A) 52 (B) 132

(C) 54 (D) 190

105. Two points on opposite sides of a tower foot are 10 metres apart on a straight level line. The elevations of the tower top are 30° and 45°. Find the tower height.

(A) $\frac{5}{2}$ (B) $\frac{5\sqrt{3}}{3}$

(C) $-5+5\sqrt{3}$ (D) 5

106. The chord of contact from P to $x^2/25-y^2/4=1$ is $x=5/2$. Find the product of the slopes of the two tangents from P.

(A) $-\frac{4}{225}$ (B) $-\frac{4}{75}$

(C) $-\frac{4}{25}$ (D) $\frac{5}{2}$

107. Evaluate $\sum_{j=0}^{3}\binom{3}{j}\binom{6}{3-j}$.

(A) 126 (B) 252

(C) 56 (D) 84

108. Circles $x^2+y^2=169$ and $(x-22)^2+y^2=169$ meet at A,B. Find $|AB|^2$.

(A) 16 (B) 192

(C) 48 (D) 676

109. Evaluate $\lim_{x\to 0}\dfrac{(1+x)^{10}-1-10x-3x^2}{x^2}$.

(A) 42 (B) 45

(C) 48 (D) 7

110. For f(x)=|x-5|/(x-10), on which maximal open intervals is f strictly decreasing?

(A) $(-\infty,10)$ (B) $(-\infty,5)$

(C) $(5,10)\cup(10,\infty)$ (D) $(5,\infty)$

111. An AP has a=9,d=7; its first n terms sum to 268. Find a_(n+1).

(A) 65 (B) 58

(C) 72 (D) 8

112. The line 3x+4y=15 cuts $x^2+y^2=36$ in chord AB. Find $|AB|^2$.

(A) 180 (B) 144

(C) 108 (D) 27

113. a=(2,0,0), b=(0,5,0), c=(0,0,5), p=a+b. Evaluate $[p\times(b\times c)]\cdot(a+c)$.

(A) −625 (B) 625

(C) 0 (D) −725

114. How many reflexive symmetric relations on a labelled 4-element set contain at least one off-diagonal ordered pair?

(A) 62 (B) 64

(C) 63 (D) 32

115. How many injective functions from {a,b,c} to a labelled 7-element set satisfy f(a)<f(b), when the codomain is naturally ordered?

(A) 210 (B) 105

(C) 35 (D) 42

116. z=6+5i. Find 1/z+1/conjugate(z).

(A) $\frac{1}{12}$ (B) 12

(C) $\frac{10}{61}$ (D) $\frac{12}{61}$

117. Let D_n be the determinant of the 3×3 matrix with diagonal entries n and off-diagonal entries 1. Find ΣD_n for n=1,...,3.

(A) 24 (B) 18

(C) 60 (D) 36

118. A rectangle has perimeter 40. Its length exceeds its width by at least 4. Find its maximum area.

(A) 16 (B) 96

(C) $4\sqrt{6}$ (D) 396

119. F′(x)=$\frac{4\left(x^2-1\right)\left(x^2+1\right)^3}{x^5}$ and F(1)=4. Find F(2)-2F(1).

(A) $-\frac{433}{16}$ (B) $\frac{369}{16}$

(C) $\frac{433}{16}$ (D) $\frac{305}{16}$

120. For integer n, consider: “If 9 divides n, then 3 divides n.” Which is its contrapositive?

(A) 9 divides n and 3 does not divide n. (B) If 3 does not divide n, then 9 does not divide n.

(C) If 3 divides n, then 9 divides n. (D) If 9 does not divide n, then 3 does not divide n.

121. The four lines ±x/18±y/14=1 form a rhombus with vertices on the coordinate axes. Find its area.

(A) 1008 (B) 504

(C) 126 (D) 252

122. $\theta \in(\pi/2,\pi)$, $\sin\theta=\frac{\sqrt{5}}{5}$. Find $\cos(\theta/2)$.

(A) $-\frac{\sqrt{50-20\sqrt{5}}}{10}$ (B) $-\frac{\sqrt{5}}{5}$

(C) $\frac{\sqrt{20\sqrt{5}+50}}{10}$ (D) $\frac{\sqrt{50-20\sqrt{5}}}{10}$

123. Let $D(t) = \det \begin{pmatrix} 7+t & 1+t \\ 1+t & 4+t \end{pmatrix}$. Find D′(0).

(A) 54 (B) -13

(C) 11 (D) 9

124. Evaluate $\int_0^\infty \log(3x)/(x^2+9)\,dx$.

(A) $\frac{\pi \log(3)}{6}$ (B) $\pi \log(3)$

(C) $\frac{\pi \log(3)}{3}$ (D) 0

125. f(x)=x^2+4x for x≤3, and f(x)=px+q for x>3. If f is differentiable at 3, find p-q.

(A) 19 (B) 21

(C) 1 (D) 10

126. For $0 \leq \theta < 12\pi$, roots of x^2-7x+(1-2sinθ)=0 have opposite signs. How many connected intervals of θ satisfy this?

(A) 12 (B) 7

(C) 1 (D) 6

127. An AP has first term 2, sum of its first 6 terms equal to 57. Find the sum of the squares of those 6 terms.

(A) 24 (B) 410

(C) 699 (D) 3249

128. For $A = \begin{bmatrix} 4 & 1 \\ 1 & 6 \end{bmatrix}$ and C=23A^{-1}, find C_{22}.

(A) 4 (B) 6

(C) -1 (D) $\frac{4}{23}$

129. Positive a,b satisfy log(a+b)=log(7)+[log a+log b]/2 in one valid base. Find $(a^4+b^4)/(a^2b^2)$.

(A) 2207 (B) 2211

(C) 47 (D) 2209

130. The system x+y+z=1, y+z=2, (k-2)z=k-7 has no solution for which k?

(A) 7 (B) 2

(C) -2 (D) 0

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. In a new unit system, the base mass, length and time units are 4 kg, 2 m and 2 s. What is the numerical value of a force of 30 N in the new system?

(A) 15 (B) 60

(C) 30 (D) $\frac{15}{2}$

132. A particle of mass 1 kg starts from rest. A force $\vec{F} = 4t\,\hat{i} + 8\,\hat{j}$ N acts for 5 s. What is the magnitude of its momentum at that instant?

(A) $90\,\text{kg m/s}$ (B) $10\sqrt{41}\,\text{kg m/s}$

(C) $20\sqrt{29}\,\text{kg m/s}$ (D) $40\,\text{kg m/s}$

133. A photon and a nonrelativistic particle have equal de Broglie wavelengths. The particle speed is $c/37$. If the photon then loses half its energy while the particle momentum is unchanged, what is the ratio of particle kinetic energy to the photon's final energy?

(A) 1 (B) 37

(C) $\frac{1}{37}$ (D) $\frac{1}{1369}$

Chemistry

134. Compare S(g)+e-→S-(g) with S-(g)+e-→S2-(g). Which statement avoids applying one sign to both steps?

(A) Both must be exothermic because sulfur forms sulfides.

(B) The second step adds an electron to an anion and is endothermic in isolation, even if the first is exothermic.

(C) Formation of a stable crystal proves both isolated gas-phase steps release energy.

(D) The second step removes an electron from neutral sulfur.

135. Dry air is taken to contain21.0% O2by volume. From 160L of this air, an isolation process recovers80.0% of its oxygen into a pure stream. All gas volumes refer to the same T,P. Find recovered O2volume in litres.

(A) $\frac{672}{25}$ (B) $\frac{168}{5}$

(C) 42 (D) $\frac{2528}{25}$

136. Ignoring special ortho effects,which decreasing acid-strength order is correct for 2-chlorobutanoic acid,3-chlorobutanoic acid,butanoic acid?

(A) butanoic >3-chlorobutanoic >2-chlorobutanoic (B) 3-chlorobutanoic >2-chlorobutanoic >butanoic

(C) 2-chlorobutanoic >3-chlorobutanoic >butanoic (D) 2-chlorobutanoic >butanoic >3-chlorobutanoic

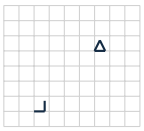
Logical Reasoning

137. Preserve the exact relation and its order: Bee : Swarm :: ? : Constellation. Choose the missing term.

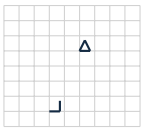
(A) Lion (B) Bee

(C) Star (D) Wolf

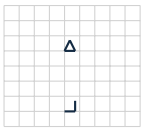
138. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



1

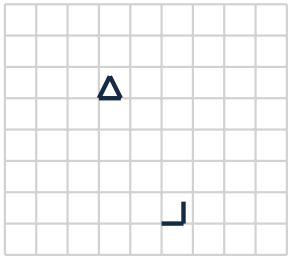


2

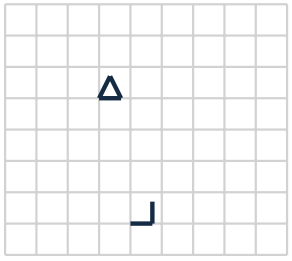


3

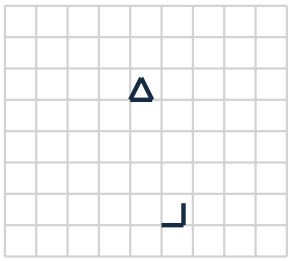
(A) Pattern T



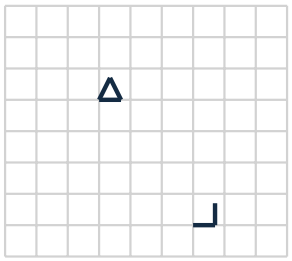
(B) Pattern R



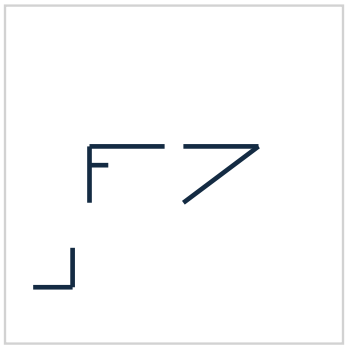
(C) Pattern V



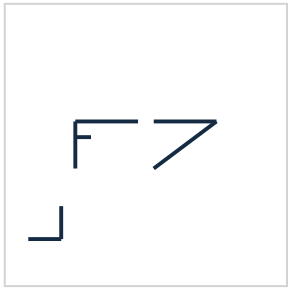
(D) Pattern Q



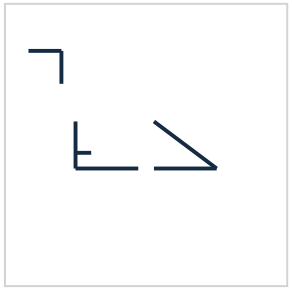
139. Choose the exact horizontal water reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



(A) Pattern R



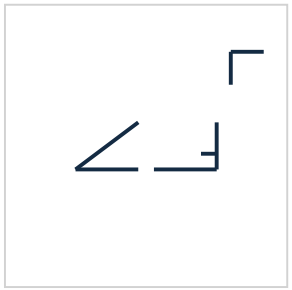
(B) Pattern Z



(C) Pattern X



(D) Pattern L



Mathematics

140. Let $z=(1+i)^9/(1-i)^4$. Find $\mathrm{Re}(z)+\mathrm{Im}(z)$.

(A) -8 (B) 32

(C) 0 (D) -128

141. For the ellipse $x^2/36+y^2/9=1$, the chord of contact from P is $12\,x/36+3\,y/9=1$. Find the square of the tangent length from P to the auxiliary circle $x^2+y^2=36$.

(A) 45 (B) 117

(C) 23373 (D) 189

142. Find the population variance of 5+4, 10+4, ..., 40+4.

(A) $\frac{21}{4}$ (B) $\frac{105}{4}$

(C) $\frac{525}{4}$ (D) $\frac{541}{4}$

BITSAT Mock 39

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. In a new unit system, the base mass, length and time units are 2 kg, 3 m and 4 s. What is the numerical value of a force of 60 N in the new system?

(A) $\frac{45}{2}$ (B) $\frac{5}{8}$

(C) 160 (D) $\frac{160}{3}$

2. A solid cylinder of mass 5 kg rolls without slipping on a horizontal rough surface. A horizontal force 30 N is applied tangentially at its top. Find its centre-of-mass acceleration; $I = MR^2/2$.

(A) $6\,\mathrm{m/s^2}$ (B) $12\,\mathrm{m/s^2}$

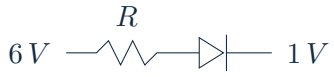
(C) $8\,\mathrm{m/s^2}$ (D) $4\,\mathrm{m/s^2}$

3. A solid sphere of radius R has uniform density $9\rho_0$. A spherical cavity of radius $R/10$ is cut with its centre at distance $R/4$ from the original centre. Find field magnitude at the cavity centre in units of $\pi G\rho_0 R$.

(A) $\frac{16}{3}$ (B) $\frac{4}{27}$

(C) 3 (D) 48

4. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 4 kΩ to a terminal at +6 V, and its cathode connects to a terminal at 1 V. Find the current from the +6 V terminal in mA.



(A) $\frac{9}{10}$ mA (B) $\frac{57}{40}$ mA

(C) $\frac{5}{4}$ mA (D) $\frac{43}{40}$ mA

5. A boat moves at 30 m/s relative to water. The current is 18 m/s and river width is 48 m. The boat lands directly opposite its starting point. Find crossing time.

(A) $\frac{8}{3}$ s (B) $\frac{5}{2}$ s

(C) $\frac{4\sqrt{34}}{17}$ s (D) 2 s

6. A particle moves with constant acceleration. It travels 15 m in the (6)th second and 17 m in the 7th second. What was its initial velocity?

(A) 5 m/s (B) 4 m/s

(C) 17 m/s (D) 3 m/s

7. A 1 kg block moving at 4 m/s compresses a spring of constant 100 N/m. During compression, a damper removes 1/3 of the block's initial kinetic energy. Find maximum compression; gravity does no work.

(A) $\frac{2\sqrt{6}}{15}$ m (B) $\frac{4\sqrt{3}}{15}$ m

(C) $\frac{2}{5}$ m (D) $\frac{8}{75}$ m

8. A thin glass lens of refractive index 3/2 has focal length 20 cm in air. It is completely immersed in a liquid of index 4/3. Find its new focal length.

(A) 5 cm (B) 20 cm

(C) 80 cm (D) $\frac{80}{3}$ cm

9. A ray in a medium of index $\frac{13}{5}$ is incident on a medium of index $\frac{13}{10}$. Find the sine of the critical-angle threshold for total internal reflection.

(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{1}{4}$ (D) $\frac{5}{13}$

10. A coil of 3 turns and area 4 m^2 carries 5 A in a uniform 2 T field. Its plane makes 30° with the field. Find torque magnitude.

(A) 120 N m (B) $60\sqrt{3}$ N m

(C) 60 N m (D) $15\sqrt{3}$ N m

11. A stretched string is divided by a bridge into lengths in the ratio 1 : 4. Tension and linear density are unchanged. The original fundamental is 40 Hz. Find the fundamental of the longer segment.

(A) $\frac{200}{3}$ Hz (B) 50 Hz

(C) 32 Hz (D) 10 Hz

12. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 40 percent. The measured increase in its absolute temperature is $\frac{2016}{5}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) 420 K (B) $\frac{2016}{5}$ K

(C) 1008 K (D) $\frac{4116}{5}$ K

13. A network is defined by $P = \mathrm{NAND}(A, B)$, $Q = \mathrm{NOR}(A, P)$ and $Y = \mathrm{AND}(Q, \overline{B})$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 1$ and $A = 1$?

(A) 2 (B) 1

(C) 0 (D) 3

14. A small-angle pendulum has period T in a stationary lift. In a vertically accelerating lift its period is 4T. Find the downward lift acceleration divided by g.

(A) $\frac{1}{16}$ (B) $\frac{15}{16}$

(C) 15 (D) $\frac{9}{16}$

15. A rigid diatomic ideal gas with vibrations frozen expands at constant pressure. It does 100 J work. Find the increase in internal energy and select its value.

(A) 100 J (B) 250 J

(C) 150 J (D) 350 J

16. An active transistor has $\alpha = \frac{49}{50}$. Its collector resistor is 5 kΩ and the measured collector-resistor drop is 40 V. Neglect leakage. What base current in mA is required?

(A) $\frac{8}{49}$ mA (B) $\frac{400}{2401}$ mA

(C) $\frac{4}{25}$ mA (D) 400 mA

17. Two short electric dipoles each moment 3 C m are located so a distant point P is at axial distance 4 m from one and equatorial distance 4 m from the other. Their fields at P are perpendicular. Find resultant field magnitude in units of k, where $k = 1/(4\pi\varepsilon_0)$.

(A) $\frac{3}{64}$ (B) $\frac{3\sqrt{2}}{64}$

(C) $\frac{3\sqrt{5}}{16}$ (D) $\frac{3\sqrt{5}}{64}$

18. A monochromatic wave of frequency 800000000 Hz enters a linear nondispersive medium with $\varepsilon_r = 25$ and $\mu_r = 4$. Its frequency is unchanged. Find its wavelength in the medium. Take $c = 3 \times 10^8$ m/s.

(A) $\frac{3}{80}$ m (B) $\frac{3}{20}$ m

(C) $\frac{3}{16}$ m (D) $\frac{3\sqrt{29}}{232}$ m

19. A metal wire has linear temperature coefficient $\frac{1}{250}$ per degree Celsius referred to 0°C. Its dimensions are fixed and it stays across a constant voltage. Find power at 80°C divided by power at 0°C.

(A) $\frac{25}{33}$ (B) $\frac{33}{25}$

(C) 1 (D) $\frac{17}{25}$

20. A hydrogen atom emits a photon in the $4 \to 3$ transition. A hydrogen-like ion of nuclear charge $Z = 4$ emits a photon of exactly the same wavelength and ends in level $n_f = 12$. What is its initial principal quantum number?

(A) 4 (B) 64

(C) 16 (D) 12

21. A uniform rod of mass 5 kg and length 3 m lies on a smooth horizontal table, free to rotate about a fixed vertical pivot at its centre. A point mass 2 kg moving perpendicular to the rod at 9 m/s strikes an end and sticks. Find the angular speed immediately afterward.

(A) $\frac{36}{11}$ rad/s (B) $\frac{18}{13}$ rad/s

(C) $\frac{36}{29}$ rad/s (D) $\frac{72}{11}$ rad/s

22. An electrostatic potential is $V(x, y) = 8x^2 - 3y$ V with x,y in metres. Find the field magnitude at $(2, 2)$ m.

(A) $\sqrt{1015}\,\mathrm{V/m}$ (B) $3\,\mathrm{V/m}$

(C) $\sqrt{265}\,\mathrm{V/m}$ (D) $\sqrt{1033}\,\mathrm{V/m}$

23. A long solid cylindrical conductor of radius 5 m carries uniformly distributed current 7 A. Find magnetic field magnitude at radial distance $R/3$ from the axis in units of $\mu_0/(2\pi)$.

(A) $\frac{7}{15}$ (B) $\frac{7}{75}$

(C) $\frac{7}{45}$ (D) $\frac{7}{5}$

24. A coil has 80 turns, area $\frac{1}{100}$ m^2, and rotates at 3 rad/s in a uniform 4 T field. At an instant the angle between its area vector and the field is 30°. Find instantaneous-emf magnitude.

(A) $\frac{3}{50}$ V (B) $\frac{8}{5}$ V

(C) $\frac{24\sqrt{3}}{5}$ V (D) $\frac{24}{5}$ V

25. A conveyor receives 7 packets each second, each of mass $\frac{3}{10}$ kg and speed 7 m/s. It accelerates them to 12 m/s without changing elevation. Find the ideal mechanical power delivered to the packets.

(A) $\frac{4053}{20}$ W (B) $\frac{399}{2}$ W

(C) $\frac{21}{2}$ W (D) $\frac{399}{4}$ W

26. 2 moles and 3 moles of the same ideal gas, initially at 400 K and 900 K, are connected in a rigid insulated system. There is no external work. Find equilibrium temperature, assuming constant molar heat capacity.

(A) 400 K (B) 700 K

(C) 900 K (D) 650 K

27. A projectile is launched at 50 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{8}{9}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ m/s^2.

(A) $\frac{400}{3}$ m (B) 80 m

(C) 160 m (D) 240 m

28. A diamond network has four outer resistors AB, BC, CD and DA, each 5 ohms, and a bridge BD also 5 ohms. The DA arm additionally contains an ideal diode with anode at D and cathode at A. A source keeps A positive relative to C. Find equivalent resistance between A and C.

(A) $\frac{5}{2}\,\Omega$ (B) $15\,\Omega$

(C) $5\,\Omega$ (D) $\frac{25}{3}\,\Omega$

29. A one-electron ion with nuclear charge $Z = 2$ absorbs a photon of energy $\frac{103}{5}$ eV. The ejected electron has maximum kinetic energy 7 eV. From which principal quantum number was the electron removed? Use $E_n = -13.6Z^2/n^2$ eV and neglect recoil.

(A) 4 (B) 2

(C) 1 (D) 3

30. A 7 kg block on a rough horizontal table is tied over an ideal pulley to a hanging 4 kg mass. The hanging mass moves downward and $\mu_k = 1/5$. Find string tension. Take $g = 10\ \text{m/s}^2$.

(A) $\frac{182}{11}$ N
(B) 70 N
(C) $\frac{336}{11}$ N
(D) 40 N

Chemistry

31. Consider the statement: NH3has a higher boiling point than PH3despite its lower molar mass. Which explanation or consequence correctly accounts for this trend rather than applying an unqualified atomic-mass rule?

(A) The ns2pair is increasingly forced into all bonds down every group.

(B) Every increase in atomic mass must increase all molecular bond strengths.

(C) Hydrogen bonding is necessarily stronger for every heavier hydride.

(D) Intermolecular hydrogen bonding is a specific anomaly to a simple molar-mass trend.

32. Limiting molar conductivities areKCl150,$K_2SO_4$300,$BaCl_2$250S cm^2/mol. A saturated $BaSO_4$ solution has solute-only conductivity $\frac{3}{400000}$S/cm. Neglect ion interactions and other ions. Ksp($BaSO_4$)is

(A) $\frac{9}{2500000000}$
(B) $\frac{9}{48400000000}$
(C) $\frac{3}{100000}$
(D) $\frac{9}{10000000000}$

33. A 1 g organic sample contains chlorine as its only halogen. Carius analysis gives $\frac{4018}{8875}$ g dry AgCl, but this is only 80.0% of the theoretical AgCl because of a known collection loss. Find chlorine mass percentage. Use Ag=108, Cl=35.5.

(A) $\frac{4018}{71}$
(B) $\frac{56}{5}$
(C) 14
(D) $\frac{35}{2}$

34. A sample of sodium propionate is treated with soda lime,heat. Which isolated product is consistent with the reagent limitation and carbon-skeleton change?

(A) methane
(B) ethane
(C) propane
(D) ethene

35. One glucoseC1 is joined to another glucoseC4. Which C1 remains a reducing-end candidate?

(A) The already glycosidically boundC1 only.
(B) C1 of the glucose whoseC4 participates in the bond.
(C) NeitherC1 because all disaccharides are nonreducing.
(D) C6 of the first residue, because it is always an aldehyde.

36. The ratio R/k is formed from the molar gas constant and Boltzmann constant. Which interpretation and units are correct?

(A) It is dimensionless and equals1.
(B) It has units joules because both constants contain energy.
(C) It is1/NA, with units mol.
(D) It is NA, with units mol-1.

37. Carbon dioxide absorbs outgoing infrared radiation. Which classification avoids confusing that process with catalytic ozone destruction by chlorine?

(A) Greenhouse gas; its infrared effect is distinct from the chlorine radical cycle.

(B) CFC; because every greenhouse gas contains chlorine.

(C) Noble gas; because it cannot absorb radiation.

(D) PAN; because all infrared absorption requires nitrate.

38. Start at an eclipsed ethane conformation and rotate only the rear carbon through180° at constant angular direction. Including the endpoint but not the start,how many staggered minima and eclipsed maxima are crossed or reached?

(A) 1 minimum and2 maxima.

(B) 3 minima and3 maxima.

(C) 2 staggered minima and1 eclipsed maximum.

(D) 2 minima and2 maxima.

39. For Tb^{4+} (Z=65),which configuration/chemical interpretation is consistent with its electron count and the commonly discussed +3 state?

(A) $4f^6$; neither accepts nor loses an electron in changing to+3.

(B) $4f^7$; reducing character toward+3.

(C) $4f^8$; oxidising character.

(D) $4f^7$; oxidising character in the stated conversion toward the common +3 state.

40. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_y; opposite facing signs; 2: p_y on one centre and p_z on the other. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (1, 1)

(B) (0, 3)

(C) (1, 2)

(D) (2, 1)

41. A first-order reaction at 340 K leaves 1/8 of its initial reactant after100 s. Its Ea/(RT) at this temperature is 12. Use ln2=0.693 and k=A exp(-Ea/RT). Which is its frequency factor A in s-1?

(A) 2079/100000 exp(-12)

(B) 2079/1000 exp(12)

(C) 2079/100000 exp(12)

(D) 100000/2079 exp(12)

42. A $\frac{1}{100}$mol iron-ion inventory contains Fe^{2+}andFe^{3+}. Fraction$\frac{1}{4}$isFe^{3+}. All iron is quantitatively reduced to metal in the stated externally driven electron-balance model,with no side reactions. Electrons accepted,in mol,are

(A) $\frac{1}{40}$

(B) $\frac{9}{400}$

(C) $\frac{11}{500}$

(D) $\frac{1}{400}$

43. Consider the statements. I. Stratospheric ozone helps absorb harmful ultraviolet radiation. II. Because ozone has that role aloft, increasing ozone at breathing level must improve air quality. Choose the correct truth pattern.

(A) I false; II true.

(B) I true; II true.

(C) I false; II false.

(D) I true; II false.

44. Atoms form ideal touching FCC spheres of radius r and cell edge a. Fractional coordinates are measured in a. Using the straight displacement between the two specified points (1/4,1/4,1/4) and (3/4,3/4,3/4), without replacing it by another periodic image, find (distance/r)^2. These points describe specified tetrahedral sites.

(A) 6
(B) 12
(C) 3
(D) 7

45. Compare neutral Li and Na. Which pair of trends is consistent with the usual alkali-metal periodic pattern?

(A) Their radii must agree because their valence-electron counts agree.
(B) Li has the larger radius and lower first ionisation energy.
(C) Na has the larger atomic radius and lower first ionisation energy.
(D) Na has the smaller radius and lower first ionisation energy.

46. A $\frac{1}{50}$ mol FeC_2O_4 inventory is analysed. A fraction $\frac{1}{4}$ of Fe^{2+} has already become Fe^{3+},but every oxalate remains unchanged. Subsequent complete oxidation gives Fe^{3+} and CO_2. Electron equivalents released,in mol,are

(A) $\frac{3}{200}$
(B) $\frac{1}{25}$
(C) $\frac{7}{200}$
(D) $\frac{11}{200}$

47. propanoic acid is treated with Br2/red P,then water. Which product satisfies carbon accounting and the reagent selectivity?

(A) 2-bromopropan-1-ol
(B) 2-bromopropanoic acid
(C) 3-bromopropanoic acid
(D) bromoethane

48. Two one-electron ions A and B satisfy Z_B/Z_A=5 and |E_B|/|E_A|=625/16 for their selected Bohr levels. In the ideal hydrogenic model E$\propto$-Z^2/n^2 and r$\propto$n^2/Z. What is r_B/r_A?

(A) -16/125
(B) 32/125
(C) 8/125
(D) 16/125

49. For a closed system ΔU(A$\rightarrow$B)=160 J and ΔU(B$\rightarrow$C)=-60 J. Two A$\rightarrow$C paths have heats 360 J and 220 J respectively. What is work on the system on the second path minus that on the first, in J?

(A) 140
(B) -20
(C) -140
(D) 300

50. For ideal gases at one T, A(g)$\rightleftharpoons$2B(g) has Kc=9 and numerical RT=3 in compatible units. What is Kp for4B(g)$\rightleftharpoons$2A(g)? Use Kp=Kc(RT)$^{\Delta n}$ with the same standard-state convention throughout.

(A) -1/729
(B) 1/729
(C) 1/1458
(D) 2/729

51. Within an additive covalent-radius model,a single X-X bond is 150 pm and an X-H bond is 105 pm. A linear H-X-X-H skeleton is assumed with collinear single bonds. Its terminal H-to-H distance,in pm,is

(A) 360
(B) 210
(C) 255
(D) 720

52. In a closed disproportionation, the only oxidation states of the redox-active element are initially Cl(0) and finally Cl(I) plus Cl(-I). No other element changes oxidation state. What fraction of the redox-active atoms ends in the higher oxidation state? Apply electron balance rather than counting other atoms.

(A) 1/4 (B) 1/2

(C) -1/2 (D) 1

53. At 1 atm compare two separate pairs: (ethanol, dimethyl ether) and (n-butane, isobutane). A student selects dimethyl ether for the first higher-boiling substance because its formula mass is unchanged, and n-butane for the second. Which assessment is consistent with intermolecular interactions?

(A) Both selections correct. (B) First selection wrong; second selection correct.

(C) First selection correct; second selection wrong. (D) Both selections wrong.

54. $\frac{279}{100}$ g aniline is acetylated to acetanilide in80.0% isolated molar yield. The isolated acetanilide is then mononitrated and gives p-nitroacetanilide in75.0% isolated molar yield. Both steps are1:1in substrate moles. Find final product mass in g. Use molar masses: aniline93, acetanilide135, p-nitroacetanilide180 g/mol.

(A) $\frac{27}{5}$ (B) $\frac{81}{20}$

(C) $\frac{81}{25}$ (D) $\frac{108}{25}$

55. A solution of one nonvolatile nonelectrolyte has exact relative vapour-pressure lowering 1/31. After removal of solvent only, the lowering becomes 1/11. The initial solvent amount was 12 mol, and Raoult's law holds without association. How many moles of solvent were removed?

(A) 42/5 (B) 4

(C) 8 (D) 8/3

56. At25°C pH=7699/1000. Use pKw=14 and log10(5)=699/1000. What is [OH-] in mol/L?

(A) -1/2000000 (B) 1/2000000

(C) 1/4000000 (D) 1/1000000

57. A five-d-orbital splitting has an upper set and lower set separated by Δ. The lower energy is -2/5Δ and the upper energy is 3/5Δ relative to the conserved barycentre. Which (upper degeneracy, lower degeneracy) is required?

(A) (2, 3) (B) (1, 4)

(C) (4, 1) (D) (3, 2)

58. A Pt(2) coordination entity has coordination number 4, exactly 1 fully bound en ligand(s), and chloride in all remaining donor sites. Take en as 2-dentate with charge 0; chloride is monodentate. Find (number of ligand entities, net charge of the coordination entity).

(A) (3, -1) (B) (4, +0)

(C) (3, +0) (D) (4, +1)

59. For an isolated ClO3- ion, use the introductory VSEPR molecular description. Which pair gives chlorine's formal oxidation state and central-atom lone-pair count? Oxygen has oxidation state-2; use the usual lowest-formal-charge Lewis picture for this main-group ion.

(A) (+5, 1) (B) (+3, 1)

(C) (+5, 2) (D) (+7, 1)

60. Evaluate I: In an ordinary molecular liquid, stronger intermolecular cohesion can impede layer rearrangement. II: A dilute gas must therefore become less viscous on heating, just like the liquid. Assume no phase change and the standard introductory regimes.

(A) I is correct; II is incorrect.
(B) Both I and II are incorrect.
(C) I is incorrect; II is correct.
(D) Both I and II are correct.

English Proficiency

61. Which ONE of the following words is misspelt?

(A) rhythm
(B) receipt
(C) accommodation
(D) priviledge

62. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. The lecturer invited questions only after finishing the entire explanation. B. She began pausing briefly after each major stage. C. Questions could then identify a difficulty while its context was still clear. D. By then, some students had lost track of the step they did not understand.

(A) By then, some students had lost track of the step they did not understand.
(B) She began pausing briefly after each major stage.
(C) Questions could then identify a difficulty while its context was still clear.
(D) The lecturer invited questions only after finishing the entire explanation.

63. Read the sentence: The supervisor was amenable to a revised schedule if the work could still be completed on time. Choose the closest meaning of 'amenable' as used in the sentence.

(A) dependent on
(B) open to
(C) responsible for
(D) already committed to

Read the passage for Question 64

The local archive once arranged every photograph under the name of the person who donated it. This was convenient for tracing ownership, but a visitor looking for an image of the old bridge needed to open many unrelated entries. The archivist therefore added descriptions of the places and objects shown, while retaining the donors' names in a separate field. No photograph was moved, and no record of ownership was discarded. This change made a different kind of search possible. It also showed why making a collection accessible need not mean simplifying away information that specialists value. The important step was to recognise that a catalogue can answer more than one question, provided that the different kinds of information are clearly distinguished.

64. In the passage, what does 'This change' refer to?

(A) Moving photographs into a new physical order and deleting donor records.
(B) Replacing every donor's name with the name of the old bridge.
(C) Adding descriptions of depicted content while keeping donor information separately.
(D) Allowing visitors to examine only photographs from a single donor.

65. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. could still identify the samples Q. because each container carried R. a separate permanent label. S. The researcher whose notes had been misplaced

(A) R-P-Q-S
(B) S-P-Q-R
(C) P-S-Q-R
(D) Q-S-P-R

66. Read the sentence: A disinterested referee was chosen because neither club had any claim on her loyalty. Choose the closest meaning of 'disinterested' as used in the sentence.

(A) unenthusiastic
(B) uninformed
(C) impartial
(D) inexperienced

67. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The charity accepts sealed packets but does not accept opened ones.

(A) The charity is accepted by sealed packets but not opened ones.
(B) Sealed packets are accepted by the charity, but opened ones are not.
(C) Opened packets are accepted by the charity, but sealed ones are not.
(D) All packets have been accepted by the charity.

68. Read the sentence: The repairs were put on hold until the missing safety report arrived. Choose the closest meaning of 'put on hold' as used here.

(A) postponed temporarily
(B) abandoned permanently
(C) continued without change
(D) completed in advance

69. She helps the parties negotiate but does not impose a binding decision upon them. What is she?

(A) mediator
(B) judge
(C) advocate
(D) arbitrator

70. Read the sentence: The elaborate explanation included several layers of detail. Choose the opposite in meaning of 'elaborate' as used in the sentence.

(A) simple
(B) careful
(C) technical
(D) lengthy

Logical Reasoning

71. Mira points to a photograph and says, "This person is the daughter of my father's only son." Mira is female, and the photographed person is female. How is the photographed person related to Mira?

(A) Son
(B) Brother
(C) Daughter
(D) Niece

72. Consider only these statements; universal statements alone do not imply existence. Some campers are hikers; No designers are campers. Conclusions: I. Some campers are not designers. II. Some designers are not hikers. Which conclusions follow necessarily?

(A) Both I and II
(B) Neither I nor II
(C) Only I
(D) Only II

73. In a row, Asha is 9th from the left and 33rd from the right. Bina is 7th from the right. Charu stands 4 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 20 (B) 5

(C) 22 (D) 21

74. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 2, 4, 13, 26, ?, 70, 79.

(A) 35 (B) 61

(C) 39 (D) 70

75. Five people occupy positions 1 to 5 from left to right, one per position, and wear five different shirt colours: red, blue, green, white and yellow. Charu is 1st from the left. Asha is 2nd from the left. Dev is 3rd from the left. Esha is 4th from the left. Bina is 5th from the left. The green shirt is in position 1. The yellow shirt is in position 2. The white shirt is in position 4. Dev does not wear red. Which shirt does Dev wear?

(A) Green (B) Blue

(C) Red (D) White

76. The age table uses the same two people throughout:

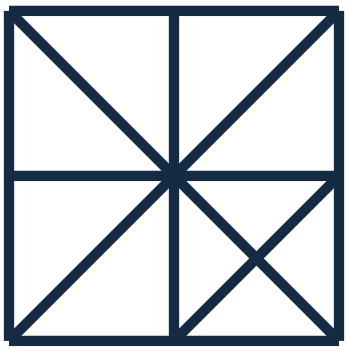
	younger	older
now	x	$x+6$
6 years ago	$x-6$	$x+0$

. The product of their ages 6 years ago was 616. What will the younger person's age be 6 years from now?

(A) 28 (B) 34

(C) 35 (D) 40

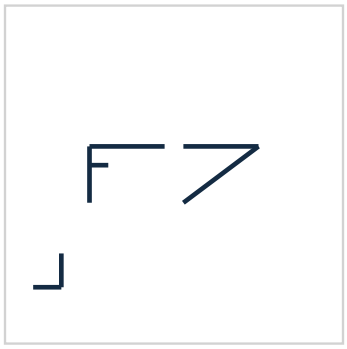
77. How many triangles are formed by all the drawn segments?



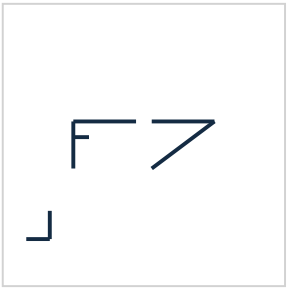
(A) 22 (B) 44

(C) 12 (D) 18

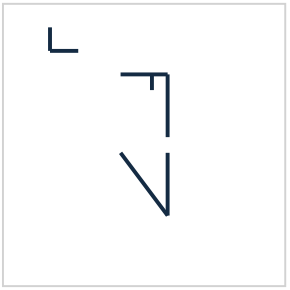
78. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



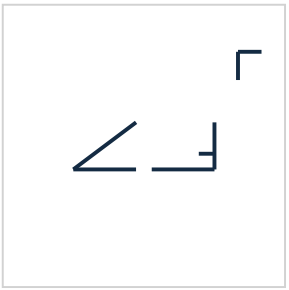
(A) Pattern X



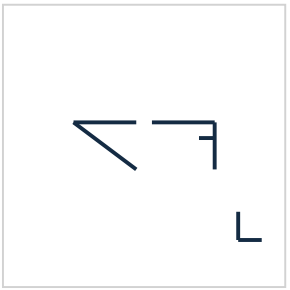
(B) Pattern R



(C) Pattern M

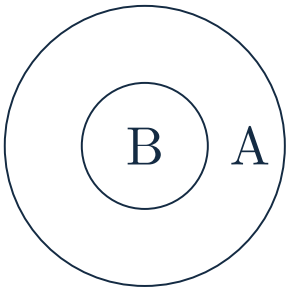


(D) Pattern Z

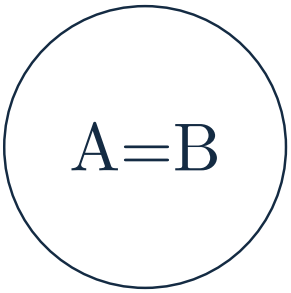


79. Choose the diagram that represents all these conditions, including the strictness information: Some A is B, some A is not B, and some B is not A.

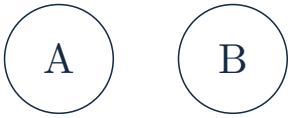
(A) Pattern Z



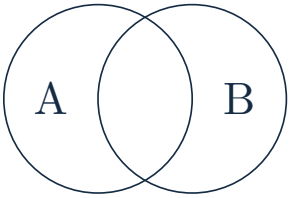
(B) Pattern V



(C) Pattern T

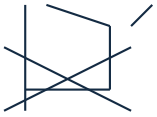


(D) Pattern N

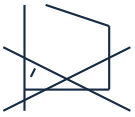


80. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

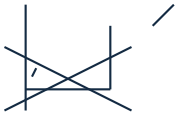




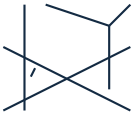
host 1



host 2



host 3



host 4

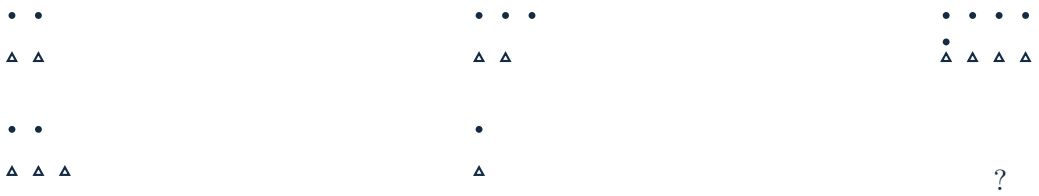
(A) 1

(B) 3

(C) 4

(D) 2

81. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern Q



(B) Pattern P



(C) Pattern L



(D) Pattern X



82. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, CHAIR is coded as FKDLU. Which word is coded as FORXG?

(A) IRUAJ (B) DMPVE

(C) CLOUD (D) BKNTC

83. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: ATE, CVH, EXK, ?, IBQ, KDT.

(A) FYM (B) HAO

(C) GZN (D) NZG

84. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?



1 2 3 4

(A) 3 (B) 2

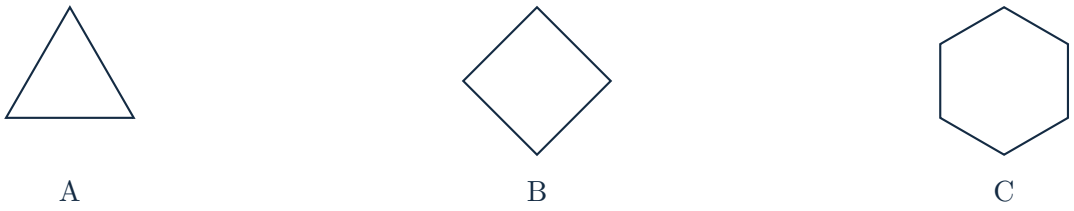
(C) 4 (D) 1

85. A walker goes 9 m north, then 7 m west, then turns right and walks 15 m. In which direction is the walker from the starting point?

(A) South-west (B) South-east

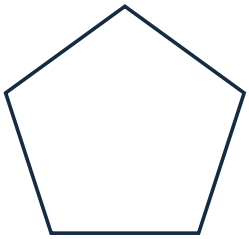
(C) North-west (D) North-east

86. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?

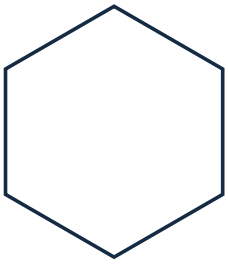


A B C

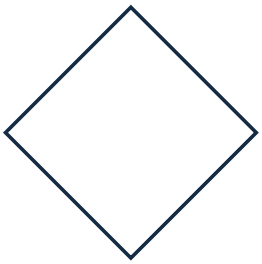
(A) Pattern T



(B) Pattern Z



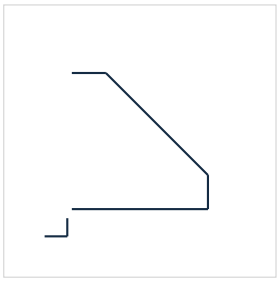
(C) Pattern R



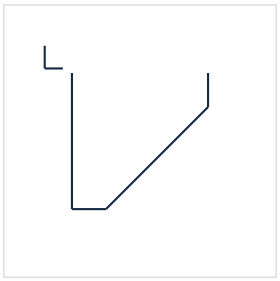
(D) Pattern W



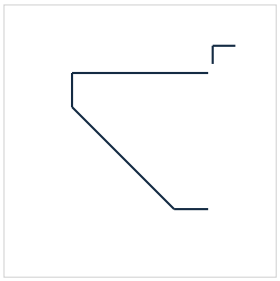
87. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1



2

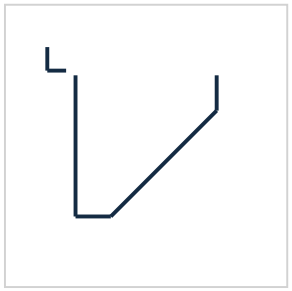


3

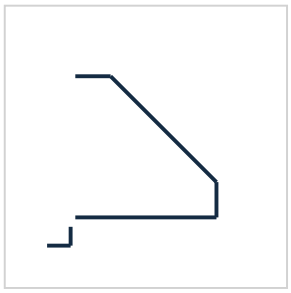
(A) Pattern L



(B) Pattern R



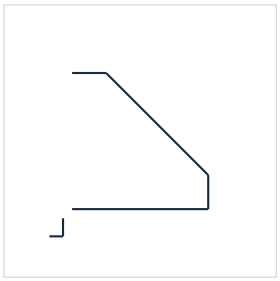
(C) Pattern N



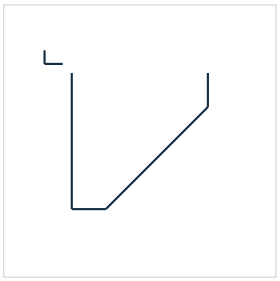
(D) Pattern X



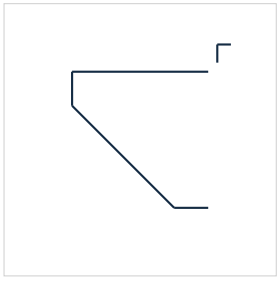
88. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

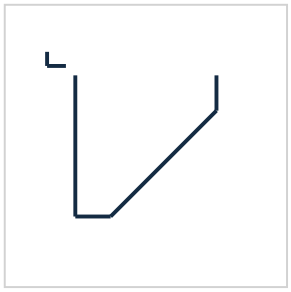


bottom left

?

bottom right

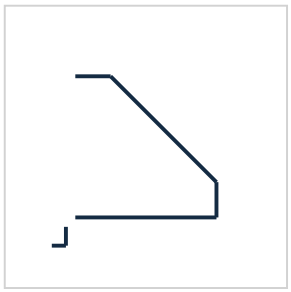
(A) Pattern Y



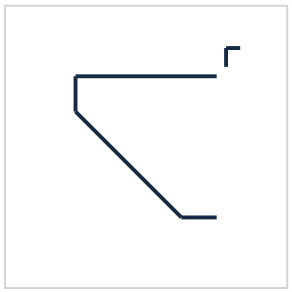
(B) Pattern V



(C) Pattern M



(D) Pattern L



89. The whole has 10 unit positions. Part A covers the first 3; part B covers 3 positions, overlapping A in exactly 1 positions as shown. How many positions remain uncovered by either part?



whole



part A positions



part B positions

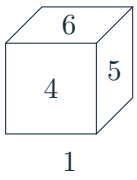
(A) 4

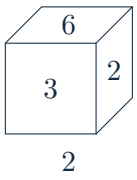
(B) 5

(C) 6

(D) 3

90. The views show the same die. Give the ordered pair: (face opposite 2, face opposite 1). Do not assume opposite labels sum to seven.





(A) $(5, 6)$
(B) $(5, 3)$
(C) $(5, 2)$
(D) $(5, 1)$

Mathematics

91. How many integers k make 12sin x+16cos x+5=k have exactly two distinct solutions in $0 \leq x < 2\pi$?

(A) 39
(B) 19
(C) 40
(D) 41

92. Find the coefficient of x^{11} in (1+2x)exp(3x).

(A) $\frac{729}{19712}$
(B) $\frac{729}{22400}$
(C) $\frac{2187}{492800}$
(D) $\frac{6561}{492800}$

93. P(A)=$\frac{7}{19}$, P(B)=$\frac{6}{19}$, P(A∩B)=$\frac{3}{19}$. Find P(neither A nor B).

(A) $\frac{13}{19}$
(B) $\frac{7}{19}$
(C) $\frac{9}{19}$
(D) $\frac{3}{19}$

94. If lim as x→0 of [sin(5x)-ax]/x^3=b is finite, find a+b.

(A) $-\frac{125}{6}$
(B) $-\frac{95}{6}$
(C) $\frac{155}{6}$
(D) 5

95. A point is equidistant from (3,0) and x=-3. Find the area between its locus, the y-axis, and y=12/2, y=12.

(A) 24
(B) 48
(C) 6
(D) 42

96. An AP has a_1=-20,d=2. How many of its first 15 terms are strictly positive?

(A) 11
(B) 3
(C) 4
(D) 12

97. How many singular 2×2 matrices with entries from{-1,0,1} have the sum of the squares of all entries equal to 3?

(A) 2
(B) 0
(C) −1
(D) 1

98. $|U|=160$, $|A|=42$, $|B|=38$, $|A\cap B|=16$. Find the number of elements belonging to both A,B or to neither.

(A) 96 (B) 112

(C) 144 (D) 48

99. $f(2x+1)=4x^2+2$. Find $\int_1^3 f(t)dt$.

(A) 12 (B) $\frac{20}{3}$

(C) $\frac{10}{3}$ (D) $\frac{40}{3}$

100. An equivalence relation has classes of sizes 10,8,6. How many unordered pairs of distinct related elements are there?

(A) 88 (B) 276

(C) 45 (D) 200

101. The letters A,A,B,C and 7 further distinct letters are arranged in a row. How many arrangements have nonadjacent As, have B,C adjacent in either order, and have exactly one A at an end?

(A) 80640 (B) 1128960

(C) 2903040 (D) 564480

102. For $(d^4y/dx^4)^3+(y')^2+x=0$, find order+degree if the degree is defined; otherwise select "Degree is not defined".

(A) 12 (B) Degree is not defined

(C) 7 (D) 8

103. L_1:r=t(1,2,0), L_2:r=(2,0,3)+s(0,1,1). Find the square of their shortest distance.

(A) 49 (B) $\frac{49}{6}$

(C) $\frac{7\sqrt{6}}{6}$ (D) 13

104. For $x=t^2$,$y=t^3$,$t>0$, find d^3y/dx^3 at $t=5$.

(A) $-\frac{3}{1000}$ (B) $-\frac{3}{10}$

(C) $\frac{3}{1000}$ (D) $\frac{3}{20}$

105. For the ellipse $x^2/36+y^2/16=1$, find the distance from its right focus to its nearer directrix.

(A) $\frac{28\sqrt{5}}{5}$ (B) $\frac{18\sqrt{5}}{5}$

(C) $\frac{8\sqrt{5}}{5}$ (D) $\frac{36\sqrt{5}}{5}$

106. A line has direction (1,6,2). A plane has normal (2,1,6). Let θ be their acute angle. Find $\tan^2\theta$.

(A) $\frac{400}{1681}$ (B) $\frac{1681}{400}$

(C) $\frac{400}{1281}$ (D) $\frac{1281}{1681}$

107. How many injective functions from{a,b,c} to{1,...,6} satisfy $f(a)<f(b)<f(c)$ and $f(a)+f(c)$ even?

(A) 4 (B) 10

(C) 20 (D) 5

108. A convergent GP has first term 5, ratio $\frac{1}{3}$. Find the sum of all terms after its first 4 terms.

(A) $\frac{5}{108}$ (B) $\frac{5}{36}$

(C) $\frac{1}{54}$ (D) $\frac{5}{54}$

109. The tangent to $y = x^3 - 3x$ at $x = 2$ meets the two coordinate axes at A and B. Find the area of triangle OAB, where O is the origin.

(A) $\frac{32}{3}$ (B) $\frac{64}{9}$

(C) $\frac{256}{9}$ (D) $\frac{128}{9}$

110. Let f(x+2)=3 x^2. Find the area enclosed between y=f(x) and y=9(x-2).

(A) $\frac{1}{2}$ (B) $\frac{9}{2}$

(C) $-\frac{27}{2}$ (D) $\frac{27}{2}$

111. X earns 8 and uses 2 labour hours; Y earns 2 and uses 3 hours. There are 100 hours and at least 5 items of Y must be made. Which model is correct?

(A) Maximize $8x + 2y$; $2x + 3y \leq 100$, $x \geq 0$, $y \geq 5$ (B) Maximize $2x + 8y$; $2x + 3y \leq 100$, $y \leq 5$

(C) Minimize $8x + 2y$; $2x + 3y \leq 100$, $y \geq 5$ (D) Maximize $8x + 2y$; $2x + 3y \geq 100$, $x, y \geq 0$

112. f(x)=x^3+5x, g=f^{-1}. Find $g''(84)$.

(A) $-\frac{24}{148877}$ (B) $\frac{24}{148877}$

(C) $\frac{1}{24}$ (D) $-\frac{24}{2809}$

113. An AP has a_1=6, d=8. Find a_1+a_3+$\cdots$+a_17.

(A) 630 (B) 702

(C) 1190 (D) 780

114. The line r=(7,4,4)+t(1,2,2) and plane 2x-y=17 are given. Find the squared minimum distance between them.

(A) $\frac{7\sqrt{5}}{5}$ (B) 0

(C) 49 (D) $\frac{49}{5}$

115. An observer at height 7 m sees the tower foot at depression 30° and its top at elevation 60°. The observer descends vertically to ground level. If the new elevation is α, find tanα.

(A) $\sqrt{3}$ (B) $\frac{\sqrt{3}}{3}$

(C) $\frac{4\sqrt{3}}{3}$ (D) $\frac{5\sqrt{3}}{3}$

116. Four observations are 10,10,x,y with population mean 10 and variance 4/2. Find xy.

(A) 100 (B) 104

(C) 192 (D) 96

117. Evaluate Σ j/2^j for j=1,...,2.

(A) 1 (B) $\frac{1}{2}$

(C) $\frac{5}{4}$ (D) 2

118. How many integers x satisfy $4^{-x} - (288)2^{-x} + 8192 \leq 0$?

(A) 9 (B) 4

(C) 3 (D) 2

119. For xy-6x-6y+36=0, find the squared length of the diagonal of the rectangle formed with the axes.

(A) 144 (B) 36

(C) 72 (D) $6\sqrt{2}$

120. 11 distinct people stand in a line. Three specified people may occupy neither end position and no two of these three may be adjacent. How many orders are possible?

(A) 20321280 (B) 8467200

(C) 1411200 (D) 3386880

121. A factory makes real quantities $x, y \geq 0$. Product X earns 4 per unit and uses 2 labour hours; Y earns 2 and uses 3 hours. At most 60 hours are available and at least 2 units of Y must be made. What is the maximum profit?

(A) 40 (B) 116

(C) 120 (D) 112

122. $A = \begin{bmatrix} 0 & 6 \\ 0 & 0 \end{bmatrix}$. Find the (1,2) entry of $(I+A)^5$.

(A) 5 (B) 7776

(C) 6 (D) 30

123. Evaluate $\int_0^{(6\pi)} \sin^4 x \, dx$.

(A) $\frac{3\pi}{4}$ (B) 3π

(C) 0 (D) $\frac{9\pi}{4}$

124. For f(x)=|x-3|/(x-5), on which maximal open intervals is f strictly decreasing?

(A) $(3,5) \cup (5,\infty)$ (B) $(3,\infty)$

(C) $(-\infty,5)$ (D) $(-\infty,3)$

125. Using digits 0,1,2,3,4,5,6,7 without repetition, how many 5-digit positive integers are divisible by 6 and have their first digit smaller than their last digit?

(A) 228 (B) 342

(C) 0 (D) 1080

126. $y'+y/(x+1)=1$ for $x \geq 0$, $y(0)=5$. Find $(8)y(7)-y(0)$.

(A) $-\frac{63}{2}$ (B) $\frac{73}{16}$

(C) $\frac{63}{2}$ (D) $\frac{73}{2}$

127. In $(2x^2 - x^{-1})^{11}$, find the ratio of the coefficient of x^7 to that of x^{10}.

(A) $\frac{7}{10}$ (B) $\frac{7}{5}$

(C) $-\frac{7}{10}$ (D) $-\frac{10}{7}$

128. F(x)=$\int_x^{(x^2)}$(t+11)^{2}dt. Find F′(8).

(A) 84224 (B) 90000

(C) 89639 (D) 90361

129. All tangents common to x^2/196+y^2/49=1 and x^2+y^2=$\frac{686}{5}$ touch the ellipse at points P_i. Find Σ(P_i,x)^2 over the four tangency points.

(A) 168 (B) $\frac{2744}{5}$

(C) 28 (D) 672

130. Find the area bounded by y=x, y=4/x, and x=8 in the first quadrant.

(A) $30 - \log(256)$ (B) 30

(C) $\log(256)$ (D) $12 - 8\log(2)$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A thin straight tube of length 5 m and internal area $\frac{1}{5000}$ m^2 is completely filled with an incompressible liquid of density 500 kg/m^3. It rotates horizontally about one end at angular speed 3 rad/s. Ignoring gravity along the tube, find the pressure difference between its outer and inner ends.

(A) 112500 Pa (B) 56250 Pa

(C) 11250 Pa (D) 18750 Pa

132. A travelling wave is $y = 3\sin[8(x - \frac{15}{2}t) + \pi/4]$ mm with x in metres and t in seconds. What is its frequency?

(A) $\frac{30}{\pi}$ Hz (B) $\frac{60}{\pi}$ Hz

(C) $\frac{\pi}{30}$ Hz (D) $\frac{240}{\pi}$ Hz

133. Two identical converging thin lenses of focal length 15 cm are separated by 45 cm. A parallel beam enters the first. Find the distance of the final image from the second lens, positive on the outgoing side.

(A) 15 cm (B) $\frac{15}{2}$ cm

(C) 30 cm (D) $\frac{45}{2}$ cm

Chemistry

134. For AgCl(s)+2NH3⇌[Ag(NH3)2]+ +Cl-, Ksp Kf=1/4. Before solid is added there is zero chloride, an unknown silver-ammine background, and free NH3=3/20 M. Measured additional dissolved AgCl is 9/272 M. All dissolved silver remains complexed; volume is fixed. What was the initial background complex concentration?

(A) 1/25 (B) 1/100

(C) 1/50 (D) -1/50

135. Two equilibrated phenolphthalein samples are colourless: one acidic, one in very strong alkali. Which comparison is chemically justified?

(A) Both must contain only the same protonated lactone because their colours agree.

(B) The acidic sample favours the lactone; the strongly alkaline sample can contain a colourless hydroxide-addition form.

(C) The strongly alkaline sample must have pH below7.

(D) The acidic sample must be pink because all lactones are coloured.

136. How many listed compounds are secondary alcohols? cyclohexanol, 2-methylpropan-1-ol, 2-methylpropan-2-ol, propan-2-ol. Count compounds,not total OH groups.

(A) 2 (B) 3

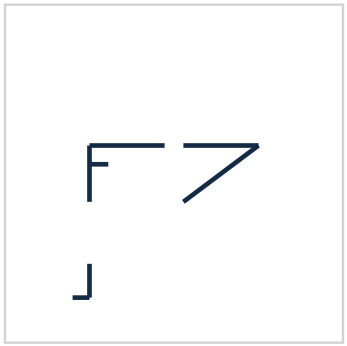
(C) 0 (D) 1

Logical Reasoning

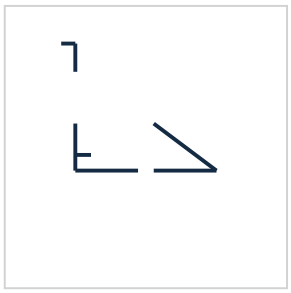
137. Preserve the exact relation and its order: Ship : Fleet :: ? : Swarm. Choose the missing term.

(A) Wolf
(B) Bee
(C) Lion
(D) Ship

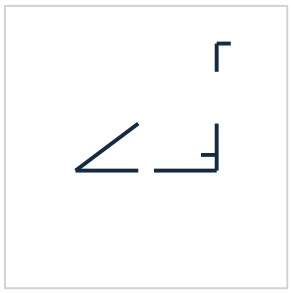
138. Choose the exact horizontal water reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



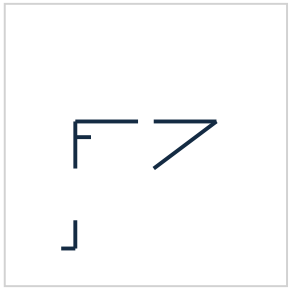
(A) Pattern R



(B) Pattern U



(C) Pattern Z



(D) Pattern Y



139. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 2 (B) 1

(C) 4 (D) 3

Mathematics

140. How many $\theta \in [0,2\pi)$ satisfy $|2\sin\theta|=1$ and $\cos\theta<0$?

(A) 4 (B) 1

(C) 2 (D) 0

141. $F'(x)=\frac{(x+1)^2}{(x+2)^4}$ and $F(0)=4$. Find $F(2)-2F(0)$.

(A) $\frac{787}{192}$ (B) $-\frac{749}{192}$

(C) $-\frac{787}{192}$ (D) $\frac{19}{192}$

142. Maximize $Z=3x+4y$ subject to $x\geq 0$, $y\geq 6$, $x+y\leq 11$, $x+2y\leq 16$.

(A) 36 (B) 38

(C) 39 (D) 32

BITSAT Mock 40

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. Two measured lengths are 3.4 cm and 0.15 cm. Their arithmetic mean is computed by division by the exact counting number 2. Using the least decimal-place precision of the measurements and retaining guard digits until the end, which result in cm should be reported?

(A) 1.9 (B) 1.7

(C) 1.775 (D) 1.8

2. Two ideal gases A and B have molar masses M and $2M$. If their rms speeds are equal, what is the ratio of their average translational kinetic energies per molecule, K_B/K_A?

(A) 2 (B) $\frac{\sqrt{2}}{2}$

(C) $\frac{1}{2}$ (D) 4

3. A stationary nucleus of mass number 220 emits an alpha particle whose kinetic energy is 4 MeV. Taking daughter and alpha masses proportional to $A-4$ and 4, find the total energy release Q, including daughter recoil.

(A) $\frac{2\sqrt{6}}{9}+4\,\text{MeV}$ (B) $\frac{110}{27}\,\text{MeV}$

(C) $4\,\text{MeV}$ (D) $220\,\text{MeV}$

4. A monochromatic wave of frequency 600000000 Hz enters a linear nondispersive medium with $\varepsilon_r = 25$ and $\mu_r = 4$. Its frequency is unchanged. Find its wavelength in the medium. Take $c = 3\times10^8$ m/s.

(A) $\frac{1}{10}\,\text{m}$ (B) $\frac{1}{4}\,\text{m}$

(C) $\frac{\sqrt{29}}{58}\,\text{m}$ (D) $\frac{1}{20}\,\text{m}$

5. Light of vacuum wavelength 400 nm passes from a medium of index 2 into one of index 3. Find the ratio of its frequencies in the second and first media.

(A) $\frac{9}{4}$ (B) $\frac{3}{2}$

(C) $\frac{1}{3}$ (D) 1

6. An active transistor has $\alpha = \frac{39}{40}$. Its collector resistor is 4 kΩ and the measured collector-resistor drop is 8 V. Neglect leakage. What base current in mA is required?

(A) $\frac{1}{20}$ mA (B) 2 mA

(C) $\frac{80}{1521}$ mA (D) $\frac{2}{39}$ mA

7. Two uniform thin rods meet at a right angle at the origin. Both have length 7 m. The rod along +x has linear density 5 times that of the rod along +y. A force parallel to the y-axis must produce pure translation. At what x-coordinate must its line of action lie?

(A) $\frac{35}{12}$ m (B) $\frac{7}{2}$ m

(C) $\frac{7}{12}$ m (D) $\frac{35}{6}$ m

8. A rod of length 6 m pivots at its lower end. A peg constrained to move horizontally stays in contact with its upper tip. At an instant the rod makes an angle with $\sin\theta = 3/5$ to the horizontal and the tip's horizontal speed has magnitude 4 m/s. Find the rod's angular-speed magnitude.

(A) $\frac{8}{15}$ rad/s (B) $\frac{2}{3}$ rad/s

(C) $\frac{2}{5}$ rad/s (D) $\frac{10}{9}$ rad/s

9. A particle has position-dependent velocity components $\vec{v} = 2x\,\hat{i} - 2y\,\hat{j}$ in SI units. Given $dx/dt = v_x$ and $dy/dt = v_y$, what is its acceleration magnitude at $(4, 4)$ m?

(A) $16\sqrt{2}\,\mathrm{m/s^2}$ (B) $16\,\mathrm{m/s^2}$

(C) $8\sqrt{2}\,\mathrm{m/s^2}$ (D) $32\sqrt{2}\,\mathrm{m/s^2}$

10. A travelling wave is $y = 4\sin[5(x - 16t) + \pi/4]$ mm with x in metres and t in seconds. What is its frequency?

(A) 80 Hz (B) $\frac{40}{\pi}$ Hz

(C) $\frac{8}{\pi}$ Hz (D) $\frac{200}{\pi}$ Hz

11. A mass 5 kg initially moves in the positive direction at 4 m/s. A force opposite to that direction rises linearly from zero to 8 N and then falls linearly to zero over a total duration 3 s. Find its signed final momentum.

(A) 14 kg m/s (B) 8 kg m/s

(C) 20 kg m/s (D) 32 kg m/s

12. A coil of 2 turns and area 2 $\mathrm{m^2}$ carries 4 A in a uniform 4 T field. Its plane makes 30° with the field. Find torque magnitude.

(A) 32 N m (B) $32\sqrt{3}$ N m

(C) $16\sqrt{3}$ N m (D) 64 N m

13. A series RC circuit is driven at fixed rms voltage. Multiplying angular frequency by 3 multiplies rms current by $\frac{9}{7}$. Find R^2/X_C^2, where X_C is the initial capacitive reactance.

(A) $\frac{20}{7}$ (B) $\frac{11}{16}$

(C) $\frac{5}{4}$ (D) 9

14. A series RLC circuit is resonant at angular frequency ω_0. Its capacitance is then multiplied by 9 while L stays fixed. What angular frequency restores resonance, as a fraction of ω_0?

(A) 1 (B) 3

(C) $\frac{1}{3}$ (D) $\frac{\sqrt{3}}{3}$

15. A parallel-plate capacitor has vacuum capacitance C_0. A dielectric of constant 5 fills a fraction $\frac{1}{3}$ of the plate area and the entire gap in that region; the remainder stays vacuum. The voltage is fixed. Find the fraction of total stored energy residing in the dielectric-filled region.

(A) $\frac{1}{11}$ (B) $\frac{5}{7}$

(C) $\frac{1}{3}$ (D) $\frac{5}{3}$

16. A particle has potential $U(x, y) = 5x^2y + 7y$ J in SI units. Find the force magnitude at $(2, 4)$ m.

(A) $\sqrt{7129}\,\mathrm{N}$ (B) $27\,\mathrm{N}$

(C) $80\,\mathrm{N}$ (D) $108\,\mathrm{N}$

17. A capacitor of capacitance 4 F is initially charged and then connected to an ideal inductor 5 H. Find the first time at which one half of the total energy is stored magnetically in the inductor.

(A) $\sqrt{5}\,\mathrm{s}$ (B) $\frac{\sqrt{5}\pi}{40}\,\mathrm{s}$

(C) $\frac{\sqrt{5}\pi}{2}\,\mathrm{s}$ (D) $2\sqrt{5}\pi\,\mathrm{s}$

18. Two identical uniform spheres touch. Their radii are multiplied by 14 while their density is divided by 14. By what factor does their gravitational attraction change?

(A) 38416 (B) 1

(C) 2744 (D) 196

19. Two coherent beams yield maximum-to-minimum interference intensity ratio $\frac{169}{121}$. Find the larger individual beam intensity divided by the smaller.

(A) 144 (B) 12

(C) $\frac{21025}{576}$ (D) 145

20. A bar magnet of moment 3 A m^2 is rotated quasistatically from 60° to 120° relative to a uniform 7 T magnetic field. Find external work.

(A) $21\sqrt{3}\,\mathrm{J}$ (B) $21\,\mathrm{J}$

(C) $\frac{21\sqrt{3}}{2}\,\mathrm{J}$ (D) $\frac{21}{2}\,\mathrm{J}$

21. A photon and a nonrelativistic particle have equal de Broglie wavelengths. The particle speed is $c/17$. If the photon then loses half its energy while the particle momentum is unchanged, what is the ratio of particle kinetic energy to the photon's final energy?

(A) $\frac{2}{17}$ (B) $\frac{1}{17}$

(C) 17 (D) 1

22. A liquid of volume 2 litres and bulk modulus 2000000000 Pa is subjected to an additional pressure of 5000000 Pa. Neglect changes in bulk modulus. Find the magnitude of volume decrease in cubic centimetres.

(A) $5\,\mathrm{cm}^3$ (B) $800000\,\mathrm{cm}^3$

(C) $\frac{1}{200}\,\mathrm{cm}^3$ (D) $5000\,\mathrm{cm}^3$

23. An astronomical telescope in normal adjustment has magnifying-power magnitude 4 and tube length 20 cm. Find the eyepiece focal length.

(A) 4 cm (B) 16 cm

(C) 5 cm (D) $\frac{4}{5}$ cm

24. A gas receives 600 J of heat while its surroundings do 50 J of work on it. It then loses 300 J of heat at fixed volume. Find its total internal-energy change.

(A) 650 J (B) 300 J

(C) 350 J (D) 950 J

25. 2 kg of water at 30°C is mixed in an insulated vessel with $\frac{252}{167}$ kg of ice at 0°C. Use $c_w = 4200$ J/(kg K), $L_f = 334000$ J/kg. Find the mass of ice remaining at equilibrium.

(A) $\frac{18459}{19205}$ kg (B) $\frac{189}{167}$ kg

(C) $\frac{126}{167}$ kg (D) $\frac{252}{167}$ kg

26. A plane wave passes through a thin prism away from edges. Separately, a pointlike aperture produces an outgoing spherical wavefront of radius 11 cm, which propagates another 6 cm in a homogeneous medium. Which pair correctly describes the prism wavefront and the later aperture wavefront?

(A) Approximately plane and unchanged in direction; spherical with radius 11 cm.

(B) Spherical; plane with infinite radius.

(C) Approximately plane and tilted; spherical with radius 17 cm.

(D) Approximately plane and tilted; spherical with radius 5 cm.

27. A uniformly charged thin rod of length 3 m and total charge 5 C lies on the x-axis. Point P is on its extension, 2 m from the nearer end. Find field magnitude at P in units of $k = 1/(4\pi\varepsilon_0)$.

(A) $\frac{5}{2}$ (B) $\frac{1}{2}$

(C) $\frac{3}{2}$ (D) $\frac{5}{6}$

28. A Newtonian liquid of viscosity $\frac{1}{5}$ Pa s has velocity profile $v(y) = 3y + 1y^2$ in SI units. Find the shear-stress magnitude at $y = 4$ m.

(A) $\frac{2}{5}$ Pa (B) $\frac{3}{5}$ Pa

(C) $\frac{11}{5}$ Pa (D) $\frac{28}{5}$ Pa

29. The quantity $Z = x^2y^1/\sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 2, 4 and 6, respectively. To first order, what is the maximum percentage error in Z?

(A) 9 (B) 5

(C) 12 (D) 11

30. A 7 kg block on a rough horizontal table is tied over an ideal pulley to a hanging 8 kg mass. The hanging mass moves downward and $\mu_k = 1/5$. Find string tension. Take $g = 10\ \text{m/s}^2$.

(A) $\frac{224}{5}$ N
(B) 64 N
(C) 80 N
(D) $\frac{576}{5}$ N

Chemistry

31. A dry mixture contains only anhydrous MgCO3 and SrCO3. Use the ordinary Group2 carbonate stability trend. In an idealised selective-heating window, the less thermally stable carbonate decomposes completely by MCO3→MO+CO2, while the more stable carbonate remains unchanged. No other mass loss occurs. This is the stated model, not a specified real temperature. Atomic masses: Mg=24, Sr=88, C=12, O=16. The initial mixture mass is 121/25 g and the solid residue after selective heating is 77/25 g. How many moles of SrCO3 remain intact after the selective stage?

(A) 121/3700
(B) 37/2200
(C) 1/100
(D) 1/25

32. For methyl benzoate,label A=carbonyl C=O,B=acyl C-O,C=alkoxy O-C(sp3). Which bond order/ length statement correctly distinguishes the two C-O single bonds shown in a Lewis formula?

(A) C has partial double-bond character,and length A < C < B.
(B) B has partial double-bond character,and length A < B < C.
(C) A is longest because oxygen is more electronegative.
(D) B and C must have identical lengths because both are drawn single.

33. A laboratory note states: V_2O_5 catalyses SO_2 oxidation in the Contact process. Which judgement and reason are consistent?

(A) Incorrect because transition metals cannot exchange electrons or adsorb reactants.

(B) Incorrect: Vanadium oxide supports redox-mediated conversion to SO3.

(C) Correct because every transition-metal species has the same d-electron count.

(D) Correct: Vanadium oxide supports redox-mediated conversion to SO3.

34. An100mL sample contains $\frac{1}{25}$ mol HA,pKa=5,and $\frac{1}{50}$ mol HCl. Add100mL $\frac{3}{5}$ M NaOH. Use usual weak-base salt hydrolysis and $\log_{10}(\frac{1}{5}) = -\frac{699}{1000}$. Final pH is

(A) $\frac{18301}{2000}$
(B) $\frac{13301}{1000}$
(C) 5
(D) 7

35. A $\frac{1}{25}$ mol FeC_2O_4 inventory is analysed. A fraction $\frac{3}{4}$ of Fe^{2+} has already become Fe^{3+},but every oxalate remains unchanged. Subsequent complete oxidation gives Fe^{3+} and CO_2. Electron equivalents released,in mol,are

(A) $\frac{1}{100}$
(B) $\frac{3}{25}$
(C) $\frac{1}{20}$
(D) $\frac{9}{100}$

36. At a fixed temperature, a diprotic acid H2A has pKa1=6, pKa2=10, and pKw=15. Find (pKb of A2-, Kb(A2-)/Kb(HA-)). Use the conjugate pair appropriate to each basic species.

(A) (5, 10000)
(B) (9, 10000)
(C) (9, 1/10000)
(D) (5, 1/10000)

37. I:An inductive effect is transmitted through sigma bonds and generally weakens with distance. II:An electromeric effect is a permanent transfer of a sigma-electron pair in an isolated alkane. Choose the truth pattern.

(A) I true;II false.
(B) Both true.
(C) Both false.
(D) I false;II true.

38. For BH_3 units in diborane,which classification and consequence are mutually consistent?

(A) Covalent; nonstoichiometry proves all hydrogen has been removed.

(B) covalent electron-deficient; multicentre bonding accommodates the electron deficiency.

(C) All hydrides contain free H+ ions in their solid lattices.

(D) Ionic; every example must be a gas composed of discrete molecules.

39. For a hypothetical MX(s),formation enthalpy is -300 kJ mol^{-1}. M atomisation is100,first ionisation500, X_2 bond dissociation200,and electron-gain enthalpy $X(g)+e^- \rightarrow X^-$ is-300,all in kJ mol^{-1}. The positive lattice dissociation enthalpy of MX is

(A) -700
(B) 600
(C) 800
(D) 700

40. In an idealised Ostwald-process material balance, $\frac{1}{20}$ mol NH_3 is fed.90.0% of its nitrogen reaches the NO/NO_2 intermediate stream, and80.0% of that stream nitrogen is recovered as HNO_3. Find HNO_3 amount in mol. Each nitrogen atom is tracked once; recycled intermediates are not double-counted.

(A) $\frac{9}{200}$
(B) $\frac{1}{25}$
(C) $\frac{9}{250}$
(D) $\frac{17}{400}$

41. I:Ethanal is generally more reactive than propanone toward HCN addition. II:Two methyl groups increase both steric shielding and electron donation at the carbonyl carbon. Select the correct assessment.

(A) Both true;II explains I.
(B) Both true;II is false as an explanation.
(C) I false;II true.
(D) Both false.

42. Which combined classification/reactivity description is correct for C6H5CH2OH?

(A) secondary because a benzene ring contains many carbons
(B) phenolic because the molecule contains a ring
(C) primary despite being benzylic
(D) tertiary because its carbon is adjacent to an aromatic system

43. For o-nitrophenol and m-nitrophenol,which is expected to show greater steam volatility on the usual intramolecular-versus-intermolecular hydrogen-bonding argument?

(A) both must have identical volatility because they are isomers
(B) neither can hydrogen-bond because both have an aromatic ring
(C) o-nitrophenol
(D) m-nitrophenol

44. Evaluate independently. I: Each resonance contributor is a different molecule rapidly interconverting. II: N_2^+ has lower bond order than N_2.

(A) Only II is correct.
(B) Neither I nor II is correct.
(C) Both I and II are correct.
(D) Only I is correct.

45. For a fixed-composition closed system dG=V dP-S dT. At a state use V=2 m3, S=5 J/K and a local path with dT/dP=0 K/Pa. Find dG/dP along this path in m3. The supplied values are a formal local thermodynamic model; 1 Pa m3=1 J.

(A) -2 (B) 2

(C) 1 (D) 4

46. At fixedT,V,A(g)→P(g)+Q(g)+R(g)is first order,and an inert gas is present. Initial totalpressure is$\frac{2}{5}$bar;complete-decomposition pressure is$\frac{4}{5}$bar. At two half-lives,totalpressure in bar is

(A) $\frac{3}{5}$ (B) $\frac{4}{5}$

(C) $\frac{7}{10}$ (D) 1

47. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_x; same facing signs; 2: parallel p_x; opposite facing signs. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (0, 3) (B) (0, 1)

(C) (1, 1) (D) (0, 2)

48. 3-bromobenzoic acid is treated in separate stages with SOCl2,excess NH3,and Br2/NaOH. Which final product and carbon-accounting description are correct?

(A) 3-bromobenzylamine;all carbons retained (B) 3-bromoaniline;the carbonyl carbon remains as CH2

(C) the original acid;amide nitrogen is never retained (D) 3-bromoaniline;one carbon fewer than the starting acid

49. A side chain is-CH2-C6H4-OH, with OH attached directly to the ring. Which amino acid and functional group match?

(A) Glycine; carboxyl side chain. (B) Serine; amide.

(C) Phenylalanine; thiol. (D) Tyrosine; phenolicOH.

50. $\frac{64}{25}$ g pure NH_4NO_2 decomposes by $NH_4NO_2 \rightarrow N_2 + 2H_2O$. Only80.0% of the theoretical dry N_2 is recovered. Find its STP volume in litres, using22.4L/mol and NH_4NO_2=64g/mol.

(A) $\frac{896}{625}$ (B) $\frac{448}{625}$

(C) $\frac{28}{25}$ (D) $\frac{112}{125}$

51. A carbon-containing organic sample contains I only among nitrogen, sulfur and halogens. It undergoes sodium fusion with sufficient sodium. Which conversion and subsequent test protocol is chemically consistent? For a silver-halide test, use a separately prepared portion boiled with nitric acid to remove cyanide/sulfide interference where present.

(A) thiocyanate; no further treatment is ever needed before adding silver nitrate.

(B) iodide; yellow AgI after nitric-acid treatment.

(C) Nitrate and sulfate in every case; all give the same Prussian-blue test.

(D) Unchanged covalent bonds; add silver nitrate directly to the untreated organic sample.

52. An orbital has 3 angular nodes and 2 radial nodes. Maximum electrons in its entire subshell are

(A) 14 (B) 18

(C) 72 (D) 10

53. Equal volumes of 3/10 M K4[Fe(CN)6] and 3/10 M glucose are mixed. The final osmotic pressure is 324/25 atm at RT=24 L atm/mol. Glucose stays molecular, and the salt dissociates only into4K+ and one intact complex anion. What fraction of salt formula units is dissociated in the final solution? Use ideal particle counting.

(A) 4/5 (B) 2/5

(C) -2/5 (D) 1/5

54. Comparep-HO-C6H4-NH2 withp-HO-C6H4-NHCOCH3. Which structural distinction should be recognised before comparing their properties?

(A) Acetylation forms an ester at oxygen while leaving the amino group free.

(B) Acetylation forms an imine whose nitrogen is directly double-bonded to carbon.

(C) Acetylation converts the amino group into a carbonyl-unconjugated amine.

(D) Acetylation converts the amino group into a carbonyl-conjugated amide.

55. For 2,3-dimethylbut-2-ene,2-methylbut-2-ene,but-1-ene,which decreasing stability order also implies increasing heat released on complete hydrogenation to the same carbon skeleton where directly comparable?

(A) tetra-substituted > mono-substituted > tri-substituted (B) tetra-substituted > tri-substituted > mono-substituted

(C) tri-substituted > tetra-substituted > mono-substituted (D) mono-substituted > tri-substituted > tetra-substituted

56. For the same nonrelativistic ion starting from rest,an accelerator changes deBroglie wavelength from $\lambda$to$\lambda/2$. Ion charge and mass are unchanged. Factor by which accelerating voltage must change is

(A) 4 (B) $\frac{1}{4}$

(C) 2 (D) $\frac{1}{2}$

57. An aldopentoseFischer projection has CHO at top, CH2OH at bottom, and OH sides at C2, C3, C4 respectively R, L, R, whereR means right andL means left. Oxidise both terminal groups to COOH without changing internal configurations. Which D/L assignment for the sugar and classification of the diacid are correct?

(A) L; meso(optically inactive). (B) D/L is fixed by the sign of rotation alone.

(C) D; chiral. (D) D; meso(optically inactive).

58. For an aldehyde acetal with aqueous acid,which product or mechanistic conclusion is correct under the stated conditions?

(A) a ketone forms by migrating an alkyl group (B) a nitrile forms by replacing oxygen with nitrogen

(C) the original aldehyde and alcohol are regenerated (D) a carboxylic acid forms without an oxidant

59. A closed system follows states A→B→C→A. For A→B, heat q=390 J and work on the system w=-210 J. For C→A, q=-295 J and w=165 J. For B→C, q=-160 J. Find work on the system on B→C in J.

(A) -70 (B) -110

(C) 110 (D) 290

60. A Ni(2) coordination entity has coordination number 6, exactly 1 fully bound en ligand(s), and chloride in all remaining donor sites. Take en as 2-dentate with charge 0; chloride is monodentate. Find (number of ligand entities, net charge of the coordination entity).

(A) (5, -2) (B) (5, -3)

(C) (6, -1) (D) (6, -2)

English Proficiency

61. Read the sentence: The cumbersome form required the same information to be entered in six different places. Choose the closest meaning of 'cumbersome' as used in the sentence.

(A) awkward to use (B) easy to lose

(C) difficult to understand in principle (D) impossible to copy

62. Read the sentence: The class was eager to begin the experiment after the long preparation. Choose the opposite in meaning of 'eager' as used in the sentence.

(A) curious (B) ready

(C) reluctant (D) impatient

63. Read the sentence: The anonymous survey was a mirror in which the organisation could examine its habits. What does the figurative expression mean in this context?

(A) It reflected information that helped the organisation assess itself. (B) It proved that appearances never differ from reality.

(C) It made every participant give the same answer. (D) It concealed the organisation's weaknesses.

64. He measures boundaries on the ground before a map or building plan is prepared. Which role fits?

(A) architectural critic (B) surveyor

(C) navigator (D) cartographer

65. Read the sentence: A clearer pattern began to emerge from the scattered observations. Choose the opposite in meaning of 'emerge' as used in the sentence.

(A) become clearer (B) disappear

(C) develop (D) appear

66. Read the sentence: Independent records corroborate the driver's account of when the road was closed. Choose the closest meaning of 'corroborate' as used in the sentence.

(A) explain in greater detail (B) establish every part beyond doubt

(C) repeat in identical words (D) support with additional evidence

67. The investigation asks how words and sentences convey meaning, rather than how speech sounds are formed. Which field is it?

(A) semantics (B) phonetics

(C) etymology (D) orthography

68. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. At a narrow bend, parked vehicles repeatedly blocked the painted lane. B. Cyclists could then use the route without being forced into the traffic at its most restricted point. C. The town painted a cycle route along the edge of a busy road. D. The council moved the parking spaces away from that bend.

(A) At a narrow bend, parked vehicles repeatedly blocked the painted lane.

(B) The council moved the parking spaces away from that bend.

(C) The town painted a cycle route along the edge of a busy road.

(D) Cyclists could then use the route without being forced into the traffic at its most restricted point.

69. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The reviewer had not examined the appendix when the first judgement was issued.

(A) The appendix had been examined before the first judgement was issued.

(B) The appendix has not been examined by the reviewer when the judgement is issued.

(C) The reviewer had not been examined by the appendix when the judgement was issued.

(D) The appendix had not been examined by the reviewer when the first judgement was issued.

70. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. will be considered for the positions Q. Only those applicants who submit R. all the required documents S. advertised in the notice.

(A) S-R-P-Q (B) Q-R-P-S

(C) P-Q-R-S (D) R-Q-P-S

Logical Reasoning

71. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 6, 24, 26, 104, ?, 424, 426.

(A) 416 (B) 106

(C) 102 (D) 424

72. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?



1 2 3 4

(A) 2 (B) 3

(C) 1 (D) 4

73. In a row, Asha is 8th from the left and 39th from the right. Bina is 10th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 27 (B) 26

(C) 24 (D) 25

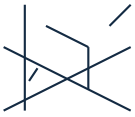
74. Consider only these statements; universal statements alone do not imply existence. All hikers are editors; No hikers are editors. Conclusions: I. No artists are hikers. II. Some artists are not hikers. Which conclusions follow necessarily?

(A) Neither I nor II (B) Only I

(C) Both I and II (D) Only II

75. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 3 (B) 1

(C) 2 (D) 4

76. The age table uses the same two people throughout:

	younger	older
now	x	$x+10$
3 years ago	$x-3$	$x+7$

. The product of their ages 3 years ago was 875. What will the older person's age be 3 years from now?

(A) 41 (B) 31

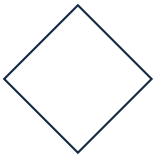
(C) 38 (D) 51

77. Preserve the exact relation and its order: Assist : Help :: ? : Quiet. Choose the missing term.

(A) Courageous (B) Silent

(C) Begin (D) Rapid

78. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

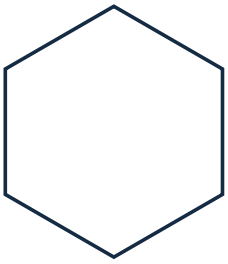


B

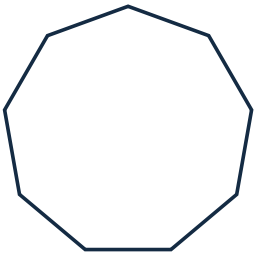


C

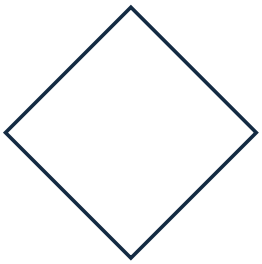
(A) Pattern P



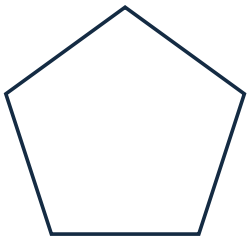
(B) Pattern R



(C) Pattern N



(D) Pattern Z



79. The whole has 9 unit positions. Part A covers the first 4; part B covers 3 positions, overlapping A in exactly 2 positions as shown. How many positions remain uncovered by either part?



whole



part A positions



part B positions

(A) 2

(B) 0

(C) 4

(D) 5

80. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

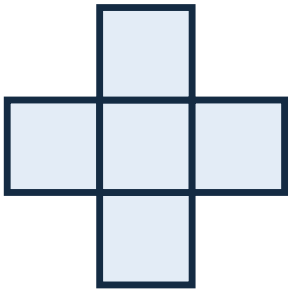


piece 1

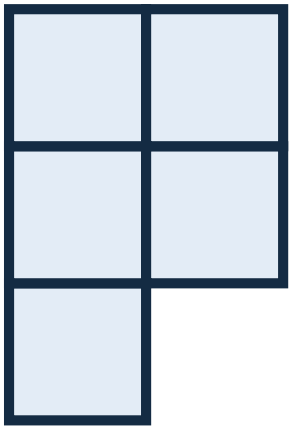


piece 2

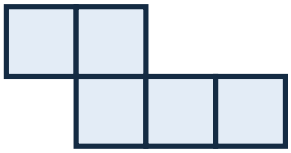
(A) Pattern U



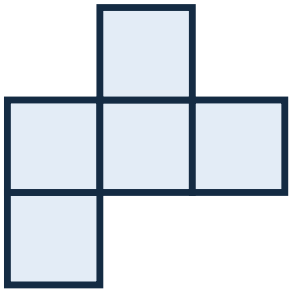
(B) Pattern N



(C) Pattern T



(D) Pattern Q



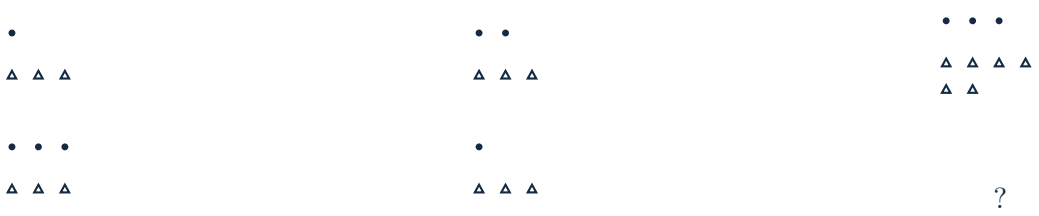
81. A walker goes 5 m west, then 8 m south, then turns left and walks 13 m. In which direction is the starting point from the walker?

(A) North-west
(B) South-west
(C) North-east
(D) South-east

82. Consider only these statements; universal statements alone do not imply existence. Some artists are campers; Some campers are artists; No artists are joggers. Conclusions: I. No joggers are campers. II. No artists are campers. Which conclusions are individually possible without contradicting the statements?

(A) Only I
(B) Neither I nor II
(C) Only II
(D) Both I and II

83. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.

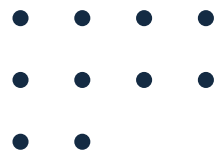


(A) Pattern U

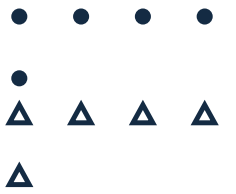
(B) Pattern X

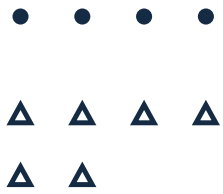


(C) Pattern V



(D) Pattern W





84. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 17$ and $5 \mapsto 32$. Which input maps to 42?

(A) 8

(B) 42

(C) 6

(D) 7

85. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: WYG, ZCK, ?, FKS, IOW, LSA.

(A) OGC

(B) DHP

(C) CGO

(D) BFN

86. Five people occupy positions 1 to 5 from left to right, one per position, and wear five different shirt colours: red, blue, green, white and yellow. Esha is 1st from the left. Asha is 2nd from the left. Charu is 3rd from the left. Dev is 4th from the left. Bina is 5th from the left. The green shirt is in position 3. The red shirt is in position 4. The blue shirt is in position 5. Asha does not wear white. Which shirt does Asha wear?

(A) Green

(B) Red

(C) Blue

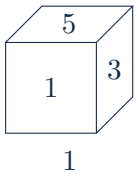
(D) Yellow

87. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, CHAIR is coded as DIBJS. Which word is coded as DMPVE?

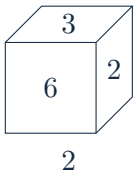
(A) BKNTC (B) ENQWF

(C) CLOUD (D) DMPVE

88. The views show the same die. Give the ordered pair: (face opposite 5, face opposite 6). Do not assume opposite labels sum to seven.



1



2

(A) $(6, 2)$ (B) $(6, 1)$

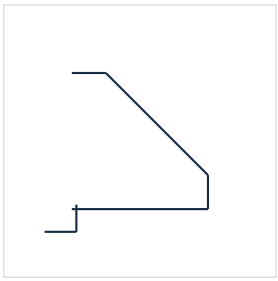
(C) $(6, 5)$ (D) $(6, 3)$

89. Lata points to a photograph and says, "This person is the son of my father's only son." Lata is female, and the photographed person is male. How is the photographed person related to Lata?

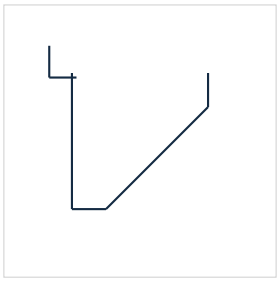
(A) Nephew (B) Brother

(C) Son (D) Daughter

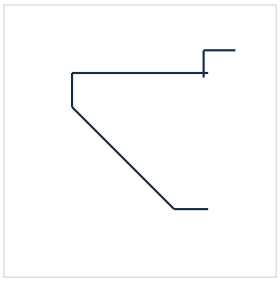
90. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1



2

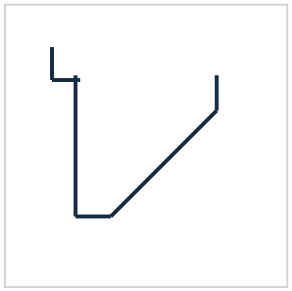


3

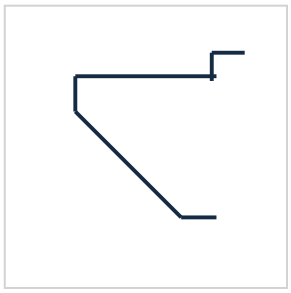
(A) Pattern P



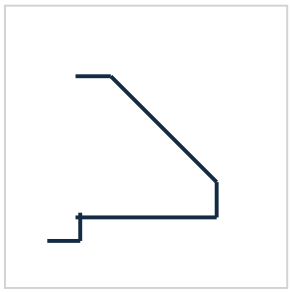
(B) Pattern Y



(C) Pattern R



(D) Pattern T



Mathematics

91. For x>0, xy′-y=x and y(1)=2. Find y(5)/5-y(1).

(A) $\log(5)$

(B) $-\log(5)$

(C) $\log(5)+2$

(D) $8+\log(3125)$

92. Evaluate the integral from 0 to 2π of (9-6 sin x)(1+cos x)/(81-36 $\sin^2$x) dx.

(A) $\frac{2\sqrt{5}\pi}{15}$

(B) $\frac{2\pi}{9}$

(C) $-\frac{\log(3)}{6}+\frac{\log(5)}{6}+\frac{2\sqrt{5}\pi}{15}$

(D) $\frac{2\sqrt{13}\pi}{39}$

93. X has mean 13, variance 36. Y=aX+2, a>0, has mean 80. Find Var(Y).

(A) 1298 (B) 7380

(C) 216 (D) 1296

94. An AP has a_1=11, d=2. Find $a_1+a_3+\cdots+a_13$.

(A) 175 (B) 322

(C) 161 (D) 126

95. How many x with $0\le x<6\pi$ satisfy $\sin^4x+\cos^4x=3/4$ and cos x>0?

(A) 24 (B) 12

(C) 15 (D) 0

96. Circles $x^2+y^2=324$ and $(x-22)^2+y^2=324$ meet at A,B. Find $|AB|^2$.

(A) 1780 (B) $2\sqrt{203}$

(C) 812 (D) 328

97. The letters A,A,B,C and 5 further distinct letters, one namedD, are arranged in a row. How many arrangements have nonadjacent As, adjacent B,C in either order, and D at an end?

(A) 7200 (B) 4800

(C) 3600 (D) 2880

98. $x=7+11(1-t^2)/(1+t^2)$, $y=4+22t/(1+t^2)$, where t is real. Which point of the containing circle is not attained?

(A) $(18, 4)$ (B) $(7, -7)$

(C) $(7, 15)$ (D) $(-4, 4)$

99. Let $f(x)=|x-3|/(x-8)$ for $x\neq 8$. Within $0<x<13$, what is the total length of the open intervals on which $f'(x)<0$?

(A) 10 (B) 3

(C) 12 (D) 13

100. A convergent GP has first term 11, ratio $\frac{1}{4}$. Find the sum of all terms after its first 6 terms.

(A) $\frac{11}{12288}$ (B) $\frac{11}{4096}$

(C) $\frac{11}{2304}$ (D) $\frac{11}{3072}$

101. Unit vectors a,b,c satisfy $|(a\times b)\cdot c|=1$. Find $|a+2b-4c|^2$.

(A) 21 (B) 49

(C) 7 (D) 20

102. L_1:r=t(1,2,0), L_2:r=(3,0,4)+s(0,1,1). Find the square of their shortest distance.

(A) $\frac{116}{5}$ (B) $\frac{50}{3}$

(C) $\frac{9}{5}$ (D) $\frac{5\sqrt{6}}{3}$

103. z=3+4i. Find 1/z+1/conjugate(z).

(A) $\frac{8}{25}$ (B) $\frac{6}{25}$

(C) $\frac{1}{6}$ (D) 6

104. A 9-element set contains distinct a,b. How many 2-element subsets contain exactly one of a,b and also contain a?

(A) 8 (B) 21

(C) 7 (D) 14

105. a=(7,0,0), b=(0,2,0), c=(0,0,4), p=a+b. Evaluate [p×(b×c)]·(a+c).

(A) −64 (B) 0

(C) −260 (D) 64

106. Find the squared distance between 6x+8y=4 and 3x+4y=9.

(A) $\frac{49}{100}$ (B) 49

(C) $\frac{7}{5}$ (D) $\frac{49}{25}$

107. r=pa+qb, a=(1,0,0), b=(1,1,0), r·a=8, r·b=13. Find $|r|^2$.

(A) 34 (B) $\sqrt{89}$

(C) 89 (D) 13

108. The chord of contact from P to $x^2/36$-$y^2/36$=1 is x=6/2. Find the product of the slopes of the two tangents from P.

(A) 3 (B) $-\frac{1}{9}$

(C) −1 (D) $-\frac{1}{3}$

109. P(A)=$\frac{1}{4}$, P(B)=$\frac{1}{2}$, P(A∩B)=$\frac{1}{8}$. Find P(neither A nor B).

(A) $\frac{5}{8}$ (B) $\frac{3}{8}$

(C) $\frac{1}{2}$ (D) $\frac{3}{4}$

110. For sqrt(d^3y/dx^3)=$(y')^2$+x, find order+degree if the degree is defined; otherwise select "Degree is not defined".

(A) Degree is not defined (B) 4

(C) 6 (D) 3

111. f(x)=sin(2x)/x+3 for x<0; f(x)=k cos x for x≥0. Find k for continuity at zero.

(A) −1 (B) 3

(C) 2 (D) 5

112. Two points on y^2=12 x have parametersu,v withuv=-3. Find the latus-rectum length of their midpoint locus.

(A) 35 (B) 37

(C) 6 (D) 72

113. F′(x)=$\frac{1}{x^2}-\frac{1}{x^2\sqrt{x^2+1}}$ and F(1)=5. Find F(5)-2F(1).

(A) $-\frac{29}{5}-\frac{\sqrt{26}}{5}+\sqrt{2}$ (B) $-\sqrt{2}+\frac{\sqrt{26}}{5}+\frac{29}{5}$

(C) $-\frac{21}{5}-\sqrt{2}+\frac{\sqrt{26}}{5}$ (D) $-\sqrt{2}+\frac{4}{5}+\frac{\sqrt{26}}{5}$

114. Two different numbers are chosen uniformly from {1,...,12}. Let X be their maximum. Find Var(X).

(A) $\frac{\sqrt{65}}{3}$ (B) $\frac{65}{9}$

(C) $\frac{26}{3}$ (D) $\frac{247}{3}$

115. An equivalence relation has classes of sizes 9,2,2. How many unordered pairs of distinct related elements are there?

(A) 38 (B) 76

(C) 13 (D) 78

116. If lim as x→0 of [sin(6x)-ax]/x^3=b is finite, find a+b.

(A) 42 (B) −36

(C) −30 (D) 6

117. Find the area between y=(x-7)(x-9) and the x-axis for 0≤x≤15.

(A) −270 (B) $\frac{818}{3}$

(C) $\frac{409}{3}$ (D) 270

118. For 3x+4y+6=0 and 4x+3y+4=0, find the squared distance from the origin to the angle bisector of positive slope.

(A) 4 (B) 2

(C) $\frac{50}{49}$ (D) $\sqrt{2}$

119. There are 12 Boolean variables p_1,...,p 12. Exactly 1 variables must be true. How many truth assignments satisfy [(p_1 implies p_2) and (p_1 implies not p_2)]?

(A) 12 (B) 10

(C) 11 (D) 1

120. $A=\begin{bmatrix}4 & 1\\ 2 & 2\end{bmatrix}$, $B=\begin{bmatrix}1 & 4\\ 0 & 1\end{bmatrix}$. Find tr(BA) without assuming AB=BA.

(A) −2 (B) 18

(C) 12 (D) 14

121. Find the area bounded by y=x, y=36/x, and x=30 in the first quadrant.

(A) $36\log(5)$ (B) $144-36\log(5)$

(C) 432 (D) $432-36\log(5)$

122. The tangent to $y = x^3 - 62x$ at $x = 8$ meets the two coordinate axes at A and B. Find the area of triangle OAB, where O is the origin.

(A) $\frac{524288}{65}$ (B) $\frac{131072}{65}$

(C) $\frac{8192}{3}$ (D) $\frac{262144}{65}$

123. If |z-11|=3, find the minimum of $|z|^2$.

(A) 112 (B) 64

(C) 196 (D) 8

124. Two distinct integers are chosen uniformly from $\{1, \ldots, 10\}$. Given that their maximum is 7, find the expected value of their minimum.

(A) $\frac{7}{2}$ (B) 7

(C) 4 (D) $\frac{11}{3}$

125. The parabola $y=x^2+px+q$ passes through (0,16). The absolute value of its vertex ordinate is minimized over $p \geq 0$. Find p^2+q^2 at the minimum.

(A) 320 (B) 576

(C) 24 (D) 80

126. Find the sum of the squares of all positive real solutions of |x-6|=4.

(A) 108 (B) 12

(C) 104 (D) 72

127. A circular sector has radius 7 and arc length $\frac{7\pi}{2}$. Find its area.

(A) $\frac{\pi}{4}$ (B) $\frac{49\pi}{4}$

(C) $\frac{7\pi^2}{8}$ (D) $\frac{49}{4}$

128. How many injective functions from{a,b,c} to{1,...,7} satisfy f(a)<f(b)<f(c) and f(a)+f(c) even?

(A) 5 (B) 35

(C) 19 (D) 90

129. Let $D(t) = \det \begin{pmatrix} 9+t & 1+t \\ 1+t & 3+t \end{pmatrix}$. Find D′(0).

(A) -10 (B) -14

(C) 14 (D) 10

130. Let $z = 5 + 7i$. Find $(1/z - 1/\overline{z})^2$.

(A) $-\frac{49}{1369}$ (B) $-\frac{7}{37}$

(C) $\frac{49}{1369}$ (D) $\frac{25}{1369}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A quantity with the dimensions of time is written $G^a h^b c^d$. Given $[G] = M^{-1}L^3T^{-2}$, $[h] = ML^2T^{-1}$, $[c] = LT^{-1}$, find $a + 2b - d$.

(A) 4
(B) $\frac{3}{2}$
(C) 3
(D) $\frac{7}{2}$

132. 1 kg of liquid is heated by 40 K at constant external pressure 5000 Pa. Its specific heat at that pressure is 4000 J/(kg K), and density falls from 1000 to 800 kg/m^3. Find internal-energy increase.

(A) $\frac{639975}{4}$ J
(B) 160000 J
(C) $\frac{639995}{4}$ J
(D) $\frac{640005}{4}$ J

133. A metal has work function 4 eV and is illuminated separately with photons of energies 12 eV and 132 eV. The fastest electrons enter the same perpendicular magnetic field. Find the second orbit radius divided by the first. Use nonrelativistic motion.

(A) $\frac{4\sqrt{6}}{3}$
(B) 4
(C) 16
(D) $\frac{\sqrt{34}}{2}$

Chemistry

134. A neutral Group1 atom K is compared with Na. What is the relevant explanation for K's easier first electron loss?

(A) Both atoms lose a neutron rather than an electron during ionisation.
(B) Its outer electron occupies a more distant and more shielded shell.
(C) K contains no core electrons.
(D) K must have a lower nuclear charge than Na.

135. Compare the elementary octahedral d-d splitting transitions of A=$[CrCl_6]^{3-}$,B=$[CrF_6]^{3-}$, C=$[Cr(H_2O)_6]^{3+}$. For this comparison use only the usual spectrochemical ordering and $E = hc/\lambda$. Which ordering of absorbed wavelengths follows?

(A) $\lambda_B > \lambda_C > \lambda_A$.
(B) $\lambda_A < \lambda_B < \lambda_C$.
(C) $\lambda_A > \lambda_B > \lambda_C$.
(D) $\lambda_A = \lambda_B = \lambda_C$ because metal oxidation states match.

136. Judge I: SCN^- can bind through S or N. II: For comparable complexes of one metal,the chelate effect can favour a multidentate ligand without changing the metal oxidation state. Which is correct?

(A) Both I and II.
(B) Only II.
(C) Only I.
(D) Neither I nor II.

Logical Reasoning

137. P is the only child of Q, and P is a parent of A. Q is the paternal grandmother of A. What can be concluded about P's gender?

(A) The stated relationships are inconsistent (B) Male

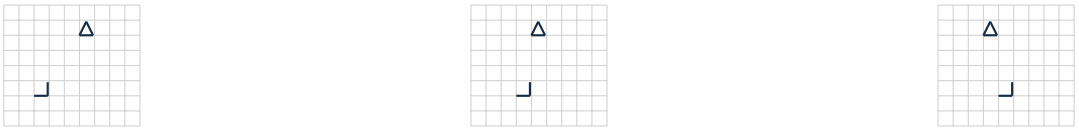
(C) Cannot be determined (D) Female

138. Each of Gita, Hari, Charu visits on one different day (Monday, Tuesday or Wednesday) and buys one different fruit (Plum, Orange, Peach). Gita visits on Wednesday, and Hari on Tuesday. The visitor on Wednesday buys peach. Hari does not buy plum. Which fruit does Hari buy?

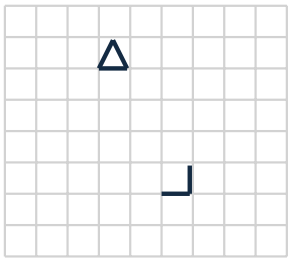
(A) Orange (B) Plum

(C) Peach (D) No unique assignment

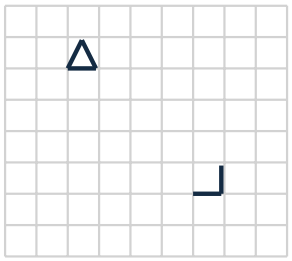
139. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



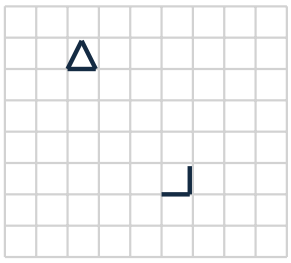
(A) Pattern P



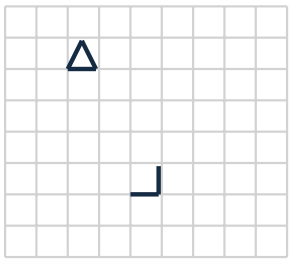
(B) Pattern R



(C) Pattern L



(D) Pattern Q



Mathematics

140. For integer n, consider: "If 6 divides n, then 2 divides n." Which is its contrapositive?

(A) If 6 does not divide n, then 2 does not divide n. (B) If 2 does not divide n, then 6 does not divide n.

(C) If 2 divides n, then 6 divides n. (D) 6 divides n and 2 does not divide n.

141. A circle passes through O=(0,0), A=(10,0), B=(0,6). Find the length of the tangent from P=(20, 0).

(A) 200 (B) 10

(C) $10\sqrt{2}$ (D) 20

142. a=(9,8,0), b=(8,-9,4). Find $|a+b|^2$-$|a-b|^2$.

(A) 0 (B) -16

(C) -32 (D) -68

BITSAT Mock 41

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

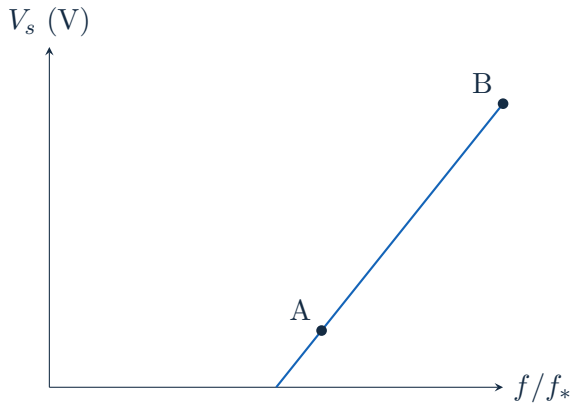
1. A gas receives 600 J of heat while its surroundings do 150 J of work on it. It then loses 300 J of heat at fixed volume. Find its total internal-energy change.

(A) 300 J
(B) 750 J
(C) 450 J
(D) 150 J

2. A one-electron ion with nuclear charge $Z = 3$ absorbs a photon of energy $\frac{987}{125}$ eV. The ejected electron has maximum kinetic energy 3 eV. From which principal quantum number was the electron removed? Use $E_n = -13.6Z^2/n^2$ eV and neglect recoil.

(A) $\frac{5}{3}$
(B) 5
(C) 6
(D) 4

3. For a metal, the stopping-potential graph has points A $(6f_*, 3\,\text{V})$ and B $(10f_*, 15\,\text{V})$. The electronic gain of the vertical recording channel is later found to be 3: its recorded voltage is 3 times the true stopping voltage. What is hf_*/e in volts?



(A) 4 V
(B) 1 V
(C) 3 V
(D) 9 V

4. Two ideal gases A and B have molar masses M and $4M$. If their rms speeds are equal, what is the ratio of their average translational kinetic energies per molecule, K_B/K_A?

(A) 1 (B) $\frac{1}{2}$

(C) 4 (D) 2

5. A uniform block of density 500 kg/m^3 floats in a liquid. Then the liquid density is reduced to $\frac{3}{4}$ times 1000 kg/m^3, while the block's volume and density are unchanged. What fraction of the block is now submerged?

(A) $\frac{1}{3}$ (B) $\frac{3}{4}$

(C) 1 (D) $\frac{2}{3}$

6. A single slit of width 4 mm is illuminated by wavelength 400 nm. A screen is 3 m away. Find the separation of the second minima on opposite sides of the central maximum, in the small-angle approximation.

(A) 1200 mm (B) $\frac{3}{10}$ mm

(C) $\frac{6}{5}$ mm (D) $\frac{2}{5}$ mm

7. A balanced Wheatstone bridge has arms AB=AD=5 ohms and BC=DC=10 ohms. A resistor of arbitrary positive resistance connects B to D. Find equivalent resistance between A and C.

(A) $\frac{25}{2}\,\Omega$ (B) $\frac{15}{2}\,\Omega$

(C) $10\,\Omega$ (D) $15\,\Omega$

8. A point charge $15q$ is placed at a vertex of a cube. Find the electric flux through the three faces not meeting that vertex, combined, in units of q/ε_0.

(A) $\frac{15}{16}$ (B) $\frac{15}{8}$

(C) $\frac{5}{8}$ (D) $\frac{15}{4}$

9. 5 kg of water at 30°C is mixed in an insulated vessel with $\frac{630}{167}$ kg of ice at 0°C. Use $c_w = 4200$ J/(kg K), $L_f = 334000$ J/kg. Find the mass of ice remaining at equilibrium.

(A) $\frac{315}{167}$ kg (B) $\frac{945}{334}$ kg

(C) $\frac{630}{167}$ kg (D) $\frac{18459}{7682}$ kg

10. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 5 percent. The measured increase in its absolute temperature is $\frac{1763}{40}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) $\frac{1763}{40}$ K (B) 17630 K

(C) $\frac{1763}{2}$ K (D) 430 K

11. The quantity $Z = x^2y^1/\sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 2, 1 and 4, respectively. To first order, what is the maximum percentage error in Z?

(A) 9 (B) 3

(C) 5 (D) 7

12. A symmetric uniform chain of total mass 5 kg hangs from equal-height supports. Each end tangent makes an angle with $\sin\theta = 3/5$, $\cos\theta = 4/5$. What is the tension at the lowest point? Take $g = 10\ \text{m/s}^2$.

(A) $\frac{100}{3}$ N (B) $\frac{75}{4}$ N

(C) 20 N (D) 25 N

13. Two ideal cells of emfs 18 V and 2 V oppose each other in a loop containing two resistors each 3 ohms. Find the current magnitude.

(A) $\frac{8}{3}$ A (B) $\frac{1}{3}$ A

(C) $\frac{10}{3}$ A (D) $\frac{20}{3}$ A

14. A conveyor receives 2 packets each second, each of mass $\frac{1}{5}$ kg and speed 2 m/s. It accelerates them to 6 m/s without changing elevation. Find the ideal mechanical power delivered to the packets.

(A) $\frac{8}{5}$ W (B) $\frac{32}{5}$ W

(C) $\frac{16}{5}$ W (D) 8 W

15. A charge in a plane vacuum wave moves at speed $c/8$ along a direction making angle θ with the magnetic field, with $\cos\theta = 1/8$. Find the ratio of the electric-force magnitude to the magnetic-force magnitude.

(A) $\frac{8\sqrt{7}}{21}$ (B) $3\sqrt{7}$

(C) $\frac{64\sqrt{7}}{21}$ (D) $\frac{3\sqrt{7}}{64}$

16. Masses 1 kg and 2 kg hang from a light inextensible string over a pulley with moment of inertia I and radius R such that $I/R^2 = 4$ kg. There is no slipping or axle friction. Find acceleration magnitude. Take $g = 10\ \text{m/s}^2$.

(A) $10\ \text{m/s}^2$ (B) $\frac{10}{7}\ \text{m/s}^2$

(C) $\frac{10}{11}\ \text{m/s}^2$ (D) $\frac{5}{2}\ \text{m/s}^2$

17. A stationary nucleus with mass number 28 and binding energy per nucleon 8 MeV splits into two equal daughters, each having total binding energy 113.0 MeV. Which statement is correct?

(A) The process absorbs 2 MeV; the daughters have larger binding energy per nucleon.

(B) No energy can be released because fission always requires a larger mass number than 28.

(C) The process releases 2 MeV; the daughters have larger binding energy per nucleon.

(D) The process releases 4 MeV; each daughter releases 2 MeV.

18. A geometrically similar solid structure is enlarged by a factor 5 in every linear dimension. Its material density is reduced by a factor 4. Its own weight is carried by a similarly scaled base. Find base compressive-stress multiplier.

(A) $\frac{5}{16}$ (B) $\frac{5}{4}$

(C) $\frac{4}{5}$ (D) 20

19. A series RC circuit is driven at fixed rms voltage. Multiplying angular frequency by 8 multiplies rms current by $\frac{8\sqrt{57}}{57}$. Find R^2/X_C^2, where X_C is the initial capacitive reactance.

(A) $8 + \frac{64\sqrt{57}}{57}$ (B) 8

(C) $\frac{64}{57}$ (D) 7

20. A solid sphere rolls without slipping from rest through vertical height 12 m. A thin ring does so through vertical height 7 m. Find the ratio of their squared centre-of-mass speeds, sphere to ring.

(A) $\frac{120}{49}$ (B) $\frac{12}{7}$

(C) $\frac{49}{120}$ (D) $\frac{12}{5}$

21. A Carnot engine initially runs between 500 K and 200 K. Its hot reservoir is raised by 60 K. By how much must the cold-reservoir temperature rise to keep efficiency unchanged?

(A) 24 K (B) 36 K

(C) $\frac{150}{7}$ K (D) 150 K

22. One mole of nonoverlapping identical spheres of radius r has total geometric material volume V. A second sample contains $\frac{9}{20}$ mole of spheres of radius $4r$. What is its total geometric material volume divided by V? Ignore packing voids.

(A) $\frac{144}{5}$ (B) $\frac{9}{5}$

(C) $\frac{9}{1280}$ (D) $\frac{36}{5}$

23. A drop of mass 1.92×10^{-17} kg is suspended in a vertical field of magnitude 400 N/C. It has excess electrons. One excess electron is removed without changing its mass. Find its immediate downward acceleration, taking $g = 10$ m/s^2 and $e = 1.6 \times 10^{-19}$ C.

(A) $\frac{10}{9}$ m/s^2 (B) 10 m/s^2

(C) $\frac{20}{3}$ m/s^2 (D) $\frac{10}{3}$ m/s^2

24. A uniform thin rod of length 2 m is hinged at one end and released from rest in a horizontal position. Find the magnitude of its initial angular acceleration. Take $g = 10$ m/s^2.

(A) 15 rad/s^2 (B) 30 rad/s^2

(C) $\frac{15}{2}$ rad/s^2 (D) $\frac{5}{2}$ rad/s^2

25. At fixed temperature and number density, the molecular collision diameter is multiplied by 5 and molecular mass by 16. Find the collision-frequency multiplier in the hard-sphere model.

(A) 25 (B) $\frac{1}{4}$

(C) $\frac{25}{16}$ (D) $\frac{25}{4}$

26. A spherical Gaussian surface encloses charges +7q and -4q. Another charge +7q is outside the surface. No charge lies on the surface. Find total outward electric flux in units of q/ε_0.

(A) 10 (B) 11

(C) 3 (D) 7

27. Current rises linearly from 0 to 2 A over 5 s and then falls linearly to 0 over another 5 s. It remains zero for the next 10 s. Find the average current over the entire 20 s interval.

(A) 2 A (B) $\frac{1}{4}$ A

(C) $\frac{2}{3}$ A (D) $\frac{1}{2}$ A

28. A 3 kg block moving at 5 m/s compresses a spring of constant 100 N/m. During compression, a damper removes 1/4 of the block's initial kinetic energy. Find maximum compression; gravity does no work.

(A) $\frac{3}{4}$ m
(B) $\frac{3\sqrt{2}}{8}$ m
(C) $\frac{\sqrt{3}}{4}$ m
(D) $\frac{\sqrt{3}}{2}$ m

29. A projectile is launched at 50 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{15}{16}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ m/s^2.

(A) 90 m
(B) 100 m
(C) 240 m
(D) 60 m

30. A long solid cylindrical conductor of radius 4 m carries uniformly distributed current 8 A. Find magnetic field magnitude at radial distance $R/2$ from the axis in units of $\mu_0/(2\pi)$.

(A) 1
(B) 2
(C) $\frac{1}{2}$
(D) 4

Chemistry

31. A primary-standard sample of pure anhydrous Na_2CO_3 weighs $\frac{159}{250}$ g. It is titrated with0.100M HCl. Find the additional acid volume in mL needed after the first carbonate-to-hydrogencarbonate equivalence point to reach complete conversion to CO_2/water. Take Na_2CO_3=106g/mol.

(A) 121
(B) 120
(C) 60
(D) 30

32. $\frac{1}{20}$ mol Cl_2 is contacted with $\frac{3}{40}$ mol KOH in cold dilute solution. Only $Cl_2 + 2KOH \rightarrow KCl + KClO + H_2O$ occurs. Final solution volume is $\frac{3}{10}$ L. Find the KClOconcentration in mol/L, assuming no side reaction or product loss.

(A) $\frac{1}{6}$
(B) $\frac{1}{4}$
(C) $\frac{1}{8}$
(D) $\frac{1}{24}$

33. A dry mixture contains only anhydrous CaCO3 and BaCO3. Use the ordinary Group2 carbonate stability trend. In an idealised selective-heating window, the less thermally stable carbonate decomposes completely by MCO3→MO+CO2, while the more stable carbonate remains unchanged. No other mass loss occurs. This is the stated model, not a specified real temperature. Atomic masses: Ca=40, Ba=137, C=12, O=16. The initial mixture mass is 791/100 g and the solid residue after selective heating is 703/100 g. What is the CaO mass fraction of the solid residue after selective heating?

(A) 88/703
(B) 200/703
(C) 16/113
(D) 112/703

34. An H_2SO_4 stock is29.0% by mass. A20.0mL aliquot is diluted to 100mL,and the diluted molarity is $\frac{18}{25}$M. H_2SO_4 M98. Stock density in g/mL is

(A) $\frac{441}{1250}$
(B) $\frac{882}{725}$
(C) $\frac{882}{3625}$
(D) $\frac{882}{145}$

35. For 2-bromobutane with one stereogenic centre,which structural classification is correct when the actual halogen-bearing carbon and molecular formula are examined?

(A) no possible optical isomerism

(B) two enantiomeric configurations

(C) a meso pair

(D) four constitutional isomers at the same connectivity

36. In an antifluorite-derived crystal, X2- occupies all FCC anion sites. A measured fraction 3/4 of occupied M cations is M2+ and the rest M+. Only tetrahedral cation vacancies compensate charge; no interstitials or anion defects occur. What fraction of all tetrahedral sites is vacant?

(A) 3/8

(B) 3/7

(C) 3/14

(D) 4/7

37. An unknown sugar has the same molecular formula as glucose but its open-chain carbonyl is at C2. What follows and what remains unproved?

(A) It must be a nonreducing disaccharide.

(B) It is a ketohexose; the molecular formula alone does not specify its full stereochemistry.

(C) It must have five carbons.

(D) It is necessarily identical to glucose at every stereocentre.

38. For the metal centre in V^{3+},Z=23. Determine the number of d electrons after formal oxidation-state assignment; use O=-2 in oxoanions.

(A) 4

(B) 3

(C) 1

(D) 2

39. An orbital has 1 angular nodes and 3 radial nodes. Maximum electrons in its entire subshell are

(A) 50

(B) 14

(C) 10

(D) 6

40. Within an additive covalent-radius model,a single X-X bond is 130 pm and an X-H bond is 105 pm. A linear H-X-X-H skeleton is assumed with collinear single bonds. Its terminal H-to-H distance,in pm,is

(A) 340

(B) 420

(C) 235

(D) 470

41. A food extract contains exactly one of the listed two-component combinations. Using the usual qualitative tests under suitable conditions, its results are: Benedict on warming: negative; iodine: positive; Biuret: negative; Sudan III: positive. Which combination matches?

(A) glucose and albumin

(B) starch and triolein

(C) starch and albumin

(D) glucose and starch

42. Propanenitrile is completely hydrolysed to its acid. The acid is reduced with LiAlH4 and worked up; the isolated alcohol is then oxidised with PCC. What is the final organic product?

(A) propanone

(B) propanoic acid

(C) ethanal

(D) propanal

43. For closed-system paths A→B→C, q_AB=420 J, w_AB=-220 J, q_BC=-180 J and w_BC=140 J, with work defined on the system. A direct C→A path has w=150 J. What is its heat q in J?

(A) -510 (B) -110

(C) 310 (D) -310

44. Rotation about the O-O sigma bond in H2O2 changes the dihedral angle without breaking bonds. Which inventory remains unchanged?

(A) Each rotated conformation must acquire an O=O pi bond. (B) A180° rotation removes two lone pairs by resonance.

(C) Three sigma bonds and four total lone pairs. (D) The number of oxygen nuclei must change.

45. For the reaction $2M + O_2 \rightarrow 2MO$, take $\Delta H^\circ = -220$ kJ and $\Delta S^\circ = -0.200$ kJ K^{-1} over the stated range. Find ΔG° in kJ at 750 K and the sign of its slope against temperature.

(A) -370 kJ; negative slope (B) -70 kJ; positive slope

(C) 70 kJ; negative slope (D) -220 kJ; zero slope

46. A saturated acyclic monocarbonyl compound X has formula C3H6O and gives iodoform but not Tollens. A separate sample is reduced with NaBH4 and protonated. Which alcohol is produced?

(A) propanal (B) ethanol

(C) propan-1-ol (D) propan-2-ol

47. For the isolated oxoanion ClO4-, choose its chlorine oxidation state and molecular geometry about chlorine.

(A) +9; trigonal planar. (B) +7; linear.

(C) +7; tetrahedral. (D) +5; tetrahedral.

48. A cubic cell has identical A atoms at all corners and at the body centre, with this arrangement repeated indefinitely. Which translation-based description is correct?

(A) Primitive cubic with a chemically distinct two-species basis.

(B) A BCC Bravais lattice; corner-to-body translation preserves the labelled arrangement.

(C) No Bravais lattice because a cell contains more than one lattice point.

(D) FCC because every cubic cell has face centring.

49. At 1 atm compare two separate pairs: (ethanol, dimethyl ether) and (1-propanol, methoxyethane). A student selects dimethyl ether for the first higher-boiling substance because its formula mass is unchanged, and 1-propanol for the second. Which assessment is consistent with intermolecular interactions?

(A) First selection wrong; second selection correct. (B) Both selections wrong.

(C) Both selections correct. (D) First selection correct; second selection wrong.

50. A carbon-containing organic sample contains N and Cl among nitrogen, sulfur and halogens. It undergoes sodium fusion with sufficient sodium. Which conversion and subsequent test protocol is chemically consistent? For a silver-halide test, use a separately prepared portion boiled with nitric acid to remove cyanide/sulfide interference where present.

(A) Unchanged covalent bonds; add silver nitrate directly to the untreated organic sample.

(B) cyanide and chloride; nitrogen test positive; chloride test after removing interfering cyanide.

(C) sulfide; no further treatment is ever needed before adding silver nitrate.

(D) Nitrate and sulfate in every case; all give the same Prussian-blue test.

51. At fixedT,V,A(g)→P(g)+Q(g)+R(g)is first order,and an inert gas is present. Initial totalpressure is$\frac{3}{5}$bar;complete-decomposition pressure is$\frac{6}{5}$bar. At two half-lives,totalpressure in bar is

(A) $\frac{9}{8}$ (B) $\frac{3}{4}$

(C) $\frac{21}{20}$ (D) $\frac{6}{5}$

52. For 2A+B→3C+4D, A also disappears by an independent channel producing no C or D, and D is independently consumed at 1/2000 M/s. Measured total A disappearance is 9/1000 M/s and net D appearance is 23/2000 M/s. What is the A disappearance rate through its independent channel, in M/s?

(A) 1/250 (B) 3/200

(C) 3/1000 (D) 13/4000

53. Compare3s and3p orbitals of a one-electron ion and of an ordinary many-electron atom in the elementary central-field picture. Which distinction is correct?

(A) They share energy in the ideal hydrogenic n-only model; screening and penetration can split them in a many-electron atom.

(B) They must be degenerate in every atom because n agrees.

(C) They must have different n because l differs.

(D) The one-electron ion has electron-electron repulsion between its only electron and itself in the elementary model.

54. An ideal H_2/N_2 mixture atT has mean molar mass15g/mol ($H_2$2,$N_2$28). Define the ensemble mean of v^2 by molecule-number weighting. Ratio of this mean to pureN_2at the sameT is

(A) $\frac{15}{2}$ (B) $\frac{\sqrt{30}}{2}$

(C) $\frac{15}{28}$ (D) $\frac{28}{15}$

55. A student uses pV=NkT with N=0.50 mol rather than a pure particle count. Which correction preserves the meaning of the variables?

(A) Either use pV=nRT with n=0.50 mol, or set N=0.50NA particles. (B) Use N=0.50/NA particles.

(C) Keep N=0.50 mol and delete k without any replacement. (D) Replace T by T^2.

56. Glucose in water can mutarotate, but a methyl glucoside with its anomericOH replaced byOCH3 does not undergo ordinary mutarotation without hydrolysis. Which structural explanation fits?

(A) The free hemiacetal can open; the glycosidic acetal locks the anomeric centre.

(B) Every ether is necessarily a free aldehyde.

(C) The methyl glucoside has no carbon atoms.

(D) Mutarotation requires cleavage of every C-C bond.

57. For ideal gases at one T, A(g)⇌2B(g) has Kc=16 and numerical RT=4 in compatible units. What is Kp for4B(g)⇌2A(g)? Use $K_p=K_c(RT)^{\Delta n}$ with the same standard-state convention throughout.

(A) 1/2048 (B) -1/4096

(C) 1/4096 (D) 1/8192

58. For benzaldehyde plus acetaldehyde in dilute base,consider the crossed addition,which product or mechanistic conclusion is correct under the stated conditions?

(A) benzaldehyde is the only enolate donor

(B) both aldehydes have identical alpha-H counts

(C) neither can participate because benzaldehyde has no alpha H

(D) benzaldehyde is the non-enolisable acceptor and acetaldehyde can be the enolate donor

59. An unknown is sucrose, maltose or lactose. It is reducing before hydrolysis; hydrolysis produces glucose plus a different aldohexose rather than a ketose. Identify the sugar and the second product.

(A) Lactose; galactose.

(B) Sucrose; fructose.

(C) Sucrose; galactose.

(D) Maltose; fructose.

60. A first-order reaction at 360 K leaves 1/16 of its initial reactant after100 s. Its Ea/(RT) at this temperature is 13. Use ln2=0.693 and k=A exp(-Ea/RT). Which is its frequency factor A in s-1?

(A) 693/25000 exp(13)

(B) 693/250 exp(13)

(C) 25000/693 exp(13)

(D) 693/25000 exp(-13)

English Proficiency

61. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. The less time the speaker spent Q. repeating familiar examples, R. for questions about the new argument. S. the more time remained

(A) P-Q-S-R

(B) R-Q-S-P

(C) Q-P-S-R

(D) S-P-Q-R

62. Read the sentence: A faint sound could be heard only after the machinery stopped. Choose the closest meaning of 'faint' as used in the sentence.

(A) unsteady

(B) unfamiliar

(C) far away

(D) barely perceptible

63. Read the sentence: The available evidence does not warrant such a confident conclusion. Choose the closest meaning of 'warrant' as used in the sentence.

(A) require everyone to accept

(B) justify

(C) suggest

(D) guarantee

64. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The staff were checking the exits while the audience waited outside.

(A) The exits were being checked by the staff while the audience waited outside.

(B) The exits had been checked by the audience while the staff waited outside.

(C) The staff were being checked by the exits while the audience waited outside.

(D) The exits are being checked by the staff while the audience waits outside.

65. Read the sentence: The organisation decided to withdraw its support for the event. Choose the opposite in meaning of 'withdraw' as used in the sentence.

(A) reconsider (B) explain

(C) offer (D) reduce

66. Use the simple present to express the stated regular procedure. Choose the best replacement: The route, which passes two schools, [carry] local buses every weekday.

(A) have carried (B) carries

(C) carrying (D) were carrying

67. Which ONE of the following words is misspelt?

(A) perserverance (B) necessary

(C) February (D) beginning

68. She watches the match from the stands but neither plays nor officiates. What is she?

(A) organiser (B) referee

(C) spectator (D) participant

69. Read the sentence: The isolated route is seldom used even in summer. Choose the opposite in meaning of 'seldom' as used in the sentence.

(A) rarely (B) occasionally

(C) briefly (D) often

70. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. during the afternoon. Q. To prevent the labels from fading, R. the curator placed the display S. away from direct sunlight

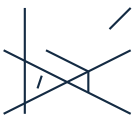
(A) S-Q-R-P (B) Q-R-S-P

(C) R-Q-S-P (D) P-R-S-Q

Logical Reasoning

71. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



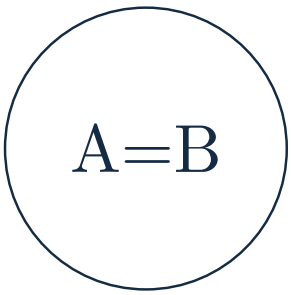
host 4

(A) 1 (B) 4

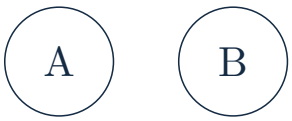
(C) 3 (D) 2

72. Choose the diagram that represents all these conditions, including the strictness information: Some A is B, some A is not B, and some B is not A.

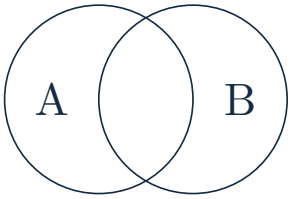
(A) Pattern M



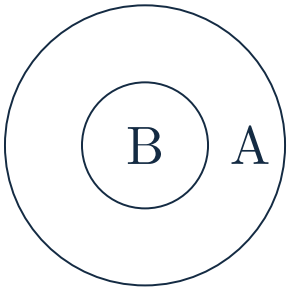
(B) Pattern P



(C) Pattern Z



(D) Pattern K



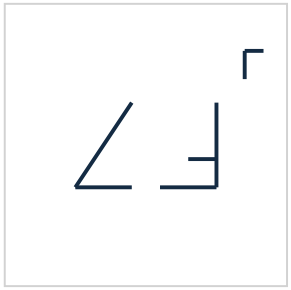
73. Preserve the exact relation and its order: Star : Constellation :: ? : Swarm. Choose the missing term.

(A) Lion
(B) Bee
(C) Wolf
(D) Ship

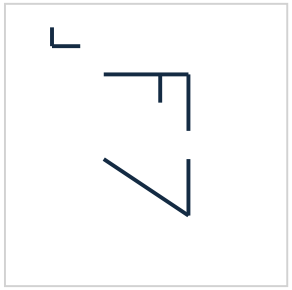
74. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



(A) Pattern R



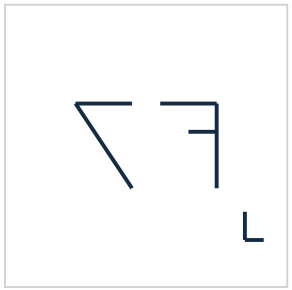
(B) Pattern Z



(C) Pattern X



(D) Pattern K



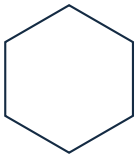
75. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

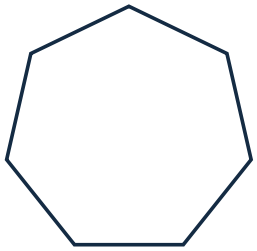


B

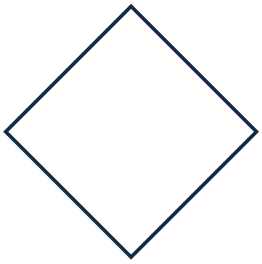


C

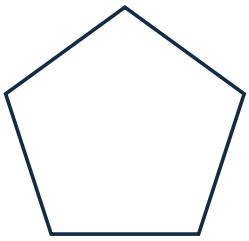
(A) Pattern L



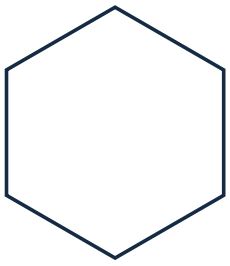
(B) Pattern W



(C) Pattern K



(D) Pattern M



76. Mira points to a photograph and says, "This person is the son of my father's only daughter." Mira is female, and the photographed person is male. How is the photographed person related to Mira?

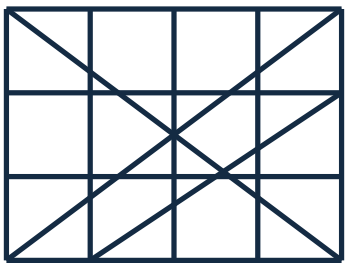
(A) Son

(B) Sister

(C) Daughter

(D) Brother

77. How many triangles are formed by all the drawn segments?



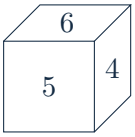
(A) 107

(B) 61

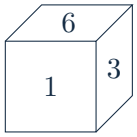
(C) 60

(D) 46

78. The views show the same die. Give the ordered pair: (face opposite 4, face opposite 6). Do not assume opposite labels sum to seven.



1



2

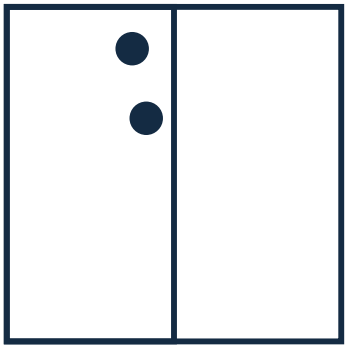
(A) $(3, 2)$

(B) $(3, 4)$

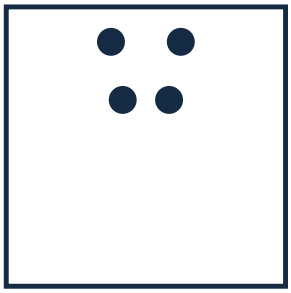
(C) $(3, 3)$

(D) $(3, 1)$

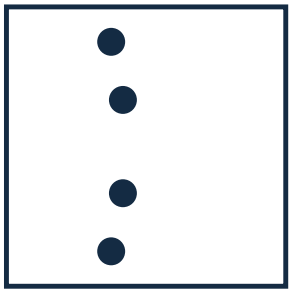
79. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



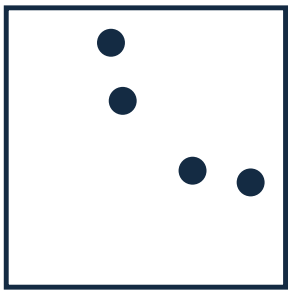
(A) Pattern U



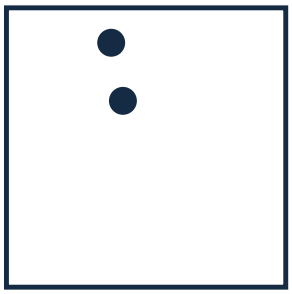
(B) Pattern N



(C) Pattern X



(D) Pattern Z



80. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 2, 6, 8, 24, ?, 78, 80.

(A) 26 (B) 72

(C) 78 (D) 74

81. In a row, Asha is 8th from the left and 35th from the right. Bina is 7th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 26 (B) 3

(C) 24 (D) 23

82. The whole has 13 unit positions. Part A covers the first 3; part B covers 3 positions, overlapping A in exactly 1 positions as shown. How many positions remain uncovered by either part?



whole part A positions part B positions

(A) 7 (B) 9

(C) 6 (D) 8

83. The age table uses the same two people throughout:

	younger	older
now	x	$x+12$
6 years ago	$x-6$	$x+6$

. The product of their ages 6 years ago was 693. What will the older person's age be 5 years from now?

(A) 38 (B) 44

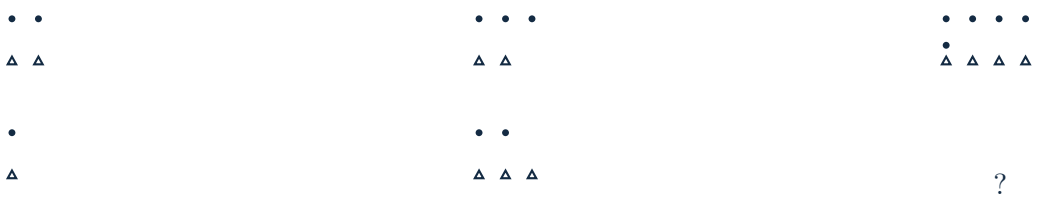
(C) 56 (D) 39

84. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: SGYN, QHAL, OICJ, ?, KKGF, ILID.

(A) HEJM (B) NKFI

(C) LIDG (D) MJEH

85. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern W



(B) Pattern U



(C) Pattern V



(D) Pattern Y



86. Consider only these statements; universal statements alone do not imply existence. All inventors are joggers; Some farmers are joggers. Conclusions: I. Some joggers are farmers. II. No farmers are inventors. Which conclusions follow necessarily?

(A) Only II

(B) Both I and II

(C) Neither I nor II

(D) Only I

87. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?



(A) 3

(B) 2

(C) 1

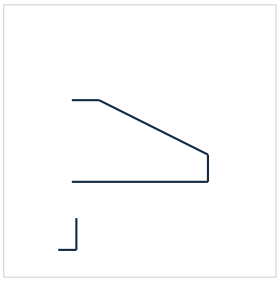
(D) 4

88. A code shifts letters cyclically through A–Z. Odd positions share one fixed shift and even positions share another. For example, CHAIR is coded as GEEFV. Which word is coded as FOMZO?

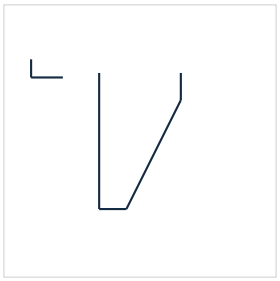
(A) CSJDL (B) JLQWS

(C) AQHBJ (D) BRICK

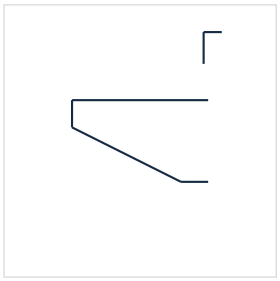
89. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

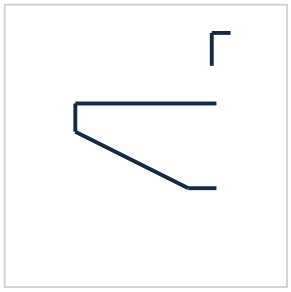


2

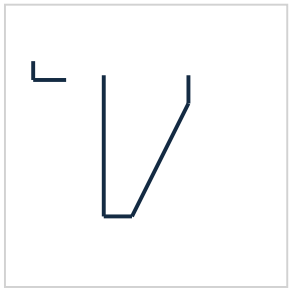


3

(A) Pattern N



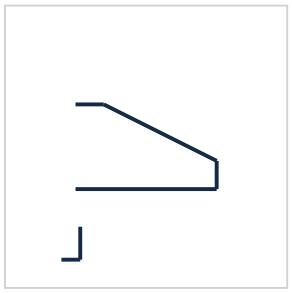
(B) Pattern R



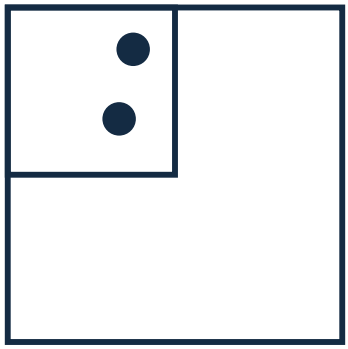
(C) Pattern T



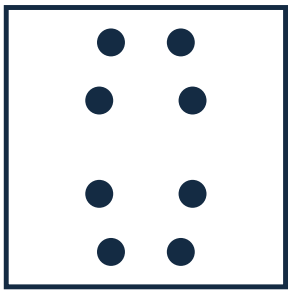
(D) Pattern Q



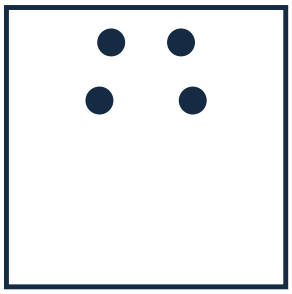
90. Starting with a unit square sheet, fold right to left along the vertical midline, then fold bottom to top along the horizontal midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



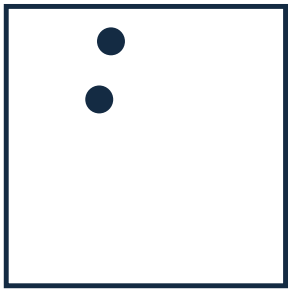
(A) Pattern V



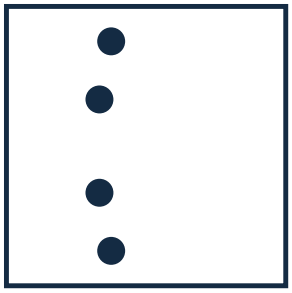
(B) Pattern X



(C) Pattern K



(D) Pattern Z



Mathematics

91. For $P(x) = \prod_{j=5}^{13}(x+j)$, find the coefficient of x^6 in $P(x)/(x+5)$.

(A) 2886 (B) 5012

(C) 2861 (D) 2506

92. α,β are roots of x^2-10x+21=0. The monic polynomial with roots α+1,β+1 is x^2-Ax+B. Find B.

(A) 22 (B) 32

(C) 23 (D) 12

93. In $(4x^2 - x^{-1})^8$, find the ratio of the coefficient of x^4 to that of x^7.

(A) $\frac{5}{16}$ (B) $-\frac{5}{16}$

(C) $-\frac{16}{5}$ (D) $\frac{5}{4}$

94. a=(2,7,0), b=(7,-2,6). Find |a+b|^2-|a-b|^2.

(A) 112 (B) 4717

(C) 0 (D) −72

95. How many solutions in $0 \leq x < 6\pi$ satisfy sin 2 x=sin x and cos x>sin x?

(A) 6 (B) 3

(C) 12 (D) 9

96. Find the mean deviation about the median for 1,2,...,13.

(A) $\frac{21}{13}$ (B) 14

(C) $\frac{42}{13}$ (D) 42

97. z=6+2i. Find 1/z+1/conjugate(z).

(A) $\frac{1}{10}$ (B) 12

(C) $\frac{1}{12}$ (D) $\frac{3}{10}$

98. An AP has a=6,d=2; its first n terms sum to 66. Find a_(n+1).

(A) 18 (B) 6

(C) 16 (D) 11

99. Find the population variance of 6+2, 12+2, ..., 42+2.

(A) 144 (B) 24

(C) 146 (D) 4

100. $A = \begin{bmatrix} 6 & 1 \\ 1 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 20 & 12 \\ 15 & 37 \end{bmatrix}$, and XA=B. Find X_{11}.

(A) $\frac{108}{35}$ (B) 3

(C) $-\frac{108}{35}$ (D) 20

101. F′(x)=$\frac{(x+1)^2}{(x+2)^4}$ and F(0)=3. Find F(5)-2F(0).

(A) $\frac{1385}{8232}$ (B) $-\frac{23311}{8232}$

(C) $\frac{26081}{8232}$ (D) $-\frac{26081}{8232}$

102. Two points on opposite sides of a tower foot are 11 metres apart on a straight level line. The elevations of the tower top are 30° and 45°. Find the tower height.

(A) $\frac{33}{2} - \frac{11\sqrt{3}}{2}$ (B) $-\frac{11}{2} + \frac{11\sqrt{3}}{2}$

(C) $\frac{11}{2}$ (D) $\frac{11}{4}$

103. Disjoint A,B have sizes 2,6. Find $|(A\times A)\cup(B\times B)|$.

(A) 52 (B) 12

(C) 40 (D) 24

104. A line has direction (1,7,2). A plane has normal (2,1,7). Let θ be their acute angle. Find $\tan^2\theta$.

(A) $\frac{529}{2916}$ (B) $\frac{529}{2387}$

(C) $\frac{2387}{2916}$ (D) $\frac{2916}{529}$

105. Circles $x^2+y^2=81$ and $(x-10)^2+y^2=81$ meet at A,B. Find $|AB|^2$.

(A) 56 (B) 124

(C) 224 (D) 324

106. $P(A)=\frac{1}{2}$, $P(B)=\frac{3}{16}$, $P(A\cap B)=\frac{3}{32}$. Find P(neither A nor B).

(A) $\frac{13}{32}$ (B) $\frac{5}{16}$

(C) $\frac{3}{32}$ (D) $\frac{13}{16}$

107. Let $\alpha=\pi/8$, $\beta=\pi/7$. Evaluate $\arccos(-\cos\alpha\cos\beta-\sin\alpha\sin\beta)$.

(A) $\frac{55\pi}{56}$ (B) $\frac{\pi}{56}$

(C) $-\frac{\pi}{56}$ (D) $\frac{57\pi}{56}$

108. $F(x)=\int_0^{x^2}(t-4)(t-49)dt$. How many local minima of F lie in [-7,7]?

(A) 3 (B) 1

(C) 5 (D) 2

109. A convergent GP has first term 8, ratio $\frac{1}{7}$. Find the sum of all terms after its first 7 terms.

(A) $\frac{2}{151263}$ (B) $\frac{1098056}{117649}$

(C) $\frac{4}{50421}$ (D) $\frac{4}{352947}$

110. $S_n=4n^2+2n$ is a sequence partial sum. Find $a_1+a_3+\cdots+a_{13}$.

(A) 210 (B) 378

(C) 390 (D) 406

111. X earns 4 and uses 2 labour hours; Y earns 3 and uses 3 hours. There are 70 hours and at least 4 items of Y must be made. Which model is correct?

(A) Minimize $4x+3y$; $2x+3y\le 70$, $y\ge 4$ (B) Maximize $4x+3y$; $2x+3y\le 70$, $x\ge 0$, $y\ge 4$

(C) Maximize $4x+3y$; $2x+3y\ge 70$, $x,y\ge 0$ (D) Maximize $3x+4y$; $2x+3y\le 70$, $y\le 4$

112. Let $\delta=\pi/8$. Evaluate $\sin\delta\ \Sigma[1/(\sin(j\delta)\sin((j+1)\delta))]$, j=2,...,3.

(A) -1 (B) $\sqrt{2}$

(C) 1 (D) $1+\sqrt{2}$

113. x=5+9(1-t^2)/(1+t^2), y=4+18t/(1+t^2), where t is real. Which point of the containing circle is not attained?

(A) $(5, -5)$ (B) $(14, 4)$

(C) $(5, 13)$ (D) $(-4, 4)$

114. Find the product of all real solutions of sqrt(13-log_3|x|)=log_3|x|-1.

(A) 0 (B) 6561

(C) -6561 (D) -81

115. y′+11y=exp(-11x), y(0)=7. Find exp(11)y(1)-7.

(A) -1 (B) 1

(C) 8 (D) $\frac{8}{e^{11}}$

116. An equivalence relation has classes of sizes 4,8,5. How many unordered pairs of distinct related elements are there?

(A) 17 (B) 30

(C) 6 (D) 44

117. Let f(x)=|x-4|/(x-10) for x≠10. Within 0<x<16, what is the total length of the open intervals on which f′(x)<0?

(A) 15 (B) 12

(C) 4 (D) 10

118. $A = \begin{bmatrix} 1 & 10 \\ 0 & 0 \end{bmatrix}$. For (I+A)^6=I+cA, find c.

(A) 5 (B) 64

(C) 63 (D) 6

119. f(x)=2+$\int_0^x$ f(x-t)dt for x≥0. Find $\int_0^1$f(t)dt.

(A) $2e$ (B) $-1 + e$

(C) 2 (D) $-2 + 2e$

120. A focal chord of y^2=24x has one endpoint parameter t=3. Find its length.

(A) 40 (B) $\frac{200}{3}$

(C) $\frac{128}{3}$ (D) $\frac{164}{3}$

121. Using digits 0,1,2,3,4,5 without repetition, how many 4-digit positive integers are divisible by 3 and have their first digit smaller than their last digit?

(A) 35 (B) 40

(C) 36 (D) 96

122. The line Re(z)=5 divides the disc |z|≤10. Find larger-segment area minus smaller-segment area.

(A) $-\frac{100\pi}{3} - 50\sqrt{3}$ (B) $25\sqrt{3} + \frac{200\pi}{3}$

(C) $-25\sqrt{3} + \frac{100\pi}{3}$ (D) $50\sqrt{3} + \frac{100\pi}{3}$

123. Under x=X+6, y=Y+2, the parabola $(y-2)^2=36(x-6)$ is normalized. Find the x coordinate of its focus in the original axes.

(A) −3 (B) 9

(C) 42 (D) 15

124. Find the area enclosed by y=x(8-x) and y=5x.

(A) $\frac{9}{2}$ (B) $-\frac{9}{2}$

(C) $\frac{129}{2}$ (D) $\frac{256}{3}$

125. If z=6-4i, find |exp(iz)|.

(A) e^6 (B) 1

(C) e^4 (D) e^{-4}

126. For t≥0 define F(t)=lim as n→ ∞ of $n^{(-3)}\Sigma$ floor(tj^2), j=1,...,n. Evaluate Σ F($(1/7)^r$), r=0,...,∞.

(A) $\frac{7}{18}$ (B) $\frac{1}{18}$

(C) $\frac{7}{6}$ (D) $\frac{1}{3}$

127. For the ellipse $x^2/49+y^2/16=1$, the chord of contact from P is 14 x/49+4 y/16=1. Find the square of the tangent length from P to the auxiliary circle $x^2+y^2=49$.

(A) 163 (B) 65

(C) 261 (D) 44895

128. Let R(x) be the remainder when x^6 is divided by (x-1)(x-2). Find R(2).

(A) 126 (B) −61

(C) 65 (D) 64

129. Unit vectors a,b,c satisfy |(a×b)·c|=1. Find $|a+2b-3c|^2$.

(A) 26 (B) 6

(C) 13 (D) 14

130. Independent events A,B have probabilities $\frac{2}{3},\frac{3}{5}$. Given A∪B, find P(A∩B).

(A) $\frac{6}{19}$ (B) $\frac{7}{13}$

(C) $\frac{6}{13}$ (D) $\frac{2}{5}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A particle moves uniformly on a circle of radius 6 m at angular speed 3 rad/s. Find the magnitude of its average velocity over an angular displacement $\pi/2$ starting at any point.

(A) $\frac{36\sqrt{2}}{\pi}$ m/s (B) $36\sqrt{2}$ m/s

(C) $\frac{72}{\pi}$ m/s (D) 18 m/s

132. A travelling wave is $y = 4\sin[5(x - 4t) + \pi/4]$ mm with x in metres and t in seconds. What is its frequency?

(A) $\frac{5}{2\pi}$ Hz (B) $\frac{10}{\pi}$ Hz

(C) $\frac{\pi}{10}$ Hz (D) $\frac{50}{\pi}$ Hz

133. A charge 2 C moves with $\vec{v} = 4\hat{i} + 3\hat{j}$ m/s in $\vec{B} = 5\hat{i} + 2\hat{j}$ T. What is the signed z-component of its magnetic force?

(A) 16 N (B) 46 N

(C) -14 N (D) 30 N

Chemistry

134. For weak-field high-spin V^{3+} and Ni^{2+},which statement follows in the spin-only approximation?

(A) They have the same moment $2\sqrt{2}$ BM despite different d-electron counts.

(B) Equal moments prove equal atomic numbers and equal ionic radii.

(C) They are both diamagnetic because ionic charge removes all d electrons.

(D) Their moments must differ because different elements cannot have equal unpaired-electron counts.

135. cyclopentanone reacts with methanol with catalytic acid,remove water. Which product correctly reflects nucleophilic addition or addition-elimination at its carbonyl rather than a different reaction class?

(A) cyclopentanol (B) the dimethyl ketal

(C) a methyl carboxylate (D) cyclopentane

136. The recurring chain segment is -O-CH2-CH2-O-CO-C6H4-CO-. Which identification and degradative bond type are correct?

(A) Natural rubber; the chain is cis-polyisoprene. (B) PTFE; only C-F and C-C bonds occur.

(C) PET; ester links can undergo hydrolysis. (D) Nylon-6,6; every link is -CO-NH-.

Logical Reasoning

137. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). The numerical component follows a constant arithmetic step. Find the missing internal term: WHR5, TJU7, QLX9, ?, KPD13, HRG15.

(A) NNA11 (B) 11ANN

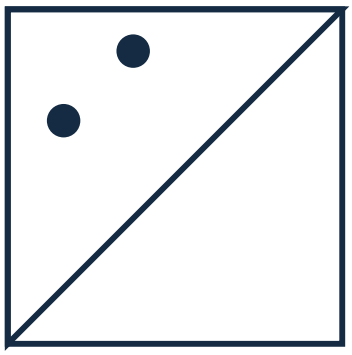
(C) OOB11 (D) MMZ11

138. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 31 and Z has code 6. Decode the separated values 16, 20, 31, 18, 12.

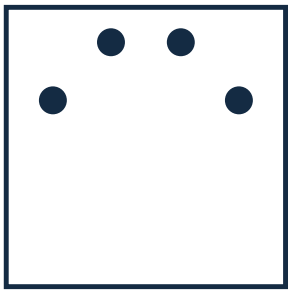
(A) OKZMS (B) TNALP

(C) PLANT (D) QMBOU

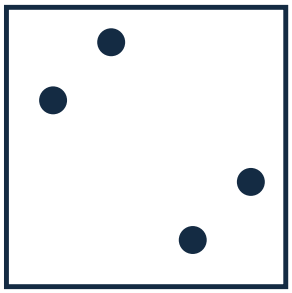
139. Starting with a unit square sheet, fold the part below the lower-left to upper-right diagonal onto the part above it. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



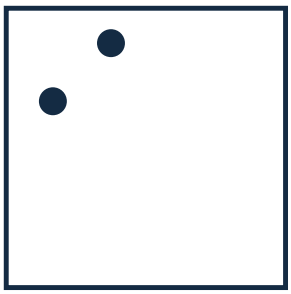
(A) Pattern U



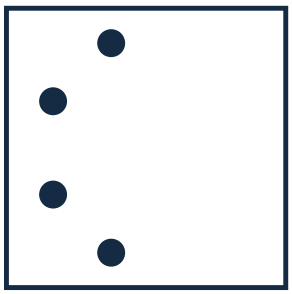
(B) Pattern L



(C) Pattern V



(D) Pattern N



Mathematics

140. 13 distinct people sit at labelled round tables of sizes 6,7. Two specified people must be at different tables, and the first specified person must be at the 6-seat table. Rotations are identical. Count the seatings.

(A) 1676505600
(B) 39916800
(C) 148262400
(D) 79833600

141. Let P(x)=(1+sqrt 2 x)^{10}+(1-sqrt 2 x)^{10}. Find P′(1).

(A) $27860\sqrt{2}$
(B) $13930\sqrt{2}+19700$
(C) 39400
(D) 247120

142. A=(4,0,2), B=(0,4,1), AP:PB=2:1 internally. Find $|AP|^2$.

(A) $\frac{44}{3}$
(B) 33
(C) 22
(D) $\frac{11}{3}$

BITSAT Mock 42

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A 3 kg mass on a string passes the top of a vertical circle of radius 2 m with string tension 210 N. Find string tension at the bottom. Take $g = 10\ \text{m/s}^2$.

(A) 390 N (B) 330 N

(C) 360 N (D) 270 N

2. A 4 kg block moves along a horizontal rough surface from $x = 0$ to $x = 5$ m under a force $F(x) = 5x$ N parallel to motion. With $\mu_k = 1/5$ and $g = 10\ \text{m/s}^2$, find the work done by the net force. The block has enough initial speed to keep moving forward.

(A) $\frac{205}{2}$ J (B) $\frac{45}{2}$ J

(C) $\frac{9}{2}$ J (D) $\frac{125}{2}$ J

3. A Newtonian liquid of viscosity $\frac{3}{10}$ Pa s has velocity profile $v(y) = 2y + 2y^2$ in SI units. Find the shear-stress magnitude at $y = 2$ m.

(A) 3 Pa (B) $\frac{9}{5}$ Pa

(C) 10 Pa (D) $\frac{3}{5}$ Pa

4. Two uniform thin rods meet at a right angle at the origin. Both have length 6 m. The rod along +x has linear density 2 times that of the rod along +y. A force parallel to the y-axis must produce pure translation. At what x-coordinate must its line of action lie?

(A) 4 m (B) 3 m

(C) 2 m (D) 1 m

5. For steady laminar flow through a capillary, the radius is multiplied by 3 and length by 4. The volume flow rate must remain unchanged and viscosity is fixed. By what factor must the pressure difference be multiplied?

(A) $\frac{1}{81}$ (B) 1

(C) $\frac{4}{9}$ (D) $\frac{4}{81}$

6. A converging thin lens forms a real image with magnification magnitude 2. The object-image separation is 9 cm. Find focal length.

(A) 2 cm (B) 18 cm

(C) $\frac{9}{4}$ cm (D) 3 cm

7. A series circuit contains resistor R, two ideal inductors L_1, L_2, and a fixed-frequency AC source of angular frequency 50 rad/s. With both inductors active, current lags by 45°; when L_2 is shorted, it lags by 30°. If $R = 60$ ohms, find L_2.

(A) $\frac{2\sqrt{3}}{5}$ H (B) $\frac{6}{5}$ H

(C) $-\frac{6}{5} + \frac{6\sqrt{3}}{5}$ H (D) $\frac{6}{5} - \frac{2\sqrt{3}}{5}$ H

8. A small-angle pendulum has period T in a stationary lift. In a vertically accelerating lift its period is 6T. Find the downward lift acceleration divided by g.

(A) $\frac{25}{36}$ (B) $\frac{1}{36}$

(C) $\frac{35}{36}$ (D) $\frac{5}{6}$

9. A square conducting loop in a constant perpendicular field 2 T has side length $\ell(t) = 4t$ m. Find induced-emf magnitude at $t = 3$ s, assuming the loop expands uniformly.

(A) 24 V (B) 192 V

(C) 96 V (D) 288 V

10. Two identical capacitors each 2 microfarads are connected in series across an ideal 40 V battery. One capacitor undergoes permanent conducting breakdown, becoming a short circuit. Find the additional charge drawn from the battery after the new steady state is reached.

(A) 80 μC (B) 20 μC

(C) 120 μC (D) 40 μC

11. A particle of mass 1 kg starts from rest and experiences $F(t) = 1t$ N along its motion for 5 s. Find the work done by the force.

(A) $\frac{125}{6}$ J (B) $\frac{625}{8}$ J

(C) $\frac{125}{2}$ J (D) $\frac{25}{2}$ J

12. A wheel initially rotates at 8 rad/s. A constant opposing angular acceleration brings it to rest after 7 s. What angle does it turn through during the last half of this stopping interval?

(A) 7 rad (B) 14 rad

(C) 56 rad (D) 28 rad

13. Two identical converging thin lenses of focal length 20 cm are separated by 80 cm. A parallel beam enters the first. Find the distance of the final image from the second lens, positive on the outgoing side.

(A) 20 cm (B) 30 cm

(C) 15 cm (D) $\frac{80}{3}$ cm

14. A charge in a plane vacuum wave moves at speed $c/12$ along a direction making angle θ with the magnetic field, with $\cos\theta = 1/12$. Find the ratio of the electric-force magnitude to the magnetic-force magnitude.

(A) $\frac{144\sqrt{143}}{143}$ (B) 144

(C) $\frac{24\sqrt{33}}{11}$ (D) $\frac{12\sqrt{143}}{143}$

15. An ideal transformer has secondary-to-primary turns ratio 3. Its primary draws 4 A at 60 V while delivering power to a resistive secondary load. Find secondary current.

(A) 4 A (B) $\frac{4}{3}$ A

(C) 36 A (D) 12 A

16. An astronomical telescope in normal adjustment has magnifying-power magnitude 8 and tube length 27 cm. Find the eyepiece focal length.

(A) $\frac{1}{3}$ cm (B) 24 cm

(C) $\frac{27}{7}$ cm (D) 3 cm

17. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 4 kΩ to a terminal at +8 V, and its cathode connects to a terminal at 0 V. Find the current from the +8 V terminal in mA.



(A) $\frac{87}{40}$ mA (B) 2 mA

(C) $\frac{73}{40}$ mA (D) $\frac{33}{20}$ mA

18. A sinusoidal source of peak $\frac{67}{10}$ V feeds a back-to-back series pair of identical Zeners through a 3 kΩ current-limiting resistor. The output is across the diode pair. Each Zener has reverse breakdown 4 V and forward drop 0.7 V. What is the peak-to-peak output voltage? Use the piecewise constant-drop model.

(A) $\frac{54}{5}$ V (B) 8 V

(C) $\frac{47}{5}$ V (D) $\frac{67}{5}$ V

19. A series RLC circuit is resonant at angular frequency ω_0. Its capacitance is then multiplied by 225 while L stays fixed. What angular frequency restores resonance, as a fraction of ω_0?

(A) $\frac{\sqrt{15}}{15}$ (B) $\frac{1}{225}$

(C) $\frac{1}{15}$ (D) 15

20. Light of vacuum wavelength 300 nm passes from a medium of index 4 into one of index 5. Find the ratio of its frequencies in the second and first media.

(A) $\frac{5}{4}$ (B) $\frac{25}{16}$

(C) $\frac{1}{5}$ (D) 1

21. A soap bubble initially of radius 6 cm is expanded quasistatically until its volume is 27 times the initial volume. Surface tension remains $\frac{3}{100}$ N/m. Find work against surface tension, in units of $\pi \times 10^{-4}$ J.

(A) $\frac{432}{25}$ (B) $\frac{1728}{25}$

(C) $\frac{864}{25}$ (D) $\frac{5616}{25}$

22. A spring of constant 30 N/m is already stretched by $\frac{1}{5}$ m. How much additional work is required to increase its extension to $\frac{3}{10}$ m quasistatically?

(A) $\frac{3}{2}$ J (B) $\frac{9}{10}$ J

(C) $\frac{3}{5}$ J (D) $\frac{3}{4}$ J

23. A uniform disc of mass 4 kg and radius 6 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) $72\,\mathrm{kg\,m^2}$ (B) $\frac{171}{2}\,\mathrm{kg\,m^2}$

(C) $\frac{135}{2}\,\mathrm{kg\,m^2}$ (D) $\frac{117}{2}\,\mathrm{kg\,m^2}$

24. A source emitting 500 Hz and a listener approach each other, each at 50 m/s relative to still air. Sound speed is 300 m/s. Find heard frequency before they pass.

(A) 750 Hz (B) 700 Hz

(C) 600 Hz (D) $\frac{1750}{3}$ Hz

25. Two ferromagnetic samples have similar remanent flux density. Their coercive fields are 1200 A/m and 4800 A/m. A reversing field of magnitude 3000 A/m will act in service. Which sample is better at retaining its original magnetisation against this field, based only on the stated coercivities?

(A) The sample with coercive field 4800 A/m. (B) The sample with coercive field 1200 A/m.

(C) Neither can have nonzero remanence. (D) Both are necessarily demagnetised equally.

26. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 20 percent. The measured increase in its absolute temperature is $\frac{968}{5}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) 440 K (B) $\frac{3168}{5}$ K

(C) $\frac{4840}{61}$ K (D) 484 K

27. A coil has 50 turns, area $\frac{1}{50}$ $\mathrm{m^2}$, and rotates at 6 rad/s in a uniform 5 T field. At an instant the angle between its area vector and the field is 30°. Find instantaneous-emf magnitude.

(A) 15 V (B) $\frac{3}{10}$ V

(C) $15\sqrt{3}$ V (D) 30 V

28. A cell supplies current $\frac{18}{5}$ A through a 6 ohm load and $\frac{18}{7}$ A through a 10 ohm load. Find its internal resistance, assuming emf is unchanged.

(A) $10\,\Omega$ (B) $\frac{2}{3}\,\Omega$

(C) $4\,\Omega$ (D) $14\,\Omega$

29. Two wires of the same material and length L, with radii r and 4r, are joined in series and stretched by a force F. Find their total stored elastic energy divided by the energy stored in the thinner wire alone under force F.

(A) 2 (B) $\frac{1}{16}$

(C) $\frac{17}{16}$ (D) 16

30. A car negotiates a level circular turn at its maximum non-slipping speed u when $\mu_s = \frac{3}{5}$. Rain lowers the coefficient to $\frac{1}{5}$ and the turn radius stays fixed. Find the new maximum speed divided by u.

(A) $\frac{\sqrt{5}}{5}$ (B) $\frac{\sqrt{3}}{3}$

(C) 3 (D) $\frac{1}{5}$

Chemistry

31. Neutral Cl and Br are compared within Group17. Which combination follows the ordinary down-group radius and first-ionisation trends?

(A) Br is larger and must have greater first ionisation energy solely because Z is larger.

(B) Br is larger and has lower first ionisation energy.

(C) Br is smaller and has higher first ionisation energy.

(D) Cl and Br are isoelectronic neutral atoms.

32. At a fixed temperature, a diprotic acid H2A has pKa1=7, pKa2=11, and pKw=14. Find (pKb of A2-, Kb(A2-)/Kb(HA-)). Use the conjugate pair appropriate to each basic species.

(A) (3, 10000) (B) (3, 1/10000)

(C) (7, 10000) (D) (7, 1/10000)

33. propan-2-ol is heated under acid-catalysed dehydration conditions. Which major product follows removal of water while preserving the stated carbon framework?

(A) propene (B) propane

(C) propyne (D) propanone

34. 1,2-dibromoethane is treated with excess NaNH2 in liquid NH3,then exactly one equivalent of methyl iodide per initially formed acetylide,and finally aqueous HgSO4/H2SO4. Work-ups are appropriate between stages. What final carbonyl product is obtained?

(A) propanal (B) ethanal

(C) propan-1-ol (D) propanone

35. Equal volumes of 2/5 M K4[Fe(CN)6] and 2/5 M glucose are mixed. The final osmotic pressure is 528/25 atm at RT=24 L atm/mol. Glucose stays molecular, and the salt dissociates only into4K+ and one intact complex anion. What fraction of salt formula units is dissociated in the final solution? Use ideal particle counting.

(A) 3/5 (B) -3/5

(C) 6/5 (D) 3/10

36. Starting with benzene,apply CH3Cl/AlCl3 (excess benzene),then Br2/FeBr3. Which organic product description correctly tracks the functional group actually present at each step?

(A) bromobenzene plus methanol
(B) mainly ortho- and para-bromotoluene
(C) benzyl bromide only
(D) mainly meta-bromotoluene

37. An orbital has 2 angular nodes and 3 radial nodes. Maximum electrons in its entire subshell are

(A) 5
(B) 14
(C) 10
(D) 72

38. For CrO_4^{2-},use four ordinary oxide oxygens. Which pair (Cr oxidation state,number of peroxide O-O links per formula entity) is correct?

(A) (+8,0).
(B) (+4,0).
(C) (+6,1).
(D) (+6,0).

39. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_x; same facing signs; 2: p_y on one centre and p_z on the other. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (0, 1)
(B) (0, 2)
(C) (0, 0)
(D) (1, 0)

40. Use octet Lewis contributors for $CO3^2$- and O3; for SO2 exclude expanded-octet contributors. For HCOO-, HCO3- and HNO3, average only the two nonprotonated central-atom-O bonds. For other species average all central-atom bonds. Find (average bond order of the first minus that of the second, sum of total lone-pair counts in one contributor of each). A lone pair on any atom is counted once.

(A) (-1/6, 14)
(B) (-1/6, 15)
(C) (-1/6, 13)
(D) (17/6, 14)

41. Equal masses of nonvolatile A (M=180) and B (M=60) are separately dissolved in equal solvent masses. A remains molecular, while B only dimerises. The measured freezing-depression ratio ΔTf(B)/ΔTf(A) is 21/10. What fraction of the original B formula units is present in dimers? All solutions obey ideal particle counting.

(A) 3/5
(B) -3/5
(C) 3/10
(D) 6/5

42. For neutral H2 and DT, take isotope mass numbers H=1, D=2, T=3 and neglect mass defects. Their gases have equal temperature. Which pair gives ($\langle v^2\rangle$_H2/$\langle v^2\rangle$_DT, number of neutrons in one DT molecule minus that in one H2 molecule)?

(A) (5/2, 5)
(B) (5/2, 2)
(C) (5/2, 3)
(D) (2/5, 3)

43. At fixed temperature a single adsorbent obeys x/m=k $p^{(1/n)}$. In experiment B the adsorbent mass is 6 times that in A and pressure is 16 times that in A. Total adsorbed amount is 48 times that in A. What is 1/n?

(A) 4/3 (B) 3/4

(C) 7/4 (D) 3/8

44. A thermal assembly contains Cu(s) and He(g) in separate inert compartments at the same temperature. Helium is heated quasistatically at constant pressure under a freely moving piston. Containers have negligible heat capacity. Use the classical Dulong-Petit solid model for Cu, neglecting its Cp-Cv difference, and the ideal monatomic model for He. No phase change occurs. Molar masses are Cu64 and He4 g/mol. Over a small common temperature rise, the measured total heat capacity C divided by total Cu-plus-He mass m and gas constant R is C/(mR)=141/640 mol/g. What is the Cu mass fraction?

(A) 33/70 (B) 7/10

(C) 3/10 (D) 37/55

45. 4-methoxybenzoic acid is treated in separate stages with SOCl2,excess NH3,and Br2/NaOH. Which final product and carbon-accounting description are correct?

(A) the original acid;amide nitrogen is never retained (B) 4-methoxyaniline;the carbonyl carbon remains as CH2

(C) 4-methoxyaniline;one carbon fewer than the starting acid (D) 4-methoxybenzylamine;all carbons retained

46. Count all distinct stereoisomers,including separately resolved enantiomers,for octahedral $[M(en)_2a_2]$ and octahedral $[Ma_3b_3]$; a,b,c,d are distinct monodentate ligands and en is symmetric bidentate ethylenediamine. What is the sum?

(A) 5 (B) 7

(C) 4 (D) 6

47. A closed fullerene cage has 70carbon atoms. Every carbon has three neighbours,and every face is a pentagon or hexagon. The cage contains12pentagons. How many hexagons are present?

(A) 35 (B) 12

(C) 58 (D) 25

48. For a closed system $\Delta U(A\to B)$=220 J and $\Delta U(B\to C)$=-30 J. Two A→C paths have heats 450 J and 250 J respectively. What is work on the system on the second path minus that on the first, in J?

(A) -200 (B) 420

(C) 200 (D) -20

49. Two open-chain aldohexoses differ at one chiral carbon but retainCHO at C1. Which label and shared class are correct?

(A) Enantiomers; all centres must agree. (B) Anomers; one must be a ketone.

(C) Epimers; both remain aldose monosaccharides. (D) Disaccharides; epimerism joins two molecules.

50. A dry mixture contains only anhydrous MgCO3 and CaCO3. Use the ordinary Group2 carbonate stability trend. In an idealised selective-heating window, the less thermally stable carbonate decomposes completely by MCO3→MO+CO2, while the more stable carbonate remains unchanged. No other mass loss occurs. This is the stated model, not a specified real temperature. Atomic masses: Mg=24, Ca=40, C=12, O=16. The selective stage gives 42/125 L CO2; subsequent stronger heating to decompose all remaining carbonate brings the cumulative CO2 volume to 112/125 L. Both dry volumes are at STP; use22.4 L/mol, with complete collection. What was the initial mass fraction of MgCO3?

(A) 63/188
(B) 33/188
(C) 3/8
(D) 75/188

51. A buffer contains 1/20 mol HA and 11/100 mol A-. After adding strong base without volume change it has pH=pKa+log10(15), with HA still present. How many moles of OH- were added? Quantitative neutralisation and the Henderson ratio model apply.

(A) 1/20
(B) 11/100
(C) 1/50
(D) 1/25

52. I:Hyperconjugation can stabilise an alkyl carbocation by donation from an adjacent C-H sigma bond. II:A resonance contributor is a separate molecule in rapid equilibrium with the other contributors. Choose the truth pattern.

(A) I false;II true.
(B) Both true.
(C) I true;II false.
(D) Both false.

53. Under the usual textbook excess-oxygen conditions,Li,Na and K favour Li_2O,Na_2O_2 and KO_2, respectively. Which oxygen-state/electronic conclusion is consistent?

(A) All three contain only O2- ions and must have identical magnetic behaviour.

(B) Larger K+ destabilises every diatomic oxygen anion,so KO2 cannot form.

(C) The peroxide ion has an odd electron and superoxide has all electrons paired.

(D) The oxygen units are oxide O^{2-},peroxide O_2^{2-} and superoxide O_2^-; KO_2 is paramagnetic.

54. Which pair correctly distinguishes cysteine and methionine side chains?

(A) Both side chains are identicalCH3 groups.
(B) Cysteine containsSH; methionine contains a thioether.
(C) Both sulphur atoms are necessarily sulphate ions.
(D) Methionine containsSH; cysteine contains no sulphur.

55. Which systematic name correctly describes the condensed connectivity HOOC-CH2-CH(OH)-CH3, including principal-group priority and numbering?

(A) 2-hydroxybutanoic acid
(B) 3-hydroxybutanoic acid
(C) butan-2-ol-4-oic acid
(D) 3-carboxybutan-2-ol

56. For AgCl(s)+2NH3⇌[Ag(NH3)2]+ +Cl-, Ksp Kf=1/4. Before solid is added there is zero chloride, an unknown silver-ammine background, and free NH3=1/5 M. Measured additional dissolved AgCl is 1/23 M. All dissolved silver remains complexed; volume is fixed. What was the initial background complex concentration?

(A) 3/200
(B) 3/50
(C) -3/100
(D) 3/100

57. Classify the two operations: I. Removal of small dissolved species by ultrafiltration. II. irreversible aggregation caused by exceeding the electrolyte coagulation threshold. Select the correct paired classification.

(A) Peptisation; dialysis

(B) Purification; coagulation

(C) Preparation by chemical reaction; Brownian motion

(D) Coagulation; purification

58. I:Two different aldehydes,each with alphaH,can in principle form two self-aldol and two crossed constitutional addition products. II:Only one aldehyde can ever serve as the enolate donor. Select the correct assessment.

(A) Both true.

(B) Both false.

(C) I false;II true.

(D) I true;II false.

59. A cubic cell has A at corners and B at the body centre. A and B are chemically distinct. Why is(a/2,a/2,a/2) not a Bravais translation?

(A) It exchanges A and B sites, so it does not preserve the chemically labelled structure.

(B) Its length is not an integer multiple of a.

(C) A primitive cubic lattice cannot have a multi-atom basis.

(D) Every vector with a fractional coordinate is forbidden in every crystal.

60. Which combined classification/reactivity description is correct for HOCH2CH2OH?

(A) no primary site because there are two OH groups

(B) two primary alcohol sites and two oxidisable carbinol carbons

(C) two secondary alcohol sites

(D) one primary and one secondary alcohol site

English Proficiency

61. Read the sentence: The editor was sceptical of the claim and asked for evidence before publishing it. Choose the closest meaning of 'sceptical' as used in the sentence.

(A) uninterested

(B) openly hostile

(C) already convinced

(D) doubtful

62. Read the sentence: The reply never refused directly, but a careful reader could read between the lines. What does 'read between the lines' mean here?

(A) infer an unstated meaning

(B) read only alternate sentences

(C) find a printing error

(D) ignore the writer's tone

63. Read the sentence: One recurring question was the thread connecting the otherwise different essays. What does the figurative expression mean in this context?

(A) It made the essays copies of one another.

(B) It created continuity among essays with different subjects.

(C) It was the only question answered in each essay.

(D) It proved that the essays had been written together.

64. He enjoys people but usually regains his energy through quiet time alone. Which word best fits the description?

(A) recluse (B) introvert

(C) extrovert (D) misanthrope

65. Read the sentence: The trustee's benevolent actions were intended to help families in difficulty. Choose the opposite in meaning of 'benevolent' as used in the sentence.

(A) protective (B) sympathetic

(C) malevolent (D) generous

66. Read the sentence: The dispute concerned a trivial detail that did not affect the outcome. Choose the opposite in meaning of 'trivial' as used in the sentence.

(A) minor (B) technical

(C) significant (D) avoidable

67. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. from several later interviews. Q. although she actually reconstructed it R. The account reads as if S. the writer had witnessed the event,

(A) P-S-Q-R (B) R-S-Q-P

(C) Q-R-S-P (D) S-R-Q-P

68. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The assistant was sorting the letters when the visitor arrived.

(A) The letters were being sorted by the assistant when the visitor arrived.

(B) The letters had been sorted by the assistant before the visitor arrived.

(C) The letters are being sorted by the assistant when the visitor arrives.

(D) The visitor was sorting the assistant when the letters arrived.

69. The work turns geographical information into carefully designed maps. Which field names this activity?

(A) typography (B) geology

(C) navigation (D) cartography

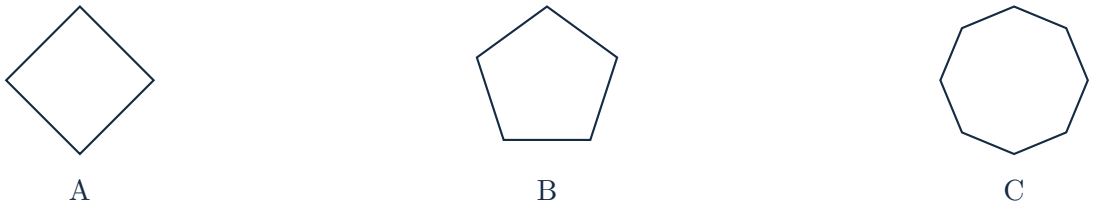
70. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. beside each unfamiliar sign. Q. the designer placed an explanation R. Instead of asking readers S. to remember a separate list of symbols,

(A) R-S-Q-P (B) P-S-Q-R

(C) Q-R-S-P (D) S-R-Q-P

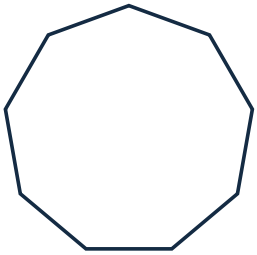
Logical Reasoning

71. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?

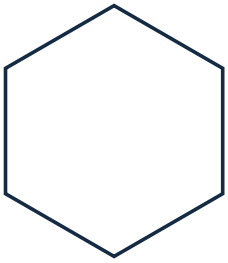


A B C

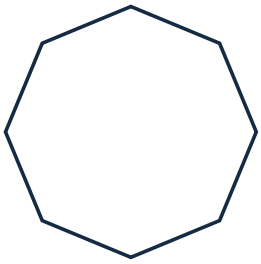
(A) Pattern P



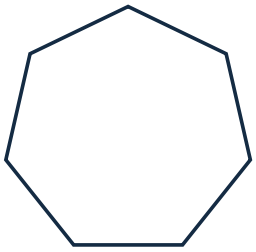
(B) Pattern U



(C) Pattern V



(D) Pattern T



72. A walker goes 12 m north, then 10 m west, then turns right and walks 11 m. In which direction is the walker from the starting point?

(A) North-west (B) South-west

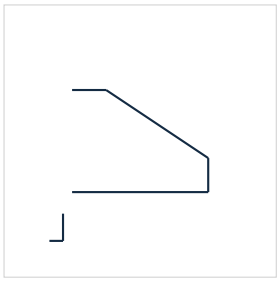
(C) South-east (D) North-east

73. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 7, 14, 21, 42, ?, 98, 105.

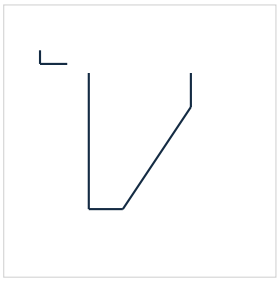
(A) 91 (B) 35

(C) 49 (D) 84

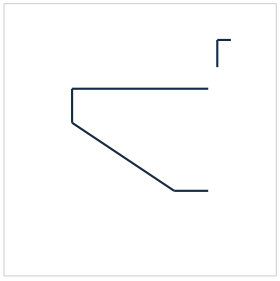
74. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right



bottom left

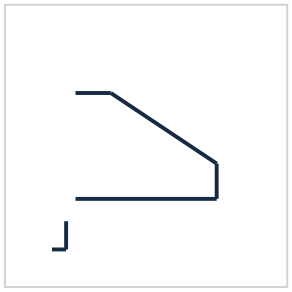
?

bottom right

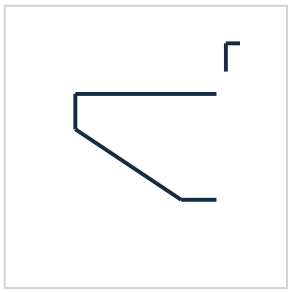
(A) Pattern Q



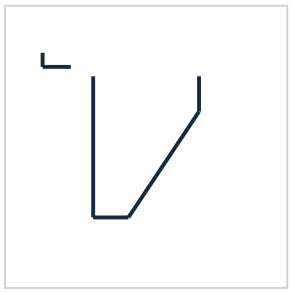
(B) Pattern L



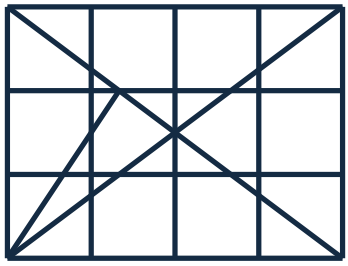
(C) Pattern V



(D) Pattern W



75. How many triangles are formed by all the drawn segments?



(A) 42 (B) 53

(C) 98 (D) 57

76. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, GRAPE is coded as LWFUJ. Which word is coded as XYTSJ?

(A) RSNMD (B) TUPOF

(C) STONE (D) CDYXO

77. The age table uses the same two people throughout:

	younger	older
now	x	$x+11$
4 years ago	$x-4$	$x+7$

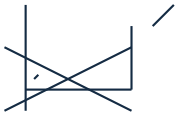
. The product of their ages 4 years ago was 782. What will the younger person's age be 2 years from now?

(A) 27 (B) 40

(C) 25 (D) 29

78. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

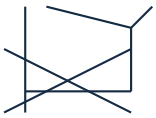




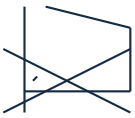
host 1



host 2



host 3



host 4

(A) 2 (B) 3

(C) 4 (D) 1

79. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?



1 2 3 4

(A) 4 (B) 1

(C) 3 (D) 2

80. The whole has 13 unit positions. Part A covers the first 4; part B covers 5 positions, overlapping A in exactly 2 positions as shown. How many positions remain uncovered by either part?

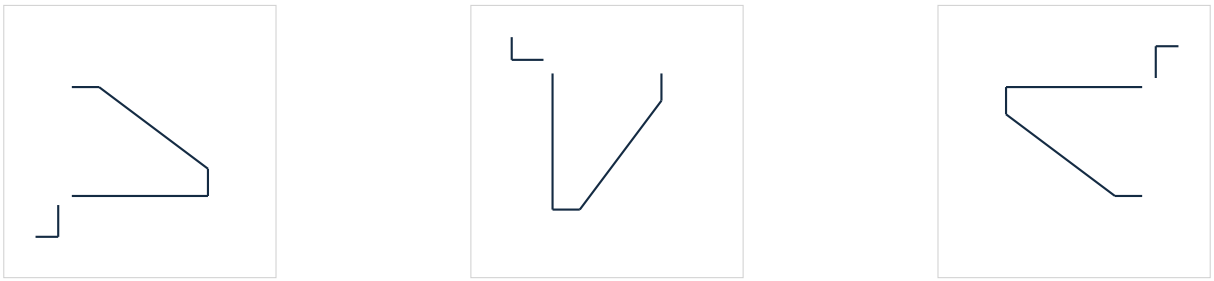


whole part A positions part B positions

(A) 6 (B) 7

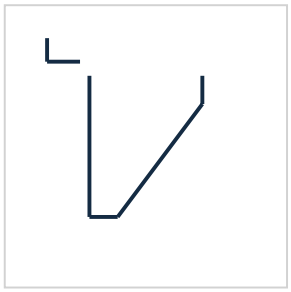
(C) 2 (D) 4

81. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1 2 3

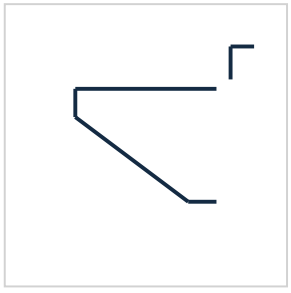
(A) Pattern M



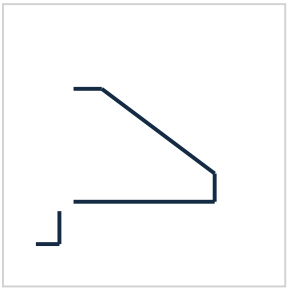
(B) Pattern X



(C) Pattern Q



(D) Pattern W



82. Kiran points to a photograph and says, "This person is the daughter of the only daughter of my father's only son." Kiran is male, and the photographed person is female. How is the photographed person related to Kiran?

(A) Daughter

(B) Brother

(C) Granddaughter

(D) Son

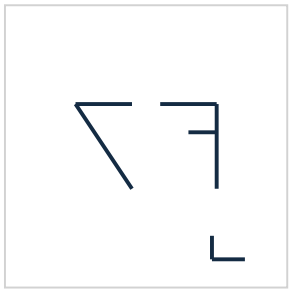
83. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



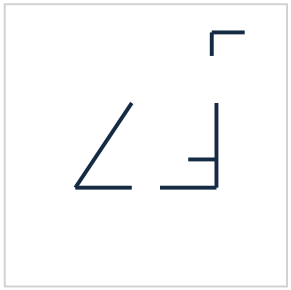
(A) Pattern N



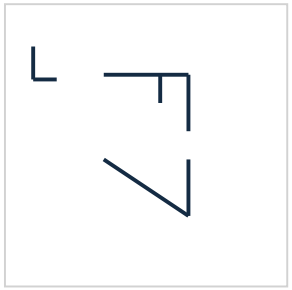
(B) Pattern V



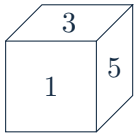
(C) Pattern Q



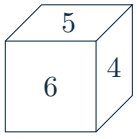
(D) Pattern M



84. The views show the same die. Give the ordered pair: (face opposite 6, face opposite 4). Do not assume opposite labels sum to seven.



1



2

(A) $(3, 2)$

(B) $(3, 4)$

(C) $(3, 3)$

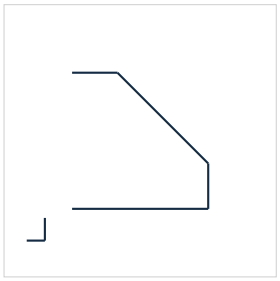
(D) $(3, 1)$

85. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: GNMK, HLKL, IJIM, ?, KFEO, LDCP.

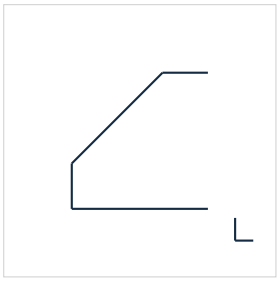
(A) IGFM (B) JHGN

(C) NGHJ (D) KIHO

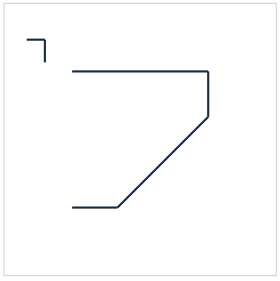
86. Select the missing bottom-right tile. Tiles reflect across the shared horizontal and vertical centre lines; there is no turning.



top left



top right

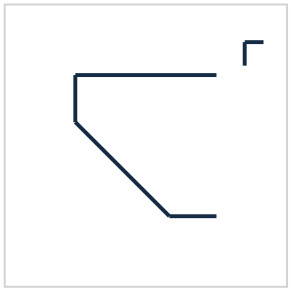


bottom left

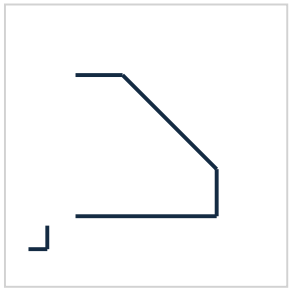
?

bottom right

(A) Pattern Y



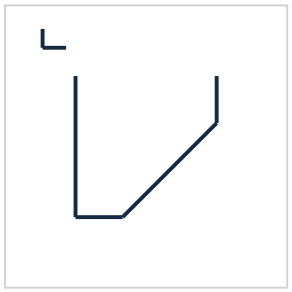
(B) Pattern M



(C) Pattern W



(D) Pattern X



87. Consider only these statements; universal statements alone do not imply existence. Some guides are not hikers; Some designers are guides. Conclusions: I. Some guides are designers. II. Some designers are not hikers. Which conclusions follow necessarily?

(A) Both I and II
(B) Only I
(C) Only II
(D) Neither I nor II

88. Consider only these statements; universal statements alone do not imply existence. All editors are designers; Some farmers are editors; Some editors are not farmers. Conclusions: I. All editors are farmers. II. Some editors are designers. Which conclusions are individually possible without contradicting the statements?

(A) Only I
(B) Both I and II
(C) Neither I nor II
(D) Only II

89. In a row, Asha is 7th from the left and 28th from the right. Bina is 6th from the right. Charu stands 4 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 18
(B) 16
(C) 17
(D) 25

90. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 29 and Z has code 4. Decode the separated values 11, 10, 15, 16, 25.

(A) RSNMD
(B) STONE
(C) TUPOF
(D) ENOTS

Mathematics

91. For $\omega \neq 1$ a cube root of unity, evaluate $\sum_{j=1}^{13}(\omega^j + \omega^{-j})^2$.

(A) 25
(B) 1
(C) 52
(D) 13

92. An AP has a_1=-44,d=4. How many of its first 14 terms are strictly positive?

(A) 14
(B) 3
(C) 13
(D) 2

93. y′-y/x=1 for x>0, y(1)=4. Find y(9)/9-y(1).

(A) $\log(9)$
(B) $-\log(9)$
(C) $18\log(3)+36$
(D) $\log(9)+4$

94. For x>0, xy′-y=x and y(1)=6. Find y(2)/2-y(1).

(A) $2\log(2)+6$
(B) $\log(2)$
(C) $\log(2)+6$
(D) $-\log(2)$

95. A convergent GP has first term 2, ratio $\frac{1}{3}$. Find the sum of all terms after its first 7 terms.

(A) $\frac{1}{2187}$
(B) $\frac{1}{243}$
(C) $\frac{1}{729}$
(D) $\frac{2}{2187}$

96. Evaluate the product of $\cos(2^j\pi/9)$ for j=0,...,2.

(A) $\frac{1}{9}$ (B) $\frac{1}{4}$

(C) $-\frac{1}{8}$ (D) $\frac{1}{8}$

97. |U|=90, |A|=22, |B|=23, |A∩B|=9. Find the number of elements belonging to both A,B or to neither.

(A) 54 (B) 63

(C) 27 (D) 81

98. Evaluate cos(4 arctan 4).

(A) $-\frac{240}{289}$ (B) $\frac{417}{289}$

(C) $\frac{225}{289}$ (D) $\frac{161}{289}$

99. In $x^2+y^2=121$, a chord through A=(11,0) has midpoint M with M_x=11/4 and M_y>0. Find its squared length.

(A) 484 (B) $\frac{363}{4}$

(C) 121 (D) 363

100. A circular sector has radius 11 and arc length $\frac{11\pi}{5}$. Find its area.

(A) 121π (B) $\frac{11\pi^2}{50}$

(C) $\frac{121\pi}{10}$ (D) $\frac{121\pi}{5}$

101. Evaluate $\int_0^{(\pi/4)} \tan^2 x \sec^4 x\, dx$.

(A) $\frac{1}{5}$ (B) $\frac{1}{3}$

(C) $\frac{2}{3}$ (D) $\frac{8}{15}$

102. What is the total length of the set satisfying (floor(x)-3)(floor(x)-7)≤0?

(A) 4 (B) 5

(C) 10 (D) 6

103. For x>1 let f(x)=arctan(2 x/(1-x^2)), with arctan in $(-\pi/2,\pi/2)$. Find the integral from 8 to 9 of $f'(x)\,dx$.

(A) $2\,\mathrm{atan}\left(\frac{1}{8}\right)$

(B) $-2\,\mathrm{atan}\,(8)+2\,\mathrm{atan}\,(9)$

(C) $-\,\mathrm{atan}\,(8)+\mathrm{atan}\,(9)$

(D) π

104. P(A)=$\frac{8}{17}$, P(B)=$\frac{4}{17}$, P(A∩B)=$\frac{2}{17}$. Find P(neither A nor B).

(A) $\frac{7}{17}$ (B) $\frac{13}{17}$

(C) $\frac{10}{17}$ (D) $\frac{5}{17}$

105. For a=(1,7,0), b=(0,1,3), c=(3,0,1), evaluate (a×c)·(a+b).

(A) −63 (B) 0

(C) 64 (D) −64

106. For $x > 0$, $xy\prime - y = x$ and $y(1) = 4$. Find the positive x at which $y = 5x$.

(A) e^5 (B) e^{-1}

(C) e (D) 1

107. There are 13 Boolean variables p_1,...,p 13. Exactly 2 variables must be true. How many truth assignments satisfy [(p_1 implies p_2) and (p_1 implies not p_2)]?

(A) 4096 (B) 55

(C) 66 (D) 8192

108. Two different numbers are chosen uniformly from {1,...,4}. Let X be their maximum. Find Var(X).

(A) $\frac{10}{3}$ (B) $\frac{\sqrt{5}}{3}$

(C) $\frac{35}{3}$ (D) $\frac{5}{9}$

109. For integer n, consider: "If 10 divides n, then 5 divides n." Which is its contrapositive?

(A) If 10 does not divide n, then 5 does not divide n. (B) 10 divides n and 5 does not divide n.

(C) If 5 does not divide n, then 10 does not divide n. (D) If 5 divides n, then 10 divides n.

110. z=10+4i. Find 1/z+1/conjugate(z).

(A) 20 (B) $\frac{1}{20}$

(C) $\frac{2}{29}$ (D) $\frac{5}{29}$

111. Evaluate the integral from 0 to 2π of (10-7 sin x)(1+cos x)/(100-49 $\sin^2$x) dx.

(A) $\frac{\sqrt{51}\pi}{102}$

(B) $\frac{2\sqrt{51}\pi}{51}$

(C) $-\frac{\log(17)}{7} + \frac{\log(10)}{7} + \frac{2\sqrt{51}\pi}{51}$

(D) $\frac{2\sqrt{149}\pi}{149}$

112. An equivalence relation has classes of sizes 9,3,3. How many unordered pairs of distinct related elements are there?

(A) 42 (B) 30

(C) 6 (D) 36

113. α,β are nonzero roots of x^2-17x+66=0. Find the sum of 1+1/α and 1+1/β.

(A) $\frac{100}{17}$ (B) $\frac{149}{66}$

(C) $\frac{17}{66}$ (D) $\frac{83}{66}$

114. How many integer k in [0,90] make $x^2 - 14x + (45 - k) = 0$ have roots of opposite signs?

(A) 45 (B) 90

(C) 44 (D) 46

115. How many injective functions from{a,b,c} to{1,...,8} satisfy f(a)<f(b)<f(c) and f(a)+f(c) even?

(A) 14 (B) 144

(C) 28 (D) 8

116. A chord of $x^2+y^2=10000$ touches $x^2/100-y^2/25=1$. Its nonzero midpoint (h,k) satisfies h=k. Find h^2+k^2.

(A) $\frac{125}{2}$ (B) $\frac{75}{4}$

(C) 75 (D) $\frac{75}{2}$

117. For x∈[0,1], find the minimum of $8(\arcsin x)^3+(\arccos x)^3$.

(A) $-\frac{25\pi^3}{343}+\frac{22\sqrt{2}\pi^3}{343}$ (B) $\frac{\pi^3}{\left(1+2\sqrt{2}\right)^2}$

(C) $\frac{\pi^3}{\left(1-2\sqrt{2}\right)^2}$ (D) $\frac{9\pi^3}{64}$

118. $f(x)=(x^2-9)/(x-3)$ for $x\neq3$, and f(3)=k. If f is differentiable at 3, find $k+f'(3)$.

(A) 4 (B) 1

(C) 7 (D) 12

119. The four lines $\pm x/18\pm y/6=1$ form a rhombus with vertices on the coordinate axes. Find its area.

(A) 54 (B) 432

(C) 216 (D) 108

120. For f(x)=|x-2|/(x-4), on which maximal open intervals is f strictly decreasing?

(A) $(2,4)\cup(4,\infty)$ (B) $(2,\infty)$

(C) $(-\infty,4)$ (D) $(-\infty,2)$

121. f(x)=10x+5, $g=f^{-1}$. Find g(g(5)).

(A) $-\frac{1}{2}$ (B) $-\frac{1}{20}$

(C) 0 (D) 5

122. For t≥0 define F(t)=lim as n→ ∞ of $n^{(-3)}\Sigma$ floor(tj^2), j=1,...,n. Evaluate Σ $F((1/8)^r)$, r=0,...,∞.

(A) $\frac{8}{21}$ (B) $\frac{8}{7}$

(C) $\frac{1}{3}$ (D) $\frac{1}{21}$

123. $A=\begin{bmatrix}3 & 1\\ 0 & 7\end{bmatrix}$. Find $\det(A^TA)$.

(A) 441 (B) 42

(C) −441 (D) 21

124. For x,y>0 with sqrt(x)+sqrt(y)=7, find d^2y/dx^2 at x=4.

(A) $-\frac{7}{16}$ (B) $\frac{7}{4}$

(C) $\frac{7}{16}$ (D) $\frac{7}{8}$

125. Two distinct integers are chosen uniformly from $\{1,\ldots,7\}$. Let U and V be their minimum and maximum. Find $\mathrm{Cov}(U,V)$.

(A) $-\frac{10}{9}$ (B) $\frac{10}{9}$

(C) $\frac{128}{9}$ (D) $\frac{46}{3}$

126. A fair coin is tossed 5 times. Given that at least one of the first-3 and last-3 windows has exactly 1 heads, find the probability that both do.

(A) $\frac{5}{24}$ (B) $\frac{5}{32}$

(C) $\frac{5}{19}$ (D) $\frac{19}{32}$

127. A line through P=(5,6) has direction (3,4). It meets 3x+4y=89 at Q. Find $|PQ|^2$.

(A) 10 (B) 100

(C) 4 (D) 50

128. An observer at height 8 m sees the tower foot at depression 45° and its top at elevation 60°. The observer descends vertically to ground level. If the new elevation is α, find $\tan\alpha$.

(A) 1 (B) $-1+\sqrt{3}$

(C) $\sqrt{3}$ (D) $1+\sqrt{3}$

129. A=(7,3,2), B=(14,3,5). Find the squared length of the component of AB perpendicular to direction (1,2,2).

(A) $\frac{353}{9}$ (B) 58

(C) $\frac{169}{9}$ (D) $\frac{161}{3}$

130. For xy-6x-5y+30=0, find the squared length of the diagonal of the rectangle formed with the axes.

(A) $\sqrt{61}$ (B) 61

(C) 121 (D) 30

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A rod of length 4 m pivots at its lower end. A peg constrained to move horizontally stays in contact with its upper tip. At an instant the rod makes an angle with $\sin\theta = 3/5$ to the horizontal and the tip's horizontal speed has magnitude 1 m/s. Find the rod's angular-speed magnitude.

(A) $\frac{3}{20}$ rad/s
(B) $\frac{5}{16}$ rad/s
(C) $\frac{5}{12}$ rad/s
(D) $\frac{1}{4}$ rad/s

132. Two parallel plane mirrors separated by 5 cm extend from axial position 0 to $\frac{45}{2}$ cm. A ray enters midway between them at axial position 0, making 45° with the normal to the mirrors. How many reflections occur before it passes the far edge? Exclude a contact exactly on the exit edge.

(A) 4
(B) 3
(C) 2
(D) 6

133. A 6 V Zener regulator feeds a constant-current load of 5 mA. Supply varies from 13 to 15 V. Regulation requires Zener current at least 1 mA, and its safe maximum is 7 mA. What is the largest permissible series resistance? Assume ideal regulation and give resistance in Ω.

(A) $\frac{1750}{3}\,\Omega$
(B) $\frac{3500}{3}\,\Omega$
(C) $7000\,\Omega$
(D) $750\,\Omega$

Chemistry

134. Across the second period,atomic radius generally decreases. For Be and B,which paired conclusion respects the first-ionisation anomaly?

(A) The two ionisation enthalpies must be equal because their nuclei are adjacent.

(B) B is smaller in the general radius trend,but Be has the greater first ionisation enthalpy.

(C) B must have the greater first ionisation enthalpy because every property follows size monotonically.

(D) Be has both the smaller radius and lower first ionisation enthalpy.

135. A solution of one nonvolatile nonelectrolyte has exact relative vapour-pressure lowering 1/27. After removal of solvent only, the lowering becomes 3/29. The initial solvent amount was 13 mol, and Raoult's law holds without association. How many moles of solvent were removed?

(A) 26/3
(B) 26/9
(C) 13/3
(D) 55/6

136. For I: uptake of gas on a finely divided solid surface; II: dissolution of a gas throughout a liquid, which classification correctly identifies surface accumulation and distinguishes it from a bulk process?

(A) Only II involves adsorption.
(B) Neither involves adsorption.
(C) I involves adsorption; II is not adsorption.
(D) Both are adsorption because both involve molecular interactions.

Logical Reasoning

137. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). The numerical component follows a constant arithmetic step. Find the missing internal term: ACPD8, XZMF11, UWJH14, ?, OQDL20, LNAN23.

(A) SUHK17 (B) RTGJ17

(C) 71JGTR (D) QSFI17

138. R is the only child of Q, and R is a parent of N. Q is the paternal grandmother of N. What can be concluded about R's gender?

(A) Male (B) Female

(C) The stated relationships are inconsistent (D) Cannot be determined

139. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 2 (B) 1

(C) 4 (D) 3

Mathematics

140. Circles $x^2+y^2=196$ and $(x-16)^2+y^2=196$ meet at A,B. Find $|AB|^2$.

(A) 272 (B) 784

(C) 528 (D) 1040

141. The line $r=(6,4,5)+t(1,2,2)$ meets $x+y+z=35$ at Q. Find $|PQ|^2$ for $P=(6,4,5)$.

(A) 413 (B) 144

(C) 12 (D) 36

142. $P(A) = 9/16$, $P(B) = 1/4$, and $P(A \cap B) = 1/8$. Find $P(A \cap B \mid A \cup B)$.

(A) $\frac{2}{11}$ (B) $\frac{2}{9}$

(C) $\frac{2}{13}$ (D) $\frac{1}{8}$

BITSAT Mock 43

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. Unpolarised light passes three ideal polarisers with axes at 0°, 30°, and 45° relative to a common reference. Find final intensity divided by incident intensity.

(A) $\frac{1}{4}$
(B) $\frac{\sqrt{3}}{8}+\frac{1}{4}$
(C) $\frac{3}{8}$
(D) $\frac{3\sqrt{3}}{32}+\frac{3}{16}$

2. Two identical uniform spheres touch. Their radii are multiplied by 2 while their density is divided by 2. By what factor does their gravitational attraction change?

(A) 16
(B) 64
(C) 4
(D) $\frac{1}{4}$

3. A point charge $23q$ is placed at a vertex of a cube. Find the electric flux through the three faces not meeting that vertex, combined, in units of q/ε_0.

(A) $\frac{23}{6}$
(B) $\frac{23}{8}$
(C) $\frac{23}{4}$
(D) $\frac{23}{16}$

4. A particle is projected vertically outward from the surface of a nonrotating planet with speed $v_e/4$, where v_e is surface escape speed. Neglect atmosphere. Find maximum height divided by planet radius.

(A) $\frac{1}{15}$
(B) $\frac{1}{4}$
(C) $\frac{16}{15}$
(D) $\frac{1}{17}$

5. A car negotiates a level circular turn at its maximum non-slipping speed u when $\mu_s=\frac{1}{2}$. Rain lowers the coefficient to $\frac{2}{5}$ and the turn radius stays fixed. Find the new maximum speed divided by u.

(A) $\frac{2\sqrt{5}}{5}$
(B) $\frac{4}{5}$
(C) $\frac{\sqrt{5}}{2}$
(D) 1

6. A charge 3 C moves with $\vec{v} = 2\hat{i} + 2\hat{j}$ m/s in $\vec{B} = 2\hat{i} + 6\hat{j}$ T. What is the signed z-component of its magnetic force?

(A) 24 N (B) 36 N

(C) 12 N (D) 48 N

7. A stationary nucleus of mass number 232 emits an alpha particle whose kinetic energy is 8 MeV. Taking daughter and alpha masses proportional to $A - 4$ and 4, find the total energy release Q, including daughter recoil.

(A) $\frac{464}{57}$ MeV (B) $\frac{236}{29}$ MeV

(C) $\frac{8\sqrt{57}}{57} + 8$ MeV (D) $\frac{228}{29}$ MeV

8. A projectile of mass 18 kg is launched with speed 15 m/s, $\cos\theta = 3/5$. At its highest point it splits into masses $M/3$ and $2M/3$. The lighter fragment moves horizontally forward at 11 m/s. Find the signed horizontal velocity of the heavier fragment immediately after the explosion.

(A) 16 m/s (B) 8 m/s

(C) 17 m/s (D) 7 m/s

9. An active transistor has $\alpha = \frac{39}{40}$. Its collector resistor is 5 kΩ and the measured collector-resistor drop is 35 V. Neglect leakage. What base current in mA is required?

(A) 7 mA (B) $\frac{7}{39}$ mA

(C) $\frac{273}{40}$ mA (D) $\frac{280}{1521}$ mA

10. A projectile is launched at 35 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{8}{9}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ m/s^2.

(A) $\frac{196}{15}$ m (B) $\frac{196}{3}$ m

(C) $\frac{588}{5}$ m (D) $\frac{196}{5}$ m

11. For a particle moving in one dimension, $v = 8 - 4x$ in SI units while $v > 0$. What is its acceleration at $x = 1$ m?

(A) -16 m/s^2 (B) 0 m/s^2

(C) 16 m/s^2 (D) 4 m/s^2

12. A balanced Wheatstone bridge has arms AB=AD=8 ohms and BC=DC=16 ohms. A resistor of arbitrary positive resistance connects B to D. Find equivalent resistance between A and C.

(A) 20 Ω (B) $\frac{16}{3}$ Ω

(C) 12 Ω (D) 24 Ω

13. Two identical converging thin lenses of focal length 30 cm are separated by 120 cm. A parallel beam enters the first. Find the distance of the final image from the second lens, positive on the outgoing side.

(A) 90 cm (B) 30 cm

(C) 15 cm (D) 45 cm

14. A reverse-biased p-n junction transports 2×10^{12} minority electrons per second from right to left and 4×10^{12} minority holes per second from left to right. It is in series with a 5 megaohm resistor across a 30 V source arranged to maintain reverse bias. In this model these carrier rates are constant. Find the resistor voltage magnitude, using $e = 1.6 \times 10^{-19}$ C.

(A) $\frac{3}{625}$ V (B) $\frac{8}{5}$ V

(C) 4800 V (D) $\frac{24}{5}$ V

15. A particle moves with constant acceleration. It travels 17 m in the (3)rd second and 23 m in the 4th second. What was its initial velocity?

(A) 5 m/s (B) 23 m/s

(C) 2 m/s (D) 6 m/s

16. The quantity $Z = x^2y^2/\sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 1, 2 and 6, respectively. To first order, what is the maximum percentage error in Z?

(A) $\sqrt{29}$ (B) 9

(C) 3 (D) 12

17. A uniform block of density 400 kg/m^3 floats in a liquid. Then the liquid density is reduced to $\frac{4}{5}$ times 1000 kg/m^3, while the block's volume and density are unchanged. What fraction of the block is now submerged?

(A) $\frac{1}{2}$ (B) $\frac{8}{25}$

(C) $\frac{2}{5}$ (D) 2

18. A spherical planet rotates uniformly. Its rotation reduces effective gravity at the equator by 7 m/s^2. By how much does rotation reduce the radial component of effective gravity at latitude 60°?

(A) $\frac{7}{2}$ m/s^2 (B) 7 m/s^2

(C) $\frac{7}{4}$ m/s^2 (D) $\frac{21}{4}$ m/s^2

19. A particle executes SHM at frequency 5 Hz. At t=0 it is at its positive extreme. In the interval $0 <= t < 1$ second, how many times does its kinetic energy equal its potential energy?

(A) 5 (B) 20

(C) 11 (D) 19

20. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 10 percent. The measured increase in its absolute temperature is $\frac{189}{2}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) $\frac{9450}{221}$ K (B) 450 K

(C) 9450 K (D) $\frac{45000}{121}$ K

21. A travelling wave is $y = 5\sin[4(x - 10t) + \pi/4]$ mm with x in metres and t in seconds. What is its frequency?

(A) $\frac{20}{\pi}$ Hz (B) 40 Hz

(C) $\frac{2}{\pi}$ Hz (D) $\frac{40}{\pi}$ Hz

22. A mass 1 kg is attached to two massless springs in series with spring constants 30 N/m and 60 N/m. Find the small-oscillation period on a smooth horizontal surface.

(A) $\frac{\sqrt{10}\pi}{5}$ s (B) $\frac{\sqrt{5}}{10}$ s

(C) $\frac{\sqrt{10}\pi}{15}$ s (D) $\frac{\sqrt{5}\pi}{5}$ s

23. The same metal is illuminated successively at wavelengths λ and $\lambda/2$. Its maximum photoelectron kinetic energies are 2 eV and 8 eV, respectively. What is the ratio of threshold wavelength to λ?

(A) $\frac{9}{2}$ (B) $\frac{3}{5}$

(C) $\frac{2}{3}$ (D) $\frac{3}{2}$

24. A coil of 6 turns and area 1 m^2 carries 6 A in a uniform 5 T field. Its plane makes 30° with the field. Find torque magnitude.

(A) $15\sqrt{3}\,\mathrm{N\,m}$ (B) $90\,\mathrm{N\,m}$

(C) $90\sqrt{3}\,\mathrm{N\,m}$ (D) $180\,\mathrm{N\,m}$

25. A particle of mass 4 kg starts from rest. A force $\vec{F} = 6t\,\hat{i} + 12\,\hat{j}$ N acts for 5 s. What is the magnitude of its momentum at that instant?

(A) $75\,\mathrm{kg\,m/s}$ (B) $60\,\mathrm{kg\,m/s}$

(C) $135\,\mathrm{kg\,m/s}$ (D) $15\sqrt{41}\,\mathrm{kg\,m/s}$

26. A uniform disc of mass 7 kg and radius 3 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) $\frac{819}{32}\,\mathrm{kg\,m^2}$ (B) $\frac{189}{8}\,\mathrm{kg\,m^2}$

(C) $\frac{63}{2}\,\mathrm{kg\,m^2}$ (D) $\frac{945}{32}\,\mathrm{kg\,m^2}$

27. A conducting rod of length 5 m moves at constant speed 5 m/s on parallel rails in a uniform perpendicular field 7 T. Total circuit resistance is 6 ohms. Neglect self-inductance. Find the external force needed to maintain the motion.

(A) $\frac{875}{6}$ N (B) $\frac{30625}{6}$ N

(C) $\frac{6125}{6}$ N (D) 175 N

28. Two uniform thin rods meet at a right angle at the origin. Both have length 2 m. The rod along +x has linear density 4 times that of the rod along +y. A force parallel to the y-axis must produce pure translation. At what x-coordinate must its line of action lie?

(A) $\frac{1}{2}$ m (B) $\frac{4}{5}$ m

(C) $\frac{1}{5}$ m (D) 1 m

29. A gas receives 700 J of heat while its surroundings do 150 J of work on it. It then loses 350 J of heat at fixed volume. Find its total internal-energy change.

(A) 350 J (B) 500 J

(C) 850 J (D) 1200 J

30. Two small identical conducting spheres initially carry positive charges $7q_0$ and $4q_0$ and are separated by distance d. They touch and are returned to the same separation. Find the final force divided by the initial force.

(A) $\frac{16}{25}$ (B) 4

(C) $\frac{121}{112}$ (D) $\frac{25}{4}$

Chemistry

31. One species A occupies every corner and every face centre of a cubic conventional cell. Which statement uses both site identity and translation symmetry?

(A) It has seven lattice points because there are six faces and one combined corner.

(B) It must be primitive cubic because only corners are valid lattice points.

(C) It is BCC with two points because face centres count as one half in total.

(D) It is an FCC Bravais lattice with four lattice points per conventional cell.

32. An ion has 10 electrons,10 neutrons and charge -1. Which combination gives its element,mass number and neutral-element group correctly?

(A) F;A=19;group17. (B) F;A=19;group18.

(C) F;A=20;group17. (D) An element with Z=10;A=20;group18.

33. For the metal centre in Co^{3+},Z=27. Determine the number of d electrons after formal oxidation-state assignment; use O=-2 in oxoanions.

(A) 7 (B) 5

(C) 8 (D) 6

34. An air sample contains fine soot particles and gaseous carbon monoxide. Which separation of classes is correct?

(A) CO is a metal aerosol and soot is a noble gas. (B) Both are carbon dioxide molecules.

(C) Both are particulate because both contain carbon. (D) Soot is particulate; CO is gaseous.

35. For the reaction $2M + O_2 \rightarrow 2MO$, take $\Delta H^\circ = -230$ kJ and $\Delta S^\circ = -0.200$ kJ K^{-1} over the stated range. Find ΔG° in kJ at 775 K and the sign of its slope against temperature.

(A) -75 kJ; positive slope (B) -230 kJ; zero slope

(C) -385 kJ; negative slope (D) 75 kJ; negative slope

36. For a neutral atom with ground-state configuration [Ar]$4s^2$,which (period,group,block) assignment is correct?

(A) (4,2,d). (B) (4,2,s).

(C) (3,2,s). (D) (4,1,s).

37. Viewed down C2-C3 of n-butane,I has a CH3-CH3 dihedral angle180°,II60°,and III0°. Which increasing potential-energy order and explanation is correct?

(A) II < I < III; the two methyl groups attract most strongly at60°.

(B) III < II < I; eclipsing minimises torsional strain.

(C) I = II < III; every staggered conformation has identical steric energy.

(D) I < II < III; I avoids methyl-methyl gauche contact and III has eclipsing methyl groups.

38. Count all distinct stereoisomers,including separately resolved enantiomers,for square-planar $[Ma_2b_2]$ and octahedral $[M(en)_2a_2]$; a,b,c,d are distinct monodentate ligands and en is symmetric bidentate ethylenediamine. What is the sum?

(A) 6 (B) 7

(C) 4 (D) 5

39. Two oxides of X contain 56 g X combined respectively with 16 g O and 24 g O. Atomic masses are X=56, O=16. Which empirical-formula conclusion jointly follows from mole ratios and the law of multiple proportions?

(A) No formula can be inferred even though both atomic masses and both compositions are supplied.

(B) First oxide X2O3; second oxide XO; fixed-X oxygen mass ratio second:first=2/3.

(C) First empirical formula XO; second empirical formula X2O3; fixed-X oxygen mass ratio second:first=3/2.

(D) Both are XO because X masses are equal.

40. P is a phenol described by a phenolic compound with6carbons,oneoxygen andsixhydrogens. Which molecular formula and Kolbe-Schmitt product pair is consistent?

(A) C7H8O;4-methylbenzoic acid (B) C6H6O2;3-hydroxybenzoic acid

(C) C6H6O;2-hydroxybenzoic acid (D) C6H12O;benzoic acid

41. For weak-field high-spin Mn^{2+} and Fe^{3+},which statement follows in the spin-only approximation?

(A) They are both diamagnetic because ionic charge removes all d electrons.

(B) Equal moments prove equal atomic numbers and equal ionic radii.

(C) They have the same moment $\sqrt{35}$ BM because both are high-spin d5 ions.

(D) Their moments must differ because different elements cannot have equal unpaired-electron counts.

42. Acidifying 1/20 mol borax theoretically gives four H3BO3 per borax formula unit. The isolated boric acid requires 4/25 mol NaOH for complete titration in the presence of glycerol, where each H3BO3 behaves as monobasic. What fraction of the theoretical boric acid was recovered?

(A) 2/5 (B) 8/5

(C) -4/5 (D) 4/5

43. A laboratory note states: A catalyst is selective if it favours one reaction pathway among competing possibilities. Which judgement and reason are consistent?

(A) Correct: Selectivity concerns the product pathway,not merely an equal increase of all product rates.

(B) Incorrect: Selectivity concerns the product pathway,not merely an equal increase of all product rates.

(C) Incorrect because transition metals cannot exchange electrons or adsorb reactants.

(D) Correct because every transition-metal species has the same d-electron count.

44. Equal masses of nonvolatile A (M=180) and B (M=60) are separately dissolved in equal solvent masses. A remains molecular, while B only dimerises. The measured freezing-depression ratio ΔTf(B)/ ΔTf(A) is 9/5. What fraction of the original B formula units is present in dimers? All solutions obey ideal particle counting.

(A) -4/5 (B) 2/5

(C) 8/5 (D) 4/5

45. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_x; opposite facing signs; 2: p_y on one centre and p_z on the other. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (0, 3) (B) (0, 1)

(C) (1, 1) (D) (0, 2)

46. A $\frac{87}{500}$ g volatile sample displaces $\frac{896}{13}$ mL wet air at 280 K and 790 mmHg. Water-vapour pressure is 30 mmHg. Using STP=273 K,760 mmHg and22.4 L/mol,the sample molar mass, rounded to the nearest whole g/mol, is

(A) 1469 (B) 115

(C) 58 (D) 60

47. Both metals in $[Co(en)_3][Cr(CN)_6]$ are formally +3. Which proposed partner preserves the total composition while exchanging ligand sets between two coordination spheres?

(A) $[Co(NH_3)_5Cl]Cl_2$ formed by adding chloride to the sample.

(B) The same salt with one ligand merely renamed; no atoms or donor attachments change.

(C) $[Cr(en)_3][Co(CN)_6]$

(D) $[Co(en)_3][Cr(CN)_6]$ with only the counter-ion positions in the written formula reversed.

48. A closed system follows states A→B→C→A. For A→B, heat q=480 J and work on the system w=-240 J. For C→A, q=-340 J and w=120 J. For B→C, q=-220 J. Find work on the system on B→C in J.

(A) 440 (B) -40

(C) 200 (D) -200

49. $\frac{16}{5}$ g pure NH_4NO_2 decomposes by $NH_4NO_2 \rightarrow N_2 + 2H_2O$. Only80.0% of the theoretical dry N_2 is recovered. Find its STP volume in litres, using22.4L/mol and NH_4NO_2=64g/mol.

(A) $\frac{28}{25}$ (B) $\frac{112}{125}$

(C) $\frac{224}{125}$ (D) $\frac{7}{5}$

50. Dry air is taken to contain21.0% O2by volume. From 180L of this air, an isolation process recovers80.0% of its oxygen into a pure stream. All gas volumes refer to the same T,P. Find recovered O2volume in litres.

(A) $\frac{189}{5}$ (B) $\frac{756}{25}$

(C) $\frac{189}{4}$ (D) $\frac{2844}{25}$

51. A student says every increase in nuclear charge must shrink a neutral atom, so I is smaller than F. Which correction applies to this within-group comparison?

(A) The student's rule is exact across every group and period.

(B) Additional occupied shells and shielding make I much larger than F despite its greater nuclear charge.

(C) I has fewer protons than F.

(D) F and I have the same occupied shells.

52. Equal volumes of 1/2 M K4[Fe(CN)6] and 1/2 M glucose are mixed. The final osmotic pressure is 156/5 atm at RT=24 L atm/mol. Glucose stays molecular, and the salt dissociates only into4K+ and one intact complex anion. What fraction of salt formula units is dissociated in the final solution? Use ideal particle counting.

(A) 8/5

(B) 4/5

(C) 2/5

(D) -4/5

53. A sample contains $\frac{7}{100}$ mol $KClO_3$ and a small amount of MnO_2 catalyst.75.0% of the chlorate decomposes by $2KClO_3 \rightarrow 2KCl + 3O_2$. The catalyst is recovered unchanged. Find O_2 formed in mol.

(A) $\frac{63}{800}$

(B) $\frac{21}{200}$

(C) $\frac{63}{400}$

(D) $\frac{21}{400}$

54. A calorimeter holding50.0g water absorbs800J and rises2.00K. Water specific heat is4.00J $g^{-1}K^{-1}$; calorimeter heat capacity is unchanged. When it holds100g water and absorbs the same heat,temperature rise in K is

(A) 2

(B) $\frac{4}{3}$

(C) 1

(D) 4

55. One mole of an acyclic monoalkene gives two moles propanone on complete reductive ozonolysis (O3,then Zn/H2O),with no other carbonyl products. Which parent and carbon-skeleton assignment fits?

(A) 2,3-dimethylbut-2-ene

(B) 2-methylpent-2-ene

(C) 3-methylpent-2-ene

(D) hex-3-ene

56. An isolated cathode compartment initially contains 1/25 mol HA and the same amount of A-, with pKa5.00. Water reduction gives one OH- per electron; there is no anode-product entry or buffer loss. What constant current for 1930 s gives final pH=5+log10(3)? F=96500 C/mol.

(A) 1

(B) 1/2

(C) 2

(D) -1

57. butan-2-ol is heated under acid-catalysed dehydration conditions. Which major product follows removal of water while preserving the stated carbon framework?

(A) mainly but-2-ene

(B) butanal

(C) but-1-ene only

(D) butane

58. A $\frac{6}{5}$ g organic sample contains chlorine as its only halogen. Carius analysis gives $\frac{5166}{8875}$ g dry AgCl, but this is only 80.0% of the theoretical AgCl because of a known collection loss. Find chlorine mass percentage. Use Ag=108, Cl=35.5.

(A) $\frac{4305}{71}$ (B) $\frac{75}{4}$

(C) 15 (D) 12

59. Salicylic acid has formulaC7H6O3. Replacing its phenolicO-H hydrogen byCOCH3 gives aspirin. What net formula change occurs?

(A) RemoveCO2, to obtainC6H6O. (B) AddC2H2O, to obtainC9H8O4.

(C) Replace allOH groups by Cl, to obtainC7H4Cl2O. (D) AddC2H3O without losingH, to obtainC9H9O4.

60. At 1 atm compare two separate pairs: (water, hydrogen sulfide) and (n-pentane, neopentane). A student selects hydrogen sulfide for the first higher-boiling substance because its formula mass is unchanged, and n-pentane for the second. Which assessment is consistent with intermolecular interactions?

(A) First selection wrong; second selection correct. (B) First selection correct; second selection wrong.

(C) Both selections wrong. (D) Both selections correct.

English Proficiency

61. Read the sentence: The auditor accepted the argument as sound because every step followed from the evidence. Choose the closest meaning of 'sound' as used in the sentence.

(A) easy to remember (B) well-founded

(C) forceful (D) widely supported

62. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The manager can extend the deadline only with the panel's consent.

(A) The deadline must be extended by the manager without the panel's consent.

(B) The deadline can have extended the manager with the panel's consent.

(C) Only the manager's consent can extend the panel's deadline.

(D) The deadline can be extended by the manager only with the panel's consent.

63. He converts geographical measurements into maps rather than measuring property boundaries on site. What is he?

(A) surveyor (B) geologist

(C) cartographer (D) navigator

64. Read the sentence: The speaker hoped to persuade the committee to support the application. Choose the opposite in meaning of 'persuade' as used in the sentence.

(A) dissuade (B) influence

(C) convince (D) inform

65. Complete: No approval was spoken, but the group treated the absence of objection as _____ consent.

(A) articulated (B) audible

(C) tacit (D) vocal

66. Read the sentence: Extra ventilation may mitigate the discomfort, though it will not remove the source of heat. Choose the closest meaning of 'mitigate' as used in the sentence.

(A) remove entirely (B) explain accurately

(C) postpone temporarily (D) make less severe

67. Read the sentence: The donation was voluntary; nobody was required to contribute. Choose the opposite in meaning of 'voluntary' as used in the sentence.

(A) compulsory (B) unpaid

(C) personal (D) generous

68. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The committee might have overlooked the final attachment.

(A) The final attachment had certainly been overlooked by the committee.

(B) The final attachment might have been overlooked by the committee.

(C) The final attachment must have been overlooked by the committee.

(D) The final attachment might overlook the committee.

69. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. The members compared the words in complete sentences rather than as isolated dictionary entries. B. Their choice then reflected the narrator's attitude as well as the basic meaning. C. Two apparently similar words produced different tones in the final paragraph. D. The language club translated a short story together.

(A) Their choice then reflected the narrator's attitude as well as the basic meaning.

(B) The language club translated a short story together.

(C) Two apparently similar words produced different tones in the final paragraph.

(D) The members compared the words in complete sentences rather than as isolated dictionary entries.

70. Complete: The guards stood _____ enough to hear the quiet exchange, but not close enough to interrupt it.

(A) near (B) nearness

(C) nearly (D) nearest

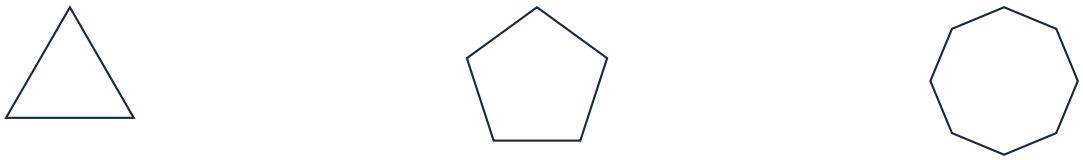
Logical Reasoning

71. Rina points to a photograph and says, "This person is the daughter of my father's only daughter." Rina is female, and the photographed person is female. How is the photographed person related to Rina?

(A) Sister (B) Brother

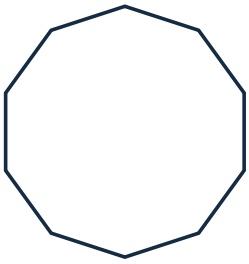
(C) Daughter (D) Son

72. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?

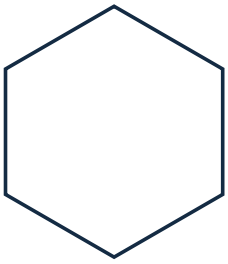


A B C

(A) Pattern K



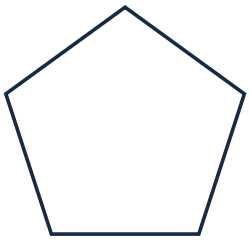
(B) Pattern Y



(C) Pattern T



(D) Pattern V



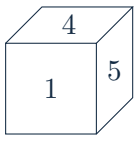
73. Consider only these statements; universal statements alone do not imply existence. No artists are joggers; All hikers are joggers. Conclusions: I. No artists are hikers. II. Some hikers are artists. Which conclusions follow necessarily?

(A) Both I and II
(B) Neither I nor II
(C) Only I
(D) Only II

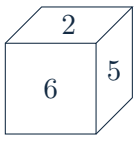
74. Preserve the exact relation and its order: Silent : Quiet :: ? : Help. Choose the missing term.

(A) Begin
(B) Silent
(C) Rapid
(D) Assist

75. The views show the same die. Give the ordered pair: (face opposite 2, face opposite 4). Do not assume opposite labels sum to seven.



1



2

(A) $(4, 4)$
(B) $(4, 1)$
(C) $(4, 2)$
(D) $(4, 3)$

76. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 4, 8, 15, 30, ?, 74, 81.

(A) 37

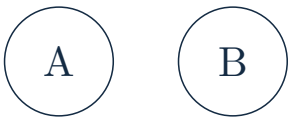
(B) 60

(C) 74

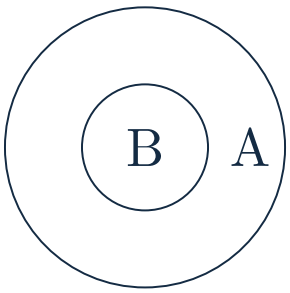
(D) 53

77. Choose the diagram that represents all these conditions, including the strictness information: All A are B, all B are A, and A is nonempty.

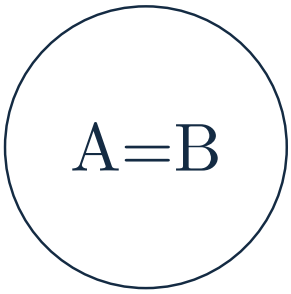
(A) Pattern P



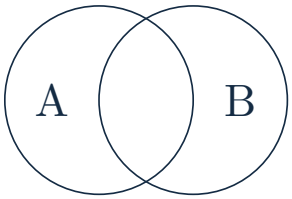
(B) Pattern V



(C) Pattern T

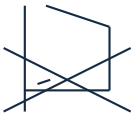


(D) Pattern Z

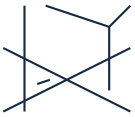


78. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





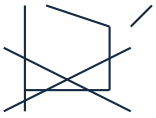
host 1



host 2



host 3



host 4

(A) 3

(B) 1

(C) 4

(D) 2

79. In a row, Asha is 6th from the left and 27th from the right. Bina is 8th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 14 (B) 16

(C) 21 (D) 15

80. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 15$ and $5 \mapsto 27$. Which input maps to 55?

(A) 12 (B) 55

(C) 11 (D) 13

81. The age table uses the same two people throughout:

	younger	older
now	x	$x + 11$
4 years ago	$x - 4$	$x + 7$

. The product of their ages 4 years ago was 570. What will the older person's age be 2 years from now?

(A) 47 (B) 34

(C) 32 (D) 36

82. Five people occupy positions 1 to 5 from left to right, one per position, and wear five different shirt colours: red, blue, green, white and yellow. Asha is 1st from the left. Esha is 2nd from the left. Bina is 3rd from the left. Charu is 4th from the left. Dev is 5th from the left. The yellow shirt is in position 1. The blue shirt is in position 2. The green shirt is in position 5. Bina does not wear white. Which shirt does Bina wear?

(A) Blue (B) Green

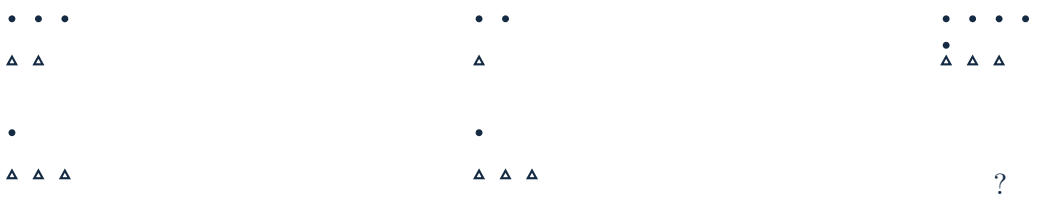
(C) White (D) Red

83. Consider only these statements; universal statements alone do not imply existence. Some bakers are designers; Some guides are designers; All designers are bakers. Conclusions: I. No bakers are guides. II. All bakers are designers. Which conclusions are individually possible without contradicting the statements?

(A) Both I and II (B) Only II

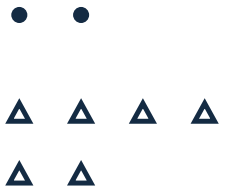
(C) Neither I nor II (D) Only I

84. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern K

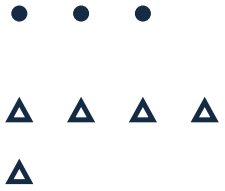
(B) Pattern L



(C) Pattern N



(D) Pattern Y





85. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, MANGO is coded as NBOHP. Which word is coded as SJWFS?

(A) QHUDQ

(B) TKXGT

(C) RIVER

(D) SJWFS

86. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

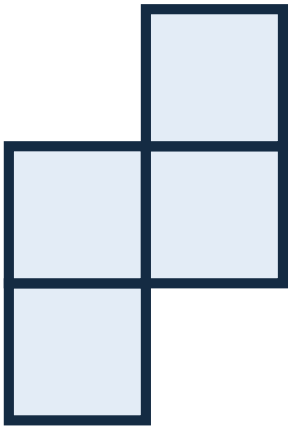


piece 1

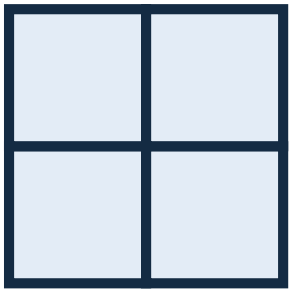


piece 2

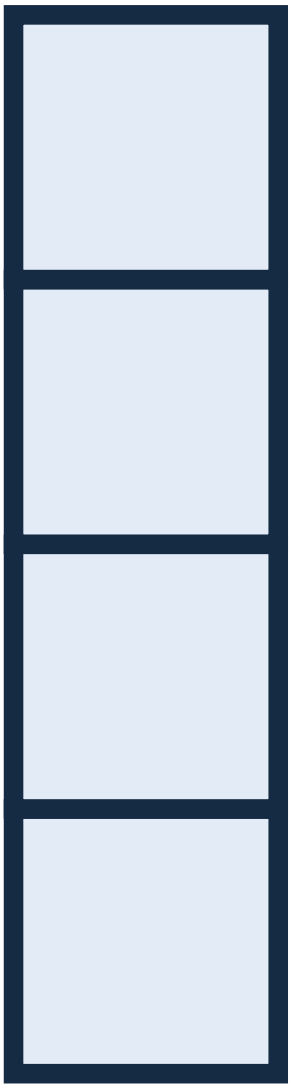
(A) Pattern N



(B) Pattern Z



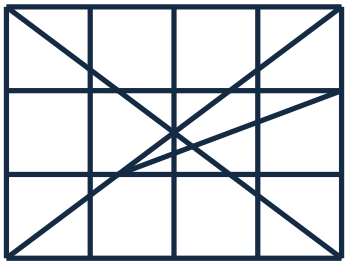
(C) Pattern U



(D) Pattern M



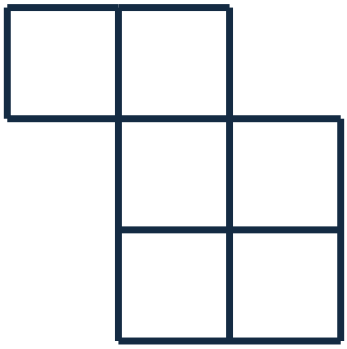
87. How many triangles are formed by all the drawn segments?



(A) 59 (B) 41

(C) 58 (D) 100

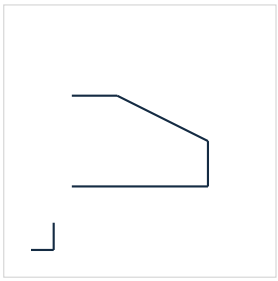
88. How many rectangles, including squares are formed by all the drawn segments?



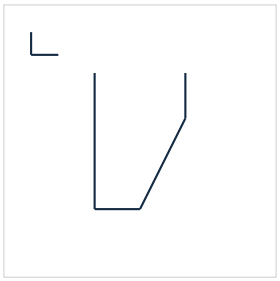
(A) 12 (B) 14

(C) 15 (D) 13

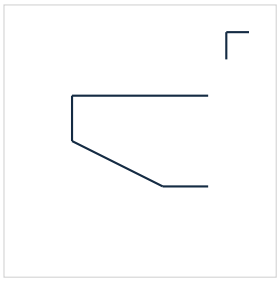
89. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

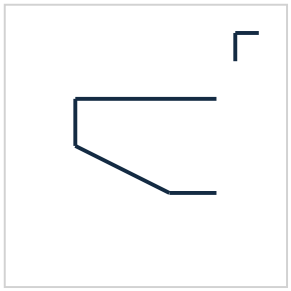


2

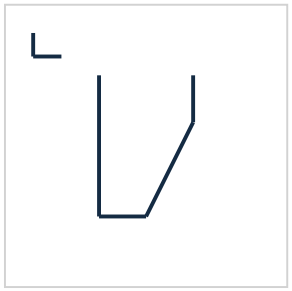


3

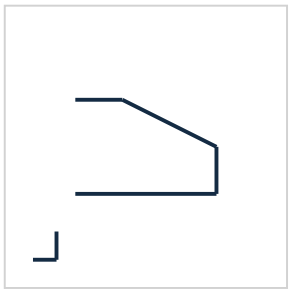
(A) Pattern U



(B) Pattern K



(C) Pattern V



(D) Pattern X



90. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?



1



2



3



4

(A) 4

(B) 1

(C) 3

(D) 2

Mathematics

91. y′+y/(x+1)=1 for x≥0, y(0)=3. Find (5)y(4)-y(0).

(A) 3 (B) 12

(C) 15 (D) −12

92. Find the product of all real solutions of sqrt(13-log_7|x|)=log_7|x|-1.

(A) 5764801 (B) −2401

(C) −5764801 (D) 0

93. Circles $x^2+y^2=256$ and $(x-22)^2+y^2=256$ meet at A,B. Find $|AB|^2$.

(A) 540 (B) 135

(C) 56 (D) 100

94. Values 8,11 have frequencies 2,k, with mean $\frac{49}{5}$. Find the population variance.

(A) $\frac{54}{5}$ (B) $\frac{3\sqrt{6}}{5}$

(C) $\frac{9}{4}$ (D) $\frac{54}{25}$

95. 6 distinct objects are distributed into three nonempty unlabelled boxes. How many distributions are possible?

(A) 270 (B) 90

(C) 540 (D) 121

96. A 10-element set contains distinct a,b. How many 3-element subsets contain exactly one of a,b and also contain a?

(A) 28 (B) 56

(C) 256 (D) 120

97. $A=\begin{bmatrix}7 & 1\\ 0 & 5\end{bmatrix}$, $P=\begin{bmatrix}1 & 6\\ 0 & 1\end{bmatrix}$. Find the (1,2) entry of $P^{-1}AP$.

(A) −11 (B) 13

(C) 12 (D) 1

98. An AP has $a_1=10$, d=4. Find $a_1+a_3+\cdots+a_9$.

(A) 180 (B) 260

(C) 150 (D) 130

99. For A={1,...,9}, how many (a,b)∈A×A satisfy a+b even and a<b?

(A) 16 (B) 40

(C) 20 (D) 32

100. Tangents at the endpoints of a focal chord of $y^2=12$ x meet atT. One endpoint has parameter 3. If F is the focus, find TF^2.

(A) 100 (B) 10

(C) $\frac{10000}{9}$ (D) 82

101. The parabola $y=x^2+px+q$ passes through (0,25). The absolute value of its vertex ordinate is minimized over $p\geq 0$. Find p^2+q^2 at the minimum.

(A) 625 (B) 725

(C) 1225 (D) 125

102. f is cubic with leading coefficient 1/3, stationary points 5,7, and lim as $x\to 0$ $[f(x)-5]/x=35$. Find f(1).

(A) $\frac{88}{3}$ (B) 40

(C) $\frac{103}{3}$ (D) 46.3333333333333

103. $F(x)=\int_0^{x^2}(t-16)(t-64)dt$. How many local minima of F lie in [-8,8]?

(A) 1 (B) 2

(C) 3 (D) 5

104. How many integers x in [-1,15] satisfy $(|x-7|-5)/(|x-7|-7)>0$?

(A) 11 (B) 13

(C) 9 (D) 17

105. Let D_n be the determinant of the 3×3 matrix with diagonal entries n and off-diagonal entries 1. Find ΣD_n for n=1,...,7.

(A) 700 (B) 2268

(C) 784 (D) 714

106. The line r=(4,6,2)+t(1,2,2) meets x+y+z=42 at Q. Find $|PQ|^2$ for P=(4,6,2).

(A) 18 (B) 54

(C) 620 (D) 324

107. a=(7,1,0), b=(t,4,1). If $a\perp b$, find $|7b|^2$.

(A) $\frac{849}{49}$ (B) 833

(C) 800 (D) 849

108. A convergent GP has first term 8, ratio $\frac{1}{3}$. Find the sum of all terms after its first 5 terms.

(A) $\frac{2}{81}$ (B) $\frac{4}{81}$

(C) $\frac{1}{162}$ (D) $\frac{968}{81}$

109. Evaluate Σ j/2^j for j=1,...,4.

(A) $\frac{27}{16}$ (B) $\frac{13}{8}$

(C) $\frac{1}{4}$ (D) 2

110. The line y=mx through O bisects the area of the triangle with vertices O,(3,2),(6,6). Assume the vertices are noncollinear. Find m.

(A) $\frac{4}{3}$ (B) $\frac{8}{9}$

(C) 2 (D) $\frac{9}{8}$

111. A circular sector has radius 3 and arc length $\frac{3\pi}{8}$. Find its area.

(A) $\frac{27\pi}{16}$ (B) $\frac{9\pi}{8}$

(C) 9π (D) $\frac{9\pi}{16}$

112. Evaluate $\sum_{j=6}^{11} \binom{j}{3}$, with the convention $\binom{j}{r} = 0$ when j<r.

(A) 480 (B) 460

(C) 495 (D) 490

113. From a point at horizontal distance 6 from a tower, its elevation is 2θ, where $\tan\theta=\frac{3}{5}$ and $0<\theta<45°$. Find the height.

(A) $\frac{45}{4}$ (B) $\frac{18}{5}$

(C) $\frac{16}{5}$ (D) $\frac{36}{5}$

114. F′(x)=$\frac{(x+1)^2}{(x+2)^4}$ and F(0)=2. Find F(5)-2F(0).

(A) $\frac{1385}{8232}$ (B) $-\frac{17849}{8232}$

(C) $-\frac{15079}{8232}$ (D) $\frac{17849}{8232}$

115. The line r=(4,5,5)+t(1,2,2) and plane 2x-y=7 are given. Find the squared minimum distance between them.

(A) 16 (B) 0

(C) $\frac{16}{5}$ (D) $\frac{4\sqrt{5}}{5}$

116. z=4+5i. Find 1/z+1/conjugate(z).

(A) $\frac{8}{41}$ (B) $\frac{10}{41}$

(C) $\frac{1}{8}$ (D) 8

117. Evaluate $\int_0^{(8\pi)} \sin^4 x\, dx$.

(A) 3π (B) 0

(C) 4π (D) π

118. How many integers x satisfy $4^{-x} - (272)2^{-x} + 4096 \leq 0$?

(A) 5 (B) 9

(C) 4 (D) 3

119. a=(10,6,0), b=(6,-10,4). Find $|a+b|^2$-$|a-b|^2$.

(A) 20672 (B) 288

(C) 0 (D) 480

120. Independent events A,B have probabilities $\frac{10}{13}, \frac{2}{3}$. Given A∪B, find P(A∩B).

(A) $\frac{4}{9}$ (B) $\frac{20}{39}$

(C) $\frac{5}{14}$ (D) $\frac{5}{9}$

121. Find the area bounded by y=x, y=81/x, and x=27 in the first quadrant.

(A) $81\log(3)$ (B) 324

(C) $324 - 81\log(3)$ (D) $162 - 81\log(3)$

122. Positivea,b satisfy log(a+b)=log(4)+[log a+log b]/2 in one valid base. Find $(a^6+b^6)/(a^3b^3)$.

(A) 194 (B) 4096

(C) 2660 (D) 2702

123. For x>1 let f(x)=arctan(2 x/(1-x^2)), with arctan in $(-\pi/2,\pi/2)$. Find the integral from 9 to 10 of $f'(x)$ dx.

(A) π

(B) $-\operatorname{atan}(9) + \operatorname{atan}(10)$

(C) $-2\operatorname{atan}(9) + 2\operatorname{atan}(10)$

(D) $2\operatorname{atan}\left(\frac{1}{9}\right)$

124. For $\omega \neq 1$ a cube root of unity, evaluate $\sum_{j=1}^{15}(\omega^j + \omega^{-j})^2$.

(A) 15 (B) 30

(C) 0 (D) 60

125. For $P(x) = \prod_{j=3}^{9}(x+j)$, find the coefficient of x^4 in $P(x)/(x+3)$.

(A) 1250 (B) 625

(C) 742 (D) 733

126. Using digits 0,1,2,3,4,5,6,7 without repetition, how many 5-digit positive integers are divisible by 4, have an even first digit, and first digit smaller than last?

(A) 320 (B) 1080

(C) 560 (D) 200

127. Let f(x+3)=4 x^2. Find the area enclosed between y=f(x) and y=16(x-3).

(A) $\frac{20}{3}$ (B) $\frac{128}{3}$

(C) $\frac{256}{3}$ (D) $-\frac{128}{3}$

128. If lim as x→0 of [sin(3x)-ax]/x^3=b is finite, find a+b.

(A) 3 (B) $-\frac{9}{2}$

(C) $-\frac{3}{2}$ (D) $\frac{15}{2}$

129. Let f(x)=|x-5|/(x-12) for x≠12. Within 0<x<19, what is the total length of the open intervals on which f′(x)<0?

(A) 18 (B) 5

(C) 19 (D) 14

130. A,B are acute, A+B=$\pi/4$ and tan A=$\frac{5}{11}$. Find tan(A-B).

(A) $\frac{7}{88}$ (B) $\frac{3}{8}$

(C) $\frac{7}{103}$ (D) $-\frac{7}{103}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. Point masses $m, m, 4m$ occupy $(0,0), (4,0), (0,4)$ m. A fourth mass m is placed at $(x,0)$ so that the centre of mass of all four is at $x_{CM} = \frac{8}{7}$ m. Find x.

(A) -4 m
(B) 4 m
(C) 8 m
(D) 0 m

132. A fixed total mass M is split between two point bodies in the ratio $8:1$, at fixed separation. Find their attraction divided by the maximum possible attraction obtainable by any split of the same M at the same separation.

(A) $\frac{32}{81}$
(B) $\frac{256}{81}$
(C) 1
(D) $\frac{4}{81}$

133. A sinusoidal source of peak $\frac{97}{10}$ V feeds a back-to-back series pair of identical Zeners through a 3 kΩ current-limiting resistor. The output is across the diode pair. Each Zener has reverse breakdown 4 V and forward drop 0.7 V. What is the peak-to-peak output voltage? Use the piecewise constant-drop model.

(A) $\frac{33}{5}$ V
(B) $\frac{47}{10}$ V
(C) $\frac{47}{5}$ V
(D) $\frac{97}{5}$ V

Chemistry

134. A gas evolved during an anion test turns limewater milky and the milkiness clears on further prolonged passage. Which gas and chemical explanation match this complete observation?

(A) CO2; calcium hydrogencarbonate accounts for the diagnostic change.
(B) HCl; elemental chlorine is deposited.
(C) NH3; molecular oxygen causes the stated change.
(D) N2; acid-base neutralisation always produces the stated colour.

135. bromoethane is taken through Mg/dry ether;CO2;acidic work-up,with suitable separate-stage work-ups. Identify the final organic product without adding or deleting carbon atoms except when the reagent itself introduces carbon.

(A) propan-1-ol
(B) propanoic acid
(C) ethanoic acid
(D) butanoic acid

136. For isopropyl ethanoate,label A=carbonyl C=O,B=acyl C-O,C=alkoxy O-C(sp3). Which bond order/length statement correctly distinguishes the two C-O single bonds shown in a Lewis formula?

(A) A is longest because oxygen is more electronegative.
(B) B and C must have identical lengths because both are drawn single.
(C) B has partial double-bond character,and length A < B < C.
(D) C has partial double-bond character,and length A < C < B.

Logical Reasoning

137. The whole has 14 unit positions. Part A covers the first 5; part B covers 5 positions, overlapping A in exactly 1 positions as shown. How many positions remain uncovered by either part?



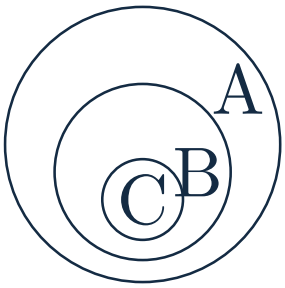
(A) 6
(B) 5
(C) 4
(D) 3

138. Five people stand in a row, one per position. Bina is at the left end and Charu at the right end. Gita is somewhere left of Farah. Which statement must be true in every arrangement satisfying the clues?

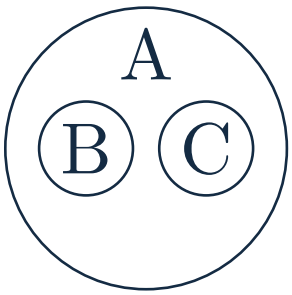
(A) Farah is immediately left of Charu.
(B) Esha is immediately right of Bina.
(C) Gita stands to the left of Farah.
(D) Gita is immediately right of Bina.

139. Choose the diagram that represents all these conditions, including the strictness information: Every C is B, every B is A; some A is not B, some B is not C, and C is nonempty.

(A) Pattern T



(B) Pattern K



(C) Pattern V



(D) Pattern Q



Mathematics

140. Evaluate the sum of r(r-1)C(8,r)2^r over all even integers r with $0 \le r \le 8$.

(A) 20440
(B) 81760
(C) 81648
(D) 81536

141. S_n=$6n^2$+2n is a sequence partial sum. Find $a_1 + a_3 + \cdots + a_{17}$.

(A) 936
(B) 504
(C) 990
(D) 968

142. Two distinguishable fair dice show X,Y. Given $X+Y \ge 11$, find the probability that $X \neq Y$ and XY is even.

(A) 1
(B) $\frac{1}{3}$
(C) $\frac{2}{3}$
(D) $\frac{1}{18}$

BITSAT Mock 44

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A particle of mass 3 kg and charge 4 C initially has speed 7 m/s. It moves through a uniform electric field 6 N/C with a net displacement 1 m along the field. A simultaneous magnetic field bends its trajectory. Neglect other forces. Find final speed.

(A) $4\,\mathrm{m/s}$ (B) $7\,\mathrm{m/s}$

(C) $\sqrt{65}\,\mathrm{m/s}$ (D) $\sqrt{57}\,\mathrm{m/s}$

2. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 25 percent. The measured increase in its absolute temperature is $\frac{1035}{4}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) $\frac{1472}{5}\,\mathrm{K}$ (B) $4140\,\mathrm{K}$

(C) $\frac{2875}{4}\,\mathrm{K}$ (D) $460\,\mathrm{K}$

3. A diamond network has four outer resistors AB, BC, CD and DA, each 10 ohms, and a bridge BD also 10 ohms. The DA arm additionally contains an ideal diode with anode at D and cathode at A. A source keeps A positive relative to C. Find equivalent resistance between A and C.

(A) $\frac{50}{3}\,\Omega$ (B) $20\,\Omega$

(C) $10\,\Omega$ (D) $15\,\Omega$

4. 4 charges each equal to q occupy vertices of a regular 4-gon of circumradius R. One charge is removed. Find the remaining field magnitude at the centre in units of kq/R^2.

(A) 1 (B) 4

(C) $\frac{1}{3}$ (D) 3

5. An ideal mixture contains 3 moles of monatomic gas and 5 moles of rigid diatomic gas. Vibrations are frozen. Find the heat-capacity ratio $\gamma = C_P/C_V$ of the mixture.

(A) $\frac{25}{17}$ (B) $\frac{5}{3}$

(C) $\frac{23}{15}$ (D) $\frac{18}{17}$

6. An object moves slowly along the axis toward a concave mirror. At an instant its distance from the mirror is 120 cm; the mirror focal-length magnitude is 30 cm. Find the image-speed magnitude divided by object-speed magnitude at that instant.

(A) $\frac{1}{9}$ (B) $\frac{4}{3}$

(C) $\frac{1}{16}$ (D) $\frac{1}{3}$

7. The accelerating potential of an X-ray tube is increased from 20 kV to 80 kV, while its tube current is reduced by a factor 4. Ignoring target heating and recoil, what is the new cutoff wavelength in nm? Use $hc = 1240$ eV nm.

(A) $\frac{31}{250}$ nm (B) $\frac{31}{125}$ nm

(C) $\frac{31}{500}$ nm (D) $\frac{31}{2000}$ nm

8. A stretched string is divided by a bridge into lengths in the ratio $1 : 4$. Tension and linear density are unchanged. The original fundamental is 50 Hz. Find the fundamental of the longer segment.

(A) 40 Hz (B) $\frac{125}{2}$ Hz

(C) 50 Hz (D) $\frac{125}{8}$ Hz

9. A long solenoid has 300 turns per metre and carries 2 A. It is completely filled by a linear magnetic core of relative permeability 2. Find magnetisation M.

(A) 300 A/m (B) 1200 A/m

(C) 600 A/m (D) 2 A/m

10. A single slit of width 2 mm is illuminated by wavelength 700 nm. A screen is 2 m away. Find the separation of the second minima on opposite sides of the central maximum, in the small-angle approximation.

(A) 2800 mm (B) $\frac{7}{5}$ mm

(C) $\frac{28}{5}$ mm (D) $\frac{14}{5}$ mm

11. A projectile is launched at 60 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{15}{16}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ m/s^2.

(A) 216 m (B) $\frac{432}{5}$ m

(C) $\frac{1728}{5}$ m (D) 144 m

12. A beam has measured power $\frac{128}{5} \times 10^{-7}$ W. A detector registers 8×10^{12} photons per second and detects a fraction $\frac{3}{4}$ of the incident photons, without altering the beam. Find its wavelength in nm. Use $e = 1.6 \times 10^{-19}$ C and $hc = 1240$ eV nm.

(A) $\frac{2480}{3}$ nm (B) 620 nm

(C) 2480 nm (D) $\frac{9920}{9}$ nm

13. A nucleus of mass number 32 splits into two equal nuclei. The parent has binding energy per nucleon $\frac{73}{10}$ MeV and each daughter has $\frac{79}{10}$ MeV. Ignoring other products, what energy is released in the split?

(A) $\frac{96}{5}$ MeV (B) $\frac{3}{5}$ MeV

(C) $\frac{1168}{5}$ MeV (D) $\frac{48}{5}$ MeV

14. A particle of mass 1 kg starts from rest. A force $\vec{F} = 4t\,\hat{i} + 8\,\hat{j}$ N acts for 4 s. What is the magnitude of its momentum at that instant?

(A) $64\,\text{kg m/s}$ (B) $8\sqrt{17}\,\text{kg m/s}$

(C) $32\sqrt{2}\,\text{kg m/s}$ (D) $32\sqrt{5}\,\text{kg m/s}$

15. Two identical converging thin lenses of focal length 20 cm are separated by 60 cm. A parallel beam enters the first. Find the distance of the final image from the second lens, positive on the outgoing side.

(A) 40 cm (B) 20 cm

(C) 10 cm (D) 30 cm

16. A solid cylinder of mass 6 kg rolls without slipping on a horizontal rough surface. A horizontal force 50 N is applied tangentially at its top. Find its centre-of-mass acceleration; $I = MR^2/2$.

(A) $\frac{100}{3}\,\text{m/s}^2$ (B) $\frac{100}{9}\,\text{m/s}^2$

(C) $\frac{50}{9}\,\text{m/s}^2$ (D) $\frac{25}{3}\,\text{m/s}^2$

17. A mass 4 kg moves with $\vec{r}(t) = 2t\hat{i} + 4t^2\hat{j}$ m. Find the z-component of angular momentum about the origin at $t = 3$ s.

(A) $96\,\text{kg m}^2\text{/s}$ (B) $288\,\text{kg m}^2\text{/s}$

(C) $144\,\text{kg m}^2\text{/s}$ (D) $576\,\text{kg m}^2\text{/s}$

18. A network is defined by $P = \text{NAND}(A, B)$, $Q = \text{NAND}(A, P)$ and $Y = \text{AND}(Q, \overline{B})$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 1$ and $P = 0$?

(A) 3 (B) 1

(C) 2 (D) 0

19. A small-angle pendulum has period T in a stationary lift. In a vertically accelerating lift its period is 3T. Find the downward lift acceleration divided by g.

(A) $\frac{10}{9}$ (B) 8

(C) $\frac{8}{9}$ (D) $\frac{1}{3}$

20. One mole of ideal gas follows $p = aV$ with positive constant a. Its volume changes from V_0 to $5V_0$. Find T_f/T_i.

(A) 25 (B) $\frac{1}{5}$

(C) 125 (D) 1

21. A particle has position-dependent velocity components $\vec{v} = 4x\,\hat{i} - 4y\,\hat{j}$ in SI units. Given $dx/dt = v_x$ and $dy/dt = v_y$, what is its acceleration magnitude at $(2, 2)$ m?

(A) $64\,\text{m/s}^2$ (B) $32\,\text{m/s}^2$

(C) $8\sqrt{2}\,\text{m/s}^2$ (D) $32\sqrt{2}\,\text{m/s}^2$

22. A direct vernier has 20 divisions equal to 19 main-scale divisions; 1 MSD is 1 mm. Its signed zero error is $\frac{1}{10}$ mm. For an object the main scale is 5 mm and the 15th vernier division coincides. Find the corrected reading.

(A) $\frac{23}{4}$ mm (B) $\frac{113}{20}$ mm

(C) $\frac{15}{4}$ mm (D) $\frac{117}{20}$ mm

23. A spring of constant 60 N/m is already stretched by $\frac{3}{10}$ m. How much additional work is required to increase its extension to $\frac{7}{10}$ m quasistatically?

(A) $\frac{84}{5}$ J (B) 12 J

(C) $\frac{27}{10}$ J (D) $\frac{36}{5}$ J

24. For steady laminar flow through a capillary, the radius is multiplied by 2 and length by 5. The volume flow rate must remain unchanged and viscosity is fixed. By what factor must the pressure difference be multiplied?

(A) $\frac{16}{5}$ (B) 1

(C) $\frac{5}{4}$ (D) $\frac{5}{16}$

25. A series RC circuit is driven at fixed rms voltage. Multiplying angular frequency by 3 multiplies rms current by $\frac{\sqrt{105}}{7}$. Find R^2/X_C^2, where X_C is the initial capacitive reactance.

(A) $\frac{15}{7}$ (B) $\frac{1}{4}$

(C) $\frac{2}{3}$ (D) 9

26. A particle is projected vertically outward from the surface of a nonrotating planet with speed $v_e/7$, where v_e is surface escape speed. Neglect atmosphere. Find maximum height divided by planet radius.

(A) $\frac{1}{49}$ (B) $\frac{1}{50}$

(C) $\frac{1}{48}$ (D) $\frac{1}{6}$

27. In an abrupt junction, the acceptor concentration is 729 times the donor concentration. Within the depletion approximation, which statement gives x_n/x_p and identifies the carriers mainly responsible for the small reverse current before breakdown?

(A) 27; majority carriers (B) 1/729; minority carriers

(C) 729; minority carriers (D) 1; majority carriers

28. At the highest point of its flight, at height 25 m, a projectile has horizontal velocity 10 m/s. It explodes into two equal fragments, both initially moving horizontally. One moves forward at 5 m/s. Find the horizontal distance travelled by the other fragment from the explosion point to the ground. Take $g = 10$ m/s^2.

(A) $25\sqrt{5}$ m (B) $\frac{15\sqrt{10}}{2}$ m

(C) $10\sqrt{5}$ m (D) $15\sqrt{5}$ m

29. A drop of mass 8.96×10^{-17} kg is suspended in a vertical field of magnitude 700 N/C. It has excess electrons. One excess electron is removed without changing its mass. Find its immediate downward acceleration, taking $g = 10$ m/s^2 and $e = 1.6 \times 10^{-19}$ C.

(A) $\frac{10}{7}$ m/s^2 (B) $\frac{5}{4}$ m/s^2

(C) $\frac{10}{9}$ m/s^2 (D) $\frac{5}{32}$ m/s^2

30. A uniform-density planet has radius R. Gravity at a depth d equals gravity at a height $(6-1)R$ above the surface. Find d/R.

(A) $\frac{1}{6}$ (B) $\frac{35}{36}$

(C) $\frac{1}{36}$ (D) $\frac{5}{6}$

Chemistry

31. A food extract contains exactly one of the listed two-component combinations. Using the usual qualitative tests under suitable conditions, its results are: Benedict on warming: negative; iodine: negative; Biuret: positive; Sudan III: positive. Which combination matches?

(A) glucose and albumin
(B) starch and albumin
(C) albumin and triolein
(D) glucose and starch

32. A Fe(3) coordination entity has coordination number 6, exactly 1 fully bound EDTA ligand(s), and chloride in all remaining donor sites. Take EDTA as 6-dentate with charge -4; chloride is monodentate. Find (number of ligand entities, net charge of the coordination entity).

(A) (2, -1)
(B) (1, -2)
(C) (6, +0)
(D) (1, -1)

33. Which combined classification/reactivity description is correct for cyclopentanol?

(A) a phenol because it is cyclic
(B) a tertiary alcohol with no carbinol H
(C) a primary alcohol with two carbinol H atoms
(D) a secondary alcohol with one carbinol H

34. A water sample contains 1/250 mol Ca(HCO3)2 and 3/1000 mol CaSO4 and no other hardness source. On quantitative boiling, Ca(HCO3)2→CaCO3(s)+CO2+H2O; CaSO4 stays dissolved. Which pair gives the CaCO3 precipitate mass and remaining hardness type? CaCO3=100 g/mol.

(A) Zero; boiling can never remove temporary hardness.
(B) 1/5 g; only temporary hardness remains.
(C) 7/10 g; no hardness remains.
(D) 2/5 g; permanent hardness remains.

35. A primary-standard sample of pure anhydrous Na_2CO_3 weighs $\frac{371}{500}$ g. It is titrated with0.100M HCl. Find the additional acid volume in mL needed after the first carbonate-to-hydrogencarbonate equivalence point to reach complete conversion to CO_2/water. Take Na_2CO_3=106g/mol.

(A) 141
(B) 70
(C) 140
(D) 35

36. A measured[A]-versus-time plot is linear,falling from $\frac{11}{100}$M at0s to $\frac{9}{100}$M at20.0s. At20s,volume is instantly doubled by solvent;the same zero-order rate constant then applies. Time after dilution to halve the new[A],in s,is

(A) 45
(B) 55
(C) $\frac{45}{2}$
(D) 20

37. An isolated NaCl cathode compartment of volume $\frac{1}{5}$L initially has $\frac{1}{50}$mol strong acid. Current5.00A flows for 772s,F=96500C/mol. Only cathodic water reduction produces OH^-,one per electron,and no anode product enters. Volume is fixed. At25C,finalpH is

(A) 1
(B) 14
(C) 7
(D) 13

38. A five-d-orbital splitting has an upper set and lower set separated by Δ. The lower energy is -3/5Δ and the upper energy is 2/5Δ relative to the conserved barycentre. Which (upper degeneracy, lower degeneracy) is required?

(A) (3, 2)
(B) (4, 1)
(C) (1, 4)
(D) (2, 3)

39. A table links one fat-soluble vitamin with normal blood-clotting processes. Which vitamin and mistaken substitution are correctly distinguished?

(A) Vitamin B2; it is retinol.
(B) Vitamin B1; it is calciferol.
(C) VitaminK; it is not the scurvy-associated vitamin C.
(D) Vitamin C; it is the same compound as phylloquinone.

40. At a fixed temperature, a diprotic acid H2A has pKa1=8, pKa2=12, and pKw=15. Find (pKb of A2-, Kb(A2-)/Kb(HA-)). Use the conjugate pair appropriate to each basic species.

(A) (7, 1/10000)
(B) (3, 10000)
(C) (7, 10000)
(D) (3, 1/10000)

41. For 2,2,4-trimethylpentane,what is the pair (number of primary carbons,number of hydrogen atoms attached to tertiary carbons)?

(A) (5,3)
(B) (5,1)
(C) (4,1)
(D) (5,0)

42. A monatomic ion contains 18 electrons and has mass number 37. Its charge is -1. Which pair gives (proton count, neutron count), without confusing ionic charge with nuclear identity?

(A) (17, 19)
(B) (17, 20)
(C) (18, 19)
(D) (16, 21)

43. In a closed disproportionation, the only oxidation states of the redox-active element are initially Cl(0) and finally Cl(V) plus Cl(-I). No other element changes oxidation state. What fraction of the redox-active atoms ends in the higher oxidation state? Apply electron balance rather than counting other atoms.

(A) 1/6
(B) 1/3
(C) -1/6
(D) 1/12

44. Which occurrence/extraction inference for KCl is chemically consistent?

(A) It occurs in sylvite; potassium occurs in ionic minerals,not as native metal.

(B) The metal is obtained by ordinary water electrolysis without preferential hydrogen evolution.

(C) The formula proves that oxygen is the only possible counter-ion in its natural deposits.

(D) The metal occurs chiefly as native metal because group1/2metals are exceptionally unreactive.

45. $\frac{96}{25}$ g pure NH_4NO_2 decomposes by $NH_4NO_2 \rightarrow N_2 + 2H_2O$. Only80.0% of the theoretical dry N_2 is recovered. Find its STP volume in litres, using22.4L/mol and NH_4NO_2=64g/mol.

(A) $\frac{168}{125}$
(B) $\frac{672}{625}$
(C) $\frac{42}{25}$
(D) $\frac{1344}{625}$

46. Evaluate independently. I: The central N formal charge in the usual nitrate contributor is zero. II: O_2^+ has higher bond order than O_2.

(A) Both I and II are correct.

(B) Neither I nor II is correct.

(C) Only II is correct.

(D) Only I is correct.

47. For one mole of propanal oxidised to propanoate by Tollens reagent,which product or mechanistic conclusion is correct under the stated conditions?

(A) one mole Ag per mole aldehyde

(B) zero Ag because aldehydes are oxidants

(C) two moles Ag per mole aldehyde

(D) four moles Ag per mole aldehyde

48. For closed-system paths A→B→C, q_AB=510 J, w_AB=-250 J, q_BC=-240 J and w_BC=230 J, with work defined on the system. A direct C→A path has w=105 J. What is its heat q in J?

(A) -355

(B) 355

(C) -95

(D) -615

49. Which coordination name keeps the bracketed sphere,counter-ion and formal metal oxidation state consistent for $K_4[Fe(CN)_6]$?

(A) potassium hexacyanidoferrate(III)

(B) potassium hexacyanidoiron(IV)

(C) potassium hexacyanidoferrate(II)

(D) tetrapotassium hexacyanidoiron(II)

50. At fixed T,P a binary R⇌P system has G=10 kJ for pure R and 30 kJ for pure P on one fixed total-amount basis. The unique minimum G=5 kJ occurs at product fraction 1/5. Which claim is consistent with the curve?

(A) The equilibrium product fraction is 1/5; moving from the minimum toward either pure endpoint raises G.

(B) The higher pure-product endpoint proves that product cannot be present at equilibrium.

(C) The equilibrium must be pure R because 10<30.

(D) The interior minimum means the two pure endpoints have equal G.

51. An iron surface is covered by an aerated water drop. In the stated differential-aeration model,the centre is oxygen-poor and the edge oxygen-rich. Which location/reaction pair is consistent?

(A) Centre:anodic $Fe \rightarrow Fe^{2+} + 2e^-$;edge:cathodic oxygen reduction generating OH^-.

(B) Both regions must be anodic because the metal is the same element.

(C) Centre:cathodic iron dissolution;edge:anodic oxygen reduction.

(D) Electrons flow through water from oxygen-rich edge to oxygen-poor centre,with no metallic current path.

52. At 1 atm compare two separate pairs: (water, hydrogen sulfide) and (n-butane, isobutane). A student selects hydrogen sulfide for the first higher-boiling substance because its formula mass is unchanged, and n-butane for the second. Which assessment is consistent with intermolecular interactions?

(A) Both selections correct.

(B) First selection correct; second selection wrong.

(C) Both selections wrong.

(D) First selection wrong; second selection correct.

53. A table labels aspirin an analgesic and diazepam a tranquilizer. Which interpretation keeps the two labels distinct?

(A) The labels prove both molecules have identical formulae.

(B) One label concerns pain relief, the other anxiety/tension-related nervous-system action.

(C) Both labels mean direct acid-base neutralisation.

(D) Both labels mean artificial sweetness.

54. An iron-ion inventory of 1/50 mol contains only Fe2+ and Fe3+. Complete externally driven reduction to metal, with no side reaction, consumes 11/250 mol electrons. What initial fraction of iron was Fe3+?

(A) 1/5 (B) 11/15

(C) 1/15 (D) 1/10

55. For a completely filled n=5 shell,how many electrons have$m_l = 0$ and$m_s = +1/2$?

(A) 10 (B) 5

(C) 25 (D) 50

56. For a hypothetical MX(s),formation enthalpy is -450 kJ mol^{-1}. M atomisation is100,first ionisation500, X_2 bond dissociation200,and electron-gain enthalpy $X(g)+e^- \rightarrow X^-$ is-300,all in kJ mol^{-1}. The positive lattice dissociation enthalpy of MX is

(A) 750 (B) 950

(C) 850 (D) -850

57. For cis-pent-2-ene,trans-pent-2-ene,pent-1-ene,which decreasing stability order also implies increasing heat released on complete hydrogenation to the same carbon skeleton where directly comparable?

(A) cis-pent-2-ene > trans-pent-2-ene > pent-1-ene (B) pent-1-ene > trans-pent-2-ene > cis-pent-2-ene

(C) trans-pent-2-ene > pent-1-ene > cis-pent-2-ene (D) trans-pent-2-ene > cis-pent-2-ene > pent-1-ene

58. Within an additive covalent-radius model,a single X-X bond is 150 pm and an X-H bond is 115 pm. A linear H-X-X-H skeleton is assumed with collinear single bonds. Its terminal H-to-H distance,in pm,is

(A) 460 (B) 530

(C) 265 (D) 380

59. A mixture contains one mole each of benzoic acid,phenol and tert-butanol. Treat with excess NaOH. What is the total amount,in mol,of NaOH consumed? Assume all stated acid-base reactions go to completion and no side reactions.

(A) 2 (B) 1/2

(C) 3/2 (D) 1

60. In a1.00 L vessel A+2B⇌2C+D. Initially A=3/5 mol, B=7/10 mol, D=0 and an unknown amount of C. At equilibrium D=1/25 mol and Kc=169/53816 using molar concentrations. What was the initial C amount in mol? All equilibrium amounts are positive.

(A) -1/20 (B) 1/20

(C) 1/10 (D) 1/40

English Proficiency

61. Both parties authorise him to hear their case and give a decision, rather than merely help them negotiate. What is he?

(A) arbitrator (B) advocate

(C) witness (D) mediator

62. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The committee will reconsider the appeal if new evidence is supplied.

(A) The appeal will be reconsidered by the committee if new evidence is supplied.

(B) The appeal was reconsidered by the committee if new evidence was supplied.

(C) New evidence will be supplied by the appeal if the committee is reconsidered.

(D) The appeal will be reconsidered whether or not new evidence is supplied.

63. When all four words are used as colour names, which does not belong with the other three?

(A) crimson (B) ruby

(C) azure (D) scarlet

64. Which ONE of the following words is misspelt?

(A) perseverance (B) forty

(C) environment (D) embarass

65. Complete: The number of errors in the two copied lists _____ smaller than it was yesterday.

(A) is (B) were

(C) have (D) are

66. Read the sentence: The additional team was brought in to accelerate the repairs. Choose the opposite in meaning of 'accelerate' as used in the sentence.

(A) speed up (B) slow down

(C) complete (D) coordinate

67. Read the sentence: The speaker's long pause was deliberate; she wanted the audience to consider the question. Choose the closest meaning of 'deliberate' as used in the sentence.

(A) hesitant (B) intentional

(C) repeated (D) unfortunate

68. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. The planners observed that route instead of simply blocking it. B. A new path was laid across the park according to the original drawing. C. Visitors nevertheless kept walking along a shorter line through the grass. D. They later adjusted the path to fit the movement the visitors had demonstrated.

(A) The planners observed that route instead of simply blocking it.

(B) They later adjusted the path to fit the movement the visitors had demonstrated.

(C) A new path was laid across the park according to the original drawing.

(D) Visitors nevertheless kept walking along a shorter line through the grass.

69. The study concerns atmospheric conditions and weather processes rather than celestial objects. Which field is it?

(A) botany (B) astronomy

(C) geology (D) meteorology

70. Choose the best opening sentence for this paragraph: A warning placed after a junction may be accurate yet useless to a walker who has already turned. A shorter cue before the junction can be more helpful. The full explanation can still follow for those who need detail.

(A) Walkers usually become lost because they refuse to read.

(B) Useful instructions must reach readers before the relevant decision.

(C) A longer description is always more reliable than a short warning.

(D) Every practical guide should contain as little information as possible.

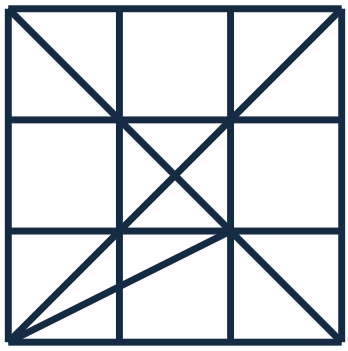
Logical Reasoning

71. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 4, 16, 18, ?, 74, 296, 298.

(A) 72 (B) 80

(C) 20 (D) 70

72. How many triangles are formed by all the drawn segments?



(A) 36 (B) 39

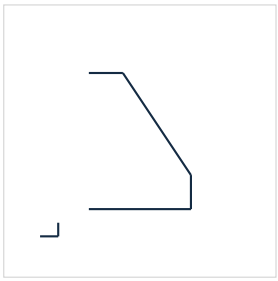
(C) 43 (D) 37

73. Aman points to a photograph and says, "This person is the daughter of the only daughter of my father's only son." Aman is male, and the photographed person is female. How is the photographed person related to Aman?

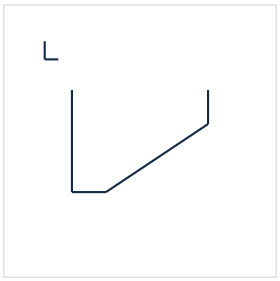
(A) Daughter (B) Granddaughter

(C) Brother (D) Son

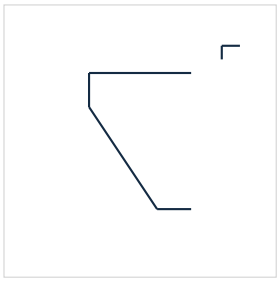
74. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

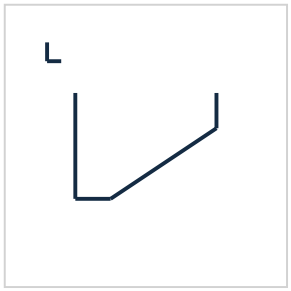


2



3

(A) Pattern T



(B) Pattern L



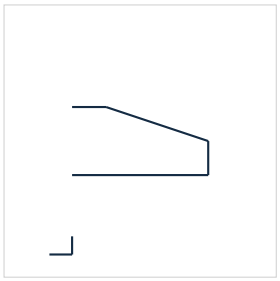
(C) Pattern W



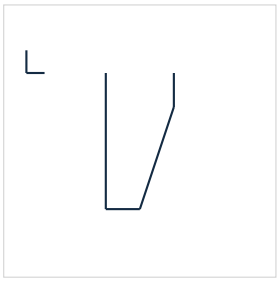
(D) Pattern Z



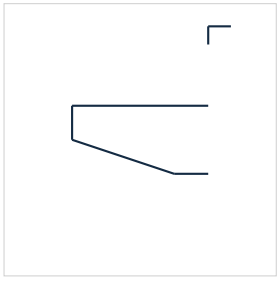
75. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

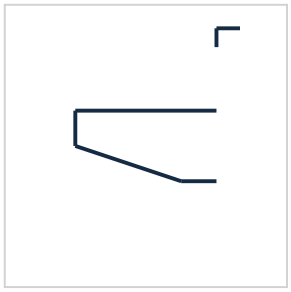


bottom left

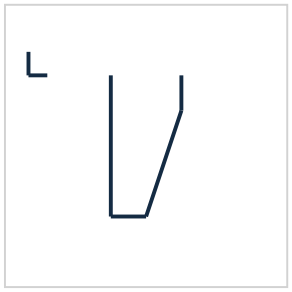
?

bottom right

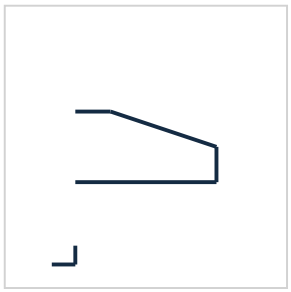
(A) Pattern Y



(B) Pattern R



(C) Pattern K

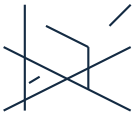


(D) Pattern L



76. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 4 (B) 1

(C) 3 (D) 2

77. In a row, Asha is 9th from the left and 37th from the right. Bina is 7th from the right. Charu stands 4 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 25 (B) 27

(C) 5 (D) 26

78. The age table uses the same two people throughout:

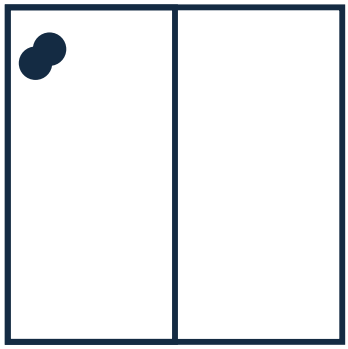
	younger	older
now	x	$x+4$
3 years ago	$x-3$	$x+1$

. The product of their ages 3 years ago was 480. What will the older person's age be 2 years from now?

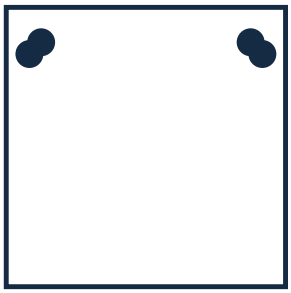
(A) 27 (B) 29

(C) 26 (D) 33

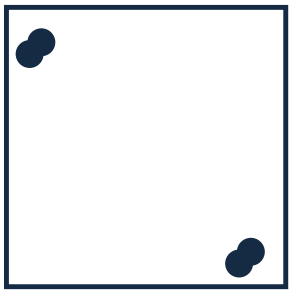
79. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



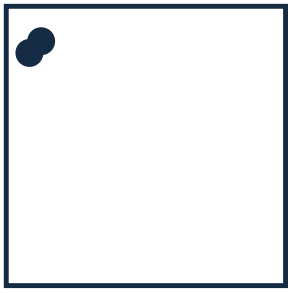
(A) Pattern M



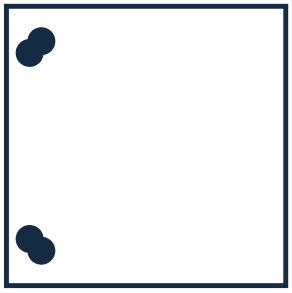
(B) Pattern P



(C) Pattern K



(D) Pattern N



80. Three pairs contain exactly one mammal and one bird. Which pair does not?

(A) Horse / Sparrow

(B) Tiger / Duck

(C) Elephant / Owl

(D) Bat / Rabbit

81. Which numbered host does NOT contain the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



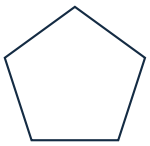
host 4

(A) 2
(B) 3
(C) 4
(D) 1

82. Preserve the exact relation and its order: Rapid : Swift :: ? : Old. Choose the missing term.

(A) Rapid
(B) Begin
(C) Silent
(D) Ancient

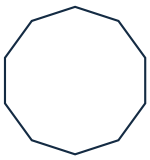
83. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

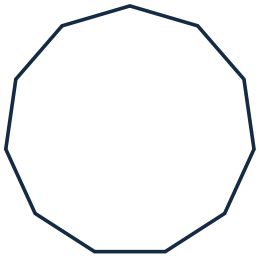


B

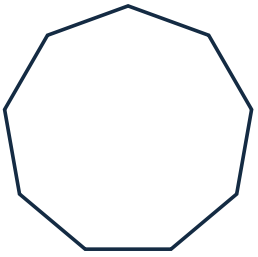


C

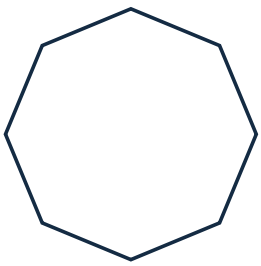
(A) Pattern V



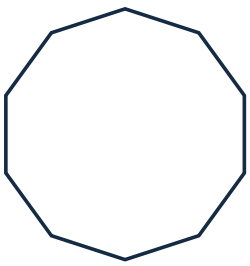
(B) Pattern U



(C) Pattern Y



(D) Pattern W



84. A code shifts letters cyclically through A–Z. Odd positions share one fixed shift and even positions share another. For example, MANGO is coded as RYSET. Which word is coded as QYRN?

(A) LAMP (B) VWWL

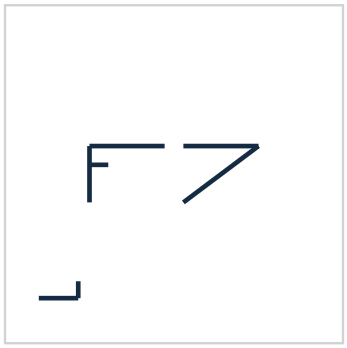
(C) KZLO (D) MBNQ

85. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: LJJF, ILMD, FNPB, ?, ZRVX, WTYV.

(A) ZSPC (B) BORY

(C) CPSZ (D) DQTA

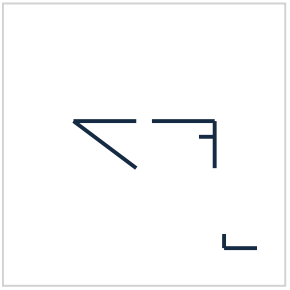
86. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



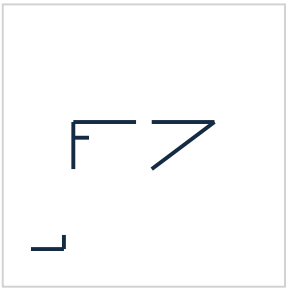
(A) Pattern W



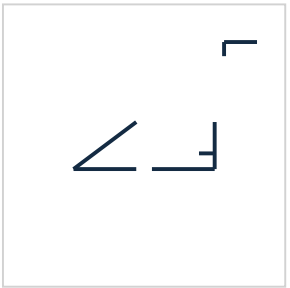
(B) Pattern Q



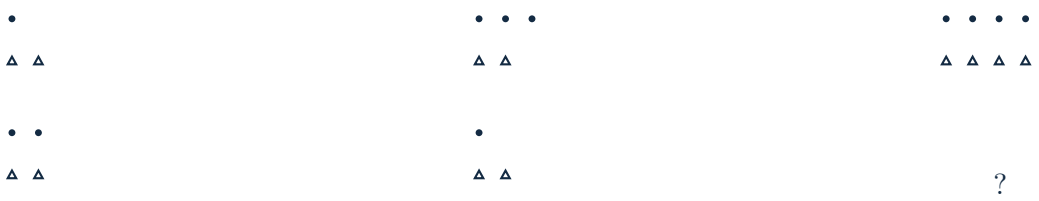
(C) Pattern T



(D) Pattern R



87. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern N

(B) Pattern M



(C) Pattern P



(D) Pattern X





88. Consider only these statements; universal statements alone do not imply existence. Some farmers are not bakers; No farmers are hikers. Conclusions: I. Some farmers are hikers. II. No hikers are farmers. Which conclusions follow necessarily?

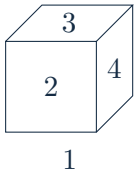
(A) Only I

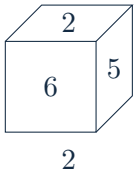
(B) Both I and II

(C) Only II

(D) Neither I nor II

89. The views show the same die. Give the ordered pair: (face opposite 4, face opposite 6). Do not assume opposite labels sum to seven.





(A) $(6, 4)$

(B) $(6, 1)$

(C) $(6, 2)$

(D) $(6, 3)$

90. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 11$ and $5 \mapsto 20$. Which input maps to 50?

(A) 50 (B) 14

(C) 16 (D) 15

Mathematics

91. Using digits 0,1,2,3,4,5,6,7,8 without repetition, how many 5-digit positive integers are divisible by 3, have first digit at most 3, and first digit smaller than last?

(A) 1680 (B) 408

(C) 1272 (D) 3780

92. Find the area bounded by y=x, y=9/x, and x=12 in the first quadrant.

(A) $\frac{135}{2}$ (B) $\log(262144)$

(C) $27 - 18\log(2)$ (D) $\frac{135}{2} - \log(262144)$

93. Evaluate the limit of $[\cos(5x)-\cos(2x)]/x^2$ as $x \to 0$.

(A) $\frac{21}{2}$ (B) $-\frac{21}{2}$

(C) -21 (D) $-\frac{3}{2}$

94. A=(5,6,5), B=(10,6,11). Find the squared length of the component of AB perpendicular to direction (1,2,2).

(A) $\frac{260}{9}$ (B) $\frac{166}{3}$

(C) 61 (D) $\frac{289}{9}$

95. Circles $x^2+y^2=324$ and $(x-20)^2+y^2=324$ meet at A,B. Find $|AB|^2$.

(A) 256 (B) 496

(C) 1696 (D) 896

96. The parabola $y=x^2+px+q$ passes through (0,36). The absolute value of its vertex ordinate is minimized over $p \geq 0$. Find p^2+q^2 at the minimum.

(A) 1440 (B) 6480

(C) 48 (D) 144

97. ABC is right angled at A with AB=12,AC=14. A vertical tower stands at the midpointM ofBC. Its top elevation from A is 60°. Find the square of the tower height.

(A) 85 (B) 507

(C) $\sqrt{255}$ (D) 255

98. A circular sector has radius 9 and arc length $\frac{3\pi}{2}$. Find its area.

(A) $\frac{243\pi}{4}$ (B) $\frac{27}{4}$

(C) $\frac{27\pi}{4}$ (D) 81π

99. Let R(x) be the remainder when x^8 is divided by (x-1)(x-2). Find R(2).

(A) 510 (B) 256

(C) 257 (D) −253

100. For R={(j,j+1):1≤j<8} on {1,...,8}, what is the minimum number of ordered pairs to add to make R transitive?

(A) 28 (B) 29

(C) 7 (D) 21

101. All tangents common to $x^2/36+y^2/9=1$ and $x^2+y^2=\frac{45}{2}$ touch the ellipse at points P_i. Find Σ(P_i, x)^2 over the four tangency points.

(A) $\frac{576}{5}$ (B) 90

(C) $\frac{144}{5}$ (D) $\frac{36}{5}$

102. Evaluate the sum of r(r-1)C(9,r)2^r over all even integers r with 0≤r≤9.

(A) 315072 (B) 314928

(C) 314784 (D) 59040

103. a=(9,4,0), b=(4,-9,4). Find $|a+b|^2$-$|a-b|^2$.

(A) −16 (B) 210

(C) −260 (D) 0

104. Positivea,b satisfy log(a+b)=log(5)+[log a+log b]/2 in one valid base. Find $(a^6+b^6)/(a^3b^3)$.

(A) 12098 (B) 12029

(C) 15625 (D) 12167

105. Evaluate the integral from 0 to 2π of (11-8 sin x)(1+cos x)/(121-64 $\sin^2$x) dx.

(A) $\frac{2\sqrt{57}\pi}{57}$

(B) $\frac{2\pi}{11}$

(C) $-\frac{\log(11)}{8}+\frac{\log(19)}{8}+\frac{2\sqrt{57}\pi}{57}$

(D) $\frac{\sqrt{57}\pi}{114}$

106. O=(0,0,0), A=(9,0,0), B=(1,4,0). Find the squared area of triangle OAB.

(A) 18 (B) 1296

(C) 324 (D) 648

107. Let f(x)=|x-6|/(x-14) for x≠14. Within 0<x<22, what is the total length of the open intervals on which f′(x)<0?

(A) 8 (B) 14

(C) 16 (D) 22

108. For x,y>0 with sqrt(x)+sqrt(y)=5, find d^2y/dx^2 at x=9.

(A) $\frac{5}{6}$ (B) $\frac{5}{54}$

(C) $-\frac{5}{54}$ (D) $\frac{5}{27}$

109. $y'+8y=\exp(-8x)$, $y(0)=6$. Find $\exp(8)y(1)-6$.

(A) 1 (B) $\frac{7}{e^8}$

(C) 7 (D) -1

110. An AP has a=2,d=4; its first n terms sum to 72. Find a_(n+1).

(A) 30 (B) 22

(C) 26 (D) 12

111. $A = \begin{bmatrix} 0 & 7 \\ 0 & 0 \end{bmatrix}$. Find the (1,2) entry of $(I+A)^8$.

(A) 5764801 (B) 56

(C) 8 (D) 7

112. Two groups of sizes 2,3 have means 5,11 and population variances 3,4. Find the combined population variance.

(A) $\frac{431}{5}$ (B) $\frac{306}{25}$

(C) $\frac{18}{5}$ (D) $\frac{7}{2}$

113. The hyperbola $x^2/49-y^2/9=1$ has eccentricity e. Find the eccentricity e′ of its conjugate hyperbola.

(A) $\frac{7\sqrt{58}}{58}$ (B) $\sqrt{58}$

(C) $\frac{\sqrt{58}}{7}$ (D) $\frac{\sqrt{58}}{3}$

114. Disjoint A,B have sizes 10,8. Find $|(A\times A)\cup(B\times B)|$.

(A) 18 (B) 164

(C) 80 (D) 244

115. The line r=(6,4,4)+t(1,2,2) and plane 2x-y=14 are given. Find the squared minimum distance between them.

(A) 0 (B) $\frac{36}{5}$

(C) $\frac{6\sqrt{5}}{5}$ (D) 36

116. How many three-element subsets of {1,2,...,9} contain both an odd and an even number?

(A) 74 (B) 80

(C) 70 (D) 40

117. For $\omega \neq 1$ a cube root of unity, evaluate $\sum_{j=1}^{20}(\omega^j+\omega^{-j})^2$.

(A) 20 (B) 38

(C) 80 (D) 4

118. If |z-8|=3, find the minimum of $|z|^2$.

(A) 5 (B) 25

(C) 55 (D) 121

119. Let f(x+4)=5 x^2. Find the area enclosed between y=f(x) and y=25(x-4).

(A) $\frac{625}{6}$ (B) $\frac{65}{6}$

(C) $\frac{125}{6}$ (D) $\frac{625}{2}$

120. The tangent toy=x^3-4 x at x=3 meets the coordinate axes at A,B. Find the area of triangleOAB.

(A) $\frac{225}{46}$ (B) $\frac{2916}{23}$

(C) 33534 (D) $\frac{1458}{23}$

121. $A = \begin{bmatrix} 10 & 3 \\ -3 & 2 \end{bmatrix}$, K=(A-$A^T$)/2. Find tr($K^2$).

(A) 18 (B) -18

(C) -9 (D) 0

122. Two circles have centres (0,0),(8,0) and radii 2,2. Find the squared minimum distance between points on the two circles.

(A) 144 (B) 4

(C) 16 (D) 48

123. An increasing positive GP has first three terms with product 1728 and first-plus-third sum 30. Find its fourth term.

(A) 12 (B) 3

(C) 24 (D) 48

124. Evaluate $\sum_{\substack{0\le r\le 10 \\ r \text{ odd}}} r\binom{10}{r}$.

(A) 5120 (B) 512

(C) 2560 (D) 23040

125. An equilateral triangle has vertex O and symmetric other vertices on y^2=20x. A circle has the side opposite O as diameter. Find the minimum distance from the parabola focus to this circle.

(A) $55 - 20\sqrt{3}$ (B) $20\sqrt{3} + 55$

(C) 55 (D) $60 - 20\sqrt{3}$

126. How many integers k make 6sin x+8cos x+4=k have exactly two distinct solutions in $0\le x<2\pi$?

(A) 9 (B) 20

(C) 19 (D) 21

127. Two X and one Y cost 11; one X and three Y cost 13. Find the cost of one X plus one Y.

(A) 13 (B) 11

(C) 24 (D) 7

128. Evaluate $\int_0^6$ x floor(x) dx.

(A) 72 (B) 15

(C) 55 (D) $\frac{125}{2}$

129. Let $D(t) = \det \begin{pmatrix} 7+t & 1+t \\ 1+t & 3+t \end{pmatrix}$. Find D′(0).

(A) 8 (B) 40

(C) 10 (D) 20

130. Independent events A,B have probabilities $\frac{5}{8}, \frac{2}{5}$. Given A∪B, find P(A∩B).

(A) $\frac{10}{31}$ (B) $\frac{1}{4}$

(C) $\frac{10}{41}$ (D) $\frac{21}{31}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A projectile of mass 12 kg is launched with speed 30 m/s, $\cos\theta = 3/5$. At its highest point it splits into masses $M/3$ and $2M/3$. The lighter fragment moves horizontally forward at 21 m/s. Find the signed horizontal velocity of the heavier fragment immediately after the explosion.

(A) 21 m/s
(B) $\frac{33}{2}$ m/s
(C) 33 m/s
(D) $\frac{69}{2}$ m/s

132. A metal wire has linear temperature coefficient $\frac{1}{500}$ per degree Celsius referred to 0°C. Its dimensions are fixed and it stays across a constant voltage. Find power at 60°C divided by power at 0°C.

(A) $\frac{25}{28}$
(B) $\frac{22}{25}$
(C) $\frac{625}{784}$
(D) $\frac{28}{25}$

133. A hydrogen atom emits a photon in the $5 \to 4$ transition. A hydrogen-like ion of nuclear charge $Z = 3$ emits a photon of exactly the same wavelength and ends in level $n_f = 12$. What is its initial principal quantum number?

(A) 8
(B) $\frac{5}{3}$
(C) 15
(D) 3

Chemistry

134. Use octet Lewis contributors for $CO3^{2}$- and HCOO-; for SO2 exclude expanded-octet contributors. For HCOO-, HCO3- and HNO3, average only the two nonprotonated central-atom-O bonds. For other species average all central-atom bonds. Find (average bond order of the first minus that of the second, sum of total lone-pair counts in one contributor of each). A lone pair on any atom is counted once.

(A) (17/6, 13)
(B) (-1/6, 13)
(C) (-1/6, 14)
(D) (-1/6, 12)

135. A thermal assembly contains Cu(s) and He(g) in separate inert compartments at the same temperature. Helium is heated quasistatically at constant pressure under a freely moving piston. Containers have negligible heat capacity. Use the classical Dulong-Petit solid model for Cu, neglecting its Cp-Cv difference, and the ideal monatomic model for He. No phase change occurs. Molar masses are Cu64 and He4 g/mol. Over a small common temperature rise, the measured total heat capacity C divided by total Cu-plus-He mass m and gas constant R is C/(mR)=49/256 mol/g. What is the Cu mass fraction?

(A) 3/4
(B) 111/154
(C) 1/4
(D) 47/84

136. Consider the statement: The relative stability of the+1state increases from the lighter Group13members toward Tl. Which explanation or consequence correctly accounts for this trend rather than applying an unqualified atomic-mass rule?

(A) Hydrogen bonding is necessarily stronger for every heavier hydride.

(B) The ns2pair is increasingly forced into all bonds down every group.

(C) The inert ns2pair increasingly resists participation in bonding.

(D) Every increase in atomic mass must increase all molecular bond strengths.

Logical Reasoning

137. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). The numerical component follows a constant arithmetic step. Find the missing internal term: WZB7, YBF11, ADJ15, ?, EHR23, GJV27.

(A) CFN19 (B) DGO19

(C) BEM19 (D) 91NFC

138. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 5 and Z has code 30. Decode the separated values 7, 12, 5, 13, 22.

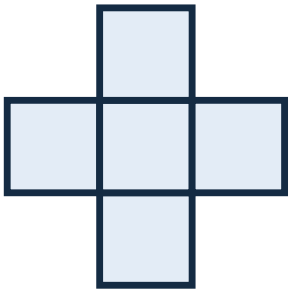
(A) BGZHQ (B) CHAIR

(C) RIAHC (D) DIBJS

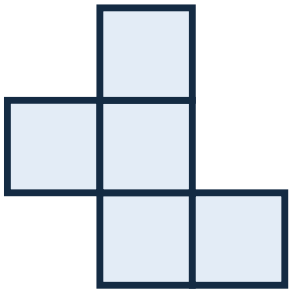
139. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



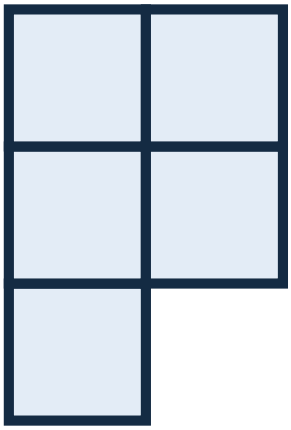
(A) Pattern K



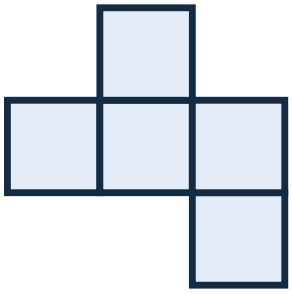
(B) Pattern R



(C) Pattern Z



(D) Pattern L



Mathematics

140. An equivalence relation has classes of sizes 3,5,6. How many unordered pairs of distinct related elements are there?

(A) 14 (B) 70

(C) 28 (D) 17

141. The image of $y^2=24x$ under reflection in $x-y=6$ has equation $(x-6)^2=C(y+6)$. Find C.

(A) 24 (B) 6

(C) 48 (D) -24

142. Event E has likelihood $\frac{4}{7}$ under source A and $\frac{3}{8}$ under source B. Its total probability is $\frac{37}{84}$. If the source is A with probability r, find r.

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$

(C) $\frac{4}{7}$ (D) $\frac{37}{84}$

BITSAT Mock 45

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A charge 2 C moves with $\vec{v} = 4\hat{i} + 3\hat{j}$ m/s in $\vec{B} = 1\hat{i} + 6\hat{j}$ T. What is the signed z-component of its magnetic force?

(A) 42 N (B) 54 N

(C) 6 N (D) 44 N

2. A 1 kg block moves along a horizontal rough surface from $x = 0$ to $x = 5$ m under a force $F(x) = 8x$ N parallel to motion. With $\mu_k = 1/5$ and $g = 10$ m/s^2, find the work done by the net force. The block has enough initial speed to keep moving forward.

(A) 18 J (B) 190 J

(C) 90 J (D) 110 J

3. A plane wave passes through a thin prism away from edges. Separately, a pointlike aperture produces an outgoing spherical wavefront of radius 13 cm, which propagates another 8 cm in a homogeneous medium. Which pair correctly describes the prism wavefront and the later aperture wavefront?

(A) Spherical; plane with infinite radius.

(B) Approximately plane and unchanged in direction; spherical with radius 13 cm.

(C) Approximately plane and tilted; spherical with radius 5 cm.

(D) Approximately plane and tilted; spherical with radius 21 cm.

4. A projectile of mass 15 kg is launched with speed 20 m/s, $\cos\theta = 3/5$. At its highest point it splits into masses $M/3$ and $2M/3$. The lighter fragment moves horizontally forward at 17 m/s. Find the signed horizontal velocity of the heavier fragment immediately after the explosion.

(A) $\frac{53}{2}$ m/s (B) $\frac{19}{2}$ m/s

(C) 19 m/s (D) 12 m/s

5. A coil has 70 turns, area $\frac{1}{100}$ m^2, and rotates at 6 rad/s in a uniform 4 T field. At an instant the angle between its area vector and the field is 30°. Find instantaneous-emf magnitude.

(A) $\frac{42}{5}$ V (B) $\frac{84}{5}$ V

(C) $\frac{7}{5}$ V (D) $\frac{42\sqrt{3}}{5}$ V

6. An ideal mixture contains 2 moles of monatomic gas and 2 moles of rigid diatomic gas. Vibrations are frozen. Find the heat-capacity ratio $\gamma = C_P/C_V$ of the mixture.

(A) $\frac{23}{15}$ (B) $\frac{9}{8}$

(C) $\frac{2}{3}$ (D) $\frac{3}{2}$

7. A particle has position-dependent velocity components $\vec{v} = 2x\,\hat{i} - 2y\,\hat{j}$ in SI units. Given $dx/dt = v_x$ and $dy/dt = v_y$, what is its acceleration magnitude at $(2, 5)$ m?

(A) $20\,\mathrm{m/s^2}$ (B) $8\,\mathrm{m/s^2}$

(C) $4\sqrt{29}\,\mathrm{m/s^2}$ (D) $28\,\mathrm{m/s^2}$

8. A uniform-density planet has radius R. Gravity at a depth d equals gravity at a height $(11-1)R$ above the surface. Find d/R.

(A) $\frac{1}{121}$ (B) $\frac{10}{11}$

(C) $\frac{120}{121}$ (D) $\frac{1330}{1331}$

9. A projectile is launched at 65 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{8}{9}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10$ $\mathrm{m/s^2}$.

(A) $\frac{676}{5}$ m (B) $\frac{676}{3}$ m

(C) $\frac{676}{15}$ m (D) $\frac{2028}{5}$ m

10. At normal incidence, an ideal absorber receives intensity I and an ideal reflector receives intensity $I/2$. Which statement gives the ratio of reflector pressure to absorber pressure, and the reason?

(A) 1/2; photons transfer momentum only on absorption.

(B) 2/2; pressure is inversely proportional to intensity.

(C) 2/2; reflection reverses the normal photon momentum.

(D) 22; reflected light transfers negative energy to the surface.

11. A galvanometer has resistance 60 ohms and full-scale current $\frac{1}{1000}$ A. Find the series resistance required to convert it into a voltmeter of range $\frac{9}{25}$ V.

(A) $360\,\Omega$ (B) $\frac{3}{500}\,\Omega$

(C) $300\,\Omega$ (D) $\frac{3}{10000}\,\Omega$

12. Two identical cells, each emf 3 V and internal resistance 2 ohms, are connected in parallel with like terminals joined and drive a load of 2 ohms. Find the current delivered by each cell.

(A) 1 A (B) $\frac{3}{4}$ A

(C) $\frac{3}{8}$ A (D) $\frac{1}{2}$ A

13. A point charge $30q$ is placed at a vertex of a cube. Find the electric flux through the three faces not meeting that vertex, combined, in units of q/ε_0.

(A) 30 (B) $\frac{15}{4}$

(C) $\frac{5}{4}$ (D) $\frac{15}{8}$

14. In an abrupt junction, the acceptor concentration is 361 times the donor concentration. Within the depletion approximation, which statement gives x_n/x_p and identifies the carriers mainly responsible for the small reverse current before breakdown?

(A) 361; minority carriers

(B) 1/361; minority carriers

(C) 19; majority carriers

(D) 1; majority carriers

15. A Newtonian liquid of viscosity $\frac{1}{2}$ Pa s has velocity profile $v(y) = 5y + 2y^2$ in SI units. Find the shear-stress magnitude at $y = 3$ m.

(A) 17 Pa

(B) 2 Pa

(C) $\frac{17}{2}$ Pa

(D) $\frac{33}{2}$ Pa

16. Three blocks of masses 3, 1 and 5 kg are stacked from top to bottom. A horizontal string holds the middle block fixed to a wall. The top block stays at rest on it, while the bottom block is pulled at constant speed along a rough horizontal floor. The lower two sliding interfaces have kinetic-friction coefficient $\frac{1}{10}$. Find required pulling force on the bottom block. Take $g = 10$ m/s^2.

(A) 16 N

(B) 4 N

(C) 5 N

(D) 13 N

17. Photons of energy 48 eV eject electrons from a surface. The fastest electrons enter a uniform magnetic field perpendicular to their velocities. In this field an electron with kinetic energy 2 eV has orbit radius r. The observed maximum orbit radius is $3r$. What is the surface work function? Neglect recoil.

(A) 66 eV

(B) 30 eV

(C) 42 eV

(D) 46 eV

18. Two identical capacitors each 4 microfarads are connected in series across an ideal 50 V battery. One capacitor undergoes permanent conducting breakdown, becoming a short circuit. Find the additional charge drawn from the battery after the new steady state is reached.

(A) 100 μC

(B) 50 μC

(C) 300 μC

(D) $\frac{200}{3}$ μC

19. A series RC circuit is driven at fixed rms voltage. Multiplying angular frequency by 4 multiplies rms current by $\frac{4\sqrt{34}}{17}$. Find R^2/X_C^2, where X_C is the initial capacitive reactance.

(A) 16

(B) 1

(C) $\frac{32}{17}$

(D) $\frac{3}{5}$

20. The accelerating potential of an X-ray tube is increased from 20 kV to 60 kV, while its tube current is reduced by a factor 3. Ignoring target heating and recoil, what is the new cutoff wavelength in nm? Use $hc = 1240$ eV nm.

(A) $\frac{31\sqrt{3}}{500}$ nm

(B) $\frac{93}{500}$ nm

(C) $\frac{31}{1500}$ nm

(D) $\frac{279}{500}$ nm

21. Two wires of the same material and length L, with radii r and 3r, are joined in series and stretched by a force F. Find their total stored elastic energy divided by the energy stored in the thinner wire alone under force F.

(A) 2

(B) $\frac{10}{9}$

(C) $\frac{9}{10}$

(D) $\frac{4}{3}$

22. A mass 1 kg initially moves in the positive direction at 3 m/s. A force opposite to that direction rises linearly from zero to 4 N and then falls linearly to zero over a total duration 6 s. Find its signed final momentum.

(A) $-9\,\mathrm{kg\,m/s}$ (B) $-3\,\mathrm{kg\,m/s}$

(C) $-21\,\mathrm{kg\,m/s}$ (D) $15\,\mathrm{kg\,m/s}$

23. A 2 kg block slides forward on a rough horizontal floor from x=0 to x=2 m. Applied force is $F(x) = 7(x-1)$ N along +x when positive. Kinetic friction coefficient is 1/5. Find net work, with $g = 10$ $\mathrm{m/s^2}$ and enough initial speed to avoid reversal.

(A) $-8\,\mathrm{J}$ (B) $-\frac{9}{2}\,\mathrm{J}$

(C) $8\,\mathrm{J}$ (D) $6\,\mathrm{J}$

24. 4 kg of water at 30°C is mixed in an insulated vessel with $\frac{504}{167}$ kg of ice at 0°C. Use $c_w = 4200$ J/(kg K), $L_f = 334000$ J/kg. Find the mass of ice remaining at equilibrium.

(A) $\frac{378}{167}$ kg (B) $\frac{36918}{19205}$ kg

(C) $\frac{504}{167}$ kg (D) $\frac{252}{167}$ kg

25. A displacement law is $x = at + bt^8$, where t is time. Which is the dimension of a^8/b?

(A) L^8T^{-8} (B) L^7

(C) L^7T^{-1} (D) T^7

26. A ray in a medium of index 2 is incident on a medium of index $\frac{3}{2}$. Find the sine of the critical-angle threshold for total internal reflection.

(A) $\frac{3}{4}$ (B) $\frac{1}{4}$

(C) $\frac{3}{2}$ (D) $\frac{1}{2}$

27. A fixed ideal gas has 5 active degrees of freedom per molecule. Assertion: its molar internal-energy change is $\frac{5}{2}R\Delta T$. Reason: its heat-capacity ratio is $\frac{7}{5}$. Choose the best statement.

(A) Both are true, but no thermodynamic relation connects them.

(B) The assertion is true and the reason is false.

(C) The assertion is false and the reason is true.

(D) Both are true, and using $C_V = R/(\gamma - 1)$ the reason explains the assertion.

28. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 15 percent. The measured increase in its absolute temperature is $\frac{6063}{40}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) $\frac{2021}{2}$ K (B) 470 K

(C) $\frac{20210}{3}$ K (D) $\frac{24863}{40}$ K

29. Unpolarised light passes three ideal polarisers with axes at 0°, 45°, and 75° relative to a common reference. Find final intensity divided by incident intensity.

(A) $\frac{1}{4} - \frac{\sqrt{3}}{8}$ (B) $\frac{3}{16}$

(C) $\frac{1}{4}$ (D) $\frac{3}{8}$

30. A uniform rod of mass 2 kg and length 4 m lies on a smooth horizontal table, free to rotate about a fixed vertical pivot at its centre. A point mass 2 kg moving perpendicular to the rod at 5 m/s strikes an end and sticks. Find the angular speed immediately afterward.

(A) $\frac{15}{26}$ rad/s (B) $\frac{5}{2}$ rad/s

(C) $\frac{15}{4}$ rad/s (D) $\frac{15}{8}$ rad/s

Chemistry

31. Glucose is converted into methyl glucoside at its anomericOH, without cleaving any carbon-carbon bond. Which change in the standard reducing test is expected before hydrolysis?

(A) The anomeric acetal no longer supplies the usual free-hemiacetal reducing end.

(B) The molecule must become galactose.

(C) The compound loses every hydroxyl group.

(D) Reduction must increase becauseOCH3 is a free CHO group.

32. For 4-methylbenzaldehyde with NaBH4,then water,which product or mechanistic conclusion is correct under the stated conditions?

(A) 4-methylbenzyl alcohol (B) 4-methylbenzoic acid

(C) 4-methoxybenzaldehyde (D) benzyl alcohol

33. For the metal centre in Ni^{2+},Z=28. Determine the number of d electrons after formal oxidation-state assignment; use O=-2 in oxoanions.

(A) 10 (B) 8

(C) 9 (D) 7

34. A lead-containing dust is trapped on a particle filter while gaseous SO2 largely passes. Which conclusion is justified?

(A) SO2 must be particulate because it came from the same stack.

(B) Lead has therefore been converted entirely to nitrogen.

(C) Particle capture proves complete removal of all chemical pollutants.

(D) The filter distinguishes physical form; it has not demonstrated removal of every gas.

35. In a closed disproportionation, the only oxidation states of the redox-active element are initially O(-I) and finally O(0) plus O(-II). No other element changes oxidation state. What fraction of the redox-active atoms ends in the higher oxidation state? Apply electron balance rather than counting other atoms.

(A) 1/2 (B) 1

(C) 1/4 (D) -1/2

36. A double-strandedDNA sample has 30%adenine among all bases. Under standardWatson-Crick pairing, what are the thymine and guanine percentages?

(A) Thymine20%; guanine30%. (B) Thymine30%; guanine40%.

(C) Thymine15%; guanine35%. (D) Thymine30%; guanine20%.

37. CO binds to an electron-rich metal carbonyl. If metal-to-CO pi back-donation increases while the ligand remains bound through carbon,which paired bond changes and donation directions are consistent?

(A) CO pi electrons are donated only from oxygen to metal,and both M-C and C-O necessarily strengthen.

(B) Metal donates into a CO bonding orbital,so C-O strengthens while M-C weakens.

(C) CO donates a carbon lone pair to metal; metal donates into CO pi-star. M-C strengthens while C-O weakens.

(D) CO is only a sigma donor; back-donation leaves the C-O force constant unchanged.

38. For an isolated ClO4- ion, use the introductory VSEPR molecular description. Which pair gives chlorine's formal oxidation state and central-atom lone-pair count? Oxygen has oxidation state-2; use the usual lowest-formal-charge Lewis picture for this main-group ion.

(A) (+9, 0) (B) (+7, 0)

(C) (+5, 0) (D) (+7, 1)

39. An isolated cathode compartment initially contains 3/50 mol HA and the same amount of A-, with pKa5.00. Water reduction gives one OH- per electron; there is no anode-product entry or buffer loss. What constant current for 2895 s gives final pH=5+log10(3)? F=96500 C/mol.

(A) 2 (B) 1/2

(C) -1 (D) 1

40. A Hinsberg separation is applied to triethylamine. Which reagent identity and class-dependent outcome are correct?

(A) benzenesulphonic acid,C6H5SO3H;all classes give the same neutral product

(B) benzenesulphonyl chloride,C6H5SO2Cl;every amine gives an alkali-soluble N-H sulphonamide

(C) benzenesulphonyl chloride,C6H5SO2Cl;no ordinary sulphonamide;the unchanged amine dissolves on acidification

(D) benzoyl chloride,C6H5COCl;no ordinary sulphonamide;the unchanged amine dissolves on acidification

41. Which proposed routes give 2-methylpropan-2-ol as the stated major alcohol? I:propanone+methylmagnesium bromide,then water;II:2-methylpropene+water/H+;III:2-methylpropanal+NaBH4. Where no work-up is written after NaBH4,assume the usual protonating work-up.

(A) I and II only (B) I andIII only

(C) all three (D) II andIII only

42. Which pair correctly describes ordinary molecular hydrogen peroxide without confusing the H-O-O-H skeleton with dioxygen?

(A) It has no O-O bond because both oxygens bind hydrogen.

(B) It has an O-O single bond and no required unpaired electron in the ordinary closed-shell Lewis description.

(C) It contains one oxygen atom and two lone pairs in total.

(D) It has an O=O double bond and the same electron count as O2.

43. For a fixed-composition closed system dG=V dP-S dT. At a state use V=6 m3, S=2 J/K and a local path with dT/dP=3 K/Pa. Find dG/dP along this path in m3. The supplied values are a formal local thermodynamic model; 1 Pa m3=1 J.

(A) -1 (B) 0

(C) 2 (D) 1

44. OHC-CH2-CH2-CONH2 is treated with NaBH4 under ordinary controlled conditions. Which product satisfies carbon accounting and the reagent selectivity?

(A) HOOC-CH2-CH2-CONH2 (B) HOCH2-CH2-CH2-CH2NH2

(C) HOCH2-CH2-CH2-CONH2 (D) OHC-CH2-CH2-CH2NH2

45. For neutral H2 and T2, take isotope mass numbers H=1, D=2, T=3 and neglect mass defects. Their gases have equal temperature. Which pair gives ($\langle v^2\rangle$_H2/$\langle v^2\rangle$_T2, number of neutrons in one T2 molecule minus that in one H2 molecule)?

(A) (1/3, 4) (B) (3, 4)

(C) (3, 3) (D) (3, 6)

46. An100mL sample contains $\frac{1}{50}$ mol HA,pKa=6,and $\frac{1}{100}$ mol HCl. Add100mL $\frac{3}{10}$ M NaOH. Use usual weak-base salt hydrolysis and $\log_{10}(\frac{1}{10})=-1$. Final pH is

(A) 6 (B) 9.5

(C) 13 (D) 7

47. For the same nonrelativistic ion starting from rest,an accelerator changes deBroglie wavelength from $\lambda$to$\lambda/3$. Ion charge and mass are unchanged. Factor by which accelerating voltage must change is

(A) 3 (B) $\frac{1}{9}$

(C) $\frac{1}{3}$ (D) 9

48. For propanone and 2,2-dimethylpropanal,which structural-reactivity comparison is correct under ordinary enolate/aldol conditions?

(A) both have alpha H (B) the aldehydic H is the alpha H

(C) only propanone has an enolisable alpha H (D) only the aldehyde has alpha H

49. A sample of sodium butanoate is treated with soda lime,heat. Which isolated product is consistent with the reagent limitation and carbon-skeleton change?

(A) ethane (B) propene

(C) propane (D) butane

50. At fixed T,P a binary R⇌P system has G=15 kJ for pure R and 40 kJ for pure P on one fixed total-amount basis. The unique minimum G=10 kJ occurs at product fraction 2/5. Which claim is consistent with the curve?

(A) The higher pure-product endpoint proves that product cannot be present at equilibrium.

(B) The interior minimum means the two pure endpoints have equal G.

(C) The equilibrium product fraction is 2/5; moving from the minimum toward either pure endpoint raises G.

(D) The equilibrium must be pure R because 15<40.

51. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_y; same facing signs; 2: parallel p_y; opposite facing signs; 3: parallel p_x; same facing signs. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (1, 3)
(B) (2, 1)
(C) (2, 2)
(D) (3, 1)

52. Ignoring special ortho effects,which decreasing acid-strength order is correct for trichloroacetic acid, dichloroacetic acid,chloroacetic acid?

(A) chloroacetic > dichloroacetic > trichloroacetic
(B) trichloroacetic > dichloroacetic > chloroacetic
(C) trichloroacetic > chloroacetic > dichloroacetic
(D) dichloroacetic > trichloroacetic > chloroacetic

53. Beam A has wavelength$4\lambda_0$ and beam B has$5\lambda_0$. A carries 2 times B's total energy. Photon-number ratioN_A/N_B is

(A) $\frac{8}{5}$
(B) $\frac{5}{2}$
(C) $\frac{4}{5}$
(D) 2

54. Particles A,B both attain$\Delta x\Delta p=\hbar/2$ in one dimension. B has4times A's mass and2times A's position uncertainty. Ratio ΔvB/ΔvA is

(A) $\frac{1}{2}$
(B) $\frac{1}{4}$
(C) 8
(D) $\frac{1}{8}$

55. A thermal assembly contains Cu(s) and He(g) in separate inert compartments at the same temperature. Helium is heated quasistatically at constant pressure under a freely moving piston. Containers have negligible heat capacity. Use the classical Dulong-Petit solid model for Cu, neglecting its Cp-Cv difference, and the ideal monatomic model for He. No phase change occurs. Molar masses are Cu64 and He4 g/mol. Over a small common temperature rise, the measured total heat capacity C divided by total Cu-plus-He mass m and gas constant R is C/(mR)=13/80 mol/g. What is the Cu mass fraction?

(A) 296/385
(B) 68/105
(C) 4/5
(D) 1/5

56. In a localised four-electron-domain model, H2O has two bond domains and two lone pairs, while NH3 has three bond domains and one lone pair. Which conclusion is correct?

(A) NH3 has no lone pair because nitrogen makes three bonds.

(B) Both must be tetrahedral molecules because their electron-domain count is four.

(C) H2O is linear because only bonded atoms count toward its electron-domain geometry.

(D) Both use an sp3 electron-domain description; their molecular shapes are bent and pyramidal respectively.

57. At a fixed temperature, a diprotic acid H2A has pKa1=9, pKa2=13, and pKw=14. Find (pKb of A2-, Kb(A2-)/Kb(HA-)). Use the conjugate pair appropriate to each basic species.

(A) (1, 10000)
(B) (5, 1/10000)
(C) (5, 10000)
(D) (1, 1/10000)

58. benzyl bromide is reacted separately with AgCN and KCN. Which pair correctly assigns the principal organic products,in that order?

(A) benzyl isocyanide,C6H5CH2-NC; benzyl cyanide,C6H5CH2-CN

(B) benzyl cyanide,C6H5CH2-CN; benzyl cyanide,C6H5CH2-CN

(C) benzyl isocyanide,C6H5CH2-NC; benzyl isocyanide,C6H5CH2-NC

(D) benzyl cyanide,C6H5CH2-CN; benzyl isocyanide,C6H5CH2-NC

59. For AgCl(s)+2NH3⇌[Ag(NH3)2]+ +Cl-, Ksp Kf=1/4. Before solid is added there is zero chloride, an unknown silver-ammine background, and free NH3=1/4 M. Measured additional dissolved AgCl is 25/464 M. All dissolved silver remains complexed; volume is fixed. What was the initial background complex concentration?

(A) 1/25 (B) 1/50

(C) 2/25 (D) -1/25

60. An H_2SO_4 stock is29.0% by mass. A20.0mL aliquot is diluted to 120mL,and the diluted molarity is $\frac{3}{5}$M. H_2SO_4 M98. Stock density in g/mL is

(A) $\frac{147}{725}$ (B) $\frac{5292}{725}$

(C) $\frac{441}{1250}$ (D) $\frac{882}{725}$

English Proficiency

61. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. it must still be checked Q. No matter how convincing R. against the relevant evidence. S. an explanation first appears,

(A) Q-S-P-R (B) S-Q-P-R

(C) P-Q-S-R (D) R-S-P-Q

62. Read the sentence: The company agreed to disclose the terms rather than keep them secret. Choose the opposite in meaning of 'disclose' as used in the sentence.

(A) explain (B) conceal

(C) publish (D) summarise

63. Read the sentence: The flexible schedule could be changed when the participants' needs changed. Choose the opposite in meaning of 'flexible' as used in the sentence.

(A) irregular (B) adjustable

(C) convenient (D) rigid

64. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. Having checked the source of the quotation, Q. before submitting the article R. to the editor. S. the writer added the reference

(A) Q-P-S-R (B) R-S-Q-P

(C) P-S-Q-R (D) S-P-Q-R

65. Complete: The first version was easier to read; _____, it omitted a condition essential to the calculation.

(A) therefore (B) however

(C) similarly (D) for example

66. The cancelled journey seemed a setback, but it gave her time to find her missing passport. It was a blessing in _____.

(A) disguise (B) cover

(C) hiding (D) concealment

67. She applies the rules during the match rather than supporting one side or merely watching. What is she?

(A) spectator (B) coach

(C) referee (D) supporter

68. Read the sentence: The revised edition will retain the useful diagrams but replace the outdated tables. Choose the closest meaning of 'retain' as used in the sentence.

(A) copy (B) protect from damage

(C) restore (D) keep

69. Use the simple present to express the stated regular procedure. Choose the best replacement: The display that shows the two schedules [update] automatically at midnight each day.

(A) are updating (B) updates

(C) have updated (D) were updated

70. Complete the question tag: I am responsible for returning the key, _____?

(A) aren't I (B) isn't I

(C) don't I (D) wasn't I

Logical Reasoning

71. The age table uses the same two people throughout:

	younger	older
now	x	$x+8$
2 years ago	$x-2$	$x+6$

. The product of their ages 2 years ago was 308. What will the younger person's age be 4 years from now?

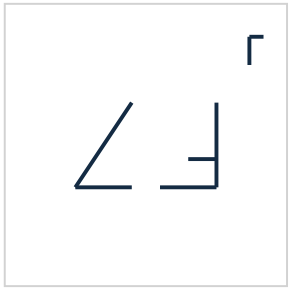
(A) 16 (B) 28

(C) 20 (D) 18

72. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



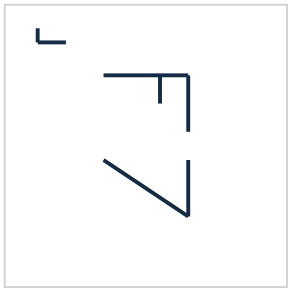
(A) Pattern W



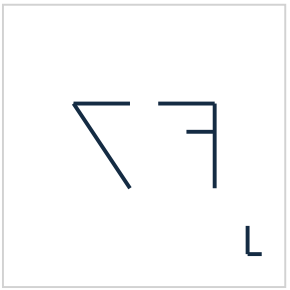
(B) Pattern P



(C) Pattern Y

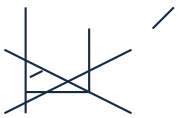


(D) Pattern L



73. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





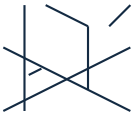
host 1



host 2



host 3



host 4

(A) 2

(B) 1

(C) 3

(D) 4

74. A code shifts letters cyclically through A–Z. Odd positions share one fixed shift and even positions share another. For example, GRAPE is coded as LQFOJ. Which word is coded as HKTTI?

(A) CLOUD (B) MJYSN

(C) BKNTC (D) DMPVE

75. Tara points to a photograph and says, "This person's only daughter is my mother." Tara is female, and the photographed person is male. How is the photographed person related to Tara?

(A) Brother (B) Son

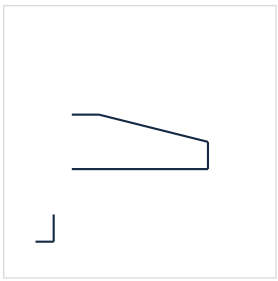
(C) Grandfather (D) Daughter

76. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 3, 12, 16, 64, ?, 272, 276.

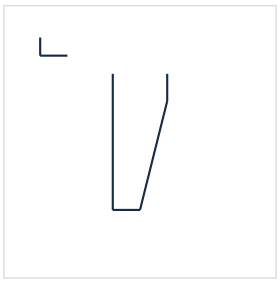
(A) 68 (B) 112

(C) 272 (D) 60

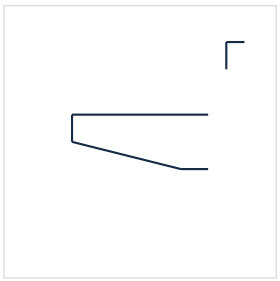
77. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

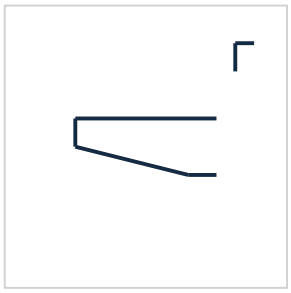


2



3

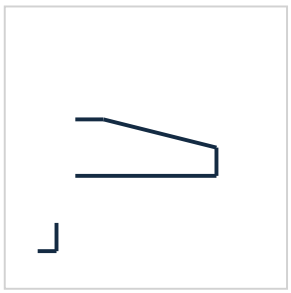
(A) Pattern P



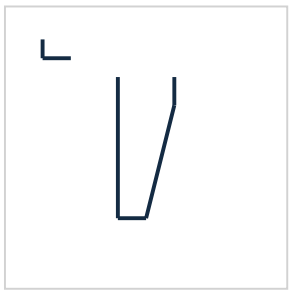
(B) Pattern Y



(C) Pattern Z



(D) Pattern U



78. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

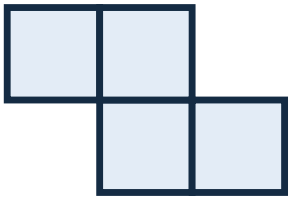


piece 1



piece 2

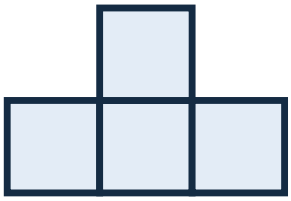
(A) Pattern W



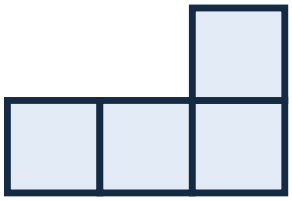
(B) Pattern T



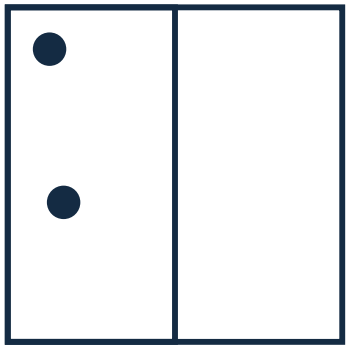
(C) Pattern Q



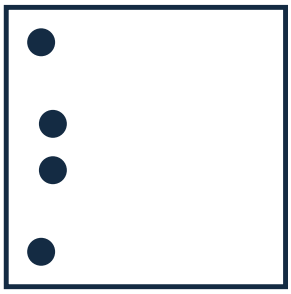
(D) Pattern Z



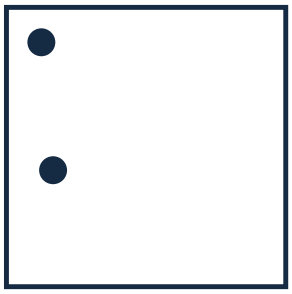
79. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



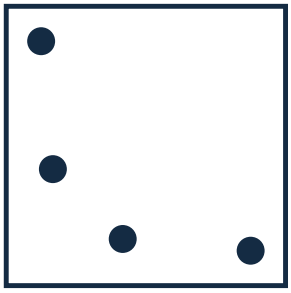
(A) Pattern Y



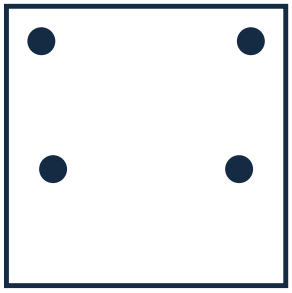
(B) Pattern P



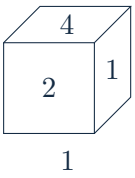
(C) Pattern T

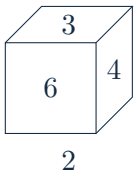


(D) Pattern V



80. The views show the same die. Give the ordered pair: (face opposite 4, face opposite 1). Do not assume opposite labels sum to seven.





(A) $(5, 2)$

(B) $(5, 6)$

(C) $(5, 1)$

(D) $(5, 3)$

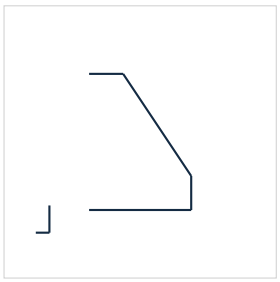
81. Consider only these statements; universal statements alone do not imply existence. Some farmers are joggers; All joggers are artists. Conclusions: I. Some farmers are not joggers. II. Some artists are joggers. Which conclusions follow necessarily?

(A) Neither I nor II
(B) Only II
(C) Only I
(D) Both I and II

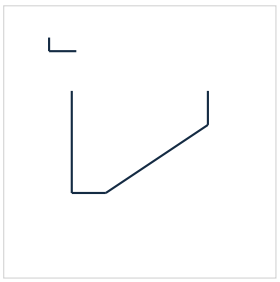
82. Three pairs contain exactly one mammal and one bird. Which pair does not?

(A) Whale / Duck
(B) Dog / Cat
(C) Elephant / Eagle
(D) Rabbit / Ostrich

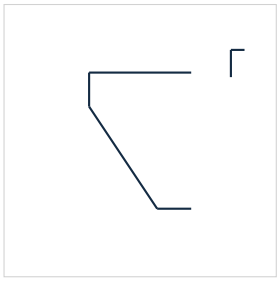
83. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

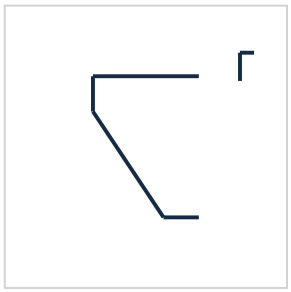


bottom left

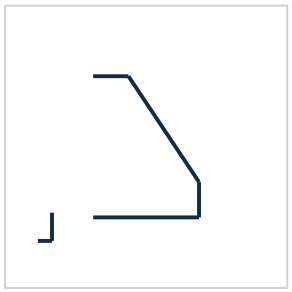
?

bottom right

(A) Pattern Y



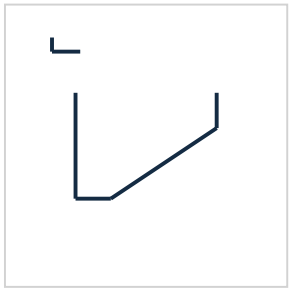
(B) Pattern P



(C) Pattern Z



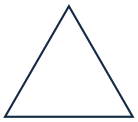
(D) Pattern V



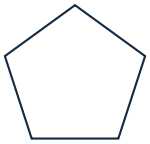
84. A walker goes 8 m east, then 11 m north, then turns left and walks 4 m. In which direction is the walker from the starting point?

(A) South-east
(B) North-west
(C) North-east
(D) South-west

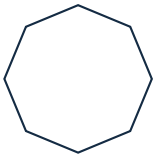
85. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

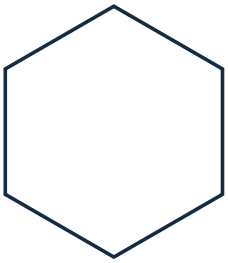


B



C

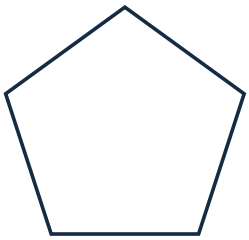
(A) Pattern Y



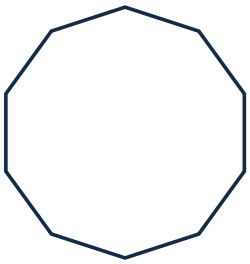
(B) Pattern P



(C) Pattern V



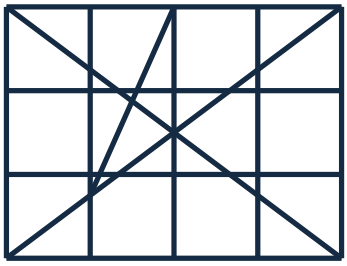
(D) Pattern L



86. Consider only these statements; universal statements alone do not imply existence. No artists are hikers; Some artists are not hikers; Some hikers are guides. Conclusions: I. No guides are artists. II. Some hikers are not guides. Which conclusions are individually possible without contradicting the statements?

(A) Both I and II
(B) Neither I nor II
(C) Only I
(D) Only II

87. How many triangles are formed by all the drawn segments?



(A) 57 (B) 41

(C) 100 (D) 59

88. In a row, Asha is 12th from the left and 33rd from the right. Bina is 7th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 7 (B) 23

(C) 21 (D) 22

89. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: MVPE, OSQC, ?, SMSY, UJTW, WGUU.

(A) QPRA (B) POQZ

(C) ARPQ (D) RQSB

90. Preserve the exact relation and its order: Begin : Commence :: ? : Old. Choose the missing term.

(A) Begin (B) Ancient

(C) Silent (D) Rapid

Mathematics

91. For x>0, xy′-y=x and y(1)=4. Find y(3)/3-y(1).

(A) $\log(3)+4$ (B) $\log(3)$

(C) $-\log(3)$ (D) $\log(27)+8$

92. Find the population variance of 3+3, 6+3, ..., 21+3.

(A) 4 (B) 39

(C) 36 (D) 12

93. For f(x)=|x-6|/(x-10), on which maximal open intervals is f strictly decreasing?

(A) $(-\infty, 10)$ (B) $(6, 10) \cup (10, \infty)$

(C) $(-\infty, 6)$ (D) $(6, \infty)$

94. Evaluate $\int$ from 0 to $\pi/2$ of $[\sin^5 x/(\sin^5 x+\cos^5 x)+\sin^5(\pi/2-x)/(\sin^5(\pi/2-x)+\cos^5(\pi/2-x))]x\,dx$.

(A) $\frac{\pi}{4}$ (B) $\frac{\pi^2}{4}$

(C) $\frac{\pi^2}{16}$ (D) $\frac{\pi^2}{8}$

95. A circular sector has radius 8 and arc length $\frac{8\pi}{3}$. Find its area.

(A) $\frac{32\pi}{3}$ (B) $\frac{\pi}{6}$

(C) $\frac{32}{3}$ (D) 64π

96. An AP has a=10,d=5; its first n terms sum to 175. Find a_(n+1).

(A) 40 (B) 45

(C) 7 (D) 25

97. f(x)=x^3+3x, g=f^{-1}. Find g″(140).

(A) $\frac{1}{30}$ (B) $-\frac{5}{79092}$

(C) $\frac{5}{79092}$ (D) $-\frac{5}{1014}$

98. Two tangents from $P = (\frac{9}{2}, 18)$ touch the parabola $y = x^2$. Each tangent and the coordinate axes bound a triangle. Find the sum of the areas of these two triangles.

(A) $\frac{243}{4}$ (B) $\frac{243}{2}$

(C) $\frac{729}{4}$ (D) $\frac{45}{2}$

99. Circles $x^2+y^2=289$ and $(x-18)^2+y^2=289$ meet at A,B. Find $|AB|^2$.

(A) $8\sqrt{13}$ (B) 1480

(C) 832 (D) 508

100. A convergent GP has first term 7, ratio $\frac{1}{6}$. Find the sum of all terms after its first 5 terms.

(A) $\frac{7}{38880}$ (B) $\frac{7}{7776}$

(C) $\frac{7}{6480}$ (D) $\frac{1}{6480}$

101. How many integers k make 24sin x+32cos x+2=k have exactly two distinct solutions in $0 \le x < 2\pi$?

(A) 79 (B) 39

(C) 81 (D) 80

102. $A = \begin{bmatrix} 0 & 4 \\ 0 & 0 \end{bmatrix}$. Find the (1,2) entry of $(I+A)^{10}$.

(A) 10 (B) 40

(C) 4 (D) 1048576

103. B needs 5 days longer than A; together they need $\frac{66}{17}$ days. A works alone for 3 days, then B joins. How many additional days finish the job?

(A) $\frac{66}{17}$ (B) 3

(C) $\frac{15}{17}$ (D) $\frac{33}{17}$

104. The chord of contact from P to $x^2/49-y^2/4=1$ is x=7/2. Find the product of the slopes of the two tangents from P.

(A) 0 (B) $\frac{4}{147}$

(C) $-\frac{4}{147}$ (D) $-\frac{4}{49}$

105. A rectangle has perimeter 44. Its length exceeds its width by at least 5. Find its maximum area.

(A) $\frac{459}{4}$ (B) 85

(C) $\frac{1911}{4}$ (D) $\frac{3\sqrt{51}}{2}$

106. y′-y/x=1 for x>0, y(1)=8. Find y(7)/7-y(1).

(A) $-\log(7)$ (B) $\log(7)+8$

(C) $7\log(7)+56$ (D) $\log(7)$

107. How many integer k in [0,20] make $x^2-7x+(10-k)=0$ have roots of opposite signs?

(A) 10 (B) 20

(C) 9 (D) 11

108. How many x with $0\leq x<7\pi$ satisfy $\sin^4x+\cos^4x=3/4$ and cos x>0?

(A) 14 (B) 16

(C) 0 (D) 28

109. X takes values 0,7,10 with probabilities 1/2,1/3,k. Find Var(X).

(A) 16 (B) $\sqrt{17}$

(C) 33 (D) 17

110. $A=\begin{bmatrix}2 & 1\\ 1 & 4\end{bmatrix}$, $B=\begin{bmatrix}8 & 4\\ 11 & 9\end{bmatrix}$, and XA=B. Find X_{11}.

(A) 8 (B) -4

(C) 4 (D) 3

111. There are 14 Boolean variables $p_1,\ldots,p_{14}$. Exactly 3 variables must be true. How many truth assignments satisfy [(p_1 implies p_2) and (p_1 implies not p_2)]?

(A) 78 (B) 66

(C) 364 (D) 286

112. Write $\sin x\cos^3x+6\sin^3x\cos x=P\sin 2x+Q\sin 4x$. Find P+2 Q.

(A) $\frac{1}{2}$ (B) $\frac{23}{8}$

(C) $\frac{9}{8}$ (D) $\frac{7}{4}$

113. $x=7+13(1-t^2)/(1+t^2)$, $y=6+26t/(1+t^2)$, where t is real. Which point of the containing circle is not attained?

(A) $(-6,6)$ (B) $(7,-7)$

(C) $(7,19)$ (D) $(20,6)$

114. $F(x)=\int_x^{(x^2)}(t+5)^2dt$. Find F′(5).

(A) 8000 (B) 9000

(C) 8900 (D) 9100

115. A vector has direction ratios (11,5,6). Find the sum of the squares of its y,z direction cosines.

(A) $\frac{121}{182}$ (B) 61

(C) $\frac{61}{182}$ (D) $\frac{\sqrt{11102}}{182}$

116. $A = \begin{bmatrix} 10 & 1 \\ 5 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix}$. Find tr(BA) without assuming AB=BA.

(A) 48 (B) 24

(C) 42 (D) -18

117. Evaluate lim as x→0 of $[(1+5x)/(1-5x)]^{(1/x)}$.

(A) 1 (B) e^5

(C) 10 (D) e^{10}

118. Let $D(t) = \det \begin{pmatrix} 5+t & 1+t \\ 1+t & 3+t \end{pmatrix}$. Find D′(0).

(A) -6 (B) 28

(C) 8 (D) 6

119. In triangle ABC, A=2 B and a/b=sqrt 3. If b=5, find the area.

(A) $-1 + \frac{25\sqrt{3}}{4}$ (B) $\frac{25\sqrt{3}}{2}$

(C) $5 + \frac{25\sqrt{3}}{2}$ (D) $1 + \frac{25\sqrt{3}}{4}$

120. For the loop $y^2=x^2(7-x)$, find its area within the strip $7/3 \le x \le 14/3$.

(A) $\frac{49\sqrt{42}}{27} + \frac{98\sqrt{21}}{27}$ (B) $\frac{392\sqrt{42}}{45}$

(C) $\frac{392\sqrt{7}}{15}$ (D) $-\frac{784\sqrt{21}}{135} + \frac{392\sqrt{42}}{45}$

121. The tangent to $y = x^3 - 77x$ at $x = 9$ meets the two coordinate axes at A and B. Find the area of triangle OAB, where O is the origin.

(A) $\frac{531441}{166}$ (B) $\frac{531441}{83}$

(C) 4374 (D) $\frac{1062882}{83}$

122. For $(d^3y/dx^3)^2/(1+x^2)+y=0$, find order+degree if the degree is defined; otherwise select "Degree is not defined".

(A) 6 (B) 3

(C) Degree is not defined (D) 5

123. Find the area enclosed by y=x(9-x) and y=7x.

(A) $\frac{4}{3}$ (B) $-\frac{4}{3}$

(C) $\frac{243}{2}$ (D) $\frac{193}{3}$

124. Two different numbers are chosen uniformly from {1,...,6}. Let X be their maximum. Find Var(X).

(A) $\frac{\sqrt{14}}{3}$ (B) $\frac{14}{3}$

(C) $\frac{14}{9}$ (D) $\frac{70}{3}$

125. In $x^2+y^2=64$, a chord through A=(8,0) has midpoint M with M_x=8/4 and M_y>0. Find its squared length.

(A) 48 (B) 192

(C) 256 (D) 64

126. The parabola $(y-2)^2=20(x-2)$ has focus F and vertex V. A circle centred at F passes through V. Find the tangent length from P=(17,12).

(A) $10\sqrt{2}$ (B) $5\sqrt{3}$

(C) $5\sqrt{7}$ (D) 175

127. How many pairs $1\le m<n\le 8$ satisfy both $5 \mid (6^m+9^n)$ and $3 \mid (m+n)$?

(A) 5 (B) 4

(C) 8 (D) 12

128. $P(A)=\frac{2}{5}$, $P(B)=\frac{3}{10}$, $P(A\cap B)=\frac{3}{20}$. Find P(neither A nor B).

(A) $\frac{9}{20}$ (B) $\frac{2}{5}$

(C) $\frac{3}{20}$ (D) $\frac{3}{10}$

129. For real t, let $A(t)=\begin{pmatrix}4+t & 1+t\\ 1+t & 6+t\end{pmatrix}$. For what t is $A(t)$ singular?

(A) $-\frac{8}{23}$ (B) $-\frac{23}{8}$

(C) $-\frac{23}{10}$ (D) $\frac{23}{8}$

130. X is binomial with n=4, p=1/3. Given $X\ge 1$, find $P(X\ge 2)$.

(A) $\frac{65}{81}$ (B) $\frac{11}{27}$

(C) $\frac{32}{65}$ (D) $\frac{33}{65}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A 2 kg mass on a string passes the top of a vertical circle of radius 1 m with string tension 100 N. Find string tension at the bottom. Take $g = 10\ \text{m/s}^2$.

(A) 220 N (B) 180 N

(C) 200 N (D) 20 N

132. A small planar detector initially faces an isotropic point source normally at distance r. It is moved to a concentric sphere whose enclosed volume is 64 times as large, and tilted so that the cosine of the angle between its normal and the incident ray is $\frac{3}{5}$. Its area is unchanged. What is the ratio of intercepted powers, final to initial?

(A) $\frac{48}{5}$ (B) $\frac{3}{20}$

(C) $\frac{3}{80}$ (D) $\frac{1}{16}$

133. A converging thin lens has focal length 20 cm in air. Object and real-image distances are positive magnitudes. Which proposed reading pair is inconsistent?

(A) $u = 40$ cm, $v = 40$ cm. (B) $u = 40$ cm, $v = 60$ cm.

(C) $u = 60$ cm, $v = 30$ cm. (D) $u = 80$ cm, $v = \frac{80}{3}$ cm.

Chemistry

134. $E^\circ(M^{4+}/M) = \frac{1}{2}$V for 4 electrons, and $E^\circ(M^{4+}/M^{2+}) = \frac{7}{10}$V for 2 electrons. A Z^{2+}/Z couple is -0.800 V. Standard cell potential for $Z + M^{2+} \rightarrow Z^{2+} + M$ is

(A) $\frac{3}{2}$ (B) $\frac{7}{5}$

(C) $\frac{11}{10}$ (D) $\frac{13}{10}$

135. For the pair ethanal and ethenol, which classification identifies the structural difference rather than merely stating that the formula agrees?

(A) geometrical isomers (B) tautomers

(C) metamers (D) conformers

136. 2-methylbut-1-ene reacts with HBr with peroxide. Which pair gives the major constitutional product and the intermediate whose greater stability fixes its orientation? Ignore stereoisomer counting.

(A) 1-bromo-2-methylbutane; tertiary radical (B) 2-bromo-2-methylbutane; primary radical

(C) 1-bromo-2-methylbutane; primary radical (D) 2-bromo-2-methylbutane; tertiary radical

Logical Reasoning

137. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). The numerical component follows a constant arithmetic step. Find the missing internal term: HHQ2, JJS5, ?, NNW11, PPY14, RRA17.

(A) LLU8

(B) MMV8

(C) 8ULL

(D) KKT8

138. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target



piece 1

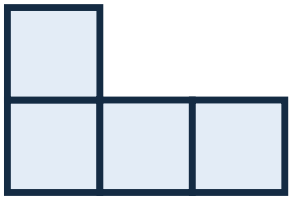


piece 2

(A) Pattern V



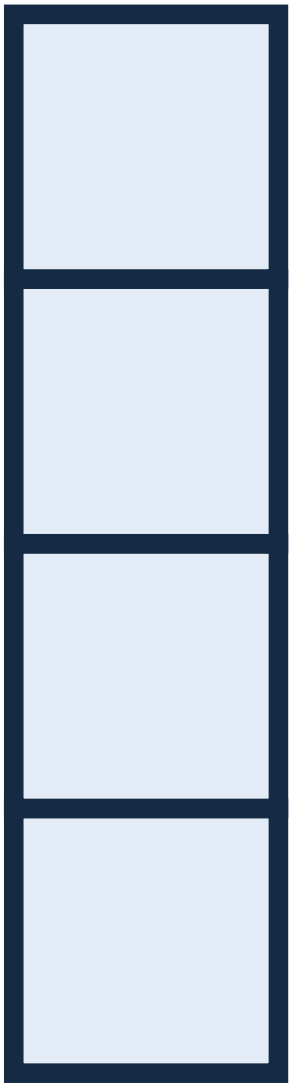
(B) Pattern T



(C) Pattern X

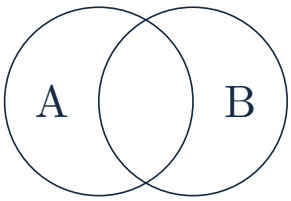


(D) Pattern K

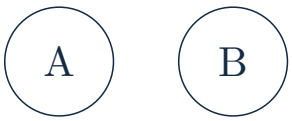


139. Choose the diagram that represents all these conditions, including the strictness information: All B are A; some A is not B, and B is nonempty.

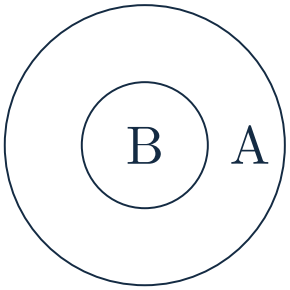
(A) Pattern N



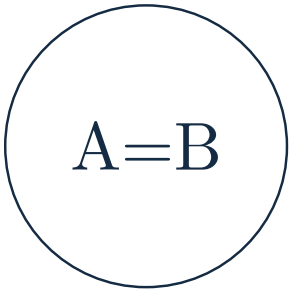
(B) Pattern V



(C) Pattern Y



(D) Pattern K



Mathematics

140. |U|=100, |A|=28, |B|=22, |A∩B|=10. Find the number of elements belonging to both A,B or to neither.

(A) 30

(B) 70

(C) 90

(D) 60

141. $F'(x)=\frac{4}{(x+1)(x+5)}$ and F(0)=2. Find F(1)-2F(0).

(A) $-2-\log(5)+\log(3)$

(B) $\log\left(\frac{5}{3}\right)$

(C) $-2+\log\left(\frac{5}{3}\right)$

(D) $\log\left(\frac{5}{3}\right)+2$

142. Two points on opposite sides of a tower foot are 12 metres apart on a straight level line. The elevations of the tower top are 30° and 45°. Find the tower height.

(A) $-6+6\sqrt{3}$

(B) $18-6\sqrt{3}$

(C) 6

(D) $6+6\sqrt{3}$

BITSAT Mock 46

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A 4 kg block moves along a horizontal rough surface from $x = 0$ to $x = 2$ m under a force $F(x) = 3x$ N parallel to motion. With $\mu_k = 1/5$ and $g = 10$ m/s^2, find the work done by the net force. The block has enough initial speed to keep moving forward.

(A) -10 J (B) 22 J

(C) 6 J (D) 10 J

2. A spring of constant 80 N/m is already stretched by $\frac{1}{5}$ m. How much additional work is required to increase its extension to $\frac{3}{10}$ m quasistatically?

(A) 2 J (B) $\frac{18}{5}$ J

(C) $\frac{12}{5}$ J (D) $\frac{2}{5}$ J

3. A plane wave passes through a thin prism away from edges. Separately, a pointlike aperture produces an outgoing spherical wavefront of radius 10 cm, which propagates another 5 cm in a homogeneous medium. Which pair correctly describes the prism wavefront and the later aperture wavefront?

(A) Approximately plane and unchanged in direction; spherical with radius 10 cm.

(B) Spherical; plane with infinite radius.

(C) Approximately plane and tilted; spherical with radius 15 cm.

(D) Approximately plane and tilted; spherical with radius 5 cm.

4. A platform of moment of inertia 2 kg m^2 carries a person of mass 2 kg at radius 2 m and rotates at 3 rad/s. The person walks to half the stated initial radius; external torque is zero. Find final angular speed. Treat the person as a point mass.

(A) $\frac{6}{5}$ rad/s (B) $\frac{3\sqrt{10}}{2}$ rad/s

(C) 12 rad/s (D) $\frac{15}{2}$ rad/s

5. A stretched string is divided by a bridge into lengths in the ratio 1 : 6. Tension and linear density are unchanged. The original fundamental is 50 Hz. Find the fundamental of the longer segment.

(A) $\frac{175}{3}$ Hz (B) 350 Hz

(C) 70 Hz (D) $\frac{300}{7}$ Hz

6. A uniformly charged thin rod of length 5 m and total charge 3 C lies on the x-axis. Point P is on its extension, 3 m from the nearer end. Find field magnitude at P in units of $k = 1/(4\pi\varepsilon_0)$.

(A) $\frac{1}{5}$ (B) $\frac{12}{121}$

(C) $\frac{3}{64}$ (D) $\frac{1}{8}$

7. A particle of mass 5 kg starts from rest. A force $\vec{F} = 4t\,\hat{i} + 8\,\hat{j}$ N acts for 3 s. What is the magnitude of its momentum at that instant?

(A) 42 kg m/s (B) 30 kg m/s

(C) $12\sqrt{13}$ kg m/s (D) 18 kg m/s

8. A bar magnet of moment 1 A m^2 is rotated quasistatically from 60° to 120° relative to a uniform 6 T magnetic field. Find external work.

(A) 2π J (B) $6\sqrt{3}$ J

(C) $3\sqrt{3}$ J (D) 6 J

9. A hydrogen atom emits a photon in the $5 \to 4$ transition. A hydrogen-like ion of nuclear charge $Z = 2$ emits a photon of exactly the same wavelength and ends in level $n_f = 8$. What is its initial principal quantum number?

(A) $\frac{5}{2}$ (B) 10

(C) 7 (D) 5

10. A stationary nucleus with mass number 20 and binding energy per nucleon 8 MeV splits into two equal daughters, each having total binding energy 84.0 MeV. Which statement is correct?

(A) No energy can be released because fission always requires a larger mass number than 20.

(B) The process absorbs 8 MeV; the daughters have larger binding energy per nucleon.

(C) The process releases 8 MeV; the daughters have larger binding energy per nucleon.

(D) The process releases 16 MeV; each daughter releases 8 MeV.

11. 5 kg of water at 40°C is mixed in an insulated vessel with $\frac{840}{167}$ kg of ice at 0°C. Use $c_w = 4200$ J/(kg K), $L_f = 334000$ J/kg. Find the mass of ice remaining at equilibrium.

(A) $\frac{420}{167}$ kg (B) $\frac{840}{167}$ kg

(C) $\frac{140700}{41917}$ kg (D) $\frac{630}{167}$ kg

12. A Carnot engine initially runs between 900 K and 200 K. Its hot reservoir is raised by 20 K. By how much must the cold-reservoir temperature rise to keep efficiency unchanged?

(A) $\frac{100}{23}$ K (B) $\frac{40}{9}$ K

(C) 90 K (D) 20 K

13. A particle executes SHM at frequency 2 Hz. At t=0 it is at its positive extreme. In the interval $0 <= t < 1$ second, how many times does its kinetic energy equal its potential energy?

(A) 8 (B) 2

(C) 7 (D) 16

14. The 2nd harmonic of an open pipe of length 2 m equals the 2nd allowed resonance of a closed pipe. Both have the same sound speed and negligible end correction. Find the closed-pipe length.

(A) 2 m (B) $\frac{3}{2}$ m

(C) $\frac{1}{2}$ m (D) 1 m

15. A ray enters one face of a prism normally, and reaches the second face at the critical angle. The prism index in air is 2. Find the prism angle in degrees.

(A) 30 degree

(B) $\frac{180\,\mathrm{atan}\,(2)}{\pi}$ degree

(C) $\frac{90}{\pi}$ degree

(D) $\frac{180\,\mathrm{atan}\left(\frac{1}{2}\right)}{\pi}$ degree

16. A particle has position-dependent velocity components $\vec{v} = 1x\,\hat{i} - 1y\,\hat{j}$ in SI units. Given $dx/dt = v_x$ and $dy/dt = v_y$, what is its acceleration magnitude at $(2, 4)$ m?

(A) $2\sqrt{5}\,\mathrm{m/s^2}$ (B) $6\,\mathrm{m/s^2}$

(C) $4\,\mathrm{m/s^2}$ (D) $2\,\mathrm{m/s^2}$

17. An astronomical telescope in normal adjustment has magnifying-power magnitude 8 and tube length 18 cm. Find the eyepiece focal length.

(A) $\frac{18}{65}$ cm (B) $\frac{18}{7}$ cm

(C) 2 cm (D) $\frac{9}{4}$ cm

18. 2 moles and 1 moles of the same ideal gas, initially at 200 K and 900 K, are connected in a rigid insulated system. There is no external work. Find equilibrium temperature, assuming constant molar heat capacity.

(A) 900 K (B) 200 K

(C) 550 K (D) $\frac{1300}{3}$ K

19. A gas receives 700 J of heat while its surroundings do 100 J of work on it. It then loses 350 J of heat at fixed volume. Find its total internal-energy change.

(A) 350 J (B) 1150 J

(C) 450 J (D) 800 J

20. A charge 3 C moves with $\vec{v} = 1\hat{i} + 3\hat{j}$ m/s in $\vec{B} = 4\hat{i} + 4\hat{j}$ T. What is the signed z-component of its magnetic force?

(A) 12 N (B) 48 N

(C) 0 N (D) -24 N

21. For a particle moving in one dimension, $v = 6 - 4x$ in SI units while $v > 0$. What is its acceleration at $x = \frac{3}{4}$ m?

(A) $3\,\mathrm{m/s^2}$ (B) $-24\,\mathrm{m/s^2}$

(C) $-12\,\mathrm{m/s^2}$ (D) $0\,\mathrm{m/s^2}$

22. A charge of $2\,\mu$C moves from potential 70 V to -70 V in an electrostatic field. Find the work done by an external agent if the charge starts and ends at rest and no other force does work.

(A) $-4\,\mu$J (B) $-280\,\mu$J

(C) $0\,\mu$J (D) $140\,\mu$J

23. A metal wire has linear temperature coefficient $\frac{1}{200}$ per degree Celsius referred to 0°C. Its dimensions are fixed and it stays across a constant voltage. Find power at 40°C divided by power at 0°C.

(A) $\frac{5}{6}$ (B) $\frac{5}{4}$

(C) $\frac{4}{5}$ (D) $\frac{6}{5}$

24. For steady laminar flow through a capillary, the radius is multiplied by 4 and length by 3. The volume flow rate must remain unchanged and viscosity is fixed. By what factor must the pressure difference be multiplied?

(A) $\frac{1}{256}$ (B) $\frac{3}{16}$

(C) 1 (D) $\frac{3}{256}$

25. A square conducting loop in a constant perpendicular field 5 T has side length $\ell(t) = 4t$ m. Find induced-emf magnitude at $t = 3$ s, assuming the loop expands uniformly.

(A) 720 V (B) 60 V

(C) 96 V (D) 480 V

26. A system has total mass 6 kg, centre-of-mass speed 3 m/s and kinetic energy 20 J measured in the CM frame. Which statement is correct?

(A) The laboratory kinetic energy is 20 J. (B) The laboratory kinetic energy is 47 J.

(C) The laboratory kinetic energy is 27 J. (D) The CM-frame kinetic energy must be zero.

27. The same metal is illuminated successively at wavelengths λ and $\lambda/2$. Its maximum photoelectron kinetic energies are 6 eV and 24 eV, respectively. What is the ratio of threshold wavelength to λ?

(A) $\frac{2}{3}$ (B) $\frac{3}{2}$

(C) 7 (D) $\frac{1}{7}$

28. Two identical uniform spheres touch. Their radii are multiplied by 7 while their density is divided by 7. By what factor does their gravitational attraction change?

(A) $\frac{1}{49}$ (B) 343

(C) 49 (D) 117649

29. A balanced Wheatstone bridge has arms AB=AD=4 ohms and BC=DC=8 ohms. A resistor of arbitrary positive resistance connects B to D. Find equivalent resistance between A and C.

(A) $24\,\Omega$ (B) $6\,\Omega$

(C) $10\,\Omega$ (D) $\frac{16}{3}\,\Omega$

30. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 2 kΩ to a terminal at +8 V, and its cathode connects to a terminal at -4 V. Find the current from the +8 V terminal in mA.



(A) 6 mA
(B) $\frac{53}{10}$ mA
(C) $\frac{113}{20}$ mA
(D) $\frac{73}{20}$ mA

Chemistry

31. $\frac{112}{25}$ g pure NH_4NO_2 decomposes by $NH_4NO_2 \rightarrow N_2 + 2H_2O$. Only80.0% of the theoretical dry N_2 is recovered. Find its STP volume in litres, using22.4L/mol and NH_4NO_2=64g/mol.

(A) $\frac{196}{125}$
(B) $\frac{49}{25}$
(C) $\frac{1568}{625}$
(D) $\frac{784}{625}$

32. For ideal gases at one T, A(g)⇌2B(g) has Kc=25 and numerical RT=5 in compatible units. What is Kp for4B(g)⇌2A(g)? Use Kp=Kc$(RT)^{\Delta n}$ with the same standard-state convention throughout.

(A) 1/15625
(B) 1/31250
(C) 2/15625
(D) -1/15625

33. Two compounds both relieve pain in a textbook example. Which further claim cannot be inferred solely from that shared effect?

(A) They necessarily have identical functional groups and molecular formula.
(B) The word analgesic can be applied to the stated effect.
(C) They share the stated pain-relieving classification.
(D) The example describes a biological effect.

34. For I: water uptake on silica-gel surfaces; II: uniform distribution of ethanol throughout water, which classification correctly identifies surface accumulation and distinguishes it from a bulk process?

(A) Neither involves adsorption.
(B) Both are adsorption because both involve molecular interactions.
(C) Only II involves adsorption.
(D) I involves adsorption; II is not adsorption.

35. Count all distinct stereoisomers,including separately resolved enantiomers,for octahedral $[Ma_4b_2]$ and square-planar [Mabcd]; a,b,c,d are distinct monodentate ligands and en is symmetric bidentate ethylenediamine. What is the sum?

(A) 5
(B) 7
(C) 4
(D) 6

36. For 1-bromo-2-methylpropane,which structural classification is correct when the actual halogen-bearing carbon and molecular formula are examined?

(A) a secondary halide
(B) an aryl halide
(C) a primary halide despite branching at the adjacent carbon
(D) a tertiary halide

37. Both metals in $[Co(NH_3)_6][Fe(CN)_6]$ are formally +3. Which proposed partner preserves the total composition while exchanging ligand sets between two coordination spheres?

(A) The same salt with one ligand merely renamed; no atoms or donor attachments change.

(B) $[Co(NH_3)_5Cl]Cl_2$ formed by adding chloride to the sample.

(C) $[Co(NH_3)_6][Fe(CN)_6]$ with only the counter-ion positions in the written formula reversed.

(D) $[Fe(NH_3)_6][Co(CN)_6]$

38. A two-step reaction has energies(kJ/mol): reactants0, first transition state40, intermediate15, second transition state55, products-20. Which pair gives(second forward activation barrier, reverse barrier from products to the second transition state)?

(A) (40, 75) kJ/mol.

(B) (55, 55) kJ/mol.

(C) (25, 75) kJ/mol.

(D) (40, 35) kJ/mol.

39. At 1 atm compare two separate pairs: (water, hydrogen sulfide) and (1-propanol, methoxyethane). A student selects hydrogen sulfide for the first higher-boiling substance because its formula mass is unchanged, and 1-propanol for the second. Which assessment is consistent with intermolecular interactions?

(A) Both selections wrong.

(B) First selection wrong; second selection correct.

(C) Both selections correct.

(D) First selection correct; second selection wrong.

40. A weak monoprotic acid HA is the only electrolyte. Conductance is $\frac{1}{50000}$S in a cell of constant2.00cm^{-1};an independent measurement gives$[H^+]=\frac{1}{10000}$M. Assume $H^+=A^-$ and negligible water ionisation. Limiting molar conductivity Λ_0(HA),in S cm^2/mol,is

(A) 800

(B) 0.4

(C) 200.0

(D) 400

41. A pure anhydrous Na2CO3 standard contains 3/500 mol carbonate. In its HCl titration, the additional acid volume from the carbonate-to-hydrogencarbonate endpoint to the final CO2/water endpoint is 40 mL. What is the HCl molarity? Assume complete stoichiometric endpoints.

(A) 3/10

(B) 3/40

(C) -3/20

(D) 3/20

42. A mixture contains one mole each of resorcinol,ethanol and benzoic acid. Treat with excess Na metal. What is the total amount,in mol,of H2? Assume all stated acid-base reactions go to completion and no side reactions.

(A) 1

(B) 1/2

(C) 3/2

(D) 2

43. An100mL sample contains $\frac{1}{25}$ mol HA,pKa=4,and $\frac{1}{50}$ mol HCl. Add100mL $\frac{3}{5}$ M NaOH. Use usual weak-base salt hydrolysis and $\log_{10}(\frac{1}{5})=-\frac{699}{1000}$. Final pH is

(A) 4

(B) $\frac{17301}{2000}$

(C) $\frac{13301}{1000}$

(D) 7

44. A laboratory note states: Every d-block ion is coloured by a d-d transition. Which judgement and reason are consistent?

(A) Correct: d0 and d10 ions have no ordinary intra-d excitation; charge-transfer colours are possible.

(B) Correct because every transition-metal species has the same d-electron count.

(C) Incorrect: d0 and d10 ions have no ordinary intra-d excitation; charge-transfer colours are possible.

(D) Incorrect because transition metals cannot exchange electrons or adsorb reactants.

45. Use octet Lewis contributors for $CO3^{2}$- and HCO3-; for SO2 exclude expanded-octet contributors. For HCOO-, HCO3- and HNO3, average only the two nonprotonated central-atom-O bonds. For other species average all central-atom bonds. Find (average bond order of the first minus that of the second, sum of total lone-pair counts in one contributor of each). A lone pair on any atom is counted once.

(A) (-1/6, 15)

(B) (-1/6, 16)

(C) (-1/6, 14)

(D) (17/6, 15)

46. Asparagine has-CH2CONH2 whereas aspartic acid has-CH2COOH. Which conversion changes the side-chain class without changing its carbon count?

(A) Replace side-chainCONH2 by COOH through amide hydrolysis.

(B) Hydrogenate a carbon-carbon triple bond.

(C) Replace the aromatic ring by Cl.

(D) Remove all carbon atoms to make glycine.

47. An ideal gas freely expands into vacuum in an insulated rigid enclosure,accessible gas volume increasing 5-fold. Which joint statement follows?

(A) q=0,w=0 and temperature is unchanged;entropy increases,so the process is irreversible.

(B) The rigid enclosure forces the gas-accessible volume to remain unchanged.

(C) q=0 and w=0 imply entropy change is zero and the expansion is reversible.

(D) The gas does work on the vacuum and its temperature necessarily falls.

48. A solution of one nonvolatile nonelectrolyte has exact relative vapour-pressure lowering 3/73. After removal of solvent only, the lowering becomes 9/79. The initial solvent amount was 14 mol, and Raoult's law holds without association. How many moles of solvent were removed?

(A) 28/3

(B) 28/9

(C) 14/3

(D) 149/15

49. One mole of an acyclic saturated hydrocarbon consumes $\frac{19}{2}$ mol O2 on complete combustion. How many hydrogen atoms are present in one molecule,and what initial relation must be used?

(A) 12; (3n+1)/2 = oxygen moles

(B) 16; 2n+2 = oxygen moles

(C) 14; (3n+1)/2 = oxygen moles

(D) 10; n+1 = oxygen moles

50. Within an additive covalent-radius model,a single X-X bond is 140 pm and an X-H bond is 105 pm. A linear H-X-X-H skeleton is assumed with collinear single bonds. Its terminal H-to-H distance,in pm,is

(A) 700

(B) 420

(C) 350

(D) 490

51. Beam A has wavelength$2\lambda_0$ and beam B has$3\lambda_0$. A carries 3 times B's total energy. Photon-number ratioN_A/N_B is

(A) $\frac{9}{2}$ (B) 2

(C) 3 (D) $\frac{2}{3}$

52. $\frac{3}{50}$ mol Cl_2 is contacted with $\frac{9}{100}$ mol KOH in cold dilute solution. Only $Cl_2 + 2KOH \rightarrow KCl + KClO + H_2O$ occurs. Final solution volume is $\frac{2}{5}$ L. Find the KClOconcentration in mol/L, assuming no side reaction or product loss.

(A) $\frac{9}{40}$ (B) $\frac{9}{80}$

(C) $\frac{3}{20}$ (D) $\frac{3}{80}$

53. An iron-ion inventory of 3/100 mol contains only Fe2+ and Fe3+. Complete externally driven reduction to metal, with no side reaction, consumes 9/125 mol electrons. What initial fraction of iron was Fe3+?

(A) 1/5 (B) 2/15

(C) 2/5 (D) 4/5

54. ethyl3-chlorobenzoate is completely hydrolysed,acidified,then treated successively with SOCl2,excess NH3,and Br2/NaOH. Which amine is obtained,assuming ring substituents survive?

(A) 3-chlorobenzylamine (B) 3-chloroaniline

(C) 4-chloroaniline (D) chlorobenzene

55. For hepta-1,3,5-triene, CH2=CH-CH=CH-CH=CH-CH3, count distinct configurational geometrical isomers. Ignore conformers and photochemical interconversion. Where reversing the entire carbon chain identifies two E/Z assignments, count them once.

(A) 5 (B) 8

(C) 4 (D) 3

56. Heating separates complementary DNA strands without cleaving their covalent backbones. Which interaction change and retained property fit?

(A) Every phosphodiester bond must be hydrolysed.

(B) All purines are necessarily converted into pyrimidines.

(C) Base-pairing interactions are disrupted; each strand's nucleotide order remains.

(D) The separated strands become proteins.

57. Evaluate independently. I: In SF_4,axial and equatorial S-F bonds must be identical by symmetry. II: XeF_4 is nonpolar despite polar Xe-F bonds.

(A) Both I and II are correct. (B) Only II is correct.

(C) Neither I nor II is correct. (D) Only I is correct.

58. A proposed H2O2 Lewis drawing has three single bonds but only one lone pair on each O. What is wrong with its electron inventory?

(A) It has four excess electrons.

(B) It accounts for only ten valence electrons rather than fourteen, and leaves the oxygens short of ordinary octets.

(C) It is forced to be the correct structure because it contains an O-O bond.

(D) It has too many nuclei because lone pairs count as atoms.

59. For AgCl(s)+2NH3⇌[Ag(NH3)2]+ +Cl-, Ksp Kf=1/4. Before solid is added there is zero chloride, an unknown silver-ammine background, and free NH3=3/10 M. Measured additional dissolved AgCl is 9/140 M. All dissolved silver remains complexed; volume is fixed. What was the initial background complex concentration?

(A) -1/20 (B) 1/10

(C) 1/40 (D) 1/20

60. A student expects persistent lithium-metal deposition during ordinary electrolysis of aqueous LiCl, based only on gas-phase first ionisation energy. Which correction to that isolation prediction is justified?

(A) Li+ hydration makes its aqueous reduction easier than cathodic water reduction.

(B) Water reduction and Li/water reaction prevent persistent Li deposition.

(C) High cathodic Li+ concentration ensures persistent Li deposition before H2.

(D) Raising the aqueous LiCl concentration removes the competing water reduction.

English Proficiency

61. Choose the correctly spelt word.

(A) successfull (B) successful

(C) succesful (D) sucessful

62. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. referred to in the catalogue. Q. only the original contained R. the handwritten note S. Although several copies were available,

(A) S-Q-R-P (B) P-Q-R-S

(C) Q-S-R-P (D) R-S-Q-P

63. Complete: We have _____ information to identify the error, but not enough to explain how it arose.

(A) a few (B) many

(C) some (D) an

64. Complete the question tag: She rarely leaves the archive unattended, _____?

(A) did she (B) doesn't she

(C) is she (D) does she

65. Complete: The lecturer referred to _____ only surviving copy, rather than to one of the later reproductions.

(A) the (B) no article

(C) an (D) a

66. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. during the previous week. Q. the books that had arrived R. By the time the doors opened, S. the volunteers had already arranged

(A) S-R-Q-P (B) R-S-Q-P

(C) Q-R-S-P (D) P-S-Q-R

67. Read the sentence: The transparent selection procedure made every criterion visible to applicants. Choose the opposite in meaning of 'transparent' as used in the sentence.

(A) public (B) simple

(C) opaque (D) fair

68. He produces a written version of a report in another language, rather than translating speech as it occurs. What is he?

(A) interpreter (B) translator

(C) transcriber (D) commentator

69. Read the sentence: The diligent archivist checked each entry against the original document instead of relying on memory. Choose the closest meaning of 'diligent' as used in the sentence.

(A) careful and hardworking (B) strict about other people's work

(C) quick to make decisions (D) naturally gifted

70. Read the sentence: Early success made the team complacent, and it stopped checking for new risks. Choose the closest meaning of 'complacent' as used in the sentence.

(A) unwilling to admit success (B) too tired to continue

(C) quietly confident after careful checking (D) too self-satisfied to remain alert

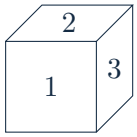
Logical Reasoning

71. Three pairs contain exactly one string instrument and one wind instrument. Which pair does not?

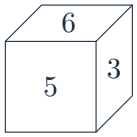
(A) Violin / Oboe (B) Ukulele / Guitar

(C) Harp / Saxophone (D) Sitar / Clarinet

72. The views show the same die. Give the ordered pair: (face opposite 2, face opposite 6). Do not assume opposite labels sum to seven.



1



2

(A) $(6, 3)$ (B) $(6, 4)$

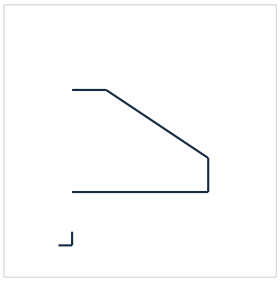
(C) $(6, 2)$ (D) $(6, 1)$

73. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 3, 6, ?, 24, 30, 60, 66.

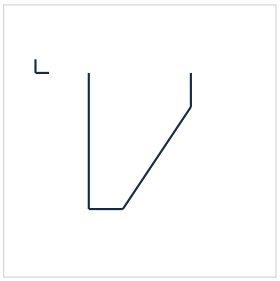
(A) 9 (B) 12

(C) 24 (D) 0

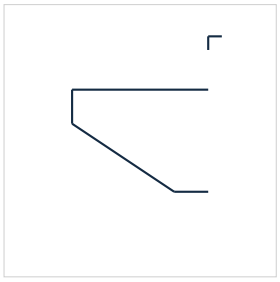
74. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right

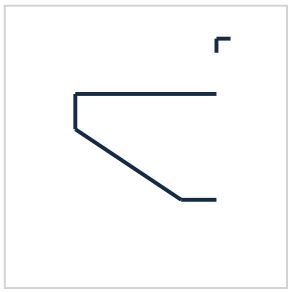


bottom left

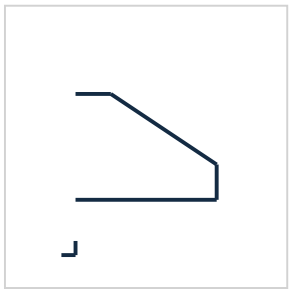
?

bottom right

(A) Pattern Y



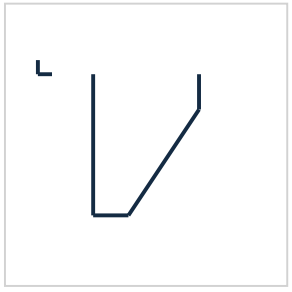
(B) Pattern V



(C) Pattern Q



(D) Pattern L



75. Consider only these statements; universal statements alone do not imply existence. All farmers are bakers; All hikers are farmers. Conclusions: I. All hikers are bakers. II. Some farmers are bakers. Which conclusions follow necessarily?

(A) Neither I nor II

(B) Only II

(C) Both I and II

(D) Only I

76. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

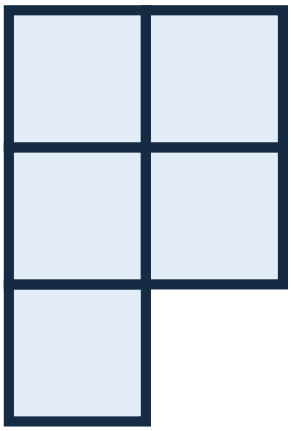


piece 1

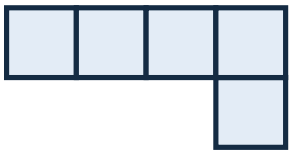


piece 2

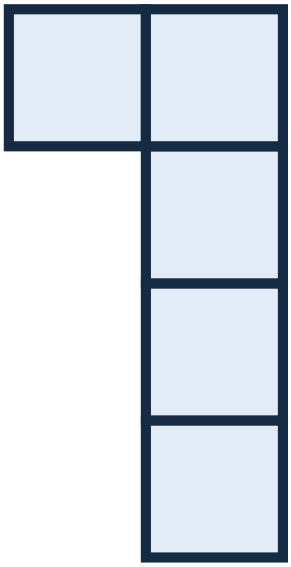
(A) Pattern N



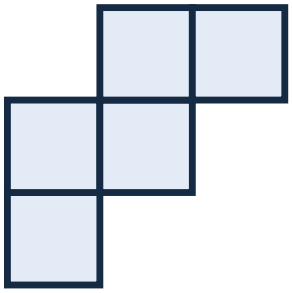
(B) Pattern T



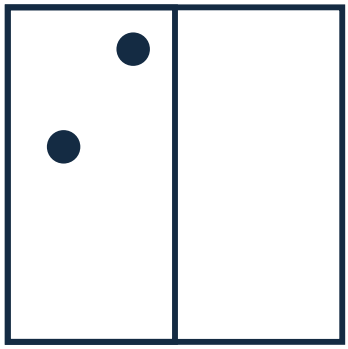
(C) Pattern U



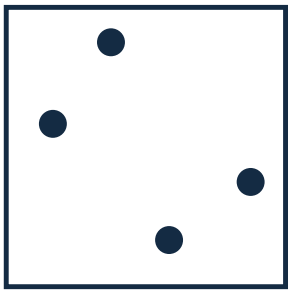
(D) Pattern P



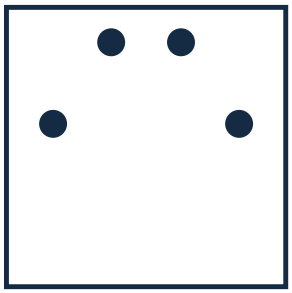
77. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



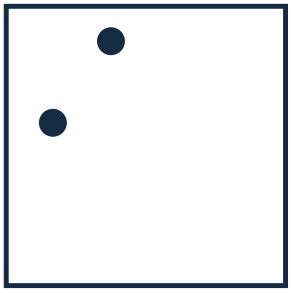
(A) Pattern X



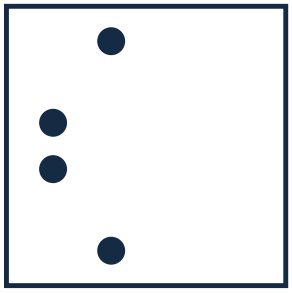
(B) Pattern K



(C) Pattern R



(D) Pattern Q



78. In a row, Asha is 7th from the left and 22nd from the right. Bina is 9th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 0

(B) 11

(C) 8

(D) 9

79. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

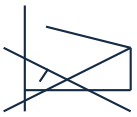




host 1



host 2



host 3



host 4

(A) 1

(B) 4

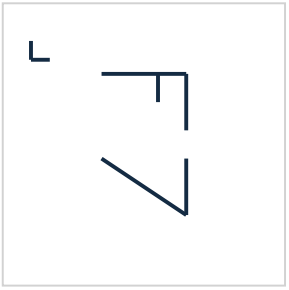
(C) 2

(D) 3

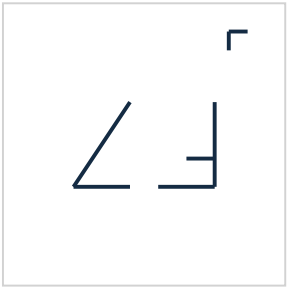
80. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



(A) Pattern M



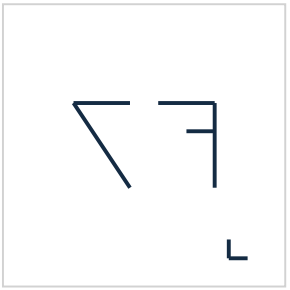
(B) Pattern U



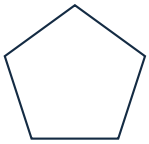
(C) Pattern L



(D) Pattern V



81. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

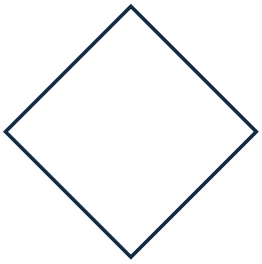


B

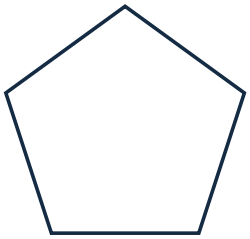


C

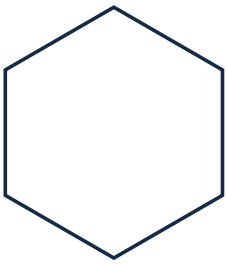
(A) Pattern Q



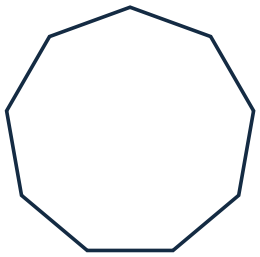
(B) Pattern R



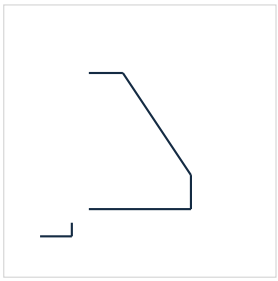
(C) Pattern U



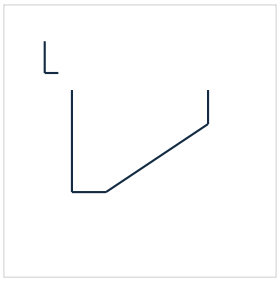
(D) Pattern P



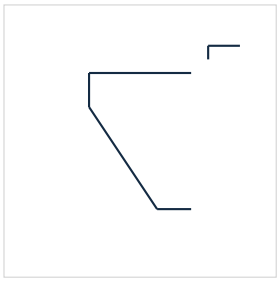
82. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1



2

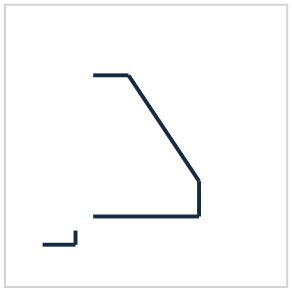


3

(A) Pattern V



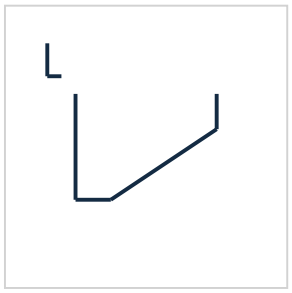
(B) Pattern T



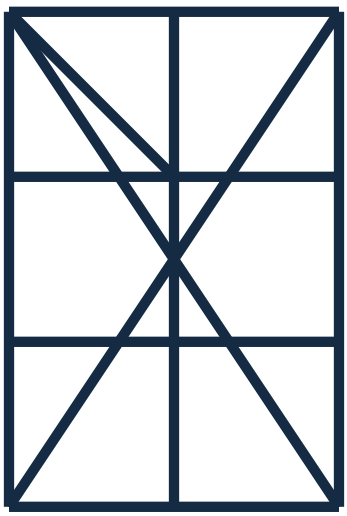
(C) Pattern U



(D) Pattern K



83. How many triangles are formed by all the drawn segments?



(A) 52

(B) 20

(C) 30

(D) 22

84. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, LAMP is coded as PEQT. Which word is coded as QERKS?

(A) NBOHP (B) LZMFN

(C) UIVOW (D) MANGO

85. Mohan points to a photograph and says, "This person's only daughter is my mother." Mohan is male, and the photographed person is female. How is the photographed person related to Mohan?

(A) Grandmother (B) Brother

(C) Son (D) Daughter

86. The whole has 13 unit positions. Part A covers the first 3; part B covers 5 positions, overlapping A in exactly 2 positions as shown. How many positions remain uncovered by either part?



whole part A positions part B positions

(A) 7 (B) 5

(C) 8 (D) 3

87. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 12$ and $5 \mapsto 21$. Which input maps to 33?

(A) 8 (B) 9

(C) 10 (D) 33

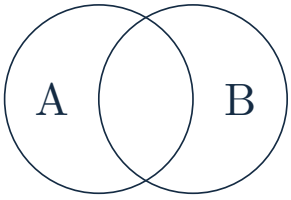
88. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: HWZT, EXAQ, ?, YZCK, VADH, SBEE.

(A) AXAM (B) NBYB

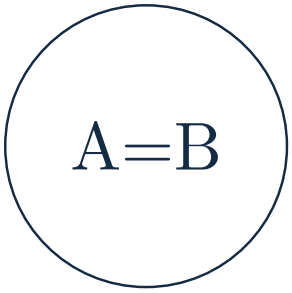
(C) BYBN (D) CZCO

89. Choose the diagram that represents all these conditions, including the strictness information: Some A is B, some A is not B, and some B is not A.

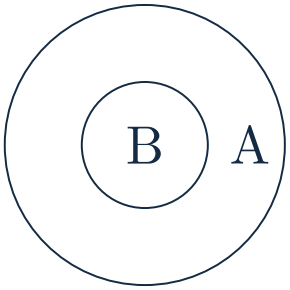
(A) Pattern U



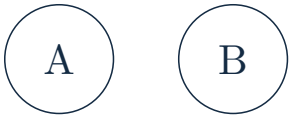
(B) Pattern X



(C) Pattern Q



(D) Pattern M



90. Three numbered triples satisfy both $y = x^2 + 7$ and $z - y = 3$. Which triple fails at least one condition? 1: (2,11,14), 2: (7,56,59), 3: (12,151,154), 4: (6,43,47).

(A) 4 (B) 1

(C) 3 (D) 2

Mathematics

91. z=2+7i. Find 1/z+1/conjugate(z).

(A) 4 (B) $\frac{1}{4}$

(C) $\frac{14}{53}$ (D) $\frac{4}{53}$

92. A=(3,0,5), B=(0,5,1), AP:PB=2:1 internally. Find $|AP|^2$.

(A) $\frac{50}{9}$ (B) $\frac{200}{9}$

(C) 50 (D) $\frac{100}{3}$

93. Find the area bounded by y=x, y=64/x, and x=24 in the first quadrant.

(A) $128 - 64\log(3)$ (B) $256 - 64\log(3)$

(C) $64\log(3)$ (D) 256

94. An AP has a_1=-88,d=8. How many of its first 15 terms are strictly positive?

(A) 12 (B) 2

(C) 3 (D) 15

95. F′(x)=$\frac{(x+1)^2}{(x+2)^4}$ and F(0)=3. Find F(3)-2F(0).

(A) $-\frac{3129}{1000}$ (B) $\frac{129}{1000}$

(C) $\frac{3129}{1000}$ (D) $-\frac{2871}{1000}$

96. Evaluate lim as n→ ∞ of (8/n)Σ|8 j/n-8/3|, j=1,...,n.

(A) $\frac{128}{9}$ (B) $\frac{160}{9}$

(C) $\frac{32}{3}$ (D) 32

97. An observer at height 9 m sees the tower foot at depression 30° and its top at elevation 60°. The observer descends vertically to ground level. If the new elevation is α, find tanα.

(A) $\frac{\sqrt{3}}{3}$ (B) $\frac{4\sqrt{3}}{3}$

(C) $\sqrt{3}$ (D) $\frac{2\sqrt{3}}{3}$

98. z_1=5+3i, z_2=8+3i, z_3=8+6i. Find the smaller angle between z_2-z_1 and z_3-z_1.

(A) $\frac{\pi}{2}$ (B) 0

(C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{4}$

99. The letters A,A,B,C and 6 further distinct letters, one namedD, are arranged in a row. How many arrangements have nonadjacent As, adjacent B,C in either order, and D at an end?

(A) 80640 (B) 30240

(C) 20160 (D) 60480

100. A 11-element set contains distinct a,b. How many 4-element subsets contain exactly one of a,b and also contain a?

(A) 36 (B) 120

(C) 84 (D) 330

101. An AP has first term 6, sum of its first 10 terms equal to 195. Find the sum of the squares of those 10 terms.

(A) 360 (B) 4545

(C) 3456 (D) 38025

102. How many reflexive symmetric relations on a labelled 3-element set contain at least one off-diagonal ordered pair?

(A) 7 (B) 4

(C) 6 (D) 8

103. Evaluate $\sum_{j=7}^{12} \binom{j}{3}$, with the convention $\binom{j}{r} = 0$ when j<r.

(A) 645 (B) 715

(C) 700 (D) 680

104. An equivalence relation has classes of sizes 6,2,6. How many unordered pairs of distinct related elements are there?

(A) 62 (B) 7

(C) 14 (D) 31

105. An AP has first term -32 and common difference 4. Find the sum of its strictly positive terms among the first 12 terms.

(A) 24 (B) 4

(C) 3 (D) -120

106. A differentiable function f satisfies $f\prime(t) = (t-9)(t-36)$ for every real t. Let $g(x) = f(x^2)$. Which is the maximal open interval containing $x = -9/2$ on which $g\prime(x) > 0$?

(A) (-infinity, -6) (B) (3, 6)

(C) (-6, -3) (D) (-3, 0)

107. A line has direction (1,8,2). A plane has normal (2,1,8). Let θ be their acute angle. Find $\tan^2\theta$.

(A) $\frac{676}{4761}$ (B) $\frac{4085}{4761}$

(C) $\frac{676}{4085}$ (D) $\frac{4761}{676}$

108. A convergent GP has first term 7, ratio $\frac{1}{5}$. Find the sum of all terms after its first 8 terms.

(A) $\frac{7}{250000}$ (B) $\frac{7}{62500}$

(C) $\frac{7}{312500}$ (D) $\frac{1}{312500}$

109. ABC is right angled at A with AB=14,AC=16. A vertical tower stands at the midpointM ofBC. Its top elevation from A is 60°. Find the square of the tower height.

(A) 339 (B) 168

(C) $113\sqrt{3}$ (D) 1356

110. In $(1+x)^7$, find the ratio of the coefficient of x^7 to the coefficient of x^6.

(A) 7 (B) $\frac{1}{7}$

(C) 1 (D) $\frac{1}{3}$

111. Find the sum of the squares of all positive real solutions of |x-2|=6.

(A) 64 (B) 80

(C) 86 (D) 8

112. A positive integer n satisfies $|\Pi(1+ji),\ j=1,\ldots,n|^2=1700$. Find n.

(A) 3 (B) 4

(C) 5 (D) 8

113. $A=\begin{bmatrix}5 & 1\\ 1 & 4\end{bmatrix}$, $B=\begin{bmatrix}22 & 10\\ 12 & 21\end{bmatrix}$, and XA=B. Find X_{11}.

(A) $-\frac{78}{19}$ (B) $\frac{78}{19}$

(C) 22 (D) 4

114. $P(A)=\frac{5}{16}$, $P(B)=\frac{7}{16}$, $P(A\cap B)=\frac{5}{32}$. Find P(neither A nor B).

(A) $\frac{3}{32}$ (B) $\frac{11}{16}$

(C) $\frac{1}{4}$ (D) $\frac{13}{32}$

115. Evaluate $\displaystyle\sum_{\substack{0\le r\le 12\\ r \text{ even}}} \frac{\binom{12}{r}}{(r+1)(r+2)}$.

(A) $\frac{8191}{182}$ (B) $\frac{4050}{91}$

(C) $\frac{16369}{182}$ (D) $\frac{4096}{13}$

116. For common tangents to $y^2=28x$ and $x^2+y^2=\frac{49}{30}$, find the sum of the squared slopes.

(A) 0 (B) 10

(C) −5 (D) 5

117. Maximize Z=3x+4y subject to $x\ge 0$, $y\ge 6$, $x+y\le 14$, $x+2y\le 19$.

(A) 47 (B) 38

(C) 48 (D) 45

118. Let A,B be the intersections of x=4 with $x^2/4-y^2/16=1$. Find the area of triangle OAB.

(A) 16 (B) $32\sqrt{3}$

(C) $4\sqrt{3}$ (D) $16\sqrt{3}$

119. a=(6,5,0), b=(5,-6,2). Find $|a+b|^2-|a-b|^2$.

(A) -44 (B) 3965

(C) 0 (D) -8

120. $y'+y/(x+1)=1$ for $x\geq 0$, $y(0)=6$. Find $(10)y(9)-y(0)$.

(A) $\frac{111}{2}$ (B) $\frac{99}{2}$

(C) $\frac{111}{20}$ (D) $-\frac{99}{2}$

121. Evaluate lim as $x\to 0$ of $[\text{sqrt}(64+x)-8-x/16+x^2/4096]/x^3$.

(A) $\frac{1}{524288}$ (B) $-\frac{1}{524288}$

(C) $-\frac{1}{4096}$ (D) $\frac{1}{8192}$

122. Let R(x) be the remainder when x^7 is divided by (x-1)(x-2). Find R(2).

(A) -125 (B) 129

(C) 128 (D) 254

123. A circular sector has radius 9 and arc length $\frac{9\pi}{4}$. Find its area.

(A) $\frac{81\pi}{8}$ (B) $\frac{\pi}{8}$

(C) 81π (D) $\frac{9\pi^2}{32}$

124. Circles $x^2+y^2=49$ and $(x-10)^2+y^2=49$ meet at A,B. Find $|AB|^2$.

(A) 16 (B) $4\sqrt{6}$

(C) 96 (D) 196

125. Find the squared distance from O to the plane through (5,0,0),(0,4,0),(0,0,2).

(A) $\frac{400}{141}$ (B) 45

(C) $\frac{20}{19}$ (D) $\frac{141}{400}$

126. How many integers k make $21\sin x+28\cos x+4=k$ have exactly two distinct solutions in $0\leq x<2\pi$?

(A) 34 (B) 70

(C) 69 (D) 71

127. For xy-7x-6y+42=0, find the squared length of the diagonal of the rectangle formed with the axes.

(A) 85 (B) $\sqrt{85}$

(C) 169 (D) 42

128. There are 15 Boolean variables p_1,...,p 15. Exactly 1 variables must be true. How many truth assignments satisfy [(p_1 implies p_2) and (p_1 implies not p_2)]?

(A) 1 (B) 15

(C) 14 (D) 32768

129. $f(x)=(x^2-49)/(x-7)$ for $x\neq 7$, and $f(7)=k$. If f is differentiable at 7, find $k+f'(7)$.

(A) 15 (B) 16

(C) 50 (D) 14

130. Let $f(x+5)=6\ x^2$. Find the area enclosed between $y=f(x)$ and $y=36(x-5)$.

(A) 648 (B) 1

(C) 432 (D) 216

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A uniform-density planet has radius R. Gravity at a depth d equals gravity at a height $(10-1)R$ above the surface. Find d/R.

(A) 1 (B) $\frac{99}{100}$

(C) $\frac{1}{10}$ (D) $\frac{9}{10}$

132. A charge $1\,\mu$C moves from +2 V to -2 V in an electrostatic field while its kinetic energy increases by 70 microjoules. Find work done by the external agent; no other force does work.

(A) $74\,\mu$J (B) $68\,\mu$J

(C) $66\,\mu$J (D) $62\,\mu$J

133. A series circuit contains resistor R, two ideal inductors L_1, L_2, and a fixed-frequency AC source of angular frequency 60 rad/s. With both inductors active, current lags by 45°; when L_2 is shorted, it lags by 30°. If $R = 60$ ohms, find L_2.

(A) $1 - \frac{\sqrt{3}}{3}$ H (B) $-1 + \sqrt{3}$ H

(C) 1 H (D) $3600 - 1200\sqrt{3}$ H

Chemistry

134. A monatomic ion contains 10 electrons and has mass number 23. Its charge is +1. Which pair gives (proton count, neutron count), without confusing ionic charge with nuclear identity?

(A) (11, 13) (B) (11, 12)

(C) (10, 13) (D) (12, 11)

135. $E^\circ(M^{4+}/M) = \frac{3}{10}$ V for 4 electrons, and $E^\circ(M^{4+}/M^{2+}) = \frac{1}{2}$ V for 2 electrons. A Z^{2+}/Z couple is -0.800 V. Standard cell potential for $Z + M^{2+} \rightarrow Z^{2+} + M$ is

(A) $\frac{6}{5}$ (B) $\frac{13}{10}$

(C) $\frac{9}{10}$ (D) $\frac{11}{10}$

136. For oxide formation, both normalised per mole of O_2, $\Delta G^\circ_A = -335 + 0.10T$ and $\Delta G^\circ_B = -430 + 0.20T$ in kJ. Ignoring kinetic limitations, at what temperatures is B metal a favourable standard-state reducing agent for oxide A?

(A) T < 950 K (B) Only T = 950 K

(C) T > 950 K (D) At all positive temperatures

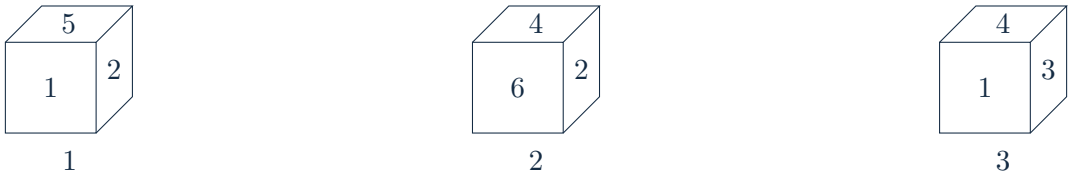
Logical Reasoning

137. Preserve the exact relation and its order: Bee : Swarm :: ? : Pack. Choose the missing term.

(A) Wolf (B) Lion

(C) Bee (D) Ship

138. The views show the same die. Give the ordered pair: (face opposite 3, face opposite 2). Do not assume opposite labels sum to seven.

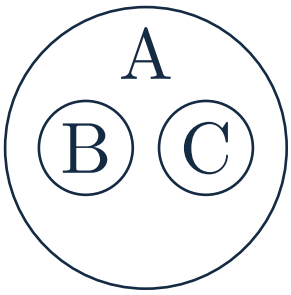


(A) $(2, 4)$ (B) $(2, 1)$

(C) $(2, 3)$ (D) $(2, 2)$

139. Choose the diagram that represents all these conditions, including the strictness information: Some A is B, some A is not B, and some B is not A. No C is A or B; C is nonempty.

(A) Pattern U



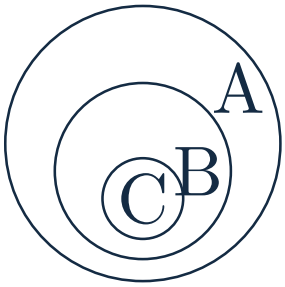
(B) Pattern Z



(C) Pattern R



(D) Pattern K



Mathematics

140. $A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$. Find tr(BA) without assuming AB=BA.

(A) 15 (B) -3

(C) 13 (D) 10

141. Evaluate Σ j/2^j for j=1,...,6.

(A) $\frac{3}{32}$ (B) $\frac{15}{8}$

(C) 2 (D) $\frac{121}{64}$

142. How many x with $0 \leq x < 8\pi$ satisfy $\sin^4 x + \cos^4 x = 3/4$ and $\cos x > 0$?

(A) 16 (B) 20

(C) 8 (D) 32

BITSAT Mock 47

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A gas receives 800 J of heat while its surroundings do 100 J of work on it. It then loses 400 J of heat at fixed volume. Find its total internal-energy change.

(A) 1300 J (B) 400 J

(C) 300 J (D) 500 J

2. A symmetric uniform chain of total mass 6 kg hangs from equal-height supports. Each end tangent makes an angle with $\sin\theta = 3/5$, $\cos\theta = 4/5$. What is the tension at the lowest point? Take $g = 10$ m/s^2.

(A) 40 N (B) 50 N

(C) $\frac{45}{2}$ N (D) 30 N

3. 2 kg of water at 20°C is mixed in an insulated vessel with $\frac{168}{167}$ kg of ice at 0°C. Use $c_w = 4200$ J/(kg K), $L_f = 334000$ J/kg. Find the mass of ice remaining at equilibrium.

(A) $\frac{21084}{34903}$ kg (B) $\frac{84}{167}$ kg

(C) $\frac{126}{167}$ kg (D) $\frac{168}{167}$ kg

4. An ideal mixture contains 4 moles of monatomic gas and 4 moles of rigid diatomic gas. Vibrations are frozen. Find the heat-capacity ratio $\gamma = C_P/C_V$ of the mixture.

(A) $\frac{3}{2}$ (B) $\frac{7}{5}$

(C) $\frac{2}{3}$ (D) $\frac{5}{3}$

5. A direct vernier has 10 divisions equal to 9 main-scale divisions; 1 MSD is 1 mm. Its signed zero error is $\frac{-1}{5}$ mm. For an object the main scale is 7 mm and the 7th vernier division coincides. Find the corrected reading.

(A) $\frac{15}{2}$ mm (B) $\frac{97}{10}$ mm

(C) $\frac{79}{10}$ mm (D) $\frac{77}{10}$ mm

6. Two resistors R and 2R are in series across an ideal 9 V source. A branch consisting of a 8 microfarad capacitor in series with a finite resistor connects across the 2R resistor. At DC steady state, find the capacitor charge.

(A) $48\,\mu$C (B) $24\,\mu$C

(C) $144\,\mu$C (D) $72\,\mu$C

7. An ideal heater of power 5000 W supplies heat to 2 kg of ice initially at 0°C in an insulated vessel for $\frac{1004}{5}$ s. Find final temperature after complete melting. Use $L_f = 334000$ J/kg and $c_w = 4200$ J/(kg K); neglect vessel heat capacity.

(A) 80 degree C (B) 40 degree C

(C) $\frac{84}{167}$ degree C (D) $\frac{4180}{21}$ degree C

8. A particle of mass 5 kg starts from rest and experiences $F(t) = 2t$ N along its motion for 2 s. Find the work done by the force.

(A) $\frac{16}{5}$ J (B) $\frac{32}{5}$ J

(C) $\frac{8}{5}$ J (D) 4 J

9. Two parallel plane mirrors separated by 2 cm extend from axial position 0 to 7 cm. A ray enters midway between them at axial position 0, making 45° with the normal to the mirrors. How many reflections occur before it passes the far edge? Exclude a contact exactly on the exit edge.

(A) 1 (B) 3

(C) 2 (D) 5

10. A wheel initially rotates at 6 rad/s. A constant opposing angular acceleration brings it to rest after 7 s. What angle does it turn through during the last half of this stopping interval?

(A) 42 rad (B) $\frac{21}{2}$ rad

(C) $\frac{21}{8}$ rad (D) $\frac{21}{4}$ rad

11. An object moves slowly along the axis toward a concave mirror. At an instant its distance from the mirror is 90 cm; the mirror focal-length magnitude is 15 cm. Find the image-speed magnitude divided by object-speed magnitude at that instant.

(A) $\frac{1}{25}$ (B) $\frac{1}{5}$

(C) $\frac{1}{49}$ (D) $\frac{1}{36}$

12. A particle has potential $U(x, y) = 6x^2y + 3y$ J in SI units. Find the force magnitude at $(4, 4)$ m.

(A) $96\sqrt{5}$ N (B) 192 N

(C) $3\sqrt{5185}$ N (D) 396 N

13. A 2 kg block moves along a horizontal rough surface from $x = 0$ to $x = 3$ m under a force $F(x) = 6x$ N parallel to motion. With $\mu_k = 1/5$ and $g = 10$ m/s^2, find the work done by the net force. The block has enough initial speed to keep moving forward.

(A) 15 J (B) 5 J

(C) 42 J (D) 27 J

14. A stretched string is divided by a bridge into lengths in the ratio 1 : 3. Tension and linear density are unchanged. The original fundamental is 30 Hz. Find the fundamental of the longer segment.

(A) 30 Hz (B) 60 Hz

(C) 40 Hz (D) 10 Hz

15. A block is placed on the vertical inner wall of a trolley. The trolley accelerates horizontally so the wall pushes the block. With coefficient of static friction $\frac{1}{5}$, what minimum trolley acceleration prevents vertical slipping? Take $g = 10\ \text{m/s}^2$.

(A) $\frac{25}{2}\ \text{m/s}^2$ (B) $10\sqrt{5}\ \text{m/s}^2$

(C) $10\ \text{m/s}^2$ (D) $50\ \text{m/s}^2$

16. A uniform thin rod of length 8 m is hinged at one end and released from rest in a horizontal position. Find the magnitude of its initial angular acceleration. Take $g = 10\ \text{m/s}^2$.

(A) $\frac{5}{8}\ \text{rad/s}^2$ (B) $120\ \text{rad/s}^2$

(C) $\frac{15}{4}\ \text{rad/s}^2$ (D) $\frac{15}{8}\ \text{rad/s}^2$

17. Unpolarised light passes three ideal polarisers with axes at 0°, 75°, and 135° relative to a common reference. Find final intensity divided by incident intensity.

(A) $-\frac{\sqrt{2}}{16} + \frac{\sqrt{6}}{16}$ (B) $\frac{1}{8} - \frac{\sqrt{3}}{16}$

(C) $\frac{1}{4}$ (D) $\frac{1}{16} - \frac{\sqrt{3}}{32}$

18. A car negotiates a level circular turn at its maximum non-slipping speed u when $\mu_s = \frac{1}{2}$. Rain lowers the coefficient to $\frac{3}{10}$ and the turn radius stays fixed. Find the new maximum speed divided by u.

(A) 1 (B) $\frac{\sqrt{30}}{10}$

(C) $\frac{\sqrt{15}}{5}$ (D) $\frac{3}{10}$

19. A projectile is launched at 15 m/s with $\sin\theta = 4/5$, $\cos\theta = 3/5$. A horizontal platform is at $\frac{8}{9}$ of its maximum height. Find the horizontal separation between its two crossings of that height. Take $g = 10\ \text{m/s}^2$.

(A) $\frac{108}{5}\ \text{m}$ (B) $\frac{36}{5}\ \text{m}$

(C) $12\ \text{m}$ (D) $\frac{72}{5}\ \text{m}$

20. A projectile of mass 6 kg is launched with speed 15 m/s, $\cos\theta = 3/5$. At its highest point it splits into masses $M/3$ and $2M/3$. The lighter fragment moves horizontally forward at 12 m/s. Find the signed horizontal velocity of the heavier fragment immediately after the explosion.

(A) $\frac{15}{2}\ \text{m/s}$ (B) $15\ \text{m/s}$

(C) $\frac{33}{2}\ \text{m/s}$ (D) $9\ \text{m/s}$

21. An ideal transformer has secondary-to-primary turns ratio 8. Its primary draws 2 A at 100 V while delivering power to a resistive secondary load. Find secondary current.

(A) $\frac{1}{4}\ \text{A}$ (B) $16\ \text{A}$

(C) $2\ \text{A}$ (D) $\frac{1}{32}\ \text{A}$

22. Two simple pendulums execute 4 and 3 small oscillations respectively in the same time at the same place. Find the first pendulum's length divided by the second's.

(A) $\frac{9}{16}$ (B) 9

(C) $\frac{4}{3}$ (D) $\frac{1}{16}$

23. An ideal gas initially has pressure 500 Pa and volume 3 m^3. It is compressed reversibly and isothermally to $V/4$. Find work done on the gas.

(A) 1125 J (B) $1500 \ln 4$ J

(C) 4500 J (D) $6000 \log(2)$ J

24. A deuteron of mass $2u$, charge $+e$, and an alpha particle of mass $4u$, charge $+2e$, start from rest. The deuteron is accelerated through potential $2V$ and the alpha particle through V. In the nonrelativistic regime, find λ_α/λ_d.

(A) 1 (B) $\frac{1}{2}$

(C) $\sqrt{2}$ (D) $\frac{\sqrt{2}}{2}$

25. A uniform disc of mass 8 kg and radius 3 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) $27\,\mathrm{kg\,m^2}$ (B) $\frac{171}{4}\,\mathrm{kg\,m^2}$

(C) $36\,\mathrm{kg\,m^2}$ (D) $\frac{117}{4}\,\mathrm{kg\,m^2}$

26. A stationary nucleus with mass number 24 and binding energy per nucleon 8 MeV splits into two equal daughters, each having total binding energy 97.0 MeV. Which statement is correct?

(A) The process absorbs 2 MeV; the daughters have larger binding energy per nucleon.

(B) The process releases 2 MeV; the daughters have larger binding energy per nucleon.

(C) The process releases 4 MeV; each daughter releases 2 MeV.

(D) No energy can be released because fission always requires a larger mass number than 24.

27. A series RC circuit is driven at fixed rms voltage. Multiplying angular frequency by 5 multiplies rms current by $\frac{5\sqrt{385}}{77}$. Find R^2/X_C^2, where X_C is the initial capacitive reactance.

(A) $\frac{125}{77}$ (B) $\frac{3}{2} + \frac{15\sqrt{385}}{154}$

(C) $\frac{3}{2}$ (D) 25

28. Three long parallel wires are equally spaced by 3 m along a straight line. Outer wires each carry 8 A in the same direction; the middle carries 5 A in the opposite direction. Find the net force per unit length on the middle wire.

(A) $\frac{40}{3\pi}$ N/m (B) 0 N/m

(C) $\frac{20}{3\pi}$ N/m (D) $\frac{16}{3\pi}$ N/m

29. Two identical uniform spheres touch. Their radii are multiplied by 13 while their density is divided by 13. By what factor does their gravitational attraction change?

(A) 2197 (B) 1

(C) 169 (D) $\frac{1}{28561}$

30. Unpolarised light of intensity I passes through an ideal polariser, then a middle polariser at unknown angle θ, and finally an analyser crossed with the first. Final intensity is $6I/64$. Find $\sin^2(2\theta)$.

(A) $\frac{1}{4}$ (B) $\frac{3}{4}$

(C) $\frac{\sqrt{3}}{2}$ (D) $\frac{3}{2}$

Chemistry

31. A tetrahedral Ni(II)d8 complex and a square-planar Ni(II)d8 cyanide complex are compared in the elementary spin model. Which fact shows why d count alone does not determine magnetism?

(A) Both geometries have paired electrons because strong-field ligands lower the d count.

(B) Tetrahedral d8 has two unpaired electrons; square-planar strong-field d8 can have none.

(C) Both geometries have two unpaired electrons because each Ni(II) centre is d8.

(D) Tetrahedral d8 has none; square-planar strong-field d8 has two unpaired electrons.

32. A Co(3) coordination entity has coordination number 6, exactly 1 fully bound dien ligand(s), and chloride in all remaining donor sites. Take dien as 3-dentate with charge 0; chloride is monodentate. Find (number of ligand entities, net charge of the coordination entity).

(A) (6, +1) (B) (5, +0)

(C) (4, -1) (D) (4, +0)

33. A $\frac{7}{5}$ g organic sample contains chlorine as its only halogen. Carius analysis gives $\frac{32144}{44375}$ g dry AgCl, but this is only 80.0% of the theoretical AgCl because of a known collection loss. Find chlorine mass percentage. Use Ag=108, Cl=35.5.

(A) 16 (B) $\frac{64}{5}$

(C) 20 (D) $\frac{4592}{71}$

34. At 1 atm compare two separate pairs: (n-pentane, neopentane) and (n-butane, isobutane). A student selects neopentane for the first higher-boiling substance because its formula mass is unchanged, and n-butane for the second. Which assessment is consistent with intermolecular interactions?

(A) Both selections correct. (B) Both selections wrong.

(C) First selection correct; second selection wrong. (D) First selection wrong; second selection correct.

35. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_y; same facing signs; 2: parallel p_y; opposite facing signs; 3: parallel p_x; opposite facing signs. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (3, 2) (B) (1, 4)

(C) (2, 3) (D) (2, 2)

36. Two oxides of X contain 48 g X combined respectively with 24 g O and 32 g O. Atomic masses are X=48, O=16. Which empirical-formula conclusion jointly follows from mole ratios and the law of multiple proportions?

(A) First oxide XO2; second oxide X2O3; fixed-X oxygen mass ratio second:first=3/4.

(B) Both are XO because X masses are equal.

(C) First empirical formula X2O3; second empirical formula XO2; fixed-X oxygen mass ratio second:first=4/3.

(D) No formula can be inferred even though both atomic masses and both compositions are supplied.

37. An aldopentoseFischer projection has CHO at top, CH2OH at bottom, and OH sides at C2, C3, C4 respectively R, L, L, whereR means right andL means left. Oxidise both terminal groups to COOH without changing internal configurations. Which D/L assignment for the sugar and classification of the diacid are correct?

(A) D/L is fixed by the sign of rotation alone.

(B) D; chiral.

(C) L; chiral.

(D) L; meso(optically inactive).

38. For a completely filled n=3 shell,how many electrons have$m_l = 1$ and$m_s = +1/2$?

(A) 4

(B) 2

(C) 18

(D) 9

39. A chain is cis-1,4-polyisoprene. Which monomer and structural detail are correctly paired?

(A) Acrylonitrile; the repeat must contain CN.

(B) Ethene; every repeat must be saturated and unbranched.

(C) 2-Methylbuta-1,3-diene; the original diene supplies the chain carbons.

(D) Styrene; the repeat must contain a phenyl ring.

40. For ethanal with NH2OH,which product or mechanistic conclusion is correct under the stated conditions?

(A) ethanenitrile,CH3CN

(B) ethanol,CH3CH2OH

(C) ethanoic acid,CH3COOH

(D) ethanal oxime,CH3CH=N-OH

41. Within an additive covalent-radius model,a single X-X bond is 130 pm and an X-H bond is 95 pm. A linear H-X-X-H skeleton is assumed with collinear single bonds. Its terminal H-to-H distance,in pm,is

(A) 225

(B) 380

(C) 190

(D) 320

42. NH3 boils above PH3 despite lower molar mass. What additional interaction invalidates a mass-only prediction?

(A) Strong intermolecular hydrogen bonding in ammonia.

(B) No attractive forces act between PH3 molecules.

(C) Greater mass always guarantees lower boiling point.

(D) A permanent covalent N-N polymer must form before ammonia can boil.

43. A compound has formula C5H10O and gives positive Tollens and no alpha hydrogen. Restrict X to the candidate structures below. Which fits every clue?

(A) pentan-2-one

(B) 2,2-dimethylpropanal

(C) 3-methylbutanal

(D) pentanal

44. A weak 1:1 electrolyte obeys exact Ka=Cα^2/(1-α). In a closed dilution its dissociation degree changes from 1/25 to 3/25, while Ka and total formula-unit amount remain constant. By what factor does its volume increase? Do not replace 1-α by 1.

(A) 108/11

(B) 9

(C) 11/108

(D) 3

45. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_y; same facing signs; 2: parallel p_y; opposite facing signs; 3: p_y on one centre and p_z on the other. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (1, 3)

(B) (2, 1)

(C) (2, 2)

(D) (3, 1)

46. A weak acid HA hasC=0.0100M,Λ0=400S cm^2/mol andsolute-only$\kappa=\frac{1}{12500}$S/cm. UseexactKa=Cα^2/(1-α),$\alpha=\Lambda m/\Lambda 0$. An equal-concentration HA solution also contains0.00100M HCl. Under imposed-H^+ common-ion approximation,[A^-]in this second solution is

(A) $\frac{1}{5000}$

(B) $\frac{1}{24500}$

(C) $\frac{1}{245000}$

(D) $\frac{1}{49000}$

47. An ideal H_2/N_2 mixture atT has mean molar mass$\frac{17}{2}$g/mol ($H_2$2,$N_2$28). Define the ensemble mean of v^2 by molecule-number weighting. Ratio of this mean to pureN_2at the sameT is

(A) $\frac{43}{4}$

(B) $\frac{17}{56}$

(C) $\frac{56}{17}$

(D) $\frac{\sqrt{43}}{2}$

48. A cubic conventional cell has A at all corners and B at all face centres, repeated with A and B distinct. Which description respects chemical labels?

(A) No periodic description can contain different chemical species.

(B) A:B=1:1 because corner and face-centre positions are both shared.

(C) An elemental FCC Bravais lattice with every occupied site equivalent.

(D) Primitive cubic translations with an A:B basis inventory1:3; face-centre shifts need not preserve labels.

49. Across the second period,atomic radius generally decreases. For N and O,which paired conclusion respects the first-ionisation anomaly?

(A) O must have the greater first ionisation enthalpy because every property follows size monotonically.

(B) The two ionisation enthalpies must be equal because their nuclei are adjacent.

(C) O is smaller in the general radius trend,but N has the greater first ionisation enthalpy.

(D) N has both the smaller radius and lower first ionisation enthalpy.

50. An ester X hydrolyses to ethanoic acid and an alcohol A of formula C3H8O. A gives a positive iodoform test. Label the ester bonds a:carbonyl C=O,b:acyl C-O,c:alkoxy O-C(sp3). Which pair correctly identifies X and the increasing bond-length order?

(A) propan-2-yl ethanoate;a < b < c
(B) propan-1-yl ethanoate;a < b < c
(C) ethyl ethanoate;a < b < c
(D) propan-2-yl ethanoate;a < c < b

51. Ce4+ accepts one electron to form Ce3+. In the elementary lanthanoid configuration description, which pair gives the change in4f occupancy and the role of Ce4+?

(A) 4f0→4f0; no electron transfer occurs.
(B) 4f7→4f8; Ce is a half-filled4f7 ion.
(C) 4f1→4f0; Ce4+ is the reducing species.
(D) 4f0→4f1; Ce4+ is the oxidising species.

52. Which coordination name keeps the bracketed sphere,counter-ion and formal metal oxidation state consistent for $[Cr(H_2O)_4Cl_2]Cl$?

(A) tetraaquadichloridochromate(III) chloride
(B) tetraaquadichloridochromium(III) chloride
(C) tetraaquadichloridochromium(II) chloride
(D) tetraaquatrichloridochromium(III)

53. In an idealised Ostwald-process material balance, $\frac{3}{50}$ mol NH_3 is fed.90.0% of its nitrogen reaches the NO/NO_2 intermediate stream, and80.0% of that stream nitrogen is recovered as HNO_3. Find HNO_3 amount in mol. Each nitrogen atom is tracked once; recycled intermediates are not double-counted.

(A) $\frac{51}{1000}$
(B) $\frac{27}{500}$
(C) $\frac{27}{625}$
(D) $\frac{6}{125}$

54. A synthetic chain contains carbon-carbon links only in its backbone, although it carries ester side groups. Which inference about backbone cleavage by simple ester hydrolysis is sound?

(A) An ester side group makes the chain a polyamide.
(B) Hydrolysing side-group esters does not itself cut the C-C backbone.
(C) Hydrolysis necessarily turns each repeat into ethene gas.
(D) Every ester anywhere guarantees cleavage of every backbone C-C bond.

55. Evaluate I: O2 is paramagnetic in elementary MO theory. II: A symmetric linear CO2 molecule has a nonzero permanent dipole because each C-O bond is polar.

(A) Both I and II are correct.
(B) Neither is correct.
(C) Only I is correct.
(D) Only II is correct.

56. A monoatomic cubic Bravais crystal is restricted to simple,BCC orFCC packing. Measured density gives$\rho a^3 NA/M=2$. Ratio atomicradius/cubeedge is

(A) $\frac{1}{2}$
(B) $\frac{\sqrt{3}}{4}$
(C) $\frac{\sqrt{2}}{4}$
(D) $\frac{1}{4}$

57. A solid oxide dissolves in acid and also in strong alkali; another molecular oxide acidifies water; a third is approximately neutral toward aqueous acid/base. Which assignment is consistent?

(A) Al_2O_3; CO_2; CO.
(B) Na2O; CO; CO2.
(C) Al2O3; CO; CO2.
(D) CO2; Al2O3; Na2O.

58. For the same nonrelativistic ion starting from rest,an accelerator changes deBroglie wavelength from $\lambda$to$\lambda/4$. Ion charge and mass are unchanged. Factor by which accelerating voltage must change is

(A) 16 (B) $\frac{1}{4}$

(C) $\frac{1}{16}$ (D) 4

59. A monatomic ion contains 10 electrons and has mass number 27. Its charge is +3. Which pair gives (proton count, neutron count), without confusing ionic charge with nuclear identity?

(A) (16, 11) (B) (10, 17)

(C) (13, 14) (D) (13, 17)

60. For a closed system ΔU(A$\rightarrow$B)=280 J and ΔU(B$\rightarrow$C)=0 J. Two A$\rightarrow$C paths have heats 540 J and 280 J respectively. What is work on the system on the second path minus that on the first, in J?

(A) 260 (B) 540

(C) -260 (D) -20

English Proficiency

61. Read the sentence: The figures are raw; duplicate entries have not yet been removed. Choose the closest meaning of 'raw' as used in the sentence.

(A) unreliable in every respect (B) not yet processed

(C) recently collected (D) difficult to obtain

62. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. To prevent these mix-ups, the manager changed both the wording and the shape of the labels. B. The distinction no longer depended on noticing a small difference in colour alone. C. The store used similar labels for two different cleaning liquids. D. Because those labels looked so alike, staff occasionally selected the wrong bottle when working quickly.

(A) The store used similar labels for two different cleaning liquids.

(B) Because those labels looked so alike, staff occasionally selected the wrong bottle when working quickly.

(C) To prevent these mix-ups, the manager changed both the wording and the shape of the labels.

(D) The distinction no longer depended on noticing a small difference in colour alone.

63. Read the sentence: She was ambivalent about moving: the new role appealed to her, but she disliked leaving her team. Choose the closest meaning of 'ambivalent' as used in the sentence.

(A) indifferent to either outcome (B) unwilling to state an opinion

(C) unable to understand the offer (D) having conflicting feelings

64. She documents word meanings and usage to prepare dictionary entries. Which word names her occupation?

(A) novelist (B) stenographer

(C) lexicographer (D) translator

65. Read the sentence: Fear of ridicule can inhibit a student's willingness to ask questions. Choose the opposite in meaning of 'inhibit' as used in the sentence.

(A) limit
(B) stimulate
(C) affect
(D) conceal

66. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The librarian lends these maps only to registered members.

(A) These maps are lent by the librarian to all visitors.
(B) Only the librarian lends these maps to registered members.
(C) These maps were lent by the librarian only to registered members.
(D) These maps are lent by the librarian only to registered members.

67. Complete: Both routes reach the same destination. One avoids steps; _____, the other is shorter but includes a staircase.

(A) as a result
(B) for instance
(C) by contrast
(D) in addition to

68. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. for visitors using wheelchairs. Q. than the earlier one, R. This arrangement is no more expensive S. but it provides a clearer route

(A) R-Q-S-P
(B) P-Q-S-R
(C) Q-R-S-P
(D) S-R-Q-P

69. Choose the correctly spelt word.

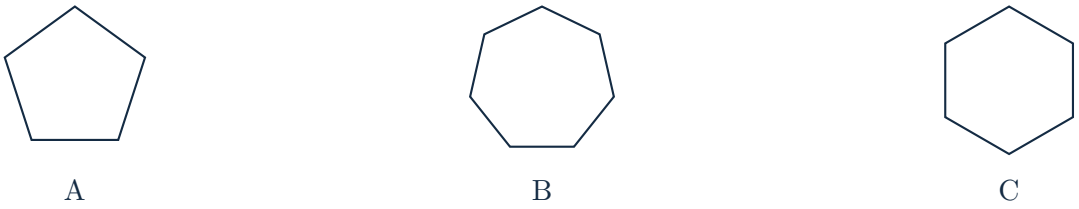
(A) accommodation
(B) accomodation
(C) accommmodation
(D) acommodation

70. The investigation uses rock layers to study the earth's physical past. Which field best names it?

(A) archaeology
(B) geology
(C) linguistics
(D) astronomy

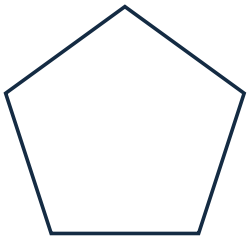
Logical Reasoning

71. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?

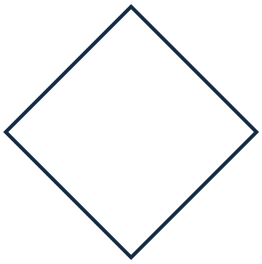


A B C

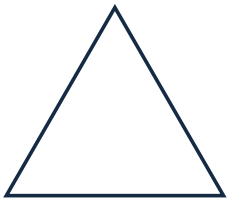
(A) Pattern U



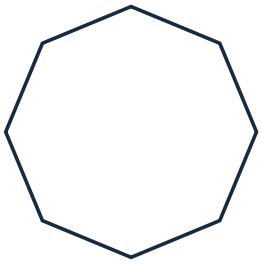
(B) Pattern W



(C) Pattern K



(D) Pattern X

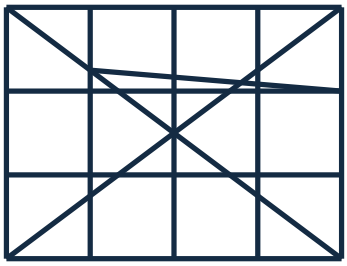


72. In a row, Asha is 11th from the left and 26th from the right. Bina is 11th from the right. Charu stands 2 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 11 (B) 12

(C) 14 (D) 13

73. How many triangles are formed by all the drawn segments?

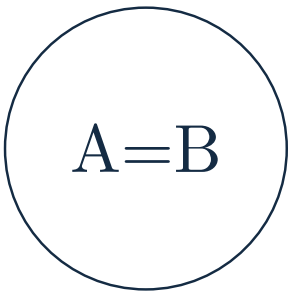


(A) 44 (B) 57

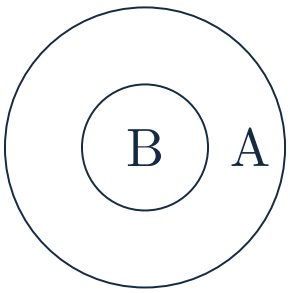
(C) 53 (D) 56

74. Choose the diagram that represents all these conditions, including the strictness information: Some A is B, some A is not B, and some B is not A.

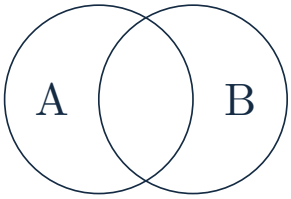
(A) Pattern K



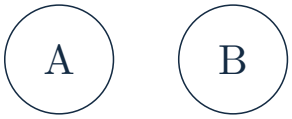
(B) Pattern P



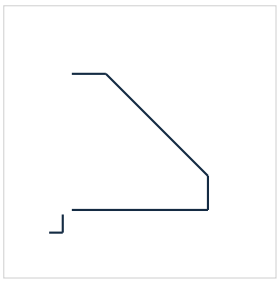
(C) Pattern W



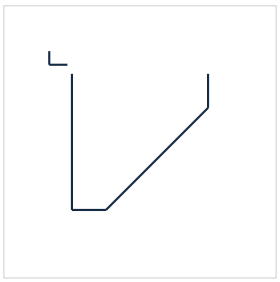
(D) Pattern Q



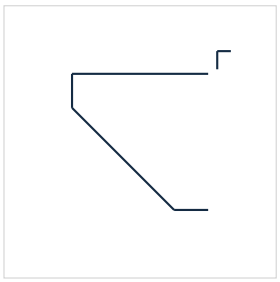
75. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

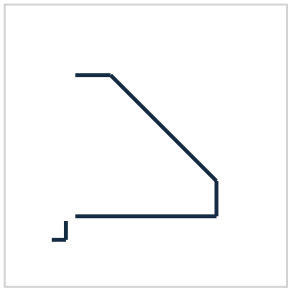


2

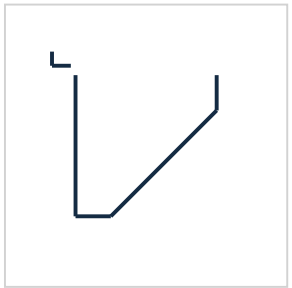


3

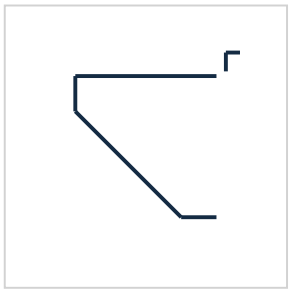
(A) Pattern N



(B) Pattern K



(C) Pattern R



(D) Pattern P



76. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





host 1



host 2



host 3



host 4

(A) 4 (B) 3

(C) 2 (D) 1

77. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

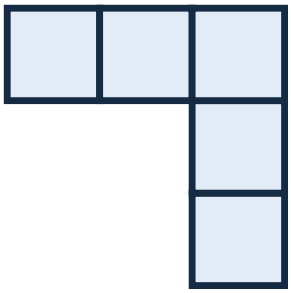


piece 1

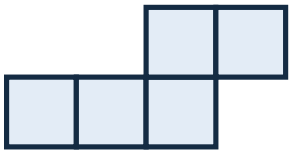


piece 2

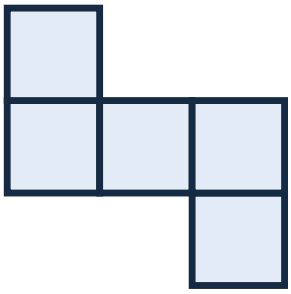
(A) Pattern V



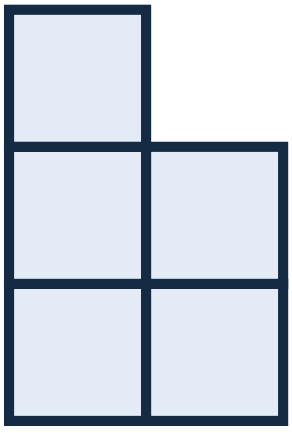
(B) Pattern K



(C) Pattern M



(D) Pattern W



78. The age table uses the same two people throughout:

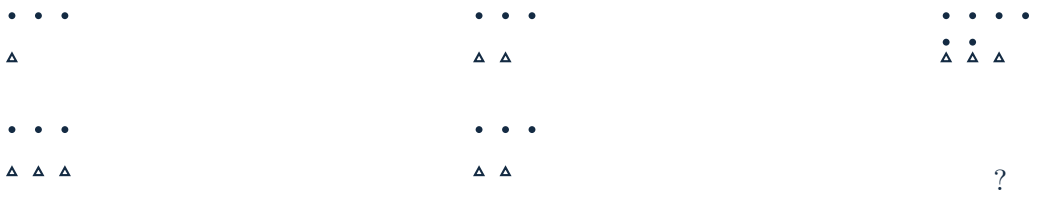
	younger	older
now	x	$x+8$
2 years ago	$x-2$	$x+6$

. The product of their ages 2 years ago was 560. What will the older person's age be 3 years from now?

(A) 33 (B) 30

(C) 31 (D) 41

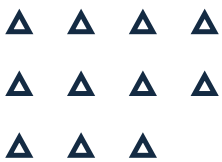
79. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.

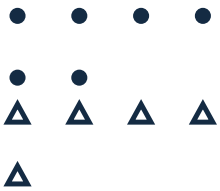


(A) Pattern U (B) Pattern L



(C) Pattern X

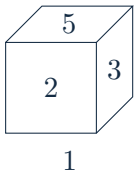




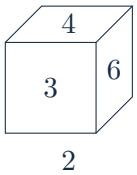
(D) Pattern W



80. The views show the same die. Give the ordered pair: (face opposite 1, face opposite 6). Do not assume opposite labels sum to seven.



1



2

(A) $(3, 2)$ (B) $(3, 1)$

(C) $(3, 5)$ (D) $(3, 3)$

81. Three pairs contain exactly one string instrument and one wind instrument. Which pair does not?

(A) Ukulele / Trumpet (B) Guitar / Harp

(C) Violin / Oboe (D) Sitar / Saxophone

82. Anu points to a photograph and says, "This person is the daughter of the only daughter of my father's only daughter." Anu is female, and the photographed person is female. How is the photographed person related to Anu?

(A) Daughter (B) Son

(C) Granddaughter (D) Brother

83. A walker goes 12 m east, then 12 m north, then turns left and walks 14 m. In which direction is the walker from the starting point?

(A) North-west (B) South-east

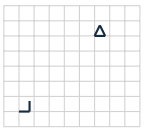
(C) North-east (D) South-west

84. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 6, 12, 16, 32, ?, 72, 76.

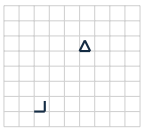
(A) 36 (B) 72

(C) 28 (D) 68

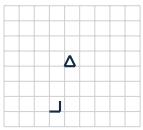
85. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



1

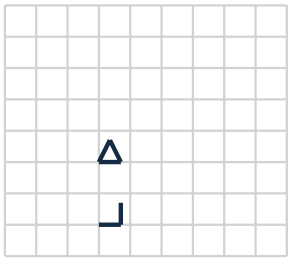


2

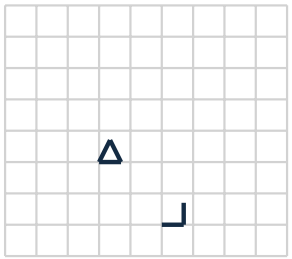


3

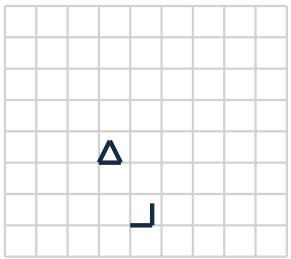
(A) Pattern W



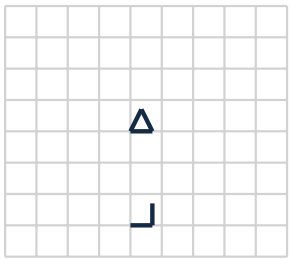
(B) Pattern N



(C) Pattern P



(D) Pattern U



86. Consider only these statements; universal statements alone do not imply existence. Some campers are not artists; Some campers are designers. Conclusions: I. Some designers are campers. II. All artists are campers. Which conclusions follow necessarily?

(A) Only I
(B) Both I and II
(C) Neither I nor II
(D) Only II

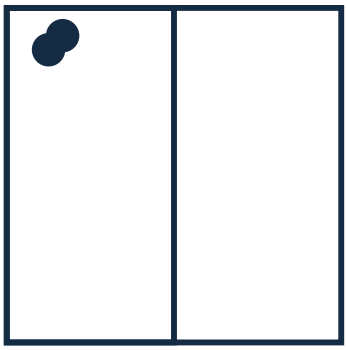
87. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: RCD, TZE, ?, XTG, ZQH, BNI.

(A) UVE
(B) FWV
(C) WXG
(D) VWF

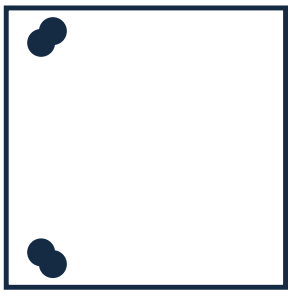
88. Preserve the exact relation and its order: Lion : Pride :: ? : Pack. Choose the missing term.

(A) Wolf
(B) Ship
(C) Lion
(D) Bee

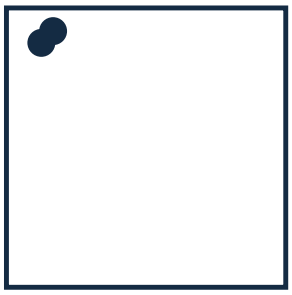
89. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



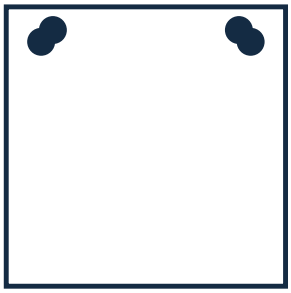
(A) Pattern X



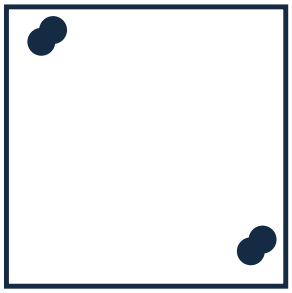
(B) Pattern K



(C) Pattern P



(D) Pattern L



90. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 13$ and $5 \mapsto 22$. Which input maps to 28?

(A) 28 (B) 8

(C) 6 (D) 7

Mathematics

91. Let f(x+6)=7 x^2. Find the area enclosed between y=f(x) and y=49(x-6).

(A) $\frac{2401}{6}$ (B) $-\frac{2401}{6}$

(C) $\frac{133}{6}$ (D) $\frac{2401}{2}$

92. Under x=X+7, y=Y+5, the parabola $(y-5)^2=16(x-7)$ is normalized. Find the x coordinate of its focus in the original axes.

(A) 23 (B) 4

(C) 3 (D) 11

93. $F'(x)=\frac{1}{x^2}-\frac{1}{x^2\sqrt{x^2+1}}$ and F(1)=5. Find F(4)-2F(1).

(A) $-\sqrt{2}+\frac{3}{4}+\frac{\sqrt{17}}{4}$ (B) $-\frac{17}{4}-\sqrt{2}+\frac{\sqrt{17}}{4}$

(C) $-\sqrt{2}+\frac{\sqrt{17}}{4}+\frac{23}{4}$ (D) $-\frac{23}{4}-\frac{\sqrt{17}}{4}+\sqrt{2}$

94. How many integers x satisfy $4^{-x}-(2176)2^{-x}+262144 \leq 0$?

(A) 12 (B) 3

(C) 5 (D) 4

95. A differentiable function f satisfies $f\prime(t) = (t-4)(t-16)$ for every real t. Let $g(x) = f(x^2)$. Which is the maximal open interval containing $x = -3$ on which $g\prime(x) > 0$?

(A) (-infinity, -4) (B) (-4, -2)

(C) (2, 4) (D) (-2, 0)

96. z=10+5i. Find 1/z+1/conjugate(z).

(A) $\frac{1}{20}$ (B) $\frac{2}{25}$

(C) $\frac{4}{25}$ (D) 20

97. An increasing positive GP has first three terms with product 9261 and first-plus-third sum 70. Find its fourth term.

(A) 63 (B) 21

(C) 189 (D) $\frac{7}{3}$

98. For R={(j,j+1):1≤j<5} on {1,...,5}, what is the minimum number of ordered pairs to add to make R transitive?

(A) 4 (B) 10

(C) 11 (D) 6

99. Find the squared distance from O to the plane through (11,0,0),(0,6,0),(0,0,6).

(A) $\frac{33}{14}$ (B) $\frac{139}{2178}$

(C) 193 (D) $\frac{2178}{139}$

100. $y'+8y=\exp(-8x)$, $y(0)=2$. Find $\exp(8)y(1)-2$.

(A) 1 (B) $\frac{3}{e^8}$

(C) 3 (D) -1

101. Evaluate $\cos(4 \arctan 5)$.

(A) $-\frac{120}{169}$ (B) $-\frac{12}{13}$

(C) $\frac{119}{169}$ (D) $-\frac{87}{13}$

102. In $(1+x)^8$, find the ratio of the coefficient of x^6 to the coefficient of x^5.

(A) $\frac{1}{2}$ (B) 2

(C) 3 (D) $\frac{4}{5}$

103. An ellipse has semiaxes 10,6. A hyperbola with the same foci has eccentricity 6. Find the ratio of the hyperbola latus-rectum length to the ellipse latus-rectum length.

(A) $\frac{175}{27}$ (B) $\frac{350}{27}$

(C) $\frac{27}{350}$ (D) 6

104. Evaluate $6^{\log_6 7} + 2^{2(\log_2 6)(\log_6 7)}$.

(A) 49 (B) 55

(C) 14 (D) 56

105. Disjoint A,B have sizes 7,3. Find $|(A\times A)\cup(B\times B)|$.

(A) 79 (B) 10

(C) 58 (D) 42

106. A vector has direction ratios (9,5,5). Find the sum of the squares of its y,z direction cosines.

(A) $\frac{50}{131}$ (B) $\frac{81}{131}$

(C) 50 (D) $\frac{5\sqrt{262}}{131}$

107. Group A has 7 observations, mean 15 and population SD 2. Group B has 8 observations, mean 20 and population SD 3. Find the population variance of the combined data.

(A) $\frac{116}{9}$ (B) 100

(C) $\frac{290}{21}$ (D) $\frac{1444}{225}$

108. $f(x)=(x^2-81)/(x-9)$ for $x\neq 9$, and $f(9)=k$. If f is differentiable at 9, find $k+f'(9)$.

(A) 18 (B) 19

(C) 36 (D) 82

109. |U|=130, |A|=37, |B|=28, |A∩B|=13. Find the number of elements belonging to both A,B or to neither.

(A) 39 (B) 117

(C) 78 (D) 91

110. f(0)=0 and f(x)=$x^2\sin(1/x)$ for x≠0. Find f′(0).

(A) 2 (B) 1

(C) 0 (D) Does not exist

111. For the ellipse $x^2/49+y^2/25=1$, find the distance from its right focus to its nearer directrix.

(A) $\frac{25\sqrt{6}}{12}$ (B) $\frac{25}{7}$

(C) $\frac{73\sqrt{6}}{12}$ (D) $7-2\sqrt{6}$

112. A=(6,0,5), B=(0,4,1), AP:PB=2:1 internally. Find $|AP|^2$.

(A) $\frac{272}{9}$ (B) 68

(C) $\frac{136}{3}$ (D) $\frac{68}{9}$

113. A circular sector has radius 5 and arc length $\frac{5\pi}{8}$. Find its area.

(A) $\frac{5\pi^2}{128}$ (B) $\frac{25}{16}$

(C) $\frac{25\pi}{16}$ (D) $\frac{25\pi}{8}$

114. 12 distinct people stand in a line. Three specified people may occupy neither end position and no two of these three may be adjacent. How many orders are possible?

(A) 261273600 (B) 121927680

(C) 21772800 (D) 43545600

115. F′(x)=xe^{6x} and F(0)=8. Find F(1)-2F(0).

(A) $\frac{289}{36}+\frac{5e^6}{36}$ (B) $-\frac{287}{36}+\frac{5e^6}{36}$

(C) $\frac{1}{36}+\frac{5e^6}{36}$ (D) $-\frac{5e^6}{36}-\frac{289}{36}$

116. Let f(x)=|x-7|/(x-16) for x≠16. Within 0<x<25, what is the total length of the open intervals on which f′(x)<0?

(A) 9 (B) 25

(C) 16 (D) 18

117. Two points on opposite sides of a tower foot are 13 metres apart on a straight level line. The elevations of the tower top are 30° and 45°. Find the tower height.

(A) $\frac{39}{2}-\frac{13\sqrt{3}}{2}$ (B) $-\frac{13}{2}+\frac{13\sqrt{3}}{2}$

(C) $\frac{13}{2}$ (D) $\frac{13}{2}+\frac{13\sqrt{3}}{2}$

118. Let z=(1+i)^{10}/(1-i)^{5}. Find Re(z)+Im(z).

(A) 0 (B) -256

(C) -4 (D) 4

119. An AP has a$_1$=10, d=6. Find a$_1$+a$_3$+$\cdots$+a_17.

(A) 1044 (B) 306

(C) 522 (D) 986

120. Let D_n be the determinant of the 3×3 matrix with diagonal entries n and off-diagonal entries 1. Find ΣD_n for n=1,...,6.

(A) 378 (B) 441

(C) 1200 (D) 390

121. $A = \begin{bmatrix} 1 & 8 \\ 0 & 0 \end{bmatrix}$. For (I+A)^{8}=I+cA, find c.

(A) 256 (B) 8

(C) 7 (D) 255

122. An AP has first term 4 and common difference 3. Find the least positive integer m for which $a_1 + a_3 + \cdots + a_{2m-1} > 300$.

(A) 11 (B) 14

(C) 10 (D) 9

123. An equivalence relation has classes of sizes 6,7,4. How many unordered pairs of distinct related elements are there?

(A) 8 (B) 42

(C) 84 (D) 136

124. Maximize Z=3x+4y subject to x≥0, y≥3, x+y≤14, x+2y≤16.

(A) 44 (B) 42

(C) 32 (D) 45

125. α,β are roots of x^{2}-15x+54=0. The monic polynomial with roots α+1,β+1 is x^{2}-Ax+B. Find B.

(A) 17 (B) 70

(C) 56 (D) 55

126. For t≥0, defineF(t)=lim as n→ ∞ of n$^{(-3)}$Σfloor(tj^{2}),j=1,...,n. EvaluateΣF((3/5)^{r}),r=1,...,∞.

(A) $\frac{5}{6}$ (B) $\frac{3}{16}$

(C) $\frac{1}{5}$ (D) $\frac{1}{2}$

127. Two different numbers are chosen uniformly from {1,...,3}. Let X be their maximum. Find Var(X).

(A) $\frac{2}{9}$ (B) $\frac{22}{3}$

(C) $\frac{\sqrt{2}}{3}$ (D) $\frac{8}{3}$

128. 7 distinct objects, includinga,b, are distributed into three nonempty unlabelled boxes. How many distributions put a and b in different boxes?

(A) 90 (B) 243

(C) 211 (D) 32

129. a=(1,0,0), a×c=(0,-3,6), a·c=3. Find $|c+a|^2$.

(A) 60 (B) 54

(C) 55 (D) 61

130. For 3x+4y+3=0 and 4x+3y+5=0, find the squared distance from the origin to the angle bisector of positive slope.

(A) 2 (B) $\sqrt{2}$

(C) 4 (D) $\frac{32}{49}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. At fixed temperature and number density, the molecular collision diameter is multiplied by 5 and molecular mass by 25. Find the collision-frequency multiplier in the hard-sphere model.

(A) 5 (B) 1

(C) $\frac{1}{5}$ (D) 25

132. A thin equiconvex lens of focal length 20 cm is cut by a plane through its principal axis into two equal halves. One half is used alone with the same optical axis. Find its focal length, ignoring diffraction and edge effects.

(A) 80 cm (B) 5 cm

(C) 20 cm (D) $20\sqrt{2}$ cm

133. In a thin-foil scattering experiment, most alpha particles are barely deflected, but 24 of 500000 incident particles undergo large-angle scattering. Which paired inference is supported without assuming the Thomson model?

(A) Most atomic volume is positively charged; electrons cause the rare large deflections.

(B) Most atomic volume is nearly empty; positive charge is concentrated in a very small central region.

(C) Most atomic mass is spread uniformly; central charge is negative.

(D) Atoms contain no electrons; positive charge is uniformly distributed.

Chemistry

134. For a reaction over300-600 K, assume temperature-independent ΔH°. The measured natural logarithm ln[K(600)/K(300)] is 5. Using ln(K2/K1)=-(ΔH°/R)(1/T2-1/T1), find ΔH°/R in K, including sign.

(A) -3000 (B) 3000

(C) 1500 (D) 6000

135. Three standard reduction potentials EA,EB,EC satisfy EB-EA=$\frac{2}{5}$V and EC-EB=$\frac{3}{10}$V. For a cell whereCis reduced andAoxidised,the standard potential in V is

(A) $\frac{7}{5}$ (B) $\frac{3}{10}$

(C) $\frac{7}{10}$ (D) $\frac{1}{10}$

136. For neutral HD and D2, take isotope mass numbers H=1, D=2, T=3 and neglect mass defects. Their gases have equal temperature. Which pair gives ($\langle v^2\rangle$_HD/$\langle v^2\rangle$_D2, number of neutrons in one D2 molecule minus that in one HD molecule)?

(A) (4/3, 1) (B) (4/3, 0)

(C) (4/3, 3) (D) (3/4, 1)

Logical Reasoning

137. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). The numerical component follows a constant arithmetic step. Find the missing internal term: FMM4, CKN7, ZIO10, ?, TEQ16, QCR19.

(A) WGP13 (B) 31PGW

(C) XHQ13 (D) VFO13

138. J is the only child of D, and J is a parent of K. D is the maternal grandmother of K. What can be concluded about J's gender?

(A) Male (B) Female

(C) The stated relationships are inconsistent (D) Cannot be determined

139. Five people occupy positions 1 to 5 from left to right, one per position, and wear five different shirt colours: red, blue, green, white and yellow. Charu is 1st from the left. Bina is 2nd from the left. Esha is 3rd from the left. Dev is 4th from the left. Asha is 5th from the left. The white shirt is in position 1. The red shirt is in position 4. The yellow shirt is in position 5. Bina does not wear blue. Which shirt does Bina wear?

(A) White (B) Blue

(C) Green (D) Red

Mathematics

140. The parabola $y=x^2+px+q$ passes through (0,49). The absolute value of its vertex ordinate is minimized over $p\geq 0$. Find p^2+q^2 at the minimum.

(A) 245 (B) 196

(C) 2597 (D) 63

141. Let k make lim as $x\to\infty$ of $[(3x^2+4x+1)/(x+5)-kx]$ finite. Find that finite limit.

(A) -11 (B) 15

(C) 4 (D) 19

142. $F'(x)=\frac{2}{(x+1)(x+3)}$ and $F(0)=2$. Find $F(1)-2F(0)$.

(A) $-2-\log(3)+\log(2)$ (B) $-2+\log\left(\frac{3}{2}\right)$

(C) $\log\left(\frac{3}{2}\right)+2$ (D) $\log\left(\frac{3}{2}\right)$

BITSAT Mock 48

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. 1 moles and 4 moles of the same ideal gas, initially at 200 K and 700 K, are connected in a rigid insulated system. There is no external work. Find equilibrium temperature, assuming constant molar heat capacity.

(A) 700 K
(B) 200 K
(C) 300 K
(D) 600 K

2. A direct vernier has 20 divisions equal to 19 main-scale divisions; 1 MSD is 1 mm. Its signed zero error is $\frac{1}{20}$ mm. For an object the main scale is 9 mm and the 3rd vernier division coincides. Find the corrected reading.

(A) $\frac{163}{20}$ mm
(B) $\frac{91}{10}$ mm
(C) $\frac{3659}{400}$ mm
(D) $\frac{59}{5}$ mm

3. A solid sphere of radius R has uniform density $7\rho_0$. A spherical cavity of radius $R/10$ is cut with its centre at distance $R/9$ from the original centre. Find field magnitude at the cavity centre in units of $\pi G\rho_0 R$.

(A) 84
(B) $\frac{12}{343}$
(C) $\frac{28}{27}$
(D) 12

4. A bob on a string of length 2 m is released from an angle θ to the downward vertical and strikes an identical stationary bob elastically at its lowest point. The second bob hangs from a string of length $\frac{2}{5}$ m and just completes a vertical circle. Find $\cos\theta$.

(A) $\frac{1}{2}$
(B) $\frac{4}{5}$
(C) $\frac{3}{5}$
(D) -1

5. A monochromatic wave of frequency 500000000 Hz enters a linear nondispersive medium with $\varepsilon_r = 4$ and $\mu_r = 1$. Its frequency is unchanged. Find its wavelength in the medium. Take $c = 3 \times 10^8$ m/s.

(A) $\frac{3}{10}$ m
(B) $\frac{3}{5}$ m
(C) $\frac{6}{5}$ m
(D) $\frac{3\sqrt{5}}{25}$ m

6. Two coaxial circular loops each radius 3 m carry equal currents 2 A in the same circulation sense. Their centres are separated by $2R$. Find field at the midpoint in units of μ_0.

(A) $\frac{\sqrt{2}}{12}$ (B) $\frac{\sqrt{2}}{6}$

(C) $\frac{2\sqrt{5}}{75}$ (D) $\frac{\sqrt{2}}{3}$

7. A particle starts from its SHM mean position moving in the positive direction with period 8 s. Find the first time its displacement equals half its positive amplitude.

(A) $\frac{1}{3}$ s (B) $\frac{4}{3}$ s

(C) $\frac{2}{3}$ s (D) 1 s

8. A uniform-density planet has radius R. Gravity at a depth d equals gravity at a height $(4-1)R$ above the surface. Find d/R.

(A) $\frac{15}{16}$ (B) $\frac{63}{64}$

(C) $\frac{1}{16}$ (D) $\frac{3}{4}$

9. A stationary nucleus with mass number 24 and binding energy per nucleon $\frac{38}{5}$ MeV splits into two equal daughters, each having total binding energy $\frac{947}{10}$ MeV. Which statement is correct?

(A) No energy can be released because fission always requires a larger mass number than 24.

(B) The process releases 14 MeV; each daughter releases 7 MeV.

(C) The process absorbs 7 MeV; the daughters have larger binding energy per nucleon.

(D) The process releases 7 MeV; the daughters have larger binding energy per nucleon.

10. A uniform disc of mass 6 kg and radius 5 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) $\frac{1425}{16}\,\mathrm{kg\,m^2}$ (B) $75\,\mathrm{kg\,m^2}$

(C) $\frac{975}{16}\,\mathrm{kg\,m^2}$ (D) $\frac{1125}{16}\,\mathrm{kg\,m^2}$

11. Unpolarised light passes three ideal polarisers with axes at 0°, 60°, and 75° relative to a common reference. Find final intensity divided by incident intensity.

(A) $\frac{\sqrt{3}}{32}+\frac{1}{16}$ (B) $\frac{1}{16}-\frac{\sqrt{3}}{32}$

(C) $\frac{\sqrt{3}}{8}+\frac{1}{4}$ (D) $\frac{\sqrt{3}}{16}+\frac{1}{8}$

12. Current rises linearly from 0 to 5 A over 2 s and then falls linearly to 0 over another 2 s. It remains zero for the next 4 s. Find the average current over the entire 8 s interval.

(A) $\frac{5}{8}$ A (B) $\frac{5}{3}$ A

(C) 5 A (D) $\frac{5}{4}$ A

13. A particle has position-dependent velocity components $\vec{v}=5x\,\hat{i}-5y\,\hat{j}$ in SI units. Given $dx/dt=v_x$ and $dy/dt=v_y$, what is its acceleration magnitude at $(6,3)$ m?

(A) $150\,\mathrm{m/s^2}$ (B) $75\sqrt{5}\,\mathrm{m/s^2}$

(C) $375\sqrt{5}\,\mathrm{m/s^2}$ (D) $15\sqrt{5}\,\mathrm{m/s^2}$

14. 3 kg of water at 30°C is mixed in an insulated vessel with $\frac{378}{167}$ kg of ice at 0°C. Use $c_w = 4200$ J/(kg K), $L_f = 334000$ J/kg. Find the mass of ice remaining at equilibrium.

(A) $\frac{55377}{38410}$ kg (B) $\frac{567}{334}$ kg

(C) $\frac{378}{167}$ kg (D) $\frac{189}{167}$ kg

15. A system has total mass 4 kg, centre-of-mass speed 3 m/s and kinetic energy 20 J measured in the CM frame. Which statement is correct?

(A) The laboratory kinetic energy is 38 J. (B) The laboratory kinetic energy is 20 J.

(C) The laboratory kinetic energy is 18 J. (D) The CM-frame kinetic energy must be zero.

16. A geometrically similar solid structure is enlarged by a factor 2 in every linear dimension. Its material density is reduced by a factor 3. Its own weight is carried by a similarly scaled base. Find base compressive-stress multiplier.

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{4}{3}$ (D) $\frac{8}{3}$

17. A parallel-plate capacitor has vacuum capacitance C_0. A dielectric of constant 6 fills a fraction $\frac{1}{3}$ of the plate area and the entire gap in that region; the remainder stays vacuum. The voltage is fixed. Find the fraction of total stored energy residing in the dielectric-filled region.

(A) $\frac{1}{4}$ (B) $\frac{3}{4}$

(C) 2 (D) 3

18. An infinite resistor ladder starts with a series resistor of 6 ohms, followed by a node having a shunt resistor of 6 ohms to the return rail and then an identical continuation of the whole ladder. Find input resistance.

(A) $3\,\Omega$ (B) $3 + 3\sqrt{5}\,\Omega$

(C) $6\,\Omega$ (D) $-3 + 3\sqrt{5}\,\Omega$

19. A thin straight tube of length 4 m and internal area $\frac{1}{2500}$ m^2 is completely filled with an incompressible liquid of density 600 kg/m^3. It rotates horizontally about one end at angular speed 7 rad/s. Ignoring gravity along the tube, find the pressure difference between its outer and inner ends.

(A) 235200 Pa (B) 33600 Pa

(C) 58800 Pa (D) 78400 Pa

20. Two short electric dipoles each moment 6 C m are located so a distant point P is at axial distance 6 m from one and equatorial distance 6 m from the other. Their fields at P are perpendicular. Find resultant field magnitude in units of k, where $k = 1/(4\pi\varepsilon_0)$.

(A) $\frac{\sqrt{5}}{6}$ (B) $\frac{\sqrt{2}}{36}$

(C) $\frac{\sqrt{5}}{36}$ (D) $\frac{1}{18}$

21. A straight conducting rod of length 5 m rotates in a plane perpendicular to a uniform 7 T field about one end at angular speed 3 rad/s. Find the emf magnitude between its ends.

(A) $\frac{525}{2}$ V (B) 105 V

(C) 525 V (D) $\frac{525}{4}$ V

22. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 30 percent. The measured increase in its absolute temperature is $\frac{1656}{5}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) $\frac{4056}{5}$ K
(B) 552 K
(C) $\frac{1656}{5}$ K
(D) 480 K

23. Charges +5 C at x=0 and +10 C at x=5 m have a nonzero net charge. The coordinate origin is shifted from x=0 to the old coordinate x=5 m, with axis direction unchanged. Find the new signed x-component of electric dipole moment.

(A) 50 C m
(B) 25 C m
(C) −25 C m
(D) −75 C m

24. A particle moves with constant acceleration. It travels 23/2 m in the (2)nd second and 33/2 m in the 3rd second. What was its initial velocity?

(A) $\frac{33}{2}$ m/s
(B) 4 m/s
(C) 5 m/s
(D) $\frac{13}{2}$ m/s

25. A network is defined by $P = \mathrm{NAND}(A, B)$, $Q = \mathrm{NOR}(A, P)$ and $Y = \mathrm{AND}(Q, \overline{B})$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 0$ and $A = 0$?

(A) 0
(B) 3
(C) 1
(D) 2

26. In a thin-foil scattering experiment, most alpha particles are barely deflected, but 39 of 500000 incident particles undergo large-angle scattering. Which paired inference is supported without assuming the Thomson model?

(A) Most atomic volume is positively charged; electrons cause the rare large deflections.

(B) Most atomic mass is spread uniformly; central charge is negative.

(C) Most atomic volume is nearly empty; positive charge is concentrated in a very small central region.

(D) Atoms contain no electrons; positive charge is uniformly distributed.

27. For steady laminar flow through a capillary, the radius is multiplied by 2 and length by 2. The volume flow rate must remain unchanged and viscosity is fixed. By what factor must the pressure difference be multiplied?

(A) 1
(B) $\frac{1}{8}$
(C) 32
(D) $\frac{1}{2}$

28. A 1 kg block moves along a horizontal rough surface from $x = 0$ to $x = 5$ m under a force $F(x) = 3x$ N parallel to motion. With $\mu_k = 1/5$ and $g = 10$ m/s^2, find the work done by the net force. The block has enough initial speed to keep moving forward.

(A) $\frac{95}{2}$ J
(B) $\frac{11}{2}$ J
(C) 65 J
(D) $\frac{55}{2}$ J

29. An object moves slowly along the axis toward a concave mirror. At an instant its distance from the mirror is 40 cm; the mirror focal-length magnitude is 10 cm. Find the image-speed magnitude divided by object-speed magnitude at that instant.

(A) $\frac{1}{9}$ (B) $\frac{1}{3}$

(C) 9 (D) $\frac{1}{25}$

30. A 6 V Zener regulator feeds a constant-current load of 2 mA. Supply varies from 14 to 16 V. Regulation requires Zener current at least 2 mA, and its safe maximum is 9 mA. What is the largest permissible series resistance? Assume ideal regulation and give resistance in Ω.

(A) 2500 Ω (B) $\frac{10000}{11}$ Ω

(C) 2000 Ω (D) 4000 Ω

Chemistry

31. An iron-ion inventory of 1/25 mol contains only Fe2+ and Fe3+. Complete externally driven reduction to metal, with no side reaction, consumes 13/125 mol electrons. What initial fraction of iron was Fe3+?

(A) 1/5 (B) 3/10

(C) 13/15 (D) 3/5

32. 2-isopropylaniline is diazotised at0-5°C,treated with CuCN,then its nitrile is fully hydrolysed and its benzylic side chain fully oxidised. Heating the resulting diacid gives which characteristic cyclic anhydride?

(A) maleic anhydride (B) phthalic anhydride

(C) isophthalic acid (D) terephthalic acid

33. For octa-1,3,5,7-tetraene, CH2=CH-CH=CH-CH=CH-CH=CH2, count distinct configurational geometrical isomers. Ignore conformers and photochemical interconversion. Where reversing the entire carbon chain identifies two E/Z assignments, count them once.

(A) 1 (B) 8

(C) 4 (D) 3

34. For benzaldehyde with concentrated NaOH followed by complete acidification,which product or mechanistic conclusion is correct under the stated conditions?

(A) phenol and methanoic acid (B) benzyl alcohol and benzoic acid

(C) benzyl alcohol and sodium benzoate as the final fully acidified pair (D) benzoic acid only

35. A closed system follows states A→B→C→A. For A→B, heat q=570 J and work on the system w=-270 J. For C→A, q=-385 J and w=75 J. For B→C, q=-280 J. Find work on the system on B→C in J.

(A) 290 (B) -10

(C) -290 (D) 590

36. 2-methylpropan-2-ol is heated under acid-catalysed dehydration conditions. Which major product follows removal of water while preserving the stated carbon framework?

(A) but-1-ene (B) 2-methylpropanal

(C) but-2-ene (D) 2-methylpropene

37. For closed-system paths A→B→C, q_AB=600 J, w_AB=-280 J, q_BC=-300 J and w_BC=320 J, with work defined on the system. A direct C→A path has w=60 J. What is its heat q in J?

(A) -400 (B) -720

(C) 400 (D) -80

38. For neutral HD and HT, take isotope mass numbers H=1, D=2, T=3 and neglect mass defects. Their gases have equal temperature. Which pair gives ($\langle v^2\rangle$_HD/$\langle v^2\rangle$_HT, number of neutrons in one HT molecule minus that in one HD molecule)?

(A) (4/3, 1) (B) (4/3, 0)

(C) (3/4, 1) (D) (4/3, 3)

39. A salt MF_2 dissolves only as $M^{2+}+2F^-$. Its pure-water molar solubility is $\frac{1}{10000}$M. A second saturated solution has imposed free $F^-=\frac{1}{1000}$M;ignore further F^- from dissolution. Dissolved M^{2+} concentration in that second solution is

(A) $\frac{1}{1000000}$ (B) $\frac{1}{100000}$

(C) $\frac{1}{2500000}$ (D) $\frac{1}{250000}$

40. Which combined classification/reactivity description is correct for 2-methylpropan-1-ol?

(A) a secondary alcohol whose product is butan-2-one (B) a tertiary alcohol whose ketone product is acetone

(C) a phenol whose oxidation is ring nitration (D) a primary alcohol whose aldehyde oxidation product is branched

41. For CH2=CH-CHO,compare contributors I:CH2=CH-CH=O; II:+CH2-CH=CH-O-; III:-CH2-CH=CH-O+. All retain the same sigma skeleton. Which decreasing contribution order is most appropriate?

(A) I > II > III (B) I > III > II

(C) II > I > III (D) III > II > I

42. NF3 and NH3 are both pyramidal. Which explanation is consistent with the smaller molecular dipole of NF3?

(A) Any three polar bonds must give the same molecular dipole.

(B) Fluorine's electronegativity alone guarantees the larger total dipole regardless of vector addition.

(C) The N-F bond-dipole resultant opposes the lone-pair-related contribution more strongly.

(D) A pyramidal molecule is necessarily nonpolar.

43. acetophenone reacts with NH2OH. Which product correctly reflects nucleophilic addition or addition-elimination at its carbonyl rather than a different reaction class?

(A) benzoic acid (B) acetophenone oxime

(C) phenylacetonitrile (D) 1-phenylethanol

44. A phenolic compound has molar mass 94 g/mol. A $\frac{47}{10}$ g sample is completely O-acetylated without ring substitution,and dry product mass is $\frac{34}{5}$ g. How many acetylatable OH groups occur per molecule? Each OH→OCOCH3 replacement adds42 g/mol.

(A) 1 (B) 3

(C) 2 (D) 4

45. For H2S2O8, take H=+1 and each sulfur=+6. Oxygen atoms are only ordinary oxide(-2) or peroxide(-1). Which pair gives (number of peroxide oxygen atoms, number of O-O peroxide links) per formula entity?

(A) (2, 1)
(B) (0, 0)
(C) (1, 1)
(D) (2, 2)

46. Compare the energy for removing one more electron from Cr^+ and from Mn^+. Which member is expected to resist that removal more strongly on the stated shell-stability basis?

(A) The ion with the smaller written charge always has the larger removal energy.

(B) They must be identical because both are positive ions.

(C) Cr^+

(D) Mn^+

47. For CH3COCH2COOCH3 and CH3COOCH3,which structural-reactivity comparison is correct under ordinary enolate/aldol conditions?

(A) the second has no alpha H
(B) neither can form an enolate
(C) the first has a more acidic methylene between two carbonyl groups
(D) both have identical two-carbonyl stabilisation

48. Two oxides of X contain 52 g X combined respectively with 16 g O and 24 g O. Atomic masses are X=52, O=16. Which empirical-formula conclusion jointly follows from mole ratios and the law of multiple proportions?

(A) Both are XO because X masses are equal.

(B) First empirical formula XO; second empirical formula X2O3; fixed-X oxygen mass ratio second:first=3/2.

(C) No formula can be inferred even though both atomic masses and both compositions are supplied.

(D) First oxide X2O3; second oxide XO; fixed-X oxygen mass ratio second:first=2/3.

49. For 2-bromo-2-methylpropane,which structural classification is correct when the actual halogen-bearing carbon and molecular formula are examined?

(A) a primary chiral halide
(B) a secondary meso halide
(C) a tertiary halide with no stereogenic carbon
(D) a vinylic halide

50. 1,2-dibromopropane is taken through excess NaNH2;water;HgSO4/H2SO4/H2O,with suitable separate-stage work-ups. Identify the final organic product without adding or deleting carbon atoms except when the reagent itself introduces carbon.

(A) propan-1-ol
(B) propanal
(C) propanone
(D) butan-2-one

51. Classify,in order,the carbon-halogen environments in CH3CH(Cl)CH3,CH2=C(Cl)CH3, C6H5CH(Cl)CH3,CH2=CHCH(Cl)CH3. Classify by the carbon directly bonded to Cl,not by whether a double bond appears somewhere in the molecule.

(A) alkyl,allyl,aryl,vinyl
(B) alkyl,vinyl,benzyl,allyl
(C) vinyl,alkyl,allyl,benzyl
(D) allyl,alkyl,vinyl,benzyl

52. A dry mixture contains only anhydrous CaCO3 and SrCO3. Use the ordinary Group2 carbonate stability trend. In an idealised selective-heating window, the less thermally stable carbonate decomposes completely by MCO3→MO+CO2, while the more stable carbonate remains unchanged. No other mass loss occurs. This is the stated model, not a specified real temperature. Atomic masses: Ca=40, Sr=88, C=12, O=16. The selective stage gives 14/25 L CO2; subsequent stronger heating to decompose all remaining carbonate brings the cumulative CO2 volume to 112/125 L. Both dry volumes are at STP; use22.4 L/mol, with complete collection. What was the initial mole fraction of SrCO3?

(A) 3/8
(B) 185/236
(C) 111/236
(D) 5/8

53. A Hinsberg separation is applied to benzylamine. Which reagent identity and class-dependent outcome are correct?

(A) benzenesulphonyl chloride,C6H5SO2Cl;every amine gives an alkali-soluble N-H sulphonamide
(B) benzenesulphonyl chloride,C6H5SO2Cl;a sulphonamide with N-H,soluble in alkali as its anion
(C) benzoyl chloride,C6H5COCl;a sulphonamide with N-H,soluble in alkali as its anion
(D) benzenesulphonic acid,C6H5SO3H;all classes give the same neutral product

54. For hypochlorous acid, which pairing of connectivity and chlorine oxidation state is correct?

(A) H-Cl-O; chlorine +3.
(B) H-O-Cl; chlorine +1.
(C) H-O-Cl; chlorine -1.
(D) H-Cl-O; chlorine +7.

55. A CO_2 sample is absorbed in 100.0 mL $Ba(OH)_2$ containing $\frac{1}{20}$ mol base. All $BaCO_3$ precipitate is filtered off; the filtrate is diluted to 200.0 mL. A 50.0 mL aliquot requires 20.0 mL of 0.500 M HCl for neutralisation. No bicarbonate forms. The absorbed CO_2 volume at STP (22.4 L mol^{-1}), in mL, is

(A) 1120
(B) 224
(C) 672
(D) 336

56. A sample of 1-bromopropane is treated with alcoholic KOH,heat. Which isolated product is consistent with the reagent limitation and carbon-skeleton change?

(A) propan-1-ol
(B) cyclopropane
(C) propyne
(D) propene

57. A sulfuric-acid stock of density 6/5 g/mL is diluted by taking20.0 mL to40.0 mL. The final molarity is 3/2 M. H2SO4 molar mass is98 g/mol. What is the mass fraction of H2SO4 in the stock?

(A) -49/200
(B) 49/400
(C) 49/200
(D) 49/100

58. In an idealised Ostwald-process material balance, $\frac{7}{100}$ mol NH_3 is fed.90.0% of its nitrogen reaches the NO/NO_2 intermediate stream, and80.0% of that stream nitrogen is recovered as HNO_3. Find HNO_3 amount in mol. Each nitrogen atom is tracked once; recycled intermediates are not double-counted.

(A) $\frac{63}{1000}$
(B) $\frac{7}{125}$
(C) $\frac{119}{2000}$
(D) $\frac{63}{1250}$

59. For A(g)⇌2B(g),the old equilibrium has Kp=4bar. Compression and a temperature change are then made before reaction can occur. Immediately afterwards,pA=2bar,pB=4bar,and the new-temperature Kp=10bar. Which conclusion is correct?

(A) Qp=8bar,so the reaction moves forward;the old Kp cannot decide the new-temperature direction.

(B) Compression always forces reverse motion,irrespective of the new-temperature Kp.

(C) Qp=8bar,so the reaction must move backward using the old Kp=4bar.

(D) Qp=2bar because the product pressure is divided by its coefficient instead of squared.

60. For the same nonrelativistic ion starting from rest,an accelerator changes deBroglie wavelength from $\lambda$to$\lambda/5$. Ion charge and mass are unchanged. Factor by which accelerating voltage must change is

(A) $\frac{1}{5}$ (B) 25

(C) $\frac{1}{25}$ (D) 5

English Proficiency

61. Read the sentence: The ambitious plan sought a major expansion rather than a small repair. Choose the opposite in meaning of 'ambitious' as used in the sentence.

(A) expensive (B) carefully timed

(C) modest in scope (D) demanding

62. She supplies the funds that make another student's education possible, rather than receiving the grant herself. What is she?

(A) trustee (B) benefactor

(C) beneficiary (D) borrower

63. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. one of its assumptions. Q. when new information changes R. it may still need revision S. However carefully a plan is prepared,

(A) R-S-Q-P (B) S-R-Q-P

(C) P-R-Q-S (D) Q-S-R-P

64. Read the sentence: The reply was equivocal, allowing both an approving and a disapproving interpretation. Choose the closest meaning of 'equivocal' as used in the sentence.

(A) hesitant (B) deliberately deceptive

(C) contradictory in every detail (D) ambiguous

65. Read the sentence: The arrangement remained stable despite repeated small disturbances. Choose the opposite in meaning of 'stable' as used in the sentence.

(A) secure (B) balanced

(C) unchanged (D) unstable

66. The study examines how organisms interact with one another and with their surroundings. Which field does this describe?

(A) anatomy (B) astronomy

(C) ecology (D) etymology

67. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. Each borrower was asked to return seeds from the plants that grew. B. A neighbourhood library started lending packets of seeds as well as books. C. In this way, a small initial collection became a shared renewable resource. D. Those returned seeds were checked and packed for the next season.

(A) A neighbourhood library started lending packets of seeds as well as books.

(B) In this way, a small initial collection became a shared renewable resource.

(C) Each borrower was asked to return seeds from the plants that grew.

(D) Those returned seeds were checked and packed for the next season.

68. Read the sentence: The repairs produced a tangible improvement: the doors now closed properly. Choose the closest meaning of 'tangible' as used in the sentence.

(A) large and permanent

(B) concrete and observable

(C) unexpected but welcome

(D) financially profitable

69. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The guide will show the original sketch only after explaining the replica.

(A) The original sketch will be shown by the guide only after the guide has explained the replica.

(B) Only the guide will show the original sketch and the replica.

(C) The replica will be shown by the guide before the original sketch is explained.

(D) The original sketch has been shown before the replica was explained.

70. Complete: The new record casts _____ on the earlier explanation, but does not by itself disprove it.

(A) doubt

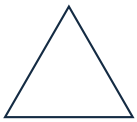
(B) a verdict

(C) a decision

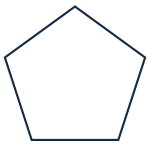
(D) a proof

Logical Reasoning

71. A changes into B by adding the same number of polygon sides on every application. Which input shape would change into C?



A

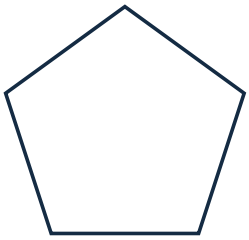


B

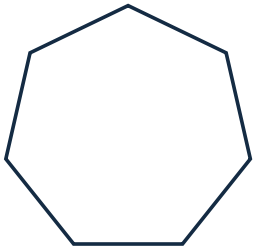


C

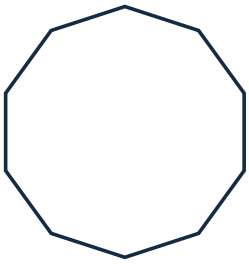
(A) Pattern M



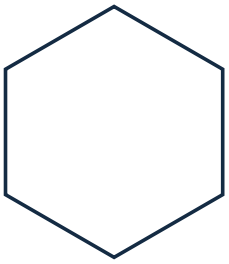
(B) Pattern Q



(C) Pattern U



(D) Pattern N



72. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 6, 12, 21, ?, 51, 102, 111.

(A) 51 (B) 42

(C) 33 (D) 30

73. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, STONE is coded as XYTSJ. Which word is coded as QFRU?

(A) VKWZ (B) LAMP

(C) KZLO (D) MBNQ

74. Three pairs contain exactly one string instrument and one wind instrument. Which pair does not?

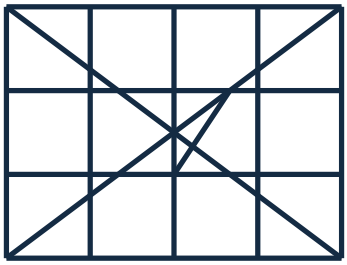
(A) Cello / Trumpet (B) Guitar / Clarinet

(C) Ukulele / Saxophone (D) Sitar / Violin

75. Vijay points to a photograph and says, "This person is the daughter of my father's only daughter." Vijay is male, and the photographed person is female. How is the photographed person related to Vijay?

(A) Niece
(B) Son
(C) Brother
(D) Daughter

76. How many triangles are formed by all the drawn segments?

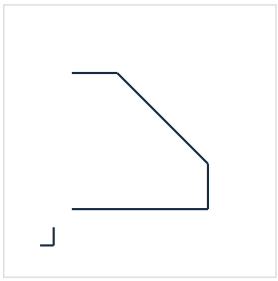


(A) 37
(B) 47
(C) 51
(D) 88

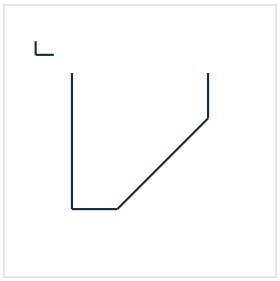
77. Consider only these statements; universal statements alone do not imply existence. Some editors are designers; No hikers are editors. Conclusions: I. No editors are designers. II. Some designers are not hikers. Which conclusions follow necessarily?

(A) Only I
(B) Both I and II
(C) Only II
(D) Neither I nor II

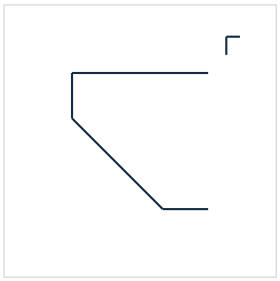
78. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1



2

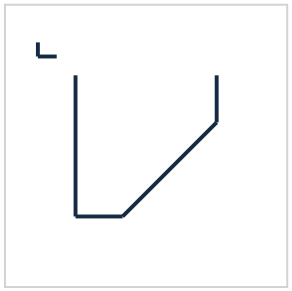


3

(A) Pattern Z



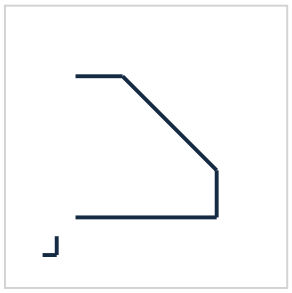
(B) Pattern Q



(C) Pattern L



(D) Pattern R



79. Consider only these statements; universal statements alone do not imply existence. Some hikers are campers; Some campers are hikers; No hikers are farmers. Conclusions: I. No campers are hikers. II. Some campers are not farmers. Which conclusions are individually possible without contradicting the statements?

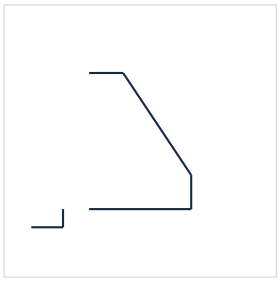
(A) Only I

(B) Both I and II

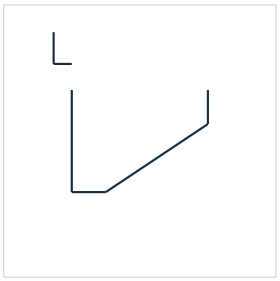
(C) Only II

(D) Neither I nor II

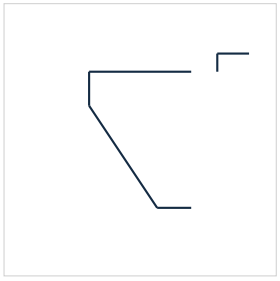
80. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right



bottom left

?

bottom right

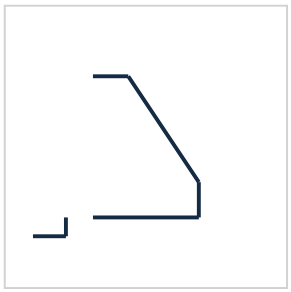
(A) Pattern P



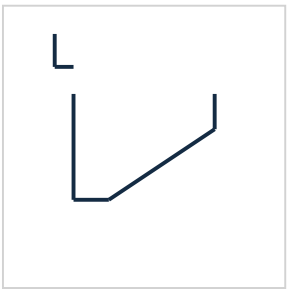
(B) Pattern Z



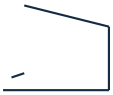
(C) Pattern Y



(D) Pattern R

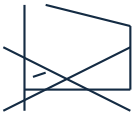


81. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

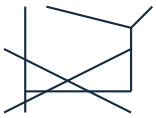




host 1



host 2



host 3



host 4

(A) 1
(B) 2
(C) 4
(D) 3

82. Preserve the exact relation and its order: Courageous : Brave :: ? : Rare. Choose the missing term.

(A) Scarce
(B) Silent
(C) Rapid
(D) Begin

83. Each letter is assigned either its forward alphabet position or its reverse alphabet position, followed by the same integer offset. A has code 32 and Z has code 7. Decode the separated values 13, 15, 32, 24, 19.

(A) TRAIN
(B) NIART
(C) USBJO
(D) SQZHM

84. The whole has 14 unit positions. Part A covers the first 4; part B covers 3 positions, overlapping A in exactly 2 positions as shown. How many positions remain uncovered by either part?



whole



part A positions



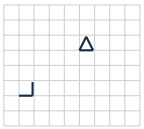
part B positions

(A) 10
(B) 7
(C) 9
(D) 5

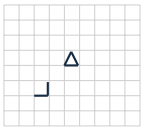
85. A word is coded as the sum of its forward alphabet positions plus 5 times its length. The code of LAM? is 62. Which letter replaces the question mark?

(A) U
(B) O
(C) Q
(D) P

86. The L-shaped unit and the triangular unit each undergo their own constant translation between frames. Neither turns or changes size. Select the next complete framed arrangement.



1

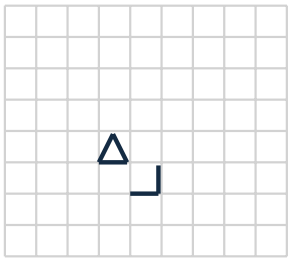


2



3

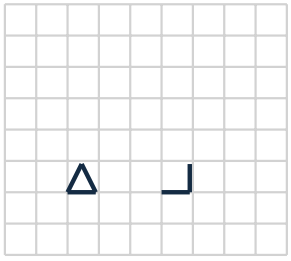
(A) Pattern L



(B) Pattern Y



(C) Pattern W



(D) Pattern X



87. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: BAO, FXS, ?, NRA, ROE, VLI.

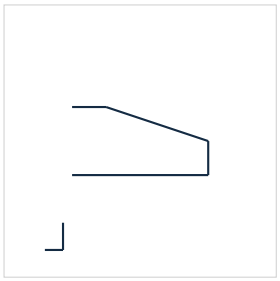
(A) KVX

(B) WUJ

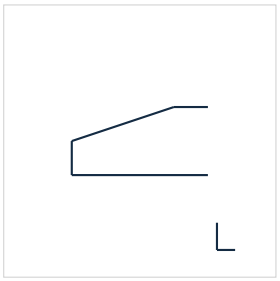
(C) ITV

(D) JUW

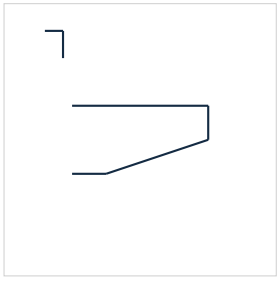
88. Select the missing bottom-right tile. Tiles reflect across the shared horizontal and vertical centre lines; there is no turning.



top left



top right

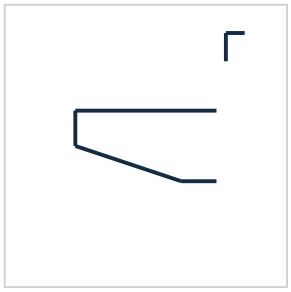


bottom left

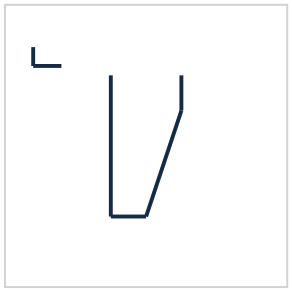
?

bottom right

(A) Pattern M



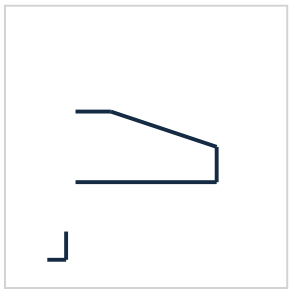
(B) Pattern T



(C) Pattern L



(D) Pattern N

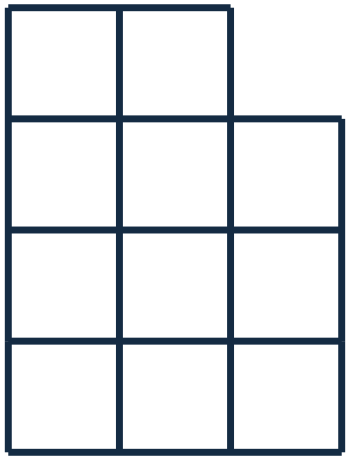


89. In a row, Asha is 9th from the left and 34th from the right. Bina is 9th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 20 (B) 21

(C) 22 (D) 27

90. How many rectangles, including squares are formed by all the drawn segments?



(A) 49
(B) 47
(C) 27
(D) 48

Mathematics

91. The line Re(z)=2 divides the disc $|z|\leq 4$. Find larger-segment area minus smaller-segment area.

(A) $4\sqrt{3}+\frac{32\pi}{3}$
(B) $-\frac{16\pi}{3}-8\sqrt{3}$
(C) $8\sqrt{3}+\frac{16\pi}{3}$
(D) $-4\sqrt{3}+\frac{16\pi}{3}$

92. $P(A)=\frac{11}{24}$, $P(B)=\frac{7}{24}$, $P(A\cap B)=\frac{7}{48}$. Find P(neither A nor B).

(A) $\frac{19}{48}$
(B) $\frac{29}{48}$
(C) $\frac{221}{576}$
(D) $\frac{13}{24}$

93. $|U|=190$, $|A|=49$, $|B|=46$, $|A\cap B|=19$. Find the number of elements belonging to both A,B or to neither.

(A) 171
(B) 133
(C) 57
(D) 114

94. Two different numbers are chosen uniformly from {1,...,11}. Let X be their maximum. Find Var(X).

(A) $\sqrt{6}$
(B) 8
(C) 70
(D) 6

95. Find the total length of $\theta\in[0,2\pi)$ satisfying $(\text{floor}(3\sin\theta)-1)(\text{floor}(3\sin\theta)-2)\leq 0$.

(A) π
(B) $2\,\text{asin}\left(\frac{1}{3}\right)$
(C) $\pi-\text{asin}\left(\frac{1}{3}\right)$
(D) $\pi-2\,\text{asin}\left(\frac{1}{3}\right)$

96. All tangents common to $x^2/36+y^2/9=1$ and $x^2+y^2=\frac{126}{5}$ touch the ellipse at points P_i. Find $\Sigma(P_i, x)^2$ over the four tangency points.

(A) $\frac{504}{5}$
(B) $\frac{864}{7}$
(C) $\frac{216}{7}$
(D) $\frac{36}{7}$

97. z=9+3i. Find 1/z+1/conjugate(z).

(A) $\frac{1}{5}$ (B) $\frac{1}{15}$

(C) $\frac{1}{18}$ (D) 18

98. How many solutions in $0 \le x < 7\pi$ satisfy sin 2 x=sin x and cos x>sin x?

(A) 14 (B) 3

(C) 7 (D) 10

99. A line through P=(7,5) has direction (3,4). It meets 3x+4y=91 at Q. Find $|PQ|^2$.

(A) 10 (B) 50

(C) 100 (D) 4

100. For f(x)=x(x-6)(x-11), find the sum of all $c \in (0,11)$ satisfying $f'(c)=0$.

(A) $\frac{34}{3}$ (B) 66

(C) 17 (D) $\frac{17}{3}$

101. An AP has a=3,d=3; its first n terms sum to 84. Find a_(n+1).

(A) 27 (B) 12

(C) 7 (D) 24

102. P=(4,3,3) has foot F on the line r=t(1,0,0). Find $|OP|^2-|OF|^2$, O being the origin.

(A) 16 (B) $3\sqrt{2}$

(C) 18 (D) 34

103. A vector has direction ratios (9,8,3). Find the sum of the squares of its y,z direction cosines.

(A) 73 (B) $\frac{81}{154}$

(C) $\frac{\sqrt{11242}}{154}$ (D) $\frac{73}{154}$

104. An equivalence relation has classes of sizes 10,2,2. How many unordered pairs of distinct related elements are there?

(A) 47 (B) 94

(C) 36 (D) 14

105. Let $f(x)=|x-8|/(x-18)$ for $x \ne 18$. Within $0<x<28$, what is the total length of the open intervals on which $f'(x)<0$?

(A) 10 (B) 20

(C) 18 (D) 27

106. Evaluate lim as x→0 of $[(1+3x)/(1-4x)]^{(1/x)}$.

(A) e^7 (B) e^3

(C) e^{-1} (D) 7

107. Find the area between y=(x-9)(x-11) and the x-axis for $0 \le x \le 14$.

(A) $\frac{1030}{3}$ (B) $-\frac{1022}{3}$

(C) $\frac{1022}{3}$ (D) $\frac{515}{3}$

108. Four observations have $\Sigma(x-9)^2=254$, $\Sigma(x-11)^2=254$. Find their population variance.

(A) $\frac{5\sqrt{10}}{2}$ (B) $\frac{125}{2}$

(C) 250 (D) $\frac{127}{2}$

109. For the ellipse $x^2/64+y^2/25=1$, the chord of contact from P is $16\,x/64+5\,y/25=1$. Find the square of the tangent length from P to the auxiliary circle $x^2+y^2=64$.

(A) $\sqrt{217}$ (B) 256

(C) 345 (D) 217

110. A convergent GP has first term 7, ratio $\frac{1}{6}$. Find the sum of all terms after its first 7 terms.

(A) $\frac{7}{1399680}$ (B) $\frac{7}{233280}$

(C) $\frac{7}{38880}$ (D) $\frac{7}{279936}$

111. a=(11,7,0), b=(7,-11,5). Find $|a+b|^2-|a-b|^2$.

(A) −25 (B) 0

(C) −50 (D) 365

112. Positivea,b satisfy log(a+b)=log(6)+[log a+log b]/2 in one valid base. Find $(a^6+b^6)/(a^3b^3)$.

(A) 46656 (B) 39100

(C) 39236 (D) 39202

113. 12 distinct people sit at labelled round tables of sizes 8,4. Two specified people must be at different tables, and the first specified person must be at the 8-seat table. Rotations are identical. Count the seatings.

(A) 14968800 (B) 116121600

(C) 3628800 (D) 7257600

114. The normal of slope 6 to $y^2=28x$ touches at P. Find the distance from P to the focus.

(A) 67081 (B) 259

(C) 245 (D) 252

115. Evaluate $\int_0^6 x^2dx+\int_0^{36}$sqrt(y)dy.

(A) 72 (B) 144

(C) 216 (D) 36

116. Evaluate $\Sigma\, j/2^j$ for j=1,...,8.

(A) $\frac{503}{256}$ (B) $\frac{1}{32}$

(C) 2 (D) $\frac{251}{128}$

117. α,β are roots of x^2-10x+24=0. The monic polynomial with roots α+1,β+1 is x^2-Ax+B. Find B.

(A) 12 (B) 25

(C) 35 (D) 26

118. y′-y/x=1 for x>0, y(1)=2. Find y(11)/11-y(1).

(A) $2 + \log(11)$ (B) $-\log(11)$

(C) $22 + 11\log(11)$ (D) $\log(11)$

119. Find the sum of the squares of all positive real solutions of |x-3|=4.

(A) 50 (B) 49

(C) 7 (D) 54

120. Event E has likelihood $\frac{7}{9}$ under source A and $\frac{2}{5}$ under source B. Its total probability is $\frac{71}{135}$. If the source is A with probability r, find r.

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{71}{135}$ (D) $\frac{7}{9}$

121. In $(1+x)^7$, find the ratio of the coefficient of x^6 to the coefficient of x^5.

(A) $\frac{1}{3}$ (B) $\frac{3}{5}$

(C) 3 (D) 2

122. How many integers k make 27sin x+36cos x+5=k have exactly two distinct solutions in $0 \le x < 2\pi$?

(A) 89 (B) 44

(C) 90 (D) 91

123. $A = \begin{bmatrix} 7 & 1 \\ 0 & 7 \end{bmatrix}$. Find $\det(A^T A)$.

(A) 49 (B) 98

(C) 2401 (D) -2401

124. Group A has 8 observations, mean 16 and population SD 2. Group B has 9 observations, mean 21 and population SD 3. Find the population variance of the combined data.

(A) $\frac{113}{17}$ (B) $\frac{538}{17}$

(C) $\frac{13}{2}$ (D) $\frac{3721}{289}$

125. $P(A) = 5/9$, $P(B) = 1/6$, and $P(A \cap B) = 1/12$. Find the probability that exactly one of A, B occurs.

(A) $\frac{5}{9}$ (B) $\frac{23}{36}$

(C) $\frac{1}{12}$ (D) $\frac{13}{18}$

126. O=(0,0,0), A=(8,0,0), B=(1,3,0). Find the squared area of triangle OAB.

(A) 12 (B) 576

(C) 288 (D) 144

127. For $(1+x)^9$, find the sum of the coefficients of all terms whose exponents are divisible by 3.

(A) 84 (B) 170

(C) 171 (D) 512

128. Evaluate the sum of r(r-1)C(11,r)2^r over all even integers r with $0 \leq r \leq 11$.

(A) 8660520 (B) 4330040

(C) 4979568 (D) 4330480

129. A rectangle has perimeter 48. Its length exceeds its width by at least 6. Find its maximum area.

(A) 36 (B) 108

(C) 135 (D) $3\sqrt{15}$

130. The tangent to $y = x^3 - 47x$ at $x = 7$ meets the two coordinate axes at A and B. Find the area of triangle OAB, where O is the origin.

(A) $\frac{117649}{25}$ (B) $\frac{4802}{3}$

(C) $\frac{117649}{50}$ (D) $\frac{117649}{100}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. An ideal heater of power 3000 W supplies heat to 4 kg of ice initially at 0°C in an insulated vessel for $\frac{1504}{3}$ s. Find final temperature after complete melting. Use $L_f = 334000$ J/kg and $c_w = 4200$ J/(kg K); neglect vessel heat capacity.

(A) $\frac{21}{167}$ degree C

(B) 10 degree C

(C) $\frac{1045}{21}$ degree C

(D) 40 degree C

132. A plane wave in vacuum travels along $+x$. At one instant its electric field is $420\hat{k}$ V/m. If the electric amplitude is 2 times this instantaneous value, which gives the instantaneous magnetic field? Use $c = 3 \times 10^8$ m/s.

(A) $-140 \times 10^{-8}\hat{j}$ T

(B) $-280 \times 10^{-8}\hat{j}$ T

(C) $+140 \times 10^{-8}\hat{j}$ T

(D) $+140 \times 10^{-8}\hat{i}$ T

133. A sinusoidal source of peak $\frac{117}{10}$ V feeds a back-to-back series pair of identical Zeners through a 1 kΩ current-limiting resistor. The output is across the diode pair. Each Zener has reverse breakdown 5 V and forward drop 0.7 V. What is the peak-to-peak output voltage? Use the piecewise constant-drop model.

(A) $\frac{57}{10}$ V

(B) $\frac{64}{5}$ V

(C) $\frac{57}{5}$ V

(D) $\frac{43}{5}$ V

Chemistry

134. At 1 atm compare two separate pairs: (n-pentane, neopentane) and (1-propanol, methoxyethane). A student selects neopentane for the first higher-boiling substance because its formula mass is unchanged, and 1-propanol for the second. Which assessment is consistent with intermolecular interactions?

(A) Both selections correct.

(B) Both selections wrong.

(C) First selection wrong; second selection correct.

(D) First selection correct; second selection wrong.

135. Why can a many-electron atom have different3s and3d orbital energies even though a hydrogenic model groups them by n=3?

(A) Both orbitals have the same angular momentum.

(B) Different penetration and shielding make the many-electron effective potential different for the two subshells.

(C) A3d orbital has a principal quantum number5.

(D) Pauli requires every orbital with the same n to have identical energy in every atom.

136. Use octet Lewis contributors for $CO3^{2}$- and HNO3; for SO2 exclude expanded-octet contributors. For HCOO-, HCO3- and HNO3, average only the two nonprotonated central-atom-O bonds. For other species average all central-atom bonds. Find (average bond order of the first minus that of the second, sum of total lone-pair counts in one contributor of each). A lone pair on any atom is counted once.

(A) (-1/6, 15)

(B) (17/6, 15)

(C) (-1/6, 16)

(D) (-1/6, 14)

Logical Reasoning

137. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). The numerical component follows a constant arithmetic step. Find the missing internal term: ZQK9, ARM13, BSO17, ?, DUS25, EVU29.

(A) CTQ21 (B) 12QTC

(C) DUR21 (D) BSP21

138. A community register uses the following three categories: editors, guides, farmers. Its complete relevant assertions are in this short paragraph; do not import everyday assumptions. Some editors are guides. Some guides are editors. No editors are farmers. Conclusions: I. No guides are farmers. II. Some guides are not farmers. Which conclusions follow necessarily?

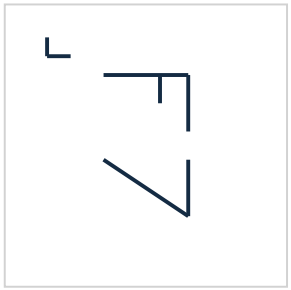
(A) Only I (B) Neither I nor II

(C) Only II (D) Both I and II

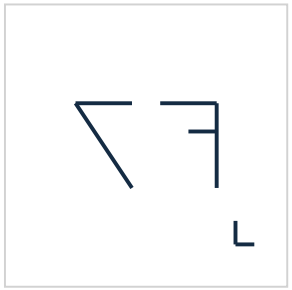
139. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



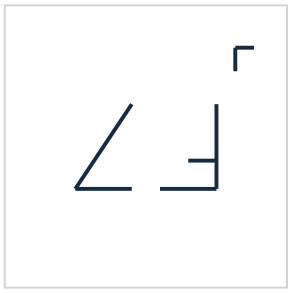
(A) Pattern K



(B) Pattern U



(C) Pattern X



(D) Pattern Y



Mathematics

140. ForA={1,...,10}, how many(a,b)∈A×A have a+b even,a<b and b-a≥4?

(A) 24 (B) 12

(C) 45 (D) 6

141. How many three-element subsets of {1,2,...,5} contain both an odd and an even number?

(A) 10 (B) 54

(C) 9 (D) 6

142. Evaluate the product of $\cos(2^j\pi/65)$ forj=0,...,4.

(A) $\frac{\sin\left(\frac{32\pi}{65}\right)}{32\sin\left(\frac{\pi}{65}\right)}$

(B) $\frac{\sin\left(\frac{32\pi}{65}\right)}{32\sin\left(\frac{\pi}{65}\right)}+1$

(C) $-1+\frac{\sin\left(\frac{32\pi}{65}\right)}{32\sin\left(\frac{\pi}{65}\right)}$

(D) $\frac{\sin\left(\frac{32\pi}{65}\right)}{16\sin\left(\frac{\pi}{65}\right)}$

BITSAT Mock 49

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. An active transistor has $\alpha = \frac{49}{50}$. Its collector resistor is 2 kΩ and the measured collector-resistor drop is 14 V. Neglect leakage. What base current in mA is required?

(A) $\frac{1}{7}$ mA (B) $\frac{7}{50}$ mA

(C) 350 mA (D) $\frac{50}{343}$ mA

2. A cell of emf 9 V and internal resistance 5 ohms drives a variable purely resistive load. Find maximum power delivered to the load.

(A) $\frac{9}{5}$ W (B) $\frac{81}{20}$ W

(C) $\frac{81}{5}$ W (D) $\frac{9}{20}$ W

3. A uniform disc of mass 6 kg and radius 3 m has a circular hole of radius $R/2$ cut out. The hole centre is $R/2$ from the original disc centre. Find the remaining plate's moment of inertia about the original central perpendicular axis.

(A) $\frac{81}{4}$ kg m^2 (B) $\frac{405}{16}$ kg m^2

(C) $\frac{351}{16}$ kg m^2 (D) 27 kg m^2

4. Three long parallel wires are equally spaced by 5 m along a straight line. Outer wires each carry 7 A in the same direction; the middle carries 6 A in the opposite direction. Find the net force per unit length on the middle wire.

(A) 0 N/m (B) $\frac{49}{20\pi}$ N/m

(C) $\frac{21}{5\pi}$ N/m (D) $\frac{21}{25\pi}$ N/m

5. Photons of energy 14 eV eject electrons from a surface. The fastest electrons enter a uniform magnetic field perpendicular to their velocities. In this field an electron with kinetic energy 2 eV has orbit radius r. The observed maximum orbit radius is $2r$. What is the surface work function? Neglect recoil.

(A) 10 eV (B) 22 eV

(C) $\frac{3}{2}$ eV (D) 6 eV

6. A circular parallel-plate capacitor of plate radius 3 cm is charged at 38 mA. Neglect fringing. What displacement current passes through the annular part of the gap between radii 2 cm and 3 cm, centred on the plate axis?

(A) $\frac{38}{3}$ mA (B) $\frac{190}{3}$ mA

(C) $\frac{190}{9}$ mA (D) $\frac{152}{9}$ mA

7. Three blocks of masses 1, 1 and 4 kg are stacked from top to bottom. A horizontal string holds the middle block fixed to a wall. The top block stays at rest on it, while the bottom block is pulled at constant speed along a rough horizontal floor. The lower two sliding interfaces have kinetic-friction coefficient $\frac{3}{10}$. Find required pulling force on the bottom block. Take $g = 10$ m/s^2.

(A) 18 N (B) 21 N

(C) 12 N (D) 24 N

8. A particle has position-dependent velocity components $\vec{v} = 2x\,\hat{i} - 2y\,\hat{j}$ in SI units. Given $dx/dt = v_x$ and $dy/dt = v_y$, what is its acceleration magnitude at $(6, 5)$ m?

(A) 20 m/s^2 (B) $4\sqrt{61}$ m/s^2

(C) 44 m/s^2 (D) 24 m/s^2

9. A particle has potential $U(x, y) = 6x^2y + 3y$ J in SI units. Find the force magnitude at $(1, 1)$ m.

(A) $6\sqrt{5}$ N (B) $3\sqrt{13}$ N

(C) 9 N (D) 15 N

10. A charge of $6\,\mu$C moves from potential 40 V to -40 V in an electrostatic field. Find the work done by an external agent if the charge starts and ends at rest and no other force does work.

(A) $0\,\mu$J (B) $-480\,\mu$J

(C) $-12\,\mu$J (D) $480\,\mu$J

11. A charge 4 C moves with $\vec{v} = 2\hat{i} + 2\hat{j}$ m/s in $\vec{B} = 1\hat{i} + 2\hat{j}$ T. What is the signed z-component of its magnetic force?

(A) 0 N (B) 24 N

(C) -8 N (D) 8 N

12. Point masses $m, m, 5m$ occupy $(0, 0), (8, 0), (0, 8)$ m. A fourth mass m is placed at $(x, 0)$ so that the centre of mass of all four is at $x_{CM} = 2$ m. Find x.

(A) 3 m (B) 13 m

(C) -8 m (D) 8 m

13. A series RC circuit is driven at fixed rms voltage. Multiplying angular frequency by 2 multiplies rms current by $\sqrt{2}$. Find R^2/X_C^2, where X_C is the initial capacitive reactance.

(A) 2 (B) 4

(C) $\frac{1}{2}$ (D) $\frac{1}{2} + \frac{\sqrt{2}}{2}$

14. A vertical uniform cylinder of mass 7 kg and cross-sectional area $\frac{3}{100}$ m^2 floats in a liquid of density 900 kg/m^3. It is pushed down slightly and released. Its ends remain on opposite sides of the free surface throughout. Find the period of vertical oscillation. Take $g = 10$ m/s^2 and neglect damping.

(A) $\frac{\sqrt{210}\pi}{45}$ s

(B) $\frac{2\sqrt{21}\pi}{9}$ s

(C) $\frac{2\sqrt{105}\pi}{45}$ s

(D) $\frac{\sqrt{210}}{90}$ s

15. A network is defined by $P = \mathrm{NOR}(A, B)$, $Q = \mathrm{NAND}(A, P)$ and $Y = \mathrm{NOR}(Q, \overline{B})$. Each of the four input pairs is applied once. How many pairs satisfy both $Y = 0$ and $P = 0$?

(A) 3

(B) 0

(C) 1

(D) 2

16. A particle satisfies $v_x = 2x$ and $v_y = 7$ in SI units, with $dx/dt = v_x$ and $dy/dt = v_y$. What angle does its acceleration make with +x when x=3 m? Differentiate the velocity components with respect to time.

(A) $\frac{180\,\mathrm{atan}\left(\frac{7}{6}\right)}{\pi}$ degree

(B) $\frac{180\,\mathrm{atan}\left(\frac{7}{12}\right)}{\pi}$ degree

(C) 0 degree

(D) 45 degree

17. A stretched string is divided by a bridge into lengths in the ratio 1 : 2. Tension and linear density are unchanged. The original fundamental is 70 Hz. Find the fundamental of the longer segment.

(A) 210 Hz

(B) 105 Hz

(C) $\frac{105}{2}$ Hz

(D) $\frac{140}{3}$ Hz

18. A long solenoid has 500 turns per metre and carries 3 A. It is completely filled by a linear magnetic core of relative permeability 5. Find magnetisation M.

(A) 12 A/m

(B) 7500 A/m

(C) 6000 A/m

(D) 1500 A/m

19. A sinusoidal source of peak $\frac{147}{10}$ V feeds a back-to-back series pair of identical Zeners through a 3 kΩ current-limiting resistor. The output is across the diode pair. Each Zener has reverse breakdown 9 V and forward drop 0.7 V. What is the peak-to-peak output voltage? Use the piecewise constant-drop model.

(A) $\frac{97}{5}$ V

(B) 18 V

(C) $\frac{83}{5}$ V

(D) $\frac{147}{5}$ V

20. A uniform-density planet has radius R. Gravity at a depth d equals gravity at a height $(14 - 1)R$ above the surface. Find d/R.

(A) $\frac{2743}{2744}$

(B) $\frac{13}{14}$

(C) $\frac{1}{14}$

(D) $\frac{195}{196}$

21. A geometrically similar solid structure is enlarged by a factor 3 in every linear dimension. Its material density is reduced by a factor 4. Its own weight is carried by a similarly scaled base. Find base compressive-stress multiplier.

(A) $\frac{3}{4}$ (B) $\frac{1}{4}$

(C) $\frac{4}{3}$ (D) $\frac{27}{4}$

22. An alpha particle begins far from a fixed nucleus of atomic number 47 with kinetic energy 8 MeV. Before its head-on approach, a retarding electric potential removes 4 MeV of kinetic energy. Find its distance of closest approach in fm, assuming only Coulomb repulsion near the nucleus. Use $ke^2 = 1.44$ MeV fm.

(A) $\frac{25}{846}$ fm (B) $\frac{423}{25}$ fm

(C) $\frac{846}{25}$ fm (D) $\frac{3384}{25}$ fm

23. The quantity $Z = x^3y^2/\sqrt{w}$ is obtained from independent measured quantities with maximum percentage errors 3, 3 and 4, respectively. To first order, what is the maximum percentage error in Z?

(A) 17 (B) 15

(C) 13 (D) 19

24. 2 moles of ideal gas with $\gamma = 7/5$ warm by 20 K in a rigid vessel. Find heat supplied in units of R.

(A) 140 (B) 100

(C) 16 (D) 50

25. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 40 percent. The measured increase in its absolute temperature is $\frac{2352}{5}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) 1176 K (B) 490 K

(C) 588 K (D) $\frac{4802}{5}$ K

26. A dipole of fixed moment 2 C m is rotated quasistatically from 60° to 120° relative to a uniform electric field 2 N/C. Find external work.

(A) 2 J (B) $\frac{4\pi}{3}$ J

(C) $4\sqrt{3}$ J (D) 4 J

27. A conducting rod of length 5 m moves at constant speed 3 m/s on parallel rails in a uniform perpendicular field 3 T. Total circuit resistance is 6 ohms. Neglect self-inductance. Find the external force needed to maintain the motion.

(A) $\frac{15}{2}$ N (B) $\frac{225}{2}$ N

(C) $\frac{75}{2}$ N (D) 45 N

28. In a series RLC circuit, ideal rms voltmeters read 40 V across R, 70 V across L and 20 V across C. Find source rms voltage.

(A) $10\sqrt{69}$ V (B) $10\sqrt{41}$ V

(C) 130 V (D) $10\sqrt{97}$ V

29. For steady laminar flow through a capillary, the radius is multiplied by 2 and length by 4. The volume flow rate must remain unchanged and viscosity is fixed. By what factor must the pressure difference be multiplied?

(A) $\frac{1}{4}$ (B) 4

(C) 2 (D) $\frac{1}{16}$

30. An object moves slowly along the axis toward a concave mirror. At an instant its distance from the mirror is 150 cm; the mirror focal-length magnitude is 25 cm. Find the image-speed magnitude divided by object-speed magnitude at that instant.

(A) $\frac{1}{49}$ (B) $\frac{1}{5}$

(C) $\frac{1}{25}$ (D) $\frac{1}{36}$

Chemistry

31. A Cr(3) coordination entity has coordination number 6, exactly 1 fully bound oxalate ligand(s), and chloride in all remaining donor sites. Take oxalate as 2-dentate with charge -2; chloride is monodentate. Find (number of ligand entities, net charge of the coordination entity).

(A) (5, -3) (B) (5, -4)

(C) (6, -2) (D) (6, -3)

32. Match glucose reactionX with bromine water and reactionY with hydroxylamine. Which product pair and distinction are correct?

(A) X: ethanol only; Y: phenol. (B) X: gluconic acid; Y: glucose oxime.

(C) X: saccharic acid; Y: nylon. (D) X: n-hexane; Y: sucrose.

33. Which set contains only purine bases, and which structural feature identifies the class?

(A) Ribose and phosphate; both are nitrogenous purines. (B) Uracil and glucose; both are bases.

(C) Adenine and guanine; fused two-ring base skeletons. (D) Cytosine and thymine; both are purines with two fused rings.

34. Consider the statements. I. A non-biodegradable pesticide may remain after its useful agricultural action is over. II. Its intended use alone proves it cannot contaminate organisms beyond the target pest. Choose the correct truth pattern.

(A) I true; II true. (B) I true; II false.

(C) I false; II true. (D) I false; II false.

35. For neutral HD and DT, take isotope mass numbers H=1, D=2, T=3 and neglect mass defects. Their gases have equal temperature. Which pair gives ($\langle v^2\rangle$_HD/$\langle v^2\rangle$_DT, number of neutrons in one DT molecule minus that in one HD molecule)?

(A) (5/3, 4) (B) (3/5, 2)

(C) (5/3, 1) (D) (5/3, 2)

36. Two paths connect the same equilibrium states. PathAhasq=300J,w=-100J using work-on signs. PathBhasq=500J. Which pair(workonB,internal-energychangeonB)is correct?

(A) (-100J,400J).
(B) (300J,200J).
(C) (-300J,0J).
(D) (-300J,200J).

37. A food extract contains exactly one of the listed two-component combinations. Using the usual qualitative tests under suitable conditions, its results are: Benedict on warming: positive; iodine: negative; Biuret: negative; Sudan III: negative. Which combination matches?

(A) starch and albumin
(B) glucose and sucrose
(C) glucose and starch
(D) glucose and albumin

38. For a neutral atom with ground-state configuration [Ar]$3d^{10}4s^{2}4p^{3}$,which (period,group,block) assignment is correct?

(A) (4,15,p).
(B) (3,15,p).
(C) (4,14,p).
(D) (4,15,d).

39. Eu2+ is converted to Eu3+ by an oxidising reagent. Which pair correctly distinguishes electron direction and f-shell count?

(A) No electron changes because the4f7 shell makes all reactions impossible.
(B) Eu2+ loses two electrons;4f7 becomes4f5.
(C) Eu2+ gains one electron;4f7 becomes4f8.
(D) Eu2+ loses one electron;4f7 becomes4f6.

40. CO binds to a metal with filled d orbitals. If metal-to-CO pi back-donation increases while the ligand remains bound through carbon,which paired bond changes and donation directions are consistent?

(A) Metal donates into a CO bonding orbital,so C-O strengthens while M-C weakens.
(B) CO is only a sigma donor; back-donation leaves the C-O force constant unchanged.
(C) CO donates a carbon lone pair to metal; metal donates into CO pi-star. M-C strengthens while C-O weakens.
(D) CO pi electrons are donated only from oxygen to metal,and both M-C and C-O necessarily strengthen.

41. A weak 1:1 electrolyte obeys exact Ka=Cα^{2}/(1-α). In a closed dilution its dissociation degree changes from 1/30 to 2/15, while Ka and total formula-unit amount remain constant. By what factor does its volume increase? Do not replace 1-α by 1.

(A) 13/232
(B) 232/13
(C) 16
(D) 4

42. For HC≡C-;CH2=CH-;CH3CH2-,which decreasing stability order and its underlying class distinction is correct? Neglect solvent-specific reversals.

(A) CH2=CH- > HC≡C- > CH3CH2-
(B) HC≡C- > CH3CH2- > CH2=CH-
(C) HC≡C- > CH2=CH- > CH3CH2-
(D) CH3CH2- > CH2=CH- > HC≡C-

43. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_y; same facing signs; 2: parallel p_x; same facing signs; 3: parallel p_x; opposite facing signs. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (1, 2)
(B) (0, 3)
(C) (1, 1)
(D) (2, 1)

44. For the reaction $2M + O_2 \rightarrow 2MO$, take $\Delta H^\circ = -240$ kJ and $\Delta S^\circ = -0.200$ kJ K^{-1} over the stated range. Find ΔG° in kJ at 800 K and the sign of its slope against temperature.

(A) -80 kJ; positive slope

(B) -400 kJ; negative slope

(C) 80 kJ; negative slope

(D) -240 kJ; zero slope

45. A student pairs thiamine with vitamin B1 but calls it fat-soluble because its name starts withth. Which correction is appropriate?

(A) Thiamine is vitaminA.

(B) Both statements are necessarily right.

(C) The name pairing is right, butB1 belongs to the water-soluble class.

(D) AllB vitamins are classified as fats.

46. For equal masses of spherical solid particles,the calibrated rate obeys r∝surface area$\times[H^+]^n$. Doubling$[H^+]$at fixed particles quadruples rate. Particle radius is then halved at fixed total solid volume, and$[H^+]$is doubled. Overall rate factor is

(A) 16

(B) 4

(C) 8

(D) 2

47. Steam is passed over red-hot carbon,then extra steam is passed with the CO-containing gas over a shift catalyst. Which species is consumed in the second step and which gas yield increases?

(A) CO2 is consumed; O2 yield increases.

(B) H2 is consumed; CO yield increases.

(C) Carbon is oxidised directly to O2 and no hydrogen is formed.

(D) CO is consumed; H_2 yield increases.

48. For A(g)⇌B(g),an equilibrium contains $\frac{1}{5}$ mol A and $\frac{2}{5}$ mol B. Remove $\frac{2}{25}$ mol B and add $\frac{1}{5}$ mol A at the same T. All remain gases in a fixed vessel. Final B amount in mol is

(A) $\frac{8}{25}$

(B) $\frac{12}{25}$

(C) $\frac{3}{5}$

(D) $\frac{9}{25}$

49. A salt MF_2 dissolves only as $M^{2+}+2F^-$. Its pure-water molar solubility is $\frac{1}{5000}$M. A second saturated solution has imposed free $F^-=\frac{1}{500}$M;ignore further F^- from dissolution. Dissolved M^{2+} concentration in that second solution is

(A) $\frac{1}{125000}$

(B) $\frac{1}{1250000}$

(C) $\frac{1}{500000}$

(D) $\frac{1}{50000}$

50. For 2A+B→3C+4D, A also disappears by an independent channel producing no C or D, and D is independently consumed at 1/2000 M/s. Measured total A disappearance is 1/100 M/s and net D appearance is 27/2000 M/s. What is the A disappearance rate through its independent channel, in M/s?

(A) 3/1000

(B) 1/250

(C) 13/4000

(D) 17/1000

51. A primary-standard sample of pure anhydrous Na_2CO_3 weighs $\frac{106}{125}$ g. It is titrated with0.100M HCl. Find the additional acid volume in mL needed after the first carbonate-to-hydrogencarbonate equivalence point to reach complete conversion to CO_2/water. Take Na_2CO_3=106g/mol.

(A) 161

(B) 160

(C) 40

(D) 80

52. For planar cyclopropenyl cation,choose the correct classification and the decisive reason. Use the explicitly stated geometry.

(A) nonaromatic; an ion can never be aromatic

(B) antiaromatic; every cyclic compound obeys 4n+2

(C) aromatic; count only the neutral double bonds and ignore charge or geometry

(D) aromatic; three continuous p orbitals contain the one double-bond pair and an empty cationic p orbital

53. For the pair propanal and propanone,which classification identifies the structural difference rather than merely stating that the formula agrees?

(A) functional isomers

(B) metamers

(C) position isomers

(D) conformers

54. Three standard reduction potentials EA,EB,EC satisfy EB-EA=$\frac{2}{5}$V and EC-EB=$\frac{2}{5}$V. For a cell whereCis reduced andAoxidised,the standard potential in V is

(A) $\frac{8}{5}$

(B) $\frac{4}{5}$

(C) 0

(D) $\frac{2}{5}$

55. A 0.0100 mol portion of Zn2+ is first fully precipitated as M(OH)2 using aqueous ammonia and is then dissolved in sufficient excess ammonia to give the usual tetraammine species. Counting only metal-containing entities, what are (moles of metal in solution at the end, moles of NH3 bound in that ammine complex)? Assume quantitative steps and no metal redox.

(A) (0.0200, 0.0400)

(B) (0, 0.0400)

(C) (0.0100, 0.0400)

(D) (0.0100, 0.0200)

56. In a simplified drug-target model, an analgesic candidate competes reversibly with another ligand for the same protein site. Increasing the other ligand reduces occupancy by the candidate. Which interpretation fits the model?

(A) The protein has necessarily polymerised into nylon.

(B) Binding-site competition, rather than proof of destruction of the protein.

(C) Every candidate molecule has irreversibly removed the site.

(D) The two ligands must be the same molecule.

57. Neutral CrO5 is represented as one terminal oxide oxygen and two peroxide O-O groups. Which pair gives(chromium oxidation state, number of peroxide oxygen atoms)?

(A) (+6, 2).

(B) (+4, 4).

(C) (+10, 0).

(D) (+6, 4).

58. $\frac{7}{100}$ mol Cl_2 is contacted with $\frac{21}{200}$ mol KOH in cold dilute solution. Only $Cl_2 + 2KOH \rightarrow KCl + KClO + H_2O$ occurs. Final solution volume is $\frac{1}{2}$ L. Find the KClOconcentration in mol/L, assuming no side reaction or product loss.

(A) $\frac{21}{200}$

(B) $\frac{21}{100}$

(C) $\frac{7}{200}$

(D) $\frac{7}{50}$

59. A thermal assembly contains Cu(s) and He(g) in separate inert compartments at the same temperature. Helium is heated quasistatically at constant pressure under a freely moving piston. Containers have negligible heat capacity. Use the classical Dulong-Petit solid model for Cu, neglecting its Cp-Cv difference, and the ideal monatomic model for He. No phase change occurs. Molar masses are Cu64 and He4 g/mol. Over a small common temperature rise, the measured total heat capacity C divided by total Cu-plus-He mass m and gas constant R is C/(mR)=61/512 mol/g. What is the Cu mass fraction?

(A) 1/8 (B) 7/8

(C) 131/168 (D) 37/44

60. Judge I: en binds through two N donors to the same metal. II: For comparable complexes of one metal,the chelate effect can favour a multidentate ligand without changing the metal oxidation state. Which is correct?

(A) Both I and II. (B) Only II.

(C) Only I. (D) Neither I nor II.

English Proficiency

61. Use an indefinite article for a previously unmentioned, nonspecific referent; no particular one has been identified. Complete: We need _____ useful explanation, not merely a long one.

(A) an (B) the

(C) a (D) no article

62. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. because it offered a safe crossing Q. Although the route was longer, R. the walkers chose it S. before the light faded.

(A) P-Q-R-S (B) Q-R-P-S

(C) S-R-P-Q (D) R-Q-P-S

63. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The team has checked every figure except the provisional total.

(A) Every figure except the provisional total has been checked by the team.

(B) The team has been checked by every figure except the provisional total.

(C) Every figure except the provisional total was being checked by the team.

(D) Every figure, including the provisional total, has been checked by the team.

64. Read the sentence: The invitation was intended to encourage participation rather than deter newcomers. Choose the opposite in meaning of 'encourage' as used in the sentence.

(A) reward (B) increase

(C) organise (D) discourage

65. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The conservator must preserve the original surface even if the stain cannot be removed.

(A) The original surface must have preserved the conservator even if the stain remains.

(B) The stain must be preserved even if the original surface cannot be removed.

(C) The original surface may be preserved if the stain is removed.

(D) The original surface must be preserved by the conservator even if the stain cannot be removed.

66. Choose the sentence that should open the following paragraph. The other sentences require context supplied earlier. A. The group combined individual notes into a single study document. B. Several sections contradicted one another because their authors had used different editions. C. The group added edition details and checked each disagreement against the texts. D. Readers could then distinguish a source difference from a simple copying error.

(A) The group combined individual notes into a single study document.

(B) Several sections contradicted one another because their authors had used different editions.

(C) Readers could then distinguish a source difference from a simple copying error.

(D) The group added edition details and checked each disagreement against the texts.

67. Read the sentence: The researcher treated the visits as discrete events rather than one continuous episode. Choose the opposite in meaning of 'discrete' as used in the sentence.

(A) recorded (B) separate

(C) continuous (D) distinct

68. Read the sentence: The team had ample time to check the result before the deadline. Choose the closest meaning of 'ample' as used in the sentence.

(A) more than sufficient (B) carefully scheduled

(C) unexpectedly available (D) exactly sufficient

69. He analyses atmospheric conditions to understand storms, not objects beyond the earth's atmosphere. What is he?

(A) astronomer (B) meteorologist

(C) surveyor (D) geologist

70. Read the sentence: Her frugal habits allowed the small research fund to last the whole year. Choose the closest meaning of 'frugal' as used in the sentence.

(A) unwilling to share anything (B) economical in use of resources

(C) unable to obtain supplies (D) indifferent to quality

Logical Reasoning

71. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 3, 6, 10, 20, ?, 48, 52.

(A) 48 (B) 24

(C) 30 (D) 36

72. Preserve the exact relation and its order: Player : Team :: ? : Pride. Choose the missing term.

(A) Bee (B) Wolf

(C) Ship (D) Lion

73. Consider only these statements; universal statements alone do not imply existence. Some designers are joggers; No inventors are designers. Conclusions: I. Some inventors are not designers. II. Some joggers are designers. Which conclusions follow necessarily?

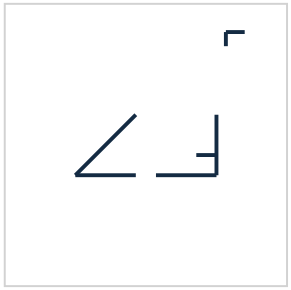
(A) Both I and II (B) Only I

(C) Neither I nor II (D) Only II

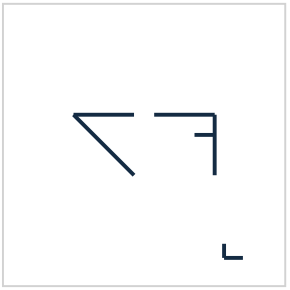
74. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



(A) Pattern Z



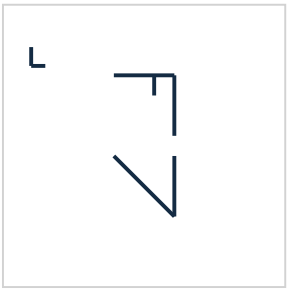
(B) Pattern W



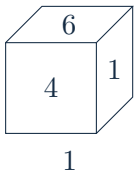
(C) Pattern P



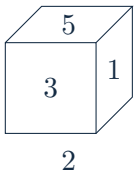
(D) Pattern L



75. The views show the same die. Give the ordered pair: (face opposite 5, face opposite 4). Do not assume opposite labels sum to seven.



1



2

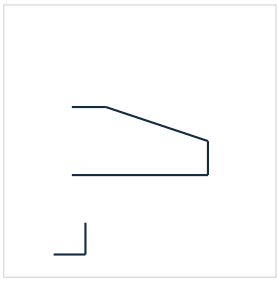
(A) $(6, 3)$

(B) $(6, 4)$

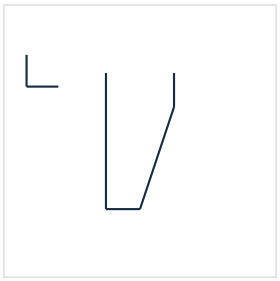
(C) $(6, 2)$

(D) $(6, 1)$

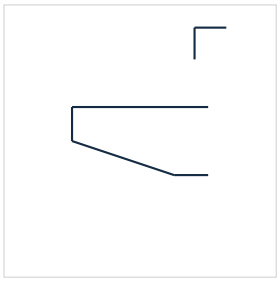
76. A single fixed quarter-turn is repeated between consecutive frames, with no reflection. Choose the next figure, preserving the attached flag.



1

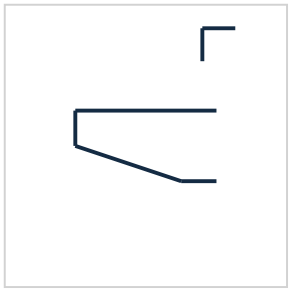


2

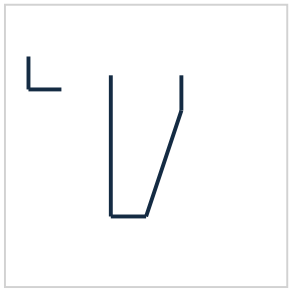


3

(A) Pattern V



(B) Pattern X



(C) Pattern L



(D) Pattern K



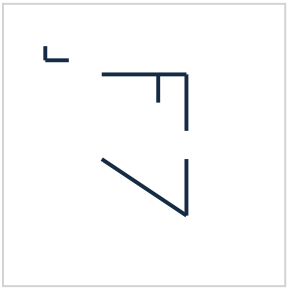
77. Choose the exact horizontal water reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



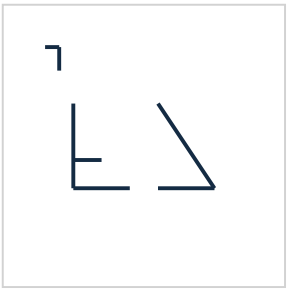
(A) Pattern M



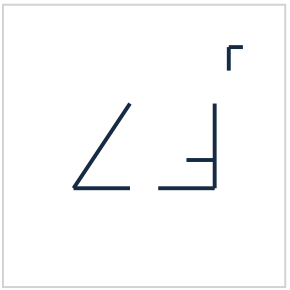
(B) Pattern P



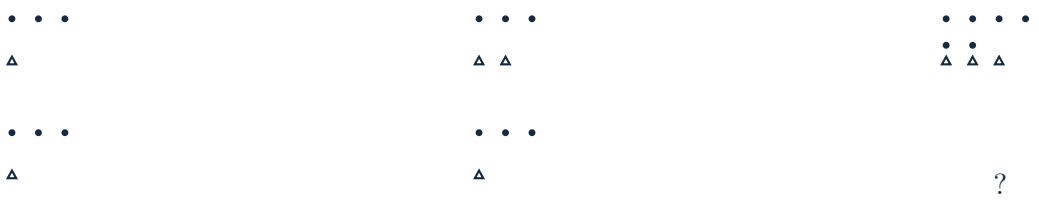
(C) Pattern W



(D) Pattern Q

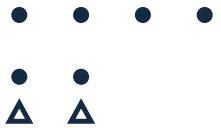


78. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern R

(B) Pattern U



(C) Pattern T



(D) Pattern Z



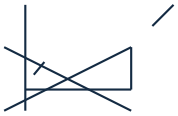


79. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.

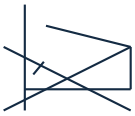




host 1



host 2



host 3



host 4

(A) 2

(B) 4

(C) 3

(D) 1

80. The age table uses the same two people throughout:

	younger	older
now	x	$x+8$
5 years ago	$x-5$	$x+3$

. The product of their ages 5 years ago was 209. What will the older person's age be 6 years from now?

(A) 25 (B) 30

(C) 38 (D) 24

81. Three pairs contain exactly one mammal and one bird. Which pair does not?

(A) Bat / Elephant (B) Rabbit / Duck

(C) Dog / Eagle (D) Horse / Sparrow

82. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

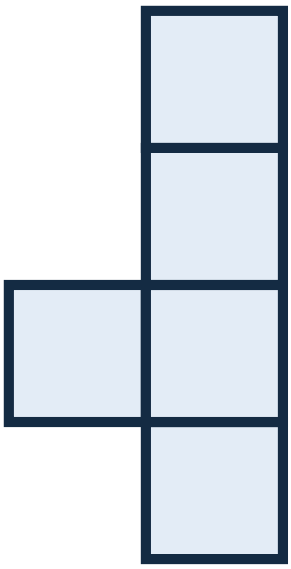


piece 1

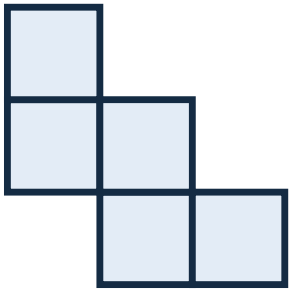


piece 2

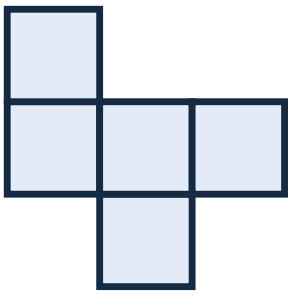
(A) Pattern Q



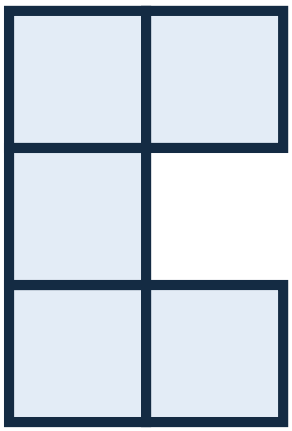
(B) Pattern M



(C) Pattern K



(D) Pattern Z

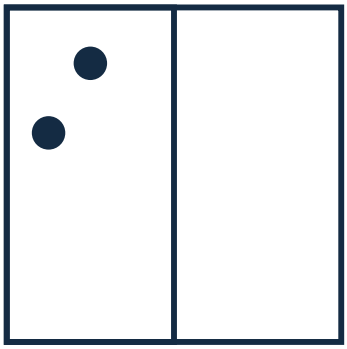


83. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 15$ and $5 \mapsto 27$. Which input maps to 47?

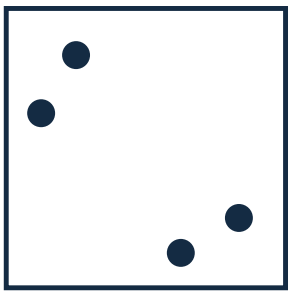
(A) 10 (B) 11

(C) 9 (D) 47

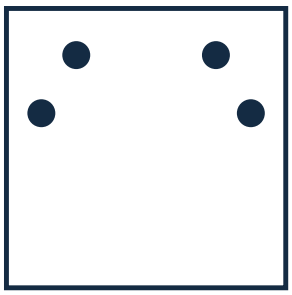
84. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



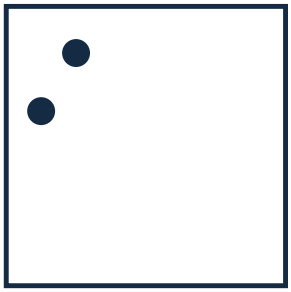
(A) Pattern Y



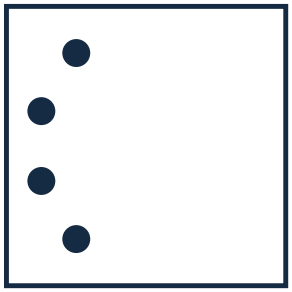
(B) Pattern T



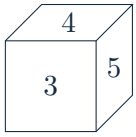
(C) Pattern N



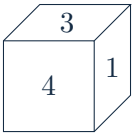
(D) Pattern L



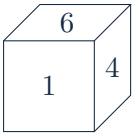
85. The views show the same die. Give the ordered pair: (face opposite 6, face opposite 2). Do not assume opposite labels sum to seven.



1



2



3

(A) $(3, 2)$ (B) $(3, 1)$

(C) $(3, 3)$ (D) $(3, 4)$

86. A walker goes 7 m west, then 4 m south, then turns right and walks 3 m. In which direction is the walker from the starting point?

(A) North-west (B) South-east

(C) North-east (D) South-west

87. Each of Charu, Farah, Esha visits on one different day (Monday, Tuesday or Wednesday) and buys one different fruit (Peach, Orange, Apple). Charu visits on Tuesday, and Farah on Monday. The visitor on Tuesday buys peach. Farah does not buy apple. Which fruit does Farah buy?

(A) Orange (B) Peach

(C) No unique assignment (D) Apple

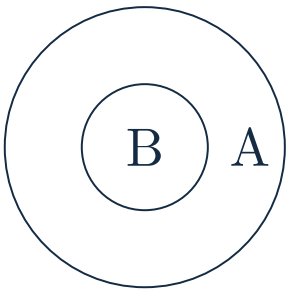
88. Mira points to a photograph and says, "This person is my mother's only son." Mira is female, and the photographed person is male. How is the photographed person related to Mira?

(A) Sister (B) Son

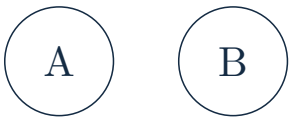
(C) Brother (D) Daughter

89. Choose the diagram that represents all these conditions, including the strictness information: All A are B, all B are A, and A is nonempty.

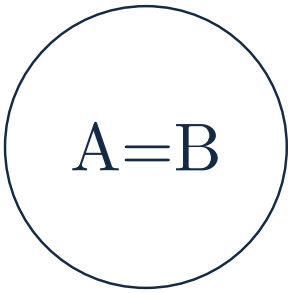
(A) Pattern M



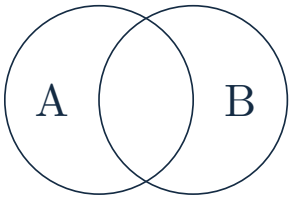
(B) Pattern Q



(C) Pattern X



(D) Pattern Z



90. In a row, Asha is 11th from the left and 35th from the right. Bina is 7th from the right. Charu stands 4 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 24 (B) 23

(C) 7 (D) 22

Mathematics

91. Find the squared distance from O to the plane through (4,0,0),(0,2,0),(0,0,6).

(A) $\frac{144}{49}$ (B) $\frac{12}{11}$

(C) 56 (D) $\frac{49}{144}$

92. $A = \begin{bmatrix} 7 & 1 \\ 0 & 4 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$. Find the (1,2) entry of $P^{-1}AP$.

(A) −8 (B) 1

(C) 9 (D) 10

93. A 12-element set contains distinct a,b. How many 2-element subsets contain exactly one of a,b and also contain a?

(A) 11 (B) 45

(C) 10 (D) 20

94. Circles $x^2+y^2=144$ and $(x-20)^2+y^2=144$ meet at A,B. Find $|AB|^2$.

(A) 44 (B) 16

(C) 576 (D) 176

95. S_n=$5n^2+4n$ is a sequence partial sum. Find $a_1 + a_3 + \cdots + a_{15}$.

(A) 639 (B) 632

(C) 352 (D) 672

96. An AP has a_1=-72,d=8. How many of its first 12 terms are strictly positive?

(A) 2 (B) 3

(C) 10 (D) 12

97. $F'(x) = -\frac{|x+1|}{(x+1)^3\sqrt{(x+1)^2+1}}$ and F(0)=2. Find F(1)-2F(0).

(A) $-\sqrt{2} + \frac{\sqrt{5}}{2} + 2$ (B) $-2 - \frac{\sqrt{5}}{2} + \sqrt{2}$

(C) $-\sqrt{2} + \frac{\sqrt{5}}{2}$ (D) $-2 - \sqrt{2} + \frac{\sqrt{5}}{2}$

98. $y'+y/(x+1)=1$ for $x\geq 0$, y(0)=5. Find (6)y(5)-y(0).

(A) $\frac{45}{2}$ (B) $\frac{35}{2}$

(C) $\frac{15}{4}$ (D) $-\frac{35}{2}$

99. $f(x)=x^3+5x$, $g=f^{-1}$. Find $g''(42)$.

(A) $-\frac{9}{512}$ (B) $-\frac{9}{16384}$

(C) $\frac{9}{16384}$ (D) $\frac{1}{18}$

100. In $(1+x)^{13}$, find the ratio of the coefficient of x^{11} to the coefficient of x^{10}.

(A) $\frac{3}{11}$ (B) 3

(C) $\frac{11}{3}$ (D) $\frac{2}{5}$

101. $x=6+8(1-t^2)/(1+t^2)$, $y=2+16t/(1+t^2)$, where t is real. Which point of the containing circle is not attained?

(A) $(6, 10)$ (B) $(14, 2)$

(C) $(-2, 2)$ (D) $(6, -6)$

102. Let $f(x+7)=8\,x^2$. Find the area enclosed between $y=f(x)$ and $y=64(x-7)$.

(A) $-\frac{2048}{3}$ (B) $\frac{88}{3}$

(C) $\frac{2048}{3}$ (D) $\frac{4096}{3}$

103. An equivalence relation has classes of sizes 2,6,5. How many unordered pairs of distinct related elements are there?

(A) 26 (B) 78

(C) 16 (D) 52

104. Let $\delta=\pi/12$. Evaluate $\sin\delta\ \Sigma[1/(\sin(j\delta)\sin((j+1)\delta))]$, $j=2,\ldots,5$.

(A) $\frac{\sqrt{3}}{3}$ (B) $-\sqrt{3}$

(C) $\sqrt{3}+2$ (D) $\sqrt{3}$

105. Evaluate $\lim_{x\to 0} \dfrac{(1+x)^{10}-1-10x-5x^2}{x^2}$.

(A) 40 (B) 50

(C) 5 (D) 45

106. How many integers k make $27\sin x+36\cos x+3=k$ have exactly two distinct solutions in $0\le x<2\pi$?

(A) 89 (B) 44

(C) 91 (D) 90

107. Evaluate the sum of $r(r-1)C(12,r)2^r$ over all even integers r with $0\le r\le 12$.

(A) 31177872 (B) 15589200

(C) 3897300 (D) 15588936

108. Evaluate $\int_0^\infty dx/[(x^2+25)(x^2+100)]$.

(A) $\frac{\pi}{1500}$ (B) $\frac{\pi}{100}$

(C) $\frac{\pi}{750}$ (D) $-\frac{\pi}{1500}$

109. A polynomial f of degree at most 2 satisfies $f'(x)+f'(-x)=8$, $f''(x)=8$, and $f(0)=2$. Find $f(2)$.

(A) 18 (B) 22

(C) 24 (D) 26

110. a=(9,6,0), b=(6,-9,3). Find $|a+b|^2$-$|a-b|^2$.

(A) -9 (B) 0

(C) 432 (D) -18

111. 8 distinct objects, includinga,b, are distributed into three nonempty unlabelled boxes. How many distributions put a and b in different boxes?

(A) 966 (B) 601

(C) 665 (D) 697

112. An observer at height 10 m sees the tower foot at depression 45° and its top at elevation 60°. The observer descends vertically to ground level. If the new elevation is α, find tanα.

(A) $10 + 10\sqrt{3}$ (B) $\sqrt{3} + 2$

(C) $1 + \sqrt{3}$ (D) $-\frac{1}{2} + \frac{\sqrt{3}}{2}$

113. Let $D(t) = \det \begin{pmatrix} 9+t & 1+t \\ 1+t & 5+t \end{pmatrix}$. Find D′(0).

(A) 16 (B) 44

(C) 12 (D) 88

114. Evaluate $7^{\log_7 5} + 2^{2(\log_2 7)(\log_7 5)}$.

(A) 32 (B) 25

(C) 10 (D) 30

115. Find the area enclosed by y=x(13-x) and y=6x.

(A) $-\frac{343}{6}$ (B) $\frac{2197}{6}$

(C) $\frac{343}{6}$ (D) $\frac{1981}{6}$

116. X has mean 13, variance 25. Y=aX+4, a>0, has mean 69. Find Var(Y).

(A) 125 (B) 4850

(C) 625 (D) 629

117. How many injective functions from{a,b,c} to{1,...,9} satisfy f(a)<f(b)<f(c) and f(a)+f(c) even?

(A) 30 (B) 84

(C) 7 (D) 44

118. The tangent to $y = x^3 - 2x$ at $x = 2$ meets the two coordinate axes at A and B. Find the area of triangle OAB, where O is the origin.

(A) $\frac{32}{5}$ (B) $\frac{64}{5}$

(C) $\frac{128}{5}$ (D) $\frac{32}{3}$

119. The parabola $y^2=8x$ has focus F. Its directrix cuts $|z-F|\leq 8$. Find the larger-segment area minus smaller-segment area.

(A) $-16\sqrt{3}+\frac{64\pi}{3}$ (B) $16\sqrt{3}+\frac{128\pi}{3}$

(C) $-32\sqrt{3}+\frac{128\pi}{3}$ (D) $32\sqrt{3}+\frac{64\pi}{3}$

120. Find the area bounded by $y=x$, $y=36/x$, and $x=24$ in the first quadrant.

(A) $108-72\log(2)$ (B) $270-72\log(2)$

(C) $72\log(2)$ (D) 270

121. There are 16 Boolean variables $p_1,\ldots,p\ 16$. Exactly 2 variables must be true. How many truth assignments satisfy [(p_1 implies p_2) and (p_1 implies not p_2)]?

(A) 91 (B) 120

(C) 105 (D) 14

122. An AP has first term 3, sum of its first 7 terms equal to 105. Find the sum of the squares of those 7 terms.

(A) 11025 (B) 63

(C) 1294 (D) 2023

123. For $A=\begin{bmatrix}5 & 1\\ 1 & 2\end{bmatrix}$ and $C=9A^{-1}$, find C_{22}.

(A) 5 (B) 2

(C) $\frac{5}{9}$ (D) -1

124. α,β are nonzero roots of $x^2-12x+32=0$. Find the sum of $1+1/\alpha$ and $1+1/\beta$.

(A) $\frac{3}{8}$ (B) $\frac{14}{3}$

(C) $\frac{19}{8}$ (D) $\frac{11}{8}$

125. For $\sin(d^4y/dx^4)+y=0$, find order+degree if the degree is defined; otherwise select "Degree is not defined".

(A) Degree is not defined (B) 8

(C) 6 (D) 7

126. For $3x+4y+9=0$ and $4x+3y+6=0$, find the squared distance from the origin to the angle bisector of positive slope.

(A) $\frac{3\sqrt{2}}{2}$ (B) $\frac{225}{98}$

(C) 9 (D) $\frac{9}{2}$

127. The four lines $\pm x/4\pm y/6=1$ form a rhombus with vertices on the coordinate axes. Find its area.

(A) 24 (B) 48

(C) 96 (D) 12

128. Evaluate $\sum_{j=4}^{8} \binom{j}{2}$, with the convention $\binom{j}{r} = 0$ when j<r.

(A) 84 (B) 80

(C) 74 (D) 83

129. Two different numbers are chosen uniformly from{1,...,9}. Let D be their absolute difference. Find Var(D).

(A) $\frac{10}{3}$ (B) $\frac{35}{9}$

(C) $\frac{40}{3}$ (D) $\frac{\sqrt{35}}{3}$

130. $F'(x)=\frac{4\left(x^2-1\right)\left(x^2+1\right)^3}{x^5}$ and F(1)=2. Find F(2)-2F(1).

(A) $\frac{337}{16}$ (B) $\frac{401}{16}$

(C) $-\frac{401}{16}$ (D) $\frac{369}{16}$

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A particle of mass 5 kg starts from rest and experiences $F(t) = 5t$ N along its motion for 4 s. Find the work done by the force.

(A) 640 J (B) 160 J

(C) $\frac{160}{3}$ J (D) 320 J

132. A uniform disc of initial mass 2 kg and radius 4 m has a concentric hole of radius $R/2$ removed. Find the remaining annulus's moment of inertia about a diameter lying in its plane.

(A) $8\,\mathrm{kg\,m^2}$ (B) $\frac{3}{2}\,\mathrm{kg\,m^2}$

(C) $\frac{15}{2}\,\mathrm{kg\,m^2}$ (D) $6\,\mathrm{kg\,m^2}$

133. For a circular-aperture telescope, observation wavelength is multiplied by 2. Its smallest resolvable angle must be reduced by a factor 3. By what factor must the objective diameter increase?

(A) 6 (B) 18

(C) 2 (D) $\frac{3}{2}$

Chemistry

134. Three standard reduction potentials EA, EB and EC use the same reference. EA = -0.20 V. A cell with C reduced and A oxidised has standard potential +0.80 V; a second cell with B reduced and C oxidised has standard potential -0.30 V. What is EB, in V, on that reference?

(A) 0.30 (B) -0.30

(C) 0.60 (D) 0.90

135. On dilution of one electrolyte,the analytical concentration falls by a factor16while measured conductivity falls by a factor4. Cell geometry is unchanged. Ratio of final to initial molar conductivity is

(A) $\frac{1}{4}$ (B) 4

(C) 16 (D) $\frac{1}{16}$

136. Consider the statements. I. An antiseptic category refers to a suitable application on living tissue. II. A compound used on a nonliving floor is automatically suitable for every living tissue at the same concentration. Choose the correct truth pattern.

(A) I false; II false. (B) I false; II true.

(C) I true; II false. (D) I true; II true.

Logical Reasoning

137. Each letter position follows its own constant cyclic alphabet step (Z is followed by A). Find the missing internal term: WFT, AHR, ?, ILN, MNL, QPJ.

(A) EJP (B) DIO

(C) PJE (D) FKQ

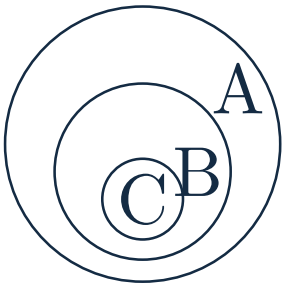
138. Consider only these statements; universal statements alone do not imply existence. All designers are bakers; All designers are campers; Some bakers are campers. Conclusions: I. No campers are designers. II. Some designers are not bakers. Which conclusions are individually possible without contradicting the statements?

(A) Neither I nor II (B) Both I and II

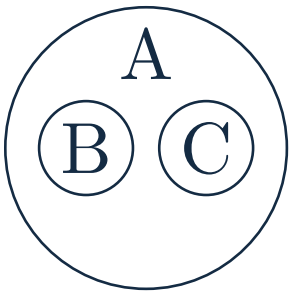
(C) Only I (D) Only II

139. Choose the diagram that represents all these conditions, including the strictness information: All B and all C are A; no B is C. B and C are nonempty, and some A belongs to neither.

(A) Pattern L



(B) Pattern T



(C) Pattern W



(D) Pattern Y



Mathematics

140. Find the maximum of $\sin^5 x \cos x$ on$[0,\pi/2]$.

(A) $\frac{\sqrt{6}}{6}$ (B) $\frac{25\sqrt{5}}{216}$

(C) 1 (D) $\frac{25\sqrt{6}}{216}$

141. For xy-2x-8y+16=0, find the squared length of the diagonal of the rectangle formed with the axes.

(A) 16 (B) 100

(C) 68 (D) $2\sqrt{17}$

142. O=(0,0,0), A=(9,0,0), B=(1,7,0). Find the squared area of triangle OAB.

(A) $\frac{3969}{4}$ (B) $\frac{63}{2}$

(C) $\frac{3969}{2}$ (D) 3969

BITSAT Mock 50

130 main questions and 12 optional bonus questions

Time: 180 minutes. Main maximum: 390 marks. Correct +3; wrong −1; unanswered 0. Select exactly one option A–D. You may move between main sections and revise answers before entering the optional bonus section. There are no section time limits. **Bonus rule.** Enter only after answering all 130 main questions and only if time remains. Entry locks every main answer. The 12 bonus questions share the original time limit and the same marking. Bonus maximum: 36 marks; combined maximum: 426. This is an original, deliberately demanding training paper using the verified 2026 format. The design target is about 10 percent more cognitive demand than the base Tejas200 mock recipes. It is not an official paper or an empirically measured difficulty increase. Use selection and revisits under the clock. Unless otherwise stated, angles are in radians and inverse trigonometric values use their principal branches. Question-specific data and approximations take priority.

Physics

1. A galvanometer of resistance 50 ohms and full-scale current $\frac{1}{1000}$ A is converted to a unidirectional voltmeter. Its branch includes a forward-biased silicon diode with constant drop 0.7 V and an added series resistor. For positive full-scale voltage $\frac{31}{20}$ V, find the added resistance.

(A) $375\,\Omega$ (B) $850\,\Omega$

(C) $800\,\Omega$ (D) $1500\,\Omega$

2. Masses 3 kg and 8 kg hang from a light inextensible string over a pulley with moment of inertia I and radius R such that $I/R^2 = 5$ kg. There is no slipping or axle friction. Find acceleration magnitude. Take $g = 10\ \mathrm{m/s^2}$.

(A) $\frac{50}{21}\ \mathrm{m/s^2}$ (B) $10\ \mathrm{m/s^2}$

(C) $\frac{25}{3}\ \mathrm{m/s^2}$ (D) $\frac{25}{8}\ \mathrm{m/s^2}$

3. A charge of $6\,\mu$C moves from potential 20 V to −20 V in an electrostatic field. Find the work done by an external agent if the charge starts and ends at rest and no other force does work.

(A) $240\,\mu$J (B) $-240\,\mu$J

(C) $-40\,\mu$J (D) $0\,\mu$J

4. A balanced Wheatstone bridge has arms AB=AD=3 ohms and BC=DC=6 ohms. A resistor of arbitrary positive resistance connects B to D. Find equivalent resistance between A and C.

(A) $4\,\Omega$ (B) $\frac{9}{2}\,\Omega$

(C) $9\,\Omega$ (D) $2\,\Omega$

5. An object moves slowly along the axis toward a concave mirror. At an instant its distance from the mirror is 125 cm; the mirror focal-length magnitude is 25 cm. Find the image-speed magnitude divided by object-speed magnitude at that instant.

(A) $\frac{1}{4}$ (B) $\frac{1}{25}$

(C) $\frac{1}{36}$ (D) $\frac{1}{16}$

6. Two masses 4 kg and 4 kg are connected by a light string over a smooth pulley and accelerate freely. A transverse pulse travels in the taut straight part of this string, of linear density μ. Compare its speed to that in a second string of density 4μ held at tension $g\,\mathrm{N}$ numerically equal to 1 kg weight. Find v_1/v_2.

(A) 16 (B) 4

(C) 2 (D) $\frac{1}{4}$

7. A coil has 20 turns, area $\frac{1}{20}\ \mathrm{m}^2$, and rotates at 6 rad/s in a uniform 2 T field. At an instant the angle between its area vector and the field is 30°. Find instantaneous-emf magnitude.

(A) 6 V (B) $6\sqrt{3}\,\mathrm{V}$

(C) $\frac{3}{10}\,\mathrm{V}$ (D) 1 V

8. A one-electron ion with nuclear charge $Z = 1$ emits when its orbit changes. The ratio of initial to final orbit radii is 36, and its final principal quantum number is 1. What is $R\lambda$ for the emitted photon? Here R is the hydrogen Rydberg constant.

(A) $\frac{36}{35}$ (B) $\frac{35}{36}$

(C) $\frac{36}{37}$ (D) 36

9. A pure radioactive sample has activity A at time 6 hours after preparation and $A/8$ at time 12 hours after preparation. At what time, measured from preparation, will its activity be one thirty-second of the activity measured at 6 hours? Assume exponential decay with a constant decay rate.

(A) $\frac{16}{3}\,\mathrm{h}$ (B) 70 h

(C) 11.0 h (D) 16 h

10. A mass 1 kg initially moves in the positive direction at 6 m/s. A force opposite to that direction rises linearly from zero to 8 N and then falls linearly to zero over a total duration 6 s. Find its signed final momentum.

(A) $-18\,\mathrm{kg\,m/s}$ (B) $30\,\mathrm{kg\,m/s}$

(C) $24\,\mathrm{kg\,m/s}$ (D) $-42\,\mathrm{kg\,m/s}$

11. A geometrically similar solid structure is enlarged by a factor 4 in every linear dimension. Its material density is reduced by a factor 4. Its own weight is carried by a similarly scaled base. Find base compressive-stress multiplier.

(A) 1 (B) $\frac{1}{4}$

(C) 16 (D) 4

12. A single slit of width 4 mm is illuminated by wavelength 500 nm. A screen is 2 m away. Find the separation of the second minima on opposite sides of the central maximum, in the small-angle approximation.

(A) $\frac{3}{4}\,\mathrm{mm}$ (B) $\frac{1}{4}\,\mathrm{mm}$

(C) 1 mm (D) $\frac{1}{2}\,\mathrm{mm}$

13. A particle has position-dependent velocity components $\vec{v} = 5x\,\hat{i} - 5y\,\hat{j}$ in SI units. Given $dx/dt = v_x$ and $dy/dt = v_y$, what is its acceleration magnitude at $(5, 5)$ m?

(A) $625\sqrt{2}\,\mathrm{m/s^2}$ (B) $250\,\mathrm{m/s^2}$

(C) $125\sqrt{2}\,\mathrm{m/s^2}$ (D) $125\,\mathrm{m/s^2}$

14. A silicon diode has a constant 0.7 V drop when forward biased and is open when reverse biased. Its anode connects through 1 kΩ to a terminal at +9 V, and its cathode connects to a terminal at 1 V. Find the current from the +9 V terminal in mA.



(A) 8 mA
(B) $\frac{33}{5}$ mA
(C) $\frac{83}{10}$ mA
(D) $\frac{73}{10}$ mA

15. A particle has potential $U(x, y) = 6x^2y + 5y$ J in SI units. Find the force magnitude at $(3, 3)$ m.
(A) $\sqrt{15145}$ N
(B) 167 N
(C) $\sqrt{6397}$ N
(D) 108 N

16. For a positive constant A, which expression describes SHM about its own equilibrium position?
(A) $x = A\sin(3t) + A\sin(6t)$.
(B) $x = A\sin^2(3t)$.
(C) $x = A\sin(3t^2)$.
(D) $x = Ae^{3t}$.

17. Two soap bubbles of radii 4 cm and 16 cm share a common thin soap film. Find the radius of curvature of the common film.
(A) 12 cm
(B) $\frac{8}{3}$ cm
(C) $\frac{32}{3}$ cm
(D) $\frac{16}{3}$ cm

18. A flat spiral coil has 30 turns whose radii are uniformly distributed from 2 cm to 5 cm. The same current 6 A flows in every turn. Approximating turns as circular and the distribution as continuous, find magnetic moment in units of $\pi \times 10^{-4}$ A m^2.
(A) 78
(B) 2205
(C) 2340
(D) 4500

19. A small planar detector initially faces an isotropic point source normally at distance r. It is moved to a concentric sphere whose enclosed volume is 125 times as large, and tilted so that the cosine of the angle between its normal and the incident ray is $\frac{1}{2}$. Its area is unchanged. What is the ratio of intercepted powers, final to initial?
(A) $\frac{2}{25}$
(B) $\frac{25}{2}$
(C) $\frac{1}{50}$
(D) $\frac{1}{25}$

20. In two states of a dilute gas, sound measurements give frequency 300 Hz and wavelength 2 m initially, and 200 Hz and 4 m finally. The heat-capacity ratio and molecular number density stay fixed, while effective collision diameter is multiplied by 4. Use $v_s = f\lambda$, $v_{rms} = \sqrt{3/\gamma}\,v_s$, and collision frequency $\nu \propto nd^2v_{rms}$. Find the collision-frequency multiplier.
(A) $\frac{16}{3}$
(B) 16
(C) $\frac{4}{3}$
(D) $\frac{64}{3}$

21. A circuit uses four ideal NAND gates: $X = A \,\text{NAND}\, B$, $U = A \,\text{NAND}\, X$, $W = B \,\text{NAND}\, X$, $Y = U \,\text{NAND}\, W$. Inputs are A=0, B=1. Output Y closes an ideal series switch only when Y=1. The switch connects a 7 ohm load to a 15 V source with internal resistance 3 ohms. Find load current.

(A) $\frac{3}{2}$ A (B) $\frac{15}{4}$ A

(C) $\frac{15}{17}$ A (D) $\frac{15}{13}$ A

22. A dipole of fixed moment 7 C m is rotated quasistatically from 60° to 120° relative to a uniform electric field 2 N/C. Find external work.

(A) 14 J (B) $7\sqrt{3}$ J

(C) $\frac{14\pi}{3}$ J (D) 7 J

23. An ideal transformer has secondary-to-primary turns ratio 5. Its primary draws 1 A at 100 V while delivering power to a resistive secondary load. Find secondary current.

(A) 25 A (B) 1 A

(C) $\frac{1}{5}$ A (D) 5 A

24. A displacement law is $x = at + bt^2$, where t is time. Which is the dimension of a^2/b?

(A) L^1 (B) L^2T^{-2}

(C) T^1 (D) L^1T^{-1}

25. The same metal is illuminated successively at wavelengths λ and $\lambda/2$. Its maximum photoelectron kinetic energies are 2 eV and 10 eV, respectively. What is the ratio of threshold wavelength to λ?

(A) $\frac{1}{3}$ (B) $\frac{4}{3}$

(C) $\frac{4}{7}$ (D) $\frac{3}{4}$

26. A satellite of mass m is moved from a circular orbit of radius R to one of radius 12R about mass M. Find the required increase in total orbital energy in units of $GMm/(2R)$.

(A) $\frac{1}{12}$ (B) $\frac{11}{12}$

(C) $\frac{13}{12}$ (D) $\frac{11}{6}$

27. A fixed-composition ideal diatomic gas is heated, and its sound speed increases by 5 percent. The measured increase in its absolute temperature is $\frac{205}{4}$ K. Find its initial absolute temperature. Neglect vibrational excitation and take the heat-capacity ratio as constant throughout this range.

(A) 1025 K (B) 500 K

(C) $\frac{1025}{2}$ K (D) 20500 K

28. 3 identical, noninteracting ideal-gas samples at the same temperature and pressure are combined without changing those conditions. Which pair both increases by a factor 3?

(A) Pressure and temperature. (B) Density and pressure.

(C) Temperature and molar heat capacity. (D) Volume and internal energy.

29. A long solenoid has 500 turns per metre and carries 3 A. It is completely filled by a linear magnetic core of relative permeability 3. Find magnetisation M.

(A) 6 A/m (B) 3000 A/m

(C) 4500 A/m (D) 1500 A/m

30. A ray enters one face of a prism normally, and reaches the second face at the critical angle. The prism index in air is $\sqrt{2}$. Find the prism angle in degrees.

(A) $\frac{180\,\text{atan}\left(\sqrt{2}\right)}{\pi}$ degree

(B) $\frac{180\,\text{atan}\left(\frac{\sqrt{2}}{2}\right)}{\pi}$ degree

(C) 45 degree

(D) $\frac{90\sqrt{2}}{\pi}$ degree

Chemistry

31. A school question classifies a compound as an analgesic. Which claim would require additional evidence rather than follow from the class name?

(A) That every amount produces only the desired effect with no other action.

(B) That a functional drug category is being used.

(C) That the intended classification involves pain relief.

(D) That the label alone does not give the molecular formula.

32. A first-order reaction is plotted as ln[A] versus time at fixedT. Which pair is correct?

(A) Slope=+k;doubling concentration doubles the half-life.

(B) Slope=-1/k;intercept is independent of initial concentration.

(C) Slope=-k;doubling initial concentration changes the intercept but not the half-life.

(D) Slope=-k;doubling initial concentration halves the half-life.

33. A carbon-containing organic sample contains S and Br among nitrogen, sulfur and halogens. It undergoes sodium fusion with sufficient sodium. Which conversion and subsequent test protocol is chemically consistent? For a silver-halide test, use a separately prepared portion boiled with nitric acid to remove cyanide/sulfide interference where present.

(A) sulfide and bromide; sulfur test positive; bromide test after removing interfering sulfide.

(B) Unchanged covalent bonds; add silver nitrate directly to the untreated organic sample.

(C) Nitrate and sulfate in every case; all give the same Prussian-blue test.

(D) thiocyanate; no further treatment is ever needed before adding silver nitrate.

34. A monohaloarene X with six carbons gives a compound P of formula C6H6O on heating with concentrated NaOH near623 K under high pressure,followed by acidification. P is converted to its sodium salt,treated with CO2 under Kolbe-Schmitt conditions,and acidified. Which pair is correct for X and the principal acid product?

(A) chlorobenzene;2-hydroxybenzoic acid

(B) chlorobenzene;3-hydroxybenzoic acid

(C) benzyl chloride;2-hydroxybenzoic acid

(D) chlorocyclohexane;benzoic acid

35. I:SOCl2 can convert a suitable aliphatic alcohol to an alkyl chloride. II:Exactly the same ordinary treatment cleanly replaces the phenolic OH of phenol by Cl through the same SN2 pathway. Choose the correct truth pattern.

(A) Both false.

(B) I false;II true.

(C) I true;II false.

(D) Both true.

36. For I: accumulation of a reactant on a solid catalyst surface; II: conversion of an indicator between its dissolved acid and base forms, which classification correctly identifies surface accumulation and distinguishes it from a bulk process?

(A) Both are adsorption because both involve molecular interactions.
(B) Only II involves adsorption.
(C) Neither involves adsorption.
(D) I involves adsorption; II is not adsorption.

37. Consider the statements. I. Warming water can reduce its equilibrium oxygen solubility under otherwise comparable conditions. II. A higher water temperature necessarily increases dissolved oxygen available to fish. Choose the correct truth pattern.

(A) I false; II true.
(B) I true; II false.
(C) I false; II false.
(D) I true; II true.

38. An intact sugar is nonreducing, but complete hydrolysis gives glucose and fructose, each positive in the standard alkaline reducing test. Which explanation fits?

(A) Fructose cannot ever give a positive alkaline reducing test.
(B) Hydrolysis frees anomeric carbonyl equivalents that were tied up in sucrose.
(C) Every glycosidic bond in every disaccharide removes both anomeric centres.
(D) Hydrolysis creates carbon atoms from water.

39. At 1 atm compare two separate pairs: (n-butane, isobutane) and (1-propanol, methoxyethane). A student selects isobutane for the first higher-boiling substance because its formula mass is unchanged, and 1-propanol for the second. Which assessment is consistent with intermolecular interactions?

(A) First selection wrong; second selection correct.
(B) Both selections wrong.
(C) Both selections correct.
(D) First selection correct; second selection wrong.

40. A closed fullerene cage has 80carbon atoms. Every carbon has three neighbours,and every face is a pentagon or hexagon. The cage contains12pentagons. How many hexagons are present?

(A) 68
(B) 40
(C) 12
(D) 30

41. At298K,a cell obeys E=0.600+$(-\frac{59}{2000})\log_{10}$Q in volts. Reverse the reaction and double every coefficient;define Q' for that new written reaction. Which pair(new E°,new slope of E versus logQ')is correct?

(A) $-\frac{3}{5}$V;$\frac{59}{4000}$V per log unit.
(B) $\frac{3}{5}$V;$-\frac{59}{2000}$V per log unit.
(C) $-\frac{3}{5}$V;$-\frac{59}{4000}$V per log unit.
(D) $-\frac{6}{5}$V;$-\frac{59}{4000}$V per log unit.

42. For homonuclear two-centre overlaps with internuclear axis x, consider independent arrangements 1: parallel p_y; same facing signs; 2: parallel p_x; same facing signs; 3: p_y on one centre and p_z on the other. The facing signs refer to the lobes after forming the final linear combination. Neglect interactions not specified. In the elementary real p-orbital model, what are (number giving a pi-symmetry MO, whether bonding or antibonding; number giving an antibonding MO with an extra intercentre node)?

(A) (1, 0)
(B) (2, 0)
(C) (0, 2)
(D) (1, 1)

43. For neutral HD and T2, take isotope mass numbers H=1, D=2, T=3 and neglect mass defects. Their gases have equal temperature. Which pair gives ($\langle v^2 \rangle$_HD/$\langle v^2 \rangle$_T2, number of neutrons in one T2 molecule minus that in one HD molecule)?

(A) (2, 2)
(B) (2, 3)
(C) (2, 5)
(D) (1/2, 3)

44. A Pt(4) coordination entity has coordination number 6, exactly 2 fully bound en ligand(s), and chloride in all remaining donor sites. Take en as 2-dentate with charge 0; chloride is monodentate. Find (number of ligand entities, net charge of the coordination entity).

(A) (5, +2)
(B) (6, +3)
(C) (4, +1)
(D) (4, +2)

45. A monatomic ion contains 18 electrons and has mass number 42. Its charge is +2. Which pair gives (proton count, neutron count), without confusing ionic charge with nuclear identity?

(A) (22, 20)
(B) (20, 22)
(C) (18, 24)
(D) (20, 24)

46. A student claims that deprotonating phenol destroys its aromaticity,so ethanol must be the stronger acid. Which correction addresses both the aromatic ring and the alcohol/phenol acidity comparison?

(A) Phenoxide retains the aromatic ring and gains resonance stabilisation of its negative charge; phenol is more acidic than ethanol.

(B) Phenol loses its ring carbons on deprotonation,so comparison is impossible.

(C) Phenoxide is antiaromatic because deprotonation adds two pi electrons to the benzene sextet.

(D) Ethoxide has the same ring-resonance stabilisation,so both acids must be equally strong.

47. An iron-ion inventory of 1/20 mol contains only Fe2+ and Fe3+. Complete externally driven reduction to metal, with no side reaction, consumes 7/50 mol electrons. What initial fraction of iron was Fe3+?

(A) 2/5
(B) 14/15
(C) 4/15
(D) 4/5

48. A compound has formula C3H6O and gives negative Tollens and positive iodoform. Restrict X to the candidate structures below. Which fits every clue?

(A) methoxyethene
(B) prop-2-en-1-ol
(C) propanal
(D) propanone

49. A sulfuric-acid stock of density 7/5 g/mL is diluted by taking20.0 mL to40.0 mL. The final molarity is 2 M. H2SO4 molar mass is98 g/mol. What is the mass fraction of H2SO4 in the stock?

(A) 14/25
(B) -7/25
(C) 7/50
(D) 7/25

50. Beam A has wavelength$3\lambda_0$ and beam B has$4\lambda_0$. A carries 3 times B's total energy. Photon-number ratioN_A/N_B is

(A) $\frac{9}{4}$ (B) 3

(C) 4 (D) $\frac{3}{4}$

51. A sample of 1,2-dibromoethane is treated with Zn in alcohol. Which isolated product is consistent with the reagent limitation and carbon-skeleton change?

(A) ethane (B) ethene

(C) ethyne (D) bromoethene

52. $\frac{93}{25}$ g aniline is acetylated to acetanilide in80.0% isolated molar yield. The isolated acetanilide is then mononitrated and gives p-nitroacetanilide in75.0% isolated molar yield. Both steps are1:1in substrate moles. Find final product mass in g. Use molar masses: aniline93, acetanilide135, p-nitroacetanilide180 g/mol.

(A) $\frac{36}{5}$ (B) $\frac{108}{25}$

(C) $\frac{27}{5}$ (D) $\frac{144}{25}$

53. At one T,ample solids establish $CaCO_3(s) \rightleftharpoons CaO(s)+CO_2(g)$,$K_p$=1 bar,and $C(s)+CO_2(g) \rightleftharpoons 2CO(g)$, K_p=16 bar. All named solids remain. He has partial pressure 2 bar. Equilibrium CO mole fraction is

(A) $\frac{1}{7}$ (B) $\frac{4}{5}$

(C) $\frac{4}{7}$ (D) $\frac{16}{19}$

54. A dry mixture contains only anhydrous SrCO3 and BaCO3. Use the ordinary Group2 carbonate stability trend. In an idealised selective-heating window, the less thermally stable carbonate decomposes completely by MCO3→MO+CO2, while the more stable carbonate remains unchanged. No other mass loss occurs. This is the stated model, not a specified real temperature. Atomic masses: Sr=88, Ba=137, C=12, O=16. The selective stage gives 28/125 L CO2; subsequent stronger heating to decompose all remaining carbonate brings the cumulative CO2 volume to 112/125 L. Both dry volumes are at STP; use22.4 L/mol, with complete collection. What was the initial mass fraction of BaCO3?

(A) 3/4 (B) 591/739

(C) 148/739 (D) 542/739

55. A crystal contains identical A atoms only at the corners of a cubic conventional cell; body and face centres are empty. Which claim follows?

(A) It is FCC because corner atoms touch an imagined face centre.

(B) It is BCC because the empty body centre counts as a lattice point.

(C) It is primitive cubic with one lattice point per cell; an edge-half translation is not a symmetry of the occupied sites.

(D) Its atom count is eight because shared corner sites are never apportioned.

56. A weak 1:1 electrolyte obeys exact Ka=Cα^2/(1-α). In a closed dilution its dissociation degree changes from 1/35 to 1/7, while Ka and total formula-unit amount remain constant. By what factor does its volume increase? Do not replace 1-α by 1.

(A) 3/85 (B) 85/3

(C) 25 (D) 5

57. Use octet Lewis contributors for $CO3^{2}$- and SO2; for SO2 exclude expanded-octet contributors. For HCOO-, HCO3- and HNO3, average only the two nonprotonated central-atom-O bonds. For other species average all central-atom bonds. Find (average bond order of the first minus that of the second, sum of total lone-pair counts in one contributor of each). A lone pair on any atom is counted once.

(A) (17/6, 14)
(B) (-1/6, 15)
(C) (-1/6, 14)
(D) (-1/6, 13)

58. An ideal H2/N2 gas mixture has a molecule-number-weighted mean of v^2 equal to7/2 times that of pure N2 at the same temperature. M(H2)=2 and M(N2)=28 g/mol. What is the H2 mole fraction? Use species $\langle v^2\rangle$=3RT/M.

(A) 5/26
(B) -5/26
(C) 5/13
(D) 5/52

59. A0.100mol/kg monoprotic weak acid solution hasΔTf=$\frac{11}{50}$K in a solvent withKf=2.00Kkg/mol. For this calibrated dilute model,take analytical molarity as0.100M,with no association. Using exact Ka=Cα^2/(1-α),Ka is

(A) $\frac{121}{1000}$
(B) $\frac{1}{10}$
(C) $\frac{1}{1000}$
(D) $\frac{1}{900}$

60. At fixed T,P a binary R$\rightleftharpoons$P system has G=20 kJ for pure R and 50 kJ for pure P on one fixed total-amount basis. The unique minimum G=15 kJ occurs at product fraction 3/5. Which claim is consistent with the curve?

(A) The equilibrium must be pure R because 20<50.
(B) The interior minimum means the two pure endpoints have equal G.
(C) The equilibrium product fraction is 3/5; moving from the minimum toward either pure endpoint raises G.
(D) The higher pure-product endpoint proves that product cannot be present at equilibrium.

English Proficiency

61. Choose the correctly spelt word.

(A) enviornment
(B) environement
(C) enviroment
(D) environment

62. Choose the passive-voice equivalent that retains the tense, participants and all stated restrictions. The technician is repairing the instrument that failed during the demonstration.

(A) Every instrument used during the demonstration is being repaired by the technician.
(B) The instrument that failed during the demonstration is being repaired by the technician.
(C) The instrument is repairing the technician who failed during the demonstration.
(D) The instrument that failed during the demonstration has been repaired by the technician.

63. He checks the final page proofs for misplaced letters and punctuation rather than deciding the book's argument. What is he?

(A) illustrator (B) proofreader

(C) publisher (D) reviewer

64. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. The report, unlike the earlier note, Q. and a decision that was approved. R. a proposal that was discussed S. distinguishes between

(A) P-S-R-Q (B) R-P-S-Q

(C) S-P-R-Q (D) Q-S-R-P

65. Wheelchair access requires a step-free passage of sufficient width. Complete: _____ of the two proposed entrances is suitable; one has steps and the other is too narrow.

(A) Neither (B) Any

(C) Every (D) Both

66. Use the standard contemporary negative adjective. The rule allowed no difference of opinion; the attitude was _____.

(A) untolerant (B) distolerant

(C) intolerant (D) imtolerant

67. Read the sentence: The inclusive rule allowed every member, not just officers, to vote. Choose the opposite in meaning of 'inclusive' as used in the sentence.

(A) broad (B) general

(C) generous (D) exclusive

68. The investigation concerns stars and planets beyond the earth's atmosphere, rather than local weather. Which field is it?

(A) astronomy (B) geology

(C) meteorology (D) geography

69. Arrange the parts to make one grammatical sentence, preserving all the given words and punctuation. P. the audience to enter Q. did the stage manager allow R. the newly repaired hall. S. Not until the lights had been tested

(A) Q-S-P-R (B) S-Q-P-R

(C) P-S-Q-R (D) R-Q-P-S

70. Read the sentence: The quiet room was conducive to concentrated reading. Choose the closest meaning of 'conducive' as used in the sentence.

(A) proof of (B) necessary for

(C) helpful to (D) designed solely for

Logical Reasoning

71. Consider only these statements; universal statements alone do not imply existence. No campers are artists; Some artists are not campers. Conclusions: I. No artists are campers. II. Some hikers are artists. Which conclusions follow necessarily?

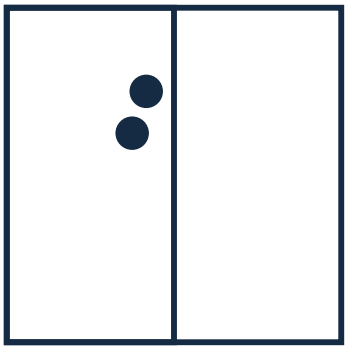
(A) Both I and II

(B) Neither I nor II

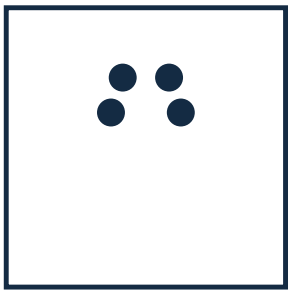
(C) Only II

(D) Only I

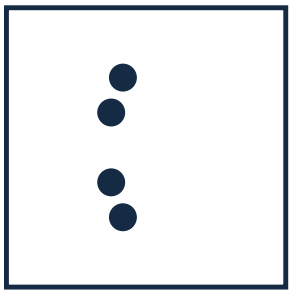
72. Starting with a unit square sheet, fold right to left along the vertical midline. The shown folded sheet is punched at both marked positions through every layer. Select the fully unfolded pattern.



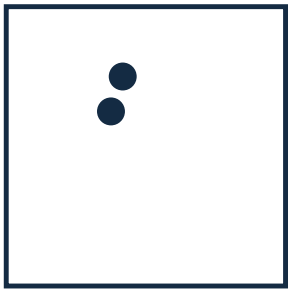
(A) Pattern K



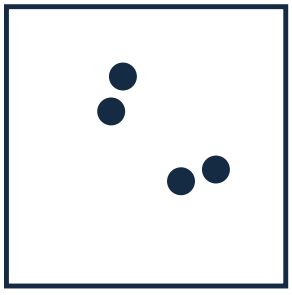
(B) Pattern P



(C) Pattern M



(D) Pattern Y



73. Lata points to a photograph and says, "This person is my mother's only son." Lata is female, and the photographed person is male. How is the photographed person related to Lata?

(A) Daughter

(B) Brother

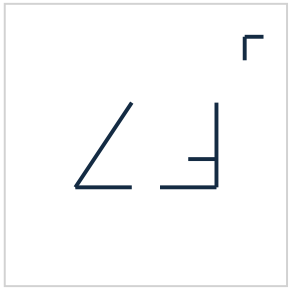
(C) Son

(D) Sister

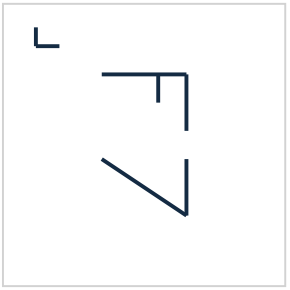
74. Choose the exact vertical mirror reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



(A) Pattern X



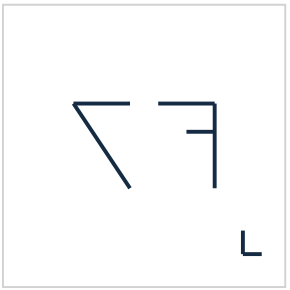
(B) Pattern N



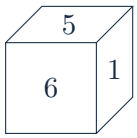
(C) Pattern Z



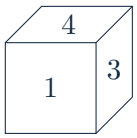
(D) Pattern L



75. The views show the same die. Give the ordered pair: (face opposite 4, face opposite 2). Do not assume opposite labels sum to seven.



1



2

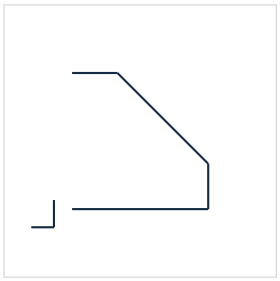
(A) $(6, 3)$

(B) $(6, 4)$

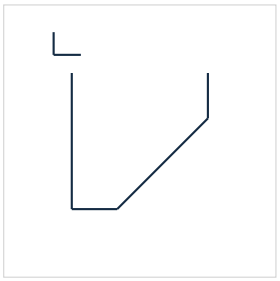
(C) $(6, 1)$

(D) $(6, 2)$

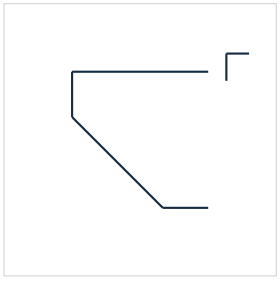
76. Select the missing bottom-right tile. The four tiles follow successive quarter-turns clockwise in the order top-left, top-right, bottom-left, bottom-right.



top left



top right



bottom left

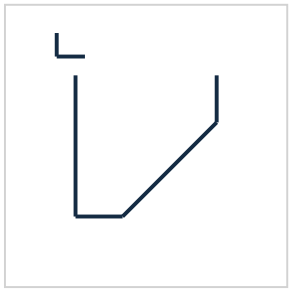
?

bottom right

(A) Pattern P



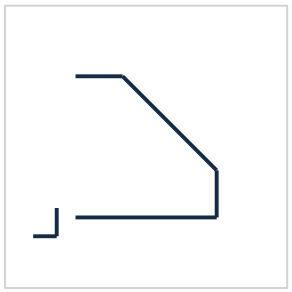
(B) Pattern Y



(C) Pattern M



(D) Pattern X

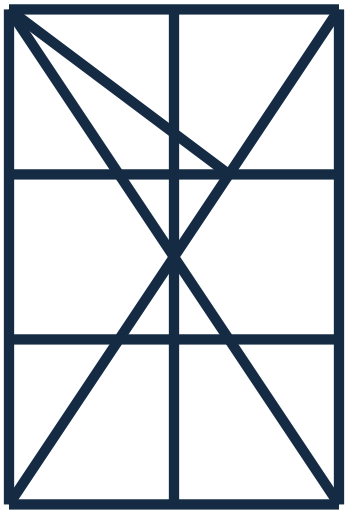


77. A code shifts letters cyclically through A–Z. The same shift is used at every position. For example, RIVER is coded as WNAJW. Which word is coded as GWNHP?

(A) AQHBJ (B) BRICK

(C) LBSMU (D) CSJDL

78. How many triangles are formed by all the drawn segments?



(A) 25 (B) 34

(C) 35 (D) 31

79. Successive steps alternate between multiplication by one fixed integer from 2 to 5 and addition of one fixed integer from 1 to 10; multiplication comes first. Find the missing term: 2, 6, 14, 42, ?, 150, 158.

(A) 50 (B) 70

(C) 126 (D) 118

80. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

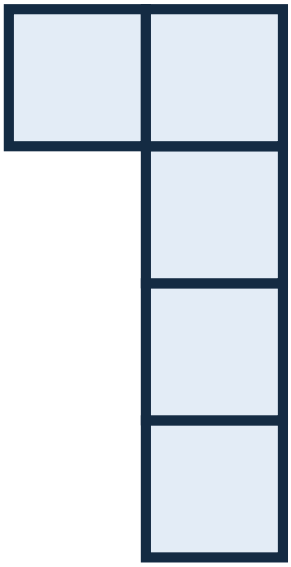


piece 1

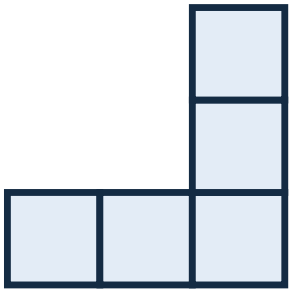


piece 2

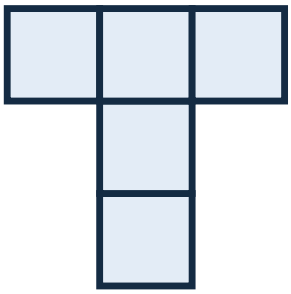
(A) Pattern T



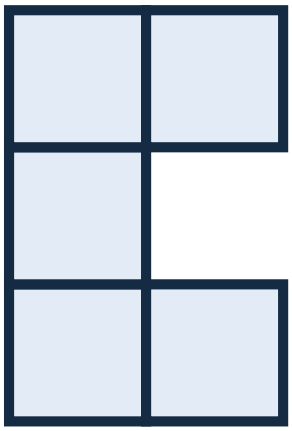
(B) Pattern P



(C) Pattern K



(D) Pattern Z



81. Which numbered host contains the complete small target, including its separate short stroke? Keep the given orientation; translation is allowed and extra host lines may be ignored.





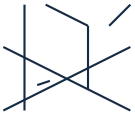
host 1



host 2



host 3



host 4

(A) 4
(B) 1
(C) 3
(D) 2

82. Three numbered figures have an even difference between their outer-polygon and inner-polygon side counts. Which one does not?



1



2



3



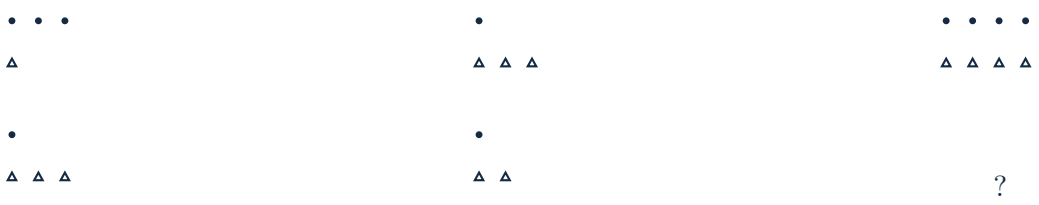
4

(A) 4
(B) 1
(C) 2
(D) 3

83. Preserve the exact relation and its order: Ship : Fleet :: ? : Team. Choose the missing term.

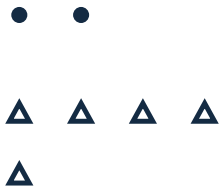
(A) Bee
(B) Lion
(C) Player
(D) Wolf

84. The last cell of each row combines the counts in its first two cells, separately for dots and triangles. Which cell fills the gap? Counts of different symbol types cannot be exchanged.



(A) Pattern R

(B) Pattern Z



(C) Pattern M



(D) Pattern Q





85. In a row, Asha is 6th from the left and 27th from the right. Bina is 9th from the right. Charu stands 3 places to the left of Bina. How many people stand strictly between Asha and Charu?

(A) 16

(B) 15

(C) 14

(D) 20

86. Five people occupy positions 1 to 5 from left to right, one per position, and wear five different shirt colours: red, blue, green, white and yellow. Bina is 1st from the left. Asha is 2nd from the left. Dev is 3rd from the left. Charu is 4th from the left. Esha is 5th from the left. The blue shirt is in position 1. The white shirt is in position 4. The yellow shirt is in position 5. Asha does not wear green. Which shirt does Asha wear?

(A) Green

(B) Red

(C) Blue

(D) White

87. A single affine rule $y = ax + b$ is used, with the same integers a, b. It gives $2 \mapsto 11$ and $5 \mapsto 20$. Which input maps to 41?

(A) 11

(B) 13

(C) 41

(D) 12

88. The age table uses the same two people throughout:

	younger	older
now	x	$x+12$
2 years ago	$x-2$	$x+10$

. The product of their ages 2 years ago was 748. What will the older person's age be 5 years from now?

(A) 41 (B) 36

(C) 39 (D) 53

89. Each letter position has its own fixed cyclic shift, applied identically in both pairs (Z wraps to A). Complete DDL : BFK :: WQK : ?.

(A) USJ (B) TRI

(C) VTK (D) ISV

90. Consider only these statements; universal statements alone do not imply existence. All editors are artists; Some designers are not editors; Some designers are artists. Conclusions: I. Some editors are not designers. II. Some artists are not designers. Which conclusions are individually possible without contradicting the statements?

(A) Only I (B) Neither I nor II

(C) Only II (D) Both I and II

Mathematics

91. A,B are acute, A+B=$\pi/4$ and tan A=$\frac{3}{4}$. Find tan(A-B).

(A) $-\frac{17}{31}$ (B) $\frac{17}{31}$

(C) $\frac{1}{7}$ (D) $\frac{17}{28}$

92. The parabola $y=x^2+px+q$ passes through(0,16), with $p\geq 0$. Its vertex ordinate has minimum possible absolute value. Find p^2+q^2.

(A) 64 (B) 320

(C) 1280 (D) 256

93. $F'(x)=\frac{(x+1)^2}{(x+2)^4}$ and F(0)=2. Find F(3)-2F(0).

(A) $\frac{2129}{1000}$ (B) $-\frac{2129}{1000}$

(C) $\frac{129}{1000}$ (D) $-\frac{1871}{1000}$

94. $f(x)=(x^2-16)/(x-4)$ for $x\neq 4$, and f(4)=k. If f is differentiable at 4, find $k+f'(4)$.

(A) 8 (B) 9

(C) 17 (D) 16

95. An increasing positive GP has first three terms with product 512 and first-plus-third sum 20. Find its fourth term.

(A) 16 (B) 2

(C) 8 (D) 32

96. Find the maximum of $\sin^6 x \cos x$ on $[0,\pi/2]$.

(A) $\frac{216\sqrt{7}}{2401}$ (B) $\frac{36\sqrt{6}}{343}$

(C) $\frac{\sqrt{7}}{7}$ (D) $\frac{\sqrt{2}}{16}$

97. $z=5+2i$. Find $1/z+1/\text{conjugate}(z)$.

(A) $\frac{1}{10}$ (B) $\frac{10}{29}$

(C) 10 (D) $\frac{4}{29}$

98. Disjoint A,B have sizes 6,2. Find $|(A\times A)\cup(B\times B)|$.

(A) 24 (B) 40

(C) 52 (D) 12

99. An AP has first term 3, sum of its first 7 terms equal to 63. Find the sum of the squares of those 7 terms.

(A) 679 (B) 63

(C) 454 (D) 3969

100. Maximize $Z=3x+4y$ subject to $x\geq 0$, $y\geq 5$, $x+y\leq 15$, $x+2y\leq 19$.

(A) 50 (B) 38

(C) 47 (D) 49

101. The tangent to $y = x^3 - 8x$ at $x = 3$ meets the two coordinate axes at A and B. Find the area of triangle OAB, where O is the origin.

(A) $\frac{1458}{19}$ (B) $\frac{729}{19}$

(C) 54 (D) $\frac{2916}{19}$

102. In $(1+x)^9$, find the ratio of the coefficient of x^7 to the coefficient of x^6.

(A) $\frac{3}{7}$ (B) $\frac{2}{3}$

(C) 3 (D) $\frac{7}{3}$

103. $S_n=7n^2+3n$ is a sequence partial sum. Find $a_1 + a_3 + \cdots + a_{19}$.

(A) 1396 (B) 1430

(C) 730 (D) 1360

104. Let $\alpha=\pi/9$, $\beta=\pi/8$. Evaluate $\arccos(-\cos\alpha\cos\beta-\sin\alpha\sin\beta)$.

(A) $\frac{\pi}{72}$ (B) $\frac{73\pi}{72}$

(C) $-\frac{\pi}{72}$ (D) $\frac{71\pi}{72}$

105. An equivalence relation has classes of sizes 6,8,5. How many unordered pairs of distinct related elements are there?

(A) 37 (B) 53

(C) 171 (D) 125

106. a=(9,6,0), b=(6,-9,5). Find $|a+b|^2$-$|a-b|^2$.

(A) -180 (B) 432

(C) 0 (D) 259

107. Using digits 0,1,2,3,4,5,6 without repetition, how many 5-digit positive integers are divisible by 4, have first digit at most 4, and first digit smaller than last?

(A) 264 (B) 840

(C) 240 (D) 36

108. How many pairs $1 \le m < n \le 10$ satisfy both $5 \mid (6^m+9^n)$ and $3 \mid (m+n)$?

(A) 7 (B) 20

(C) 6 (D) 12

109. Two points on $y^2=8x$ have parameters with product -1. Find the latus-rectum length of their midpoint locus.

(A) 8 (B) 2

(C) 16 (D) 4

110. The hyperbola $x^2/9-y^2/36=1$ has eccentricity e. Find the eccentricity e' of its conjugate hyperbola.

(A) $\frac{\sqrt{5}}{5}$ (B) $3\sqrt{5}$

(C) $\frac{\sqrt{5}}{2}$ (D) $\sqrt{5}$

111. For $t \ge 0$, defineF(t)=lim as $n \to \infty$ of $n^{(-3)}\Sigma$floor(tj^2),j=1,...,n. Evaluate$\Sigma F((4/6)^r)$,r=1,...,∞.

(A) $\frac{2}{3}$ (B) 0

(C) 1 (D) 2

112. $y'+10y=\exp(-10x)$, $y(0)=7$. Find $\exp(10)y(1)-7$.

(A) 1 (B) 8

(C) -1 (D) $\frac{8}{e^{10}}$

113. Evaluate $\int_0^8 x \,\text{floor}(x)\, dx$.

(A) 154 (B) $\frac{512}{3}$

(C) 28 (D) 140

114. Find the area bounded by $y=x$, $y=49/x$, and $x=28$ in the first quadrant.

(A) $147 - 98\log(2)$ (B) $98\log(2)$

(C) $\frac{735}{2} - 98\log(2)$ (D) $\frac{735}{2}$

115. $A = \begin{bmatrix} 3 & 2 \\ -2 & 6 \end{bmatrix}$, K=(A-$A^T$)/2. Find tr($K^2$).

(A) 0 (B) -8

(C) 8 (D) -4

116. How many injective functions from{a,b,c} to{1,...,10} satisfy f(a)<f(b)<f(c) and f(a)+f(c) even?

(A) 30 (B) 120

(C) 320 (D) 60

117. From a point at horizontal distance 4 from a tower, its elevation is 2θ, where $\tan\theta=\frac{2}{3}$ and $0<\theta<45°$. Find the height.

(A) $\frac{16}{3}$ (B) $\frac{5}{3}$

(C) $\frac{48}{5}$ (D) $\frac{8}{3}$

118. Four observations have $\Sigma(x-12)^2=254$, $\Sigma(x-14)^2=254$. Find their population variance.

(A) $\frac{127}{2}$ (B) 250

(C) $\frac{125}{2}$ (D) $\frac{5\sqrt{10}}{2}$

119. The line y=mx through O bisects the area of the triangle with vertices O,(3,6),(5,5). Assume the vertices are noncollinear. Find m.

(A) $\frac{8}{11}$ (B) $\frac{11}{8}$

(C) $\frac{5}{3}$ (D) $-\frac{1}{2}$

120. O=(0,0,0), A=(7,0,0), B=(1,7,0). Find the squared area of triangle OAB.

(A) $\frac{49}{2}$ (B) 2401

(C) $\frac{2401}{2}$ (D) $\frac{2401}{4}$

121. Circles $x^2+y^2=121$ and $(x-6)^2+y^2=121$ meet at A,B. Find $|AB|^2$.

(A) 112 (B) 484

(C) 256 (D) 448

122. Find the squared distance from O to the plane through (3,0,0),(0,5,0),(0,0,5).

(A) $\frac{15}{11}$ (B) 59

(C) $\frac{225}{43}$ (D) $\frac{43}{225}$

123. Let $z=(1+i)^{11}/(1-i)^6$. Find Re(z)+Im(z).

(A) 0 (B) 8

(C) $4\sqrt{2}$ (D) -512

124. A circular sector has radius 5 and arc length $\frac{5\pi}{3}$. Find its area.

(A) $\frac{25\pi}{3}$ (B) 25π

(C) $\frac{25}{6}$ (D) $\frac{25\pi}{6}$

125. Under x=X+2, y=Y+6, the parabola $(y-6)^2=32(x-2)$ is normalized. Find the x coordinate of its focus in the original axes.

(A) 10 (B) -6

(C) 8 (D) 34

126. There are 17 Boolean variables p_1,...,p 17. Exactly 3 variables must be true. How many truth assignments satisfy [(p_1 implies p_2) and (p_1 implies not p_2)]?

(A) 560 (B) 455

(C) 680 (D) 131072

127. Let $D(t) = \det \begin{pmatrix} 5+t & 1+t \\ 1+t & 5+t \end{pmatrix}$. Find D′(0).

(A) 8 (B) 48

(C) -12 (D) -8

128. P(A)=$\frac{11}{19}$, P(B)=$\frac{6}{19}$, P(A∩B)=$\frac{3}{19}$. Find P(neither A nor B).

(A) $\frac{14}{19}$ (B) $\frac{11}{19}$

(C) $\frac{5}{19}$ (D) $\frac{8}{19}$

129. Two points on opposite sides of a tower foot are 14 metres apart on a straight level line. The elevations of the tower top are 30° and 45°. Find the tower height.

(A) $\frac{7}{2}$ (B) $-7+7\sqrt{3}$

(C) $\frac{7\sqrt{3}}{3}$ (D) $21-7\sqrt{3}$

130. For $(d^4y/dx^4)^2+(y')^2+x=0$, find order+degree if the degree is defined; otherwise select "Degree is not defined".

(A) 8 (B) 7

(C) Degree is not defined (D) 6

Optional bonus section

Stop before this section unless you are choosing bonus entry. All 130 main questions must already have answers. You cannot revise main answers after entering. No extra time is allowed.

Physics

131. A particle satisfies $v_x = 3x$ and $v_y = 6$ in SI units, with $dx/dt = v_x$ and $dy/dt = v_y$. What angle does its acceleration make with +x when x=5 m? Differentiate the velocity components with respect to time.

(A) 90 degree

(B) 180 degree

(C) 0 degree

(D) 45 degree

132. Unpolarised light passes three ideal polarisers with axes at 0°, 15°, and 45° relative to a common reference. Find final intensity divided by incident intensity.

(A) $\frac{3}{8}$

(B) $\frac{3\sqrt{3}}{32} + \frac{3}{16}$

(C) $\frac{\sqrt{3}}{8} + \frac{1}{4}$

(D) $\frac{\sqrt{3}}{16} + \frac{1}{8}$

133. A stationary nucleus with mass number 32 and binding energy per nucleon $\frac{38}{5}$ MeV splits into two equal daughters, each having total binding energy $\frac{1241}{10}$ MeV. Which statement is correct?

(A) The process releases 5 MeV; the daughters have larger binding energy per nucleon.

(B) No energy can be released because fission always requires a larger mass number than 32.

(C) The process absorbs 5 MeV; the daughters have larger binding energy per nucleon.

(D) The process releases 10 MeV; each daughter releases 5 MeV.

Chemistry

134. Evaluate I: BF3 is trigonal planar in the elementary VSEPR model. II: NH3 is trigonal planar because nitrogen makes three bonds.

(A) Neither is correct.

(B) Only I is correct.

(C) Only II is correct.

(D) Both I and II are correct.

135. Two sulfides MS and NS have Ksp values10^{-20} and10^{-16}. Initially both free metal-ion concentrations are0.0100 M. For at least99.9% precipitation of M with no N precipitation, which free sulfide interval works? Ignore complexes and volume change.

(A) $10^{-15} \leq$[S2-]$<10^{-14}$ M.

(B) $10^{-18} \leq$[S2-]$<10^{-16}$ M.

(C) [S2-]$<10^{-18}$ M for both quantitative precipitation and no precipitation.

(D) [S2-]$\geq 10^{-14}$ M with no upper limit.

136. A nucleotide contains a pentose bearingOH at the 2-prime carbon and uracil as its base. Which identification and contrast are correct?

(A) A disaccharide; uracil is glucose.

(B) A standardDNA thymidine unit; uracil is thymine.

(C) AnRNA building block; DNA normally hasH instead of that2-primeOH.

(D) A peptide; OH proves an amide bond.

Logical Reasoning

137. Choose the additional piece that, together with BOTH given pieces, covers the target exactly once. Quarter-turns and translations are allowed; flipping is forbidden.



target

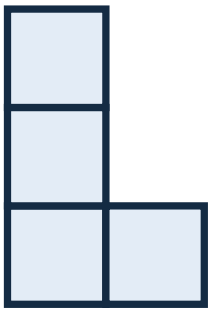


piece 1



piece 2

(A) Pattern U



(B) Pattern T



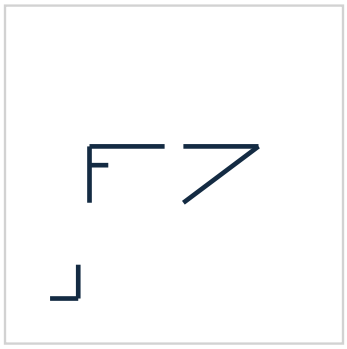
(C) Pattern Z



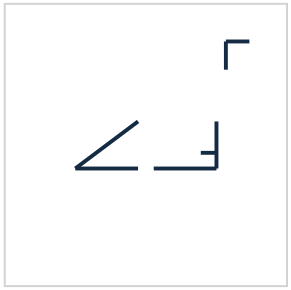
(D) Pattern P



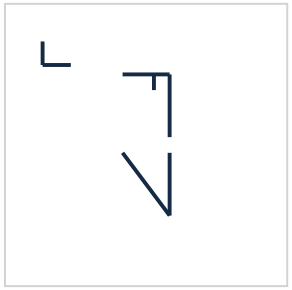
138. Choose the exact horizontal water reflection of the shown figure. The entire flag is part of the object; the outer grey square is only a fixed reference frame.



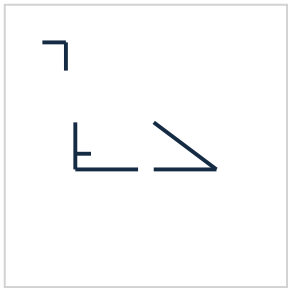
(A) Pattern Z



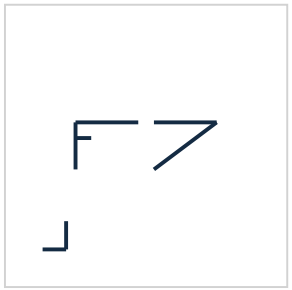
(B) Pattern W



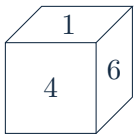
(C) Pattern R



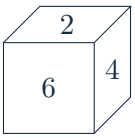
(D) Pattern U



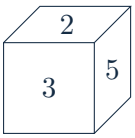
139. The views show the same die. Give the ordered pair: (face opposite 6, face opposite 5). Do not assume opposite labels sum to seven.



1



2



3

(A) $(3, 3)$

(B) $(3, 4)$

(C) $(3, 2)$

(D) $(3, 1)$

Mathematics

140. For parabola $(y-5)^2=36(x-7)$ with focus abscissa p, find the maximum of $|z-7|$ when $|z-p|=36$.

(A) 45 (B) 36

(C) 52 (D) 27

141. Event E has likelihood $\frac{2}{7}$ under source A and $\frac{5}{7}$ under source B. Its total probability is $\frac{4}{7}$. If the source is A with probability r, find r.

(A) $\frac{2}{3}$ (B) $\frac{2}{7}$

(C) $\frac{1}{3}$ (D) $\frac{4}{7}$

142. How many integers k make $18\sin x+24\cos x+6=k$ have exactly two distinct solutions in $0\leq x<2\pi$?

(A) 60 (B) 59

(C) 61 (D) 29

BITSAT Mock 01

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**B**	27	**D**	53	**B**	79	**B**	105	**B**
2	**B**	28	**C**	54	**A**	80	**D**	106	**C**
3	**D**	29	**A**	55	**A**	81	**D**	107	**C**
4	**C**	30	**A**	56	**B**	82	**A**	108	**B**
5	**D**	31	**C**	57	**D**	83	**C**	109	**D**
6	**D**	32	**B**	58	**B**	84	**B**	110	**D**
7	**A**	33	**A**	59	**C**	85	**B**	111	**A**
8	**C**	34	**A**	60	**B**	86	**C**	112	**A**
9	**A**	35	**A**	61	**D**	87	**A**	113	**D**
10	**B**	36	**D**	62	**A**	88	**D**	114	**D**
11	**D**	37	**C**	63	**B**	89	**D**	115	**D**
12	**A**	38	**B**	64	**B**	90	**B**	116	**C**
13	**C**	39	**B**	65	**C**	91	**D**	117	**D**
14	**C**	40	**C**	66	**A**	92	**B**	118	**A**
15	**B**	41	**A**	67	**A**	93	**B**	119	**A**
16	**C**	42	**D**	68	**D**	94	**C**	120	**C**
17	**A**	43	**D**	69	**B**	95	**A**	121	**C**
18	**D**	44	**C**	70	**C**	96	**C**	122	**A**
19	**B**	45	**A**	71	**B**	97	**B**	123	**A**
20	**A**	46	**D**	72	**D**	98	**D**	124	**C**
21	**C**	47	**A**	73	**A**	99	**D**	125	**C**
22	**B**	48	**D**	74	**A**	100	**D**	126	**B**
23	**D**	49	**B**	75	**C**	101	**B**	127	**A**
24	**B**	50	**C**	76	**C**	102	**A**	128	**C**
25	**A**	51	**C**	77	**A**	103	**B**	129	**B**
26	**B**	52	**D**	78	**C**	104	**A**	130	**B**

Optional bonus questions 131–142

131	**C**	132	**A**	133	**B**	134	**A**	135	**C**	136	**B**
137	**A**	138	**B**	139	**C**	140	**B**	141	**C**	142	**A**

BITSAT Mock 02

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**B**	27	**C**	53	**A**	79	**D**	105	**C**
2	**D**	28	**B**	54	**A**	80	**A**	106	**A**
3	**A**	29	**A**	55	**D**	81	**D**	107	**B**
4	**B**	30	**A**	56	**C**	82	**A**	108	**C**
5	**C**	31	**B**	57	**B**	83	**A**	109	**A**
6	**B**	32	**A**	58	**B**	84	**D**	110	**D**
7	**B**	33	**C**	59	**C**	85	**C**	111	**B**
8	**B**	34	**C**	60	**B**	86	**B**	112	**C**
9	**A**	35	**B**	61	**C**	87	**A**	113	**C**
10	**C**	36	**D**	62	**B**	88	**B**	114	**A**
11	**B**	37	**D**	63	**C**	89	**C**	115	**B**
12	**C**	38	**C**	64	**A**	90	**C**	116	**D**
13	**D**	39	**A**	65	**D**	91	**A**	117	**A**
14	**A**	40	**D**	66	**B**	92	**C**	118	**C**
15	**D**	41	**B**	67	**D**	93	**D**	119	**D**
16	**C**	42	**A**	68	**A**	94	**D**	120	**B**
17	**D**	43	**D**	69	**A**	95	**A**	121	**A**
18	**C**	44	**A**	70	**B**	96	**C**	122	**A**
19	**D**	45	**C**	71	**D**	97	**A**	123	**B**
20	**A**	46	**B**	72	**A**	98	**D**	124	**C**
21	**A**	47	**D**	73	**B**	99	**B**	125	**B**
22	**D**	48	**C**	74	**D**	100	**B**	126	**B**
23	**C**	49	**D**	75	**B**	101	**A**	127	**C**
24	**B**	50	**A**	76	**C**	102	**D**	128	**D**
25	**A**	51	**B**	77	**B**	103	**B**	129	**C**
26	**D**	52	**A**	78	**C**	104	**D**	130	**D**

Optional bonus questions 131–142

131	**A**	132	**B**	133	**C**	134	**A**	135	**C**	136	**B**
137	**B**	138	**A**	139	**C**	140	**B**	141	**C**	142	**A**

BITSAT Mock 03

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**B**	27	**D**	53	**B**	79	**C**	105	**B**
2	**D**	28	**B**	54	**A**	80	**C**	106	**A**
3	**C**	29	**D**	55	**C**	81	**A**	107	**D**
4	**D**	30	**B**	56	**C**	82	**A**	108	**C**
5	**C**	31	**B**	57	**D**	83	**C**	109	**A**
6	**A**	32	**A**	58	**A**	84	**A**	110	**C**
7	**B**	33	**C**	59	**A**	85	**C**	111	**A**
8	**C**	34	**C**	60	**B**	86	**D**	112	**C**
9	**B**	35	**D**	61	**B**	87	**D**	113	**C**
10	**A**	36	**C**	62	**B**	88	**B**	114	**B**
11	**A**	37	**D**	63	**C**	89	**B**	115	**A**
12	**A**	38	**C**	64	**D**	90	**D**	116	**B**
13	**B**	39	**C**	65	**B**	91	**B**	117	**A**
14	**D**	40	**B**	66	**A**	92	**C**	118	**A**
15	**D**	41	**B**	67	**C**	93	**D**	119	**C**
16	**A**	42	**A**	68	**A**	94	**D**	120	**D**
17	**B**	43	**A**	69	**A**	95	**B**	121	**B**
18	**C**	44	**B**	70	**D**	96	**D**	122	**A**
19	**C**	45	**B**	71	**D**	97	**A**	123	**C**
20	**D**	46	**D**	72	**B**	98	**D**	124	**D**
21	**A**	47	**B**	73	**C**	99	**D**	125	**C**
22	**B**	48	**D**	74	**A**	100	**C**	126	**D**
23	**C**	49	**D**	75	**A**	101	**B**	127	**A**
24	**A**	50	**A**	76	**D**	102	**B**	128	**A**
25	**A**	51	**D**	77	**B**	103	**B**	129	**B**
26	**C**	52	**A**	78	**B**	104	**D**	130	**C**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**A**	135	**B**	136	**C**
137	**B**	138	**C**	139	**A**	140	**C**	141	**A**	142	**B**

BITSAT Mock 04

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**A**	53	**B**	79	**B**	105	**B**
2	**C**	28	**A**	54	**B**	80	**B**	106	**B**
3	**D**	29	**D**	55	**B**	81	**C**	107	**A**
4	**C**	30	**C**	56	**D**	82	**B**	108	**B**
5	**C**	31	**B**	57	**D**	83	**A**	109	**B**
6	**D**	32	**B**	58	**A**	84	**B**	110	**B**
7	**B**	33	**C**	59	**A**	85	**A**	111	**D**
8	**B**	34	**C**	60	**B**	86	**D**	112	**B**
9	**A**	35	**C**	61	**A**	87	**A**	113	**B**
10	**C**	36	**B**	62	**B**	88	**D**	114	**C**
11	**A**	37	**A**	63	**B**	89	**D**	115	**A**
12	**B**	38	**C**	64	**D**	90	**D**	116	**D**
13	**A**	39	**C**	65	**A**	91	**D**	117	**D**
14	**C**	40	**D**	66	**A**	92	**C**	118	**C**
15	**A**	41	**D**	67	**D**	93	**C**	119	**A**
16	**B**	42	**D**	68	**C**	94	**C**	120	**C**
17	**B**	43	**B**	69	**B**	95	**D**	121	**C**
18	**B**	44	**A**	70	**C**	96	**D**	122	**B**
19	**A**	45	**D**	71	**A**	97	**A**	123	**C**
20	**B**	46	**D**	72	**B**	98	**D**	124	**D**
21	**D**	47	**A**	73	**D**	99	**A**	125	**B**
22	**C**	48	**C**	74	**C**	100	**A**	126	**C**
23	**A**	49	**A**	75	**C**	101	**D**	127	**B**
24	**D**	50	**A**	76	**A**	102	**D**	128	**A**
25	**B**	51	**C**	77	**C**	103	**A**	129	**C**
26	**D**	52	**A**	78	**C**	104	**A**	130	**A**

Optional bonus questions 131–142

131	**A**	132	**C**	133	**B**	134	**B**	135	**C**	136	**A**
137	**B**	138	**A**	139	**C**	140	**C**	141	**A**	142	**B**

BITSAT Mock 05

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**A**	27	**D**	53	**D**	79	**A**	105	**B**
2	**B**	28	**A**	54	**A**	80	**D**	106	**D**
3	**A**	29	**C**	55	**D**	81	**B**	107	**A**
4	**C**	30	**C**	56	**A**	82	**D**	108	**B**
5	**D**	31	**B**	57	**D**	83	**B**	109	**A**
6	**B**	32	**B**	58	**B**	84	**A**	110	**A**
7	**A**	33	**B**	59	**A**	85	**B**	111	**D**
8	**A**	34	**C**	60	**B**	86	**C**	112	**C**
9	**A**	35	**C**	61	**B**	87	**C**	113	**D**
10	**B**	36	**B**	62	**B**	88	**D**	114	**D**
11	**C**	37	**B**	63	**A**	89	**A**	115	**C**
12	**D**	38	**A**	64	**D**	90	**A**	116	**A**
13	**D**	39	**A**	65	**A**	91	**B**	117	**B**
14	**B**	40	**D**	66	**A**	92	**B**	118	**D**
15	**C**	41	**A**	67	**C**	93	**C**	119	**C**
16	**B**	42	**C**	68	**D**	94	**D**	120	**A**
17	**D**	43	**D**	69	**B**	95	**D**	121	**A**
18	**B**	44	**D**	70	**C**	96	**C**	122	**D**
19	**D**	45	**C**	71	**D**	97	**A**	123	**C**
20	**D**	46	**B**	72	**C**	98	**C**	124	**A**
21	**B**	47	**C**	73	**D**	99	**B**	125	**B**
22	**A**	48	**C**	74	**A**	100	**B**	126	**A**
23	**C**	49	**C**	75	**C**	101	**D**	127	**D**
24	**B**	50	**A**	76	**C**	102	**B**	128	**B**
25	**C**	51	**D**	77	**B**	103	**A**	129	**C**
26	**A**	52	**A**	78	**B**	104	**C**	130	**C**

Optional bonus questions 131–142

131	**C**	132	**B**	133	**A**	134	**A**	135	**B**	136	**C**
137	**B**	138	**C**	139	**A**	140	**A**	141	**C**	142	**B**

BITSAT Mock 06

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**B**	53	**B**	79	**C**	105	**A**
2	**B**	28	**A**	54	**D**	80	**A**	106	**C**
3	**B**	29	**C**	55	**D**	81	**C**	107	**D**
4	**B**	30	**D**	56	**B**	82	**C**	108	**C**
5	**C**	31	**B**	57	**D**	83	**B**	109	**B**
6	**A**	32	**A**	58	**A**	84	**D**	110	**B**
7	**B**	33	**B**	59	**C**	85	**B**	111	**C**
8	**C**	34	**A**	60	**A**	86	**A**	112	**C**
9	**D**	35	**C**	61	**B**	87	**B**	113	**B**
10	**A**	36	**C**	62	**C**	88	**A**	114	**A**
11	**A**	37	**C**	63	**D**	89	**A**	115	**C**
12	**C**	38	**B**	64	**B**	90	**D**	116	**D**
13	**C**	39	**D**	65	**C**	91	**D**	117	**D**
14	**D**	40	**B**	66	**A**	92	**B**	118	**A**
15	**B**	41	**B**	67	**D**	93	**D**	119	**B**
16	**B**	42	**C**	68	**A**	94	**D**	120	**D**
17	**A**	43	**D**	69	**A**	95	**C**	121	**A**
18	**A**	44	**D**	70	**B**	96	**D**	122	**D**
19	**D**	45	**A**	71	**D**	97	**C**	123	**A**
20	**C**	46	**C**	72	**C**	98	**A**	124	**B**
21	**D**	47	**D**	73	**B**	99	**B**	125	**B**
22	**D**	48	**A**	74	**C**	100	**B**	126	**A**
23	**C**	49	**B**	75	**D**	101	**A**	127	**A**
24	**B**	50	**C**	76	**A**	102	**A**	128	**C**
25	**A**	51	**A**	77	**D**	103	**C**	129	**D**
26	**A**	52	**A**	78	**B**	104	**C**	130	**B**

Optional bonus questions 131–142

131	**B**	132	**C**	133	**A**	134	**B**	135	**A**	136	**C**
137	**C**	138	**B**	139	**A**	140	**A**	141	**C**	142	**B**

BITSAT Mock 07

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**B**	53	**C**	79	**D**	105	**D**
2	**B**	28	**C**	54	**D**	80	**D**	106	**C**
3	**A**	29	**A**	55	**B**	81	**B**	107	**A**
4	**C**	30	**B**	56	**B**	82	**C**	108	**A**
5	**B**	31	**C**	57	**A**	83	**A**	109	**C**
6	**C**	32	**A**	58	**C**	84	**A**	110	**C**
7	**A**	33	**A**	59	**D**	85	**C**	111	**B**
8	**D**	34	**D**	60	**B**	86	**A**	112	**B**
9	**A**	35	**C**	61	**D**	87	**D**	113	**C**
10	**D**	36	**B**	62	**C**	88	**C**	114	**D**
11	**C**	37	**B**	63	**B**	89	**A**	115	**D**
12	**B**	38	**B**	64	**C**	90	**A**	116	**B**
13	**B**	39	**A**	65	**A**	91	**A**	117	**A**
14	**D**	40	**A**	66	**D**	92	**D**	118	**D**
15	**B**	41	**A**	67	**A**	93	**B**	119	**D**
16	**C**	42	**A**	68	**B**	94	**D**	120	**A**
17	**A**	43	**C**	69	**A**	95	**C**	121	**C**
18	**B**	44	**A**	70	**B**	96	**D**	122	**A**
19	**D**	45	**C**	71	**B**	97	**B**	123	**B**
20	**A**	46	**D**	72	**B**	98	**D**	124	**A**
21	**C**	47	**B**	73	**B**	99	**B**	125	**C**
22	**A**	48	**D**	74	**C**	100	**C**	126	**A**
23	**A**	49	**D**	75	**C**	101	**D**	127	**B**
24	**D**	50	**D**	76	**D**	102	**B**	128	**C**
25	**D**	51	**B**	77	**D**	103	**A**	129	**A**
26	**C**	52	**C**	78	**B**	104	**B**	130	**C**

Optional bonus questions 131–142

131	**A**	132	**B**	133	**C**	134	**A**	135	**B**	136	**C**
137	**B**	138	**C**	139	**A**	140	**B**	141	**A**	142	**C**

BITSAT Mock 08

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**A**	27	**D**	53	**D**	79	**A**	105	**D**
2	**D**	28	**C**	54	**B**	80	**A**	106	**C**
3	**D**	29	**C**	55	**B**	81	**C**	107	**D**
4	**C**	30	**A**	56	**D**	82	**C**	108	**A**
5	**A**	31	**D**	57	**D**	83	**D**	109	**A**
6	**B**	32	**A**	58	**B**	84	**B**	110	**C**
7	**A**	33	**A**	59	**D**	85	**A**	111	**C**
8	**C**	34	**C**	60	**A**	86	**B**	112	**D**
9	**D**	35	**B**	61	**C**	87	**C**	113	**C**
10	**C**	36	**D**	62	**A**	88	**B**	114	**C**
11	**B**	37	**C**	63	**D**	89	**D**	115	**D**
12	**C**	38	**D**	64	**D**	90	**D**	116	**A**
13	**B**	39	**A**	65	**A**	91	**B**	117	**D**
14	**A**	40	**B**	66	**B**	92	**B**	118	**C**
15	**D**	41	**B**	67	**B**	93	**B**	119	**D**
16	**D**	42	**A**	68	**C**	94	**B**	120	**D**
17	**B**	43	**C**	69	**B**	95	**B**	121	**C**
18	**B**	44	**C**	70	**A**	96	**A**	122	**A**
19	**C**	45	**A**	71	**D**	97	**A**	123	**B**
20	**B**	46	**B**	72	**C**	98	**A**	124	**D**
21	**B**	47	**C**	73	**A**	99	**B**	125	**D**
22	**A**	48	**C**	74	**C**	100	**A**	126	**C**
23	**A**	49	**A**	75	**B**	101	**C**	127	**B**
24	**A**	50	**A**	76	**B**	102	**C**	128	**A**
25	**D**	51	**B**	77	**A**	103	**A**	129	**D**
26	**B**	52	**C**	78	**D**	104	**B**	130	**B**

Optional bonus questions 131–142

131	**C**	132	**B**	133	**A**	134	**A**	135	**B**	136	**C**
137	**C**	138	**A**	139	**B**	140	**A**	141	**B**	142	**C**

BITSAT Mock 09

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**A**	27	**B**	53	**D**	79	**C**	105	**D**
2	**B**	28	**B**	54	**D**	80	**D**	106	**D**
3	**C**	29	**A**	55	**D**	81	**C**	107	**A**
4	**A**	30	**A**	56	**B**	82	**D**	108	**B**
5	**B**	31	**A**	57	**C**	83	**A**	109	**D**
6	**D**	32	**A**	58	**D**	84	**B**	110	**D**
7	**C**	33	**B**	59	**C**	85	**C**	111	**A**
8	**B**	34	**B**	60	**D**	86	**B**	112	**C**
9	**C**	35	**B**	61	**A**	87	**A**	113	**B**
10	**D**	36	**B**	62	**B**	88	**B**	114	**A**
11	**B**	37	**A**	63	**D**	89	**C**	115	**C**
12	**C**	38	**A**	64	**A**	90	**D**	116	**C**
13	**D**	39	**B**	65	**B**	91	**B**	117	**C**
14	**A**	40	**C**	66	**C**	92	**C**	118	**C**
15	**B**	41	**C**	67	**D**	93	**C**	119	**C**
16	**C**	42	**B**	68	**C**	94	**A**	120	**B**
17	**D**	43	**B**	69	**A**	95	**B**	121	**A**
18	**B**	44	**A**	70	**B**	96	**B**	122	**A**
19	**A**	45	**C**	71	**B**	97	**C**	123	**C**
20	**A**	46	**A**	72	**D**	98	**D**	124	**D**
21	**D**	47	**C**	73	**C**	99	**D**	125	**B**
22	**A**	48	**D**	74	**D**	100	**D**	126	**D**
23	**D**	49	**C**	75	**B**	101	**A**	127	**B**
24	**D**	50	**A**	76	**A**	102	**A**	128	**B**
25	**C**	51	**A**	77	**A**	103	**A**	129	**B**
26	**C**	52	**D**	78	**A**	104	**A**	130	**D**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**B**	135	**C**	136	**A**
137	**A**	138	**B**	139	**C**	140	**A**	141	**C**	142	**B**

BITSAT Mock 10

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**C**	27	**B**	53	**B**	79	**C**	105	**A**
2	**B**	28	**D**	54	**C**	80	**B**	106	**D**
3	**B**	29	**C**	55	**A**	81	**D**	107	**B**
4	**A**	30	**C**	56	**A**	82	**D**	108	**B**
5	**C**	31	**A**	57	**C**	83	**A**	109	**A**
6	**A**	32	**B**	58	**B**	84	**C**	110	**D**
7	**A**	33	**D**	59	**A**	85	**A**	111	**D**
8	**B**	34	**D**	60	**C**	86	**B**	112	**D**
9	**A**	35	**B**	61	**A**	87	**B**	113	**D**
10	**D**	36	**B**	62	**A**	88	**B**	114	**C**
11	**C**	37	**B**	63	**C**	89	**A**	115	**A**
12	**D**	38	**B**	64	**A**	90	**C**	116	**C**
13	**A**	39	**D**	65	**C**	91	**A**	117	**A**
14	**D**	40	**A**	66	**B**	92	**D**	118	**D**
15	**C**	41	**D**	67	**B**	93	**A**	119	**D**
16	**B**	42	**C**	68	**B**	94	**C**	120	**C**
17	**D**	43	**C**	69	**D**	95	**A**	121	**D**
18	**D**	44	**D**	70	**D**	96	**B**	122	**C**
19	**D**	45	**A**	71	**C**	97	**C**	123	**B**
20	**C**	46	**A**	72	**A**	98	**C**	124	**A**
21	**A**	47	**D**	73	**B**	99	**B**	125	**B**
22	**B**	48	**B**	74	**A**	100	**D**	126	**B**
23	**A**	49	**D**	75	**D**	101	**B**	127	**B**
24	**B**	50	**C**	76	**C**	102	**C**	128	**B**
25	**B**	51	**A**	77	**D**	103	**A**	129	**C**
26	**A**	52	**C**	78	**D**	104	**A**	130	**C**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**C**	135	**A**	136	**B**
137	**C**	138	**B**	139	**A**	140	**A**	141	**C**	142	**B**

BITSAT Mock 11

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**B**	27	**D**	53	**B**	79	**A**	105	**A**
2	**B**	28	**A**	54	**D**	80	**C**	106	**D**
3	**D**	29	**D**	55	**A**	81	**A**	107	**C**
4	**A**	30	**C**	56	**C**	82	**C**	108	**B**
5	**A**	31	**D**	57	**C**	83	**C**	109	**D**
6	**C**	32	**A**	58	**B**	84	**D**	110	**C**
7	**A**	33	**A**	59	**D**	85	**C**	111	**D**
8	**D**	34	**A**	60	**B**	86	**D**	112	**B**
9	**C**	35	**B**	61	**B**	87	**A**	113	**D**
10	**D**	36	**D**	62	**C**	88	**B**	114	**A**
11	**D**	37	**A**	63	**D**	89	**B**	115	**C**
12	**B**	38	**A**	64	**B**	90	**D**	116	**A**
13	**B**	39	**B**	65	**D**	91	**D**	117	**C**
14	**B**	40	**C**	66	**A**	92	**D**	118	**C**
15	**A**	41	**A**	67	**A**	93	**A**	119	**C**
16	**B**	42	**C**	68	**A**	94	**A**	120	**C**
17	**A**	43	**D**	69	**B**	95	**A**	121	**A**
18	**B**	44	**C**	70	**C**	96	**B**	122	**D**
19	**C**	45	**C**	71	**B**	97	**B**	123	**A**
20	**A**	46	**D**	72	**B**	98	**B**	124	**D**
21	**C**	47	**B**	73	**A**	99	**A**	125	**D**
22	**A**	48	**C**	74	**A**	100	**C**	126	**B**
23	**C**	49	**B**	75	**C**	101	**B**	127	**B**
24	**C**	50	**A**	76	**D**	102	**B**	128	**B**
25	**B**	51	**B**	77	**B**	103	**C**	129	**A**
26	**D**	52	**D**	78	**D**	104	**D**	130	**C**

Optional bonus questions 131–142

131	**A**	132	**B**	133	**C**	134	**B**	135	**A**	136	**C**
137	**B**	138	**A**	139	**C**	140	**A**	141	**C**	142	**B**

BITSAT Mock 12

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**C**	27	**C**	53	**A**	79	**B**	105	**D**
2	**C**	28	**A**	54	**C**	80	**D**	106	**B**
3	**C**	29	**D**	55	**B**	81	**D**	107	**D**
4	**B**	30	**A**	56	**A**	82	**D**	108	**C**
5	**B**	31	**A**	57	**A**	83	**C**	109	**B**
6	**A**	32	**D**	58	**C**	84	**A**	110	**C**
7	**D**	33	**B**	59	**C**	85	**A**	111	**A**
8	**D**	34	**D**	60	**D**	86	**C**	112	**B**
9	**C**	35	**B**	61	**B**	87	**A**	113	**D**
10	**B**	36	**B**	62	**B**	88	**B**	114	**A**
11	**B**	37	**D**	63	**D**	89	**D**	115	**C**
12	**A**	38	**B**	64	**A**	90	**C**	116	**B**
13	**C**	39	**B**	65	**D**	91	**A**	117	**D**
14	**D**	40	**C**	66	**C**	92	**C**	118	**C**
15	**C**	41	**B**	67	**A**	93	**A**	119	**D**
16	**A**	42	**B**	68	**B**	94	**B**	120	**C**
17	**A**	43	**D**	69	**A**	95	**A**	121	**D**
18	**B**	44	**A**	70	**C**	96	**B**	122	**D**
19	**D**	45	**C**	71	**D**	97	**A**	123	**A**
20	**A**	46	**A**	72	**C**	98	**C**	124	**A**
21	**A**	47	**C**	73	**C**	99	**C**	125	**A**
22	**B**	48	**C**	74	**B**	100	**C**	126	**A**
23	**B**	49	**D**	75	**A**	101	**C**	127	**B**
24	**D**	50	**A**	76	**B**	102	**B**	128	**D**
25	**B**	51	**D**	77	**B**	103	**D**	129	**D**
26	**D**	52	**A**	78	**A**	104	**B**	130	**B**

Optional bonus questions 131–142

131	**B**	132	**C**	133	**A**	134	**C**	135	**B**	136	**A**
137	**C**	138	**B**	139	**A**	140	**B**	141	**A**	142	**C**

BITSAT Mock 13

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**A**	53	**D**	79	**C**	105	**A**
2	**A**	28	**D**	54	**D**	80	**C**	106	**B**
3	**D**	29	**C**	55	**A**	81	**A**	107	**D**
4	**B**	30	**D**	56	**B**	82	**B**	108	**A**
5	**D**	31	**A**	57	**D**	83	**C**	109	**B**
6	**B**	32	**D**	58	**D**	84	**D**	110	**C**
7	**B**	33	**A**	59	**A**	85	**D**	111	**B**
8	**A**	34	**D**	60	**C**	86	**B**	112	**A**
9	**A**	35	**C**	61	**B**	87	**A**	113	**D**
10	**C**	36	**C**	62	**A**	88	**A**	114	**C**
11	**A**	37	**A**	63	**D**	89	**A**	115	**B**
12	**D**	38	**A**	64	**B**	90	**B**	116	**C**
13	**C**	39	**B**	65	**C**	91	**D**	117	**B**
14	**A**	40	**C**	66	**A**	92	**A**	118	**B**
15	**C**	41	**B**	67	**D**	93	**C**	119	**A**
16	**B**	42	**B**	68	**C**	94	**C**	120	**C**
17	**D**	43	**B**	69	**B**	95	**A**	121	**D**
18	**C**	44	**C**	70	**A**	96	**D**	122	**B**
19	**B**	45	**B**	71	**D**	97	**B**	123	**C**
20	**B**	46	**A**	72	**C**	98	**C**	124	**D**
21	**A**	47	**B**	73	**D**	99	**D**	125	**B**
22	**C**	48	**D**	74	**B**	100	**D**	126	**C**
23	**B**	49	**C**	75	**A**	101	**A**	127	**A**
24	**C**	50	**C**	76	**B**	102	**D**	128	**B**
25	**B**	51	**A**	77	**C**	103	**A**	129	**D**
26	**A**	52	**B**	78	**D**	104	**A**	130	**C**

Optional bonus questions 131–142

131	**A**	132	**B**	133	**C**	134	**B**	135	**C**	136	**A**
137	**A**	138	**C**	139	**B**	140	**C**	141	**B**	142	**A**

BITSAT Mock 14

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**A**	27	**C**	53	**C**	79	**B**	105	**B**
2	**A**	28	**B**	54	**A**	80	**A**	106	**A**
3	**C**	29	**D**	55	**C**	81	**C**	107	**B**
4	**D**	30	**C**	56	**C**	82	**C**	108	**C**
5	**A**	31	**A**	57	**D**	83	**D**	109	**C**
6	**D**	32	**D**	58	**B**	84	**A**	110	**D**
7	**D**	33	**D**	59	**A**	85	**B**	111	**B**
8	**A**	34	**D**	60	**C**	86	**B**	112	**A**
9	**B**	35	**A**	61	**D**	87	**D**	113	**B**
10	**B**	36	**A**	62	**C**	88	**A**	114	**A**
11	**B**	37	**D**	63	**B**	89	**C**	115	**B**
12	**D**	38	**D**	64	**D**	90	**D**	116	**C**
13	**B**	39	**B**	65	**A**	91	**A**	117	**B**
14	**D**	40	**B**	66	**C**	92	**C**	118	**C**
15	**B**	41	**B**	67	**B**	93	**D**	119	**D**
16	**A**	42	**B**	68	**A**	94	**C**	120	**C**
17	**C**	43	**B**	69	**A**	95	**A**	121	**B**
18	**B**	44	**A**	70	**B**	96	**D**	122	**D**
19	**C**	45	**A**	71	**D**	97	**D**	123	**B**
20	**D**	46	**B**	72	**D**	98	**D**	124	**C**
21	**C**	47	**A**	73	**C**	99	**D**	125	**D**
22	**A**	48	**B**	74	**B**	100	**D**	126	**A**
23	**B**	49	**D**	75	**B**	101	**A**	127	**A**
24	**C**	50	**C**	76	**A**	102	**A**	128	**A**
25	**A**	51	**C**	77	**A**	103	**C**	129	**C**
26	**A**	52	**C**	78	**C**	104	**B**	130	**B**

Optional bonus questions 131–142

131	**C**	132	**A**	133	**B**	134	**C**	135	**A**	136	**B**
137	**C**	138	**A**	139	**B**	140	**A**	141	**B**	142	**C**

BITSAT Mock 15

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**C**	27	**A**	53	**A**	79	**A**	105	**A**
2	**D**	28	**D**	54	**C**	80	**D**	106	**D**
3	**B**	29	**B**	55	**B**	81	**B**	107	**A**
4	**B**	30	**B**	56	**C**	82	**D**	108	**C**
5	**D**	31	**D**	57	**D**	83	**C**	109	**A**
6	**B**	32	**D**	58	**B**	84	**B**	110	**C**
7	**C**	33	**A**	59	**C**	85	**A**	111	**C**
8	**A**	34	**A**	60	**C**	86	**A**	112	**B**
9	**B**	35	**A**	61	**B**	87	**B**	113	**D**
10	**A**	36	**A**	62	**B**	88	**A**	114	**C**
11	**A**	37	**D**	63	**D**	89	**C**	115	**D**
12	**A**	38	**B**	64	**B**	90	**C**	116	**A**
13	**B**	39	**B**	65	**C**	91	**B**	117	**B**
14	**C**	40	**B**	66	**D**	92	**D**	118	**A**
15	**B**	41	**A**	67	**A**	93	**D**	119	**B**
16	**C**	42	**D**	68	**C**	94	**D**	120	**C**
17	**A**	43	**B**	69	**A**	95	**C**	121	**A**
18	**C**	44	**B**	70	**A**	96	**A**	122	**A**
19	**D**	45	**D**	71	**B**	97	**C**	123	**B**
20	**D**	46	**D**	72	**D**	98	**D**	124	**B**
21	**A**	47	**C**	73	**D**	99	**B**	125	**C**
22	**C**	48	**B**	74	**C**	100	**D**	126	**D**
23	**D**	49	**A**	75	**B**	101	**B**	127	**A**
24	**D**	50	**C**	76	**C**	102	**A**	128	**B**
25	**A**	51	**A**	77	**A**	103	**B**	129	**C**
26	**C**	52	**C**	78	**D**	104	**D**	130	**C**

Optional bonus questions 131–142

131	**C**	132	**A**	133	**B**	134	**B**	135	**A**	136	**C**
137	**C**	138	**B**	139	**A**	140	**C**	141	**A**	142	**B**

BITSAT Mock 16

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**B**	53	**A**	79	**B**	105	**B**
2	**B**	28	**D**	54	**A**	80	**D**	106	**A**
3	**A**	29	**A**	55	**D**	81	**B**	107	**C**
4	**A**	30	**B**	56	**B**	82	**B**	108	**C**
5	**C**	31	**B**	57	**C**	83	**C**	109	**B**
6	**B**	32	**B**	58	**C**	84	**D**	110	**A**
7	**A**	33	**B**	59	**D**	85	**D**	111	**D**
8	**A**	34	**B**	60	**C**	86	**B**	112	**D**
9	**C**	35	**A**	61	**B**	87	**B**	113	**D**
10	**B**	36	**D**	62	**A**	88	**A**	114	**B**
11	**B**	37	**A**	63	**B**	89	**C**	115	**C**
12	**A**	38	**A**	64	**C**	90	**D**	116	**B**
13	**C**	39	**A**	65	**C**	91	**B**	117	**D**
14	**D**	40	**B**	66	**B**	92	**B**	118	**C**
15	**C**	41	**D**	67	**A**	93	**D**	119	**C**
16	**B**	42	**C**	68	**D**	94	**B**	120	**B**
17	**C**	43	**B**	69	**A**	95	**A**	121	**C**
18	**C**	44	**A**	70	**D**	96	**A**	122	**A**
19	**C**	45	**C**	71	**C**	97	**A**	123	**B**
20	**D**	46	**D**	72	**A**	98	**D**	124	**D**
21	**A**	47	**B**	73	**A**	99	**D**	125	**C**
22	**D**	48	**C**	74	**A**	100	**C**	126	**A**
23	**D**	49	**C**	75	**D**	101	**B**	127	**C**
24	**D**	50	**A**	76	**C**	102	**A**	128	**A**
25	**A**	51	**D**	77	**C**	103	**C**	129	**D**
26	**B**	52	**D**	78	**A**	104	**A**	130	**D**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**B**	135	**C**	136	**A**
137	**A**	138	**C**	139	**B**	140	**A**	141	**C**	142	**B**

BITSAT Mock 17

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**A**	27	**A**	53	**B**	79	**D**	105	**B**
2	**B**	28	**A**	54	**C**	80	**B**	106	**C**
3	**A**	29	**C**	55	**D**	81	**B**	107	**D**
4	**A**	30	**B**	56	**A**	82	**D**	108	**C**
5	**C**	31	**C**	57	**B**	83	**A**	109	**A**
6	**D**	32	**B**	58	**A**	84	**C**	110	**C**
7	**C**	33	**D**	59	**A**	85	**D**	111	**D**
8	**B**	34	**D**	60	**B**	86	**B**	112	**B**
9	**C**	35	**B**	61	**D**	87	**A**	113	**C**
10	**B**	36	**A**	62	**A**	88	**D**	114	**A**
11	**B**	37	**D**	63	**B**	89	**A**	115	**D**
12	**C**	38	**B**	64	**B**	90	**C**	116	**A**
13	**B**	39	**D**	65	**D**	91	**D**	117	**C**
14	**A**	40	**C**	66	**A**	92	**A**	118	**A**
15	**A**	41	**C**	67	**B**	93	**A**	119	**A**
16	**B**	42	**B**	68	**C**	94	**A**	120	**D**
17	**D**	43	**A**	69	**C**	95	**B**	121	**C**
18	**D**	44	**D**	70	**A**	96	**D**	122	**D**
19	**C**	45	**B**	71	**B**	97	**B**	123	**C**
20	**B**	46	**C**	72	**A**	98	**B**	124	**B**
21	**D**	47	**D**	73	**C**	99	**B**	125	**D**
22	**A**	48	**C**	74	**A**	100	**C**	126	**A**
23	**C**	49	**C**	75	**D**	101	**B**	127	**C**
24	**D**	50	**A**	76	**C**	102	**B**	128	**D**
25	**D**	51	**A**	77	**C**	103	**C**	129	**A**
26	**D**	52	**A**	78	**B**	104	**B**	130	**D**

Optional bonus questions 131–142

131	**C**	132	**B**	133	**A**	134	**B**	135	**C**	136	**A**
137	**B**	138	**C**	139	**A**	140	**A**	141	**B**	142	**C**

BITSAT Mock 18

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**A**	27	**D**	53	**D**	79	**D**	105	**B**
2	**D**	28	**B**	54	**B**	80	**B**	106	**A**
3	**A**	29	**B**	55	**D**	81	**B**	107	**D**
4	**B**	30	**D**	56	**C**	82	**D**	108	**C**
5	**A**	31	**B**	57	**D**	83	**D**	109	**B**
6	**D**	32	**C**	58	**C**	84	**C**	110	**D**
7	**B**	33	**A**	59	**A**	85	**C**	111	**D**
8	**D**	34	**C**	60	**A**	86	**A**	112	**D**
9	**A**	35	**B**	61	**D**	87	**B**	113	**C**
10	**B**	36	**A**	62	**A**	88	**D**	114	**C**
11	**B**	37	**A**	63	**B**	89	**A**	115	**D**
12	**C**	38	**C**	64	**A**	90	**A**	116	**C**
13	**C**	39	**D**	65	**C**	91	**C**	117	**B**
14	**C**	40	**B**	66	**C**	92	**A**	118	**C**
15	**C**	41	**C**	67	**B**	93	**C**	119	**D**
16	**B**	42	**D**	68	**D**	94	**B**	120	**B**
17	**A**	43	**B**	69	**B**	95	**A**	121	**D**
18	**C**	44	**D**	70	**A**	96	**D**	122	**B**
19	**A**	45	**B**	71	**C**	97	**A**	123	**B**
20	**C**	46	**A**	72	**B**	98	**D**	124	**C**
21	**B**	47	**B**	73	**D**	99	**B**	125	**D**
22	**A**	48	**B**	74	**A**	100	**A**	126	**A**
23	**D**	49	**A**	75	**C**	101	**C**	127	**C**
24	**C**	50	**C**	76	**B**	102	**A**	128	**A**
25	**D**	51	**A**	77	**C**	103	**A**	129	**B**
26	**A**	52	**D**	78	**A**	104	**B**	130	**A**

Optional bonus questions 131–142

131	**C**	132	**A**	133	**B**	134	**C**	135	**A**	136	**B**
137	**C**	138	**B**	139	**A**	140	**C**	141	**B**	142	**A**

BITSAT Mock 19

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**B**	27	**A**	53	**D**	79	**D**	105	**D**
2	**D**	28	**C**	54	**B**	80	**B**	106	**C**
3	**B**	29	**D**	55	**B**	81	**C**	107	**A**
4	**B**	30	**B**	56	**C**	82	**C**	108	**B**
5	**B**	31	**B**	57	**D**	83	**A**	109	**D**
6	**D**	32	**A**	58	**A**	84	**D**	110	**B**
7	**C**	33	**C**	59	**D**	85	**C**	111	**A**
8	**A**	34	**A**	60	**D**	86	**A**	112	**C**
9	**C**	35	**C**	61	**A**	87	**C**	113	**B**
10	**B**	36	**C**	62	**B**	88	**A**	114	**D**
11	**D**	37	**C**	63	**D**	89	**A**	115	**A**
12	**A**	38	**B**	64	**B**	90	**A**	116	**C**
13	**A**	39	**B**	65	**B**	91	**A**	117	**B**
14	**A**	40	**A**	66	**D**	92	**A**	118	**B**
15	**D**	41	**A**	67	**C**	93	**C**	119	**C**
16	**C**	42	**B**	68	**A**	94	**D**	120	**A**
17	**D**	43	**D**	69	**A**	95	**D**	121	**C**
18	**C**	44	**B**	70	**C**	96	**B**	122	**A**
19	**B**	45	**B**	71	**B**	97	**C**	123	**C**
20	**B**	46	**A**	72	**B**	98	**D**	124	**B**
21	**C**	47	**A**	73	**D**	99	**D**	125	**D**
22	**C**	48	**A**	74	**D**	100	**D**	126	**B**
23	**A**	49	**D**	75	**B**	101	**C**	127	**A**
24	**A**	50	**D**	76	**C**	102	**D**	128	**C**
25	**D**	51	**C**	77	**D**	103	**A**	129	**A**
26	**A**	52	**C**	78	**B**	104	**B**	130	**B**

Optional bonus questions 131–142

131	**A**	132	**B**	133	**C**	134	**A**	135	**B**	136	**C**
137	**A**	138	**C**	139	**B**	140	**B**	141	**C**	142	**A**

BITSAT Mock 20

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**B**	27	**D**	53	**D**	79	**D**	105	**A**
2	**A**	28	**D**	54	**D**	80	**D**	106	**B**
3	**A**	29	**B**	55	**A**	81	**B**	107	**D**
4	**A**	30	**A**	56	**D**	82	**D**	108	**D**
5	**C**	31	**A**	57	**C**	83	**D**	109	**D**
6	**B**	32	**B**	58	**C**	84	**B**	110	**B**
7	**C**	33	**B**	59	**C**	85	**B**	111	**D**
8	**D**	34	**A**	60	**A**	86	**C**	112	**D**
9	**D**	35	**B**	61	**B**	87	**B**	113	**C**
10	**A**	36	**B**	62	**C**	88	**A**	114	**D**
11	**C**	37	**D**	63	**C**	89	**C**	115	**A**
12	**D**	38	**C**	64	**A**	90	**B**	116	**D**
13	**D**	39	**A**	65	**A**	91	**D**	117	**B**
14	**B**	40	**C**	66	**D**	92	**C**	118	**A**
15	**A**	41	**D**	67	**D**	93	**B**	119	**D**
16	**C**	42	**B**	68	**B**	94	**B**	120	**A**
17	**C**	43	**A**	69	**A**	95	**B**	121	**A**
18	**B**	44	**A**	70	**B**	96	**A**	122	**B**
19	**C**	45	**B**	71	**C**	97	**A**	123	**B**
20	**D**	46	**B**	72	**A**	98	**C**	124	**C**
21	**A**	47	**B**	73	**C**	99	**B**	125	**D**
22	**B**	48	**C**	74	**D**	100	**C**	126	**C**
23	**B**	49	**D**	75	**A**	101	**A**	127	**C**
24	**C**	50	**D**	76	**A**	102	**B**	128	**A**
25	**A**	51	**C**	77	**C**	103	**C**	129	**A**
26	**B**	52	**A**	78	**A**	104	**C**	130	**C**

Optional bonus questions 131–142

131	**B**	132	**C**	133	**A**	134	**C**	135	**B**	136	**A**
137	**B**	138	**C**	139	**A**	140	**C**	141	**B**	142	**A**

BITSAT Mock 21

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**C**	27	**B**	53	**B**	79	**A**	105	**A**
2	**A**	28	**B**	54	**C**	80	**B**	106	**D**
3	**B**	29	**A**	55	**A**	81	**B**	107	**C**
4	**D**	30	**A**	56	**D**	82	**A**	108	**B**
5	**A**	31	**B**	57	**B**	83	**B**	109	**A**
6	**C**	32	**D**	58	**D**	84	**A**	110	**B**
7	**A**	33	**A**	59	**C**	85	**C**	111	**A**
8	**B**	34	**C**	60	**A**	86	**D**	112	**D**
9	**C**	35	**D**	61	**B**	87	**D**	113	**A**
10	**B**	36	**B**	62	**C**	88	**D**	114	**B**
11	**D**	37	**A**	63	**A**	89	**B**	115	**C**
12	**C**	38	**A**	64	**A**	90	**C**	116	**B**
13	**D**	39	**C**	65	**D**	91	**D**	117	**D**
14	**D**	40	**A**	66	**B**	92	**B**	118	**A**
15	**B**	41	**D**	67	**C**	93	**A**	119	**B**
16	**A**	42	**C**	68	**D**	94	**D**	120	**C**
17	**A**	43	**A**	69	**B**	95	**C**	121	**D**
18	**C**	44	**A**	70	**A**	96	**D**	122	**C**
19	**C**	45	**D**	71	**C**	97	**D**	123	**B**
20	**D**	46	**C**	72	**A**	98	**C**	124	**C**
21	**C**	47	**D**	73	**D**	99	**D**	125	**D**
22	**D**	48	**C**	74	**A**	100	**C**	126	**B**
23	**D**	49	**B**	75	**B**	101	**A**	127	**A**
24	**B**	50	**B**	76	**C**	102	**A**	128	**B**
25	**A**	51	**B**	77	**C**	103	**C**	129	**A**
26	**B**	52	**B**	78	**D**	104	**B**	130	**C**

Optional bonus questions 131–142

131	**A**	132	**C**	133	**B**	134	**B**	135	**A**	136	**C**
137	**A**	138	**C**	139	**B**	140	**B**	141	**A**	142	**C**

BITSAT Mock 22

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**C**	27	**C**	53	**B**	79	**D**	105	**B**
2	**A**	28	**D**	54	**A**	80	**C**	106	**A**
3	**D**	29	**B**	55	**B**	81	**A**	107	**B**
4	**C**	30	**D**	56	**A**	82	**A**	108	**A**
5	**C**	31	**C**	57	**C**	83	**C**	109	**B**
6	**B**	32	**B**	58	**B**	84	**B**	110	**C**
7	**D**	33	**D**	59	**C**	85	**D**	111	**C**
8	**B**	34	**B**	60	**A**	86	**C**	112	**D**
9	**D**	35	**A**	61	**D**	87	**A**	113	**C**
10	**A**	36	**C**	62	**C**	88	**C**	114	**A**
11	**C**	37	**A**	63	**A**	89	**D**	115	**A**
12	**A**	38	**A**	64	**B**	90	**D**	116	**B**
13	**B**	39	**A**	65	**B**	91	**A**	117	**C**
14	**C**	40	**A**	66	**D**	92	**B**	118	**D**
15	**A**	41	**D**	67	**C**	93	**A**	119	**C**
16	**B**	42	**B**	68	**B**	94	**D**	120	**A**
17	**A**	43	**C**	69	**A**	95	**C**	121	**D**
18	**B**	44	**D**	70	**A**	96	**D**	122	**A**
19	**C**	45	**D**	71	**D**	97	**B**	123	**D**
20	**A**	46	**D**	72	**B**	98	**B**	124	**A**
21	**D**	47	**B**	73	**A**	99	**D**	125	**B**
22	**B**	48	**D**	74	**A**	100	**B**	126	**C**
23	**D**	49	**D**	75	**B**	101	**D**	127	**D**
24	**B**	50	**C**	76	**B**	102	**C**	128	**C**
25	**A**	51	**B**	77	**B**	103	**D**	129	**A**
26	**A**	52	**C**	78	**C**	104	**B**	130	**C**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**A**	135	**B**	136	**C**
137	**B**	138	**A**	139	**C**	140	**A**	141	**B**	142	**C**

BITSAT Mock 23

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**B**	27	**D**	53	**A**	79	**C**	105	**B**
2	**C**	28	**A**	54	**C**	80	**C**	106	**B**
3	**A**	29	**C**	55	**A**	81	**B**	107	**A**
4	**C**	30	**B**	56	**D**	82	**A**	108	**B**
5	**B**	31	**C**	57	**B**	83	**D**	109	**D**
6	**C**	32	**C**	58	**A**	84	**D**	110	**D**
7	**D**	33	**C**	59	**D**	85	**C**	111	**D**
8	**C**	34	**D**	60	**B**	86	**A**	112	**D**
9	**D**	35	**B**	61	**B**	87	**D**	113	**A**
10	**B**	36	**D**	62	**A**	88	**B**	114	**B**
11	**C**	37	**B**	63	**C**	89	**A**	115	**C**
12	**A**	38	**D**	64	**A**	90	**C**	116	**D**
13	**A**	39	**A**	65	**D**	91	**C**	117	**B**
14	**D**	40	**B**	66	**A**	92	**D**	118	**A**
15	**B**	41	**B**	67	**B**	93	**D**	119	**C**
16	**D**	42	**B**	68	**B**	94	**A**	120	**B**
17	**B**	43	**D**	69	**D**	95	**D**	121	**D**
18	**C**	44	**D**	70	**C**	96	**C**	122	**A**
19	**A**	45	**A**	71	**B**	97	**A**	123	**B**
20	**A**	46	**A**	72	**B**	98	**D**	124	**A**
21	**D**	47	**C**	73	**B**	99	**C**	125	**A**
22	**A**	48	**C**	74	**A**	100	**B**	126	**B**
23	**D**	49	**A**	75	**D**	101	**C**	127	**C**
24	**B**	50	**C**	76	**A**	102	**C**	128	**A**
25	**B**	51	**A**	77	**C**	103	**B**	129	**C**
26	**A**	52	**B**	78	**D**	104	**C**	130	**A**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**C**	135	**A**	136	**B**
137	**C**	138	**B**	139	**A**	140	**B**	141	**A**	142	**C**

BITSAT Mock 24

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**A**	27	**B**	53	**D**	79	**C**	105	**C**
2	**A**	28	**D**	54	**C**	80	**A**	106	**C**
3	**C**	29	**C**	55	**D**	81	**B**	107	**B**
4	**A**	30	**B**	56	**C**	82	**D**	108	**D**
5	**A**	31	**D**	57	**A**	83	**D**	109	**C**
6	**A**	32	**C**	58	**B**	84	**B**	110	**A**
7	**C**	33	**B**	59	**B**	85	**D**	111	**D**
8	**D**	34	**A**	60	**D**	86	**C**	112	**D**
9	**D**	35	**A**	61	**C**	87	**C**	113	**D**
10	**C**	36	**D**	62	**A**	88	**B**	114	**D**
11	**C**	37	**B**	63	**D**	89	**B**	115	**A**
12	**A**	38	**B**	64	**A**	90	**A**	116	**A**
13	**A**	39	**C**	65	**A**	91	**B**	117	**C**
14	**B**	40	**A**	66	**B**	92	**C**	118	**D**
15	**B**	41	**B**	67	**D**	93	**B**	119	**B**
16	**B**	42	**A**	68	**B**	94	**C**	120	**D**
17	**B**	43	**C**	69	**C**	95	**B**	121	**A**
18	**D**	44	**D**	70	**B**	96	**A**	122	**B**
19	**B**	45	**B**	71	**D**	97	**C**	123	**D**
20	**D**	46	**B**	72	**A**	98	**C**	124	**C**
21	**B**	47	**A**	73	**D**	99	**A**	125	**D**
22	**A**	48	**D**	74	**B**	100	**A**	126	**B**
23	**C**	49	**A**	75	**C**	101	**A**	127	**D**
24	**D**	50	**A**	76	**C**	102	**A**	128	**C**
25	**C**	51	**C**	77	**A**	103	**B**	129	**A**
26	**D**	52	**C**	78	**A**	104	**B**	130	**B**

Optional bonus questions 131–142

131	**A**	132	**C**	133	**B**	134	**A**	135	**C**	136	**B**
137	**B**	138	**A**	139	**C**	140	**B**	141	**C**	142	**A**

BITSAT Mock 25

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**C**	27	**D**	53	**B**	79	**A**	105	**C**
2	**B**	28	**C**	54	**A**	80	**C**	106	**D**
3	**C**	29	**C**	55	**B**	81	**B**	107	**D**
4	**B**	30	**C**	56	**C**	82	**D**	108	**D**
5	**B**	31	**A**	57	**D**	83	**B**	109	**B**
6	**D**	32	**D**	58	**A**	84	**B**	110	**B**
7	**A**	33	**B**	59	**D**	85	**C**	111	**C**
8	**A**	34	**A**	60	**A**	86	**A**	112	**B**
9	**B**	35	**C**	61	**A**	87	**C**	113	**D**
10	**A**	36	**B**	62	**A**	88	**D**	114	**A**
11	**B**	37	**B**	63	**C**	89	**D**	115	**C**
12	**A**	38	**D**	64	**B**	90	**A**	116	**D**
13	**D**	39	**C**	65	**A**	91	**C**	117	**A**
14	**A**	40	**C**	66	**C**	92	**B**	118	**C**
15	**A**	41	**C**	67	**D**	93	**A**	119	**A**
16	**D**	42	**D**	68	**D**	94	**B**	120	**D**
17	**C**	43	**A**	69	**B**	95	**B**	121	**C**
18	**B**	44	**B**	70	**B**	96	**A**	122	**A**
19	**A**	45	**A**	71	**C**	97	**B**	123	**D**
20	**C**	46	**B**	72	**D**	98	**C**	124	**C**
21	**B**	47	**D**	73	**B**	99	**C**	125	**D**
22	**D**	48	**A**	74	**D**	100	**A**	126	**B**
23	**A**	49	**D**	75	**B**	101	**A**	127	**A**
24	**D**	50	**C**	76	**C**	102	**B**	128	**B**
25	**B**	51	**B**	77	**A**	103	**A**	129	**D**
26	**D**	52	**C**	78	**A**	104	**D**	130	**C**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**A**	135	**C**	136	**B**
137	**A**	138	**B**	139	**C**	140	**A**	141	**C**	142	**B**

BITSAT Mock 26

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**C**	27	**B**	53	**C**	79	**A**	105	**B**
2	**B**	28	**A**	54	**C**	80	**A**	106	**D**
3	**C**	29	**D**	55	**A**	81	**A**	107	**B**
4	**C**	30	**A**	56	**D**	82	**B**	108	**D**
5	**C**	31	**D**	57	**A**	83	**D**	109	**A**
6	**D**	32	**B**	58	**D**	84	**C**	110	**C**
7	**B**	33	**B**	59	**C**	85	**D**	111	**B**
8	**C**	34	**A**	60	**A**	86	**B**	112	**C**
9	**B**	35	**D**	61	**C**	87	**D**	113	**A**
10	**A**	36	**A**	62	**B**	88	**C**	114	**D**
11	**D**	37	**C**	63	**B**	89	**A**	115	**B**
12	**C**	38	**A**	64	**A**	90	**B**	116	**A**
13	**D**	39	**A**	65	**D**	91	**B**	117	**A**
14	**B**	40	**B**	66	**A**	92	**D**	118	**D**
15	**D**	41	**A**	67	**C**	93	**D**	119	**D**
16	**B**	42	**C**	68	**D**	94	**A**	120	**C**
17	**A**	43	**B**	69	**A**	95	**B**	121	**D**
18	**A**	44	**B**	70	**B**	96	**C**	122	**C**
19	**D**	45	**B**	71	**A**	97	**C**	123	**C**
20	**A**	46	**C**	72	**B**	98	**C**	124	**A**
21	**B**	47	**D**	73	**D**	99	**A**	125	**D**
22	**B**	48	**B**	74	**D**	100	**C**	126	**D**
23	**C**	49	**D**	75	**C**	101	**A**	127	**B**
24	**A**	50	**C**	76	**B**	102	**A**	128	**C**
25	**A**	51	**D**	77	**C**	103	**B**	129	**A**
26	**D**	52	**B**	78	**C**	104	**B**	130	**B**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**A**	135	**B**	136	**C**
137	**C**	138	**A**	139	**B**	140	**A**	141	**B**	142	**C**

BITSAT Mock 27

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**C**	27	**A**	53	**D**	79	**A**	105	**C**
2	**C**	28	**C**	54	**B**	80	**A**	106	**D**
3	**D**	29	**D**	55	**D**	81	**D**	107	**D**
4	**B**	30	**D**	56	**C**	82	**A**	108	**D**
5	**A**	31	**B**	57	**A**	83	**B**	109	**B**
6	**D**	32	**C**	58	**B**	84	**B**	110	**A**
7	**D**	33	**A**	59	**D**	85	**C**	111	**B**
8	**A**	34	**D**	60	**B**	86	**C**	112	**A**
9	**A**	35	**C**	61	**D**	87	**A**	113	**D**
10	**A**	36	**B**	62	**B**	88	**A**	114	**C**
11	**C**	37	**B**	63	**B**	89	**D**	115	**C**
12	**A**	38	**D**	64	**C**	90	**D**	116	**B**
13	**D**	39	**C**	65	**B**	91	**C**	117	**D**
14	**B**	40	**A**	66	**D**	92	**A**	118	**C**
15	**B**	41	**A**	67	**A**	93	**C**	119	**A**
16	**D**	42	**C**	68	**C**	94	**D**	120	**D**
17	**A**	43	**A**	69	**A**	95	**A**	121	**C**
18	**B**	44	**D**	70	**A**	96	**A**	122	**B**
19	**C**	45	**D**	71	**C**	97	**D**	123	**A**
20	**B**	46	**A**	72	**C**	98	**B**	124	**A**
21	**B**	47	**B**	73	**C**	99	**C**	125	**B**
22	**C**	48	**C**	74	**B**	100	**B**	126	**A**
23	**B**	49	**A**	75	**B**	101	**A**	127	**B**
24	**A**	50	**A**	76	**D**	102	**D**	128	**B**
25	**B**	51	**C**	77	**D**	103	**C**	129	**C**
26	**C**	52	**B**	78	**B**	104	**B**	130	**D**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**B**	135	**C**	136	**A**
137	**B**	138	**A**	139	**C**	140	**B**	141	**C**	142	**A**

BITSAT Mock 28

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**B**	53	**B**	79	**C**	105	**C**
2	**C**	28	**B**	54	**C**	80	**D**	106	**B**
3	**C**	29	**C**	55	**B**	81	**A**	107	**C**
4	**A**	30	**A**	56	**A**	82	**A**	108	**D**
5	**A**	31	**D**	57	**B**	83	**C**	109	**C**
6	**C**	32	**D**	58	**B**	84	**A**	110	**C**
7	**D**	33	**A**	59	**C**	85	**B**	111	**D**
8	**D**	34	**D**	60	**B**	86	**C**	112	**C**
9	**C**	35	**C**	61	**D**	87	**B**	113	**A**
10	**C**	36	**A**	62	**C**	88	**D**	114	**A**
11	**D**	37	**C**	63	**A**	89	**B**	115	**D**
12	**A**	38	**A**	64	**B**	90	**B**	116	**A**
13	**A**	39	**D**	65	**D**	91	**D**	117	**B**
14	**B**	40	**B**	66	**A**	92	**B**	118	**A**
15	**B**	41	**D**	67	**A**	93	**B**	119	**C**
16	**A**	42	**C**	68	**B**	94	**D**	120	**D**
17	**D**	43	**A**	69	**B**	95	**A**	121	**B**
18	**D**	44	**B**	70	**C**	96	**D**	122	**B**
19	**B**	45	**B**	71	**A**	97	**B**	123	**A**
20	**B**	46	**A**	72	**C**	98	**D**	124	**A**
21	**A**	47	**D**	73	**D**	99	**D**	125	**B**
22	**A**	48	**C**	74	**A**	100	**C**	126	**B**
23	**D**	49	**A**	75	**C**	101	**A**	127	**B**
24	**B**	50	**D**	76	**D**	102	**A**	128	**C**
25	**B**	51	**C**	77	**D**	103	**C**	129	**A**
26	**C**	52	**A**	78	**B**	104	**D**	130	**C**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**B**	135	**C**	136	**A**
137	**A**	138	**B**	139	**C**	140	**B**	141	**A**	142	**C**

BITSAT Mock 29

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**C**	27	**A**	53	**B**	79	**D**	105	**C**
2	**D**	28	**B**	54	**A**	80	**A**	106	**D**
3	**A**	29	**C**	55	**C**	81	**C**	107	**C**
4	**D**	30	**B**	56	**A**	82	**B**	108	**C**
5	**A**	31	**A**	57	**D**	83	**B**	109	**A**
6	**C**	32	**D**	58	**B**	84	**D**	110	**B**
7	**D**	33	**A**	59	**C**	85	**A**	111	**C**
8	**A**	34	**B**	60	**D**	86	**B**	112	**D**
9	**A**	35	**A**	61	**B**	87	**B**	113	**B**
10	**D**	36	**A**	62	**A**	88	**D**	114	**A**
11	**C**	37	**B**	63	**C**	89	**A**	115	**A**
12	**D**	38	**B**	64	**D**	90	**B**	116	**B**
13	**A**	39	**D**	65	**B**	91	**A**	117	**C**
14	**B**	40	**A**	66	**C**	92	**B**	118	**B**
15	**C**	41	**C**	67	**A**	93	**D**	119	**D**
16	**B**	42	**C**	68	**A**	94	**A**	120	**C**
17	**C**	43	**D**	69	**D**	95	**C**	121	**A**
18	**D**	44	**C**	70	**B**	96	**D**	122	**A**
19	**B**	45	**D**	71	**D**	97	**D**	123	**C**
20	**A**	46	**B**	72	**C**	98	**A**	124	**D**
21	**B**	47	**A**	73	**A**	99	**A**	125	**D**
22	**B**	48	**B**	74	**A**	100	**D**	126	**B**
23	**D**	49	**C**	75	**C**	101	**A**	127	**D**
24	**A**	50	**D**	76	**C**	102	**B**	128	**C**
25	**B**	51	**B**	77	**C**	103	**C**	129	**B**
26	**C**	52	**C**	78	**D**	104	**B**	130	**B**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**B**	135	**A**	136	**C**
137	**B**	138	**C**	139	**A**	140	**A**	141	**B**	142	**C**

BITSAT Mock 30

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**A**	27	**B**	53	**A**	79	**C**	105	**B**
2	**B**	28	**D**	54	**B**	80	**C**	106	**B**
3	**A**	29	**B**	55	**A**	81	**A**	107	**D**
4	**C**	30	**D**	56	**C**	82	**C**	108	**A**
5	**D**	31	**B**	57	**A**	83	**D**	109	**C**
6	**B**	32	**A**	58	**C**	84	**D**	110	**A**
7	**C**	33	**D**	59	**D**	85	**D**	111	**A**
8	**C**	34	**B**	60	**A**	86	**C**	112	**B**
9	**C**	35	**A**	61	**C**	87	**B**	113	**B**
10	**B**	36	**D**	62	**B**	88	**C**	114	**C**
11	**A**	37	**D**	63	**B**	89	**B**	115	**C**
12	**C**	38	**D**	64	**B**	90	**B**	116	**A**
13	**D**	39	**C**	65	**C**	91	**D**	117	**A**
14	**A**	40	**C**	66	**D**	92	**A**	118	**D**
15	**C**	41	**B**	67	**A**	93	**B**	119	**D**
16	**D**	42	**C**	68	**D**	94	**A**	120	**C**
17	**B**	43	**B**	69	**A**	95	**B**	121	**D**
18	**A**	44	**D**	70	**A**	96	**D**	122	**B**
19	**B**	45	**C**	71	**A**	97	**D**	123	**A**
20	**B**	46	**A**	72	**B**	98	**A**	124	**C**
21	**A**	47	**C**	73	**A**	99	**A**	125	**C**
22	**A**	48	**D**	74	**B**	100	**C**	126	**B**
23	**D**	49	**B**	75	**A**	101	**C**	127	**D**
24	**C**	50	**B**	76	**A**	102	**B**	128	**B**
25	**D**	51	**B**	77	**D**	103	**D**	129	**D**
26	**A**	52	**A**	78	**D**	104	**C**	130	**C**

Optional bonus questions 131–142

131	**A**	132	**C**	133	**B**	134	**B**	135	**C**	136	**A**
137	**A**	138	**C**	139	**B**	140	**B**	141	**A**	142	**C**

BITSAT Mock 31

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**D**	53	**D**	79	**B**	105	**B**
2	**B**	28	**D**	54	**A**	80	**B**	106	**A**
3	**D**	29	**C**	55	**D**	81	**D**	107	**A**
4	**A**	30	**A**	56	**C**	82	**C**	108	**B**
5	**B**	31	**A**	57	**B**	83	**A**	109	**C**
6	**A**	32	**B**	58	**C**	84	**B**	110	**C**
7	**C**	33	**D**	59	**A**	85	**D**	111	**A**
8	**B**	34	**C**	60	**D**	86	**A**	112	**C**
9	**C**	35	**B**	61	**B**	87	**C**	113	**D**
10	**D**	36	**C**	62	**B**	88	**D**	114	**A**
11	**A**	37	**B**	63	**C**	89	**C**	115	**A**
12	**C**	38	**B**	64	**B**	90	**C**	116	**A**
13	**A**	39	**D**	65	**D**	91	**A**	117	**C**
14	**B**	40	**A**	66	**D**	92	**B**	118	**C**
15	**C**	41	**B**	67	**A**	93	**B**	119	**B**
16	**A**	42	**A**	68	**C**	94	**B**	120	**D**
17	**D**	43	**A**	69	**A**	95	**A**	121	**C**
18	**C**	44	**C**	70	**A**	96	**C**	122	**D**
19	**D**	45	**A**	71	**C**	97	**C**	123	**C**
20	**C**	46	**C**	72	**D**	98	**A**	124	**B**
21	**A**	47	**C**	73	**B**	99	**D**	125	**B**
22	**B**	48	**B**	74	**A**	100	**D**	126	**D**
23	**B**	49	**B**	75	**D**	101	**B**	127	**C**
24	**B**	50	**D**	76	**B**	102	**D**	128	**A**
25	**A**	51	**A**	77	**A**	103	**D**	129	**D**
26	**B**	52	**D**	78	**A**	104	**B**	130	**D**

Optional bonus questions 131–142

131	**C**	132	**A**	133	**B**	134	**A**	135	**C**	136	**B**
137	**C**	138	**A**	139	**B**	140	**B**	141	**C**	142	**A**

BITSAT Mock 32

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**A**	53	**D**	79	**B**	105	**C**
2	**A**	28	**B**	54	**A**	80	**D**	106	**A**
3	**B**	29	**D**	55	**C**	81	**C**	107	**C**
4	**B**	30	**B**	56	**C**	82	**A**	108	**D**
5	**A**	31	**C**	57	**A**	83	**C**	109	**A**
6	**C**	32	**C**	58	**A**	84	**B**	110	**C**
7	**C**	33	**D**	59	**C**	85	**A**	111	**D**
8	**C**	34	**A**	60	**B**	86	**A**	112	**B**
9	**A**	35	**A**	61	**B**	87	**C**	113	**B**
10	**A**	36	**C**	62	**A**	88	**B**	114	**A**
11	**D**	37	**A**	63	**B**	89	**D**	115	**B**
12	**A**	38	**D**	64	**A**	90	**A**	116	**C**
13	**C**	39	**A**	65	**B**	91	**C**	117	**A**
14	**D**	40	**B**	66	**C**	92	**A**	118	**B**
15	**A**	41	**D**	67	**C**	93	**C**	119	**B**
16	**B**	42	**B**	68	**A**	94	**D**	120	**D**
17	**C**	43	**D**	69	**D**	95	**D**	121	**D**
18	**D**	44	**B**	70	**D**	96	**B**	122	**B**
19	**D**	45	**D**	71	**B**	97	**D**	123	**C**
20	**C**	46	**B**	72	**D**	98	**B**	124	**D**
21	**C**	47	**B**	73	**D**	99	**A**	125	**C**
22	**B**	48	**B**	74	**C**	100	**A**	126	**B**
23	**A**	49	**B**	75	**B**	101	**A**	127	**A**
24	**D**	50	**D**	76	**A**	102	**A**	128	**B**
25	**B**	51	**C**	77	**C**	103	**D**	129	**C**
26	**B**	52	**A**	78	**D**	104	**D**	130	**C**

Optional bonus questions 131–142

131	**A**	132	**B**	133	**C**	134	**B**	135	**C**	136	**A**
137	**B**	138	**A**	139	**C**	140	**C**	141	**B**	142	**A**

BITSAT Mock 33

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**A**	27	**B**	53	**A**	79	**C**	105	**C**
2	**C**	28	**B**	54	**A**	80	**B**	106	**C**
3	**C**	29	**D**	55	**B**	81	**A**	107	**C**
4	**C**	30	**A**	56	**C**	82	**C**	108	**B**
5	**A**	31	**A**	57	**D**	83	**D**	109	**A**
6	**B**	32	**C**	58	**B**	84	**A**	110	**B**
7	**D**	33	**B**	59	**A**	85	**C**	111	**D**
8	**A**	34	**B**	60	**A**	86	**D**	112	**B**
9	**B**	35	**C**	61	**A**	87	**A**	113	**C**
10	**D**	36	**D**	62	**C**	88	**C**	114	**B**
11	**A**	37	**A**	63	**B**	89	**C**	115	**C**
12	**D**	38	**B**	64	**A**	90	**A**	116	**D**
13	**C**	39	**D**	65	**D**	91	**D**	117	**B**
14	**A**	40	**D**	66	**B**	92	**D**	118	**D**
15	**C**	41	**D**	67	**D**	93	**B**	119	**A**
16	**C**	42	**D**	68	**B**	94	**A**	120	**A**
17	**B**	43	**B**	69	**A**	95	**D**	121	**B**
18	**A**	44	**B**	70	**C**	96	**A**	122	**A**
19	**B**	45	**C**	71	**B**	97	**A**	123	**D**
20	**D**	46	**C**	72	**B**	98	**D**	124	**B**
21	**B**	47	**A**	73	**B**	99	**A**	125	**A**
22	**B**	48	**C**	74	**D**	100	**D**	126	**C**
23	**A**	49	**D**	75	**D**	101	**C**	127	**B**
24	**C**	50	**B**	76	**B**	102	**C**	128	**B**
25	**D**	51	**A**	77	**D**	103	**C**	129	**C**
26	**D**	52	**C**	78	**A**	104	**A**	130	**D**

Optional bonus questions 131–142

131	**A**	132	**C**	133	**B**	134	**A**	135	**B**	136	**C**
137	**B**	138	**C**	139	**A**	140	**B**	141	**C**	142	**A**

BITSAT Mock 34

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**D**	53	**A**	79	**A**	105	**C**
2	**A**	28	**B**	54	**B**	80	**A**	106	**C**
3	**B**	29	**A**	55	**C**	81	**A**	107	**D**
4	**B**	30	**C**	56	**A**	82	**C**	108	**C**
5	**B**	31	**C**	57	**B**	83	**C**	109	**D**
6	**D**	32	**D**	58	**A**	84	**B**	110	**B**
7	**D**	33	**D**	59	**B**	85	**A**	111	**A**
8	**A**	34	**B**	60	**A**	86	**B**	112	**A**
9	**C**	35	**B**	61	**A**	87	**D**	113	**B**
10	**D**	36	**A**	62	**B**	88	**C**	114	**A**
11	**B**	37	**D**	63	**A**	89	**D**	115	**C**
12	**C**	38	**A**	64	**C**	90	**D**	116	**B**
13	**A**	39	**C**	65	**D**	91	**B**	117	**D**
14	**A**	40	**C**	66	**D**	92	**A**	118	**A**
15	**D**	41	**B**	67	**C**	93	**D**	119	**B**
16	**B**	42	**A**	68	**B**	94	**C**	120	**D**
17	**B**	43	**D**	69	**A**	95	**A**	121	**D**
18	**A**	44	**D**	70	**B**	96	**A**	122	**B**
19	**B**	45	**C**	71	**D**	97	**B**	123	**C**
20	**C**	46	**D**	72	**B**	98	**B**	124	**C**
21	**A**	47	**A**	73	**C**	99	**A**	125	**D**
22	**A**	48	**B**	74	**B**	100	**A**	126	**D**
23	**C**	49	**D**	75	**A**	101	**B**	127	**C**
24	**C**	50	**B**	76	**C**	102	**D**	128	**D**
25	**C**	51	**C**	77	**D**	103	**A**	129	**C**
26	**D**	52	**C**	78	**B**	104	**B**	130	**C**

Optional bonus questions 131–142

131	**B**	132	**C**	133	**A**	134	**B**	135	**A**	136	**C**
137	**B**	138	**A**	139	**C**	140	**A**	141	**B**	142	**C**

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Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**A**	53	**D**	79	**A**	105	**D**
2	**D**	28	**D**	54	**A**	80	**A**	106	**A**
3	**B**	29	**C**	55	**C**	81	**B**	107	**C**
4	**B**	30	**C**	56	**A**	82	**D**	108	**A**
5	**D**	31	**B**	57	**D**	83	**B**	109	**C**
6	**B**	32	**D**	58	**C**	84	**A**	110	**B**
7	**A**	33	**D**	59	**C**	85	**C**	111	**D**
8	**D**	34	**A**	60	**B**	86	**C**	112	**B**
9	**D**	35	**D**	61	**D**	87	**B**	113	**A**
10	**C**	36	**C**	62	**A**	88	**A**	114	**D**
11	**C**	37	**B**	63	**D**	89	**C**	115	**B**
12	**A**	38	**C**	64	**A**	90	**C**	116	**A**
13	**B**	39	**B**	65	**B**	91	**C**	117	**D**
14	**B**	40	**B**	66	**C**	92	**D**	118	**B**
15	**A**	41	**C**	67	**B**	93	**D**	119	**C**
16	**A**	42	**A**	68	**C**	94	**A**	120	**A**
17	**A**	43	**B**	69	**A**	95	**B**	121	**D**
18	**B**	44	**B**	70	**B**	96	**C**	122	**B**
19	**C**	45	**B**	71	**B**	97	**D**	123	**C**
20	**B**	46	**A**	72	**C**	98	**A**	124	**B**
21	**A**	47	**A**	73	**B**	99	**A**	125	**C**
22	**B**	48	**A**	74	**D**	100	**C**	126	**C**
23	**A**	49	**C**	75	**A**	101	**B**	127	**C**
24	**D**	50	**A**	76	**D**	102	**D**	128	**B**
25	**C**	51	**D**	77	**D**	103	**A**	129	**A**
26	**C**	52	**D**	78	**D**	104	**B**	130	**D**

Optional bonus questions 131–142

131	**B**	132	**C**	133	**A**	134	**B**	135	**C**	136	**A**
137	**A**	138	**B**	139	**C**	140	**C**	141	**A**	142	**B**

BITSAT Mock 36

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**C**	53	**B**	79	**C**	105	**B**
2	**D**	28	**A**	54	**C**	80	**B**	106	**A**
3	**A**	29	**B**	55	**A**	81	**A**	107	**C**
4	**D**	30	**B**	56	**A**	82	**C**	108	**B**
5	**A**	31	**D**	57	**A**	83	**B**	109	**D**
6	**A**	32	**D**	58	**A**	84	**D**	110	**B**
7	**A**	33	**A**	59	**C**	85	**B**	111	**C**
8	**D**	34	**C**	60	**A**	86	**D**	112	**D**
9	**C**	35	**D**	61	**B**	87	**A**	113	**A**
10	**C**	36	**A**	62	**B**	88	**D**	114	**C**
11	**B**	37	**B**	63	**A**	89	**A**	115	**B**
12	**D**	38	**D**	64	**A**	90	**C**	116	**B**
13	**A**	39	**C**	65	**D**	91	**D**	117	**D**
14	**B**	40	**B**	66	**A**	92	**B**	118	**C**
15	**C**	41	**C**	67	**D**	93	**C**	119	**A**
16	**B**	42	**D**	68	**C**	94	**D**	120	**B**
17	**A**	43	**D**	69	**B**	95	**B**	121	**B**
18	**D**	44	**B**	70	**C**	96	**A**	122	**A**
19	**B**	45	**D**	71	**B**	97	**D**	123	**C**
20	**B**	46	**B**	72	**B**	98	**A**	124	**D**
21	**A**	47	**B**	73	**D**	99	**A**	125	**C**
22	**B**	48	**B**	74	**D**	100	**C**	126	**D**
23	**C**	49	**C**	75	**A**	101	**C**	127	**A**
24	**C**	50	**A**	76	**A**	102	**A**	128	**D**
25	**D**	51	**C**	77	**C**	103	**B**	129	**A**
26	**C**	52	**B**	78	**C**	104	**C**	130	**D**

Optional bonus questions 131–142

131	**A**	132	**B**	133	**C**	134	**C**	135	**A**	136	**B**
137	**A**	138	**B**	139	**C**	140	**A**	141	**C**	142	**B**

BITSAT Mock 37

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**B**	27	**B**	53	**C**	79	**A**	105	**B**
2	**C**	28	**A**	54	**C**	80	**C**	106	**B**
3	**C**	29	**D**	55	**A**	81	**A**	107	**C**
4	**C**	30	**D**	56	**C**	82	**C**	108	**A**
5	**D**	31	**D**	57	**C**	83	**C**	109	**A**
6	**A**	32	**C**	58	**B**	84	**A**	110	**A**
7	**B**	33	**A**	59	**D**	85	**D**	111	**D**
8	**B**	34	**D**	60	**B**	86	**B**	112	**D**
9	**B**	35	**B**	61	**C**	87	**A**	113	**C**
10	**B**	36	**A**	62	**B**	88	**D**	114	**A**
11	**C**	37	**D**	63	**A**	89	**B**	115	**D**
12	**A**	38	**A**	64	**D**	90	**A**	116	**C**
13	**C**	39	**D**	65	**C**	91	**B**	117	**D**
14	**C**	40	**B**	66	**A**	92	**D**	118	**A**
15	**B**	41	**C**	67	**B**	93	**C**	119	**B**
16	**A**	42	**B**	68	**D**	94	**B**	120	**C**
17	**C**	43	**D**	69	**A**	95	**A**	121	**A**
18	**D**	44	**A**	70	**B**	96	**A**	122	**B**
19	**D**	45	**A**	71	**D**	97	**B**	123	**D**
20	**A**	46	**D**	72	**B**	98	**C**	124	**C**
21	**A**	47	**A**	73	**B**	99	**C**	125	**A**
22	**A**	48	**A**	74	**B**	100	**A**	126	**D**
23	**B**	49	**B**	75	**D**	101	**C**	127	**B**
24	**D**	50	**B**	76	**D**	102	**D**	128	**C**
25	**D**	51	**C**	77	**C**	103	**D**	129	**B**
26	**A**	52	**B**	78	**C**	104	**B**	130	**D**

Optional bonus questions 131–142

131	**A**	132	**C**	133	**B**	134	**C**	135	**B**	136	**A**
137	**A**	138	**C**	139	**B**	140	**A**	141	**C**	142	**B**

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Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**B**	27	**C**	53	**C**	79	**B**	105	**C**
2	**D**	28	**B**	54	**A**	80	**C**	106	**B**
3	**D**	29	**A**	55	**A**	81	**C**	107	**D**
4	**A**	30	**D**	56	**A**	82	**B**	108	**B**
5	**A**	31	**A**	57	**A**	83	**D**	109	**A**
6	**B**	32	**D**	58	**D**	84	**C**	110	**C**
7	**C**	33	**A**	59	**D**	85	**B**	111	**A**
8	**A**	34	**B**	60	**C**	86	**C**	112	**C**
9	**B**	35	**C**	61	**D**	87	**A**	113	**A**
10	**A**	36	**D**	62	**B**	88	**A**	114	**C**
11	**A**	37	**A**	63	**B**	89	**A**	115	**B**
12	**B**	38	**B**	64	**D**	90	**B**	116	**D**
13	**C**	39	**B**	65	**A**	91	**C**	117	**A**
14	**B**	40	**C**	66	**A**	92	**C**	118	**B**
15	**C**	41	**C**	67	**C**	93	**B**	119	**D**
16	**D**	42	**B**	68	**A**	94	**D**	120	**B**
17	**D**	43	**D**	69	**B**	95	**C**	121	**B**
18	**C**	44	**B**	70	**C**	96	**A**	122	**D**
19	**D**	45	**D**	71	**A**	97	**D**	123	**D**
20	**B**	46	**D**	72	**D**	98	**B**	124	**C**
21	**D**	47	**B**	73	**C**	99	**D**	125	**A**
22	**C**	48	**B**	74	**D**	100	**A**	126	**D**
23	**B**	49	**A**	75	**B**	101	**D**	127	**C**
24	**A**	50	**C**	76	**D**	102	**A**	128	**A**
25	**A**	51	**B**	77	**D**	103	**C**	129	**A**
26	**C**	52	**C**	78	**A**	104	**B**	130	**B**

Optional bonus questions 131–142

131	**A**	132	**B**	133	**C**	134	**B**	135	**A**	136	**C**
137	**C**	138	**A**	139	**B**	140	**A**	141	**B**	142	**C**

BITSAT Mock 39

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**C**	27	**B**	53	**B**	79	**D**	105	**C**
2	**C**	28	**D**	54	**C**	80	**D**	106	**C**
3	**C**	29	**B**	55	**C**	81	**C**	107	**B**
4	**D**	30	**C**	56	**B**	82	**C**	108	**D**
5	**D**	31	**D**	57	**A**	83	**C**	109	**D**
6	**B**	32	**D**	58	**C**	84	**B**	110	**D**
7	**A**	33	**C**	59	**A**	85	**C**	111	**A**
8	**C**	34	**B**	60	**A**	86	**A**	112	**A**
9	**A**	35	**B**	61	**D**	87	**A**	113	**A**
10	**B**	36	**D**	62	**D**	88	**B**	114	**D**
11	**B**	37	**A**	63	**B**	89	**B**	115	**C**
12	**A**	38	**C**	64	**C**	90	**A**	116	**D**
13	**C**	39	**D**	65	**B**	91	**A**	117	**A**
14	**B**	40	**A**	66	**C**	92	**A**	118	**B**
15	**B**	41	**C**	67	**B**	93	**C**	119	**C**
16	**A**	42	**B**	68	**A**	94	**B**	120	**B**
17	**D**	43	**D**	69	**A**	95	**D**	121	**D**
18	**A**	44	**A**	70	**A**	96	**C**	122	**D**
19	**A**	45	**C**	71	**D**	97	**B**	123	**D**
20	**C**	46	**D**	72	**C**	98	**B**	124	**A**
21	**A**	47	**B**	73	**D**	99	**B**	125	**B**
22	**D**	48	**D**	74	**A**	100	**A**	126	**C**
23	**A**	49	**A**	75	**B**	101	**B**	127	**C**
24	**D**	50	**B**	76	**B**	102	**C**	128	**C**
25	**D**	51	**A**	77	**A**	103	**B**	129	**D**
26	**B**	52	**B**	78	**D**	104	**A**	130	**A**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**C**	135	**B**	136	**A**
137	**B**	138	**A**	139	**C**	140	**C**	141	**B**	142	**A**

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Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**B**	53	**B**	79	**C**	105	**A**
2	**A**	28	**C**	54	**D**	80	**B**	106	**D**
3	**B**	29	**D**	55	**B**	81	**A**	107	**C**
4	**D**	30	**A**	56	**A**	82	**A**	108	**D**
5	**D**	31	**C**	57	**D**	83	**D**	109	**B**
6	**D**	32	**B**	58	**C**	84	**D**	110	**B**
7	**A**	33	**D**	59	**C**	85	**C**	111	**D**
8	**D**	34	**A**	60	**A**	86	**D**	112	**C**
9	**A**	35	**D**	61	**A**	87	**C**	113	**C**
10	**B**	36	**A**	62	**C**	88	**C**	114	**B**
11	**B**	37	**A**	63	**A**	89	**A**	115	**A**
12	**B**	38	**B**	64	**B**	90	**A**	116	**C**
13	**C**	39	**D**	65	**B**	91	**A**	117	**B**
14	**C**	40	**C**	66	**D**	92	**A**	118	**B**
15	**B**	41	**A**	67	**A**	93	**D**	119	**C**
16	**A**	42	**C**	68	**C**	94	**C**	120	**D**
17	**C**	43	**C**	69	**D**	95	**B**	121	**D**
18	**D**	44	**A**	70	**B**	96	**C**	122	**D**
19	**A**	45	**B**	71	**B**	97	**A**	123	**B**
20	**B**	46	**C**	72	**B**	98	**D**	124	**A**
21	**B**	47	**B**	73	**D**	99	**A**	125	**A**
22	**A**	48	**D**	74	**B**	100	**D**	126	**C**
23	**A**	49	**D**	75	**C**	101	**A**	127	**B**
24	**C**	50	**B**	76	**A**	102	**B**	128	**C**
25	**C**	51	**B**	77	**B**	103	**B**	129	**D**
26	**C**	52	**A**	78	**D**	104	**C**	130	**A**

Optional bonus questions 131–142

131	**A**	132	**C**	133	**B**	134	**B**	135	**C**	136	**A**
137	**B**	138	**A**	139	**C**	140	**B**	141	**C**	142	**A**

BITSAT Mock 41

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**C**	27	**D**	53	**A**	79	**A**	105	**C**
2	**B**	28	**A**	54	**A**	80	**A**	106	**A**
3	**B**	29	**D**	55	**A**	81	**C**	107	**A**
4	**C**	30	**A**	56	**A**	82	**D**	108	**A**
5	**D**	31	**C**	57	**C**	83	**B**	109	**D**
6	**C**	32	**C**	58	**D**	84	**D**	110	**B**
7	**B**	33	**D**	59	**A**	85	**B**	111	**B**
8	**B**	34	**B**	60	**A**	86	**D**	112	**C**
9	**A**	35	**B**	61	**A**	87	**C**	113	**D**
10	**D**	36	**B**	62	**D**	88	**D**	114	**C**
11	**D**	37	**B**	63	**B**	89	**C**	115	**B**
12	**A**	38	**D**	64	**A**	90	**A**	116	**D**
13	**A**	39	**D**	65	**C**	91	**D**	117	**B**
14	**B**	40	**A**	66	**B**	92	**B**	118	**C**
15	**C**	41	**B**	67	**A**	93	**B**	119	**D**
16	**B**	42	**D**	68	**C**	94	**C**	120	**B**
17	**C**	43	**D**	69	**D**	95	**A**	121	**C**
18	**B**	44	**C**	70	**B**	96	**C**	122	**D**
19	**B**	45	**B**	71	**B**	97	**D**	123	**D**
20	**A**	46	**D**	72	**C**	98	**A**	124	**A**
21	**A**	47	**C**	73	**B**	99	**A**	125	**C**
22	**A**	48	**B**	74	**D**	100	**A**	126	**A**
23	**D**	49	**A**	75	**C**	101	**B**	127	**A**
24	**C**	50	**B**	76	**A**	102	**B**	128	**D**
25	**D**	51	**C**	77	**B**	103	**C**	129	**D**
26	**C**	52	**C**	78	**A**	104	**B**	130	**C**

Optional bonus questions 131–142

131	**A**	132	**B**	133	**C**	134	**A**	135	**B**	136	**C**
137	**A**	138	**C**	139	**B**	140	**B**	141	**C**	142	**A**

BITSAT Mock 42

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**A**	27	**A**	53	**D**	79	**A**	105	**D**
2	**B**	28	**C**	54	**B**	80	**A**	106	**C**
3	**A**	29	**C**	55	**B**	81	**B**	107	**C**
4	**C**	30	**B**	56	**D**	82	**C**	108	**D**
5	**D**	31	**B**	57	**B**	83	**B**	109	**C**
6	**A**	32	**A**	58	**D**	84	**D**	110	**D**
7	**D**	33	**A**	59	**A**	85	**B**	111	**B**
8	**C**	34	**D**	60	**B**	86	**A**	112	**A**
9	**B**	35	**A**	61	**D**	87	**B**	113	**B**
10	**D**	36	**B**	62	**A**	88	**D**	114	**A**
11	**B**	37	**C**	63	**B**	89	**C**	115	**C**
12	**A**	38	**D**	64	**B**	90	**B**	116	**D**
13	**B**	39	**C**	65	**C**	91	**A**	117	**B**
14	**A**	40	**A**	66	**C**	92	**D**	118	**C**
15	**B**	41	**A**	67	**B**	93	**A**	119	**C**
16	**D**	42	**C**	68	**A**	94	**B**	120	**A**
17	**C**	43	**B**	69	**D**	95	**C**	121	**A**
18	**C**	44	**B**	70	**A**	96	**D**	122	**A**
19	**C**	45	**C**	71	**D**	97	**B**	123	**A**
20	**D**	46	**A**	72	**A**	98	**D**	124	**C**
21	**B**	47	**D**	73	**C**	99	**D**	125	**B**
22	**D**	48	**C**	74	**A**	100	**C**	126	**C**
23	**D**	49	**C**	75	**D**	101	**D**	127	**B**
24	**B**	50	**A**	76	**C**	102	**B**	128	**D**
25	**A**	51	**D**	77	**D**	103	**B**	129	**A**
26	**A**	52	**C**	78	**C**	104	**A**	130	**B**

Optional bonus questions 131–142

131	**C**	132	**A**	133	**B**	134	**B**	135	**A**	136	**C**
137	**B**	138	**A**	139	**C**	140	**C**	141	**B**	142	**A**

BITSAT Mock 43

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**C**	53	**A**	79	**D**	105	**D**
2	**C**	28	**B**	54	**B**	80	**A**	106	**D**
3	**B**	29	**B**	55	**A**	81	**D**	107	**D**
4	**A**	30	**C**	56	**A**	82	**D**	108	**B**
5	**A**	31	**D**	57	**A**	83	**B**	109	**B**
6	**A**	32	**A**	58	**C**	84	**A**	110	**B**
7	**A**	33	**D**	59	**B**	85	**C**	111	**D**
8	**B**	34	**D**	60	**A**	86	**B**	112	**A**
9	**B**	35	**A**	61	**B**	87	**A**	113	**A**
10	**D**	36	**B**	62	**D**	88	**B**	114	**C**
11	**A**	37	**D**	63	**C**	89	**D**	115	**C**
12	**C**	38	**D**	64	**A**	90	**A**	116	**A**
13	**D**	39	**C**	65	**C**	91	**B**	117	**A**
14	**D**	40	**C**	66	**D**	92	**C**	118	**A**
15	**C**	41	**C**	67	**A**	93	**A**	119	**C**
16	**B**	42	**D**	68	**B**	94	**D**	120	**D**
17	**A**	43	**A**	69	**B**	95	**B**	121	**C**
18	**C**	44	**D**	70	**A**	96	**A**	122	**D**
19	**B**	45	**B**	71	**C**	97	**B**	123	**C**
20	**B**	46	**C**	72	**B**	98	**D**	124	**B**
21	**A**	47	**C**	73	**C**	99	**A**	125	**B**
22	**D**	48	**C**	74	**D**	100	**A**	126	**D**
23	**D**	49	**B**	75	**C**	101	**B**	127	**B**
24	**C**	50	**B**	76	**A**	102	**C**	128	**C**
25	**D**	51	**B**	77	**C**	103	**C**	129	**D**
26	**A**	52	**B**	78	**B**	104	**A**	130	**C**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**A**	135	**B**	136	**C**
137	**B**	138	**C**	139	**A**	140	**B**	141	**A**	142	**C**

BITSAT Mock 44

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**C**	27	**C**	53	**B**	79	**A**	105	**A**
2	**D**	28	**D**	54	**A**	80	**D**	106	**C**
3	**A**	29	**B**	55	**B**	81	**B**	107	**C**
4	**A**	30	**B**	56	**C**	82	**D**	108	**B**
5	**A**	31	**C**	57	**D**	83	**B**	109	**A**
6	**A**	32	**D**	58	**D**	84	**A**	110	**C**
7	**D**	33	**D**	59	**A**	85	**C**	111	**B**
8	**B**	34	**D**	60	**B**	86	**B**	112	**B**
9	**C**	35	**B**	61	**A**	87	**C**	113	**D**
10	**D**	36	**C**	62	**A**	88	**C**	114	**B**
11	**B**	37	**D**	63	**C**	89	**A**	115	**B**
12	**A**	38	**A**	64	**D**	90	**D**	116	**C**
13	**A**	39	**C**	65	**A**	91	**C**	117	**B**
14	**C**	40	**B**	66	**B**	92	**D**	118	**B**
15	**A**	41	**B**	67	**B**	93	**B**	119	**A**
16	**B**	42	**B**	68	**C**	94	**A**	120	**D**
17	**B**	43	**A**	69	**D**	95	**D**	121	**B**
18	**D**	44	**A**	70	**B**	96	**A**	122	**C**
19	**C**	45	**B**	71	**A**	97	**D**	123	**D**
20	**A**	46	**C**	72	**C**	98	**C**	124	**C**
21	**D**	47	**C**	73	**B**	99	**B**	125	**A**
22	**B**	48	**A**	74	**C**	100	**D**	126	**C**
23	**B**	49	**C**	75	**D**	101	**A**	127	**D**
24	**D**	50	**A**	76	**D**	102	**C**	128	**D**
25	**C**	51	**A**	77	**A**	103	**D**	129	**A**
26	**C**	52	**D**	78	**B**	104	**A**	130	**A**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**B**	135	**A**	136	**C**
137	**A**	138	**B**	139	**C**	140	**C**	141	**A**	142	**B**

BITSAT Mock 45

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**A**	27	**D**	53	**A**	79	**D**	105	**A**
2	**C**	28	**B**	54	**D**	80	**B**	106	**D**
3	**D**	29	**B**	55	**C**	81	**B**	107	**A**
4	**B**	30	**D**	56	**D**	82	**B**	108	**A**
5	**A**	31	**A**	57	**A**	83	**C**	109	**D**
6	**D**	32	**A**	58	**A**	84	**C**	110	**C**
7	**C**	33	**B**	59	**A**	85	**A**	111	**D**
8	**C**	34	**D**	60	**D**	86	**A**	112	**A**
9	**A**	35	**A**	61	**A**	87	**D**	113	**A**
10	**C**	36	**D**	62	**B**	88	**D**	114	**C**
11	**C**	37	**C**	63	**D**	89	**A**	115	**C**
12	**D**	38	**B**	64	**C**	90	**B**	116	**C**
13	**B**	39	**D**	65	**B**	91	**B**	117	**D**
14	**A**	40	**C**	66	**A**	92	**C**	118	**D**
15	**C**	41	**A**	67	**C**	93	**B**	119	**B**
16	**D**	42	**B**	68	**D**	94	**D**	120	**D**
17	**B**	43	**B**	69	**B**	95	**A**	121	**B**
18	**A**	44	**C**	70	**A**	96	**B**	122	**D**
19	**B**	45	**B**	71	**C**	97	**B**	123	**A**
20	**C**	46	**B**	72	**D**	98	**A**	124	**C**
21	**B**	47	**D**	73	**C**	99	**C**	125	**B**
22	**A**	48	**C**	74	**A**	100	**C**	126	**C**
23	**A**	49	**C**	75	**C**	101	**A**	127	**B**
24	**D**	50	**C**	76	**A**	102	**B**	128	**A**
25	**B**	51	**B**	77	**B**	103	**D**	129	**B**
26	**A**	52	**B**	78	**D**	104	**C**	130	**D**

Optional bonus questions 131–142

131	**A**	132	**C**	133	**B**	134	**C**	135	**B**	136	**A**
137	**A**	138	**B**	139	**C**	140	**B**	141	**C**	142	**A**

BITSAT Mock 46

Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**A**	27	**B**	53	**C**	79	**D**	105	**A**
2	**A**	28	**C**	54	**B**	80	**D**	106	**C**
3	**C**	29	**B**	55	**C**	81	**B**	107	**C**
4	**D**	30	**C**	56	**C**	82	**C**	108	**C**
5	**A**	31	**D**	57	**B**	83	**C**	109	**A**
6	**D**	32	**A**	58	**B**	84	**D**	110	**B**
7	**B**	33	**A**	59	**D**	85	**A**	111	**A**
8	**D**	34	**D**	60	**B**	86	**A**	112	**B**
9	**B**	35	**A**	61	**B**	87	**B**	113	**B**
10	**C**	36	**C**	62	**A**	88	**C**	114	**D**
11	**A**	37	**D**	63	**C**	89	**A**	115	**A**
12	**B**	38	**A**	64	**D**	90	**A**	116	**B**
13	**A**	39	**B**	65	**A**	91	**D**	117	**D**
14	**B**	40	**D**	66	**B**	92	**B**	118	**D**
15	**A**	41	**D**	67	**C**	93	**B**	119	**C**
16	**A**	42	**D**	68	**B**	94	**C**	120	**B**
17	**C**	43	**B**	69	**A**	95	**D**	121	**A**
18	**D**	44	**C**	70	**D**	96	**B**	122	**C**
19	**C**	45	**A**	71	**B**	97	**B**	123	**A**
20	**D**	46	**A**	72	**C**	98	**D**	124	**C**
21	**C**	47	**A**	73	**B**	99	**D**	125	**A**
22	**B**	48	**A**	74	**C**	100	**C**	126	**C**
23	**A**	49	**C**	75	**D**	101	**B**	127	**A**
24	**D**	50	**C**	76	**A**	102	**A**	128	**C**
25	**D**	51	**B**	77	**B**	103	**D**	129	**A**
26	**B**	52	**B**	78	**D**	104	**D**	130	**D**

Optional bonus questions 131–142

131	**B**	132	**C**	133	**A**	134	**B**	135	**C**	136	**A**
137	**A**	138	**C**	139	**B**	140	**C**	141	**B**	142	**A**

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Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**C**	53	**C**	79	**B**	105	**C**
2	**A**	28	**B**	54	**B**	80	**C**	106	**A**
3	**B**	29	**C**	55	**C**	81	**B**	107	**A**
4	**A**	30	**B**	56	**B**	82	**C**	108	**B**
5	**C**	31	**B**	57	**A**	83	**A**	109	**D**
6	**A**	32	**D**	58	**A**	84	**A**	110	**C**
7	**B**	33	**A**	59	**C**	85	**C**	111	**A**
8	**C**	34	**D**	60	**A**	86	**A**	112	**A**
9	**B**	35	**D**	61	**B**	87	**D**	113	**C**
10	**D**	36	**C**	62	**A**	88	**A**	114	**B**
11	**A**	37	**C**	63	**D**	89	**C**	115	**B**
12	**C**	38	**B**	64	**C**	90	**D**	116	**D**
13	**A**	39	**C**	65	**B**	91	**A**	117	**B**
14	**C**	40	**D**	66	**D**	92	**D**	118	**A**
15	**D**	41	**D**	67	**C**	93	**B**	119	**C**
16	**D**	42	**A**	68	**A**	94	**C**	120	**D**
17	**D**	43	**B**	69	**A**	95	**B**	121	**D**
18	**C**	44	**A**	70	**B**	96	**C**	122	**C**
19	**B**	45	**B**	71	**B**	97	**C**	123	**B**
20	**A**	46	**B**	72	**B**	98	**D**	124	**B**
21	**A**	47	**A**	73	**B**	99	**D**	125	**B**
22	**A**	48	**D**	74	**C**	100	**A**	126	**D**
23	**B**	49	**C**	75	**D**	101	**C**	127	**A**
24	**D**	50	**A**	76	**D**	102	**A**	128	**C**
25	**D**	51	**D**	77	**D**	103	**B**	129	**D**
26	**B**	52	**B**	78	**A**	104	**D**	130	**A**

Optional bonus questions 131–142

131	**A**	132	**C**	133	**B**	134	**B**	135	**C**	136	**A**
137	**A**	138	**B**	139	**C**	140	**C**	141	**A**	142	**B**

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Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**D**	27	**B**	53	**B**	79	**C**	105	**B**
2	**B**	28	**D**	54	**B**	80	**A**	106	**A**
3	**C**	29	**A**	55	**C**	81	**B**	107	**A**
4	**A**	30	**C**	56	**D**	82	**A**	108	**B**
5	**A**	31	**D**	57	**C**	83	**A**	109	**D**
6	**B**	32	**B**	58	**D**	84	**C**	110	**B**
7	**C**	33	**D**	59	**A**	85	**D**	111	**B**
8	**A**	34	**B**	60	**B**	86	**B**	112	**D**
9	**D**	35	**A**	61	**C**	87	**D**	113	**C**
10	**C**	36	**D**	62	**B**	88	**A**	114	**B**
11	**A**	37	**A**	63	**B**	89	**B**	115	**C**
12	**D**	38	**A**	64	**D**	90	**D**	116	**D**
13	**B**	39	**D**	65	**D**	91	**C**	117	**C**
14	**D**	40	**D**	66	**C**	92	**A**	118	**D**
15	**A**	41	**A**	67	**A**	93	**B**	119	**B**
16	**B**	42	**C**	68	**B**	94	**D**	120	**A**
17	**B**	43	**B**	69	**A**	95	**D**	121	**A**
18	**B**	44	**A**	70	**A**	96	**B**	122	**A**
19	**A**	45	**A**	71	**D**	97	**A**	123	**C**
20	**C**	46	**C**	72	**B**	98	**C**	124	**D**
21	**A**	47	**C**	73	**B**	99	**C**	125	**A**
22	**D**	48	**B**	74	**D**	100	**A**	126	**D**
23	**C**	49	**C**	75	**A**	101	**D**	127	**B**
24	**B**	50	**C**	76	**C**	102	**C**	128	**B**
25	**D**	51	**B**	77	**C**	103	**D**	129	**C**
26	**C**	52	**A**	78	**C**	104	**A**	130	**C**

Optional bonus questions 131–142

131	**B**	132	**A**	133	**C**	134	**C**	135	**B**	136	**A**
137	**A**	138	**C**	139	**B**	140	**B**	141	**C**	142	**A**

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Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**A**	27	**B**	53	**A**	79	**C**	105	**A**
2	**B**	28	**B**	54	**B**	80	**B**	106	**A**
3	**C**	29	**A**	55	**C**	81	**A**	107	**B**
4	**A**	30	**C**	56	**B**	82	**D**	108	**A**
5	**D**	31	**A**	57	**D**	83	**A**	109	**D**
6	**C**	32	**B**	58	**A**	84	**B**	110	**B**
7	**D**	33	**C**	59	**B**	85	**D**	111	**C**
8	**B**	34	**B**	60	**A**	86	**D**	112	**C**
9	**D**	35	**D**	61	**C**	87	**A**	113	**C**
10	**B**	36	**D**	62	**B**	88	**C**	114	**D**
11	**D**	37	**B**	63	**A**	89	**C**	115	**C**
12	**D**	38	**A**	64	**D**	90	**B**	116	**C**
13	**C**	39	**D**	65	**D**	91	**A**	117	**D**
14	**A**	40	**C**	66	**A**	92	**D**	118	**B**
15	**A**	41	**B**	67	**C**	93	**C**	119	**D**
16	**C**	42	**C**	68	**A**	94	**D**	120	**B**
17	**B**	43	**C**	69	**B**	95	**B**	121	**C**
18	**C**	44	**A**	70	**B**	96	**A**	122	**D**
19	**A**	45	**C**	71	**B**	97	**D**	123	**A**
20	**D**	46	**C**	72	**D**	98	**B**	124	**C**
21	**A**	47	**D**	73	**D**	99	**B**	125	**A**
22	**C**	48	**B**	74	**B**	100	**A**	126	**D**
23	**A**	49	**A**	75	**A**	101	**C**	127	**B**
24	**B**	50	**A**	76	**C**	102	**C**	128	**B**
25	**B**	51	**D**	77	**C**	103	**A**	129	**B**
26	**D**	52	**D**	78	**A**	104	**D**	130	**A**

Optional bonus questions 131–142

131	**B**	132	**C**	133	**A**	134	**A**	135	**B**	136	**C**
137	**A**	138	**C**	139	**B**	140	**B**	141	**C**	142	**A**

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Answer key

Main questions 1–130

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	**C**	27	**B**	53	**C**	79	**A**	105	**B**
2	**D**	28	**D**	54	**B**	80	**D**	106	**C**
3	**B**	29	**B**	55	**C**	81	**B**	107	**C**
4	**B**	30	**C**	56	**B**	82	**B**	108	**C**
5	**D**	31	**A**	57	**C**	83	**C**	109	**D**
6	**B**	32	**C**	58	**A**	84	**A**	110	**C**
7	**A**	33	**A**	59	**D**	85	**C**	111	**A**
8	**A**	34	**A**	60	**C**	86	**B**	112	**A**
9	**D**	35	**C**	61	**D**	87	**D**	113	**A**
10	**A**	36	**D**	62	**B**	88	**A**	114	**C**
11	**A**	37	**B**	63	**B**	89	**A**	115	**B**
12	**C**	38	**B**	64	**A**	90	**D**	116	**D**
13	**C**	39	**A**	65	**A**	91	**B**	117	**C**
14	**D**	40	**D**	66	**C**	92	**B**	118	**C**
15	**A**	41	**C**	67	**D**	93	**D**	119	**B**
16	**B**	42	**A**	68	**A**	94	**B**	120	**D**
17	**D**	43	**B**	69	**B**	95	**D**	121	**D**
18	**C**	44	**D**	70	**C**	96	**A**	122	**C**
19	**C**	45	**B**	71	**D**	97	**B**	123	**B**
20	**D**	46	**A**	72	**A**	98	**B**	124	**D**
21	**A**	47	**D**	73	**B**	99	**A**	125	**A**
22	**A**	48	**D**	74	**D**	100	**C**	126	**A**
23	**C**	49	**D**	75	**C**	101	**A**	127	**A**
24	**A**	50	**A**	76	**C**	102	**A**	128	**C**
25	**B**	51	**B**	77	**B**	103	**D**	129	**B**
26	**B**	52	**B**	78	**C**	104	**D**	130	**D**

Optional bonus questions 131–142

131	**C**	132	**B**	133	**A**	134	**B**	135	**A**	136	**C**
137	**A**	138	**C**	139	**B**	140	**A**	141	**C**	142	**B**